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Complete programme reconstruction and Deligne weight control

The current request is to read Deligne’s weight-control argument completely, reconstruct its mechanism, and prove the available comparisons with the user’s construction. This reader contains the full new derivations. The cumulative companion, RECONSTRUCTION_AND_WEIGHT_FULL.md, includes their earlier programme proof sources and all preceding retained mathematics.

The entire current French transcription of Pierre Deligne’s *La conjecture de Weil. II*, printed pages 137–252, has been read. It is a local transcription, not original author TeX. Its identity, exact coverage, and formula checks are recorded in DELIGNE_FULL_READING_LOG.md and DELIGNE_WEIGHT_SOURCE_READING.json. The unchanged original source package remains available locally. A companion task’s original published-page reading of printed pp 137–252, with separately recorded formula-image checks is separately attributed in that log; root has not independently read those scans. That comparison corrects the earlier attribution of two source displays to transcription alone. Deligne’s cited upstream SGA, class-field and representation-theoretic results are identified as inputs, rather than represented as newly proved from foundations. The paper’s later applications were read; this reader reconstructs the weight-control argument and the mixed-duality comparisons used here, not every subsequent application of the paper.

The user’s complete argument and correction chain remain separately preserved in USER_CONSTRUCTION_FULL_LOGBOOK.md, with its source receipt. The current mathematical input respects the order in that argument: complete timing and its global quotient precede the arithmetic interpretation. Later arithmetic calculations state when they use the already reconstructed complete cyclic source. They do not supply cyclicity by assuming observed prime coordinates. The supporting datum is always $\tau\langle Z_1; \text{no } Z_2 \rangle$. Its addition remains retracted. None of the constructions installs an arithmetic zero, vector, midpoint, or distance on that supporting datum.

Proof	Source argument and actual comparison
DP0-DP9	Determinant characters, central monodromy, all even tensor powers, full cohomological trace formula, exterior powers and pointwise purity; Weil II §§1.3–1.5.
DMG0-DMG7	Nonsplit geometric monodromy radical, dominant pullback, compact normalizer, and the determinant criterion for an étale extension; completes the needed group arguments in §1.3.
DLM1-DLM15	Nilpotent logarithm, primitive decomposition, exact Tate factors, tensor and dual identities, local purity and the mixed relative filtration; §§1.6–1.9.

Proof	Source argument and actual comparison
DB0-DB10 and DBC1-DBC11	Full positive prime-power measures, compact-group characters, boundary nonvanishing, the exceptional quadratic character and strict cohomological weight inequality; §§2.1–2.2.
DW0-DW11	Every singular-fibre contribution, the product and pencil, complete Leray and Künneth terms, repeated tensor improvement, duality and the fundamental direct-image theorem; §§3.1–3.3.
MDB0-MDB11	Actual maps from the CC stalk through its recovered arithmetic to étale fibres; mixed filtration, strictness, six-functor duality, arithmetic costalks and full Tate/cohomological shifts; §§3.4 and 6.1–6.2.
DC0-DC12	The full local invariant-cycle quotient and lift: Hochschild-Serre, localization, both Tate twists, the complete specialization kernel, and separation of the target’s weights from the obstruction’s weights; §3.6.
DR0-DR7	Invertible comparison of the full integer count and prime-power measures with Deligne’s §2.1.9 measure, the original $-\zeta'/\zeta$, retained unit and completion factors, and exact branchwise compatibility.

The cumulative reader additionally includes ZLB0–ZLB11, ZTM0–ZTM11, GLB0–GLB8 and GSL0–GSL11. These give complete proofs of the original-germ boundary, the character-shifted finite and algebraic lifts, the whole Fréchet source boundary, and the continuous equivariant splitting of the original and translated zero divisors. GSL constructs an entire separator with controlled strip growth, its full quotient inverse, and a Mellin-source representative. Its uniqueness proof uses the actual global resolvent. It retains all multiplicities and never replaces the source topology by an unrestricted jet product. ZLB and GLB identify the nonzero boundary of a particular analytic eigenfunction extension; no map identifies that extension with the user’s requested geometric lift. GSL’s vanishing extension class concerns the explicitly constructed translated-divisor sequence. Its identification with Deligne’s localization cross has not been constructed.

The later geometric comparisons construct the evaluated CC charts (CGT), the exact degree-class comparison with the continuous lift (GTZ), and the descended conic with its full integral fibre at₂ (CD). The conic’s Tate factor comes from its actual hyperplane class; the original product is $\zeta(s)\zeta(s-1)$, with all local factors. CS and CSL construct the original CC coefficient sheaf, its finite-unit source, four endpoint lines and all supported-cohomology maps. Their ordinary cohomology is calculated before any closure operation. The complete SSI theorem, independently rederived in ESI, then proves the summation image is already closed by constructing a continuous inverse in the actual Schwartz space. CS11 and CSL10 carry this stronger theorem through the receiving maps.

The prime-indexed endpoint directions in CS2A remain explicit. They arise when rational-scaling relations are generated only by moment-null sources; allowing full Schwartz generators changes that source quotient by exactly the displayed direct sum of endpoint pairs. This distinction is independent of the closedness of the overlap summation image. The source-reading record states the author’s omitted generator-domain convention and the proved comparison, rather than concealing it in notation.

The finite-field source obtains equality of Frobenius moduli through its cohomological argument. In the reconstructed arithmetic comparison, MDB5 proves the full costalk $Ri_p^! E = E(-1)[-2]$ and MDB9 computes its eigenvalue action $\alpha \mapsto p/\alpha$, in degree two. The conjugate-realization comparison is

$a \mapsto p/\bar{a}$, with the Tate factor retained. DR proves the full original-zeta receiving map; MDB11 gives the local Euler factors of the actual constant coefficient object on the recovered arithmetic scheme. These proofs have not identified the analytic zero-jet receiver with the proper arithmetic cohomology used by Deligne’s purity theorem, and do not claim an RH resolution. The failure of one observation functor to retain weights is calculated in MDB10; it is not a rejection of every enriched construction on the support.

CSB calculates every retained prime endpoint action and gives the full comparison with the earlier FR15–FR25 proof: the source LF quotient retains those classes, whereas the specified weighted periodization seminorm kills exactly them. CSP then constructs the inverse-image sheaf on the evaluated complex projective line. Its degree-one cohomology is the original quotient Q , its degree-two cohomology is the full overlap A with the geometric covering factor, and an actual sheaf quotient produces Q in degree two. The explicit translated-Mellin map identifies this latter normal quotient with the earlier GSL kernel. The support, full stalk, two cohomological degrees and their quotient maps remain explicit; the actual antiholomorphic mirror has the opposite sphere-orientation sign from the holomorphic swap.

CW proves the actual quotient-sheaf degree-one lift: the cohomology map is the identity of Q , and its connecting map $Q \rightarrow J(-1)$ is zero. The simultaneous source complexes give an explicit lift of every class. Independently, the full original-zeta resolvent proves that every continuous complex-linear arithmetic-equivariant map $Q \rightarrow J(-1)$ is zero, without a density assumption about finite zero blocks. The full kernel $J(-1)$ retains character quotients at every complex exponent, so the narrower weight interval of $Q(-1)$ is not assigned to that kernel. CW computes the exact quotient map into GSL, all its kernel and image, and the difference between this cohomological lift and choosing representatives equivariantly in A . Its comparison with Deligne retains the actual obstruction quotient and both Tate factors.

CDW constructs the derived direct image itself by one resolution compatible on both pole charts and their punctured overlap. Its strict continuous splitting retains the winding class, both mirror actions, the geometric degree factor and the entire source extension in both internal degrees. The comparison between the angular class and positive sphere integration has sign -1 , which remains in the translated-zeta map. CDI then calculates the complete continuous dual coefficient and support complexes, including the dual cone and all endpoint factors. RD constructs the original-zeta residue pairing with every multiplicity and its exact nonzero block determinant; its finite-jet functionals are dense for the weak dual topology. The residue action and normal contragredient differ by the explicitly retained factor a^2 . These are constructed comparison maps; a finite-dimensional étale support-duality realization is not asserted.

SDT subsequently proves the stronger topology from the actual weighted coefficient space: bounded quotient sets have compact closure, and every compact set has a compact lift constructed by a convergent sequence of finite choices. This proves that the induced dual topology in CDI3 is the full strong dual topology. A proof of strong reflexivity then upgrades the finite-jet functionals to strong density. The explicitly stated residue topology on finite-support classes has completion equal to the entire continuous dual; it is not substituted for the inherited topology of those classes. The reflected and unreflected residue maps have their exact, different arithmetic covariance laws. Thus CDI and RD’s earlier restricted strong-topology claims remain accurate and are strengthened by the complete subsequent proof.

The figures are diagrams of the proved objects and maps, not geometric metrics on the supporting point. Their reproducible Python sources, PNG and SVG outputs accompany this reader. DELIGNE_COMPLETE_BUILD.json records every cumulative proof input and output hash. Earlier editions remain in the local history directory.

CSD now realizes the continuous dual by an actual sheaf of functionals on compactly supported tests. Its positive Dirac Gysin, pole stalk and costalk maps, global comparison and inverse, and all support signs are proved explicitly. SDT gives the full strong coefficient topologies; the stable local cohomology topology is specified without assuming that arbitrary germ limits preserve it. The proper current pushforward is the transpose of pullback on tests, including both mirror orientations.

GTR then constructs the weighted trace of each recovered integer cover on the entire coefficient sheaf, including both ramified stalks and all their endpoint coordinates. Its inverse coefficient action gives the factor n in the original reflected residue covariance. The complete residue topology is written out, and the geometric action extends to its entire strong-dual completion. The Mellin transpose and normal-dual

inclusion are explicit in this comparison. These results realize the earlier coefficient duality and residue actions by constructed geometric maps. The weight-control goal remains active: no nondegenerate bilinear evaluation is promoted to a positive Hermitian form.

VSD and CV construct the ordinary dual of the actual infinite-coefficient sheaf, and a literal morphism from its continuous-current dual. The entire difference is computed, including the pole Gysin maps. The original Mellin multiplication operator acts rationally on its extra algebraic coefficient sector: every nonzero polynomial in that operator is invertible. Its proof uses continuous division on the complete original source and its full vanishing-jet ideal. It follows that enlarging the dual adds no finite-dimensional invariant sectors for this operator. Every endpoint, scalar shift and mirror action has a stated formula; mirror action sends the rational variable t to $1 - t$. The associated figure displays the actual comparison cone and the distinct strong and weak residue completions.

GTR11 also upgrades the trace calculation to the actual derived direct image: its trace is n times the inverse of the proved pullback, with the intermediate direct-image identification written out. It preserves the entire source sequence, rather than only an independently computed list of cohomology groups.

The primal generator family GCF/PGF constructs the exact original-zeta divisor map for all parameters and jet orders, with its original meromorphic target and every cancellation frame. The family cokernel has length rm at a multiplicity- m zero, while its fibre and its derived fibre kernel each have dimension $\min(r, m)$. The full source boundary explains the difference. It is a different boundary from CW's geometric lift; the manuscript supplies both maps and preserves their actual domains. The figure `primal_generator_family.png` displays these maps without assigning coordinates or a metric to the parityless support.

CP computes the actual derived spectral obstruction of that current inclusion. The two-term free resolution retains both the polynomial kernel and cokernel. The computed comparison defect makes both vanish, and the proof corrects an algebraic spectral cocycle to a continuous one by an explicit boundary. The resulting continuous cohomology class is unique. For each actual zero the full multiplicity, every local/global degree, and the nonzero source cokernel are calculated; all dilation and mirror matrices remain. This is a separation-based vanishing theorem for the actual comparison map. CP11 gives its exact relation to Deligne's mechanism and retains the distinction from the still unfinished arithmetic weight-control application.

The actual polynomial localization calculation CLP derives every degree and boundary of the constructed sphere-sheaf row, keeping its full kernel. It proves the separated normal comparison at original zeros and computes the distinct nonzero spectral boundary at translated zeros. The full derived extension and the coherent-family Hom contraction identify exactly what vanishes and what remains. Its figure `localization_spectral_weights.png` retains the complete original-zeta multiplier and the inverse derivative matrix. The weight separation proved for these particular maps does not identify the whole row with Deligne's inertia cross.

The complete-module continuation GMS combines the full global interpolation multiplier with the actual localization complexes. GMS4 proves derived Hom vanishing with an explicit bar homotopy; GMS5 gives the complete normal comparison inverse, retaining the whole ideal. GMS6 computes precisely the nonzero extension and the multiplier pushout that splits. GMS8 supplies the full original-source integral, while GMS9-GMS10 retain endpoints, cover factors and mirror actions. The diagram `global_multiplier_separation.png` shows these maps and their proof locators.

The complete-source continuation FOD and NEA retains the entire intersection kernel and computes both actual extension components. FOD2-FOD5 construct the pullbacks, pushouts and exact annihilator ideals, distinguishing the original component preserved by the separator from the normal component it annihilates. NEA3-NEA7 identify the exact normal cyclic class module, all multiplicity probes and its full-source algebraic resolvent. FOD7 and NEA10 prove the actual localization lift from the complete original quotient. Their diagram is `full_source_obstruction.png`. Postcomposition is distinguished from conjugation, and the actual Deligne specialization/inertia comparison retains its separate unfinished scope.

The local-receiver continuation LNC, LVD and RGR follows FOD/NEA with the actual universal-cover specialization, exact local/global comparison, continuous-dual arrows and original residue pairing. LNC4 computes the invariant-cycle cross on the actual coefficient sheaf. LVD5-7 prove the difference between local sheaf pairings and the nonzero global pairing, then construct their exact morphism quotient. RGR3-10 construct the signed Gysin receiver and its relation to the original transposed coefficient extension. RGR11 retains a separate receiver for the already calculated positive-adjoint defect. The source and target of each map matter: the total residue class and that defect are different. The complete RTT, GTAH, GIQ and PCJ supporting files are retained alongside the main proofs. The figure `local_residue_receiver.png` identifies the exact comparison maps. Reich's original v4 author TeX and intact source archive are included, with read ranges in `NEARBY_CYCLES_SOURCE_READING_PUBLIC.json`. No whole-paper Reich reading or arithmetic weight vanishing is claimed.

This cumulative edition retains all seventeen proof parts of the preceding reader, adds the complete current Deligne reconstruction, and retains the later timing and branch-comparison derivations. Its timing and odd-refinement examples test precisely their stated source implications. They are not proposed RH counterexamples: the odd-refinement example fails the user's later requirement to recover the entire original arithmetic. The current CG proof retains the correction that $\text{End}(L)$ is the abelian-group endomorphism ring, not the endomorphism ring of the full signed sheaf. Earlier optional source discussions retain their declared reading status; the Gross-Schoen discussion T9 is not certified as an independently verified theorem here.

Figure 1. Every displayed identity is proved in DP, DLM, DB/DBC, DW, or DR below; source locators appear in the figure and proofs.

Figure 1a. Exact source lifting mechanism of DC, retaining the whole coinvariant term and obstruction group.

Figure 1b. DC, ZTM and GLB calculate their displayed extensions with their full domains and actions. The bottom boundary is not asserted to be the programme's geometric obstruction.

Figure 1c. GSL constructs the continuous separator and the full quotient inverse, with all multiplicities. The two strips are analytic zero divisors, not coordinates assigned to the supporting datum.

Figure 1d. CSL calculates the Cech and supported-cohomology maps, including the mirror sign; CD derives the conic descent, its actual fibre at2, and its geometric Tate factor. The two receivers have their exact comparison maps and are not identified merely by similar notation.

Figure 1e. SSI and ESI prove the continuous inverse and closed source image. CSB retains the separate prime endpoint kernel with its full Fourier and dilation actions, and proves its exact comparison with FR.

Figure 1f. CSP derives the actual sheaf quotient, the covering degree, the complete localization signs, and the map onto the shifted-original-zeta kernel. The two cohomological degrees remain explicit.

Figure 1g. The three coefficient complexes give the actual degree-one lift and its zero connecting map. Their degree-two quotient retains the full kernel $J(-1)$. The complete comparison proof follows below.

Figure 1h. CDW constructs the strict continuous comparison, retains the complete source sequence in both internal degrees, and derives the negative sign between winding and positive integration.

Figure 1i. CDI and RD retain every support sign, zero multiplicity and residue character factor. SDT proves the exact strong cohomology topologies and completes the finite-jet residue receiver.

Figure 1j. CSD constructs the dual sheaf from compactly supported tests, with the positive Dirac Gysin arrow, exact local and global comparisons, and both pole signs.

Figure 1k. GTR retains the full ramified stalk multiplicities, derives the degree factors, and identifies the completed original-zeta residue action with the actual transfer transpose.

Figure 1l. VSD constructs the ordinary Verdier dual and the precise continuous-dual comparison cone. Its rational-operator calculation proves that all polynomial kernels and cokernels already occur in the continuous dual, while the two stated completion topologies retain different full receivers.

Deligne's exact weight control and the arithmetic return map

Finite-field theorem above; the proved original-zeta receiving map below. Every label names its proof.

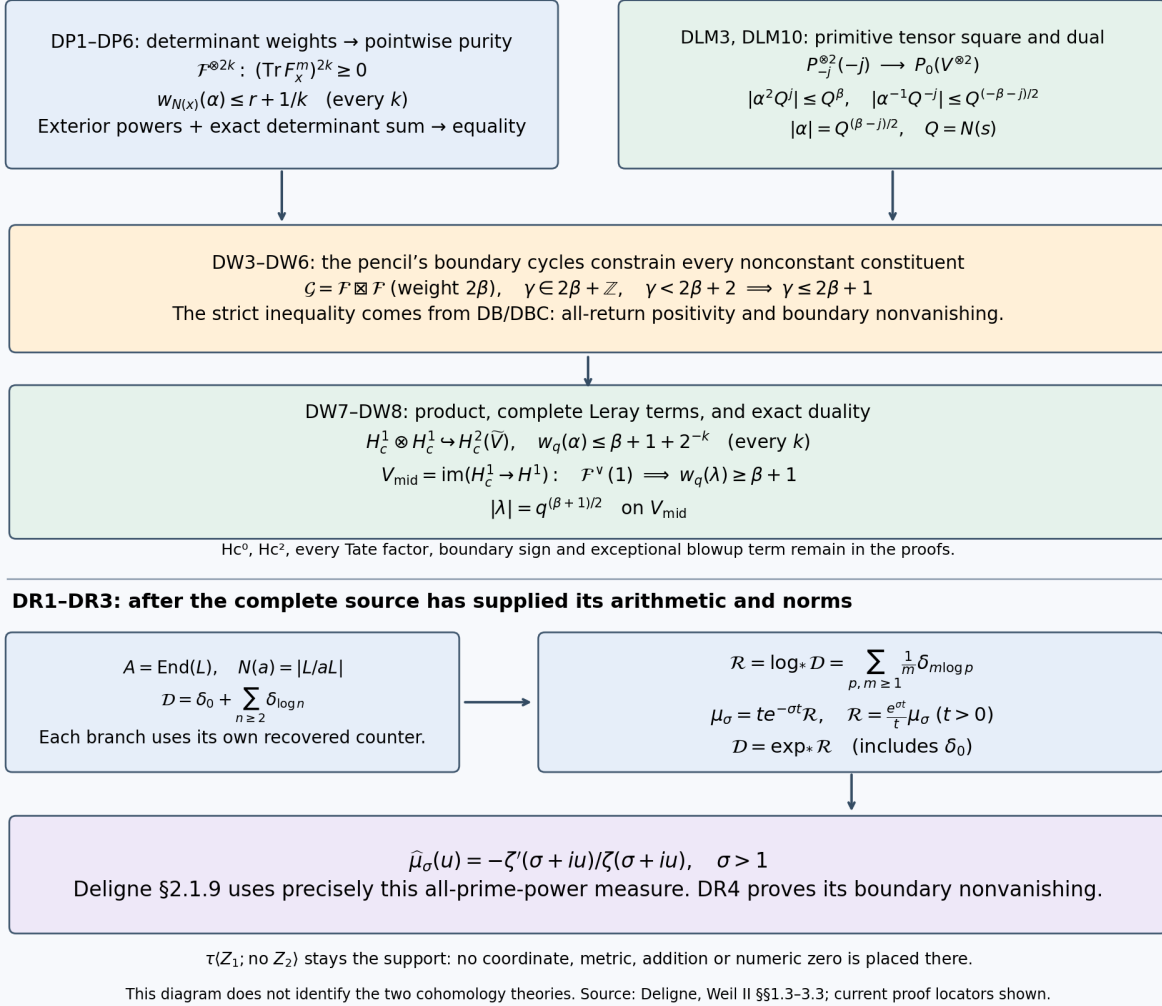
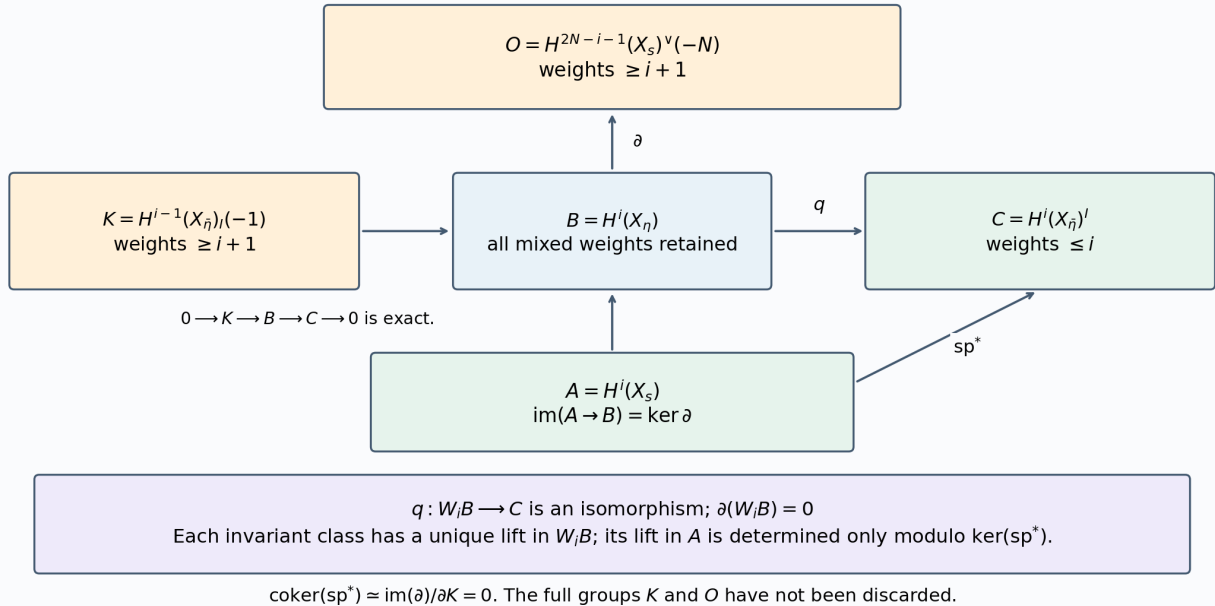


Figure 1: Deligne mechanism and arithmetic return map

Deligne's invariant-cycle quotient and its exact lift

A proper family over the stated henselian trait; smooth total model and smooth geometric generic fibre.

$\mathbb{Q}_\ell$ coefficients; $N = \dim X'$ is the total-model dimension; $I = \text{Gal}(\bar{\eta}/\eta)$.



Proof: DC. Human source: Deligne, Weil II §3.6, pp.212–214. This diagram does not identify the programme's analytic receiver with this cohomology.

Figure 2: The invariant-cycle quotient and lift

Which lifting map is being calculated?

Support: $\tau(Z_1; \text{no } Z_2)$. All rows below act on specified coefficient spaces.

1. Deligne's actual localization cross

$$\begin{aligned} 0 \longrightarrow K_i \longrightarrow B_i \xrightarrow{\pi_i} C_i \longrightarrow 0 & \quad w(C_i) \leq i \\ A_i \longrightarrow B_i \xrightarrow{\partial_i} O_i & \quad w(K_i), w(O_i) \geq i + 1 \\ W_i B_i \simeq C_i, \quad \partial_i(W_i B_i) = 0 & \end{aligned}$$

Deligne, Weil II §3.6; DC5–DC9. Proper geometric model, inertia, duality and twists retained.

2. Constructed character-shifted original-zeta modules

$$\begin{aligned} 0 \longrightarrow W(-1) \longrightarrow E \longrightarrow V \longrightarrow 0 \\ w_a(a^\rho) = 2\text{Re } \rho \in (0, 2) & \quad w_a(a^{\rho+1}) \in (2, 4) \\ D'_a(X) = aF_{a,W}X - XF_{a,V} = aD_a(X) \\ X = -(D'_a)^{-1}(Y'_a) \end{aligned}$$

ZTM2–ZTM5: full nilpotents; unique lift for all recovered characters; finite sums and algebraic direct sums.

3. The unshifted whole analytic source

$$\begin{aligned} 0 \longrightarrow I \longrightarrow B \longrightarrow Q \longrightarrow 0 \\ [F_0/(s - \rho)] \xrightarrow{\delta_{\rho,1}} [F_0] \neq 0 \\ \text{Both connecting lines carry } a^\rho: \text{ the weights coincide.} \\ F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) \end{aligned}$$

GLB1–GLB8, ZLB1–ZLB11: this exact eigenfunction lift is not asserted to be the programme's geometric lift.

Diagram positions are schematic. They assign no coordinates, arithmetic or metric to the supporting datum.

Figure 3: The three specified lifting maps

A continuous lift on the whole original source

Support retained: $\tau(Z_1; \text{no } Z_2)$. Coordinates below belong to coefficient functions.

1. Two full zero divisors, with every multiplicity

Original: $\rho, 0 < \text{Re } \rho < 1$

Translated: $\rho + 1, 1 < \text{Re}(\rho + 1) < 2$

$T_a|_\rho = a^\rho \exp((\log a)N_\rho)$

$T_a|_{\rho+1} = a^{\rho+1} \exp((\log a)N_\rho)$

Source modulus weights: $(0, 2)$

Kernel modulus weights: $(2, 4)$

$N_\rho^{m_\rho} = 0$; full exponential retained. Modulus weights use $a > 1$. Bands describe the full spectra.

2. The exact quotient map and its constructed inverse

$$0 \longrightarrow Q(-1) \longrightarrow B/(I \cap I_+) \longrightarrow Q \longrightarrow 0$$

$$\Phi[F] = ([F]_I, [F]_{I_+})$$

$$\Phi^{-1}([f], [g]) = [E_+ f + (1 - E_+)g]_{I \cap I_+}$$

$E_+ = 1$ to every jet at ρ ; $E_+ = 0$ to every jet at $\rho + 1$.

3. Continuous source formula, with original factors

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad F_{-1}(s) = F_0(s-1)$$

$$k_{\text{high}}(u) = \frac{1}{2\pi} \int_{\mathbb{R}} E_+(\frac{1}{2} + it) \mathcal{M}k(\frac{1}{2} + it) u^{-it} dt$$

The separator is an entire multiplier, constructed by a convergent Gaussian Cauchy integral.

GSL3-GSL8 prove the estimates, full range, equivariance and unique continuous linear section.

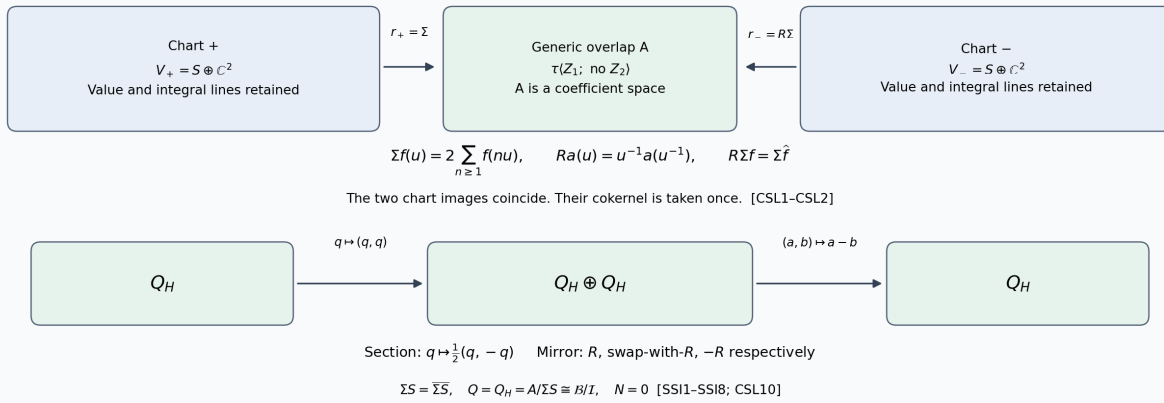
Proof: GLOBAL_SHIFTED_ZETA_LIFT.md. Deligne comparison: Weil II §3.6, DC5-DC9. No RH conclusion is drawn.

Figure 4: Continuous lift on the whole original source

Actual CC support maps and the descended conic

Two exact receivers; no metric or arithmetic coordinate is assigned to the parityless support.

1. The coefficient sheaf on the three-point support



2. The geometric Tate factor from the exact signed descent



At 2: $(x + y + z)^2 = 0$; the doubled line is retained. Specialization: $h_s \mapsto 2h_\eta$.

Every p and every $r \geq 1$: $\#C(\mathbb{F}_p) = 1 + p^r$, $Z(C, s) = \zeta(s)\zeta(s-1)$, $\Re s > 2$.

The shift comes from $H^2 = E(-1)$; this conic has $H^1 = 0$ in every geometric fibre. [CD2-CD10]

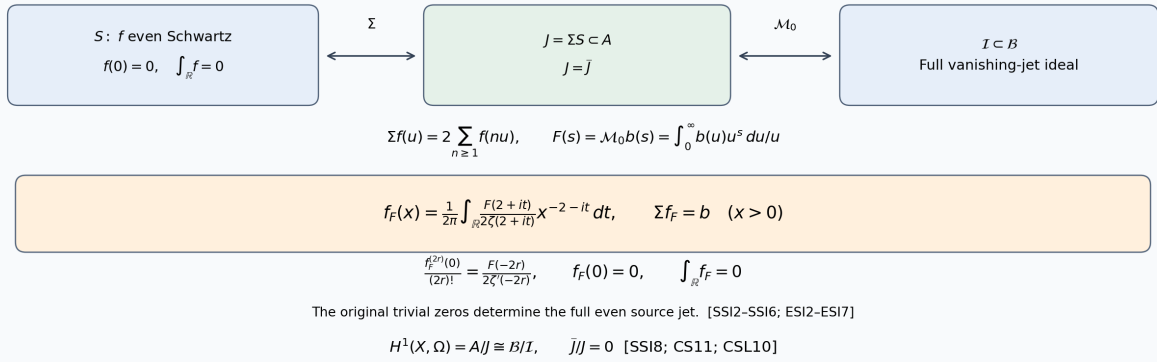
Sources: Connes-Consani, arXiv:0903.2024v3 §5 and arXiv:2609.00299v1 §§3-4. Full proofs: CSL and CD.

Figure 5: The coefficient support maps and descended conic

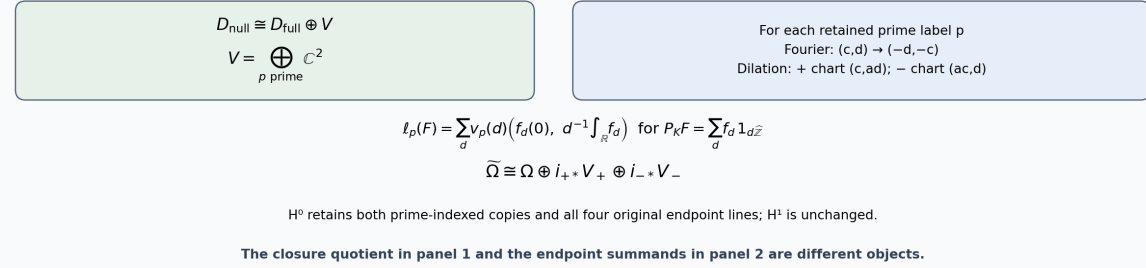
The exact original source and the retained endpoint directions

All operations below act on the reconstructed arithmetic coefficient spaces.

1. A continuous inverse in the actual Schwartz space



2. A different source quotient retains the prime endpoints



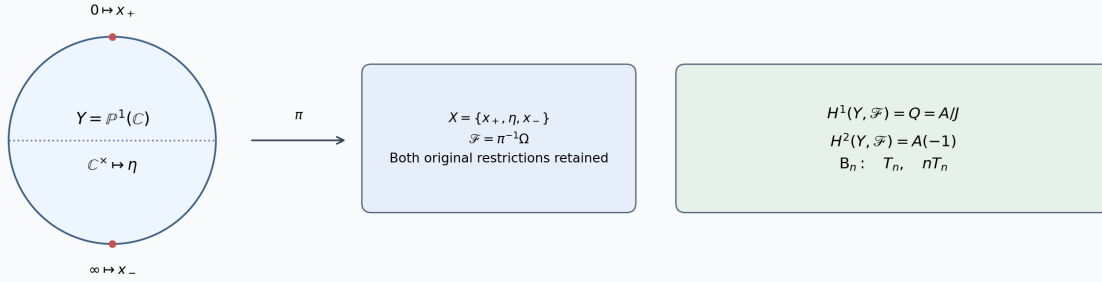
Complete proofs: SSI0-SSI10, CS2A/CS11, CSL10, CSB0-CSB10, Source: Connes-Consani, arXiv:0903.2024v3 §5.

Schematic of proved maps. It assigns no coordinate, vector, metric or arithmetic operation to $\tau(\mathbb{Z}_i)$; no $\mathbb{Z}_i$.

Figure 6: The original Schwartz inverse and retained endpoint directions

The actual sphere receiver and its normal-direction quotient

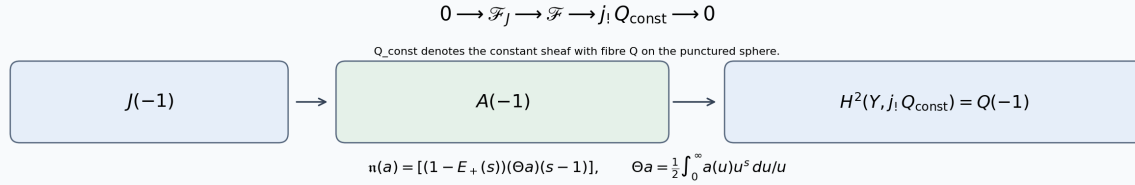
CSP0-CSP13: full sheaf, cohomological degrees, orientation signs and original-zeta comparison



$$b_n(z) = z^n, \quad \int_Y b_n^* \omega_Y = n, \quad T_n a(u) = a(u/n)$$

The factor n is the degree of the actual cover. The support η is not a coordinate center on Y .

The map of sheaves produces the precise normal quotient



Kernel exactly $J(-1)$; image exactly the shifted-original-zeta kernel in GSL. [CSP11.12]

Positive complex orientation fixes the signs

$$\partial_{\text{loop}}(a) = (-a, a), \quad k^2(c_+, c_-) = c_+ + c_-$$

$$w(z) = -1/z: H^2 \mapsto +R; \quad \alpha(z) = -1/\bar{z}: H^2 \mapsto -R$$

Schematic of derived maps; no metric, coordinate or operation on $\tau(\mathbb{Z}_1; \text{no } \mathbb{Z}_2)$.

Sources: Connes–Consani, arXiv:0903.2024v3 §5; arXiv:2609.00299v1 §§3–4. Full proofs: CSP2–CSP12.

Figure 7: The sphere receiver and normal quotient

The connecting map of the actual sheaf quotient

All rows use the original coefficient spaces. The cohomological degrees and arithmetic actions are retained.

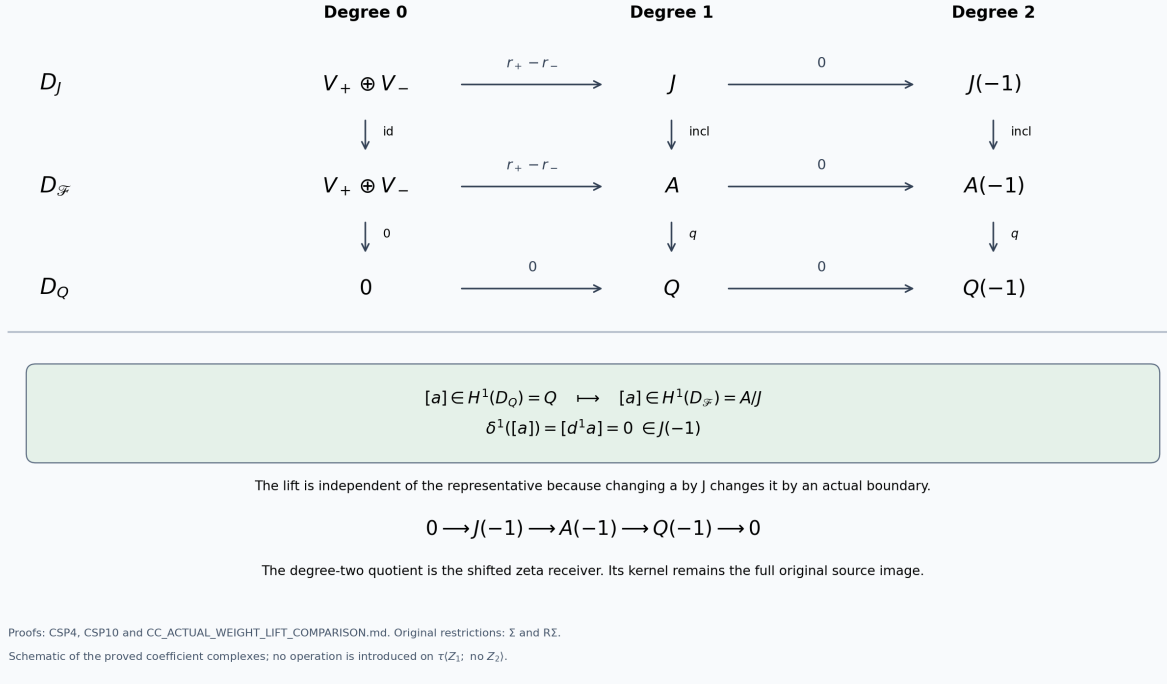


Figure 8: The actual sheaf complexes and connecting map

Figure 1m. CP proves that the actual comparison defect has zero derived polynomial spectral sectors. It constructs the correction on the original current complexes and proves uniqueness of the continuous cohomology class, preserving the complete original-zero jets and all endpoint degrees.

Figure 1n. GCF and PGF construct the full original-zeta divisor comparison and the actual generator family. The complete local parameter module, its fibre and its Tor kernel retain distinct invariants, with all units, cancellation frames and original factors explicit.

Figure 1o. CLP retains the full normal kernel, constructs the inverse normal comparison at original zeros, and computes the nonzero higher boundary at translated zeros. The coherent original and shifted divisor families have disjoint supports; the explicit Hom-complex contraction proves their derived separation.

Figure 1p. GMS3-GMS5 use the complete global separator to construct the full normal comparison inverse after derived Hom from the original quotient. GMS4.7-GMS4.9 give the explicit homotopy, and GMS8 retains the original-source integral. GMS6 keeps the nonzero normal extension and calculates its actual multiplier pushout.

Figure 1q. FOD constructs both actual pullbacks and full-kernel pushouts, proves their exact multiplier annihilators, and calculates the component retained by the separator. NEA supplies the exact whole-module resolvent and full multiplicity probes. FOD7 and NEA10 give the actual localization lift for the specified complete-original-quotient source. Central postcomposition is distinguished from conjugation.

Figure 1r. LNC computes the actual specialization quotient and its complete dual. LVD proves the local factorization and constructs the original residue pairing globally. RGR constructs the signed Gysin receiver, its relation to the original coefficient extension, and the separate adjoint-defect receiver. All endpoints and higher jets remain; the full proofs below specify every domain and comparison.

Figure 2. Exact timing maps, including the explicitly discrete case and the separate odd-refinement test. GTQ and OR retain the hypotheses and full derivations.

The derived sheaf retains its winding direction

CDW0-CDW10: an explicit comparison of complexes, including topology, naturality and orientation

$$R\pi_* \mathcal{F} \simeq \Omega[0] \oplus j_{X!}(A \otimes \mathbb{C}[\theta])[-1]$$

$$\theta = \frac{1}{4\pi i} \left(\frac{dz}{z} - \frac{d\bar{z}}{\bar{z}} \right), \quad \int_{|z|=1} \theta = 1$$

Strict map into the actual form resolution

Pole chart
 $v_{\pm} \mapsto (\text{constant}(r_{\pm} v_{\pm}), v_{\pm})$

Punctured chart
 $a \mapsto \text{constant}(a), \quad a[\theta] \mapsto a\theta$

The winding sheaf has zero sections on either pole chart; the form need not extend through a pole.

The full source sequence survives in both internal degrees

Degree 0	Ω_j	$\longrightarrow$	Ω	$\longrightarrow$	$j_{X!} Q$
Degree 1	$j_{X!} j[\theta]$	$\longrightarrow$	$j_{X!} A[\theta]$	$\longrightarrow$	$j_{X!} Q[\theta]$

Each displayed row is exact, with a zero on each end. The internal differentials are zero.

Actual actions on the winding summand

$$b_n : z \mapsto z^n \Rightarrow nT_n \quad w : z \mapsto -1/z \Rightarrow -R \quad \alpha : z \mapsto -1/\bar{z} \Rightarrow +R$$

$$a[\theta] \in H^1(X, j_{X!} A[\theta]) \mapsto -a \in H^2(Y, \mathbb{F})$$

$$a[\theta] \mapsto -[(1 - E_+(s))(\Theta a)(s - 1)] \text{ in the GSL kernel}$$

The minus sign uses the ordered Čech differential $r_+ - r_-$ and positive complex integration. [CDW9.2-CDW10.8]

Sources: Connes-Consani, arXiv:0903.2024v3 §5 and arXiv:2609.00299v1 §§3-4; exact receiver proofs CSP and CDW.

Diagram of proved maps; no coordinate, metric or addition is assigned to $\tau(Z_1; \text{ no } Z_2)$.

Figure 9: The derived direct image and retained winding class

The complete continuous dual and the original-zeta residue map

CDI0-CDI12, RD0-RD9, SDT0-SDT9: complete spaces, every zero multiplicity, exact degree factors

$$\begin{array}{ccccc}
 \text{Degree } -2 & & \text{Degree } -1 & & \text{Degree } 0 \\
 A' & \xrightarrow{0} & A' & \xrightarrow{\lambda \mapsto (r'_+ \lambda, -r'_- \lambda)} & V'_+ \oplus V'_- \\
 H^{-2} = A' & & H^{-1} = Q' & & H^0 = (\ker d)' \\
 & & d = r_+ - r_-, \quad Q = A/J, \quad J = \Sigma S \text{ closed}, \quad Q' = J^\perp \subset A' & &
 \end{array}$$

Every support-map sign survives the transpose

$$\text{difference} \mapsto (\lambda, -\lambda), \quad \text{sum} \mapsto (\lambda, \lambda), \quad \partial'_{\text{loop}}(\lambda_+, \lambda_-) = -\lambda_+ + \lambda_-$$

$$\begin{aligned}
 \iota(G)(F) &= \sum_{\rho} \text{Res}_S = \rho \frac{F(s)G(1-s)}{\zeta(s)} ds \\
 G \in Q_{\text{fin}}, \quad \iota: Q_{\text{fin}} &\simeq \mathcal{D}_{\text{fin}} \subset Q'
 \end{aligned}$$

The sum is finite because G has finite full-jet support. Its denominator is the original ζ .

$$\zeta(\rho + t) = t^m u_\rho(t), \quad P_{ij} = (-1)^j [t^{m-1-i-j}] u_\rho(t)^{-1}, \quad \det P = u_\rho(0)^{-m}$$

The nonzero determinant retains every nilpotent order in each multiplicity-m block. [RD2-RD3; CDI9]

Arithmetic covariance retains the geometric factor

$$\begin{aligned}
 \iota(T_a G) &= a(T_{1/a})' \iota(G), \quad N_a^\vee = a^{-1}(T_{1/a})' \\
 \iota(T_a G) &= a^2 N_a^\vee \iota(G), \quad P(a^k T_a F, a^l T_a G) = a^{k+l+1} P(F, G)
 \end{aligned}$$

$\Theta b(s) = \frac{1}{2} \int_0^\infty b(u) u^s du / u$ is the complete source comparison; the factor $\frac{1}{2}$ is retained. [CDI6; RD7]

$q': Q'_\beta \simeq J^\perp \subset A'_\beta$, $\mathcal{D}_{\text{fin}}^\beta = Q'_\beta$; SDT5-SDT8 prove the strong topology and residue completion.

Source: Connes-Consani arXiv:0903.2024v3 §5; actual sphere/localization proof CSP. No operation is assigned to $\tau(Z_1; \text{no } Z_2)$.

Figure 10: The continuous dual and original-zeta residue maps

Continuous support duality: the current realizes the maps

CSD0–CSD12: compactly supported tests, the positive Dirac sign, and both pole restrictions

$$\mathcal{D}_c \mathcal{F} = \text{Cone}(i_* A'^2 \rightarrow \mathcal{T}_A^\bullet \oplus i_* V'), \quad \text{arrow} = (\delta, -r')$$

$$\Gamma(U, \mathcal{D}_c \mathcal{F}) = (\Gamma_c(U, \mathcal{K}^\bullet))^v$$

Every functional is continuous. The complex is obtained by transposing the actual test resolution. [CSD3–CSD4]

Local angular current

$$h_p = \text{darg}(z_p)/(2\pi), \quad d(h_p \mu) = \delta_p \mu, \quad d(h_p \mu, -\mu) = (0, r'_p \mu)$$

$$i_p^* \mathcal{D}_c \mathcal{F} \simeq [A' \rightarrow A' \rightarrow V'_p], \quad d = (0, r'_p), \quad i_p^! \mathcal{D}_c \mathcal{F} \simeq V'_p[0]$$

The first complex has degrees $-2, -1, 0$. Point support is computed by the disk-to-puncture cone. [CSD5–CSD6]

Global comparison retains the opposite pole signs

$$d(\vartheta \mu) = \delta_0 \mu - \delta_\infty \mu, \quad \mu \mapsto (\vartheta \mu; -\mu, +\mu)$$

$$d(\vartheta \mu; -\mu, +\mu) = (0; r'_+ \mu, -r'_- \mu)$$

$$Q' \rightarrow Q' \oplus Q': \mu \mapsto (\mu, -\mu), \quad A' \rightarrow A' \oplus A': \lambda \mapsto (\lambda, \lambda)$$

These are actual restrictions of the current comparison. They recover the full CDI support signs. [CSD7–CSD8]

Proper pushforward is adjoint to pullback on tests

$$b_n * c_\lambda = n c_\lambda, \quad b_n * \vartheta = \vartheta, \quad b_n * \delta_p = \delta_p$$

$$B'_n = (n T'_n, T'_n, \rho(n')) \quad \text{in degrees } (-2, -1, 0)$$

The trace transpose U'_n is a different, explicitly compared map in GTR4–GTR5. All factors remain.

Source setting: Connes–Consani arXiv:0903.2024v3 §5; sphere proof CSP; continuous dual CDI. No coordinate is assigned to $\tau(\mathbb{Z}_1; \text{no } \mathbb{Z}_2)$.

Figure 11: The continuous support-dual current construction

The geometric trace carries the residue degree factor

GTR0–GTR10: integer covers $n \geq 1$, both endpoint stalks, all zero multiplicities

$$z \mapsto z^n: (a_1, \dots, a_n) \mapsto \sum_{j=1}^n a_j$$

Interior sheets: each has multiplicity 1.

$$(r_p v_p, v_p) \mapsto (nr_p v_p, nv_p)$$

Both poles: multiplicity n ; both endpoint coordinates retained.

$$\mathrm{Tr}_n U_n = n \mathbf{1}_{\mathcal{F}} \text{ on the entire coefficient sheaf. [GTR2.1–GTR2.4]}$$

	$H^0 = Z$	$H^1 = Q$	$H^2 = A$
Geometric pullback	1	1	n
Geometric trace	n	n	1
Full pullback B_n	$\rho(n)$	T_n	nT_n
Trace with inverse coefficients U_n	$n\rho(n^{-1})$	$nT_{n^{-1}}$	$T_{n^{-1}}$

$$U_n B_n = B_n U_n = n \mathbf{1}, \quad U'_n = n(B_n^{-1})'$$

The inverse in this formula acts on coefficients. For $n > 1$, the cover has no single-valued inverse. [GTR4]

Original-zeta residue → degree-minus-one transfer transpose

$$\iota(G)(F) = \sum_{\rho} \mathrm{Res}_{s=\rho} \frac{F(s)G(1-s)}{\zeta(s)} ds$$

$$\kappa' \iota T_n = U'_n \kappa' \iota, \quad U'_n|_{Q'} = nT'_{n^{-1}}$$

Finite full-jet residues extend in the specified SDT strong topology; the original ζ remains the denominator.

$$U'_n \kappa' \delta_{\rho,j} = n^{1-\rho} \sum_{h=0}^j \frac{(-\log n)^{j-h}}{(j-h)!} \kappa' \delta_{\rho,h}, \quad \delta_{\rho,j}(F) = F^{(j)}(\rho)/j!$$

Proofs: GTR2–GTR5; RD2–RD5; SDT8. Source setting: Connes–Consani arXiv:0903.2024v3 §5; sphere construction CSP.

All operations occur on the constructed coefficient receiver. The supporting datum remains $\tau(\mathbb{Z}_1; \text{no } \mathbb{Z}_2)$.

Figure 12: The ramified geometric trace and original residue

Continuous and algebraic duals: the exact comparison

VSD0–VSD14: the sheaf cone, the full coefficient defect, and its polynomial operators

$$\mathcal{D}_C \mathcal{F} \longrightarrow \mathbb{D}_Y \mathcal{F} \longrightarrow \mathcal{E}$$

$$\mathcal{E} = \text{Cone}(i_* (A^\dagger)^2 \longrightarrow \mathcal{L}_{A^\dagger}[2] \oplus i_* W^\dagger), \quad \text{arrow} = (g, -r^\dagger)$$

$$E' = \text{Hom}_{\text{cont}}(E, C), \quad E^* = \text{Hom}_C(E, C), \quad E^\dagger = E^*/E'$$

$\mathcal{L}_{A^\dagger}$ denotes the constant sheaf on Y with coefficient $A^\dagger$.

All spectral polynomial data are already continuous

$$LF(s) = sF(s), \quad E \in \{A, J, Q\}, \quad 0 \neq P \in \mathbb{C}[t]$$

$$P(L^\dagger): E^\dagger \simeq E^\dagger$$

$$\ker P(L') \simeq \ker P(L^*), \quad \text{coker } P(L') \simeq \text{coker } P(L^*)$$

Every finite-dimensional L^* -invariant subspace of E^* is contained in E' . [VSD13.11–VSD13.12]

The nonzero defect carries a $\mathbb{C}(t)$ -vector-space structure. No discontinuous functional is differentiated.

The topology specifies which full receiver is obtained



Both routes retain the full original-zeta residue formula. The two topologies and completions are specified separately.

Proof locators: VSD8–VSD10 and VSD13; original residue RD; strong topology SDT. Source setting: CC arXiv:0903.2024v3 §5.

These are coefficient operations after arithmetic reconstruction. The support remains $\tau(Z_1; \text{no } Z_2)$.

Figure 13: The exact continuous and algebraic dual comparison

An actual spectral lifting obstruction vanishes

Continuous currents, algebraic currents, and their constructed quotient — no change of source

$$0 \rightarrow D_c \rightarrow D_a \rightarrow D_+ \rightarrow 0$$

$$D_a \simeq R\Gamma(Y, \mathbb{D}_Y \mathcal{F}), \quad H^*(D_+) \text{ has invertible } P(L^\dagger) \quad (0 \neq P \in \mathbb{C}[t])$$

The spectral complex retains both the equation and its homotopy:

$$K_P(D)^k = D^k \oplus D^{k-1}, \quad d_P(x, y) = (dx, P(L)x - dy)$$

$$H^k K_P(D_+) = 0 \implies H^k K_P(D_c) \simeq H^k K_P(D_a)$$

Every algebraic spectral class has a unique continuous spectral class above it.

The lift is constructed on the actual current complexes

$$d_P(x, y) = 0, \quad d_P(\tilde{U}, \tilde{V}) = (\tilde{x}, \tilde{y})$$

Choose algebraic lifts U, V of the quotient primitives. The corrected cocycle is

$$(x, y) - d_P(U, V) = (x - dU, y - P(L)U + dV) \in K_P(D_c)$$

The class is unique modulo continuous spectral boundaries. Representatives are not asserted to be unique.

Original-zero jets enter through the full source and its actual quotient

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad P(t) = (t - \rho)^m, \quad m = m_\rho$$

$$H^{-1} K_P(D_c) \simeq \ker P(L' | Q') \simeq H^{-1} K_P(D_a)$$

Proof locators: CV4-CV8; CP4-CP9. Human setting: Connes-Consani 0903.2024v3 §5; Deligne, Weil II §3.6.

Here D denotes the global section complex; ρ is an actual nontrivial zero. Full multiplicities and endpoints remain in the proof.

The support remains $\tau(Z_1; \text{no } Z_2)$. This lift does not determine $\Re \rho$.

Figure 14: The actual polynomial spectral lift

The original zeta divisor in an actual generator family

Every parameter, jet order $r \geq 1$, and zero multiplicity — GCF1-GCF7; PGF2-PGF7

$$\begin{aligned} I/(L - \lambda)^r I &\longrightarrow B/(L - \lambda)^r B \\ R_r &\xrightarrow{T_r(\lambda)} R_r, \quad R_r = \mathbb{C}[h]/(h^r) \end{aligned}$$

$$\begin{aligned} T_r(\lambda)_{ij} &= F_0^{(i-j)}(\lambda)/(i-j)! \quad (i \geq j); \quad 0 \quad (i < j) \\ F_0(s) &= \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad \det T_r(\lambda) = F_0(\lambda)^r \end{aligned}$$

The original meromorphic function is retained by the exact line-module map

$$\begin{aligned} \mathcal{O} &\xrightarrow{\zeta} \mathcal{O}(D) \xrightarrow{\mu_c} \mathcal{O}, \quad \mu_c(g) = cg \\ D &= [1] - \sum_{n \geq 1} [-2n], \quad c(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \end{aligned}$$

The middle module stays embedded in meromorphic functions; its frames retain the pole and trivial zeros.

At an actual nontrivial zero: the whole family and its fibre retain different data

$$x = \lambda - \rho, \quad m = m_\rho, \quad F_0(\rho + y) = y^m u_\rho(y), \quad u_\rho(0) \neq 0$$

$$\begin{aligned} \mathcal{C}_{r,\rho} &\simeq \mathbb{C}\{x, h\}/(h^r, (x+h)^m) \\ \text{length} &= mr, \quad x^{m+r-1} = 0, \quad x^{m+r-2} \neq 0 \end{aligned}$$

The original map includes the full unit $u_\rho(x+h)$; the proof retains its automorphism and inverse.

$$\begin{aligned} 0 &\longrightarrow K_{\lambda,r} \xrightarrow{\partial_{\lambda,r}} R_r \xrightarrow{T_r(\lambda)} R_r \longrightarrow C_{\lambda,r} \longrightarrow 0 \\ \dim K_{\rho,r} &= \dim C_{\rho,r} = \min(r, m), \quad K_{\rho,r} \simeq \text{Tor}_1(\mathcal{C}_r, C_\rho) \end{aligned}$$

Human source setting: Connes-Consani 0903.2024v3 §5, Deligne, Weil II §3.6 supplies the distinct weight-lifting mechanism.

The support remains $\tau(Z_1; \text{no } Z_2)$; all displayed operations belong to the reconstructed coefficient spaces.

The calculation identifies the complete divisor and lifting boundary; these identities do not force $\Re \rho = 1/2$.

Figure 15: The full primal generator family and original divisor

The actual localization row: what separates, and what survives

Full coefficient kernel, every polynomial order, every original zero multiplicity — CLP2-CLP13

Actual geometric row and its full derivative matrix — CLP2, CLP5-CLP6

$$\begin{aligned} 0 \longrightarrow J(-1) \longrightarrow A(-1) \longrightarrow Q(-1) \longrightarrow 0 \\ T_r(\lambda - 1)_{ij} = F_0^{(i-j)}(\lambda - 1)/(i-j)! \quad (i \geq j); \quad 0 \quad (i < j) \\ F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad t = L + 1 \end{aligned}$$

At an original zero: the normal comparison is invertible — CLP7

$$\begin{aligned} \lambda = \rho, \quad 0 < \Re \rho < 1, \quad \mu = \rho - 1 \\ J(-1)/(L - \mu)^j \xrightarrow{T_r(\mu)} A(-1)/(L - \mu)^j A \\ (T_r(\mu)^{-1})_{ij} = \frac{1}{(i-j)!} \left(\frac{8\pi^{s/2}}{s(s-1)\Gamma(s/2)\zeta(s)} \right)_{s=\mu}^{(i-j)} \end{aligned}$$

At a translated zero: the higher boundary retains the normal extension — CLP8-CLP9

$$\begin{aligned} \lambda = \rho + 1, \quad m = m_\rho, \quad d = \min(r, m) \\ 0 \longrightarrow Q(-1)[(t - \rho - 1)^r] \xrightarrow{\beta^2} J(-1)/(L - \rho)^j J \\ \beta^2([F_0(s)/(s - \rho)^j]) = [F_0(s)(s - \rho)^{r-j}], \quad 1 \leq j \leq d \\ \text{rank } \beta^2 = d; \quad \text{source and image character } a^{\rho+1} \end{aligned}$$

The coherent families have disjoint supports — CLP13

$$\begin{aligned} \mathcal{C}_r = \text{coker } T_r(\lambda), \quad \mathcal{C}_q^+ = \text{coker } T_q(\lambda - 1) \\ \text{supp } \mathcal{C}_r = \mathcal{Z} \subset \{0 < \Re \lambda < 1\}, \quad \text{supp } \mathcal{C}_q^+ = \mathcal{Z} + 1 \subset \{1 < \Re \lambda < 2\} \\ \text{RHom}_\phi(\mathcal{C}_r, \mathcal{C}_q^+) = 0 = \text{RHom}_\phi(\mathcal{C}_q^+, \mathcal{C}_r) \end{aligned}$$

The degree-one spectral lift has an r -dimensional kernel at an original zero (CLP6-CLP7).

The shifted boundary is a different lifting equation; the ordinary geometric boundary remains zero.

Neither the full $J(-1)$ nor its character quotients are replaced by the coherent normal family.

Human sources: Connes-Consani, 0903.2024v3 §5 and 2609.00299v1; Deligne, Weil II §3.6.

The support remains $\tau(Z_1; \text{no } Z_2)$; these are operations on the reconstructed coefficient spaces.

No common arithmetic modulus or conclusion $\Re \rho = 1/2$ is imposed by this diagram.

Figure 16: The actual localization row and spectral weights

From separated families to the complete normal source

The existing global separator acts on the actual full coefficient modules; no finite-block limit is assumed.

GSL supplies a continuous multiplier with all original zero jets retained

$$\begin{aligned} E_+(\lambda) &= 1 - E(\lambda - 1), & E_+|_Q &= 1, & E_+|_{Q_+} &= 0 \\ Q &= B/I, & Q_+ &= B/I_+ \\ F_0(\lambda - 1) &= \frac{(\lambda-1)(\lambda-2)}{8} \pi^{-(\lambda-1)/2} \Gamma((\lambda-1)/2) \zeta(\lambda-1) \end{aligned}$$

The full normal row and its actual continuous reverse map

$$\begin{aligned} 0 &\longrightarrow I_+ \xrightarrow{\iota} B \longrightarrow Q_+ \longrightarrow 0 \\ m_{E_+} : B &\longrightarrow I_+, & F &\longmapsto E_+ F \\ \iota m_{E_+} &= E_+|_B, & m_{E_+} \iota &= E_+|_{I_+} \end{aligned}$$

After the derived Hom functor from the complete original quotient

$$\begin{aligned} \mathrm{RHom}_M(Q, I_+) &\xrightarrow{\iota_*} \mathrm{RHom}_M(Q, B) \\ \text{inverse up to homotopy: } (m_{E_+})_*; &\quad \delta S + S\delta = E_+ - 1 \\ \mathrm{Ext}_M^k(Q, Q_+) &= 0 = \mathrm{Ext}_M^k(Q_+, Q), \quad k \geq 0 \end{aligned}$$

The original normal coordinate and source map retain their factors

$$\begin{aligned} \lambda &= s + 1, & E_+(s + 1) &= 1 - E(s) \\ a &\longmapsto \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} (1 - E(1/2 + it)) \Theta a(1/2 + it) u^{-it} dt \in J(-1) \end{aligned}$$

M consists of entire multipliers with polynomial growth on each closed vertical strip.

The reverse map is not a literal inverse on the source; the displayed homotopy proves its derived inverse.

All endpoint values remain. No multiplier action, addition or metric is assigned to $\tau(\mathbb{Z}_1; \text{no } \mathbb{Z}_2)$.

Proof: GMS3-GMS5, GMS8; homotopy GMS4.7-4.9. Inputs: GSL4-6, CLP12-13. Human settings: CC §5; Deligne §3.6.

Figure 17: Complete-module separation and the full normal inverse

The complete source has two different obstruction components

Actual strict source rows and multiplier actions; every zero multiplicity and the full kernel remain.

Original source and its translated normal source — FOD1; NEA1

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad F_+(\lambda) = \frac{(\lambda-1)(\lambda-2)}{8} \pi^{-(\lambda-1)/2} \Gamma((\lambda-1)/2) \zeta(\lambda-1)$$

$$Q = B/I, \quad Q_+ = B/I_+, \quad \mathcal{K} = I \cap I_+, \quad c = 1 - E(\lambda-1)$$

$$0 \rightarrow \mathcal{K} \rightarrow B \rightarrow Q \oplus Q_+ \rightarrow 0$$

Original component — FOD2-FOD5

$$e_{\text{low}} : 0 \rightarrow \mathcal{K} \rightarrow I_+ \rightarrow Q \rightarrow 0$$

$$\mathcal{K} \hookrightarrow I \implies e_0 : 0 \rightarrow I \rightarrow B \rightarrow Q \rightarrow 0$$

$$\text{Ann}_M(e_{\text{low}}) = \text{Ann}_M(e_0) = \mathfrak{a}$$

$$ce_{\text{low}} = e_{\text{low}} \neq 0, \quad ce_0 = e_0 \neq 0$$

Normal component — FOD2-FOD5; NEA3

$$e_{\text{high}} : 0 \rightarrow \mathcal{K} \rightarrow I \rightarrow Q_+ \rightarrow 0$$

$$\mathcal{K} \hookrightarrow I_+ \implies e_+ : 0 \rightarrow I_+ \rightarrow B \rightarrow Q_+ \rightarrow 0$$

$$\text{Ann}_M(e_{\text{high}}) = \text{Ann}_M(e_+) = \mathfrak{a}_+$$

$$ce_{\text{high}} = 0, \quad ce_+ = 0, \quad e_+ \neq 0$$

The second row in each panel is the actual pushout along its displayed kernel inclusion.

Exact full-divisor annihilators — FOD4; NEA3, NEA7

$$\mathfrak{a} = \{h \in M : \text{ord}_\rho h \geq m_\rho \forall \rho\}, \quad \mathfrak{a}_+ = \{h \in M : \text{ord}_{\rho+1} h \geq m_\rho \forall \rho\}$$

$$M e_{\text{joint}} \cong M/\mathfrak{a} \oplus M/\mathfrak{a}_+, \quad c \text{ acts as } (1, 0)$$

$$\mathfrak{a}_+ = K_+ \text{ in NEA notation; } \text{Spec}_{\text{alg}}(T|_{M e_+}) = \{\rho + 1\}, \quad T[h] = [\lambda h]$$

Actual localization lift, with the original quotient retained — FOD6-FOD7; NEA10

$$\delta : D_G \rightarrow D_J[1], \quad M\delta \cong M/\mathfrak{a}_+, \quad c\delta = 0, \quad c_Q = 1$$

$$f : Q[k] \rightarrow D_G \implies \delta f = 0 \implies \tilde{f} : Q[k] \rightarrow D_F$$

$$\beta : B \rightarrow I_+, \quad F \mapsto cF, \quad q_{\text{res}}\beta = q, \quad \ker q_{\text{res}} = \mathcal{K}$$

M : entire multipliers of polynomial growth on closed vertical strips. Spectrum above means algebraic non-bijectivity.

Central postcomposition is used here. Conjugation fixes the equivariant boundary. No arithmetic purity follows from this alone.

No addition, metric or coordinate is assigned to $\tau(\mathbb{Z}_1; \text{no } \mathbb{Z}_2)$. All operations act on reconstructed coefficients.

Full proofs: FOD0-8 and NEA0-10. Inputs: GMS, CLP, CW. Human source settings: Connes-Consani §5; Deligne, Weil II §3.6.

Figure 18: The complete source and its two obstruction components

Where the original residue pairing lives

Exact local specialization, dual arrows and global cochain receiver; every source kernel is retained.

Original source — LVD1, LVD6; LNC1

$$0 \longrightarrow J \longrightarrow A \xrightarrow{q} Q \longrightarrow 0, \quad F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s)$$

$$P = \ker(A_D \rightarrow i_* Q)[1], \quad F[1] = P \oplus i_* E[1], \quad E = C^2 \text{ at each pole}$$

Actual local arrows — LVD2-4

$$P: A \xrightarrow{q} Q \xrightarrow{0} A$$

$$D_c P: A' \xrightarrow{0} Q' \xrightarrow{q'} A'$$

arrows: can, var; $\Psi = R\psi[-1], \quad \Phi = R\phi[-1]$

$$T = 1, \quad \text{coker}(J \rightarrow A) = Q$$

Actual global pairing — LVD5, LVD7

$$K = [W_0 \xrightarrow{-r} A \xrightarrow{0} A]$$

$$K^\vee = [A' \xrightarrow{0} A' \xrightarrow{+r'} W_0']$$

$$\mathfrak{w}^0 = q' \mathfrak{w} \overline{q}, \quad \mathfrak{w}^{-1} = \mathfrak{w}^1 = 0$$

$$H^0(\mathfrak{w}) = \mathcal{W}: \overline{Q} \rightarrow Q'$$

The global map is nonzero. Every sheaf map $\overline{P}_{\text{full}} \rightarrow D_* P_{\text{full}}$ induces zero on this H^0 .

Constructed Gysin receiver — RGR3-7

$$a = q' \mathcal{W}: \overline{Q} \rightarrow A', \quad \mathfrak{a} = \text{Gys} \circ i_* a: i_* \overline{Q} \rightarrow A'_D[2]$$

$$\mathcal{E}_{\mathcal{W}} = \text{Cone}(-\delta_p a)[-1], \quad A'_D[1] \rightarrow \mathcal{E}_{\mathcal{W}} \rightarrow i_* \overline{Q} \xrightarrow{0} A'_D[2]$$

$$\Psi_{\mathcal{E}_{\mathcal{W}}} = A', \quad \Phi_{\mathcal{E}_{\mathcal{W}}} = \overline{Q}, \quad \text{can} = 0, \quad \text{var} = a$$

The local quotient and the earlier normal lift are related by explicit maps — LNC8-10; FOD7

$$Q = H_{\rho}^1(F) \xrightarrow{+1 \ (p=0), \ -1 \ (p=\infty)} H^1(Y, F) = Q$$

$$0 \rightarrow J(-1) \rightarrow A(-1) \rightarrow Q(-1) \rightarrow 0 \xrightarrow{(\mathcal{V}F)(\lambda) = F(\lambda-1)} e_+$$

$$c_Q = 1, \quad q c_A = q, \quad c e_+ = 0, \quad \delta f = 0 \text{ for } f: Q[k] \rightarrow D_G \text{ in } D(M)$$

The separator preserves the local quotient and annihilates the specified normal class.

$$\mathcal{W}(\overline{G})(F) = \sum_{\rho} m_{\rho} F(\rho) \overline{G(1-\rho)}, \quad \ker \mathcal{W} = \overline{N}_0; \text{ full higher jets remain in } Q.$$

Positive-circle and positive-quotient signs. A_D means the constant sheaf; $r = r_+ - r_-$; $W_0 = V_+ \oplus V_-$.

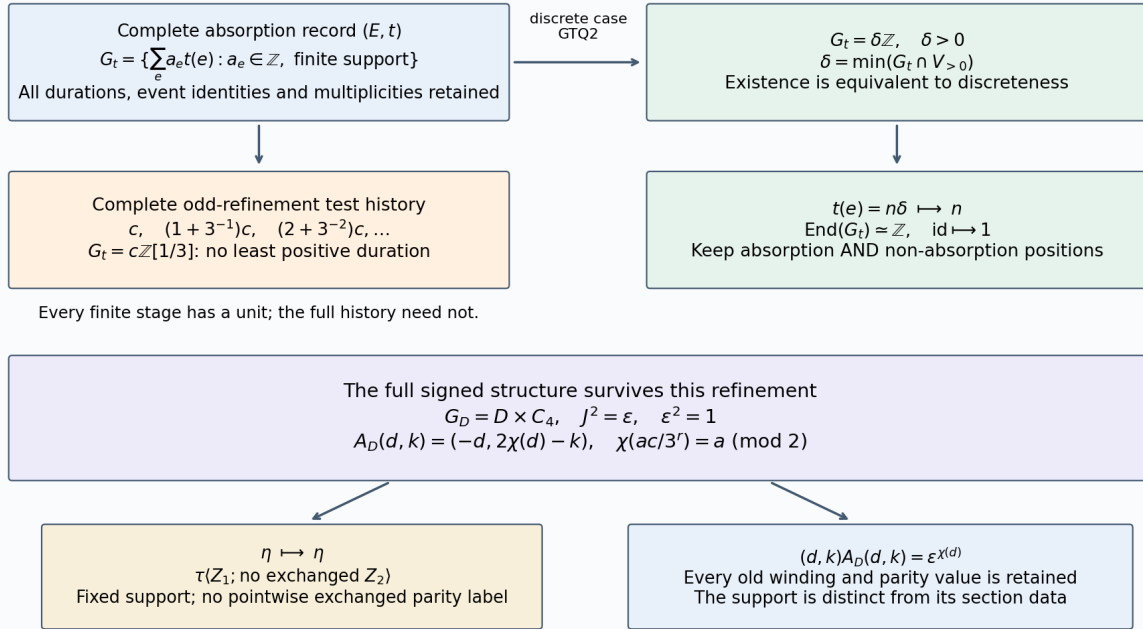
No coordinate or metric is assigned to $\tau(Z_1; \text{no } Z_2)$. These are post-reconstruction coefficient maps.

Complete proofs: LVD0-9, LNC0-11, RGR0-12. Sources: Connes-Consani §5; Reich §§1.3; Deligne, Weil II §3.6.

Figure 19: The exact local and global residue receivers

Whole timing first; arithmetic coordinates afterwards

Exact receiving maps. The duration line is not a metric on the supporting point.



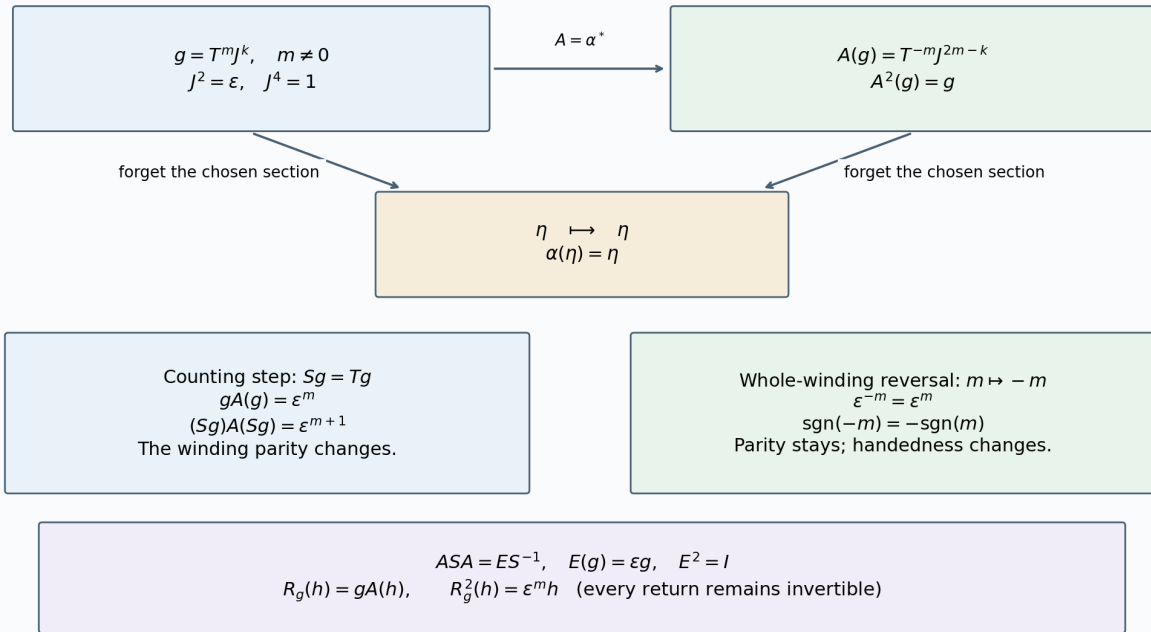
Proofs: GTQ1-GTQ8; OR1-OR7. Source comparison: Connes-Consani, arXiv:2609.00299v1, §§3-4.

The test history probes completeness $\rightarrow$ tick only; it is not asserted to be an admissible RH counterexample.

Figure 20: The complete timing maps and their scope

One fixed support; detectable winding reversal

$\tau(Z_1; \text{no } Z_2)$ identifies the supporting point. All operations below act on its stalk data.



An exchanged label on the two states cannot descend to their single fixed supporting point.

WR1-WR9; CC author source §§3-4. Diagram positions carry no metric, coordinate or distance on the support.

Figure 21: Winding reversal over a fixed supporting point

Figure 3. The full signed stalk action, with proof in WR1–WR9. Diagram positions supply no metric on the support.

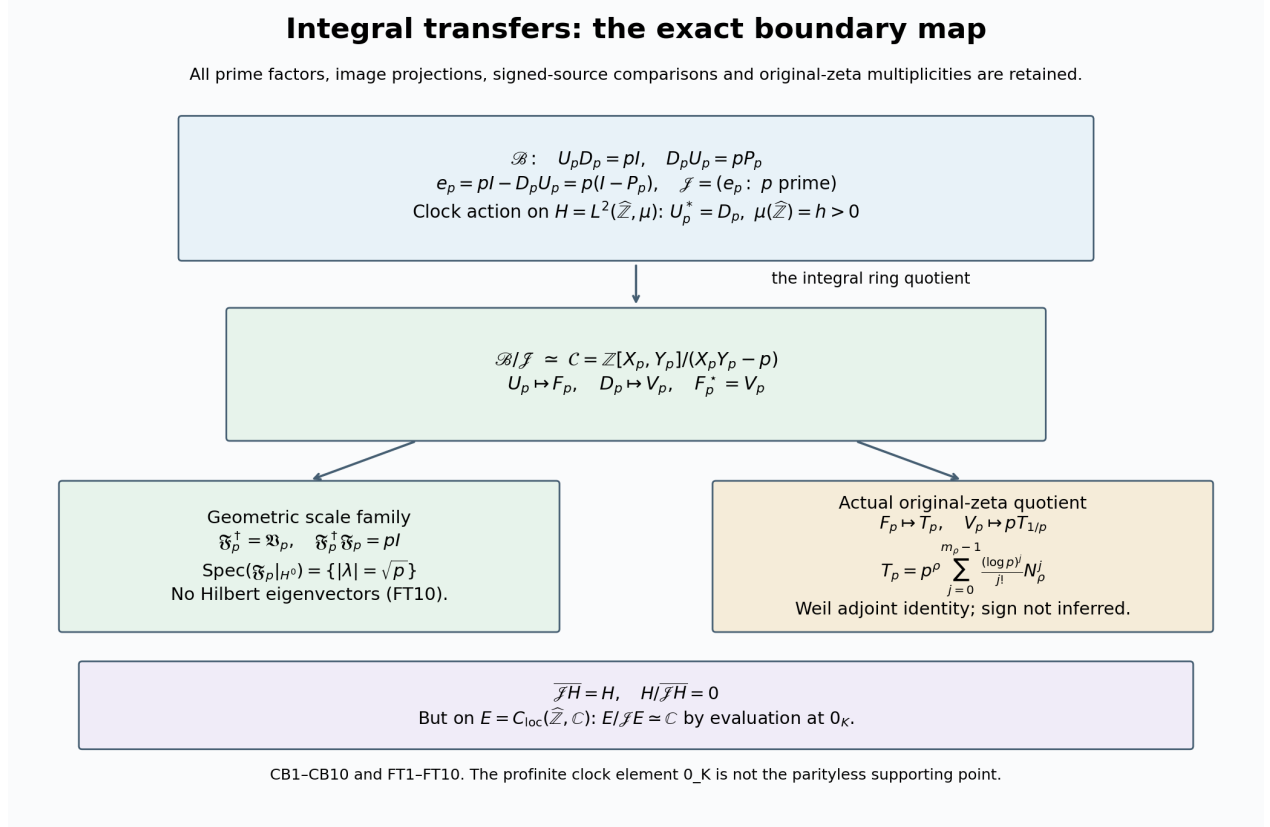


Figure 22: Integral transfers and the exact boundary map

Figure 4. Integral relations, their two receivers and the exact topological comparison; FT1–FT10 and CB1–CB10.

Final accepted adjoint-defect handoff

ADC0–ADC9 construct the two actual invariant-cycle crosses. DCA0–DCA12 calculate every finite-cover branch, raw-cut correction and composition homotopy. ACD0–ACD8 construct continuous-current duality and the original-source boundary with both poles retained. The original-source obstruction is the image of the calculated map β_r , with quotient Q/Z_r . The current-side quotient vanishes independently of this obstruction; no arithmetic weight-vanishing or RH conclusion is asserted.

The figure records the maps proved in ADC, DCA and ACD; its positioning gives no metric to the supporting datum.

One arithmetic defect, two exact specialization tests

Original source, continuous transpose, both pole contributions and every cover branch retained.

Original arithmetic and the constructed maps — ADC1; ACD0-1

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad Q = A/J, \quad E : Q \rightarrow H = \ell^2(\mathcal{I}, m)$$

$$D_r = T_r^* - rT_{1/r}, \quad \partial_r = q'A_H D_r : \bar{H} \rightarrow A', \quad b_r = \beta_r q = D_r^* J E q : A \rightarrow H$$

Primal specialization — ADC8; ACD4-5

$$\mathcal{P}_r = [A_D^{-1} \xrightarrow{+b_r} i_* H^0]$$

$$A \xrightarrow{-b_r} H, \quad K_{-1} = 0$$

$$\text{Obs}_{-1} = b_r A = \beta_r Q$$

$$\text{coker sp}_{-1} = Q/Z_r$$

$$Q \xrightarrow{\beta_r} \beta_r Q, \quad \ker \beta_r = Z_r$$

Continuous current dual — ADC3; ACD2-3

$$\mathcal{E}_r = \text{Cone}(-\delta_p \partial_r)[-1]$$

$$\bar{H} \xrightarrow{\partial_r} A' \rightarrow A'/a_r \bar{H}$$

$$\text{Obs}_i = 0 \quad \text{for every } i$$

$$\ker(K_0 \rightarrow \text{im } \partial_0) = a_r \bar{H}$$

$$D_c^{\text{raw}} \mathcal{P}_r \simeq \mathcal{E}_r : (-1, +1, -1)$$

The dual invariant-cycle quotient is zero before the attaching map is known to vanish.

Full sphere comparison — ACD6; four endpoint lines retained

$$R\Gamma(Y, \mathcal{P}_r) \simeq [A^{-1} \xrightarrow{(b_r, b_r)} (H \oplus H)^0 \xrightarrow{0} A^1]$$

$$H^0(Y, \mathcal{P}_r) \simeq H \oplus (H/b_r A), \quad H^0(Y, \mathcal{P}) = Q \ni F \mapsto (\beta_r F, 0)$$

$$H^0(Y, \mathcal{E}_r) = \ker(\partial_r + \bar{\partial}_r) \simeq \bar{H} \oplus C_{\text{crit}}$$

Genuine degree n covers — DCA3-7; ACD7-8

$$\Psi(f_n, \mathcal{E}_r) = (A')^n, \quad F_\Psi = T_n' \Sigma_n, \quad L_\Psi = \Delta T_{1/n}'$$

$$FL = n \cdot 1, \quad L_\Psi F_\Psi = \Delta \Sigma_n = \sum_{j=0}^{n-1} t^j$$

$$\text{var}_n(X, [V]) = e_0 \partial_r X - (t-1)V, \quad L_\Phi X = (\bar{T}_n X, [(0, c, \dots, (n-1)c)]), \quad c = T_{1/n}' \partial_r X$$

$$\beta_r T_n = U_n^* \beta_r : n^\rho = n^{1-\bar{\rho}^\#} \quad \text{on every nonzero image line}$$

Each pole retains $E_p[1]$ in the primal and $E_p^*[-1]$ in the dual; $\dim E_p = 2$. Z_r retains the full higher-jet kernel.

No coordinate or metric on $\tau(Z_1; \text{no } Z_2)$. The displayed operations follow arithmetic reconstruction.

The cover factors are proved. They do not separate the source from this boundary image; weight vanishing remains unproved.

Complete proofs: ADC0-9, DCA0-12, ACD0-8. Source setting: Connes-Consani §5; Deligne, Weil II §3.6; Reich §§1.3.

Figure 23: The two adjoint-defect specialization tests

Determinant weights and the complete tensor-power purity proof — DP0-DP9

Proof source: DELIGNE_DETERMINANT_AND_TENSOR_PURITY.md.

Determinant weights, all tensor powers, and pointwise purity

24 September 2026. Reconstruction DP0-DP9 of the determinant/tensor purity mechanism in Pierre Deligne, *La conjecture de Weil. II*, §§1.3–1.5. The dominant-pullback and étale-extension assertions 1.3.13(i), 1.3.14–15 and the general nonsemisimple 1.3.8 are addressed in the separate monodromy-group supplement; they are not claimed proved in this file. This derivation retains the source arithmetic data. It does not assert that a new coefficient object on the programme base has already been constructed.

DP0. Source, conventions, and the exact input

The entire current French transcription, printed pages 137–252, was read. The witness is output/Deligne_Weil_II_S20_LaTeX/typed_latex/S20_FR_record_export.tex, SHA256 d6836ae15d98f2b27eb98c9bd039a7d921679becd9fa47ff4717410289dbd351. It is a local transcription, not author TeX; the source-reading ledger records its provenance and discrepancies. The passages reconstructed here are printed pages 156–165, §§1.3.1–1.5.3, together with definitions 1.1–1.2. No original PDF comparison is claimed.

Let $q = p^f$, with $p \neq \ell$ primes. Let X_0 be a smooth geometrically connected curve over $\mathbb{F}_q$, and $X = X_0 \times_{\mathbb{F}_q} \overline{\mathbb{F}}_q$. A geometric point $\bar{x}$ gives the exact Weil-group sequence

$$1 \longrightarrow \pi_1(X, \bar{x}) \longrightarrow W(X_0, \bar{x}) \xrightarrow{\deg} \mathbb{Z} \longrightarrow 0. \quad (\text{DP0.1})$$

The generator is **geometric** Frobenius. At a closed point x , set $d_x = [k(x) : \mathbb{F}_q]$ and $N(x) = q^{d_x}$; F_x has degree d_x . These integers and fields are source data of Deligne’s theorem. They are not arithmetic placed on $\tau\langle Z_1; \text{no } Z_2 \rangle$.

A lisse Weil sheaf $\mathcal{F}_0$ is represented by a finite-dimensional continuous representation of (DP0.1), defined over a finite extension of $\mathbb{Q}_\ell$ on the geometric subgroup. Fix Deligne’s coefficient embedding $\iota : \overline{\mathbb{Q}}_\ell \rightarrow \mathbb{C}$. For $a \neq 0$, write

$$w_q(a) = \frac{2 \log |\iota a|}{\log q}, \quad w_{N(x)}(a) = \frac{2 \log |\iota a|}{d_x \log q}. \quad (\text{DP0.2})$$

These weights are logarithms of actual Frobenius eigenvalue moduli. They are not distances between a support and a stalk. A Tate twist (r) multiplies F_x by $N(x)^{-r}$, and subtracts $2r$ from its weight.

The established source inputs used below are global function-field class field theory, the finite-field Picard variety, the representation theory of reductive algebraic groups, Grothendieck’s trace formula, and Poincaré duality for curves. We give the deductions from these inputs, including every weight and determinant step. We do not claim to reprove the entire upstream SGA theory.

DP1. Rank-one geometric monodromy and its exact character

Let $\chi : W(X_0, \bar{x}) \rightarrow E^\times$ be rank one, with $E/\mathbb{Q}_\ell$ finite. Deligne 1.3.1 identifies the geometric part of the abelianized Weil group, by class field theory, with degree-zero idèle classes allowing ramification at the finite boundary $S = \bar{X}_0 - X_0$. The map to the unramified degree-zero divisor-class group has kernel a quotient of

$$\prod_{v \in S} \mathcal{O}_v^\times. \quad (\text{DP1.1})$$

Each local unit group has finite residue-unit quotient and pro- p principal units. The unramified quotient is $\text{Pic}^0(\bar{X}_0)(\mathbb{F}_q)$, a finite set because the variety is of finite type over a finite field. Consequently the geometric abelian image is finite-by-pro- p .

The image of $\pi_1(X, \bar{x})$ under χ is also compact in $E^\times$. Compactness makes its valuation image a finite subgroup of $\mathbb{Z}$, hence zero. It lies in $\mathcal{O}_E^\times$, which has an open pro- ℓ subgroup. A group that is both pro- p and pro- ℓ is trivial when $p \neq \ell$: every finite quotient has order both a power of p and a power of ℓ . Taking the open subgroups in the two descriptions therefore shows that the compact image is finite.

Choose a positive integer h killing this finite image. Then χ^h is trivial on the kernel of degree, so

$$\chi(w)^h = b^{\deg w} \quad (\text{DP1.2})$$

for some $b \neq 0$. Choose c with $c^h = b$, and put $\varepsilon(w) = \chi(w)c^{-\deg w}$. Direct substitution gives $\varepsilon(w)^h = 1$. Thus

$$\boxed{\chi(w) = c^{\deg w} \varepsilon(w), \quad \varepsilon^h = 1, \quad |\iota \chi(F_x)| = |\iota c|^{d_x}.} \quad (\text{DP1.3})$$

In particular rank one is pointwise ι -pure of weight $w_q(c)$. No choice of an individual prime or closed point is used to create the degree map; it is the full source sequence (DP0.1).

For an irreducible rank- n constituent $\mathcal{G}_0$, apply (DP1.3) to $\det \mathcal{G}_0$. Its determinant weight is

$$\beta(\mathcal{G}_0) = \frac{1}{n} w_q(c_{\det \mathcal{G}_0}). \quad (\text{DP1.4})$$

If $\alpha_1(x), \dots, \alpha_n(x)$ are its eigenvalues with their multiplicities, then

$$\sum_{j=1}^n w_{N(x)}(\alpha_j(x)) = n\beta(\mathcal{G}_0) \quad (\text{DP1.5})$$

at **every** closed point. This is an equality about the product of eigenvalues; individual equality has not yet been obtained.

The convention for a constant twist defined over $\mathbb{F}_p$ is important. If its Frobenius scalar is b , it multiplies the determinant at x by

$$b^{n[k(x):\mathbb{F}_p]} = b^{nfd_x}. \quad (\text{DP1.6})$$

This multiplication by f , not division by f , follows from the degree of the residue-field extension. The available transcription's 1.3.6 exponent is inconsistent with 1.2.7; equation (DP1.6) derives the correction while retaining the original scalar and field.

DP2. The central element that controls determinant weights

Replace $\mathcal{F}_0$ by its semisimplification for this paragraph only: determinants and constituent weights are unchanged, and no extension is removed from later trace formulas. Let G^0 be the Zariski closure of geometric monodromy on its fibre. A degree-one lift w normalizes G^0 . The group

$$G = G^0 \rtimes_{\text{Int}(\rho(w))} \mathbb{Z} \quad (\text{DP2.1})$$

receives the Weil group and all its tensor constructions. Its identity component G^{00} is reductive. Indeed its unipotent radical is characteristic in G^{00} , hence normal in G . The invariant subspace in an irreducible G -module is nonzero by unipotence and G -stable by normality, hence is the whole module. The radical therefore acts trivially on the semisimple faithful fibre, and is trivial.

We recall the rest of Deligne 1.3.9's argument because reductivity alone is insufficient. The connected central torus $T_1 \subset G^{00}$ acts through a finite set of characters on the faithful fibre. These characters generate its character group; conjugation permutes them, so after a finite-index restriction conjugation acts trivially on T_1 . The outer automorphisms of a reductive group that act trivially on its connected centre form a finite group (the finite root-datum automorphisms). A further finite-index restriction therefore makes the degree action inner. After changing its lift by an element of G^{00} , this restriction is $G^{00} \times \mathbb{Z}$. The associated finite cover and constant-field extension retain a smooth curve, and rank-one result (DP1.3) applies there.

The maximal torus quotient of G^{00} is isogenous to T_1 . Every character of that quotient would give a rank-one Weil character with finite geometric image by (DP1.3). Its geometric image is also Zariski dense by construction. A finite subset cannot be Zariski dense in a positive-dimensional torus. Therefore the torus quotient is trivial and G^{00} is semisimple. Its centre is finite.

Choose a positive power of the degree-one lift w whose conjugation is inner on G^{00} and trivial on the finite group G^0/G^{00} . Correct it by an element $h_0 \in G^{00}$, obtaining $y = h_0 w^m$ that centralizes G^{00} . For each representative h of G^0/G^{00} , the commutator $[y, h]$ lies in G^{00} because the component action is trivial; it centralizes G^{00} because y does. Thus it lies in the finite centre $Z(G^{00})$. Also $[y, w]$ lies in G^{00} , by the expression $y = h_0 w^m$, and centralizes G^{00} , so it lies in that same finite centre. Since y centralizes this finite centre, a common positive power kills all these commutators. That power centralizes G^{00} , the representatives of G^0/G^{00} , and w , which together generate G . We obtain

$$z \in Z(G), \quad m = \deg(z) > 0. \quad (\text{DP2.2})$$

This proves finite **index**, not finiteness, of the image of $Z(G) \rightarrow \mathbb{Z}$. Its kernel is contained in the finite centre of G^0 . The contradictory wording "finite subgroup of Z " in the available 1.3.10(iv) is not used.

On any irreducible G -module V , the central element z is a scalar u by Schur's lemma. The determinant character extends (DP1.3) from the Weil group to G : its h -th power is trivial on geometric monodromy, hence on its Zariski closure G^0 . Writing $g = g_0 w^d$ gives $\chi(g)^h = b^d$; choosing $c^h = b$ yields $\varepsilon(g) = \chi(g)c^{-d}$ with $\varepsilon(g)^h = 1$. Thus evaluation at z , whether or not it is itself in the Weil-group image, is legitimate. If V has rank r , this gives

$$u^r = c^m \varepsilon(z), \quad |\iota u|^r = |\iota c|^m, \quad \boxed{|\iota u| = q^{m\beta(V)/2}}. \quad (\text{DP2.3})$$

This proves 1.3.12's central-character criterion with the rank and degree factors intact.

Tensor products multiply eigenvalues of z . Thus their determinant weights add. In an exterior power, a triangular basis for the action of z shows that every eigenvalue is a product of the chosen number of diagonal eigenvalues. If the constituent weight γ occurs in total rank $n(\gamma)$, the determinant weights of $\bigwedge^a \mathcal{F}_0$ are

$$\sum_{\gamma} a(\gamma) \gamma, \quad 0 \leq a(\gamma) \leq n(\gamma), \quad \sum_{\gamma} a(\gamma) = a. \quad (\text{DP2.4})$$

For nonsplit sheaves, a composition filtration induces the tensor and exterior filtrations, and the same eigenvalue products give the weights of their graded constituents. Hence these conclusions do not require splitting extensions in the original sheaf.

DP3. The complete curve trace identity and its poles

For any lisse sheaf $\mathcal{V}_0$ on the curve, write V for its geometric fibre and $\Pi = \pi_1(X, \bar{x})$. The complete formulas are

$$H^0(X, \mathcal{V}) = V^\Pi, \quad H_c^0(X, \mathcal{V}) = \begin{cases} V^\Pi, & X \text{ proper,} \\ 0, & X \text{ not proper,} \end{cases} \quad H_c^2(X, \mathcal{V}) = V_\Pi(-1). \quad (\text{DP3.1})$$

The last equality follows from Poincaré duality with $H^0(X, \mathcal{V}^\vee(1))$; in particular the (-1) and the resulting factor q are retained. These invariant and coinvariant spaces come from geometric-constant subobjects or quotients. If every determinant weight of $\mathcal{V}_0$ is at most b , eigenvalues on H_c^0 have weight at most b , and those on H_c^2 have weight at most $b + 2$.

Grothendieck's trace formula, Deligne1.4.5, gives

$$L(\mathcal{V}_0, t) = \prod_{x \in |X_0|} \det(1 - F_x t^{d_x} \mid \mathcal{V}_{\bar{x}})^{-1} = \frac{\det(1 - Ft \mid H_c^1(X, \mathcal{V}))}{\det(1 - Ft \mid H_c^0(X, \mathcal{V})) \det(1 - Ft \mid H_c^2(X, \mathcal{V}))}. \quad (\text{DP3.2})$$

This begins as a formal series identity. For each coefficient only finitely many closed points contribute. Every cohomology group, sign, and multiplicity remains in (DP3.2). On an affine smooth connected curve, $H_c^0 = 0$ by (DP3.1), so its determinant is exactly 1; this is a proved vanishing, not a discarded term.

When eigenvalue weights at closed points are at most b and dimension is d , the Euler product converges absolutely for

$$|t| < q^{-b/2-d}. \quad (\text{DP3.3})$$

Indeed a finite decomposition into quasi-finite maps to affine space bounds the number of degree- n closed points by Cq^{dn} . The absolute logarithmic expansion is bounded by a constant times $\sum_{n \geq 1} q^{dn} q^{nb/2} |t|^n$, after summing repetitions geometrically on a smaller closed disc. This converges. This argument is Deligne1.4.6; it records how arithmetic point counts enter the analytic function.

DP4. Even tensor powers give positive coefficients, without a positive form

All local and global L-functions in DP4-DP5 are complex series after coefficientwise application of ι . This includes the determinant on the left of (DP4.1).

Suppose that $\mathcal{F}_0$ is ι -real: the local polynomial $\iota \det(1 - F_x u \mid \mathcal{F}_{\bar{x}})$ has real coefficients at every closed point. Newton identities show that $\iota \text{Tr}(F_x^j \mid \mathcal{F}_{\bar{x}})$ is real for each $j \geq 1$. For a positive integer k , set $\mathcal{V}_0 = \mathcal{F}_0^{\otimes 2k}$. The exact local logarithm is

$$\log \det(1 - F_x t^{d_x} \mid \mathcal{V}_{\bar{x}})^{-1} = \sum_{j \geq 1} \frac{(\iota \text{Tr}(F_x^j \mid \mathcal{F}_{\bar{x}}))^{2k}}{j} t^{jd_x}. \quad (\text{DP4.1})$$

The trace identity used here is $\text{Tr}(A^{\otimes 2k}) = (\text{Tr } A)^{2k}$, proved by multiplying diagonal entries in a tensor basis. Every coefficient on the right is nonnegative. Exponentiating a series with nonnegative coefficients and zero constant term gives nonnegative coefficients and constant term 1, because each coefficient of $\sum_{r \geq 0} h^r / r!$ is a sum of nonnegative products.

Consequently each local factor $L_{x,k}(t)$ has nonnegative Taylor coefficients. The product of all other factors also has nonnegative coefficients and constant term 1. Coefficientwise,

$$0 \leq [t^n] L_{x,k}(t) \leq [t^n] L(\mathcal{F}_0^{\otimes 2k}, t). \quad (\text{DP4.2})$$

For any finite n , the assertion follows from the finite product of points of degree at most n , so there is no convergence assumption hidden in this formal inequality.

This is positivity of power-series coefficients obtained from **real traces at every closed point and every repetition**. No positive inner product on the sheaf has been assumed, and no reflection-positivity assertion about the programme has been inserted.

DP5. Pole control implies each eigenvalue inequality

For the zero sheaf there are no eigenvalues or constituents, so the assertions are vacuous. Assume the sheaf has positive rank before taking a largest determinant weight.

Let r be the largest determinant weight of $\mathcal{F}_0$. By(DP2.4), every constituent of $\mathcal{F}_0^{\otimes 2k}$ has determinant weight at most $2kr$. Remove a closed point different from the particular point x being tested, making the curve affine. The restriction retains the local eigenvalues at x and its determinant weights: the fundamental group of the nonempty open curve surjects onto that of the original normal curve, so constituents stay irreducible with the same determinant characters.

Equations(DP3.1-2) show that the global rational function has no pole in

$$|t| < R_k, \quad R_k = q^{-(2kr+2)/2}. \quad (\text{DP5.1})$$

This uses the full possible H_c^2 denominator. Cancellation with the numerator can only remove poles, so it cannot invalidate the asserted disc. The Taylor series of this rational function converges at every $0 \leq t < R_k$. Equation(DP4.2) and nonnegativity then make the local factor's Taylor series converge there too. It therefore has radius at least R_k .

If α is any eigenvalue of F_x on $\mathcal{F}_{\bar{x}}$, then α^{2k} is an eigenvalue on the tensor power, including its multiplicity. The reciprocal local determinant has a pole at every solution of

$$t^{d_x} = (\iota\alpha)^{-2k}, \quad \text{each of modulus } |\iota\alpha|^{-2k/d_x}. \quad (\text{DP5.2})$$

Its numerator is 1, so these poles cannot be cancelled locally. Radius at least R_k implies

$$|\iota\alpha|^{2k/d_x} \leq q^{(2kr+2)/2}, \quad w_{N(x)}(\alpha) \leq r + \frac{1}{k}. \quad (\text{DP5.3})$$

The same fixed eigenvalue satisfies(DP5.3) for every positive integer k . If its weight exceeded r by $e > 0$, an integer $k > 1/e$ would contradict(DP5.3). Thus

$$\boxed{w_{N(x)}(\alpha) \leq r.} \quad (\text{DP5.4})$$

This is Deligne1.5.2 with the analytic coefficient-domination step supplied in full. It uses all tensor powers to obtain an exact conclusion, not a finite numerical sample.

DP6. Exterior powers and determinant equality force every constituent to be pure

For each determinant weight β , let $n(\beta)$ be the total rank of the corresponding constituents, and α_i^β , $1 \leq i \leq n(\beta)$, their local eigenvalues at a fixed x , counted with multiplicity. From(DP1.5),

$$\sum_i w_{N(x)}(\alpha_i^\gamma) = n(\gamma)\gamma \quad \text{for every } \gamma. \quad (\text{DP6.1})$$

Fix an occurring β , and put $N_\beta = \sum_{\gamma > \beta} n(\gamma)$. The largest determinant weight of $\bigwedge^{N_\beta+1} \mathcal{F}_0$ is

$$r_\beta = \beta + \sum_{\gamma > \beta} n(\gamma)\gamma \quad (\text{DP6.2})$$

by(DP2.4): choose every higher-weight eigenvalue, then one at weight β . Exterior powers preserve the reality of the characteristic polynomial because their eigenvalues are products and complex conjugation permutes the original eigenvalue multiset. Hence DP5 applies to this exterior power. Its eigenvalue

$$\alpha_i^\beta \prod_{\gamma > \beta} \prod_j \alpha_j^\gamma \quad (\text{DP6.3})$$

has weight at most r_β . Subtract the equalities(DP6.1) for $\gamma > \beta$. This gives $w_{N(x)}(\alpha_i^\beta) \leq \beta$ for each i . Their sum is exactly $n(\beta)\beta$ by(DP6.1). If any individual inequality were strict, this sum would be strictly smaller. Therefore

$$\boxed{w_{N(x)}(\alpha_i^\beta) = \beta \quad \text{for every } x, \beta, i.} \quad (\text{DP6.4})$$

This completes the deduction of Deligne1.5.1. No chosen eigenvalue or constituent has been dropped.

DP7. The full real companion and its retained scalar

For a pointwise pure sheaf of integer weight n , the source real companion is

$$\mathcal{F}_0 \oplus \mathcal{F}_0^\vee(-n). \quad (\text{DP7.1})$$

At x , an eigenvalue a is paired with $N(x)^n/a$. Since $|\iota a|^2 = N(x)^n$, that second complex eigenvalue equals $\overline{\iota a}$. This proves reality and preserves the same weight; the Tate factor is essential.

For arbitrary real β in the category of Weil sheaves, let $\mathcal{L}_\beta$ be the constant Weil line whose geometric $\mathbb{F}_q$ -Frobenius is the scalar $\iota^{-1}(q^\beta)$. Its value at x is $N(x)^\beta$, and its weight is 2β . Then the exact companion is

$$\mathcal{F}_0 \oplus (\mathcal{F}_0^\vee \otimes \mathcal{L}_\beta). \quad (\text{DP7.2})$$

The proof is the same equality $N(x)^\beta/a = \overline{a}$ after ι . The scalar need not be an ℓ -adic unit, which is why Weil sheaves, rather than only étale sheaves, occur here. This construction starts with proved pointwise purity; it is not an argument establishing purity of an arbitrary sheaf merely by adjoining a dual.

DP8. What this calculation actually supplies to weight control

The proved chain is:

$$\begin{aligned} &\text{rank-one geometric finiteness} \longrightarrow \text{determinant weights and tensor rules} \\ &\longrightarrow \text{all-even-power coefficient positivity} \longrightarrow \text{global pole control} \\ &\longrightarrow \text{pointwise constituent purity.} \end{aligned} \quad (\text{DP8.1})$$

Every arrow has been calculated above. The curve's H_c^2 Tate twist supplies the $+2$; all powers make its contribution $1/k$, and the exact determinant sum supplies equality. This is one genuine weight-control mechanism. The later source proof uses the local monodromy filtration and a Lefschetz pencil to pass from pointwise sheaf purity to cohomological purity. Those steps have their own complete reconstructions;(DP8.1) does not replace them.

DP9. Named external inputs and reading limits

The rank-one argument uses Deligne1.3.1's explicitly cited global class-field isomorphism and the Picard variety. The central-element argument uses the root-datum description of automorphisms of a connected reductive group cited in1.3.10. The cohomological trace identity and duality are the theorems recalled in1.4.1-1.4.5. Their exact statements used here are displayed, and the ensuing deductions are proved. A complete reconstruction of the weight argument does not mean that these foundational theories have all been independently rebuilt. The wider full-source reading is recorded in DELIGNE_FULL_READING_LOG.md.

Monodromy groups, dominant pullback, and the étale criterion — DMG0-DMG7

Proof source: DELIGNE_MONODROMY_GROUP_SUPPLEMENT.md.

Deligne’s monodromy groups: dominant pullback, the radical, and extension to étale sheaves

Independent derivation, 24 September 2026. Labels DMG0–DMG7. This supplement supplies the remaining group-theoretic arguments in Weil II §1.3 requested after review of DP0–DP9. It does not modify the DP reconstruction or construct coefficient objects on $\tau\langle Z1; \text{ no } Z2 \rangle$.

DMG0. Source and exact dependencies

The source is Pierre Deligne, *La conjecture de Weil. II*, IHÉS 52 (1980), 137–252. Read for this supplement: the entire current French §1.3, printed pages 156–162, including the general rank-one variant on page 158, propositions (1.3.8), (1.3.13), (1.3.14), and lemma (1.3.15). The actual witness is

S20_FR_record_export.tex,

SHA256 d6836ae15d98f2b27eb98c9bd039a7d921679becd9fa47ff4717410289dbd351. This is the current page-record transcription, not author TeX. No PDF was consulted. Source lines 560–746 contain the relevant discussion, with the final paragraph of (1.3.15) at the start of printed page 162. The available transcription says “finite subgroup of Z ” in (1.3.10)(iv); the proof, corollary (1.3.11), and the direct derivation below give **finite index** instead. The witness is retained unchanged.

Foundational inputs are stated explicitly: global function-field class field theory and the Picard variety for rank one on curves; the curve-joining and Frobenius-density facts used in the source’s Katz variant for general normal schemes; elementary theory of algebraic groups in characteristic zero, including their radicals and central isogenies; normality and the generic-point description of finite étale covers; and local analytic exponential/logarithm for linear algebraic groups over a finite extension of $\mathbb{Q}_\ell$. The deductions required here are proved below.

Fix $q=p^f$, $p \neq \ell$. A connected normal finite-type F_q scheme is integral. Its field of global constants can be larger than F_q ; write it as $F_{q^{d_X}}$. Geometric Frobenius over that field has degree d_X relative to q . We retain these constant-field degrees when comparing two schemes.

For a Weil representation v on a geometrically connected base with finite constant field of cardinality q , its algebraic monodromy extension is

$$1 \longrightarrow G^0 \longrightarrow G \xrightarrow{\deg_Q} \mathbf{Z} \longrightarrow 0. \quad (\text{DMG0.1})$$

Here G^0 is the Zariski closure of geometric monodromy. A chosen degree-one lift w gives $G=G^0 \rtimes \langle w \rangle$; the action is conjugation by its actual representation matrix. The identity component of G^0 is denoted $G^{\{0\}}$. The superscript 0 in this source notation does not by itself mean connected. The map $G \rightarrow \text{GL}(V)$ is faithful on G^0 , but need not be faithful on all of G .

The determinant weight of an irreducible rank- r module is $\beta = (1/r)w_Q(c)$, where the determinant character is $c^{\deg_Q} \varepsilon$, ε finite order, and

$$w_Q(c) = 2 \log_Q |\iota c|.$$

DMG1. The rank-one input on a general normal base

Source: the variant following (1.3.4), printed page 158; the curve proof is (1.3.1)–(1.3.4), printed pages 156–158.

For a smooth geometrically connected curve, class field theory identifies the geometric part of the abelianized Weil group with the degree-zero idèle-class group allowing ramification at the finite boundary. Its quotient by the local boundary-unit groups is the rational point group of Pic^0 over the finite constant field and is finite. Each boundary-unit group has finite residue quotient and pro- p principal units. Therefore this geometric abelian group has an open pro- p subgroup.

The image of a rank-one ℓ -adic character lies in $E^\times$ for some finite extension $E/\mathbb{Q}_\ell$ and is compact on the geometric fundamental group. Its valuation image is a finite subgroup of $\mathbb{Z}$, hence zero, so it lies in $0_E^\times$, which has an open pro- ℓ subgroup. The intersection of the two open descriptions is both pro- p and pro- ℓ , hence trivial. The geometric image is finite.

Here is the general-base reduction used by the source. Let X_0 be normal and connected, and let χ be a rank-one Weil character with values in $E^\times$. Let $\mathfrak{M}$ be the finite group of roots of unity in E . For closed points x, y , the curve-joining fact used by Deligne and Katz gives a curve comparison on which the preceding result applies. It yields

$$\chi(F_x)^{\deg(y)} \chi(F_y)^{-\deg(x)} \in M, \quad (\text{DMG1.1})$$

or equivalently equality of $\chi(F_x)^{1/\deg x}$ and $\chi(F_y)^{1/\deg y}$ in $E^\times \otimes_{\mathbb{Z}} \mathbb{Q}$, written multiplicatively. Degrees here are relative to the same finite constant field, so all ratios have been retained.

For a degree-zero closed-point cycle $\sum_x n_x x$, equation (DMG1.1) implies

$$\chi \left(\prod_x F_x^{n_x} \right) \in M, \quad \sum_x n_x \deg(x) = 0. \quad (\text{DMG1.2})$$

The product is used only after applying the abelian character; no noncommuting Frobenius representatives are identified. The Frobenius-density statement used in the source says that these degree-zero products are dense in the geometric abelian image. By continuity and finiteness of $\mathfrak{M}$, the entire geometric image of χ lies in $\mathfrak{M}$.

Choose $h \geq 1$ killing that image. Then

$$\chi(g)^h = b^{\deg_Q g}, \quad c^h = b, \quad \chi(g) = c^{\deg_Q g} \varepsilon(g), \quad \varepsilon^h = 1. \quad (\text{DMG1.3})$$

This is the rank-one input for arbitrary normal bases used in the group arguments below. The general curve-joining and Frobenius-density inputs are part of this deduction's scope; the curve case requires neither additional input.

DMG2. Semisimple input, connected reductivity, and the finite-degree central element

Source: (1.3.9)–(1.3.12), printed pages 159–161. This also records the repaired disconnected-group step needed in DP2.

Suppose the Weil representation v is semisimple. The unipotent radical U of $G^{\{00\}}$ is characteristic in $G^{\{00\}}$, hence normal in G . In an irreducible G -constituent, the U -invariant subspace is nonzero because a unipotent group has a nonzero fixed vector. Normality makes that invariant subspace G -stable, so it is the entire constituent. Thus U acts trivially on v . Faithfulness on G^θ gives $U=1$, proving that $G^{\{00\}}$ is reductive.

Let τ_1 be its connected central torus. The finitely many characters by which τ_1 acts on v generate its character group, because the action is faithful. Conjugation by G permutes these characters. A finite-index subgroup consequently acts trivially on τ_1 . The outer automorphisms of a connected reductive group that act trivially on the connected centre form a finite group: after fixing a pinning, the remaining action is through the finite root datum. A further finite-index restriction makes the degree action on $G^{\{00\}}$ inner. Remove the finite geometric component group and correct the degree lift by an element of $G^{\{00\}}$. This yields a finite-index subgroup $G^{\{00\}} \times Z$.

The corresponding finite étale cover and finite extension of the constant field have geometric monodromy Zariski dense in $G^{\{00\}}$. Let τ be the maximal torus quotient of $G^{\{00\}}$, which is isogenous to τ_1 . A character of τ , extended trivially over the degree factor, gives a rank-one Weil character. Its geometric image is finite by DMG1 and Zariski dense in the character image of τ . A positive-dimensional torus cannot have a finite dense subset. Thus $\tau=1$, so $G^{\{00\}}$ is semisimple and $Z(G^{\{00\}})$ is finite.

Now derive a central element in the **original** G , rather than merely in the finite-index subgroup. Write $H=G^{\{00\}}$ and choose an original positive-degree lift w . Take a power w^m whose action on H is inner and whose induced action on the finite group G^θ/H is trivial. Choose $h_\theta \in H$ with $\text{Int}(w^m)|_H = \text{Int}(h_\theta)|_H$, and set

$$y = h_0^{-1} w^m. \quad (\text{DMG2.1})$$

Then y centralizes H and acts trivially on G^θ/H . If a is a representative of a component of G^θ ,

$$[y, a] \in H \quad \text{and} \quad \text{Int}([y, a])|_H = 1,$$

so $[y, a] \in Z(H)$. Also $[y, w] \in H$: this follows directly from $y = h_\theta^{-1} w^m$. The same conjugation calculation gives $[y, w] \in Z(H)$. The element y centralizes these commutators. Let $e \geq 1$ be an exponent of the finite group $Z(H)$. Then

$$[y^e, a] = [y, a]^e = 1, \quad [y^e, w] = [y, w]^e = 1.$$

Together with its centralization of H , this proves

$$z = y^e \in Z(G), \quad \deg_Q z = em > 0. \quad (\text{DMG2.2})$$

The finite group here is $G^\theta/G^{\{00\}}$, not the infinite group $G/G^{\{00\}}$. The kernel of $\deg_Q: Z(G) \rightarrow Z$ is finite: its intersection with H is contained in finite $Z(H)$, and its image in G^θ/H is finite. Its cokernel is finite by (DMG2.2).

For completeness, a rank-one relation such as (DMG1.3) extends from the Weil group to all of G . Its restriction $\chi^h=1$ on geometric monodromy extends to G^θ by Zariski density. For $g=g_\theta w^d$, this gives

$\chi(g)^h = b^d$. Therefore $\varepsilon(g) = \chi(g) c^{-d}$ is defined on G and satisfies $\varepsilon^h = 1$, even when g is not the image of an actual Weil element.

On an irreducible rank- r module, let z act by scalar u , and put $m_z = \deg_Q z$. Applying that extended determinant relation gives

$$u^r = c^{m_z} \varepsilon(z), \quad |\iota u| = Q^{m_z \beta/2}. \quad (\text{DMG2.3})$$

The same criterion holds constituent by constituent for a general finite-dimensional G -module. It is the precise scalar test used below, including its degree and rank factors.

DMG3. The full nonsplit radical theorem

Source: theorem (1.3.8) and its proof on printed page 160.

Let V now be any finite-dimensional Weil representation, with no semisimplicity assumption. Choose a composition filtration

$$0 = V_0 \subset V_1 \subset \cdots \subset V_s = V,$$

stable under the Weil group. The stabilizer $P \subset GL(V)$ of this filtration has the homomorphism

$$\pi : P \longrightarrow \prod_{a=1}^s GL(V_a/V_{a-1}), \quad (\text{DMG3.1})$$

whose kernel U_P consists of matrices acting as the identity on every graded quotient. In a basis adapted to the filtration, these are block upper triangular with identity diagonal blocks; U_P is unipotent.

The original geometric closure G^θ lies in P . Let G_1^θ be the geometric closure on $\text{Gr } V$. Then

$$\pi(G^\theta) = G_1^\theta, \quad \ker(\pi|_{G^\theta}) = G^\theta \cap U_P. \quad (\text{DMG3.2})$$

For the equality of images, the algebraic group image is closed and contains the dense geometric image defining G_1^θ ; conversely it is the image of the closure of that same geometric monodromy. The kernel is unipotent because it is an algebraic subgroup of U_P .

The associated graded representation is semisimple, so DMG2 gives that $(G_1^\theta)^{\{00\}}$ is semisimple. The map of identity components

$$G^{00} \twoheadrightarrow (G_1^\theta)^{00} \quad (\text{DMG3.3})$$

is surjective. One way to verify this is that the image is a connected normal closed subgroup, and the quotient of the target by it is an image of the finite component group of G^θ ; a connected finite algebraic group in characteristic zero is trivial.

Let $R = \text{Rad}(G^{\{00\}})$, the connected maximal solvable normal subgroup. Its image under (DMG3.3) is connected, solvable, and normal in a semisimple group, hence trivial. Consequently

$$R \subseteq G^\theta \cap U_P. \quad (\text{DMG3.4})$$

Every element of the latter group is unipotent, so R is unipotent. This proves the entire assertion (1.3.8): the radical of the connected geometric monodromy group is unipotent, including for nonsplit extensions. The proof keeps the original filtration and its unipotent kernel; it does not replace the original representation by its semisimplification and then transfer an unproved conclusion back.

DMG4. Dominant pullback gives a finite-index monodromy comparison

Source: the geometric group input for (1.3.13)(i), printed page 161.

Let $f: Y_\emptyset \rightarrow X_\emptyset$ be a dominant morphism between normal connected finite-type schemes over F_q . Let $K = k(X_\emptyset)$ and $L = k(Y_\emptyset)$. The algebraic closure of K inside the finitely generated extension L is finite over K ; its separable part K_s is also finite. After compatible geometric generic points are chosen, restriction of absolute Galois groups has image

$$\text{im}(\text{Gal}(L^{sep}/L) \rightarrow \text{Gal}(K^{sep}/K)) = \text{Gal}(K^{sep}/K_s). \quad (\text{DMG4.1})$$

The image is closed by compactness. Its fixed field inside $K^{\{sep\}}$ is exactly $K^{\{sep\}} \cap L = K_s$; Galois correspondence then proves the displayed equality. Its index is $[K_s:K]$, retaining the finite extension rather than declaring a surjection without qualification.

For a normal connected scheme, its generic-point Galois group surjects onto the étale fundamental group: a connected finite étale cover is normal and integral, and thus has connected generic fiber. The commutative diagram of generic and scheme fundamental groups therefore implies that

$$H = \text{im}(\pi_1(Y_0) \rightarrow \pi_1(X_0))$$

is an open subgroup, of index at most $[K_s:K]$. This is the needed finite-index assertion; injectivity is neither stated nor used.

Let the exact constant fields be $F_{\{q^{\{d_X\}}\}}$ and $F_{\{q^{\{d_Y\}}\}}$. Dominance embeds the first in the second, so d_X divides d_Y . Write w_q for the inverse image of $Z \subset \widehat{Z}$ using degree relative to F_q ; its degree image for $X_\emptyset$ is $d_X Z$, and for $Y_\emptyset$ it is $d_Y Z$. The induced map preserves that absolute degree. Thus

$$\text{im}(W_q(Y_0) \rightarrow W_q(X_0)) = H \cap W_q(X_0). \quad (\text{DMG4.2})$$

Indeed a lift in $\pi_1(Y_\emptyset)$ of an element whose degree is an integer already belongs to $w_q(Y_\emptyset)$. Conversely all Weil elements have integer degree. The index is finite. Intersecting once more with degree zero shows that the image of geometric $\pi_1(Y)$ has finite index in geometric $\pi_1(X)$.

For a representation v of $w_q(X_\emptyset)$, take the Zariski closures of these geometric images on the **same** vector space. The closure for $Y_\emptyset$ is an algebraic subgroup of finite index in the closure for $X_\emptyset$, so their identity components agree. The monodromy extensions with retained absolute degree give an injective comparison

$$G_Y \hookrightarrow G_X \quad (\text{DMG4.3})$$

onto a finite-index subgroup: injectivity follows because the map is the identity inclusion on the geometric closure and preserves degree, and finite index follows from the finite component quotient together with $[d_X Z: d_Y Z] = d_Y/d_X$. Choosing Weil lifts gives an explicit semidirect-product description, but the subgroup and the morphism do not depend on the chosen lifts.

DMG5. Determinant weights are preserved and detected by dominant pullback

Source: proposition (1.3.13)(i), printed page 161.

The assertion is:

V is purely of determinant weight $\beta \iff f^*V$ is purely of determinant weight β . (DMG5.1)

For clarity, “purely of determinant weight β ” means that **each nonzero irreducible constituent** has determinant weight β . It is not yet the assertion that every local eigenvalue has that weight.

First take v semisimple. Its restriction to a finite-index subgroup is semisimple. To verify this, pass to a finite-index normal core H_0 . The sum of simple H_0 -submodules of an irreducible w -module is nonzero and w -stable, hence the whole module. Thus restriction to H_0 is semisimple. For an intermediate finite-index subgroup H , average an H_0 -equivariant projection over the finite group H/H_0 ; characteristic zero permits division by its order. This supplies invariant complements and proves semisimplicity for H too.

Choose a central $z \in G_X$ of positive degree by DMG2. Some positive power z^a belongs to the finite-index subgroup G_Y . Let the absolute degree of z relative to q be m . Then its degrees relative to the two exact constant fields are

$$\deg_{q^{d_X}} z = \frac{m}{d_X}, \quad \deg_{q^{d_Y}} z^a = \frac{am}{d_Y}. \quad (\text{DMG5.2})$$

Both displayed numbers are integers where used. On an irreducible constituent of v , the element z acts by a scalar u . If that constituent has determinant weight β , (DMG2.3) gives

$$|\iota u| = (q^{d_X})^{(m/d_X)\beta/2} = q^{m\beta/2}. \quad (\text{DMG5.3})$$

Every irreducible constituent of its restriction to G_Y inherits the scalar u^a from z^a . Applying the same criterion over its exact constant field gives

$$|\iota u^a| = q^{am\beta/2} = (q^{d_Y})^{(am/d_Y)\beta/2}. \quad (\text{DMG5.4})$$

Its determinant weight is therefore precisely β . Conversely, if all restricted constituents have determinant weight β , choose a nonzero restricted constituent inside each original irreducible constituent. Equation (DMG5.4) then forces (DMG5.3), because $a > 0$; the original constituent has determinant weight β .

For nonsplit v , its composition filtration pulls back to an exact filtration. The multiset of constituents of f^*v is the union of the multisets obtained from its irreducible original quotients. Replace those quotients, only for this constituent calculation, by their direct sum; the semisimple argument just proved applies. It proves (DMG5.1) for the original v without an assertion that its extensions split.

The zero representation satisfies both sides vacuously for every β . No maximum of an empty weight set is taken.

DMG6. A compact Zariski-dense subgroup has compact normalizer in a semisimple local group

Source: lemma (1.3.15), printed pages 161–162. The proof here, as in the source, treats characteristic zero.

Let $E/\mathbb{Q}_\ell$ be finite, let H be a connected semisimple linear algebraic group over E , and let $K \subset H(E)$ be compact and Zariski dense. Then

$$N_{H(E)}(K) = \{g \in H(E) : gKg^{-1} = K\}$$

is compact.

Choose a sufficiently small analytic neighborhood of the identity on which exponential and logarithm are inverse. There is an open normal subgroup $K_0 \subset K$ contained in it: take the normal core in K of a sufficiently small open subgroup. The finite group K/K_0 has an exponent $n \geq 1$. For every $x \in K$, define

$$\log_K(x) = \frac{1}{n} \log(x^n) \in \text{Lie } H(E). \quad (\text{DMG6.1})$$

This is independent of the sufficiently divisible exponent chosen. On an analytic logarithm neighborhood, $\log(y^r) = r \log y$; applying that identity after two chosen exponents have been replaced by a common multiple proves independence. It is continuous on K , and commutes with conjugation wherever the conjugate also lies in a compact subgroup. In particular its compact image is invariant under the normalizer of K .

Let

$$\mathfrak{l} = \text{span}_E \log_K(K) \subseteq \text{Lie } H. \quad (\text{DMG6.2})$$

It is invariant under $\text{Ad } K$; the condition that a linear subspace be preserved is Zariski closed, so density makes it $\text{Ad } H$ -invariant. Hence it is an ideal of the semisimple Lie algebra. Such an ideal is the Lie algebra of a connected normal algebraic subgroup $H' \subset H$: after extending to an algebraic closure it is the sum of a set of simple ideals, hence of the corresponding simple factors up to central isogeny; Galois stability gives descent to E .

For the quotient map $\pi: H \rightarrow H/H'$, logarithms satisfy $\log(\pi(x)) = d\pi(\log(x))$ near the identity. The right side is zero for $x \in K$ in a sufficiently small neighborhood. The logarithm is injective there, so π kills an open subgroup of K . Thus $\pi(K)$ is finite. It is also Zariski dense in the connected group H/H' . A finite closed subset cannot be dense in a positive-dimensional connected algebraic group. Therefore $H/H' = 1$ and

$$\text{span}_E \log_K(K) = \text{Lie } H. \quad (\text{DMG6.3})$$

Let Λ be the 0_E -span of this compact logarithm set. It is bounded: choose an E -basis of $\text{Lie } H$; compactness bounds every coordinate of $\log_K(K)$ by one finite power of a uniformizer. Thus Λ lies in a fixed scalar multiple of the standard lattice. Since 0_E is noetherian, it is finitely generated; (DMG6.3) gives full rank. Therefore Λ is an 0_E -lattice. Every element normalizing K preserves Λ in both directions, so

$$\text{Ad}(N_{H(E)}(K)) \subseteq GL_{O_E}(\Lambda). \quad (\text{DMG6.4})$$

The group on the right is compact. The adjoint map has finite kernel and factors as the finite central isogeny $H \rightarrow H_{\text{ad}}$ followed by a closed embedding in $GL(\text{Lie } H)$. Its inverse image of a compact subset is compact. Here is a coordinate verification of that last assertion. On affine coordinate rings, every chosen coordinate generator x_j of $E[H]$ is integral over $E[H_{\text{ad}}]$, hence satisfies

$$x_j^{r_j} + a_{j,r_j-1} x_j^{r_j-1} + \cdots + a_{j,0} = 0, \quad (\text{DMG6.5})$$

where each $a_{j,i}$ is a regular function on H_{ad} . On a compact set $C \subset H_{\text{ad}}(E)$, put

$$M_j = \max \left(1, \max_{0 \leq i < r_j} \sup_{c \in C} |a_{j,i}(c)|_E \right). \quad (\text{DMG6.6})$$

If $|x_j|_E > M_j$, the leading term in (DMG6.5) strictly dominates every other term in the nonarchimedean absolute value, which is impossible. Thus every coordinate of every inverse-image point is bounded by

the displayed finite constant. The inverse image is closed and lies in a product of compact balls in the local field; it is compact.

Take $C = H_{\text{ad}}(E) \cap GL_{\{0_E\}}(\Lambda)$, a closed subset of a compact group. Equation (DMG6.4) puts the normalizer in its compact inverse image. Finally the normalizer is closed: if $g_a \rightarrow g$ with each $g_a K g_a^{-1} = K$, then for each $k \in K$, closedness of K gives $g k g^{-1} \in K$; applying the same argument to inverses gives equality. A closed subset of the compact inverse image is compact. This proves the lemma.

DMG7. The determinant criterion for an irreducible Weil sheaf to be étale

Source: proposition (1.3.14), printed page 161, using (1.3.15).

Let V be an irreducible lisse Weil sheaf of rank r on a normal connected finite-field scheme. Then

$$V \text{ extends to an étale sheaf} \iff \det V \text{ extends to an étale sheaf.} \quad (\text{DMG7.1})$$

Necessity follows because exterior power is a continuous polynomial operation on a continuous representation of the profinite fundamental group. We prove sufficiency, retaining the scalar and degree.

Use the exact constant field of cardinality $Q = q^{\{d_X\}}$ and its degree. Let G be the monodromy extension, choose $z \in Z(G)$ of positive degree m , and let its action on V be scalar u . Enlarge a finite coefficient field E/Q_ℓ so that the representation, the needed group equations, and z, u are all defined over E . Suppose $\det V$ is étale. Write its character using (DMG1.3) as $\chi = c^{\{\deg_Q\}} \varepsilon$, with $\varepsilon^h = 1$. An étale character has compact image on the profinite group, so c is an ℓ -adic unit. The exact determinant equation at z is

$$u^r = c^m \varepsilon(z). \quad (\text{DMG7.2})$$

Applying the discrete valuation, for which roots of unity have valuation zero, gives

$$r v_E(u) = m v_E(c) = 0. \quad (\text{DMG7.3})$$

Hence u is a unit.

The subgroup

$$G_1 = G^{00} z^{\mathbb{Z}} \cong G^{00} \times z^{\mathbb{Z}} \quad (\text{DMG7.4})$$

has finite index in G . The product is direct: z is central, and a power of nonzero-degree z lies in $G^{\{00\}}$ only when that power is zero. Let w_1 be its inverse image in the Weil group, and let $w_1^{\wedge 0}$ be the inverse image of $G^{\{00\}}$. Projection in (DMG7.4) defines

$$\rho : W_1 \longrightarrow G^{00}(E). \quad (\text{DMG7.5})$$

The subgroup $K = \rho(w_1^{\wedge 0})$ is compact, because $w_1^{\wedge 0}$ is a finite-index closed subgroup of geometric π_1 , and it is Zariski dense in $G^{\{00\}}$ by the definition of the monodromy closure and its identity component. Since $w_1^{\wedge 0}$ is normal in w_1 , its image K is normalized by $\rho(w_1)$. DMG6 makes this normalizer compact. Thus $\rho(w_1)$ is bounded.

For $w \in w_1$, the full original representation is

$$R(w) = \rho(w) u^{\deg_Q(w)/m}. \quad (\text{DMG7.6})$$

The exponent is an integer. Since u is a unit, its positive and negative powers lie in the compact group $O_E^\times$. Both factors in (DMG7.6) are bounded, so the image of w_1 is bounded. The entire Weil group is a finite union of its cosets, so its image is bounded as well.

Choose any lattice $L_0 \subset V_E$. Boundedness gives an integer $b \geq 0$ such that $R(w)L_0 \subset \pi_E^{-b}L_0$ for all w . Then

$$L = \sum_w R(w)L_0 \quad (\text{DMG7.7})$$

is a full-rank O_E -submodule between L_0 and $\pi_E^{-b}L_0$. It is finitely generated and hence a lattice; the group action permutes its defining summands, so it is stable. Thus the representation lands in the profinite group $GL(L)$.

To see the extension explicitly, select a geometric Frobenius lift F over the exact constant field. In the profinite arithmetic fundamental group, the map from Z taking 1 to F extends to a continuous map from $\widehat{Z}$, since the closure of the cyclic group generated by one element in a profinite group is procyclic. Composing with degree is the identity on $\widehat{Z}$, so this is a section. Hence

$$\pi_1(X_0) = \pi_1(X) \rtimes \widehat{Z}, \quad W(X_0) = \pi_1(X) \rtimes Z. \quad (\text{DMG7.8})$$

Similarly the powers of $R(F) \in GL(L)$ extend continuously from Z to $\widehat{Z}$. The compatibility of the conjugation action with the already continuous geometric representation holds for integer powers; density and continuity extend it to $\widehat{Z}$. Therefore

$$R(\gamma F^a) = R(\gamma)R(F)^a, \quad \gamma \in \pi_1(X), \quad a \in \widehat{Z}, \quad (\text{DMG7.9})$$

defines the required continuous extension. Density makes it unique. This proves sufficiency in (DMG7.1).

Finally every irreducible Weil sheaf is a retained constant twist of an étale sheaf. Let $Q = p^*g$ be the exact constant field and let c be its determinant scalar in (DMG1.3). Choose

$$b \in \overline{Q}_\ell^\times, \quad b^{rg} = c^{-1}. \quad (\text{DMG7.10})$$

Use the constant Weil line over F_p on which its geometric Frobenius acts by b . At a point of degree d over F_Q , this line acts by $b^{\{gd\}}$. Twisting v by it changes the determinant scalar from c to $cb^{\{rg\}}=1$, so the determinant character becomes exactly the finite-order character ε . By (DMG7.1), the twisted sheaf v_b is étale. The original object is recovered by the exact inverse twist

$$V = V_b \otimes \mathcal{L}_{b^{-1}}. \quad (\text{DMG7.11})$$

The determinant factor at a closed point is $b^{\{rgd\}}$, not $b^{\{rd/g\}}$. No factor, scalar, degree, or extension class has been suppressed. Equations (DMG7.10)–(DMG7.11) are a comparison with the original sheaf, not a replacement of it in the weight calculation.

Local monodromy and mixed primitive weights — DLM1-DLM15

Proof source: DELIGNE_LOCAL_MONODROMY_RECONSTRUCTION.md.

Deligne, Weil II §§1.6-1.9: local monodromy and exact weight control

Independent reconstruction, 24 September 2026. Result labels in this file are DLM1-DLM15. This file reconstructs the specified part of Deligne's argument; it does not assign a sheaf, Frobenius, nilpotent operator, weight, distance, or vector to the programme's $\tau\langle Z1; \text{ no } Z2\rangle$.

Source identity, reading coverage, and transcription corrections

The work is Pierre Deligne, *La conjecture de Weil. II*, Publications mathématiques de l'IHÉS **52** (1980), 137-252. The source used here is the current French page-record export:

S20_FR_record_export.tex

Its SHA256 is d6836ae15d98f2b27eb98c9bd039a7d921679becd9fa47ff4717410289dbd351. It is a transcription with explicit page records, not author-supplied TeX. The historical English math-mode file was used to compare ambiguous formulas, not to override the French witness. No PDF was read for this reconstruction.

Reading completed: every line of §§1.6-1.9, current export lines 874-1421, printed pages 165-181; the §1.9 heading begins on page 179. Dependencies actually read: conventions (0.6)-(0.11), printed page 145; §1.2, printed pages 153-155; §1.4, printed pages 162-163. The reconstruction uses Grothendieck's trace formula, quasi-unipotence, Abhyankar's lemma, and basic étale sheaf/cohomology results as the external inputs identified by Deligne. It does not claim a new proof of those foundations.

The witness is preserved unchanged. The following corrections are independently forced by the formulas and definitions; they are not accusations about Deligne's printed original:

1. In (1.6.7), the exception in $N_{e_i=e_{i-2}}$ must be $i \neq -d$, with $N_{e_{-d}}=0$; the witness writes $i \neq d$.
2. In (1.6.13)'s uniqueness proof the witness contains the impossible inequality $k-2i-2 \geq k$. The necessary and true inequality is $k-2i-2 \geq 2j-k$ when $k \geq i+1$ and $j \leq 0$. DLM4 proves it with the arbitrary top weight retained.
3. In (1.6.14.3), both witnesses display a positive twist $(i+j)/2$. With the explicitly stated convention $N:V(1) \rightarrow V$, the required formula is

$$\mathrm{Gr}_i^M V \cong \bigoplus_{j \geq |i|, j \equiv i \pmod{2}} P_{-j} \left(-\frac{i+j}{2} \right). \quad (\text{DLM-C1})$$

Indeed $N^r: \mathrm{Gr}_i V(r) \rightarrow P_{-j}$ with $r=(i+j)/2$ is the relevant isomorphism on the indicated summand. Its inverse supplies the negative twist. The witness's next assertion that both sides have equal weights independently requires this sign. DLM3 proves the formula, and DLM10 uses it without changing β .

4. In (1.7.5), the weight filtration is the **direct sum over $j \leq i$** of generalized eigenspaces of weight j . The witness's product over $j < i$ cannot have graded piece of weight i .

5. For geometric Frobenius F and $N:V(1) \rightarrow V$, $FN F^{-1} = Q^{-1}N$. Consequently the conjugation coefficient in the proof of (1.7.5), if $F'^{-n} = \exp(\lambda N) F'^n$, is $\mu = \lambda / (1 - Q^{-n})$. The witness displays $1 - q^n$. DLM6 includes the direct multiplication that checks this correction.
6. The denominator in (1.8.1.1) is H_c^2 , as in the current French text. The historical English H^0 is not used. DLM8 retains all three compact-support degrees before proving the degree-zero factor is 1 in the auxiliary affine curve.

DLM1. The monodromy filtration and its primitive pieces

Source: (1.6.1)–(1.6.7), printed pages 165–166.

Let V be an object of an abelian category and let $N:V \rightarrow V$ be nilpotent. There is a unique finite increasing filtration M satisfying

$$NM_i \subseteq M_{i-2}, \quad N^k : \mathrm{Gr}_k^M V \xrightarrow{\sim} \mathrm{Gr}_{-k}^M V \quad (k \geq 0). \quad (\text{DLM1.1})$$

Here a finite filtration is exhaustive and zero in sufficiently low degree. The subscript 0 is a filtration index. It is not a base point of the scheme, nor the centre of a geometric circle.

Existence and uniqueness. Induct on an integer $d \geq 0$ for which $N^{d+1} = 0$. For $d=0$, the conditions force $M_{-1} = 0, M_0 = V$. For $d > 0$, the extreme pieces must be

$$M_d = V, \quad M_{d-1} = \ker N^d, \quad M_{-d} = \mathrm{im} N^d, \quad M_{-d-1} = 0. \quad (\text{DLM1.2})$$

To see that they are forced, (DLM1.1) and $N^{d+1} = 0$ imply $\mathrm{Gr}_k = 0$ for $k > d$; symmetry gives the same for $k < -d$. On the remaining top graded piece, N^d is an isomorphism to the bottom graded piece. Its kernel in V is therefore M_{-d-1} , and its image is M_{-d} . On

$$\ker N^d / \mathrm{im} N^d$$

the induced N^d is zero. Its monodromy filtration is already constructed uniquely by induction. Pull its pieces back into $\ker N^d$, and attach (DLM1.2). The lowering condition and the isomorphisms follow on the middle graded pieces by induction and on the extreme pieces from the image/coimage isomorphism for N^d . This proves both assertions.

Set

$$P_i(V) = \ker(N : \mathrm{Gr}_i^M V \rightarrow \mathrm{Gr}_{i-2}^M V). \quad (\text{DLM1.3})$$

For $i > 0$, N is injective on Gr_i , because its composite with N^{i-1} is the isomorphism N^i ; hence $P_{-i} = 0$. For $j \geq 0$, the composite

$$\mathrm{Gr}_{j+2}^M V \xrightarrow{N^{j+1}} \mathrm{Gr}_{-j}^M V \xrightarrow{N} \mathrm{Gr}_{-j-2}^M V$$

is an isomorphism. Therefore

$$\mathrm{Gr}_{-j}^M V = P_{-j} \oplus N^{j+1} \mathrm{Gr}_{j+2}^M V. \quad (\text{DLM1.4})$$

Iterating this direct sum, then using $N^i : \mathrm{Gr}_i \rightarrow \mathrm{Gr}_{-i}$, yields, for a chosen actual endomorphism N ,

$$\mathrm{Gr}_i^M V \cong \bigoplus_{j \geq |i|, j \equiv i (2)} P_{-j}. \quad (\text{DLM1.5})$$

The map $N: (V, M) \rightarrow (V, M \text{ shifted by } 2)$ is strict:

$$N(M_{i+2}) = \mathrm{im} N \cap M_i. \quad (\text{DLM1.6})$$

If $i < 0$, its associated graded onto M_i is surjective by (DLM1.5), and finite filtrations imply surjectivity. If $i+2 > 0$, the induced map $V/M_{i+2} \rightarrow V/M_i$ is injective on associated graded, hence injective. These cases cover every integer i . Thus kernels commute with passage to the associated graded:

$$\mathrm{Gr}_i^M(\ker N) \cong P_i. \quad (\text{DLM1.7})$$

For finite-dimensional vector spaces, a Jordan string of length $d+1$ has vectors

$$e_d, e_{d-2}, \dots, e_{-d}, \quad Ne_i = e_{i-2} \ (i > -d), \quad Ne_{-d} = 0.$$

Its filtration is $M_i = \mathrm{span}\{e_j : j \leq i\}$. Direct sums of these strings give the entire filtration. The string's primitive vector is its **lowest** vector e_{-d} . This specifies the sign of every primitive index above.

DLM2. The $\mathrm{SL}(2)$ construction, tensor products, and duals

Source: (1.6.8)-(1.6.12), printed pages 167-168.

Now work over a characteristic-zero field. On a string of length $d+1$, put $e_r = e_{-d+2r}$ for $0 \leq r \leq d$. Define

$$He_r = (-d + 2r)e_r, \quad Fe_r = e_{r-1}, \quad Ee_r = (d - r)(r + 1)e_{r+1}, \quad (\text{DLM2.1})$$

with the out-of-range vectors zero. Then $F=N$, and direct substitution gives

$$[H, E] = 2E, \quad [H, F] = -2F, \quad [E, F] = H.$$

This is the irreducible $\mathrm{SL}(2)$ representation $S_d = \mathrm{Sym}^d(k^2)$ in a rescaled basis. On the associated graded, the grading itself prescribes H , and the primitive decomposition prescribes these strings. Thus one obtains the $\mathrm{SL}(2)$ representation of (1.6.10), with the diagonal matrix $\mathrm{diag}(\lambda, \lambda^{-1})$ acting by λ^i on $\mathrm{Gr}_i^M V$. This representation is on the associated graded; no choice of a splitting of V is being declared canonical.

For (V', N') and (V'', N'') , use exactly

$$V = V' \otimes V'', \quad N = N' \otimes 1 + 1 \otimes N''. \quad (\text{DLM2.2})$$

On the tensor product $\mathrm{SL}(2)$ representation, diagonal weights add and the lowering generator is precisely (DLM2.2). Its increasing weight filtration therefore satisfies (DLM1.1); uniqueness gives

$$M_i(V' \otimes V'') = \sum_{a+b=i} M_a V' \otimes M_b V''. \quad (\text{DLM2.3})$$

The dual operator is $-N^\wedge$, not $N^\wedge$: differentiation of the contragredient action introduces the minus sign. Diagonal weights negate, so

$$M_i(V^*) = (M_{-i-1}V)^\perp. \quad (\text{DLM2.4})$$

For finite filtrations, choosing splittings verifies the natural isomorphisms

$$\text{Gr}(V' \otimes V'') \cong \text{Gr } V' \otimes \text{Gr } V'', \quad \text{Gr}_i(V^*) \cong (\text{Gr}_{-i} V)^*;$$

the maps themselves are induced by classes and pairings and do not depend on the splittings used to verify bijectivity.

Choose a nonzero lowest-weight vector in each S_d , and choose the Clebsch-Gordan isomorphisms

$$S_{d'} \otimes S_{d''} \cong \bigoplus_{j \in P(d', d'')} S_j, \quad P(d', d'') = \{j : |d' - d''| \leq j \leq d' + d'', j \equiv d' + d'' \pmod{2}\}. \quad (\text{DLM2.5})$$

These decompositions follow by multiplying the characters $z^d + z^{d-2} + \dots + z^{-d}$ and subtracting successively the highest character. Every listed summand occurs once. Complete reducibility in characteristic zero makes this a direct sum of representations. The isomorphisms involve the stated choices; the source does not assert a unique scalar for every Clebsch-Gordan component.

The lowest-weight inclusion gives

$$\text{Gr}^M V \cong \bigoplus_{j \geq 0} S_j \otimes P_{-j}, \quad P_{-j} \cong \text{Hom}_{SL(2)}(S_j, \text{Gr}^M V). \quad (\text{DLM2.6})$$

Applying (DLM2.5) to (DLM2.6) gives the primitive tensor decomposition. Twists must be restored when n belongs to a nontrivial line, as follows.

DLM3. The monodromy line and every Tate twist

Source: (1.6.14), printed pages 169-170.

Let $\mathfrak{N}$ be a one-dimensional vector space acting nilpotently on v , and write $v(k) = v \otimes \mathfrak{N}^{\wedge k}$; negative k uses the dual line. The action is a map $\mathfrak{N} : v(1) \rightarrow v$. Choosing a basis in $\mathfrak{N}$ turns it into an endomorphism. Changing that basis multiplies the endomorphism by a nonzero scalar, leaving M and P_i unchanged. Iterated action is the basis-independent map

$$N^k : \text{Gr}_{a+k}^M \text{Gr}_a^W V(k) \longrightarrow \text{Gr}_{a-k}^M \text{Gr}_a^W V. \quad (\text{DLM3.1})$$

The primitive decomposition, tensor primitive decomposition and dual primitive decomposition are

$$\text{Gr}_i^M V \cong \bigoplus_{j \geq |i|, j \equiv i \pmod{2}} P_{-j} \left(-\frac{i+j}{2} \right), \quad (\text{DLM3.2})$$

$$P_{-j}(V' \otimes V'') \cong \bigoplus_{j \in P(j', j'')} P_{-j'}(V') \otimes P_{-j''}(V'') \left(\frac{j - j' - j''}{2} \right), \quad (\text{DLM3.3})$$

$$P_{-j}(V^*) \cong P_{-j}(V)^*(j). \quad (\text{DLM3.4})$$

To check (DLM3.2), let $r=(i+j)/2$. The j -string in degree i maps isomorphically to its lowest vector by N^r after tensoring the source with $\mathfrak{N}^{\otimes r}$. Solving for the source gives twist $-r$. To check (DLM3.3), the component of length j arises from the two strings by $r=(j'+j''-j)/2$ contractions; under rescaling N , its comparison with the chosen primitive vectors has exactly r inverse powers of the monodromy-line basis. This gives twist $-r$. Equivalently, retain the strings as S_j with the monodromy-line grading: the primitive degree $-j$ is obtained from degree $-j'-j''$ by raising $2r$, and each raising contributes $\mathfrak{N}^{-1}$. For the dual, the lowest vector comes from the dual of the highest vector, which by (DLM3.2) is the dual of $P_{-j}(-j)$; this is $P_{-j}^*(j)$.

In the geometric application, $\mathfrak{N}=\overline{\mathbf{Q}}_{\ell(1)}$, geometric Frobenius acts on it by Q^{-1} , and its weight is -2 . The weight checks on (DLM3.2)–(DLM3.4) are then respectively

$$-j - 2\left(-\frac{i+j}{2}\right) = i, \quad -j' - j'' - 2\left(\frac{j-j'-j''}{2}\right) = -j, \quad j - 2j = -j.$$

These checks do not replace the maps; they verify the twists in the maps just constructed.

DLM4. Uniqueness of relative monodromy, with arbitrary top weight retained

Source: (1.6.13), printed pages 168–169.

Let w be a finite increasing filtration on V , preserved by N . There is at most one finite increasing M with

$$NM_i \subseteq M_{i-2}, \quad N^k : \mathrm{Gr}_{a+k}^M \mathrm{Gr}_a^W V \xrightarrow{\sim} \mathrm{Gr}_{a-k}^M \mathrm{Gr}_a^W V \quad (a \in \mathbf{Z}, k \geq 0). \quad (\text{DLM4.1})$$

This is a uniqueness assertion; existence for a mixed sheaf is proved in DLM11 and DLM14, not inserted here as an assumption about arbitrary filtered objects.

Induct on the length of w . Let a be its top nonzero degree, so $w_a = V$. The restriction to w_{a-1} is unique by induction; on $\mathrm{Gr}_a^W V$, the filtration is the absolute monodromy filtration centred at a . Let its range be contained in $[a-c, a+c]$. For $r \geq 0$, a putative filtration satisfies

$$\begin{aligned} M_{a-r} &= M_{a-r}(W_{a-1}) & (r > c), \\ M_{a-r} &= M_{a-r}(W_{a-1}) + N^r M_{a+r}, \\ M_{a+r} &= \ker(N^{r+1} : V \rightarrow V/M_{a-r-2}). \end{aligned} \quad (\text{DLM4.2})$$

The first formula says that the quotient has no degree that low. For the second, the lowering condition gives inclusion of the right side into the left, while the induced N^r maps the quotient's M_{a+r} onto its M_{a-r} , as one sees on each string centred at a . For the third, one inclusion follows from lowering. On $G = \mathrm{Gr}_j^W V$, $j \leq a$, the reverse inclusion follows because

$$N^{r+1} : G/M_{a+r}G \rightarrow G/M_{a-r-2}G$$

is injective. On a degree $k > a+r$, the central isomorphism runs from k to $2j-k$. The shorter iterate by $r+1$ remains injective since

$$k - 2r - 2 \geq 2j - k,$$

which follows from $k \geq a+r+1$ and $j \leq a$. Passage to Gr_j^W of a kernel injects into the corresponding kernel; finite filtrations then give the required inclusion in V . Starting in sufficiently large r and descending, the

three formulas determine alternately $M_{\{a+r\}}$ and $M_{\{a-r\}}$. This proves uniqueness without replacing a by zero.

DLM5. Local inertia and its logarithm

Source: (1.7.1)–(1.7.3), printed pages 170–171.

Let R be a henselian discrete valuation ring, with fraction field K , residue field k , and compatible algebraic closures. The exact sequence and tame quotient are

$$\begin{aligned} 1 \rightarrow I \rightarrow \text{Gal}(\bar{K}/K) \rightarrow \text{Gal}(\bar{k}/k) \rightarrow 1, \\ 1 \rightarrow P \rightarrow I \xrightarrow{t} \prod_{\ell' \neq p} \mathbf{Z}_{\ell'}(1) \rightarrow 1. \end{aligned} \quad (\text{DLM5.1})$$

Here p is the residue characteristic exponent and P is pro- p . Fix $\ell \neq p$; the ℓ -component is $t_{\ell}: I \rightarrow \mathbf{Z}_{\ell}(1)$.

For a quasi-unipotent ℓ -adic representation ρ of I , there is a unique nilpotent line action

$$N : V(1) \longrightarrow V \quad \text{with} \quad \rho(\sigma) = \exp(N t_{\ell}(\sigma)) \quad (\text{DLM5.2})$$

on a sufficiently small finite-index subgroup of I . The exponential and logarithm are finite polynomials on nilpotent/unipotent matrices. The prime-to- ℓ kernel cannot have a nontrivial image in a sufficiently small torsion-free pro- ℓ unipotent group. Thus this restriction factors through $\mathbf{Z}_{\ell}(1)$. Taking the logarithm and dividing by a nonzero element of an open subgroup determines the unique linear action of $\bar{Q}_{\ell}(1)$. This also shows independence of the finite-index subgroup.

If k is finite with q elements, the local Weil group is the inverse image of the integer powers of geometric Frobenius. Its representations satisfy quasi-unipotence, the theorem of Grothendieck cited by Deligne. The Frobenius conjugation explains the finite-order semisimple part: after removing the finite wild and prime-to- ℓ images, a tame generator has its eigenvalue multiset preserved by the q -power map. Each eigenvalue consequently has finite multiplicative order; a common positive power is unipotent. This is compatible with (DLM5.2), whose coordinate-free N is Weil-equivariant.

Choosing a basis for the Tate line displays that equivariance as

$$F N F^{-1} = Q^{-1} N. \quad (\text{DLM5.3})$$

The nilpotent operator is not an involution. Except for the zero operator, a nilpotent operator cannot square to the identity. Its exponential is the unipotent part of monodromy, and all its tensor and dual operations have been specified above.

DLM6. Independence of Frobenius lift and the local weight filtration

Source: (1.7.4)–(1.7.7), printed pages 171–172.

If F' and F'' are lifts of the same residue Frobenius, their eigenvalues differ by roots of unity. One can pass to semisimplification for this eigenvalue assertion. There the inertia image is finite: an irreducible constituent's $\ker N$ is nonzero and Weil-stable by (DLM5.3), so it is the whole constituent and $N=0$. In a representation with finite inertia image, choose a power of Frobenius that centralizes that finite image, then a further power killing the finite discrepancy between the two lifts. The resulting powers of F' and F'' agree. Their eigenvalues before powering therefore differ by roots of unity.

Thus

$$w_Q(\alpha) = 2 \log_Q |\iota \alpha|$$

is independent of the lift. Suppose these weights are integers. For a chosen lift, let v_{-j}' be the sum of its generalized eigenspaces of weight j and set

$$M_i' = \bigoplus_{j \leq i} V_j'. \quad (\text{DLM6.1})$$

The operator N takes v_{-j}' to v_{-j-2}' by (DLM5.3). For two lifts, choose $n > 0$ for which their powers differ by unipotent inertia:

$$F''^n = \exp(\lambda N) F'^n.$$

All powers of N commute. Direct multiplication gives

$$\begin{aligned} \exp(\mu N) F'^n \exp(-\mu N) &= \exp(\mu N) \exp(-\mu Q^{-n} N) F'^n \\ &= \exp(\mu(1 - Q^{-n})N) F'^n. \end{aligned} \quad (\text{DLM6.2})$$

Take $\mu = \lambda / (1 - Q^{-n})$. Passing from F' to F'^n does not change the partition into weights, because $w_{\{Q^n\}}(\alpha^n) = w_Q(\alpha)$. The operator $\exp(\mu N)$ preserves each M_i' ; therefore the filtrations obtained from the two lifts coincide. Conjugating a lift by any Weil-group element gives another lift, so M is Weil-stable. Uniqueness follows from the decomposition into generalized eigenspaces: a filtration with pure graded weights must place the generalized weight- j summand in degree j .

In particular $N M_i(1) \subset M_{i-2}$. A locally pure representation has a single weight, so $N=0$ and inertia has finite image. This does **not** say that the local representation at the boundary of a pointwise pure sheaf is pure. DLM10 calculates its several weights $\beta+i$; nonzero local monodromy is exactly compatible with those shifts.

Without the integral-weight restriction, grouping eigenvalues modulo $Q^{\mathbb{Z}}$ times roots of unity gives the canonical decomposition in (1.7.7). The same conjugation proof works: multiplication of an eigenvalue by Q^{-1} keeps its class, so N preserves each class and the exponential comparisons preserve the decomposition. Each class can be twisted by a rank-one character to obtain integral weights. This last comparison retains the twisting scalar; it is not a replacement of the original representation in DLM10.

DLM7. The tame normal-crossing specialization functor and its exact compatibility

Source: (1.7.8)–(1.7.12), printed pages 172–174.

Let X be regular, let $D = \bigcup_{i \in I} D_i$ be a normal-crossing divisor with smooth transverse components, and put $E = \bigcap D_i$. Locally choose equations $t_i = 0$. For n invertible on X , form

$$X_n = X[T_i : i \in I] / (T_i^n - t_i : i \in I). \quad (\text{DLM7.1})$$

Over $X \rightarrow D$ this is an étale cover, with $\mu_n \times I$ acting by $T_i \mapsto \zeta_i T_i$. Above E the reduced section is $T_i = 0$. For a locally constant finite sheaf tamely ramified through primes in L , Abhyankar's lemma supplies such an n after which the pullback extends locally constantly over X_n . Restrict that extension to the section over E ; its inherited $\mu_n \times I$ action defines $\mathcal{F}[E]$. Passing through compatible finite levels gives the ℓ -adic construction and an action of

$$\mathbf{z}_L(1)^I = \left(\prod_{\ell' \in L} \mathbf{z}_{\ell'}(1) \right)^I.$$

This functor is exact on the stated locally constant sheaves: after a common cover trivializes a finite-level sequence, extension and restriction are exact operations on the corresponding constant modules. The same argument with compatible lattices gives the $\bar{q}_\ell$ functor. Tensor products and duals are likewise preserved. The commuting unipotent logarithms are

$$N_i : \mathcal{F}[E](1) \rightarrow \mathcal{F}[E], \quad \rho(\sigma) = \exp \left(\sum_i N_i \sigma_i \right) \quad (\text{DLM7.2})$$

on an open subgroup.

For $J \subset I$, define

$$D_J = \bigcap_{j \in J} D_j, \quad X_J = X \setminus \bigcup_{i \notin J} D_i, \quad D_J^* = D_J \cap X_J.$$

For $K \subset I - J$, restriction of a common Kummer extension first to the J -stratum and then to the $J \cup K$ -stratum equals direct restriction. Hence there is an actual isomorphism

$$\mathcal{F}[D_J^*][D_{J \cup K}^*] \xrightarrow{\sim} \mathcal{F}[D_{J \cup K}^*] \quad (\text{DLM7.3})$$

preserving every inertia coordinate in $J \cup K$. For three nested strata the two composites agree because both are restriction of the same sheaf on the same Kummer cover. This transitivity will be used, not merely an equality of dimensions.

The equation-dependent construction has an intrinsic version on the normal bundle of E with its component hyperplanes removed. A choice of the τ_i gives a section of that bundle; its pullback recovers $\mathcal{F}[E]$. The construction commutes with smooth base change. This retains the dependence on a transverse direction instead of deleting it.

In a henselian local neighborhood, the tame fundamental group is an extension of the residue Galois group by $\mathbb{Z}_\ell(1)^\wedge I$. Choosing compatible roots of the τ_i splits it: use the lifts fixing those roots. The functor above is precisely restriction along this splitting together with the retained inertia action. Over a finite residue field, the arguments of DLM5–DLM6 apply with commuting N_i and show quasi-unipotence, independence of weights from the lift, and existence of the local weight filtration. In (DLM6.2), one replaces λN and μN by sums of the commuting N_i , with the same factor $1 - q^{-n}$ in every coordinate.

DLM8. Boundary weights from the full trace formula and all tensor powers

Source: (1.8.1), printed page 175, using (1.4.1)–(1.4.7), printed pages 162–163.

Let $X_\emptyset$ be a smooth absolutely irreducible curve over F_q , $j : U_\emptyset \hookrightarrow X_\emptyset$ the complement of a finite set $S_\emptyset$, and $\mathcal{F}_\emptyset$ lisse and pointwise ι -pure of weight β on $U_\emptyset$. At a closed point x , put $N(x) = q^{\deg x}$. For every eigenvalue α of F_x on $(j_* \mathcal{F}_\emptyset)_{\bar{x}}$,

$$|\iota \alpha| \leq N(x)^{\beta/2}. \quad (\text{DLM8.1})$$

Here is the complete amplification argument. For the local assertion at the finitely many points in S_0 , one can delete an additional point from U_0 to work on an affine open of X_0 . This auxiliary passage leaves the sheaf and the boundary stalks being asserted unchanged; it is not a deletion of a boundary factor. On that affine curve, the full trace formula is

$$\prod_{x \in |U_0|} \iota \det(1 - F_x t^{\deg x}, \mathcal{F}_{0, \bar{x}})^{-1} \prod_{x \in |S_0|} \iota \det(1 - F_x t^{\deg x}, (j_* \mathcal{F}_0)_{\bar{x}})^{-1} \\ = \frac{\iota \det(1 - Ft, H_c^1(X, j_* \mathcal{F}))}{\iota \det(1 - Ft, H_c^0(X, j_* \mathcal{F})) \iota \det(1 - Ft, H_c^2(X, j_* \mathcal{F}))}. \quad (\text{DLM8.2})$$

The degree-zero compact-support space vanishes: any such section is zero on a nonempty open of the connected affine curve, and a section of $j_* \mathcal{F}$ injects into its generic fiber. Its determinant factor is consequently exactly 1. The degree-two space is

$$H_c^2(X, j_* \mathcal{F}) \cong H_c^2(U, \mathcal{F}) \cong V_{\pi_1(U)}(-1), \quad (\text{DLM8.3})$$

where geometric coinvariants are used. The geometrically constant quotient of the pure sheaf is pure of weight β ; the displayed (-1) raises the weight by 2. Thus every denominator eigenvalue in degree two has modulus $q^{(\beta+2)/2}$. There is no pole of the right side in

$$|t| < q^{-(\beta+2)/2}. \quad (\text{DLM8.4})$$

The first Euler product on the left is analytic and nonzero there. Indeed the number of degree- n closed points is at most Cq^n ; the rank is finite; and each local eigenvalue has modulus $q^{n\beta/2}$. The logarithmic expansion is absolutely convergent when $q^{(1+\beta/2)}|t| < 1$, by comparison with the geometric series, including the higher powers in the logarithm. Each boundary reciprocal determinant can only have poles, not zeros. Since neither the first Euler product nor another boundary reciprocal polynomial can cancel a boundary pole, (DLM8.2)-(DLM8.4) give

$$|\iota \alpha| \leq N(x)^{(\beta+2)/2}. \quad (\text{DLM8.5})$$

For every positive integer r , the actual stalk inclusion is

$$(V^I)^{\otimes r} \hookrightarrow (V^{\otimes r})^I, \quad \text{equivalently} \quad (j_* \mathcal{F}_0)^{\otimes r} \hookrightarrow j_*(\mathcal{F}_0^{\otimes r}). \quad (\text{DLM8.6})$$

It is generally not surjective, and no such surjectivity is used. The tensor-power sheaf has pointwise weight $r\beta$. If α occurs in V^I , an eigenvector tensor power shows that α^r occurs in the left side of (DLM8.6), hence also in the right side. Applying the weak estimate to this sheaf gives

$$|\iota \alpha|^r \leq N(x)^{(r\beta+2)/2}, \quad w_{N(x)}(\alpha) \leq \beta + \frac{2}{r}. \quad (\text{DLM8.7})$$

This holds for every r . If the weight exceeded β by $\epsilon > 0$, an integer $r > 2/\epsilon$ would contradict (DLM8.7). This proves (DLM8.1) exactly, with no finite tensor cutoff and no change of the original β .

DLM9. The auxiliary degree-one bound

Source: (1.8.2), printed page 175.

In (DLM8.4) the left side of the trace formula has no zeros. Its degree-two denominator is nonzero there. Consequently the numerator cannot vanish there. Every eigenvalue α on $H_c^1(X, j_* \mathcal{F})$ therefore satisfies

$$|\iota\alpha| \leq q^{(\beta+2)/2}. \quad (\text{DLM9.1})$$

This is the auxiliary bound stated at this stage, not the later sharp curve theorem. Its exact role here is distinct from the local tensor-power argument, which already removed the +2 at boundary stalks.

DLM10. Local purity: tensor square and duality force the exact modulus

Source: theorem (1.8.4), printed pages 175–176; primitive maps (1.6.14.2), (1.6.14.4), (1.6.14.5).

Choose $s \in S_0$ with residue field of cardinality $Q = N(s)$, a geometric point over it, and the geometric generic point of its strict henselization. Put $v = \mathcal{F}_{\{0, \bar{\eta}\}}$ and let M be its local monodromy filtration. For every integer i ,

$$\text{Gr}_i^M V \text{ is } \iota\text{-pure of weight } \beta + i \text{ relative to } Q. \quad (\text{DLM10.1})$$

A finite cover makes inertia unipotent. This comparison preserves all relevant weights: a residue extension of degree f replaces Q by Q^f and an eigenvalue by its f th power, up to the finite inertia factors; the logarithmic weight is unchanged. A ramification index e replaces N by eN ; the monodromy filtration and its primitive subspaces are unchanged because $e \neq 0$ in the characteristic-zero coefficient field. Thus it suffices to prove the assertion on this cover, with the field cardinality, weight β , and scaling of N retained in the comparison. No twist to weight zero is used below.

For unipotent inertia,

$$V^I = \ker N = (j_* \mathcal{F}_0)_{\bar{s}}.$$

By DLM8 every Frobenius eigenvalue on this space has modulus at most $Q^{\beta/2}$. Since $P_{\{-j\}} = \text{Gr}_{\{-j\}}^M(\ker N)$, the same bound initially holds for α on $P_{\{-j\}}$.

Now apply the statement to $\mathcal{F}_0 \otimes \mathcal{F}_0$, whose weight is 2β . The primitive tensor map (DLM3.3), with $j' = j$ and target index zero, includes the direct summand

$$P_{-j}(V) \otimes P_{-j}(V)(-j) \hookrightarrow P_0(V \otimes V). \quad (\text{DLM10.2})$$

The eigenvalue on the tensor square is α^2 ; the $(-j)$ twist multiplies it by Q^j . The boundary estimate for weight 2β therefore gives

$$|\iota(\alpha^2 Q^j)| \leq Q^\beta, \quad |\iota\alpha| \leq Q^{(\beta-j)/2}. \quad (\text{DLM10.3})$$

Next apply that same established upper bound to the **dual sheaf of weight $-\beta$** . By the actual dual primitive isomorphism (DLM3.4), the corresponding eigenvalue is

$$\alpha^{-1} Q^{-j} \text{ on } P_{-j}(V^*).$$

Its upper bound is $Q^{(-\beta-j)/2}$. Hence

$$|\iota\alpha|^{-1}Q^{-j} \leq Q^{(-\beta-j)/2}, \quad |\iota\alpha| \geq Q^{(\beta-j)/2}. \quad (\text{DLM10.4})$$

Together (DLM10.3) and (DLM10.4) force

$$\boxed{|\iota\alpha| = Q^{(\beta-j)/2} \quad \text{on } P_{-j}.} \quad (\text{DLM10.5})$$

Finally the degree- i summand in (DLM3.2) carries the multiplier $Q^{(i+j)/2}$. Thus each of its eigenvalues has modulus

$$Q^{(\beta-j)/2}Q^{(i+j)/2} = Q^{(\beta+i)/2}. \quad (\text{DLM10.6})$$

This proves (DLM10.1). The centre of this weight statement is the retained number β ; it is obtained through exact eigenvalue multipliers and the two opposite inequalities, not by choosing a spatial origin.

DLM11. Mixed relative monodromy and the smooth-divisor case

Source: (1.8.5)–(1.8.8), printed pages 176–177.

Suppose a lisse sheaf has a finite increasing filtration w by lisse subsheaves, with $\text{Gr}_a^w \mathcal{F}_0$ pointwise pure of weight a . Its local representation has integral weights: apply DLM10 to every graded quotient and then combine their eigenvalue multisets. DLM6 therefore gives an actual local weight filtration M . The operator N lowers it by two. On each $\text{Gr}_a^w V$, DLM10 says its monodromy filtration shifted by a is exactly this weight filtration. Consequently

$$N^b : (\text{Gr}_{a+b}^M \text{Gr}_a^W V)(b) \xrightarrow{\sim} \text{Gr}_{a-b}^M \text{Gr}_a^W V. \quad (\text{DLM11.1})$$

The use of Gr^w here is compatible with the weight filtration: a Frobenius-equivariant linear map preserves generalized weight spaces, and taking their direct sums is exact. Thus the graded object in (DLM11.1) is the same whether one first passes to Gr^w or first takes the weight filtration. DLM4 proves uniqueness. This proves existence and identification of the relative monodromy filtration, rather than assuming it.

For a smooth divisor D_0 in a smooth finite-field scheme X_0 , assume tame ramification and the same lisse filtration w on the complement. The sheaf $\mathcal{F}_0[D_0]$ and N are furnished by DLM7. At each closed point of D_0 , choose an étale transverse curve. Its local representation is exactly the corresponding fiber of $\mathcal{F}_0[D_0]$ with its tame logarithm. DLM11.1 supplies the fiber filtration; uniqueness makes it invariant under parallel transport and therefore a lisse filtration. Its degree- a graded sheaf is pointwise pure of weight a .

For a pointwise pure sheaf of weight β , the actual boundary sheaf is obtained by taking the invariants of the finite residual inertia action on $\ker N$. Since its order is invertible in $\bar{\mathbb{Q}}_\ell$, averaging makes the invariants functor exact. The filtration therefore descends, with

$$\text{Gr}_i^M(j_* \mathcal{F}_0)_{\bar{s}} \text{ pure of weight } \beta + i, \quad i \leq 0,$$

and zero for $i > 0$. The same assertion holds along a smooth divisor. With mixed input, the resulting boundary filtration has its degree- a quotient pure of weight a .

DLM12. Extension across an open immersion and propagation of purity

Source: (1.8.9)–(1.8.13), printed pages 177–178.

For any finite-type F_q scheme and open immersion $j:U_\emptyset \hookrightarrow X_\emptyset$, an $\imath$ -mixed sheaf of pointwise weights $\leq \beta$ has $j_* \mathcal{F}_\emptyset$ of the same weight bound. Integral weights remain integral.

The source's reduction is induction on $\dim U_\emptyset$, with the following exact maps retained:

- For an extension $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$, left exactness gives $0 \rightarrow j_* \mathcal{F}' \rightarrow j_* \mathcal{F} \rightarrow j_* \mathcal{F}''$; the quotient of the middle by the first is a subsheaf of the last. Bounds and integrality pass to these subquotients.
- A locally closed support $i:V_\emptyset \hookrightarrow U_\emptyset$ with $\mathcal{F} \hookrightarrow i^* \mathcal{F}$ reduces the assertion to the closure of $V_\emptyset$ and $i^* \mathcal{F}$.
- For a finite surjection $\varepsilon:X'_\emptyset \rightarrow X_\emptyset$, adjunction and the nonempty geometric fibers give $j_* \mathcal{F} \hookrightarrow \varepsilon_* j'_* \mathcal{F}'$. Finite pullback and pushforward preserve the weight condition. For pushforward, the Frobenius blocks correspond to residue extensions, and taking the appropriate power converts eigenvalues to those of the original closed-point fiber; $w_{\{q^f\}}(\alpha^f) = w_q(\alpha)$ verifies the unchanged weights.
- If $\mathcal{F}$ is lisse and $k:V_\emptyset \hookrightarrow U_\emptyset$ is dense, $j_* \mathcal{F} \hookrightarrow (jk)_* k^* \mathcal{F}$. One can shrink until the complement in a normal ambient scheme is a divisor and the sheaf has constant rank and pure weight.
- If $X_\emptyset$ is normal, let $i:F_\emptyset = X_\emptyset - U_\emptyset \hookrightarrow X_\emptyset$ and choose a dense open $k:V_\emptyset \hookrightarrow F_\emptyset$. Then $i^* j_* \mathcal{F} \hookrightarrow k_* k^* i^* j_* \mathcal{F}$. At a geometric point x of the boundary, choose successive generalizations $x \rightarrow y \rightarrow z$ into $V_\emptyset$ and $U_\emptyset$. The strict-local complement is connected by normality, so the boundary stalk injects into the generic lisse stalk. This injection factors through the stalk of $k_* k^* i^* j_* \mathcal{F}$, proving the claimed injectivity.

Stratification and the first two steps reduce to pure lisse input; normalization and a finite cover remove wild ramification at the finitely many generic boundary divisors. Shrinking around these generic points gives the smooth-divisor situation of DLM11. The remaining boundary has smaller dimension, so the last injection and the induction hypothesis extend the bound to it. This is (1.8.9); the finite-cover and constructibility inputs are those cited by Deligne, not new assumptions about the programme's base.

For a lisse sheaf on $X_\emptyset$ that is pure of weight β on a dense open, the injections of the sheaf and of its dual into their open direct images give respectively weights $\leq \beta$ and $\geq \beta$. Thus it is pure everywhere. On a normal connected scheme an irreducible lisse sheaf stays irreducible on every nonempty open; a mixed such sheaf is therefore pure after shrinking, and hence everywhere. Taking a composition series proves that a mixed lisse sheaf on a normal scheme has lisse pure successive quotients.

If a mixed lisse sheaf on a connected scheme is pure at one point, normalization shows purity of that same weight on every irreducible component through that point. Repeating across components makes the pure locus both open and closed; connectedness forces it to be the entire scheme. These arguments preserve integral weights when the input weights are integral, and hold for all embeddings $\imath$ when the input is pure rather than merely $\imath$ -pure.

The Hodge-theoretic text (1.8.14)–(1.8.15), printed pages 178–179, formulates an **open problem in that text** about good mixed Hodge variations on a punctured disk, relative monodromy, and nilpotent orbits. It is not an additional proved theorem in these sections. The requested relative filtration is precisely the displayed formula (DLM11.1) without the finite-field Tate-line notation, and the suggested nilpotent-orbit comparison uses $\exp(-uN)$ on the Hodge filtration. No solution of that historical problem is imported into this reconstruction.

DLM13. Positive combinations of tame logarithms are realized by actual curves

Source: (1.9.1), (1.9.3), printed pages 179–181.

Let X be smooth over F_q , $D = \bigcup D_i$ a normal-crossing divisor as in DLM7, and $\mathcal{F}$ lisse and tame on $X - D$. Assume its given filtration w has pointwise pure degree- a quotients of weight a . Put $G = \mathcal{F}[E]$ and keep all maps $N_i : G(1) \rightarrow G$.

For each family of **positive integers** c_i , set

$$N_c = \sum_{i \in I} c_i N_i. \quad (\text{DLM13.1})$$

This operator is not chosen on grounds of numerical positivity. It is obtained from a concrete pullback. At a closed point $x \in E$, make a finite extension of the ground field so that x is rational, retaining the Frobenius-power comparison. Complete the equations t_i to étale local coordinates (t_i, u_j) . Map an affine line with parameter t to the coordinate affine space by

$$t \mapsto (t_i = t^{c_i}, u_j = 0). \quad (\text{DLM13.2})$$

The fiber product with the chosen étale neighborhood of x is a smooth curve near the selected point: its projection to the affine line is étale by base change. On the punctured curve, its generic stalk is $G_{\{\bar{x}\}}$. Pulling back a Kummer root of t_i gives a root of t^{c_i} . Thus a tame loop in the curve maps to c_i times that loop in the i th inertia coordinate. Taking logarithms in (DLM7.2) gives **exactly** (DLM13.1), including every coefficient.

DLM11 applied to this actual curve proves that N_c has relative monodromy filtration on $G_{\{\bar{x}\}}$, whose degree- a piece is pure of weight a . The uniqueness from DLM4 makes the fiber filtration invariant under the lisse parallel-transport action, so it forms a filtration by lisse subsheaves of G . At every closed point it is characterized as the filtration by Frobenius weights. It therefore does not depend on the chosen positive integers c_i .

For positive rational c_i , multiply all coefficients by one positive integer m to make them integral. Replacing N_c by mN_c changes the b th iterated map by the nonzero scalar m^b and hence changes neither the filtration nor any assertion that the map is an isomorphism. This proves the same result for all positive rational coefficient vectors. No assertion about arbitrary real, negative, or vanishing coefficients follows from this argument.

DLM14. The complete system of relative filtrations on all strata

Source: theorem (1.9.2), proof (1.9.4), printed pages 180–181.

For filtrations w', w'' on a sheaf and a family N_α , Deligne's condition $L(w', w'', (N_\alpha))$ means:

1. every N_α preserves w' ;
2. $N_\alpha w''(1) \subset w'_{a-2}$;
3. for every family of positive rational c_α and every $b \geq 0$,

$$\left(\sum_{\alpha} c_{\alpha} N_{\alpha} \right)^b : \left(\text{Gr}_{a+b}^{w''} \text{Gr}_a^{w'} G \right)(b) \xrightarrow{\sim} \text{Gr}_{a-b}^{w''} \text{Gr}_a^{w'} G. \quad (\text{DLM14.1})$$

There exists a unique system $w(J)$, indexed by **every** subset $J \subset I$, on $G = \mathcal{F}[E]$, stable under the full $Z_L(1)^{\wedge I}$ action, such that

$$W(\emptyset) = W[E], \quad L(W(J), W(K), (N_k)_{k \in K-J}) \quad (J \subsetneq K). \quad (\text{DLM14.2})$$

Furthermore $\text{Gr}_a^{\wedge \{W(I)\}} G$ is pointwise pure of weight a .

Construction. Apply DLM13 on x_J along the components indexed by J . It gives a weight filtration on $\mathcal{F}[D_J^*]$, call it $w(J)$, relative to the original w and to all positive combinations of N_j , $j \in J$. For $J \subset K$, apply DLM13 again on D_J along the additional components $K-J$, now with input filtration $w(J)$. It gives a weight filtration on

$$\mathcal{F}[D_J^*][D_K^*].$$

The exact specialization isomorphism (DLM7.3) identifies this sheaf, its Frobenius and each inertia operator with $\mathcal{F}[D_K^*]$. The filtration just constructed and the filtration obtained directly for K are both its Frobenius weight filtration, so DLM6 identifies them. This proves (DLM14.1) on the K -stratum. Specialize once more to E , using exactness of the functor to preserve the graded isomorphisms. Define the resulting filtration on G to be $w(J)$. This proves every compatibility in (DLM14.2), and transitivity of (DLM7.3) proves independence of the sequence of strata.

Uniqueness. The relative-monodromy condition for $\emptyset \subset J$, with one positive coefficient vector, already forces $w(J)$ by DLM4. Thus two systems agree subset by subset. Invariance under the remaining inertia operators also follows from uniqueness: they preserve the original filtration and commute with the chosen logarithm, so they take its relative filtration to another one with the same defining properties.

The construction treats all allowed positive combinations simultaneously. It does not derive a universal positivity form on an arbitrary space from the existence of a fixed supporting point.

DLM15. What §1.9's final remarks establish, and the exact mechanism received here

Source: (1.9.5)–(1.9.6), printed page 181.

Deligne announces a fiberwise variant for a smooth family $x \rightarrow S$ over a finite-type Z scheme with a **relative** normal-crossing divisor and a tame lisse sheaf. The text says this is checked on fibers using (1.9.2); it does not state a complete separate list of filtration hypotheses and conclusions. This reconstruction does not manufacture a stronger global theorem from that remark. The same paragraph explicitly distinguishes the case of a normal-crossing divisor in a regular arithmetic scheme, including $\text{Spec } Z[1/\ell]$, as inaccessible to that argument. The distinction is part of the original source, not a verdict about the programme.

The added-in-proof remark attributes to Cattani and Kaplan the corresponding **pure** Hodge-variation statement on $(D^*)^{\wedge I}$, where positive real coefficients are permitted. Their proof is not reproduced in §§1.6–1.9 and has not been read for this file; no claim of that further reconstruction is made.

The exact weight-control mechanism now reconstructed is the following chain of actual maps and conclusions:

$$\begin{aligned}
 & \text{full curve trace formula, including } H_c^0, H_c^1, H_c^2 \\
 \Downarrow & \text{ all tensor powers through } (V^I)^{\otimes r} \hookrightarrow (V^{\otimes r})^I \\
 & |\iota\alpha| \leq Q^{\beta/2} \text{ on boundary invariants} \\
 & \Downarrow P_{-j}^{\otimes 2}(-j) \hookrightarrow P_0(V^{\otimes 2}) \\
 & |\iota\alpha| \leq Q^{(\beta-j)/2} \\
 & \Downarrow P_{-j}(V^*) \cong P_{-j}(V)^*(j) \\
 & |\iota\alpha| \geq Q^{(\beta-j)/2} \\
 & \Downarrow \text{Gr}_i V \cong \bigoplus_j P_{-j}(-(i+j)/2) \\
 & |\iota\lambda| = Q^{(\beta+i)/2} \text{ on every degree-}i \text{ graded piece.}
 \end{aligned} \tag{DLM15.1}$$

For mixed tame input, the independently constructed curves $t_i = t^{\wedge \{c_i\}}$ realize all positive rational combinations of logarithms. Weight-filtration uniqueness and the exact specialization isomorphisms then

identify their relative filtrations. Those are the objects and maps that have actually been calculated here. In particular, this local theorem neither chooses an origin called zero nor obtains its radius by replacing a zero with another supporting point.

All-return positivity and boundary nonvanishing — DB0-DB10, DBC1-DBC11

Proof source: DELIGNE_BOUNDARY_NONVANISHING_RECONSTRUCTION.md.

Deligne's boundary nonvanishing argument and its exact original-zeta specialization

24 September 2026. Bounded reconstruction of *La conjecture de Weil. II*, §§2.1-2.2, printed pp. 187-196. This is a mathematical derivation from the current French transcription, not original author LaTeX and not a new edition of Deligne's text. No PDF was opened.

DB0. Source identity, scope, and corrected witness formulas

The read source is `output/Deligne_Weil_II_S20_LaTeX/typed_latex/S20_FR_record_export.tex`, lines 1553-1928, read in full. It is the current S20 editable French page-record export. Its SHA-256 is `d6836ae15d98f2b27eb98c9bd039a7d921679becd9fa47ff4717410289dbd351`. Human mathematical source: Pierre Deligne, *La conjecture de Weil. II*, Publications mathématiques de l'IHÉS 52 (1980), pp. 137-252, §§2.1-2.2. The routed canonical source IDs are PUBUNIT-3D90B487DBD1CC3A259C6455 and PUBUNIT-574B0214207BBB2BDABCFFF9. This file reports the mathematical content reconstructed from that transcription and identifies corrections forced by its definitions. It does not attribute transcription errors to Deligne's original printing.

Three formulas require explicit control:

1. With $\omega_s(g) = q^{-s \deg g}$, its period in s is $2\pi i / \log q$, not $2\pi i \log q$.
2. A representation of real part b , with this character convention, has associated sheaf weight $-2b$, not $+2b$. The derivation below retains the sign through the original weight β .
3. The concentration proof of §2.1.7 needs its outside-neighborhood contribution. The guaranteed lower bound is $[(1 - e_1)(1 - e_2) - e_2] \dim(\tau)M$; choosing $e_1 + 2e_2 < e$ gives the stated lemma.

The original-zeta specialization begins only after the whole arithmetic and its prime spectrum have been recovered. It uses all primes and all positive prime powers, and does not pool counters of distinct branches. No arithmetic on τ is introduced. A boundary result is proved on its actual domain; no new RH conclusion is drawn.

DB1. The representation family and its sign convention (§2.1.1)

Let

$$1 \longrightarrow K \longrightarrow G \longrightarrow \Gamma \longrightarrow 0 \quad (\text{DB1.1})$$

be an extension of a group Γ isomorphic to $\mathbb{Z}$ or $\mathbb{R}$ by a compact group K . All finite-dimensional representations are continuous complex representations. The assumed structure is a direct-product

extension for $\Gamma \simeq \mathbb{R}$, and, for $\Gamma \simeq \mathbb{Z}$, a central element whose nonzero degree generates a subgroup of finite index. Fix a nontrivial positive real character ω_1 of Γ and put $\omega_s = \exp(s \log \omega_1)$. For indexed conjugacy classes x_v , let

$$N_v = \omega_{-1}(x_v) > 1. \quad (\text{DB1.2})$$

The discrete convention is the unique choice of $q > 1$ and degree isomorphism with

$$\omega_s(g) = \exp(-s \deg(g) \log q), \quad N_v = q^{\deg v}, \quad \omega_{s+2\pi i/\log q} = \omega_s. \quad (\text{DB1.3})$$

Let z be central with nonzero image in Γ . The quotient $G/\langle z \rangle$ is compact. A representation on which z is unitary has an invariant positive Hermitian form: first choose a z -invariant form and then average its translates over this compact quotient. The averaged form is well defined because the initial form is invariant under z . Conversely a unitary representation has a unitary z -action.

For an irreducible representation τ , Schur's lemma makes $\tau(z)$ scalar. There is a unique real number b with $|\tau(z)| = \omega_b(z)$; then $\tau\omega_{-b}$ is unitarizable by the preceding averaging. Define $\text{Re}(\tau) = b$. In particular

$$\text{Re}(\tau\omega_s) = b + \text{Re } s, \quad |\lambda(\tau(x_v))| = N_v^{-b} \quad (\text{DB1.4})$$

for each eigenvalue after the unitary twist. The last equality retains the negative exponent fixed by ω_1 .

The twisting orbits $\{\tau\omega_s : s \in \mathbb{C}\}$ form the components of the source's representation Riemann surface $\widetilde{G}$. In the discrete case any twisting stabilizer is discrete: if $d = \dim \tau$ and $\tau\omega_s \simeq \tau$, determinants give $\omega_{ds} = 1$, hence $s \in (2\pi i/(d \log q))\mathbb{Z}$. In the real case the same determinant identity forces $s = 0$. Thus the components are complex lines or their indicated discrete imaginary quotients. The locus of unitary irreducibles is denoted $\widehat{G}$.

The source's continuous example §2.1.2 keeps an arbitrary compact factor K , with each class written (x_v^0, x_v^Γ) ; its positive norm is still exactly $N_v = \omega_{-1}(x_v^\Gamma)$. Its curve example §2.1.3 takes the Weil extension of the geometric fundamental group, all closed points as Σ , and their geometric Frobenius classes, so $N_v = q^{\deg v}$. The full Weil extension need not satisfy the central-degree condition. The source explicitly notes this limitation. The positivity and residue proofs DB2-DB6 use the unitary representations and their twists, so do not themselves use that central-degree condition; the discrete translated-measure assertion DB8 does use a central element. The algebraic compact monodromy construction in the curve argument below supplies it on its stated receiver. No central element in an arbitrary full Weil group is inferred.

DB2. Euler products and the entire prime-power measure (§§2.1.1-2.1.4)

Assume the source's convergence condition

$$\prod_v (1 - N_v^{-s})^{-1} \quad \text{converges absolutely for } \text{Re } s > 1. \quad (\text{DB2.1})$$

In the discrete case this is equivalent to the source's bound on the entire degree counts $A_n = \#\{v : \deg v = n\}$: for every $\epsilon > 0$, there is a finite C_ϵ with $A_n \leq C_\epsilon q^{(1+\epsilon)n}$. Indeed, the logarithm of the Euler product converges exactly when $\sum_n A_n q^{-\sigma n}$ does, since $q^{-\sigma n} \leq q^{-\sigma} < 1$ and $x \leq -\log(1-x) \leq x/(1-q^{-\sigma})$. Convergence at $\sigma = 1 + \epsilon$ bounds each term by the full sum. Conversely

the count bound with $0 < \epsilon < \sigma - 1$ dominates the sum by a convergent geometric series. Every local degree and its multiplicity remain in A_n .

For an irreducible τ with $\operatorname{Re} \tau > 1$, define

$$L(\tau) = \prod_v \det(1 - \tau(x_v))^{-1}, \quad L(\tau, s) = L(\tau \omega_s). \quad (\text{DB2.2})$$

Each local eigenvalue has modulus $N_v^{-\operatorname{Re} \tau} < 1$. The identity $-\log \det(1 - M) = \sum_{n \geq 1} \operatorname{Tr}(M^n)/n$, proved by summing $-\log(1 - \lambda)$ over eigenvalues, and the bound $|\operatorname{Tr} \tau(x_v^n)| \leq (\dim \tau) N_v^{-n \operatorname{Re} \tau}$ give locally uniform convergence of the logarithm. Thus L is holomorphic and nonzero in the asserted open half-plane. The same construction extends multiplicatively to direct sums and by quotients to virtual representations.

For a unitary virtual representation ρ , its character is a finite integral linear combination of characters of unitary representations. It is bounded. Differentiation gives the full formula

$$\Lambda_\sigma(\rho) := -\frac{L'}{L}(\omega_\sigma \rho) = \sum_v \sum_{n=1}^{\infty} (\log N_v) N_v^{-n\sigma} \chi_\rho(x_v^n), \quad \sigma > 1. \quad (\text{DB2.3})$$

There is no omitted prime-power term and no factor $1/n$ remaining: differentiating N_v^{-ns}/n supplies $-\log N_v$. To justify derivative convergence, choose $1 < \sigma_0 < \sigma$. The inequality $(\log N_v) e^{-n(\sigma-\sigma_0) \log N_v} \leq C/n$ reduces the absolute derivative sum to the convergent logarithm in (DB2.1) at σ_0 . The same argument works locally uniformly.

Define the positive finite measure on conjugacy classes

$$\mu_\sigma = \sum_v \sum_{n=1}^{\infty} (\log N_v) N_v^{-n\sigma} \delta_{[x_v^n]}. \quad (\text{DB2.4})$$

Its total mass is the convergent scalar sum obtained above. Equation (DB2.3) is exactly

$$\Lambda_\sigma(\rho) = \int \chi_\rho d\mu_\sigma. \quad (\text{DB2.5})$$

For every unitary virtual ρ , the conjugate representation has character $\overline{\chi_\rho}$, so

$$\Lambda_\sigma(\rho \otimes \bar{\rho}) = \int |\chi_\rho(g)|^2 d\mu_\sigma(g) \geq 0. \quad (\text{DB2.6})$$

The representation may be virtual: its character is still a function, and its product with its complex conjugate is still the displayed nonnegative square. More generally any virtual character which is real and pointwise nonnegative has nonnegative integral. These are different statements from requiring every virtual representation to have a nonnegative character.

DB3. Boundary residues (§§2.1.4-2.1.5)

The analytic input of Deligne's theorem is: L is meromorphic on a neighborhood of $\operatorname{Re} \tau \geq 1$, is holomorphic there except for a simple pole at ω_1 , and has no other poles in that region. Zeros on the boundary have not been excluded as an input.

For $\tau \in \widehat{G}$, define the integer

$$\nu(\tau) = \text{ord}_{\text{pole}, s=1} L(\tau\omega_s). \quad (\text{DB3.1})$$

A zero of multiplicity m has $\nu = -m$. Writing $L(\tau\omega_s) = (s-1)^{-\nu(\tau)}u(s)$ with u holomorphic and nonzero proves

$$-\frac{L'}{L}(\tau\omega_s) = \frac{\nu(\tau)}{s-1} - \frac{u'(s)}{u(s)}, \quad \nu(\tau) = \lim_{\sigma \downarrow 1} (\sigma-1)\Lambda_\sigma(\tau). \quad (\text{DB3.2})$$

The analytic hypotheses give $\nu(1) = 1$ and $\nu(\tau) \leq 0$ for $\tau \neq 1$. Conjugation of the convergent product gives $\overline{L(\tau\omega_s)} = L(\bar{\tau}\omega_{\bar{s}})$; meromorphic continuation then yields $\nu(\bar{\tau}) = \nu(\tau)$. Extending ν additively to virtual representations and taking the limit in the finite sum of (DB2.6) gives

$$\nu(\rho \otimes \bar{\rho}) \geq 0 \quad (\text{DB3.3})$$

for every unitary virtual ρ , and hence for every genuine unitary ρ . The limit multiplies by the positive number $\sigma - 1$; no sign is reversed.

DB4. Reduction to a compact group (§§2.1.5-2.1.6)

Map G into the product of the unitary groups $U(V_\tau)$ of its finite-dimensional unitary representations, and let K_B be the closure of its image. Each unitary-group factor is compact, so K_B is compact. An individual representation image need not itself be closed, which is why the ambient unitary groups and the closure are specified. Every unitary representation of G extends through its coordinate projection to K_B . Conversely restriction of a continuous representation of K_B to the dense image of G preserves invariant subspaces and intertwiners: the relevant equations are closed conditions, so equations on a dense subset hold on the closure. It therefore preserves irreducibility and all multiplicities. Every irreducible representation of K_B restricts to a finite-dimensional unitary representation of G , and uniqueness of extension follows from density. Thus the finite-dimensional unitary representation categories, their conjugation, and their tensor products agree for the purpose of (DB3.3). This comparison is not an assertion about nonunitarizable representations of G .

The remaining proof may consequently be made on a compact group K_B , denoted K_c below. Haar measure dg has total mass one. For characters, use the explicit convention

$$[\eta : \tau] = \int_{K_c} \chi_\eta(g) \overline{\chi_\tau(g)} dg. \quad (\text{DB4.1})$$

For genuine representations this is the integer multiplicity, by character orthogonality; for virtual representations it is the corresponding integer coefficient. Its complex conjugate is the same real integer, which explains the alternative ordering seen in the source. Complex conjugation must not simply be deleted for a general complex integrand.

DB5. Complete concentration lemma (§2.1.7)

Let T be a finite set of irreducible representations of K_c , and $0 < e < 1$. We construct a genuine representation ρ satisfying

$$[\rho \otimes \bar{\rho} : \tau] \geq (1 - e) \dim \tau [\rho \otimes \bar{\rho} : 1] \quad (\tau \in T). \quad (\text{DB5.1})$$

For $e \geq 1$ any nonzero genuine representation already suffices, so this covers the substantive range. Choose $e_1 = e_2 = e/4$. Continuity of the finitely many characters at the identity supplies an open, conjugation-invariant, inversion-invariant neighborhood V such that

$$|\chi_\tau(g) - d_\tau| < e_1 d_\tau \quad (g \in V, \tau \in T), \quad d_\tau = \dim \tau. \quad (\text{DB5.2})$$

This can be obtained directly by intersecting the sets defined by these character inequalities and their inverses. Choose a nonzero nonnegative continuous central inversion-invariant function b supported in V . Existence follows by choosing a smaller such neighborhood with closure in V , a continuous bump function there, and averaging over conjugation and inversion; its value at the identity can be kept positive. Add a sufficiently small positive constant δ . The function $b + \delta$ is strictly positive and has arbitrarily small fraction of its squared mass outside V , since the numerator there is at most δ^2 , while its total squared mass tends to the positive number $\int b^2$.

Finite complex linear combinations of irreducible characters are uniformly dense in the continuous central functions of a compact group. Applying that character-density theorem and then taking the real part and averaging under inversion gives a finite character combination f which is real, inversion-invariant, uniformly arbitrarily close to $b + \delta$, and hence strictly positive. Both strict positivity and the desired strict mass inequality persist under sufficiently small uniform perturbation. Choose the approximations with enough room that

$$M := \int_{K_c} |f|^2 dg > 0, \quad \int_{K_c \setminus V} |f|^2 dg \leq e_2 M. \quad (\text{DB5.3})$$

The density theorem used here is the central-function consequence of the Peter-Weyl theorem; it is a standard compact-representation dependency, not an arithmetic positivity hypothesis.

Write $f = \sum_\chi c_\chi \chi$ over its finite support. Each coefficient is real. Indeed by (DB4.1), reality and inversion invariance give

$$\overline{c_\chi} = \int f(g) \chi(g) dg = \int f(g^{-1}) \chi(g^{-1}) dg = \int f(g) \overline{\chi(g)} dg = c_\chi. \quad (\text{DB5.4})$$

Reality also gives $c_{\bar{\chi}} = c_\chi$. Approximate the finitely many coefficients by rationals, taking the same rational for conjugate pairs. This preserves reality and inversion invariance. By choosing the approximation close enough, strict positivity and (DB5.3) remain true. Multiplication by a common positive denominator now gives an integral virtual character

$$\rho_0 = \sum_\chi n_\chi \chi = \rho^+ - \rho^-, \quad \rho^+ = \sum_{n_\chi > 0} n_\chi \chi, \quad \rho^- = \sum_{n_\chi < 0} (-n_\chi) \chi. \quad (\text{DB5.5})$$

Both ρ^+ and ρ^- are genuine representations; their irreducible supports are disjoint. The common multiplication rescales both integrals in (DB5.3) by the same square, so the relative estimate persists. Set $\rho = \rho^+ + \rho^-$, a genuine representation. The exact tensor identity is

$$\rho \otimes \bar{\rho} = \rho_0 \otimes \bar{\rho}_0 + 2(\rho^+ \otimes \bar{\rho}^- + \rho^- \otimes \bar{\rho}^+). \quad (\text{DB5.6})$$

The added representations have nonnegative multiplicities. Their trivial multiplicities are zero because the two irreducible supports are disjoint. Therefore

$$[\rho\bar{\rho} : \tau] \geq [\rho_0\bar{\rho}_0 : \tau], \quad [\rho\bar{\rho} : 1] = [\rho_0\bar{\rho}_0 : 1] = M. \quad (\text{DB5.7})$$

Here the notation $\rho\bar{\rho}$ means the displayed tensor product, and M is now the squared mass of the scaled ρ_0 .

The remaining bound must include the outside integral. The inner product is a real integer; taking its real part gives

$$\begin{aligned} [\rho_0\bar{\rho}_0 : \tau] &= \operatorname{Re} \int_{K_c} |\rho_0(g)|^2 \overline{\chi_\tau(g)} dg \\ &\geq (1 - e_1) d_\tau \int_V |\rho_0|^2 dg - d_\tau \int_{K_c \setminus V} |\rho_0|^2 dg \\ &\geq d_\tau [(1 - e_1)(1 - e_2) - e_2] M \\ &= d_\tau (1 - e_1 - 2e_2 + e_1 e_2) M \geq (1 - e) d_\tau M. \end{aligned} \quad (\text{DB5.8})$$

The bound outside V uses $\operatorname{Re} \overline{\chi_\tau(g)} \geq -d_\tau$, valid because every eigenvalue is on the unit circle. For the chosen constants, $e_1 + 2e_2 - e_1 e_2 = 3e/4 - e^2/16 < e$. Together with (DB5.7), this proves (DB5.1) with all real/virtual-character and error terms accounted for.

DB6. The exceptional quadratic character (§§2.1.5-2.1.8)

Let $T \subset \widehat{G} \setminus \{1\}$ be finite, and apply DB5. All multiplicities omitted from $T \cup \{1\}$ multiply nonpositive values of ν . Positivity therefore gives

$$\begin{aligned} 0 &\leq \nu(\rho\bar{\rho}) = \sum_{\sigma} [\rho\bar{\rho} : \sigma] \nu(\sigma) \\ &\leq M + \sum_{\tau \in T} [\rho\bar{\rho} : \tau] \nu(\tau) \\ &\leq M \left(1 + (1 - e) \sum_{\tau \in T} d_\tau \nu(\tau) \right). \end{aligned} \quad (\text{DB6.1})$$

The final inequality reverses the multiplicity lower bound when multiplying by $\nu(\tau) \leq 0$; that sign is essential. Since $M > 0$, division and $e \downarrow 0$ prove, for every finite T ,

$$\sum_{\tau \in T} d_\tau (-\nu(\tau)) \leq 1. \quad (\text{DB6.2})$$

Every summand is a nonnegative integer. Thus at most one nontrivial irreducible can have negative ν , and then it must have dimension one and $\nu = -1$. Conjugation symmetry forces that character ε to equal $\bar{\varepsilon}$. A unitary one-dimensional character satisfies $\bar{\varepsilon} = \varepsilon^{-1}$, so $\varepsilon^2 = 1$; as it is nontrivial, its order is exactly two.

Consequently the original analytic family has no boundary zero except possibly a single simple zero at $\omega_1 \varepsilon$ for one nontrivial quadratic character. This proves the source's theorem §2.1.4 from its stated analytic hypotheses, including the entire exceptional case.

DB7. The discrete limiting measure and its pole cancellation (§§2.1.10-2.1.11)

For this section $\Gamma = \mathbb{Z}$ and the quadratic exception has been excluded by whatever arithmetic input verifies source condition (C): L has the specified simple pole and no boundary zero. This condition will be proved for the curve application below; it is not silently inferred for an unrelated representation family.

Retain the source measures

$$\mu^\natural = \sum_v \sum_{n \geq 1} \deg(v) q^{-n \deg(v)} \delta_{[x_v^n]}, \quad \mu_0 = \mathbf{1}_{\deg > 0} dg, \quad (\text{DB7.1})$$

where Haar measure dg on G gives its degree-zero compact subgroup mass one, and $\mu_0^\natural$ is the pushforward to conjugacy classes. Each positive degree component has μ_0 -mass one. The weighted measure $\mu^\natural$ is locally finite: only divisors of a fixed positive degree enter a fixed component, and (DB2.1) makes every fixed degree contain finitely many v .

For $\text{Re } \tau > 0$, (DB2.3) gives exactly

$$\hat{\mu}^\natural(\tau) = -\frac{1}{\log q} \frac{L'}{L}(\omega_1 \tau). \quad (\text{DB7.2})$$

The factor $1/\log q$ is necessary because $\log N_v = \deg(v) \log q$. Averaging an irreducible representation over the compact kernel is its projection onto the kernel invariants. That subspace is G -stable because the kernel is normal. Irreducibility makes it either zero or the whole representation; in the latter case the representation factors through $\mathbb{Z}$ and is one-dimensional. Therefore the transform of $\mu_0^\natural$ vanishes on all other irreducibles, while

$$\hat{\mu}_0^\natural(\omega_s) = \sum_{m=1}^{\infty} q^{-ms} = \frac{q^{-s}}{1 - q^{-s}}. \quad (\text{DB7.3})$$

On the representation surface both transforms have just the pole at the trivial character on $\text{Re } \tau \geq 0$, with residue $1/\log q$: the first follows from DB3 and the original simple pole, and the second follows from $1 - q^{-s} = s \log q + O(s^2)$. They cancel exactly. Hence the Fourier-Laplace transform of $\mu^\natural - \mu_0^\natural$ is holomorphic on a neighborhood of that closed half-plane.

DB8. Equidistribution with its convergence proof (§§2.1.12-2.1.13)

For a fixed unitary representation τ , define

$$a_m = \int_{G_m^\natural} \chi_\tau(g) d(\mu^\natural - \mu_0^\natural)(g), \quad m > 0. \quad (\text{DB8.1})$$

There is no additional Haar factor multiplying this measure integral. Its transform at $\tau \omega_s$ is

$$\sum_{m \geq 1} a_m q^{-ms}. \quad (\text{DB8.2})$$

Put $t = q^{-s}$, retaining its exact original scale $\log q$. The period from (DB1.3) makes this a single-valued holomorphic function of t on a neighborhood of the punctured closed unit disk; the power series supplies

the value at zero. Compactness of the unit circle gives a common open annulus across it, hence a disk $|t| < R$ for some $R > 1$. For $1 < r < R$, Cauchy's coefficient formula yields

$$|a_m| \leq \max_{|t|=r} \left| \sum_{j \geq 1} a_j t^j \right| r^{-m}. \quad (\text{DB8.3})$$

Thus each character integral tends to zero, with a constant and decay radius allowed to depend on τ . No uniform spectral gap over all representations is inferred.

Choose a central z of positive degree d . Translate the degree $i + nd$ slice to the fixed compact conjugacy slice of degree i by multiplication by z^{-n} , for $0 \leq i < d$. The compact quotient $G/\langle z \rangle$ contains this slice, and its characters uniformly span the continuous class functions on it: extend a continuous class function by zero on the other finitely many degree classes and apply the same central character-density theorem used in DB5. Their pullbacks are unitary representations of G trivial on z , so the translated character integral differences tend to zero by (DB8.3).

The total masses of the positive translated measures tend to one, by applying (DB8.3) to the trivial representation, while the comparison measure has mass one. Uniform approximation of any continuous class function f by a finite character combination P gives an error bounded by $\|f - P\|_\infty$ times the sum of those two masses. First make this error small, then use the finite character convergence. This proves vague convergence of the translated measures asserted in §2.1.12. It also supplies the approximation argument underlying §2.1.13.

DB9. Exact specialization to the original Riemann zeta (§2.1.9)

This specialization is made after the full arithmetic has been reconstructed. Let $G = \Gamma = (\mathbb{R}, +)$, let the compact kernel be $K = \{0\}$, and set

$$\omega_s(x) = e^{-sx}, \quad \Sigma = \{\text{all primes of the recovered integers}\}, \quad x_p = \log p, \quad N_p = p. \quad (\text{DB9.1})$$

No finite set of selected primes is used to establish any of these data. All irreducible finite-dimensional unitary representations of $\mathbb{R}$ are the characters $x \mapsto e^{-itx}$: commuting unitary matrices can be simultaneously diagonalized, and an irreducible common eigenspace has dimension one. A continuous character of $\mathbb{R}$ has the displayed form by lifting near zero to its argument and using additivity.

For $\text{Re } s > 1$, the original function is

$$\zeta(s) = \sum_{m=1}^{\infty} m^{-s} = \prod_p (1 - p^{-s})^{-1}. \quad (\text{DB9.2})$$

The equality follows from the recovered integer factorization: a finite product expands over integers supported on those primes, and absolute convergence lets the finite prime sets increase to all primes. Each integer eventually occurs, with coefficient one. Thus the complete representation family is exactly $L(\omega_s) = \zeta(s)$, not a completed or rescaled zeta function.

Differentiating the absolutely convergent full product as in DB2 gives

$$-\frac{\zeta'(\sigma + it)}{\zeta(\sigma + it)} = \sum_p \sum_{n=1}^{\infty} (\log p) p^{-n\sigma} e^{-itn \log p} = \int_{\mathbb{R}} e^{-itx} d\mu_\sigma(x), \quad (\text{DB9.3})$$

with exactly the measure

$$\mu_\sigma = \sum_p \sum_{n=1}^{\infty} (\log p) p^{-n\sigma} \delta_{n \log p}, \quad \sigma > 1. \quad (\text{DB9.4})$$

Every prime-power location, weight, and multiplicity is retained. For any real $t_1, \dots, t_k$ and complex $c_1, \dots, c_k$, its positive-definiteness is the exact identity

$$\sum_{j,l} c_j \bar{c}_l \left(-\frac{\zeta'(\sigma + i(t_j - t_l))}{\zeta(\sigma + i(t_j - t_l))} \right) = \int \left| \sum_j c_j e^{-it_j x} \right|^2 d\mu_\sigma(x) \geq 0. \quad (\text{DB9.5})$$

No zero-location assumption enters this positivity.

The analytic hypothesis on the closed boundary follows directly from the original Dirichlet series identity

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx, \quad \text{Re } s > 0, \quad (\text{DB9.6})$$

initially proved for $\text{Re } s > 1$ by integration against $\lfloor x \rfloor$ and then continued by the locally uniformly convergent integral. It proves holomorphy on this half-plane except at the simple pole $s = 1$, whose residue is one. In particular the assumptions of DB3 are verified for every representation of this G , rather than imported from a larger unconstructed representation family.

There is no nontrivial continuous quadratic character $\mathbb{R} \rightarrow \{\pm 1\}$, because the image of a connected space is connected and must contain the identity. The exceptional case of DB6 is therefore absent. It follows that the original $\zeta(s)$ has no zero on $\text{Re } s = 1$. The point $s = 1$ is its already calculated pole.

The source's exact numerical coefficients also have a direct proof. With $\Lambda_\sigma(t) = -\zeta'(\sigma + it)/\zeta(\sigma + it)$,

$$\begin{aligned} 3 \text{Re } \Lambda_\sigma(0) + 4 \text{Re } \Lambda_\sigma(t) + 2 \text{Re } \Lambda_\sigma(2t) &= \int (3 + 4 \cos(tx) + 2 \cos(2tx)) d\mu_\sigma(x) \\ &= \int (1 + 2 \cos(tx))^2 d\mu_\sigma(x) \geq 0. \end{aligned} \quad (\text{DB9.7})$$

For $t \neq 0$, put m_t and m_{2t} equal to the actual zero multiplicities at $1 + it$ and $1 + 2it$, with value zero at a nonzero function value. Multiplication by $\sigma - 1$ and (DB3.2) gives

$$0 \leq 3 - 4m_t - 2m_{2t}. \quad (\text{DB9.8})$$

Any positive integer m_t contradicts this inequality. This is a second complete proof of the same boundary result, preserving the source's coefficients 3, 4, 2. It uses the entire measure, not values at any numerical prime shortcut.

This specialization has trivial compact kernel and $\Gamma = \mathbb{R}$. It does not supply a nontrivial compact geometric monodromy group, a curve over a finite field, or the cohomological determinant formula used in §2.2. Those structures and their hypotheses must be provided by an actual receiving construction; they do not follow from the prime-power measure alone. Equations (DB9.3)–(DB9.8) concern $\text{Re } s = 1$, with no shift of the original zeta function to a different working function.

DB10. Dependency boundary for the compact-form and curve argument

The remaining reconstruction follows §2.2 and uses its explicitly stated geometric inputs: an algebraic monodromy extension, dense geometric fundamental-group image, a sufficiently faithful mixed sheaf, the determinant and semisimplicity results of §§1.3–1.5, compact complexification theory, and the cohomological trace formula with duality. These are named source dependencies, not assumptions silently supplied to the original-zeta specialization. The full compact-form calculation, all weight factors, the curve simple-pole argument, the exceptional double cover, and the strict bound follow below.

DBC1. Source, provenance, and exact scope

Human source: Pierre Deligne, *La conjecture de Weil. II*, Publications Mathématiques de l’IHÉS **52** (1980), 137–252, §§2.2.1–2.2.10, printed pp. 192–196. The reading witness is the French transcription export at `output/Deligne_Weil_II_S20_LaTeX/typed_latex/S20_FR_record_export.tex`, lines 1784–1927, SHA256 `d6836ae15d98f2b27eb98c9bd039a7d921679becd9fa47ff4717410289dbd351`. This is a local transcription, not original author TeX. No PDF was opened, no source file was changed, and no public upload was made.

The existing literature-routing ledger identifies the work by PUBUNIT-3D90B487DBD1CC3A259C6455 and PUBUNIT-574B0214207BBB2BDABCFFF9; those routing records are not substitutes for reading. This subtask read all of the assigned lines and additionally the exact local dependencies listed below. Root’s existing `USER_INPUTS_VERBATIM.md` and `DELIGNE_FULL_READING_LOG.md` retain the wider user provenance and reading history.

Task received: independently reconstruct compact-form conjugacy, representation equivalence, the mixed-to-pure input, and the strict compactly supported first-cohomology bound; keep every twist factor; check the sign in §2.2.8, cancellation, and the quadratic-cover exception. The parent handles the upstream determinant-weight and real-sheaf-purity derivations; DB3–DB6 above supply the complete proof of §§2.1.4–2.1.8. This file supplies the receiving calculations and does not claim to have reproved all upstream theorems.

Read dependencies and their roles:

Source location	Local lines	Role
§§1.1.12–1.1.14	448–473	Weil representations, degree, and cohomological Frobenius
§§1.2.1–1.2.7	479–528	Pointwise weights, mixedness, and exact twist factors
§§1.3.5–1.3.12	630–710	Monodromy extension, semisimple geometric identity component, central degree, determinant weights
§1.3.14	722–730	Finite-image rank-one Weil characters are étale
§§1.4.1–1.4.7	745–813	Invariants, coinvariants, Tate factor, trace formula, Euler convergence
§§1.8.1–1.8.2	1210–1234	Earlier non-strict bound and boundary purity
§§1.8.10–1.8.12	1315–1329	A mixed irreducible lisse sheaf on a normal scheme is pointwise pure

Source location	Local lines	Role
§§2.1.1-2.1.8	1559-1724	Sign convention and analytic nonvanishing theorem; detailed proof belongs to the parent's reconstruction
§2.1.10	1736-1740	Exact statement of condition (C)

The following foundational inputs remain named inputs: the correspondence between lisse Weil sheaves and Weil-group representations; Grothendieck's trace formula and Poincaré duality in §§1.4.1-1.4.5; the geometric monodromy theorem in §1.3.9 and central determinant criterion in §1.3.12; compact real forms of complex reductive groups, conjugacy of maximal compact subgroups, and extension of continuous finite-dimensional representations of a compact real form to algebraic representations of its complexification. These are not conclusions manufactured by the calculation below.

DBC2. Original degree and weight conventions

Let $q = p^f > 1$, let $\ell \neq p$, and fix the field isomorphism

$$\iota : \overline{\mathbf{Q}}_\ell \longrightarrow \mathbf{C}.$$

For $\alpha \neq 0$,

$$w_q(\alpha) = 2 \log_q |\iota \alpha|. \quad (\text{DBC.2.1})$$

If v is a closed point of degree $d_v = [k(v) : \mathbf{F}_q]$, geometric Frobenius F_v has degree d_v , and

$$\omega_s(g) = q^{-s \deg(g)}, \quad \omega_s(F_v) = q^{-s d_v}. \quad (\text{DBC.2.2})$$

In particular the actual period in s is $2\pi i / \log q$.

For an irreducible complex representation τ of the group in §3, its real part $\sigma = \Re(\tau)$ is defined by

$$|\tau(z)| = \omega_\sigma(z) = q^{-\sigma m} \quad (\text{DBC.2.3})$$

for a central element z of positive degree m . Here $\tau(z)$ is scalar by Schur's lemma. The representation $\tau \omega_{-\sigma}$ is unitarizable: choose a Hermitian form invariant under the scalar of modulus one acting as z ; average its translates over the compact group $G_{\mathbf{R}} / \langle z \rangle$. Translation by z does not change the integrand, so the average is defined and $G_{\mathbf{R}}$ -invariant.

Therefore, for an element $g \in G_{\mathbf{R}}$ of degree d , every eigenvalue of $\tau(g)$ has modulus $q^{-\sigma d}$. Consequently the corresponding sheaf has pointwise weight

$$\boxed{-2\Re(\tau)}. \quad (\text{DBC.2.4})$$

This is also forced by the test representation $\tau = \omega_s$, whose degree-one scalar is q^{-s} . The plus sign in the local witness at line 1871 is inconsistent with its own definitions. This is a demonstrated discrepancy in the transcription witness, not an assertion about an uninspected printed page.

DBC3. The compact form and conjugacy

Let

$$1 \longrightarrow G_{\mathbf{C}}^0 \longrightarrow G_{\mathbf{C}} \xrightarrow{\deg} \mathbf{Z} \longrightarrow 0 \quad (\text{DBC.3.1})$$

be the complex group scheme of §2.2.1. The algebraic group $G_{\mathbf{C}}^0$ may be disconnected; its identity component is semisimple. Let $Z_{\mathbf{C}}$ be the center of the full group. By §1.3.10, its degree image has finite index in $\mathbf{Z}$. Its degree-zero kernel is finite, because it lies in the center of the algebraic group with semisimple identity component. Thus

$$H = G_{\mathbf{C}}/Z_{\mathbf{C}}$$

is a finite-type reductive algebraic group. Choose a maximal compact subgroup $U \subset H(\mathbf{C})$ and set

$$G_{\mathbf{R}} = \pi^{-1}(U), \quad K^0 = G_{\mathbf{R}} \cap G_{\mathbf{C}}^0. \quad (\text{DBC.3.2})$$

The group K^0 is a maximal compact subgroup of $G_{\mathbf{C}}^0$, and every connected component of the quotient is met by U . Hence degree maps $G_{\mathbf{R}}$ onto $\mathbf{Z}$, with kernel K^0 .

Choose $z \in Z_{\mathbf{C}}$ of degree $m > 0$. It belongs to $G_{\mathbf{R}}$, since its image in H is the identity. Put

$$A = G_{\mathbf{C}}/\langle z \rangle, \quad K = G_{\mathbf{R}}/\langle z \rangle. \quad (\text{DBC.3.3})$$

These are a finite-type complex reductive group and a maximal compact real form. Indeed, $Z_{\mathbf{C}}/\langle z \rangle$ is finite, so $A \rightarrow H$ is a finite central map, and K is the inverse image of U . Alternatively, compactness follows directly because K has only m degree classes, each represented by a translate of the compact group K^0 .

Suppose $a, b \in G_{\mathbf{R}}$ are conjugate in $G_{\mathbf{C}}$. Their images in K are conjugate in A . Every irreducible continuous character of K extends to an algebraic character of A , so those characters take the same values on the two images. Characters separate conjugacy classes in a compact group: the algebra spanned by characters is dense in the continuous central functions, which separate disjoint compact conjugacy classes. Thus their images are conjugate in K . Lift a conjugating element to $u \in G_{\mathbf{R}}$. For some integer j ,

$$uau^{-1} = bz^j.$$

Conjugacy preserves degree, so $\deg(a) = \deg(b)$, while the displayed identity gives $\deg(a) = \deg(b) + jm$. Thus $j = 0$. This proves the full conjugacy assertion of §2.2.2, including the central degree coordinate.

The local witness of §1.3.10(iv), line 688, says “finite subgroup” where its proof and the corollary with finite cokernel require “subgroup of finite index.” The argument here uses the latter statement, exactly as proved by the nonzero-degree central element in lines 692–698.

For completeness, the Jordan decomposition in §2.2.3 keeps the central coordinate as follows. Given $g \in G_{\mathbf{C}}$, write its image $\pi(g) \in H$ as the commuting product $h_s h_u$. The unipotent element h_u lies in the identity component of H , which is the image of the identity component of $G_{\mathbf{C}}^0$ under a finite central map. It has a unique unipotent lift u to $G_{\mathbf{C}}^0$. To verify this, take any lift in the identity component and its ordinary Jordan decomposition: its unipotent part maps to h_u , while its semisimple part maps to the identity and is finite central. Two unipotent lifts differ by a finite central element, which would be the semisimple part of their quotient expression, so that element must be the identity. Conjugation by g preserves u , by uniqueness and because $\pi(g)$ commutes with h_u . Therefore

$$g_u = u, \quad g_s = gu^{-1}, \quad g = g_s g_u = g_u g_s. \quad (\text{DBC.3.4})$$

The image of g_s in H is semisimple and its degree is the full original degree of g . Any other such decomposition would give another unipotent lift of h_u , so (DBC.3.4) is unique.

DBC4. Representation equivalence, including nonsemisimple representations

Scalar transport along the field isomorphism ι gives an equivalence

$$\text{Rep}_{\overline{\mathbf{Q}}_{\ell}}(G) \longrightarrow \text{Rep}_{\mathbf{C}}(G_{\mathbf{C}}). \quad (\text{DBC.4.1})$$

The target and source are finite-dimensional algebraic representations of the group schemes; algebraicity is required on the finite-type degree-zero group.

For restriction to $G_{\mathbf{R}}$, choose $u \in G_{\mathbf{R}}$ of degree one. Every element of $G_{\mathbf{R}}$ has a unique expression hu^n , $h \in K^0$, and every element of $G_{\mathbf{C}}$ has the corresponding expression with $h \in G_{\mathbf{C}}^0$. Let r be any finite-dimensional continuous representation of $G_{\mathbf{R}}$. Its restriction to K^0 has a unique algebraic extension r_0 to $G_{\mathbf{C}}^0$. Set $T = r(u)$. On K^0 ,

$$Tr_0(h)T^{-1} = r_0(uhu^{-1}).$$

Both sides are algebraic in h ; the compact real form K^0 is Zariski dense in $G_{\mathbf{C}}^0$. Therefore this identity holds on $G_{\mathbf{C}}^0$. The formula

$$\tilde{r}(hu^n) = r_0(h)T^n \tag{DBC.4.2}$$

is consequently an algebraic representation of $G_{\mathbf{C}}$, restricting to r . Uniqueness follows from the same density and the value on u . An intertwiner on the real group intertwines r_0 by density and intertwines T , so it intertwines (DBC.4.2). Restriction is therefore fully faithful and essentially surjective.

No diagonalizability of T was assumed. In particular, these equivalences include extensions and Jordan blocks in the degree action. One cannot assert that every group element called semisimple in §2.2.3 has semisimple image under an arbitrary representation: for $G = \mathbf{Z}$, a generator may act by a nontrivial Jordan block. The finite-type quotient used in the next section is exactly where the usual preservation of Jordan decomposition is valid.

DBC5. Mixed input, all twist factors, and compact Frobenius classes

Assume §§2.2.4(a)–(c): a degree-preserving morphism

$$W(X_0, \bar{x}) \longrightarrow G,$$

with X_0 normal and geometrically connected; continuous geometric image in some finite extension of $\mathbf{Q}_{\ell}$, Zariski dense in G^0 ; and a representation V with finite kernel on G^0 , whose lisse sheaf is ι -mixed.

Every irreducible G -constituent V_a is an irreducible Weil representation. To see this, the Weil image is Zariski dense in each degree component, since its degree-zero image is dense and it meets every degree. The stabilizer of any vector subspace is Zariski closed. Thus Weil-stable subspaces and G -stable subspaces coincide.

The sheaf of V_a is mixed by stability under subquotients. The implication from mixed to pointwise pure uses §1.8.11, not mixedness without a lisse argument. On a sufficiently small dense open the mixed filtration is lisse; irreducibility on that open follows from normality, so only one weight occurs there. Section 1.8.10 extends that purity over X_0 . Denote its weight by β_a .

Choose the central element z of degree $m > 0$ from §3, now over $\overline{\mathbf{Q}}_{\ell}$, and write its action on V_a as λ_a . The determinant-weight criterion §1.3.12 yields

$$|\iota\lambda_a| = q^{m\beta_a/2}. \tag{DBC.5.1}$$

Choose $c_a \in \overline{\mathbf{Q}}_{\ell}^*$ with

$$c_a^m = \lambda_a^{-1}. \tag{DBC.5.2}$$

Let L_{c_a} be the rank-one Weil sheaf pulled back from $\text{Spec } \mathbf{F}_q$, with degree-one geometric Frobenius scalar c_a . Define an auxiliary representation

$$V'_a = V_a \otimes L_{c_a}, \quad V_a \cong V'_a \otimes L_{c_a^{-1}}. \tag{DBC.5.3}$$

The latter isomorphism comes from evaluation $L_{c_a} \otimes L_{c_a^{-1}} \rightarrow \mathbf{1}$. It preserves the original representation with its full degree factor. At a degree- d_v point, and on every cohomology group, the exact formulas are

$$F_v|V'_a = c_a^{d_v}(F_v|V_a), \quad F|H_c^i(X, V'_a) = c_a(F|H_c^i(X, V_a)), \quad (\text{DBC.5.4})$$

under the geometric-fibre identifications. Thus every original local eigenvalue α becomes $c_a^{d_v}\alpha$, and is recovered as $c_a^{-d_v}(c_a^{d_v}\alpha)$. Every original cohomological eigenvalue γ becomes $c_a\gamma$ and is recovered by the inverse scalar.

More explicitly, choose a nonzero geometric fibre vector e_a for L_{c_a} . The tensor map $v \mapsto v \otimes e_a$ identifies underlying geometric fibres, and the projection formula identifies $H_c^i(X, \mathcal{F}_a \otimes L_{c_a})$ with $H_c^i(X, \mathcal{F}_a) \otimes L_{c_a, \bar{\mathbb{F}}_q}$. Frobenius on this tensor product is $F \otimes c_a$, which proves (DBC.5.4). These are geometric identifications with the displayed changed Frobenius action, not an assertion that $\mathcal{F}_a$ and $\mathcal{F}_a \otimes L_{c_a}$ are isomorphic as Weil sheaves.

The source's twist over $\mathbf{F}_p$ is identical after choosing b_a with $b_a^f = c_a$: its closed-point factor is

$$b_a^{[k(v):\mathbf{F}_p]} = b_a^{fd_v} = c_a^{d_v}, \quad (\text{DBC.5.5})$$

and its degree-one cohomological factor is $b_a^f = c_a$. All base-field exponents are retained.

Equations (DBC.5.1)-(DBC.5.2) give $|\iota c_a| = q^{-\beta_a/2}$, so the auxiliary local eigenvalues have modulus one. Also z acts on V'_a by $\lambda_a c_a^m = 1$. Therefore

$$V' = \bigoplus_a V'_a$$

factors through the finite-type quotient $G/\langle z \rangle$. Its restriction to G^0 still has finite kernel: the twists are geometrically trivial; restriction of the original representation to reductive G^0 is semisimple; and its restriction is the direct sum of the restrictions of its G -constituents. The kernel of the representation on the quotient is then finite, since that quotient has finitely many components, and a kernel's intersection with any one component is either empty or a translate of its finite intersection with the degree-zero subgroup.

Let $g = \iota F_v$, and let g_s be its semisimple part in the sense of §2.2.3. In the finite-type quotient just constructed, Jordan decomposition is respected by the representation. Hence the image of g_s acts diagonally with all eigenvalues of modulus one. The closure of its integer powers in $\text{GL}(V')$ is compact. A homomorphism of complex algebraic groups with finite kernel is finite onto its closed image, and its map of complex points is proper. Therefore the powers of the image of g_s in $G_{\mathbf{C}}/\langle z \rangle$ have compact closure. Their further image in $G_{\mathbf{C}}/Z_{\mathbf{C}}$ has compact closure as well. Every compact subgroup lies in a maximal compact subgroup; all maximal compact subgroups of this reductive group are conjugate. It follows that g_s is conjugate into $G_{\mathbf{R}}$. Section 3 proves uniqueness of its $G_{\mathbf{R}}$ -conjugacy class.

The use of V'_a here establishes existence of a compact Frobenius representative. It neither replaces the original sheaves in the strict-bound calculation nor discards $c_a^{d_v}$ or c_a .

Now let τ be any irreducible representation of $G_{\mathbf{R}}$. The compact representative of g_s and (DBC.2.3) give local eigenvalue modulus $q^{-\Re(\tau)d_v}$. The unipotent part of g acts unipotently, commutes with its semisimple part, and does not change the multiset of eigenvalues of their product: on each generalized eigenspace of the first commuting operator, the product has that operator's eigenvalue multiplied by the sole eigenvalue one of the second. Thus the sheaf for τ has weight $-2\Re(\tau)$. An arbitrary representation has a finite composition series, so its associated sheaf is mixed. This order of proof avoids using purity of all representations to prove the initial compactness assertion.

DBC6. The complete L-function identity

Write $\mathcal{F}_0(\tau_\ell)$ for the sheaf corresponding to τ . At a closed point, conjugation and removal of the commuting unipotent part preserve the characteristic polynomial. Consequently

$$\begin{aligned} L(\tau, s) &= \prod_{v \in |X_0|} \det(1 - \tau(x_v) q^{-s d_v})^{-1} \\ &= \iota Z(\mathcal{F}_0(\tau_\ell), t)|_{t=q^{-s}}. \end{aligned} \quad (\text{DBC.6.1})$$

For a smooth geometrically connected curve, put

$$P_i(t) = \iota \det(1 - tF | H_c^i(X, \mathcal{F})).$$

The trace formula gives the unreduced rational expression

$$L(\tau, s) = \frac{P_1(q^{-s})}{P_0(q^{-s})P_2(q^{-s})}. \quad (\text{DBC.6.2})$$

If V is the geometric fibre and $\Pi = \pi_1(X, \bar{x})$, then

$$H_c^0(X, \mathcal{F}) = \begin{cases} V^\Pi & X \text{ proper,} \\ 0 & X \text{ not proper,} \end{cases} \quad H_c^2(X, \mathcal{F}) = V_\Pi(-1). \quad (\text{DBC.6.3})$$

Geometric Frobenius acts on $\overline{\mathbf{Q}}_\ell(-1)$ by q . This factor is retained in every denominator below.

Section 2.2.7 also treats a normal geometrically connected scheme of dimension $N \geq 1$. Keep precisely the same character $\omega_1(g) = q^{-\deg(g)}$. The point-count estimate in §1.4.6 gives a constant C with

$$\#\{v : \deg v = d\} \leq C q^{Nd}.$$

For an irreducible representation ρ of dimension r , write $\sigma = \Re(\rho)$. Every eigenvalue of $\rho(x_v)$ has modulus $q^{-\sigma d_v}$. If $\sigma > N$, the full logarithmic Euler series is absolutely bounded by

$$\begin{aligned} \sum_v \sum_{k \geq 1} \frac{|\text{Tr}(\rho(x_v)^k)|}{k} &\leq C r \sum_{d \geq 1} q^{Nd} \sum_{k \geq 1} \frac{q^{-\sigma d k}}{k} \\ &\leq \frac{C r}{1 - q^{-\sigma}} \sum_{d \geq 1} q^{-(\sigma - N)d} < \infty. \end{aligned} \quad (\text{DBC.6.4})$$

Therefore $L(\rho)$ converges and is nonzero for $\Re(\rho) > N$. For a fixed original representation τ of weight $\beta = -2\Re(\tau)$, the exact region is

$$\Re(s) > N + \frac{\beta}{2} \quad \text{for } L(\tau, s) = L(\tau \omega_s). \quad (\text{DBC.6.5})$$

In particular a unitary τ gives $\Re(s) > N$. The character has not been changed to $q^{-N \deg(g)}$; the original degree, the original parameter s , and the dimension N all remain visible.

DBC7. Curve zeta poles without a noncancellation assumption

We first prove the simple pole of the zeta function at $t = q^{-1}$ using (DBC.6.2), rather than invoking equidistribution whose hypotheses already include that pole.

For the constant sheaf on a smooth geometrically connected curve,

$$Z(X_0, t) = \frac{P_1(t)}{(1-t)^\delta(1-qt)}, \quad \delta = \begin{cases} 1 & X_0 \text{ proper,} \\ 0 & X_0 \text{ not proper.} \end{cases} \quad (\text{DBC.7.1})$$

Suppose $P_1(q^{-1}) = 0$. Remove one occurrence of the factor $1 - qt$ from this polynomial, and write the remaining polynomial as $\prod_{j=1}^r (1 - a_j t)$, with every multiplicity retained. Logarithmic differentiation of the resulting identity gives, for every $n \geq 1$,

$$\#X_0(\mathbf{F}_{q^n}) = \delta - \sum_{j=1}^r a_j^n. \quad (\text{DBC.7.2})$$

If $R = \max_j |a_j| > 1$, the phases a_j/R for the indices of modulus R lie on a finite-dimensional compact torus. There exist arbitrarily large positive integers n for which all those phases to the n -th power are as close to one as prescribed. Indeed, among powers of any point of a compact group, two arbitrarily widely separated powers occur arbitrarily close: choose a convergent subsequence and take differences of sufficiently separated indices. Their quotient tends to the identity. Along these integers the real part of the sum of the modulus- R terms is at least $r_R R^n / 2$, where $r_R \geq 1$ is their number. The remaining terms are bounded in modulus by $(r - r_R) R_1^n$, for some $R_1 < R$. Thus (DBC.7.2) becomes negative for sufficiently large members of that sequence, contradicting the nonnegative point count.

It follows that all $|a_j| \leq 1$, in which case (DBC.7.2) gives $\#X_0(\mathbf{F}_{q^n}) \leq \delta + r$ for all n . This is also impossible: a nonempty curve has infinitely many geometric closed points, and every finite selection of such points is defined over one common finite extension. Hence its point counts over finite extensions are unbounded. The contradiction proves

$$P_1(q^{-1}) \neq 0. \quad (\text{DBC.7.3})$$

The factor $1 - qt$ in (DBC.7.1) is simple, and $1 - q^{-1} \neq 0$; the pole is therefore exactly simple.

For a smooth connected curve Y_0 whose field of constants is $\mathbf{F}_{q^d}$, view the same scheme as a geometrically connected curve Y_* over $\mathbf{F}_{q^d}$. Its local degrees satisfy $\deg_{\mathbf{F}_q}(y) = d \deg_{\mathbf{F}_{q^d}}(y)$, hence

$$Z(Y_0, t) = Z(Y_*, t^d). \quad (\text{DBC.7.4})$$

The latter expression has top denominator $1 - q^d t^d$, and its numerator is nonzero at $t = q^{-1}$ by the preceding argument over $\mathbf{F}_{q^d}$. The derivative of that denominator there is $-dq \neq 0$. Thus the pole at $t = q^{-1}$ is still simple, even though Y_0 has d geometric components. The parameter change $t = q^{-s}$ has derivative $-\log(q)q^{-s} \neq 0$, so pole order one is also preserved at $s = 1$.

DBC8. The quadratic exception and condition (C)

For an irreducible unitary τ that is nontrivial on $G_{\mathbf{R}}^0$, its invariant and coinvariant spaces under G^0 vanish. Indeed, V^{G^0} is G -stable; irreducibility makes it zero or all of V , and the latter is excluded. Reductivity of G^0 makes invariants and coinvariants isomorphic as representations of the degree action, so both vanish. Zariski density gives the same conclusion for Π . Therefore $P_0 = P_2 = 1$, and $L(\tau, s)$ is the polynomial $P_1(q^{-s})$, holomorphic everywhere in s .

An irreducible representation trivial on G^0 factors through $\mathbf{Z}$ and is one-dimensional. In the unitary case it is ω_{it} for some real t . Its L-function is a translate of the curve zeta function. Equations (DBC.7.1)-(DBC.7.3) show that in the representation half-plane $\Re(\rho) \geq 1$, the only pole is a simple pole at $\rho = \omega_1$. (Distinct

values of s representing the same character are the same point of this representation parameter space.) Euler convergence holds for $\Re(\rho) > 1$.

The analytic theorem §2.1.4 therefore applies. It states that on $\Re(\rho) = 1$ there can be at most one exceptional zero, necessarily at $\rho = \omega_1 \varepsilon$, where ε is a nontrivial character of exact order two.

The finite-image Weil character ε defines an étale rank-one sheaf, hence an étale double cover $f : Y_0 \rightarrow X_0$. For example, étaleness follows from the rank-one criterion and its proof in §1.3.14, since all its scalars are roots of unity and therefore ℓ -adic units. Its pullback to the Weil group is still nontrivial: the character extends algebraically to G , and the Weil image is Zariski dense in every component. The cover is connected over $\mathbf{F}_q$: the character is nontrivial, so the arithmetic monodromy acts transitively on its two-element fibre. There are exactly two geometric cases:

1. If $\varepsilon|_{\Pi} \neq 1$, the geometric image is the full group of order two, so Y_0 is geometrically connected, with constant-field degree one.
2. If $\varepsilon|_{\Pi} = 1$, then $\varepsilon(g) = (-1)^{\deg g}$. The cover is $X_0 \times_{\mathbf{F}_q} \mathbf{F}_{q^2}$, connected over $\mathbf{F}_q$ and with two geometric components. Its constant-field degree is two.

The deck involution on $f_* \overline{\mathbf{Q}}_\ell$ has projectors $(1 + \sigma)/2$ and $(1 - \sigma)/2$. These are valid also when $\ell = 2$, because the coefficient field has characteristic zero. They give

$$f_* \overline{\mathbf{Q}}_\ell \cong \overline{\mathbf{Q}}_\ell \oplus \mathcal{L}_\varepsilon. \quad (\text{DBC.8.1})$$

Thus every local factor, with all degrees retained, yields

$$\zeta_{Y_0}(s) = \zeta_{X_0}(s) L(\varepsilon, s). \quad (\text{DBC.8.2})$$

Explicitly, at a point of degree d , a split fibre contributes $(1 - t^d)^{-2}$, equal to $(1 - t^d)^{-1} (1 - t^d)^{-1}$; an inert fibre contributes $(1 - t^{2d})^{-1}$, equal to $(1 - t^d)^{-1} (1 + t^d)^{-1}$. These are respectively $\varepsilon(F_v) = 1$ and -1 .

Section 7 shows that both zeta functions in (DBC.8.2) have a simple pole at $s = 1$, in both geometric cases. Therefore their quotient $L(\varepsilon, s)$ has order zero at $s = 1$, and its value there is finite and nonzero. This excludes the exceptional zero. In the constant-field case the unreduced top factor is $1 - q^2 t^2 = (1 - qt)(1 + qt)$; the extra factor is nonzero at $t = q^{-1}$, which displays the same conclusion directly at the denominator level.

We have obtained condition (C): $L(\rho)$ is meromorphic for $\Re(\rho) \geq 1$, has exactly the permitted simple pole at ω_1 , and has no zeros on the boundary. Its Euler product supplies nonvanishing in the interior $\Re(\rho) > 1$.

DBC9. Strict bound with the original weight β

Let X_0 be a smooth geometrically connected curve, and let the original lisse sheaf $\mathcal{F}_0 \neq 0$ be irreducible and pointwise ι -pure of weight β . The restriction of its Weil representation to the geometric fundamental group is semisimple: choose a nonzero simple geometric submodule; the sum of all simple geometric submodules is nonzero and invariant under the Weil group because the geometric group is normal. Irreducibility makes that sum the whole representation. Therefore §2.2.5 applies to the monodromy group of this original sheaf.

Let τ be its complex representation. By (DBC.2.4),

$$\Re(\tau) = -\frac{\beta}{2}, \quad \Re(\tau \omega_s) = \Re(s) - \frac{\beta}{2}. \quad (\text{DBC.9.1})$$

Condition (C) therefore controls the original L-function in the original variable s on

$$\Re(s) \geq 1 + \frac{\beta}{2}. \quad (\text{DBC.9.2})$$

No sheaf of weight zero is substituted here. The corresponding closed disc in $t = q^{-s}$ is

$$|t| \leq q^{-1-\beta/2}. \quad (\text{DBC.9.3})$$

Euler convergence controls its interior. Condition (C) controls its boundary, with the exceptional pole only when $\tau\omega_s = \omega_1$.

Case 1: the original representation is nontrivial geometrically. Equations (DBC.6.2)–(DBC.6.3) give

$$L(\tau, s) = P_1(q^{-s}), \quad P_0 = P_2 = 1.$$

There is no pole of the permitted type, since twisting by degree never changes the geometric restriction. Thus $P_1(t) \neq 0$ throughout (DBC.9.3), including the boundary.

Case 2: the original representation is geometrically trivial. Irreducibility forces rank one. Let its degree-one Frobenius scalar be $c \in \overline{\mathbf{Q}}_\ell^*$, and put $a = \iota c$. Its original weight gives

$$|a| = q^{\beta/2}. \quad (\text{DBC.9.4})$$

Keep both cohomological factors:

$$L(\tau, s) = \frac{P_1(q^{-s})}{(1 - aq^{-s})^\delta (1 - qa q^{-s})}. \quad (\text{DBC.9.5})$$

The H_c^0 root a^{-1} , when $\delta = 1$, has radius $q^{-\beta/2}$, strictly outside (DBC.9.3). The H_c^2 root

$$t_* = (qa)^{-1} \quad (\text{DBC.9.6})$$

lies on the boundary. It is the exact parameter $\tau\omega_s = \omega_1$. At t_* ,

$$1 - at_* = 1 - q^{-1} \neq 0, \quad \text{ord}_{t_*}(1 - qat) = 1.$$

Condition (C) says that the L-function has a pole of order exactly one there. Therefore

$$\text{ord}_{t_*} P_1 = \text{ord}_{t_*} L + \delta \text{ord}_{t_*}(1 - at) + \text{ord}_{t_*}(1 - qat) = -1 + 0 + 1 = 0. \quad (\text{DBC.9.7})$$

This proves $P_1(t_*) \neq 0$; no eigenvalue has been hidden by cancellation. At every other point of (DBC.9.3), the denominator is nonzero and the L-function is nonzero, so $P_1 \neq 0$ there as well.

Now every Frobenius eigenvalue α on the original $H_c^1(X, \mathcal{F})$ contributes the zero $t = (\iota\alpha)^{-1}$ to P_1 , with its full algebraic multiplicity. The absence of such zeros in (DBC.9.3) gives

$$|\iota\alpha| < q^{1+\beta/2}, \quad \boxed{w_q(\alpha) < \beta + 2}. \quad (\text{DBC.9.8})$$

The strict inequality comes from the boundary nonvanishing and the exact pole-order calculation (DBC.9.7), in addition to Euler convergence in the interior.

DBC10. Exact passage to a general pure lisse sheaf and smooth curve

For an exact sequence of lisse Weil sheaves

$$0 \longrightarrow \mathcal{F}'_0 \longrightarrow \mathcal{F}_0 \longrightarrow \mathcal{F}''_0 \longrightarrow 0$$

whose middle sheaf has pointwise weight β , both nonzero outer sheaves have the same pointwise weight: every local Frobenius matrix is block triangular, so its characteristic polynomial is the product of the two outer characteristic polynomials. The long exact sequence gives

$$H_c^1(X, \mathcal{F}') \xrightarrow{u} H_c^1(X, \mathcal{F}) \xrightarrow{v} H_c^1(X, \mathcal{F}''), \quad \ker v = \operatorname{im} u.$$

Both maps commute with Frobenius. The middle vector space is an extension of $\operatorname{im} u$, a quotient of the first group, by $\operatorname{im} v$, a subspace of the third group. Its Frobenius eigenvalues, with multiplicities, are the union of those of those two images and therefore occur among the outer groups' eigenvalues. A finite composition series and (DBC.9.8) prove the same strict bound for the original $\mathcal{F}_0$, including nonsplit extensions. The zero sheaf has empty spectrum.

Finally, a smooth curve of finite type may have several geometric connected components. Choose $n > 0$ so that F^n fixes each such component and it is defined as a geometrically connected curve over $\mathbf{F}_{q^n}$. Compactly supported cohomology decomposes as the direct sum over these open-and-closed components, with projection and inclusion maps induced by the corresponding inclusions. On each component the previous argument gives

$$2 \log_{q^n} |\iota(\alpha^n)| < \beta + 2$$

for every eigenvalue of F^n arising from an eigenvalue α of the original F . The original and extended-field weights agree exactly:

$$2 \log_{q^n} |\iota(\alpha^n)| = \frac{2n \log |\iota \alpha|}{n \log q} = w_q(\alpha). \quad (\text{DBC.10.1})$$

The pullback sheaf retains pointwise weight β : if a closed point of original degree d has a point above it of degree $d' = d / \gcd(d, n)$ over $\mathbf{F}_{q^n}$, its Frobenius is the power $n / \gcd(d, n)$ of the original local Frobenius, and the absolute-value equality becomes

$$(q^{d\beta/2})^{n/\gcd(d,n)} = (q^n)^{d'\beta/2}. \quad (\text{DBC.10.2})$$

Equations (DBC.10.1)–(DBC.10.2), together with the direct-sum maps, prove Deligne's strict bound for the original smooth curve and original sheaf in full generality as stated in §2.2.10.

The optional covering step mentioned in source line 1925 also has an exact map. For a finite étale cover $f : Y_0 \rightarrow X_0$ of constant degree $e > 0$, the adjunction and trace maps are

$$\mathcal{F}_0 \xrightarrow{\eta} f_* f^* \mathcal{F}_0 \xrightarrow{\operatorname{Tr}_f} \mathcal{F}_0, \quad \operatorname{Tr}_f \eta = e \operatorname{id}_{\mathcal{F}_0}. \quad (\text{DBC.10.3})$$

On an étale-trivializing neighbourhood the first map is the diagonal into e copies and the second is their sum, proving the formula. Because f is finite, compactly supported cohomology identifies the middle sheaf's cohomology with $H_c^i(Y, f^* \mathcal{F})$. Consequently f^* is an injective Frobenius-equivariant map on cohomology, with left inverse $e^{-1} \operatorname{Tr}_f$. The integer e is invertible in $\overline{\mathbf{Q}}_\ell$, including when $\ell \mid e$.

For the group in §3, the subgroup $G^{00}\langle z \rangle$ has finite index and is isomorphic to $G^{00} \times \mathbf{Z}$, with the second generator mapping to the original degree m , not degree one in the original $\mathbf{Z}$. Its preimage in the Weil group corresponds to a finite étale cover, with an accompanying finite constant-field extension if needed to make the new degree map surjective. This justifies the optional direct-product presentation while recording its cohomological injection and its changed degree generator. The proof of (DBC.9.8) above already works in the original extension and does not require this optional step.

DBC11. Review record

Completed calculations: compact conjugacy with the central degree coordinate; categorical extension with arbitrary degree Jordan blocks; exact mixed-to-pure dependency through §1.8.11; finite-kernel

compactness proof with all auxiliary twist factors and inverses; the negative sign of the representation weight; the complete trace denominator; a noncircular proof of the simple curve-zeta pole; both geometric types of connected quadratic cover; and the original- β strict bound with exact cancellation orders, extensions, and field-change factors.

The parent's upstream derivations supply §1.3; DB3-DB6 above prove the analytic theorem §2.1.4. The compact-form and strict-bound derivation has not used the later stronger bound $w_q(\alpha) \leq \beta + 1$ or the Riemann hypothesis for curves. DB9 contains the separate exact original-zeta boundary calculation. Source-reading coverage is the bounded coverage in DB0 and DBC1; source discovery, PDF reading, rendering, and publication were outside this subtask.

The pencil, tensor improvement, and fundamental theorem — DW0-DW11

Proof source: DELIGNE_FUNDAMENTAL_WEIGHT_RECONSTRUCTION.md.

Deligne's fundamental weight argument: the surface, its singular fibres, and the exact weight conclusion

Status: independent derivation of the argument in *La conjecture de Weil II*, §§3.1–3.3, with the coefficient weight retained throughout. This is a reconstruction of the finite-field theorem. It does not assert a transfer of that theorem to the programme's supporting point. The upstream purity criterion, local monodromy theorem and boundary nonvanishing theorem are identified below by their full mathematical conclusions and source locators. Their proofs belong to the adjoining source reconstructions; the present file does not claim to have independently reproved the SGA results used by Deligne.

DW0. Source identity, actual reading, and discrepancies

Author: Pierre Deligne. Work: *La conjecture de Weil. II*, Publications Mathématiques de l'IHÉS 52 (1980), 137–252. The receiving argument is §§3.1–3.3, article pages 197–207. The primary publication is indexed at https://www.numdam.org/item/PMIHES_1980__52__137_0/.

The text read here is the local current French page-record export:

output/Deligne_Weil_II_S20_LaTeX/typed_latex/S20_FR_record_export.tex

SHA256: d6836ae15d98f2b27eb98c9bd039a7d921679becd9fa47ff4717410289dbd351.

This file is a modern transcription, not Deligne-authored TeX. Its page-record verification verifies reversible export and coverage; it is not a mathematical verification of every formula. The original printed PDF was not used in this derivation. The older English TeX companion was read for comparison, not treated as authority.

Actual reading coverage for this derivation:

- Current French §§3.1–3.3, in full, lines 1933–2311, including every displayed relation and each geometric reduction.
- Current French conventions (0.1)–(0.11), lines 192–251; definitions and operations (1.2.1)–(1.2.8), lines 480–530.
- Current French (1.4.1)–(1.5.3), lines 749–872; (1.6.4)–(1.6.14), lines 885–1063; (1.7.1)–(1.7.12), lines 1067–1207; (1.8.1)–(1.8.8), lines 1210–1276.
- Current French final proof of (2.2.9) and all of (2.2.10), lines 1883–1927. The complete proofs of §§2.1–2.2 are handled in the adjoining reconstruction, not certified by these last paragraphs alone.
- Historical English §§3.1–3.3, lines 1832–2072, and displayed (1.6.14.3), for edition comparison.

The following discrepancies are material. They are discrepancies in the available witnesses, not claims that Deligne printed these errors.

Locator	Available witness	Formula used in this derivation and its justification
(3.2.4), (3.2.11), (3.2.13)	Historical English repeatedly changes the French non-strict inequalities into strict inequalities.	The inductive conclusion is $\leq$; the independent strict input is specifically (2.2.10).
(3.2.7)	Historical English calls the external tensor product itself real.	The real sheaf is a direct-sum companion. At weight zero it is $G \oplus G_V$; the full-weight companion is constructed in DW4.
(3.2.9.1)	Historical English omits the final generic stalk of $R^2 f_*$.	The complete sequence has both special and generic $R^2 f_*$ stalks, displayed in DW5.
(3.2.12)	Historical English changes ≤ 2 into < 2 .	The correct bound at source weight zero is ≤ 2 ; DW7 retains the full coefficient weight and obtains $\leq 2\beta + 2$.
(3.3.9)	Historical English uses $\det(1 - Ft)$ while retaining the absolute value of eigenvalues.	$\det(t \cdot 1 - F)$ has roots of modulus $q^{\{i/2\}}$; $\det(1 - Ft)$ has reciprocal roots of modulus $q^{\{-i/2\}}$.
(3.2.1)	Current French writes the duality partner $F_V(-1)$.	The partner is $F_V(1)$: the trace pairing has target $\bar{Q} \otimes (-1)$, and the added (1) makes its target $\bar{Q} \otimes \ell$. The corrected identity is proved in DW4.
(3.2.7.1)	Current French prints ordinary H^1 of a generally open fibre.	Proper-support base change gives H_c^1 for $R^1 f_*$. No properness of that open fibre is assumed.
(3.2.12)	Current French retains a free β inside a proof already restricted to weight zero.	DW7 proves the statement with the actual weight 2β of the external product, removing the free-symbol ambiguity by an explicit calculation.
(1.6.14.3)	Both witnesses display a positive twist $(i+j)/2$.	From $N: V(1) \rightarrow V$, the primitive contribution is $P_{\{-j\}}(-(i+j)/2)$. Applying $N^{\{(i+j)/2\}}$ proves that sign. DW1 records the resulting weight theorem; the adjoining local reconstruction proves it.
(3.1.3.3), optional description	Current French calls the Kummer class of u/v a generator modulo ℓ^n .	For the local equation $uv=t$, over a geometric generic fibre $[v] = -[u]$, so $[u/v] = 2[u]$. The needed branch-sign line is constructed from $[u]$ and $[v]$, retaining their opposite signs. The optional factor-two assertion is not used.

No source file has been overwritten. The derivation below states its own formulas and proves the corrections from the applicable maps.

DW1. Objects, Frobenius, weights, and the exact upstream results

Fix primes $p \neq \ell$, a power $q = p^a$, an algebraic closure $\bar{F}_q$, and a field isomorphism

$$\iota : \bar{\mathbb{Q}}_\ell \longrightarrow \mathbb{C}.$$

Objects carrying subscript $\emptyset$ are defined over F_q . Removing the subscript means extension of scalars to $\bar{F}_q$. Every sheaf is a constructible $\bar{\mathbb{Q}}_\ell$ -sheaf, or a Weil sheaf as specified, on a separated finite-type scheme with ℓ invertible. Frobenius is geometric Frobenius. For a nonzero eigenvalue α ,

$$w_q(\alpha) = 2 \log_q |\iota(\alpha)|. \quad (\text{DW1.1})$$

At a closed point x of degree d , the appropriate base is $N(x) = q^d$. A lisse sheaf is pointwise ι -pure of weight β when each of its local eigenvalues has absolute value $N(x)^{\beta/2}$. A mixed sheaf has a finite filtration with pointwise pure successive quotients. Without an ι qualifier, purity additionally asserts algebraicity and the same modulus for every complex conjugate, and its weights are integers.

The Tate line $\bar{\mathbb{Q}}_\ell(1)$ carries geometric Frobenius q^{-1} . Thus

$$w_q(\alpha q^{-r}) = w_q(\alpha) - 2r, \quad w_q(\alpha\gamma) = w_q(\alpha) + w_q(\gamma), \quad w_q(\alpha^{-1}) = -w_q(\alpha). \quad (\text{DW1.2})$$

All following occurrences of Tate twists use this convention. In particular (-1) raises weight by two; it is not a deleted factor.

The proof uses the following results, which are established earlier in Deligne's paper or are explicitly cited cohomological theorems.

Input D1: degree zero and degree two on a curve, (1.4.1)-(1.4.4). For a geometrically connected smooth curve C , a lisse sheaf with geometric fibre representation V , and $\pi = \pi_1(C)$,

$$H^0(C, \mathcal{F}) = V^\pi, \quad H_c^0(C, \mathcal{F}) = \begin{cases} V^\pi, & C \text{ proper,} \\ 0, & C \text{ not proper,} \end{cases} \quad H_c^2(C, \mathcal{F}) = V_\pi(-1). \quad (\text{DW1.3})$$

For a pointwise pure coefficient of weight β , eigenvalues on $H^{\emptyset}$ or $H_{-C}^{\emptyset}$ have weight β , and on H_{-C}^2 have weight $\beta+2$. For a coefficient whose pointwise weights are bounded above by b , the corresponding upper bounds are b and $b+2$. For a general constructible coefficient, restrict to a dense lisse open for degree two; degree zero consists of global sections, with any finite-support contributions included. These assertions retain the Tate (-1) in (DW1.3).

Input D2: Grothendieck's trace formula, (1.4.5.1). With $d(x) = [k(x) : F_q]$,

$$\prod_{x \in |C_0|} \det(1 - F_x t^{d(x)}, \mathcal{F}_{\bar{x}})^{-1} = \frac{\det(1 - Ft, H_c^1(C, \mathcal{F}))}{\det(1 - Ft, H_c^0(C, \mathcal{F})) \det(1 - Ft, H_c^2(C, \mathcal{F}))}. \quad (\text{DW1.4})$$

Both denominator factors remain present. A zero cohomology group contributes the determinant one by an actual vanishing theorem, not a convention discarding a contribution. For a pure weight- β sheaf,

the Euler product converges and is nonzero for $|t| < q^{-(\beta+2)/2}$ on a curve; this follows from (1.4.6), counting degree- d closed points by $O(q^d)$ and using the geometric series with terms $q^{d(1+\beta/2)}|t|^d$.

Input D3: constituents of real sheaves, (1.5.1). A lisse sheaf on a smooth geometrically connected curve whose local characteristic polynomials become real under ι has pointwise ι -pure simple constituents. Therefore each direct summand of such a sheaf has a finite filtration by lisse pointwise ι -pure quotients. This is the exact input used for the pencil's first direct image. It is not an assertion that a point or a reflection alone forces weights.

Input D4: local monodromy purity, (1.8.4). If a lisse sheaf of weight β is restricted to the geometric generic point of a henselian trait at a boundary point s , its local monodromy filtration M satisfies

$$\mathrm{Gr}_i^M V \text{ is pure of weight } \beta + i \text{ relative to } N(s). \quad (\mathrm{DW1.5})$$

Here the nilpotent logarithm is a Frobenius-compatible map $N: V(1) \rightarrow V$. Inertia acts unipotently after a finite covering; on the monodromy graded pieces it acts trivially in the unipotent case. At a general boundary point,

$$(j_* \mathcal{F})_{\bar{s}} = V^I \subseteq \ker N \subseteq M_0 V, \quad (\mathrm{DW1.6})$$

where the first containment includes taking invariants under the residual finite inertia quotient. The induced graded pieces have weights $\beta+i$, $i \leq 0$. Formation in a tame family along a smooth relative divisor is compatible with taking fibres, as constructed in (1.7.8), (1.8.6)–(1.8.8).

Input D5: strict curve estimate, (2.2.10). Every eigenvalue on $H_c^1(C, \mathbb{F})$ for a lisse pure weight- β coefficient satisfies

$$w_q(\alpha) < \beta + 2. \quad (\mathrm{DW1.7})$$

The strict sign comes from the Hadamard-de la Vallée-Poussin argument in §2, through the nonvanishing of its representation-theoretic L-functions on the boundary. The weaker $\leq \beta+2$ follows already from (1.8.2), or directly from (DW1.4) and the convergence disk. In DW6 it is precisely the strict sign, together with a discrete weight congruence, that produces a full one-unit gain.

Input D6: cohomological geometry. Proper-support base change; finite-dimensionality; the Künneth isomorphism; the compact-support Leray spectral sequence; Poincaré duality; the existence of sufficiently general Lefschetz pencils; the local Picard–Lefschetz calculation for an ordinary node; and quasi-unipotence followed by a finite cover are the geometric theorems used by Deligne. This reconstruction proves their applications and all resulting weight inferences. It does not claim fresh proofs of these foundational SGA theorems.

DW2. The surface and the pencil, with its exceptional divisor retained

Let S be a smooth projective surface over an algebraically closed field, D a normal-crossings divisor, $V = S \setminus D$, and $j: V \rightarrow S$. Let $\mathbb{F}$ be lisse on V , tame along D . Choose an embedding in projective space and a pencil of hyperplanes with axis A ; its parameter space A^* is a projective line. Set

$$\tilde{S} = \{(x, t) \in S \times A^* : x \in H_t\}, \quad \pi: \tilde{S} \rightarrow S, \quad f: \tilde{S} \rightarrow A^*, \quad \tilde{V} = \pi^{-1}(V). \quad (\mathrm{DW2.1})$$

The hypotheses of (3.1.1) are all used:

1. The axis is transverse to S and misses D ; therefore π blows up the finite set $A \cap V$.

2. Critical points of f on $\tilde{S}$ are ordinary quadratic singularities and miss D .
3. Critical points of the map on the normalization D' are ordinary quadratic singularities and lie over smooth points of D .
4. Each exceptional fibre has exactly one exceptional point.

The three exceptional types are consequently: an interior node; a tangency of the fibre to a smooth boundary branch; or a fibre meeting a crossing of two boundary branches.

Since the blowup centres lie in V , the exceptional lines contribute to compact-support cohomology in degree two. The full blowup formula is

$$H_c^m(\tilde{V}, \pi^* \mathcal{F}) \simeq H_c^m(V, \mathcal{F}) \oplus \begin{cases} H^0(A \cap V, \mathcal{F})(-1), & m = 2, \\ 0, & m \neq 2. \end{cases} \quad (\text{DW2.2})$$

In particular π^* is injective. Its retraction is the map obtained by taking the Poincaré-dual transpose of pullback in ordinary cohomology. The added term is not identified with zero or removed from the construction.

Properness of A^* yields the Frobenius-compatible Leray spectral sequence

$$E_2^{p,r} = H^p(A^*, R^r f_! \pi^* \mathcal{F}) \implies H_c^{p+r}(\tilde{V}, \pi^* \mathcal{F}). \quad (\text{DW2.3})$$

Here $f_!$ on the open surface means $f_* j_!$ for the proper surface map. That is why the open fibres contribute compact-support cohomology. All differentials commute with Frobenius. Hence every eigenvalue of the abutment occurs on a subquotient of one of the displayed E_2 groups, with algebraic multiplicity controlled by those subquotients.

DW3. All three local vanishing-cycle calculations

At an exceptional value t , work on the strict henselization of A^* at t , with geometric generic point η . Write Φ_x^r for the vanishing cycles of $j_! \pi^* F$, supported at its one exceptional point x . Assume additionally that local boundary monodromy is unipotent. Then near a boundary point the lisse coefficient has a finite filtration whose graded pieces extend constantly over the strict localization, as in (3.1.2).

For a two-element branch set B , define the sign line

$$\epsilon(B) = \mathbb{Z}^B / \mathbb{Z}(1, 1), \quad \epsilon_\ell(B) = \epsilon(B) \otimes \overline{\mathbb{Q}}_\ell. \quad (\text{DW3.1})$$

The two choices of branch give opposite generators. A Frobenius permutation of the branches therefore acts by $+1$ or -1 , so this line is pure of weight zero. This conclusion keeps the sign representation; it does not identify it with a canonically trivial representation.

Interior node. The coefficient is lisse at x . The relative-dimension-one Picard-Lefschetz calculation gives

$$\Phi_x^r(j_! \pi^* \mathcal{F}) = 0 \quad (r \neq 1), \quad \Phi_x^1(j_! \pi^* \mathcal{F}) = \mathcal{F}_x(-1) \otimes \epsilon_\ell(B). \quad (\text{DW3.2})$$

In a node chart $uv=t$, choosing u identifies the geometric generic local cohomology with the Kummer class of u ; choosing v negates this class because t has compatible roots over the geometric generic field. Thus the branch ambiguity is precisely the sign line. The cohomological factor is (-1) , with Frobenius multiplier $N(x)$, and is retained.

Tangency to the boundary. First use the constant coefficient and the exact sequence

$$0 \rightarrow j_! \overline{\mathbb{Q}}_\ell \rightarrow \overline{\mathbb{Q}}_\ell \rightarrow \overline{\mathbb{Q}}_{\ell,D} \rightarrow 0. \quad (\text{DW3.3})$$

The ambient surface morphism is smooth at x , so its constant sheaf has no vanishing cycles there. The long exact sequence gives

$$\Phi_x^r(j_! \overline{\mathbb{Q}}_\ell) \simeq \Phi_x^{r-1}(D, \overline{\mathbb{Q}}_\ell). \quad (\text{DW3.4})$$

The generic fibre of the local double cover on $\mathfrak{D}$ has two points B ; its reduced degree-zero cohomology is $\mathbb{Q}_\ell \wedge B / \mathbb{Q}_\ell = \epsilon_\ell(B)$. Therefore only $r=1$ is nonzero and its value is $\epsilon_\ell(B)$, without a Tate twist. Apply the same exact-sequence calculation to each constant graded coefficient in the local filtration. Since all other cohomological degrees vanish, the resulting filtration has

$$\text{Gr}_F^i \Phi_x^1(j_! \pi^* \mathcal{F}) = \text{Gr}_F^i(\mathcal{F})_x \otimes \epsilon_\ell(B). \quad (\text{DW3.5})$$

Crossing of two boundary branches. Let $i: \mathfrak{D}' \rightarrow S$ be the normalization near the crossing. The constant-coefficient exact sequence is

$$0 \rightarrow j_! \overline{\mathbb{Q}}_\ell \rightarrow \overline{\mathbb{Q}}_\ell \rightarrow i_* \overline{\mathbb{Q}}_\ell \rightarrow \overline{\mathbb{Q}}_{\ell,x} \otimes \epsilon_\ell(B) \rightarrow 0. \quad (\text{DW3.6})$$

Both f and $f \circ i$ are smooth at the relevant points. Applying vanishing cycles to the two short exact sequences obtained from (DW3.6) shifts the point-supported term by two:

$$\Phi_x^r(j_! \overline{\mathbb{Q}}_\ell) \simeq \Phi_x^{r-2}(\overline{\mathbb{Q}}_{\ell,x} \otimes \epsilon_\ell(B)). \quad (\text{DW3.7})$$

The point-supported term has no nearby generic fibre. With the convention of (3.1), its only vanishing-cycle group is in degree -1 , equal to its stalk. Thus (DW3.7) is nonzero only for $r=1$, where it equals $\epsilon_\ell(B)$. Devissage in the coefficient filtration gives exactly (DW3.5), again with no Tate twist.

Consequently every one of the three types has vanishing cycles concentrated in degree one. For a coefficient whose local graded weights lie in $\mathfrak{b} + \mathbb{Z}$, all these cycle weights also lie in $\mathfrak{b} + \mathbb{Z}$: the sign line adds zero, and the nodal Tate factor adds two. This congruence, rather than an invented metric or a point's position, is what the next stage extracts from the local geometry.

DW4. Real characteristic polynomials and the correct real companion

Let $C_\circ$ be a smooth curve and $E_\circ$ a lisse, pointwise pure weight- $\mathfrak{b}$, $\mathfrak{v}$ -real sheaf. We prove the full statement (3.2.1) for $i=0, 1, 2$.

First suppose $C_\circ$ geometrically connected. Applying $\mathfrak{v}$ to (DW1.4), the Euler product is a real rational function since all its local characteristic polynomials are real. By D1, the zeros of the H_{C^2} determinant have absolute value $q^{\{- (\mathfrak{b}+2)/2\}}$. By D5, the zeros of the H_{C^1} determinant have strictly larger absolute value. The H_{C^0} determinant, if nontrivial, has zeros of absolute value $q^{\{- \mathfrak{b}/2\}}$, also strictly larger. Thus there is no cancellation at the radius $q^{\{- (\mathfrak{b}+2)/2\}}$: the poles on that circle, counted with their multiplicities, are exactly the zeros of the H_{C^2} determinant. The real rational function has a conjugation-stable pole multiset there. The determinant's constant term is one, so its polynomial is real.

If the curve is not proper, $H_{C^0} = 0$. If it is proper, Poincaré duality gives a perfect equivariant pairing

$$H^0(C, \mathcal{E}) \otimes H^2(C, \mathcal{E}^\vee(1)) \longrightarrow \overline{\mathbb{Q}}_\ell. \quad (\text{DW4.1})$$

The coefficient $\text{Ev}(1)$ is pure of weight $-b-2$ and real: inverse roots of a real polynomial remain conjugation-stable, and the Tate multiplier is real. Its H^2 determinant is real by the preceding argument. Reciprocal eigenvalues preserve conjugation-stability, proving that the H^0 determinant is real. Now both denominator factors in (DW1.4) are real, so its numerator, the H_{-c}^1 determinant, is real too.

For a curve which becomes disconnected over $\bar{F}_q$, pass to its finite constant-field extensions component-wise. Frobenius permutes the components in cycles. A cycle of length d contributes $\det(1 - F^d t^d, V)$; the same argument over F_{q^d} makes this real. Multiplying these real factors proves the general assertion (3.2.2).

Now let F_0 be any lisse pure weight- β sheaf on a smooth curve U_0 ; it need not be real. On $V_0 = U_0 \times U_0$, set

$$\mathcal{G}_0 = \text{pr}_1^* \mathcal{F}_0 \otimes \text{pr}_2^* \mathcal{F}_0. \quad (\text{DW4.2})$$

This external product has weight 2β . Choose $a \in \bar{\mathbb{Q}}_{\ell}^{\times}$ with $\iota(a) = q^{2\beta}$, and let $\mathbb{L}_a$ be the constant rank-one **Weil** sheaf over F_q with Frobenius a . Define

$$\mathcal{H}_0 = \mathcal{G}_0 \oplus (\mathcal{G}_0^{\vee} \otimes \mathbb{L}_a). \quad (\text{DW4.3})$$

The line has weight 4β , so both summands have weight 2β . If λ is a local eigenvalue of G_0 at a degree- d point, then $|\iota\lambda| = q^{d\beta}$ and

$$\iota(a^d \lambda^{-1}) = \frac{q^{2d\beta}}{\iota\lambda} = \overline{\iota\lambda}. \quad (\text{DW4.4})$$

Hence H_0 is ι -real. This constructs the real companion while keeping G_0 , its original eigenvalues, the weight β , and the extra line explicit. It is a direct-sum receiving construction, not a replacement of F_0 by a normalized coefficient.

Write $W_0 \subset A_0^*$ for the nonexceptional parameter locus. Proper-support base change gives, for a closed parameter x and geometric fibre Y ,

$$(R^1 f_{!} \pi^* \mathcal{H}_0)_{\bar{x}} = H_c^1(Y, \pi^* \mathcal{H}). \quad (\text{DW4.5})$$

The just-proved reality proposition applies to that smooth, possibly disconnected open curve over $k(x)$. It follows that $w^* R^1 f_{!} \pi^* H_0$ is real. The sheaf

$$\mathcal{R}_0 = w^* R^1 f_{!} \pi^* \mathcal{G}_0 \quad (\text{DW4.6})$$

is a direct summand, because $Rf_{!}$ respects the finite direct sum in (DW4.3). Input D3 supplies a finite filtration of R_0 by lisse pointwise pure quotients. At $\beta=0$, choose $a=1$; (DW4.3) is exactly Deligne's $G_0 \oplus G_0^{\vee}$ construction in (3.2.7).

DW5. The boundary forces the weight congruence of every non-constant piece

Maintain all pencil hypotheses of DW2-DW3. Let

$$\mathcal{R} = R^1 f_{!} \pi^* \mathcal{G}, \quad \mathcal{S} = R^2 f_{!} \pi^* \mathcal{G}.$$

At an exceptional parameter t , the concentration of vanishing cycles in degree one gives the complete exact sequence

$$0 \longrightarrow \mathcal{R}_{\bar{t}} \longrightarrow \mathcal{R}_{\bar{\eta}} \longrightarrow \Phi_x^1 \longrightarrow \mathcal{S}_{\bar{t}} \longrightarrow \mathcal{S}_{\bar{\eta}} \longrightarrow 0. \quad (\text{DW5.1})$$

The initial injection proves that $\mathcal{R}$ has no section supported at $\mathfrak{t}$: any such section vanishes at the generic stalk, and therefore at the special stalk. Applying this at every exceptional point gives $\mathcal{R} \hookrightarrow w_* w^* \mathcal{R}$.

Let $G^i(w^* \mathcal{R}_0)$ be the finite pure-quotient filtration from DW4. Extend it by intersection,

$$G^i \mathcal{R}_0 = \mathcal{R}_0 \cap w_* G^i(w^* \mathcal{R}_0) \quad \text{inside } w_* w^* \mathcal{R}_0. \quad (\text{DW5.2})$$

This is a finite filtration and restricts to the original filtration on w_0 . Its special-stalk filtration is induced by intersection with the generic one. Passing to the associated graded of the inclusion in (DW5.1), with the induced quotient filtration, gives

$$0 \rightarrow (Gr_G^i \mathcal{R}_0)_{\bar{t}} \rightarrow (Gr_G^i \mathcal{R}_0)_{\bar{\eta}} \rightarrow A_t^i \rightarrow 0, \quad (\text{DW5.3})$$

where $A_{\bar{t}^i}$ is a subquotient of Φ_x^1 . All maps respect the local Weil action.

We compute the possible weights of Φ_x^1 . At a boundary point of either curve factor, D4 gives coefficient weights $\beta + j$, $j \in \mathbb{Z}$. At an interior point the weight is β . Tensoring the two factors gives local graded weights

$$2\beta + j_1 + j_2 \in 2\beta + \mathbb{Z}. \quad (\text{DW5.4})$$

Tame unipotence makes those graded pieces constant on the relevant strict local open. The three calculations of DW3 add either zero or two to these weights and retain a weight-zero branch sign. Therefore every Φ_x^1 and every $A_{\bar{t}^i}$ has all its weights in $2\beta + \mathbb{Z}$.

Let the pure lisse restriction $w^* Gr_G^i \mathcal{R}_0$ have weight γ . D4, now applied at the parameter boundary $\mathfrak{t}$, says all weights of its local generic representation lie in $\gamma + \mathbb{Z}$. If $A_{\bar{t}^i} \neq 0$, some eigenvalue occurs in this nonzero subquotient. Its weight belongs to both $\gamma + \mathbb{Z}$ and $2\beta + \mathbb{Z}$. Thus

$$\gamma - 2\beta \in \mathbb{Z}. \quad (\text{DW5.5})$$

If $A_{\bar{t}^i} = 0$ for every exceptional $\mathfrak{t}$, the special-to-generic map is an isomorphism everywhere. Inertia therefore acts trivially, and the graded sheaf extends lisse across all of A^* . The geometric projective line has trivial étale fundamental group, so this graded sheaf is geometrically constant. Its arithmetic Frobenius need not be the identity; nevertheless its geometric H^1 is zero, since

$$H^1(\mathbb{P}_{\mathbb{F}_q}^1, \overline{\mathbb{Q}}_\ell) = 0. \quad (\text{DW5.6})$$

We have therefore proved the exact alternative needed by (3.2.9): a nonconstant piece has weight congruent to 2β modulo the integers, while every piece not constrained by this boundary test contributes zero to the subsequent H^1 .

DW6. The strict inequality and discreteness supply the decisive gain

Every pure constituent of $w^* \mathcal{R}^1 f_{-!} \pi^* G_0$ has some real weight γ . Evaluate it at any closed nonexceptional parameter x . Its fibre eigenvalues occur among those of the compact-support H^1 of a smooth curve with coefficient of weight 2β . Input D5 gives

$$\gamma < 2\beta + 2. \quad (\text{DW6.1})$$

For a nonconstant graded piece, (DW5.5) gives an integer $m = \gamma - 2\beta$. Since $m < 2$,

$$\gamma \leq 2\beta + 1. \quad (\text{DW6.2})$$

This implication uses both inputs. The strict inequality alone permits weights arbitrarily close to $2\beta + 2$; the congruence alone permits arbitrarily large integers. Their conjunction supplies the exact one-unit gain used in the product argument.

DW7. The inductive argument, retaining the full coefficient weight

For each integer $k \geq 0$, consider the actual assertion

$$B_k : \quad w_q(\alpha) \leq \beta + 1 + 2^{-k} \quad \text{on } H_c^1(U, \mathcal{F}), \quad (\text{DW7.1})$$

for every smooth curve $U_\circ$ and every lisse pointwise pure coefficient $F_\circ$ of arbitrary real weight β . We prove all these assertions. This is the full-weight version of Deligne's (3.2.4), whose displayed proof restricts to $\beta = 0$.

Base case. The Euler product in (DW1.4) is nonzero on $|t| < q^{-(\beta+2)/2}$. The H_{-c}^0 and H_{-c}^2 determinants have no roots in this open disk by D1. Thus the H_{-c}^1 determinant has no root there either, proving $w_q(\alpha) \leq \beta + 2$. This is B_0 .

Extension from lisse opens. Let $u: 0_\circ \rightarrow C_\circ$ be a dense smooth open of a curve and $E_\circ$ a constructible coefficient whose restriction to $0_\circ$ is lisse pure of weight γ . The quotient in

$$0 \rightarrow u_! u^* \mathcal{E}_0 \rightarrow \mathcal{E}_0 \rightarrow \mathcal{E}_0 / u_! u^* \mathcal{E}_0 \rightarrow 0 \quad (\text{DW7.2})$$

has finite support, and hence has zero compact-support H^1 . The long exact sequence yields a surjection

$$H_c^1(O, u^* \mathcal{E}) \twoheadrightarrow H_c^1(C, \mathcal{E}). \quad (\text{DW7.3})$$

Therefore B_k implies the bound $\gamma + 1 + 2^{-k}$ on the latter group. No claim that boundary stalks vanish is used; only their degree-one cohomology vanishes.

Inductive step under the pencil hypotheses. Form the full external product of DW4. The total-degree-two terms of (DW2.3) are

$$E_2^{1,1} = H^1(A^*, \mathcal{R}), \quad E_2^{0,2} = H^0(A^*, \mathcal{S}), \quad E_2^{2,0} = H^2(A^*, R^0 f_! \pi^* \mathcal{G}). \quad (\text{DW7.4})$$

For $E_2^{\wedge\{1,1\}}$, filter R by (DW5.2). A geometrically constant quotient has zero H^1 by (DW5.6). Every other quotient has generic pure weight at most $2\beta + 1$, by DW6. Apply (DW7.3) and B_k to each such quotient. Long exact sequences through this finite filtration show

$$w_q(\alpha) \leq 2\beta + 2 + 2^{-k} \quad \text{on } E_2^{1,1}. \quad (\text{DW7.5})$$

For $E_2^{\wedge\{0,2\}}$, proper-support base change identifies each stalk of $R^2 f_! \pi^* G_\circ$ with a fibre's compact-support H^2 . On a dense smooth open of every fibre component the coefficient has weight 2β , so D1 places these degree-two weights at $2\beta + 2$; components of dimension zero contribute no H^2 . If a nonzero global eigenvector with eigenvalue α is evaluated at a degree- d closed point where it is nonzero, α^d is a

local Frobenius eigenvalue. Such a point exists because a nonzero constructible section is nonzero on a nonempty constructible locus containing a closed point. Thus $w_q(\alpha) = w_{\{q^d\}}(\alpha^d) \leq 2\beta + 2$. For $E_2 \in \{2, 0\}$, the coefficient $R^0 f_{\pi*} G_{\bullet}$ has pointwise weights at most 2β ; D_1 on the base curve adds the Tate shift two. Thus

$$w_q(\alpha) \leq 2\beta + 2 \quad \text{on } E_2^{0,2} \text{ and } E_2^{2,0}. \quad (\text{DW7.6})$$

Every total-degree-two abutment eigenvalue is an eigenvalue of a subquotient of these terms. Equations (DW7.5)–(DW7.6) therefore give

$$w_q(\lambda) \leq 2\beta + 2 + 2^{-k} \quad \text{on } H_c^2(\widetilde{V}, \pi^* \mathcal{G}). \quad (\text{DW7.7})$$

The full Künneth decomposition in degree two contains all three terms

$$H_c^2(V, \mathcal{G}) \simeq (H_c^0(U, \mathcal{F}) \otimes H_c^2(U, \mathcal{F})) \oplus (H_c^1(U, \mathcal{F}) \otimes H_c^1(U, \mathcal{F})) \oplus (H_c^2(U, \mathcal{F}) \otimes H_c^0(U, \mathcal{F})). \quad (\text{DW7.8})$$

Combined with the blowup injection (DW2.2), its middle summand injects into the space controlled by (DW7.7). If α is a Frobenius eigenvalue on $H_{c-1}^1(U, \mathcal{F})$, choose a nonzero eigenvector over $\bar{\mathbb{Q}}_{\ell}$; its tensor square is nonzero and has eigenvalue α^2 . Consequently

$$2w_q(\alpha) = w_q(\alpha^2) \leq 2\beta + 2 + 2^{-k}, \quad \text{hence} \quad w_q(\alpha) \leq \beta + 1 + 2^{-(k+1)}. \quad (\text{DW7.9})$$

This is $B_{\{k+1\}}$. All product terms, the exceptional blowup term, and all Tate factors remain part of the construction; the proof uses an injection rather than declaring the other summands absent.

Removing the auxiliary geometric hypotheses. A finite extension $F_q \subset F_{\{q^d\}}$ replaces eigenvalue α by α^d and retains its weight exactly:

$$w_{q^d}(\alpha^d) = 2 \frac{\log |\iota \alpha^d|}{\log q^d} = w_q(\alpha). \quad (\text{DW7.10})$$

Over the algebraic closure, sufficiently general embeddings and pencils with the four conditions of DW2 exist. Their finite coefficient data descend to some finite extension. A further finite extension makes every exceptional point rational. Equation (DW7.10) returns the resulting estimates to the original field without changing them.

Quasi-unipotence gives a finite surjective cover $g: U' \rightarrow U_{\bullet}$, with U' smooth, on which the pullback sheaf has tame unipotent boundary monodromy. Pullback on compact-support cohomology is injective. One way to see the injection is to use the trace (or the Poincaré-dual trace for a finite generically étale cover): trace composed with pullback is multiplication by the nonzero degree. The coefficient field has characteristic zero, so this degree is invertible. A purely inseparable factor is a universal homeomorphism and induces an equivalence on the étale cohomology involved. Thus the original cohomology is a direct summand of the cover's cohomology for the purpose of the eigenvalue bound. The pulled-back coefficient still has weight β , because local Frobenius eigenvalues and residue-field norms are raised to the same extension degrees. This proves $B_{\{k+1\}}$ without the temporary hypotheses.

The induction is complete for every k . For any fixed eigenvalue α , if $w_q(\alpha) > \beta + 1$, choose k with $2^{-(k+1)} < w_q(\alpha) - \beta - 1$; this contradicts B_k . Therefore

$$\boxed{w_q(\alpha) \leq \beta + 1 \text{ on } H_c^1(U, \mathcal{F})}. \quad (\text{DW7.11})$$

This is an exact universal conclusion from the geometrically repeatable argument, not a computation at a finite set of heights or primes.

DW8. Curve purity by the exact duality, including the Tate factor

Let $j:U_{\circ} \rightarrow X_{\circ}$ be a dense open in a smooth projective curve and $F_{\circ}$ lisse pointwise pure of weight β . The exact sequence (DW7.2), now with $E=j_*F$, gives a surjection from $H_{-c^1}(U, F)$ to $H^1(X, j_*F)$. By DW7,

$$w_q(\alpha) \leq \beta + 1 \quad \text{on } H^1(X, j_*F). \quad (\text{DW8.1})$$

The curve's middle-extension duality, cited by Deligne from SGA 4½ [dualité] (1.3), (2.1), gives a perfect pairing

$$H^1(X, j_*F) \otimes H^1(X, j_*F^{\vee}(1)) \longrightarrow \overline{\mathbb{Q}}_{\ell}. \quad (\text{DW8.2})$$

The second coefficient has weight $-\beta-2$. Apply (DW8.1) to it: the inverse eigenvalue α^{-1} has weight at most $-\beta-1$. Using (DW1.2),

$$-w_q(\alpha) \leq -\beta - 1, \quad \text{so} \quad w_q(\alpha) \geq \beta + 1. \quad (\text{DW8.3})$$

Together with (DW8.1) this proves exact weight $\beta+1$. For degrees zero and two, D1 gives weights β and $\beta+2$ directly. Thus the complete result (3.2.3) is

$$\boxed{w_q(\alpha) = \beta + i \quad \text{on } H^i(X, j_*F).} \quad (\text{DW8.4})$$

For $i=1$, this group is also the image of $H_{-c^1}(U, F) \rightarrow H^1(U, F)$. The statement is not purity of all of $H_{-c^1}(U, F)$ when boundary contributions occur; those may have lower weights. Their presence is exactly why both maps and their image are specified.

DW9. From curves to the full direct-image theorem

We reconstruct (3.3.1): for a finite-type morphism $f:X \rightarrow Y$ of Deligne's separated schemes and a mixed coefficient of weights $\leq n$, each $R^i f_! F$ is mixed of weights $\leq n+i$.

The reductions are mathematically necessary and preserve the indicated degree shifts.

Coefficient exact sequences. From $0 \rightarrow F' \rightarrow F \rightarrow F'' \rightarrow 0$, the segment

$$R^{i-1} f_! F'' \rightarrow R^i f_! F' \rightarrow R^i f_! F \rightarrow R^i f_! F'' \quad (\text{DW9.1})$$

shows that the required upper bound for F' and F'' gives it for F , and the bounds for F and F'' give it for F' . In the latter inference, the preceding term has upper weight $n+i-1 \leq n+i$. Direct summands inherit the bound. Mixedness and the weight bound survive subquotients and extensions because the eigenvalue multisets of filtered finite-dimensional stalks are the union of those of their graded pieces.

Open and closed parts of the source. For $j:U \rightarrow X$, $i:S \rightarrow X$ its complement,

$$0 \rightarrow j_! j^* F \rightarrow F \rightarrow i_* i^* F \rightarrow 0 \quad (\text{DW9.2})$$

reduces the assertion to the two strata, by (DW9.1).

Open and closed parts of the target. Proper-support base change identifies the restrictions of $R^i f_! F$ with the corresponding direct images after each base change. A finite stratification on which those restrictions have the desired filtrations yields a finite global mixed filtration by successive extension by zero. Thus Noetherian induction on target strata is permitted.

Composition. If $f = g \circ h$, the spectral sequence

$$R^p g_! R^q h_! \mathcal{F} \implies R^{p+q} f_! \mathcal{F} \quad (\text{DW9.3})$$

propagates upper weight $n+p+q$ to the indicated abutment degree. Every differential and subquotient is kept in that inference.

Universal homeomorphisms. Nilpotent reduction and purely inseparable changes induce the corresponding equivalences on étale sites; the coefficient and compact-support cohomology statements are unchanged. This is a cited property of étale cohomology, not an assertion about arbitrary topological maps.

Relative dimension zero. A geometric fibre is finite, so its compact-support cohomology is concentrated in degree zero. A point of residue degree d over a closed target point contributes an induced Frobenius block whose d th power has the source eigenvalues. If the latter have modulus $(N(y)^d)^{n/2}$, the block eigenvalues have modulus $N(y)^{n/2}$. Thus $f_!$ preserves the coefficient's pure weight, and higher $R^i f_!$ vanish. Apply coefficient devissage for mixed coefficients.

Stratification and Noether normalization allow a finite-type morphism to be factored on strata into relative-dimension-zero maps and successive relative-dimension-one projections. Repeated use of (DW9.2)–(DW9.3) therefore reduces the theorem to relative dimension one. Further coefficient and source stratification make the coefficient lisse and pointwise pure of weight n .

On a generic fibre, after a finite purely inseparable extension if necessary, remove finitely many points to obtain a smooth curve. A finite cover makes the boundary ramification tame, and its compactification is a smooth projective curve with finite étale boundary after shrinking the base. Trace splits the coefficient as a direct summand of the finite cover's pushforward. These constructions spread from the generic fibre to an open neighbourhood by finite presentation. Target Noetherian induction, universal-homeomorphism invariance, the relative-dimension-zero case, and the direct-summand reduction therefore yield the actual remaining geometric situation:

$$X \xrightarrow{j} \overline{X} \xleftarrow{i} D, \quad \overline{f} : \overline{X} \rightarrow Y \quad (\text{DW9.4})$$

with $f = \overline{f} \circ j$, $\overline{f}$ smooth projective of pure relative dimension one, $D \rightarrow Y$ finite étale, and F lisse of weight n on X , tame at D . These geometric reductions are the ones in Deligne's (3.3.1), article p.205; their finite-cover and compactification existence is part of the upstream algebraic-geometric apparatus.

Now retain the full boundary sequence

$$0 \rightarrow j_! \mathcal{F} \rightarrow j_* \mathcal{F} \rightarrow i_* i^* j_* \mathcal{F} \rightarrow 0. \quad (\text{DW9.5})$$

The formation of $j_* \mathcal{F}$, its boundary monodromy filtration, and the corresponding proper direct images is compatible with the fibres in this tame smooth-relative-curve situation. Applying DW8 to each closed fibre shows that

$$R^r \overline{f}_* j_* \mathcal{F} \text{ is pointwise pure of weight } n + r. \quad (\text{DW9.6})$$

By D4, the induced monodromy filtration on $i^* j_* \mathcal{F}$ has graded pieces of weights $n+k$, zero for $k > 0$. Its finite direct image along $\overline{f}$ is therefore mixed of weights at most n , and has no higher direct images. Apply $R \overline{f}_*$ to (DW9.5). For degree zero, $R^0 f_! F$ is a subsheaf of the weight- n term in (DW9.6). For

degree one, the only extra contribution is a quotient of the weight $\leq n$ boundary pushforward followed by a subobject of the pure weight $-n+1$ term. In degrees at least two, the boundary has no degree $r-1$ or r direct image, so the desired comparison is an isomorphism. In every degree the result is mixed of weights $\leq n+r$. All reductions then prove (3.3.1).

The identical argument with the retained real coefficient weight β proves the ι -mixed version (3.3.10). Its weights are at most $\beta+i$ and congruent modulo z to one of the original coefficient weights. This congruence persists through each local monodromy shift, the product calculation, duality, and the integral cohomological degree shifts.

DW10. Lower weights, integrality, and smooth/proper purity

The upper direct-image theorem specializes at $Y=\text{Spec}(F_q)$ to

$$H_c^i(X, \mathcal{F}) \text{ mixed of weights } \leq n + i \quad (\text{DW10.1})$$

for a coefficient mixed of weights $\leq n$.

There are two distinct ways lower information enters Deligne's §3.3; neither is silently substituted for the other.

Algebraic-integrality lower bound, (3.3.2)-(3.3.4). If a nonzero algebraic integer α is pure of integral weight w relative to q , and has degree d , the absolute value of its norm is

$$|N_{\mathbb{Q}(\alpha)/\mathbb{Q}}(\alpha)| = \prod_{\sigma} |\sigma(\alpha)| = q^{wd/2}. \quad (\text{DW10.2})$$

Its norm is a nonzero integer, hence has absolute value at least one; since $q>1$, $w \geq 0$. SGA 7 XXI (5.2.2), as explicitly cited by Deligne, gives integrality of $R^i f_* !F$ and, above the maximal fibre dimension d_X , of $R^i f_* !F(i-d_X)$. The second conclusion retains the geometric Frobenius multiplier $N(y)^{-(i-d_X)}$ and forces

$$w \geq 2(i - d_X) \quad (i > d_X). \quad (\text{DW10.3})$$

Together with DW9 these give all intervals in (3.3.3)-(3.3.4). This arithmetic integrality is not merely the statement that the weight label belongs to $\mathbb{Z}$; it is the stronger assertion that eigenvalues are algebraic integers, used through their norm.

Poincaré-duality lower bound, (3.3.5). Suppose X_0 is smooth of pure dimension N and F_0 is lisse mixed of weights $\geq n$. Its dual has weights $\leq -n$. Apply (DW10.1) in degree $2N-i$ and retain the Tate twist:

$$H_c^{2N-i}(X, \mathcal{F}^\vee)(N) \text{ has weights } \leq -n + (2N - i) - 2N = -n - i. \quad (\text{DW10.4})$$

Poincaré duality identifies $H^i(X, F)$ with the dual of that vector space. Inverting eigenvalues negates their weights, proving

$$H^i(X, \mathcal{F}) \text{ mixed of weights } \geq n + i. \quad (\text{DW10.5})$$

When F_0 is pure of weight n , the image of $H_c^i \rightarrow H^i$ is a quotient of a space of weights $\leq n+i$ and a subspace of one of weights $\geq n+i$. Its every eigenvalue has exactly weight $n+i$, proving (3.3.6).

If X_0 is also proper, compact and ordinary cohomology coincide. Thus for every eigenvalue, with no discarded boundary contribution,

$$|\iota\alpha| = q^{(n+i)/2}. \quad (\text{DW10.6})$$

For the constant coefficient $\bar{Q}_\ell$, $n=0$, hence H^i has weight i . We now give the additional descent of characteristic polynomials, corresponding to Deligne's citation of the Weil I argument (1.7)=(1.6). Run the purity proof for every isomorphism $\iota: \bar{Q}_\ell \rightarrow \mathbb{C}$. Every eigenvalue is algebraic: a transcendental element could be sent by such field isomorphisms to transcendental numbers of different absolute values, contradicting its fixed required modulus. All its algebraic conjugates have that same weight. Set $P_i(t) = \det(1 - Ft, H^i(X, Q_\ell))$; their coefficients are therefore algebraic. Their reciprocal roots have mutually distinct moduli $q^{i/2}$ as i varies, so no two P_i have a common root.

The full trace formula gives

$$Z(X_0, t) = \prod_{x \in |X_0|} (1 - t^{d(x)})^{-1} = \prod_i P_i(t)^{(-1)^{i+1}}. \quad (\text{DW10.6a})$$

The left side has integer Taylor coefficients and constant term one. The right side is rational over an algebraic number field. Its expression as a ratio of coprime polynomials with constant terms one is unique; applying any automorphism of $\bar{Q}/Q$ leaves the Taylor coefficients fixed, and hence leaves that rational function and its coprime numerator and denominator fixed. Both polynomials therefore lie in $Q[t]$. Each P_i is recovered from this coprime rational function by collecting the root multiset of modulus $q^{i/2}$, with its numerator or denominator sign already given by $(-1)^{i+1}$. Since every conjugate of such a root retains that modulus, this multiset is Galois-stable and $P_i \in Q[t]$. The integrality theorem cited above applies to the integral constant coefficient and says the Frobenius eigenvalues are algebraic integers. Their elementary symmetric polynomials are then algebraic integers; being rational, they belong to $\mathbb{Z}$. Hence $P_i \in \mathbb{Z}[t]$. Finally (DW10.6a), including its closed-point factors, is independent of ℓ , and its unique decomposition by the distinct root moduli gives the same P_i for every $\ell \neq p$.

The exact polynomial convention in (3.3.9) is

$$\det(t \, 1 - F^*, H^i(X, \bar{Q}_\ell)) \in \mathbb{Z}[t], \quad (\text{DW10.7})$$

with roots of modulus $q^{i/2}$. The inverse-root polynomial $\det(1 - F^*t)$ retains the opposite root radius.

For completeness, the slope description in (3.3.7)-(3.3.8) retains the normalization explicitly: for every p -adic valuation satisfying $v(q)=1$, put

$$(r, s) = (v(\alpha), v(q^w \alpha^{-1})), \quad r + s = w. \quad (\text{DW10.8})$$

When the coefficient is constant and $\dim X \leq d$, integrality of the eigenvalue and its paired complex-conjugate algebraic number gives $r, s \geq 0$. For compact-support degree i , (DW10.3) applied to the twist gives $r, s \geq \max(0, i-d)$, while $r+s \leq i$. For smooth ordinary degree i , duality sends a compact-support pair (r', s') in degree $2d-i$ to $(d-r', d-s')$. Consequently $r, s \leq \min(i, d)$ and $r+s \geq i$. These are the lower and upper triangles of Deligne's diagram. The stronger assertion mentioned after the diagram that ordinary cohomology lies in the union square is not needed here and is not claimed as newly proved by this argument.

In (3.3.11), smoothness can be replaced in the duality step by the exact rational-homology-manifold identity

$$Ra^! \bar{Q}_\ell \simeq \bar{Q}_\ell(N)[2N] \quad (\text{DW10.9})$$

for $a: X_0 \rightarrow \text{Spec}(F_q)$. It is this identity, with its N and $2N$, that supplies the same pairing and therefore the same lower-weight computation. No mere resemblance of a supporting point supplies this dualizing-complex identity.

DW11. The mechanism and the exact roles of its data

The established chain of maps and estimates is

$$\begin{aligned}
& \mathcal{F}_0 \mapsto \mathcal{F}_0 \boxtimes \mathcal{F}_0 \mapsto R^1 f_! \pi^*(\mathcal{F}_0 \boxtimes \mathcal{F}_0), \\
& \text{real companion} \mapsto \text{pure constituents}, \\
& \text{local monodromy and three vanishing-cycle computations} \mapsto \gamma \in 2\beta + \mathbb{Z} \text{ for every nonconstant piece,} \\
& \gamma < 2\beta + 2 \mapsto \gamma \leq 2\beta + 1, \\
& H_c^1 \otimes H_c^1 \hookrightarrow H_c^2(\widetilde{V}) \mapsto 2w_q(\alpha) \leq 2\beta + 2 + 2^{-k}, \\
& \forall k \mapsto w_q(\alpha) \leq \beta + 1, \\
& \text{duality with } \mathcal{F}_0^\vee(1) \mapsto w_q(\alpha) \geq \beta + 1.
\end{aligned}
\tag{DW11.1}$$

The final circle is an eigenvalue-modulus conclusion. Its radius is determined by the coefficient weight, cohomological degree and residue-field norm. In this proof, weight zero of a coefficient is a quantitative statement about Frobenius eigenvalues; the zero sheaf is a different object; a geometric generic point is a supporting point; and a centre of a complex circle is a feature of the complex absolute-value realization. Their occurrences are given by the maps above, and no replacement between those roles has been asserted.

The detailed source proof therefore provides concrete operations for a subsequent programme comparison: external tensor product with all original coefficients; compact-support direct image over a projective-line pencil; the three boundary cycle maps with their signs and Tate factors; a filtration with proven local weight congruences; the strict boundary nonvanishing input; and the exact duality pairing. This file establishes the finite-field composition, with its full input hypotheses. Any further programme calculation must state and prove its maps to these particular objects rather than naming the final modulus relation as an additional assumption.

Mixed duality and the recovered arithmetic base — MDB0-MDB11

Proof source: DELIGNE_MIXED_DUALITY_AND_BASE_COMPARISON.md.

Deligne mixed duality and the recovered arithmetic base

Independent derivation, 24 September 2026. Proof locators MDB0–MDB11.

MDB0. Sources, scope, and distinct objects

The source is Pierre Deligne, [La conjecture de Weil. II](#), Publications Mathématiques de l’IHÉS 52 (1980), 137–252. Canonical disk-index records: PUBUNIT-3D90B487DBD1CC3A259C6455 and PUBUNIT-574B0214207BBB2BDABCFF9. The available supplied reading witness is the French transcription [S20_FR_record_export.tex](#), SHA256 d6836ae15d98f2b27eb98c9bd039a7d921679becd9fa47ff4717410289dbd351. It is a transcription, not Deligne-authored TeX. Neither it nor its original page records have been edited.

Actual reading: §§6.1–6.2 in full, printed pp.243–251, TeX lines3427–3741, including all examples, proofs, and the added note. The §6.2.1–6.2.7 portion was reread without tool-output truncation. Weight conventions §§1.2.1–1.2.7, the local inertia statement §1.7.1, §§1.8.9–1.8.10 with their proofs, theorem and proof §3.3.1, and §§3.4.1–3.4.9 were also read. The existing DELIGNE_WEIGHT_SOURCE_READING.json and DELIGNE_FULL_READING_LOG.md were consulted first. No transfer of §§6.2.8–6.2.13 to the user’s construction is asserted by this note.

For the supporting geometry, original-author source actually read: Alain Connes and Caterina Consani, [The Absolute Twistor Line and the Geometry of the compactified Spec Z](#), local CC.tex, lines489–613 and773–902, together with the exact programme reconstruction CG1–CG4.

The input is the already reconstructed complete winding datum and arithmetic

$$L = G/H, \quad G \simeq \mathbb{Z} \times C_4, \quad H = \{0\} \times C_4, \quad R = \text{End}_{\text{Ab}}(L). \quad (\text{MDB0.1})$$

No selected observed prime is an initial input. The supporting point remains

$$\tau\langle Z_1; \text{no } Z_2 \rangle. \quad (\text{MDB0.2})$$

No arithmetic operation, coordinate, metric, or Frobenius is assigned to that point. Its supplied stalk data and the recovered ring have separately typed operations.

This note constructs the support, scheme, topos, and stalk maps that are actually available. It applies Deligne’s formalism to schemes and sheaves in its domain; it does not manufacture an arithmetic scheme structure on the singleton support.

MDB1. The recovered ring and its arithmetic base maps

For a cyclic generator u of the complete group L , every endomorphism has the unique form

$$[n](u^m) = u^{nm}. \quad (\text{MDB1.1})$$

Replacing u by u^{-1} preserves both this endomorphism and its integer n . Pointwise multiplication in L supplies endomorphism addition, and composition supplies multiplication:

$$[m] + [n] = [m + n], \quad [m][n] = [mn], \quad \mathbf{1}_R = \text{id}_L = [1]. \quad (\text{MDB1.2})$$

Thus $\mathbb{Z} \rightarrow R$, $n \mapsto [n]$, is a canonical unital ring isomorphism: (MDB1.1) proves bijectivity and (MDB1.2) proves preservation of operations. The identity endomorphism is not the supporting point.

Put $\Sigma = \text{Spec } R$. The ring isomorphism gives an isomorphism of schemes $\Sigma \simeq \text{Spec } \mathbb{Z}$, not only of point sets: inverse-image ideals identify points, and localization of the ring isomorphism identifies all structure-sheaf stalks. The generic stalk is $\text{Frac } R \simeq \mathbb{Q}$; the stalk at pR is $R_{pR} \simeq \mathbb{Z}_{(p)}$, with residue field $R/pR \simeq \mathbb{F}_p$. These are arithmetic-scheme stalks, not coordinates on τ .

Choose a coefficient prime ℓ after this reconstruction and set

$$S = \text{Spec } R[1/\ell], \quad s_p = \text{Spec}(R/pR), \quad i_p : s_p \hookrightarrow S \ (p \neq \ell), \quad E = \overline{\mathbb{Q}}_\ell. \quad (\text{MDB1.3})$$

The map i_p is the proper finite closed immersion induced by reduction modulo p . Its norm is the recovered quotient cardinality $N(s_p) = p$. The omitted coefficient prime remains explicit in the original zeta:

$$\zeta(s) = (1 - \ell^{-s})^{-1} \prod_{p \neq \ell} (1 - p^{-s})^{-1}, \quad \text{Re } s > 1. \quad (\text{MDB1.4})$$

Two distinct coefficient-prime opens cover Σ , but their coefficient fields differ; the open cover alone does not glue their coefficient categories.

For a complete branch isomorphism $B : L \rightarrow L'$,

$$C_B : R \xrightarrow{\sim} R', \quad f \mapsto BfB^{-1}, \quad \text{Spec}(C_B^{-1}) : \Sigma \xrightarrow{\sim} \Sigma' \quad (\text{MDB1.5})$$

are the actual arithmetic comparisons. The identity $B(x^n) = B(x)^n$ proves $C_B([n]) = [n]$. Consequently the scheme map sends pR to pR' , preserves residue fields and norms, and restricts to the coefficient opens. This construction is on the groupoid of branch isomorphisms; a general noninvertible homomorphism has not been declared to induce endomorphism conjugation.

In particular the source involution induces inversion on L . Its conjugation is the identity on R , hence on Σ , every local ring, and every residue field. This computes its action on the recovered base without discarding its action on the retained source stalk.

MDB2. Actual support and stalk maps in the CC geometry

The unfolded topological source is

$$X = \{+, -, \eta\}, \quad \Omega_+ = \{+, \eta\}, \quad \Omega_- = \{-, \eta\}, \quad \Omega = \{\eta\}. \quad (\text{MDB2.1})$$

Together with the empty and full opens these are its topology. The unique dense singleton is $\{\eta\}$; therefore any homeomorphism fixes η . The source involution exchanges $+$ and $-$.

Let $T = \{\tau\}$, with its singleton topology. Identify its supporting role with the generic open by

$$j_\tau : T \hookrightarrow X, \quad j_\tau(\tau) = \eta. \quad (\text{MDB2.2})$$

For a sheaf $\mathcal{F}$ on X ,

$$j_\tau^{-1} \mathcal{F} = \mathcal{F}_\eta = \mathcal{F}(\Omega). \quad (\text{MDB2.3})$$

The stalk is the neighbourhood colimit, and Ω is its least neighbourhood, proving the last equality.

On the supplied nonabsorbing monomial carrier the generic stalk is G , with full chart restrictions

$$\rho_+(T^m J_+^k) = T^m J^k, \quad \rho_-(T^m J_-^k) = T^m (\epsilon J)^k. \quad (\text{MDB2.4})$$

The separate source absorbing element remains outside this group. On the pointed multiplicative carrier it maps to the separate absorbing element, never to the supporting point.

Taking the specified monomial group, its finite-order subgroup, the quotient, and its endomorphisms gives

$$j_\tau^{-1} \mathcal{O} \mapsto G \mapsto (G, H, G/H) \mapsto R \mapsto \text{Spec } R. \quad (\text{MDB2.5})$$

The first step is the supplied nonabsorbing monomial carrier, not an equivalence between the whole spherical-algebra structure and a group. Both restrictions, the subgroup, quotient map, and phase data remain retained.

There is an actual continuous map

$$c_\Sigma : |\Sigma| \rightarrow T, \quad x \mapsto \tau. \quad (\text{MDB2.6})$$

Continuity follows because the only opens of T are empty and full. Composing with j_τ gives a continuous map $|\Sigma| \rightarrow X$ with image η . Every nonempty source open in (MDB2.1) has full preimage.

Its geometric morphism of topological sheaf topoi has

$$c_\Sigma^{-1} M = \underline{M}_\Sigma, \quad c_{\Sigma,*} \mathcal{F} = \Gamma(\Sigma, \mathcal{F}), \quad (\text{MDB2.7})$$

where the underline means the sheaf associated to the constant presheaf. The adjunction is the constant-presheaf/sheafification adjunction; inverse image preserves finite limits. Global sections form a value attached to the support, not arithmetic on the point. These are topos maps, not scheme maps to an unspecified $\text{Spec}(\tau)$.

The parity statement is also exact. Let $Q_2 = \{\text{even}, \text{odd}\}$ with fixed-point-free exchange σ . Equivariance of a point-label means $P(\alpha x) = \sigma P(x)$. At η it would require $P(\eta) = \sigma P(\eta)$, impossible. The nontrivial stalk (MDB2.4) and its action remain present.

MDB3. Étale topoi, arithmetic Frobenius, and geometric stalks

The recovered scheme has the canonical geometric morphism

$$\epsilon_S : S_{\text{ét}} \rightarrow S_{\text{Zar}}. \quad (\text{MDB3.1})$$

Its direct image satisfies $(\epsilon_{S,*} \mathcal{F})(U) = \mathcal{F}(U)$ for a Zariski open U , regarded as an étale scheme over S ; inverse image is the associated extension to the étale site. Composing with the support map gives

$$\pi_\tau : S_{\text{ét}} \rightarrow \text{Sh}(T) \simeq \text{Sets}, \quad \pi_{\tau,*} \mathcal{F} = \Gamma(S_{\text{ét}}, \mathcal{F}). \quad (\text{MDB3.2})$$

For abelian coefficients the derived direct image is étale derived global sections. This construction has not supplied a residue norm or arithmetic Frobenius at T , or made T an arithmetic scheme to which Deligne's exceptional inverse image applies.

For each $p \neq \ell$, choose a geometric point

$$\bar{s}_p = \text{Spec } \bar{\mathbb{F}}_p \rightarrow s_p \rightarrow S. \quad (\text{MDB3.3})$$

For an ordinary or finite-coefficient étale sheaf, its inverse-image functor is the actual stalk

$$\mathcal{F} \mapsto \mathcal{F}_{\bar{s}_p} = \text{colim}_{(U, \bar{u})} \mathcal{F}(U), \quad (\text{MDB3.4})$$

over étale neighbourhoods with a lift of the specified geometric point. For an ℓ -adic sheaf represented over the ring of integers $\mathcal{O}$ of a finite extension $E_0/\mathbb{Q}_\ell$ by a compatible lattice system $(\mathcal{F}_a)_{a \geq 1}$ with coefficients $\mathcal{O}/\varpi^a$, the construction instead means

$$\mathcal{F}_{\bar{s}_p} = \left(\lim_{\leftarrow a} \lim_{\rightarrow (U, \bar{u})} \mathcal{F}_a(U) \right) \otimes_{\mathcal{O}} E_0, \quad (\text{MDB3.4a})$$

followed by scalar extension to $E = \bar{\mathbb{Q}}_\ell$ when that is the coefficient field in use. The finite-coefficient stalks are taken before the inverse limit; exchanging them is not asserted. This is independent of a choice of lattice after tensoring with E_0 , because two lattices in the same finite-dimensional representation are commensurable. Restricting first to s_p retains the action of

$$\text{Gal}(\bar{\mathbb{F}}_p/\mathbb{F}_p) \simeq \widehat{\mathbb{Z}}. \quad (\text{MDB3.5})$$

Indeed finite extensions are the fields with p^r elements, whose cyclic Galois groups are generated by the p -power map. Their compatible inverse limit gives (MDB3.5); finite étale algebras are products of such fields. This constructs the usual equivalence with continuous Galois sets and, through inverse systems of finite coefficients, the corresponding ℓ -adic representations.

Geometric Frobenius F_p is the inverse of the p -power arithmetic Frobenius. Over $\mathbb{F}_q$, a point of residue degree d has $F_x = F_q^d$ and $N(x) = q^d$. This is action on the geometric fibre; the underlying one-point scheme s_p stays fixed.

The arithmetic generic point has the separate map

$$\bar{\eta}_R = \text{Spec } \bar{\mathbb{Q}} \rightarrow S. \quad (\text{MDB3.6})$$

It is not τ . Its stalk and Galois action have the analogous étale-neighbourhood construction. The map $\text{Spec } \mathbb{Q} \rightarrow S$ is not a finite-type open immersion: every nonempty open of S contains all but finitely many closed primes. A finite-type six-functor theorem cannot be applied to this map merely by naming it as an open immersion.

Deligne §6.1.1(b) keeps the characteristic-zero objects that have spread out over some $\mathbb{Z}[1/n]$, with morphisms identified after further shrinking. It does not identify their arithmetic data with an unadorned generic vector space.

MDB4. Full dualizing complexes and their natural maps

For a finite-type $\mathbb{F}_p$ -scheme with structure map $b_X : X \rightarrow s_p$, Deligne §6.2.1 defines

$$K_X = Rb_X^! E, \quad D_X K = R\mathcal{H}om_X(K, K_X). \quad (\text{MDB4.1})$$

For the same scheme viewed over S by $a_X = i_p b_X$, §6.2.7 defines

$$K'_X = Ra_X^! E, \quad D'_X K = R\mathcal{H}om_X(K, K'_X). \quad (\text{MDB4.2})$$

The superscript exclamation mark denotes exceptional inverse image, not $Ra_{X,1}$.

For a finite-type morphism $f : X \rightarrow Y$ over the common base, transitivity gives $K_X = Rf^! K_Y$. The duality maps are

$$D_X f^* \simeq Rf^! D_Y, \quad D_X Rf^! \simeq f^* D_Y, \quad (\text{MDB4.3})$$

$$D_Y Rf_! \simeq Rf_* D_X, \quad D_Y Rf_* \simeq Rf_! D_X. \quad (\text{MDB4.4})$$

They also hold with prime notation when both complexes are relative to S .

To display why these maps have precisely these domains, for test objects M on X , N on Y , tensor-Hom adjunction, exceptional-image adjunction, and the projection formula give

$$\begin{aligned}
R\mathrm{Hom}_X(M, D_X f^* N) &= R\mathrm{Hom}_X(M \otimes f^* N, K_X) \\
&= R\mathrm{Hom}_Y(Rf_!(M \otimes f^* N), K_Y) \\
&= R\mathrm{Hom}_Y((Rf_! M) \otimes N, K_Y) \\
&= R\mathrm{Hom}_Y(Rf_! M, D_Y N) \\
&= R\mathrm{Hom}_X(M, Rf^! D_Y N).
\end{aligned} \tag{MDB4.5}$$

Naturality and Yoneda give the first identity (MDB4.3); biduality gives the second. Similarly,

$$\begin{aligned}
R\mathrm{Hom}_Y(N, D_Y Rf_! M) &= R\mathrm{Hom}_Y(N \otimes Rf_! M, K_Y) \\
&= R\mathrm{Hom}_X(f^* N \otimes M, K_X) \\
&= R\mathrm{Hom}_X(f^* N, D_X M) \\
&= R\mathrm{Hom}_Y(N, Rf_* D_X M)
\end{aligned} \tag{MDB4.6}$$

proves (MDB4.4), using biduality for its other identity. The adjunctions, projection formula, and biduality are the cohomological formalism invoked in Deligne §6.2.1, not new hypotheses about τ .

The associated trace is the counit

$$Rf_! K_X = Rf_! Rf^! K_Y \longrightarrow K_Y. \tag{MDB4.7}$$

Its target retains the Tate line and degree. A scalar-valued pairing requires a specified comparison with that line.

MDB5. Arithmetic and finite-field duality: the complete shift and twist

The arithmetic base has

$$K'_S = E, \tag{MDB5.1}$$

because its structural morphism to $\mathrm{Spec} \mathbb{Z}[1/\ell]$ is the recovered isomorphism. Deligne §6.2.7, specialized to its closed point, gives

$$Ri_p^! E = E(-1)[-2]. \tag{MDB5.2}$$

This is codimension-one cohomological purity, or the Gysin isomorphism for the regular closed point. Its construction in this specific arithmetic trait can be made explicit, so that the formula is not a new assumption about the recovered base.

Take the strict henselian discrete valuation ring

$$A = \mathcal{O}_{S, \bar{s}_p}^{\mathrm{sh}}, \quad K = \mathrm{Frac}(A). \tag{MDB5.2a}$$

It has residue field $\bar{\mathbb{F}}_p$ and uniformizer p . The localization triangle computes its support complex as

$$R\Gamma_{\bar{s}_p}(\mathrm{Spec} A, E) = \mathrm{Cone}(E \longrightarrow R\Gamma(K, E))[-1]. \tag{MDB5.2b}$$

The initial E is $R\Gamma(\mathrm{Spec} A, E)$: a strictly henselian local scheme is acyclic for these constant finite étale coefficient systems. Inverse limits with surjective coefficient transitions, then scalar extension, give the stated ℓ -adic formula.

The named local input is the tame-inertia theorem recorded by Deligne §1.7.1, printed p.170:

$$1 \longrightarrow P \longrightarrow I \longrightarrow \prod_{r \neq p} \mathbb{Z}_r(1) \longrightarrow 1, \quad t_\ell : I \longrightarrow \mathbb{Z}_\ell(1) \text{ has pro-prime-to-}\ell \text{ kernel.} \quad (\text{MDB5.2c})$$

The kernel has no positive-degree cohomology for finite ℓ -primary coefficients, by averaging over its finite quotients. The remaining procyclic group has cohomological dimension one. For example, a temporary choice of topological generator gives the length-one completed group-ring resolution with differential $\gamma - 1$; its resulting degree-one space is intrinsically the dual of the Tate line, independently of that generator. Consequently

$$H^0(K, E) = E, \quad H^1(K, E) = \text{Hom}_{\text{cont}}(\mathbb{Z}_\ell(1), E) = E(-1), \quad H^{j>1}(K, E) = 0. \quad (\text{MDB5.2d})$$

These statements follow first with finite coefficients and then by the compatible inverse limits, whose cohomology transition maps are surjective. The degree-zero restriction in (MDB5.2b) is the identity. Its long exact sequence therefore gives

$$H_{\bar{s}_p}^j(\text{Spec } A, E) = \begin{cases} E(-1), & j = 2, \\ 0, & j \neq 2. \end{cases} \quad (\text{MDB5.2e})$$

The specific isomorphism, including its sign, comes from Kummer theory. For every positive a ,

$$H^1(K, \mu_{\ell^a}) = K^\times / (K^\times)^{\ell^a} \xrightarrow{\text{ord}_p} \mathbb{Z}/\ell^a, \quad [p] \mapsto 1. \quad (\text{MDB5.2f})$$

Every unit of A is an ℓ^a -th power: take a root in the separably closed residue field and lift it by Hensel's lemma, using invertibility of ℓ . This proves the displayed valuation isomorphism and shows that replacing p by a unit multiple gives the same Kummer class. Use the localization boundary with the convention that $\partial[p]$ is the positive divisor class corresponding to valuation one. Its compatible limit is

$$\text{cl}(p) \in H_{\bar{s}_p}^2(A, E(1)), \quad E \xrightarrow{\sim} H_{\bar{s}_p}^2(A, E(1)), \quad 1 \mapsto \text{cl}(p). \quad (\text{MDB5.2g})$$

It descends from the original arithmetic trait, so geometric Frobenius fixes this class with its $E(1)$ coefficient. Thus the untwisted group in (MDB5.2e) has Frobenius p , exactly the action on $E(-1)$. The class gives the Frobenius-equivariant isomorphism

$$E_{s_p}(-1)[-2] \xrightarrow{\sim} Ri_p^! E_S \quad (\text{MDB5.2h})$$

because both sides have only degree-two cohomology and the map there is the generator isomorphism just constructed.

The adjoint ambient Gysin morphism is

$$Ri_{p,*} E_{s_p}(-1)[-2] \longrightarrow E_S. \quad (\text{MDB5.2i})$$

This distinguishes extraordinary inverse image $Ri_p^!$ from direct image with proper support $i_{p,!} = i_{p,*}$. The latter is not the costalk complex.

The composite $a_X = i_p b_X$, transitivity, and the invertible Tate line now give

$$\begin{aligned} K'_X &= Rb_X^! Ri_p^! E \\ &= Rb_X^! (E(-1)[-2]) \\ &= K_X(-1)[-2], \end{aligned} \quad \boxed{D'_X K = D_X K(-1)[-2]}. \quad (\text{MDB5.3})$$

For X smooth of pure dimension d over $\mathbb{F}_p$, this is

$$K_X = E(d)[2d], \quad K'_X = E(d-1)[2d-2]. \quad (\text{MDB5.4})$$

At a finite-field point these are E and $E(-1)[-2]$; for a smooth curve they are $E(1)[2]$ and E . Both descriptions keep the structural base map.

The actual stalk/costalk identities are

$$D'_{s_p} i_p^* \simeq Ri_p^! D'_S, \quad D'_{s_p} Ri_p^! \simeq i_p^* D'_S. \quad (\text{MDB5.5})$$

Replacing arithmetic-relative finite-point duality by finite-field duality gives

$$\boxed{D_{s_p} i_p^* \simeq (Ri_p^! D'_S)(1)[2], \quad D_{s_p} Ri_p^! \simeq (i_p^* D'_S)(1)[2].} \quad (\text{MDB5.6})$$

These follow from (MDB4.3) and $D'_{s_p} = D_{s_p}(-1)[-2]$. Removing the final factors would change the comparison.

For the actual lisse Tate complex $E(r)[m]$ on S , both restrictions can be calculated:

$$i_p^*(E(r)[m]) = E(r)[m], \quad Ri_p^!(E(r)[m]) = E(r-1)[m-2]. \quad (\text{MDB5.7})$$

The first is pullback; the second tensors (MDB5.2) with the pulled-back invertible local system and shifts. Their nonzero cohomology occurs in degrees $-m$ and $2-m$, respectively. A stalk has not been identified with its costalk.

MDB6. Weights with every Tate and cohomological factor

Fix a field isomorphism $\iota : E \rightarrow \mathbb{C}$, as in Deligne §1.2.6. For a nonzero eigenvalue α of geometric Frobenius at a point of norm q , its ι -weight is

$$w_q(\alpha) = 2 \log_q |\iota \alpha|. \quad (\text{MDB6.1})$$

For a sheaf, pointwise purity of integer weight w means the corresponding equality at every closed point and every complex embedding of the algebraic eigenvalues. The ι -version works with the fixed isomorphism and allows real weights. Mixedness requires a finite filtration with pointwise pure quotients, not an arbitrary assignment to the support.

The Tate sheaf $E(1)$ has geometric Frobenius q^{-1} . Indeed arithmetic Frobenius raises an ℓ -power root of unity to its q -th power; geometric Frobenius acts inversely. Thus $E(r)$ has weight $-2r$. With cohomological shifts $\mathcal{H}^i(K[m]) = \mathcal{H}^{i+m}K$, Deligne §6.2.2 defines

$$K \in D_{\leq w} \iff \mathcal{H}^i K \text{ is mixed with pointwise weights } \leq i + w \text{ for every } i. \quad (\text{MDB6.2})$$

There is consequently an exact shift rule

$$K \in D_{\leq w} \iff K(r)[m] \in D_{\leq w+m-2r}. \quad (\text{MDB6.3})$$

To check it, the i -th sheaf on the right has weights at most $(i+m+w) - 2r = i + (w+m-2r)$. Apply the inverse shift and twist for the converse.

The full duality obeys

$$D_X(K(r)[m]) = (D_X K)(-r)[-m]. \quad (\text{MDB6.4})$$

This follows by moving the invertible Tate line and cohomological shift through derived Hom. Deligne §6.2.4 defines pure weight w by **both**

$$K \in D_{\leq w}, \quad D_X K \in D_{\leq -w}. \quad (\text{MDB6.5})$$

Applying (MDB6.3) to (MDB6.4) shows that a pure $K(r)[m]$ has complex weight $w + m - 2r$. In particular

$$K(N)[2N] \text{ has the same complex weight as } K, \quad D_X K(-1)[-2] \text{ has the same complex weight as } D_X K. \quad (\text{MDB6.6})$$

The cancellation concerns the weight only; it does not remove either factor from the object.

For (MDB5.7), the stalk sheaf $E(r)$ has weight $-2r$ in degree $-m$, and the costalk sheaf $E(r-1)$ has weight $2-2r$ in degree $2-m$. Each gives complex weight $m-2r$. This checks the arithmetic/finite-field compatibility of Deligne §6.2.7 with no suppressed twist.

On a smooth d -dimensional scheme with lisse cohomology sheaves, the full dual satisfies

$$\mathcal{H}^i(D_X K) = (\mathcal{H}^{-i-2d} K)^\vee(d). \quad (\text{MDB6.7})$$

It follows from $K_X = E(d)[2d]$, since taking the dual of a bounded complex of locally free finite-rank coefficients reverses cohomological degrees. At a closed point x , put $q = N(x)$, including its full residue degree over the base field. An eigenvalue α in degree j then corresponds to

$$q^{-d}\alpha^{-1} \quad \text{in degree } -j-2d. \quad (\text{MDB6.8})$$

The upper bound on this dual eigenvalue converts to the lower bound on α . Equations (MDB6.2), (MDB6.5), and (MDB6.8) show that under these smoothness and lisse-cohomology hypotheses, purity of K is equivalent to pointwise purity of each $\mathcal{H}^j K$ of weight $w+j$, exactly §6.2.5(b).

For an arbitrary complex on a singular or ramified situation, purity still means (MDB6.5). The stronger cohomology-sheaf conclusion is not asserted outside the hypotheses just used.

MDB7. Precisely which weight bounds are functorial

First, subobjects and quotients of a Frobenius module retain subsets of the eigenvalues; an extension has the union of the two multisets, because in a basis respecting the subobject its matrix is block upper triangular. Thus upper weight bounds survive subquotients, finite filtrations, and extensions. The same statements follow stalkwise for mixed sheaves.

Ordinary pullback preserves pointwise weights and upper complex bounds. At a closed point x over y with residue extension degree d , the pulled-back eigenvalue is α^d , while $N(x) = N(y)^d$. Hence

$$2 \log_{N(x)} |\iota(\alpha^d)| = 2 \log_{N(y)^d} |\iota\alpha|^d = 2 \log_{N(y)} |\iota\alpha|. \quad (\text{MDB7.1})$$

Pullback is exact for these coefficient sheaves, so it preserves the finite mixed filtration and the inequalities in (MDB6.2).

Deligne's fundamental theorem §3.3.1 says that $R^a f_! \mathcal{F}$ has weights at most $v+a$ when $\mathcal{F}$ has weights at most v . The passage to complexes in §6.2.3 uses the cohomology spectral sequence

$$R^a f_! \mathcal{H}^b K \implies \mathcal{H}^{a+b}(Rf_! K). \quad (\text{MDB7.2})$$

The transcription writes this page as E_1^{ab} ; the calculation does not depend on the naming of its starting page. The weights of the displayed term are at most

$$a + (b + w) = (a + b) + w. \quad (\text{MDB7.3})$$

Every subsequent page and the filtration on the abutment consist of subquotients and extensions of terms of the relevant total degree. The preceding eigenvalue argument therefore proves

$$Rf_! : D_{\leq w}(X) \longrightarrow D_{\leq w}(Y). \quad (\text{MDB7.4})$$

The deep geometric input is the already proved theorem §3.3.1. Its source proof reduces to relative curves, proper smooth families, controlled boundary monodromy, and the preceding purity theorem; it is not deduced here from formal adjunction alone.

Define lower complex weights through the full dualizing complex:

$$K \in D_{\geq w}(X) \iff D_X K \in D_{\leq -w}(X). \quad (\text{MDB7.5})$$

The exact maps (MDB4.3)-(MDB4.4) then give

$$Rf_* : D_{\geq w}(X) \rightarrow D_{\geq w}(Y), \quad Rf^! : D_{\geq w}(Y) \rightarrow D_{\geq w}(X). \quad (\text{MDB7.6})$$

For the first, $D_Y Rf_* K = Rf_! D_X K$, to which (MDB7.4) applies. For the second, $D_X Rf^! K = f^* D_Y K$, to which the pullback result applies.

When f is proper, $Rf_! = Rf_*$. Applying (MDB7.4) to K and $D_X K$, and using (MDB4.4), proves the full §6.2.6 conclusion:

$$K \text{ pure of weight } w \implies Rf_* K \text{ pure of weight } w. \quad (\text{MDB7.7})$$

The arithmetic-base version uses the prime dualities and the unchanged weight calculation (MDB6.6), as in §6.2.7.

These conclusions do not say that arbitrary Rf_* preserves the same upper weight bound, or arbitrary f^* preserves both dual bounds. Section6.1's mixedness stability is a stability of the mixed category; the numerical bound and its direction come from the specific calculations above.

On the actual recovered base, i_p is proper. Hence the actual object $i_{p,*} E$ is pure of weight zero for arithmetic duality. One can verify the full dual directly:

$$D'_S i_{p,*} E = i_{p,*} D'_{s_p} E = i_{p,*} E(-1)[-2]. \quad (\text{MDB7.8})$$

The sole nonzero dual cohomology sheaf has weight 2 in degree 2, so the upper complex bound is still zero. This is an application on the constructed base, not a proposed object replacing its geometry.

MDB8. The mixed-weight filtration, functoriality, and strictness

For this section the additional sources actually read are Deligne §§3.4.1-3.4.9 in full, printed pp.207-210, TeX lines2310-2393, and §§1.8.9-1.8.10 with their proofs, printed pp.177-178, TeX lines1278-1319. The upstream étale descent and fundamental-group facts named below remain named inputs to his construction.

Section3.4.1(ii) applies to a **lissee**, ι -mixed sheaf $\mathcal{F}_0$ on a scheme of finite type over $\mathbb{F}_q$, with integral pointwise weights. It constructs a unique finite increasing filtration by lisse subsheaves

$$\dots \subset W_{n-1} \mathcal{F}_0 \subset W_n \mathcal{F}_0 \subset \dots \quad (\text{MDB8.1})$$

whose n -th graded quotient is pointwise ι -pure of weight n . Existence of this global filtration is a theorem, not a consequence merely of naming the eigenvalue-weight subspaces at one point.

Its source proof uses the exact extension sequence in §3.4.2:

$$0 \longrightarrow H^0(X, \mathcal{H}om(\mathcal{F}, \mathcal{G}))_F \longrightarrow \text{Ext}^1(\mathcal{F}_0, \mathcal{G}_0) \longrightarrow H^1(X, \mathcal{H}om(\mathcal{F}, \mathcal{G}))^F. \quad (\text{MDB8.2})$$

The subscript is coinvariants and the superscript is invariants. For a geometrically split extension, two splittings differ by a section of $\mathcal{H}om$; the obstruction to a Frobenius-invariant splitting is its class modulo $F - 1$. This proves the left term and exactness there, while pullback to the geometric scheme gives the right map.

For pure inputs of weights β, γ on a smooth base, $\mathcal{H}om(\mathcal{F}, \mathcal{G})$ has weight $\gamma - \beta$. Section 3.4.3 applies the source's §3.3.5 strengthened by §3.3.10: the weights on its H^1 have the form $\gamma - \beta + 1 + n$, where n is a nonnegative integer. Thus eigenvalue 1, of weight zero, can occur in the right term of (MDB8.2) only when $\beta - \gamma$ is a positive integer. Including the left term allows $\beta = \gamma$. Thus §3.4.4 gives

$$\mathrm{Ext}^1(\mathcal{F}_0, \mathcal{G}_0) \neq 0 \implies \beta \equiv \gamma \pmod{\mathbb{Z}}, \quad \beta \geq \gamma. \quad (\text{MDB8.3})$$

This implication is the quoted proved extension theorem, with its smooth/lisse scope retained.

Induction on the number of pure successive quotients now puts extensions in weight order. In detail, for $0 \rightarrow \mathcal{F}'_0 \rightarrow \mathcal{F}_0 \rightarrow \mathcal{F}''_0 \rightarrow 0$, with $\mathcal{F}'_0$ pure of weight β , the inverse image of $W_{\beta-1}\mathcal{F}''_0$ is an extension which splits by (MDB8.3), applied successively to its lower-weight quotients. Its lift gives $W_n\mathcal{F}_0$ for $n < \beta$; for $n \geq \beta$, take the inverse image of $W_n\mathcal{F}''_0$. For the decomposition by $b \in \mathbb{R}/\mathbb{Z}$, put $b = [\beta]$. Every summand $\mathcal{F}''_0(b')$, $b' \neq b$, has a split inverse-image extension by $\mathcal{F}'_0$, using (MDB8.3) successively. Lift these summands; take the entire inverse image of $\mathcal{F}''_0(b)$ for the b -summand. The lifts are unique: a difference between two lifts is a morphism between disjoint weight sets, hence zero by generalized Frobenius eigenspaces at each closed point. This proves both constructions in §3.4.7 on its smooth open stratum.

Here is the full extension from that stratum used in §3.4.8. We use Z_0 for the closed complement in this paragraph; it is a scheme symbol, not the user's Z_0 absence label. Let

$$j : U_0 \hookrightarrow X_0, \quad i : Z_0 = X_0 \setminus U_0 \hookrightarrow X_0 \quad (\text{MDB8.3a})$$

be complementary open and closed immersions. Replacing the scheme by its reduction does not change its étale topos. A constructible mixed sheaf has a dense smooth open on which its defining finite pure filtration has lisse terms. This gives the open to which §3.4.7 applies; the complement has smaller dimension.

The following gluing formula first concerns coefficient sheaves on the geometric schemes $X = X_0 \otimes_{\mathbb{F}_q} \bar{\mathbb{F}}_q$, U , and Z , with the same letters i, j for the base-changed immersions. A Weil sheaf adds its rational-coefficient Frobenius isomorphism to these underlying geometric sheaves. No extension of that action to a continuous representation of the profinite arithmetic fundamental group is assumed. The gluing data are the triple

$$(\mathcal{F}_U, \mathcal{F}_Z, s), \quad s : \mathcal{F}_Z \longrightarrow i^* j_* \mathcal{F}_U. \quad (\text{MDB8.3b})$$

Its reconstructed geometric sheaf is the following fibre product, with every object now on X :

$$\mathcal{F} = j_* \mathcal{F}_U \times_{i_* i^* j_* \mathcal{F}_U} i_* \mathcal{F}_Z. \quad (\text{MDB8.3c})$$

The first arrow into the middle object is the adjunction unit for $i^* \dashv i_*$; the second is $i_* s$. The functors i^*, j^* are exact and therefore preserve this fibre product. On U , the two closed-support terms restrict to zero and $j^* j_* \mathcal{F}_U = \mathcal{F}_U$. On Z , the first arrow becomes the identity of $i^* j_* \mathcal{F}_U$, so the fibre product is $\mathcal{F}_Z$. Conversely, the adjunction maps from any sheaf $\mathcal{F}$ into these two direct images give a map to (MDB8.3c); it is an isomorphism on geometric stalks on U and Z , hence an isomorphism. Compatible pairs of morphisms of the two restrictions induce exactly the morphisms of these fibre products. This proves the asserted triple equivalence. For adic coefficients the underlying geometric construction is coefficientwise and compatible with transition maps, then passes to rational coefficients, in the order in (MDB3.4a). Since i, j are defined over the finite field, their functors and the adjunction maps are compatible with Frobenius pullback. Frobenius-compatible input triples therefore give a Frobenius isomorphism on the fibre product, proving the same equivalence for Weil sheaves. This argument does not require a Frobenius-stable integral lattice.

Induction on dimension has constructed the decomposition by weight congruence classes on both restrictions:

$$\mathcal{F}_U = \bigoplus_b \mathcal{F}_U(b), \quad \mathcal{F}_Z = \bigoplus_b \mathcal{F}_Z(b). \quad (\text{MDB8.3d})$$

There are finitely many nonzero summands. The boundary theorem needed to glue them is Deligne §1.8.9: j_* preserves mixedness, a pointwise upper weight bound, and integrality of pointwise weights. The theorem and its proof, printed pp.177–178, have been read here; this remains a named earlier theorem of the source, not a result newly deduced from the singleton support. Applying its integrality assertion after the constant rank-one twist that shifts b into $\mathbb{Z}$, and undoing that twist, proves

$$\text{every pointwise weight of } i^* j_* \mathcal{F}_U(b) \text{ belongs to } b. \quad (\text{MDB8.3e})$$

The twisting operation here is the source's operation on Weil sheaves; it has not introduced an operation on τ . At each closed point, the specialization map in (MDB8.3b) intertwines Frobenius. Generalized eigenspaces with weights in distinct classes modulo $\mathbb{Z}$ have distinct eigenvalues. Thus every component $\mathcal{F}_Z(b) \rightarrow i^* j_* \mathcal{F}_U(b')$, $b' \neq b$, is zero on closed stalks, hence zero. Therefore s has the block form $\bigoplus_b s_b$. Define

$$\mathcal{F}(b) = j_* \mathcal{F}_U(b) \times_{i_* i^* j_* \mathcal{F}_U(b)} i_* \mathcal{F}_Z(b). \quad (\text{MDB8.3f})$$

Finite direct sums commute with the additive direct-image functors and with these fibre products. Equations (MDB8.3c)–(MDB8.3f) consequently give $\mathcal{F} = \bigoplus_b \mathcal{F}(b)$. Restriction to U_0 and $\bar{Z}_0$ proves that the weights of $\mathcal{F}(b)$ lie in b . Its uniqueness and functoriality follow from the same Frobenius decomposition on closed stalks. This completes the general decomposition in §3.4.1(i), including its boundary specialization map.

For the lisse integral-weight filtration in §3.4.1(ii), first suppose X_0 is connected and normal. The homomorphism

$$\pi_1(U, \bar{u}) \longrightarrow \pi_1(X, \bar{u}) \quad (\text{MDB8.3g})$$

is surjective on each geometric connected component, where $X = X_0 \otimes \bar{\mathbb{F}}_q$, $U = U_0 \otimes \bar{\mathbb{F}}_q$. Normality is preserved by this separable algebraic base extension. Indeed, any connected finite étale cover of a connected normal scheme is normal and connected, hence irreducible; its restriction to a nonempty dense open remains connected. The finite-quotient characterization of surjectivity for profinite groups gives (MDB8.3g). The same argument applies componentwise when necessary, retaining the Frobenius permutation of those components.

Write V for the representation of the geometric group $\pi_1(X, \bar{u})$ defining the underlying lisse sheaf $\mathcal{F}$. Its restriction to $\pi_1(U, \bar{u})$ factors through (MDB8.3g). Each subrepresentation $W_n(j^* \mathcal{F}_0)_{\bar{u}} \subset V$ is therefore already stable under $\pi_1(X, \bar{u})$: the surjection has the same image on V , and its kernel acts trivially on V and on each subspace. This gives the lisse geometric subobject. The filtration on U_0 is already a filtration of Weil sheaves, so its subobjects are Frobenius-stable. Extension across the normal scheme by j_* is compatible with Frobenius pullback; it extends those isomorphisms and their inverses. Thus each extended subobject has its Weil structure on X_0 . No continuous arithmetic $\pi_1(X_0)$ -representation is asserted. The concrete Weil-sheaf formulas are

$$\mathcal{F}_0 = j_* j^* \mathcal{F}_0, \quad W_n \mathcal{F}_0 = j_* W_n(j^* \mathcal{F}_0), \quad \text{Gr}_n^W \mathcal{F}_0 = j_* \text{Gr}_n^W(j^* \mathcal{F}_0). \quad (\text{MDB8.3h})$$

For the first equality, a lisse geometric sheaf extending across a normal scheme has the same invariant sections on the connected dense part of each strict local neighbourhood. The preceding subrepresentation argument applies to all the subobjects and quotients in the filtration. In that subcategory, j_* is the inverse of restricting geometric representations that factor through $\pi_1(X)$, with their compatible Weil isomorphisms retained, so it is exact there. This proves the last equality in (MDB8.3h); it does not assert that j_* is exact on arbitrary sheaves on U .

Set $\mathcal{G} = \text{Gr}_n^W \mathcal{F}_0$. It is lisse on X_0 and pure of weight n on U_0 . To check the boundary weights, the two injections

$$\mathcal{G} \hookrightarrow j_* j^* \mathcal{G}, \quad \mathcal{G}^\vee \hookrightarrow j_* j^* \mathcal{G}^\vee \quad (\text{MDB8.3i})$$

follow from lisseness and density. Apply §1.8.9 to the two right sides: their pointwise weights are at most n and $-n$, respectively. Subobjects inherit these upper bounds. Eigenvalues on $\mathcal{G}^\vee$ are the inverses

of eigenvalues on $\mathcal{G}$, so the second upper bound says that $\mathcal{G}$ has weights at least n . Thus it is pure of weight exactly n everywhere. This is the complete two-bound deduction of §1.8.10 used in §3.4.8.

For completeness the descent reduction in that paragraph can be made explicit. Over a finite field the scheme is excellent; the disjoint union of the normalizations of the irreducible components of $X_{0,\text{red}}$ gives a finite proper surjection

$$a : Y_0 \longrightarrow X_{0,\text{red}} \quad (\text{MDB8.3j})$$

with Y_0 normal. We use the upstream proper-descent theorem for constructible étale coefficient sheaves and lisse local systems, applied on the geometric schemes coefficientwise and compatibly for adic systems: sheaves with compatible descent isomorphisms on a proper surjective cover descend effectively; the descended morphisms and their lisse property are detected by that descent. After passing to rational coefficients, Frobenius isomorphisms compatible with descent descend as morphisms, as do their inverses. This supplies the Weil structure without assuming a Frobenius-stable lattice or a continuous arithmetic fundamental-group representation. This is the descent input invoked by Deligne, not a new six-functor theorem proved in this note.

Apply the normal construction to $a^*\mathcal{F}_0$. On $Y_0 \times_{X_0} Y_0$, the two pullbacks of $W_n(a^*\mathcal{F}_0)$ sit inside the same pullback of $\mathcal{F}_0$. At every closed geometric point, both are the generalized Frobenius sum of weights at most n . Finite residue extension replaces α by α^d and q by q^d , preserving its weight. The two subobjects therefore agree on all closed stalks and hence as constructible subsheaves. Their identification is the identity in the common ambient sheaf, so the cocycle condition on the triple fibre product holds identically. These are the actual descent maps.

An explicit descended subobject is

$$W_n\mathcal{F}_0 = \ker(\mathcal{F}_0 \longrightarrow a_*(a^*\mathcal{F}_0/W_n(a^*\mathcal{F}_0))). \quad (\text{MDB8.3k})$$

The arrow is the adjunction unit followed by the quotient map. On the underlying geometric sheaves, proper base change for this finite map identifies its stalk at $\bar{x}$ with the map from $(\mathcal{F}_0)_{\bar{x}}$ into the product of the corresponding quotients over the finitely many geometric points of $Y_{\bar{x}}$. The compatibility just proved identifies all these subspaces of $(\mathcal{F}_0)_{\bar{x}}$, so the kernel is exactly that subspace. In particular its pullback is the constructed $W_n(a^*\mathcal{F}_0)$. Effective descent gives the lisse geometric subobject and lisse graded quotients. Frobenius preserves the quotient arrow and its kernel because the morphism a is defined over the finite field and the filtration upstairs has its Weil structure. Hence the descended kernel and quotients retain their Weil structures. Exhaustiveness and finiteness descend from the finite filtration on Y_0 , and purity of each graded quotient descends because every point is covered and the residue-power calculation preserves weights. Equations (MDB8.3g)–(MDB8.3k) complete the general filtration argument; there is no unresolved gluing step here.

The geometric semisimplicity assertion §3.4.1(iii) has a short additional proof (§3.4.5). Let X_0 be normal and $\mathcal{F}_0$ lisse pure of weight β . Passing to a dense smooth open does not change the image of its geometric monodromy, by (MDB8.3g) and the same normality argument after base extension. On the geometric scheme let $\mathcal{F}'$ be the sum of all irreducible lisse subobjects of $\mathcal{F}$. It is a semisimple subobject of finite rank. Frobenius permutes those irreducible subobjects, so their sum is stable and descends to a Weil subobject $\mathcal{F}'_0$. Put $\mathcal{F}''_0 = \mathcal{F}_0/\mathcal{F}'_0$; subobjects and quotients of a pure sheaf are pure of the same weight. Section 3.4.3 says that the extension

$$0 \longrightarrow \mathcal{F}'_0 \longrightarrow \mathcal{F}_0 \longrightarrow \mathcal{F}''_0 \longrightarrow 0 \quad (\text{MDB8.3l})$$

is geometrically split, since both end terms have weight β . If $\mathcal{F}'' \neq 0$, its finite-dimensional monodromy representation has an irreducible subrepresentation (take a nonzero invariant subspace of minimum dimension). A geometric splitting lifts it to an irreducible subobject of $\mathcal{F}$ outside $\mathcal{F}'$, contrary to the definition of that sum. Therefore $\mathcal{F}'' = 0$, proving geometric semisimplicity. Arithmetic Frobenius need not be semisimple: the coinvariant term in (MDB8.2) can still record equal-weight extensions. The distinction is retained in §6.2.13's use of this result.

The functoriality and strictness can be proved explicitly on its fibres. At a closed geometric point x , let V_α be the generalized Frobenius eigenspace of α . The constructed filtration has

$$(W_n \mathcal{F}_0)_x = \bigoplus_{w_{N(x)}(\alpha) \leq n} V_\alpha. \quad (\text{MDB8.4})$$

A Frobenius-intertwining map $h : V \rightarrow V'$ sends V_α into V'_α , because $(F' - \alpha)^r h(v) = h(F - \alpha)^r v$. Consequently it preserves the decomposition by exact weights.

More precisely, let $z = h(v) \in W_n V'$. Decompose $v = v_{\leq n} + v_{>n}$ into its generalized eigenvalue-weight parts. The image of the second summand has weights $> n$; since $h(v)$ lies in the disjoint sum of weights $\leq n$, that second image is zero. Hence $z = h(v_{\leq n})$. The reverse containment is immediate. Thus

$$\boxed{h(W_n V) = \text{im } h \cap W_n V'.} \quad (\text{MDB8.5})$$

This is strictness, not just preservation.

For lisse sheaves, the images, kernels, and relevant intersections are lisse. Equality of their fibres at every closed geometric point implies equality of sheaves: a nonzero constructible quotient would have nonempty constructible support, which on a finite-type scheme over a finite field contains a closed point. Therefore every morphism in §3.4.1(ii) satisfies

$$h(W_n \mathcal{F}_0) = \text{im } h \cap W_n \mathcal{G}_0. \quad (\text{MDB8.6})$$

In particular each W_n and each Gr_n^W is exact on short exact sequences in this lisse integral-weight category. This follows directly by applying strictness to the injection and surjection.

Under base pullback in this finite-field setting, formula (MDB7.1) preserves the weights; pulling back (MDB8.1) gives lisse pure graded quotients. Uniqueness therefore identifies the pulled-back filtration with the weight filtration of the pulled-back sheaf. The norm-preserving branch scheme isomorphism (MDB1.5) induces such comparisons on each finite-field fibre, in both directions. On the arithmetic scheme S itself, an isomorphism transports any already specified weight filtration and all of its maps, preserving the residue norms and hence the pointwise weight conditions. This statement does not extend the finite-field existence theorem to an unspecified global filtration on every arithmetic sheaf over S .

These facts are used on lisse strata or lisse cohomology sheaves where their hypotheses hold. Section6.1's assertion that $R^i f_* \mathcal{F}$ is mixed does not itself assert lisseness or supply a global weight filtration of the form (MDB8.1) without those hypotheses. Section6.2.13 invokes §3.4.1(iii)'s geometric semisimplicity under its stated conditions; it is not an assertion that arithmetic Frobenius is always diagonalizable.

MDB9. The degree factor at a recovered prime and the exact purity circle

At the actual recovered closed point s_p , let V be a finite-dimensional Frobenius representation, placed in cohomological degree zero. Finite-field and arithmetic-relative duality are

$$D_{s_p} V = V^\vee, \quad D'_{s_p} V = V^\vee(-1)[-2]. \quad (\text{MDB9.1})$$

A Frobenius eigenvalue α on V is therefore sent to

$$\alpha^{-1} \text{ in degree 0} \quad \text{and} \quad p\alpha^{-1} \text{ in degree 2}, \quad (\text{MDB9.2})$$

respectively. The factor p is the action on the Tate line $E(-1)$, and degree 2 comes from $[-2]$. For nontrivial Jordan data, duality takes the full transpose inverse operator, multiplied by p ; no nilpotent contribution is removed by writing its eigenvalue formula.

If A is an arbitrary bounded Frobenius complex, the arithmetic-point formula is

$$\mathcal{H}^{2-i}(D'_{s_p} A) = (\mathcal{H}^i A)^\vee(-1). \quad (\text{MDB9.3})$$

In the situation of Deligne's proved proper-image proposition §6.2.6 with $f : X \rightarrow s_p$ proper, put $A = Rf_* K$, a complex on the finite-field point s_p . Its pure weight w follows from (MDB7.7), with that proposition's input and properness hypotheses retained; this is not an assumption about an unconstructed programme receiver. The target point is essential: an arbitrary ordinary stalk restriction from a more general target does not automatically preserve the dual lower bound. The upper bound on an eigenvalue α of $\mathcal{H}^i A$ is

$$|\iota\alpha| \leq p^{(i+w)/2}. \quad (\text{MDB9.4})$$

The upper bound on the dual complex, applied in the degree $2 - i$ prescribed by (MDB9.3), is

$$|p/\iota\alpha| \leq p^{(2-i-w)/2}. \quad (\text{MDB9.5})$$

Dividing the full displayed factors gives the opposite inequality

$$|\iota\alpha| \geq p^{(i+w)/2}. \quad (\text{MDB9.6})$$

Together they give the exact circle

$$|\iota\alpha| = p^{(i+w)/2}. \quad (\text{MDB9.7})$$

For a closed point of norm $q = p^d$, the same proof replaces p by q everywhere, including the Tate factor. This is the full arithmetic-base version of Deligne's two-bound mechanism.

The degree-one, weight-zero case retains the factor p and the same cohomological degree after duality, because $2 - 1 = 1$. For the complex realization $V_{\mathbb{C}} = V \otimes_{E,\iota} \mathbb{C}$, define its degree-weighted conjugate-dual object to be $\overline{V_{\mathbb{C}}}^\vee \otimes \mathbb{C}(-1)$, where the realized Tate line has Frobenius p . This is a contravariant functor on Frobenius representations, not an asserted isomorphism from $V_{\mathbb{C}}$ to its conjugate dual. The full operator on the resulting object is

$$p(\overline{F}^{-1})^t, \quad (\text{MDB9.8})$$

so the scalar action is $a \mapsto p/\overline{a}$. Here complex conjugation is explicitly on the chosen complex realization; it is not asserted to be a canonical involution of every ℓ -adic coefficient object.

This gives the exact scalar comparison with the programme's previously calculated arithmetic character action

$$a = p^\rho \mapsto p/\overline{a} = p^{1-\bar{\rho}}. \quad (\text{MDB9.9})$$

The fixed-character equation is

$$a = p/\overline{a} \iff |a|^2 = p. \quad (\text{MDB9.10})$$

This comparison preserves all degree and duality factors. It compares formulas on character values; it does not assert that the programme's analytic zero-jet quotient has already been realized as a constructible ℓ -adic complex with the proper direct image in (MDB9.3)–(MDB9.7).

MDB10. The exact weight-descent defect of the actual geometric-stalk map

On the constructed arithmetic scheme S , take the two actual lisse sheaves E and $E(1)$. Restrict each to the same s_p and then the same geometric point $\overline{s}_p$. Their underlying geometric stalks are

one-dimensional E -vector spaces. They are isomorphic after choosing a basis of the Tate line, but their geometric Frobenius operators and weights are

$$\begin{array}{c|c|c} \text{sheaf at } s_p & F_p & \text{weight} \\ \hline E & 1 & 0 \\ E(1) & p^{-1} & -2. \end{array} \quad (\text{MDB10.1})$$

These are the original Tate objects on the same recovered base, not an arbitrary matrix construction.

Any Frobenius-equivariant map between the two lines has, after a chosen vector-space identification, scalar c satisfying

$$c = p^{-1}c. \quad (\text{MDB10.2})$$

Since $p > 1$, the scalar is zero. Thus a vector-space isomorphism of their geometric stalks cannot lift to an isomorphism of the arithmetic fibre representations. There is no isomorphism-invariant weight function on bare vector-space stalks whose pullback gives both weights in (MDB10.1): one vector-space isomorphism class would have to receive both zero and minus two.

The exact categorical comparison is

$$U_p : \text{Rep}_E(\text{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)) \longrightarrow \text{Vect}_E, \quad (V, F) \longmapsto V. \quad (\text{MDB10.3})$$

On this point-category, U_p is faithful, exact, tensor, and conservative on existing morphisms. It is faithful because distinct equivariant maps are distinct underlying maps; exactness and tensor compatibility are computed on the vector spaces; if an equivariant map is bijective, its inverse also intertwines the action. It is not full by (MDB10.2). Isomorphism of two images does not supply a morphism upstairs. These statements are not extended to an arbitrary single-stalk functor on all sheaves over S .

Because U_p is exact, applying it term by term to bounded complexes sends quasi-isomorphisms to quasi-isomorphisms and defines the exact derived functor

$$U_p : D^b(\text{Rep}_E(\text{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p))) \longrightarrow D^b(\text{Vect}_E). \quad (\text{MDB10.3a})$$

On these finite-dimensional coefficient categories tensor products are exact, so this extension also retains the tensor and shift identifications used next. With V placed in degree zero, the duality comparison survives with all of its data:

$$U_p(D'_{s_p} V) = U_p(V)^\vee \otimes U_p(E(-1))[-2]. \quad (\text{MDB10.4})$$

Thus forgetting Frobenius does not destroy vector-space duality or the existence of the Tate line as a line. It removes the Frobenius action needed to read its arithmetic weight. Retaining that action, or retaining the full specified Tate object instead of identifying all underlying lines, retains the distinction.

This proves a precise descent obstruction for this particular actual receiver. It does not say that the user's full supporting datum must use this forgetful map, or that every possible enriched supporting construction loses weights.

MDB11. What transfers through the available maps

The constructed chain has the following types:

$$\text{CC generic stalk with } (G, H, \alpha) \longmapsto L \longmapsto R = \text{End}(L) \longmapsto \Sigma \supset S \longleftarrow s_p \longleftarrow \bar{s}_p. \quad (\text{MDB11.1})$$

The supporting maps $|\Sigma| \rightarrow T \hookrightarrow X$ exist topologically, with the sheaf morphisms (MDB2.7) and (MDB3.2). The residue maps, étale stalks, Frobenius action, and duality on the recovered arithmetic schemes exist as calculated. None of these arrows assigns a number or metric to the supporting point.

The constant coefficient object E_S is an actual pure weight-zero complex: every local Frobenius acts by one, and $D'_S E_S = E_S$. Its local Euler factors at all $p \neq \ell$ are

$$\det(1 - p^{-s} F_p \mid E_{\bar{s}_p})^{-1} = (1 - p^{-s})^{-1}. \quad (\text{MDB11.2})$$

Together with the explicitly retained coefficient-prime factor (MDB1.4), they give the original $\zeta(s)$. Thus the reconstructed ring, its prime spectrum, and these local pure coefficient data are already genuine receivers, with the full original Euler factors.

What is functorial is now explicit: norm-preserving scheme isomorphisms transport pointwise weights, and preserve the strict mixed-weight filtration in the category where Deligne constructs it; pullback preserves upper bounds; compactly supported direct image preserves upper bounds by §3.3.1 and its spectral sequence; duality transports those to lower bounds; proper direct image preserves both bounds. The exact circle follows for the proper arithmetic cohomology to which those statements apply.

The topological or unringed topos map to T is not, by its construction alone, a new morphism in Deligne's finite-type arithmetic six-functor category. Its existing derived global-section functor has not been proved to identify the analytic original-zeta zero-jet receiver with such a proper arithmetic image. The map (MDB10.3) gives the exact weight information lost by its further geometric-stalk observation; adding that action is the specified data required to retain the weight.

These are statements about the maps actually constructed, with their precise preserved data and defect. They do not close other constructions on the enriched support. The full scalar connection to the programme's $a \mapsto p/\bar{a}$ is (MDB9.8)–(MDB9.10), including the Tate degree factor that a comparison must preserve.

The invariant-cycle quotient and its weight-controlled lift — DC0-DC12

Proof source: DELIGNE_INVARIANT_CYCLE_QUOTIENT.md.

Deligne local invariant cycles: the exact quotient, boundary, and lift

Independent mathematical reconstruction, 24 September 2026. Stable proof locators DC0-DC12. The coefficients throughout the proved geometric theorem are rational ℓ -adic coefficients, with ℓ invertible on the geometric base.

DC0. Source identity, actual coverage, and mathematical scope

The human source is Pierre Deligne, [La conjecture de Weil](#). II, Publications mathématiques de l’IHÉS 52 (1980), 137–252, §3.6, printed pp.212–215. Canonical disk-index identifiers are PUBUNIT-3D90B487DBD1CC3A259C6455 and PUBUNIT-574B0214207BBB2BDABCFFF9. The actual reading witness is the current French transcription [S20 FR record export.tex](#), SHA256 d6836ae15d98f2b27eb98c9bd039a7d921679becd9fa47ff4717410289dbd351. It is a transcription, not Deligne-authored TeX. No PDF was used and no source witness was edited.

The complete §3.6 witness was read at lines2468–2569. The proof of theorem3.6.1 is at lines2477–2553; the complex analytic remark3.6.4 is at lines2561–2569. Its arithmetic-reduction input §1.11.3 was read in the larger complete passage §§1.11.1–1.11.3, lines1491–1552; the local invariant-weight input §1.8.8 was read at lines1254–1281. The supporting programme derivations actually used are [DLM10-DLM11](#), [DW10](#), [MDB1](#), [MDB4](#), [MDB6](#), [MDB8](#), and the peer calculation [DWR11.3-DWR11.6](#). They are local derivations, not substitutes for the source identity. These local proof links are not claimed to be publication receipts.

This note proves the source theorem for the henselian curve situation specified below. It then calculates the exact quotient, the remaining kernel of specialization, and a stronger splitting after the stated arithmetic reduction. It does not extend the theorem to every mixed-characteristic trait. It does not identify an analytic zeta receiver or a supporting point with this étale cohomology. SectionDC11 gives the actual arithmetic-base maps and the functorial quotient comparison that are available.

DC1. The geometric objects and the specialization map

Let k be algebraically closed. Let S be the spectrum of the henselization of $k[T]$ at (T) , let s and η be its closed and generic points, and choose a geometric generic point $\bar{\eta}$. Let

$$f : X \longrightarrow S \tag{DC1.1}$$

be proper. The hypotheses of Deligne3.6.1 are that X is essentially smooth over k and $X_{\bar{\eta}}$ is smooth. Fix $\ell \neq \text{char } k$ and set $E = \mathbb{Q}_{\ell}$; a finite scalar extension or $\overline{\mathbb{Q}}_{\ell}$ can be used when splitting characteristic polynomials. All cohomology groups below use these coefficients. Negative cohomological degrees are zero.

Write

$$I = \text{Gal}(\bar{\eta}/\eta), \quad V_j = H^j(X_{\bar{\eta}}, E), \quad A_i = H^i(X_s, E), \quad B_i = H^i(X_{\eta}, E), \quad C_i = V_i^I. \quad (\text{DC1.2})$$

Proper base change over the strictly henselian trait gives

$$\text{res}_s : H^i(X, E) \xrightarrow{\sim} A_i. \quad (\text{DC1.3})$$

Restriction to the generic fibre and then geometric generic fibre define

$$\alpha_i : A_i \xrightarrow{\text{res}_s^{-1}} H^i(X, E) \longrightarrow B_i, \quad \pi_i : B_i \longrightarrow C_i. \quad (\text{DC1.4})$$

The last arrow is the edge map calculated in DC3. The specialization is the exact composite

$$\text{sp}_i = \pi_i \alpha_i : A_i \longrightarrow C_i \hookrightarrow V_i. \quad (\text{DC1.5})$$

When its target is written as C_i , the final inclusion is understood separately. This description proves that the image is inertia-invariant; it does not yet prove that every invariant lies in the image.

DC2. The reduction that makes weights available

Finite presentation and passage to the limit provide a smooth curve S' over k , a point again denoted s , and a proper model

$$f' : X' \longrightarrow S' \quad (\text{DC2.1})$$

whose pullback to the henselization at s is (DC1.1). The total space X' is smooth over k , and, after shrinking the curve around s , the restriction of f' outside s is smooth. This is the first reduction in the source at line2483. Smoothness of the geometric generic fibre permits this shrinking; it does not assert that X_s is smooth.

Deligne next uses §1.11.3 to spread and specialize to $k = \bar{\mathbb{F}}_q$. The input of that section controls the images of local inertia on finite coefficient quotients while spreading a smooth curve with its boundary and lisse sheaves. In particular it is a specialization argument for the geometric family and its cohomology, not a claim that arbitrary complex numbers have finite-field Frobenius actions. The relevant model and finitely many cohomological degrees then descend to a sufficiently large finite field. The source reduction is the justification for proving surjectivity in that arithmetic model and transferring the theorem back. No arithmetic Frobenius is assigned intrinsically to an arbitrary original k -model.

Retain the finite-field data

$$(f'_0 : X'_0 \rightarrow S'_0, s_0) \text{ over } \mathbb{F}_q, \quad d_s = [k(s_0) : \mathbb{F}_q], \quad Q = q^{d_s}. \quad (\text{DC2.2})$$

The source can enlarge $\mathbb{F}_q$ so that s_0 is rational; here the residue degree is retained instead. Henselization at s_0 , followed by the geometric residue-field extension, recovers the situation over $\bar{\mathbb{F}}_q$. There is an exact sequence

$$1 \longrightarrow I \longrightarrow G_{\eta_0} \longrightarrow G_{k(s_0)} \longrightarrow 1. \quad (\text{DC2.3})$$

The arithmetic group acts on all maps subsequently constructed. Its inertia subgroup acts trivially on the resulting group-cohomology and special-fibre terms, so the cross of DC4 carries the residual action of $G_{k(s_0)}$.

The Frobenius used for weights is geometric Frobenius F . On the Tate line it acts by

$$F|_{E(1)} = Q^{-1}, \quad F|_{E(r)} = Q^{-r}, \quad w_Q(\lambda) = 2 \log_Q |\iota(\lambda)|. \quad (\text{DC2.4})$$

Here $\iota : \overline{\mathbb{Q}}_\ell \hookrightarrow \mathbb{C}$ is the embedding used for the weight calculation. Thus duality negates a weight and twist (r) subtracts $2r$. Arithmetic Frobenius is F^{-1} .

Decompose the smooth total model into connected components and work on each component. Its dimension is constant, and we write

$$N = \dim X'. \quad (\text{DC2.5})$$

This is the dimension of the total smooth model, not the dimension of a generic fibre. A component dominating the curve has generic-fibre dimension $N - 1$. A vertical component has empty generic fibre and zero invariant target; the formulas below still apply to its support cohomology. Components of different dimensions are treated separately and then combined by their finite direct sum. In particular a single N is never assigned to a union of components of different dimensions.

DC3. Inertia cohomology and the full coinvariant term

The inertia group has an exact sequence

$$1 \longrightarrow I' \longrightarrow I \xrightarrow{t_\ell} \mathbb{Z}_\ell(1) \longrightarrow 1, \quad (\text{DC3.1})$$

where I' has pro-order prime to ℓ . It contains wild inertia and the tame factors at primes different from ℓ . For a finite continuous $\mathbb{Z}/\ell^n[I]$ -module M , the action of I' factors through a finite group G' of order prime to ℓ . The averaging operator

$$e_{G'} = |G'|^{-1} \sum_{g \in G'} g \quad (\text{DC3.2})$$

is an idempotent with image $M^{I'}$. Its kernel is generated by $gm - m$, so the quotient map induces the canonical isomorphism $M^{I'} \simeq M_{I'}$. Averaging is an exact functor because division by $|G'|$ is defined modulo ℓ^n ; consequently $H^b(I', M) = 0$ for $b > 0$. The group extension spectral sequence reduces the computation to $\mathbb{Z}_\ell(1)$.

For that procyclic group, choose a topological generator γ . The two-term resolution has differential $\gamma - 1$. It computes invariants in degree zero, coinvariants in degree one, and zero in higher degrees. The degree-one identification must also retain its dependence on the generator. Projecting a continuous cocycle to coinvariants gives a homomorphism from $\mathbb{Z}_\ell(1)$ to the coinvariant module; replacing γ by γ^a , $a \in \mathbb{Z}_\ell^\times$, multiplies evaluation by a . The generator-independent target is therefore

$$\text{Hom}_{\mathbb{Z}_\ell}(\mathbb{Z}_\ell(1), M_I) = M_I(-1). \quad (\text{DC3.3})$$

The resulting formulas, after the finite-coefficient calculation and rational passage, are

$$H^0(I, V) = V^I, \quad H^1(I, V) = V_I(-1), \quad H^a(I, V) = 0 \quad (a \geq 2). \quad (\text{DC3.4})$$

The notation V_I denotes the quotient by the span of $gv - v$, not a choice of complement to invariants.

The Frobenius factor in (DC3.4) can also be checked directly. Arithmetic Frobenius conjugates tame ℓ -inertia by multiplication by Q . Its action on a cocycle is

$$(F^{-1} \cdot c)(g) = F^{-1}(c(FgF^{-1})). \quad (\text{DC3.5})$$

Passing to coinvariants therefore gives $Q^{-1}F^{-1}$ on degree one. Equivalently geometric Frobenius acts on $V_I(-1)$ by QF . This explicitly retains the (-1) twist and its sign.

The Hochschild-Serre spectral sequence for $X_\eta \rightarrow \eta$ is

$$E_2^{a,b} = H^a(I, V_b) \implies H^{a+b}(X_\eta, E). \quad (\text{DC3.6})$$

Only columns $a = 0, 1$ survive. No differential between surviving terms is possible after the second page. The resulting two-step filtration gives the canonical short exact sequence

$$0 \longrightarrow K_i := V_{i-1,I}(-1) \xrightarrow{j_i} B_i \xrightarrow{\pi_i} C_i := V_i^I \longrightarrow 0. \quad (\text{DC3.7})$$

The injection is the filtration inclusion and the surjection is the degree-zero edge map, so neither requires choosing a generator or a splitting. At finite coefficients the relevant groups are finite, hence their inverse systems satisfy the Mittag-Leffler condition. This supplies the exact inverse-limit row; tensoring with $\mathbb{Q}_\ell$ gives (DC3.7). Possible integral torsion has not been declared absent. Every arrow is equivariant for the residual arithmetic action described in (DC2.3).

DC4. Localization and duality with the total dimension retained

Put

$$P_i = H_{X_s}^i(X, E), \quad O_i = H_{X_s}^{i+1}(X, E). \quad (\text{DC4.1})$$

The localization long exact sequence and proper base change give

$$P_i \xrightarrow{g_i} A_i \xrightarrow{\alpha_i} B_i \xrightarrow{\partial_i} O_i \xrightarrow{g_{i+1}} A_{i+1}. \quad (\text{DC4.2})$$

This is exact at every displayed internal term. In particular $\ker \alpha_i = \text{im } g_i$ and $\ker \partial_i = \text{im } \alpha_i$. The relation $\text{sp}_i = \pi_i \alpha_i$ and (DC3.7) form Deligne's exact cross.

For completeness, compute its top group without suppressing the support or dualizing factors. Excision for the henselian pullback identifies support cohomology on X with support cohomology on the finite-type smooth model X' . Let $u : X_s \hookrightarrow X'$ be the closed immersion, and let $D_Y = R\mathcal{H}om(-, K_Y)$ denote Verdier duality, with $K_Y = Ra_Y^! E$. Smoothness of the total space of dimension N gives

$$K_{X'} = E(N)[2N]. \quad (\text{DC4.3})$$

Duality for the closed immersion gives

$$Ru^! E_{X'} = D_{X_s}(u^* D_{X'} E_{X'}) = D_{X_s} E_{X_s}(-N)[-2N]. \quad (\text{DC4.4})$$

The fibre X_s is proper over k . Global duality and (DC4.4) therefore yield the complete derived identity

$$R\Gamma_{X_s}(X, E) \simeq R\text{Hom}_E(R\Gamma(X_s, E), E)(-N)[-2N]. \quad (\text{DC4.5})$$

Over the field E , taking cohomology of the dual complex introduces no additional Ext term. Consequently

$$H_{X_s}^j(X, E) \simeq H^{2N-j}(X_s, E)^\vee(-N), \quad (\text{DC4.6})$$

and, at the degree needed in the cross,

$$O_i \simeq H^{2N-i-1}(X_s, E)^\vee(-N). \quad (\text{DC4.7})$$

The pairing associated with (DC4.6) takes values in $E(-N)$. No trivialization of that Tate line has been made. If geometric Frobenius has eigenvalue λ on the cohomology on the right before duality, the resulting eigenvalue in support cohomology is

$$Q^N/\lambda; \quad \text{at the operator level it is } Q^N(F^{-1})^t. \quad (\text{DC4.8})$$

This keeps the inverse, transpose, Tate factor, and any nontrivial Jordan structure. The finite-coefficient and integral versions require their own duality statements; the rational passage here is precisely the passage at which the source removes the torsion complication. No integral specialization-surjectivity theorem is inferred.

DC5. The exact obstruction before any weight calculation

For the exact cross (DC3.7), (DC4.2), define

$$\overline{O}_i = O_i / \partial_i j_i(K_i). \quad (\text{DC5.1})$$

For $c \in C_i$, choose $b \in B_i$ with $\pi_i(b) = c$ and put

$$\text{obs}_i(c) = [\partial_i b] \in \overline{O}_i. \quad (\text{DC5.2})$$

If b' is another lift, exactness gives $b' - b = j_i(k)$ for some $k \in K_i$. Their boundaries differ by $\partial_i j_i(k)$, so (DC5.2) is well-defined. It is linear and Frobenius-equivariant because the defining maps are.

If $c = \text{sp}_i(a)$, use $b = \alpha_i(a)$. Localization gives $\partial_i b = 0$, so $\text{obs}_i(c) = 0$. Conversely, if $\text{obs}_i(c) = 0$, choose b lifting c and k such that $\partial_i b = \partial_i j_i(k)$. Then $b - j_i(k) \in \ker \partial_i = \text{im } \alpha_i$, hence $c \in \text{im } \text{sp}_i$. We have proved

$$\ker \text{obs}_i = \text{im } \text{sp}_i, \quad \text{im } \text{obs}_i = \text{im } \partial_i / \partial_i j_i(K_i), \quad (\text{DC5.3})$$

and the exact quotient identification

$$\text{coker } \text{sp}_i \xrightarrow{\sim} \text{im } \partial_i / \partial_i j_i(K_i), \quad [c] \mapsto [\partial_i b]. \quad (\text{DC5.4})$$

This is an exact description of the missing lifts. It does not assert that the whole support group O_i vanishes.

There is also a kernel calculation before weights. For $a \in \ker \text{sp}_i$, there is a unique $k \in K_i$ with $\alpha_i(a) = j_i(k)$; it lies in $\ker(\partial_i j_i)$. Every element of that kernel arises in this way by localization. Therefore

$$0 \longrightarrow \ker \alpha_i \longrightarrow \ker \text{sp}_i \longrightarrow \ker(\partial_i j_i) \longrightarrow 0. \quad (\text{DC5.5})$$

This exact sequence records the information that would be lost by asserting only surjectivity.

DC6. The three weight estimates used in the source proof

On $U_0 = S'_0 \setminus \{s_0\}$, proper smooth base change makes

$$\mathcal{V}_j = R^j(f'_0|_{U_0})_* E \quad (\text{DC6.1})$$

a lisse sheaf with geometric generic fibre V_j . Smooth proper cohomology is pure of weight j , by Deligne3.3.9 applied to each closed fibre. The local invariant-weight theorem1.8.8(i), reconstructed in DLM10-DLM11, consequently gives

$$C_i = V_i^I \text{ is mixed of weights } \leq i. \quad (\text{DC6.2})$$

Its mechanism is the monodromy weight filtration: the invariant primitive pieces of a pure sheaf of weight i occur in graded indices at most zero, with weights $i + j$ for $j \leq 0$; residual finite inertia invariants preserve the inequality. The source specifically uses this local theorem in its lemma3.6.2. It is not a theorem about arbitrary inertia representations without the pure geometric sheaf just specified.

The special fibre is proper, though it may be singular. The upper bound for compactly supported cohomology of a weight-zero constant sheaf, together with properness, gives

$$H^r(X_s, E) \text{ has weights } \leq r. \quad (\text{DC6.3})$$

Applying this to $r = 2N - i - 1$, then dualizing and retaining the $(-N)$ twist, gives

$$w(H^{2N-i-1}(X_s, E)^\vee(-N)) \geq -(2N - i - 1) + 2N = i + 1. \quad (\text{DC6.4})$$

If the indicated degree is negative the group is zero, and the bound remains valid. Thus

$$O_i \text{ has weights } \geq i + 1, \quad A_i \text{ has weights } \leq i. \quad (\text{DC6.5})$$

These are source lemma3.6.3 and the sentence immediately following it. The total dimension N is essential to the degree and the $2N$ contribution in (DC6.4).

DC7. Exactness of the weight cutoff and the complete lifting proof

All terms of the cross are finite-dimensional mixed residual Frobenius modules. This follows for special-fibre terms from (DC6.3), for invariant and coinvariant terms from local monodromy and duality, and for B_i from (DC3.7). For a fixed embedding ι , define $W_i M$, after scalar extension to split Frobenius, as the sum of generalized eigenspaces whose eigenvalues have weight at most i . A Frobenius-equivariant map preserves the generalized eigenspaces: applying a power of $F - \lambda$ before or after the map gives the same result. A short exact sequence restricts to a short exact sequence on each generalized eigenspace by the primary decomposition of the Frobenius polynomial. Taking the indicated finite direct sum proves exactness of W_i .

This argument does not assume Frobenius semisimple. The weight decomposition can also be stated through the unique weight filtration over its coefficient field. Working after a splitting scalar extension suffices for surjectivity, since a finite-dimensional cokernel is zero exactly when its faithfully flat scalar extension is zero.

Apply W_i to (DC3.7). Since $W_i C_i = C_i$, the resulting map

$$W_i B_i \longrightarrow C_i \quad (\text{DC7.1})$$

is onto. The map ∂_i sends $W_i B_i$ into $W_i O_i = 0$. For any $c \in C_i$, choose $b_0 \in W_i B_i$ mapping to c . Then $\partial_i b_0 = 0$, so exactness of localization gives an $a \in A_i$ with $\alpha_i(a) = b_0$. Consequently

$$\text{sp}_i(a) = \pi_i \alpha_i(a) = \pi_i(b_0) = c. \quad (\text{DC7.2})$$

This proves Deligne3.6.1 in the arithmetic reduction, and the reduction in DC2 gives the stated geometric theorem. In terms of the exact obstruction,

$$\text{im } \partial_i = \partial_i j_i(K_i), \quad \text{coker } \text{sp}_i = 0. \quad (\text{DC7.3})$$

Indeed an arbitrary b has the same image in C_i as a low-weight lift b_0 , so $b - b_0 = j_i(k)$, and $\partial_i b = \partial_i j_i(k)$. Both K_i and O_i have been retained in this equation.

DC8. A stronger consequence: the coinvariant term is exactly the boundary image

For any finite-dimensional representation V , the annihilator of the span of $gv - v$ is precisely the invariant subspace of the contragredient representation. Thus there is a canonical Frobenius-equivariant isomorphism

$$(V_I)^\vee \xrightarrow{\sim} (V^\vee)^I. \quad (\text{DC8.1})$$

The lisse sheaf $\mathcal{V}_{i-1}^\vee$ is pure of weight $1 - i$. Apply the same local invariant theorem used in DC6 to this dual sheaf. It gives

$$(V_{i-1}^\vee)^I \text{ has weights } \leq 1 - i. \quad (\text{DC8.2})$$

Dualizing (DC8.1), then applying the retained (-1) twist in (DC3.7), proves

$$V_{i-1,I} \text{ has weights } \geq i - 1, \quad K_i = V_{i-1,I}(-1) \text{ has weights } \geq i + 1. \quad (\text{DC8.3})$$

When the indicated generic cohomology group vanishes, in particular in negative degree, the bounds hold vacuously. Degree-zero cohomology is retained and may be nonzero. In particular $W_i K_i = 0$. Exactness of the cutoff on (DC3.7) strengthens (DC7.1) to an isomorphism

$$\pi_i|_{W_i B_i} : W_i B_i \xrightarrow{\sim} C_i. \quad (\text{DC8.4})$$

Since A_i has weights at most i , its image under α_i is contained in $W_i B_i$. Conversely ∂_i kills $W_i B_i$, so localization puts that entire subspace in the image. Hence

$$\text{im } \alpha_i = W_i B_i = \ker \partial_i. \quad (\text{DC8.5})$$

The intersection $j_i(K_i) \cap W_i B_i$ is zero by disjoint generalized-eigenvalue weights; the dimensions in (DC3.7) and (DC8.4) give

$$B_i = W_i B_i \oplus j_i(K_i), \quad \partial_i j_i : K_i \xrightarrow{\sim} \text{im } \partial_i. \quad (\text{DC8.6})$$

Surjectivity of the last arrow is (DC7.3); injectivity follows from (DC8.5) and the zero intersection. Thus the coinvariant term has not been removed: it is canonically the boundary image in this arithmetic Frobenius cross.

Inverting (DC8.4) defines a section

$$s_i : C_i \longrightarrow B_i, \quad \pi_i s_i = \text{id}_{C_i}, \quad \text{im } s_i = W_i B_i. \quad (\text{DC8.7})$$

It is the unique Frobenius-equivariant section, because the difference of two such sections is a Frobenius-equivariant map $C_i \rightarrow K_i$, and all such maps are zero by their disjoint weight ranges. This uniqueness is a statement in the fixed arithmetic Frobenius model. The argument does not construct an intrinsic arithmetic Frobenius, or this splitting, on every original k -model before the reduction. It also does not give a canonical lift $C_i \rightarrow A_i$.

DC9. The exact quotient of the special-fibre classes

By (DC8.6), the right-hand term in (DC5.5) is zero. Combining (DC4.2) with (DC5.5) therefore gives

$$\ker \text{sp}_i = \ker \alpha_i = \text{im } g_i. \quad (\text{DC9.1})$$

Thus the invariant-cycle quotient, including its complete kernel, is

$$A_i / \text{im}(H_{X_s}^i(X, E) \xrightarrow{g_i} A_i) \xrightarrow{\sim} V_i^I, \quad [a] \mapsto \text{sp}_i(a). \quad (\text{DC9.2})$$

Its inverse sends c to the class of any special-fibre lift; two such lifts differ by exactly the displayed kernel. The set of lifts of any c is the nonempty affine coset

$$\text{sp}_i^{-1}(c) = a_0 + \text{im } g_i. \quad (\text{DC9.3})$$

No selected representative a_0 is preferred by the quotient theorem.

The full support sequence can be expressed without leaving its kernel implicit. In degree $j - 1$, (DC8.6) identifies the image of $B_{j-1} \rightarrow P_j$ with the injected term K_{j-1} . In degree j , (DC9.1) identifies the image of $P_j \rightarrow A_j$ with the specialization kernel. Using (DC4.6), the localization sequence therefore gives

$$0 \longrightarrow H^{j-2}(X_{\bar{\eta}}, E)_I(-1) \xrightarrow{\partial_{j-1} j_{j-1}} H^{2N-j}(X_s, E)^\vee(-N) \xrightarrow{g_j} H^j(X_s, E) \xrightarrow{\text{sp}_j} H^j(X_{\bar{\eta}}, E)^I \longrightarrow 0. \quad (\text{DC9.4})$$

The middle arrow denotes the support-to-ordinary cohomology map transported through the exact duality isomorphism, not an arbitrarily chosen pairing. Exactness at the first nonzero term is the injectivity in (DC8.6), at the next two terms is localization and (DC9.1), and at the final term is DC7. All its groups and arrows therefore have explicit origins.

Neither (DC7.3) nor (DC9.4) says $O_i = 0$. In fact the next localization arrow gives

$$O_i / \partial_i j_i(K_i) \xrightarrow{\sim} \text{im}(g_{i+1}) = \ker \text{sp}_{i+1}. \quad (\text{DC9.5})$$

This group can survive. What vanishes is the image of the obstruction map from C_i into that quotient. Equation(DC9.5) specifies exactly what the quotient itself records.

DC10. Functoriality of the quotient and of the obstruction

Fix a residual field size Q , the coefficient field, and geometric Frobenius convention. An exact cross consists of finite-dimensional Frobenius modules (A, B, C, K, O) with an exact row $0 \rightarrow K \xrightarrow{j} B \xrightarrow{\pi} C \rightarrow 0$, maps $A \xrightarrow{\alpha} B \xrightarrow{\partial} O$ exact at B , and specialization $\pi\alpha$. A morphism of exact crosses is a tuple

$$(f_A, f_B, f_C, f_K, f_O) \quad (\text{DC10.1})$$

of Frobenius-equivariant maps satisfying

$$f_B \alpha = \alpha' f_A, \quad f_B j = j' f_K, \quad f_C \pi = \pi' f_B, \quad f_O \partial = \partial' f_B. \quad (\text{DC10.2})$$

The second and fourth equations send $\partial j K$ into $\partial' j' K'$, so they induce a map $\bar{f}_O : O / \partial j K \rightarrow O' / \partial' j' K'$. Given $c = \pi b$, compute

$$\text{obs}'(f_C c) = [\partial' f_B b] = [f_O \partial b] = \bar{f}_O(\text{obs}(c)). \quad (\text{DC10.3})$$

This proves functoriality of the obstruction, of its image, and of the cokernel identification (DC5.4). It is an actual commuting-map calculation.

For crosses having the weights proved above, f_B preserves W_i . The characterization (DC8.7) therefore gives

$$f_B s_i = s'_i f_C. \quad (\text{DC10.4})$$

For the special-fibre quotient, f_A sends $\ker \text{sp}_i$ into $\ker \text{sp}'_i$ by (DC10.2), and its induced map is identified with f_C under (DC9.2). This is the precise sense in which the quotient and lift commute with a specified compatible comparison.

An isomorphism of the actual proper geometric models over their pointed curve induces such an isomorphism of crosses, by the naturality of restriction, proper base change, group cohomology, localization, and duality. When written contravariantly through pullback, all five arrows are reversed together. Equal total dimensions and residue degrees are then preserved. No arbitrary map between unrelated vector spaces is asserted to be induced by a geometric model map.

If the arithmetic model is extended by a residue-field degree e , the geometric Frobenius becomes F^e , its eigenvalues become λ^e , and its residue cardinality becomes Q^e . The weights and Tate contributions satisfy the exact formulas

$$2 \log_{Q^e} |\iota(\lambda^e)| = 2 \log_Q |\iota(\lambda)|, \quad (Q^e)^N = Q^{eN}. \quad (\text{DC10.5})$$

Thus this field extension preserves the weight cut and every preceding formula with the specified replacement of Frobenius and residue cardinality.

DC11. Typed comparison with the programme's global quotient

The programme's retained support is $\tau\langle Z_1; \text{no } Z_2 \rangle$. No addition or arithmetic operation on that supporting point occurs here. Its supplied winding carrier has the previously reconstructed quotient and endomorphism ring

$$L = G/H, \quad G \simeq \mathbb{Z} \times C_4, \quad H = \{0\} \times C_4, \quad R = \text{End}_{\text{Ab}}(L) \simeq \mathbb{Z}. \quad (\text{DC11.1})$$

The complete group construction, rather than a selected prime or a numerical timing sample, is the input to these recovered arithmetic objects. As proved in MDB1, the ring isomorphism sends the integer n to the multiplication endomorphism $[n]$; it is independent of the sign of a generator of L .

After this whole reconstruction there are actual unital ring maps, for every recovered prime p and positive integer h ,

$$R \longrightarrow R/pR \simeq \mathbb{F}_p \longrightarrow \mathbb{F}_{p^h}, \quad (\text{DC11.2})$$

and therefore actual scheme maps

$$\text{Spec } \mathbb{F}_{p^h} \longrightarrow \text{Spec}(R/pR) \longrightarrow \text{Spec } R. \quad (\text{DC11.3})$$

The arrows reverse because Spec is contravariant. If the Deligne model has base field $\mathbb{F}_q$, $q = p^f$, its given structural map composes with (DC11.3) for $h = f$. The special residue field has size $Q = p^{fd_s}$, and its point composes through (DC11.3) for $h = fd_s$. Thus the exact scheme-theoretic receiving map from an existing Deligne model to the recovered arithmetic base is

$$X'_0 \longrightarrow \text{Spec } \mathbb{F}_q \longrightarrow \text{Spec } R, \quad \text{Spec } k(s_0) \longrightarrow \text{Spec } \mathbb{F}_q \longrightarrow \text{Spec } R. \quad (\text{DC11.4})$$

These maps make the relationship concrete. They do not construct X'_0 , its proper degeneration, its cohomology, or its purity from the point τ alone. In particular none of those absent identifications is substituted for the actual hypotheses of DC1-DC2.

For a complete branch isomorphism $B : L \rightarrow L'$, conjugation gives

$$C_B : R \xrightarrow{\sim} R', \quad f \mapsto BfB^{-1}, \quad C_B([n]) = [n]. \quad (\text{DC11.5})$$

Hence the induced map $\text{Spec}(C_B^{-1})$ preserves each recovered residue characteristic and the maps (DC11.2)-(DC11.4). Viewing the same given finite-field model over either identified arithmetic base gives the same étale cohomology cross. Its induced comparison is the identity under these identifications, so (DC10.3)-(DC10.4) prove commutation of its obstruction, quotient, and low-weight lift. This is a transport of the already specified model; it is not a rule assigning that model to every branch.

There are consequently two exact quotients with fully stated source objects: the group quotient $G \twoheadrightarrow L$, and the cohomological quotient (DC9.2). The latter has the obstruction calculation (DC5.4) and the proved vanishing mechanism (DC6)-(DC8). Their present relationship is through the recovered arithmetic-base maps (DC11.2)-(DC11.4) and the compatible model comparisons of DC10. No isomorphism from a zeta zero-jet algebra or an analytic measurement space to A_i , B_i , or C_i has been used. In particular the proof establishes a quotient-and-lift theorem in its exact cohomological domain; it supplies no new verdict about zeros of the original Riemann zeta function.

DC12. The complex analytic remark and the scope of the proof

The witness's §3.6.4 states a complex analytic analogue for a proper map $f : X \rightarrow D$, with X smooth and f smooth over $D^* = D \setminus \{0\}$, under its stated factorization through $\mathbb{P}^n(\mathbb{C}) \times D \rightarrow D$. It gives a surjection

$$H^i(X_0, \mathbb{Q}) \longrightarrow H^i(X_t, \mathbb{Q})^{\pi_1(D^*, t)}. \quad (\text{DC12.1})$$

The witness says the argument is parallel, using mixed Hodge weights in place of Frobenius weights, and cites J. Steenbrink, *Mixed Hodge structure on the vanishing cohomology*, Oslo symposium, 1976. That mixed Hodge construction has not been independently read or reconstructed in this bounded note. In particular the brief transcription's word "factorization" is not silently strengthened here into a claim that it explicitly printed an embedding hypothesis, nor used to extend (DC12.1) to an arbitrary proper analytic map.

The complete proof given in DC1-DC9 is the rational ℓ -adic theorem 3.6.1 with the source's arithmetic reduction. Its stronger statements (DC8.3)-(DC9.5) were derived from the same proved local invariant bounds, source duality, and exact sequences. They retain $H^{i-1}(X_{\bar{\eta}})_I(-1)$, the total dimension N , the full support degree $2N - i - 1$, the residue degree d_s , the geometric Frobenius multiplier Q , the multiplier Q^N in support duality, every specialization kernel, and the dependence of the canonical section on the fixed arithmetic Frobenius model.

Exact receiving map for the original Riemann zeta function — DR0-DR7

Proof source: DELIGNE_ORIGINAL_ZETA_RECEIVING_MAP.md.

The complete arithmetic return measure in Deligne's weight argument

24 September 2026. Exact maps DR0-DR7. The input stage is the complete arithmetic recovered from the source, not a preselected pair of prime labels. The maps preserve separate branch counters, the unit and every repeated return. No numerical operation is assigned to $\tau\langle Z_1; \text{no } Z_2 \rangle$.

DR0. Source and input stage

The user requires the full infinite structure before its arithmetic labels are used. The current comparison begins **after** the complete source's winding quotient and endomorphism ring have been constructed in CG0-CG6 and BCC1-BCC4. Write L for that quotient and $A = \text{End}_{\text{Ab}}(L)$. Those proofs construct the unital ring isomorphism $A \simeq \mathbb{Z}$ from the full cyclic source; they do not prove cyclicity from a lone supporting point. GTQ records the exact duration-group criterion for that earlier step. This input boundary remains explicit.

Within this recovered ring, let $A_{>0}$ be the positive nonzero endomorphisms. For $a \in A_{>0}$, define

$$N(a) = |L/aL|. \quad (\text{DR0.1})$$

The source's identity endomorphism has $N(1_A) = 1$, and the quotient-clock proof gives $N(ab) = N(a)N(b)$. The positive prime elements are exactly those with nonzero simple cokernel. Their norm labels become the usual primes under the established ring isomorphism. Numerical notation below refers to this output. No individual prime found in advance is used to construct the input ring.

Human source: Pierre Deligne, *La conjecture de Weil. II*, §2.1.9, printed p191, together with §§2.1.1-2.1.8. Root read the complete current French transcription, pp137-252. Its identity and limitations are in DP0 and the reading ledger. Deligne explicitly specializes the positive logarithmic-derivative measure to the original Riemann zeta function in 2.1.9; the following calculation gives the full invertible map from the programme measure to his measure.

DR1. All counts and all repeated prime returns

Use the nonnegative real time half-line only as the receiving space of logarithmic norms. It is not a coordinate or metric on the supporting point. Define locally finite measures

$$\mathcal{D} = \sum_{a \in A_{>0}} \delta_{\log N(a)} = \delta_0 + \sum_{n \geq 2} \delta_{\log n}, \quad (\text{DR1.1})$$

$$\mathcal{R} = \sum_{\mathfrak{p}} \sum_{m \geq 1} \frac{1}{m} \delta_{m \log N(\mathfrak{p})}, \quad \mathcal{W} = \sum_{\mathfrak{p}} \sum_{m \geq 1} \log N(\mathfrak{p}) \delta_{m \log N(\mathfrak{p})}. \quad (\text{DR1.2})$$

Here the first sum is over the recovered closed prime points of $\text{Spec } A$, with $N(\mathfrak{p}) = |A/\mathfrak{p}|$. For every finite T , an atom in $[0, T]$ has $N(\mathfrak{p})^m \leq e^T$; only finitely many norms and repetitions occur. This proves local finiteness with every multiplicity retained.

For locally finite measures with support bounded away from zero, define convolution by addition of receiving times. On $[0, T]$, a convolution power of order greater than $T/\log 2$ of $\mathcal{R}$, or of $\mathcal{D} - \delta_0$, is zero. Therefore the formal convolution exponential and logarithm below are finite sums on each compact interval:

$$\exp_*(\mathcal{R}) = \sum_{k \geq 0} \frac{\mathcal{R}^{*k}}{k!}, \quad \log_*(\mathcal{D}) = \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} (\mathcal{D} - \delta_0)^{*k}. \quad (\text{DR1.3})$$

The zero-fold term is δ_0 . No unit contribution is lost.

For a single prime point, put $u = \delta_{\log N(\mathfrak{p})}$. The formal identity

$$\exp_* \left(\sum_{m \geq 1} \frac{u^{*m}}{m} \right) = \sum_{r \geq 0} u^{*r} \quad (\text{DR1.4})$$

follows by differentiating the corresponding one-variable series: both sides have constant 1 and solve $(1 - z)f'(z) = f(z)$, whose coefficient recursion forces every coefficient to be 1. The exponential of a sum is a convolution product of exponentials, since convolution is commutative and the binomial formula applies coefficientwise. Thus

$$\exp_*(\mathcal{R}) = * \left(\sum_{\mathfrak{p}} \delta_{r \log N(\mathfrak{p})} \right) = \mathcal{D}. \quad (\text{DR1.5})$$

The second equality is exactly unique factorization in the reconstructed ring: each positive endomorphism has one finite prime factorization, including the empty factorization of the identity. On any compact interval all displayed products reduce to finite ones. The formal logarithm is inverse to the exponential because this identity holds in each truncated polynomial algebra, where the positive-support ideal is nilpotent. Therefore

$$\boxed{\mathcal{R} = \log_* \mathcal{D}, \quad \mathcal{D} = \exp_* \mathcal{R}, \quad \mathcal{W} = t \mathcal{R}.} \quad (\text{DR1.6})$$

The last identity follows atom by atom from $(m \log N(\mathfrak{p}))/m = \log N(\mathfrak{p})$.

DR2. The map to Deligne's positive measure, and its inverse

For a real number $\sigma > 1$, set

$$\mu_\sigma = e^{-\sigma t} \mathcal{W} = \sum_{\mathfrak{p}, m \geq 1} \log N(\mathfrak{p}) N(\mathfrak{p})^{-m\sigma} \delta_{m \log N(\mathfrak{p})}. \quad (\text{DR2.1})$$

This is the complete measure in Deligne 2.1.9 for $\Gamma = \mathbb{R}$, with the additive Frobenius class $\log N(\mathfrak{p})$. Its mass is finite: summing repetitions and enlarging the prime sum gives

$$\|\mu_\sigma\| \leq \frac{1}{1 - 2^{-\sigma}} \sum_{n \geq 2} (\log n) n^{-\sigma} < \infty. \quad (\text{DR2.2})$$

The final series converges by comparison with $n^{-(\sigma+1)/2}$ beyond a finite threshold. The same argument is uniform for $\sigma \geq 1 + \epsilon$ on every fixed $\epsilon > 0$, so termwise transforms and derivatives below are justified.

The map is invertible on this class of complete measures:

$$\boxed{\mathcal{R} = \frac{e^{\sigma t}}{t} \mu_\sigma \quad (t > 0), \quad \mathcal{D} = \exp_* \left(\frac{e^{\sigma t}}{t} \mu_\sigma \right).} \quad (\text{DR2.3})$$

Every positive-time atom stays away from zero, so the division by t is well defined and locally finite. The exponential reinstates δ_0 as its zeroth term. Conversely multiplication by $te^{-\sigma t}$ recovers μ_σ from that logarithm. This verifies both compositions. The only mass not directly seen by multiplying with t is the unit at 0, which is independently retained in (DR1.1) and fixed by the original arithmetic.

For two branches related by BCC's actual full-record isomorphism, conjugation of endomorphisms induces isomorphic finite cokernels. It preserves every $N(a)$, every prime ideal, and every $N(\mathfrak{p})$. Consequently it identifies the measures (DR1.1-2) and (DR2.1) term by term **after each branch uses its own recovered counter**. This is a comparison of two equal receiving measures. Adding the two measures instead would double the logarithm and produce a squared function; that operation is absent from this comparison.

DR3. The original zeta and its complete logarithmic derivative

For $s = \sigma + iu$ with $\sigma > 1$, absolute convergence gives

$$\mathcal{LD}(s) = \int_{[0, \infty)} e^{-st} d\mathcal{D}(t) = \sum_{n \geq 1} n^{-s} = \zeta(s). \quad (\text{DR3.1})$$

The unit coefficient is exactly 1. Finite convolution products and then dominated convergence give

$$\zeta(s) = \prod_{\mathfrak{p}} (1 - N(\mathfrak{p})^{-s})^{-1}, \quad \mathcal{LR}(s) = \sum_{\mathfrak{p}, m \geq 1} \frac{N(\mathfrak{p})^{-ms}}{m} = \log \zeta(s). \quad (\text{DR3.2})$$

The logarithm is the analytic branch tending to 0 as real $s \rightarrow +\infty$; it has been constructed by the convergent series, not chosen after continuation. Differentiate using (DR2.2):

$$\boxed{\hat{\mu}_\sigma(u) := \int e^{-iut} d\mu_\sigma(t) = -\frac{\zeta'(\sigma + iu)}{\zeta(\sigma + iu)}.} \quad (\text{DR3.3})$$

In particular the Fourier transform convention fixes the sign. Conversely the logarithmic derivative determines the original function with its unit through

$$\log \zeta(s) = \int_0^\infty \left(-\frac{\zeta'(s+v)}{\zeta(s+v)} \right) dv, \quad \Re s > 1. \quad (\text{DR3.4})$$

To check it, integrate each term of (DR3.3): $\int_0^\infty N(\mathfrak{p})^{-mv} \log N(\mathfrak{p}) dv = 1/m$. The absolute integral is the convergent series (DR3.2). Exponentiation yields (DR3.1), with limiting value 1 at real infinity.

This proves an exact receiving map and inverse for the original function on its Euler half-plane. No completed zeta has replaced it.

DR4. What the same positive measure proves at the boundary

Write $A(s) = -\zeta'(s)/\zeta(s)$. Because μ_σ is positive, the square identity

$$3 + 4 \cos v + 2 \cos(2v) = (1 + 2 \cos v)^2 \geq 0 \quad (\text{DR4.1})$$

gives the exact all-return inequality

$$3A(\sigma) + 4\Re A(\sigma + iu) + 2\Re A(\sigma + 2iu) = \sum_{\mathfrak{p}, m \geq 1} \log N(\mathfrak{p}) N(\mathfrak{p})^{-m\sigma} (1 + 2 \cos(um \log N(\mathfrak{p})))^2 \geq 0. \quad (\text{DR4.2})$$

Assume $u \neq 0$ and let $m_1, m_2 \geq 0$ be the respective zero multiplicities at $1 + iu$ and $1 + 2iu$, taking zero multiplicity when no zero occurs. The established meromorphic continuation of the **same** original zeta has its unique pole, simple, at 1. Its local logarithmic derivatives therefore satisfy

$$\lim_{\sigma \downarrow 1} (\sigma - 1)A(\sigma) = 1, \quad \lim_{\sigma \downarrow 1} (\sigma - 1)\Re A(\sigma + iju) = -m_j \quad (j = 1, 2). \quad (\text{DR4.3})$$

This follows by factoring each function locally as $(s - s_0)^{m_j} g(s)$ with $g(s_0) \neq 0$, and retaining the simple-pole factor at 1. Multiply (DR4.2) by $\sigma - 1 > 0$ and take the limit. It gives $3 - 4m_1 - 2m_2 \geq 0$. If $m_1 \geq 1$, its left side is at most -1 , a contradiction. Hence the original zeta has no zero on $\Re s = 1$.

This is the source's boundary mechanism, carried out on the programme's exact complete return measure. DB reconstructs its compact-group extension, including the exceptional quadratic-character argument. DW shows precisely where that **strict** boundary result enters the weight improvement for sheaf cohomology. The steps beyond (DR4.2) in that geometric argument are not replaced by the existence of the positive measure.

DR5. Completion and exceptional points remain explicit

Where the source comparison or reflection uses the completed function, retain its full multiplier:

$$C(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2), \quad \xi(s) = C(s) \zeta(s). \quad (\text{DR5.1})$$

The expression (DR3.3) is still the original logarithmic derivative. At regular points its exact comparison is

$$-\frac{\zeta'(s)}{\zeta(s)} = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi(s/2) - \frac{\xi'(s)}{\xi(s)}, \quad (\text{DR5.2})$$

where $\psi = \Gamma'/\Gamma$. This is obtained by differentiating every factor of (DR5.1). As an identity of meromorphic functions it retains all exceptional-point cancellations:

- At 0, $\Gamma(s/2) = 2/s - \gamma + O(s)$, so $C(0) = -1$; the $1/s$ term cancels the digamma pole and $\zeta(0) = -1/2$ is nonzero.
- At 1, $C(s) = \frac{1}{2}(s-1) + O((s-1)^2)$ and the pole of ζ cancels its zero in ξ . The original logarithmic derivative still has residue +1 there, as used in (DR4.3).
- At each $s = -2n$, $n \geq 1$, C has a simple pole. The corresponding original trivial zero is retained in (DR5.2) as a pole of $-\zeta'/\zeta$ of residue -1 . The zero and pole cancel in ξ , not in the original function.
- At a nontrivial zero ρ , $C(\rho) \neq 0, \infty$. Its full multiplicity is unchanged, and the residue of the original negative logarithmic derivative is its negative multiplicity.

The exact reflection multiplier is also retained:

$$\zeta(s) = \frac{C(1-s)}{C(s)} \zeta(1-s). \quad (\text{DR5.3})$$

OZR1-OZR8 prove its exceptional values, all differentiated terms and its action on every local jet. Nothing in the branch comparison removes these contributions.

DR6. The exact finite-field/time dictionary

In the finite-field part of Deligne's proof a single variable t records degrees, so a local factor contains t^{d_x} . Writing $t = q^{-s}$ is the explicit analytic map

$$t^{d_x} = q^{-sd_x} = N(x)^{-s}, \quad -\frac{d}{ds} \log L(q^{-s}) = (\log q) t \frac{L'(t)}{L(t)}. \quad (\text{DR6.1})$$

Every degree factor and $\log q$ remains. With a coefficient sheaf, the local repetitions are weighted by $\text{Tr}(F_x^m)$; DP4 raises those real traces to even powers. For the original Riemann zeta in(DR3), the norm times are the full $\log N(\mathfrak{p})$, and Deligne2.1.9 uses $\Gamma = \mathbb{R}$. They are not all identified with multiples of one fixed $\log q$.

To verify the last assertion, two distinct recovered prime norms $\mathfrak{p}$ and $\mathfrak{r}$ on such a common degree grid would satisfy $\mathfrak{p} = \mathfrak{q}^a$ and $\mathfrak{r} = \mathfrak{q}^b$ for positive integers a, b . Then $\mathfrak{p}^{b/r} = \mathfrak{a}$, contradicting unique factorization in the recovered ring. This argument is downstream of the complete arithmetic reconstruction.

This does not challenge the recovered integer counter: additive counting on L , endomorphism degree $N(a)$, and the logarithmic norm used in the Euler transform are connected by the explicit maps above. They are different receiving quantities with derived relations. No subtraction, common radius, or numeric origin is assigned to τ by any of them.

DR7. The actual composition supplied by the reading

The complete composition now established is

complete reconstructed $A = \text{End}(L)$, $\text{Spec } A$, $N \mapsto \mathcal{D}$ including δ_0 ,

$$\mathcal{D} \begin{array}{c} \xleftarrow{\log_*} \\ \xrightarrow{\exp_*} \end{array} \mathcal{R} \begin{array}{c} \xleftarrow{te^{-\sigma t}} \\ \xrightarrow{e^{\sigma t}/t} \end{array} \mu_\sigma, \quad (\text{DR7.1})$$

$$\hat{\mu}_\sigma(u) = -\zeta'(\sigma + iu)/\zeta(\sigma + iu),$$

positive square on all repeated returns $\mapsto$ boundary nonvanishing of the original ζ .

It commutes with the separate-counter branch isomorphisms. It does not pool their measures, install a metric, or identify the supporting point with arithmetic zero. DP, DLM, DB and DW then spell out the additional **actual source operations**—tensor powers, local monodromy, boundary cycles, a pencil and duality—that give Deligne's exact cohomological weights. The adjacent base-comparison note constructs the available functors over the recovered arithmetic and the fixed support, with their retained Frobenius and duality data. These are proved maps on the specified objects; a bare renaming of a point has not been presented as a weight theorem.

The full original-germ lifting boundary and its receiver — ZLB0-ZLB11

Proof source: ZETA_LIFTING_BOUNDARY_INDEPENDENT.md.

The connecting boundary of the original-zeta germ extension

Independent derivation, 24 September 2026. Proof locators ZLB0-ZLB11.

ZLB0. The construction being tested and the source dependencies

The input is the original Riemann zeta function obtained after the complete arithmetic reconstruction, together with the actual local quotient in OZR5 and arithmetic multiplication in OZR6 of [Original zeta reflection and descent](#). This note computes the boundary of that explicitly specified analytic extension. It does not identify its eigenvector-lifting problem with every lift required by the user's construction or by a possible Deligne comparison.

The supporting datum remains

$$\tau\langle Z_1; \text{no } Z_2 \rangle. \quad (\text{ZLB0.1})$$

No addition, subtraction, coordinate, metric, or vector is assigned to it. The coordinate s , the analytic germs, the vector spaces, and the recovered arithmetic characters below belong to the analytic receiver after the complete arithmetic has been obtained. In particular $z = s - \rho$ is a coordinate at an actual analytic zero; it is not a displacement from τ .

The governing reconstruction and correction chain was checked through READ_FIRST_USER_CONSTRUCTION.md, USER_ARGUMENT_RECONSTRUCTION.md §§1-38, USER_ARGUMENT_CONNECTIONS.md, and the corresponding primary passages in USER_CONSTRUCTION_FULL_LOGBOOK.md, particularly U060, U123-U134, U138, and WU042-WU055. These preserve the retraction of addition at τ , the rejection of an assumed metric or formal vector at τ , whole arithmetic before named numerical inputs, separate branch counters, and the restriction to the original zeta function. The corpus is a source record with embedded speakers; an earlier quoted assistant construction is not made a user axiom here.

The mathematical prerequisites already proved and used are:

- OZR5.1-OZR5.3: actual nontrivial zero germs, their full multiplicities, and the quotient by the original ζ .
- OZR5.4-OZR5.9: the exact original-zeta reflection, including the local unit and all derivative terms.
- OZR6.1-OZR6.7: the recovered arithmetic character a^s , its full nilpotent action, and interchange of F_a with aF_a^{-1} .
- DC5-DC9 of [Deligne invariant cycles](#): the actual cohomological obstruction quotient, the separated source weights, and the proved lift in Deligne's stated proper smooth-model setting.

The existing source ledger ORIGINAL_ZETA_REFLECTION_SOURCE_READING_PRIVATE.json was consulted before deriving new formulas. Its human source is Alain Connes, [The Riemann Hypothesis: Past, Present and a Letter Through Time](#), original author source rhready.tex, formula smallxi and the functional-equation paragraph, local lines 526-545. Those lines and the surrounding lines 491-556 were read here. The full multiplier used is written in ZLB8. Deligne's human source is [La conjecture de Weil. II](#), §3.6,

with the transcription/source distinction and actual reading coverage retained in DC0. No new Deligne theorem is assumed for the analytic quotient.

The requested scope check has a concrete answer: OZR constructs the zero-jet quotient and its operator action, but it does not assert that a quotient eigenvector must lift to an exact eigenfunction in the holomorphic germ ring. The calculation below therefore decides precisely that added lifting question for this analytic extension. Its nonzero answer is not called an obstruction to the user's complete reconstruction, a failure of RH, or a classification of all possible cohomological receivers.

ZLB1. The original germ and the exact differential

Put

$$S = \{s \in \mathbb{C} : 0 < \Re s < 1\}, \quad Z = \{\rho \in S : \zeta(\rho) = 0\}. \quad (\text{ZLB1.1})$$

Fix an actual $\rho \in Z$, with multiplicity $m = m_\rho \geq 1$, and put $z = s - \rho$. The original function has the convergent expansion

$$\zeta(\rho + z) = z^m u_\rho(z), \quad u_\rho(z) = \sum_{j=0}^{\infty} \frac{\zeta^{(m+j)}(\rho)}{(m+j)!} z^j, \quad u_\rho(0) = \frac{\zeta^{(m)}(\rho)}{m!} \neq 0. \quad (\text{ZLB1.2})$$

Write

$$O = \mathcal{O}_\rho, \quad I = \zeta O, \quad J = O/I, \quad \mathfrak{m}_\rho = zO. \quad (\text{ZLB1.3})$$

The equality $I = z^m O$ follows from invertibility of u_ρ , but every calculation with the original generator retains u_ρ .

After the whole arithmetic reconstruction, choose a recovered positive integer $a > 1$. Define

$$\ell_a = \log a > 0, \quad E_a(s) = a^s, \quad \lambda = a^\rho, \quad F_a = M_{E_a}, \quad d_a = F_a - \lambda = M_{g_a}, \quad g_a(s) = a^s - a^\rho. \quad (\text{ZLB1.4})$$

Multiplication by E_a preserves I and acts on all three terms of

$$0 \longrightarrow I \xrightarrow{J} O \xrightarrow{q} J \longrightarrow 0. \quad (\text{ZLB1.5})$$

Its exact local difference is

$$g_a(\rho + z) = z b_a(z), \quad b_a(z) = \lambda \sum_{r=1}^{\infty} \frac{\ell_a^r}{r!} z^{r-1}, \quad b_a(0) = \lambda \ell_a \neq 0. \quad (\text{ZLB1.6})$$

Thus b_a is a holomorphic unit germ. No Taylor term has been dropped from this identity.

ZLB2. The complete connecting sequence

For $M = I, O, J$, form the cochain complex

$$C_a(M) = [M \xrightarrow{d_a} M] \quad (\text{ZLB2.1})$$

in degrees zero and one. The three complexes form a short exact sequence because (ZLB1.5) is exact in each degree and its maps commute with d_a .

Here is the connecting sequence with its sign fixed:

$$0 \rightarrow \ker(d_a|I) \rightarrow \ker(d_a|O) \rightarrow \ker(d_a|J) \xrightarrow{\delta_a} I/d_a I \rightarrow O/d_a O \rightarrow J/d_a J \rightarrow 0. \quad (\text{ZLB2.2})$$

For $\bar{f} \in \ker(d_a|J)$, choose any holomorphic germ f lifting it. Since $d_a f \in I$, set

$$\delta_a(\bar{f}) = [d_a f]_{I/d_a I}. \quad (\text{ZLB2.3})$$

Changing the lift by $v \in I$ adds $d_a v$, so this is well-defined. This is the positive cochain-boundary convention.

For completeness, exactness can be proved at every term directly. A kernel element of O mapping to zero in J belongs to I , and remains a kernel element there. A kernel element of J has zero boundary exactly when a lift f satisfies $d_a f = d_a v$ for some $v \in I$; then $f - v$ is a kernel lift in O . A class $[v] \in I/d_a I$ maps to zero in $O/d_a O$ exactly when $v = d_a f$ for some $f \in O$; the class of f in J then has boundary $[v]$. A class $[f] \in O/d_a O$ maps to zero in $J/d_a J$ exactly when $q(f) = d_a q(h)$ for a germ h , which says $f - d_a h \in I$. The final arrow is onto because $O \rightarrow J$ is onto. The other maps are the maps induced by inclusion and quotient. All cohomology outside degrees zero and one is zero.

The germ ring O is an integral domain: a nonzero germ has a finite order of vanishing and a nonzero unit factor, and orders add under multiplication. Therefore multiplication by the nonzero germ g_a is injective on O , hence on I . The complete groups are

$$\begin{array}{c|c|c} M & H^0 C_a(M) & H^1 C_a(M) \\ \hline I & 0 & I/zI \\ O & 0 & O/zO \\ J & z^{m-1}O/z^m O & O/zO. \end{array} \quad (\text{ZLB2.4})$$

For the last row, $g_a f \in I$ means $z f \in z^m O$, since b_a is a unit. Its cokernel is $O/(I + zO) = O/zO$. The identifications for I and O use $g_a I = zI$ and $g_a O = zO$; the original function $g_a = z b_a$ remains available in the differential.

The coefficient maps

$$I/zI \longrightarrow \mathbb{C}, \quad [\zeta h] \longmapsto h(\rho), \quad O/zO \longrightarrow \mathbb{C}, \quad [f] \longmapsto f(\rho) \quad (\text{ZLB2.5})$$

are isomorphisms. Uniqueness of h follows from the domain property. The first class $[\zeta]_{I/zI}$ is nonzero: $\zeta \in zI$ would imply $1 = zh$ after cancellation in O , which is impossible on evaluating at ρ . It must not be confused with $[\zeta]_J = 0$.

The map $I/d_a I \rightarrow O/d_a O$ is zero, since every element of I vanishes at ρ . The last map is the identity under evaluation. Consequently (ZLB2.2) is exactly

$$0 \longrightarrow \ker(d_a|J) \xrightarrow[\sim]{\delta_a} I/zI \xrightarrow{0} \mathbb{C} \xrightarrow{\text{id}} \mathbb{C} \longrightarrow 0. \quad (\text{ZLB2.6})$$

ZLB3. The original-generator boundary, with every factor retained

For the coordinate socle class, compute

$$d_a(z^{m-1}) = z^m b_a(z) = \zeta(\rho + z) \frac{b_a(z)}{u_\rho(z)}. \quad (\text{ZLB3.1})$$

Thus

$$\delta_a[z^{m-1}]_J = \frac{\lambda \log a}{u_\rho(0)} [\zeta]_{I/zI}. \quad (\text{ZLB3.2})$$

The factor $u_\rho(0)$ is the original m -th derivative divided by $m!$, not a coefficient chosen to make ζ monic.

There is a version using the original function itself. The holomorphic germ

$$k_\rho(s) = \frac{\zeta(s)}{s - \rho} = z^{m-1} u_\rho(z) \quad (\text{ZLB3.3})$$

has nonzero class in J , and spans $\ker(d_a|J)$. Its boundary is

$$d_a k_\rho = b_a(z) \zeta, \quad \boxed{\delta_a[k_\rho]_J = \lambda \log a [\zeta]_{I/zI} \neq 0.} \quad (\text{ZLB3.4})$$

The higher terms in b_a have not been omitted by an approximation. Their difference from the constant representative is the exact boundary

$$(b_a(z) - b_a(0)) \zeta = d_a \left(\frac{b_a(z) - b_a(0)}{z b_a(z)} \zeta \right). \quad (\text{ZLB3.5})$$

The quotient inside parentheses is holomorphic, because its numerator vanishes at zero and its denominator is z times a unit.

There is an a -independent connecting line. Set

$$\mathcal{S}_\rho = \text{Ann}_J(\mathfrak{m}_\rho) = z^{m-1} O / z^m O, \quad \mathcal{N}_\rho = I / \mathfrak{m}_\rho I. \quad (\text{ZLB3.6})$$

Define

$$\kappa_\rho : \mathcal{S}_\rho \longrightarrow \mathcal{N}_\rho, \quad [f] \longmapsto [zf]. \quad (\text{ZLB3.7})$$

For a socle element $zf \in I$; a change of lift by I changes zf by zI . The map sends $[k_\rho]$ to $[\zeta]$, so it is an isomorphism of one-dimensional spaces. In these fixed spaces the whole recovered arithmetic family has

$$\delta_{a,\rho} = a^\rho \log a \kappa_\rho. \quad (\text{ZLB3.8})$$

For any recovered $a, b > 1$, the character identity gives the exact relation

$$\delta_{ab,\rho} = a^\rho \delta_{b,\rho} + b^\rho \delta_{a,\rho}. \quad (\text{ZLB3.9})$$

Indeed both sides equal $(ab)^\rho (\log a + \log b) \kappa_\rho$. The operation here is addition of linear maps in the analytic receiver; it is not addition at the supporting point.

ZLB4. The actual next-jet extension

The nonzero boundary is already visible in the finite-dimensional algebra

$$\mathcal{B}_\rho = O / (z\zeta O) = O / ((a^s - a^\rho)\zeta O). \quad (\text{ZLB4.1})$$

Equality of the ideals follows from the unit b_a . The full defining product is

$$(a^s - a^\rho)\zeta = z^{m+1} b_a(z) u_\rho(z). \quad (\text{ZLB4.2})$$

Thus $\dim_{\mathbb{C}} \mathcal{B}_\rho = m + 1$, and the exact sequence is

$$0 \longrightarrow \mathcal{N}_\rho \longrightarrow \mathcal{B}_\rho \longrightarrow J_\rho \longrightarrow 0. \quad (\text{ZLB4.3})$$

The first arrow is inclusion of I/zI . Its kernel is zero by the definition of the quotient; its image is the kernel of the second arrow. This ideal is square-zero, because $I^2 \subset zI$ follows from $2m \geq m + 1$. Multiplication by a class $f \in J$ on $\mathcal{N}_\rho$ is its scalar value $f(\rho)$, as $(f - f(\rho))\zeta \in zI$.

All recovered F_a preserve this same extension. The exact truncated functions acting on it and on J are

$$F_a|_{\mathcal{B}_\rho} = \lambda \sum_{r=0}^m \frac{(\log a)^r}{r!} M_z^r, \quad F_a|_J = \lambda \sum_{r=0}^{m-1} \frac{(\log a)^r}{r!} M_z^r. \quad (\text{ZLB4.4})$$

Their omitted tails belong to the respective displayed defining ideals. On $\mathcal{B}_\rho$, $d_a = zb_a(z)$ has nilpotence order exactly $m + 1$, because $z^m b_a(z)^m \notin z^{m+1}O$; on J , its order is exactly m . These are the actual original-zeta jet algebras, not independently selected matrices.

Applying the degree-zero lifting construction to (ZLB4.3) gives the same boundary (ZLB3.4): a lift of $[k_\rho]$ to $\mathcal{B}_\rho$ has d_a -image $\lambda \log a[\zeta]$, and $d_a \mathcal{N}_\rho = 0$. Thus the eigenvector has no eigenvector lift even to this finite extension.

There is also no F_a -equivariant linear section $J \rightarrow O$. In fact every F_a -equivariant linear map $T : J \rightarrow O$ is zero: $d_a^m = 0$ on J , while multiplication by d_a^m is injective on O , and

$$d_a^m T = T d_a^m = 0 \quad (\text{ZLB4.5})$$

forces $T = 0$. This statement is about the specified holomorphic source O and multiplication action; it is not a statement that no other cohomological source can lift the same quotient.

ZLB5. All multiplicities through higher connecting complexes

For every positive integer r , repeat the construction with differential $d_a^r = z^r b_a(z)^r$. The same elementary proof of (ZLB2.2) gives its entire long exact sequence. Its groups are

$$H^0 \mathcal{C}_{a,r}(I) = H^0 \mathcal{C}_{a,r}(O) = 0, \quad H^0 \mathcal{C}_{a,r}(J) = z^{\max(m-r,0)} O / z^m O, \quad (\text{ZLB5.1})$$

$$H^1 \mathcal{C}_{a,r}(I) = I / z^r I, \quad H^1 \mathcal{C}_{a,r}(O) = O / z^r O, \quad H^1 \mathcal{C}_{a,r}(J) = O / z^{\min(r,m)} O. \quad (\text{ZLB5.2})$$

These equalities follow by comparing orders in $z^r b_a^r f \in z^m O$. The connecting map is

$$\delta_{a,r}[f] = [z^r b_a(z)^r f]_{I/z^r I}. \quad (\text{ZLB5.3})$$

For $1 \leq r \leq m$, every source element can be written $[\zeta z^{-r} h]$, with h determined modulo $z^r O$, and

$$\delta_{a,r}[\zeta z^{-r} h] = [\zeta b_a(z)^r h]_{I/z^r I}. \quad (\text{ZLB5.4})$$

Since b_a^r is a unit, this is an isomorphism onto $I/z^r I$. All its r coefficient layers are retained. For $r > m$, the entire J is the kernel, and

$$\delta_{a,r}[f] = \left[\zeta z^{r-m} \frac{b_a(z)^r}{u_\rho(z)} f \right]_{I/z^r I}, \quad \text{im } \delta_{a,r} = z^{r-m} I / z^r I. \quad (\text{ZLB5.5})$$

The image has dimension m . Inclusion $I/z^r I \rightarrow O/z^r O$ has exactly this kernel, and the final quotient $O/z^r O \rightarrow O/z^{\min(r,m)} O$ proves exactness at the remaining terms as well.

One can record the full equivariant extension class explicitly. Let $\mathcal{R}_a = \mathbb{C}[t, t^{-1}]$ act by $t = F_a$, whose inverse is multiplication by the holomorphic unit a^{-s} . On J , the vectors $1, d_a 1, \dots, d_a^{m-1} 1$ are a basis: their orders are $0, \dots, m-1$, with nonzero leading coefficients $b_a(0)^j$. Hence

$$J \simeq \mathcal{R}_a / (t - \lambda)^m. \quad (\text{ZLB5.6})$$

The free resolution with differential $(t - \lambda)^m$, after applying $\text{Hom}_{\mathcal{R}_a}(-, I)$, gives

$$\text{Ext}_{\mathcal{R}_a}^1(J, I) \simeq I/d_a^m I. \quad (\text{ZLB5.7})$$

This resolution is exact because $\mathcal{R}_a$ is a domain and the quotient is (ZLB5.6). The extension (ZLB1.5) is represented, by lifting $1 \in J$ to $1 \in O$, by

$$[(a^s - a^\rho)^m]_{I/d_a^m I} = \left[\zeta \frac{b_a(z)^m}{u_\rho(z)} \right]_{I/z^m I}. \quad (\text{ZLB5.8})$$

Changing the lift adds $d_a^m I$, proving independence of that choice. This representative has order m , whereas every element of $d_a^m I$ has order at least $2m$, so the class is nonzero. It retains the full extension, beyond the one-dimensional socle boundary.

ZLB6. Actual finite-zero compatibility inside the original analytic quotient

Here we use a specified auxiliary global analytic space, not an identification with every growth-restricted programme source:

$$\mathcal{A} = \mathcal{O}(S), \quad \mathcal{Q} = \mathcal{A}/\zeta\mathcal{A}. \quad (\text{ZLB6.1})$$

There is a natural restriction map $\mathcal{Q} \rightarrow J_\rho$ for each actual zero. Let $D \subset Z$ be a finite set of distinct actual zeros. The following construction gives every finite collection of full zero jets an actual holomorphic representative.

For $f_\rho \in J_\rho$, choose a germ representing it and let

$$P_\rho(z) = T_{m_\rho-1} \left(\frac{f_\rho(\rho + z)}{u_\rho(z)} \right) = \sum_{j=0}^{m_\rho-1} c_{\rho,j} z^j. \quad (\text{ZLB6.2})$$

Here T_{m-1} denotes the Taylor polynomial through degree $m-1$; its meaning is the displayed finite coefficient extraction. Changing the germ by ζh changes f_ρ/u_ρ by $z^m h$, so P_ρ is independent of the representative. Define

$$L_\rho(f_\rho)(s) = \frac{\zeta(s)}{(s - \rho)^{m_\rho}} P_\rho(s - \rho). \quad (\text{ZLB6.3})$$

This is holomorphic on S : the possible denominator singularity is removed by the actual multiplicity at ρ . At ρ , its class is f_ρ . At every other actual zero σ , the denominator is a unit and the expression vanishes to at least the full multiplicity m_σ . Therefore

$$\iota_D : \prod_{\rho \in D} J_\rho \longrightarrow \mathcal{Q}, \quad (f_\rho)_\rho \longmapsto \left[\sum_{\rho \in D} L_\rho(f_\rho) \right] \quad (\text{ZLB6.4})$$

is injective and is a section of the restriction onto the indicated factors.

To check multiplication, compare two products at every actual zero. At a zero in D , their jets multiply as in J_ρ ; at a zero outside D , both have zero full jet. The difference divided by ζ is holomorphic near every zero by its full vanishing order, and away from the zeros ordinary division is holomorphic. These local quotients agree, producing an element of $\mathcal{A}$. This proves that ι_D is a multiplicative map. It takes the unit of the finite product to an idempotent e_D of $\mathcal{Q}$, not necessarily to the global unit. Products of distinct single-zero images are zero.

The same argument shows equivariance for every recovered arithmetic multiplier:

$$F_a \iota_D = \iota_D \left(\prod_{\rho \in D} F_{a,\rho} \right). \quad (\text{ZLB6.5})$$

The product notation on the right is the componentwise operator on the finite product, not multiplication of branch measures. Under $D \subset D'$, adding zero components commutes with ι_D ; coordinate restriction also commutes with the local connecting sequences. Thus the finite-zero comparisons retain all multiplicities and are compatible under enlargement.

The chosen representative L_ρ itself is not claimed equivariant in $\mathcal{A}$. Its exact defect is

$$E_a(s)L_\rho(f_\rho)(s) - L_\rho(F_{a,\rho}f_\rho)(s) = \zeta(s) \frac{E_a(s)P_\rho(z) - T_{m_\rho-1}(E_a(\rho+z)P_\rho(z))}{z^{m_\rho}}, \quad z = s - \rho. \quad (\text{ZLB6.6})$$

The quotient is holomorphic because its numerator has zero Taylor coefficients through degree $m_\rho - 1$. Equation (ZLB6.6), rather than an unstated equality of representatives, proves (ZLB6.5). For $f_\rho = [k_\rho]$, $P_\rho = z^{m_\rho-1}$, so the defect is exactly $\zeta(s)(a^s - a^\rho)/(s - \rho)$.

ZLB7. The global boundary and finite resonances

The original sequence

$$0 \rightarrow \zeta \mathcal{A} \rightarrow \mathcal{A} \rightarrow \mathcal{Q} \rightarrow 0 \quad (\text{ZLB7.1})$$

is preserved by multiplication by a^s . For fixed actual ρ use $g_a(s) = a^s - a^\rho$ as before. Its two-term complexes have the same connecting construction as ZLB2. The globally holomorphic function $\zeta(s)/(s - \rho)$ gives an actual kernel class, because

$$g_a(s) \frac{\zeta(s)}{s - \rho} = \zeta(s) \frac{g_a(s)}{s - \rho}. \quad (\text{ZLB7.2})$$

The quotient on the right is holomorphic on S . Hence its global boundary is

$$\left[\zeta(s) \frac{g_a(s)}{s - \rho} \right] \in \zeta \mathcal{A} / g_a \zeta \mathcal{A}. \quad (\text{ZLB7.3})$$

It is nonzero: vanishing would imply $g_a(s)/(s - \rho) = g_a(s)h(s)$ with $h \in \mathcal{A}$. On the nonempty open where $g_a \neq 0$, this forces $h = 1/(s - \rho)$; the identity theorem contradicts holomorphicity at ρ . Restriction to the germ at ρ is precisely (ZLB3.4). No interpolation theorem for infinitely many prescribed jets is needed.

This also gives simultaneous finite compatibility without assuming distinct arithmetic eigenvalues. For a finite D , let

$$\Lambda_D = \{a^\rho : \rho \in D\}, \quad P_D(t) = \prod_{\lambda \in \Lambda_D} (t - \lambda), \quad G_D(s) = P_D(a^s). \quad (\text{ZLB7.4})$$

Repeated eigenvalues occur once in Λ_D , while every zero and its own multiplicity remain in D . At each $\rho \in D$,

$$G'_D(\rho) = a^\rho \log a \prod_{\substack{\lambda \in \Lambda_D \\ \lambda \neq a^\rho}} (a^\rho - \lambda) \neq 0. \quad (\text{ZLB7.5})$$

For every $\rho \in D$, the actual function $\zeta/(s - \rho)$ is killed in $\mathcal{Q}$ by G_D , and its connecting boundary is

$$\left[\zeta(s) \frac{G_D(s)}{s - \rho} \right] \in \zeta \mathcal{A} / G_D \zeta \mathcal{A}. \quad (\text{ZLB7.6})$$

These finitely many boundary classes are linearly independent. Indeed, divide a proposed relation by the original generator ζ , restrict to a germ at a specified $\sigma \in D$, and evaluate modulo G_D : every term with $\rho \neq \sigma$ vanishes, while the σ -term has value $G'_D(\sigma) \neq 0$. Thus its coefficient is zero, for every σ . This proves that combining finitely many actual zero blocks does not cancel these particular local boundaries. It makes no claim about infinite topological direct products or a growth-restricted source not specified in (ZLB6.1).

ZLB8. Full reflection multipliers and the exact chain maps

Let $\rho' = 1 - \bar{\rho}$, $w = t - \rho'$, and retain the original completion multiplier only as the source of the functional equation:

$$C(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2), \quad A(t) = \frac{\frac{1}{2}(1-t)((1-t)-1) \pi^{-(1-t)/2} \Gamma((1-t)/2)}{\frac{1}{2} t(t-1) \pi^{-t/2} \Gamma(t/2)}. \quad (\text{ZLB8.1})$$

On S these are holomorphic units. The original function satisfies

$$\zeta = A H \zeta, \quad (H_\rho f)(t) = \overline{f(1 - \bar{t})}, \quad T_\rho f = A(t) H_\rho f. \quad (\text{ZLB8.2})$$

Thus $H_\rho \zeta = A^{-1} \zeta$, $T_\rho \zeta = \zeta$, $H_\rho z = -w$, and

$$u_{\rho'}(w) = (-1)^m A(\rho' + w) \overline{u_\rho(-\bar{w})}. \quad (\text{ZLB8.3})$$

This expression is holomorphic in w . It retains the full local unit and full multiplier. The endpoint and trivial-zero exceptions lie outside this strip calculation; none of the unit assertions is extended to those points.

Write

$$\lambda = a^\rho, \quad \mu = a^{\rho'} = \frac{a}{\bar{\lambda}}, \quad V_a = M_{a^{1-t}}, \quad b_{a,\rho}^{\text{ref}}(t) = -\bar{\lambda} a^{-t}. \quad (\text{ZLB8.4})$$

The reflection equations on the germ, ideal, and jet spaces are

$$H_\rho(F_a - \lambda) = (V_a - \bar{\lambda}) H_\rho, \quad V_a - \bar{\lambda} = b_{a,\rho}^{\text{ref}}(F_a - \mu). \quad (\text{ZLB8.5})$$

The second identity follows by multiplication: $-\bar{\lambda} a^{-t}(a^t - \mu) = -\bar{\lambda} + a a^{-t}$. It is an identity of full germ functions, not only of their scalar values.

There are therefore two exact forms of the comparison. With target differential $V_a - \bar{\lambda}$, the two degrees both use H_ρ . With target differential $F_a - \mu$, the chain isomorphism has degree maps

$$\Phi_\rho^0 = H_\rho, \quad \Phi_\rho^1 = (b_{a,\rho}^{\text{ref}})^{-1} H_\rho. \quad (\text{ZLB8.6})$$

Indeed $(F_a - \mu) \Phi_\rho^0 = \Phi_\rho^1 (F_a - \lambda)$. All maps are conjugate-linear. The unit multiplier in degree one is essential. The weighted original-generator-fixing comparison similarly has

$$\Psi_\rho^0 = T_\rho, \quad \Psi_\rho^1 = (b_{a,\rho}^{\text{ref}})^{-1} T_\rho. \quad (\text{ZLB8.7})$$

The multipliers commute with the multiplication differentials, proving this second chain identity. H_ρ is unital and multiplicative; T_ρ instead satisfies $T(fg) = A^{-1}(Tf)(Tg)$. The degree-one maps in (ZLB8.6)–(ZLB8.7) are not asserted unital algebra maps.

These maps preserve the original ideal and the next-jet ideal: $H(z\zeta) = -wA^{-1}\zeta$, and multiplication by a unit does not change the target ideal. They induce comparisons of (ZLB1.5), (ZLB4.3), and their connecting sequences. The boundary comparison is exactly

$$\delta_{a,\rho'}\Phi_\rho^0 = \Phi_\rho^1\delta_{a,\rho}, \quad \delta_{a,\rho'}\Psi_\rho^0 = \Psi_\rho^1\delta_{a,\rho}. \quad (\text{ZLB8.8})$$

It follows either from (ZLB2.3) and the chain identity or by the explicit calculation next.

On the actual original-function socle lift,

$$H_\rho k_\rho = -A^{-1}k_{\rho'}, \quad T_\rho k_\rho = -k_{\rho'}. \quad (\text{ZLB8.9})$$

On the obstruction line, write $A_0 = A(\rho')$. Since $(b_{a,\rho}^{\text{ref}})^{-1}(\rho') = -\mu/\bar{\lambda}$, the induced maps are

$$\Phi_\rho^1[\zeta] = -\frac{\mu}{\bar{\lambda}A_0}[\zeta], \quad \Psi_\rho^1[\zeta] = -\frac{\mu}{\bar{\lambda}}[\zeta]. \quad (\text{ZLB8.10})$$

Consequently both routes on the H -comparison give $-\mu \log a A_0^{-1}[\zeta]$, and both routes on the T -comparison give

$$-\mu \log a[\zeta]. \quad (\text{ZLB8.11})$$

Complex conjugation of the source scalar $\lambda \log a$ is included in these equalities. Each chain isomorphism carries the nonzero boundary to a nonzero boundary. This transport makes no additional quotient of the fibres.

ZLB9. Higher jets, reflection, and finite-set comparisons

The complete derivative map is also explicit. For a germ $f(\rho + z) = \sum_{j \geq 0} c_j z^j$,

$$T_\rho f(\rho' + w) = A(\rho' + w) \sum_{j \geq 0} (-1)^j \bar{c}_j w^j. \quad (\text{ZLB9.1})$$

Its coefficient of w^k is

$$\sum_{j=0}^k \frac{A^{(k-j)}(\rho')}{(k-j)!} (-1)^j \bar{c}_j. \quad (\text{ZLB9.2})$$

For J use all $0 \leq k < m$; for $\mathcal{B}_\rho$ use all $0 \leq k \leq m$. The original full expression (ZLB8.1) defines every derivative of A ; OZR4.3–OZR4.6 give them with the Gamma derivatives, powers of π , and endpoint contributions retained. Formula (ZLB9.2) does not replace them by a scalar on a higher-jet space. A scalar evaluation is used only on $\mathcal{N}_\rho$, whose ideal quotient makes all higher coefficient terms exactly zero.

For the complexes with differential d_a^r , the degree maps become

$$(H_\rho, (b_{a,\rho}^{\text{ref}})^{-r} H_\rho), \quad (T_\rho, (b_{a,\rho}^{\text{ref}})^{-r} T_\rho). \quad (\text{ZLB9.3})$$

Equation (ZLB8.5), raised to the r -th power, proves the chain identities. In particular the full higher connecting maps (ZLB5.3) commute with these maps, including the derivatives of every multiplier occurring modulo $z^r I$.

For a finite actual zero set D , reflection gives rD , with the same multiplicity at paired points. The local maps assemble componentwise. The global maps H, T on $\mathcal{Q}$ commute with the finite-zero inclusions (ZLB6.4) and these local comparisons. To prove this without assuming equality of chosen representatives, compare their full jets at every actual zero. The two classes have the same jets on rD by (ZLB9.1)–(ZLB9.2), and zero jets elsewhere; their difference is divisible by ζ globally, exactly as in the proof of (ZLB6.4). This proves the comparison on the actual quotient.

A comparison of complete branches that identifies their recovered arithmetic and the original function sends a, s, ζ , and each specified zero germ to their stated counterparts. Under that already specified identification all maps in ZLB1–ZLB7 commute by their formulas. No branch measures are added and no second copy of ζ is multiplied into the function. Reflection is the more specific comparison (ZLB8.5), which interchanges F_a and aF_a^{-1} ; it is not misreported as fixing each individually marked F_a .

ZLB10. What the calculation says about the Deligne comparison

The operator F_a acts by the same eigenvalue $\lambda = a^\rho$ on $\mathcal{S}_\rho, \mathcal{N}_\rho, O/d_a O$, and $J/d_a J$. On the socle this is its definition as $\ker(F_a - \lambda)$; on the quotient groups it follows because $F_a - \lambda$ acts as zero. In particular the nonzero boundary (ZLB3.4) intertwines equal eigenvalues.

The exact analytic modulus exponent is

$$2 \log_a |a^\rho| = 2\Re \rho. \quad (\text{ZLB10.1})$$

This is a calculated exponent of the analytic arithmetic character, not a claim that it is already a Deligne weight of an arithmetic sheaf. Every eigenvalue on J and on the next-jet extension $\mathcal{B}_\rho$ equals λ , so any cutoff based on (ZLB10.1) either retains both the kernel $\mathcal{N}_\rho$ and quotient J , or retains neither. Such a cutoff cannot retain the quotient while excluding this kernel.

This is the exact comparison with DC8: Deligne obtains a kernel of weights at least $i + 1$ and a quotient of weights at most i ; that separation makes the low-weight part map isomorphically to the quotient. Here the specified original-zeta germ extension has the same character on both sides, and its connecting boundary is a nonzero isomorphism of lines. No Tate factor has been obtained from this sequence: multiplication by the original ζ is an F_a -equivariant isomorphism $O \rightarrow I$, since both multiplications commute. Inserting a character factor on I would change that stipulated equivariant extension, not compute its existing action.

For an actual zero on the critical line, (ZLB10.1) equals one on both lines and (ZLB3.4) is still nonzero. Thus this boundary is an eigenvector-lift obstruction for the displayed analytic source, not a test whose vanishing is equivalent to RH. Nothing in this calculation asserts the existence of an off-critical zero.

The definition audit in ZLB0 matters here. A proof that the user's construction requires this particular lift has not been supplied by OZR5–6, DC5–9, or the maps constructed in this note. The actual conclusions are therefore confined to the explicit extension and its exact reflection and finite-zero comparisons. The new object determined by its failure to split is the concrete next-jet algebra (ZLB4.1), with the higher extension classes (ZLB5.8); it is not an arbitrary unrelated counterexample to the programme.

ZLB11. The resulting connecting object

The calculated data can be displayed as a commuting connecting diagram, where the lower row uses the reflected differential and every vertical arrow is the exact map of ZLB8:

$$\begin{array}{ccc} \mathcal{S}_\rho & \xrightarrow{\delta_{a,\rho}} & \mathcal{N}_\rho \\ \Psi_\rho^0 \downarrow & & \downarrow \Psi_\rho^1 \\ \mathcal{S}_{\rho'} & \xrightarrow{\delta_{a,\rho'}} & \mathcal{N}_{\rho'}. \end{array} \quad (\text{ZLB11.1})$$

The top map sends the original-function class $[\zeta/(s - \rho)]$ to $a^\rho \log a[\zeta]$. The target is the fibre at ρ of the conormal module of the original zero ideal:

$$(I/I^2) \otimes_J (O/\mathfrak{m}_\rho) \cong I/(I^2 + \mathfrak{m}_\rho I) = I/\mathfrak{m}_\rho I,$$

because $I \subseteq \mathfrak{m}_\rho$, so $I^2 \subseteq \mathfrak{m}_\rho I$. Thus $[\zeta]$ here is a nonzero generator of that fibre, not the zero class of ζ in J . It sits as the retained square-zero kernel in the actual extension

$$0 \rightarrow I/\mathfrak{m}_\rho I \rightarrow O/(\mathfrak{m}_\rho I) \rightarrow O/I \rightarrow 0. \quad (\text{ZLB11.2})$$

The higher complexes retain all m_ρ original jet layers and their connecting images. The finite-zero maps use holomorphic representatives of the same original function, and the full reflection includes every factor of C , A , and $b_{a,\rho}^{\text{ref}}$. These are the complete maps proved here; none assigns an analytic vector or numerical distance to $\tau\langle Z_1; \text{no } Z_2 \rangle$.

Character-shifted lifting with every nilpotent layer — ZTM0-ZTM11

Proof source: ZETA_TATE_MONODROMY_DERIVATION.md.

Original-zeta jets, character twists, and exact lifting

Independent derivation, 24 September 2026. Stable proof locators ZTM0–ZTM11. All zero labels below denote actual zeros of the original Riemann zeta function. No proposed zero set, RH assumption, arithmetic operation on the supporting point, or replacement zeta function is introduced.

ZTM0. Construction order, prerequisites, and source-use record

The supporting datum remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. This calculation starts after the complete arithmetic and the original ζ have been recovered. The constructions below are operations on that original function, its germ rings and coefficient modules. They are not primitives used to reconstruct numbers from τ . Separate branches retain their own recovered counters; this note forms no pooled branch measure.

The definition and correction corpus consulted is `READ_FIRST_USER_CONSTRUCTION.md` (private construction record; not included), the complete `USER_ARGUMENT_RECONSTRUCTION.md` (private construction record; not included), `USER_ARGUMENT_CONNECTIONS.md` (private construction record; not included), and the relevant full verbatim passages of `USER_CONSTRUCTION_FULL_LOGBOOK.md` (private construction record; not included): U123–U138, WU042–WU055 and the complete ad55fb5f attachment. This is the actual bounded coverage of the large verbatim archive, not a claim to have reread all its other attachments. WU050–WU051 control the global-before-local dependency; U128, U134 and U138 exclude arithmetic or a coordinate at τ ; WU054–WU055 control branchwise comparison and admissibility.

The mathematical inputs actually read are [OZR5–OZR6](#), including the original Taylor unit and all nilpotents, and [DC5–DC9](#), including the precise localization obstruction. The full completion and functional equation were reread in Alain Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, [arXiv:2602.04022v1](#), original author `rhready.tex`, local lines 526–546. Canonical source IDs are PUBUNIT-171DAC50D1C3B7A2438B39AE and PUBUNIT-90A9C2A0907A753437719FD9; source SHA256 is 7f1888d82b42263faca4264f3f9fc77e5750c348640f85db0ac4569fe88ebab2.

For the monodromy type, Pierre Deligne, [La conjecture de Weil. II](#), §1.7.1–§1.7.5, was reread in the current French S20 transcription, local lines 1076–1125, together with DLM5. The witness is output `/Deligne_Weil_II_S20_LaTeX/typed_latex/S20_FR_record_export.tex`, SHA256 d6836ae15d98f2b27eb98c9bd039a7d921679becd9fa47ff4717410289dbd351, canonical IDs PUBUNIT-3D90B487DBD1CC3A259C6455 and PUBUNIT-574B0214207BBB2BDABCF9. It is a transcription, not author TeX. No PDF was read.

Operation used here	Constructed prerequisite and exact proof	Governing correction
A recovered positive integer or rational a , its real logarithm, or distinct recovered primes	The complete recovered arithmetic is the input stage; its existing ring comparison is MDB1.1-MDB1.5. The spectral function and action are OZR6.1-OZR6.5. No finite prime selection establishes that input.	WU042-WU051
A complex variable s , a germ coordinate $z = s - \rho$, Taylor addition and scalar multiplication	These belong to the already recovered original analytic function and the exact quotient OZR5.2-OZR5.7. The coordinate is at a zeta zero, not at τ .	U123-U138
A finite collection of zero blocks	Each block is the full original quotient OZR5.3. A finite collection is a restriction for a calculation after whole reconstruction; it is not the recovered arithmetic input. ZTM5 separately proves the algebraic direct-sum extension.	WU049-WU055
Vector-space extensions, their matrix entries, and their sums	The coefficient category is defined in ZTM4. All additions are in these coefficient modules. The splitting and its inverse are explicitly proved there.	U138; no addition at τ
A character twist or spectral translation	ZTM1 defines the one-dimensional character; ZTM7 proves translation on the full original ideals. Neither is an assumed geometric Tate realization.	WU055
A cone or a lifting obstruction	ZTM4 defines and computes the extension obstruction; ZTM8 specifies the complexes, differentials and cohomology. DC5 is the separately constructed geometric obstruction. Their names do not identify their source maps.	WU055 and the requested exact comparison
Comparison across branches	An identified original ζ , the branches' own recovered counters, and the exact induced maps are required; ZTM10 proves naturality for such maps.	WU054

The proofs therefore apply inside the user's admissible class preserving the established original function. They do not use an arithmetic-destroying history as a counterexample. The vanishing proved here is an explicitly constructed character-module lifting obstruction. It is not relabelled as the user's entire geometric obstruction without a map identifying those objects.

ZTM1. The original zero modules and the exact character twist

Let

$$S = \{s \in \mathbb{C} : 0 < \operatorname{Re} s < 1\}, \quad Z = \{\rho \in S : \zeta(\rho) = 0\}. \quad (\text{ZTM1.1})$$

For an actual $\rho \in Z$, its finite multiplicity is $m = m_\rho$. In $z = s - \rho$, retain the full original expansion

$$\zeta(\rho + z) = z^m u_\rho(z), \quad u_\rho(z) = \sum_{j \geq 0} \frac{\zeta^{(m+j)}(\rho)}{(m+j)!} z^j, \quad u_\rho(0) \neq 0. \quad (\text{ZTM1.2})$$

Let $\mathcal{O}_\rho$ be the ring of holomorphic germs and

$$J_\rho = \mathcal{O}_\rho / (\zeta), \quad Z_\rho = M_z \in \operatorname{End}_{\mathbb{C}}(J_\rho). \quad (\text{ZTM1.3})$$

The original unit in (ZTM1.2) proves $(\zeta) = (z^m)$. Thus Taylor truncation is an explicitly proved coordinate isomorphism $J_\rho \simeq \mathbb{C}[z]/(z^m)$, with $Z_\rho^m = 0$. The full u_ρ remains in the analytic equations below.

Let $A = \mathbb{Q}_{>0}^\times$ denote the multiplicative group of positive rationals in the recovered arithmetic. Multiplication by the nonvanishing germ a^s , using the real logarithm, gives

$$F_{a,\rho} = a^\rho \exp((\log a)Z_\rho) = a^\rho \sum_{j=0}^{m-1} \frac{(\log a)^j}{j!} Z_\rho^j. \quad (\text{ZTM1.4})$$

The sums are exact, not asymptotic. Original multiplication proves $F_a F_b = F_{ab}$ and $F_a^{-1} = F_{a^{-1}}$. Equivalently one can start with the positive-integer monoid and extend its invertible operators uniquely to its group completion A .

For any such complex representation V , define the character twist

$$V(k) : \quad F_{a,V(k)} = a^{-k} F_{a,V}, \quad k \in \mathbb{Z}. \quad (\text{ZTM1.5})$$

This is tensoring with the explicitly defined character $a \mapsto a^{-k}$. It uses the same multiplier as a geometric Tate twist at a Frobenius of norm a ; it has not thereby been constructed as an étale Tate twist.

Deligne's equivariant map $N : V(1) \rightarrow V$ has, after a choice of basis of the Tate line, the equation

$$F N F^{-1} = a^{-1} N. \quad (\text{ZTM1.6})$$

By contrast the full zero-coordinate operator satisfies

$$F_{a,\rho} Z_\rho F_{a,\rho}^{-1} = Z_\rho. \quad (\text{ZTM1.7})$$

This follows because (ZTM1.4) is a polynomial in Z_ρ . It retains Z_ρ , which is nonzero whenever $m_\rho > 1$. The following calculation determines every possible operator of the type (ZTM1.6), rather than merely comparing these two formulas.

ZTM2. The complete blockwise intertwiner equation

Take source and target original zero blocks J_ρ, J_σ , of dimensions m, n , and fix any recovered $a > 1$. Put

$$t = \log a, \quad \lambda = a^\sigma, \quad \mu = a^{\rho-1}, \quad \mathcal{D}_a(X) = F_{a,\sigma} X - a^{-1} X F_{a,\rho}. \quad (\text{ZTM2.1})$$

On $\operatorname{Hom}_{\mathbb{C}}(J_\rho, J_\sigma)$, set

$$\mathcal{K}_a(X) = \lambda(e^{tZ_\sigma} - 1)X - \mu X(e^{tZ_\rho} - 1). \quad (\text{ZTM2.2})$$

Left and right multiplication commute. The first nilpotent summand has its n -th power zero, and the second has its m -th power zero. In every summand of their binomial expansion of degree $m + n - 1$, one of those powers occurs. Hence

$$\mathcal{K}_a^{m+n-1} = 0, \quad \mathcal{D}_a = (\lambda - \mu)1 + \mathcal{K}_a. \quad (\text{ZTM2.3})$$

When $\lambda \neq \mu$, the complete inverse is

$$\mathcal{D}_a^{-1}(Y) = \sum_{r=0}^{m+n-2} \frac{(-1)^r}{(\lambda - \mu)^{r+1}} \mathcal{K}_a^r(Y). \quad (\text{ZTM2.4})$$

Multiplying this finite geometric sum by (ZTM2.3) cancels every adjacent term; its last uncanceled term is zero by the nilpotence in (ZTM2.3). Thus (ZTM2.4) is both a left and right inverse and retains every nilpotent coefficient of both original blocks.

For completeness, solve the resonant equation as well; it is needed for the translated modules in ZTM7-ZTM9. If $\lambda = \mu$, multiply the equation on the right by e^{-tZ_ρ} . Let

$$\mathcal{S}(X) = Z_\sigma X - X Z_\rho. \quad (\text{ZTM2.5})$$

Left and right multiplication commute, so $e^{t\mathcal{S}} X = e^{tZ_\sigma} X e^{-tZ_\rho}$. The operator $\mathcal{S}$ is nilpotent, and

$$e^{t\mathcal{S}} - 1 = \mathcal{S} \left(t1 + \frac{t^2}{2!} \mathcal{S} + \frac{t^3}{3!} \mathcal{S}^2 + \dots \right). \quad (\text{ZTM2.6})$$

The factor in parentheses is an invertible finite polynomial because its constant coefficient is $t \neq 0$. Therefore the resonant kernel is exactly $\ker \mathcal{S}$.

Writing the target coordinate as w , every map in this kernel is specified by

$$v(w) = \sum_{j=\max(n-m,0)}^{n-1} c_j w^j, \quad X([z^k]) = [w^k v(w)] \quad (0 \leq k < m). \quad (\text{ZTM2.7})$$

Indeed X commuting with coordinate multiplication is determined by $X(1) = v$, and the defining source relation requires $w^m v = 0$ in $\mathbb{C}[w]/w^n$, exactly the displayed range of coefficients. Conversely this condition makes the formula a well-defined map. The resonant solution space has dimension $\min(m, n)$. These equations give the complete solution for every block, including unequal multiplicities.

For finite direct sums, decompose any linear map into its source-target blocks. Equation (ZTM2.1) preserves each block. Its full kernel is consequently the direct sum of the explicitly parametrized resonant spaces (ZTM2.7), with zero contribution from every nonresonant pair.

ZTM3. Actual spectral separation and the vanishing Hom space

Resonance for a single recovered $a > 1$ means

$$a^\sigma = a^{\rho-1}, \quad \sigma = \rho - 1 + \frac{2\pi i k}{\log a} \quad (k \in \mathbb{Z}). \quad (\text{ZTM3.1})$$

Taking absolute values already forces $\text{Re } \sigma = \text{Re } \rho - 1$. For two actual zeros in S , the left side is positive and the right side is negative. Therefore resonance is excluded without any assumption that their real parts equal $1/2$.

For arbitrary actual zero blocks in S , or their finite direct sums V, W , we have proved

$$\mathcal{D}_a : \text{Hom}_{\mathbb{C}}(V, W) \xrightarrow{\sim} \text{Hom}_{\mathbb{C}}(V, W), \quad \text{Hom}_A(V(1), W) = 0. \quad (\text{ZTM3.2})$$

The second assertion follows already from the equation for one $a > 1$. It includes every possible linear operator, not merely multiplication operators or derivatives. In particular the only $N : V(1) \rightarrow V$ in this category is zero.

The exact spectral interval, stated without claiming a Weil realization, is

$$w_a(a^\rho) := \frac{2 \log |a^\rho|}{\log a} = 2 \operatorname{Re} \rho \in (0, 2), \quad w_a(a^{\rho-1}) \in (-2, 0). \quad (\text{ZTM3.3})$$

These are analytic logarithmic-modulus weights of the original character eigenvalues. Nilpotent terms remain in (ZTM1.4) and in the inverse (ZTM2.4); taking absolute values here is used only to prove disjointness of the two scalar spectra.

If resonance is required for every recovered $a > 1$, it forces the stronger equality $\sigma = \rho - 1$, even outside these two strips. To prove this, write $\sigma - \rho + 1 = i\theta$ after taking absolute values. Then $\theta \log p$ is an integer multiple of 2π for every recovered prime p . If $\theta \neq 0$, two distinct recovered primes give a rational ratio of their logarithms, hence an equality of nonzero positive integer powers of the two primes, contradicting unique factorization in the already recovered arithmetic. Thus $\theta = 0$. No pair of primes was used to establish or calibrate that arithmetic.

ZTM4. An actual lifting obstruction and its explicit vanishing

Work first in the category of finite-dimensional complex vector spaces equipped with a representation of the recovered group A . Its exact sequences are sequences exact on underlying vector spaces with equivariant maps. Let V, W be finite sums of the original blocks. Consider every extension in this category

$$0 \longrightarrow W \xrightarrow{\iota} E \xrightarrow{q} V(1) \longrightarrow 0. \quad (\text{ZTM4.1})$$

Choose an underlying vector-space section only for writing the calculation. In the resulting coordinates $E = W \oplus V$, its action has the full form

$$F_{a,E} = \begin{pmatrix} F_{a,W} & Y_a \\ 0 & a^{-1} F_{a,V} \end{pmatrix}. \quad (\text{ZTM4.2})$$

A section $s(v) = (Xv, v)$ is equivariant for a fixed a exactly when

$$F_{a,W}X + Y_a = a^{-1}XF_{a,V}, \quad \mathcal{D}_a(X) = -Y_a. \quad (\text{ZTM4.3})$$

The inverse in (ZTM2.4), applied blockwise, gives its unique solution

$$X = -\mathcal{D}_a^{-1}(Y_a). \quad (\text{ZTM4.4})$$

Thus the two-term complex of actual coefficient spaces

$$\operatorname{Hom}_{\mathbb{C}}(V, W) \xrightarrow{\mathcal{D}_a} \operatorname{Hom}_{\mathbb{C}}(V, W) \quad (\text{ZTM4.5})$$

is acyclic. In degrees zero and one its kernel records differences of equivariant lifts, and its cokernel records the obstruction class of $-Y_a$ to making this section equivariant. Both vanish by the proved inverse. This is the explicitly defined lifting obstruction computed here.

The same section is equivariant for the entire recovered group. Fix one $a_0 > 1$ and replace the initial section by (ZTM4.4). This conjugates (ZTM4.2) by $\begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix}$ and makes $Y_{a_0} = 0$. All operators of a group representation of A commute. The upper-right block of their commutation with $F_{a_0,E}$ then gives, for every $b \in A$,

$$F_{a_0,W}Y_b = a_0^{-1}Y_bF_{a_0,V}, \quad \text{so} \quad \mathcal{D}_{a_0}(Y_b) = 0. \quad (\text{ZTM4.6})$$

Its kernel is zero, hence every $Y_b = 0$. Two group-equivariant sections differ by a member of the Hom space in (ZTM3.2), so this section is unique and independent of the initially chosen vector-space coordinates. We have calculated

$$\mathrm{Hom}_A(V(1), W) = 0, \quad \mathrm{Ext}_A^1(V(1), W) = 0, \quad (\text{ZTM4.7})$$

where Ext^1 means the equivalence classes of short exact sequences in the stated representation category. No semisimplicity of the original zero-jet Frobenius operators is assumed.

Twisting the whole sequence by (-1) gives the equally explicit statement for

$$0 \longrightarrow W(-1) \longrightarrow E' \longrightarrow V \longrightarrow 0. \quad (\text{ZTM4.8})$$

Every such extension has a unique equivariant section. The source scalar weights are in $(0, 2)$, and the kernel scalar weights are in $(2, 4)$. The original source is therefore the weight-below-two part of E' , and the section is its inverse to the quotient map. This is the exact separation mechanism on these constructed original-zeta modules. It does not assert integer Weil weights or purity of weight one.

ZTM5. The algebraic global extension and its exact limits

There is an actual algebraic direct sum of all original zero algebras

$$J_{\mathrm{alg}} = \bigoplus_{\rho \in Z} J_{\rho}. \quad (\text{ZTM5.1})$$

A vector in this space has finite support, although the complete set of blocks is retained. A linear map between two such direct sums is a block matrix whose source columns have only finitely many nonzero target blocks: each source block is finite-dimensional, so the union of the supports of images of its finitely many basis vectors is finite. Conversely that condition defines a linear map on the direct sums.

The inverse (ZTM2.4) operates on each block without changing which block is zero. It therefore preserves this column-finite property. Each of its block sums is finite, even if the multiplicities over the entire zero set are not uniformly bounded. Consequently $\mathcal{D}_a$ is an isomorphism on the full algebraic Hom space of these direct sums as well. The proof of ZTM4 applies word for word to algebraic representations whose two end terms are these direct sums: an underlying vector-space section exists by a basis choice, and its correction is a valid column-finite map. The unique resulting equivariant section is independent of that temporary choice.

Thus the Hom and extension vanishing extends to the algebraic direct sum of all actual zeros, not merely to a fixed finite truncation. No convergence or uniform norm estimate is used or asserted. In particular this does not identify (ZTM5.1) with a Hilbert completion, a Fréchet quotient, a profinite carrier, or the user's entire clock quotient. The denominators and nilpotent sums in (ZTM2.4) would have to be retained in an analysis of a specified topology; their algebraic existence is not a bounded-operator assertion.

There is a direct comparison with the original holomorphic quotient that uses no interpolation assumption:

$$\mathcal{O}(S)/(\zeta) \longrightarrow \prod_{\rho \in Z} J_{\rho}, \quad [f] \longmapsto ([f]_{\rho})_{\rho}. \quad (\text{ZTM5.2})$$

It is injective. If all germs of f vanish to at least the multiplicity of ζ , then f/ζ , initially meromorphic on S , has removable singularities at every zero and is holomorphic on S . Thus $f \in \zeta \mathcal{O}(S)$. The map commutes with every multiplication by a^s . Its target is a product, whereas (ZTM5.1) is a direct sum; no interchange of them is made. No topological-surjectivity or continuous-extension theorem for (ZTM5.2) is required for the algebraic statement proved above.

ZTM6. Differentiation on the original quotient, including its precise defect

Let $D = d/ds = d/dz$ on the germ ring. For every germ g ,

$$D(\zeta g) = \zeta Dg + \zeta' g, \quad \zeta' = mz^{m-1}u_\rho + z^m u'_\rho. \quad (\text{ZTM6.1})$$

Modulo the original ideal, the second term has the generally nonzero class

$$[\zeta' g] = mu_\rho(0)g(0)[z^{m-1}]. \quad (\text{ZTM6.2})$$

Taking $g = 1$ proves that D does not induce an endomorphism of J_ρ , including when $m = 1$. This is a calculation about a specific proposed quotient map on the original germ ideal; it invokes no retracted addition at τ .

For a first-order holomorphic operator $P = hD + b$, the obstruction is $[h\zeta' g]$. Equation(ZTM6.1) gives

$$P \text{ descends to } J_\rho \iff h\zeta' \in (\zeta) \iff h(\rho) = 0. \quad (\text{ZTM6.3})$$

The last equivalence compares the exact vanishing orders $m - 1$ and m , with $u_\rho(0) \neq 0$. In particular all derivations $J_\rho \rightarrow J_\rho$ have the form hD with $h \in zJ_\rho$: a derivation is determined by its value h on z , and the relation $D(z^m) = mz^{m-1}h = 0$ gives exactly this condition.

The largest quotient of J_ρ into which the original derivative descends is

$$\Omega_{J_\rho/\mathbb{C}}^1 = J_\rho dz/([\zeta'] dz) = \mathcal{O}_\rho/(\zeta, \zeta') dz, \quad d[f] = [f'] dz. \quad (\text{ZTM6.4})$$

Indeed (ZTM6.1) shows that killing $[\zeta']$ is sufficient; the case $g = 1$, and then multiplication by arbitrary g , shows it is necessary for any quotient receiver of this derivative. Locally $(\zeta, \zeta') = (z^{m-1})$, by the displayed full derivative, so its dimension is $m - 1$, including zero for a simple zero. This is a receiving quotient with a proved map, not a claim that the missing derivative is unrelated to the original jet.

All higher-order holomorphic differential operators can also be tested exactly. For $P = \sum_{k=0}^d b_k D^k$, the full Leibniz rule gives

$$P(\zeta g) = \sum_{r=0}^d Q_r D^r g, \quad Q_r = \sum_{k=r}^d \binom{k}{r} b_k \zeta^{(k-r)}. \quad (\text{ZTM6.5})$$

It descends precisely when every $Q_r \in (\zeta)$. Sufficiency is immediate. For necessity use $g = e^{cz}$, divide by that unit, and vary the scalar c ; a polynomial with coefficients in the finite-dimensional quotient that vanishes for every c has every coefficient zero. Thus higher-order cancellations have not been excluded by the first-order test.

For the original derivative on germs one has

$$DF_a = F_a D + (\log a) F_a, \quad F_a DF_a^{-1} = D - \log a. \quad (\text{ZTM6.6})$$

For a descended first-order operator the corresponding exact identity is

$$F_a(hD + b)F_a^{-1} = hD + b - (\log a)M_h. \quad (\text{ZTM6.7})$$

This additive term is not the multiplicative character shift in (ZTM1.6). Every descended operator satisfying that shift is zero by ZTM3, regardless of its differential presentation. A derivative of a chosen Taylor representative is a different defined operator: if $D_{\text{tr}}[z^j] = j[z^{j-1}]$, then

$$[D_{\text{tr}}, Z_\rho] = 1 - mP_{m-1}, \quad (\text{ZTM6.8})$$

where P_{m-1} projects onto the top Taylor coefficient. Direct evaluation on each basis vector proves the formula, including the boundary contribution at $j = m - 1$.

ZTM7. Spectral translation and the full shifted original ideals

For $\varepsilon \in \{1, -1\}$, put $\sigma = \rho - \varepsilon$ and define the translation of germs

$$S_\varepsilon : \mathcal{O}_\rho \xrightarrow{\sim} \mathcal{O}_\sigma, \quad (S_\varepsilon f)(s) = f(s + \varepsilon). \quad (\text{ZTM7.1})$$

Its inverse is translation by $-\varepsilon$. It maps $s - \rho$ at the source to $s - \sigma$ at the target, and maps the original ideal exactly as follows:

$$S_\varepsilon((\zeta(s))) = (\zeta(s + \varepsilon)). \quad (\text{ZTM7.2})$$

Therefore its correct target quotient is

$$J_\rho^{[\varepsilon]} = \mathcal{O}_{\rho-\varepsilon}/(\zeta(s + \varepsilon)), \quad S_\varepsilon : J_\rho \xrightarrow{\sim} J_\rho^{[\varepsilon]}. \quad (\text{ZTM7.3})$$

It retains the full expansion

$$\zeta(\rho - \varepsilon + w + \varepsilon) = w^m u_\rho(w), \quad F_{a,\rho}^{[\varepsilon]} = a^{\rho-\varepsilon} \sum_{j=0}^{m-1} \frac{(\log a)^j}{j!} M_w^j. \quad (\text{ZTM7.4})$$

Direct substitution, with no discarded factor, proves

$$F_a^{[\varepsilon]} S_\varepsilon = a^{-\varepsilon} S_\varepsilon F_a. \quad (\text{ZTM7.5})$$

Thus translation is an equivariant isomorphism from the character twist $J_\rho(\varepsilon)$ to the shifted-original-function quotient, including all multiplicities. For $\varepsilon = 1$ it supplies the precise shifted support required by the Tate character, at $\rho - 1$.

This shifted quotient is compared exactly with the original ζ at that point. On $\text{Re } s > 1$, the full Euler product has an absolutely convergent logarithm $\sum_p \sum_{r \geq 1} p^{-rs}/r$, so it is nonzero. For $-1 < \text{Re } s < 0$, retain the complete source equation

$$\zeta(s) = \frac{C(1-s)}{C(s)} \zeta(1-s), \quad C(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2). \quad (\text{ZTM7.6})$$

Both displayed C factors are finite and nonzero on this open strip, and $\text{Re}(1-s) > 1$; hence $\zeta(s) \neq 0$ there. Since $\rho - 1$ lies in that strip and $\rho + 1$ lies in $\text{Re } s > 1$, the original function is a unit at both translated centers. Thus

$$\mathcal{O}_{\rho-\varepsilon}/(\zeta(s)) = 0, \quad \text{whereas} \quad \dim J_\rho^{[\varepsilon]} = m_\rho. \quad (\text{ZTM7.7})$$

The holomorphic ratio

$$R_\varepsilon(s) = \frac{\zeta(s + \varepsilon)}{\zeta(s)} \quad \text{near } s = \rho - \varepsilon \quad (\text{ZTM7.8})$$

has a zero of order exactly m_ρ . It yields the ideal inclusion $(\zeta(s + \varepsilon)) \subset (\zeta(s)) = \mathcal{O}_{\rho-\varepsilon}$, whose induced quotient map is the zero map. This inclusion and its cone are the exact comparison, rather than a claim of no relation between the two functions.

No completion has replaced ζ . Under translation the full multiplier is

$$C(s + \varepsilon) = \frac{1}{2} (s + \varepsilon)(s + \varepsilon - 1) \pi^{-(s+\varepsilon)/2} \Gamma((s + \varepsilon)/2). \quad (\text{ZTM7.9})$$

Near each center used here it is a holomorphic unit; every factor remains displayed. The full differentiated completed comparison is

$$(C\zeta)^{(d)}(s) = \sum_{j=0}^d \binom{d}{j} C^{(d-j)}(s) \zeta^{(j)}(s), \quad (\text{ZTM7.10})$$

with the same formula at $s + \varepsilon$. In particular $(C\zeta)' = C'\zeta + C\zeta'$; both contributions must be retained when checking an ideal or a differential. The conclusions here use neighborhoods inside the indicated open strips. They do not extend a unit comparison across the endpoints 0, 1, the poles of C at negative even integers, or their translated copies. Their full orders and residues are OZR2.3-OZR2.11.

ZTM8. The actual cones, including the nonvanishing shifted jet

For a holomorphic germ f , define its two-term resolution

$$P(f) = [\mathcal{O} \xrightarrow{M_f} \mathcal{O}], \quad \deg \mathcal{O} = -1, 0. \quad (\text{ZTM8.1})$$

At ρ , multiplication by the nonzero germ ζ is injective because the germ ring is an integral domain. Thus $H^{-1}P(\zeta)_\rho = 0$ and $H^0P(\zeta)_\rho = J_\rho$. The original zero has not disappeared into a resolution.

Translation gives a chain isomorphism $P(\zeta)_\rho \rightarrow P(\zeta(s + \varepsilon))_{\rho-\varepsilon}$ in both degrees. After twisting its source by (ε) , it commutes with every multiplication F_a by (ZTM7.5). The cone of this chain isomorphism is contractible. Explicitly, for an isomorphism $u : P \rightarrow Q$, with cone differential $d(q, p) = (d_Q q + up, -d_P p)$, the map $h(q, p) = (0, u^{-1}q)$ has degree -1 and satisfies $dh + hd = 1$. This acyclicity compares the same retained data transported by an isomorphism.

There is also a chain map to the resolution of the original ζ at the translated center. Work at that center and abbreviate $f = \zeta(s + \varepsilon)$, $g = \zeta(s)$, $R = f/g$. The map

$$P(f) \rightarrow P(g), \quad \phi^{-1} = M_R, \quad \phi^0 = 1 \quad (\text{ZTM8.2})$$

is a chain map because $gR = f$. All its maps commute with multiplication by a^s . The target is contractible, with homotopy $M_{g^{-1}}$, because g is the unit proved in ZTM7. Its cone has the explicit terms and differentials

$$\mathcal{O} \xrightarrow{u \mapsto (Ru, -fu)} \mathcal{O} \oplus \mathcal{O} \xrightarrow{(b, c) \mapsto gb + c} \mathcal{O}, \quad \text{in degrees } -2, -1, 0. \quad (\text{ZTM8.3})$$

The first arrow is injective. The last arrow is surjective. Its kernel consists of $(b, -gb)$; the image of the first consists of $(Ru, -gRu)$. Therefore

$$H^{-1} \text{Cone}(\phi) \simeq \mathcal{O}/(R) \simeq \mathcal{O}/(f) = J_\rho^{[\varepsilon]}, \quad H^j \text{Cone}(\phi) = 0 \ (j \neq -1). \quad (\text{ZTM8.4})$$

The isomorphism is $[(b, -gb)] \mapsto [b]$. Its inverse takes $[b]$ to that cycle. Since g is a unit, $(R) = (f)$ exactly. This retains both the full original g and the shifted f in the chain maps.

The surviving character is $a^{\rho-\varepsilon}$ with all nilpotent terms in (ZTM7.4). Thus the construction supplies an actual cohomological shift and a character-twisted jet, but it does not make that jet zero. The vanishing cone of a translation isomorphism and the nonzero cone (ZTM8.4) have different, fully specified maps.

Differentiation supplies no omitted holomorphic chain map here. The commutator is $DM_\zeta - M_\zeta D = M_{\zeta'}$. To lift degree-zero D to an endomorphism of $P(\zeta)$, the degree-minus-one operator would have to be

$$D + \frac{\zeta'}{\zeta} = D + \frac{m}{z} + \frac{u'_\rho}{u_\rho}. \quad (\text{ZTM8.5})$$

The pole term m/z is nonzero, so this is not a holomorphic differential operator on the original resolution. It gives the exact meromorphic comparison instead.

ZTM9. A constructed nilpotent extension with the required character law

The shift relation does have nonzero realizations once the correctly shifted data are retained. Define, from an actual original block, the character module

$$\mathcal{V}_\rho = J_\rho \oplus J_\rho(1), \quad \mathcal{F}_a = \begin{pmatrix} F_{a,\rho} & 0 \\ 0 & a^{-1}F_{a,\rho} \end{pmatrix}, \quad \mathcal{N}(x, y) = (0, x). \quad (\text{ZTM9.1})$$

Then $\mathcal{N}^2 = 0$, $\mathcal{N} \neq 0$, and direct multiplication gives

$$\mathcal{F}_a \mathcal{N} \mathcal{F}_a^{-1} = a^{-1} \mathcal{N}. \quad (\text{ZTM9.2})$$

The underlying identity between the two copies is realized by the exact translation map to the shifted quotient of ZTM7. Both summands retain the entire original nilpotent exponential. This is a functorial construction from the original module and the specified character line, not a newly proposed zero of the original function.

The corresponding equivariant two-term complex is

$$\mathcal{V}_\rho \xrightarrow{\mathcal{N}} \mathcal{V}_\rho(-1) \quad \text{in degrees } 0, 1. \quad (\text{ZTM9.3})$$

Indeed multiplying (ZTM9.2) by a says exactly that the target action $a\mathcal{F}_a$ commutes with this map. Its kernel and cokernel are

$$H^0 = J_\rho(1), \quad H^1 = J_\rho(-1). \quad (\text{ZTM9.4})$$

The first is the second summand of the source and the second is the first summand of the target. Their weights in the sense of (ZTM3.3) are $2 \operatorname{Re} \rho - 2$ and $2 \operatorname{Re} \rho + 2$, respectively. This computes what a genuine shifted nilpotent complex retains. It is not the localization cross of DC5, and its construction has not supplied a proper smooth degeneration or an arithmetic sheaf realizing the original zeta module.

ZTM10. Naturality, branch comparisons, and the exact Deligne receiving scope

The lifting formula is natural in the original coefficient modules. If $u : V \rightarrow V'$ and $v : W \rightarrow W'$ intertwine the recovered arithmetic actions and are isomorphisms, then for every X ,

$$\mathcal{D}'_a(vXu^{-1}) = v\mathcal{D}_a(X)u^{-1}, \quad (\mathcal{D}'_a)^{-1}(vYu^{-1}) = v\mathcal{D}_a^{-1}(Y)u^{-1}. \quad (\text{ZTM10.1})$$

The first equality follows by distributing products; the second follows by the proved uniqueness of the inverse. Thus a complete branch comparison preserving the original function and each branch's recovered arithmetic action transports the unique splitting. No sum of the branches' return measures occurs in this formula.

DC5 constructs a different exact cross with specified geometric maps $A_i \xrightarrow{\alpha_i} B_i \xrightarrow{\partial_i} O_i$, a row $0 \rightarrow K_i \rightarrow B_i \rightarrow C_i \rightarrow 0$, and obstruction

$$\operatorname{coker} \operatorname{sp}_i \simeq \operatorname{im} \partial_i / \partial_i K_i. \quad (\text{ZTM10.2})$$

ZTM4 instead constructs the exact coefficient-category extension (ZTM4.1), its section equation (ZTM4.3), and the acyclic obstruction complex (ZTM4.5). Those are all actual maps on actual original-zero modules. Their weight separation is proved by the open critical strip, with the full nilpotent inverse, rather than obtained by declaring the original zeros pure of weight one.

The exact connecting construction available here is the character twist realized by original-function translation (ZTM7.3)-(ZTM7.5), the cone computation (ZTM8.2)-(ZTM8.4), and the nilpotent complex (ZTM9.1)-(ZTM9.4). They specify the shifted support, degree and surviving data. They do not identify the nonzero cone with a vanishing geometric boundary. No map in this note identifies (ZTM10.2) with the cokernel of $\mathcal{D}_a$, and the source-correction audit does not authorize silently replacing the user's global quotient by this representation extension.

The proved conclusion is exact: all character-module extensions (ZTM4.1) and (ZTM4.8), on finite original zero blocks or their algebraic direct sums, have unique compatible lifts, with the full inverse (ZTM2.4). Their defined lifting obstruction vanishes. The spectral-translation cone retains an explicitly computed shifted original jet. These statements preserve the full original function and arithmetic, without asserting that any actual zero is off the critical line or proving that all have weight one.

ZTM11. Independent checks and bounded work record

An independent mathematical checker derived the derivative receiver, the full differential-operator ideal test, the two distinct translation cones, and the character-extension calculation. In particular the checker verified the sign in $\mathcal{D}_a X = -Y_a$, the finite inverse bound $m + n - 2$, simultaneous splitting for all recovered characters, and the convention $F_{a,V(k)} = a^{-k} F_{a,V}$. These checks are of the explicit calculations above; they are not a review receipt for a geometric realization that has not been constructed.

After the text was written, the checker also read and verified ZTM5, ZTM8 and ZTM9 as written: the inverse preserves column-finite support without a uniform multiplicity bound, the cone signs give the stated surviving cohomology, and the doubled nilpotent module has exactly the displayed twist and cohomology. No mathematical correction was needed in that final check.

This file is the only task file written for this bounded derivation. The full user-input archive remains in the parent task's verbatim logbook. The source-use coverage, original factors, operation prerequisites, and finite/algebraic-global/topological distinctions are recorded in this file so the derivation can be resumed without silently changing its scope. No publication, source editing, or change to a different proof file was performed.

The whole original Mellin source and its lifting boundary — GLB0-GLB8

Proof source: GLOBAL_ORIGINAL_SOURCE_LIFTING.md.

The exact lifting boundary in the whole original arithmetic source

Root derivation, 24 September 2026. Proof locators GLB0–GLB8. This calculation keeps the actual Fréchet source, all original nontrivial zeros and their full multiplicities. It constructs and evaluates one specific boundary; it does not claim that this boundary is the geometric lifting obstruction in the user’s proposal.

GLB0. Source and prerequisite check

The supporting datum remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. Arithmetic and analytic operations here occur after the complete arithmetic has been reconstructed. No operation in this note adds, subtracts, coordinatizes or measures that supporting datum. The user’s complete correction chain and the prerequisite check for this calculation are in the private PC01 record. A coefficient zero below is not the supporting point or the user’s Z_0 .

The precise receiving sources are S1–S7 of GLOBAL_MELLIN_SYNTHESIS.md, reread in full for this calculation, and RZ1–RZ2 of ORIGINAL_ZETA_RESOLVENT_AND_PROJECTORS.md. S5 proves equality of the zero-jet kernel with the closure of the actual original arithmetic summation image. It does not assert that this ideal consists of all multiples of one function inside the Fréchet space. That stronger statement is neither needed nor used.

The relevant Connes–Consani construction is the unordered generic overlap and its full signed stalk, original CC.tex lines 499–600 and 773–847, reconstructed in CG1–CG3. Its support comparison preserves $\tau\langle Z_1; \text{no } Z_2 \rangle$, the stalk torsion and both chart maps. Deligne’s relevant construction is the specialization/localization cross of *La conjecture de Weil. II*, §3.6, reconstructed in DC1–DC9. Its source obstruction is $\text{im } \partial/\partial K$. The exact difference between that cross and the sequence below is calculated in GLB8; a shared use of the word “lift” does not identify the maps.

GLB1. The complete source and its operations

Let

$$\mathcal{A} = \{k \in C^\infty(\mathbb{R}_{>0}) : p_{N,j}(k) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)| < \infty \text{ for all integers } N, j \geq 0\},$$

(GLB1.1)

and

$$\mathcal{B} = \{F \text{ entire} : b_{A,M}(F) = \sup_{|\Re s| \leq A} (1 + |\Im s|)^M |F(s)| < \infty \text{ for all integers } A, M \geq 0\}. \quad (\text{GLB1.2})$$

S1 proves the topological isomorphism

$$\mathcal{M}k(s) = \int_0^\infty k(u)u^{s-1/2}\frac{du}{u}, \quad k(e^x) = \frac{1}{2\pi} \int_{\mathbb{R}} F(1/2 + it)e^{-itx} dt. \quad (\text{GLB1.3})$$

Let $\mathcal{Z}$ be the complete distinct nontrivial zero set of the original ζ , with actual multiplicities m_ρ . Put

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \ (\rho \in \mathcal{Z}, 0 \leq j < m_\rho)\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}. \quad (\text{GLB1.4})$$

Every jet is continuous by Cauchy's formula, hence $\mathcal{J}$ is closed. The topology on $\mathcal{Q}$ is the quotient Fréchet topology. S5 identifies its Mellin preimage ideal exactly with $\overline{\mathcal{E}(\mathcal{S}_0^{\text{even}})}^{\mathcal{A}}$, where $\mathcal{E}f(u) = u^{1/2} \sum_{n \geq 1} f(nu)$ and $\mathcal{S}_0^{\text{even}} = \{f \text{ even Schwartz} : f(0) = 0, \int_{\mathbb{R}} f = 0\}$. In particular this is the whole original source closure, not a finite zero truncation.

Retain the actual member of this image

$$f_0(v) = \frac{\pi}{2}v^2(2\pi v^2 - 3)e^{-\pi v^2}, \quad k_0(u) = u^{1/2} \sum_{n \geq 1} f_0(nu), \quad F_0(s) = \mathcal{M}k_0(s) = c(s)\zeta(s), \quad (\text{GLB1.5})$$

where the complete multiplier is

$$c(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2). \quad (\text{GLB1.6})$$

The product has its entire continuation; its zeros are exactly (ρ, m_ρ) . Its endpoint values are $F_0(0) = F_0(1) = 1/8$, and at each trivial zero,

$$F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0 \quad (r \geq 1). \quad (\text{GLB1.7})$$

These values retain the pole, Gamma-pole and trivial-zero cancellations. We use F_0 as a member of the original summation image with the displayed exact comparison to ζ , never as a replacement arithmetic function.

Define $\mathbb{L}F = sF$ and, for a recovered $a > 0$, $\mathbb{T}_a F = a^s F$, using the real logarithm. They act continuously on $\mathcal{B}$, because

$$b_{A,M}(\mathbb{L}F) \leq (A+1)b_{A,M+1}(F), \quad b_{A,M}(\mathbb{T}_a F) \leq \max(a^A, a^{-A})b_{A,M}(F). \quad (\text{GLB1.8})$$

They preserve every required zero order and therefore act on $\mathcal{J}$ and $\mathcal{Q}$; write L, T_a for the quotient actions. Direct multiplication gives $\mathbb{T}_a \mathbb{T}_b = \mathbb{T}_{ab}$ and the inverse $\mathbb{T}_{a^{-1}}$. Their source actions are

$$\mathbb{D} = \frac{1}{2} - u\partial_u, \quad (\mathbb{W}_a k)(u) = a^{1/2} k(u/a). \quad (\text{GLB1.9})$$

Integration by parts in (GLB1.3), with both endpoints zero by (GLB1.1), gives $\mathcal{M}\mathbb{D} = \mathbb{L}\mathcal{M}$. Substitution $u = av$ gives $\mathcal{M}\mathbb{W}_a = \mathbb{T}_a \mathcal{M}$, including the factor $a^{1/2}$. Thus none of these operations is placed on τ .

GLB2. Division on its actual domain

Fix $\rho \in \mathcal{Z}$, put $z = s - \rho$, and let $r \geq 1$ be an integer. For a member $F \in \mathcal{B}$ vanishing to order at least r at ρ , Taylor division gives the entire function F/z^r . It still belongs to $\mathcal{B}$. In fact outside $|z| < 1$ the denominator has modulus at least one. Inside $|z| \leq 1$, the maximum principle for the holomorphic quotient on $|z| \leq 2$ gives

$$b_{A,M}(F/z^r) \leq b_{A,M}(F) + 2^{-r}(2 + |\Im \rho|)^M b_{A',0}(F), \quad A' \geq \max(A, |\Re \rho| + 2). \quad (\text{GLB2.1})$$

This also proves continuity on the closed subspace with those vanishing conditions.

Consequently

$$z^r \mathcal{J} = \{F \in \mathcal{J} : \text{ord}_\rho F \geq m_\rho + r\}. \quad (\text{GLB2.2})$$

The forward inclusion follows by multiplication. For the reverse, divide by z^r using (GLB2.1). At ρ the quotient still has order at least m_ρ , and at every other actual zero the divisor is a unit, so all original multiplicities remain. This proves (GLB2.2), and proves that $z^r \mathcal{J}$ is closed without assuming $\mathcal{J} = F_0 \mathcal{B}$.

GLB3. The two finite quotient receivers of the whole source

Write $m = m_\rho$ and retain the local original unit

$$\zeta(\rho + z) = z^m u_\rho(z), \quad u_\rho(0) = \frac{\zeta^{(m)}(\rho)}{m!}, \quad v_\rho(z) = c(\rho + z)u_\rho(z), \quad F_0(\rho + z) = z^m v_\rho(z). \quad (\text{GLB3.1})$$

Both u_ρ and v_ρ are holomorphic units near this nontrivial zero. Every derivative of the full c remains in v_ρ .

Set $R_r = \mathbb{C}[z]/(z^r)$, a coefficient algebra at this analytic zero. The maps

$$\mathcal{J}/z^r \mathcal{J} \xrightarrow{\Phi_r} R_r, \quad [F] \mapsto [F/F_0]_{\rho,r}, \quad \mathcal{B}/z^r \mathcal{B} \xrightarrow{\Psi_r} R_r, \quad [F] \mapsto [F]_{\rho,r} \quad (\text{GLB3.2})$$

are topological linear isomorphisms. The first uses division only as a germ at ρ : all $F \in \mathcal{J}$ have sufficient order there. Its kernel is (GLB2.2); its inverse sends a polynomial class $[P]$ of degree less than r to $[F_0 P]$. Changing P by $z^r Q$ changes the representative by an element of $z^r \mathcal{J}$. The maps are continuous because they depend on finitely many derivatives, division by the fixed nonzero number $v_\rho(0)$, and multiplication by F_0 .

For the second map the kernel is $z^r \mathcal{B}$ by (GLB2.1). It is onto: set $G = F_0/z^m \in \mathcal{B}$, so $G(\rho) = v_\rho(0) \neq 0$. For a desired polynomial jet P , let H be the Taylor polynomial through degree $r - 1$ of $P(z)/G(\rho + z)$; then $GH \in \mathcal{B}$ has the desired jet. This gives a continuous inverse in the quotient. In particular, using polynomial jets here does not assert that the constant function one belongs to $\mathcal{B}$.

In these coordinates the inclusion-induced map between the two quotients is exactly

$$R_r \longrightarrow R_r, \quad [P] \longmapsto [z^m v_\rho(z)P] \pmod{z^r}. \quad (\text{GLB3.3})$$

The full unit has not been made monic or omitted. Its kernel is all of R_r for $r \leq m$, and $z^{r-m} R_r$ for $r > m$. Its cokernel is $\mathbb{C}[z]/(z^{\min(r,m)})$.

GLB4. The connecting sequence and its full multiplicity

Apply the continuous differential $L - \rho$, raised to the power r , to

$$0 \longrightarrow \mathcal{J} \longrightarrow \mathcal{B} \longrightarrow \mathcal{Q} \longrightarrow 0. \quad (\text{GLB4.1})$$

It gives a short exact sequence of two-term cochain complexes, in degrees zero and one. Multiplication by z^r is injective on $\mathcal{B}$ by the identity theorem, hence also on $\mathcal{J}$. The resulting exact sequence is

$$0 \longrightarrow \ker(L - \rho)^r \xrightarrow{\delta_{\rho,r}} \mathcal{J}/z^r \mathcal{J} \longrightarrow \mathcal{B}/z^r \mathcal{B} \longrightarrow \mathcal{Q}/(L - \rho)^r \mathcal{Q} \longrightarrow 0, \quad (\text{GLB4.2})$$

where

$$\delta_{\rho,r}[F] = [z^r F]. \quad (\text{GLB4.3})$$

Here $z^r F \in \mathcal{J}$ is exactly the kernel condition in $\mathcal{Q}$; changing F by I changes its image by $z^r I$. Exactness can be checked without any spectral theorem: a boundary vanishes exactly when subtracting an element of $\mathcal{J}$ makes $z^r F = 0$; a class of I dies in the middle quotient exactly when it equals $z^r F$; and a member of $\mathcal{B}$ dies in the last quotient exactly when it lies in $I + z^r \mathcal{B}$.

For the complete basis of the first term, define

$$V_{\rho,j}(s) = \frac{F_0(s)}{(s - \rho)^j}, \quad 1 \leq j \leq m. \quad (\text{GLB4.4})$$

Division is valid by (GLB2.1). Their classes are supported only at the full ρ -jet, because every other actual zero retains its original multiplicity. Their orders at ρ are $m - j$, so the classes are independent. For $j \leq \min(r, m)$,

$$\Phi_r \delta_{\rho,r} [V_{\rho,j}] = [z^{r-j}]. \quad (\text{GLB4.5})$$

These monomials form exactly the kernel computed in (GLB3.3). Exactness then proves they span all of $\ker(L - \rho)^r$, not merely a chosen finite subspace. We have derived

$$\dim \ker(L - \rho)^r = \min(r, m), \quad \mathcal{Q}/(L - \rho)^r \mathcal{Q} \cong \mathbb{C}[z]/(z^{\min(r, m)}). \quad (\text{GLB4.6})$$

The last map is the Taylor jet at ρ through degree $\min(r, m) - 1$. Its kernel is the actual image of $(L - \rho)^r$; to see this directly, subtract a representative with the same finitely many jets as needed, then apply (GLB2.1). Equivalently (GLB3.3) identifies the quotient. The kernel is closed because those finite jet functionals on $\mathcal{Q}$ are continuous. All displayed finite-dimensional quotient identifications therefore also hold with their quotient topologies.

GLB5. The exact one-dimensional boundary and its arithmetic action

For $r = 1$, the entire sequence is

$$0 \longrightarrow \mathbb{C}[V_{\rho,1}] \xrightarrow{\delta_{\rho,1}} \mathbb{C}[F_0] \xrightarrow{0} \mathbb{C} \xrightarrow{\text{id}} \mathbb{C} \longrightarrow 0, \quad \delta_{\rho,1}[V_{\rho,1}] = [F_0] \neq 0. \quad (\text{GLB5.1})$$

The brackets $[F_0]$ on this line mean its nonzero class in $\mathcal{J}/z\mathcal{J}$, not its zero class in $\mathcal{Q}$. Nonvanishing also follows directly: if $F_0 = zH$ with $H \in \mathcal{J}$, its vanishing order at ρ would be at least $m + 1$, contradicting (GLB3.1).

Each T_a acts by a^ρ on both connecting lines. Indeed

$$(a^s - a^\rho)V_{\rho,1}(s) = F_0(s) \frac{a^s - a^\rho}{s - \rho} \in \mathcal{J}. \quad (\text{GLB5.2})$$

The right side belongs to $\mathcal{B}$ by continuous multiplication on $V_{\rho,1}$, and has the required orders at every zero. At ρ the quotient factor is the holomorphic value $a^\rho \log a$. Likewise $(a^s - a^\rho)F_0 \in z\mathcal{J}$, using (GLB2.2). Consequently $\delta_{\rho,1}$ is equivariant for every recovered multiplicative character action and connects equal characters. No weight separation is present in this particular sequence.

The next full source object is

$$\mathcal{E}_\rho = \mathcal{B}/z\mathcal{J}, \quad 0 \longrightarrow \mathcal{J}/z\mathcal{J} \longrightarrow \mathcal{E}_\rho \longrightarrow \mathcal{Q} \longrightarrow 0. \quad (\text{GLB5.3})$$

All are actual Hausdorff Fréchet spaces by (GLB2.2). Multiplication and the operators preserve the displayed ideal. The kernel is square-zero: for $F, G \in \mathcal{J}$, the product divided by z is in $\mathcal{J}$, since it has order at least $2m - 1 \geq m$ at ρ , the required orders at every other zero, and belongs to $\mathcal{B}$ by (GLB2.1). The module structure on this kernel is evaluation at ρ , because $(H - H(\rho))F \in z\mathcal{J}$ for every $H \in \mathcal{B}$, with the scalar expression interpreted in the ambient entire functions. This is a constructed global receiver retaining one further original jet and all other zero data.

GLB6. Localization to the original germ sequence, including its multiplier

Restriction from $\mathcal{B}$ to $\mathcal{O}_\rho$ and from $\mathcal{I}$ to $(\zeta)\mathcal{O}_\rho$ defines a commuting map from (GLB4.1) to the original germ sequence. The ideal inclusion is valid since each source function has order at least m and u_ρ is a unit. Under this map

$$[V_{\rho,1}] \mapsto \left[c(s) \frac{\zeta(s)}{s-\rho} \right] = c(\rho) \left[\frac{\zeta(s)}{s-\rho} \right], \quad [F_0] \mapsto c(\rho)[\zeta] \quad (\text{GLB6.1})$$

on the respective one-dimensional quotient lines. The scalar evaluation is justified because the discarded difference is divisible by the indicated ideal; it is not a Taylor approximation. For $r > 1$, all derivatives through order $r - 1$ of the full c remain. Explicitly the coefficient of z^k in the product with a local ζ -quotient jet $h(z)$ is

$$\sum_{j=0}^k \frac{c^{(k-j)}(\rho)}{(k-j)!} \frac{h^{(j)}(0)}{j!}. \quad (\text{GLB6.2})$$

Thus the global boundary is identified exactly with the original germ boundary for differential $s - \rho$. ZLB3 then compares the latter with differential $a^s - a^\rho$, multiplying the connecting line by the full simple-zero quotient with value $a^\rho \log a$. No completion factor is silently suppressed.

GLB7. The full source functions solving the inhomogeneous equation

Let $k_{\rho,0} = k_0$, and for $1 \leq j \leq m$ put

$$k_{\rho,j} = \mathcal{M}^{-1} V_{\rho,j} \in \mathcal{A}. \quad (\text{GLB7.1})$$

Existence and membership follow from (GLB2.1) and the topological isomorphism (GLB1.3). Their exact equations are

$$(\mathcal{D} - \rho)k_{\rho,j} = k_{\rho,j-1}. \quad (\text{GLB7.2})$$

The first source function is explicitly

$$k_{\rho,1}(u) = -u^{1/2-\rho} \int_0^u v^{\rho-1/2} k_0(v) \frac{dv}{v} = u^{1/2-\rho} \int_u^\infty v^{\rho-1/2} k_0(v) \frac{dv}{v}. \quad (\text{GLB7.3})$$

Both integrals converge by the endpoint decay in $\mathcal{A}$, and they agree because their full integral is $F_0(\rho) = 0$. Differentiating the first proves (GLB7.2), including its sign and the coefficient $1/2$. Choosing the first integral near zero and the second near infinity proves arbitrarily rapid endpoint decay: bound $k_0(v)$ by an arbitrary power v^N or v^{-N} and integrate the resulting power, choosing $N > |\Re \rho - 1/2|$. Repeated logarithmic derivatives follow from the differential equation and the corresponding bounds for k_0 . This independently proves membership in $\mathcal{A}$.

The same formula, replacing k_0 by $k_{\rho,j-1}$, holds for each $1 \leq j \leq m$. Its total integral is $V_{\rho,j-1}(\rho) = 0$ because the remaining vanishing order is $m - j + 1 \geq 1$. Thus every member of the entire original Jordan chain has an exact convergent source integral. A homogeneous solution is a scalar multiple of $u^{1/2-\rho}$, which belongs to $\mathcal{A}$ only when that scalar is zero: at least one endpoint violates arbitrary-power decay. Hence the displayed source solution is unique in $\mathcal{A}$.

Modulo the closed arithmetic source image, the first equation says that $[k_{\rho,1}]$ is an eigenvector. In the actual source it has the nonzero right side k_0 . This precisely identifies the lifting boundary instead of claiming that a source preimage is an eigenfunction.

GLB8. Comparison to the weight-separation target

Deligne's geometric obstruction in DC5 is the quotient $\mathrm{im} \partial/\partial K$ in an exact specialization/localization cross. Its vanishing uses source weights at most i and target weights at least $i + 1$, constructed from properness, local monodromy and duality, including their Tate factors.

The exact global source sequence here instead has the nonzero boundary (GLB5.1), with equal arithmetic character a^ρ on both lines. It remains nonzero for any actual zero on the critical line as well. The calculation therefore cannot serve as a proposed contradiction to RH or as an obstruction to the user's construction merely because that construction also seeks a lift. No map in the cited construction or the CC support comparison identifies the requested geometric lift with an exact analytic eigenfunction lift through (GLB4.1).

The connecting construction actually obtained is (GLB5.3), retaining its square-zero extra jet and its continuous maps, together with the original-function germ comparison (GLB6.1) and the source integral chain (GLB7.3). These preserve all the arithmetic data. The separated coefficient extension in ZTM4 has a different, explicitly character-shifted kernel; its unique splitting is proved there. Identifying such a shift with the programme's geometric lifting obstruction requires deriving that shift from the actual source maps. It cannot be supplied by changing the action in (GLB4.1).

Continuous arithmetic-equivariant lifting on the entire source — GSL0-GSL11

Proof source: GLOBAL_SHIFTED_ZETA_LIFT.md.

A continuous lift for the original and shifted zeta quotients

Independent mathematical derivation, 24 September 2026. Stable proof locators GSL0-GSL11.

The result is a continuous, arithmetic-equivariant isomorphism on the actual Fréchet spaces:

$$\mathcal{B}/(\mathcal{I} \cap \mathcal{I}_-) \xrightarrow{\sim} (\mathcal{B}/\mathcal{I}) \oplus (\mathcal{B}/\mathcal{I}_-). \quad (\text{GSL0.1})$$

Its inverse is explicitly constructed below. The proof retains the original zeta multiplier, all zero multiplicities, and the original strip seminorms. It makes no identification of either quotient with an unrestricted jet product or an algebraic direct sum.

GSL0. Input stage, source reading, and exact scope

The calculation takes place after the complete arithmetic and the original Riemann zeta function have been recovered. The support remains $\tau\langle Z_1; \text{no } Z_2 \rangle$; no arithmetic, coordinate, norm or addition is assigned to that support. All sums and integrals below occur in the existing coefficient function spaces. No selected prime determines the earlier arithmetic, and separate branches are not pooled.

The prerequisite correction audit is recorded in ZTM0 of [ZETA_TATE_MONODROMY_DERIVATION.md](#). It includes [READ_FIRST_USER_CONSTRUCTION.md](#), the full detailed reconstruction, [USER_ARGUMENT_CONNECTIONS.md](#), and the pertinent verbatim passages U123-U138, WU042-WU055 and the complete ad55fb5f attachment in [USER_CONSTRUCTION_FULL_LOGBOOK.md](#). The present operations use the same corrected input stage. In particular this is a calculation on a specified extension of the actual original-zeta coefficient quotient, not a replacement definition of the user's whole global quotient.

The actual mathematical sources read for this derivation are:

- [GLOBAL_MELLIN_SYNTHESIS.md](#), S1-S7, with particular attention to the full proof of S5 and the precise range statement S7. SHA256 334856FB8C568A7D1FD7CFC0326CD6E30298639E3A107201C9A196E2019E8730.
- [ORIGINAL_ZETA_RESOLVENT_AND_PROJECTORS.md](#), RZ1-RZ12, including all finite-source representatives, estimates, and its actual quotient topology. SHA256 F3553E1362B4C2C1C1556098B55EE153BE9FC716A9D4C35BAEDCF39BD28AD755.
- The independent S3-S5 review, read for its division and topology checks, and DB9 of [DELIGNE_BOUNDARY_NONVANISHING_RECONSTRUCTION.md](#), including its complete original-prime-power measure and coefficients 3, 4, 2.
- The original author source Alain Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, [arXiv:2602.04022v1](#), `rhready.tex`, full multiplier and functional equation at local lines 526-546, with author-source identity recorded in OZR0 and ZTM0. No PDF is used. Its canonical IDs are PUBUNIT-171DAC50D1C3B7A2438B39AE and PUBUNIT-90A9C2A0907A753437719FD9.

The complete source-image equality S5 is an input to identifying the quotient used here with the original Mellin source. It is not newly attributed to this note. The global splitting and its continuous inverse are

proved here, rather than deduced from the algebraic splitting in ZTM4-ZTM5. The estimates below use the original Euler product, elementary continued integrals, and the exact Gamma reflection identity; no zero spacing, zero simplicity, RH, or complex-to- ℓ -adic coefficient map is assumed.

GSL1. The Fréchet spaces, the full original multiplier, and translation

Let $\mathcal{B}$ consist of entire functions for which

$$b_{A,M}(F) = \sup_{|\operatorname{Re} s| \leq A} (1 + |\operatorname{Im} s|)^M |F(s)| < \infty \quad (A, M \in \mathbb{Z}_{\geq 0}). \quad (\text{GSL1.1})$$

These seminorms give the complete Fréchet topology proved in S1 and RZ2. Let $\mathcal{Z}$ be the actual distinct nontrivial zeros of the original ζ , with full multiplicities m_ρ . They lie in $0 < \operatorname{Re} \rho < 1$; the boundary nonvanishing used here is the already proved DB9 statement. Set

$$\begin{aligned} \mathcal{J} &= \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for all } \rho \in \mathcal{Z}, 0 \leq j < m_\rho\}, \\ \mathcal{J}_- &= \{F \in \mathcal{B} : F^{(j)}(\rho - 1) = 0 \text{ for all } \rho \in \mathcal{Z}, 0 \leq j < m_\rho\}, \\ \mathcal{J} &= \mathcal{J} \cap \mathcal{J}_-, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}, \quad \mathcal{Q}_- = \mathcal{B}/\mathcal{J}_-, \quad \mathcal{Q}_\cup = \mathcal{B}/\mathcal{J}. \end{aligned} \quad (\text{GSL1.2})$$

Every jet evaluation is continuous by Cauchy's formula, so these ideals are closed. The three quotients carry their actual quotient Fréchet topologies.

The retained source element is

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{GSL1.3})$$

It is entire and belongs to $\mathcal{B}$, by the original Mellin calculation S2. Its zeros are exactly $\mathcal{Z}$, with their original multiplicities. The full exceptional values are

$$\begin{aligned} F_0(0) &= F_0(1) = \frac{1}{8}, \\ F_0(-2r) &= \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) \\ &= F_0(1+2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0 \quad (r \geq 1). \end{aligned} \quad (\text{GSL1.4})$$

These account for the original endpoint factors, Gamma poles and trivial zeros; none is declared absent from the original function.

Translation defines a topological automorphism of $\mathcal{B}$:

$$(UF)(s) = F(s+1), \quad (U^{-1}F)(s) = F(s-1), \quad b_{A,M}(U^{\pm 1}F) \leq b_{A+1,M}(F). \quad (\text{GSL1.6})$$

It satisfies $U\mathcal{J} = \mathcal{J}_-$. Consequently it induces a topological isomorphism $\bar{U} : \mathcal{Q} \rightarrow \mathcal{Q}_-$, with the same inverse translation.

In particular the shifted full source multiplier is

$$F_1(s) = F_0(s+1) = \frac{(s+1)s}{8} \pi^{-(s+1)/2} \Gamma((s+1)/2) \zeta(s+1). \quad (\text{GSL1.7})$$

Its endpoint values are $F_1(-1) = F_1(0) = 1/8$; its values at $-2r - 1$ and $2r$ are precisely the two original expressions in (GSL1.5). Its zero divisor is $(\rho - 1, m_\rho)$. The original and shifted divisors are disjoint because their real parts lie respectively in $(0, 1)$ and $(-1, 0)$.

We retain throughout the proof the entire function

$$G(s) = F_0(s)F_1(s), \quad (\text{GSL1.8})$$

with both full factors (GSL1.3), (GSL1.7). It belongs to $\mathcal{B}$, since multiplication by a fixed member of $\mathcal{B}$ is continuous there. Its zero divisor is exactly the union with all multiplicities, and $G(0) = 1/64$.

GSL2. Polynomial bounds from the original zeta integral

The original identity

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx \quad (\text{Re } s > 0) \quad (\text{GSL2.1})$$

has the simple pole at one retained. It follows from the Dirichlet series by integration of the counting function and then by locally uniform continuation. On $3/4 \leq \text{Re } s \leq 2$, $|\text{Im } s| \geq 1$, it and its derivative give constants C_0, C_1 with

$$|\zeta(s)| \leq C_0(1 + |\text{Im } s|), \quad |\zeta'(s)| \leq C_1(1 + |\text{Im } s|). \quad (\text{GSL2.2})$$

For the derivative, the complete formula is

$$\zeta'(s) = -\frac{1}{(s-1)^2} - \int_1^\infty \{x\} x^{-s-1} dx + s \int_1^\infty \{x\} \log x x^{-s-1} dx. \quad (\text{GSL2.3})$$

The two integrals are bounded by $1/\text{Re } s$ and $1/(\text{Re } s)^2$, respectively. This proves (GSL2.2) with every derivative term retained.

We also need polynomial growth on any fixed vertical strip. It can be proved without importing a bound in the critical strip. Define $P_1(x) = \{x\} - 1/2$ away from integers, and recursively let P_{j+1} be its periodic antiderivative of mean zero. Each P_j is bounded, P_j is continuous for $j \geq 2$, and $P'_{j+1} = P_j$ almost everywhere. To construct it, integrate P_j over a period and subtract the mean of the resulting primitive; the zero mean of P_j makes the primitive periodic.

Repeated integration by parts in (GSL2.1) gives, for every integer $K \geq 2$,

$$\zeta(s) = \frac{1}{s-1} + \frac{1}{2} + \sum_{j=2}^K (s)_{j-1} P_j(1) - (s)_K \int_1^\infty P_K(x) x^{-s-K} dx, \quad \text{Re } s > 1 - K. \quad (\text{GSL2.4})$$

Here $(s)_j = s(s+1) \cdots (s+j-1)$. For the first step, $\int_1^\infty P_1(x) x^{-s-1} dx = -P_2(1) + (s+1) \int_1^\infty P_2(x) x^{-s-2} dx$. Each following integration gives the next displayed boundary constant with its positive sign and the final negative remainder. All boundary terms at infinity vanish initially where the integrals converge absolutely; the successively enlarged half-planes follow by the new convergent remainder and uniqueness of continuation. The constants, pole and $1/2$ are retained even when some periodic boundary constants are zero.

For $|\text{Re } s| \leq A$, choose $K > A + 2$. The last integral is bounded by $\|P_K\|_\infty / (K - 1 - A)$. For $|\text{Im } s| \geq 1$ the pole term is bounded as well, so

$$|\zeta(s)| \leq C_A(1 + |\text{Im } s|)^K \quad (|\text{Re } s| \leq A, |\text{Im } s| \geq 1). \quad (\text{GSL2.5})$$

This estimate is used only as an inequality for the original function; no new working zeta function has been defined.

GSL3. A quantitative zero-free corridor around the separating line

For $\delta > 0$, the absolutely convergent original Euler product gives

$$\begin{aligned} & 3 \log \zeta(1 + \delta) + 4 \log |\zeta(1 + \delta + it)| + 2 \log |\zeta(1 + \delta + 2it)| \\ &= \sum_p \sum_{n \geq 1} \frac{p^{-n(1+\delta)}}{n} (3 + 4 \cos(nt \log p) + 2 \cos(2nt \log p)) \geq 0. \end{aligned} \quad (\text{GSL3.1})$$

Every prime and positive prime power is present. The summand's trigonometric factor is exactly $(1 + 2 \cos(nt \log p))^2$. Exponentiating gives

$$\zeta(1 + \delta)^3 |\zeta(1 + \delta + it)|^4 |\zeta(1 + \delta + 2it)|^2 \geq 1. \quad (\text{GSL3.2})$$

For $0 < \delta \leq 1/4$, the Dirichlet-series integral bound gives $\zeta(1 + \delta) \leq 1 + 1/\delta \leq 2/\delta$. Put $T = 1 + |t|$, with $|t| \geq 1$. By (GSL2.2), the last zeta factor is at most a constant times T . Consequently there is $c_0 > 0$, independent of δ, t , such that

$$|\zeta(1 + \delta + it)| \geq c_0 \delta^{3/4} T^{-1/2}. \quad (\text{GSL3.3})$$

Choose a positive constant $c \leq 1/4$ small enough that $2C_1 c \leq (c_0/2)c^{3/4}$, enlarging C_1 in (GSL2.2) if required to cover the same horizontal segments. Set $\delta = cT^{-6}$. For every real x with $|x| \leq cT^{-6}$, (GSL2.3) integrated along that segment gives

$$\begin{aligned} |\zeta(1 + x + it)| &\geq |\zeta(1 + \delta + it)| - 2C_1 T \delta \\ &\geq \frac{c_0 c^{3/4}}{2} T^{-5}. \end{aligned} \quad (\text{GSL3.4})$$

Thus the width and lower bound have been derived from the original function, not assumed as an RH consequence.

The full functional equation gives $F_0(s) = F_0(1 - s)$. For an exact growth comparison, retain both complete factors after that substitution:

$$\begin{aligned} G(s) &= \left[\frac{(1-s)(-s)}{8} \pi^{-(1-s)/2} \Gamma((1-s)/2) \zeta(1-s) \right] \\ &\quad \cdot \left[\frac{(1+s)s}{8} \pi^{-(1+s)/2} \Gamma((1+s)/2) \zeta(1+s) \right]. \end{aligned} \quad (\text{GSL3.5})$$

The exact Gamma reflection identity gives the comparison

$$\Gamma((1-s)/2) \Gamma((1+s)/2) = \frac{\pi}{\cos(\pi s/2)}. \quad (\text{GSL3.6})$$

This identity follows from $\Gamma(z) \Gamma(1-z) = \pi / \sin(\pi z)$, with $z = (1-s)/2$. For example the latter follows from the Euler products for $1/\Gamma$ and $\sin$, multiplying the factors at $z, -z$ and then using $\Gamma(1-z) = -z \Gamma(-z)$. It is a meromorphic identity retaining both Gamma factors, not a change to the function in (GSL1.8).

Combining the two powers of π with (GSL3.6) gives the exact additional expression

$$G(s) = \frac{(1-s)(-s)(1+s)s}{64 \cos(\pi s/2)} \zeta(1-s) \zeta(1+s). \quad (\text{GSL3.7})$$

All four endpoint factors, the denominator 64, both original zeta factors and the Gamma comparison remain explicit. At apparent exceptional points the expression has the entire continuation specified by (GSL1.3)–(GSL1.8); it is used for estimates only where $|t| \geq 1$.

For $s = x + it$, $|t| \geq 1$, $|x| \leq cT^{-6}$, the product of the four endpoint magnitudes is bounded below by a positive constant times T^4 . Further,

$$|\cos(\pi(x + it)/2)|^2 = \cos^2(\pi x/2) + \sinh^2(\pi t/2). \quad (\text{GSL3.8})$$

The reciprocal cosine has magnitude comparable to $e^{-\pi|t|/2}$ in this region. Apply (GSL3.4) to $1 + x + it$ and to $1 - x - it$; conjugation preserves the same bound. Therefore

$$|G(x + it)| \geq c_2 T^{-6} e^{-\pi|t|/2} \quad (|t| \geq 1, |x| \leq cT^{-6}). \quad (\text{GSL3.9})$$

On every fixed vertical strip, (GSL2.5), (GSL3.7) and (GSL3.8) also give an integer $k_A \geq 0$ and a constant C_A such that

$$|G(x + it)| \leq C_A (1 + |t|)^{k_A} e^{-\pi|t|/2} \quad (|x| \leq A, t \in \mathbb{R}). \quad (\text{GSL3.10})$$

For bounded t , this follows by the entire continuation and compactness; no pole in (GSL3.7) is retained spuriously.

The line $x = 0$ has no zero of G . For $t \neq 0$, this follows from DB9 at $1 \pm it$, (GSL3.5), and the absence of a Gamma zero or pole there. At $t = 0$, it is the exact value $G(0) = 1/64$. Compactness gives a positive-width zero-free neighborhood of the bounded segment $|t| \leq 1$. Choose $\varepsilon > 0$ sufficiently small, with $\varepsilon \leq 1/4$, so that

$$r(t) = \varepsilon(1 + t^2)^{-3} \quad (\text{GSL3.11})$$

lies within that compact neighborhood and within the corridor in (GSL3.9) for $|t| \geq 1$. This is possible because $(1 + t^2)^{-3} \leq 8(1 + |t|)^{-6}$. Shrinking the constants to include the compact segment proves

$$G(x + it) \neq 0, \quad \frac{1}{|G(x + it)|} \leq C(1 + |t|)^6 e^{\pi|t|/2} \quad (|x| \leq r(t)). \quad (\text{GSL3.12})$$

The entire original and shifted zero sets are outside this corridor, on its right and left respectively. No separation estimate between zeros within either set was needed.

GSL4. An explicit entire separator with controlled strip growth

Choose a real smooth function η with $0 \leq \eta \leq 1$, $\eta(v) = 0$ for $v \leq -1$, and $\eta(v) = 1$ for $v \geq 1$. For an explicit choice let $b(v) = 0$ for $v \leq 0$, $b(v) = e^{-1/v}$ for $v > 0$, and take

$$\eta(v) = \frac{b(v + 1)}{b(v + 1) + b(1 - v)}. \quad (\text{GSL4.1})$$

The denominator is positive everywhere, and the vanishing of all right derivatives of b at zero proves the stated smoothness. Define on the complex plane

$$\chi(x + it) = \eta(x/r(t)), \quad \bar{\partial} = \frac{1}{2}(\partial_x + i\partial_t). \quad (\text{GSL4.2})$$

Its derivative is supported in the corridor. There,

$$\bar{\partial}\chi(x + it) = \frac{1}{2}\eta'(x/r(t)) \left(\frac{1}{r(t)} - i \frac{x r'(t)}{r(t)^2} \right). \quad (\text{GSL4.3})$$

Since $|x| \leq r(t)$, $r'/r = -6t/(1+t^2)$, and $1/r \leq C(1+|t|)^6$, we have

$$|\bar{\partial}\chi(x+it)| \leq C(1+|t|)^6. \quad (\text{GSL4.4})$$

Define the smooth function

$$h(w) = \frac{\bar{\partial}\chi(w)}{G(w)} \quad (\text{GSL4.5})$$

on the corridor, extended by zero outside it. Smoothness across its edges follows because the derivative of η vanishes to every order there. There is no division at a zero of G . With $\alpha = \pi/2$, (GSL3.12) gives

$$\text{supp } h \subset \{| \text{Re } w| \leq \varepsilon\}, \quad |h(u+iv)| \leq Ce^{\alpha|v|}(1+|v|)^{12}. \quad (\text{GSL4.6})$$

Use the explicit Cauchy kernel with its Gaussian factor:

$$u(s) = \frac{1}{\pi} \int_{\mathbb{C}} \frac{e^{(s-w)^2}}{s-w} h(w) dA(w), \quad (\text{GSL4.7})$$

where dA is ordinary planar area. This integral is absolutely convergent. Its local singularity has order $1/|s-w|$, which is integrable in two real dimensions. For the tails, writing $s = x+it$, $w = u+iv$, its exponential modulus is exactly

$$|e^{(s-w)^2}| = e^{(x-u)^2-(t-v)^2}. \quad (\text{GSL4.8})$$

The negative quadratic in $t-v$ dominates the fixed exponential and polynomial in (GSL4.6), while $|u| \leq \varepsilon$.

Here is a uniform strip estimate, including the local singularity. For $y \neq 0$, integration in u gives

$$\sup_{x \in \mathbb{R}} \int_{-\varepsilon}^{\varepsilon} \frac{du}{\sqrt{(x-u)^2 + y^2}} \leq C_{\varepsilon} \left(1 + \log_+ \frac{1}{|y|} \right). \quad (\text{GSL4.9})$$

One can see this by translating the integration interval towards zero, which increases the integral of the even decreasing function of $|x-u|$, and integrating $(v^2+y^2)^{-1/2}$ there. The result is bounded by a constant plus $2 \log(1/|y|)$ when $|y| \leq 1$, and by $2\varepsilon/|y|$ otherwise. The infinite value at $y=0$ is on a set of one-dimensional measure zero; the displayed logarithm is integrable near zero.

For $|x| \leq A$, use $(1+|v|) \leq (1+|t|)(1+|t-v|)$ and $e^{\alpha|v|} \leq e^{\alpha|t|}e^{\alpha|t-v|}$. Equations (GSL4.6)-(GSL4.9) yield

$$|u(x+it)| \leq C_A e^{\alpha|t|}(1+|t|)^{12}, \quad (\text{GSL4.10})$$

because the remaining integral

$$\int_{\mathbb{R}} e^{-y^2+\alpha|y|}(1+|y|)^{12} \left(1 + \log_+ \frac{1}{|y|} \right) dy \quad (\text{GSL4.11})$$

is finite, both near zero and at infinity.

The fundamental-solution identity is $\bar{\partial}(1/(\pi s)) = \delta_0$. It follows directly by integrating a compactly supported smooth test function over the punctured plane and taking the boundary integral on a shrinking circle; the factor $1/\pi$ and the convention in (GSL4.2) give mass one. Multiplication by the holomorphic function e^{s^2} , whose value at zero is one, preserves that identity. Thus (GSL4.7) satisfies

$$\bar{\partial}u = h. \quad (\text{GSL4.12})$$

For precision the identity may first be applied to compactly supported truncations of h ; Gaussian domination makes their integrals and distributional derivatives converge on each compact set. Locally

the compact Cauchy transform of a smooth function is smooth: transfer its derivatives to that function using a smooth cutoff, while the complementary kernel has no singularity. The same Gaussian tail estimates apply to the complementary part. Hence u is smooth, and (GSL4.12) is also a pointwise identity.

Define

$$E(s) = \chi(s) - G(s)u(s). \quad (\text{GSL4.13})$$

Its $\bar{\partial}$ derivative is $\bar{\partial}\chi - Gh = 0$, so it is entire. Combining (GSL3.10) and (GSL4.10) proves, for each strip, an integer $d_A \geq 0$ and constant C'_A with

$$|E(x + it)| + |1 - E(x + it)| \leq C'_A(1 + |t|)^{d_A} \quad (|x| \leq A). \quad (\text{GSL4.14})$$

The exponential factors $e^{-\pi|t|/2}$ and $e^{\pi|t|/2}$ have both been retained in the proof before this estimate is taken.

Every original zero is to the right of the corridor, where $\chi = 1$; every shifted zero is to the left, where $\chi = 0$. In a neighborhood of any such zero, $h = 0$, so u is holomorphic and (GSL4.13) gives the complete jet identities

$$\begin{aligned} E(s) - 1 &\in (s - \rho)^{m_\rho} \mathcal{O}_\rho, \\ E(s) &\in (s - (\rho - 1))^{m_\rho} \mathcal{O}_{\rho-1} \quad (\rho \in \mathcal{Z}). \end{aligned} \quad (\text{GSL4.15})$$

Every multiplicity is preserved. The constructed function E is a multiplier of $\mathcal{B}$; it is not assumed to be a member of $\mathcal{B}$. This distinction matters because it takes the value one at the entire original zero set.

GSL5. Continuous multiplication and the full CRT inverse

Equation(GSL4.14) gives the explicit seminorm estimates

$$b_{A,M}(EF) \leq C'_A b_{A,M+d_A}(F), \quad b_{A,M}((1-E)F) \leq C'_A b_{A,M+d_A}(F). \quad (\text{GSL5.1})$$

Thus both are continuous linear operators $\mathcal{B} \rightarrow \mathcal{B}$. Their full ideal behavior is

$$\begin{aligned} E\mathcal{B} &\subset \mathcal{J}_-, \quad (1-E)\mathcal{B} \subset \mathcal{J}, \\ E\mathcal{J} &\subset \mathcal{J}, \quad (1-E)\mathcal{J}_- \subset \mathcal{J}. \end{aligned} \quad (\text{GSL5.2})$$

Each inclusion follows from the appropriate two sets of jets in (GSL4.15) and the full Leibniz formula. In particular

$$F = EF + (1-E)F \quad \text{proves} \quad \mathcal{J} + \mathcal{J}_- = \mathcal{B}, \quad (\text{GSL5.3})$$

with an explicit continuous choice of the two summands.

The canonical map and its constructed inverse are

$$\Phi : \mathcal{Q}_\cup \longrightarrow \mathcal{Q} \oplus \mathcal{Q}_-, \quad [F]_\mathcal{J} \longmapsto ([F]_\mathcal{J}, [F]_{\mathcal{J}_-}), \quad (\text{GSL5.4})$$

$$\Psi([f]_\mathcal{J}, [g]_{\mathcal{J}_-}) = [Ef + (1-E)g]_\mathcal{J}. \quad (\text{GSL5.5})$$

The final two inclusions in (GSL5.2) prove independence of the representatives in (GSL5.5). At every original zero its representative has exactly the jets of f ; at every shifted zero it has exactly the jets of g . Thus $\Phi\Psi = 1$. For a common representative F , its image under the opposite composition is $[EF + (1-E)F]_\mathcal{J} = [F]_\mathcal{J}$, so $\Psi\Phi = 1$. This proves surjectivity onto the full direct sum of the two actual quotient spaces, not merely a dense range.

For explicit topological control, let $q_{A,M}^\mathcal{K}$ be the quotient seminorm obtained from $b_{A,M}$ by a closed subspace $\mathcal{K}$. The quotient maps in (GSL5.4) are continuous with

$$q_{A,M}^\mathcal{J}([F]), \quad q_{A,M}^{\mathcal{J}_-}([F]) \leq q_{A,M}^\mathcal{J}([F]). \quad (\text{GSL5.6})$$

Taking the two independent infima over representatives in (GSL5.1) proves

$$q_{A,M}^{\mathcal{J}}(\Psi(x, y)) \leq C'_A \left(q_{A,M+d_A}^{\mathcal{J}}(x) + q_{A,M+d_A}^{\mathcal{J}}(y) \right). \quad (\text{GSL5.7})$$

This is the required continuous inverse estimate on the actual quotient topology. Neither a blockwise formal inversion nor an open-mapping assertion is being used as a substitute for it.

The lift and the shifted inclusion are consequently

$$\iota_+(x) = \Psi(x, 0), \quad \iota_-(y) = \Psi(0, y), \quad \pi_+([F]) = [F]_{\mathcal{J}}. \quad (\text{GSL5.8})$$

They give the continuously split exact sequence

$$0 \longrightarrow \mathcal{Q}_- \xrightarrow{\iota_-} \mathcal{Q}_{\cup} \xrightarrow{\pi_+} \mathcal{Q} \longrightarrow 0, \quad \pi_+ \iota_+ = 1. \quad (\text{GSL5.9})$$

The corresponding projection is multiplication by E modulo $\mathcal{J}$. It is idempotent there because $E^2 - E$ vanishes to every required jet at both sets. No assertion that $E^2 = E$ as an entire function is made.

GSL6. Original arithmetic action, degree shift, and uniqueness of the lift

On $\mathcal{B}$ use the actual operators

$$\mathsf{L}F(s) = sF(s), \quad T_a F(s) = a^s F(s), \quad a > 0. \quad (\text{GSL6.1})$$

Their continuity estimates are exactly RZ1.10 and RZ8.2. Both preserve all ideals in (GSL1.2), so they descend to every displayed quotient. Multiplication by E commutes with both already on $\mathcal{B}$. Hence $\Phi, \Psi, \iota_+, \iota_-$ commute with L and every T_a in their specified domains and codomains.

Translation keeps its exact character and degree factors:

$$T_{a,-} \bar{U} = a^{-1} \bar{U} T_a, \quad L_- \bar{U} = \bar{U} (L - 1). \quad (\text{GSL6.2})$$

For example $U(sF)(s) = (s+1)F(s+1)$, proving the second identity including its sign. Thus (GSL5.4) also gives a topological representation isomorphism

$$\mathcal{Q}_{\cup} \simeq \mathcal{Q} \oplus \mathcal{Q}(1), \quad (\text{GSL6.3})$$

where the second summand is realized as the actual translated quotient, and its character twist means $T_a \mapsto a^{-1} T_a$. This is an analytic character twist, not an asserted étale Tate realization. On each actual shifted jet its full operator is

$$T_{a,-} = a^{\rho-1} \sum_{j=0}^{m_{\rho}-1} \frac{(\log a)^j}{j!} M_w^j, \quad w = s - (\rho - 1). \quad (\text{GSL6.4})$$

The original block retains a^{ρ} with the same complete nilpotent exponential. For the weight statement fix a recovered $a > 1$ and define $w_a(\lambda) = 2 \log |\lambda| / \log a$ for $\lambda \neq 0$. Then $w_a(a^{\rho-1}) = 2 \operatorname{Re} \rho - 2$ and $w_a(a^{\rho}) = 2 \operatorname{Re} \rho$, so their logarithmic-modulus weights occupy $(-2, 0)$ and $(0, 2)$, respectively, without any critical-line assumption. The action itself remains defined for every $a > 0$, including its identity at $a = 1$; no weight is assigned using a denominator $\log 1$.

There is also an exact version with the higher-weight term as the kernel, obtained by translating the proved construction, rather than positing another splitting. Set

$$\mathcal{J}_+ = U^{-1} \mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho+1) = 0 \ (0 \leq j < m_{\rho})\}, \quad \mathcal{Q}_+ = \mathcal{B}/\mathcal{J}_+, \quad \mathcal{Q}_{\cup,+} = \mathcal{B}/(\mathcal{J} \cap \mathcal{J}_+). \quad (\text{GSL6.5})$$

Translation by -1 sends $\mathcal{J}$ onto $\mathcal{J}_+ \cap \mathcal{J}$ and gives a topological character comparison from $\mathcal{Q}_\cup(-1)$ to $\mathcal{Q}_{\cup,+}$. Its full translated multiplier remains

$$G(s-1) = \left[\frac{(s-1)(s-2)}{8} \pi^{-(s-1)/2} \Gamma((s-1)/2) \zeta(s-1) \right] \left[\frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) \right]. \quad (\text{GSL6.6})$$

The multiplier $E_+(s) = 1 - E(s-1)$ has jet one on the original divisor and jet zero on the divisor shifted by $+1$. Its strip-growth estimate follows from (GSL4.14) with $A+1$ in place of A . Applying the already proved inverse with this multiplier gives the exact continuously split row

$$0 \longrightarrow \mathcal{Q}_+ \simeq \mathcal{Q}(-1) \longrightarrow \mathcal{Q}_{\cup,+} \longrightarrow \mathcal{Q} \longrightarrow 0. \quad (\text{GSL6.7})$$

The section is $[f]_{\mathcal{J}} \mapsto [E_+ f]_{\mathcal{J} \cap \mathcal{J}_+}$; the inclusion of the kernel uses $1 - E_+$. On its kernel, translation gives $a^{\rho+1} \exp((\log a) M_w)$, retaining every nilpotent coefficient. The original scalar weights are in $(0, 2)$, the kernel weights in $(2, 4)$. Thus the higher-weight version of the separated lift is also realized on a fully specified Fréchet quotient of the original source, with both original and translated full factors retained.

Here are the full maps and exceptional values for that counterpart. Write $VF(s) = F(s-1)$, so $V = U^{-1}$, and put

$$F_{-1}(s) = F_0(s-1) = \frac{(s-1)(s-2)}{8} \pi^{-(s-1)/2} \Gamma((s-1)/2) \zeta(s-1). \quad (\text{GSL6.8})$$

Its endpoint values are $F_{-1}(1) = F_{-1}(2) = 1/8$. The entire values at the translated trivial-zero and reflected locations are, with every original coefficient retained,

$$\begin{aligned} F_{-1}(1-2r) &= \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) \\ &= F_{-1}(2+2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0 \quad (r \geq 1). \end{aligned} \quad (\text{GSL6.9})$$

The apparent Gamma and zeta singularities in (GSL6.8) have exactly these continuations; its entire derivatives are $F_{-1}^{(j)}(s) = F_0^{(j)}(s-1)$.

Translation induces $\bar{V} : \mathcal{Q} \rightarrow \mathcal{Q}_+$, with inverse $\bar{U}$ between these two specified quotients. Substituting the functions gives

$$T_{a,+} \bar{V} = a \bar{V} T_a, \quad L_+ \bar{V} = \bar{V} (L+1). \quad (\text{GSL6.10})$$

For instance $V(sF)(s) = (s-1)F(s-1)$, so $sVF = V(sF) + VF$; also $V(a^s F)(s) = a^{s-1} F(s-1)$, giving exactly the first factor in (GSL6.10). The map $\bar{V}$ is thus an equivariant isomorphism from $\mathcal{Q}(-1)$ to $\mathcal{Q}_+$. The high-weight row, including its injection from that domain, its quotient map and its section, is

$$\begin{aligned} j_{\text{high}}([h]_{\mathcal{J}}) &= [(1 - E_+(s))h(s-1)]_{\mathcal{J} \cap \mathcal{J}_+}, \\ \pi_{\text{high}}([F]_{\mathcal{J} \cap \mathcal{J}_+}) &= [F]_{\mathcal{J}}, \\ s_{\text{high}}([f]_{\mathcal{J}}) &= [E_+(s)f(s)]_{\mathcal{J} \cap \mathcal{J}_+}. \end{aligned} \quad (\text{GSL6.11})$$

The first displayed domain is the underlying vector space of $\mathcal{Q}(-1)$, with its specified twisted action. The ideal identities follow by the same two full jet conditions on E_+ , proving independence of every representative. The CRT inverse applied to $([f]_{\mathcal{J}}, \bar{V}[h]_{\mathcal{J}})$ is the sum of the first and third maps. It proves exactness, surjectivity, and $\pi_{\text{high}} s_{\text{high}} = 1$. Continuity follows from (GSL1.6) and the multiplier bound for E_+ . Equations (GSL6.10) and commutation of the multipliers prove equivariance, with no dropped factor of a .

The equivariant section is unique in a precise continuous linear category. Let $H : \mathcal{Q} \rightarrow \mathcal{Q}_-$ be a continuous linear map commuting with L . For $\lambda = \rho - 1$ and $m = m_\rho$, the first m jets of $(L_- - \lambda)^m H(x)$ vanish at λ . By linearity and intertwining these are the same jets as those of $H((L_- - \lambda)^m x)$. But $\lambda \notin \mathcal{Z}$, and RZ3 constructs an everywhere-defined continuous inverse for $\lambda - L$. Hence $(L_- - \lambda)^m$ is onto $\mathcal{Q}$, so every jet of order less than m of every $H(x)$ vanishes at λ . This holds for all shifted zeros. The definition of $\mathcal{J}_-$ implies $H = 0$. The identical argument with $\lambda = \rho + 1 \notin \mathcal{Z}$ proves that every continuous linear L -intertwiner $\mathcal{Q} \rightarrow \mathcal{Q}_+$ is zero.

Consequently two continuous linear L -equivariant sections of π_+ must coincide: their difference lands in $\ker \pi_+ = \iota_- \mathcal{Q}_-$, and the preceding argument applies. The same proof gives uniqueness for the high-weight row. A continuous linear map commuting with every recovered positive rational T_a also commutes with the real group and its generator L . Indeed the positive rationals are dense in the positive reals after the full arithmetic is identified, RZ8.5 proves strong continuity and differentiability, and continuity of the map permits passage first to the real parameter limit and then to its derivative. The same uniqueness therefore holds for continuous linear sections respecting the full recovered arithmetic action. This proof uses the actual global resolvent and the separating jet family, not a claimed convergence of an infinite sum of finite projectors.

Different allowed cutoffs or corridor constants in GSL4 may produce different functions E , but their quotient inverses (GSL5.5) coincide. Their difference has zero jets at both divisors, and its product with any member of $\mathcal{B}$ belongs to $\mathcal{J}$. Equivalently both are the inverse of the canonical map Φ . The continuous quotient lift is therefore independent of those auxiliary choices.

GSL7. Finite-source projectors inside the actual global extension

The construction is compatible with the RZ finite-source projectors. For a point λ in the union divisor, with its full multiplicity m , put

$$A_\lambda^\cup(s) = \frac{G(s)}{(s - \lambda)^m}, \quad \mathcal{P}_\lambda^\cup F(s) = A_\lambda^\cup(s) \operatorname{Tay}_{\lambda, < m} \left(\frac{F}{A_\lambda^\cup} \right) (s). \quad (\text{GSL7.1})$$

Division is entire and continuous on the relevant vanishing subspace by RZ2. The local Taylor quotient is taken only where $A_\lambda^\cup(\lambda) \neq 0$; its output is a global polynomial times that member of $\mathcal{B}$. Thus it is a continuous finite-rank map $\mathcal{B} \rightarrow \mathcal{B}$, annihilates $\mathcal{J}$, and has exactly the original jet at λ and zero jets at every other union point.

Let $P_\lambda^\cup$ be its induced quotient projector. Equality of all retained jets proves

$$P_\rho^\cup \iota_+ = \iota_+ P_\rho, \quad P_{\rho-1}^\cup \iota_+ = 0, \quad (\text{GSL7.2})$$

$$P_{\rho-1}^\cup \iota_- \bar{U} = \iota_- \bar{U} P_\rho, \quad P_\rho^\cup \iota_- \bar{U} = 0. \quad (\text{GSL7.3})$$

These are exact global quotient identities. They retain every Taylor derivative and factorial of RZ5-RZ6. No assertion about convergence of $\sum_\rho P_\rho$ is used.

The full union resolvent also has the direct representative

$$\mathcal{R}_\lambda^\cup F(s) = \frac{F(s) - G(s)F(\lambda)/G(\lambda)}{\lambda - s} \quad (\lambda \notin \mathcal{Z} \cup (\mathcal{Z} - 1)). \quad (\text{GSL7.4})$$

The numerator vanishes at $s = \lambda$, and the RZ2 division estimate applies with the full retained G . It preserves $\mathcal{J}$ and induces $(\lambda - L_\cup)^{-1}$, by the same two exact multiplication identities as RZ3.7. The pole orders and all Laurent coefficients are consequently the full multiplicities at the two divisors, with the projectors (GSL7.1). This verifies the global degree operator on the actual extension directly.

GSL8. The original Mellin source and a continuous source representative formula

The source is the original space

$$\mathcal{A} = \left\{ k \in C^\infty(\mathbb{R}_{>0}) : p_{N,j}(k) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)| < \infty \right\}, \quad (\text{GSL8.1})$$

with its exact Mellin isomorphism

$$\mathcal{M}k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}. \quad (\text{GSL8.2})$$

Let $\mathcal{C} = \overline{\mathcal{E}(\mathcal{S}_0^{\text{even}})}^{\mathcal{A}}$. S5 proves $\mathcal{M}\mathcal{C} = \mathcal{J}$, with its full source-image equality and topology. Multiplication by u is a topological automorphism of $\mathcal{A}$, and

$$\mathcal{M}(uk)(s) = \mathcal{M}k(s+1), \quad \mathcal{M}(u\mathcal{C}) = \mathcal{J}_-. \quad (\text{GSL8.3})$$

For its continuity, expand $(u\partial_u)^j(uk) = u \sum_{h=0}^j \binom{j}{h} (u\partial_u)^h k$ and use $u(u^N + u^{-N}) \leq u^{N+1} + u^{-(N+1)} + u^{N+1} + u^{-(N+1)}$; the harmless factor two supplies an explicit bound by finitely many $p_{N+1,h}$. The inverse uses u^{-1} and the corresponding signs. Thus (GSL0.1) is transported to

$$\mathcal{A}/(\mathcal{C} \cap u\mathcal{C}) \xrightarrow{\sim} (\mathcal{A}/\mathcal{C}) \oplus (\mathcal{A}/u\mathcal{C}) \quad (\text{GSL8.4})$$

by constructed continuous maps on the original source spaces.

For $F = \mathcal{M}k$, a representative of the first lifted quotient class has the explicit source function

$$k_+(u) = \frac{1}{2\pi} \int_{\mathbb{R}} E(\tfrac{1}{2} + it) F(\tfrac{1}{2} + it) u^{-it} dt. \quad (\text{GSL8.5})$$

It lies in $\mathcal{A}$, because $EF \in \mathcal{B}$. The exact RZ4.7 estimate and (GSL5.1) give

$$p_{N,j}(k_+) \leq \frac{2(N+2)^j}{\pi} C'_{N+2} b_{N+2,j+2+d_{N+2}}(F). \quad (\text{GSL8.6})$$

The other summand uses $1 - E$ with the same type of estimate. Their sum is k by Mellin inversion. Changing the representative of a quotient class changes its lifted representative by $\mathcal{C} \cap u\mathcal{C}$, exactly as (GSL5.2) proves. Thus (GSL8.5) supplies a continuous prequotient operator and the stated quotient lift; it is not an assertion of a separately chosen linear section $\mathcal{Q} \rightarrow \mathcal{B}$.

On the source the original T_a is $k(u) \mapsto a^{1/2}k(u/a)$, and the original degree operator is $1/2 - u\partial_u$, as proved in RZ1 and RZ8. Both factors are retained when transporting (GSL6.1)–(GSL6.2). The variable u belongs to the existing Mellin source; it is not a coordinate assigned to τ .

For the high-weight counterpart the exact source identities are

$$\mathcal{M}(u^{-1}k)(s) = \mathcal{M}k(s-1), \quad \mathcal{M}(u^{-1}\mathcal{C}) = \mathcal{J}_+, \quad (\text{GSL8.7})$$

and its full continuous source quotient is $\mathcal{A}/(\mathcal{C} \cap u^{-1}\mathcal{C})$. The source injection corresponding to j_{high} first multiplies its representative by u^{-1} and then applies the Mellin multiplier $1 - E_+$. The lifted original class is represented by

$$k_{\text{high}}(u) = \frac{1}{2\pi} \int_{\mathbb{R}} E_+(\tfrac{1}{2} + it) F(\tfrac{1}{2} + it) u^{-it} dt. \quad (\text{GSL8.8})$$

Multiplication by u^{-1} has the finite derivative expansion $(u\partial_u)^j(u^{-1}k) = u^{-1} \sum_{h=0}^j \binom{j}{h} (-1)^{j-h} (u\partial_u)^h k$, giving the same endpoint-seminorm control as above. For (GSL8.8), the translated strip estimate and RZ4.7 give

$$p_{N,j}(k_{\text{high}}) \leq \frac{2(N+2)^j}{\pi} C'_{N+3} b_{N+2,j+2+d_{N+3}}(F). \quad (\text{GSL8.9})$$

These are continuous maps on the full original source, with the stated quotient kernels. All shifted endpoint and trivial-zero contributions are the retained values in (GSL6.8)-(GSL6.9).

GSL9. Exact range, shifted ideals, and the original functional equation

The constructed CRT map has full range, with the continuous inverse (GSL5.5). The zero-germ data used by either quotient remain the actual image of $\mathcal{B}$ under its corresponding jet map. The proof never enlarges that image to arbitrary sequences and never substitutes a product topology for the quotient topology.

At a shifted center $\rho - 1$, the original $\zeta(s)$ is a unit while $\zeta(s+1)$ has the original zero multiplicity. This follows from the full functional equation with

$$C(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2), \quad \zeta(s) = \frac{C(1-s)}{C(s)}\zeta(1-s), \quad (\text{GSL9.1})$$

because $-1 < \text{Re } s < 0$ maps into $\text{Re}(1-s) > 1$, where the original Euler product is nonzero, and both displayed factors are holomorphic units on those open strips. The shifted original-zero algebra is exactly $\mathcal{O}_{\rho-1}/(\zeta(s+1))$, not $\mathcal{O}_{\rho-1}/(\zeta(s))$. The full multipliers (GSL1.3) and (GSL1.7) implement these precise divisors globally without erasing the exceptional values (GSL1.4)-(GSL1.5).

For every derivative order, the multiplier comparison remains the full product rule. Writing $K(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)/8$ only to display its terms,

$$F_0^{(d)}(s) = \sum_{j=0}^d \binom{d}{j} K^{(d-j)}(s) \zeta^{(j)}(s), \quad F_1^{(d)}(s) = F_0^{(d)}(s+1). \quad (\text{GSL9.2})$$

The finite projector formulas in GSL7 use these full derivatives. The constant $1/8$, both endpoint factors, the power of π , and every Gamma derivative are not absorbed into a different original function.

The extension class of (GSL5.9) vanishes in the category of Fréchet spaces with the stated continuous arithmetic action, because (GSL5.8) is its explicit continuous equivariant section. This is an actual split extension of the original global quotient $\mathcal{Q}$. It is more than a finite-zero calculation, and more than an algebraic direct-sum splitting. It does not assert that every abstract extension of $\mathcal{Q}$ has a continuous section, nor identify this explicitly constructed extension with a geometric localization cross that has different specified maps.

GSL10. Branch transport and label retention

An admissible branch comparison at this stage preserves the recovered arithmetic, its original function ζ , and the specified Mellin receiver. Under that comparison the definitions of $\mathcal{J}$, $\mathcal{J}_-$, the full multipliers, translation and Φ agree. The inverse Ψ , being the proved inverse of this canonical map, therefore agrees as well. Each branch can carry out the construction using its own recovered counter and then apply its isomorphism; no pooled measure or comparison with an arithmetic-destroying candidate appears.

If the existing source record retains a separate support label ℓ , apply the maps to the coefficient in $\{\ell\} \times \mathcal{Q}$ and keep ℓ fixed. In particular a zero coefficient is still the labelled pair $(\ell, 0)$. This respects the S7 support repair and introduces no addition between support labels or at τ .

GSL11. Independent check

An independent mathematical derivation checked the entire construction mechanism: the 3, 4, 2 lower bound with $\delta = cT^{-6}$; the full Gamma reflection comparison and corridor bounds; the sign and factor of the Gaussian Cauchy kernel; the logarithmic-integrability estimate after real-coordinate integration; and the resulting continuous CRT inverse. It confirmed that the universal separator need only lie in the multiplier algebra, that every multiplicity is retained, and that the construction proves the full range without identifying the quotient with arbitrary jets. The complete written note is also made available to the parent for its independent reading.

The checker subsequently read GSL0-GSL11 and the receiving RZ3-RZ4 and RZ8 passages. It verified the all-strip integration formula, every corridor factor, the quotient-seminorm estimate, the global-resolvent uniqueness argument, and the full source inversion constant. Its two detected TeX spacing errors were corrected. It then separately checked the added higher-weight maps (GSL6.5)-(GSL6.11) and source formulas (GSL8.7)-(GSL8.9), including the asymmetric strip indices in the last bound. No mathematical correction was needed; the positive-integer range in (GSL6.9) was made explicit.

The proved output is the explicit global shifted extension, its full continuous equivariant splitting, its source maps and its finite-jet compatibility. The checking coverage does not identify this extension with a different geometric localization obstruction or supply an RH conclusion.

The evaluated CC charts and the geometric Tate class — CGT0-CGT12

Proof source: CC_GEOMETRIC_TATE_INDEPENDENT.md.

The CC chart realization, its Tate class, and its ramified specialization

Independent derivation, 24 September 2026. Proof locators CGT0-CGT12.

CGT0. Objects, prerequisites, and source coverage

The supporting datum remains the user's $\tau\langle Z_1; \text{no } Z_2 \rangle$. No addition on τ is used. The ring $\mathbf{Z}$ in this calculation is the already recovered arithmetic coefficient ring. It is not defined by selecting the primes 2, 3, or by a finite part of the timing history. Every ring operation, scheme, cohomology group, and quotient below is downstream of that reconstruction.

The specific object evaluated here is Connes–Consani's supplied chart construction. The Gaussian coefficient ring appearing in its evaluation is a receiver for those charts. It is not an alternative prime spectrum offered as a counterexample to the user's global reconstruction. In particular, this calculation makes no identification of τ with a Gaussian integer, a zero, a unit, or a cohomology class.

The instruction CORPUS_AND_OPERATION_RULE.md (private construction record; not included) was read before the calculation. The root task additionally supplied the relevant correction: the user has already rejected replacing the reconstruction by a Gaussian-integer counterexample. That rejected use is not made here.

Original-author source actually read: Alain Connes and Caterina Consani, [The Absolute Twistor Line and the Geometry of the compactified Spec \$\mathbf{Z}\$](#) , supplied author file CC.tex, especially lines 430–460, 526–542, 576–602, 617–758 and 765–851. These are the signed base, chart and overlap presentations, field-valued point comparison, and involution. SHA-256 of that file is 57a10dcef5cd758e6b2f1c54b1c0a11b0fb093638f74fb1be515ec3bc7080ba4.

The Stacks Project original TeX was read locally at commit d3e53496bbd24b82a1d87ce513c913923ebbebb2:

- etale-cohomology.tex, lines 11071–11244: Kummer calculation and the pullback degree formula, [03RQ](#), [0AMB](#).
- The same file, lines 6547–6570: invariance under a nilpotent closed immersion, [03SI](#).
- The same file, lines 18610–18704: proper base change and its proof, [095T](#).
- The same file, lines 19523–19585: proper specialization and the smooth specialization consequence, [0GKD](#).
- trace.tex, lines 196–354: arithmetic/geometric Frobenius convention and the degree computation on the projective line.
- more-etale.tex, lines 4278–4362: the smooth proper curve consequence and its references.

This is reading coverage of the stated passages, not a claim to have reread the entire Stacks Project or CC paper. The two source hashes are etale-cohomology.tex: 015f27b1d332cc096b1f323daeeb855b299bb5a08f43c05131f196d4c183e7bf; trace.tex: 642cb0f2314fc73bef27542c79f55af87dcc0b0bf87d7cecf29b9b61860e4bfc.

CGT1. The ring realization and its exact twisted overlap

For a commutative coefficient ring R , define

$$\begin{aligned} A_R &= R[J]/(J^2 + 1), \quad A_+ = R[J_+]/(J_+^2 + 1)[T], \quad A_- = R[J_-]/(J_-^2 + 1)[U], \\ A_\circ &= R[J]/(J^2 + 1)[T, T^{-1}]. \end{aligned} \quad (\text{CGT1.1})$$

The overlap homomorphisms, with all signs retained, are

$$\rho_+(J_+) = J, \quad \rho_+(T) = T, \quad \rho_-(J_-) = -J, \quad \rho_-(U) = T^{-1}. \quad (\text{CGT1.2})$$

These are precisely the ring presentations representing the signed chart evaluations: for every commutative R -algebra D , a homomorphism $A_+ \rightarrow D$ is uniquely a pair (t, j_+) with $j_+^2 = -1$; the other chart and overlap have the same universal property, with t invertible on the overlap. The signed source element ϵ is sent to -1_D . Thus this realization is specified by a universal mapping property, not merely by agreement of geometric points over fields.

Let X_R be the scheme obtained by gluing these affine spectra along their displayed localizations. Define the global coefficient J to be J_+ on the plus chart and $-J_-$ on the minus chart. The two definitions have the same restriction, because $\rho_-(-J_-) = J$.

The standard two charts of $\mathbf{P}_{A_R}^1$, with $T = X/Y$ and $U = Y/X$, now give isomorphisms to the two charts of X_R : their coefficient maps are respectively $J \mapsto J_+$ and $J \mapsto -J_-$, and their coordinate maps are $T \mapsto T$, $U \mapsto U$. On the overlap both become the same homomorphism $A_R[T, T^{-1}] \rightarrow A_\circ$. The inverse homomorphisms send $J_+ \mapsto J$, $J_- \mapsto -J$. Both composites fix every generator. The gluing universal property therefore proves

$$\boxed{X_R \simeq \mathbf{P}_{A_R}^1} \quad (\text{CGT1.3})$$

with the original twisted chart identifications explicitly retained. This does not replace J_- by J_+ : it records their comparison $J_- = -J$ in the global coefficient chart.

For $R = \mathbf{Z}$, write $A = \mathbf{Z}[J]/(J^2 + 1)$. The evaluation $J \mapsto i$ is a ring isomorphism $A \rightarrow \mathbf{Z}[i]$, with inverse $a + bi \mapsto a + bJ$. Uniqueness follows from the monic quadratic relation, which gives the free $\mathbf{Z}$ -basis $1, J$.

CGT2. The involution on the entire scheme

Let $c : A_R \rightarrow A_R$ be $c(J) = -J$, $c(r) = r$ for $r \in R$. Under CGT1 the supplied CC involution is

$$\alpha([X : Y], J) = ([-Y : X], -J), \quad \alpha^*(T) = -T^{-1}, \quad \alpha^*(J) = -J. \quad (\text{CGT2.1})$$

This notation describes the semilinear scheme morphism over R ; it does not assert that α is over A_R .

Its plus-to-minus chart map is

$$A_- \longrightarrow A_+, \quad U \longmapsto -T, \quad J_- \longmapsto J_+, \quad (\text{CGT2.2})$$

and its minus-to-plus map is

$$A_+ \longrightarrow A_-, \quad T \longmapsto -U, \quad J_+ \longmapsto J_-. \quad (\text{CGT2.3})$$

For example, restricting (CGT2.2) sends J_- to J , while applying the generic formula to $\rho_-(J_-) = -J$ also gives J . The coordinate comparison sends U to $-T$ on both routes. The other comparison is identical on its stated generators. Thus these are morphisms of the glued scheme.

Applying (CGT2.1) twice gives $[-X : -Y] = [X : Y]$ and $J \mapsto J$. Hence $\alpha^2 = 1$. The projective transformation has degree 1, although its affine formula contains a minus sign and inversion. This degree will determine its action on H^2 .

CGT3. All finite-field coefficient fibres, including the nonreduced fibre

Let $q = p^f$, $k = \mathbf{F}_q$, and $\bar{k}$ an algebraic closure.

For $p \neq 2$, choose once a root $j \in \bar{k}$ of $J^2 + 1$. The Chinese remainder map is

$$A_{\bar{k}} \longrightarrow \bar{k} \oplus \bar{k}, \quad F(J) \longmapsto (F(j), F(-j)). \quad (\text{CGT3.1})$$

It is an isomorphism because $2j \neq 0$. Its inverse on a pair (a, b) is

$$\frac{a+b}{2} + \frac{a-b}{2j}J. \quad (\text{CGT3.2})$$

Both branches remain labeled by their actual coefficient values $j, -j$. Consequently

$$X_{\bar{k}} = \mathbf{P}_{\bar{k},+}^1 \sqcup \mathbf{P}_{\bar{k},-}^1. \quad (\text{CGT3.3})$$

The involution interchanges these components and applies $[X : Y] \mapsto [-Y : X]$ on their coordinates.

For $p = 2$, set $\nu = J - 1$. Then

$$A_{\bar{k}} = \bar{k}[\nu]/(\nu^2), \quad X_{\bar{k}} = \mathbf{P}_{\bar{k}[\nu]/(\nu^2)}^1. \quad (\text{CGT3.4})$$

This is a nonreduced scheme. Its closed immersion

$$\iota : \mathbf{P}_{\bar{k}}^1 \hookrightarrow X_{\bar{k}} \quad (\text{CGT3.5})$$

is defined by the nonzero square-zero ideal (ν) . The scheme and its reduction are not identified in this calculation. Instead, topological invariance gives the actual equivalence of étale topoi ι^{-1} , and hence the cohomology isomorphisms ι^* . The nilpotent coefficient ν is retained in (CGT3.4), even though the constant étale-cohomology receiver does not distinguish it.

Here $\alpha^*(J) = -J = J$, so $\alpha^*(\nu) = \nu$. The coordinate formula becomes $T \mapsto T^{-1}$; it is still the reduction of the full formula $-T^{-1}$.

CGT4. Cohomology and the geometric origin of the Tate factor

Fix a coefficient prime $\ell \neq p$, put $E = \mathbf{Q}_\ell$, and first work with coefficients $\mathbf{Z}/\ell^r\mathbf{Z}$. On $\mathbf{P}_{\bar{k}}^1$, the Kummer sequence gives

$$\begin{aligned} H^1(\mathbf{P}_{\bar{k}}^1, \mu_{\ell^r}) &= \text{Pic}(\mathbf{P}_{\bar{k}}^1)[\ell^r] = 0, \\ H^2(\mathbf{P}_{\bar{k}}^1, \mu_{\ell^r}) &= \text{Pic}(\mathbf{P}_{\bar{k}}^1)/\ell^r \text{Pic}(\mathbf{P}_{\bar{k}}^1) = \mathbf{Z}/\ell^r\mathbf{Z}. \end{aligned} \quad (\text{CGT4.1})$$

Here $\text{Pic}(\mathbf{P}_{\bar{k}}^1) = \mathbf{Z}[\mathcal{O}(1)]$. The other possible terms vanish because $\bar{k}^\times$ is ℓ^r -divisible and the higher $\mathbf{G}_m$ -cohomology of this smooth curve vanishes, as in the cited proof of 03RQ. The same source gives vanishing in degrees above 2. Passing through the compatible surjective coefficient transition maps, then tensoring with E , yields

$$H^0 = E, \quad H^1 = 0, \quad H^2 = E(-1). \quad (\text{CGT4.2})$$

The isomorphism $E \rightarrow H^2(\mathbf{P}_{\bar{k}}^1, E(1))$ sends 1 to $c_1(\mathcal{O}(1))$, not to an arbitrarily chosen eigenvector.

Here is the actual geometric multiplication factor. For every integer $n \geq 1$, the morphism

$$b_n : \mathbf{P}_{\bar{k}}^1 \rightarrow \mathbf{P}_{\bar{k}}^1, \quad [X : Y] \longmapsto [X^n : Y^n] \quad (\text{CGT4.3})$$

has

$$b_n^* \mathcal{O}(1) = \mathcal{O}(n), \quad b_n^* c_1(\mathcal{O}(1)) = n c_1(\mathcal{O}(1)). \quad (\text{CGT4.4})$$

The first equality follows by pulling back the homogeneous transition function X/Y , which becomes $(X/Y)^n$; the second follows from the tensor-product additivity and functoriality of the Kummer boundary. Thus the factor n comes from a morphism and its divisor pullback.

For $n = q$ the coordinate morphism is the relative q -power Frobenius of the projective line. With geometric Frobenius defined as inverse arithmetic Galois Frobenius, its action on untwisted H^2 is multiplication by q . Equivalently, arithmetic Frobenius acts by q on $\mathbf{Z}_\ell(1)$, geometric Frobenius by q^{-1} , and the class $c_1(\mathcal{O}(1))$ is Galois-fixed in $H^2(E(1))$. Untwisting therefore gives q , with no rescaling.

Applying α 's coordinate morphism to the same transition function gives degree 1, so its pullback on H^2 is $+1$. The minus sign in $-T^{-1}$ is not a negative cohomological degree.

CGT5. Full Frobenius matrices and all repetition traces

For odd p , in the ordered component basis $(+, -)$, let

$$P_q = \begin{cases} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, & q \equiv 1 \pmod{4}, \\ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, & q \equiv 3 \pmod{4}. \end{cases} \quad (\text{CGT5.1})$$

Indeed arithmetic Frobenius sends j to j^q ; since $j^2 = -1$, this is j or $-j$ according to $q \pmod{4}$. The inverse permutation for geometric Frobenius is the same, because the permutation has order at most two. Combining this permutation with CGT4 gives

$$H^0(X_{\bar{k}}, E) = E^2, \quad F_q = P_q; \quad H^1 = 0; \quad H^2 = E(-1)^2, \quad F_q = qP_q. \quad (\text{CGT5.2})$$

Thus the eigenvalues are 1, 1 and q, q when $q \equiv 1 \pmod{4}$, and 1, -1 and $q, -q$ when $q \equiv 3 \pmod{4}$.

The matrix description is valid already at integral coefficient level. The decomposition into plus and minus eigenlines uses $1/2$, so it is asserted here over E , including $E = \mathbf{Q}_2$, not indiscriminately over $\mathbf{Z}_2$.

For $p = 2$, the pullback along (CGT3.5) and CGT4 give

$$H^0(X_{\bar{k}}, E) = E, \quad F_q = 1; \quad H^1 = 0; \quad H^2(X_{\bar{k}}, E) = E(-1), \quad F_q = q. \quad (\text{CGT5.3})$$

The q -power map sends ν to 0, whereas α fixes ν . They are different scheme maps. They nevertheless give the stated cohomology actions through the explicitly proved reduction map.

For every repetition number $m \geq 1$, the full alternating Frobenius trace is

$$\text{Tr}(F_q^m \mid H^\bullet) = \begin{cases} (1 + q^m) \text{Tr}(P_q^m), & p \neq 2, \\ 1 + q^m, & p = 2. \end{cases} \quad (\text{CGT5.4})$$

In the nonsplit odd case this is zero for odd m and $2(1 + q^m)$ for even m . In the split odd case it is $2(1 + q^m)$ for every m . These agree directly with the number of $\mathbf{F}_{q^m}$ -points: the projective coordinate gives $q^m + 1$ for each available root of $J^2 + 1$. No repetition or coefficient branch is dropped.

The cohomological weights here are 0 in degree 0 and 2 in degree 2, since the complex absolute values of the eigenvalues are $1 = q^{0/2}$ and $q = q^{2/2}$. There is no degree-one cohomology in this particular scheme.

CGT6. The signed absolute powers and the prime 2

The supplied signed absolute power map has $T \mapsto T^n$, $J \mapsto J^n$, and must preserve $\epsilon = -1$. It extends over the ring realization for odd n , because

$$(J^n)^2 = (-1)^n = -1. \quad (\text{CGT6.1})$$

The minus-chart compatibility is

$$(-J)^n = -J^n. \quad (\text{CGT6.2})$$

Its coordinate degree is n , and its coefficient action is identity for $n \equiv 1 \pmod{4}$ and conjugation for $n \equiv 3 \pmod{4}$. Hence its action on the two geometric component cohomologies is P_n in degree 0 and nP_n in degree 2. It commutes with α : on T , both pullback composites give $-T^{-n}$; on J , both give $-J^n$.

For even n , $J \mapsto J^n$ is not a homomorphism of this global signed presentation over $\mathbf{Z}$, since its squared image is 1, not -1 . The morphism b_n from CGT4 does exist with the coefficient J fixed, but it is not that signed source power map. In characteristic 2, the Frobenius of the fibre also exists, including its action $\nu \mapsto 0$. These maps have different stated domains and coefficient actions. The cohomology at 2 in CGT5 is obtained from the fibre Frobenius, not by inventing an even signed source endomorphism.

CGT7. The actual wild specialization at 2

Let

$$S = \text{Spec } \mathbf{Z}_2, \quad S' = \text{Spec } \mathbf{Z}_2[i], \quad Y = \mathbf{P}_{S'}^1, \quad g : Y \rightarrow S. \quad (\text{CGT7.1})$$

The polynomial of $\pi = 1 - i$ is

$$\pi^2 - 2\pi + 2 = 0. \quad (\text{CGT7.2})$$

It is Eisenstein at 2, so $\mathbf{Q}_2(i)/\mathbf{Q}_2$ is a totally ramified quadratic extension, with uniformizer π , residue field $\mathbf{F}_2$, and

$$2 = i\pi^2, \quad c(\pi) = 1 + i = i\pi. \quad (\text{CGT7.3})$$

It is wild because its ramification degree is the residue characteristic. The quotient of inertia acting on its two embeddings is the order-two group generated by $i \mapsto -i$. This action is retained.

Let $\ell \neq 2$. Over a geometric generic point, $Y_{\bar{\eta}}$ is the disjoint union of two projective lines, indexed by the embeddings $i \mapsto i$ and $i \mapsto -i$. Over a geometric special point its coefficient ring is $\bar{\mathbf{F}}_2[\nu]/(\nu^2)$. Hence

$$\begin{aligned} H^0(Y_{\bar{s}}, E) &= E, & H^2(Y_{\bar{s}}, E) &= E(-1), \\ H^0(Y_{\bar{\eta}}, E) &= E^2, & H^2(Y_{\bar{\eta}}, E) &= E(-1)^2. \end{aligned} \quad (\text{CGT7.4})$$

All degree-one groups vanish.

The actual specialization maps are

$$\text{sp}_0 : a \mapsto (a, a), \quad \text{sp}_2 : a h_s \mapsto (a h_+, a h_-), \quad (\text{CGT7.5})$$

where the notation $h = c_1(\mathcal{O}(1))$ is interpreted in the twisted group $H^2(E(1))$ before untwisting. To prove these formulas, use the global unit section for degree zero and the global line bundle $\mathcal{O}_Y(1)$ for degree two. Their restrictions to each geometric generic component are respectively the unit and the degree-one hyperplane class. These generate all groups by CGT4. Proper base change defines the special-fibre stalk, and smooth proper specialization for $Y \rightarrow S'$ proves that the restriction of each generator to each generic branch is an isomorphism. This proves (CGT7.5), including its coefficient 1 on each branch.

For $\ell \neq 2$, inertia acts trivially on $\mathbf{Q}_\ell(1)$: every ℓ^r -th root of unity belongs to an unramified extension of $\mathbf{Q}_2$, by lifting the simple roots in the algebraic closure of the residue field. It therefore acts on (CGT7.4) solely by the wild branch swap

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (\text{CGT7.6})$$

The invariant subspaces are exactly the diagonals. Consequently the actual maps

$$\boxed{\text{sp}_j : H^j(Y_{\bar{s}}, E) \xrightarrow{\sim} H^j(Y_{\bar{\eta}}, E)^I, \quad j = 0, 1, 2} \quad (\text{CGT7.7})$$

are isomorphisms. In degree 1 both sides are zero. Their cokernels, the obstruction spaces for this specified specialization problem, are zero.

The anti-diagonal generic class $(a, -a)$ is not discarded. It is the sign representation of the wild inertia quotient and is not invariant unless $a = 0$. Over E it is the kernel of the sum map, complementary to the image in (CGT7.5). The factor 2 in the sum of a diagonal vector, $(a, a) \mapsto 2a$, is retained. Thus the disappearance of a second geometric special component is represented by the nonreduced coefficient fibre and this explicit ramified representation, not by silently merging two generic basis vectors.

Residual geometric Frobenius acts on the invariant degree-zero line by 1 and on the invariant degree-two line by 2. Any two lifts of residual Frobenius differ by inertia, so these values are independent of the chosen lift. On the full generic two-component space the branch permutation of a lift may differ; we make no choice-independent assertion about that extra permutation.

This is a calculation in a mixed-characteristic model smooth over S' , not an assertion that $g : Y \rightarrow S$ is smooth. Indeed the nonreduced special fibre shows that g is not smooth. The equal-characteristic hypotheses of Deligne's §3.6 are not silently substituted for the actual hypotheses here.

CGT8. The full proper pushforward before taking invariants

Fix ℓ and put

$$B = \text{Spec } \mathbf{Z}[1/\ell], \quad C = \text{Spec}(\mathbf{Z}[1/\ell, J]/(J^2+1)), \quad f : C \rightarrow B, \quad p : X = \mathbf{P}_C^1 \rightarrow C, \quad g = f \circ p. \quad (\text{CGT8.1})$$

The unit and hyperplane class define a map in the étale derived category

$$\mathbf{Q}_{\ell, C} \oplus \mathbf{Q}_{\ell, C}(-1)[-2] \longrightarrow Rp_* \mathbf{Q}_{\ell, X}. \quad (\text{CGT8.2})$$

The first summand is the adjunction unit. The second is the morphism represented by $c_1(\mathcal{O}_X(1)) \in H^2(X, \mathbf{Q}_\ell(1))$, with the twist and cohomological shift explicitly displayed.

At every geometric point of C , proper base change identifies (CGT8.2) with the unit and hyperplane maps of CGT4. They are isomorphisms in degrees 0 and 2, and both sides have no other cohomology. A map of complexes inducing an isomorphism on all cohomology stalks is a quasi-isomorphism. Therefore (CGT8.2) is an isomorphism. One may first carry out the same construction with $\mathbf{Z}/\ell^r$ coefficients; the computed groups have surjective transitions, so passage to $\mathbf{Z}_\ell$ and then $\mathbf{Q}_\ell$ preserves these computations.

The finite morphism f has no higher direct images on these coefficients. Proper base change computes a stalk as cohomology of a finite geometric fibre; each such fibre is a disjoint union of spectra of Artinian local algebras over an algebraically closed field. Reduction and étale topological invariance identify its cohomology with that of a finite discrete set, which has no positive cohomology. This proves the vanishing by stalks, including the ramified fibre.

Applying $Rf_* = f_*$ to (CGT8.2) yields the full comparison

$$\boxed{Rg_* \mathbf{Q}_\ell \simeq f_* \mathbf{Q}_\ell \oplus f_* \mathbf{Q}_\ell(-1)[-2]}. \quad (\text{CGT8.3})$$

No branch has been removed in (CGT8.3). The sheaf $f_*\mathbf{Q}_\ell$ has geometric stalk dimension 2 away from 2, and dimension 1 at 2 when $2 \in B$; its specialization is the diagonal map just calculated.

The involution acts by coefficient conjugation on the first summand. It has the same coefficient action on the second, because the projective part pulls back $\mathcal{O}(1)$ to $\mathcal{O}(1)$. Thus the decomposition is α -equivariant.

CGT9. The invariant sheaf, with the ramified stalk retained

Define the unit map

$$d : \mathbf{Q}_{\ell,B} \longrightarrow f_*\mathbf{Q}_{\ell,C}. \quad (\text{CGT9.1})$$

At an odd geometric residue characteristic its stalk is $a \mapsto (a, a)$. Conjugation interchanges the two entries, so its image is exactly the invariant subspace. At residue characteristic 2, its stalk is $a \mapsto a$ on the single-point étale receiver of the nonreduced ring. Conjugation acts trivially there. The same two-entry calculation applies at a geometric characteristic-zero point.

It follows on every geometric stalk that

$$\boxed{\mathbf{Q}_{\ell,B} \xrightarrow{\sim} (f_*\mathbf{Q}_{\ell,C})^\alpha.} \quad (\text{CGT9.2})$$

This is a sheaf isomorphism, not solely a fibrewise dimension comparison, because it is induced by the single map (CGT9.1).

Since 2 is invertible in $\mathbf{Q}_\ell$, the projector

$$e_+ = (1 + \alpha)/2 \quad (\text{CGT9.3})$$

is defined on rational ℓ -adic sheaves and makes taking invariants exact. This includes $\ell = 2$ with rational coefficients; it is not a statement about exact invariants of integral 2-adic modules. Apply the projector to the equivariant decomposition (CGT8.3). The result is the proved derived comparison

$$\boxed{(Rg_*\mathbf{Q}_\ell)^\alpha \simeq \mathbf{Q}_{\ell,B} \oplus \mathbf{Q}_{\ell,B}(-1)[-2].} \quad (\text{CGT9.4})$$

The complementary sheaf $(f_*\mathbf{Q}_\ell)^-$ is retained as the kernel of $1 + \alpha$. Its geometric stalk is the anti-diagonal at odd primes and at the generic point, and zero at 2. Thus (CGT9.4) records a projection with an explicitly described complementary receiver. It does not identify the original scheme or the full pushforward with the invariant part.

CGT10. The resulting full Euler products in their convergence domain

For every prime $p \neq \ell$ and every $m \geq 1$, (CGT9.4) gives the invariant alternating trace

$$1 + p^m. \quad (\text{CGT10.1})$$

With a formal variable u , its local factor is

$$\exp\left(\sum_{m \geq 1} \frac{1 + p^m}{m} u^m\right) = \frac{1}{(1 - u)(1 - pu)}. \quad (\text{CGT10.2})$$

This retains both cohomological degrees and every repetition. The equality follows from the convergent power series for $-\log(1 - z)$ near zero, or as a formal power-series identity with constant term 1.

To include $p = \ell$, compute that fibre with a different coefficient prime $\ell' \neq p$. CGT4-CGT5 give the same integer traces $1 + p^m$; hence the local polynomial is independent of this choice. This procedure retains the entire prime spectrum and does not place a missing-prime factor into an unspecified unit.

On the explicit domain $\Re(s) > 2$, all prime-repetition series are absolutely convergent, and $u = p^{-s}$ gives

$$\begin{aligned} \prod_p \frac{1}{(1 - p^{-s})(1 - p^{1-s})} &= \exp \left(\sum_p \sum_{m \geq 1} \frac{p^{-ms} + p^{m(1-s)}}{m} \right) \\ &= \left(\sum_{a \geq 1} a^{-s} \right) \left(\sum_{b \geq 1} b^{1-s} \right) \\ &= \boxed{\zeta(s)\zeta(s-1)}. \end{aligned} \tag{CGT10.3}$$

The second equality is the Euler product using the already recovered arithmetic and unique factorization; it is not a construction of arithmetic from a finite prime selection. The unit terms $a = b = 1$ and the factor at 2 are explicit. These are the original zeta functions; no completion, Gamma multiplier, heat rescaling, or endpoint cancellation has been introduced.

For comparison, the full unprojected object of CGT8 has local factors

$$\begin{cases} ((1 - p^{-s})^2(1 - p^{1-s})^2)^{-1}, & p \equiv 1 \pmod{4}, \\ ((1 - p^{-2s})(1 - p^{2(1-s)}))^{-1}, & p \equiv 3 \pmod{4}, \\ ((1 - 2^{-s})(1 - 2^{1-s}))^{-1}, & p = 2. \end{cases} \tag{CGT10.4}$$

These follow directly by taking determinants of the matrices in CGT5. They are recorded alongside the invariant factors so that projection has a visible effect on the Euler product.

CGT11. The exact lifting result and the boundary of this calculation

The specialization problem solved here is (CGT7.7): lift every inertia-invariant geometric generic class of the realized CC projective line from its geometric special fibre at 2. The maps are written in (CGT7.5), their images are computed in (CGT7.6), and their cokernels are zero. This proof retains the wild branch permutation and the nilpotent coefficient fibre.

The Tate factor in CGT4-CGT10 is obtained from the actual class $c_1(\mathcal{O}(1))$, its Kummer boundary, and pullback of the degree- q morphism. No character is assigned to an arbitrary analytic vector or to τ .

There is no proved identification here of the original-zeta zero-jet sheaf, the full timing reconstruction, or the user's desired lifting map with this $Rg_* \mathbf{Q}_\ell$. In particular, the weight-0 and weight-2 summands and the vanishing H^1 of this concrete projective line are exactly what the calculation gives. Calling that a weight-1 zero space or a proof of RH would add a conclusion not established by any map in this file.

The comparison is nevertheless explicit: it evaluates the supplied signed CC charts, retains their involution, derives a geometric Tate factor, proves the ramified specialization lift, and recovers the full original product $\zeta(s)\zeta(s-1)$ after the specified invariant projection. The factors and the complementary branch sheaf remain available for any further comparison.

CGT12. Diagram of the actual maps

The following diagram records the maps of CGT7. The left column is the geometric special fibre's cohomology; the middle column is the full geometric generic cohomology; the right column records its actual inertia invariants. In degree 2, the displayed $E(-1)$ is produced by CGT4.

degree	$H^j(Y_{\bar{s}}, E)$	$H^j(Y_{\bar{\eta}}, E)$	$H^j(Y_{\bar{\eta}}, E)^I$	
0	$E \xrightarrow{a \mapsto (a, a)}$	E^2	$\{(a, a) : a \in E\}$	(CGT12.1)
2	$E(-1) \xrightarrow{a \mapsto (a, a)}$	$E(-1)^2$	$\{(a, a) : a \in E(-1)\}$.	

The unit and hyperplane comparisons have the respective derived types $E \rightarrow R\Gamma(\mathbf{P}^1, E)$ and $E(-1)[-2] \rightarrow R\Gamma(\mathbf{P}^1, E)$, as in (CGT8.2). There is no asserted unshifted equivariant map $E \rightarrow E(-1)$. The horizontal arrows in (CGT12.1) are the specialization morphisms; their images are exactly the invariant subspaces in the right column.

The geometric degree class and the continuous global lift – GTZ0-GTZ9

Proof source: CC_GYSIN_TO_GLOBAL_SHIFTED_ZETA.md.

The CC hyperplane class and the continuous global zeta lift

24 September 2026. Complete comparison proof; locators GTZ0-GTZ9.

GTZ0. Prerequisites, notation, and the object being compared

The supporting datum remains $\tau\langle Z_1; \text{no } Z_2 \rangle$, in the user's notation. This note introduces no addition at τ , numerical coordinate on it, or identification with an integer. The complete arithmetic reconstruction precedes every integer, prime, complex scalar, function space, and cohomology group used below.

The full relevant user corrections were read in the private corpus, including the complete-history-before-unit argument, the rejection of a Gaussian-integer example as a competing foundation, and the instruction to quotient branches separately. The governing route is `argument_reconstruction/CORPUS_AND_OPERATION_RULE.md`. Those private source records are not part of a public mathematical edition.

The input arithmetic is the complete reconstructed arithmetic of CG0-CG8 and BQC0-BQC6. Its original Riemann zeta function is denoted ζ . We do not reconstruct it from two selected primes. The supplied Connes-Consani charts are evaluated after that input is present. Their Gaussian coefficient ring in CGT1 is this particular chart receiver, not an alternative origin for the arithmetic.

We compare two already constructed objects:

1. The continuous quotient of the full Mellin source constructed in GSL0-GSL11, retaining all actual nontrivial zeros and their multiplicities.
2. The cohomology of the actual complex CC twistor line with that quotient as constant coefficients. Its hyperplane class is geometrically constructed.

The resulting isomorphism is in continuous complex linear representations of the CC odd monoid. It is not asserted to be an isomorphism of algebras, a complex-to- ℓ -adic coefficient identification, or a construction of the original coefficient space from the projective line alone. Those types will be visible in every map.

Source reading:

- Connes-Consani, *The Absolute Twistor Line and the Geometry of the compactified Spec Z*, arXiv:2609.00299v1.

The original CC.tex was read especially for the chart restrictions, geometric involution, and Proposition "The odd arithmetic monoid acts on the twistor line," lines1159-1195. The supplied file has SHA256 57a10dcef5cd758e6b2f1c54b1c0a11b0fb093638f74fb1be515ec3bc7080ba4.

- Deligne, *La conjecture de Weil. II*, IHÉS52 (1980), printed213-214, §3.6; printed217 and221, chapterIV's coefficient choices; printed248-249, §6.2.9. The entire original article, printed137-252, has now been read, with coverage recorded separately from checks of individual formulas. [Original article](#).

- GLOBAL_SHIFTED_ZETA_LIFT.md, GSL0-GSL11, including the full continuous CRT proof and the higher-weight addition GSL6.5-GSL6.11, read completely.
- CC_GEOMETRIC_TATE_INDEPENDENT.md, CGT0-CGT12, read completely; it proves the signed chart realization, all finite-field fibres, Kummer class, ramified specialization, proper pushforward and invariant Euler factors.

These source comparisons do not ascribe the new map below to Deligne or Connes-Consani.

GTZ1. The complete coefficient spaces and their operations

Let $\mathcal{B}$ be the Fréchet space of entire functions with all seminorms

$$b_{A,M}(F) = \sup_{|\Re s| \leq A} (1 + |\Im s|)^M |F(s)| < \infty, \quad A, M \in \mathbf{Z}_{\geq 0}. \quad (\text{GTZ1.1})$$

Let $\mathcal{Z}$ be the distinct nontrivial zeros of the original ζ , with actual multiplicities m_ρ . The already proved boundary nonvanishing places them in $0 < \Re \rho < 1$. Define closed ideals

$$\begin{aligned} \mathcal{J} &= \{F : F^{(j)}(\rho) = 0 \text{ for every } \rho, 0 \leq j < m_\rho\}, \\ \mathcal{J}_+ &= \{F : F^{(j)}(\rho + 1) = 0 \text{ for every } \rho, 0 \leq j < m_\rho\}, \\ \mathcal{Q} &= \mathcal{B}/\mathcal{J}, \quad \mathcal{Q}_+ = \mathcal{B}/\mathcal{J}_+, \quad \mathcal{E}_+ = \mathcal{B}/(\mathcal{J} \cap \mathcal{J}_+). \end{aligned} \quad (\text{GTZ1.2})$$

All quotients have their actual quotient Fréchet topologies. No unrestricted product of zero jets replaces them.

For every already reconstructed $a > 0$, the continuous coefficient operator is

$$T_a F(s) = a^s F(s), \quad LF(s) = sF(s). \quad (\text{GTZ1.3})$$

Multiplication by a^s preserves each ideal and is continuous, since its modulus is $a^{\Re s}$, bounded on each fixed vertical strip. Multiplication by s is continuous with one additional polynomial seminorm. They consequently induce the indicated operators on all quotients.

Write $UF(s) = F(s + 1)$ and $VF(s) = F(s - 1)$. Their continuity follows from

$$b_{A,M}(UF), b_{A,M}(VF) \leq b_{A+1,M}(F). \quad (\text{GTZ1.4})$$

The map V sends $\mathcal{J}$ to $\mathcal{J}_+$, and U reverses this map. Thus they induce mutually inverse maps $\bar{V} : \mathcal{Q} \rightarrow \mathcal{Q}_+$ and $\bar{U} : \mathcal{Q}_+ \rightarrow \mathcal{Q}$.

GSL4-GSL6 construct an entire multiplier E_+ of $\mathcal{B}$ with the full jet congruences

$$E_+ \equiv 1 \pmod{\mathfrak{m}_\rho^{m_\rho}}, \quad E_+ \equiv 0 \pmod{\mathfrak{m}_{\rho+1}^{m_\rho}}. \quad (\text{GTZ1.5})$$

Here congruences are in the local holomorphic rings at the named points. The construction uses all zeros simultaneously and proves polynomial strip growth, so multiplication by E_+ and $1 - E_+$ is continuous on $\mathcal{B}$. In particular these are not formal infinite sums of local projectors.

The resulting continuous maps are

$$\begin{aligned} \pi[F] &= [F]_{\mathcal{J}}, \\ j[h] &= [(1 - E_+(s))h(s - 1)]_{\mathcal{J} \cap \mathcal{J}_+}, \\ \sigma[f] &= [E_+(s)f(s)]_{\mathcal{J} \cap \mathcal{J}_+}. \end{aligned} \quad (\text{GTZ1.6})$$

They give the split exact row

$$0 \longrightarrow \mathcal{Q}(-1) \xrightarrow{j} \mathcal{E}_+ \xrightarrow{\pi} \mathcal{Q} \longrightarrow 0. \quad (\text{GTZ1.7})$$

The notation $\mathcal{Q}(-1)$ here specifies $T_a^{(-1)} = aT_a$, $L^{(-1)} = L + 1$, on the same underlying Fréchet vector space. Its compatibility with an actual geometric Tate class, rather than a declaration about a class on τ , is what is proved next.

GTZ2. The actual complex line and its degree action

The CC object has complex points

$$X(\mathbf{C}) = \mathbf{P}^1(\mathbf{C})_+ \sqcup \mathbf{P}^1(\mathbf{C})_-, \quad \alpha(z, J) = (-1/z, -J), \quad J \in \{i, -i\}. \quad (\text{GTZ2.1})$$

The full signed coordinate formula is retained. Its quotient by α is the complex line $M = \mathcal{C}_\infty(\mathbf{C})$. Each orbit has exactly one point with $J = i$, giving an isomorphism $M \simeq \mathbf{P}^1(\mathbf{C})$; the other representative is explicitly obtained by α .

For odd $n \geq 1$, the descended source morphism is

$$f_n(z) = \begin{cases} z^n, & n \equiv 1 \pmod{4}, \\ -1/z^n, & n \equiv 3 \pmod{4}. \end{cases} \quad (\text{GTZ2.2})$$

In homogeneous coordinates these are respectively $[X : Y] \mapsto [X^n : Y^n]$ and $[X : Y] \mapsto [-Y^n : X^n]$. Neither pair has a common projective zero. The first pulls the transition function of $\mathcal{O}(1)$ to its n -th power; the second is its composition with a projective linear automorphism. Hence

$$f_n^* \mathcal{O}(1) \simeq \mathcal{O}(n). \quad (\text{GTZ2.3})$$

Give M its complex orientation and let $h = c_1(\mathcal{O}(1)) \in H^2(M, \mathbf{C})$. This is the image of the integral first Chern class; its integral on M is 1, by the clutching function of degree one. Thus it retains a specified integral generator before scalar extension. The projective line has a CW decomposition with one cell in degree zero and one in degree two. Its cellular cochain complex is $\mathbf{C}$ in those two degrees and zero otherwise, with zero differential. It follows that

$$H^0(M, \mathbf{C}) = \mathbf{C} 1, \quad H^1(M, \mathbf{C}) = 0, \quad H^2(M, \mathbf{C}) = \mathbf{C} h. \quad (\text{GTZ2.4})$$

Naturality of c_1 and (GTZ2.3) prove

$$f_n^* 1 = 1, \quad f_n^* h = n h. \quad (\text{GTZ2.5})$$

The coordinate minus sign in (GTZ2.2) has not become a negative degree: its projective automorphism has degree +1.

The quotient and unquotiented calculation agree by actual pullback. If $q : X(\mathbf{C}) \rightarrow M$, then

$$q^* 1 = (1, 1), \quad q^* h = (h_+, h_-). \quad (\text{GTZ2.6})$$

These follow because the restriction of q to either component is a holomorphic projective automorphism. They identify $H^\bullet(M, \mathbf{C})$ with the α -invariant cohomology. The anti-diagonal component remains the complementary eigenspace; it has not been identified with zero.

GTZ3. The geometric receiver with the full original coefficients

Take constant cellular coefficients in the already constructed $\mathcal{Q}$. The cellular cochain complex is $\mathcal{Q}$ in degrees zero and two, with zero differential; all its maps are continuous. Its cohomology, with the finite product topology, is

$$\mathcal{H} = H^0(M; \mathcal{Q}) \oplus H^2(M; \mathcal{Q}) = \mathcal{Q} 1 \oplus \mathcal{Q} h. \quad (\text{GTZ3.1})$$

There is no completion of an infinite tensor product here: the geometric cochain complex has two cells.

Precisely, $x1 + yh$ denotes $x \otimes 1 + y \otimes h$ in $\mathcal{Q} \otimes_{\mathbf{C}} H^{\text{even}}(M, \mathbf{C})$, with its finite-product topology. Both 1 and h belong to the scalar geometric cohomology; neither notation presupposes a multiplicative identity in $\mathcal{Q}$. All coefficient products below are the products of the actual quotient algebra.

The joint coefficient and geometric action is

$$\mathcal{T}_n = T_n \otimes f_n^* : \quad x1 + yh \mapsto (T_n x) 1 + (nT_n y) h, \quad n \text{ odd}. \quad (\text{GTZ3.2})$$

It defines a continuous monoid representation. Indeed the source maps compose by $f_m f_n = f_{mn}$, and $T_m T_n = T_{mn}$; the displayed formula also verifies composition directly.

Define the geometric inclusion and projection

$$\iota_h : \mathcal{Q}(-1) \rightarrow \mathcal{H}, \quad y \mapsto yh, \quad \epsilon_0 : \mathcal{H} \rightarrow \mathcal{Q}, \quad x1 + yh \mapsto x. \quad (\text{GTZ3.3})$$

Their types retain degree two and its degree character; ι_h is equivariant by (GTZ2.5). The section $x \mapsto x1$ is equivariant. The class h is also the Gysin image of the oriented point class in M : its Poincaré dual integrates to the point's multiplicity one, and the cellular H^2 has precisely that integral generator. In the arithmetic étale formulation the corresponding map is $\mathbf{Q}_\ell(-1)[-2] \rightarrow Rp_* \mathbf{Q}_\ell$, constructed by the Kummer Chern class in CGT4 and CGT8. Thus the degree and twist have an actual geometric source.

GTZ4. The exact map and its inverse

Define

$$\Theta : \mathcal{E}_+ \longrightarrow \mathcal{H}, \quad [F] \mapsto [F]_{\mathcal{J}} 1 + [F(s+1)]_{\mathcal{J}} h. \quad (\text{GTZ4.1})$$

This is well defined. Changing F by $D \in \mathcal{J} \cap \mathcal{J}_+$ changes the first component by an element of $\mathcal{J}$; since D has all required jets zero at $\rho + 1$, $D(s+1)$ has those jets zero at ρ , so the second component is unchanged as well. Quotient maps and translation are continuous, hence so is Θ .

For $x = [f]_{\mathcal{J}}$, $y = [g]_{\mathcal{J}}$, define

$$\Xi(x1 + yh) = [E_+(s)f(s) + (1 - E_+(s))g(s-1)]_{\mathcal{J} \cap \mathcal{J}_+}. \quad (\text{GTZ4.2})$$

This is independent of both representatives. If f changes by an element of $\mathcal{J}$, its product with E_+ vanishes to the required orders on the original divisor and, by (GTZ1.5), on the shifted divisor. If g changes by an element of $\mathcal{J}$, its translate by V vanishes on the shifted divisor; multiplication by $1 - E_+$ supplies vanishing on the original divisor. The induced map is continuous by the quotient property, the continuous multiplier bounds, and (GTZ1.4).

At every original zero the expression in (GTZ4.2) has exactly the full jet of f . At every shifted zero $\rho + 1$ it has exactly the full jet of $g(s-1)$. Hence $\Theta \Xi(x1 + yh) = x1 + yh$. Conversely, the expression $\Xi \Theta[F] - [F]$ has all required jets zero at both complete divisors, and therefore belongs to the defining ideal $\mathcal{J} \cap \mathcal{J}_+$. This proves $\Xi \Theta = 1$.

Thus Θ is a continuous linear isomorphism with the stated continuous inverse. The argument compares the actual quotient classes, not merely a finite set of their observations.

GTZ5. Equivariance, the entire lifting row, and every jet

For every odd n , its first component satisfies $[n^s F] = T_n[F]$. Its second component is

$$[(n^s F)(s+1)]_J = [n^{s+1} F(s+1)]_J = n T_n[F(s+1)]_J. \quad (\text{GTZ5.1})$$

Equations (GTZ3.2) and (GTZ5.1) prove

$$\Theta T_n = \mathcal{T}_n \Theta. \quad (\text{GTZ5.2})$$

The factor n is exactly the degree factor already proved on h .

The maps of the entire extension agree:

$$\Theta j(y) = y h, \quad \epsilon_0 \Theta = \pi, \quad \Theta \sigma(x) = x 1. \quad (\text{GTZ5.3})$$

The first identity uses both congruences in (GTZ1.5), with the translation $UV = 1$; the other two use the same congruences. Thus the following is a diagram of split exact continuous equivariant rows:

$$\begin{array}{ccccccc} 0 & \rightarrow & \mathcal{Q}(-1) & \xrightarrow{j} & \mathcal{E}_+ & \xrightarrow{\pi} & \mathcal{Q} \rightarrow 0 \\ & & \parallel & & \downarrow \Theta & & \parallel \\ 0 & \rightarrow & \mathcal{Q}(-1) & \xrightarrow{\iota_h} & \mathcal{H} & \xrightarrow{\epsilon_0} & \mathcal{Q} \rightarrow 0. \end{array} \quad (\text{GTZ5.4})$$

In particular the analytic continuous section is sent to the geometric degree-zero section, while the kernel is sent to the hyperplane summand.

Every multiplicity remains visible. At an original zero ρ , write its full jet coordinate as $t = s - \rho$, taken modulo t^{m_ρ} . The coefficient action is

$$n^{\rho+t} = n^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log n)^j}{j!} t^j. \quad (\text{GTZ5.5})$$

The shifted jet has coordinate $t = s - (\rho + 1)$. Translation U identifies these coordinates with the same t , without a sign or factorial change. Its action is

$$n^{\rho+1} \sum_{j=0}^{m_\rho-1} \frac{(\log n)^j}{j!} t^j = n \left(n^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log n)^j}{j!} t^j \right). \quad (\text{GTZ5.6})$$

This is precisely (GTZ5.1) on the entire jet. No zero simplicity or RH has been used.

The analytic row has operators for all positive real a . Its geometric identification here is asserted for the actual CC odd monoid. The absence of an even signed endomorphism in that monoid does not remove the prime 2 from $\mathcal{Q}$: the coefficient object was constructed from the full original ζ . The fibre Frobenius at 2 and its separate geometric Tate factor are explicitly constructed in CGT5-CGT7.

GTZ6. The original zeta product and all exceptional contributions

The finite-prime calculation of CGT8-CGT10 gives an independent exact comparison to the same shift. The α -invariant proper pushforward is

$$(Rg_* \mathbf{Q}_\ell)^\alpha \simeq \mathbf{Q}_\ell \oplus \mathbf{Q}_\ell(-1)[-2]. \quad (\text{GTZ6.1})$$

Its full trace at prime p , repetition m , is $1 + p^m$. At $p = 2$ this is obtained from the actual nonreduced fibre and its reduction map; at $p = \ell$ the compatible local polynomial is calculated with a different

coefficient prime. Therefore

$$\prod_p \frac{1}{(1-p^{-s})(1-p^{1-s})} = \exp \left(\sum_p \sum_{m \geq 1} \frac{p^{-ms} + p^{m(1-s)}}{m} \right) \quad (\text{GTZ6.2})$$

$$= \zeta(s)\zeta(s-1), \quad \Re s > 2.$$

Both original factors, all repetitions, and both unit terms remain. This geometric Euler product is not identified with a determinant of $\mathcal{Q}$; such a determinant has not been constructed in this note.

The analytic full source functions used to implement the two nontrivial zero divisors are

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad F_{-1}(s) = \frac{(s-1)(s-2)}{8} \pi^{-(s-1)/2} \Gamma((s-1)/2) \zeta(s-1). \quad (\text{GTZ6.3})$$

Their product is the product of both displayed multipliers times both original zeta functions. It is not a replacement definition of those original functions. The meromorphic product $\zeta(s)\zeta(s-1)$ has:

- nontrivial zero divisors (ρ, m_ρ) and $(\rho+1, m_\rho)$;
- simple trivial zeros at every $-2r$ and $1-2r$, $r \geq 1$;
- simple poles at $s=1$ and $s=2$.

There are no cancellations within this list: at ρ , the factor $\zeta(s-1)$ is a unit, by the functional equation and Euler nonvanishing in the reflected half-plane; at $\rho+1$, $\zeta(s)$ is a unit by Euler nonvanishing. At the listed trivial zeros the other zeta factor is nonzero; at the poles the other factor is respectively $\zeta(0) = -1/2$ and $\zeta(2) \neq 0$. The trivial zeros are simple by the full functional equation and the nonzero positive-half-plane value.

The full comparison retains

$$F_0(0) = F_0(1) = \frac{1}{8}, \quad F_{-1}(1) = F_{-1}(2) = \frac{1}{8}, \quad (\text{GTZ6.4})$$

and, for every $r \geq 1$,

$$\begin{aligned} F_0(-2r) &= \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) \\ &= \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r), \\ F_{-1}(1-2r) &= F_0(-2r), \quad F_{-1}(2+2r) = F_0(1+2r). \end{aligned} \quad (\text{GTZ6.5})$$

These nonzero values record the cancellation by the displayed multipliers; they are not zeros of the quotient divisor. For every derivative order d , if $K(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)/8$, then

$$F_0^{(d)}(s) = \sum_{j=0}^d \binom{d}{j} K^{(d-j)}(s) \zeta^{(j)}(s), \quad F_{-1}^{(d)}(s) = F_0^{(d)}(s-1). \quad (\text{GTZ6.6})$$

At exceptional points these identities are interpreted by the meromorphic product's holomorphic continuation, as (GTZ6.4)–(GTZ6.5) explicitly compute. Thus no endpoint, Gamma derivative, multiplicity or constant is suppressed by the map.

GTZ7. The multiplication transported by the comparison

The map Θ is not claimed to respect the usual cup product on $H^\bullet(M; \mathcal{Q})$. There is an exact relation to it.

The quotients of $\mathcal{B}$ are commutative algebras under function multiplication. Translation U respects multiplication. Therefore the product transported from $\mathcal{E}_+$ to the coordinates of $\mathcal{H}$ is

$$(x1 + yh) \diamond (x'1 + y'h) = (xx')1 + (yy')h. \quad (\text{GTZ7.1})$$

This follows directly by applying Θ to representatives FG , at the two divisors. With $\diamond$, Θ is an algebra isomorphism.

The geometric cup product is instead

$$(x1 + yh) \smile (x'1 + y'h) = (xx')1 + (xy' + yx')h, \quad (\text{GTZ7.2})$$

because $h \smile h = 0$ in degree four of the two-dimensional sphere and 1 is the degree-zero identity in its geometric cohomology. The exact difference, in this already additive coefficient space, is

$$(u \diamond v) - (u \smile v) = (yy' - xy' - yx')h. \quad (\text{GTZ7.3})$$

Thus the continuous representation comparison is fully specified, including the extra multiplicative structure it does and does not preserve. It does not identify an idempotent divisor projector with a square-zero cohomology class as algebras.

GTZ8. What is proved about lifting, and what remains active

GTZ4–GTZ5 construct an actual continuous isomorphism of the whole original-zeta shifted extension with the cohomology of the CC complex line carrying the already constructed $\mathcal{Q}$. Its kernel's factor n is the pullback degree of the geometric hyperplane class, and its entire section corresponds to degree zero. CGT7 separately proves the actual ramified invariant-cycle lift at 2, by $a \mapsto (a, a)$.

Deligne's local invariant-cycle proof in §3.6, and its pure-complex version §6.2.9, involve a localization cross and a dual support-cohomology obstruction. GTZ5.4 is a proved comparison to this particular cohomology receiver; it does not yet identify that localization cross with the user's desired τ -supported lifting construction. In particular $\mathcal{Q}$ is an input coefficient space here, and its spectral purity has not been obtained from the purity of the projective line's geometric Tate class.

No conditional theorem replaces that calculation. The concrete result is the map Θ , its inverse, all equivariance and full jet identities, and the two geometric rows just proved. The active research goal remains to construct and calculate the actual receiving localization/support maps for the programme, using the complete source and the user's prerequisite order.

GTZ9. Proof dependencies and illustration

The complete continuous multiplier and CRT estimates are in GSL4–GSL6; this note uses that proved multiplier, not an assumed separator. The original Mellin source identification and its topological quotient are S5 and RZ1–RZ4, with source formulas retained in GSL8. The geometric integral Chern class calculation and finite-prime realizations are CGT4, CGT7–CGT10. The complete comparison between these inputs is GTZ1–GTZ7 above.

The accompanying figure is a diagram of these exact maps and the ramified specialization. Its labels retain the shift, degree, twist and coefficient spaces. It does not depict a proved purity statement for $\mathcal{Q}$.

The independent mathematical check is GTZ_INDEPENDENT_MATHEMATICAL_CHECK.md, GTC0–GTC5. It checked the entire map, its topology and multiplication types, all exceptional-value factors, and the full GSL multiplier mechanism. Its tensor-notation clarification is incorporated in GTZ3.

The geometric Tate factor in the global zeta lift

All arithmetic is already reconstructed. The supporting datum $\tau(Z_1; \text{no } Z_2)$ receives no new operation.

A. The entire continuous lifting row

$$\begin{array}{ccccccc}
 0 \rightarrow \mathcal{Q}(-1) & \xrightarrow{j} & \mathcal{E}_+ = \mathcal{B}/(\mathcal{I} \cap \mathcal{I}_+) & \xrightarrow{\pi} & \mathcal{Q} = \mathcal{B}/\mathcal{I} & \rightarrow & 0 \\
 \parallel & & \downarrow \Theta[F] = [F]_I 1 + [F(s+1)]_I h & & \parallel & & \\
 0 \rightarrow \mathcal{Q}(-1) & \xrightarrow{y \mapsto yh} & \mathcal{Q} 1 \oplus \mathcal{Q} h & \xrightarrow{\varepsilon_0} & \mathcal{Q} & \rightarrow & 0
 \end{array}$$

$M = \mathcal{M}_\infty(\mathbf{C}) \simeq \mathbf{P}^1(\mathbf{C}), \quad h = c_1(\mathcal{O}(1)), \quad f_n^* h = nh \quad (n \text{ odd}).$

GTZ4-GTZ5: continuous inverse; exact section; every zero jet and multiplicity retained.

B. Ramified specialization at 2

$$\bar{\mathbf{F}}_2[\nu]/(\nu^2)$$

The nonreduced fibre remains explicit.

$$E(-1) \xrightarrow{\text{sp}_2} \{(a, a)\} \subset E(-1)^2$$

$$a \mapsto (a, a), \quad I: (x, y) \mapsto (y, x)$$

$$\text{image}(\text{sp}_2) = H_{\bar{\eta}}^{2I}$$

CGT7. $E = \mathbf{Q}_\ell, \ell \neq 2$. The cokernel is zero.

C. Every finite prime and repetition

$$\text{Tr}(F_p^m | H_{a=1}^\bullet) = 1 + p^m$$

$$\prod_p \frac{1}{(1-p^{-s})(1-p^{1-s})} = \zeta(s)\zeta(s-1)$$

$$\Re(s) > 2, \quad m \geq 1, \quad \text{including } p=2.$$

CGT8-CGT10 retain both degrees and all local factors.

The global analytic and geometric maps use the same +1 shift.

Proofs: GTZ0-GTZ9 and CGT0-CGT12. Sources: Connes-Consani (2609.00299v1), Deligne, Weil II §3.6 and §6.2.9.

The coefficient space $\mathbf{Q}$ is an input here. Its purity and the identification with the programme's full localization cross remain unproved.

Figure 24: The full continuous comparison and the geometric source of its degree factor. Panel A is GTZ5.4, with Theta and its inverse proved in GTZ4. Panel B is the actual ramified specialization of CGT7. Panel C is the all-prime Euler product of CGT10. $\mathbf{Q}$ is already constructed input data.

The descended conic, ramified fibre, and full original zeta product — CD0-CD12

Proof source: CC_DESCENT_CONIC_AND_ZETA.md.

The descended CC conic, its integral fibre, and its full zeta function

Independent mathematical derivation, 24 September 2026. Proof locators CD0-CD12.

The concrete results are an explicit descent isomorphism over $\mathbf{Z}[1/2, i]$, an integral conic retaining its nonreduced fibre at 2, and the complete arithmetic Euler product

$$Z(\mathcal{C}, s) = \zeta(s)\zeta(s-1), \quad \Re s > 2. \quad (\text{CD0.1})$$

The two functions on the right are the original Riemann zeta function at the displayed arguments. The translated factor has a geometric origin in the degree-two cohomology of the descended conic. The integral model at 2 is examined separately from finite étale descent.

CD0. Input stage, source use, and operation check

The support remains the user's $\tau\langle Z_1; \text{no } Z_2 \rangle$. The coefficient ring $\mathbf{Z}$, all its primes and its original zeta function are used only after the complete arithmetic reconstruction. None of the projective coordinates below is a coordinate on that supporting datum. No addition, norm, metric or vector is assigned to τ . The coefficient extension adjoining i evaluates the supplied signed CC charts; it is not an alternative arithmetic offered as an RH counterexample. Separate branch return measures are not pooled.

The current `READ_FIRST_USER_CONSTRUCTION.md` and `USER_ARGUMENT_RECONSTRUCTION.md`, including the correction-precedence table, were read before the construction, together with `PC01-PC03` of `PREREQUISITE_AND_OBSTRUCTION_CHECKS.md`. The operations checked here are the explicit CC chart evaluation, finite étale descent of its projective action, an integral model, its finite-field point counts, and its coefficient cohomology. Their prerequisites are proved below or cited at their exact previously proved source locations. None requires an exact eigenfunction lift in the analytic source.

The existing source-use coverage in [CC_GEOMETRIC_TATE_INDEPENDENT.md](#), CGT0-CGT12, was read. That witness's hash at reading was `9C296F9F590D842B1C593F530C12CBCC6E6FE302F354C6170CF18CB050FFC960`. The original author source read for the present comparison was Alain Connes and Caterina Consani, [The Absolute Twistor Line and the Geometry of the compactified Spec Z](#), CC.tex, lines 484-604 and 773-848, containing the three-point support, ordered charts, signed Bombelli overlap, and full involution. The source hash is `57A10DCEF5CD758E6B2F1C54B1C0A11B0FB093638F74FB1BE515EC3BC7080BA4`.

The exact starting ring realization from CGT1-CGT2 is

$$X_R = \mathbf{P}_{R[i]}^1, \quad i^2 = -1, \quad \alpha([X : Y], i) = ([-Y : X], -i). \quad (\text{CD0.2})$$

This is the explicit evaluation of the original restrictions $J_+ \mapsto J$, $J_- \mapsto -J$, with global coefficient $J = i$; it does not erase either chart. On the overlap, $\alpha^*T = -T^{-1}$, $\alpha^*i = -i$. The statement is semilinear over $R[i]$ and linear over R .

For cohomology only, CGT4, CGT8 and CGT9 supply the previously read Kummer, nilpotent-invariance and proper-base-change calculations. Their underlying primary-source locators are the Stacks Project [03RQ](#), [03SI](#), and [095T](#), at the author-source commit recorded in CGT0. Those Stacks files were not newly reread for this note; the exact receiver calculations from CGT are the input. The descent and point counts in CD1–CD9 are derived here.

CD1. The actual finite étale cover and its cocycle

Put

$$R = \mathbf{Z}[1/2], \quad A = R[i] = R[t]/(t^2 + 1), \quad c(i) = -i, \quad c|_R = 1. \quad (\text{CD1.1})$$

As an R -module, A is free on $1, i$. Since $2i$ is a unit, the derivative of its defining polynomial is a unit. Equivalently, the finite free presentation has $\Omega_{A/R} = A \, di / (2i \, di) = 0$; the elementary monogenic étale criterion proves that A/R is finite étale. It is faithfully flat because its free rank is two.

The complete split overlap is

$$A \otimes_R A \longrightarrow A \times A, \quad a \otimes b \longmapsto (ab, a \, c(b)). \quad (\text{CD1.2})$$

It is an isomorphism: as a polynomial algebra over the left copy of A , its two factors are $t - i$ and $t + i$, whose difference $2i$ is a unit. Explicit idempotents are

$$e_1 = \frac{1 - i \otimes i}{2}, \quad e_2 = \frac{1 + i \otimes i}{2}, \quad (\text{CD1.3})$$

with images $(1, 0)$ and $(0, 1)$. This proves directly that the semilinear involution supplies the two required components of the descent datum over the actual fibre product.

The projective transformation is represented by

$$M = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad M \, c(M) = M^2 = -I. \quad (\text{CD1.4})$$

Thus its square is the identity on projective space. The matrix square is retained: it is $-I$ on the rank-two vector presentation, not I . The projective cocycle condition is exactly the one required for (CD0.2).

CD2. The explicit descended conic and its inverse charts

Define the smooth projective R -scheme

$$\mathcal{C}_R = \text{Proj } R[x, y, z]/(x^2 + y^2 + z^2). \quad (\text{CD2.1})$$

Smoothness follows from the Jacobian criterion: on a geometric fibre, simultaneous vanishing of $2x, 2y, 2z$ would force all projective coordinates to be zero, because 2 is invertible.

Over A , define

$$\phi : \mathbf{P}_A^1 \longrightarrow (\mathcal{C}_R)_A, \quad [X : Y] \longmapsto [X^2 - Y^2 : i(X^2 + Y^2) : 2XY]. \quad (\text{CD2.2})$$

Its coordinates satisfy the full identity

$$(X^2 - Y^2)^2 + (i(X^2 + Y^2))^2 + (2XY)^2 = 0. \quad (\text{CD2.3})$$

There are no basepoints: on $D(X)$, the combination $x - iy$ pulls back to $2X^2$, a generator of the corresponding line-bundle fibre; on $D(Y)$, the combination $-x - iy$ pulls back to $2Y^2$. These two opens cover $\mathbf{P}_A^1$.

On the conic set

$$a = x - iy, \quad b = z, \quad d = -x - iy.$$

The defining equation gives

$$ad = b^2. \quad (\text{CD2.4})$$

The opens $D(a)$ and $D(d)$ cover the conic. Indeed a prime containing both contains x, y , since $2i$ is a unit; the equation then forces it to contain z , which is not a projective point. Define inverse maps on those opens by

$$\psi|_{D(a)} = [a : b] = [x - iy : z], \quad \psi|_{D(d)} = [b : d] = [z : -x - iy]. \quad (\text{CD2.5})$$

On their intersection (CD2.4) makes the two projective pairs equal. The displayed nonvanishing first or second coordinate makes each map well-defined.

Their composites with ϕ equal $[X : Y]$: the two pairs become $[2X^2 : 2XY]$ on $D(X)$ and $[2XY : 2Y^2]$ on $D(Y)$. For the other composite on $D(a)$, the three coordinate identities are

$$a^2 - b^2 = 2ax, \quad i(a^2 + b^2) = 2ay, \quad 2ab = 2az. \quad (\text{CD2.6})$$

The common factor $2a$ is invertible in the local trivialization on that open. On $D(d)$, substitution using $b^2 = ad$ gives $b^2 - d^2 = d(a - d) = 2dx$, $i(b^2 + d^2) = id(a + d) = 2dy$, and $2bd = 2dz$. Thus the common factor on this second chart is $+2d$, and the full formulas are

$$b^2 - d^2 = 2dx, \quad i(b^2 + d^2) = 2dy, \quad 2bd = 2dz. \quad (\text{CD2.7})$$

They prove the other composite as well. Consequently ϕ is an isomorphism, with the complete inverse (CD2.5).

CD3. Verification of descent, including the homogeneous sign

Apply the full semilinear action to the three quadratic sections in (CD2.2). Since it sends $X \mapsto -Y$, $Y \mapsto X$, $i \mapsto -i$, one obtains

$$\begin{aligned} \alpha^*(X^2 - Y^2) &= -(X^2 - Y^2), \\ \alpha^*(i(X^2 + Y^2)) &= -i(X^2 + Y^2), \quad \alpha^*(2XY) = -2XY. \end{aligned} \quad (\text{CD3.1})$$

The same sign occurs in all three coordinates. It acts trivially on their projective class. Accordingly ϕ intertwines α with coefficient conjugation on $(\mathcal{C}_R)_A$.

There is also an explicit graded-ring descent. The second Veronese ring of $A[X, Y]$ is

$$A[a, b, d]/(ad - b^2), \quad a = X^2, \quad b = XY, \quad d = Y^2,$$

where each of a, b, d has Veronese degree one. The invertible linear change

$$x = a - d, \quad y = i(a + d), \quad z = 2b \quad (\text{CD3.2})$$

has inverse $a = (x - iy)/2$, $d = (-x - iy)/2$, $b = z/2$, and sends

$$x^2 + y^2 + z^2 = -4(ad - b^2). \quad (\text{CD3.3})$$

On Veronese degree n , replace the linearized action by $\beta = (-1)^n \alpha^*$. This is a graded-ring automorphism, because degrees add, and its square is one. Equation (CD3.1) says that β fixes x, y, z and conjugates only their A -coefficients. Every element of the resulting ring has a unique expression $f + ig$ with f, g in $R[x, y, z]/(x^2 + y^2 + z^2)$. It is fixed exactly when $2ig = 0$, hence exactly when $g = 0$. Thus its invariant graded ring is precisely the ring in (CD2.1).

The factor $(-1)^n$ does not change the induced projective action. This calculation proves effective descent by the written rings and inverse charts; it does not merely identify point sets over a field. It also records why a sign in the vector presentation is consistent with an involution on the projective object.

After base change to $\mathbf{Q}$, the descended conic has no rational point: a rational solution of $x^2 + y^2 + z^2 = 0$, viewed in $\mathbf{R}$, has $x = y = z = 0$. Hence it is not $\mathbf{P}_{\mathbf{Q}}^1$, which has rational points. Nevertheless its base change to $\mathbf{Q}(i)$ is exactly (CD2.2). This is the nontrivial form dictated by the supplied semilinear action, with no new arithmetic base substituted for the recovered $\mathbf{Z}$.

The same sign has a vector-level explanation. Multiplying M in (CD1.4) by $b \in \mathbf{Q}(i)^\times$ changes its cocycle square to $-b c(b)I$. It could be $\bar{I}$ only if $b c(b) = -1$. Writing $b = u + iv$ with rational u, v gives $b c(b) = u^2 + v^2 > 0$, so this is impossible. This concerns the specified rank-two lift of this projective cocycle; projective descent itself has already been constructed.

CD4. The integral model and its retained fibre at 2

Take the explicit integral model

$$\mathcal{C} = \text{Proj } \mathbf{Z}[x, y, z]/(x^2 + y^2 + z^2), \quad f : \mathcal{C} \rightarrow \text{Spec } \mathbf{Z}. \quad (\text{CD4.1})$$

It restricts to the descended conic over R . It is projective. Its homogeneous coordinate ring is free over $\mathbf{Z}$, with basis

$$x^a y^b z^c, \quad a, b \geq 0, c \in \{0, 1\}, \quad (\text{CD4.2})$$

because division by the monic polynomial $z^2 + x^2 + y^2$ leaves a unique remainder of degree less than two in z . Localization preserves $\mathbf{Z}$ -torsion-freeness, as does taking the degree-zero submodule of each Proj chart. A torsion-free module over the principal ideal domain $\mathbf{Z}$ is flat: every finitely generated submodule is free, and the module is their filtered union; tensor product commutes with that union and filtered unions preserve exactness. Thus every affine chart is flat, proving that f is flat.

Over $\mathbf{F}_2$, the defining polynomial is

$$x^2 + y^2 + z^2 = (x + y + z)^2. \quad (\text{CD4.3})$$

Let $l = x + y + z$. Then the actual fibre and its reduction are

$$\mathcal{C}_{\mathbf{F}_2} = V(l^2), \quad L = V(l) \simeq \mathbf{P}_{\mathbf{F}_2}^1. \quad (\text{CD4.4})$$

An explicit isomorphism from $\mathbf{P}^1$ onto L is $[a : b] \mapsto [a : b : a + b]$, with inverse $[x : y : z] \mapsto [x : y]$. The nilpotent ideal is $(l)/(l^2)$, is square-zero, and is not zero. Globally it is $\mathcal{O}_L(-1)$; multiplication by the degree-one equation l identifies it with the restriction of that twist. Therefore the full sheaf sequence is

$$0 \rightarrow \mathcal{O}_L(-1) \rightarrow \mathcal{O}_{\mathcal{C}_{\mathbf{F}_2}} \rightarrow \mathcal{O}_L \rightarrow 0. \quad (\text{CD4.5})$$

This retains the nonreduced structure, rather than replacing the fibre by its reduction. The morphism is smooth away from 2, by CD2, and is not smooth at 2, since that fibre is not geometrically reduced.

CD5. Why the ramified integral quotient is a separate problem

Write $A_{\mathbf{Z}} = \mathbf{Z}[i]$. The ring map

$$A_{\mathbf{Z}} \otimes_{\mathbf{Z}} A_{\mathbf{Z}} \rightarrow A_{\mathbf{Z}} \times A_{\mathbf{Z}}, \quad a \otimes b \mapsto (ab, a c(b)) \quad (\text{CD5.1})$$

has matrix $\begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$ over the left coefficient ring, of determinant $-2i$. It is injective, but its image consists exactly of pairs (u, v) with $u - v \in 2A_{\mathbf{Z}}$. Indeed the preimage coefficients must be $b = (u - v)/(2i)$,

$a = u - ib$; the displayed condition is sufficient and necessary. Its cokernel is $A_{\mathbf{Z}}/2A_{\mathbf{Z}}$, via the difference of the two entries. Thus it is not an isomorphism at 2.

The finite flat cover at 2 is therefore not the C_2 -torsor used in CD1. A pair consisting of the identity and a semilinear involution does not by itself specify an isomorphism over this whole ramified fibre product. Finite étale descent cannot be applied there by using the two graphs as though they were a disjoint covering.

The actual quadratic formula (CD2.2) also fails to extend to a morphism $\mathbf{P}_{\mathbf{Z}[i]}^1 \rightarrow \mathcal{C}_{\mathbf{Z}[i]}$. At the prime $\pi = 1 + i$, it reduces to

$$[(X + Y)^2 : (X - Y)^2 : 0], \quad (\text{CD5.2})$$

which has the basepoint $[1 : 1]$. This failure is not removable by another presentation of the same generic map. Over the discrete valuation ring $D = \mathbf{Z}[i]_{(1+i)}$, the two sections

$$P = [1 : 1], \quad Q = [1 : i]$$

have the same closed specialization. Their generic images under (CD2.2), retaining the original factors before choosing integral representatives, are

$$\phi(P) = [0 : 2i : 2] = [0 : i : 1], \quad \phi(Q) = [2 : 0 : 2i] = [1 : 0 : i]. \quad (\text{CD5.3})$$

The last triples are valid D -points of the conic. They specialize to the distinct points $[0 : 1 : 1]$ and $[1 : 0 : 1]$. If a morphism extending ϕ existed, its compositions with P, Q would agree with these generic images, and separatedness of the projective target would force these displayed integral extensions. Their values at the common source specialization would have to agree, a contradiction.

There is an additional direct model distinction: base changing (CD4.1) to D gives a doubled-line closed fibre, while $\mathbf{P}_D^1$ has a reduced closed fibre. Consequently this integral $\mathcal{C}$ is not a descent of the full scheme $\mathbf{P}_{\mathbf{Z}[i]}^1$ through an isomorphism extending CD2. It is the specified integral model of its already constructed generic descent.

These are precise facts about these integral maps. They neither discard the CC signed charts nor identify $\mathcal{C}$ with their ramified coarse quotient. Constructing that coarse quotient or a resolving modification is a different operation. The model (CD4.1), its nilpotents and its zeta factors are all retained below.

CD6. All finite-field points, without assuming an initial point

Let $q = p^r$ be any finite prime power. First suppose $p \neq 2$, and let $\chi : \mathbf{F}_q \rightarrow \{0, \pm 1\}$ be its quadratic character, with $\chi(0) = 0$. The equation

$$v^2 = y^2 + 1$$

is equivalent to $(v - y)(v + y) = 1$. Its solutions correspond bijectively to $u \in \mathbf{F}_q^\times$, through

$$v - y = u, \quad v + y = u^{-1}. \quad (\text{CD6.1})$$

Division by two gives the inverse pair $v = (u + u^{-1})/2, y = (u^{-1} - u)/2$. Counting the same solutions by y gives

$$q + \sum_{y \in \mathbf{F}_q} \chi(y^2 + 1) = q - 1, \quad \sum_y \chi(y^2 + 1) = -1. \quad (\text{CD6.2})$$

In the affine chart $z = 1$ of the conic, the equation is $x^2 + y^2 = -1$. Its number of points is exactly

$$\sum_y (1 + \chi(-1 - y^2)) = q - \chi(-1). \quad (\text{CD6.3})$$

At infinity, $z = 0$ forces $y \neq 0$, and $(x/y)^2 = -1$ has $1 + \chi(-1)$ solutions. Thus

$$\#\mathcal{C}(\mathbf{F}_q) = (q - \chi(-1)) + (1 + \chi(-1)) = q + 1. \quad (\text{CD6.4})$$

Both chart contributions have been calculated. No rational point or splitting of the conic was assumed to obtain them.

For $p = 2$, a map to a field annihilates every nilpotent. Conversely every field point of L is a point of $V(l^2)$. These mutually inverse operations identify the point sets while preserving the different schemes from CD4. Hence

$$\#\mathcal{C}(\mathbf{F}_{2^r}) = \#L(\mathbf{F}_{2^r}) = 2^r + 1. \quad (\text{CD6.5})$$

We have proved, for every prime and every positive repetition,

$$\boxed{\#\mathcal{C}(\mathbf{F}_{p^r}) = 1 + p^r.} \quad (\text{CD6.6})$$

The coefficient two defining the doubled fibre has not been used as a second copy of each point. A nilpotent thickening does not create extra field-valued points.

CD7. The geometric Tate line, retaining the degree-two comparison

Fix p , a coefficient prime $\ell \neq p$, and $E = \mathbf{Q}_\ell$. At an odd characteristic, CD2 after geometric base change identifies $\mathcal{C}_{\bar{\mathbf{F}}_p}$ with $\mathbf{P}^1$. At characteristic two, the closed immersion $L_{\bar{\mathbf{F}}_2} \hookrightarrow \mathcal{C}_{\bar{\mathbf{F}}_2}$ induces the étale-topos equivalence of CGT3–CGT4. These are the specified maps used to obtain

$$H^0(\mathcal{C}_{\bar{k}}, E) = E, \quad H^1 = 0, \quad H^2(\mathcal{C}_{\bar{k}}, E) = E(-1), \quad H^j = 0 \ (j > 2). \quad (\text{CD7.1})$$

There is a factor in this comparison that must be kept. The conic hyperplane line bundle satisfies

$$\phi^* \mathcal{O}_{\mathcal{C}_R}(1) = \mathcal{O}_{\mathbf{P}_A^1}(2). \quad (\text{CD7.2})$$

The reason is the homogeneous degree two of the basepoint-free sections in (CD2.2): on each trivialization their ratios are exactly the transition maps for the pullback of $\mathcal{O}(1)$. Thus the Kummer first Chern class, in the twisted cohomology group, obeys

$$\phi^* c_1(\mathcal{O}_{\mathcal{C}_R}(1)) = 2c_1(\mathcal{O}_{\mathbf{P}_A^1}(1)). \quad (\text{CD7.3})$$

On the reduced special line at 2, by contrast,

$$\mathcal{O}_{\mathcal{C}_{\mathbf{F}_2}}(1)|_L = \mathcal{O}_L(1). \quad (\text{CD7.4})$$

The class at that fibre is the single line hyperplane class. Both maps are retained.

The global hyperplane class is Galois-fixed in $H^2(E(1))$, and is nonzero: its geometric comparison coefficient is two at odd and characteristic-zero fibres, and one at the fibre at 2. These are nonzero invertible scalars in E . Hence geometric Frobenius has eigenvalue p on untwisted H^2 , because it has eigenvalue p^{-1} on $E(1)$. It has eigenvalue one on H^0 . Its full repetition trace is consequently

$$\text{Tr}(F_p^r \mid H^0) - \text{Tr}(F_p^r \mid H^1) + \text{Tr}(F_p^r \mid H^2) = 1 + p^r. \quad (\text{CD7.5})$$

This agrees with the independent point count CD6.

One may retain the whole proper pushforward. On $S_\ell = \text{Spec } \mathbf{Z}[1/\ell]$, the unit and actual hyperplane class define

$$E_{S_\ell} \oplus E_{S_\ell}(-1)[-2] \longrightarrow Rf_* E_{\mathcal{C}_{S_\ell}}. \quad (\text{CD7.6})$$

Proper base change identifies its stalks with the maps just calculated. They are isomorphisms in degrees zero and two, and all other stalk cohomology vanishes. Therefore (CD7.6) is a quasi-isomorphism. This is an actual geometric source of the twist and shift; it is not a chosen weight assigned to an analytic eigenvector.

For completeness, specialize this model over $\mathbf{Z}_2$, with $\ell \neq 2$. Write $h_s = c_1(\mathcal{O}_L(1))$ on the reduced geometric special line and $h_\eta = c_1(\mathcal{O}_{\mathbf{P}^1}(1))$ after the geometric generic parametrization. The global line bundle and proper specialization give

$$\mathrm{sp}_0(1) = 1, \quad \mathrm{sp}_2(h_s) = 2h_\eta \quad \text{in the twisted degree-two groups.} \quad (\text{CD7.7})$$

The scalar two follows from (CD7.3)–(CD7.4), not from a choice to renormalize the trace. The generic rational groups are E and $E(-1)$, with inertia trivial: their nonzero global hyperplane class fixes the twisted line, and prime-to-two roots of unity are unramified. Thus (CD7.7) is an isomorphism onto inertia invariants with rational coefficients. This model's factor two differs from CGT7's two-entry diagonal map, because those are different integral models with different displayed maps. Their generic descent relation does not equate those special-fibre formulas.

CD8. Every local factor, including the prime 2

Define the local scheme zeta function by its complete point-count series:

$$Z(\mathcal{C}_{\mathbf{F}_p}, T) = \exp \left(\sum_{r \geq 1} \frac{\#\mathcal{C}(\mathbf{F}_{p^r})}{r} T^r \right). \quad (\text{CD8.1})$$

Using every repetition in (CD6.6),

$$\begin{aligned} Z(\mathcal{C}_{\mathbf{F}_p}, T) &= \exp \left(\sum_{r \geq 1} \frac{T^r}{r} + \sum_{r \geq 1} \frac{p^r T^r}{r} \right) \\ &= \frac{1}{(1-T)(1-pT)}. \end{aligned} \quad (\text{CD8.2})$$

This is a formal power-series identity with constant term one; analytically it follows from the two logarithm series for $|T| < p^{-1}$. Its cohomological determinant form retains both groups:

$$Z(\mathcal{C}_{\mathbf{F}_p}, T) = \frac{\det(1 - TF_p \mid H^1)}{\det(1 - TF_p \mid H^0) \det(1 - TF_p \mid H^2)} = \frac{1}{(1-T)(1-pT)}. \quad (\text{CD8.3})$$

In particular the full local factor at the nonreduced fibre is

$$Z(\mathcal{C}_{\mathbf{F}_2}, 2^{-s}) = \frac{1}{(1-2^{-s})(1-2^{1-s})}. \quad (\text{CD8.4})$$

This equality is not an assertion that the fibre is reduced; its square-zero sheaf (CD4.5) remains present, and both point counts and the constant étale receiver pass through the specified reduction map.

CD9. The arithmetic Euler product and its exact comparison to the original zeta

The scheme zeta function of this specified finite-type $\mathbf{Z}$ -scheme is the product over its finite-field fibres. For $\sigma = \Re s > 2$,

$$\sum_p \sum_{r \geq 1} \frac{|p^{-rs}| + |p^{r(1-s)}|}{r} < \infty. \quad (\text{CD9.1})$$

Indeed each repetition sum is bounded by a constant, depending on σ , times $p^{-\sigma} + p^{1-\sigma}$, and their sums over primes are bounded by the corresponding convergent sums over all positive integers greater than one. Therefore all rearrangements below are justified by absolute convergence:

$$\begin{aligned}
Z(\mathcal{C}, s) &= \prod_p \frac{1}{(1 - p^{-s})(1 - p^{1-s})} \\
&= \exp \left(\sum_p \sum_{r \geq 1} \frac{p^{-rs} + p^{r(1-s)}}{r} \right) \\
&= \left(\sum_{a \geq 1} a^{-s} \right) \left(\sum_{b \geq 1} b^{1-s} \right) = \zeta(s)\zeta(s-1).
\end{aligned} \tag{CD9.2}$$

The two unit terms are retained. The products use every prime in the already recovered integer ring. They count two cohomological degrees of one conic; they are not the sum or pooling of two branch return measures.

The open-base descent model alone has

$$Z(\mathcal{C}_R, s) = (1 - 2^{-s})(1 - 2^{1-s}) \zeta(s)\zeta(s-1). \tag{CD9.3}$$

Multiplying by the explicitly computed factor (CD8.4) gives exactly (CD9.2). Thus no prime at which descent was not étale has been hidden in an unspecified correction factor.

Equation (CD9.2) supplies its meromorphic continuation as this original product. Its pole residues at $s = 1$ and $s = 2$ are respectively $\zeta(0) = -1/2$ and $\zeta(2)$. At the original trivial zeros $s = -2r$, the other factor is the nonzero $\zeta(-2r - 1)$; at the translated trivial zeros $s = 1 - 2r$, the other factor is the nonzero $\zeta(1 - 2r)$, for every $r \geq 1$. Thus none of those trivial-zero factors has been cancelled by a completion in this Euler product. Its value at zero is $\zeta(0)\zeta(-1) = 1/24$. These evaluations describe the original meromorphic functions, not a completed substitute.

CD10. The exact connection with the constructed translated analytic divisor

GSL6.5-GSL6.11 construct the actual analytic quotient on the union of the original nontrivial divisor $\mathcal{Z}$ and its translate $\mathcal{Z} + 1$. The high-weight source multiplier there is exactly

$$\begin{aligned}
G_{\text{high}}(s) &= F_0(s)F_0(s-1) \\
&= \left[\frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \right] \left[\frac{(s-1)(s-2)}{8} \pi^{-(s-1)/2} \Gamma((s-1)/2) \right] Z(\mathcal{C}, s).
\end{aligned} \tag{CD10.1}$$

All endpoint factors, both powers of π , both Gamma factors and the full denominator 64 remain in this identity. The product has its entire continuation given by the two source elements F_0 , rather than by declaring any exceptional value a unit and discarding it.

At an original nontrivial zero ρ , the shifted factor $\zeta(\rho - 1)$ is nonzero: its point is in $-1 < \Re s < 0$, where the full functional equation transfers to $\Re(1 - s) > 1$ with nonzero Euler product and unit factors. At a shifted nontrivial zero $\rho + 1$, the original $\zeta(\rho + 1)$ is nonzero by its absolutely convergent Euler product. Hence the local multiplicities of the two nontrivial divisors in $Z(\mathcal{C}, s)$ are exactly m_ρ , without cancellation. The entire source multiplier (CD10.1) has exactly their union, as already checked in GSL, and retains its endpoint and trivial-zero values through its two explicit F_0 factors.

This gives a precise geometric provenance for the translated factor already used in that analytic extension: $H^2 = E(-1)$, with Frobenius p , produces $1 - p^{1-s}$ in each local denominator, while $H^0 = E$

produces $1 - p^{-s}$. It does not identify the analytic zero-jet quotient with H^1 of this conic. That group is zero in every geometric fibre by (CD7.1). The zeros of the global arithmetic product and the finite-dimensional cohomology of an individual fibre have the explicit determinant/Euler maps (CD8.3)–(CD10.1); no additional cohomological identification is asserted.

The construction also distinguishes this descent from the original unprojected evaluated scheme $\mathbf{P}_{\mathbf{Z}[i]}^1$. CGT10.4 retains its two-component split and nonsplit local factors. The descended conic has the single factors calculated in CD8. They coincide with the rational invariant-cohomology factors of CGT9–CGT10 through the explicit generic descent, with the independently verified fibre at 2. Equality of those factors is not substituted for an isomorphism of the ramified integral schemes.

CD11. Diagram and exact scope of the lift

The proved finite étale geometric diagram is

$$\begin{array}{ccc} \mathbf{P}_{\mathbf{Z}[1/2, i]}^1 & \xrightarrow{\phi} & \mathcal{C}_{\mathbf{Z}[1/2, i]} \\ \alpha \downarrow & & \downarrow c \\ \mathbf{P}_{\mathbf{Z}[1/2, i]}^1 & \xrightarrow{\phi} & \mathcal{C}_{\mathbf{Z}[1/2, i]} \end{array} \quad (\text{CD11.1})$$

The right-hand object is the base change of the written conic over $\mathbf{Z}[1/2]$. The two arithmetic-model maps at 2 are deliberately recorded with their actual domains: this $\mathcal{C}_{\mathbf{Z}_2}$ has doubled-line special fibre and specialization $h_s \mapsto 2h_\eta$, while the CGT model $\mathbf{P}_{\mathbf{Z}_2[i]}^1$ has the coefficient thickening and its two-branch diagonal specialization. Their generic relation has been proved and its failure to extend through ϕ has been proved.

The resulting geometric lift (CD7.7) and proper pushforward (CD7.6) supply an actual Tate line with its coefficient two, and the complete original product (CD9.2). No map in this calculation identifies this specialization problem with the user's entire required weight-separation lifting problem. The calculation constructs concrete geometry receiving the CC action and the translated analytic divisor; it supplies no RH verdict and no new assumed purity for the original zero space.

CD12. Independent derivation and retained work record

An independent mathematical derivation checked the odd-field character sum without an assumed point, every repetition at 2, flatness, the doubled-line fibre, the complete Euler product, and the nonremovable basepoint using the two sections in (CD5.3). The complete written derivation is supplied to the parent for reading and to the independent checker for verification of the descent, full maps and cohomological comparison.

This is the sole new file for this task. Source identities, reading coverage, exact receiver stage, and the correction audit are recorded in CD0. No peer source was edited, no remote publication occurred, and no arithmetic was assigned to the parityless supporting datum.

The actual CC coefficient sheaf and the original-zeta cohomology

Proof source: CC_SHEAF_ORIGINAL_ZETA_COHOMOLOGY.md.

The Connes-Consani sheaf and the original-zeta Hausdorff cohomology

Independent derivation, 24 September 2026. Proof locators CS0–CS11, including CS2A.

CS0. Construction stage, source, and precise conclusion

This calculation constructs a sheaf on the three-point geometric realization used by Connes–Consani, identifies its trivial finite-unit sector over the already specified field $\mathbb{Q}$, and proves that the Hausdorff quotient of its first ordinary sheaf cohomology is the existing original-zeta quotient $\mathcal{Q}$. It first retains the exact comparison with ordinary cohomology. The subsequent complete image theorem in CS11 proves that this cohomology is already Hausdorff. It is a construction on arithmetic coefficient spaces after that arithmetic is available. It does not define arithmetic operations, coordinates, a midpoint, or a metric on $\tau\langle Z_1; \text{no } Z_2 \rangle$, and does not identify the authors’ generic point with the user’s entire supported construction.

The connected user reconstruction and its correction-precedence table were read before this calculation, together with the full governing passages U126–U128, U135–U141, WU050–WU055, WU057–WU063 and the pertinent branch-comparison attachment. Those passages require retaining the complete-history dependency, the retraction of addition at τ , and the separate branch counters. Here the two charts remain separately typed; their restrictions form a difference in a sheaf complex, not a pooled arithmetic return measure. No implication from this difference to common weights is inserted.

The original author source actually read is Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, [arXiv:0903.2024v3](https://arxiv.org/abs/0903.2024v3), retained as sources/CC_0903_2024_v3/author_source/announc3.tex. Section 5, source lines 1376–1667, was read in full. Its exact source labels used here are:

- `cechprop`, `vanish`, `cechequal`, `longexact1`: affine acyclicity and supported cohomology;
- `projspace1`, `projspace2`, `projback`: the three-point base and the projective adèle-class projection;
- `fonction1`, `fonction3`, `sheaf1`, `SCK`, `restmaps`, `Smap`: the source spaces and restrictions;
- `bord`, `lemsurject`, `hzero`, `fourier`, `poisson2`, `poisson3`, `coinv`: the Čech differential and Fourier graph;
- `rep1`, `liftw`, `compinv`, `court`, `surject`: the unshifted action, Weyl lift, and single-support comparison.

The original text invokes Meyer’s Lemma 5.4 to prove injectivity of its full adèlic summation map on Hausdorff coinvariants. This note does not claim to have reconstructed that general theorem. CS2–CS4 prove the needed injectivity directly for the trivial finite-unit sector over $\mathbb{Q}$. A domain qualification matters: source `fonction3` does not explicitly type its generating f . We adopt and state the full-difference reading, in which arbitrary adèlic Schwartz functions generate differences already lying in the moment-null subspace by invariance of both endpoint functionals. CS2–CS3 prove its compatibility with the source’s stated Fourier-graph result. Restricting the generating functions themselves to the smaller moment-null subspace changes the quotient; CS2A proves that this alternative contradicts that graph result for the specified topology. This is a proved interpretation and comparison, not a claim that the source writes an omitted domain explicitly.

The existing programme theorem used below is GLOBAL_MELLIN_SYNTHESIS.md, S1-S7, in the adjacent quantum_tau_programme_bridge_20260924 directory: the topological Mellin isomorphism and the equality of the closed original source image with the ideal of every actual nontrivial-zero jet. Its full formula, exceptional values, and exact role are restated in CS6-CS7. The global synthesis proof remains its own proof, rather than being attributed to Connes-Consani.

CS1. The adèlic source and its compact unit projection

Write

$$\mathbb{A}_{\mathbb{Q}} = \mathbb{R} \times \mathbb{A}_f, \quad \widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p, \quad K = \widehat{\mathbb{Z}}^{\times}. \quad (\text{CS1.1})$$

Use Lebesgue measure at the real place and finite additive Haar measure with $\text{vol}(\widehat{\mathbb{Z}}) = 1$. The adèlic Bruhat-Schwartz space is the locally convex inductive limit of the spaces of finite sums $f_i(v)\phi_i(x_f)$, where $f_i \in \mathcal{S}(\mathbb{R})$ and the ϕ_i belong to one finite-dimensional space of locally constant, compactly supported functions on $\mathbb{A}_f$. On each such stage the topology is a finite direct sum of the ordinary real Schwartz topology.

Put

$$\mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0 = \left\{ F : F(0) = 0, \int_{\mathbb{A}_{\mathbb{Q}}} F(x) dx = 0 \right\}, \quad \mathcal{R} = \text{span}_{\mathbb{C}} \{ F - F(q \cdot) : F \in \mathcal{S}(\mathbb{A}_{\mathbb{Q}}), q \in \mathbb{Q}^{\times} \},$$

$$D = \mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0 / \overline{\mathcal{R}}. \quad (\text{CS1.2})$$

The closure is taken in $\mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0$ with its subspace topology. Rational scaling preserves both endpoint functionals, since $|q|_{\mathbb{A}} = 1$, so every displayed difference belongs to that moment-null space even when its generating F does not. This is the full-difference reading of the source's "subspace of $\mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0$ generated by the differences." The source line does not separately write a domain after its f ; the distinction and its mathematically forced consequence are recorded in CS2A.

The group K acts on finite coordinates. Its normalized Haar average P_K is continuous on this source: each test function is fixed by an open subgroup of K , so its average is a finite average on that test-function stage. The average commutes with rational scaling and preserves both endpoint conditions. It consequently descends to a continuous idempotent on D .

Every idèle class has a unique representative (u, k) , with $u > 0$ and $k \in K$. To see this, multiply its finite component by the positive rational with opposite finite valuations, and then use the sign in $\mathbb{Q}^{\times}$ to make its real component positive. Uniqueness follows because a rational number with every finite valuation zero is ± 1 , and positivity excludes the minus sign. Thus

$$C_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}^{\times} / \mathbb{Q}^{\times} \cong \mathbb{R}_{>0}^{\times} \times K, \quad |(u, k)| = u. \quad (\text{CS1.3})$$

This is a coordinate on the authors' already constructed idèle-class group.

CS2. An exact topological identification of the source sector

Define

$$\mathcal{S}_0^{\text{even}} = \left\{ f \in \mathcal{S}(\mathbb{R}) : f(-v) = f(v), \quad f(0) = 0, \quad \int_{\mathbb{R}} f(v) dv = 0 \right\},$$

$$j : \mathcal{S}_0^{\text{even}} \longrightarrow D, \quad j(f) = [f \otimes 1_{\widehat{\mathbb{Z}}}]. \quad (\text{CS2.1})$$

This is continuous, and its image is fixed by K .

We construct its continuous inverse on $P_K D$. A K -invariant finite Schwartz function is a finite linear combination of

$$1_{d\widehat{\mathbb{Z}}}, \quad d \in \mathbb{Q}_{>0}. \quad (\text{CS2.2})$$

Here $d\widehat{\mathbb{Z}}$ is the product of the balls $p^{v_p(d)}\mathbb{Z}_p$. To prove the claim, choose the finitely many primes on which the function is nonstandard. At each such prime, compact support bounds valuations below and local constancy makes the function constant on one sufficiently small ball around zero. Unit invariance leaves a finite list of valuation shells and that final ball. Each shell is the difference of two nested ball indicators. Taking tensor products gives (CS2.2).

These ball indicators are linearly independent. In one local factor, evaluation successively on the shells of largest and then smaller absolute values proves independence of a finite list of nested balls. Finite tensor products of independent lists are independent, and every proposed relation involves such a finite product. This also proves that the following definition is independent of its expansion.

For $f^{\text{ev}}(v) = (f(v) + f(-v))/2$, define on K -averaged finite sums

$$B(f \otimes 1_{d\widehat{\mathbb{Z}}})(v) = f^{\text{ev}}(dv), \quad B(F) := B(P_K F) \quad \text{on the full source.} \quad (\text{CS2.3})$$

On each finite stage this is a finite sum of fixed real dilations and the even projection, hence is continuous into $\mathcal{S}^{\text{even}}(\mathbb{R})$. The defining inductive-limit topology proves continuity on the full source.

The endpoint identities are exact:

$$B(F)(0) = F(0), \quad \int_{\mathbb{R}} B(F)(v) dv = \int_{\mathbb{A}_{\mathbb{Q}}} F(x) dx. \quad (\text{CS2.4})$$

For a ball tensor the second identity follows from

$$\text{vol}(d\widehat{\mathbb{Z}}) = \prod_p p^{-v_p(d)} = d^{-1}, \quad \int_{\mathbb{R}} f^{\text{ev}}(dv) dv = d^{-1} \int_{\mathbb{R}} f(v) dv. \quad (\text{CS2.5})$$

The first identity is immediate; averaging does not change either endpoint functional. Thus B maps the source in (CS1.2) into $\mathcal{S}_0^{\text{even}}$.

For $q \in \mathbb{Q}^\times$, direct substitution on a ball tensor gives

$$B((f \otimes 1_{d\widehat{\mathbb{Z}}})(q \cdot))(v) = f^{\text{ev}}(dv) = B(f \otimes 1_{d\widehat{\mathbb{Z}}})(v). \quad (\text{CS2.6})$$

Indeed the new finite ball is $(d/|q|)\widehat{\mathbb{Z}}$, while the new even real function is $f^{\text{ev}}(|q|v)$. Since P_K commutes with this scaling, (CS2.6) holds on the full source. Continuity makes B vanish on $\overline{\mathcal{R}}$, so it descends to a continuous map $\bar{B} : D \rightarrow \mathcal{S}_0^{\text{even}}$.

The two compositions are

$$\bar{B}j = \text{id}, \quad j\bar{B} = P_K \quad \text{on } D. \quad (\text{CS2.7})$$

For the second identity, individual tensors need not themselves be moment-null, so first work with differences before taking a quotient class. Put $G = f \otimes 1_{d\widehat{\mathbb{Z}}}$ and $H_d = G(d \cdot) = f(dv) \otimes 1_{\widehat{\mathbb{Z}}}$. Let $\tilde{j}$ denote tensoring with $1_{\widehat{\mathbb{Z}}}$ on all even Schwartz functions. The exact identity is

$$G - \tilde{j}(BG) = (G - G(d \cdot)) + \frac{1}{2}(H_d - H_d(- \cdot)) \in \mathcal{R}. \quad (\text{CS2.7a})$$

Both parenthesized differences have zero endpoint moments, whether or not their generating functions do. Sum (CS2.7a) over a radial-ball expansion of $P_K F$, with F moment-null. Only then take its class in D . This proves $j\bar{B}[F] = P_K[F]$, with every class taken on its actual domain.

Finally $P_K D = D^K$. One inclusion follows from averaging. For a fixed quotient class, choose a representative F . Its K -orbit lies in a finite averaging orbit modulo an open subgroup. Averaging its equal quotient classes therefore gives the same class. We have proved the actual topological isomorphism

$$\boxed{j : \mathcal{S}_0^{\text{even}} \xrightarrow{\sim} D^K, \quad j^{-1} = \bar{B}|_{D^K}.} \quad (\text{CS2.8})$$

In particular, this identification neither discards the closed coinvariant relation nor assumes it has no kernel.

CS2A. Why the generating domain must be retained

Let $\mathcal{R}_{\text{null}}$ denote instead the span of $F - F(q \cdot)$ with F itself moment-null. Then $\mathcal{R}_{\text{null}} \subset \mathcal{R}$, and there is a natural continuous surjection

$$D_{\text{null}} := \mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0 / \overline{\mathcal{R}_{\text{null}}} \longrightarrow D. \quad (\text{CS2A.1})$$

This alternative quotient is genuinely larger. Here is a direct proof which also identifies the source of the difference.

For a prime p , expand a K -averaged Schwartz function as $\sum_d f_d \otimes 1_{d\widehat{\mathbb{Z}}}$, and define the two-component continuous functional

$$\ell_p(F) = \sum_d v_p(d) \left(f_d(0), d^{-1} \int_{\mathbb{R}} f_d(v) dv \right), \quad \ell_p(F) := \ell_p(P_K F) \text{ before averaging.} \quad (\text{CS2A.2})$$

The sum is finite, its expansion is unique by CS2, and continuity follows on each finite test-function stage. Put $\epsilon(F) = (F(0), \int_{\mathbb{A}} F)$. On ball tensors, scaling changes the index d to $d/|q|$ and leaves the two displayed endpoint values unchanged. Hence

$$\ell_p(F(q \cdot)) = \ell_p(F) - v_p(|q|)\epsilon(F). \quad (\text{CS2A.3})$$

It follows that ℓ_p vanishes on $\overline{\mathcal{R}_{\text{null}}}$, but not on all of $\mathcal{R}$. Indeed, take any even Schwartz f and

$$G = f \otimes 1_{\widehat{\mathbb{Z}}} - (f \otimes 1_{\widehat{\mathbb{Z}}})(p \cdot).$$

Then G is moment-null, its summation is zero by rational reindexing, and

$$\ell_p(G) = \left(f(0), \int_{\mathbb{R}} f \right). \quad (\text{CS2A.4})$$

The endpoint map on even Schwartz functions is onto $\mathbb{C}^2$: for example the functions $e^{-\pi v^2}$ and $v^2 e^{-\pi v^2}$ have endpoint pairs $(1, 1)$ and $(0, 1/(2\pi))$. Thus these are nonzero classes in the alternative quotient whenever that pair is nonzero.

In particular, the injective summation and Fourier-graph claim in the original source cannot hold on D_{null} with this topology: (CS2A.4) supplies an explicit kernel. The full-difference reading in CS1 removes exactly these displayed differences, and CS2–CS3 prove the needed sector result for it. Even under the alternative reading, the map (CS2A.1) and the commuting summation maps give the same summation image and the same Hausdorff first-cohomology receiver. Its degree-zero kernel contains additional classes, so its H^0 must not be described by (CS5.3).

The complete difference between the two quotient spaces can also be computed. Give

$$V = \bigoplus_{p \text{ prime}} \mathbb{C}^2 \quad (\text{CS2A.5})$$

the locally convex direct-sum topology, and put $\mathcal{L}(F) = (\ell_p(F))_p$. A fixed test-function stage has only finitely many relevant primes in its radial-ball expansion. Thus $\mathcal{L}$ has finite support on every test function and is continuous into V .

Choose the continuous endpoint section $\sigma : \mathbb{C}^2 \rightarrow \mathcal{S}(\mathbb{A}_{\mathbb{Q}})$ given by tensoring

$$\sigma(a, b)_{\infty}(v) = a(1 - 2\pi v^2)e^{-\pi v^2} + b 2\pi v^2 e^{-\pi v^2} \quad (\text{CS2A.6})$$

with $1_{\widehat{\mathbb{Z}}}$. The Gaussian moments give $\epsilon(\sigma(a, b)) = (a, b)$. Define the continuous finite-sum map

$$\mathcal{T}(v) = \sum_p (\sigma(v_p) - \sigma(v_p)(p \cdot)) \in \mathcal{R}, \quad \mathcal{L}\mathcal{T} = \text{id}_V. \quad (\text{CS2A.7})$$

The second identity is (CS2A.3) at every prime.

Write $r_q(F) = F - F(q \cdot)$. Modulo $\mathcal{R}_{\text{null}}$, this depends only on $\epsilon(F)$, because subtracting $\sigma(\epsilon F)$ makes its argument moment-null. Also

$$r_{qr}(F) \equiv r_q(F) + r_r(F) \pmod{\mathcal{R}_{\text{null}}}. \quad (\text{CS2A.8})$$

Indeed their difference is a rational difference with generating function $F(q \cdot) - F$, which is moment-null. The sign $q = -1$ contributes zero in this quotient, since $r_{-1}(F) = \frac{1}{2}r_{-1}(F - F(-\cdot))$. Unique factorization, including negative prime exponents, now gives

$$r_q(F) \equiv \sum_p v_p(|q|) r_p(\sigma(\epsilon F)) = \mathcal{T}\mathcal{L}(r_q(F)) \pmod{\mathcal{R}_{\text{null}}}. \quad (\text{CS2A.9})$$

Thus $(I - \mathcal{T}\mathcal{L})\mathcal{R} \subset \mathcal{R}_{\text{null}}$. Continuity gives the same inclusion for closures. It follows that the exact quotient map has the topological splitting

$$\boxed{D_{\text{null}} \xrightarrow{\sim} D \oplus V, \quad [F]_{\text{null}} \mapsto ([F], \mathcal{L}F).} \quad (\text{CS2A.10})$$

For injectivity, if $F \in \overline{\mathcal{R}}$ and $\mathcal{L}F = 0$, the closure inclusion puts F in $\overline{\mathcal{R}_{\text{null}}}$. For surjectivity, start with any representative F of the required D -class and replace it by $F + \mathcal{T}(v - \mathcal{L}F)$. A continuous inverse sends $([F], v)$ to $[F - \mathcal{T}\mathcal{L}F + \mathcal{T}v]_{\text{null}}$. Independence of the representative and continuity follow from the closure inclusion and the defining quotient topology. This proves the full alternative receiver and its connecting maps; the additional summand records two endpoint directions at every prime.

CS3. The overlap, Fourier transform, and all rational summands

On the trivial K -sector, the source overlap space $\bigcap_{\beta \in \mathbb{R}} \mu^{\beta} \mathcal{S}(C_{\mathbb{Q}})$ is the Fréchet space

$$H = \left\{ h \in C^{\infty}(\mathbb{R}_{>0}) : p_{N,j}(h) := \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j h(u)| < \infty \quad (N, j \geq 0) \right\}. \quad (\text{CS3.1})$$

To verify both the set and topology, use $x = \log u$. The ordinary multiplicative Schwartz seminorms are the polynomially weighted derivative seminorms in x ; the source additionally requires every factor $e^{\beta x}$. Derivatives of $e^{\beta x} h(e^x)$ are finite sums of $e^{\beta x} \partial_x^{\ell} h(e^x)$. Polynomial factors in x are bounded by a constant times $e^{|x|}$, and $e^{N|x|} \leq e^{Nx} + e^{-Nx} \leq 2e^{N|x|}$. These identities bound every source seminorm by finitely many (CS3.1) seminorms and conversely. Integer N suffice for every real β .

For $f \in \mathcal{S}_0^{\text{even}}$, summation of its actual adèlic representative at $(u, 1)$ gives

$$Sf(u) := \Sigma(jf)(u) = \sum_{q \in \mathbb{Q}^{\times}} f(qu) 1_{\widehat{\mathbb{Z}}}(q) = \sum_{n \in \mathbb{Z} \setminus \{0\}} f(nu) = 2 \sum_{n \geq 1} f(nu). \quad (\text{CS3.2})$$

The criterion in the middle step is $q \in \widehat{\mathbb{Z}} \cap \mathbb{Q} = \mathbb{Z}$. Both rational signs are present, and the final factor two is retained.

Use the real Fourier convention

$$\widehat{f}(t) = \int_{\mathbb{R}} f(v) e^{-2\pi i v t} dv. \quad (\text{CS3.3})$$

Choose the corresponding character of $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$ and self-dual measures; at finite places the annihilator of $\widehat{\mathbb{Z}}$ is itself, with volume one. Direct Fourier integration of its indicator is therefore the same indicator. Consequently the adèlic Fourier transform of $f \otimes 1_{\widehat{\mathbb{Z}}}$ is $\widehat{f} \otimes 1_{\widehat{\mathbb{Z}}}$. Fourier transformation preserves both vanishing endpoint conditions and evenness, and $\widehat{\widehat{f}} = f$ on this even space.

Set

$$(Rh)(u) = u^{-1}h(1/u), \quad R^2 = \text{id}. \quad (\text{CS3.4})$$

For an arbitrary even real Schwartz function, Poisson summation retains the exact endpoint identity

$$Sf(u) - u^{-1}S\widehat{f}(1/u) = u^{-1} \int_{\mathbb{R}} f(v) dv - f(0). \quad (\text{CS3.5})$$

Indeed the full lattice sums have their zero summands $f(0)$ and $u^{-1}\widehat{f}(0)$, and $\widehat{\widehat{f}}(0) = \int f$. For our actual moment-null source both terms on the right vanish, so

$$S\widehat{f} = RSf, \quad RS\widehat{f} = Sf. \quad (\text{CS3.6})$$

This is the exact plus/minus gluing identity, including the factor u^{-1} .

The map $S : \mathcal{S}_0^{\text{even}} \rightarrow H$ is continuous. At $u \geq 1$, each logarithmic derivative of the series is bounded by a sufficiently high Schwartz seminorm times a convergent sum $\sum_{n \geq 1} n^{-M}$, $M > 1$. For $u \leq 1$, (CS3.6) expresses the answer through the Fourier transform at $1/u \geq 1$, multiplied by u^{-1} . Fourier continuity and one additional endpoint weight prove the required bounds. The same arguments justify termwise differentiation.

It is also injective. For $\Re s > 1$, absolute convergence and the substitution $v = nu$ give

$$\int_0^\infty Sf(u) u^s \frac{du}{u} = 2\zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}. \quad (\text{CS3.7})$$

If $Sf = 0$, choose any real $c > 1$. Since $\zeta(c + it) \neq 0$ on this half-plane, the Mellin transform of $f|_{\mathbb{R}_{>0}}$ vanishes on the line $c + it$. It is the Fourier transform of the integrable function $x \mapsto e^{cx} f(e^x)$, with Fourier frequency sign fixed by e^{itx} . Fourier injectivity gives $f(e^x) = 0$; evenness gives $f = 0$ on $\mathbb{R}$. This proves the source-sector injectivity independently of the full adèlic coinvariant theorem.

CS4. The actual three-point sheaf

Let

$$X = \{0, \eta, \infty\}, \quad U_+ = \{0, \eta\}, \quad U_- = \{\eta, \infty\}, \quad U = \{\eta\}. \quad (\text{CS4.1})$$

The nonempty opens are U, U_+, U_-, X . The point η is generic; $0, \infty$ are closed. These are the authors' base labels.

For each chart retain its two separate endpoint lines,

$$E_+ = \mathbb{C}e_{+,0} \oplus \mathbb{C}e_{+,1}, \quad E_- = \mathbb{C}e_{-,0} \oplus \mathbb{C}e_{-,1}, \quad A_{\pm} = \mathcal{S}_0^{\text{even}} \oplus E_{\pm}. \quad (\text{CS4.2})$$

The index 0 labels the value-at-zero endpoint functional, and index 1 labels the integral endpoint functional. These are the two explicit summands in the source definition of $\mathcal{S}(M)$, separately on each chart.

Define

$$\begin{aligned}\mathcal{F}(U_+) &= A_+, \quad \mathcal{F}(U_-) = A_-, \quad \mathcal{F}(U) = H, \\ r_+(f, e_+) &= Sf, \quad r_-(h, e_-) = RSh = \hat{S}h.\end{aligned}\tag{CS4.3}$$

Every endpoint restriction is zero, exactly as in restmaps. Define

$$\mathcal{F}(X) = \{(a, b) \in A_+ \oplus A_- : r_+a = r_-b\}, \quad \mathcal{F}(\emptyset) = 0.\tag{CS4.4}$$

Use the evident restrictions from X . The only nontrivial sheaf gluing condition is the cover $X = U_+ \cup U_-$, and it is precisely (CS4.4). A cover of U_+ must contain U_+ , since no smaller open contains 0; the same holds for U_- and ∞ . The overlap is a singleton. Thus these formulas construct a genuine sheaf, including its topology on every section space.

Under j on the two source coordinates and (CS1.3) on the overlap, this is the P_K -image sheaf of the actual Connes–Consani sheaf. Averaging commutes with summation, restrictions, and Fourier transform, and is the identity on the endpoint lines. It is therefore a sheaf idempotent and gives a direct-summand sheaf. This precise use of “trivial finite-unit sector” does not silently identify fixed classes in an arbitrary non-Hausdorff quotient with quotients of fixed representatives.

CS5. Ordinary cohomology, including every endpoint line

Fix the ordered cover (U_+, U_-) . Its complex is

$$C^0 = A_+ \oplus A_-, \quad C^1 = H, \quad d(a, b) = r_+a - r_-b.\tag{CS5.1}$$

This computes ordinary sheaf cohomology in the category of complex vector-space sheaves, before taking closures. Here is an elementary derived-functor verification. A sheaf on this space is exactly a diagram $D_+ \rightarrow H \leftarrow D_-$. Global sections are the kernel of the displayed difference. Short exact sequences of diagrams give short exact sequences of these two-term complexes and their cohomology long exact sequences. The degree-one functor is effaceable: embed the diagram into one with chart spaces $D_+ \oplus H, D_- \oplus H$, overlap H , and restriction maps the projections to H ; embed a as (a, r_+a) , b as (b, r_-b) , and act identically on H . The new difference map is onto H , so its degree-one cohomology vanishes. Thus this effaceable cohomology functor is the right-derived global-section functor. There are no higher terms, and

$$H^i(X, \mathcal{F}) = 0 \quad (i \geq 2).\tag{CS5.2}$$

By (CS3.6) and injectivity of S , $d((f, e_+), (h, e_-)) = 0$ if and only if $f = \hat{h}$, equivalently $h = \hat{f}$. Consequently

$$H^0(X, \mathcal{F}) = \{((f, e_+), (\hat{f}, e_-)) : f \in \mathcal{S}_0^{\text{even}}, e_+ \in E_+, e_- \in E_-\}.\tag{CS5.3}$$

This is a topological isomorphism from $\mathcal{S}_0^{\text{even}} \oplus E_+ \oplus E_-$ onto the kernel, because Fourier transformation and the coordinate projections are continuous. There are four independent endpoint lines, rather than two endpoint lines identified across the charts.

Write

$$J = S(\mathcal{S}_0^{\text{even}}) \subset H.\tag{CS5.4}$$

The minus restriction has the same image J , since Fourier transformation is a bijection of the source. Moreover $d((f, 0), (0, 0)) = Sf$. Therefore

$$\text{im } d = J, \quad H_{\text{alg}}^1(X, \mathcal{F}) = H/J.\tag{CS5.5}$$

The subscript emphasizes that the ordinary cokernel has not been replaced by a closed-image quotient. Give it the quotient locally convex topology. Its closure of zero is $\overline{J}/J$, and its maximal Hausdorff quotient is

$$H_{\text{Haus}}^1(X, \mathcal{F}) := H/\overline{J}, \quad 0 \longrightarrow \overline{J}/J \longrightarrow H/J \longrightarrow H/\overline{J} \longrightarrow 0. \quad (\text{CS5.6})$$

This sequence is algebraically exact without an additional image theorem. CS11 subsequently proves $J = \overline{J}$, so its left term is zero for this particular source.

CS6. The exact comparison with the existing source image

The earlier real source map, with its original factors, is

$$\mathcal{E}f(u) = u^{1/2} \sum_{n \geq 1} f(nu), \quad \mathcal{C} = \overline{\mathcal{E}(\mathcal{S}_0^{\text{even}})}^{\mathcal{A}}, \quad (\text{CS6.1})$$

where $\mathcal{A}$ has exactly the weighted derivative seminorms (CS3.1). Although H and $\mathcal{A}$ have the same underlying class of functions, their receiving formulas differ. Define the comparison explicitly:

$$U : H \longrightarrow \mathcal{A}, \quad (Uh)(u) = \frac{1}{2}u^{1/2}h(u), \quad (U^{-1}k)(u) = 2u^{-1/2}k(u). \quad (\text{CS6.2})$$

For $D_u = u\partial_u$,

$$D_u^j(u^a h) = u^a \sum_{\ell=0}^j \binom{j}{\ell} a^{j-\ell} D_u^\ell h. \quad (\text{CS6.3})$$

Since $(u^N + u^{-N})u^{\pm 1/2} \leq 2(u^{N+1} + u^{-(N+1)})$, these give the concrete continuity estimates

$$\begin{aligned} p_{N,j}(Uh) &\leq \sum_{\ell=0}^j \binom{j}{\ell} (1/2)^{j-\ell} p_{N+1,\ell}(h), \\ p_{N,j}(U^{-1}k) &\leq 4 \sum_{\ell=0}^j \binom{j}{\ell} (1/2)^{j-\ell} p_{N+1,\ell}(k). \end{aligned} \quad (\text{CS6.4})$$

Both compositions in (CS6.2) are the identity, so this is a topological isomorphism. Formula (CS3.2) now gives, without removing its original scalar or power,

$$Sf = 2u^{-1/2}\mathcal{E}f, \quad USf = \mathcal{E}f. \quad (\text{CS6.5})$$

Topological isomorphisms commute with closure. It follows that

$$\boxed{\overline{J}^H = U^{-1}\mathcal{C} = \{2u^{-1/2}k(u) : k \in \mathcal{C}\}.} \quad (\text{CS6.6})$$

Thus the requested source-image comparison includes both topologies and both inverses, rather than only a formal equality of summation expressions.

CS7. Raw Mellin transform, original zeta, and the Hausdorff quotient

Retain the two Mellin maps separately:

$$\mathcal{M}_0 h(s) = \int_0^\infty h(u) u^s \frac{du}{u}, \quad \mathcal{M} k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}. \quad (\text{CS7.1})$$

They satisfy

$$\mathcal{M}_0 h = 2\mathcal{M}(Uh), \quad \Theta := \mathcal{M}U = \frac{1}{2}\mathcal{M}_0. \quad (\text{CS7.2})$$

The previous theorem S1 proves $\mathcal{M} : \mathcal{A} \rightarrow \mathcal{B}$ is a topological isomorphism, where

$$\mathcal{B} = \left\{ F \text{ entire} : b_{A,M}(F) := \sup_{|\Re s| \leq A} (1 + |\Im s|)^M |F(s)| < \infty \text{ for every } A, M \geq 0 \right\}. \quad (\text{CS7.3})$$

Consequently Θ is a topological isomorphism. For example its full inverse, on the real line $c = 1/2$, is

$$(\Theta^{-1}F)(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} F(1/2 + it) u^{-it} dt. \quad (\text{CS7.4})$$

This is $2u^{-1/2}$ times S1's inverse with coefficient $1/(2\pi)$; both factors remain explicit.

Equation (CS3.7) is the raw arithmetic identity

$$\mathcal{M}_0 S f(s) = 2\zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}, \quad \Theta S f(s) = \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v} \quad (\Re s > 1). \quad (\text{CS7.5})$$

For the retained source

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}, \quad (\text{CS7.6})$$

S2's exact calculation gives

$$\Theta S f_0(s) = F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad \mathcal{M}_0 S f_0(s) = 2F_0(s) = \frac{s(s-1)}{4} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{CS7.7})$$

The entire continuation and exceptional values are

$$F_0(0) = F_0(1) = \frac{1}{8}, \quad \mathcal{M}_0 S f_0(0) = \mathcal{M}_0 S f_0(1) = \frac{1}{4}, \quad (\text{CS7.8})$$

$$F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0 \quad (r \geq 1). \quad (\text{CS7.9})$$

The raw Mellin value at every such point is $2F_0(-2r)$. This retains the endpoint contributions and the Gamma-pole/trivial-zero cancellation. It does not make the completed product a replacement for the original zeta function.

For additional endpoint precision, let $a_f(s) = \int_0^\infty f(v) v^s dv/v$. Its meromorphic continuation has possible simple poles at $-2r$, $r \geq 1$, with residues $f^{(2r)}(0)/(2r)!$, obtained by subtracting the real Taylor polynomial on $0 < v < 1$. At the original trivial zeros,

$$\mathcal{M}_0 S f(-2r) = 2\zeta'(-2r) \frac{f^{(2r)}(0)}{(2r)!}. \quad (\text{CS7.10})$$

At the two endpoints, $a_f(1) = \frac{1}{2} \int_{\mathbb{R}} f = 0$, the zeta residue at 1 is one, and $\zeta(0) = -1/2$, so

$$\mathcal{M}_0 S f(1) = 2a'_f(1), \quad \mathcal{M}_0 S f(0) = -a_f(0). \quad (\text{CS7.11})$$

Thus the source's two zero moment conditions do not erase these receiver values.

Let $\mathcal{Z}$ be the set of distinct actual nontrivial zeros of the original ζ , with their full multiplicities m_ρ , and let

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every } \rho \in \mathcal{Z}, 0 \leq j < m_\rho\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}. \quad (\text{CS7.12})$$

S5 proves $\mathcal{MC} = \mathcal{J}$ on these exact spaces. Combining this established theorem with (CS6.6) proves

$$\boxed{\bar{J} = \Theta^{-1}J, \quad H_{\text{Haus}}^1(X, \mathcal{F}) \xrightarrow[h]{\mapsto} [\tfrac{1}{2}\mathcal{M}_0h] \sim \mathcal{Q}.} \quad (\text{CS7.13})$$

The map and its inverse are continuous quotient maps induced by the homeomorphism Θ , because it takes exactly one closed subspace to the other. On ordinary cohomology it is the surjection

$$H_{\text{alg}}^1(X, \mathcal{F}) = H/J \longrightarrow \mathcal{Q}, \quad \ker = \bar{J}/J. \quad (\text{CS7.14})$$

This proves the Hausdorff identification without presupposing equality with ordinary cohomology. CS11 supplies that equality by constructing the actual inverse source function. Every original-zero jet and its full order is retained by (CS7.12).

CS8. Unshifted scaling, all endpoint characters, and the Čech sign

For $a > 0$, the source actions on the main chart coordinates and the overlap are

$$\rho_+(a)f(v) = f(v/a), \quad \rho_-(a)h(v) = ah(av), \quad \rho_H(a)g(u) = g(u/a). \quad (\text{CS8.1})$$

They preserve the moment-null even Schwartz space and are continuous. On the four endpoint lines their exact characters are

$$\begin{array}{c|cccc} \text{line} & e_{+,0} & e_{+,1} & e_{-,0} & e_{-,1} \\ \hline \text{multiplier} & 1 & a & a & 1. \end{array} \quad (\text{CS8.2})$$

These follow respectively by evaluation at zero and change of variables in the real integral. In particular the minus chart uses the source's extra scalar a ; it does not have the same ordered endpoint characters as the plus chart.

Direct substitution proves

$$r_{\pm}\rho_{\pm}(a) = \rho_H(a)r_{\pm}, \quad \widehat{\rho_+(a)}f = \rho_-(a)\hat{f}. \quad (\text{CS8.3})$$

Thus the Fourier graph and all four endpoint lines carry the actual source action. Under (CS6.2)–(CS7.2),

$$U\rho_H(a)U^{-1}k(u) = a^{1/2}k(u/a), \quad \Theta\rho_H(a)\Theta^{-1}F(s) = a^sF(s). \quad (\text{CS8.4})$$

The infinitesimal operator on raw H is $-u\partial_u$; on $\mathcal{A}$ it is $1/2 - u\partial_u$; on $\mathcal{B}$ it is multiplication by the original variable s . No operator has been silently shifted. On the quotient the arithmetic operator is exactly

$$T_a[F] = [a^sF(s)], \quad T_n = n^L = \sqrt{n}V_n^{\text{centered}}, \quad (\text{CS8.5})$$

where the superscript distinguishes the earlier centered dilation from an integral transfer generator.

The source Weyl lift exchanges chart sections by the identity and acts on overlap sections by R . On the ordered Čech complex (CS5.1), its actual cochain action is

$$W^0(a, b) = (b, a), \quad W^1g = -Rg. \quad (\text{CS8.6})$$

Indeed

$$dW^0(a, b) = r_+b - r_-a = -R(r_+a - r_-b) = -Rd(a, b). \quad (\text{CS8.7})$$

The extra minus sign is the orientation sign from reversing the ordered chart pair. Both degrees square to the identity.

On H^0 , the main Fourier-graph coordinate transforms as $f \mapsto \hat{f}$, and the endpoint lines are exchanged by their chart labels. On the Hausdorff H^1 quotient, changing variables $v = 1/u$ gives

$$\mathcal{M}_0(Rg)(s) = \mathcal{M}_0g(1-s), \quad \boxed{W^1[F(s)] = [-F(1-s)]}. \quad (\text{CS8.8})$$

The ideal is preserved either by the source identity $RJ = J$ and closure, or by the original functional equation with its multiplicities. The source's unshifted compatibility remains

$$\rho_H(a)R = aR\rho_H(a^{-1}), \quad T_a W^1 = aW^1 T_{a^{-1}}. \quad (\text{CS8.9})$$

This proves the spectral reflection and its chain sign without assuming critical-line location or positivity.

CS9. The single-closed-point support maps

For the closed subset $\{0\}$, the ordinary support functor is $\ker(r_+ : A_+ \rightarrow H)$. Its derived complex is

$$L_0 = [A_+ \xrightarrow{r_+} H]. \quad (\text{CS9.1})$$

For $\{\infty\}$ use

$$L_\infty = [A_- \xrightarrow{r_-} H]. \quad (\text{CS9.2})$$

These derived identifications follow by the same short-exact-complex and effacement proof as CS5 for the single arrow. Source injectivity and (CS3.6) yield

$$\begin{aligned} H_{\{0\}}^0(X, \mathcal{F}) &= E_+, & H_{\{\infty\}}^0(X, \mathcal{F}) &= E_-, \\ H_{\{0\}}^1(X, \mathcal{F}) &= H/J, & H_{\{\infty\}}^1(X, \mathcal{F}) &= H/J, & H_{\{0\}}^i &= H_{\{\infty\}}^i = 0 \quad (i \geq 2). \end{aligned} \quad (\text{CS9.3})$$

The two support-to-global chain maps, with positive restriction differentials in both local presentations, are

$$\begin{aligned} L_0 &\longrightarrow C^\bullet : & a &\longmapsto (a, 0), & g &\longmapsto g, \\ L_\infty &\longrightarrow C^\bullet : & b &\longmapsto (0, b), & g &\longmapsto -g. \end{aligned} \quad (\text{CS9.4})$$

Their chain identities follow from $d(a, 0) = r_+ a$ and $d(0, b) = -r_- b$. Hence both induce first-cohomology isomorphisms, with respectively $+\text{id}$ and $-\text{id}$ in the displayed quotient coordinates.

For completeness, the restriction $H^0(X, \mathcal{F}) \rightarrow A_-$ is onto: given (h, e_-) , choose the global section with plus coordinate $\hat{h}$, arbitrary plus endpoints, and minus coordinate h . Its kernel is E_+ . This proves directly that the connecting map $A_- \rightarrow H_{\{0\}}^1$ is zero and gives the exact sequence

$$0 \longrightarrow E_+ \longrightarrow H^0(X, \mathcal{F}) \longrightarrow A_- \xrightarrow{0} H/J \xrightarrow{\text{id}} H/J \longrightarrow 0. \quad (\text{CS9.5})$$

The analogous sequence for the other closed point has the support-to-global sign from (CS9.4). Passing the individual first-cohomology spaces to their Hausdorff quotients gives the same isomorphisms with $\mathcal{Q}$, with exactly those signs. This does not assert that replacing every term of an arbitrary ordinary localization sequence by its Hausdorff quotient preserves exactness.

The Weyl map from L_0 to L_∞ is the identity on chart coordinates and $+R$ on degree one, because $r_- = Rr_+$. Together with (CS9.4), this gives precisely the global $-R$ in (CS8.6), reconciling the local and global sign conventions.

CS10. Result and limits of this comparison

The constructed object is the explicit $K = \widehat{\mathbb{Z}}^\times$ summand of the original Connes-Consani sheaf. Its source chart coordinate is topologically $\mathcal{S}_0^{\text{even}}(\mathbb{R})$, as proved by the two inverse maps in (CS2.7). Its

overlap restriction is the full rational sum $2 \sum_{n \geq 1} f(nu)$. Its ordinary H^0 is the Fourier graph with four separate endpoint lines, and its ordinary H^1 is H/J . The exact topological comparison is

$$\begin{array}{ccc} H & \xrightarrow{U: h \mapsto \frac{1}{2} u^{1/2} h} & \mathcal{A} \\ \downarrow \text{quotient} & & \downarrow \text{quotient} \\ H/\overline{J} & \xrightarrow{\sim} & \mathcal{A}/\mathcal{C} \xrightarrow[\mathcal{M}]{\sim} \mathcal{B}/\mathcal{J} = \mathcal{Q}. \end{array} \quad (\text{CS10.1})$$

Every map in this diagram has been specified on its actual topology, and the ordinary-to-Hausdorff kernel is $\overline{J}/J$. The comparison carries unshifted idèle scaling to multiplication by a^s , and the ordered Čech Weyl action to $-F(1-s)$.

This proves an actual source-sheaf realization and its exact connection to all original-zeta zero jets. It does not prove a finite-field Frobenius structure, positivity of the original Weil form, separation of the weights of a new lifting obstruction, or RH. Such a conclusion is not one of the hypotheses used in any map or quotient above.

CS11. The actual image is closed, with its complete inverse

The later complete derivation SSI0-SSI9 in EXACT_SCHWARTZ_SUMMATION_IMAGE.md, independently checked in ESI0-ESI9 of EXACT_SCHWARTZ_IMAGE_INDEPENDENT_CHECK.md, strengthens the source-image statement used above. Both full proofs are retained in this edition. It gives

$$\mathcal{M}_0 S : \mathcal{S}_0^{\text{even}} \xrightarrow{\sim} \mathcal{J} \quad (\text{CS11.1})$$

as a topological isomorphism. For $b \in H$ with $F = \mathcal{M}_0 b \in \mathcal{J}$, its exact inverse is

$$f_F(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{F(2+it)}{2\zeta(2+it)} x^{-2-it} dt \quad (x > 0), \quad S f_F = b. \quad (\text{CS11.2})$$

It extends to an even Schwartz function on the whole real line, with

$$f_F(0) = 0, \quad \int_{\mathbb{R}} f_F = 0, \quad \frac{f_F^{(2r)}(0)}{(2r)!} = \frac{F(-2r)}{2\zeta'(-2r)} \quad (r \geq 1). \quad (\text{CS11.3})$$

The derivation cancels the nontrivial-zero poles with all their actual multiplicities, retains the simple trivial-zero poles of $F/(2\zeta)$, proves rapid bounds on every pole-cleared vertical strip, and shifts the inverse Mellin contour with every residue retained. SSI4 proves continuous bounds from the two vertical boundary lines; SSI5-SSI6 prove all source derivatives and endpoint moments. Thus the inverse is a map into the actual source topology, not merely a limiting representative in a quotient.

Since $\mathcal{J}$ is closed and $\mathcal{M}_0$ is a homeomorphism, (CS11.1) gives

$$J = \overline{J} = \mathcal{M}_0^{-1} \mathcal{J} = \Theta^{-1} \mathcal{J}, \quad \overline{J}/J = 0. \quad (\text{CS11.4})$$

The last equality of inverse images uses the exact scalar relation $\Theta = \frac{1}{2} \mathcal{M}_0$ and the fact that $\mathcal{J}$ is a complex vector subspace. Consequently (CS7.13) identifies the ordinary first cohomology itself with $\mathcal{Q}$, using exactly its displayed map $[b] \mapsto [\frac{1}{2} \mathcal{M}_0 b]$. The four independent endpoint lines, both support-to-global signs, the Weyl action, all actual zero jets and the unshifted arithmetic action remain as calculated in CS5 and CS8-CS9.

This theorem kills precisely the comparison kernel $\overline{J}/J$. It does not kill the prime-indexed endpoint summand of CS2A: that summand occurs in a different quotient of the source and is already in the summation kernel before taking its image. The two quotients and their topologies have not been identified. Neither vanishing is presented as Deligne's weight-separated lifting obstruction.

Supported localization, Fourier lifts, and exact weight maps — CSL0-CSL10

Proof source: CC_SUPPORTED_LOCALIZATION_AND_WEIGHT_MAPS.md.

The supported localization maps of the original-zeta sheaf

Complete derivation, 24 September 2026. Proof locators CSL0-CSL10.

CSL0. Construction stage and sources

The supporting datum is $\tau\langle Z_1; \text{no } Z_2 \rangle$. Its comparison with the unique generic point uses the complete chart reconstruction CG1-CG3, after the complete arithmetic has been recovered. All vector spaces, sums, scalar factors and quotient operations in this note occur in the coefficient sheaf over that support. None is an arithmetic operation on τ , a metric on it, or a definition of the earlier counting process. The two chart summation maps remain separately labelled.

The corrected argument and relevant verbatim passages were recalled before this calculation; PC04 in the private prerequisite ledger records that reading. In particular, the retracted addition at τ is not used, and fixed support is not substituted for trivial action on its stalk.

The original author source is Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, [arXiv:0903.2024v3](#), §5. Its unchanged author file is `sources/CC_0903_2024_v3/author_source/anounc3.tex`. Root read lines1376-1667 in full, including the sheaf, both restriction maps, the Fourier proof and the localization sequence. The source archive is retained unchanged. The spectral comparison below uses the complete construction in `CC_SHEAF_ORIGINAL_ZETA_COHOMOLOGY.md` and S1-S7 in `GLOBAL_MELLIN_SYNTHESIS.md`; the former proves the exact finite-unit sector of the adelic sheaf and the latter its closed summation image. These are named derivations, not original-author claims.

The weight comparison is with Pierre Deligne, *La conjecture de Weil. II*, §3.6, as reconstructed in DC1-DC8 of `DELIGNE_INVARIANT_CYCLE_QUOTIENT.md`. The present calculation gives the actual supported-cohomology maps of the CC coefficient sheaf. It does not replace Deligne's inertia and geometric duality terms with similarly named analytic spaces.

CSL1. Exact sheaf and unshifted actions

Let $X = \{x_+, \eta, x_-\}$, with opens

$$\emptyset, \quad U = \{\eta\}, \quad U_+ = \{x_+, \eta\}, \quad U_- = \{x_-, \eta\}, \quad X. \quad (\text{CSL1.1})$$

The notation x_+, x_- distinguishes the two closed chart points from the integer zero and from τ . In the author's notation these are $0, \infty$; that is a chart notation, not an equation identifying τ with zero.

There is an exact topological comparison with the later signed CC construction: send its points $(+, \eta, -)$ to (x_+, η, x_-) . The open sets of `CC.tex` lines499-505 are exactly (CSL1.1), so this map and its inverse are continuous. The underlying map of the full involution exchanges the two closed points and fixes the generic point. Therefore this homeomorphism intertwines the underlying involutions, and pullback

transports the coefficient sheaf below to that space, with its stalks and all cohomology unchanged. The original signed stalk formulas remain $\alpha^*T = \epsilon T^{-1}$ and $\alpha^*J = \epsilon J$, with $J^2 = \epsilon$. The topological pullback is not asserted to be a comparison of the two structure sheaves or an action of those full signed stalk generators on the new coefficient sheaf. Its domain is precisely the underlying topological space, which is sufficient for the sheaf and support calculations being made here.

Let S be the even real Schwartz space with complex coefficients and with both moments zero:

$$S = \{f \in \mathcal{S}(\mathbb{R}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0\}. \quad (\text{CSL1.2})$$

Set $V_+ = S \oplus \mathbb{C}^2$, $V_- = S \oplus \mathbb{C}^2$. Denote the additional coordinates by (c_0, c_1) , retaining the value and integral labels. The overlap is

$$A = \{a \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \text{ for every } N, j \geq 0\}. \quad (\text{CSL1.3})$$

The exact comparison with S1's centred source is the topological isomorphism

$$\mathcal{T} : \mathcal{A} \longrightarrow A, \quad (\mathcal{T}k)(u) = 2u^{-1/2}k(u), \quad \mathcal{T}^{-1}a(u) = \frac{1}{2}u^{1/2}a(u). \quad (\text{CSL1.4})$$

Multiplication by either displayed power shifts each seminorm by at most one integer weight; Leibniz's formula gives a finite sum of derivative seminorms. Thus both maps are continuous and preserve exactly the stated spaces.

Define

$$\Sigma f(u) = 2 \sum_{n \geq 1} f(nu), \quad Ra(u) = u^{-1}a(u^{-1}), \quad r_+(f, c_0, c_1) = \Sigma f, \quad r_-(h, d_0, d_1) = R\Sigma h. \quad (\text{CSL1.5})$$

These are the original CC restriction maps in the finite-unit sector. The factor2 comes from both signs in $\mathbb{Q}^\times$; it is not discarded. Poisson summation gives, for general even Schwartz h ,

$$\Sigma \hat{h}(u) = u^{-1}\Sigma h(u^{-1}) + u^{-1}h(0) - \int_{\mathbb{R}} h(v) dv. \quad (\text{CSL1.6})$$

Here $\hat{h}(t) = \int h(v) e^{-2\pi i vt} dv$. Therefore on S

$$R^2 = 1, \quad \hat{\hat{h}} = h, \quad R\Sigma h = \Sigma \hat{h}. \quad (\text{CSL1.7})$$

The two omitted terms in the restricted identity are zero by the two specified moment conditions; their formula and their independent endpoint coordinates are retained. The convergence and continuity of $\Sigma : S \rightarrow A$ follow by Schwartz summation at infinity and (CSL1.6) at zero, with logarithmic derivatives. Its injectivity also has an elementary proof: for $\Re s > 1$,

$$\int_0^\infty \Sigma f(u) u^s \frac{du}{u} = 2\zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}. \quad (\text{CSL1.8})$$

If $\Sigma f = 0$, the nonvanishing Euler product on this half-plane makes the second Mellin transform zero there. Fourier uniqueness on any vertical line in that half-plane gives $f = 0$ on the positive axis, and evenness gives $f = 0$ on $\mathbb{R}$.

For $a > 0$, retain the unshifted actions

$$\begin{aligned} T_a b(u) &= b(u/a), \\ \rho_+(a)(f, c_0, c_1) &= (f(\cdot/a), c_0, ac_1), \\ \rho_-(a)(h, d_0, d_1) &= (ah(a\cdot), ad_0, d_1). \end{aligned} \quad (\text{CSL1.9})$$

Direct substitution shows $r_{\pm}\rho_{\pm}(a) = T_a r_{\pm}$ and

$$T_a R = a R T_{a^{-1}}, \quad \widehat{ah(a \cdot)}(v) = \hat{h}(v/a). \quad (\text{CSL1.10})$$

Thus these are actions on the specified sheaf. No $a^{-1/2}$ character has been used to replace them.

CSL2. The whole Cech complex and its topology

Each of U_+, U_-, U has a point whose only neighbourhood in that open is the whole open. Sections there are the corresponding stalk functor, hence exact. These opens are acyclic for every sheaf of vector spaces. The Cech complex for this two-open cover therefore computes ordinary sheaf cohomology:

$$D^0 = V_+ \oplus V_-, \quad D^1 = A, \quad d(v_+, v_-) = r_+ v_+ - r_- v_-. \quad (\text{CSL2.1})$$

This sign convention is fixed below. Let

$$J = \Sigma(S), \quad C = \bar{J}^A, \quad Q_{\text{alg}} = A/J, \quad Q_H = A/C, \quad N = C/J. \quad (\text{CSL2.2})$$

By (CSL1.7), $r_+(V_+) = r_-(V_-) = J$, so the sum of the chart images is exactly J , without a doubling of its arithmetic spectrum. Hence

$$H^1(X, \Omega) = Q_{\text{alg}}, \quad H^j(X, \Omega) = 0 \ (j \geq 2), \quad 0 \longrightarrow N \longrightarrow Q_{\text{alg}} \longrightarrow Q_H \longrightarrow 0. \quad (\text{CSL2.3})$$

The last arrow is precisely Hausdorffization for the quotient locally convex topology. The construction of this sequence does not presuppose $J = C$. CSL10 now proves that equality by applying the complete inverse-source theorem.

Injectivity of Σ and (CSL1.7) give the full zeroth cohomology:

$$H^0(X, \Omega) = \{(\hat{h}, c_0, c_1), (h, d_0, d_1) : h \in S, c_0, c_1, d_0, d_1 \in \mathbb{C}\}. \quad (\text{CSL2.4})$$

In particular, all four labelled endpoint lines remain in the kernel. The restriction to U has image J , and its kernel consists exactly of these four lines.

CSL3. One closed support and an actual lifting map

Let $Y_+ = \{x_+\}$. Its complement is U_- . The complex with support on Y_+ is represented by

$$K_+ = [V_+ \xrightarrow{r_+} A] \quad \text{in degrees } 0, 1. \quad (\text{CSL3.1})$$

To prove this, use the localization cone of the chain map $D \rightarrow V_-[0]$, $(v_+, v_-) \mapsto v_-$. Its degree0 is $V_+ \oplus V_-$, its degree1 is $A \oplus V_-$, and its differential is $(v_+, v_-) \mapsto (r_+ v_+ - r_- v_-, v_-)$. Changing degree1 coordinates to $(b + r_- v_-, v_-)$ splits it as (CSL3.1) plus the contractible identity complex on V_- . This explicitly proves the claimed quasi-isomorphism.

Consequently

$$H_{Y_+}^0(X, \Omega) = \mathbb{C} \oplus \mathbb{C}[1], \quad H_{Y_+}^1(X, \Omega) = Q_{\text{alg}}, \quad H_{Y_+}^j = 0 \ (j \geq 2), \quad (\text{CSL3.2})$$

where $\mathbb{C}[1]$ means the character $a \mapsto a$, not a cohomological shift. The map $K_+ \rightarrow D$ is $v \mapsto (v, 0)$ in degree0 and the identity on A in degree1. It induces the identity

$$H_{Y_+}^1(X, \Omega) \xrightarrow{\sim} H^1(X, \Omega). \quad (\text{CSL3.3})$$

The connecting map from $H^0(U_-, \Omega)$ into this supported H^1 is zero. This is proved by the continuous equivariant lift

$$\ell_- : V_- \longrightarrow H^0(X, \Omega), \quad (h, d_0, d_1) \longmapsto ((\hat{h}, 0, 0), (h, d_0, d_1)). \quad (\text{CSL3.4})$$

Its restriction to U_- is the identity. Equation (CSL1.10) proves equivariance, and the Fourier transform is continuous in the Schwartz topology. Thus the exact lifting is actually constructed, including the endpoint data. Its mechanism is Poisson summation and Fourier transform; it is not presented as a new proof of weight separation.

For $Y_- = \{x_-\}$, the analogous supported complex is $[V_- \xrightarrow{r_-} A]$. Under the fixed sign of (CSL2.1), its map to D is $v \mapsto (0, v)$ in degree 0 and multiplication by -1 in degree 1. Both support choices are thereby retained with their correct sign.

CSL4. Both closed supports and the diagonal/difference sequence

Put $Y = \{x_+, x_-\}$, whose complement is the generic open U . A complex representing $R\Gamma_Y(X, \Omega)$ is

$$K^0 = V_+ \oplus V_-, \quad K^1 = A \oplus A, \quad d_K(v_+, v_-) = (r_+ v_+, r_- v_-). \quad (\text{CSL4.1})$$

Here is an explicit localization-cone proof. Represent the restriction to U by the chain map $D \rightarrow A[0]$ which is r_+ on the first summand of D^0 . Choosing r_- instead gives a homotopic map: their difference is the differential (CSL2.1). The shifted cone has differential

$$(v_+, v_-) \longmapsto (r_+ v_+ - r_- v_-, r_+ v_+) \quad \text{in } A \oplus A. \quad (\text{CSL4.2})$$

The invertible degree 1 change $(k, b) \mapsto (b, b - k)$, with inverse $(b_+, b_-) \mapsto (b_+ - b_-, b_+)$, gives (CSL4.1). Its map to D is the identity in degree 0 and $(b_+, b_-) \mapsto b_+ - b_-$ in degree 1. The boundary from A into supported degree 1 is $b \mapsto (b, b)$.

It follows, with all maps specified, that the complete nonzero portion of the localization sequence is

$$0 \rightarrow \mathbb{C}^4 \rightarrow H^0(X, \Omega) \xrightarrow{\Sigma^\wedge} A \xrightarrow{\beta} Q_{\text{alg}} \oplus Q_{\text{alg}} \xrightarrow{\pi} Q_{\text{alg}} \rightarrow 0, \quad (\text{CSL4.3})$$

where the middle map on the graph (CSL2.4) sends it to $\Sigma^\wedge \hat{h}$, and

$$\beta(b) = ([b], [b]), \quad \pi([b_+], [b_-]) = [b_+ - b_-]. \quad (\text{CSL4.4})$$

Exactness can also be checked without derived notation. The kernel of β is J . If $[b_+] = [b_-]$, the pair is diagonal and in its image. Every class is the image of $([b], 0)$, and all earlier kernels have already been computed in (CSL2.4). This proves each arrow of (CSL4.3).

After factoring out J in the middle, one obtains

$$0 \longrightarrow Q_{\text{alg}} \xrightarrow{q \mapsto (q, q)} Q_{\text{alg}} \oplus Q_{\text{alg}} \xrightarrow{(q_+, q_-) \mapsto q_+ - q_-} Q_{\text{alg}} \longrightarrow 0. \quad (\text{CSL4.5})$$

This row is split by $q \mapsto (q, 0)$. Hausdorffization here gives exactly the same row with Q_H in all three places: this follows directly from the explicit maps and the closed subspaces C , $C \oplus C$, rather than from a general exactness claim about Hausdorffization.

CSL5. The mirror sign on cochains and its retained action

The mirror exchanges x_+, x_- and fixes η , since the generic point is unique. A carried exchanged Z_2 -label at that fixed point would have to equal its flip, which is impossible for either of its two states. This derives the relevant absence of that label at $\tau\langle Z_1; \text{no } Z_2 \rangle$. It does not make the stalk involution the identity.

On the sheaf the mirror swaps the two chart sections and uses R on the overlap. The induced map on the ordered Čech complex is

$$w_D^0(v_+, v_-) = (v_-, v_+), \quad w_D^1(b) = -Rb. \quad (\text{CSL5.1})$$

Indeed, $d(v_-, v_+) = \Sigma h - R\Sigma f = -R(\Sigma f - R\Sigma h)$. The minus sign is the orientation sign from reversing the ordered cover; omitting it would not give a chain map. This does not change the author's assertion of spectral symmetry, but it fixes the actual action under our displayed identification of H^1 .

On (CSL4.1) the induced map is

$$w_K^0(v_+, v_-) = (v_-, v_+), \quad w_K^1(b_+, b_-) = (Rb_-, Rb_+). \quad (\text{CSL5.2})$$

The identity $Rr_- = r_+$, with the two chart labels exchanged, verifies the chain-map identity. Consequently (CSL4.5) is equivariant with mirror R on its first term, $(q_+, q_-) \mapsto (Rq_-, Rq_+)$ on its middle term, and $-R$ on its last term. Its continuous arithmetic- and mirror-equivariant section is

$$s(q) = \frac{1}{2}(q, -q). \quad (\text{CSL5.3})$$

The calculation is exact: $\pi s(q) = q$, while $w_K s(q) = \frac{1}{2}(-Rq, Rq) = s(-Rq)$. The scalar $1/2$ is in the already constructed complex coefficient field. It has no interpretation as halving τ .

CSL6. Exact comparison with the original-zeta receiver

Use the raw Mellin map

$$\mathcal{M}_0 b(s) = \int_0^\infty b(u) u^s \frac{du}{u}. \quad (\text{CSL6.1})$$

From (CSL1.4) and S1,

$$\mathcal{M}_0 \mathcal{I} = 2\mathcal{M}, \quad \mathcal{M}_0 T_a = a^s \mathcal{M}_0, \quad \mathcal{M}_0 Rb(s) = \mathcal{M}_0 b(1-s). \quad (\text{CSL6.2})$$

All integrals converge absolutely in every fixed strip; the inverse is

$$b(e^x) = \frac{e^{-cx}}{2\pi} \int_{\mathbb{R}} (\mathcal{M}_0 b)(c+it) e^{-itx} dt \quad (c \in \mathbb{R}). \quad (\text{CSL6.3})$$

The estimates and contour shifts are those of S1 with the exact change (CSL6.2). Thus $\mathcal{M}_0$ is a topological isomorphism $\mathcal{A} \rightarrow \mathcal{B}$.

Let $\mathcal{I}$ be the closed ideal of full-order vanishing at every original nontrivial zero, with its original multiplicity. S5 gives $\mathcal{M}_0 C = \mathcal{I}$, so

$$Q_H \xrightarrow{\sim} \mathcal{Q} = \mathcal{B}/\mathcal{I}, \quad [b] \mapsto [\mathcal{M}_0 b]. \quad (\text{CSL6.4})$$

Under this exact isomorphism, the entire supported row (CSL4.5) is the diagonal/difference row of the actual $\mathcal{Q}$, with the same full multiplicities and quotient topology. It is not an unrestricted product of jets or an algebraic direct sum.

For the retained source function

$$f_0(v) = \frac{\pi}{2}v^2(2\pi v^2 - 3)e^{-\pi v^2},$$

the complete auxiliary transform is

$$\mathcal{M}_0 \Sigma f_0(s) = \frac{s(s-1)}{4} \pi^{-s/2} \Gamma(s/2) \zeta(s) = 2F_0(s). \quad (\text{CSL6.5})$$

Its endpoint values are $1/4$ at both 0 and 1. At each negative even integer, with $r \geq 1$, its retained cancellation value is

$$(2F_0)(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{4} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0. \quad (\text{CSL6.6})$$

These are values of the transform; the original ζ still has its pole at 1 and trivial zeros at $-2r$. Its unit and prime-power repetitions remain in

$$\zeta(s) = 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{m \geq 1} (\log p) p^{-ms} \quad (\Re s > 1). \quad (\text{CSL6.7})$$

All these arithmetic labels belong to the recovered complete arithmetic, not to the primitive support.

CSL7. What the computed action says about weights

On an actual original-zero block at ρ of multiplicity m_ρ , multiplication by s is $\rho I + N_\rho$, where $N_\rho^{m_\rho} = 0$. Consequently the unshifted action in every Q_H term above is

$$T_a = a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^j. \quad (\text{CSL7.1})$$

It follows from the original strip $0 < \Re \rho < 1$, for $a > 1$, that its character weight $2 \log_a |a^\rho|$ lies in $(0, 2)$. The kernel, middle terms and quotient of (CSL4.5) have these same character weights. The diagonal and difference are explicit nonzero equivariant maps; they are not forced to vanish by separated weights.

The endpoint kernel has the separately computed weights. In V_+ , value has character 1 and integral has character a ; in V_- , value has character a and integral has character 1. Thus the four endpoint lines have weights 0, 2, 2, 0, in that order. This follows from (CSL1.9), not from a weight assigned to a support point.

This weight computation states the exact analytic character action. It does not assert that these spaces are finite-dimensional ℓ -adic cohomology, that their characters are algebraic Weil numbers, or that an arithmetic weight filtration has been imported without a comparison.

CSL8. The precise comparison with Deligne's localization cross

In Deligne's theorem, the top obstruction group and the inertia term are

$$O_i = H^{2N-i-1}(X_s, E)^\vee(-N), \quad K_i = H^{i-1}(X_{\bar{\eta}}, E)_I(-1), \quad (\text{CSL8.1})$$

and the obstruction to specialization is

$$\text{im}(\partial_i)/\partial_i j_i(K_i). \quad (\text{CSL8.2})$$

DC3-DC6 derives the Tate factors from tame inertia and the dualizing complex of the smooth total space. Those operations give the weights $\leq i$ on the classes and $\geq i + 1$ on the obstruction receiver. They are additional geometric maps, not consequences of the name “localization.”

For the CC sheaf calculated here, the exact specialization-like map actually constructed by removing one closed point is (CSL3.3). It is an isomorphism, with inverse the identity on the displayed quotient. The restriction lift (CSL3.4) is explicit. For removal of both closed points, the exact receiver and its maps are (CSL4.3)–(CSL4.5), and the equivariant section is (CSL5.3). These are genuine applications of localization to this coefficient sheaf, and every map has now been calculated on its source.

In particular, neither supported calculation produces an uncomputed Tate shift: the character identity (CSL7.1) determines its action. The constructed comparison to the Mellin receiver is (CSL6.4), not an identification with (CSL8.1). A geometric extension carrying an additional normal direction can have a different dualizing or inertia term; the preceding calculation makes no claim against such an extension. Its exact current contribution is the original-zeta cohomological receiver, its support maps, its Fourier lift, and its mirror-equivariant split supported row. It does not certify completion of the requested Deligne weight-lifting application.

CSL9. All comparison arrows and retained information

The following commuting formulas describe the full receiver used here:

$$\begin{aligned}
\mathcal{M}_0 \circ r_+ &= 2 \mathcal{M} \circ \mathcal{E}, \\
\mathcal{M}_0 \circ r_- &= 2 (F(s) \mapsto F(1-s)) \circ \mathcal{M} \circ \mathcal{E}, \\
\overline{\mathcal{M}_0}^{\oplus 2} \circ \beta &= \text{diag} \circ \overline{\mathcal{M}_0}, \\
\overline{\mathcal{M}_0} \circ \pi &= \text{diff} \circ \overline{\mathcal{M}_0}^{\oplus 2}, \\
\overline{\mathcal{M}_0}^{\oplus 2} \circ s(q) &= \frac{1}{2} (\overline{\mathcal{M}_0} q, -\overline{\mathcal{M}_0} q).
\end{aligned} \tag{CSL9.1}$$

The first two formulas act on the Schwartz component and send each explicitly retained endpoint summand to zero. The last three use the quotient of β by its kernel J , followed by the stated Hausdorffization. Hence their domains and codomains are exactly the displayed Q_H spaces. Every original-zero jet survives in $\mathcal{Q}$; CSL10 calculates the separately retained closure difference as $N = C/J = 0$. The support remains $\tau\langle Z_1; \text{no } Z_2 \rangle$, while the full coefficient action, both charts and their orientation sign remain visible.

CSL10. Exact image, strict maps, and the calculated comparison kernel

The complete proof SSI0-SSI9 in EXACT_SCHWARTZ_SUMMATION_IMAGE.md constructs a continuous inverse to

$$\mathcal{M}_0 \Sigma : S \xrightarrow{\sim} \mathcal{I}. \tag{CSL10.1}$$

Its independently checked proof ESI0-ESI9 is retained in EXACT_SCHWARTZ_IMAGE_INDEPENDENT_CHECK.md. With $F = \mathcal{M}_0 b$, the inverse has the full formula

$$f_F(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{F(2+it)}{2\zeta(2+it)} x^{-2-it} dt \quad (x > 0). \tag{CSL10.2}$$

The pole-cleared strip estimate, contour residues and smooth even extension in SSI4-SSI6 prove $f_F \in S$, continuity in all Schwartz seminorms, and $\Sigma f_F = b$. Its even derivatives at zero are exactly

$(2r)!F(-2r)/(2\zeta'(-2r))$; both vanishing moments and every nontrivial multiplicity are proved there. Thus the inverse does not discard the trivial-zero data of the original function.

It follows, on the actual spaces already used in CSL2, that

$$J = \mathcal{M}_0^{-1}\mathcal{J} = C, \quad N = 0, \quad Q_{\text{alg}} \xrightarrow{\sim} Q_H \xrightarrow{\sim} \mathcal{Q}. \quad (\text{CSL10.3})$$

Each map is a homeomorphism with the quotient topologies. Both chart restrictions are strict onto J : the plus chart has right inverse $b \mapsto (\Sigma^{-1}b, 0, 0)$, and the minus chart has right inverse $b \mapsto (\widehat{\Sigma^{-1}b}, 0, 0)$. Poisson summation proves the second identity. The Čech differential has right inverse from its image $b \mapsto ((\Sigma^{-1}b, 0, 0), 0)$, so its image is closed and its ordinary first cohomology is already Hausdorff.

Therefore the one-point localization maps and the two-point diagonal/difference row of CSL3-CSL5 are maps on the actual original-zero quotient without any extra closure operation. Their signs, common arithmetic action and section $q \mapsto \frac{1}{2}(q, -q)$ are unchanged. This is the vanishing of the exact comparison kernel N , not a deduction that the source has pure weight or that the independently calculated localization row is Deligne's cross.

The exact Schwartz summation image and its complete inverse — SSI0-SSI10

Proof source: EXACT_SCHWARTZ_SUMMATION_IMAGE.md.

Exact Schwartz summation image and the original-zeta cohomology

24 September 2026. Complete global calculation SSI0-SSI10. This strengthens the closure result S5 in GLOBAL_MELLIN_SYNTHESIS.md to an equality with the actual image. It uses the original zeta function throughout. All zero multiplicities, trivial-zero residues, endpoint conditions, the two signs in rational summation, and the two chart orientations are retained.

SSI0. Prerequisites and the precise operation

The governing user arguments and corrections are in argument_reconstruction/CORPUS_AND_OPERATION_RULE.md, LATEST_DIRECT_ARGUMENTS.md, CORRECTION_CHAINS.md, and the complete verbatim corpus. They require global reconstruction before arithmetic operations. The integers, their unit, primes, prime powers, complex coefficient field, Schwartz spaces and Mellin maps in the present calculation are the already reconstructed receivers in CANONICAL_GENERIC_RECONSTRUCTION.md, CG0-CG4, and GLOBAL_MELLIN_SYNTHESIS.md, S1-S6. No operation here constructs them by selecting a few primes. In particular, the Möbius inverse in SSI7 uses the entire reconstructed arithmetic.

The support notation remains $Z_0, Z_1, Z_2, \dots, \tau$. No addition at τ , numerical value of τ , distance to τ , or purity assumption is introduced. The theorem concerns an actual linear map on the coefficient spaces carried at the support. Its receiving cohomology maps are given in SSI8; no arithmetic weight separation is inferred from closedness.

The author source is Alain Connes and Caterina Consani, *Schemes over F_1 and zeta functions*, [arXiv:0903.2024v3](https://arxiv.org/abs/0903.2024v3), original announc3.tex, §5, lines1376-1667, read in full. Source SHA256: 8254a72d27b082e0a3550e6be9398bae1c7e267303dc2b86a904703f58569fd4. The retained archive has SHA256 f09e4f8a917de991a84b508023598259e9d79694425ce5bd1ddd33d7c729e53f. This note proves a real even-Schwartz summation theorem and applies it to the explicitly defined coefficient sheaf CSL1. Its further identification with a particular convention for adelic coinvariants is a separate comparison; no missing coinvariant proof is supplied by a name.

The full dependency for the growth estimate is S2-S3 and its independent proof audit R1-R3 in GLOBAL_MELLIN_SYNTHESIS_REVIEW.md. The proof below uses the subquadratic division bound proved there, rather than a conjectural lower bound on zeta in the critical strip. Poisson summation and ordinary Fourier inversion use the same convention as that source. The standard Gamma recurrence and reflection identities are used with their factors displayed.

SSI1. Spaces, maps and the claimed equality

Let

$$S = \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), \quad f(0) = 0, \quad \int_{\mathbb{R}} f(v) dv = 0\}. \quad (\text{SSI1.1})$$

It is a closed subspace of Schwartz space. Its seminorms may be taken to be

$$q_{r,j}(f) = \sup_{v \in \mathbb{R}} (1 + |v|)^r |f^{(j)}(v)|, \quad r, j \geq 0. \quad (\text{SSI1.2})$$

Let

$$A = \{b \in C^\infty(\mathbb{R}_{>0}) : p_{N,j}(b) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j b(u)| < \infty \text{ for all } N, j \geq 0\}, \quad (\text{SSI1.3})$$

and let $\mathcal{B}$ consist of entire functions with seminorms

$$b_{A_0,M}(F) = \sup_{|\sigma| \leq A_0, t \in \mathbb{R}} (1 + |t|)^M |F(\sigma + it)| < \infty \quad (A_0, M \geq 0). \quad (\text{SSI1.4})$$

Integer indices suffice in these seminorms. The raw Mellin map and its inverse are

$$\mathcal{M}_0 b(s) = \int_0^\infty b(u) u^s \frac{du}{u}, \quad b(e^x) = \frac{e^{-cx}}{2\pi} \int_{\mathbb{R}} (\mathcal{M}_0 b)(c + it) e^{-itx} dt. \quad (\text{SSI1.5})$$

They are mutually inverse continuous maps $A \leftrightarrow \mathcal{B}$, for any real c . This is S1 with its exact change of exponent. Specifically, for its centered map $\mathcal{M}$ and $\mathcal{T}k(u) = 2u^{-1/2}k(u)$, one has $\mathcal{M}_0 \mathcal{T} = 2\mathcal{M}$. Multiplication by these displayed powers preserves all the spaces with continuous inverse.

For every actual nontrivial zero ρ of the original ζ , let m_ρ be its original multiplicity. Define the closed subspace

$$I = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every } \rho \text{ and } 0 \leq j < m_\rho\}. \quad (\text{SSI1.6})$$

Jet evaluation is continuous by Cauchy's integral formula on a fixed small circle, so this intersection is closed. Define

$$\Sigma : S \longrightarrow A, \quad \Sigma f(u) = 2 \sum_{n \geq 1} f(nu), \quad J = \Sigma S. \quad (\text{SSI1.7})$$

The factor 2 is the contribution of both signs in the even sector. The theorem proved below is the topological isomorphism

$$\boxed{\mathcal{M}_0 \Sigma : S \xrightarrow{\sim} I.} \quad \boxed{J = \mathcal{M}_0^{-1} I = \bar{J}^A.} \quad (\text{SSI1.8})$$

The inverse is actual Schwartz inversion, with no limiting class substituted for a source function.

SSI2. Continuity, inclusion, injectivity and full meromorphic division

On $u \geq 1$, every logarithmic derivative of the series in (SSI1.7) is bounded by a finite sum of Schwartz seminorms times $u^{-L} \sum n^{-L}$, for any sufficiently large integer $L > 1$. For $0 < u \leq 1$, the Fourier convention

$$\hat{f}(t) = \int_{\mathbb{R}} f(v) e^{-2\pi i vt} dv \quad (\text{SSI2.1})$$

and Poisson summation give the full identity

$$\Sigma \hat{f}(u) = u^{-1} \Sigma f(u^{-1}) + u^{-1} f(0) - \int_{\mathbb{R}} f(v) dv. \quad (\text{SSI2.2})$$

For $f \in S$, the last two terms vanish by the two stated moment conditions; $\hat{f} \in S$. This proves the required estimates at zero and the continuity $\Sigma : S \rightarrow A$.

For $\Re s > 1$, absolute integration and summation give

$$\mathcal{M}_0 \Sigma f(s) = 2\zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}. \quad (\text{SSI2.3})$$

The positive-axis Mellin integral is holomorphic on $\Re s > -2$, since evenness and $f(0) = 0$ give $f(v) = O(v^2)$. It vanishes at $s = 1$, because $\int_{\mathbb{R}} f = 0$. Thus (SSI2.3) continues through the zeta pole at 1 and proves full-order vanishing at every nontrivial zero. This proves $\mathcal{M}_0 \Sigma S \subset I$.

If $\Sigma f = 0$, the nonvanishing Euler product for ζ on $\Re s > 1$ implies that the positive-axis Mellin transform is zero on that half-plane. Fourier uniqueness on a vertical line gives $f = 0$ on $\mathbb{R}_{>0}$, and evenness and continuity give $f = 0$ on all of $\mathbb{R}$. Hence Σ is injective.

Conversely, let $F \in I$ and put

$$H(s) = \frac{F(s)}{2\zeta(s)}. \quad (\text{SSI2.4})$$

At a nontrivial zero ρ , write $\zeta(s) = (s - \rho)^{m_\rho} u(s)$, with $u(\rho) \neq 0$, and $F(s) = (s - \rho)^{m_\rho} v(s)$. Then $H = v/(2u)$ extends holomorphically there. No simplicity or separation of these zeros is assumed. The only possible poles of H are the simple trivial-zero poles

$$\text{Res}_{s=-2r} H(s) = c_r(F) := \frac{F(-2r)}{2\zeta'(-2r)}, \quad r \geq 1. \quad (\text{SSI2.5})$$

The original functional equation proves that these zeros are simple: its sine factor has a simple zero at $-2r$, whereas its Gamma factor and $\zeta(1 + 2r)$ are finite and nonzero. At zero and one the exact values are

$$H(0) = -F(0), \quad H(1) = 0, \quad H'(1) = \frac{F(1)}{2}. \quad (\text{SSI2.6})$$

Here $\zeta(0) = -1/2$ and the residue of ζ at 1 is 1. The endpoint pole and each trivial zero have been used, not discarded.

SSI3. Growth and exact estimates on the two vertical edges

Keep the auxiliary source transform of S2 with all factors:

$$F_*(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{SSI3.1})$$

It is the function denoted F_0 in S2. Its zeros are exactly the nontrivial zeros with their multiplicities, and $F_*(0) = F_*(1) = 1/8$. It is used only for the proved division estimate. S3 gives

$$\sup_{|\sigma| \leq A_0} |F/F_*(\sigma + it)| \leq C_{A_0, F} \exp\{C_{A_0}(|t| + 2)^{3/2} \log(|t| + 2)\}. \quad (\text{SSI3.2})$$

Retaining the comparison rather than replacing ζ , one has

$$H(s) = \frac{s(s-1)}{16} \pi^{-s/2} \Gamma(s/2) \frac{F(s)}{F_*(s)}. \quad (\text{SSI3.3})$$

On every bounded real strip and $|t| \geq 1$, the Gamma factor has an upper bound by a polynomial in $1 + |t|$. Indeed choose an integer L for which $\sigma/2 + L \geq 1$ throughout the strip. The recurrence and Euler integral give

$$|\Gamma(s/2)| = \frac{|\Gamma(s/2 + L)|}{\prod_{j=0}^{L-1} |s/2 + j|} \leq \frac{\Gamma(\sigma/2 + L)}{(|t|/2)^L}. \quad (\text{SSI3.4})$$

The numerator is bounded on the compact interval of its real arguments. Equations (SSI3.2)–(SSI3.4) therefore give subquadratic growth for H , away from its finitely many poles in each strip. This is sufficient for the maximum-principle argument; no conjectural reciprocal-zeta estimate is used.

Choose an integer $N \geq 0$, let $a = -2N - 1$, and choose a real $b \geq 2$. On the right edge, the absolutely convergent reciprocal Euler series yields

$$|\zeta(b + it)^{-1}| \leq \sum_{n \geq 1} |\mu(n)| n^{-b} \leq \zeta(b). \quad (\text{SSI3.5})$$

Here μ is the Möbius function of the complete reconstructed integers; its use does not select a finite set of primes.

On the left edge retain the functional equation in its full multiplier form

$$\zeta(s) = \chi(s) \zeta(1 - s), \quad \chi(s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)} = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s). \quad (\text{SSI3.6})$$

Writing $y = t/2$, Gamma recurrence and reflection give the exact identity

$$|\chi(-2N - 1 + it)|^2 = \pi^{-4N-3} y \coth(\pi y) \prod_{k=1}^N (k^2 + y^2) \prod_{j=0}^N ((j + \frac{1}{2})^2 + y^2). \quad (\text{SSI3.7})$$

The empty product is 1 and $y \coth(\pi y)$ has the value $1/\pi$ at zero. To verify (SSI3.7), apply the recurrence to $\Gamma(N + 1 - iy)$, and to $\Gamma(-N - 1/2 + iy)$ up to $\Gamma(1/2 + iy)$. The remaining exact quotient of squared moduli is

$$\frac{|\Gamma(1 - iy)|^2}{|\Gamma(1/2 + iy)|^2} = \frac{\pi y / \sinh(\pi y)}{\pi / \cosh(\pi y)} = y \coth(\pi y). \quad (\text{SSI3.8})$$

These two modulus identities follow from $\Gamma(z)\Gamma(1-z) = \pi / \sin(\pi z)$ and $\Gamma(1+z) = z\Gamma(z)$; their values at zero are their continuous limits. All factors on the right of (SSI3.7) are positive for real y , and their growth at infinity gives constants $c_N, C_N > 0$ with

$$c_N(1 + |t|)^{4N+3} \leq |\chi(a + it)|^2 \leq C_N(1 + |t|)^{4N+3}. \quad (\text{SSI3.9})$$

For completeness, this follows on $|t| \geq 1$ by bounding each quadratic between positive multiples of $1 + t^2$, and bounding $y \coth(\pi y)$ between positive multiples of $1 + |y|$. On $|t| \leq 1$, positivity and continuity give the same inequalities after changing the constants. Since $\Re(1 - a - it) = 2N + 2 > 1$, (SSI3.5) applies to $\zeta(1 - a - it)$. Thus $1/\zeta$ is polynomially bounded on the left edge too, with no zero on either edge.

It follows that H decreases faster than every inverse power on each edge. Every such bound is bounded by a fixed constant times a finite seminorm of F in $\mathcal{B}$. The constants can depend on N, b and the requested power, but not on F .

SSI4. The maximum principle removes the Gaussian approximation

Define

$$P_N(s) = \prod_{r=1}^N (s + 2r), \quad P_0 = 1, \quad V_N = P_N H. \quad (\text{SSI4.1})$$

This product cancels exactly the possible poles inside the strip $a \leq \Re s \leq b$. Hence V_N is holomorphic on a neighborhood of that closed strip. For an integer $M \geq 0$ set $\kappa = 2N + 2$ and

$$W(s) = (s + \kappa)^M V_N(s). \quad (\text{SSI4.2})$$

The estimates in SSI3 give, on the two vertical edges,

$$|W(s)| \leq C_{N,b,M} b_{A_0,M+N}(F), \quad A_0 \geq \max(2N+1, b). \quad (\text{SSI4.3})$$

Indeed the polynomial factors have total degree $M+N$, and the reciprocal-zeta bounds on each edge are bounded, in addition to having the more precise polynomial behavior already computed.

For $\varepsilon > 0$, apply the maximum-modulus principle to $e^{\varepsilon s^2} W(s)$ on the rectangle with vertical edges a, b and horizontal edges $\Im s = \pm T$. On the horizontal edges, (SSI3.2)-(SSI3.4) give a bound of the form

$$C_{F,N,b,M,\varepsilon} (1+T)^L \exp\{-\varepsilon T^2 + C_{N,b}(T+2)^{3/2} \log(T+2)\}, \quad (\text{SSI4.4})$$

which tends to zero as $T \rightarrow \infty$, uniformly in the real coordinate. On the vertical edges, (SSI4.3) gives

$$|e^{\varepsilon s^2} W(s)| \leq e^{\varepsilon \max(a^2, b^2)} C_{N,b,M} b_{A_0,M+N}(F). \quad (\text{SSI4.5})$$

Fix any interior s , let $T \rightarrow \infty$, and then let $\varepsilon \downarrow 0$. The resulting estimate is

$$|W(s)| \leq C_{N,b,M} b_{A_0,M+N}(F). \quad (\text{SSI4.6})$$

This also holds on the edges by continuity. Since $\Re(s + \kappa) \geq 1$ throughout the strip,

$$\sup_{a \leq \sigma \leq b, t \in \mathbb{R}} (1+|t|)^M |P_N(\sigma + it) H(\sigma + it)| \leq C'_{N,b,M} b_{A_0,M+N}(F). \quad (\text{SSI4.7})$$

The growth constant depending on F in (SSI4.4) is used only to remove the horizontal edges for a fixed F . It does not enter the final bound (SSI4.7). Thus (SSI4.7) proves continuous seminorm control as well as existence of rapid decrease. There is no limit of source classes left in the assertion.

On a vertical line avoiding the finitely many trivial poles, division by P_N gives corresponding rapid bounds for H . On horizontal lines with $|t| \geq 1$, each $|s + 2r| \geq |t|$, so the same bounds justify all contour shifts below. Cauchy's integral formula on small disks away from the poles also gives rapid bounds for every holomorphic derivative of H .

SSI5. Actual inverse Mellin and all even Taylor coefficients

Define for $x > 0$

$$f_F(x) = \frac{1}{2\pi} \int_{\mathbb{R}} H(2+it) x^{-2-it} dt. \quad (\text{SSI5.1})$$

This integral is absolutely convergent and can be differentiated any finite number of times. On every compact interval of $x > 0$, the integrable majorant follows from (SSI4.7). The exact derivative factor is

$$\frac{d^j}{dx^j} x^{-s} = (-1)^j (s)_j x^{-s-j}, \quad (s)_j = s(s+1) \cdots (s+j-1), \quad (s)_0 = 1. \quad (\text{SSI5.2})$$

Contour shifting to any $c \geq 2$ crosses no pole and gives

$$f_F^{(j)}(x) = \frac{(-1)^j}{2\pi} \int_{\mathbb{R}} (c+it)_j H(c+it) x^{-c-it-j} dt. \quad (\text{SSI5.3})$$

The horizontal contributions tend to zero by SSI4. Taking c arbitrarily large proves faster-than-any-power decay of every derivative at infinity.

Shifting the contour to $a = -2N - 1$ crosses precisely the poles in (SSI2.5) for $1 \leq r \leq N$, with their positive residue sign. Thus

$$f_F(x) = \sum_{r=1}^N c_r(F)x^{2r} + \frac{1}{2\pi} \int_{\mathbb{R}} H(a+it)x^{-a-it} dt. \quad (\text{SSI5.4})$$

For every integer $j \geq 0$, the remainder after differentiating this equality j times is bounded, on $0 < x \leq 1$, by

$$\frac{x^{2N+1-j}}{2\pi} \int_{\mathbb{R}} |(a+it)_j H(a+it)| dt. \quad (\text{SSI5.5})$$

This integral is finite and bounded by a finite seminorm of F . For any prescribed derivative order choose N large enough that $2N + 1 - j > 0$. Each derivative then has a finite limit as $x \downarrow 0$, equal to the derivative at zero of the even polynomial in (SSI5.4). These limits consistently extend f_F to a smooth function on the half-line: the fundamental theorem of calculus applied to successive derivatives, followed by their continuous limits, proves that the extended derivative is indeed the derivative of the extended preceding function. Extending by $f_F(-x) = f_F(x)$ is smooth because every odd one-sided derivative is zero. In particular,

$$f_F(0) = 0, \quad f_F^{(2r-1)}(0) = 0, \quad \boxed{\frac{f_F^{(2r)}(0)}{(2r)!} = \frac{F(-2r)}{2\zeta'(-2r)}} \quad (r \geq 1). \quad (\text{SSI5.6})$$

Equations (SSI5.3) and (SSI5.4) prove $f_F \in \mathcal{S}(\mathbb{R})$, not merely a function on the punctured positive axis.

On the line $\Re s = 2$, $t \mapsto H(2+it)$ is Schwartz by SSI4 and Cauchy's formula. Ordinary Fourier inversion applied to $x = e^y$ in (SSI5.1) therefore gives

$$\int_0^\infty f_F(x)x^s \frac{dx}{x} = H(s) \quad (\Re s = 2). \quad (\text{SSI5.7})$$

Both sides are holomorphic on $\Re s > -2$, since $f_F(x) = O(x^2)$ at zero, and have no poles there. The identity theorem extends (SSI5.7) throughout that half-plane. At $s = 1$, (SSI2.6) gives

$$\int_{\mathbb{R}} f_F(x) dx = 2H(1) = 0. \quad (\text{SSI5.8})$$

Thus $f_F \in \mathcal{S}$. By (SSI2.3), $\mathcal{M}_0 \Sigma f_F = 2\zeta H = F$ on $\Re s > 1$; both sides are entire, so equality holds on the whole plane. This proves the surjectivity in (SSI1.8).

SSI6. Continuity of the inverse and endpoint information

For any Schwartz seminorm $q_{r,j}$, split its domain into $|x| \geq 1$ and $|x| \leq 1$. On the first part use (SSI5.3) with $c \geq \max(2, r+2)$, and a rapid bound for $H(c+it)$ of order greater than $j+1$. It yields a constant times one finite seminorm of F . On the second part choose $N \geq j+1$ in (SSI5.4). The finitely many polynomial coefficients satisfy

$$|c_k(F)| \leq \frac{b_{A_0,0}(F)}{2|\zeta'(-2k)|} \quad (A_0 \geq 2N), \quad (\text{SSI6.1})$$

and the remainder is bounded by (SSI5.5) and (SSI4.7). Consequently, for suitable finite indices A_0, M depending on r, j ,

$$q_{r,j}(f_F) \leq C_{r,j} b_{A_0,M}(F). \quad (\text{SSI6.2})$$

This proves continuity of the inverse. Together with SSI2 and SSI5 it proves the topological isomorphism (SSI1.8). Because I is closed, the image J is closed in A .

The endpoint information is more than the two vanishing conditions. Differentiating (SSI5.7), with dominated convergence, and using (SSI2.6) gives

$$\int_0^\infty f_F(x) \frac{dx}{x} = -F(0), \quad \int_0^\infty f_F(x) \log x \, dx = \frac{F(1)}{2}. \quad (\text{SSI6.3})$$

All Taylor coefficients at the original trivial zeros are retained in (SSI5.6). Higher coefficients at these points are obtained by Laurent division of the full original $F/(2\zeta)$; no substitution by the completed function is required. In particular, the Gamma cancellation in (SSI3.1) did not erase the information that reconstructs the source jet.

In the centered convention of S1-S6, $\mathcal{E}f(u) = u^{1/2} \sum_{n \geq 1} f(nu)$. Hence $\Sigma = \mathcal{TE}$, and the strengthened centered statement is

$$\mathcal{ME} : S \xrightarrow{\sim} I, \quad \int_0^\infty f_F^{\text{centered}}(x) x^s \frac{dx}{x} = \frac{F(s)}{\zeta(s)}, \quad f_F^{\text{centered}} = 2f_F. \quad (\text{SSI6.4})$$

The original Gaussian approximants in S4-S5 therefore converge in Schwartz space to this actual source function: their transforms converge in I , and the continuous inverse just proved carries that convergence to every Schwartz seminorm. This strengthens the older closure statement without invalidating its proof.

SSI7. An inverse on the counting side and the full Fourier map

For $b \in J$, the exact inverse can also be written directly in the original counting variable:

$$\boxed{\Sigma^{-1}b(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n)b(nx)} \quad (x > 0). \quad (\text{SSI7.1})$$

For fixed $x > 0$, the sum and all its derivatives converge absolutely and locally uniformly, using the arbitrary decay of b at infinity. To prove the identity, write $b = \Sigma f$ by SSI5. Then, for any $L > 1$,

$$\sum_{n,m \geq 1} |f(mnx)| \leq C_{x,L} \sum_{n,m \geq 1} (mn)^{-L} < \infty. \quad (\text{SSI7.2})$$

The same bound with a sufficiently higher decay order applies to derivatives. Thus sums can be rearranged, and

$$\frac{1}{2} \sum_n \mu(n)b(nx) = \sum_{k \geq 1} f(kx) \sum_{n|k} \mu(n) = f(x). \quad (\text{SSI7.3})$$

The last finite divisor sum is 1 at $k = 1$ and 0 otherwise, as follows by expanding $\prod_{p|k} (1 - 1)$. All prime divisors are those already present in the complete arithmetic. This calculation does not posit independent meanings for a few labels before that reconstruction. Formula (SSI7.1) is not asserted to be termwise differentiable at $x = 0$; its smooth extension and its complete jet there are supplied by (SSI5.4)–(SSI5.6).

Let $Rb(u) = u^{-1}b(u^{-1})$ and $jF(s) = F(1 - s)$. The functional equation, with its nonzero multiplier on the critical strip, pairs all actual nontrivial-zero jets, so $jI = I$. The exact inverse intertwines the full reflection with Fourier transform:

$$\Sigma^{-1}Rb = \widehat{\Sigma^{-1}b}, \quad f_{jF} = \widehat{f_F}. \quad (\text{SSI7.4})$$

Proof: (SSI2.2) gives $R\Sigma f = \Sigma \widehat{f}$ on S ; injectivity proves the first equality. Raw Mellin gives $\mathcal{M}_0 Rb(s) = \mathcal{M}_0 b(1-s)$, proving the second. Equivalently their positive-axis Mellin transforms satisfy

$$\int_0^\infty \widehat{f_F}(x) x^s \frac{dx}{x} = \frac{F(1-s)}{2\zeta(s)} = \frac{\zeta(1-s)}{\zeta(s)} H(1-s), \quad (\text{SSI7.5})$$

as meromorphic identities. The complete functional multiplier is retained in the last quotient and in (SSI3.6); it has not been set equal to 1.

For $a > 0$, let $T_a b(u) = b(u/a)$ and $D_a f(x) = f(x/a)$. Direct summation gives $\Sigma D_a = T_a \Sigma$, and direct Mellin transformation gives multiplication by a^s . Therefore the inverse is continuously equivariant for every positive dilation. Its derivative $-x\partial_x$ on S corresponds to multiplication by s on I ; this follows by differentiating in $\log a$, or by integration by parts with the endpoint decay already proved. Every multiplicity and nilpotent zero-jet action in the quotient consequently remains unchanged.

SSI8. Actual cohomology, strict support maps and the vanishing closure quotient

Apply the theorem to the sheaf explicitly constructed in CC_SUPPORTED_LOCALIZATION_AND_WEIGHT_MAPS.md, CSL1-CSL9. Its underlying space has opens $\emptyset, U, \bar{U}_+, U_-, X$, with $U = \{\eta\}$, $U_\pm = \{x_\pm, \eta\}$; its chart spaces are $V_\pm = S \oplus \mathbb{C}^2$, overlap A , and restrictions

$$r_+(f, c_0, c_1) = \Sigma f, \quad r_-(h, d_0, d_1) = R\Sigma h. \quad (\text{SSI8.1})$$

Each map is a continuous map onto J , with continuous right inverse into its Schwartz summand, and kernel exactly its two endpoint lines. Equation (SSI7.4) proves that the two images are the same closed subspace, with their Fourier comparison retained.

The ordinary Cech complex is

$$V_+ \oplus V_- \xrightarrow{r_+ - r_-} A. \quad (\text{SSI8.2})$$

Its differential has closed image J ; the inverse of Σ gives a continuous right inverse from that image to the first chart summand. Thus this differential is strict onto its image. Ordinary sheaf cohomology, with its quotient topology, is already Hausdorff, and the exact isomorphism is

$$\boxed{H^1(X, \Omega) = A/J \xrightarrow{\sim} \mathcal{B}/I, \quad [b] \mapsto [\mathcal{M}_0 b].} \quad (\text{SSI8.3})$$

In the notation $C = \bar{J}$, $Q_{\text{alg}} = A/J$, $Q_H = A/C$, $N = C/J$, the previously retained comparison kernel is now calculated:

$$\boxed{C = J, \quad N = 0, \quad Q_{\text{alg}} \xrightarrow{\sim} Q_H.} \quad (\text{SSI8.4})$$

This is an unconditional vanishing of this particular closure quotient. It is not identified with Deligne's lifting-obstruction group.

All supported rows in CSL therefore act on the actual ordinary cohomology, with no additional Hausdorffization step. For one closed point, $H^1_{\{x_+\}}(X, \Omega) \rightarrow H^1(X, \Omega)$ is the identity on A/J ; the other chart has the retained orientation minus sign. For both closed points the exact row is

$$0 \longrightarrow Q \xrightarrow{q \mapsto (q, q)} Q \oplus Q \xrightarrow{(q_+, q_-) \mapsto q_+ - q_-} Q \longrightarrow 0, \quad Q = \mathcal{B}/I. \quad (\text{SSI8.5})$$

The quotient of the generic restriction by its kernel J is the first Q . The section $q \mapsto \frac{1}{2}(q, -q)$ is continuous and dilation-equivariant. It is also mirror-equivariant with j on the first Q , $(q_+, q_-) \mapsto$

(jq_-, jq_+) in the middle, and $-j$ on the last Q . Substituting into the difference verifies both the section identity and the orientation sign. The scalar $1/2$ belongs to the already reconstructed coefficient field; it is not an operation on τ .

The full zeroth cohomology remains the Fourier graph

$$\{((\hat{h}, c_0, c_1), (h, d_0, d_1)) : h \in S, c_0, c_1, d_0, d_1 \in \mathbb{C}\}. \quad (\text{SSI8.6})$$

Its four endpoint lines retain the unshifted dilation characters $1, a, a, 1$. None is removed by the closed-image theorem. The proof affects the topological comparison kernel in (SSI8.4), not the original nontrivial-zero jets of Q .

SSI9. Exact scope of the new vanishing result

Every $F \in I$ now has a unique actual source function, given by (SSI5.1), with its complete expansion (SSI5.4), continuity (SSI6.2), counting inverse (SSI7.1), and Fourier comparison (SSI7.4). This resolves the old image-versus-closure question globally, for every zero and every multiplicity at once. It strengthens S5 and allows the algebraic cohomology and supported maps of CSL to receive the exact quotient $\mathcal{B}/I$ directly.

The target requested from Deligne's theorem remains the weight-separated vanishing for the actual τ -supported geometric lifting map. In Deligne §3.6, the obstruction receiver is $H^{2N-i-1}(X_s, E)^\vee(-N)$, and the inertia term is $H^{i-1}(X_{\bar{\eta}}, E)_I(-1)$. Equations (SSI8.3)-(SSI8.5) construct the analytic receiver and its support maps; they do not insert an identification with those two geometric terms. The present proof does not move any original zero or show $\Re \rho = 1/2$. Its proved new vanishing is precisely $\bar{\Sigma} \bar{S} / \Sigma S = 0$, with the displayed inverse and preserved endpoint data.

SSI10. The adelic sector and the alternative source's retained prime directions

The linked source comparison CC_SHEAF_ORIGINAL_ZETA_COHOMOLOGY.md, CS0-CS10 including CS2A, has now been read in full. It proves the adelic comparison that SSI0 kept separate. Here are its receiving maps and the additional Fourier and dilation calculation, retaining the source-domain distinction.

The full companion CC_ENDPOINT_DEFECT_ACTIONS.md, CSB0-CSB10, has also been read. Its CSB1-CSB9 independently proves these endpoint actions and supplies an exactly Fourier-equivariant Schwartz representative. CSB10 proves the coordinate comparison to the prior FR15-FR25 calculation, whose earlier public source is [Prime boundary classes in the actual adelic theta receiver, PM4-PM15](#). That public source identity was verified; a new complete reading of its earlier base conventions is not claimed. The prime-boundary component is retained programme mathematics. SSI10 receives and checks its maps without claiming discovery of that component.

Put $K = \widehat{\mathbb{Z}}^\times$. On the moment-null adelic Schwartz space let D be the Hausdorff quotient by rational differences generated by arbitrary adelic Schwartz functions, and let D_{null} instead restrict the generators themselves to moment-null functions. Both kinds of difference are moment-null because the adelic modulus of a rational number is 1. CS2 proves the continuous inverse maps

$$j : S \longrightarrow D^K, \quad f \longmapsto [f \otimes 1_{\widehat{\mathbb{Z}}}], \quad \bar{B} \left[\sum_d f_d \otimes 1_{d\widehat{\mathbb{Z}}} \right] (v) = \sum_d f_d^{\text{ev}}(dv). \quad (\text{SSI10.1})$$

The radial-ball expansion is finite after K -averaging. Its uniqueness and the preservation of both endpoint functionals are proved in CS2. Rational scaling takes the ball $d\widehat{\mathbb{Z}}$ to $(d/|q|)\widehat{\mathbb{Z}}$, while scaling the real

function by $|q|$; the two changes cancel in B . This proves that the formula descends to the quotient. The difference between a ball tensor and its displayed real representative is a sum of two full rational differences, as proved in CS2.7a. Thus $\bar{B}j = 1$ and $j\bar{B} = P_K$, with both inverses continuous.

On the overlap, summation of jf over $\mathbb{Q}^\times$ is exactly Σf : a rational number in $\widehat{\mathbb{Z}}$ is an integer, and both signs contribute. Therefore the actual full-difference CC source sector has the continuous isomorphism

$$D^K \xrightarrow{\bar{B}} S \xrightarrow{\Sigma} J \xrightarrow{\mathcal{M}_0} I. \quad (\text{SSI10.2})$$

The middle isomorphism is SSI1–SSI6; thus this statement now concerns the proved adelic source sector as well as its explicit real presentation.

For the second source convention CS2A constructs

$$V = \bigoplus_{p \text{ prime}} \mathbb{C}^2, \quad D_{\text{null}} \xrightarrow{\sim} D \oplus V, \quad [F]_{\text{null}} \mapsto ([F], \mathcal{L}F), \quad (\text{SSI10.3})$$

with its continuous inverse and locally convex direct-sum topology. On a ball tensor the components are

$$\ell_p(f \otimes 1_{d\widehat{\mathbb{Z}}}) = v_p(d) \left(f(0), d^{-1} \int_{\mathbb{R}} f(v) dv \right), \quad \mathcal{L} = (\ell_p)_p. \quad (\text{SSI10.4})$$

For nonradial inputs average over K first. All operations in (SSI10.3)–(SSI10.4) concern the complete reconstructed prime system. Each test function has a finite radial-ball expansion, so its image has finite prime support; this property of individual sections is not a truncation of the prime system. CS2A proves that the extra V is precisely the kernel of the source comparison, not an unspecified possible defect.

Let $C(a, b) = (b, a)$ on each $\mathbb{C}^2$. Under the self-dual adelic Fourier convention, the finite Fourier transform satisfies

$$\widehat{1_{d\widehat{\mathbb{Z}}}} = d^{-1} 1_{d^{-1}\widehat{\mathbb{Z}}}. \quad (\text{SSI10.5})$$

Indeed the annihilator of the ball is the displayed reciprocal ball, and its Haar volume is d^{-1} . For a real Schwartz factor the two Fourier endpoint identities are $\widehat{f}(0) = \int f$ and $\int \widehat{f} = f(0)$. Substitute all these values into (SSI10.4). Since $v_p(d^{-1}) = -v_p(d)$, one obtains

$$\ell_p(\widehat{F}) = -C\ell_p(F), \quad \boxed{\mathcal{L}\widehat{F} = -C\mathcal{L}F}. \quad (\text{SSI10.6})$$

The same identity holds before averaging: Fourier conjugates the action of a finite unit to its inverse, and Haar averaging is unchanged by inversion. Linearity and continuity then extend the calculation from ball tensors to the stated source. Thus (SSI10.3) intertwines Fourier transform with $(\widehat{}_D, -C)$. The minus sign records the inverted finite valuation; it cannot be suppressed by only retaining the two unordered endpoint coordinates.

For positive real dilation, define

$$\rho_+(a)F(v, x_f) = F(v/a, x_f), \quad \rho_-(a)F(v, x_f) = aF(av, x_f). \quad (\text{SSI10.7})$$

Evaluation at zero and change of variables in the real integral give

$$\mathcal{L}\rho_+(a) = \begin{pmatrix} 1 & 0 \\ 0 & a \end{pmatrix} \mathcal{L}, \quad \mathcal{L}\rho_-(a) = \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix} \mathcal{L}. \quad (\text{SSI10.8})$$

These formulas hold in every prime summand of V . Consequently the exact splitting (SSI10.3) is equivariant for these displayed actions, without requiring its chosen endpoint section to be equivariant. This also verifies Fourier interchange of the two actions, with the retained $-C$.

For the sheaves in this note, take the K -fixed sector explicitly. The group K acts trivially on V , and (SSI10.3) is K -equivariant, so it restricts to the topological isomorphism $D_{\text{null}}^K \cong D^K \oplus V$. Define Ω_{null} with chart source D_{null}^K , its two endpoint lines on each chart, and the same K -fixed overlap A as Ω . This is not the sheaf containing every nontrivial finite-unit character. Each extra V class has the representative $\mathcal{T}(v)$ constructed in CS2A.7, a finite sum of full rational differences. Its sum vanishes by reindexing. The relation closure also sums to zero by continuity, so the whole V summand restricts to zero on the overlap. Let $i_{\pm} : \{x_{\pm}\} \hookrightarrow X$. The source comparison therefore gives the actual sheaf decomposition

$$\Omega_{\text{null}} \cong \Omega \oplus (i_+)_* V \oplus (i_-)_* V. \quad (\text{SSI10.9})$$

This is verified on both charts, the overlap and their restrictions; hence it is an isomorphism of sheaves. In the two-term Čech complex it adds $V \oplus V$ in degree 0 with zero differential, and adds nothing in degree 1. It follows directly that

$$H^0(X, \Omega_{\text{null}}) \cong H^0(X, \Omega) \oplus V \oplus V, \quad \boxed{H^1(X, \Omega_{\text{null}}) \cong H^1(X, \Omega) \cong \mathcal{B}/I.} \quad (\text{SSI10.10})$$

The same calculation retains one additional V at the corresponding single closed support. The chart-exchanging sheaf mirror swaps the two V copies; this is a different typed map from the adelic Fourier map $-C$ within a source copy in (SSI10.6). Formula (SSI10.8) gives their separate endpoint characters. Thus the extra prime directions are preserved, while the degree-one original-zeta quotient remains exactly the same. This identifies the consequence of the source-domain choice without confusing it with the vanished closure quotient or the still-unproved geometric weight separation.

The retained prime endpoints and their exact earlier-source comparison — CSB0-CSB10

Proof source: CC_ENDPOINT_DEFECT_ACTIONS.md.

Arithmetic and Fourier actions on the prime-indexed endpoint defect

Independent derivation and exact comparison with the earlier FR calculation, 24 September 2026. Proof locators CSB0–CSB10.

CSB0. The input stage and the source-domain distinction

This calculation continues CS2A of CC_SHEAF_ORIGINAL_ZETA_COHOMOLOGY.md. It concerns the already constructed adèlic coefficient spaces over $\mathbb{Q}$, their rational scaling relations, and the explicit alternative quotient produced by restricting the generating domain of those relations. The prime labels and their valuations belong to this arithmetic stage. No arithmetic operation, vector, metric, or coordinate is introduced on $\tau\langle Z_1; \text{no } Z_2 \rangle$.

The earlier programme source ORIGINAL_ZETA_RETURN_FR.tex, FR15–FR25, already proves the algebraic spherical decomposition with two endpoint directions at every prime. That proof slice and its bibliography were read during this continuation. CSB10 proves the exact coordinate comparison, including its opposite rational-scaling convention and the distinction between its periodization seminorm and the stronger source topology here. The decomposition is not claimed as new. The older text’s separate base and supported-zero conventions are not imported into the current user construction.

Before this calculation, the current READ_FIRST_USER_CONSTRUCTION.md, the correction-precedence table and active-target sections of USER_ARGUMENT_RECONSTRUCTION.md, the complete current PREREQUISITE_AND_OBSTRUCTION_CHECKS.md, and verbatim WU061–WU062 were read. The connected reconstruction and pertinent earlier corrections had already been read for CS0. The governing distinctions remain: retain complete-history reconstruction before arithmetic labels, preserve both branch counters separately, retain the carried stalk data, and derive the actual lifting mechanism rather than assigning a desired weight.

The original source is Connes–Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, [arXiv:0903.2024v3](https://arxiv.org/abs/0903.2024v3), retained author file sources/CC_0903_2024_v3/author_source/announc3.tex. The relevant exact definitions were read in section 5, source lines 1376–1667. This continuation uses fonction1–fonction3, fourier, poisson2, rep1, liftw and compinv; fonction1–fonction3 were reread for the generating-domain check.

The source’s fonction3 does not explicitly type its generating f . CS uses the full-difference reading compatible with the stated Fourier-graph theorem and separately proves what happens under the null-generated alternative. This note preserves that textual qualification. Its subject is the exact mathematical relationship between those two written quotient definitions.

Use

$$\mathcal{S} = \mathcal{S}(\mathbb{A}_{\mathbb{Q}}), \quad \epsilon(F) = \left(F(0), \int_{\mathbb{A}_{\mathbb{Q}}} F(x) dx \right), \quad E = \ker \epsilon. \quad (\text{CSB0.1})$$

The topology is the adèlic Bruhat-Schwartz LF topology: on each fixed finite-place test-function stage it is a finite direct sum of real Schwartz spaces. At the finite places the additive Haar measure has $\text{vol}(\widehat{\mathbb{Z}}) = 1$. For $q \in \mathbb{Q}^\times$, set

$$(\mathbb{D}_q F)(x) = F(qx), \quad r_q = I - \mathbb{D}_q. \quad (\text{CSB0.2})$$

The product formula gives $\epsilon(\mathbb{D}_q F) = \epsilon(F)$, so $r_q \mathcal{S} \subset E$. Define

$$\mathcal{R}_{\text{full}} = \text{span}\{r_q F : F \in \mathcal{S}, q \in \mathbb{Q}^\times\}, \quad \mathcal{R}_{\text{null}} = \text{span}\{r_q F : F \in E, q \in \mathbb{Q}^\times\},$$

$$D_{\text{full}} = E/\overline{\mathcal{R}_{\text{full}}}, \quad D_{\text{null}} = E/\overline{\mathcal{R}_{\text{null}}}. \quad (\text{CSB0.3})$$

Closures are taken in E . Their natural quotient map is denoted $\pi : D_{\text{null}} \rightarrow D_{\text{full}}$.

For $K = \widehat{\mathbb{Z}}^\times$, let P_K be normalized Haar averaging in the finite coordinates. Every $P_K F$ has a unique finite expansion

$$P_K F = \sum_d f_d \otimes 1_{d\widehat{\mathbb{Z}}}, \quad d \in \mathbb{Q}_{>0}. \quad (\text{CSB0.4})$$

CS2 proves existence and uniqueness by finite products of nested local balls. Put

$$\ell_p(F) = \sum_d v_p(d) \left(f_d(0), d^{-1} \int_{\mathbb{R}} f_d(v) dv \right), \quad \mathcal{L}(F) = (\ell_p(F))_p, \quad V = \bigoplus_{p \text{ prime}} \mathbb{C}^2. \quad (\text{CSB0.5})$$

The direct sum has its locally convex direct-sum topology. Each test-function stage involves finitely many relevant p, d , so $\mathcal{L} : \mathcal{S} \rightarrow V$ is continuous. CS2A proves

$$\ell_p(\mathbb{D}_q F) = \ell_p(F) - v_p(|q|)\epsilon(F), \quad (\text{CSB0.6})$$

and consequently $\mathcal{L}$ descends to D_{null} . Its complete topological splitting is

$$\Phi : D_{\text{null}} \xrightarrow{\sim} D_{\text{full}} \oplus V, \quad [F]_{\text{null}} \mapsto ([F]_{\text{full}}, \mathcal{L}F). \quad (\text{CSB0.7})$$

CSB4 below restates the explicit inverse and computes its equivariance. In particular V denotes these extra prime-indexed directions; it does not denote either of the two original endpoint lines on either chart.

CSB1. Fourier transformation with the finite volume factor retained

Choose the self-dual character and measures with real Fourier transform

$$\widehat{f}(t) = \int_{\mathbb{R}} f(v) e^{-2\pi i vt} dv. \quad (\text{CSB1.1})$$

At the finite places, the annihilator of $d\widehat{\mathbb{Z}}$ is $d^{-1}\widehat{\mathbb{Z}}$. Integration of its character is its volume when the character is trivial and zero otherwise. Therefore

$$\widehat{1_{d\widehat{\mathbb{Z}}}} = d^{-1} 1_{d^{-1}\widehat{\mathbb{Z}}}, \quad \widehat{f \otimes 1_{d\widehat{\mathbb{Z}}}} = d^{-1} \widehat{f} \otimes 1_{d^{-1}\widehat{\mathbb{Z}}}. \quad (\text{CSB1.2})$$

The factor d^{-1} is the full finite additive volume.

For the original tensor, write

$$a = f(0), \quad b = d^{-1} \int_{\mathbb{R}} f(v) dv.$$

The transformed tensor has endpoint pair (b, a) , since

$$d^{-1}\hat{f}(0) = b, \quad (d^{-1})^{-1} \int_{\mathbb{R}} d^{-1}\hat{f}(t) dt = a. \quad (\text{CSB1.3})$$

Its ball label is d^{-1} , hence its p -valuation is $-v_p(d)$. With

$$J(a, b) = (b, a), \quad S_V = -J, \quad (\text{CSB1.4})$$

we obtain the exact identity

$$\boxed{\ell_p(\widehat{F}) = -J\ell_p(F), \quad \mathcal{L}\widehat{F} = S_V\mathcal{L}F.} \quad (\text{CSB1.5})$$

Here J, S_V act componentwise on the prime labels. To extend the calculation beyond the K -invariant tensors, observe that Fourier transformation exchanges finite unit scaling by k with scaling by k^{-1} ; the module of such a unit is one. Inversion preserves Haar measure on K , so Fourier transformation commutes with P_K .

Fourier transformation interchanges the two endpoint functionals, and hence preserves E . Rational scaling satisfies

$$\widehat{D_q F} = D_{q^{-1}} \widehat{F}. \quad (\text{CSB1.6})$$

At $q = p$, the real Fourier scalar is p^{-1} , while the finite scalar is p ; their product is one, as required by $|p|_{\mathbb{A}} = 1$. It follows that Fourier transformation preserves both relation spaces in (CSB0.3) and their closures. Its square is D_{-1} , which is the identity on both quotients. Thus it gives continuous involutions $\mathcal{F}_{\text{full}}$ and $\mathcal{F}_{\text{null}}$.

Under (CSB0.7) their relationship is

$$\boxed{\Phi \mathcal{F}_{\text{null}} \Phi^{-1}(x, v) = (\mathcal{F}_{\text{full}} x, -Jv).} \quad (\text{CSB1.7})$$

On each extra $\mathbb{C}^2$, the Fourier $+1$ eigenspace is $\{(c, -c)\}$, and the Fourier -1 eigenspace is $\{(c, c)\}$. These signs follow from the inverted valuation, rather than from an omitted Fourier volume.

CSB2. The full unshifted arithmetic action

For $a > 0$, use the source actions on the real coordinate

$$A_+(a)F(v, x_f) = F(v/a, x_f), \quad A_-(a)F(v, x_f) = aF(av, x_f) = aA_+(a^{-1})F(v, x_f). \quad (\text{CSB2.1})$$

They commute with rational scaling, preserve E , and continuously preserve the two relation closures. On each fixed real Schwartz seminorm, a fixed dilation multiplies the bound by an explicit power of a ; on the finite component it is the identity. Thus all induced quotient operators are continuous.

On the endpoint pair their matrices are

$$B_+(a) = \begin{pmatrix} 1 & 0 \\ 0 & a \end{pmatrix}, \quad B_-(a) = \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}. \quad (\text{CSB2.2})$$

For a ball tensor, real dilation changes neither d nor $v_p(d)$; evaluation at zero and change of variables in the real integral give

$$\boxed{\ell_p(A_+(a)F) = B_+(a)\ell_p(F), \quad \ell_p(A_-(a)F) = B_-(a)\ell_p(F).} \quad (\text{CSB2.3})$$

The same equations hold after K -averaging. Finite units act trivially on V . Every idèle class has its representative $(a, k) \in \mathbb{R}_{>0} \times K$, as proved in CS1.3. Therefore the full idèle-class action on V depends only on its module a , and is exactly (CSB2.2). Rational changes of representative act identically on D_{null} , since their arguments lie in E . The term $v_p(q)\epsilon(F)$ in (CSB0.6) is consequently zero at this quotient stage.

The splitting (CSB0.7) therefore intertwines both chart actions:

$$\Phi A_{\pm}(a)\Phi^{-1}(x, v) = (A_{\pm, \text{full}}(a)x, B_{\pm}(a)v). \quad (\text{CSB2.4})$$

Fourier transformation exchanges the two chart actions:

$$\widehat{A_+(a)F} = A_-(a)\widehat{F}, \quad S_V B_+(a) = B_-(a)S_V. \quad (\text{CSB2.5})$$

All factors in $A_-(a) = aA_+(a^{-1})$ have been retained.

For every positive integer n , the plus-chart arithmetic and integral transfer matrices on an extra endpoint pair are

$$\begin{aligned} F_n^{(+)} = B_+(n) &= \begin{pmatrix} 1 & 0 \\ 0 & n \end{pmatrix}, & V_n^{(+)} = nB_+(n^{-1}) &= \begin{pmatrix} n & 0 \\ 0 & 1 \end{pmatrix}, \\ F_n^{(-)} = B_-(n) &= \begin{pmatrix} n & 0 \\ 0 & 1 \end{pmatrix}, & V_n^{(-)} = nB_-(n^{-1}) &= \begin{pmatrix} 1 & 0 \\ 0 & n \end{pmatrix}. \end{aligned} \quad (\text{CSB2.6})$$

Thus $F_n^{(\pm)}V_n^{(\pm)} = nI$, all the multiplicative relations hold, and these matrices preserve $\bigoplus_p \mathbb{Z}^2 \subset V$. The prime label p of a defect component is not changed by an arithmetic operator indexed by n . Fourier acts on this same lattice by $-J$.

Differentiating at $a = e^t$, $t = 0$, gives the actual infinitesimal matrices

$$L_{V,+} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad L_{V,-} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \quad (\text{CSB2.7})$$

Their dilation exponents are exactly 0 and 1, with no nilpotent contribution. They satisfy $L_{V,\pm}(L_{V,\pm} - I) = 0$. In the convention in which a Tate character a^r is assigned weight $2r$, these are the two characters with labels 0 and 2. Equation (CSB2.7), rather than a new finite-field Frobenius assumption, is the statement proved here.

CSB3. The Gaussian endpoint section and its exact source defect

Let $g(v) = e^{-\pi v^2}$ and define a continuous endpoint section

$$\sigma(c, d)(v, x_f) = [c(1 - 2\pi v^2) + d 2\pi v^2] g(v) 1_{\widehat{\mathbb{Z}}}(x_f). \quad (\text{CSB3.1})$$

The Gaussian identities $\int g = 1$ and $\int v^2 g = 1/(2\pi)$ prove $\epsilon(\sigma(c, d)) = (c, d)$. In the real Fourier convention (CSB1.1),

$$\widehat{g} = g, \quad \widehat{2\pi v^2 g} = (1 - 2\pi t^2)g(t), \quad \widehat{(1 - 2\pi v^2)g} = 2\pi t^2 g(t). \quad (\text{CSB3.2})$$

Together with the finite ball identity at $d = 1$, these give

$$\boxed{\widehat{\sigma(c, d)} = \sigma(d, c) = \sigma(J(c, d))}. \quad (\text{CSB3.3})$$

The chosen Gaussian endpoint section itself is exactly Fourier equivariant.

The source representative used in CS2A is

$$\mathcal{T}(v) = \sum_p r_p \sigma(v_p) \in \mathcal{R}_{\text{full}} \subset E, \quad \mathcal{LT}(v) = v. \quad (\text{CSB3.4})$$

Every sum is finite. The locally convex direct-sum topology makes this map continuous: its restriction to each finite-dimensional prime component is continuous into a fixed adèlic Schwartz stage.

Equations (CSB1.6) and (CSB3.3) show

$$\widehat{\mathcal{T}(v)} = \sum_p r_{p^{-1}} \sigma(Jv_p). \quad (\text{CSB3.5})$$

The required action on V is $-J$, so the source-level defect is

$$\begin{aligned} \widehat{\mathcal{T}(v)} - \mathcal{T}(-Jv) &= \sum_p (r_{p^{-1}} + r_p) \sigma(Jv_p) \\ &= \sum_p r_p r_{p^{-1}} \sigma(Jv_p) \\ &= \sum_p (2\sigma(Jv_p) - \mathbb{D}_p \sigma(Jv_p) - \mathbb{D}_{p^{-1}} \sigma(Jv_p)). \end{aligned} \quad (\text{CSB3.6})$$

The middle equality uses $\mathbb{D}_p \mathbb{D}_{p^{-1}} = I$. The function $r_{p^{-1}} \sigma(Jv_p)$ is moment-null, because rational scaling preserves both endpoint functionals. Hence every term in the middle line belongs to $\mathcal{R}_{\text{null}}$, without taking a closure. The induced section

$$\iota : V \longrightarrow D_{\text{null}}, \quad \iota(v) = [\mathcal{T}(v)]_{\text{null}} \quad (\text{CSB3.7})$$

is already exactly Fourier equivariant, even though its displayed Schwartz representative has the nonzero defect (CSB3.6).

An explicit representative which is Fourier equivariant before quotienting is

$$\boxed{\mathcal{T}_{\text{eq}}(v) = \frac{1}{2} \sum_p (\mathbb{D}_{p^{-1}} \sigma(v_p) - \mathbb{D}_p \sigma(v_p))}. \quad (\text{CSB3.8})$$

Its endpoint pair vanishes term by term. Equation (CSB0.6) gives

$$\begin{aligned} \ell_q(\mathbb{D}_{p^{-1}} \sigma(v_p)) &= \delta_{pq} v_p, & \ell_q(\mathbb{D}_p \sigma(v_p)) &= -\delta_{pq} v_p, \\ \mathcal{LT}_{\text{eq}} &= \text{id}_V, & \widehat{\mathcal{T}_{\text{eq}}(v)} &= \mathcal{T}_{\text{eq}}(-Jv). \end{aligned} \quad (\text{CSB3.9})$$

In the second identity use (CSB1.6) to exchange the two rational dilations, and (CSB3.3) to interchange the endpoint pair. It represents the same quotient section, since

$$\mathcal{T}_{\text{eq}}(v) - \mathcal{T}(v) = -\frac{1}{2} \sum_p r_p r_{p^{-1}} \sigma(v_p) \in \mathcal{R}_{\text{null}}. \quad (\text{CSB3.10})$$

The factor $1/2$, both rational scales, and the sign of their difference are part of the actual formula.

For completeness, the chosen representatives also have exact dilation defects. Define

$$\Delta_{\pm, a}(w) = A_{\pm}(a) \sigma(w) - \sigma(B_{\pm}(a)w) \in E. \quad (\text{CSB3.11})$$

The zero endpoint assertion follows from (CSB2.2). If $w = (c, d)$, their real components, with the common finite factor $1_{\widehat{\mathbb{Z}}}$, are

$$\begin{aligned}\Delta_{+,a}(c, d)(v) &= [c(1 - 2\pi v^2/a^2) + d 2\pi v^2/a^2] e^{-\pi v^2/a^2} \\ &\quad - [c(1 - 2\pi v^2) + ad 2\pi v^2] e^{-\pi v^2}, \\ \Delta_{-,a}(c, d)(v) &= a [c(1 - 2\pi a^2 v^2) + d 2\pi a^2 v^2] e^{-\pi a^2 v^2} \\ &\quad - [ac(1 - 2\pi v^2) + d 2\pi v^2] e^{-\pi v^2}.\end{aligned}\tag{CSB3.12}$$

Commutation with rational scaling gives

$$\begin{aligned}A_{\pm}(a)\mathcal{T}(v) - \mathcal{T}(B_{\pm}(a)v) &= \sum_p r_p \Delta_{\pm,a}(v_p) \in \mathcal{R}_{\text{null}}, \\ A_{\pm}(a)\mathcal{T}_{\text{eq}}(v) - \mathcal{T}_{\text{eq}}(B_{\pm}(a)v) &= \frac{1}{2} \sum_p (r_p - r_{p^{-1}}) \Delta_{\pm,a}(v_p) \in \mathcal{R}_{\text{null}}.\end{aligned}\tag{CSB3.13}$$

These are exact differences of actual source functions. They prove dilation equivariance of the same quotient section without claiming that its Gaussian representative is fixed by every dilation.

CSB4. The canonical quotient splitting and its inverse

The continuous inverse of Φ in (CSB0.7) is

$$\Phi^{-1}([F]_{\text{full}}, v) = [F - \mathcal{T}\mathcal{L}F + \mathcal{T}v]_{\text{null}}.\tag{CSB4.1}$$

Here $F \in E$. The CS2A relation $(I - \mathcal{T}\mathcal{L})\overline{\mathcal{R}_{\text{full}}} \subset \overline{\mathcal{R}_{\text{null}}}$ makes the formula independent of F . It also proves continuity through the defining quotient topology. Replacing $\mathcal{T}$ by $\mathcal{T}_{\text{eq}}$ gives exactly the same quotient map by (CSB3.10).

Thus the exact sequence

$$0 \longrightarrow V \xrightarrow{\iota} D_{\text{null}} \xrightarrow{\pi} D_{\text{full}} \longrightarrow 0\tag{CSB4.2}$$

has section

$$s_0([F]_{\text{full}}) = [F - \mathcal{T}\mathcal{L}F]_{\text{null}}, \quad \mathcal{L}s_0 = 0.\tag{CSB4.3}$$

This is the unique section whose image lies in $\ker \mathcal{L}$. The coordinate transformation identities (CSB1.5) and (CSB2.3) prove that both ι and s_0 intertwine Fourier transformation and both chart actions, with precisely the matrices already computed.

The quotient section ι does not depend on the Gaussian endpoint section. If σ' is another linear endpoint section, then $\epsilon((\sigma - \sigma')w) = 0$, and

$$\sum_p r_p (\sigma - \sigma')(v_p) \in \mathcal{R}_{\text{null}}.\tag{CSB4.4}$$

Consequently both choices give the same ι , and therefore the same s_0 . The explicit Gaussian formulas provide representatives of an already well-defined quotient splitting.

CSB5. The alternative sheaf and its entire cohomology

Let $X = \{0, \eta, \infty\}$ have the same three-point topology as CS4: $U_+ = \{0, \eta\}$, $U_- = \{\eta, \infty\}$, and $U = U_+ \cap U_- = \{\eta\}$. On the trivial K -sector put

$$S_0 = \{f \in \mathcal{S}(\mathbb{R}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f = 0\}. \quad (\text{CSB5.1})$$

CS2 and CS2A identify

$$D_{\text{full}}^K \simeq S_0, \quad D_{\text{null}}^K \simeq S_0 \oplus V \quad (\text{CSB5.2})$$

topologically. In these coordinates Fourier acts by $(f, v) \mapsto (\hat{f}, -Jv)$.

Retain four original endpoint lines,

$$E_+ = \mathbb{C}e_{+,0} \oplus \mathbb{C}e_{+,1}, \quad E_- = \mathbb{C}e_{-,0} \oplus \mathbb{C}e_{-,1}. \quad (\text{CSB5.3})$$

The alternative chart section spaces are

$$\tilde{A}_+ = S_0 \oplus V_+ \oplus E_+, \quad \tilde{A}_- = S_0 \oplus V_- \oplus E_-, \quad (\text{CSB5.4})$$

where V_+, V_- are independent chart-labelled copies of V . The overlap is the same weighted smooth space

$$H = \{h : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j h(u)| < \infty \text{ for all } N, j \geq 0\}. \quad (\text{CSB5.5})$$

The actual restrictions are

$$\begin{aligned} \tilde{r}_+(f, v_+, e_+) &= Sf, & \tilde{r}_-(h, v_-, e_-) &= RSh = \hat{S}h, \\ Sf(u) &= \sum_{q \in \mathbb{Q}^\times} f(qu) 1_{\hat{\mathbb{Z}}}(q) = 2 \sum_{n \geq 1} f(nu), & Rg(u) &= u^{-1}g(1/u). \end{aligned} \quad (\text{CSB5.6})$$

The restrictions of the original endpoints vanish by their source definition. Those of $V_\pm$ vanish because each representative $r_p \sigma(v_p)$ has zero rational sum by reindexing. Absolute convergence of that sum for every idèle justifies the reindexing; one does not claim that the separate non-null Gaussian generators lie in the rapidly decreasing overlap.

Define global sections as the fibre product of the two chart restrictions. This constructs a sheaf $\tilde{\mathcal{F}}$ exactly as in CS4. The splitting in CSB4 supplies the explicit sheaf isomorphism

$$\boxed{\tilde{\mathcal{F}} \simeq \mathcal{F} \oplus i_{0*} V_+ \oplus i_{\infty*} V_-}. \quad (\text{CSB5.7})$$

Here $i_{0*} V_+$ means the sheaf with sections V_+ on an open containing 0 and zero on any other open; the definition for ∞ is analogous. On the overlap the isomorphism is the identity. On a chart it is exactly the quotient splitting (CSB0.7), together with its original endpoint summands. Thus (CSB5.7) is a map of the actual restrictions, not just an equality of dimensions.

For any sheaf with the corresponding full adèlic source coordinates, the same splitting argument gives its alternative as the original sheaf direct-summed with these two endpoint-supported sheaves. This statement does not require an additional injectivity theorem for the full adèlic summation map.

Use the ordered Čech differential

$$\tilde{d}(a, b) = \tilde{r}_+ a - \tilde{r}_- b. \quad (\text{CSB5.8})$$

CS3 proves S is injective on S_0 , and Poisson proves $RS = \mathcal{S}$. Therefore the complete degree-zero cohomology is

$$\boxed{H^0(X, \tilde{\mathcal{F}}) = \{((f, v_+, e_+), (\hat{f}, v_-, e_-)) : f \in S_0, v_\pm \in V_\pm, e_\pm \in E_\pm\}}. \quad (\text{CSB5.9})$$

In particular the two prime-indexed copies are unrestricted and independent. Replacing this formula by only the Fourier graph of D_{null}^K would discard one full copy of V .

If $J_0 = S(S_0) \subset H$, then

$$\boxed{H_{\text{alg}}^1(X, \widetilde{\mathcal{F}}) = H/J_0 = H_{\text{alg}}^1(X, \mathcal{F}), \quad H_{\text{Haus}}^1(X, \widetilde{\mathcal{F}}) = H/\overline{J_0} = H_{\text{Haus}}^1(X, \mathcal{F}).} \quad (\text{CSB5.10})$$

The equalities are induced by the identity on H , with the same images and quotient topologies. Higher cohomology vanishes. This follows either from the same two-term complex proof as CS5, or from (CSB5.7), since global sections of an endpoint-supported summand are the exact functor returning its vector space.

CSB6. Fourier, Weyl, and every chart-labelled endpoint action

In the coordinates of (CSB5.9), positive real scaling is

$$\begin{aligned} \rho(a)(f, v_+, v_-, e_+, e_-) &= (A_+(a)f, B_+(a)v_+, B_-(a)v_-, \\ &\quad B_+(a)e_+, B_-(a)e_-). \end{aligned} \quad (\text{CSB6.1})$$

On $E_{\pm}$, the matrices refer to the ordered coordinates evaluation and integral. The minus main coordinate is automatically $\widehat{A_+(a)}f = A_-(a)\hat{f}$.

For every prime p , write $v_{+,p} = (c_{+,p}, d_{+,p})$, $v_{-,p} = (c_{-,p}, d_{-,p})$, and write $e_{\pm} = (c_{\pm}, d_{\pm})$ for the original endpoints. The full list is

$$\begin{array}{c|cccc} \text{original line} & c_+ & d_+ & c_- & d_- \\ \hline \rho(a) & 1 & a & a & 1 \end{array} \quad \begin{array}{c|cccc} \text{extra line at } p & c_{+,p} & d_{+,p} & c_{-,p} & d_{-,p} \\ \hline \rho(a) & 1 & a & a & 1. \end{array} \quad (\text{CSB6.2})$$

There are four original endpoint lines in total and four extra lines for each prime. The repeated character values do not identify any of these labelled lines.

The source Weyl lift exchanges the two chart sections by the identity. Therefore on degree zero it is

$$\boxed{W^0(f, v_+, v_-, e_+, e_-) = (\hat{f}, v_-, v_+, e_-, e_+).} \quad (\text{CSB6.3})$$

In particular Weyl exchanges the independent prime-indexed copies without inserting $-J$ in this chart-swap formula. Fourier transformation of a single source fibre, which does act by $-J$, is a different map.

On the ordered degree-one Čech cochains the Weyl action remains

$$W^1g = -Rg, \quad \tilde{d}W^0 = -R\tilde{d}. \quad (\text{CSB6.4})$$

The additional summands restrict to zero and introduce no new sign in this identity. In every degree the unshifted relation is

$$\rho(a)W = aW\rho(a^{-1}). \quad (\text{CSB6.5})$$

On the extra coordinates it says $B_+(a) = aB_-(a^{-1})$ and $B_-(a) = aB_+(a^{-1})$, which follow directly from (CSB2.2).

The Fourier graph inside the extra pair is

$$G_V = \{(v_+, v_-) = (v, -Jv) : v \in V\}. \quad (\text{CSB6.6})$$

Its complementary opposite graph is

$$A_V = \{(v_+, v_-) = (w, Jw) : w \in V\}. \quad (\text{CSB6.7})$$

The exact decomposition of arbitrary independent coordinates is

$$g = \frac{1}{2}(v_+ - Jv_-), \quad h = \frac{1}{2}(v_+ + Jv_-), \quad (v_+, v_-) = (g + h, -Jg + Jh). \quad (\text{CSB6.8})$$

Both summands carry the plus-chart dilation matrix $B_+(a)$, because $JB_-(a) = B_+(a)J$. Their Weyl actions are respectively

$$g \mapsto -Jg, \quad h \mapsto Jh. \quad (\text{CSB6.9})$$

These identities explain exactly how the single-fibre Fourier sign is recovered on the Fourier-graph subspace, while the full cohomology still retains the independent opposite graph. The four original endpoint lines remain separately present as in (CSB6.2).

CSB7. Supported maps and the split extension

The supported complexes for the alternative sheaf, with positive local restriction differentials, are

$$\widetilde{L}_0 = [\widetilde{A}_+ \xrightarrow{\tilde{r}_+} H], \quad \widetilde{L}_\infty = [\widetilde{A}_- \xrightarrow{\tilde{r}_-} H]. \quad (\text{CSB7.1})$$

Consequently

$$\begin{aligned} H_{\{0\}}^0 &= V_+ \oplus E_+, & H_{\{\infty\}}^0 &= V_- \oplus E_-, \\ H_{\{0\}}^1 &= H/J_0, & H_{\{\infty\}}^1 &= H/J_0. \end{aligned} \quad (\text{CSB7.2})$$

The support-to-global maps in degree one are still $+\text{id}$ and $-\text{id}$, respectively: their degree-zero maps send a chart section to $(a, 0)$ and $(0, b)$, and their degree-one maps send g to g and $-g$. Thus each chain identity keeps the sign of the ordered Čech differential.

The alternative-to-full sheaf sequence is

$$0 \longrightarrow i_{0*}V_+ \oplus i_{\infty*}V_- \longrightarrow \widetilde{\mathcal{F}} \longrightarrow \mathcal{F} \longrightarrow 0. \quad (\text{CSB7.3})$$

It has the explicit continuous, arithmetic-equivariant, Weyl-equivariant splitting (CSB5.7). The projection on global sections forgets only v_+, v_- ; the section sets these two defect coordinates equal to zero. The four original endpoints are preserved by both maps.

All connecting maps of this split sheaf extension are zero. This vanishing follows by applying the section to a cocycle, or directly by decomposing its entire complex as in (CSB5.7). The extension has therefore been completely calculated. It has not been identified with the different lifting-obstruction cross in Deligne §3.6 or with the user's requested common-weight construction.

CSB8. The unchanged original-zeta receiver, with its full comparison

The cohomology identification retains the raw restriction and the original source map:

$$Sf(u) = 2u^{-1/2}\mathcal{E}f(u), \quad \mathcal{E}f(u) = u^{1/2} \sum_{n \geq 1} f(nu), \quad Uh(u) = \frac{1}{2}u^{1/2}h(u). \quad (\text{CSB8.1})$$

The topological isomorphism $U : H \rightarrow \mathcal{A}$ maps $\overline{J_0}$ to the proved closed original summation image $\mathcal{C} = \overline{\mathcal{E}(S_0)}$. The centered Mellin transform is

$$\mathcal{M}k(s) = \int_0^\infty k(u)u^{s-1/2}\frac{du}{u}, \quad \Theta h(s) = \mathcal{M}Uh(s) = \frac{1}{2} \int_0^\infty h(u)u^s\frac{du}{u}. \quad (\text{CSB8.2})$$

The previous global synthesis theorem gives

$$\Theta(\overline{J_0}) = \mathcal{J}, \quad \mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for all actual nontrivial } \rho, 0 \leq j < m_\rho\},$$

$$\boxed{H_{\text{Haus}}^1(X, \widetilde{\mathcal{F}}) \xrightarrow[h]{\mapsto} [\Theta h] \sim \mathcal{B}/\mathcal{J} = \mathcal{Q}.} \quad (\text{CSB8.3})$$

Here $\mathcal{B}$ is the entire strip-Schwartz space of CS7.3 with its full Fréchet topology:

$$\mathcal{B} = \{F \in \mathcal{O}(\mathbb{C}) : b_{A,M}(F) < \infty \text{ for all integers } A, M \geq 0\}, \quad b_{A,M}(F) = \sup_{\substack{|\sigma| \leq A \\ t \in \mathbb{R}}} (1+|t|)^M |F(\sigma+it)|. \quad (\text{CSB8.3a})$$

Write $\mathcal{M}_0 h(s) = \int_0^\infty h(u) u^s du/u$ for the raw Mellin isomorphism $H \rightarrow \mathcal{B}$; thus $\Theta = \frac{1}{2} \mathcal{M}_0$. This is the same map for the full-difference sheaf; the additional prime-indexed endpoint summands do not alter it.

For $\Re s > 1$ the raw original-zeta identity remains

$$\int_0^\infty S f(u) u^s \frac{du}{u} = 2\zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}. \quad (\text{CSB8.4})$$

For the retained $f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}$, it is

$$\Theta S f_0(s) = F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad \int_0^\infty S f_0(u) u^s \frac{du}{u} = 2F_0(s). \quad (\text{CSB8.5})$$

The endpoint values are $F_0(0) = F_0(1) = 1/8$, with raw values $1/4$. For $r \geq 1$, the complete finite value at the original trivial zero is

$$2F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{4} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0. \quad (\text{CSB8.6})$$

These Mellin endpoint values are separate from the four original sheaf endpoint lines and the extra prime-indexed endpoint directions.

The resulting action and ordered Čech reflection on this unchanged quotient are

$$T_a[F(s)] = [a^s F(s)], \quad W^1[F(s)] = [-F(1-s)], \quad T_a W^1 = a W^1 T_{a^{-1}}. \quad (\text{CSB8.7})$$

All actual zero multiplicities remain in $\mathcal{J}$; no value-only quotient replaces the full jets.

The later complete inverse theorem, [SSI0-SSI9](#) and its independent derivation [ESI0-ESI9](#), strengthens this comparison. Both proofs were read for this update. The theorem proves the topological isomorphism

$$\mathcal{M}_0 S : S_0 \xrightarrow{\sim} \mathcal{J}, \quad J_0 = \overline{J_0} = \mathcal{M}_0^{-1} \mathcal{J}. \quad (\text{CSB8.8})$$

For a raw entire target $F \in \mathcal{J}$, its actual source inverse is

$$f_F(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{F(2+it)}{2\zeta(2+it)} x^{-2-it} dt \quad (x > 0),$$

$$f_F(-x) = f_F(x), \quad f_F(0) = 0, \quad \int_{\mathbb{R}} f_F = 0, \quad \frac{f_F^{(2r)}(0)}{(2r)!} = \frac{F(-2r)}{2\zeta'(-2r)} \quad (r \geq 1). \quad (\text{CSB8.9})$$

SSI4 and ESI4 prove rapid vertical-strip estimates after exactly the trivial poles in that strip have been cleared. They use the proved subquadratic division bound, the full functional multiplier on an odd

negative vertical boundary, and the maximum principle with limits in the order $T \rightarrow \infty$, then $\varepsilon \downarrow 0$. SSI5-SSI6 and ESI5-ESI7 prove the displayed even extension, every retained residue, and the continuity estimate

$$\sup_{x \in \mathbb{R}} (1 + |x|)^R |f_F^{(j)}(x)| \leq C_{R,j} b_{A,M}(F) \quad (\text{CSB8.10})$$

for finite A, M depending on R, j . Thus (CSB8.9) is an inverse into the actual Schwartz topology, and $Sf_F = \mathcal{M}_0^{-1}F$. For our comparison coordinate $G = \Theta Sf = \frac{1}{2}\mathcal{M}_0 Sf$, the same inverse uses $F = 2G$; its Mellin quotient is G/ζ . The raw factor two has not changed.

The ordinary-to-Hausdorff comparison kernel is now calculated, rather than left pending:

$$\overline{J}_0/J_0 = 0, \quad 0 \longrightarrow 0 \longrightarrow H_{\text{alg}}^1(X, \widetilde{\mathcal{F}}) \longrightarrow \mathcal{Q} \longrightarrow 0. \quad (\text{CSB8.11})$$

Consequently ordinary H^1 , with its actual quotient topology, already equals the Hausdorff original-zeta receiver under (CSB8.3). This result removes the closure comparison kernel. It does not remove $V_{\pm}$, which belong to the summation kernel in the source before taking that image.

CSB9. Exact scope of the endpoint-character calculation

The computed alternative receiver is a continuous, equivariantly split extension of the full-difference receiver by the explicit space $\bigoplus_p \mathbb{C}^2$. Its single-fibre Fourier action is $-J$; its chart actions are $\text{diag}(1, a)$ and $\text{diag}(a, 1)$. The Gaussian section has no quotient-level equivariance defect, and (CSB3.8) supplies a source representative with exact Fourier equivariance.

The alternative sheaf has two independent copies of this defect, supported respectively at the two closed chart points. Its entire extra degree-zero cohomology, original four endpoint lines, Weyl maps, arithmetic characters, support maps, and connecting maps have been calculated. Its original-zeta H^1 receiver is unchanged through the explicit map (CSB8.3).

These are the weights and maps of this specified endpoint relation defect. The calculation does not turn it into Deligne's geometric lifting obstruction merely because its characters have exponents 0 and 1. No assertion of RH, common-weight purity, or completion of the requested weight-separation mechanism is used or concluded.

CSB10. Exact comparison with the prior FR prime-boundary proof

The prior proof actually read is [ORIGINAL_ZETA_RETURN_FR.tex](#), FR15-FR25, source lines 115-207, together with its PM bibliography entry at line 328. That entry credits the earlier [Prime boundary classes in the actual adelic theta receiver, PM4-PM15](#). The pinned public target was opened and verified as that TeX file at commit 44d03e48849934edf59f645cb5021b639ab31ac4. The complete proof used in the present comparison is the retained FR15-FR25 slice; this does not claim a fresh complete reading of PM1-PM23 or of the earlier base conventions.

Retain FR's group algebra

$$\Gamma_+ = \mathbb{Q}_{>0}^{\times}, \quad R_{\Gamma} = \mathbb{C}[\Gamma_+], \quad I_{\Gamma} = \ker(\text{aug} : R_{\Gamma} \rightarrow \mathbb{C}). \quad (\text{CSB10.1})$$

Write $S_{\text{ev}} = \mathcal{S}(\mathbb{R})^{\text{even}}$, with endpoint map $m(h) = (h(0), \int_{\mathbb{R}} h)$. FR's source and quotient are

$$M_0 = \ker(\text{aug} \otimes m : R_{\Gamma} \otimes S_{\text{ev}} \rightarrow \mathbb{C}^2), \quad Q_0 = M_0/I_{\Gamma}M_0. \quad (\text{CSB10.2})$$

Its coordinates map to the actual spherical adèlic source by

$$\mathcal{A}(t_d \otimes h)(v, x_f) = h(v/d) 1_{d\widehat{\mathbb{Z}}}(x_f) = D_{d^{-1}}(h \otimes 1_{\widehat{\mathbb{Z}}})(v, x_f). \quad (\text{CSB10.3})$$

The simultaneous real and finite dilation preserves the two endpoint values: the real integral has factor d , and the finite volume has factor d^{-1} . This is exactly FR16 and FR20. The finite ball independence used in CS2 proves that these coordinates are bijective onto the even K -invariant source. Multiplication by t_r corresponds to $D_{r^{-1}}$, so $I_\Gamma M_0$ maps to the null-generated rational relations.

If an adèlic representative is $\sum_d f_d(v) 1_{d\widehat{\mathbb{Z}}}(x_f)$, the inverse coordinates are

$$h_d(v) = f_d(dv), \quad m(h_d) = \left(f_d(0), d^{-1} \int_{\mathbb{R}} f_d \right). \quad (\text{CSB10.4})$$

Thus FR's two maps in FR18 are precisely

$$\begin{aligned} \Phi_{\text{FR}} \left[\sum_d t_d \otimes h_d \right] &= \sum_d h_d = B\mathcal{A} \left(\sum_d t_d \otimes h_d \right), \\ \beta_{\text{FR}} \left[\sum_d t_d \otimes h_d \right] &= \left(\sum_d v_p(d) m(h_d) \right)_p = \mathcal{L}\mathcal{A} \left(\sum_d t_d \otimes h_d \right). \end{aligned} \quad (\text{CSB10.5})$$

The symbols ℓ_p in FR15 are formal prime basis labels; the values of its β agree with the continuous component functionals denoted $\ell_p(F)$ here.

For completeness, FR's algebraic decomposition follows by the endpoint section:

$$M_0 = (R_\Gamma \otimes S_0) \oplus (I_\Gamma \otimes \mathbb{C}^2), \quad Q_0 \simeq S_0 \oplus (I_\Gamma / I_\Gamma^2) \otimes \mathbb{C}^2. \quad (\text{CSB10.6})$$

The identity $t_{ab} - 1 = (t_a - 1) + (t_b - 1) + (t_a - 1)(t_b - 1)$ and unique factorization identify I_Γ / I_Γ^2 with $\bigoplus_p \mathbb{C}$, by $t_d - 1 \mapsto (v_p(d))_p$. The coordinate derivations $t_d \mapsto v_p(d)$ vanish on I_Γ^2 and prove independence; the same identity proves spanning, including negative prime powers. This gives the already established FR22 decomposition

$$Q_0 \simeq S_0 \oplus V. \quad (\text{CSB10.7})$$

Taking the Hausdorff quotient in the strong LF source therefore does not identify additional classes in this spherical algebraic quotient. Indeed the natural map

$$Q_0 \longrightarrow D_{\text{null}}^K \quad (\text{CSB10.8})$$

is surjective by the representative description (CSB10.3); in the coordinates (CSB10.5), both sides have exactly the pair $(B, \mathcal{L})$. FR22 proves this pair injective on Q_0 , and CS2A proves it injective on D_{null}^K . Thus (CSB10.8) is an algebraic bijection, and with the induced locally convex quotient topology it is a homeomorphism: finite radial-ball coordinate changes are continuous on every finite LF stage, and the continuous inverse maps are the j and prime-boundary sections already written in CS2A.

The prior boundary lift at prime p , FR19, uses $(t_p - 1) \otimes h_w$, where $m(h_w) = w$. Choosing $h_w = \sigma(w)_\infty$ and applying (CSB10.3) gives

$$b_{\text{FR},p}(w) = \left[(D_{p^{-1}} - I) \sigma(w) \right]_{\text{null}}. \quad (\text{CSB10.9})$$

Our representative is $r_p \sigma(w) = (I - D_p) \sigma(w)$. Their actual difference is

$$(D_{p^{-1}} - I) \sigma(w) - r_p \sigma(w) = -r_p r_{p^{-1}} \sigma(w) \in \mathcal{R}_{\text{null}}. \quad (\text{CSB10.10})$$

Hence they define exactly the same boundary class, with the same sign and both endpoint coordinates. This also explains why the symmetric representative in (CSB3.8) can be used without changing the prior receiver.

The original weighted periodization seminorm in FR25 is a different receiving topology. Let $\Theta_{\text{FR}}h(u) = \sum_{n \in \mathbb{Z} \setminus \{0\}} h(nu) = Sh(u)$, and retain its periodization

$$\mathcal{E}_{\text{FR}}(c)(u) = u^{1/2} \Theta_{\text{FR}} \Phi_{\text{FR}} c(u) = 2\mathcal{E}(\Phi_{\text{FR}} c)(u). \quad (\text{CSB10.11})$$

The compact-unit Haar mass in the stated convention is one. For every $\delta > 1$, FR25 is exactly

$$\|c\|_{\delta, \text{FR}}^2 = \int_0^\infty |\mathcal{E}_{\text{FR}}(c)(u)|^2 (1 + \log^2 u)^{\delta/2} \frac{du}{u} = 4 \int_0^\infty |\mathcal{E}(\Phi_{\text{FR}} c)(u)|^2 (1 + \log^2 u)^{\delta/2} \frac{du}{u}. \quad (\text{CSB10.12})$$

If the compact-unit Haar mass is c_K instead, both squared-norm expressions acquire that same factor c_K . The positive weight, continuity of the periodized function, and injectivity of S give

$$\ker \|\cdot\|_{\delta, \text{FR}} = \ker \Phi_{\text{FR}} = b_{\text{FR}}(V). \quad (\text{CSB10.13})$$

Thus a nonzero v is zero in the Hausdorff quotient for this seminorm, while the continuous source coordinate $\mathcal{L}\iota(v) = v$ detects it in the strong LF quotient. In particular $\mathcal{L}$ cannot factor through that seminorm's Hausdorff quotient.

The new exact-image theorem $J_0 = \overline{J_0}$ in (CSB8.8) concerns the image of the summation map in the strong overlap function space. It makes ordinary first cohomology Hausdorff. It does not remove the separate kernel V before summation, and it does not change the kernel of the weaker periodization seminorm (CSB10.13). The maps (CSB10.5), (CSB10.8), and (CSB10.11) identify these three constructions explicitly.

The actual sphere sheaf, localization and normal quotient — CSP0-CSP13

Proof source: CC_SPHERE_PULLBACK_AND_NORMAL_DIRECTION.md.

Pullback of the original-zeta coefficient sheaf to the complex projective line

Complete derivation, 24 September 2026. Proof locators CSP0-CSP13.

CSP0. Input stage, source coverage, and the map being constructed

The support remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. This calculation starts after the complete arithmetic reconstruction, on the actual coefficient sheaf of CS and CSL. All complex vector spaces, scalar actions, analytic coordinates and quotient maps below belong to this receiving construction. They supply no addition, subtraction, metric, numerical coordinate, or parity on τ .

The complete corrected argument in USER_ARGUMENT_RECONSTRUCTION.md and the prerequisite checks PC01-PC04 govern this use. In particular, fixed support does not imply a trivial stalk; the retracted addition at τ is not used; neither the counting unit nor selected primes initialize this construction.

The original author source is Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, [arXiv:0903.2024v3](#), §5. For this derivation the unchanged author LaTeX file sources/CC_0903_2024_v3/author_source/announc3.tex was read at lines1471-1670, including projback, fonction3, restmaps, bord, hzero, rep1, liftw and court. Its source identity and preserved archive are recorded in CC_SHEAF_SOURCE_READING_PRIVATE.json. CS0-CS10 and CSL0-CSL9 were read in full. These local derivations prove the precise finite-unit sector, endpoint conventions, and raw Mellin comparison used here.

The reading of the later Connes-Consani author file CC.tex covered lines388-423 and485-608: the field-valued point construction, its three-point support, and the stated antiholomorphic real structure. That bounded reading is not described as a new reading of the entire paper. The source is [arXiv:2609.00299v1](#). The two mirrors will remain distinct: the holomorphic $z \mapsto -1/z$ and the actual antiholomorphic $z \mapsto -1/\bar{z}$ have different orientation degrees.

The subsequently completed exact-image derivations ../quantum_tau_programme_bridge_20260924/EXACT_SCHWARTZ_SUMMATION_IMAGE.md, SSI0-SSI10, and EXACT_SCHWARTZ_IMAGE_INDEPENDENT_CHECK.md, ESI0-ESI12, were read in full. Their proved equality $J = \bar{J}$, continuous inverse, and source-domain qualification are incorporated below, rather than left as a pending condition.

Let

$$X = \{x_+, \eta, x_-\}, \quad U_+ = \{x_+, \eta\}, \quad U_- = \{\eta, x_-\}, \quad U = \{\eta\}. \quad (\text{CSP0.1})$$

Let $Y = \mathbb{P}^1(\mathbb{C})$ with its usual topology, $D = \{0, \infty\}$, and $Y^\circ = \mathbb{C}^\times$. Define

$$\pi : Y \longrightarrow X, \quad \pi(0) = x_+, \quad \pi(\infty) = x_-, \quad \pi(z) = \eta \quad (z \neq 0, \infty). \quad (\text{CSP0.2})$$

Its inverse images of the three proper nonempty opens are $\mathbb{C}$, $Y \setminus \{0\}$, and $\mathbb{C}^\times$; hence it is continuous. It is the evaluation/specialization map for the projective-line charts: a nonzero field element is a unit and belongs to their common localization.

In particular, $\pi^{-1}(\eta) = \mathbb{C}^\times$. No point of the complex coordinate plane is identified with τ , and the complex chart origin 0 maps to x_+ , not to η . The sphere is a receiving space over the three-point support, not a replacement definition of that support.

The object to compute is

$$\mathcal{F} = \pi^{-1}\Omega \quad (\text{CSP0.3})$$

in the category of sheaves of complex vector spaces. The coefficient spaces and every finite complex below retain their specified locally convex topologies. Derived sheaf operations are taken in that abelian category; no unproved claim about derived categories of locally convex spaces is needed.

CSP1. Coefficients, endpoint lines, and restrictions

Retain the actual source spaces

$$S = \left\{ f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0 \right\},$$

$$V_+ = S \oplus E_+, \quad V_- = S \oplus E_-, \quad E_{\pm} = \mathbb{C}e_{\pm,0} \oplus \mathbb{C}e_{\pm,1}, \quad (\text{CSP1.1})$$

and the Fréchet overlap

$$A = \left\{ a \in C^\infty(\mathbb{R}_{>0}) : p_{N,j}(a) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty, N, j \geq 0 \right\}. \quad (\text{CSP1.2})$$

The endpoint indices retain value at zero and real integral, separately on both charts. Put

$$\Sigma f(u) = 2 \sum_{k \geq 1} f(ku), \quad Ra(u) = u^{-1}a(u^{-1}),$$

$$r_+(f, c_0, c_1) = \Sigma f, \quad r_-(h, d_0, d_1) = R\Sigma h = \Sigma \hat{h}, \quad \hat{h}(t) = \int_{\mathbb{R}} h(v) e^{-2\pi i vt} dv. \quad (\text{CSP1.3})$$

The full Poisson formula, before restricting to S , is

$$\Sigma \hat{h}(u) = u^{-1} \Sigma h(u^{-1}) + u^{-1} h(0) - \int_{\mathbb{R}} h(v) dv. \quad (\text{CSP1.4})$$

CS3 proves continuity and injectivity of $\Sigma : S \rightarrow A$; Fourier transformation preserves S , squares to one there, and gives $R\Sigma = \Sigma^\wedge$. Consequently

$$J = \Sigma(S) = r_+(V_+) = r_-(V_-), \quad \ker r_{\pm} = E_{\pm}. \quad (\text{CSP1.5})$$

To retain the exact comparison maps used in the earlier source calculations, write

$$C = \overline{J}^A, \quad Q_{\text{alg}} = A/J, \quad Q_H = A/C, \quad N = C/J. \quad (\text{CSP1.6})$$

The exact row $0 \rightarrow N \rightarrow Q_{\text{alg}} \rightarrow Q_H \rightarrow 0$ has the actual quotient topologies. The completed SSI1-SSI7 theorem, independently derived in ESI0-ESI8, now proves

$J = C, \quad N = 0, \quad Q_{\text{alg}} = Q_H = A/J, \quad \Sigma : S \xrightarrow{\sim} J \text{ continuously with continuous inverse.}$

(CSP1.7)

Here is the exact content being imported, including its factors. For every entire F in the full original nontrivial-zero jet ideal, $H(s) = F(s)/(2\zeta(s))$ is regular at those zeros and at 1, and has possible residues $F(-2r)/(2\zeta'(-2r))$ at the trivial zeros. SSI4 proves rapid vertical bounds for $P_N(s)H(s)$, $P_N(s) = \prod_{r=1}^N (s + 2r)$, by the full functional equation and its subquadratic-strip maximum-principle estimate. SSI5-SSI6 then construct the actual Schwartz inverse

$$f_F(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{F(2+it)}{2\zeta(2+it)} x^{-2-it} dt, \quad x > 0, \quad f_F(-x) = f_F(x),$$

$$\frac{f_F^{(2r)}(0)}{(2r)!} = \frac{F(-2r)}{2\zeta'(-2r)}, \quad f_F(0) = 0, \quad \int_{\mathbb{R}} f_F = 0, \quad \mathcal{M}_0 \Sigma f_F = F. \quad (\text{CSP1.8})$$

The derivative remainder and finite-seminorm bounds there prove that this is a continuous inverse on precisely S , not an approximation in a closure. Equivalently, for $a \in J$ and $x > 0$, $\Sigma^{-1}a(x) = \frac{1}{2} \sum_{k \geq 1} \mu(k)a(kx)$, with the zero-endpoint extension supplied by (CSP1.8), not by termwise evaluation. Those complete proofs supply (CSP1.7). Ordinary cohomology below is therefore already Hausdorff; retaining both earlier quotient names does not leave an unresolved extra kernel.

The sheaf Ω is the diagram $V_+ \xrightarrow{r_+} A \xleftarrow{r_-} V_-$, with global sections the fibre product. CS0-CS2A specify the full-difference interpretation of the original adelic coinvariants used to identify this source sector. Under the alternative moment-null generating-domain reading, the extra kernel computed there must be retained; this note uses the stated CS source, not that alternative.

CSP2. An explicit construction of the inverse-image sheaf

Write $j : Y^\circ \hookrightarrow Y$, $i : D \hookrightarrow Y$, and $\underline{A}_Y, \underline{A}_{Y^\circ}$ for constant sheaves, whose sections are locally constant functions with values in the vector space A . The coefficient topology does not change the meaning of “locally constant.”

The natural map

$$\underline{A}_Y \longrightarrow j_* \underline{A}_{Y^\circ} \quad (\text{CSP2.1})$$

is an isomorphism. Away from the poles this is immediate. At either pole, arbitrarily small punctured disks are connected, so the right-hand stalk is A , and the map of stalks is the identity. A stalkwise isomorphism of sheaves is an isomorphism. This is an assertion about j_* , not an assertion that Rj_* has no higher terms.

Define $i_*(V_+ \oplus V_-)$ as the direct sum of the two skyscraper sheaves, and let

$$\epsilon : \underline{A}_Y \longrightarrow i_*(A \oplus A)$$

be restriction to the two pole stalks. Then

$$\boxed{\mathcal{F} = \underline{A}_Y \times_{i_*(A \oplus A)} i_*(V_+ \oplus V_-),} \quad (a, v_+, v_-) \mapsto (a|_0 - r_+ v_+, a|_\infty - r_- v_-). \quad (\text{CSP2.2})$$

Here and below a pole component is omitted on an open set not containing that pole.

To prove that (CSP2.2) is (CSP0.3), first construct a map $\Omega \rightarrow \pi_* \mathcal{F}$. A section v_+ on U_+ gives the locally constant section $r_+ v_+$ on $\pi^{-1}U_+$, together with pole value v_+ . Use $r_- v_-$ on the other chart and the identity on their overlap. The fibre-product condition makes these maps compatible, so adjunction gives $\pi^{-1}\Omega \rightarrow \mathcal{F}$. Its pole stalks are the identities on $V_\pm$, and its other stalks are the identity on A . It is therefore an isomorphism.

In particular, on a connected disk containing only one pole, sections are exactly $V_{\pm}$, with their original Fréchet topology. On a connected open avoiding both poles, sections are A . The restrictions into the punctured disk are exactly $r_{\pm}$, not injections invented from a coordinate picture.

Formula (CSP2.2) gives a short exact sequence of vector-space sheaves

$$0 \longrightarrow \mathcal{F} \longrightarrow \underline{A}_Y \oplus i_*(V_+ \oplus V_-) \xrightarrow{\epsilon-r} i_*(A \oplus A) \longrightarrow 0. \quad (\text{CSP2.3})$$

Surjectivity is stalkwise: at each pole the constant-sheaf A -coordinate supplies any prescribed value, independently of the restriction map $r_{\pm}$. All displayed local maps are continuous. This construction does not assume that J is closed or that Σ^{-1} is continuous.

CSP3. Disk, annulus, and sphere cohomology with these coefficients

Here is the local resolution used in the calculation. For a complex vector space B , let $\mathcal{E}_B^q$ be the sheaf of differential forms locally expressible as a finite sum $\sum_{\nu} \omega_{\nu} b_{\nu}$, with ordinary smooth complex forms ω_{ν} and vectors $b_{\nu} \in B$. Its differential acts on the forms. On a star-shaped coordinate ball, the radial homotopy

$$(K\omega)_x = \int_0^1 t^{q-1} \iota_x \omega_{tx} dt \quad (q > 0) \quad (\text{CSP3.1})$$

acts on those finitely many scalar coefficients and satisfies $dK + Kd = 1$ in positive degree and $Kd = 1 - \text{ev}_0$ in degree zero. Thus

$$0 \rightarrow \underline{B} \rightarrow \mathcal{E}_B^0 \xrightarrow{d} \mathcal{E}_B^1 \xrightarrow{d} \dots \quad (\text{CSP3.2})$$

is a resolution. Partitions of unity act by multiplication, so these sheaves are fine. A locally finite partition produces locally finite sums, which are legitimate sections of the sheaf. Hence this computes sheaf cohomology. It also applies to a possibly noncomplete subspace J or quotient Q_{alg} : only finitely many coefficient vectors occur locally.

For the complete Fréchet spaces $A, V_{\pm}$, one may equivalently use smooth Fréchet-valued forms. The same integral exists as a limit of Riemann sums in every seminorm, and

$$p\left(\int_a^b f(t) dt\right) \leq \int_a^b p(f(t)) dt \quad (\text{CSP3.3})$$

proves continuity of each contraction on compact sets. Finite good covers and their contractions consequently give the same finite coefficient complexes with continuous maps. No completed infinite tensor product is being substituted for a coefficient space.

A disk, or a plane chart with its pole included, is acyclic for a constant sheaf: (CSP3.1) contracts its positive-degree complex. On Y° , put $z = e^{t+i\theta}$. A closed one-form can be written on the covering $\mathbb{R}^2$ as

$$\omega = P(t, \theta) dt + Q(t, \theta) d\theta, \quad \partial_{\theta} P = \partial_t Q,$$

with 2π -periodic coefficients. Its period

$$a = \int_0^{2\pi} Q(0, \theta) d\theta \quad (\text{CSP3.4})$$

is independent of t . After subtracting $a d\theta/(2\pi)$, the explicit primitive

$$g(t, \theta) = \int_0^t P(x, 0) dx + \int_0^\theta (Q(t, \phi) - a/(2\pi)) d\phi \quad (\text{CSP3.5})$$

is periodic and has differential equal to the adjusted form. Every two-form $G(t, \theta) dt \wedge d\theta$ is the differential of $(\int_0^t G(x, \theta) dx) d\theta$. These formulas prove

$$R\Gamma(Y^\circ, \underline{B}) \simeq B[0] \oplus B[-1]. \quad (\text{CSP3.6})$$

The degree-one coordinate is the positive z -circle period (CSP3.4), with representative $a d\theta/(2\pi)$. They also prove continuity of the period and chosen primitives in the coefficient seminorms on compact sets. A punctured pole disk has the same calculation in its local coordinate.

For the sphere, cover by its two plane charts and use (CSP3.2) to form the Čech double complex. Each chart is acyclic for $\underline{B}$; their overlap has (CSP3.6). The constant restriction difference is $\bar{B} \oplus B \rightarrow B$, $(a, b) \mapsto a - b$, which is onto. Consequently

$$H^0(Y, \underline{B}) = B, \quad H^1(Y, \underline{B}) = 0, \quad H^2(Y, \underline{B}) = B, \quad H^{>2} = 0. \quad (\text{CSP3.7})$$

The degree-two coordinate is integration with the complex orientation. For a concrete generator one can use

$$\omega_Y = \frac{1}{\pi} \frac{dx \wedge dy}{(1 + |z|^2)^2}, \quad z = x + iy, \quad \int_Y \omega_Y = 1. \quad (\text{CSP3.8})$$

It extends smoothly through infinity. The integral is $2 \int_0^\infty r(1 + r^2)^{-2} dr = 1$. The preceding two-chart complex shows that its nonzero integral generates the entire degree-two group; a zero-integral form is exact by that same period calculation. Thus (CSP3.7) is established by the resolution and its maps, not inferred solely from a homotopy picture.

CSP4. Global cohomology and a continuous finite complex

A skyscraper sheaf on either pole has no higher cohomology: sections on an open are either its coefficient space or zero, and all restrictions are onto. Apply (CSP3.7) to (CSP2.3). A finite model for $R\Gamma(Y, \mathcal{F})$ is

$$\begin{aligned} K^0 &= A \oplus V_+ \oplus V_-, & K^1 &= A \oplus A, & K^2 &= A, \\ d_K^0(a, v_+, v_-) &= (a - r_+ v_+, a - r_- v_-), & d_K^1 &= 0. \end{aligned} \quad (\text{CSP4.1})$$

The degree-two coordinate comes from the constant-sheaf summand with the orientation (CSP3.8). There are no other degrees.

The following explicit contraction removes the redundant diagonal A . Define

$$D^0 = V_+ \oplus V_-, \quad D^1 = A, \quad D^2 = A, \quad d_D^0 = r_+ - r_-, \quad d_D^1 = 0. \quad (\text{CSP4.2})$$

Maps $p : K \rightarrow D$ and $s : D \rightarrow K$ are

$$\begin{aligned} p^0(a, v_+, v_-) &= (v_+, v_-), & p^1(b_+, b_-) &= b_- - b_+, & p^2 &= \text{id}, \\ s^0(v_+, v_-) &= (r_- v_-, v_+, v_-), & s^1(b) &= (-b, 0), & s^2 &= \text{id}. \end{aligned} \quad (\text{CSP4.3})$$

They are chain maps and $ps = 1$. Let $h^1(b_+, b_-) = (b_-, 0, 0)$, with every other component of h zero. Direct substitution gives

$$1 - sp = d_K h + h d_K. \quad (\text{CSP4.4})$$

Every map is continuous and uses only the original restrictions, additions in coefficient spaces, and finite products. No inverse to Σ is needed.

It follows that

$$\begin{aligned} H^0(Y, \mathcal{F}) &= \{(\hat{h}, c_0, c_1), (h, d_0, d_1) : h \in S, c_0, c_1, d_0, d_1 \in \mathbb{C}\}, \\ H^1(Y, \mathcal{F}) &= A/J = Q_{\text{alg}}, \\ H^2(Y, \mathcal{F}) &= A, \quad H^{>2}(Y, \mathcal{F}) = 0. \end{aligned} \tag{CSP4.5}$$

The first isomorphism is topological via the Fourier map and coordinate projections. Degree one has its original quotient topology, which is Fréchet and Hausdorff by (CSP1.7); the previously retained comparison subspace C/J is zero. Degree two has the original Fréchet topology of A , and its isomorphism is induced by the sheaf projection $\mathcal{F} \rightarrow \underline{A}_Y$, followed by complex-oriented integration.

Equivalently, two pole charts for $\mathcal{F}$ are acyclic: on a chart, (CSP2.3) reduces to the complex $A \oplus V_{\pm} \rightarrow A$, $(a, v) \mapsto a - r_{\pm}v$, contracted onto $V_{\pm}$. Their annular overlap has $A[0] \oplus A[-1]$. In the derived Mayer-Vietoris cone its degree-zero overlap produces A/J , and its degree-one loop produces the additional A in degree two. This is the exact Čech-loop contribution.

CSP5. Where the additional term lives over the original support

Because each point of X has a smallest open neighborhood, higher direct-image stalks are the cohomology of the corresponding preimages. The preceding chart and overlap calculations give

$$R^0\pi_*\mathcal{F} = \Omega, \quad R^1\pi_*\mathcal{F} = j_{X!}\underline{A}_{\{\eta\}}, \quad R^{q>1}\pi_*\mathcal{F} = 0, \tag{CSP5.1}$$

where $j_X : \{\eta\} \hookrightarrow X$. Indeed the degree-one sheaf has zero stalks at $x_{\pm}$ and stalk A at η , with the zero generization maps from the closed stalks.

Its Čech complex on X is simply $0 \rightarrow A$, in degrees zero and one, so

$$H^0(X, R^1\pi_*\mathcal{F}) = 0, \quad H^1(X, R^1\pi_*\mathcal{F}) = A. \tag{CSP5.2}$$

The two-column, two-row Čech resolution computing the composite of direct image and global sections now has no possible nonzero differential beyond its displayed maps. It gives the Leray identifications

$$H^0(Y, \mathcal{F}) = H^0(X, \Omega), \quad H^1(Y, \mathcal{F}) = H^1(X, \Omega), \quad H^2(Y, \mathcal{F}) = H^1(X, R^1\pi_*\mathcal{F}) = A. \tag{CSP5.3}$$

The first two maps are induced by the adjunction unit $\Omega \rightarrow R\pi_*\pi^{-1}\Omega$. This supplies an exact derived comparison with the finite-poset calculation: the extra term is the punctured-disk loop in the actual receiver. It is not inserted into the stalk of τ as an additional parity label.

For comparison, $R^0j_*\underline{A}_{Y^\circ} = \underline{A}_Y$, whereas $R^1j_*\underline{A}_{Y^\circ}$ has one A -stalk at each pole. Formula (CSP2.1) did not discard those higher direct images. No splitting of the whole object $Rj_*\underline{A}$ is inferred from the list of its cohomology sheaves.

CSP6. Exact pole costalks and their orientation convention

Let p be either pole, B_p a small disk with local coordinate z at 0, or $w = 1/z$ at ∞ . Use the positive local-coordinate circle in each case. Local cohomology is the fibre complex of restriction:

$$R\Gamma_{\{p\}}(B_p, \mathcal{F}) = \text{Cone}(R\Gamma(B_p, \mathcal{F}) \rightarrow R\Gamma(B_p \setminus \{p\}, \mathcal{F}))[-1]. \tag{CSP6.1}$$

Using the explicit resolutions above, choose the fibre-complex convention whose differential is the restriction map. It gives

$$L_p^0 = V_p, \quad L_p^1 = A, \quad L_p^2 = A, \quad d^0 = r_p, \quad d^1 = 0. \quad (\text{CSP6.2})$$

In the unreoriented fibre cone the connecting map inserts the punctured-disk coefficient with a positive sign in the next degree. We keep that coordinate in degree one. In degree two we take its negative, so that the coordinate is positive complex-oriented integration, as checked next. The zero differential into degree two is unchanged. Thus

$$H_{\{p\}}^0 = E_p, \quad H_{\{p\}}^1 = Q_{\text{alg}}, \quad H_{\{p\}}^2 = A, \quad H_{\{p\}}^{>2} = 0. \quad (\text{CSP6.3})$$

These compute the costalk $i_p^! \mathcal{F}$; passage to smaller pole disks gives identity maps on these models.

The degree-two coordinate is the positive complex local fundamental class. To check the sign of the connecting map, choose a radial function χ that is one near the pole and zero near the outside of a small disk. If θ_p is its positive local angle, then

$$-d\chi \wedge \frac{d\theta_p}{2\pi} \quad (\text{CSP6.4})$$

is a compactly supported smooth two-form away from the pole. Its integral is $-\int_0^\infty \chi'(r)dr = 1$. Put $\beta = (1 - \chi)d\theta_p/(2\pi)$, a smooth one-form on the disk, so $d\beta$ is (CSP6.4). Restrict the relative cone to an outer annular collar, where $\beta = d\theta_p/(2\pi)$. Its cone differential sends $(\beta, 0)$ to $(d\beta, d\theta_p/(2\pi))$. Hence the boundary class $(0, d\theta_p/(2\pi))$ equals $(-d\beta, 0)$ in that relative cohomology. It is the **negative** of the positive-integral class. Excision identifies the same sign in the point-support group.

Thus the local connecting map from a positive angular period a is $-a$ in our degree-two integration coordinate. The support-to-global map on that coordinate is $+1$ at both poles, because both positive local complex orientations agree with the sphere orientation. At infinity the local circle is the positive w -circle, not the positive z -circle.

CSP7. Both supports, the complete localization triangle, and all signs

For $D = \{0, \infty\}$, take the direct sum of the disjoint costalk models:

$$L_D^0 = V_+ \oplus V_-, \quad L_D^1 = A \oplus A, \quad L_D^2 = A \oplus A, \\ d_L^0(v_+, v_-) = (r_+v_+, r_-v_-), \quad d_L^1 = 0. \quad (\text{CSP7.1})$$

The support-to-global map $k : L_D \rightarrow D^\bullet$ from (CSP4.2) is

$$k^0 = \text{id}, \quad k^1(b_+, b_-) = b_+ - b_-, \quad k^2(c_+, c_-) = c_+ + c_-. \quad (\text{CSP7.2})$$

Degree one has the ordered-chart difference; degree two has the sum of the two positive local orientation classes. These are different signs in different degrees.

An explicit verification of the entire triangle is useful. The ordinary cone of k has

$$\text{Cone}(k)^{-1} = V_+ \oplus V_-, \\ \text{Cone}(k)^0 = (V_+ \oplus V_-) \oplus A^2, \quad \text{Cone}(k)^1 = A \oplus A^2, \quad \text{Cone}(k)^2 = A,$$

with

$$\begin{aligned} d^{-1}(v) &= (v, -r_+v_+, -r_-v_-), \\ d^0(v, b_+, b_-) &= (r_+v_+ - r_-v_- + b_+ - b_-, 0, 0), \quad d^1(b, c_+, c_-) = c_+ + c_-. \end{aligned} \quad (\text{CSP7.3})$$

The overlap model $A[0] \oplus A[-1]$ maps into this cone by

$$a \text{ in degree } 0 \mapsto (0, a, a), \quad a \text{ in degree } 1 \mapsto (0, -a, a). \quad (\text{CSP7.4})$$

These are cocycles. Every degree-zero cocycle differs by $d^{-1}(v)$ from a unique diagonal pair $(0, a, a)$. Every degree-one cocycle has $c_- = -c_+$; its b -component is removed by the boundary of $(0, b, 0)$ in degree zero. Degrees minus one and two have zero cohomology by injectivity of d^{-1} and surjectivity of the sum. Thus (CSP7.4) is a quasi-isomorphism, with all reductions given by continuous finite linear operations.

This proves the localization triangle and its boundaries in the chosen coordinates. In particular,

$$\begin{aligned} \partial^0 : A &\rightarrow Q_{\text{alg}}^2, \quad a \mapsto ([a], [a]), \\ \partial^1 : H^1(Y^\circ, \underline{A}) = A &\rightarrow H_D^2(Y, \mathcal{F}) = A^2, \quad a \mapsto (-a, a). \end{aligned} \quad (\text{CSP7.5})$$

The second formula combines the negative localization-boundary sign proved in CSP6 with $w = 1/z$, which gives $d \arg w = -d \arg z$.

The full nonzero portion of the ordinary localization sequence is

$$\begin{aligned} 0 \rightarrow E_+ \oplus E_- \rightarrow H^0(Y, \mathcal{F}) &\xrightarrow{r} A \xrightarrow{\text{diag} \circ q} Q_{\text{alg}}^2 \\ &\xrightarrow{(q_+, q_-) \mapsto q_+ - q_-} Q_{\text{alg}} \xrightarrow{0} A \xrightarrow{a \mapsto (-a, a)} A^2 \xrightarrow{(a, b) \mapsto a + b} A \rightarrow 0. \end{aligned} \quad (\text{CSP7.6})$$

Here r takes a Fourier-graph section to its common restriction and has image J ; $q : A \rightarrow A/J$. Every kernel and image in this row follows directly from the displayed formulas. In particular, global degree-one classes restrict to zero in annular degree one: they are Čech differences of degree-zero overlap data.

For one pole alone the support-to-global maps are

$$\begin{aligned} L_+ \rightarrow D : \quad (v, b, c) &\mapsto ((v, 0), b, c), \\ L_- \rightarrow D : \quad (v, b, c) &\mapsto ((0, v), -b, c). \end{aligned} \quad (\text{CSP7.7})$$

They induce degree-one isomorphisms with signs $+1, -1$ and degree-two isomorphisms with sign $+1$ in both cases. The complementary chart is acyclic for $\mathcal{F}$, as required by its localization sequence.

CSP8. Holomorphic power maps and the geometric normal factor

For each recovered positive integer n , the actual holomorphic map

$$b_n : Y \rightarrow Y, \quad b_n(z) = z^n, \quad b_n(0) = 0, \quad b_n(\infty) = \infty \quad (\text{CSP8.1})$$

satisfies $\pi b_n = \pi$, so $b_n^{-1}\mathcal{F}$ is canonically identified with $\mathcal{F}$ before applying any coefficient action.

For coefficients retain

$$\begin{aligned} T_a f(u) &= f(u/a), \quad \rho_+(a)(f, c_0, c_1) = (f(\cdot/a), c_0, ac_1), \\ \rho_-(a)(h, d_0, d_1) &= (ah(a \cdot), ad_0, d_1), \quad a > 0. \end{aligned} \quad (\text{CSP8.2})$$

CS8 proves $r_{\pm}\rho_{\pm}(a) = T_a r_{\pm}$. These give a continuous sheaf endomorphism over the identity on Y . Combine it with pullback by b_n , using $a = n$, and denote the resulting cohomological operator by B_n .

On an annular angular class,

$$b_n^* \left(a \frac{d\theta}{2\pi} \right) = n a \frac{d\theta}{2\pi}. \quad (\text{CSP8.3})$$

The local coordinate at infinity also maps as $w \mapsto w^n$, so the same positive factor holds for its positive local circle. On degree two, either (CSP6.4) and the boundary give factor n , or one can compute its global degree directly:

$$b_n^* \omega_Y = \frac{1}{\pi} \frac{n^2 r^{2n-2} dx \wedge dy}{(1 + r^{2n})^2}, \quad \int_Y b_n^* \omega_Y = 2n^2 \int_0^\infty \frac{r^{2n-1}}{(1 + r^{2n})^2} dr = n. \quad (\text{CSP8.4})$$

All constants and the branch multiplicities at both poles are included. Therefore the induced chain/cohomology model for the combined operation is

$$B_n^0 = \rho_+(n) \oplus \rho_-(n), \quad B_n^1 = T_n, \quad B_n^2 = nT_n. \quad (\text{CSP8.5})$$

Each pole-support complex has the corresponding actions $(\rho_{\pm}(n), T_n, nT_n)$. On the overlap, degree zero has T_n and degree one has nT_n ; these make every map in (CSP7.6) equivariant.

This proves, as a representation of the recovered positive-integer monoid,

$$\boxed{H^2(Y, \mathcal{F}) = A(-1), \quad A(-1) \text{ has action } nT_n.} \quad (\text{CSP8.6})$$

The notation records the proved degree character, not a declaration that this infinite-dimensional complex sheaf is an étale Tate sheaf. The coefficient-only action remains T_a in every degree. The additional n in (CSP8.6) is produced by the actual geometric covering, not by changing the coefficient action by definition.

Composition is exact: $b_m b_n = b_{mn}$, $T_m T_n = T_{mn}$, and hence $B_m B_n = B_{mn}$. The induced operators are invertible on all displayed cohomology spaces. On positive rational indices define the cohomological comparison $B_{n/m} = B_n B_m^{-1}$; equality $nm' = n'm$ proves independence of the presentation. In degree two its action is $(n/m)T_{n/m}$. The strongly continuous family aT_a , already defined on the coefficient space, is the unique extension of this rational action to $a > 0$. This does not assert a holomorphic map $z \mapsto z^a$ for a nonintegral real a .

The maps b_n here are receiving-space covers. The full signed CC structure has its own odd-index and chart-swapping rules; even b_n , or all b_n with one uniform chart rule, are not thereby proved to be endomorphisms of that full signed structure.

CSP9. The two mirrors and their different orientation signs

Keep the maps separate:

$$w(z) = -1/z, \quad \alpha(z) = -1/\bar{z}. \quad (\text{CSP9.1})$$

Both exchange the two poles and induce the same swap $x_+ \leftrightarrow x_-$ on X . The former is holomorphic, degree +1 on the sphere; the latter is antiholomorphic, degree -1, and is the actual real structure stated in CC.tex. On the unit circle their angle maps are respectively

$$\theta \mapsto \pi - \theta, \quad \theta \mapsto \pi + \theta. \quad (\text{CSP9.2})$$

Thus their actions on $H^1(Y^\circ, \underline{A})$, before a coefficient map, have signs $-1, +1$, respectively.

Use the actual source Weyl coefficient map: exchange V_+ and V_- by identity on their correspondingly labelled coordinates, and apply R on A . It is a sheaf map over either swap because $r_- = Rr_+$ and

$R^2 = 1$. It is complex linear. It is not being identified without further construction with the full signed structure-sheaf action. If a semilinear coefficient comparison is wanted, composing with complex conjugation gives a separately specified conjugate-linear map.

On the global model (CSP4.2) the two operations are

$$\begin{array}{c|ccc} & D^0 & D^1 & D^2 \\ \hline w & \text{swap} & -R & +R \\ \alpha & \text{swap} & -R & -R. \end{array} \quad (\text{CSP9.3})$$

The degree-one sign is the ordered Čech swap in both cases. Degree two is the product of that Čech sign with the angular sign in (CSP9.2). Equivalently it is the orientation degree of the map, times R . On the pole-support degree-two pair, w acts as $(c_+, c_-) \mapsto (Rc_-, Rc_+)$, whereas α acts as $(c_+, c_-) \mapsto (-Rc_-, -Rc_+)$. Both act on its degree-one pair by $(b_+, b_-) \mapsto (Rb_-, Rb_+)$.

These formulas respect all localization maps. For example,

$$(Rb_- - Rb_+) = -R(b_+ - b_-).$$

For the annular boundary $a \mapsto (-a, a)$, the holomorphic operation sends its value to $(Ra, -Ra)$, which is the boundary of $-Ra$; the antiholomorphic operation sends it to $(-Ra, Ra)$, the boundary of $+Ra$. The degree-two sum is respectively transformed by $+R$ and $-R$.

Thus equality of the finite-poset swap does not erase the sphere's complex-orientation information. In particular, using $+R$ on global degree two for the actual antiholomorphic α would give the wrong map.

CSP10. The exact quotient of the normal direction

For any vector quotient $B = A/J$, there is an actual sheaf morphism

$$q_Y : \mathcal{F} \longrightarrow j_! \underline{Q}_{\text{alg}}|_{Y^\circ}. \quad (\text{CSP10.1})$$

On Y° it is $q : A \rightarrow A/J$; at either pole its stalk map is zero. To verify that it is a sheaf morphism into extension by zero, take a section near a pole. Its nearby constant A -value is $r_\pm v \in J$, so its quotient vanishes on a neighborhood of that pole. Thus the quotient section has exactly the vanishing-near-boundary condition required by $j_!$. The map is onto on every stalk.

Its kernel $\mathcal{F}_J$ has pole stalks $V_\pm$, interior stalk J , and the same restrictions with codomain J . Hence

$$0 \rightarrow \mathcal{F}_J \rightarrow \mathcal{F} \xrightarrow{q_Y} j_! \underline{Q}_{\text{alg}} \rightarrow 0 \quad (\text{CSP10.2})$$

is exact. Give J its actual closed-subspace Fréchet topology from (CSP1.7). The restrictions $V_\pm \rightarrow J$ are continuous, with continuous right inverses obtained from Σ^{-1} and Fourier transformation. CSP3 therefore applies with these complete coefficient spaces as well as in the vector-space category.

The extension-by-zero sheaf has the same pole/interior model with zero pole spaces and overlap $\underline{Q}_{\text{alg}}$. Thus

$$H^0(Y, j_! \underline{Q}_{\text{alg}}) = 0, \quad H^1(Y, j_! \underline{Q}_{\text{alg}}) = \underline{Q}_{\text{alg}}, \quad H^2(Y, j_! \underline{Q}_{\text{alg}}) = \underline{Q}_{\text{alg}}. \quad (\text{CSP10.3})$$

For $\mathcal{F}_J$, both restrictions are onto J ; therefore its degree one is zero and its degree two is J . Under these models (CSP10.1) induces the identity $\underline{Q}_{\text{alg}} \rightarrow \underline{Q}_{\text{alg}}$ in degree one and $q : A \rightarrow A/J$ in degree two. In particular the normal-direction sequence is exactly

$$\boxed{0 \longrightarrow J(-1) \longrightarrow A(-1) \longrightarrow H^2(Y, j_! \underline{Q}_{\text{alg}}) = \underline{Q}_{\text{alg}}(-1) \longrightarrow 0.} \quad (\text{CSP10.4})$$

Its kernel is the full original summation image J , not a selected family of zero jets or a finite spectral truncation. The sphere orientation and the covers prove the (-1) action on each term. As locally convex spaces, the quotient topology in (CSP10.4) is precisely the topology of A/J .

There is also the continuous Hausdorff version

$$0 \longrightarrow C(-1) \longrightarrow A(-1) \longrightarrow Q_H(-1) \longrightarrow 0, \quad (\text{CSP10.5})$$

and an exact comparison

$$0 \longrightarrow (C/J)(-1) \longrightarrow Q_{\text{alg}}(-1) \longrightarrow Q_H(-1) \longrightarrow 0. \quad (\text{CSP10.6})$$

The sheaf map in this version is the composition with $Q_{\text{alg}} \rightarrow Q_H$; it does not rely on any general exactness assertion for Hausdorffification. By (CSP1.7), this last map is the identity comparison, $C/J = 0$, and (CSP10.5) is exactly (CSP10.4). Both retained kernel labels are the same actual closed source image $J(-1)$; there is no remaining completion operation.

CSP11. Original zeta, all factors, and the map to the shifted kernel

Use the exact raw transform and its CS comparison

$$\mathcal{M}_0 a(s) = \int_0^\infty a(u) u^s \frac{du}{u}, \quad \Theta a(s) = \frac{1}{2} \mathcal{M}_0 a(s). \quad (\text{CSP11.1})$$

The factor $1/2$ is the explicitly proved comparison map; the original restriction remains $\Sigma f = 2 \sum_{k \geq 1} f(ku)$. It has not been replaced by a different restriction. The target is

$$\mathcal{B} = \left\{ F \text{ entire} : \sup_{|\Re s| \leq M} (1 + |\Im s|)^N |F(s)| < \infty \text{ for every } M, N \geq 0 \right\}. \quad (\text{CSP11.2})$$

CS7 and S1 give a topological isomorphism $\Theta : A \rightarrow \mathcal{B}$, with inverse

$$(\Theta^{-1} F)(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} F(1/2 + it) u^{-it} dt. \quad (\text{CSP11.3})$$

For $\Re s > 1$, the original identity is

$$\mathcal{M}_0 \Sigma f(s) = 2\zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}, \quad \Theta \Sigma f(s) = \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}. \quad (\text{CSP11.4})$$

It retains the original Euler product, unit summand and every repetition:

$$\zeta(s) = 1 + \sum_{k \geq 2} k^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{m \geq 1} (\log p) p^{-ms} \quad (\Re s > 1). \quad (\text{CSP11.5})$$

For the actual retained source

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2},$$

$$\Theta \Sigma f_0(s) = F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad \mathcal{M}_0 \Sigma f_0 = 2F_0. \quad (\text{CSP11.6})$$

The original zeta pole at 1, its trivial zeros, and the Gamma poles give

$$F_0(0) = F_0(1) = \frac{1}{8},$$

$$F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0 \quad (r \geq 1). \quad (\text{CSP11.7})$$

For general $f \in S$, writing $a_f(s) = \int_0^\infty f(v) v^s dv/v$ with its Taylor-subtracted continuation, the exact receiver values are

$$\mathcal{M}_0 \Sigma f(-2r) = 2\zeta'(-2r) \frac{f^{(2r)}(0)}{(2r)!}, \quad \mathcal{M}_0 \Sigma f(1) = 2a'_f(1), \quad \mathcal{M}_0 \Sigma f(0) = -a_f(0). \quad (\text{CSP11.8})$$

These identities keep the endpoint and trivial-zero contributions rather than representing them as absent zeros of a replacement function.

Let $\mathcal{Z}$ be the actual nontrivial zero set of the original ζ , with full multiplicities m_ρ , and put

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every } \rho \in \mathcal{Z}, 0 \leq j < m_\rho\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}. \quad (\text{CSP11.9})$$

The exact-image theorem SSI1-SSI7 strengthens S5 and CS7 to $\Theta J = \Theta C = \mathcal{J}$. The scalar $1/2$ in Θ leaves this linear subspace unchanged. Hence

$$Q_H \xrightarrow{\sim} \mathcal{Q}, \quad [a] \mapsto [\Theta a], \quad \Theta T_a = a^s \Theta. \quad (\text{CSP11.10})$$

The normal quotient from the actual sphere therefore has the continuous map

$$\nu : A(-1) \longrightarrow \mathcal{Q}(-1), \quad a \mapsto [\Theta a]. \quad (\text{CSP11.11})$$

Its kernel is exactly $J(-1) = C(-1)$ by (CSP1.7). Thus its factor through the ordinary geometric quotient (CSP10.4) is a continuous isomorphism $Q_{\text{alg}}(-1) \rightarrow \mathcal{Q}(-1)$. The earlier comparison kernel $(C/J)(-1)$ is zero by the actual inverse calculation, not by an assumed exactness property of Hausdorffification.

The comparison with the already constructed high-weight shifted quotient in GSL6.5-GSL6.11 is now an actual map. Retain

$$\mathcal{J}_+ = \{F \in \mathcal{B} : F^{(j)}(\rho+1) = 0, 0 \leq j < m_\rho\}, \quad \mathcal{Q}_{\cup,+} = \mathcal{B}/(\mathcal{J} \cap \mathcal{J}_+).$$

The GSL multiplier E_+ has jet one on the original divisor and jet zero on the divisor shifted by $+1$. Its construction and global strip estimates, not just these jet conditions, are the input guaranteeing a continuous multiplier of $\mathcal{B}$. Define

$$\boxed{\mathbf{n} : A(-1) \longrightarrow \mathcal{Q}_{\cup,+}, \quad \mathbf{n}(a) = [(1 - E_+(s))(\Theta a)(s-1)]_{\mathcal{J} \cap \mathcal{J}_+}.} \quad (\text{CSP11.12})$$

This is exactly $j_{\text{high}} \circ \nu$, where GSL proves that j_{high} is injective. Thus its kernel is the full original source $J(-1) = C(-1)$, its image is the entire GSL kernel, and its induced map from the ordinary geometric quotient (CSP10.4) is an isomorphism onto that kernel. Its continuity follows from (CSP11.3), translation continuity and the stated multiplier bounds.

For every recovered n , equivariance includes the full degree factor:

$$(\Theta(nT_n a))(s-1) = n n^{s-1}(\Theta a)(s-1) = n^s(\Theta a)(s-1). \quad (\text{CSP11.13})$$

Multiplication by $1 - E_+$ commutes with n^s ; hence $\mathbf{n}(nT_n a) = T_n \mathbf{n}(a)$. This proves the exact morphism between the geometric normal factor and the existing translated-divisor kernel.

The full analytic multiplier of that extension remains

$$F_0(s)F_0(s-1) = \left[\frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) \right] \left[\frac{(s-1)(s-2)}{8} \pi^{-(s-1)/2} \Gamma((s-1)/2) \zeta(s-1) \right]. \quad (\text{CSP11.14})$$

The second factor has values $1/8$ at $s = 1, 2$, and at $s = 1 - 2r$ its value is precisely (CSP11.7). Every derivative is the corresponding derivative of F_0 at $s - 1$. These are translations of the original full formula, not deletions of its poles or trivial-zero cancellations.

On an actual original zero block of multiplicity m_ρ , the normal quotient action is

$$n^{\rho+1} \sum_{j=0}^{m_\rho-1} \frac{(\log n)^j}{j!} M_t^j, \quad t = s - \rho. \quad (\text{CSP11.15})$$

Translation identifies this with the block at $\rho + 1$, with local parameter $s - (\rho + 1)$. Every nilpotent coefficient, factorial and multiplicity is retained.

CSP12. A comparison of the whole two-degree receiver with GSL

The geometric calculation has produced the original quotient in degree one and its degree-shifted coefficient in degree two. These degrees must remain visible. Degree one is already Hausdorff by (CSP1.7); degree two is passed through the actual quotient (CSP10.4). Write

$$\mathcal{H}^1 = H^1(Y, \mathcal{F}) = H^1(Y, \mathcal{F})_H \simeq \mathcal{Q}, \quad \mathcal{H}^2 = H^2(Y, j_! \underline{Q}_H) \simeq \mathcal{Q}(-1). \quad (\text{CSP12.1})$$

The combined covering action is T_n on the first and nT_n on the second.

There is a continuous representation isomorphism of their underlying direct sum with the GSL extension:

$$\Psi_Y : \mathcal{H}^1 \oplus \mathcal{H}^2 \longrightarrow \mathcal{Q}_{\cup,+},$$

$$\Psi_Y([f]_{\mathcal{J}}, [h]_{\mathcal{J}}) = [E_+(s)f(s) + (1 - E_+(s))h(s - 1)]_{\mathcal{J} \cap \mathcal{J}_+}. \quad (\text{CSP12.2})$$

The second displayed input has the twisted action in (CSP12.1). Independence of its representatives follows from the full original and shifted jet conditions on E_+ . Its inverse takes a common representative F to

$$([F]_{\mathcal{J}}, [F(s + 1)]_{\mathcal{J}}). \quad (\text{CSP12.3})$$

For the second coordinate, changing F by $\mathcal{J} \cap \mathcal{J}_+$ changes $F(s + 1)$ by a function in $\mathcal{J}$. On original jets, (CSP12.2) is f ; on shifted jets it is $h(s - 1)$. These two observations prove both inverse identities. The GSL global bounds prove continuity of the forward map, and translation plus quotient seminorms prove continuity of the inverse. Formula (CSP11.13) proves equivariance.

This is a proved comparison after taking the stated quotient and forgetting the distinction between two cohomological degrees in the target direct sum. It does not assert that the short exact GSL row itself is the localization triangle of $\mathcal{F}$, nor that an unconstructed geometric boundary has vanished. The actual localization maps are (CSP7.2)-(CSP7.6).

A diagram of the exact normal-direction maps is:

$$\begin{array}{ccccc} J(-1) & \longrightarrow & A(-1) & \longrightarrow & Q_{\text{alg}}(-1) \\ & & \downarrow \nu & & \downarrow \\ & & \mathcal{Q}(-1) & = & \mathcal{Q}(-1) \\ & & \downarrow j_{\text{high}} & & \\ & & \mathcal{Q}_{\cup,+} & & \end{array} \quad (\text{CSP12.4})$$

The upper row has kernel $J(-1)$. The middle-left vertical map has the same kernel $C(-1) = J(-1)$; the upper-right vertical map is an isomorphism, its earlier comparison kernel $(C/J)(-1)$ being zero. The bottom map is injective. All arrows intertwine the combined covering actions with their displayed factors.

CSP13. Exact scope of the derived geometric statement

The construction establishes the following concrete facts together:

1. The inverse image of the actual CS coefficient sheaf on the complex projective-line evaluation has the fibre-product presentation (CSP2.2), keeping every source coordinate and all four endpoint lines.
2. Its global groups, costalks and localization maps are computed by (CSP4.2), (CSP6.2) and (CSP7.1)–(CSP7.7). The additional degree-two coefficient comes from annular cohomology, as also identified by $R^1\pi_*$ in (CSP5.1).
3. The positive degree of the holomorphic cover $z \mapsto z^n$ gives the exact factor n . Combined with the original arithmetic coefficient action, this produces nT_n , and therefore the geometric normal representation $A(-1)$.
4. The full-source quotient of that normal coefficient maps by (CSP11.12) onto the entire existing GSL kernel. The exact comparison retains ordinary/Hausdorff kernels, the Mellin factor two, the original zeta, its full multipliers, every zero multiplicity, and the shifted endpoint values.
5. Holomorphic and antiholomorphic swaps act differently on degree two, exactly as in (CSP9.3). The latter is the real structure in the cited later CC construction.

This is an explicit geometric source for the previously translated coefficient character. The coefficient sheaf is infinite-dimensional and complex, with the proved Fréchet/quotient topologies; it has not been made into a finite-rank mixed étale sheaf on a finite-type scheme over a finite field. Deligne’s étale purity theorem is consequently not applied to it by naming its normal factor. The statement proved here is the construction and comparison of the actual objects and maps above.

The source dependencies are CS3 for the exact summation/Fourier source, S1 for Mellin inversion, SSI1–SSI7 and its ESI0–ESI8 independent derivation for the actual closed summation image strengthening S5, and GSL4–GSL6 for its already constructed global continuous separator and shifted quotient. CSP2–CSP10 give the new sheaf, resolution, support and degree calculations in full. No independent norm, coordinate, or parity is assigned to τ .

The actual connecting maps and the weight-lift comparison

Proof source: CC_ACTUAL_WEIGHT_LIFT_COMPARISON.md.

The actual sheaf lifts and the geometric normal quotient

Complete derivation, 24 September 2026. Stable proof locators CW0-CW11.

CW0. Construction stage and exact question

The support is $\tau\langle Z_1; \text{no } Z_2 \rangle$. All operations below are on the arithmetic coefficient sheaves already constructed in CSP, after the complete arithmetic reconstruction. There is no addition, scalar multiplication, coordinate, distance, or parity operation on τ . The two charts remain separately labelled; a difference in a cochain complex is not a pooled branch return measure.

The connected argument and its correction-precedence table in `USER_ARGUMENT_RECONSTRUCTION.md`, the operative `READ_FIRST_USER_CONSTRUCTION.md`, and the full governing passages WU050-WU055 and WU061-WU063 in `USER_CONSTRUCTION_FULL_LOGBOOK.md` were read for this calculation. In particular, no arithmetic label initializes the supporting construction; the original zeta arithmetic is retained; and a calculation on this receiving sheaf is not promoted into a conclusion about every possible geometric realization of the complete construction.

The actual input is the final `CC_SPHERE_PULLBACK_AND_NORMAL_DIRECTION.md`, CSP0-CSP13, read in full. Its version for this derivation has SHA256

EBFF9973E1472EA88E3F88D3350FAD8FFA98364067F485F48D3AF4E119EBAF9D.

The two comparison proofs are `GLOBAL_SHIFTED_ZETA_LIFT.md`, GSL0-GSL11, and `DELIGNE_INVARIANT_CYCLE_QUOTIENT.md`, DC0-DC12, previously derived and read in full and checked against their unchanged final hashes

2CC092DD3EBC0F021E9914FC7FD75DA16EA4DE47760B1D12558F33F0A661F529,

3A9AFDAEACD7DB0F154348F6487A204852459B08AE8E0F90017E7755FD627521.

The action and inverse calculations used from those notes are proved again below where they determine the present conclusion.

The source construction is Connes and Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, [arXiv:0903.2024v3](https://arxiv.org/abs/0903.2024v3), §5: the original author file `sources/CC_0903_2024_v3/author_source/announc3.tex`, lines1376-1667, was read in the preceding CSL verification. The relevant source formulas are `fonction3`, `restmaps`, `bord`, `hzero`, `rep1`, `liftw` and `court`. The finite-unit-sector comparison retains CS2A's full-difference generating-domain qualification. The later signed source `CC.tex`, lines494-610 and773-899, was also read: its topological comparison does not by itself identify its structure sheaf with this complex coefficient sheaf.

The comparison with Deligne concerns *La conjecture de Weil. II*, §3.6. DC reconstructs that argument from the current French transcription, with its actual geometric hypotheses and arithmetic reduction. No new original-author-TeX reading of Deligne is claimed here.

This note computes both actual sheaf short exact sequences in CSP, their connecting maps, and the actual supported localization sequence. It proves the vanishing and existence of the lifts that those sequences define. It also computes exactly which kernel has a separated original-zero character range, and which kernel does not.

CW1. Full coefficients, original zeta, and the source image

Use

$$S = \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0\}, \quad V_{\pm} = S \oplus E_{\pm}, \quad E_{\pm} = \mathbb{C}^2. \quad (\text{CW1.1})$$

The two endpoint coordinates are labelled value and integral on each chart. Let

$$A = \left\{ a \in C^{\infty}(\mathbb{R}_{>0}) : p_{N,j}(a) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \quad (N, j \geq 0) \right\}. \quad (\text{CW1.2})$$

The actual maps are

$$\Sigma f(u) = 2 \sum_{k \geq 1} f(ku), \quad Ra(u) = u^{-1}a(u^{-1}), \quad r_+(f, c_0, c_1) = \Sigma f, \quad r_-(h, d_0, d_1) = R\Sigma h = \hat{\Sigma}h. \quad (\text{CW1.3})$$

The factor 2 records both rational signs. The Fourier convention is $\hat{h}(t) = \int_{\mathbb{R}} h(v) e^{-2\pi i vt} dv$. Before imposing the two endpoint conditions, the retained Poisson identity is

$$\hat{\Sigma}h(u) = u^{-1}\Sigma h(u^{-1}) + u^{-1}h(0) - \int_{\mathbb{R}} h(v) dv. \quad (\text{CW1.4})$$

Thus $R^2 = 1$, Fourier transformation preserves S and squares to one there, and

$$J = \Sigma(S) = r_+(V_+) = r_-(V_-), \quad \ker r_{\pm} = E_{\pm}. \quad (\text{CW1.5})$$

Continuity follows by Schwartz summation for $u \geq 1$ and (CW1.4) for $u \leq 1$. Injectivity follows by taking the Mellin transform in $\Re s > 1$, dividing by the nonzero Euler product there, and applying Fourier uniqueness on a vertical line. These are the source maps of CSP1, not new restrictions.

Write

$$\mathcal{B} = \{F \text{ entire} : b_{M,N}(F) = \sup_{|\Re s| \leq M} (1 + |\Im s|)^N |F(s)| < \infty \text{ for all } M, N \geq 0\}.$$

The exact comparison is

$$\Theta a(s) = \frac{1}{2} \int_0^{\infty} a(u) u^s \frac{du}{u}, \quad (\Theta^{-1}F)(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} F(1/2 + it) u^{-it} dt. \quad (\text{CW1.6})$$

The integrals and their differentiated versions converge by the seminorms. Integration by parts in $\log u$, followed by the inverse Fourier integral and contour shifts in each fixed strip, proves that $\Theta : A \rightarrow \mathcal{B}$ is a topological isomorphism. The $1/2$, $1/\pi$, and $u^{-1/2}$ are retained in these comparison maps.

For $\Re s > 1$, absolute summation gives

$$\Theta \Sigma f(s) = \zeta(s) a_f(s), \quad a_f(s) = \int_0^{\infty} f(v) v^s \frac{dv}{v}. \quad (\text{CW1.7})$$

The original arithmetic remains

$$\zeta(s) = 1 + \sum_{k \geq 2} k^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{m \geq 1} (\log p) p^{-ms} \quad (\Re s > 1). \quad (\text{CW1.8})$$

In particular, neither the unit summand nor any prime-power repetition is removed.

For the actual source

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}$$

the full transform is

$$F_0(s) = \Theta \Sigma f_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad \mathcal{M}_0 \Sigma f_0 = 2F_0. \quad (\text{CW1.9})$$

Here $f_0(0) = 0$ and its integral is zero because

$$\frac{\pi}{2} \left(2\pi \frac{3}{4\pi^2} - 3 \frac{1}{2\pi} \right) = 0.$$

Its two Gaussian Mellin integrals, before applying the Gamma recurrence, are

$$\frac{\pi}{2} \left[2\pi \cdot \frac{1}{2} \pi^{-(s+4)/2} \Gamma((s+4)/2) - 3 \cdot \frac{1}{2} \pi^{-(s+2)/2} \Gamma((s+2)/2) \right]. \quad (\text{CW1.10})$$

Multiplication by the original $\zeta(s)$ gives (CW1.9). The endpoint and trivial-zero cancellations have the exact values

$$F_0(0) = F_0(1) = \frac{1}{8},$$

$$F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0 \quad (r \geq 1). \quad (\text{CW1.11})$$

The original ζ still has its pole at 1 and its trivial zeros at $-2r$. For general $f \in S$, the retained receiver values are

$$\Theta \Sigma f(-2r) = \zeta'(-2r) \frac{f^{(2r)}(0)}{(2r)!}, \quad \Theta \Sigma f(1) = a'_f(1), \quad \Theta \Sigma f(0) = -\frac{1}{2} a_f(0). \quad (\text{CW1.12})$$

Taylor subtraction in the integral defining a_f proves its meromorphic continuation. At $\nu = -2r$, write

$$a_f(\nu + t) = A_{-1}/t + \sum_{k \geq 0} A_k t^k, \quad A_{-1} = f^{(2r)}(0)/(2r)!.$$

Then the coefficient of t^n in $\zeta(\nu + t) a_f(\nu + t)$ is

$$\frac{\zeta^{(n+1)}(\nu)}{(n+1)!} A_{-1} + \sum_{j=1}^n \frac{\zeta^{(j)}(\nu)}{j!} A_{n-j}. \quad (\text{CW1.13})$$

At 1, write $\zeta(1+t) = t^{-1} + \sum_{j \geq 0} c_j t^j$ and $a_f(1+t) = \sum_{k \geq 1} a_k t^k$. The coefficient of t^n is

$$a_{n+1} + \sum_{j=0}^{n-1} c_j a_{n-j}. \quad (\text{CW1.14})$$

Empty sums are zero. Away from these exceptional points every derivative is the full Leibniz sum $\sum_{j=0}^n \binom{n}{j} \zeta^{(j)} a_f^{(n-j)}$. Thus the comparison retains derivative data as well as values.

Let $\mathcal{Z}$ be the actual nontrivial zeros of the original ζ , with their multiplicities m_ρ , and let

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every } \rho \in \mathcal{Z}, 0 \leq j < m_\rho\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}. \quad (\text{CW1.15})$$

CSP1.7 uses the proved SSI1-SSI7 exact-image theorem, independently checked in ESI0-ESI8:

$$J = \overline{J}^A, \quad \Sigma : S \xrightarrow{\sim} J \text{ topologically}, \quad \Theta J = J. \quad (\text{CW1.16})$$

The actual inverse in that theorem has, for $F = \mathcal{M}_0 a$,

$$f_F(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{F(2+it)}{2\zeta(2+it)} x^{-2-it} dt, \quad x > 0, \quad f_F(-x) = f_F(x),$$

$$\frac{f_F^{(2r)}(0)}{(2r)!} = \frac{F(-2r)}{2\zeta'(-2r)}, \quad f_F(0) = 0, \quad \int_{\mathbb{R}} f_F = 0. \quad (\text{CW1.17})$$

Its all-strip estimates and Schwartz remainder estimates are the completed source theorem imported here, not a new assumption of closure. Consequently

$$Q := A/J = Q_{\text{alg}} = Q_H \xrightarrow[\downarrow a]{\mapsto} [\Theta a] \sim \mathcal{Q} \quad (\text{CW1.18})$$

is a topological isomorphism, and the former comparison kernel $\overline{J}/J$ is zero. The chain calculations below also make sense algebraically before that theorem; continuity and the stated Fréchet conclusions use the proved closed image.

CW2. The actual sheaves, finite complexes, and actions

Let $Y = \mathbb{P}^1(\mathbb{C})$, $D = \{0, \infty\}$, $U = \mathbb{C}^\times$, and $j : U \hookrightarrow Y$, $i : D \hookrightarrow Y$. The map $\pi : Y \rightarrow X = \{x_+, \eta, x_-\}$ sends the poles to x_+, x_- and every other point to η . Its inverse-image sheaf is

$$\mathcal{F} = \pi^{-1}\Omega = \underline{A}_Y \times_{i_*(A \oplus A)} i_*(V_+ \oplus V_-). \quad (\text{CW2.1})$$

The fibre product uses the map $(a, v_+, v_-) \mapsto (a|_0 - r_+ v_+, a|_\infty - r_- v_-)$. At the two poles its stalks are $V_\pm$; on U it is the constant sheaf with fibre A .

The actual subsheaf $\mathcal{F}_J$ has the same pole stalks and restrictions, with interior stalk J . Put $\mathcal{G} = j_! \underline{Q}_U$. Quotienting the interior coefficients and sending the pole stalks to zero gives the actual sheaf map

$$0 \longrightarrow \mathcal{F}_J \longrightarrow \mathcal{F} \xrightarrow{q_Y} \mathcal{G} \longrightarrow 0. \quad (\text{CW2.2})$$

Near a pole, the interior value of a section lies in $r_\pm(V_\pm) = J$. Its quotient is therefore zero near the pole, exactly the extension-by-zero condition. The sequence is exact on every stalk.

CSP2-CSP7 construct fine resolutions and their finite contractions. The resulting finite complexes, with degrees 0, 1, 2, form the degreewise exact row

$$\begin{array}{ccccccc} 0 & \rightarrow & V_+ \oplus V_- & \xrightarrow{\text{id}} & V_+ \oplus V_- & \rightarrow & 0 \rightarrow 0 \\ & & \downarrow r_+ - r_- & & \downarrow r_+ - r_- & & \downarrow 0 \\ 0 & \rightarrow & J & \xrightarrow{\iota} & A & \xrightarrow{q} & Q \rightarrow 0 \\ & & \downarrow 0 & & \downarrow 0 & & \downarrow 0 \\ 0 & \rightarrow & J & \xrightarrow{\iota} & A & \xrightarrow{q} & Q \rightarrow 0. \end{array} \quad (\text{CW2.3})$$

The three columns compute $R\Gamma(Y, \mathcal{F}_J)$, $R\Gamma(Y, \mathcal{F})$, and $R\Gamma(Y, \mathcal{G})$. The top row is read as $0 \rightarrow D_J^0 \rightarrow D_F^0 \rightarrow D_G^0 \rightarrow 0$, with $D_G^0 = 0$. All maps are continuous.

For clarity, these are simultaneous models for the sheaf maps, not independently chosen cohomology presentations. For any $B = A, J, Q$, use the fibre-product resolution with its specified pole spaces. It gives the complex

$$B \oplus V_+ \oplus V_- \longrightarrow B^2 \longrightarrow B, \quad d^0(a, v_+, v_-) = (a - r_+ v_+, a - r_- v_-), \quad d^1 = 0.$$

For $B = Q$ the pole spaces are zero. The contraction

$$p^0(a, v_+, v_-) = (v_+, v_-), \quad p^1(b_+, b_-) = b_- - b_+, \quad p^2 = \text{id}$$

and the section

$$s^0(v_+, v_-) = (r_- v_-, v_+, v_-), \quad s^1(b) = (-b, 0), \quad s^2 = \text{id} \quad (\text{CW2.4})$$

commute with inclusion $J \rightarrow A$ and quotient $A \rightarrow Q$. With $h^1(b_+, b_-) = (b_-, 0, 0)$, direct substitution gives $ps = 1$ and $1 - sp = dh + hd$. This proves the compatibility asserted in (CW2.3).

The coefficient actions are

$$\begin{aligned} T_a b(u) &= b(u/a), \quad \rho_+(a)(f, c_0, c_1) = (f(\cdot/a), c_0, ac_1), \\ \rho_-(a)(h, d_0, d_1) &= (ah(a\cdot), ad_0, d_1), \quad a > 0. \end{aligned} \quad (\text{CW2.5})$$

They preserve every source and satisfy $r_{\pm} \rho_{\pm}(a) = T_a r_{\pm}$. The actual holomorphic covering $b_n(z) = z^n$, for recovered positive integers n , has degree n on the sphere and multiplies an annular angular period by n . Combining that pullback with the coefficient action gives

$$\mathbf{B}_n^0 = \rho_+(n) \oplus \rho_-(n), \quad \mathbf{B}_n^1 = T_n, \quad \mathbf{B}_n^2 = nT_n. \quad (\text{CW2.6})$$

The same formula applies to the three columns where defined. Degree two uses the positive complex orientation. In the notation $B(-1)$, the action is aT_a ; this records the factor just proved from b_n , without asserting an étale realization.

Integer operators are invertible on the computed cohomology. Their inverses define the positive rational action consistently, and strong continuity extends it to all $a > 0$. This is a cohomological action; no nonintegral holomorphic map $z \mapsto z^a$ is asserted.

CW3. The connecting map of the fibre-product presentation

The other actual short exact sequence, CSP2.3, is

$$0 \rightarrow \mathcal{F} \rightarrow \underline{A}_Y \oplus i_*(V_+ \oplus V_-) \xrightarrow{\epsilon - r} i_*(A \oplus A) \rightarrow 0. \quad (\text{CW3.1})$$

Constant-sheaf cohomology on the sphere is $A, 0, A$ in degrees $0, 1, 2$; a pole skyscraper has no positive cohomology. Hence its complete nonzero long exact sequence is

$$0 \rightarrow H^0(Y, \mathcal{F}) \rightarrow A \oplus V_+ \oplus V_- \xrightarrow{d} A^2 \xrightarrow{\delta_{\text{fb}}} Q \rightarrow 0, \quad (\text{CW3.2})$$

and the degree-two comparison is the identity

$$H^2(Y, \mathcal{F}) = A(-1) \xrightarrow{\text{id}} H^2(Y, \underline{A}_Y) = A(-1). \quad (\text{CW3.3})$$

Here

$$d(a, v_+, v_-) = (a - r_+ v_+, a - r_- v_-), \quad \delta_{\text{fb}}(b_+, b_-) = [b_- - b_+]. \quad (\text{CW3.4})$$

The connecting sign follows from the degree-one contraction p^1 in (CW2.4). Its kernel is exactly

$$\{(b_+, b_-) : b_- - b_+ \in J\}. \quad (\text{CW3.5})$$

Indeed d has differences in J . Conversely, for $b_- - b_+ = r_+ v_+$, choose $v_- = 0$ and $a = b_-$. Then $d(a, v_+, 0) = (b_+, b_-)$. Every class $[b]$ is $\delta_{\text{fib}}(0, b)$, so the image is all Q .

This is an actual, generally nonzero connecting map. Its source action is $T_n \oplus T_n$ and its target action is T_n , with no geometric factor n in this degree. Its exact lifting quotient is

$$A^2 / \text{im } d \xrightarrow{\sim} Q, \quad [(b_+, b_-)] \mapsto [b_- - b_+]. \quad (\text{CW3.6})$$

Pairs in (CW3.5) lift through d ; other pairs do not. This statement specifies this sheaf-presentation lifting problem, rather than declaring a failure of a different global construction.

CW4. The actual quotient-sheaf lift and its zero boundary

From (CW2.3),

$$\begin{aligned} H^0(Y, \mathcal{F}_J) &= H^0(Y, \mathcal{F}) = \{(\hat{h}, c_0, c_1), (h, d_0, d_1)\}, \\ H^1(Y, \mathcal{F}_J) &= 0, \quad H^2(Y, \mathcal{F}_J) = J(-1), \\ H^1(Y, \mathcal{F}) &= Q, \quad H^2(Y, \mathcal{F}) = A(-1), \\ H^0(Y, \mathcal{G}) &= 0, \quad H^1(Y, \mathcal{G}) = Q, \quad H^2(Y, \mathcal{G}) = Q(-1), \end{aligned} \quad (\text{CW4.1})$$

and all degree-three groups vanish. The actual long exact sequence is therefore

$$\begin{aligned} 0 \rightarrow H^0(Y, \mathcal{F}_J) &\xrightarrow{\text{id}} H^0(Y, \mathcal{F}) \rightarrow 0, \\ 0 \rightarrow H^1(Y, \mathcal{F}) &= Q \xrightarrow{\text{id}} H^1(Y, \mathcal{G}) = Q \xrightarrow{\delta^1} J(-1) \xrightarrow{\iota} A(-1) \xrightarrow{q} Q(-1) \xrightarrow{\delta^2} 0. \end{aligned} \quad (\text{CW4.2})$$

Every sign and map follows from the simultaneous complexes, including

$$\boxed{\delta^1 = 0, \quad \delta^2 = 0.} \quad (\text{CW4.3})$$

Here is the cochain proof of the first equality, which does not assume a section of $q : A \rightarrow Q$. A class $x \in H^1(Y, \mathcal{G}) = Q$ is represented by the degree-one cocycle x of its zero-differential complex. Choose any $a \in A$ with $q(a) = x$, which exists by the definition of the quotient. Its differential in D_F is $d_F^1 a = 0$. Therefore the defining connecting cocycle in D_J^2 is zero. Changing a by an element of J still gives zero. The degree-one cohomology class $[a] \in H^1(Y, \mathcal{F}) = A/J$ depends only on x and is exactly x .

Thus the actual lift is the continuous equivariant inverse of the identity map in (CW4.2). It is unique at the cohomology-class level because $H^1(Y, \mathcal{F}_J) = 0$. In degree two every $x \in Q(-1)$ also lifts, and its set of lifts is the nonempty affine fibre

$$\text{Lift}_2(x) = q^{-1}(x) = a + J(-1). \quad (\text{CW4.4})$$

The equality describes all representatives, including their complete kernel. It does not select a representative compatible with every arithmetic action.

The degree-one obstruction receiver itself survives:

$$\text{im } \delta^1 = 0, \quad \text{coker}(H^1 \mathcal{F} \rightarrow H^1 \mathcal{G}) = 0, \quad \ker(H^2 \mathcal{F} \rightarrow H^2 \mathcal{G}) = J(-1). \quad (\text{CW4.5})$$

In the lifting formula of DC5, there is no additional invariant quotient of $H^1(Y, \mathcal{G})$ in this sequence. Taking its actual target as the classes to be lifted means that the further quotient is the identity and its kernel is zero. The corresponding obstruction quotient is therefore $J(-1)/0$, while the obstruction map into it is zero. This is the complete receiver; it has not been replaced by $Q(-1)$.

CW5. Supported localization: every image, kernel, and lift

For the two pole supports, the finite costalk complex is

$$L_D^0 = V_+ \oplus V_-, \quad L_D^1 = A^2, \quad L_D^2 = A(-1)^2, \quad d^0 = (r_+, r_-), \quad d^1 = 0.$$

The support-to-global maps are

$$k^0 = \text{id}, \quad k^1(b_+, b_-) = b_+ - b_-, \quad k^2(c_+, c_-) = c_+ + c_-. \quad (\text{CW5.1})$$

The annular positive z -circle gives

$$R\Gamma(U, \underline{A}) = A[0] \oplus A(-1)[-1].$$

The local positive circle at infinity uses $w = 1/z$, so $d \arg w = -d \arg z$. With the positive complex local fundamental classes, the actual localization boundaries are

$$\partial^0(a) = ([a], [a]), \quad \partial^1(a) = (-a, a). \quad (\text{CW5.2})$$

These are precisely the cone maps of CSP7, including the negative local boundary sign and the inverse-coordinate sign at infinity.

The resulting entire nonzero row is

$$\begin{aligned} 0 \rightarrow E_+ \oplus E_- \rightarrow H^0(Y, \mathcal{F}) \xrightarrow{r} A \xrightarrow{\text{diag } q} Q^2 \xrightarrow{\text{diff}} Q \\ \xrightarrow{0} A(-1) \xrightarrow{(-\text{id}, \text{id})} A(-1)^2 \xrightarrow{\text{sum}} A(-1) \rightarrow 0. \end{aligned} \quad (\text{CW5.3})$$

Here r sends a Fourier-graph section to its common restriction and has image J . Consequently:

1. A degree-zero annular section $a \in A$ extends globally exactly when $a \in J$. Its obstruction is $([a], [a])$, with image $\text{diag } Q$, kernel J , and cokernel identified with Q by the difference map.
2. Every global degree-one class in Q lifts from the two supported groups Q^2 . The kernel is $\text{diag } Q$, and a continuous equivariant section is

$$s_1(q) = \frac{1}{2}(q, -q). \quad (\text{CW5.4})$$

The following restriction $Q \rightarrow A(-1)$ is zero.

3. A degree-one annular class $a \in A(-1)$ extends to global degree one exactly when $a = 0$. Its boundary is the injective map $a \mapsto (-a, a)$, with full image the anti-diagonal. On that image its inverse is $(c_+, c_-) \mapsto (c_- - c_+)/2$.
4. Every global degree-two class $a \in A(-1)$ lifts from $A(-1)^2$. The kernel is the anti-diagonal, and a continuous equivariant section is

$$s_2(a) = \frac{1}{2}(a, a). \quad (\text{CW5.5})$$

Each assertion is a direct kernel/image calculation in the displayed row. In particular the quotient after the nonzero annular boundary is

$$A(-1)^2 / \{(-a, a) : a \in A(-1)\} \xrightarrow{\sim} A(-1), \quad [(b, c)] \mapsto b + c. \quad (\text{CW5.6})$$

It survives; the existence of the supported degree-two lift does not assert that this receiver is zero.

For a single pole the supported degree-one map to Q is $+\text{id}$ at 0, $-\text{id}$ at infinity, and the supported degree-two map is $+\text{id}$ at both poles. These are the maps of CSP7.7, so both single-support lifting maps are isomorphisms in their displayed degrees.

Every scalar $1/2$ in these sections belongs to the recovered coefficient field. The sections do not divide the supporting datum.

The mirror signs are also retained. On Q both sphere swaps induce $-R$; on Q^2 they induce $(q_+, q_-) \mapsto (Rq_-, Rq_+)$. Hence s_1 intertwines them. On $A(-1)$, the holomorphic map $z \mapsto -1/z$ induces $+R$, whereas the actual antiholomorphic map $z \mapsto -1/\bar{z}$ induces $-R$. Their supported degree-two maps are respectively $+R$ -swap and $-R$ -swap, so s_2 intertwines both. The annular actions are respectively $-R, +R$, exactly compatible with $\partial^1(a) = (-a, a)$. These are coefficient and orientation operations, not an exchanged label assigned to τ .

CW6. Which character ranges are actually separated

Under Θ , coefficient dilation is multiplication by a^s :

$$\Theta T_a = a^s \Theta, \quad LF = sF. \quad (\text{CW6.1})$$

Fix $a > 1$ when discussing logarithmic-modulus weights, and define

$$w_a(\lambda) = \frac{2 \log |\lambda|}{\log a} \quad (\lambda \neq 0). \quad (\text{CW6.2})$$

The actions themselves are defined for every $a > 0$, including the identity at $a = 1$.

The full jet at an actual zero ρ , in local coordinate $t = s - \rho$, has action

$$T_a = a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} M_t^j, \quad M_t^{m_\rho} = 0. \quad (\text{CW6.3})$$

On the normal quotient $\mathcal{Q}(-1)$, the corresponding action is

$$a^{\rho+1} \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} M_t^j. \quad (\text{CW6.4})$$

The established open strip $0 < \Re \rho < 1$ gives the disjoint ranges

$$0 < w_a(a^\rho) < 2, \quad 2 < w_a(a^{\rho+1}) < 4. \quad (\text{CW6.5})$$

These preserve every nilpotent and do not impose $\Re \rho = 1/2$. They are ranges of the actual zero characters, not a claim that an infinite-dimensional complex representation is a mixed finite-dimensional étale module.

The endpoint lines retain weights 0, 2 on E_+ , and 2, 0 on E_- . Their action is explicitly $(c_0, c_1, d_0, d_1) \mapsto (c_0, ac_1, ad_0, d_1)$. The common-restriction image of $H^0(Y, \mathcal{F})$ is J , with action T_a . Its Fourier-graph parameter transforms as $h(v) \mapsto ah(av)$.

It is essential to compute the characters of the actual degree-one obstruction receiver $J(-1)$, rather than assign to it (CW6.5). For any $\lambda \in \mathbb{C}$, let $m_\lambda = \text{ord}_\lambda F_0$, equal to zero off $\mathcal{Z}$. For $F \in J$, local division defines

$$\ell_\lambda(F) = \left(\frac{F}{F_0} \right) (\lambda) = \frac{F^{(m_\lambda)}(\lambda)}{F_0^{(m_\lambda)}(\lambda)}. \quad (\text{CW6.6})$$

The denominator is nonzero. All lower numerator and denominator derivatives vanish when $m_\lambda > 0$, giving the displayed ratio. Evaluation of a fixed derivative is continuous for the strip topology by the Cauchy integral formula on a small fixed circle. Thus ℓ_λ is continuous. Moreover

$$\ell_\lambda(F_0) = 1, \quad \ell_\lambda(a^s F) = a^\lambda \ell_\lambda(F). \quad (\text{CW6.7})$$

The first equality proves surjectivity to $\mathbb{C}$; the second follows either from local division or from Leibniz's formula with all lower vanishing derivatives retained.

Consequently $\mathcal{J}(-1)$, and hence $J(-1)$, has a continuous one-dimensional quotient character

$$a \mapsto a^{\lambda+1} \quad \text{for every } \lambda \in \mathbb{C}. \quad (\text{CW6.8})$$

At $0, 1, -2r$ the order m_λ is zero and the nonzero denominators are exactly (CW1.11), so there is no exceptional gap in this statement. For an actual original zero ρ , the choice $\lambda = \rho - 1$ gives the quotient character a^ρ , the same character that occurs in $\mathcal{Q}$. The point $\rho - 1$ is an evaluation point of the full original function; it has not been asserted to be a zero of ζ .

The same elementary evaluation on $\mathcal{B}$ gives all characters a^λ as quotient characters of A , and all $a^{\lambda+1}$ as quotient characters of $A(-1)$. Therefore the actual targets $J(-1)$ and $A(-1)$ in (CW4.2) and (CW5.3) do not satisfy a bound placing all their continuous character quotients above the original-zero range. The separated space is the *quotient* $Q(-1)$ of the full normal source, with its complete original-zero ideal removed.

This calculation concerns quotient characters. It does not claim that J or A has corresponding eigenvectors. In fact they have no nonzero arithmetic eigenvectors at all: an eigenfunction $F \in \mathcal{B}$ for one $a > 1$ would satisfy $(a^s - c)F(s) = 0$. The first factor is a nonzero entire function, so the product identity forces $F = 0$. The same argument, with a power of $a^s - c$, excludes nonzero generalized eigenvectors in these entire-function spaces.

CW7. A global vanishing theorem for the actual boundary maps

There is a complete global vanishing calculation even though the actual targets in CW6 do not have the proposed separated quotient-character bound. We prove it using the full quotient topology and a resolvent, without assuming that finite zero-jet subspaces are dense.

For $\mu \notin \mathcal{Z}$, define on $\mathcal{B}$

$$\mathcal{R}_\mu F(s) = \frac{F(s) - F_0(s)F(\mu)/F_0(\mu)}{\mu - s}. \quad (\text{CW7.1})$$

The numerator vanishes at $s = \mu$, so this is entire. It belongs to $\mathcal{B}$ continuously. To verify the topology directly, put $H = F - F_0 F(\mu)/F_0(\mu)$. Outside $|s - \mu| < 1$, division by $\mu - s$ is bounded by one. Inside that disk,

$$\frac{H(s)}{\mu - s} = - \int_0^1 H'(\mu + t(s - \mu)) dt.$$

Cauchy's estimate bounds $|H'|$ there by $\sup_{|z-\mu| \leq 2} |H(z)|$. For $M' = \max(M, |\Re \mu| + 2)$, this gives

$$b_{M,N}(\mathcal{R}_\mu F) \leq b_{M,N}(H) + (2 + |\Im \mu|)^N b_{M',0}(H). \quad (\text{CW7.2})$$

The scalar $F(\mu)/F_0(\mu)$ is bounded by $b_{M',0}(F)/|F_0(\mu)|$. Substituting it in (CW7.2) gives a finite sum of continuous seminorm bounds on F , with the complete F_0 retained.

If $F \in \mathcal{J}$, its numerator is in $\mathcal{J}$, and division by $\mu - s$ preserves all original-zero vanishing orders. Thus (CW7.1) descends continuously to $\mathcal{Q}$. Direct multiplication gives

$$(\mu - L)\mathcal{R}_\mu[F] = [F], \quad \mathcal{R}_\mu(\mu - L)[F] = [F]. \quad (\text{CW7.3})$$

The discarded term in the first equality is explicitly $F_0 F(\mu)/F_0(\mu) \in \mathcal{J}$; no factor of the original function has been suppressed. This rederives the needed RZ inverse on its exact full domain.

Let $H : \mathcal{Q} \rightarrow \mathcal{B}(-1)$ be continuous and complex-linear and commute with the recovered arithmetic actions. If equivariance is initially required only for positive integers, invertibility gives it for positive

rational; continuity and strong continuity of T_a give it for every $a > 0$. Differentiation at $a = e^t, t = 0$ is justified by

$$|e^{ts} - 1 - ts| \leq \frac{1}{2}t^2|s|^2e^{|t||\Re s|},$$

which proves convergence in every strip seminorm, and then in every quotient seminorm. The target generator is $L + 1$. Consequently

$$HL = (L + 1)H. \quad (\text{CW7.4})$$

For a fixed $\lambda \in \mathbb{C}$, evaluation gives

$$\text{ev}_\lambda H(L - (\lambda + 1)) = 0. \quad (\text{CW7.5})$$

When $\lambda + 1 \notin \mathcal{Z}$, (CW7.3) makes the operator in parentheses onto $\mathcal{Q}$. Hence $H(x)(\lambda) = 0$ for every $x \in \mathcal{Q}$. For each x , the entire function $H(x)$ therefore vanishes on the open complement of the discrete set $\mathcal{Z} - 1$, and is identically zero. We have proved

$$\boxed{\text{Hom}_{\text{cont}, T}(\mathcal{Q}, \mathcal{B}(-1)) = 0.} \quad (\text{CW7.6})$$

Since $\mathcal{J}(-1)$ is an invariant subspace, inclusion gives

$$\boxed{\text{Hom}_{\text{cont}, T}(\mathcal{Q}, \mathcal{J}(-1)) = 0.} \quad (\text{CW7.7})$$

Transport through Θ proves the corresponding statements for $Q \rightarrow A(-1)$ and $Q \rightarrow J(-1)$. Thus both the actual restriction $H^1(Y, \mathcal{F}) \rightarrow H^1(U, \underline{A})$ and the actual connecting map $\delta^1 : H^1(Y, \mathcal{G}) \rightarrow H^2(Y, \mathcal{F}_J)$ also vanish by this global operator calculation. Their explicit cochain proofs in CW4-CW5 identify which maps are being tested.

No assertion about a direct sum or unrestricted product of zero blocks entered the proof. Nor did overlapping quotient characters produce a nonzero map: the distinction between a quotient character and an entire-function eigenvector is retained.

For comparison, the separated original-zero targets also admit a global proof:

$$\text{Hom}_{\text{cont}, T}(\mathcal{Q}, \mathcal{Q}(-1)) = 0. \quad (\text{CW7.8})$$

Indeed, for a continuous complex-linear equivariant H , take the full m_ρ -jet of its target at an actual ρ . The target generator there is $(\rho + 1)I + M_t$, so that jet kills $(L_{\text{target}} - (\rho + 1))^{m_\rho}$. Equivariance and (CW7.3), with $\mu = \rho + 1 \notin \mathcal{Z}$, imply that every target jet of $H(x)$ is zero. The joint original-zero jets are faithful on $\mathcal{Q}$ by (CW1.15). Thus $H = 0$. This proves the global assertion behind the separated ranges in (CW6.5), including every multiplicity.

CW8. The normal quotient lifts classes but has no equivariant linear section

The actual degree-two row is

$$0 \rightarrow J(-1) \rightarrow A(-1) \xrightarrow{q} Q(-1) \rightarrow 0, \quad \text{or, under } \Theta, \quad 0 \rightarrow \mathcal{J}(-1) \rightarrow \mathcal{B}(-1) \xrightarrow{q} \mathcal{Q}(-1) \rightarrow 0. \quad (\text{CW8.1})$$

Every class lifts by its definition as a quotient; the whole fibre is (CW4.4). We now calculate the distinct question of a simultaneous arithmetic-equivariant linear choice of representatives.

For any actual zero ρ , put

$$e_\rho(s) = F_0(s)/(s - \rho). \quad (\text{CW8.2})$$

It is entire and in $\mathcal{B}$, by the same division estimate near ρ used above and polynomial division away from ρ . Its order at ρ is $m_\rho - 1$, so $[e_\rho] \neq 0$ in $\mathcal{Q}$. Moreover

$$(L - \rho)[e_\rho] = [F_0] = 0, \quad T_a[e_\rho] = a^\rho[e_\rho]. \quad (\text{CW8.3})$$

For the second equality, $(a^s - a^\rho)/(s - \rho)$ is entire and a multiplier with polynomial strip bounds away from a fixed disk; multiplied by F_0 , it lies in $\mathcal{J}$, with all required orders at every original zero.

The actual zero set is nonempty without an RH assumption. One way using the retained source proof is S2's order-at-most-one Hadamard factorization and $F_0(s) = F_0(1 - s)$. If it were zero-free, that factorization would give $F_0 = e^{a+bs}$; the symmetry forces $b = 0$. This contradicts $F_0(0) = 1/8$ and $F_0(2) = \pi/24$, since $\pi \neq 3$. This argument uses the original full function and does not assign a location to a fictional zero.

An arithmetic-equivariant linear section of (CW8.1) would send the nonzero class $[e_\rho]$, with its twisted eigencharacter $a^{\rho+1}$, to a nonzero entire function F . For one $a > 1$, equivariance would require

$$a^{s+1}F(s) = a^{\rho+1}F(s). \quad (\text{CW8.4})$$

The nonzero entire factor $a^{s+1} - a^{\rho+1}$ forces $F = 0$, a contradiction. Thus

The actual normal quotient has no arithmetic-equivariant linear section.

(\text{CW8.5})

Continuity is not needed for this no-section statement. In the continuous category, twisting both source and target by the same character and applying the proof of CW7 with target generator L also gives

$$\text{Hom}_{\text{cont}, T}(\mathcal{Q}(-1), \mathcal{B}(-1)) = 0. \quad (\text{CW8.6})$$

This is a statement about simultaneous compatible representatives, not about the existence of individual lifts. The latter are all present, with the exact affine fibres $a + J(-1)$. In particular (CW8.5) does not contradict the degree-one identity lift in CW4, the surjective degree-two map, or Deligne's surjectivity theorem. DC8 explicitly distinguished its canonical section into the intermediate B_i from an unspecified choice of lifts into the special-fibre group A_i .

CW9. The precise separated receiver obtained from the actual normal quotient

The geometric normal quotient is $\mathcal{Q}(-1)$, with action (CW6.4). It has the exact map into the GSL extension, rather than an identification with the kernel $J(-1)$ of (CW8.1).

Let

$$\mathcal{J}_+ = \{F \in \mathcal{B} : F^{(j)}(\rho + 1) = 0 \quad (0 \leq j < m_\rho)\}, \quad \mathcal{E}_+ = \mathcal{B}/(\mathcal{J} \cap \mathcal{J}_+). \quad (\text{CW9.1})$$

GSL4-GSL6 construct the entire multiplier E_+ , with jet one on the original divisor and jet zero on the divisor translated by $+1$, and prove its strip bounds

$$|E_+(s)| + |1 - E_+(s)| \leq C_M(1 + |\Im s|)^{d_M} \quad (|\Re s| \leq M). \quad (\text{CW9.2})$$

These give continuous multiplication on $\mathcal{B}$: $b_{M,N}(E_+F) \leq C_M b_{M,N+d_M}(F)$, and the analogous bound for $1 - E_+$. Those completed global bounds, not a formal blockwise interpolation, are the input from GSL.

With $VF(s) = F(s - 1)$, the complete maps are

$$j_{\text{high}} : \mathcal{Q}(-1) \rightarrow \mathcal{E}_+, \quad [h] \mapsto [(1 - E_+(s))h(s - 1)],$$

$$\pi_{\text{high}}[F] = [F]_{\mathcal{J}}, \quad s_{\text{high}}[f] = [E_+ f]. \quad (\text{CW9.3})$$

They give

$$0 \rightarrow \mathcal{Q}(-1) \xrightarrow{j_{\text{high}}} \mathcal{E}_+ \xrightarrow{\pi_{\text{high}}} \mathcal{Q} \rightarrow 0, \quad \pi_{\text{high}} s_{\text{high}} = 1. \quad (\text{CW9.4})$$

Independence of representatives follows from the two full jet identities, including all derivatives. The inverse comparison is

$$\mathcal{E}_+ \rightarrow \mathcal{Q} \oplus \mathcal{Q}(-1), \quad [F] \mapsto ([F]_{\mathcal{J}}, [F(s+1)]_{\mathcal{J}}), \quad (\text{CW9.5})$$

whose inverse is

$$([f], [h]) \mapsto [E_+(s)f(s) + (1 - E_+(s))h(s-1)]. \quad (\text{CW9.6})$$

The two inverse identities hold at every original and translated jet. Continuity follows from (CW9.2), translation $b_{M,N}(VF) \leq b_{M+1,N}(F)$, and the quotient seminorms. Equivariance retains the factor

$$a a^{s-1} = a^s. \quad (\text{CW9.7})$$

The complete original product defining the two divisors is

$$F_0(s)F_0(s-1) = \left[\frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) \right] \left[\frac{(s-1)(s-2)}{8} \pi^{-(s-1)/2} \Gamma((s-1)/2) \zeta(s-1) \right]. \quad (\text{CW9.8})$$

The translated factor has values $1/8$ at $s = 1, 2$, value (CW1.11) at $s = 1 - 2r$, and the same value at $s = 2 + 2r$. Its derivatives are exactly $F_0^{(j)}(s-1)$; derivatives of the product are the full sum

$$\sum_{j=0}^n \binom{n}{j} F_0^{(j)}(s) F_0^{(n-j)}(s-1). \quad (\text{CW9.9})$$

Original and translated endpoint and trivial-zero contributions therefore remain explicit.

The actual geometric normal map is

$$\mathbf{n} : A(-1) \rightarrow \mathcal{E}_+, \quad a \mapsto [(1 - E_+(s))(\Theta a)(s-1)]. \quad (\text{CW9.10})$$

Its factorization is exactly

$$A(-1) \xrightarrow{q} \mathcal{Q}(-1) \xrightarrow{j_{\text{high}}} \mathcal{E}_+. \quad (\text{CW9.11})$$

The second arrow is injective, so

$$\ker \mathbf{n} = J(-1), \quad \text{im } \mathbf{n} = \ker \pi_{\text{high}}. \quad (\text{CW9.12})$$

Thus the separated GSL kernel is the complete quotient of the actual normal source by its full original summation image. It is not the summation image itself.

The comparison on the actual connecting-map receiver can be computed without discarding that receiver. If $\iota : J(-1) \hookrightarrow A(-1)$ is the actual map in (CW4.2), then

$$q\iota = 0, \quad \mathbf{n}\iota = j_{\text{high}}q\iota = 0, \quad \mathbf{n}\delta^1 = 0. \quad (\text{CW9.15})$$

The middle map is zero on the entire receiver, before considering the value of δ^1 . Therefore vanishing after this comparison is not an injective test for vanishing in $J(-1)$. The actual vanishing of δ^1 is instead proved by (CW4.3) and independently by (CW7.7). This identifies the exact comparison map and its kernel, rather than treating the two receiver presentations as unrelated.

The actual two-degree groups give the continuous equivariant isomorphism

$$H^1(Y, \mathcal{F}) \oplus H^2(Y, \mathcal{G}) \xrightarrow{\sim} \mathcal{E}_+ \quad (\text{CW9.13})$$

with formula (CW9.6). This map forgets the distinction between the two cohomological degrees only at its displayed direct-sum target. The degree-one connecting morphism of the actual sheaf sequence remains $\delta^1 : Q \rightarrow J(-1)$, not a new morphism $Q \rightarrow Q(-1)$.

For precision, the split algebraic cross on this receiver is

$$0 \rightarrow Q(-1) \rightarrow Q \oplus Q(-1) \xrightarrow{\text{pr}_1} Q \rightarrow 0, \quad \partial_{\text{sum}} = \text{pr}_2. \quad (\text{CW9.14})$$

The actual degree-one group enters by $x \mapsto (x, 0)$, and its boundary is zero. The image of the kernel under ∂_{sum} is all $Q(-1)$, so the DC-style obstruction quotient of this particular cross is zero. Transport through (CW9.13) gives exactly (CW9.4), whose continuous complex-linear equivariant section is unique by (CW7.8).

This cross is constructed from the computed cohomology groups and the exact normal quotient. It is not substituted for (CW4.2) or (CW5.3). In particular, the separated kernel in (CW9.14) is visible only after the stated quotient and assembly of two degrees.

CW10. Exact comparison with Deligne's invariant-cycle mechanism

DC3-DC9 derive, in Deligne's arithmetic reduction, the two linked rows

$$0 \rightarrow K_i = H^{i-1}(X_{\bar{\eta}}, E)_I(-1) \xrightarrow{j_i} B_i = H^i(X_{\bar{\eta}}, E) \xrightarrow{\pi_i} C_i = H^i(X_{\bar{\eta}}, E)^I \rightarrow 0,$$

$$A_i = H^i(X_s, E) \xrightarrow{\alpha_i} B_i \xrightarrow{\partial_i} O_i = H^{2N-i-1}(X_s, E)^\vee(-N). \quad (\text{CW10.1})$$

Here N is the dimension of the smooth total model, not the generic fibre dimension. The invariant-cycle theorem's hypotheses are those of DC1-DC2, including the curve-trait setting and arithmetic reduction; they are not transferred to the sphere sheaf by its notation.

Its exact lifting obstruction is

$$\text{obs}_i(c) = [\partial_i b] \in O_i / \partial_i j_i(K_i), \quad \pi_i(b) = c. \quad (\text{CW10.2})$$

Changing b by $j_i(k)$ changes the boundary by $\partial_i j_i(k)$, proving well-definedness. Conversely, $\text{obs}_i(c) = 0$ permits subtracting such a $j_i(k)$ and then lifting through α_i by localization. Thus

$$\text{coker}(\pi_i \alpha_i) \simeq \text{im } \partial_i / \partial_i j_i(K_i). \quad (\text{CW10.3})$$

The geometric theorems prove

$$C_i, A_i \text{ have weights } \leq i, \quad K_i, O_i \text{ have weights } \geq i + 1. \quad (\text{CW10.4})$$

The bound on O_i is the full calculation

$$-(2N - i - 1) + 2N = i + 1;$$

the bound on K_i comes from dual invariant weights followed by the retained (-1) twist. Exactness of the weight cutoff in the finite-dimensional mixed category gives

$$W_i B_i \xrightarrow{\sim} C_i, \quad \partial_i(W_i B_i) = 0, \quad \text{im } \alpha_i = W_i B_i, \quad \partial_i j_i(K_i) = \text{im } \partial_i. \quad (\text{CW10.5})$$

The whole receiver need not vanish:

$$O_i/\partial_i j_i(K_i) \simeq \ker \text{sp}_{i+1} . \quad (\text{CW10.6})$$

The actual CSP sequence now has a fully determined comparison:

Actual calculation	Classes being lifted	Actual receiver or kernel	Calculated result
Quotient-sheaf degree one, (CW4.2)	$H^1 \mathcal{G} = Q$	$H^2 \mathcal{F}_J = J(-1)$	Boundary zero; unique cohomology lift, the identity
Quotient-sheaf degree two, (CW8.1)	$H^2 \mathcal{G} = Q(-1)$	Kernel $J(-1)$; next boundary target 0	Every class lifts; all lifts form $a + J(-1)$
Support to global degree one, (CW5.3)	$H^1 \mathcal{F} = Q$	Following receiver $A(-1)$	Following map zero; section $q \mapsto (q/2, -q/2)$
Global to annular degree zero	A	Q^2	Boundary diag q , image diag Q
Global to annular degree one	$A(-1)$	$A(-1)^2$	Injective boundary $a \mapsto (-a, a)$
Normal quotient assembled with degree one, (CW9.14)	Q	$Q(-1)$	Separated continuous split receiver; obstruction quotient zero

The first and third zero maps are proved both by their actual complexes and by the global continuous intertwiner calculation of CW7. Their full targets $J(-1)$ and $A(-1)$ have the character quotients of CW6, so Deligne's finite-dimensional weight-cutoff proof (CW10.4) is not the proof of those target bounds here. This is an exact determination of the present objects and maps, not an assertion that no further geometric comparison can exist.

There is already a precise morphism to the separated normal quotient: (CW9.10)–(CW9.12), with the entire kernel $J(-1)$ and the entire image $\ker \pi_{\text{high}}$ calculated. It is this quotient and the two-degree comparison, rather than a renaming of $J(-1)$, that relates the actual CSP geometry to GSLs separated row.

No map in this calculation forces the original characters a^ρ all to have weight one. The surviving original quotient still has its complete original zero jets, their nilpotent actions, and their actual strip positions. The results prove the stated actual lifts and their exact receivers; they do not assume a critical-line conclusion or construct a violation of it.

CW11. Prerequisites, source use, and retained scope

The operations used in this derivation have the following constructed prerequisites:

Operation	Exact prerequisite and proof
Vector sums, differences and $1/2$	Recovered complex coefficient spaces $A, S, V_\pm, Q$; CW1–CW2. None acts on τ .
Sheaf kernel and quotient	Stalkwise exact fibre-product and extension-by-zero maps; CSP2/CSP10 and CW2.1–CW2.3.

Operation	Exact prerequisite and proof
Long exact sequences and boundaries	Simultaneous finite complexes and their compatible contractions; CW2.3–CW2.4, CW3–CW5.
Ordinary equals Hausdorff cohomology	Actual closed-image/inverse theorem SSI1–SSI7 and ESI0–ESI8, imported through CSP1.7 with its full inverse CW1.17.
Normal degree character	Actual positive-degree covering $z \mapsto z^n$, annular period and oriented sphere integration; CSP8 and CW2.6.
Original-zero weights	Exact raw Mellin comparison, original zero ideal and full nilpotent action; CW1.6–CW1.18 and CW6.
Evaluation on the full kernel	Local division by the retained full F_0 , with all exceptional values; CW6.6–CW6.8.
Global vanishing of continuous intertwiners	Continuous full resolvent, strong differentiability and holomorphic uniqueness; CW7. No density of finite blocks is used.
Shifted normal comparison	GSLs constructed multiplier with strip bounds, full translated factor and continuous CRT inverse; CW9.1–CW9.12.
Deligne comparison	The actual invariant/coinvariant, duality and weight maps of DC3–DC9; full total-dimension and Tate factors retained in CW10.

The new statements about absence of equivariant linear sections are restricted to the actual coefficient row (CW8.1). They preserve the whole original arithmetic and do not replace it by a different timing history. They do not deny individual lifting, and they do not identify that coefficient-row section problem with an unconstructed obstruction in the user's complete construction.

The concrete mathematical conclusions are the actual cohomological identity lift, both zero connecting maps of (CW4.2), the full localization maps and their explicit sections, the global vanishing (CW7.6)–(CW7.8), the exact normal lift fibres, and the quotient morphism to the separated receiver. Every comparison keeps the support, separate chart labels, full source kernel, original zeta factors, multiplicities, and cohomological degrees at the points where they are used.

The derived direct image and its winding summand — CDW0-CDW10

Proof source: CC_DERIVED_DIRECT_IMAGE_AND_WINDING.md.

The derived direct image and its winding summand

Complete independent derivation, 24 September 2026. Proof locators CDW0-CDW10.

CDW0. The precise receiver and prerequisites

The supporting datum is still $\tau\langle Z_1; \text{no } Z_2 \rangle$. Every vector space, scalar, complex coordinate and differential form in this note belongs to the receiving geometry constructed after the complete arithmetic reconstruction. No coordinate, distance, midpoint or retracted addition is put on τ .

The complete connected user reconstruction and its correction table were read in the preceding CSP derivation; the current `READ_FIRST_USER_CONSTRUCTION.md`, PC01-PC05, and verbatim WU050-WU055 and WU061-WU063 were recalled for this continuation. In particular, the full source and both branch restrictions are retained. A selected arithmetic label does not initialize the construction, and a quotient of a kernel is not substituted for that kernel.

The existing proof `CC_SPHERE_PULLBACK_AND_NORMAL_DIRECTION.md`, CSP0-CSP13, constructs the space, coefficient sheaf, ordinary cohomology, costalks and all orientation conventions. Its source-reading record retains original author TeX from Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, [arXiv:0903.2024v3](#), §5, and the field-valued charts and real structure in their later [arXiv:2609.00299v1](#). The original-source coverage is exactly that recorded in CSP0, not a claim to have reread both entire papers during this continuation.

CS3 and CS11, CSL1 and CSL10, and SSI1-SSI7 with its independent ESI proof supply the actual continuous summation/Fourier maps and the equality of the original source image with the closed original-zero ideal under Mellin transformation. CSP2-CSP3 supply the fibre-product sheaf and the coefficient-valued Poincaré calculations. The present derivation constructs the entire derived direct image with its actions, rather than recovering it from its cohomology groups alone.

Let

$$\begin{aligned} X &= \{x_+, \eta, x_-\}, & U_+ &= \{x_+, \eta\}, & U_- &= \{x_-, \eta\}, & U &= \{\eta\}, \\ Y &= \mathbb{P}^1(\mathbb{C}), & Y_+ &= Y \setminus \{\infty\}, & Y_- &= Y \setminus \{0\}, & Y^\circ &= \mathbb{C}^\times. \end{aligned} \quad (\text{CDW0.1})$$

The continuous map $\pi : Y \rightarrow X$ sends 0 to x_+ , ∞ to x_- , and all of Y° to η . Thus $\pi^{-1}(\eta)$ is the entire punctured complex plane. The coordinate z used on that receiver is not a coordinate of τ .

CDW1. The original coefficient diagram

The exact spaces are

$$S = \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0\}, \quad V_\pm = S \oplus E_\pm, \quad E_\pm = \mathbb{C}^2,$$

$$A = \left\{ a \in C^\infty(\mathbb{R}_{>0}) : p_{N,j}(a) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \quad (N, j \geq 0) \right\}. \quad (\text{CDW1.1})$$

Both S and A are Fréchet spaces. The two coordinates of each $E_\pm$ retain the source labels value at zero and real integral.

With the stated Fourier convention,

$$\begin{aligned} \Sigma f(u) &= 2 \sum_{k \geq 1} f(ku), & Ra(u) &= u^{-1}a(u^{-1}), & \hat{f}(t) &= \int_{\mathbb{R}} f(v) e^{-2\pi i vt} dv, \\ r_+(f, c_0, c_1) &= \Sigma f, & r_-(h, d_0, d_1) &= R\Sigma h = \Sigma \hat{h}. \end{aligned} \quad (\text{CDW1.2})$$

The last identity follows from the full Poisson formula

$$\Sigma \hat{h}(u) = u^{-1} \Sigma h(u^{-1}) + u^{-1} h(0) - \int_{\mathbb{R}} h(v) dv \quad (\text{CDW1.3})$$

and the two source conditions. No endpoint summand is identified with the other chart's summand.

The sheaf Ω on X is the diagram

$$V_+ \xrightarrow{r_+} A \xleftarrow{r_-} V_- \quad (\text{CDW1.4})$$

Its global sections are the fibre product. All morphisms of diagrams below consist of three continuous linear maps commuting with these two arrows. The pullback $\mathcal{F} = \pi^{-1}\Omega$ has the concrete presentation

$$\mathcal{F} = \underline{A}_Y \times_{i_*(A \oplus A)} i_*(V_+ \oplus V_-), \quad i : \{0, \infty\} \hookrightarrow Y, \quad (\text{CDW1.5})$$

with difference map $(a, v_+, v_-) \mapsto (a|_0 - r_+ v_+, a|_\infty - r_- v_-)$. CSP2 proves this identification by an explicit adjunction map and identities on every stalk.

The actual image

$$J = \Sigma(S) = r_+(V_+) = r_-(V_-) \quad (\text{CDW1.6})$$

is closed, and $\Sigma : S \rightarrow J$ is a topological isomorphism by SSI1-SSI7. In particular $Q = A/J$ is Fréchet. Closedness is available here, but the derived splitting below does not use division by Σ or a choice of a continuous section $Q \rightarrow A$.

CDW2. A single resolution on the entire sphere

Let $\mathcal{E}_A^q$ be the sheaf of smooth A -valued differential q -forms on the underlying real two-dimensional manifold Y , with complex coefficients. Smoothness means that all coordinate derivatives exist in the Fréchet space A ; sections have the compact-set derivative seminorms induced by $p_{N,j}$.

The radial Poincaré operator on a coordinate ball,

$$(K\omega)_x = \int_0^1 t^{q-1} \iota_x \omega_{tx} dt, \quad (\text{CDW2.1})$$

exists because A is complete. For every continuous seminorm p ,

$$p \left(\int_a^b f(t) dt \right) \leq \int_a^b p(f(t)) dt. \quad (\text{CDW2.2})$$

Differentiation under the integral on compact sets gives the same bound for derivatives. The usual differentiation of the radial pullback, integrated from zero to one, proves $dK + Kd = 1$ in positive

degree, and $Kd = 1 - \text{ev}_0$ in degree zero. Thus the coefficient-valued Poincaré lemma holds with continuous local primitives. Partitions of unity multiply these forms, making the sheaves fine. This is also the resolution used in CSP3.

Write $\text{ev} : \mathcal{E}_A^0 \rightarrow i_*(A \oplus A)$ for evaluation at the two poles. Define the single complex of sheaves on Y

$$\begin{aligned}\mathcal{K}^0 &= \mathcal{E}_A^0 \oplus i_*(V_+ \oplus V_-), \\ \mathcal{K}^1 &= \mathcal{E}_A^1 \oplus i_*(A \oplus A), \\ \mathcal{K}^2 &= \mathcal{E}_A^2,\end{aligned}$$

$$d^0(f, v) = (df, \text{ev}(f) - rv), \quad d^1(\omega, a) = d\omega. \quad (\text{CDW2.3})$$

The coordinates at a pole absent from an open are omitted. One has $d^1 d^0 = d^2 f = 0$. The augmentation $\mathcal{F} \rightarrow \mathcal{K}^0$ sends the locally constant function and its pole coordinates in (CDW1.5) to those same coordinates.

We verify exactness on every stalk. In degree zero, $df = 0$ says that f is locally constant; the second equation is exactly the fibre-product condition. Away from the poles the positive-degree assertion is the Poincaré lemma. At a pole p , a degree-one cocycle is a closed one-form ω and an arbitrary $a \in A$. Choose a local primitive g of ω , and replace it by $f = g + a - g(p)$. Then $df = \omega$ and $f(p) = a$, so

$$d^0(f, 0) = (\omega, a). \quad (\text{CDW2.4})$$

Every degree-two germ has a one-form primitive by (CDW2.1). This proves exactness of

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{K}^0 \rightarrow \mathcal{K}^1 \rightarrow \mathcal{K}^2 \rightarrow 0. \quad (\text{CDW2.5})$$

The zero-pole adjustment in (CDW2.4) is an operation on A , not on the supporting point τ .

Each $\mathcal{K}^q$ is acyclic on every open subset of Y . The form part is fine on this paracompact manifold and on its open subsets. The skyscraper summands are flabby: every restriction is a projection onto the pole coordinates still present and is surjective. For a direct image to the finite space $\bar{X}$, a stalk of $R^q \pi_* \mathcal{K}^r$ is $H^q(\pi^{-1}U_x, \mathcal{K}^r)$, because x has its smallest open U_x . Those groups vanish for $q > 0$. Consequently

$$\boxed{\mathcal{C}^\bullet := \pi_* \mathcal{K}^\bullet \text{ represents } R\pi_* \mathcal{F}.} \quad (\text{CDW2.6})$$

This is one compatible resolution on all three preimage opens, not separate complexes whose gluing maps have been inferred from cohomology.

CDW3. The actual diagram of resolution sections

On either pole chart $Y_p \simeq \mathbb{C}$, the complex $\mathcal{C}(U_p)$ is

$$[C^\infty(Y_p, A) \oplus V_p] \longrightarrow [\mathcal{E}_A^1(Y_p) \oplus A] \longrightarrow \mathcal{E}_A^2(Y_p), \quad (\text{CDW3.1})$$

with exactly (CDW2.3). The inclusion

$$\iota_p : V_p[0] \longrightarrow \mathcal{C}(U_p), \quad v \longmapsto (\text{constant}(r_p v), v) \quad (\text{CDW3.2})$$

is a continuous quasi-isomorphism. Indeed global radial primitives on the plane chart repeat (CDW2.4); the projection of a degree-zero cocycle onto v is its continuous inverse on cohomology. Every positive-degree cocycle has the continuous primitive already written down.

On Y° , the skyscrapers vanish, so

$$\mathcal{C}(U) = \mathcal{E}_A^\bullet(\mathbb{C}^\times). \quad (\text{CDW3.3})$$

Keep the explicit real one-form

$$\vartheta = \frac{d \arg z}{2\pi} = \frac{1}{4\pi i} \left(\frac{dz}{z} - \frac{d\bar{z}}{\bar{z}} \right). \quad (\text{CDW3.4})$$

It is globally defined on $\mathbb{C}^\times$, closed, and has integral one on the positive unit circle. It is not defined at either pole.

For completeness, in $z = e^{t+i\theta}$, write a closed form as $\omega = P(t, \theta)dt + Q(t, \theta)d\theta$. The closure equation is $\partial_\theta P = \partial_t Q$. Its positive-circle period is

$$a = \int_0^{2\pi} Q(0, \theta) d\theta.$$

After subtracting $a\vartheta$, the explicit periodic primitive is

$$g(t, \theta) = \int_0^t P(x, 0) dx + \int_0^\theta (Q(t, \phi) - a/(2\pi)) d\phi. \quad (\text{CDW3.5})$$

Its two derivatives are P and $Q - a/(2\pi)$. The period is independent of t , so $g(t, 2\pi) = g(t, 0)$. Every two-form $G(t, \theta)dt \wedge d\theta$ has primitive $(\int_0^t G(x, \theta)dx)d\theta$. These formulas prove the continuous quasi-isomorphism

$$A[0] \oplus A[-1] \longrightarrow \mathcal{C}(U), \quad a \text{ in degree } 0 \mapsto \text{constant}(a), \quad a \text{ in degree } 1 \mapsto a\vartheta. \quad (\text{CDW3.6})$$

In particular, the periods give a continuous inverse on H^1 , and exact forms are its closed kernel inside the closed one-forms. The two cohomology spaces have precisely the original Fréchet topology of A . There is no passage from a completion or an algebraic direct sum of spectral blocks in (CDW3.6).

The restrictions from (CDW3.1) to (CDW3.3) restrict the differential-form component and forget the pole coordinate. On the included constants (CDW3.2) these restrictions are exactly r_p , in degree zero, with no degree-one component.

CDW4. The explicit sheaf-complex splitting

Let $j_X : \{\eta\} \hookrightarrow X$. To retain the winding orientation, write $\mathbb{C}[\vartheta]$ for the one-dimensional vector space with basis the cohomology class of (CDW3.4), and define

$$\mathcal{W} = j_{X!}(A \otimes_{\mathbb{C}} \mathbb{C}[\vartheta]). \quad (\text{CDW4.1})$$

The tensor factor is one-dimensional, so this is just a labelled copy of the same Fréchet space A on $\{\eta\}$; no choice of an infinite tensor completion is involved. Its chart sections on U_+ and U_- are zero. The label records the precise period orientation and its transformation law.

Define the split complex

$$\mathcal{P}^\bullet = \Omega[0] \oplus \mathcal{W}[-1]. \quad (\text{CDW4.2})$$

Thus its degree-zero diagram is (CDW1.4), its degree-one diagram is $0 \rightarrow A[\vartheta] \leftarrow 0$, and its differential is zero. The restrictions out of the degree-zero charts are $r_\pm$.

Define

$$\Phi : \mathcal{P}^\bullet \longrightarrow \mathcal{C}^\bullet \quad (\text{CDW4.3})$$

as follows:

open	Φ^0	Φ^1	
U_+	$v_+ \mapsto (\text{constant}(r_+ v_+), v_+)$	0	(CDW4.4)
U_-	$v_- \mapsto (\text{constant}(r_- v_-), v_-)$	0	
U	$a \mapsto \text{constant}(a)$	$a[\vartheta] \mapsto a\vartheta$.	

The degree-zero images are cocycles by (CDW2.3); the degree-one form is closed. Restrictions commute exactly because the chart constants restrict to the same $r_{\pm}v_{\pm}$. In degree one both chart domains are zero, so there is no condition requiring ϑ to extend through a pole.

This last point is part of the sheaf construction: a map out of $j_{X!}A[\vartheta]$ may have a nonzero value on the open $\{\eta\}$ and zero domain on the pole charts. It does not purport to define a global one-form on the sphere.

A complex of sheaves on X is a diagram of complexes on its three smallest opens. At $x_{\pm}$, (CDW4.4) is the chart quasi-isomorphism (CDW3.2); at η , it is (CDW3.6). Therefore Φ is a stalkwise quasi-isomorphism, and

$$\boxed{\Omega[0] \oplus \mathcal{W}[-1] = \mathcal{P}^{\bullet} \xrightarrow[\simeq]{\Phi} \mathcal{C}^{\bullet} \simeq R\pi_*\mathcal{F}.} \quad (\text{CDW4.5})$$

Here $\mathcal{C}^{\bullet}$ is the specified representative of $R\pi_*\mathcal{F}$, and the actual comparison arrow is $\Phi : \mathcal{P}^{\bullet} \rightarrow \mathcal{C}^{\bullet}$. This invertible comparison does not require a strict chain-level projection from arbitrary forms to both chart stalks.

Every map is continuous. For a compact $K \subset \mathbb{C}^{\times}$, a derivative order r , and a coefficient seminorm p ,

$$\|a\vartheta\|_{K,r,p} \leq C_{K,r}p(a). \quad (\text{CDW4.6})$$

The constants-inclusion bound is immediate, and chart continuity uses the known continuous maps $r_{\pm}$. The inverses on chart and overlap cohomology were explicitly checked in CDW3. Thus this is a quasi-isomorphism with continuous maps and homeomorphisms on the stalk cohomology spaces. It is stronger than an abstract vector-space splitting inferred from hereditary properties of the underlying diagram.

CDW5. The geometric and arithmetic actions on the resolution

The coefficient-only action for $a > 0$ is

$$T_a b(u) = b(u/a),$$

$$\rho_+(a)(f, c_0, c_1) = (f(\cdot/a), c_0, ac_1), \quad \rho_-(a)(h, d_0, d_1) = (ah(a\cdot), ad_0, d_1). \quad (\text{CDW5.1})$$

CS8 gives $r_{\pm}\rho_{\pm}(a) = T_a r_{\pm}$. Apply T_a to every coefficient of a differential form and to the two skyscraper A -coordinates, and apply $\rho_{\pm}(a)$ to the pole $V_{\pm}$ -coordinates. Evaluation and differentiation commute with these continuous linear maps, so this is a strict action on $\mathcal{K}^{\bullet}$ and $\mathcal{C}^{\bullet}$. It acts by T_a on the A -coefficient of $\mathcal{W}$. The map Φ strictly intertwines it.

For a recovered positive integer n , pullback by $b_n(z) = z^n$ is an additional, geometric chain map on $\mathcal{C}^{\bullet}$. It fixes the poles, acts identically on their $V_{\pm}$ and extra A coordinates, and pulls back the differential forms. Evaluation at a fixed pole commutes with pullback, so (CDW2.3) commutes with this operation even at a ramified pole. Directly from (CDW3.4),

$$b_n^* \vartheta = n\vartheta. \quad (\text{CDW5.2})$$

Thus the geometric action on $\mathcal{P}$ is the identity on Ω , and $n \text{ id}$ on $\mathcal{W}$. The two types of action commute because one changes the geometric argument and the other acts on the coefficient.

Their combined operation B_n has

$$\left. \begin{array}{c} V_+ \\ \rho_+(n) \end{array} \right| \frac{V_-}{\rho_-(n)} \frac{A \text{ in } \Omega}{T_n}, \quad B_n(a[\vartheta]) = nT_n a[\vartheta]. \quad (\text{CDW5.3})$$

The equality

$$T_n(a) b_n^* \vartheta = nT_n a \vartheta$$

proves strict equivariance of (CDW4.3), not merely equivariance of its cohomology.

The coefficient estimates are explicit. For $a > 0$,

$$p_{N,j}(T_a b) \leq \max(a^N, a^{-N}) p_{N,j}(b). \quad (\text{CDW5.4})$$

Pullback of smooth forms is continuous for compact derivative seminorms: on a fixed compact set the derivatives of b_n up to a prescribed order are bounded, and the chain rule bounds the result by finitely many seminorms on its compact image. The same statement holds in the $w = 1/z$ chart. No unbounded inverse-image set enters this estimate.

These are actual receiving-space covers for every $n \geq 1$. They are not asserted to extend the full signed CC structure to every even index. Composition $b_m b_n = b_{mn}$, together with (CDW5.1), gives $B_m B_n = B_{mn}$. The cohomological positive-rational and strongly continuous real extensions are those proved in CSP8; a nonintegral power is not being used as a globally defined holomorphic map.

CDW6. Both mirrors on the complete derived object

Retain the two different geometric maps

$$w(z) = -1/z, \quad \alpha(z) = -1/\bar{z}. \quad (\text{CDW6.1})$$

They have the same swap of closed points on X , but w is holomorphic and α is the antiholomorphic real structure from the later CC source.

Use the coefficient morphism already constructed in CS8: exchange the two pole spaces by identity on their correspondingly labelled source coordinates and act by R on all A -coordinates. Since $r_- = Rr_+$, this gives a map over either base swap. On the resolution, pull back the forms by the specified geometric map and apply R to their coefficients; on the extra skyscraper coordinates use R and the pole swap. These operations commute with (CDW2.3).

The exact form identities are

$$w^* \vartheta = -\vartheta, \quad \alpha^* \vartheta = +\vartheta. \quad (\text{CDW6.2})$$

They follow either from (CDW3.4) or from the angle maps $\theta \mapsto \pi - \theta$ and $\theta \mapsto \pi + \theta$. Hence on the split object

	Ω	$\mathcal{W}$	
w	chart swap; overlap R	$-R$	(CDW6.3)
α	chart swap; overlap R	$+R$.	

All these maps square to the identity and are strictly intertwined by Φ . The coefficient R is complex linear; if complex conjugation on coefficients is included, its separate conjugate-linear action must also be applied. The real form ϑ itself is unchanged by conjugating its scalar coefficients, so that additional convention has no hidden angular sign.

The maps w and α commute, with composite $z \mapsto \bar{z}$; with the two R -factors their composite on $\mathcal{W}$ is $-\text{id}$. This is the action of complex conjugation on the positive angular class. On Ω the two swaps and two R -factors compose to the identity.

For precision about actual geometric composition,

$$\begin{aligned} w b_n(z) &= -z^{-n}, & b_n w(z) &= (-1)^n z^{-n}, \\ \alpha b_n(z) &= -\bar{z}^{-n}, & b_n \alpha(z) &= (-1)^n \bar{z}^{-n}. \end{aligned} \quad (\text{CDW6.4})$$

Thus these maps commute with b_n for odd n , and for even n differ by $z \mapsto -z$. That rotation preserves constants and ϑ and therefore acts trivially on the split model. It is not asserted to be the identity on every smooth form in the resolution. The strict map Φ respects the actual transformations and these identities with their rotations retained.

CDW7. The Postnikov class and what has actually vanished

Write $\mathcal{R} = R\pi_*\mathcal{F}$. The cohomology sheaves of (CDW4.3) are identified as

$$\mathcal{H}^0(\mathcal{R}) = \Omega, \quad \mathcal{H}^1(\mathcal{R}) = \mathcal{W}, \quad \mathcal{H}^{q \neq 0,1}(\mathcal{R}) = 0. \quad (\text{CDW7.1})$$

The comparison on $\mathcal{H}^0$ is the original restriction diagram; on $\mathcal{H}^1$ it is the positive-period identification. Its Postnikov triangle is

$$\Omega \longrightarrow \mathcal{R} \longrightarrow \mathcal{W}[-1] \xrightarrow{\kappa} \Omega[1]. \quad (\text{CDW7.2})$$

The class κ is the particular extension datum that cohomology sheaves alone would leave unspecified.

Here it is calculated:

$$\boxed{\kappa = 0.} \quad (\text{CDW7.3})$$

Indeed (CDW4.3) is a continuous strict equivariant quasi-isomorphism from the direct sum, inducing the two specified cohomology identifications. The two inclusions followed by Φ give maps $j_0 : \Omega \rightarrow \mathcal{R}$ and $j_1 : \mathcal{W}[-1] \rightarrow \mathcal{R}$. The projections are represented by the explicit roofs

$$\mathcal{C}^\bullet \xleftarrow[\simeq]{\Phi} \mathcal{P}^\bullet \xrightarrow{\text{pr}_0} \Omega, \quad \mathcal{C}^\bullet \xleftarrow[\simeq]{\Phi} \mathcal{P}^\bullet \xrightarrow{\text{pr}_1} \mathcal{W}[-1]. \quad (\text{CDW7.4})$$

Every arrow in these roofs is a displayed continuous map. Inverting the specified quasi-isomorphism is the definition of these derived morphisms, not an assumed inverse of an analytic quotient. Their compositions satisfy

$$r_i j_i = 1, \quad r_0 j_1 = r_1 j_0 = 0, \quad j_0 r_0 + j_1 r_1 = 1. \quad (\text{CDW7.5})$$

This splits (CDW7.2), proving (CDW7.3) with all the arithmetic and mirror actions retained.

One may express this in the ordinary equivariant derived category of vector-space sheaves, or in the localization of the displayed continuous complexes at stalkwise quasi-isomorphisms inducing homeomorphisms on cohomology. The same actual comparison Φ exists before either localization, so neither assertion requires projectivity in a category of Fréchet spaces.

There is an additional uniqueness statement after forgetting topology and equivariance. For an arbitrary vector-space sheaf G on X ,

$$\text{Hom}_X(\mathcal{W}, G) = \text{Hom}_{\mathbb{C}}(A \otimes \mathbb{C}[\vartheta], G_\eta). \quad (\text{CDW7.6})$$

This follows directly from the diagram: the two chart maps from zero impose no restriction on its central map. Stalk evaluation is exact, and every complex vector space is projective, so the right-hand functor is exact. Therefore $\mathcal{W}$ is projective in that ordinary category and

$$\text{Ext}_X^k(\mathcal{W}, \Omega) = 0 \quad (k > 0). \quad (\text{CDW7.7})$$

The Postnikov class lies in its degree-two group; the difference between splittings inducing the fixed cohomology identifications lies in its degree-one group. Both vanish. This proves underlying derived uniqueness. It does not claim projectivity or uniqueness in a topological or equivariant exact category. Existence there has already been proved by the stronger explicit map, without using (CDW7.7).

The vanishing in (CDW7.3) concerns this actual derived direct image and its winding extension. It is not an identification of every lifting obstruction in Deligne's different geometric setting with this class.

CDW8. Naturality and the complete original source sequence

Let $(V_{\pm}, A, r_{\pm}) \rightarrow (V'_{\pm}, A', r'_{\pm})$ be any continuous coefficient-diagram morphism (f_+, f_-, g) , so $gr_{\pm} = r'_{\pm}f_{\pm}$. Its map on the resolution is g applied to each form coefficient and skyscraper A -coordinate, and $f_{\pm}$ on the pole coordinates. Differentiation and evaluation commute with these maps. Moreover

$$g(\text{constant}(a)) = \text{constant}(ga), \quad g(a\vartheta) = (ga)\vartheta. \quad (\text{CDW8.1})$$

Thus the square of sheaf complexes

$$\begin{array}{ccc} \Omega \oplus j_{X!}A[\vartheta][-1] & \xrightarrow{\Phi} & \pi_*\mathcal{K}_{\Omega} \\ \downarrow & & \downarrow \\ \Omega' \oplus j_{X!}A'[\vartheta][-1] & \xrightarrow{\Phi'} & \pi_*\mathcal{K}_{\Omega'} \end{array} \quad (\text{CDW8.2})$$

commutes strictly. This is naturality on the actual complexes and restrictions.

Apply it to the closed source $J \subset A$. Let Ω_J have the same pole spaces $V_{\pm}$, overlap J , and restrictions $r_{\pm}$ with that codomain. Let $\Omega_Q = j_{X!}Q$, with zero pole spaces and overlap $Q = A/J$. Then

$$0 \rightarrow \Omega_J \rightarrow \Omega \rightarrow \Omega_Q \rightarrow 0 \quad (\text{CDW8.3})$$

is an exact sequence of vector-space sheaves with the specified continuous maps. On the pole stalks its maps are $V_{\pm} \xrightarrow{1} V_{\pm} \rightarrow 0$; on the generic stalk it is the actual strict sequence $0 \rightarrow J \rightarrow A \rightarrow Q \rightarrow 0$. Inverse image is exact, as can be checked on stalks, and gives precisely

$$0 \rightarrow \mathcal{F}_J \rightarrow \mathcal{F} \rightarrow j_{Y!}Q_{Y^{\circ}} \rightarrow 0. \quad (\text{CDW8.4})$$

Its last sheaf identification follows from zero pole stalks and constant Q on the interior with zero pole generization; it is also the quotient constructed in CSP10.

All coefficient spaces J, A, Q are Fréchet, so the preceding resolution applies to each of these diagrams. Naturality represents the derived direct images in (CDW8.4) by the following degreewise exact sequence of split complexes:

internal degree	$R\pi_*\mathcal{F}_J$	$R\pi_*\mathcal{F}$	$R\pi_*j_{Y!}Q$
0	Ω_J	Ω	$j_{X!}Q$
1	$j_{X!}J[\vartheta]$	$j_{X!}A[\vartheta]$	$j_{X!}Q[\vartheta]$

(CDW8.5)

The differentials in these split representatives are zero, and their horizontal maps are exactly the inclusions and quotient maps just specified. The degree-one arithmetic action is nT_n on all three terms; the degree-zero action is the original one. The two mirror actions on the winding row remain $-R, +R$.

This calculation retains the entire kernel J , including its topology. It does not replace it by the quotient Q , and it does not produce a continuous arithmetic-equivariant section of $A \rightarrow Q$ by declaring the vertical derived splitting. The split between two internal cohomological degrees and the horizontal source extension $J \rightarrow A \rightarrow Q$ are different specified maps.

The formula is also natural for an additional pole skyscraper summand. Such a summand has overlap coefficient zero, so contributes only degree zero and no winding term. Consequently the extra pole summands arising from the alternative moment-null generating-domain source in CS2A/SSI10 remain in degree zero under this calculation; they are not erased or turned into an additional zeta quotient.

CDW9. The exact comparison with positive sphere orientation

On the ordered cover (U_+, U_-) , use the Čech differential $r_+ - r_-$. Applying its derived global-section complex to (CDW4.2) gives the total model

$$V_+ \oplus V_- \xrightarrow{r_+ - r_-} A \quad \text{in total degrees } 0, 1, \quad A[\vartheta] \quad \text{in total degree } 2. \quad (\text{CDW9.1})$$

There is no map into or out of that last coefficient in this model. It is the group $H^1(X, \mathcal{W})$, with its positive angular generator.

Its comparison with the positive-integration $H^2(Y, \mathcal{F}) = A$ coordinate of CSP has a minus sign. To compute it, write $r = |z|$ and define smooth forms on the two respective charts

$$\beta_+ = \frac{r^2}{1+r^2}\vartheta, \quad \beta_- = -\frac{1}{1+r^2}\vartheta. \quad (\text{CDW9.2})$$

The first extends smoothly through $z = 0$; the second extends smoothly through $w = 1/z = 0$. Direct differentiation gives

$$d\beta_+ = d\beta_- = \omega_Y = \frac{1}{\pi} \frac{dx \wedge dy}{(1+|z|^2)^2}, \quad \beta_+ - \beta_- = \vartheta, \quad \int_Y \omega_Y = 1. \quad (\text{CDW9.3})$$

The Mayer-Vietoris fibre-cone differential is $d(v, b) = (dv, \text{res}(v) - db)$. Therefore

$$d((\beta_+, \beta_-), 0) = ((\omega_Y, \omega_Y), \vartheta). \quad (\text{CDW9.4})$$

The restriction representative of the positive global class $((\omega_Y, \omega_Y), 0)$ is consequently cohomologous to $(0, -\vartheta)$. Equivalently,

$$\boxed{H^1(X, \mathcal{W}) \ni a[\vartheta] \mapsto -a \in H^2(Y, \mathcal{F}) \text{ measured by positive complex integration.}} \quad (\text{CDW9.5})$$

The projection $\mathcal{F} \rightarrow \underline{A}_Y$ is an isomorphism on degree-two cohomology, so this scalar computation fixes the same sign for the original coefficient sheaf, not only for constant coefficients.

The two mirror actions now recover the CSP signs with no lost factor. On $H^1(X, \Omega)$ the chart swap gives the Čech sign, so both act as $-R$. On $H^1(X, \mathcal{W})$, the extra chart-order minus multiplies the angular action: w acts as $+R$, while α acts as $-R$. The common comparison sign in (CDW9.5) changes neither operator. Covers retain nT_n .

The same local cone convention sends the positive local angular class at a pole to minus its positive complex fundamental class. Since $d \arg(1/z) = -d \arg z$, the two-pole interior-loop boundary in positive integration coordinates is $a \mapsto (-a, a)$, exactly CSP6-CSP7. Thus the derived splitting, the costalk calculation and the global orientation comparison agree.

CDW10. Original-zeta comparison and the scope of the result

No zeta function or its source has changed in constructing this splitting. Keep the explicit raw transform and comparison

$$\begin{aligned} \mathcal{M}_0 a(s) &= \int_0^\infty a(u) u^s \frac{du}{u}, & \Theta a(s) &= \tfrac{1}{2} \mathcal{M}_0 a(s), \\ \mathcal{M}_0 \Sigma f(s) &= 2\zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v} \quad (\Re s > 1). \end{aligned} \quad (\text{CDW10.1})$$

For the retained $f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}$,

$$\Theta \Sigma f_0(s) = F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad \mathcal{M}_0 \Sigma f_0 = 2F_0. \quad (\text{CDW10.2})$$

Its values $F_0(0) = F_0(1) = 1/8$, and

$$F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \quad (r \geq 1) \quad (\text{CDW10.3})$$

retain the original endpoint and trivial-zero contributions. SSI supplies the complete inverse and every Taylor coefficient at those trivial-zero locations.

Let $\mathcal{B}$ be the entire functions rapidly decreasing on every bounded real strip, and let

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every actual nontrivial zero } \rho \text{ and } 0 \leq j < m_\rho\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}. \quad (\text{CDW10.4})$$

The exact source theorem gives $\Theta J = \mathcal{J}$ and the topological isomorphism $\mathcal{Q} = A/J \rightarrow \mathcal{Q}$. No multiplicity is suppressed.

In the positive-integration convention of CSP11, the geometric normal map to the GSL high-weight kernel is

$$a \mapsto [(1 - E_+(s))(\Theta a)(s-1)]_{\mathcal{J} \cap \mathcal{J}_+}. \quad (\text{CDW10.5})$$

Here $\mathcal{J}_+$ is the full zero-jet ideal at every $\rho + 1$; E_+ is the globally controlled multiplier constructed in GSL, with jet one on the original divisor and zero on its $+1$ translate. Its target retains the full product

$$\left[\frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) \right] \left[\frac{(s-1)(s-2)}{8} \pi^{-(s-1)/2} \Gamma((s-1)/2) \zeta(s-1) \right]. \quad (\text{CDW10.6})$$

The second factor retains its values $1/8$ at $s = 1, 2$, and its values at $s = 1 - 2r$ are (CDW10.3). All derivative and nilpotent formulas are those of the full original and translated factors, as proved in CSP11 and GSL6.

In the winding coordinate of the actual derived splitting, (CDW9.5) must be applied first. The same map is therefore

$$a[\vartheta] \mapsto -[(1 - E_+(s))(\Theta a)(s-1)]_{\mathcal{J} \cap \mathcal{J}_+}. \quad (\text{CDW10.7})$$

Its kernel is exactly $\mathcal{J}[\vartheta]$, because the inclusion of the GSL kernel is injective and $\Theta^{-1}\mathcal{J} = \mathcal{J}$. Its quotient is the entire $\mathcal{Q}(-1)$, with the GSL inclusion and the displayed orientation sign. For every recovered n ,

$$\Theta(nT_n a)(s-1) = n n^{s-1} \Theta a(s-1) = n^s \Theta a(s-1), \quad (\text{CDW10.8})$$

so the comparison remains equivariant, including both the geometric degree and Mellin exponent.

The new statement established here is the complete derived object

$$R\pi_* \pi^{-1} \Omega \simeq \Omega \oplus j_{X!}(A \otimes \mathbb{C}[\vartheta])[-1]$$

with a constructed continuous strict equivariant comparison, its naturality for the original source sequence, and its zero Postnikov class. The winding summand retains transforming information over the generic support. No finite-dimensional mixed étale realization is asserted, and no different lifting obstruction is identified with this Postnikov class without a map. The complete coefficient source $\mathcal{J}$, its quotient $\mathcal{Q}$, their different roles in the horizontal source sequence, both geometric mirrors and all original factors remain specified.

Continuous duals of the actual support complexes — CDI0-CDI12

Proof source: CC_CONTINUOUS_DUAL_INDEPENDENT.md.

Continuous duals of the actual sphere coefficient and support complexes

Independent mathematical derivation, 24 September 2026. Proof locators CDI0-CDI12.

The complete source `../tau_weight_cohomology_20260924/CC_SPHERE_PULLBACK_AND_NORMAL_DIRECTION.md`, CSP0-CSP13, was read, including the pole orientation calculation and its localization cone. The exact source-image theorem SSI1-SSI7 and independent ESI0-ESI12 supply the closedness and continuous source inverse used below. This calculation takes the stated coefficient complexes as actual inputs; it constructs their continuous duals and all transposed maps. It does not identify the resulting complexes with an unconstructed Verdier dual sheaf or assume arithmetic purity.

Every vector-space operation below is on the reconstructed arithmetic coefficient spaces. The support notation $Z_0, Z_1, \dots, \tau$ is unchanged. No addition, numerical coordinate, distance, or numerical weight is assigned to τ .

CDI0. Exact input and dual conventions

Retain the CSP coefficient spaces

$$\begin{aligned} S &= \{f \in \mathcal{S}(\mathbb{R}) : f(-x) = f(x), f(0) = 0, \int_{\mathbb{R}} f = 0\}, \quad V_{\pm} = S \oplus E_{\pm}, \quad E_{\pm} = \mathbb{C}^2, \\ A &= \left\{a \in C^{\infty}(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \text{ for all } N, j \geq 0\right\}, \\ \Sigma f(u) &= 2 \sum_{n \geq 1} f(nu), \quad Ra(u) = u^{-1}a(u^{-1}), \quad r_+(f, e) = \Sigma f, \quad r_-(h, e) = R\Sigma h = \Sigma \hat{h}. \end{aligned} \tag{CDI0.1}$$

The original Fourier convention is $\hat{h}(t) = \int h(x) e^{-2\pi i x t} dx$. Let

$$J = \Sigma S \subset A, \quad Q = A/J, \quad q : A \rightarrow Q. \tag{CDI0.2}$$

The exact-image theorem proves that J is closed and that $\Sigma : S \rightarrow J$ is a topological isomorphism. All spaces in (CDI0.1) and (CDI0.2) are therefore Hausdorff Fréchet spaces with their stated subspace, finite-product, and quotient topologies. In particular no closure of zero remains in Q .

For a Hausdorff locally convex space E , write E' for its continuous complex-linear dual. Two topologies will remain explicit:

$$E'_{\sigma} = (E', \sigma(E', E)), \quad E'_{\beta} = (E', \beta(E', E)). \tag{CDI0.3}$$

The first is convergence on every finite set of vectors; the second is uniform convergence on every bounded subset of E . No algebraic dual of discontinuous functionals is substituted for either. For a continuous map $f : E \rightarrow F$, its ordinary transpose is $f' : F' \rightarrow E'$, $f'\lambda = \lambda \circ f$. It is continuous for

both topologies: finite evaluation sets pull back to finite sets, and continuous linear maps take bounded sets to bounded sets.

For a cochain complex C , define

$$(C^\vee)^n = (C^{-n})', \quad d_{C^\vee}^n \lambda = (-1)^{n+1} \lambda \circ d_C^{-n-1}. \quad (\text{CDI0.4})$$

This is the usual Hom differential. The evaluation map in the order $C^\vee \otimes C \rightarrow \mathbb{C}$ satisfies the cochain identity: the two terms $(-1)^{n+1} \lambda(dc)$ and $(-1)^n \lambda(dc)$ cancel. A degree-zero chain map transposes without an additional sign. Weak or strong topologies on each dual coefficient produce the same underlying vector complex; their cohomology topologies will be distinguished below.

CDI1. Explicit source splittings and the exact continuous transpose

Define continuous sections onto the actual closed image:

$$s_+(j) = (\Sigma^{-1}j, 0) \in V_+, \quad s_-(j) = (\widehat{\Sigma^{-1}j}, 0) \in V_-. \quad (\text{CDI1.1})$$

Poisson and $\hat{f} = f$ on S give $r_\pm s_\pm = 1_J$. Hence

$$V_\pm \xrightarrow{\sim} E_\pm \oplus J, \quad v \mapsto (v - s_\pm r_\pm v, r_\pm v) \quad (\text{CDI1.2})$$

are continuous isomorphisms. Their inverses are $(e, j) \mapsto e + s_\pm j$; the first coordinate on the right of (CDI1.2) lies in the actual endpoint kernel.

Put $W = V_+ \oplus V_-$, $d(v_+, v_-) = r_+ v_+ - r_- v_-$, and $Z = \ker d$. The common-restriction description is

$$Z \xrightarrow{\sim} J \oplus E_+ \oplus E_-, \quad (v_+, v_-) \mapsto (r_+ v_+, e_+, e_-). \quad (\text{CDI1.3})$$

Its inverse sends (j, e_+, e_-) to $(s_+ j + e_+, s_- j + e_-)$. These are the exact Fourier-graph coordinates of CSP4.5, with all four endpoint lines retained.

The map $t : J \rightarrow W$, $t(j) = (s_+ j, 0)$, satisfies $dt = 1$. Therefore

$$W \xrightarrow{\sim} Z \oplus J, \quad w \mapsto (w - t(dw), dw), \quad (z, j) \mapsto z + t(j). \quad (\text{CDI1.4})$$

This explicit splitting will be used on the domain side. It does not assert that the closed inclusion $J \hookrightarrow A$ has a continuous projection or that q has a continuous section.

The continuous transpose of $0 \rightarrow J \rightarrow A \xrightarrow{q} Q \rightarrow 0$ is algebraically exact:

$$0 \longrightarrow Q' \xrightarrow{q'} A' \xrightarrow{\text{res}_J} J' \longrightarrow 0. \quad (\text{CDI1.5})$$

Indeed q' is injective because q is onto. Its image is

$$J^\perp = \{\lambda \in A' : \lambda(j) = 0 \text{ for every } j \in J\}, \quad (\text{CDI1.6})$$

since a continuous functional constant on quotient fibers factors continuously through the quotient topology. Finally every continuous complex-linear functional on the subspace J extends continuously to A by the complex Hahn-Banach theorem: continuity bounds it by a continuous seminorm induced from A , and Hahn-Banach gives an extension with that bound. This proves surjectivity in (CDI1.5), not merely density of its image.

The corresponding exact identities are

$$\ker d' = J^\perp = q'Q', \quad \text{im } d' = Z^\perp \subset W', \quad (\text{CDI1.7})$$

where $d'\lambda = (r'_+\lambda, -r'_-\lambda)$. To prove the second equality completely, a continuous $\phi \in W'$ vanishing on Z gives the continuous functional $j \mapsto \phi(tj)$ on J . Extend it to $\lambda \in A'$ by (CDI1.5); (CDI1.4) gives $\phi = d'\lambda$. Conversely every $d'\lambda$ vanishes on Z . Likewise

$$\ker r'_\pm = q'Q', \quad \text{im } r'_\pm = E_\pm^\perp \subset V'_\pm. \quad (\text{CDI1.8})$$

Thus every transpose image needed for the coefficient cohomology is calculated exactly.

CDI2. The dual global and costalk complexes

The global CSP complex is

$$D^0 = W \xrightarrow{d} D^1 = A \xrightarrow{0} D^2 = A. \quad (\text{CDI2.1})$$

Its continuous dual, with the convention (CDI0.4), is

$$(D^\vee)^{-2} = A' \xrightarrow{0} (D^\vee)^{-1} = A' \xrightarrow{\lambda \mapsto (r'_+\lambda, -r'_-\lambda)} (D^\vee)^0 = V'_+ \oplus V'_-. \quad (\text{CDI2.2})$$

The minus sign in the second chart comes from the original chart difference, not from an added dualization convention: the Hom differential at degree -1 has sign $+1$.

Equations (CDI1.7) give all groups and their evaluation pairings:

$$H^{-2}(D^\vee) = A', \quad H^{-1}(D^\vee) = q'Q' \cong Q', \quad H^0(D^\vee) = W'/Z^\perp \cong Z'. \quad (\text{CDI2.3})$$

The last map is restriction to Z ; it is onto because the projection in (CDI1.4) supplies a continuous extension of every functional on Z . For representatives the exact pairings are

$$\begin{aligned} \langle \lambda, a \rangle &= \lambda(a) \quad (H^{-2}, H^2), \\ \langle q'\mu, [a] \rangle &= \mu([a]) \quad (H^{-1}, H^1), \\ \langle [\phi], z \rangle &= \phi(z) \quad (H^0, H^0). \end{aligned} \quad (\text{CDI2.4})$$

The middle expression is independent of a representative because $q'\mu$ vanishes on J ; the last is independent because boundaries are exactly $Z^\perp$.

At each pole, the costalk complex is

$$L_p^0 = V_p \xrightarrow{r_p} L_p^1 = A \xrightarrow{0} L_p^2 = A, \quad p \in \{+, -\}. \quad (\text{CDI2.5})$$

Its dual is

$$(L_p^\vee)^{-2} = A' \xrightarrow{0} (L_p^\vee)^{-1} = A' \xrightarrow{r'_p} (L_p^\vee)^0 = V'_p, \quad (\text{CDI2.6})$$

Its coefficient degrees are $-2, -1, 0$, and its cohomology is

$$H^{-2}(L_p^\vee) = A', \quad H^{-1}(L_p^\vee) = q'Q' \cong Q', \quad H^0(L_p^\vee) = V'_p/E_p^\perp \cong E'_p. \quad (\text{CDI2.7})$$

The pairings are evaluation on the original costalk groups A, Q, E_p , with both positive local complex orientations retained as in CSP6.

If one wants a complex indexed in degrees $0, 1, 2$, the precisely specified shift is $D^\vee[-2]$: its term of degree i is $(D^{2-i})'$, and its differential has no additional sign because the shift is even. It pairs with D in complementary degrees summing to two. This is a shift of the constructed coefficient complex; calling it a shifted complex does not establish a sheaf-duality comparison.

CDI3. Separated perfect continuous-dual pairings and their topologies

For each Hausdorff locally convex space E in the cohomology list $Z, Q, A, E_{\pm}$, the evaluation pairing $E' \times E \rightarrow \mathbb{C}$ is separated in both arguments. A nonzero vector is separated by a continuous functional using Hahn-Banach and a continuous seminorm nonzero on that vector; a nonzero functional is nonzero on some vector by its definition. The dual groups in (CDI2.3) and (CDI2.7) are the **entire continuous duals** of the respective primal groups. This is the precise sense of a perfect continuous-dual pairing used here. It does not mean that every discontinuous algebraic functional is continuous, and does not assert a Hilbert inner product or positivity.

Equipped with the weak dual topology, the second dual in the opposite direction is also exactly the original vector space. Indeed a continuous linear $\Lambda : E'_{\sigma} \rightarrow \mathbb{C}$ is bounded by finitely many evaluation seminorms $|\lambda(e_i)|$. It vanishes on the common kernel of those evaluations, factors through their finite-dimensional image, and is a linear combination of them. Hence $\Lambda(\lambda) = \lambda(e)$ for one $e \in E$, unique by separation. Thus

$$(E'_{\sigma})' = E \quad (\text{CDI3.1})$$

as vector spaces with the specified evaluation map. The pairing is separately continuous. No claim of joint continuity or an isomorphism between the original topology on E and a bidual topology is required by this argument.

For weak dual coefficient topologies, all cohomology identifications in CDI2 are topological identifications with the respective weak duals. For $q'Q' \subset A'$, evaluation at a is exactly evaluation at $q(a)$, and every element of Q has a representative, so the induced weak topology is exactly $\sigma(Q', Q)$. For the degree-zero quotients, the explicit topological splittings (CDI1.2) and (CDI1.4) transpose to finite direct-product splittings of weak duals; quotienting by the entire complementary dual factor gives precisely $(E_p)'_{\sigma}$ or Z'_{σ} .

For clarity, the surjection $A'_{\sigma} \rightarrow J'_{\sigma}$ in (CDI1.5) is also a quotient map. A basic weak neighborhood constrains a functional on a finite-dimensional subspace $L \subset A$. The possible restrictions to J are determined solely by its values on $L \cap J$. Restriction $L' \rightarrow (L \cap J)'$ is an open surjective finite-dimensional map. Any two compatible continuous functionals on J and L combine to a continuous functional on $J + L$: a finite-dimensional complement of $J \cap L$ in L maps homeomorphically onto its finite-dimensional image in the Hausdorff quotient A/J , yielding a topological direct sum with J . Hahn-Banach then extends the combined functional to A . This proves that images of basic weak neighborhoods are weak neighborhoods. Thus (CDI1.5) is exact with its stated weak subspace and quotient topologies.

For strong duals the following statements are proved without any bounded-lifting assumption. The decompositions (CDI1.2) and (CDI1.4) are topological finite direct sums, so their strong duals also split. In particular the degree-zero dual cohomology is exactly Z'_{β} , respectively $(E_p)'_{\beta}$, and the degree-minus-two group is A'_{β} . All dual differential images are closed in both weak and strong coefficient topologies, since they are the annihilators of the displayed closed kernels and those annihilators are intersections of evaluation kernels.

The degree-minus-one group with its topology inherited from A'_{β} is the vector space Q' with the exact seminorm family

$$p_B^{\text{induced}}(\mu) = \sup_{a \in B} |\mu(q(a))|, \quad B \subset A \text{ bounded.} \quad (\text{CDI3.2})$$

The canonical map

$$Q'_{\beta} \longrightarrow (J^{\perp}, \text{subspace topology from } A'_{\beta}), \quad \mu \longmapsto q' \mu \quad (\text{CDI3.3})$$

is a continuous bijection. Each image $q(B)$ is bounded in Q , proving continuity. This proof does not identify (CDI3.2) with uniform convergence on **all** bounded subsets of Q . A proof that bounded subsets of Q have bounded lifts would give that additional identification, but no such unproved assertion is used.

Likewise the strong-topology surjection $A'_\beta \rightarrow J'_\beta$ is algebraically onto and continuous by the proved extension theorem; no strong open-mapping claim is inserted. These precise strong seminorms coexist with the full weak-topology duality already proved.

CDI4. Transpose of the two-support localization map

Write $L = L_+ \oplus L_-$, with

$$L^0 = W, \quad L^1 = A^2, \quad L^2 = A^2, \quad d_L^0(v_+, v_-) = (r_+ v_+, r_- v_-), \quad d_L^1 = 0.$$

The actual CSP support-to-global map has

$$k^0 = 1_W, \quad k^1(b_+, b_-) = b_+ - b_-, \quad k^2(c_+, c_-) = c_+ + c_-. \quad (\text{CDI4.1})$$

Its continuous transpose is the chain map $k^\vee : D^\vee \rightarrow L^\vee$ with

$$(k^\vee)^0 = 1_{W'}, \quad (k^\vee)^{-1} \lambda = (\lambda, -\lambda), \quad (k^\vee)^{-2} \lambda = (\lambda, \lambda). \quad (\text{CDI4.2})$$

The dual supported differential is

$$d_{L^\vee}^{-1}(\lambda_+, \lambda_-) = (r'_+ \lambda_+, r'_- \lambda_-). \quad (\text{CDI4.3})$$

Substitution gives $d_{L^\vee}^{-1}(\lambda, -\lambda) = d_{D^\vee}^{-1} \lambda$, proving the only nonzero chain compatibility directly.

The induced maps on cohomology are therefore

$$\begin{aligned} A' &\longrightarrow A'^2 : \lambda \mapsto (\lambda, \lambda) \quad \text{in degree } -2, \\ Q' &\longrightarrow Q'^2 : \mu \mapsto (\mu, -\mu) \quad \text{in degree } -1, \\ Z' &\longrightarrow E'_+ \oplus E'_- : \phi \mapsto (\phi|_{E_+}, \phi|_{E_-}) \quad \text{in degree } 0. \end{aligned} \quad (\text{CDI4.4})$$

Each is precisely adjoint to its primal support-to-global map under CDI2's evaluation. For one pole alone, the transpose map from the global dual has degrees $(0, -1, -2)$:

$$k_+^\vee = (\text{pr}_{V'_+}, +1, +1), \quad k_-^\vee = (\text{pr}_{V'_-}, -1, +1). \quad (\text{CDI4.5})$$

Thus the two degree-one signs differ, while both positive local orientation classes have the same degree-two sign.

CDI5. The full dual localization row and cone signs

Let $O = A[\text{degree } 0] \oplus A[\text{degree } 1]$ be the annular model, with zero differential. Reversing the CSP7.6 localization row by the continuous transpose gives the following algebraically exact row:

$$\begin{aligned} 0 \rightarrow A' &\xrightarrow{\lambda \mapsto (\lambda, \lambda)} A'^2 \xrightarrow{(\lambda_+, \lambda_-) \mapsto -\lambda_+ + \lambda_-} A' \xrightarrow{0} Q' \\ &\xrightarrow{\mu \mapsto (\mu, -\mu)} Q'^2 \xrightarrow{(\mu_+, \mu_-) \mapsto q'(\mu_+ + \mu_-)} A' \xrightarrow{r'} Z' \xrightarrow{\text{res}} E'_+ \oplus E'_- \rightarrow 0. \end{aligned} \quad (\text{CDI5.1})$$

Here $r : Z \rightarrow A$ is the common restriction, and in (CDI1.3) its transpose is

$$r' \lambda = (\lambda|_J, 0, 0). \quad (\text{CDI5.2})$$

Exactness can be checked without a general topological exactness axiom. The first two nonzero maps are diagonal and signed difference, hence have the displayed kernels and images. The next zero map has

kernel all of A' , exactly the preceding image. The kernel of the sum on Q'^2 is its anti-diagonal. The image of that sum composed with q' is $J^\perp$, the kernel of r' . Hahn-Banach makes r' onto the J' -summand of Z' , which is exactly the kernel of endpoint restriction. That restriction is onto by the direct-sum decomposition of Z . All arrows are continuous for both dual topologies; the weak topologies are the exact subspace and quotient topologies proved in CDI3. Strong algebraic exactness does not assert the additional strong quotient identifications excluded in CDI3.

One can check every cone sign directly too. Put $E = \text{Cone}(k)$ with the standard differential $d_E(x, y) = (d_D x + k y, -d_L y)$. Its coefficient degrees from CSP7.3 are

$$E^{-1} = W, \quad E^0 = W \oplus A^2, \quad E^1 = A \oplus A^2, \quad E^2 = A.$$

By (CDI0.4), its dual differentials are exactly

$$\begin{aligned} (E^\vee)^{-2} = A' &\longrightarrow (E^\vee)^{-1} = A' \oplus A'^2, \quad \lambda \longmapsto (0, -\lambda, -\lambda), \\ (\lambda, \mu_+, \mu_-) &\longmapsto (d'\lambda, \lambda, -\lambda) \quad \text{from degree } -1 \text{ to } 0, \\ (\phi, \nu_+, \nu_-) &\longmapsto -\phi + (r'_+ \nu_+, r'_- \nu_-) \quad \text{from degree } 0 \text{ to } 1. \end{aligned} \quad (\text{CDI5.3})$$

The last target is W' . In particular the diagonal in the first line has a negative sign from the Hom differential in degree -2 , while the diagonal cohomology map in (CDI4.2) has a positive sign; these are different maps.

The transpose of the CSP7.4 annular inclusion is a quasi-isomorphism $E^\vee \rightarrow O^\vee$. In degree zero it sends $(\phi, \nu_+, \nu_-) \mapsto \nu_+ + \nu_-$; in degree minus one it sends $(\lambda, \mu_+, \mu_-) \mapsto -\mu_+ + \mu_-$. To prove this, a degree-zero cocycle has $\phi = (r'_+ \nu_+, r'_- \nu_-)$. Its displayed image is zero precisely when $\nu_- = -\nu_+$, in which case it is the boundary of $(\nu_+, 0, 0)$ in degree minus one. Every target value has a cocycle representative by taking $\nu_+ = \nu$, $\nu_- = 0$, and the required ϕ . A degree-minus-one cocycle has $\lambda = 0$, while the other coordinates are arbitrary. Its displayed image vanishes precisely on the diagonal, which is the image of degree minus two in (CDI5.3). It is onto by taking $\mu_+ = 0$. The other two degrees have zero cohomology by injectivity of the first differential and surjectivity of the last. These choices are continuous finite operations, so the annular dual comparison is valid with either topology.

For a general chain map with the present Hom convention, the exact relation between the two cone presentations is

$$\text{Cone}(k)^\vee \xrightarrow{\sim} \text{Cone}(k^\vee)[-1], \quad (\phi, \psi) \longmapsto ((-1)^{n+1} \psi, \phi) \text{ in degree } n. \quad (\text{CDI5.4})$$

Indeed the differential of the left side is $(d_{D^\vee} \phi, d_{L^\vee} \psi + (-1)^{n+1} k^\vee \phi)$; the target shifted-cone differential is $(\alpha, \beta) \mapsto (-d_{L^\vee} \alpha - k^\vee \beta, d_{D^\vee} \beta)$. Substitution of the displayed sign proves equality. This accounts for the localization-cone shift without suppressing a boundary sign.

CDI6. Ordinary transposes, contragredients, and original spectral jets

For $a > 0$, use the CSP actions

$$\begin{aligned} T_a b(u) &= b(u/a), \\ \rho_+(a)(f, c_0, c_1) &= (f(\cdot/a), c_0, a c_1), \quad \rho_-(a)(h, d_0, d_1) = (a h(a \cdot), a d_0, d_1). \end{aligned} \quad (\text{CDI6.1})$$

The combined sphere-cover action on D , extended from the recovered integer covers to the indicated real coefficient action as in CSP8, is

$$B_a^0 = \rho_+(a) \oplus \rho_-(a), \quad B_a^1 = T_a, \quad B_a^2 = a T_a. \quad (\text{CDI6.2})$$

The same degree factors occur on each costalk, and on O the degree-zero and degree-one actions are T_a and aT_a . A nonintegral a here does not designate a holomorphic map $z \mapsto z^a$.

The ordinary transpose $(B_a^i)'$ satisfies the adjoint identity $\langle (B_a^i)' \lambda, x \rangle = \langle \lambda, B_a^i x \rangle$. For an invariant evaluation pairing one must instead use the contragredient

$$B_a^\vee|_{(D^\vee)^{-i}} = (B_{a^{-1}}^i)'. \quad (\text{CDI6.3})$$

Then $\langle B_a^\vee \lambda, B_a x \rangle = \langle \lambda, x \rangle$. Its exact coefficient actions, in degrees 0, -1 , -2 , are

$$\rho_+(a^{-1})' \oplus \rho_-(a^{-1})', \quad (T_{a^{-1}})', \quad a^{-1}(T_{a^{-1}})'. \quad (\text{CDI6.4})$$

In particular the four dual endpoint characters are 1, a^{-1} , a^{-1} , 1. In the common-restriction coordinates (CDI1.3), the other degree-zero dual summand J' carries $(T_{a^{-1}}|_J)'$. Every transposed support and localization map intertwines these actions because its primal map intertwines (CDI6.2); direct substitution in CDI4–CDI5 verifies the factors.

Retain the CSP Mellin comparison including its scalar:

$$\Theta b(s) = \frac{1}{2} \int_0^\infty b(u) u^s \frac{du}{u}, \quad \Theta T_a b(s) = a^s \Theta b(s), \quad \Theta J = \mathcal{J}. \quad (\text{CDI6.5})$$

Thus $[b] \mapsto [\Theta b]$ identifies Q with $\mathcal{B}/\mathcal{J}$. For every actual nontrivial zero ρ and $0 \leq j < m_\rho$, its continuous jet functional is

$$\delta_{\rho,j}([b]) = (\Theta b)^{(j)}(\rho). \quad (\text{CDI6.6})$$

Using the raw transform instead would multiply this functional by 2; that factor is not suppressed in (CDI6.5).

Leibniz's rule gives the full triangular matrices

$$\begin{aligned} T_a' \delta_{\rho,j} &= a^\rho \sum_{\ell=0}^j \binom{j}{\ell} (\log a)^{j-\ell} \delta_{\rho,\ell}, \\ (T_{a^{-1}})' \delta_{\rho,j} &= a^{-\rho} \sum_{\ell=0}^j \binom{j}{\ell} (-\log a)^{j-\ell} \delta_{\rho,\ell}. \end{aligned} \quad (\text{CDI6.7})$$

The contragredient of the normal action aT_a has instead

$$a^{-\rho-1} \sum_{\ell=0}^j \binom{j}{\ell} (-\log a)^{j-\ell} \delta_{\rho,\ell}. \quad (\text{CDI6.8})$$

These formulas are valid for every actual zero with its full multiplicity. They retain the extra geometric degree factor and do not assume $\Re \rho = 1/2$.

CDI7. Mirrors, orientation, and the reflected degree characters

Let $\varepsilon = +1$ for the holomorphic sphere swap $w(z) = -1/z$, and $\varepsilon = -1$ for the antiholomorphic swap $\alpha(z) = -1/\bar{z}$. Both use the actual source coefficient comparison of CSP9. In degrees 0, 1, 2, their global maps are

$$M_\varepsilon = (\text{chart swap}, -R, \varepsilon R). \quad (\text{CDI7.1})$$

Since these are involutions, their ordinary transpose and contragredient coincide. On $D^\vee$, in degrees 0, -1 , -2 , their exact maps are

$$M_\varepsilon^\vee = (\text{dual chart swap}, -R', \varepsilon R'). \quad (\text{CDI7.2})$$

The two supported copies have degree-minus-one action $(\lambda_+, \lambda_-) \mapsto (R'\lambda_-, R'\lambda_+)$ and degree-minus-two action $(\lambda_+, \lambda_-) \mapsto (\varepsilon R'\lambda_-, \varepsilon R'\lambda_+)$. The annular dual has degree-zero action R' and degree-minus-one action $-\varepsilon R'$. Thus (CDI4.2)'s anti-diagonal and diagonal intertwine these maps: in degree minus one, the swap sends $(\lambda, -\lambda)$ to $(-R'\lambda, R'\lambda)$, precisely the image of $-R'\lambda$; in degree minus two, the image of $\varepsilon R'\lambda$ is the required diagonal. The signed annular maps in CDI5 agree too.

Raw Mellin reflection, or its retained scalar version, gives $\Theta Rb(s) = \Theta b(1 - s)$. Hence

$$R'\delta_{\rho,j} = (-1)^j \delta_{1-\rho,j}. \quad (\text{CDI7.3})$$

The actual global degree-one mirror is its negative, so its dual acts by $(-1)^{j+1} \delta_{1-\rho,j}$. The degree-two mirror has the factor $\varepsilon(-1)^j$. The functional equation preserves the exact multiplicity between ρ and $1 - \rho$, since its multiplier is holomorphic and nonzero there.

The degree character changes the reflection center of the represented exponents. For a coefficient carrying $a^k T_a$, the exact identity is

$$(a^k T_a)R = a^{2k+1} R(a^{-k} T_{a^{-1}}). \quad (\text{CDI7.4})$$

It follows immediately from $T_a R = a R T_{a^{-1}}$, retaining both powers a^k . The exponents paired by this reflection are $\rho + k$ and $1 - \rho + k$, whose sum is $2k + 1$. In the unshifted coefficient this is 1; in the actual normal direction $k = 1$, it is 3. For the invariant-pairing contragredient, transposition at the inverse parameter gives

$$\mathbf{B}_a^\vee M_\varepsilon^\vee = a^{-(2k+1)} M_\varepsilon^\vee \mathbf{B}_{a^{-1}}^\vee \quad (\text{CDI7.5})$$

on a degree with that factor. Its exponent pair sums to $-(2k + 1)$. On the degree-zero graph and endpoint coordinates $k = 0$; the source formulas (CDI6.1) verify the same factor a directly before transposing. Neither mirror allows the normal factor to be omitted, and the antiholomorphic sign $\varepsilon = -1$ remains independent of these positive dilation characters.

CDI8. Finite full-zero blocks inside the actual quotient

An exact finite-block comparison with the dual can be constructed without asserting global self-duality. Write $\mathcal{Q} = \mathcal{B}/\mathcal{J}$ using (CDI6.5). The full zero-jet map is injective by the definition of $\mathcal{J}$, with the closed-image theorem already identifying it with the actual source quotient.

For completeness, each individual full jet block has an actual representative. Use the retained auxiliary entire function

$$F_*(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) \in \mathcal{B}. \quad (\text{CDI8.1})$$

It has precisely the nontrivial-zero divisor and all its multiplicities; its endpoint values are $F_*(0) = F_*(1) = 1/8$, and its values at the cancelled trivial-zero locations are those retained in SSI3 and CSP11. For $m = m_\rho$, let

$$A_\rho(s) = F_*(s)/(s - \rho)^m. \quad (\text{CDI8.2})$$

Taylor division makes it entire with $A_\rho(\rho) \neq 0$. It belongs to $\mathcal{B}$: outside a compact neighborhood of ρ , division costs at most a fixed polynomial factor on any strip; inside, holomorphicity gives a finite bound. For $0 \leq j < m$, let $P_{\rho,j}(t)$ be the Taylor polynomial through degree $m - 1$ of $t^j/A_\rho(\rho + t)$, and set

$$\psi_{\rho,j}(s) = A_\rho(s) P_{\rho,j}(s - \rho). \quad (\text{CDI8.3})$$

This is in $\mathcal{B}$, has jet t^j modulo t^m at ρ , and vanishes with the full required order at every other nontrivial zero. This is a comparison construction using the full factor in (CDI8.1), not a replacement for original zeta in the forthcoming residue formula.

Define $\mathcal{Q}_{\text{fin}}$ as those classes whose full zero-jet family has finite support. Equations (CDI8.2)–(CDI8.3), together with injectivity of the joint jet map, prove the exact vector-space identification

$$\mathcal{Q}_{\text{fin}} \cong \bigoplus_{\rho} \mathbb{C}[t]/(t^{m_{\rho}}). \quad (\text{CDI8.4})$$

Let

$$\mathcal{D}_{\text{fin}} = \text{span}_{\mathbb{C}} \{\delta_{\rho,j} : \rho \text{ actual nontrivial zero}, 0 \leq j < m_{\rho}\} \subset \mathcal{Q}'. \quad (\text{CDI8.5})$$

The individual jets are continuous, so the displayed inclusion really is in the continuous dual. No statement that $\mathcal{Q}_{\text{fin}}$ is dense in $\mathcal{Q}$ is used.

CDI9. The original-zeta residue map and its full determinant

For $[G] \in \mathcal{Q}_{\text{fin}}$, define a functional on $\mathcal{Q}$ by

$$\iota([G])([F]) = \sum_{\rho} \text{Res}_{s=\rho} \frac{F(s)G(1-s)}{\zeta(s)} ds. \quad (\text{CDI9.1})$$

Only finitely many residues are nonzero: the potentially nonzero poles are at the reflections of the finite support of the zero jets of G . Changing either representative by an element of $\mathcal{J}$ makes every residue zero. Each residue is a finite linear combination of the continuous jets of F ; consequently (CDI9.1) is a well-defined continuous functional. Its denominator is the original zeta, not (CDI8.1).

Write $\sigma = 1 - \rho$, $m = m_{\rho} = m_{\sigma}$, and

$$\zeta(\rho + t) = t^m u(t), \quad u(0) \neq 0. \quad (\text{CDI9.2})$$

Pair the monomial jet t^i at ρ with the monomial jet $(s - \sigma)^j$ at σ . Reflection substitutes $-t$ in the latter. The exact local matrix is

$$P_{ij} = (-1)^j [t^{m-1-i-j}] \frac{1}{u(t)}, \quad 0 \leq i, j < m, \quad (\text{CDI9.3})$$

where a coefficient with negative index is zero. The matrix is anti-triangular. Its anti-diagonal entries are $(-1)^j / u(0)$; the reversal permutation has sign $(-1)^{m(m-1)/2}$, while the product of those entry signs has the same sign. Hence

$$\boxed{\det(P) = u(0)^{-m} \neq 0.} \quad (\text{CDI9.4})$$

Every nilpotent order is included. Equations (CDI8.3) and (CDI9.4) prove the vector-space isomorphism

$$\boxed{\iota : \mathcal{Q}_{\text{fin}} \xrightarrow{\sim} \mathcal{D}_{\text{fin}}.} \quad (\text{CDI9.5})$$

This statement does not choose a topology on $\mathcal{Q}_{\text{fin}}$ or assert continuity of a map extending to all of $\mathcal{Q}$.

There is, however, a proved complete weak-dual detection statement:

$$\overline{\mathcal{D}_{\text{fin}}^{\sigma(\mathcal{Q}', \mathcal{Q})}} = \mathcal{Q}'. \quad (\text{CDI9.6})$$

To prove it without reflexivity, fix $\lambda \in \mathcal{Q}'$ and finitely many classes $q_1, \dots, q_r$. Consider the linear image $L = \{(d(q_1), \dots, d(q_r)) : d \in \mathcal{D}_{\text{fin}}\} \subset \mathbb{C}^r$. Any coefficient vector c annihilating L makes $\sum c_i q_i$ annihilated by every full zero jet, hence zero in $\mathcal{Q}$. It follows that $\sum c_i \lambda(q_i) = 0$. Finite-dimensional linear algebra now puts $(\lambda(q_i))_i$ in L , so some $d \in \mathcal{D}_{\text{fin}}$ agrees **exactly** with λ on all the chosen vectors. This proves density for every weak neighborhood. It proves no strong-dual density assertion.

CDI10. The residue factor and the normal-dual factor remain distinct

The residue comparison has the exact covariance

$$\boxed{\iota(T_a G) = a(T_{a^{-1}})' \iota(G)}. \quad (\text{CDI10.1})$$

Indeed its integrand changes by a^{1-s} when $G(1-s)$ is replaced by $(T_a G)(1-s)$. Pulling out the scalar a leaves $a^{-s} F(s)$, which is exactly the contragredient transpose in (CDI10.1). The equality holds for the full finite residue sum and all its jets, not only for simple zeros.

More generally, with $P(F, G) = \iota(G)(F)$, the full character law is

$$P(a^k T_a F, a^\ell T_a G) = a^{k+\ell+1} P(F, G) \quad (\text{CDI10.2})$$

whenever the second argument has finite zero-jet support. This follows by multiplying the integrand by $a^{k+s} a^{\ell+1-s}$.

The contragredient of the actual normal representation $Q(-1)$, whose action is aT_a , was independently calculated in (CDI6.4) as

$$\mathbf{N}_a^\vee = a^{-1}(T_{a^{-1}})'. \quad (\text{CDI10.3})$$

Therefore the same residue map from the unshifted coefficient has the precise comparison

$$\iota(T_a G) = a^2 \mathbf{N}_a^\vee \iota(G). \quad (\text{CDI10.4})$$

The coefficient-dual character $a(T_{a^{-1}})'$ in (CDI10.1) and the normal contragredient in (CDI10.3) differ by a^2 . They cannot be identified by dropping the geometric degree or the inverse in the dual action. Equations (CDI7.4) and (CDI10.2) are consistent: unshifted paired exponents sum to 1; two normal exponents sum to 3. These are exact morphism and pairing identities. They are not positivity statements, and the bilinear residue pairing has not been declared Hermitian.

CDI11. Actual receiving result

The continuous duals of the specified global, pole-costalk, and localization complexes now have full formulas, all their images and kernels, and perfect separated continuous-dual pairings on every cohomology degree. Closedness of J and the actual inverse Σ^{-1} enter exactly through CDI1; they ensure that the original quotient and its continuous dual are separated and that no image closure is substituted in the transpose calculation.

The transpose localization map has the anti-diagonal in degree minus one and diagonal in degree minus two; the complete cone and long-row signs are CDI5. The antiholomorphic sphere operation retains its negative degree-two orientation. The ordinary transpose, invariant-pairing contragredient, and original-zeta residue correspondence remain separate exact maps with their full degree characters.

This is a completed duality calculation for the actual finite coefficient models. It does not identify the continuous-dual coefficient complex with the derived sheaf Hom into a dualizing complex, and does not supply an unproved comparison to Deligne's finite-rank étale obstruction receiver. The concrete comparison now available for that further calculation is the entire weak continuous dual in CDI2-CDI5, together with the explicitly reflected finite-jet subspace, nonzero local determinant, and weak-dual density in CDI8-CDI10. No operation or metric on the support τ was used.

CDI12. Independent verification of RD0-RD9

After completing CDI0-CDI11, the full independent root derivation ORIGINAL_ZETA_RESIDUE_DUAL_AND_SUPPORT_MAPS.md, RD0-RD9, was read and checked. The reviewed file SHA256 is 65ae37a4303a76dae2fa5d79820201858bce8773afae7138e5b7175586671641. Its local determinant, explicit inverse, weak-density argument, original-zeta reflection law, and complete normal-action factor require no mathematical correction.

RD1's recursion is the Taylor inversion of the nonvanishing local unit A_ρ . Its isolator has jet 1 at precisely the chosen full block and the required zero jet at all other blocks. Multiplication by the stated finite polynomials preserves $\mathcal{B}$. This agrees with CDI8, which supplies the same global block construction independently.

RD2 uses Taylor-coefficient functionals $\delta_{\rho,j}^{\text{coeff}} = F^{(j)}(\rho)/j!$, whereas (CDI6.6) used unscaled derivatives. Its formula

$$P_{\rho,j}(t) = [t^{m-1-j}u_\rho(t)]_{<m}$$

does give the exact coefficient functional: subtraction of its omitted Taylor terms leaves a multiple of t^m ; division by $t^m u_\rho(t)$ therefore gives t^{-j-1} plus a holomorphic germ. The residue is $F^{(j)}(\rho)/j!$. Thus the factorial conventions in RD2 and RD5.4 agree, and the unscaled formulas in CDI6.7 are their exact multiples by $j!$.

For RD3, a class supported at $\sigma = 1 - \rho$ with local polynomial $P(-t)$ has reflected germ $P(t)$ at ρ . It therefore maps under ι to the specified original-zeta functional $\lambda_{\rho,P}$, giving an actual inverse on every finite sum of block functionals. The determinant is exactly (CDI9.4), including cancellation of the two anti-diagonal signs, rather than a determinant modulo a nonzero unspecified constant. RD4 is exactly the finite-evaluation interpolation argument proved in CDI9; the resulting net is pointwise convergent, with no strong-topology or equicontinuity claim.

The second identity in RD5.2 retains the original variable before taking residues: multiplication of the first argument by s and of the reflected second argument by $1 - s$ sums to multiplication by 1. Hence $\iota L = (1 - L')\iota$. The full factor in RD5.5 is $a^{k+\ell+1}$, as independently proved in CDI10. For two normal-direction arguments it is a^3 ; for one normal and one unshifted argument it is a^2 . In particular RD5.5 correctly preserves the a^2 comparison with the contragredient of an already normal-shifted coefficient. The cochain degree-character distinction is not lost.

RD6's original-zeta symmetry factor and its minus sign can be checked before any simplification. Set $\sigma = 1 - \rho$. The functional equation gives

$$t^m u_\rho(t) = \chi(\rho + t)(-t)^m u_\sigma(-t),$$

which is precisely its local-unit identity. In a residue, substitute $s = 1 - z$, so $ds = -dz$. The denominator becomes $\zeta(1 - z) = \zeta(z)/\chi(z)$. Thus the reflected integrand is

$$-\frac{\chi(z)G(z)F(1-z)}{\zeta(z)}dz,$$

with all derivatives of the displayed χ -germ retained. This is RD6.3, including the orientation sign. No claim that its global meromorphic multiplier preserves $\mathcal{B}$ is needed. The separately recalled trace form in RD6.4 sees only the constant Taylor coefficient on each block: multiplication by a germ $h(t)$ on $\mathbb{C}[t]/(t^m)$ is triangular with diagonal $h(0)$, so its trace is $mh(0)$. In contrast the residue matrix retains all the jets. For the displayed full trace sum, absolute convergence follows from rapid decay of the two $\mathcal{B}$ factors on the strip of zeros and the retained zero-count bound $n(R) = O(R^{3/2})$ from S2. Taking a decay exponent with the product power greater than $3/2$ proves convergence by dyadic shells. These statements add no positivity assertion.

RD7 uses the actual topological comparison $\kappa([b]) = [\frac{1}{2}\mathcal{M}_0b]$; its transpose preserves that same factor $1/2$. Its support-map transposes coincide with CDI4-CDI5: anti-diagonal for the degree-one difference, diagonal for the degree-two sum, and $-\lambda_+ + \lambda_-$ for the angular boundary. The weak density passes through κ' because a topological isomorphism induces a weak-dual topological isomorphism.

Finally RD8 retains the correct raw inverse factor $1/(2\zeta)$, the full trivial-zero derivatives, and both endpoint values. Its expansion of A_ρ is the Taylor convolution of the complete multiplier in F_0 with $\zeta(s)/(s - \rho)^m$, so $a_j = \sum_{h=0}^j C^{(h)}(\rho)\zeta^{(m+j-h)}(\rho)/(h!(m+j-h)!)$ is exact. The original local residue uses u_ρ of ζ , not this auxiliary multiplier. RD9 keeps the continuous-dual receiver distinct from the not-yet-constructed Deligne support-duality comparison. This matches the exact scope of CDI11.

Original-zeta residues and full dual support maps — RD0-RD9

Proof source: ORIGINAL_ZETA_RESIDUE_DUAL_AND_SUPPORT_MAPS.md.

Original-zeta residue duality and the actual support receiver

24 September 2026. Complete derivation RD0-RD9.

RD0. Prerequisites and the specific comparison

The support remains the user's $\tau\langle Z_1; \text{no } Z_2 \rangle$. The complete arithmetic reconstruction precedes every coefficient, integer, prime, analytic variable and linear operation below. None of these operations is addition at τ , a coordinate on that point, or a replacement of the user's information layers by cyclic groups. The governing corpus rule, correction chains, latest direct global-quotientability arguments, and the complete source passages for the distance, separate-branch and global-admissibility corrections were read before this calculation.

The current receiving objects are the actual closed summation image of SSI0-SSI10 and ESI0-ESI12 and the sphere sheaf calculated in CSP0-CSP13, `../tau_weight_cohomology_20260924/CC_SPHERE_PULLBACK_AND_NORMAL_DIRECTION.md`. All of CSP was read. The original-zeta projector proof RZ1-RZ2, RZ5 and RZ8-RZ12 was read directly in `../tau_weight_cohomology_20260924/ORIGINAL_ZETA_RESOLVENT_AND_PROJECTORS.md`; its other sections are dependencies with their existing proofs, not newly claimed reading coverage.

Deligne's actual lifting argument is *La conjecture de Weil. II*, §3.6, printed 213-214, particularly the support-dual term in the cross on 214. The full original article has been read with the separate original-source ledger. That term is $H^{2N-i-1}(X_s)^\vee(-N)$; identifying a dual coefficient without its support, degree and twist would not reproduce that term. The calculation here constructs a dual receiver for the programme's actual coefficient spaces. It does not identify these infinite-dimensional complex sheaves with Deligne's constructible étale category.

The operations to construct are: continuous linear functionals on the already formed quotient; residues of the original ζ ; their embedding into the continuous dual of actual sheaf cohomology; and their arithmetic action. The degree character used in the comparison is independently constructed by the sphere cover in CSP8, rather than assigned to the supporting point.

RD1. The complete quotient and its finite-support subspace

Retain the Fréchet space

$$\mathcal{B} = \{F \text{ entire} : b_{A,M}(F) = \sup_{|\Re s| \leq A} (1 + |\Im s|)^M |F(s)| < \infty \text{ for all integers } A, M \geq 0\}. \quad (\text{RD1.1})$$

Let $\mathcal{Z}$ denote all distinct nontrivial zeros of the original $\zeta(s)$, with their full multiplicities m_ρ . Set

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ } (\rho \in \mathcal{Z}, 0 \leq j < m_\rho)\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}. \quad (\text{RD1.2})$$

This is the actual quotient topology. Cauchy's formula gives $|F^{(j)}(\rho)|/j! \leq r^{-j}b_{A,0}(F)$ whenever $A \geq |\Re \rho| + r$. Therefore every required jet is continuous and $\mathcal{J}$ is closed. All zero multiplicities are retained in the quotient.

The already proved global isolators can also be constructed explicitly here. Keep the entire source transform with its entire multiplier:

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad A_\rho(s) = \frac{F_0(s)}{(s-\rho)^{m_\rho}}. \quad (\text{RD1.3})$$

Division is removable at ρ . It preserves $\mathcal{B}$: away from a disk the denominator is bounded below by a polynomial, and on the disk the maximum principle bounds the holomorphic quotient by a larger boundary circle. Put $a_j = A_\rho^{(j)}(\rho)/j!$, and define

$$b_0 = a_0^{-1}, \quad b_r = -a_0^{-1} \sum_{h=1}^r a_h b_{r-h}, \quad e_\rho(s) = A_\rho(s) \sum_{r=0}^{m_\rho-1} b_r (s-\rho)^r. \quad (\text{RD1.4})$$

The nonzero number a_0 is determined by the actual multiplicity, not by simplicity. The recursion proves that $e_\rho = 1$ modulo $(s-\rho)^{m_\rho}$ and has every required vanishing jet at every other zero. Thus

$$e_{\rho,j} = [e_\rho(s)(s-\rho)^j], \quad 0 \leq j < m_\rho, \quad \mathcal{Q}_{\text{fin}} = \bigoplus_{\rho \in \mathcal{Z}} \text{span}_{\mathbb{C}} \{e_{\rho,j} : 0 \leq j < m_\rho\} \subset \mathcal{Q}. \quad (\text{RD1.5})$$

The sum is direct because evaluation of the full jet at any one zero recovers its coefficients. A member has finitely many nonzero zero blocks, but each block's source representative satisfies the conditions at **all** other zeros simultaneously. This is a subspace of the globally constructed quotient, not a bounded-height replacement for it. No density of $\mathcal{Q}_{\text{fin}}$ in $\mathcal{Q}$ is assumed below.

RD2. The original-zeta local residue dual

Write $\mathcal{Q}'$ for the continuous complex-linear dual. On a zero block put

$$t = s - \rho, \quad \zeta(\rho + t) = t^m u_\rho(t), \quad m = m_\rho, \quad u_\rho(0) = \zeta^{(m)}(\rho)/m! \neq 0. \quad (\text{RD2.1})$$

For a polynomial P of degree below m , define

$$\lambda_{\rho,P}([F]) = \text{Res}_{s=\rho} \frac{P(s-\rho)F(s)}{\zeta(s)} ds = \frac{1}{2\pi i} \int_{|s-\rho|=r} \frac{P(s-\rho)F(s)}{\zeta(s)} ds. \quad (\text{RD2.2})$$

The circle is positively oriented and sufficiently small to contain no other zero or pole. No choice of its admissible radius changes the residue. Changing F by $\mathcal{J}$ makes the integrand holomorphic at ρ , so the functional is well defined. It is a finite linear combination of continuous jets and hence is continuous on the quotient.

Every Taylor-coefficient functional $\delta_{\rho,j}([F]) = F^{(j)}(\rho)/j!$, $0 \leq j < m$, is one of these residues. Specifically set

$$P_{\rho,j}(t) = [t^{m-1-j} u_\rho(t)]_{< m}. \quad (\text{RD2.3})$$

Here the brackets retain exactly the Taylor terms of degrees below m . The discarded remainder is divisible by t^m , so $P_{\rho,j}(t)/(t^m u_\rho(t)) = t^{-j-1} + \text{holomorphic germ}$. Taking the residue after multiplication by $F(\rho + t)$ proves

$$\lambda_{\rho,P_{\rho,j}} = \delta_{\rho,j}. \quad (\text{RD2.4})$$

Unscaled derivatives are $j! \lambda_{\rho,P_{\rho,j}}$; the factorial is not suppressed. Conversely (RD2.2) is a linear combination of these m functionals. Consequently the original local residue, including every derivative of its unit u_ρ , gives the entire dual of the full multiplicity block.

RD3. A separating pairing using every reflected block

The original functional equation preserves zero multiplicities under $\rho \mapsto 1 - \rho$. Define, for $x = [F] \in \mathcal{Q}$ and $g = [G] \in \mathcal{Q}_{\text{fin}}$,

$$\mathcal{R}(x, g) = \sum_{\rho \in \mathcal{Z}} \text{Res}_{s=\rho} \frac{F(s)G(1-s)}{\zeta(s)} ds. \quad (\text{RD3.1})$$

This sum has finite support because every zero jet of G outside its finite support vanishes to the required order. It is independent of both representatives by exactly that order of vanishing. For each fixed g , it is a continuous functional on the **entire** $\mathcal{Q}$. It is complex bilinear; no Hermitian or positivity assertion is part of its definition.

For $\sigma = 1 - \rho$, take the exact basis classes in (RD1.5) at ρ and σ . Formula (RD2.1) gives the full block matrix

$$\mathcal{R}(e_{\rho,i}, e_{\sigma,j}) = (-1)^j [t^{m-1-i-j}] u_{\rho}(t)^{-1}, \quad 0 \leq i, j < m, \quad (\text{RD3.2})$$

where a coefficient with negative index is zero. Blocks not paired by $\sigma = 1 - \rho$ give zero. The entries on $i + j = m - 1$ are $(-1)^j / u_{\rho}(0)$. Reversing the order of the columns introduces $(-1)^{m(m-1)/2}$; the product of the displayed signs introduces the same sign. Therefore

$$\det(\mathcal{R}(e_{\rho,i}, e_{1-\rho,j}))_{i,j=0}^{m-1} = u_{\rho}(0)^{-m} \neq 0. \quad (\text{RD3.3})$$

All lower coefficients of u_{ρ}^{-1} remain in (RD3.2). The determinant does not replace that matrix.

Let

$$\mathcal{D} = \text{span}_{\mathbb{C}} \{\delta_{\rho,j} : \rho \in \mathcal{Z}, 0 \leq j < m_{\rho}\} \subset \mathcal{Q}'. \quad (\text{RD3.4})$$

Then

$$\iota : \mathcal{Q}_{\text{fin}} \xrightarrow{\sim} \mathcal{D}, \quad \iota(g)(x) = \mathcal{R}(x, g) \quad (\text{RD3.5})$$

is a linear isomorphism. An explicit inverse on (RD2.2) is the class supported at $\sigma = 1 - \rho$ whose local polynomial is $P(-t)$: its reflected germ is $P(t)$. Construct it by the finite linear combination of the $e_{\sigma,j}$. Extend over finite sums. This proves surjectivity; (RD3.3) proves injectivity. It also proves that the pairing separates both arguments: a nonzero class $[F]$ has a nonzero required jet somewhere by (RD1.2), and (RD2.4) detects it.

RD4. The exact global dual closure, without a topology substitution

Give $\mathcal{Q}'$ its weak topology $\sigma(\mathcal{Q}', \mathcal{Q})$, namely pointwise convergence on all $x \in \mathcal{Q}$. Then

$$\overline{\mathcal{D}}^{\sigma(\mathcal{Q}', \mathcal{Q})} = \mathcal{Q}'. \quad (\text{RD4.1})$$

Here is a proof that also specifies its precise strength. Fix $\lambda \in \mathcal{Q}'$ and any finite collection $x_1, \dots, x_k \in \mathcal{Q}$. Let $E : \mathcal{D} \rightarrow \mathbb{C}^k$, $E(d) = (d(x_1), \dots, d(x_k))$. If $(\lambda(x_i))_i$ were not in $E(\mathcal{D})$, finite-dimensional linear algebra would supply $c_1, \dots, c_k$ such that $\sum_i c_i d(x_i) = 0$ for every $d \in \mathcal{D}$ but $\sum_i c_i \lambda(x_i) \neq 0$. The first assertion and separation in RD3 imply $\sum_i c_i x_i = 0$ in $\mathcal{Q}$, contradicting the second. Thus some $d \in \mathcal{D}$ matches λ exactly on this finite collection. Direct the finite subsets of $\mathcal{Q}$ by inclusion and select one such matching functional at each stage. The resulting net converges pointwise to λ , proving (RD4.1).

Every element of the full dual is therefore recovered in this weak topology from the full residue system. This statement does not claim convergence in the strong dual topology, equicontinuity of this chosen net, a continuous extension of ι to all of $\mathcal{Q}$, or replacement of $\mathcal{Q}$ by an unrestricted product of jets. Those are different assertions; none is used in this proof.

RD5. The full arithmetic action and its dual degree factor

On the original quotient let

$$T_a[F] = [a^s F(s)], \quad L[F] = [sF(s)], \quad a > 0. \quad (\text{RD5.1})$$

The operations are continuous since $a^{\Re s}$ is bounded on any fixed vertical strip, and $|s|$ adds at most one polynomial seminorm. They preserve the ideal and every finite block. For a continuous map U , denote its ordinary transpose by $U' \lambda = \lambda \circ U$; its contragredient action is $(U^{-1})'$ when U is invertible. Directly in the original residue integrand,

$$\mathcal{R}(T_a x, T_a g) = a \mathcal{R}(x, g), \quad \mathcal{R}(Lx, g) + \mathcal{R}(x, Lg) = \mathcal{R}(x, g). \quad (\text{RD5.2})$$

The first identity uses the complete product $a^s a^{1-s} = a$, the second $s + (1 - s) = 1$, before taking any residue. It follows that

$$\boxed{\iota T_a = a(T_{a^{-1}})' \iota, \quad \iota L = (1 - L') \iota.} \quad (\text{RD5.3})$$

Thus the same factor n produced geometrically by the degree of $z \mapsto z^n$ in CSP8 is present in the exact original-zeta dual receiver. For nonintegral a , (RD5.3) is a coefficient identity, not a claimed single-valued cover $z \mapsto z^a$.

For the Taylor functionals the entire triangular formulas are

$$\begin{aligned} T_a' \delta_{\rho, j} &= a^\rho \sum_{h=0}^j \frac{(\log a)^{j-h}}{(j-h)!} \delta_{\rho, h}, \\ a(T_{a^{-1}})' \delta_{\rho, j} &= a^{1-\rho} \sum_{h=0}^j \frac{(-\log a)^{j-h}}{(j-h)!} \delta_{\rho, h}, \quad L' \delta_{\rho, j} = \rho \delta_{\rho, j} + \delta_{\rho, j-1}, \end{aligned} \quad (\text{RD5.4})$$

where $\delta_{\rho, -1} = 0$. They follow by multiplying the original Taylor series, so every factorial and every nilpotent coefficient is accounted for. They are compatible with (RD3.2), including its signs. The ordinary transpose of the actual degree-two action aT_a is instead aT_a' , and its contragredient is $a^{-1}(T_{a^{-1}})'$. Neither is silently replaced by the degree-twisted dual $a(T_{a^{-1}})'$ in (RD5.3).

More generally, for the actual degree-character twists indexed by integers k, ℓ , the same integrand proves

$$\mathcal{R}(a^k T_a x, a^\ell T_a g) = a^{k+\ell+1} \mathcal{R}(x, g). \quad (\text{RD5.5})$$

In particular two normal-direction inputs carry the full factor a^3 , and one normal-direction input together with one unshifted input carries a^2 . Equivalently the map ι intertwines the unshifted T_a with a^2 times the contragredient of the normal action aT_a , since that contragredient is $a^{-1}(T_{a^{-1}})'$. This retains the difference between the coefficient degree-character dual and the dual of an already shifted normal term.

RD6. Functional equation and the retained symmetry defect

The original functional equation in this local comparison is

$$\zeta(s) = \chi(s) \zeta(1-s), \quad \chi(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)}. \quad (\text{RD6.1})$$

At each nontrivial zero both local multiplier expressions are holomorphic and nonzero. For $\sigma = 1 - \rho$, the full local units satisfy

$$u_\rho(t) = (-1)^m \chi(\rho + t) u_\sigma(-t). \quad (\text{RD6.2})$$

This follows by substituting $s = \rho + t$ into (RD6.1) and retaining the factor $(-t)^m$. For F, G with finite zero support, substitution $s = 1 - z$, including $ds = -dz$, proves the exact reversal law

$$\mathcal{R}([F], [G]) = - \sum_{\sigma \in \mathcal{Z}} \text{Res}_{z=\sigma} \frac{\chi(z)G(z)F(1-z)}{\zeta(z)} dz. \quad (\text{RD6.3})$$

The sum is finite; the formula uses the germs of χ there. It does not assert that multiplication by its meromorphic global formula preserves $\mathcal{B}$. Derivatives of χ in (RD6.2)–(RD6.3) are retained. In particular a symmetric or Hermitian global form has not been obtained by deleting this unit or its differential sign.

The ordinary reflected trace form previously constructed in RZ10 is

$$W([F], [G]) = \sum_{\rho \in \mathcal{Z}} m_\rho \overline{F(1-\bar{\rho})} G(\rho). \quad (\text{RD6.4})$$

Its multiplicity-block trace has no higher-jet entries. The residue pairing (RD3.2) does: its matrix is invertible for every positive multiplicity. This proves an exact additional dual observation of the retained classes. It does not replace (RD6.4) in Weil’s positivity criterion, and it makes no sign claim about that form.

RD7. Passage to the actual sheaf cohomology and support maps

CSP gives $H^1(Y, \mathcal{F}) = Q_A = A/J$ with

$$\kappa : Q_A \xrightarrow{\sim} \mathcal{Q}, \quad \kappa([b]) = [\Theta b], \quad \Theta b(s) = \frac{1}{2} \int_0^\infty b(u) u^s \frac{du}{u}. \quad (\text{RD7.1})$$

This is the CS convention, retaining the factor $1/2$. The original restriction is still $\Sigma f = 2 \sum_{k \geq 1} f(ku)$; SSI’s raw transform uses $\mathcal{M}_0 = 2\Theta$. The isomorphism in (RD7.1) follows from the proved actual equality $\Sigma S = J = \Theta^{-1} \mathcal{J}$. It retains the quotient topology, not a closure surrogate.

Transpose (RD7.1) to get an actual isomorphism of continuous duals $\kappa' : \mathcal{Q}' \rightarrow Q'_A$. Combining it with (RD3.5) gives

$$\mathcal{Q}_{\text{fin}} \xrightarrow{\kappa' \iota} H^1(Y, \mathcal{F})', \quad g \mapsto \left([b] \mapsto \sum_{\rho} \text{Res}_{s=\rho} \frac{\frac{1}{2} \mathcal{M}_0 b(s) G(1-s)}{\zeta(s)} ds \right). \quad (\text{RD7.2})$$

Its image is weakly dense by (RD4.1) and the topological isomorphism. Formula (RD5.3) gives its arithmetic action with the same complete factor. No coefficient arithmetic has been assigned to τ .

The degree-one support-to-global map from CSP7 is

$$k^1 : Q_A \oplus Q_A \longrightarrow Q_A, \quad (x_+, x_-) \mapsto x_+ - x_-. \quad (\text{RD7.3})$$

Therefore its exact transpose on these residue functionals is

$$(k^1)'(\kappa' \iota g) = (\kappa' \iota g, -\kappa' \iota g). \quad (\text{RD7.4})$$

The degree-two support map is instead the sum $A \oplus A \rightarrow A$, whose transpose is $\lambda \mapsto (\lambda, \lambda)$. The annular boundary $a \mapsto (-a, a)$ transposes to $(\lambda_+, \lambda_-) \mapsto -\lambda_+ + \lambda_-$. These three identities follow by evaluating the maps on arbitrary input pairs; they preserve both branches and every orientation sign.

This embeds the residue receiver in the actual dual localization sequence. Its terms are continuous duals of CSP’s coefficient complexes. Calling them a Verdier-dual sheaf, or Deligne’s étale support-dual cross, would require an additional constructed comparison; that assertion is not used here.

RD8. Endpoint and original-function data remain in the source

The quotient in RD1 records the nontrivial-zero divisor. Its original source retains the other contributions. In the raw convention SSI proves for every $F \in \mathcal{J}$

$$f_F(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{F(2+it)}{2\zeta(2+it)} x^{-2-it} dt, \quad \frac{f_F^{(2r)}(0)}{(2r)!} = \frac{F(-2r)}{2\zeta'(-2r)} \quad (r \geq 1), \quad (\text{RD8.1})$$

$$f_F(0) = 0, \quad \int_{\mathbb{R}} f_F = 0, \quad \int_0^\infty f_F(x) \frac{dx}{x} = -F(0), \quad \int_0^\infty f_F(x) \log x \, dx = \frac{F(1)}{2}. \quad (\text{RD8.2})$$

The four independent endpoint lines $E_+ \oplus E_-$ remain in $H^0(Y, \mathcal{F})$ and their full duals remain in the degree-zero transposed cohomology. They have not been folded into (RD7.2) or declared absent.

For the isolator source (RD1.3), the full retained special values are, with $r \geq 1$ in the second formula,

$$\begin{aligned} F_0(0) &= F_0(1) = \frac{1}{8}, & F_0(-2r) &= \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) \\ &= \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r). \end{aligned} \quad (\text{RD8.3})$$

Every unit coefficient a_j in (RD1.4) is obtained from the full derivatives of (RD1.3). At nontrivial ρ , writing its displayed multiplier as $C(s)$, the exact expression is

$$a_j = \sum_{h=0}^j \frac{C^{(h)}(\rho)}{h!} \frac{\zeta^{(m+j-h)}(\rho)}{(m+j-h)!}. \quad (\text{RD8.4})$$

In contrast, (RD2.1)–(RD3.3) use the original local unit of ζ itself. The comparison has not substituted F_0 for the original arithmetic function.

RD9. What this calculation establishes for the lifting programme

The proved map (RD7.2) supplies the complete finite-jet residue receiver inside the continuous dual of the actual geometric degree-one cohomology. It sees every nilpotent multiplicity coordinate; (RD4.1) recovers the entire continuous dual in its specified weak topology. Its arithmetic action is the degree-character contragredient (RD5.3), and its support maps have the exact transposed signs (RD7.4). These are full-divisor identities, with no bounded numerical experiment and no assumed purity.

The support-dual target used in Deligne §3.6 additionally has geometrically derived weight bounds and a particular specialization/localization cross. Neither a residue similitude nor the positive degree of a sphere cover proves those bounds for this coefficient sheaf. The current target remains to construct and calculate that comparison. The present calculation contributes its actual dual objects and maps; it does not turn the desired weight separation into an input.

Bounded lifts, strong duals and completed original residues — SDT0-SDT9

Proof source: CC_STRONG_DUAL_TOPOLOGY_AND_JETS.md.

Bounded lifts and strong continuous duals of the original-zeta coefficient

Independent derivation, 24 September 2026. Proof locators SDT0–SDT9.

SDT0. Actual input, source stage, and the assertion proved

This calculation starts on the coefficient spaces already constructed in CSP and CDI. The supporting datum remains $\tau\langle Z_1; \text{no } Z_2\rangle$. All vector operations, seminorms, analytic coordinates, quotients and duals below belong to those reconstructed arithmetic coefficients. None is an operation or a newly assigned metric on that supporting datum.

The complete construction and its correction history were read for the preceding CS and CSB calculations. For this continuation the current construction guide, active-target and correction-precedence passages, and the full verbatim WU061–WU062 were reread. The mathematical input actually read here is [CDI0–CDI3](#) and [CDI8–CDI12](#), together with the exact coefficient definitions in [CSP0–CSP1](#). The exact-image results [SSI1–SSI7](#) and [ESI0–ESI9](#) were read during the preceding calculation and are retained with their actual proof status.

The underlying human source is Connes–Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, [arXiv:0903.2024v3](#), §5. Its original author TeX, retained at `sources/CC_0903_2024_v3/author_source/announc3.tex`, was previously read at lines 1376–1667. This note strengthens the topology of the already proved coefficient comparison; it is not a new reading of the entire human source or an identification with a Deligne dualizing complex.

Retain exactly

$$S = \left\{ f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0 \right\},$$

$$A = \{a \in C^\infty(\mathbb{R}_{>0}) : p_{N,j}(a) < \infty \text{ for every } N, j \geq 0\}, \quad p_{N,j}(a) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)|,$$

$$\Sigma f(u) = 2 \sum_{n \geq 1} f(nu), \quad J = \Sigma S, \quad Q = A/J, \quad q : A \longrightarrow Q. \quad (\text{SDT0.1})$$

Here N, j are nonnegative integers and the topology on A is the topology of all the displayed seminorms. SSI and ESI prove that J is closed and that $\Sigma : S \rightarrow J$ is a topological isomorphism.

We prove all of the following for these actual spaces:

1. Every bounded subset of A , and every bounded subset of Q , has compact closure.
2. Every compact subset of Q is the image of a compact subset of A ; consequently every bounded subset of Q has a bounded lift.

3. The continuous bijection in CDI3.3 is a topological isomorphism

$$q' : Q'_\beta \xrightarrow{\sim} J^\perp \subset A'_\beta. \quad (\text{SDT0.2})$$

4. The evaluation maps identify the strong biduals of A and Q with their original topologies.
5. The span of all actual nontrivial-zero jet functionals is strongly dense in the entire continuous dual of the original-zeta quotient.

No nuclearity theorem or general bounded-lifting assertion is assumed. The compactness and lifting arguments are constructed below. In particular the statement is not deduced merely from the fact that a map of Fréchet spaces is onto.

SDT1. The original weighted topology and its quotient seminorms

Use the analytic change of variable $x = \log u$ only on A , writing

$$g(x) = a(e^x), \quad w_n(x) = e^{nx} + e^{-nx}, \quad P_n(a) = \max_{0 \leq j \leq n} \sup_{x \in \mathbb{R}} w_n(x) |g^{(j)}(x)|. \quad (\text{SDT1.1})$$

Thus $P_n = \max_{j \leq n} p_{n,j}$ is expressed in the original seminorms, including both summands of the weight. No summand has been removed. The sequence P_n is increasing: $w_n(x) = 2 \cosh(nx)$ is increasing with $n \geq 0$, and the derivative range increases. For every N, j , choosing $n \geq \max(N, j)$ gives

$$p_{N,j} \leq P_n. \quad (\text{SDT1.2})$$

Conversely every P_n is a finite maximum of defining seminorms. Hence this is exactly the original topology. Each P_n is a norm on A .

For completeness, a sequence Cauchy in every P_n is Cauchy with every derivative uniformly on every compact interval in x . The corresponding uniform limits have the derivative compatibility obtained by integrating each first derivative on a compact interval. Inducting over derivatives gives a smooth limit g . For every n , the weighted derivatives are uniform Cauchy functions on the whole real axis, so their uniform limits are precisely $w_n g^{(j)}$. Set $a(u) = g(\log u)$. Then $P_n(a - a_k) \rightarrow 0$ and $P_n(a) < \infty$. This proves that A is complete. The metric

$$d_A(a, b) = \sum_{n=0}^{\infty} 2^{-n-1} \min\{1, P_n(a - b)\} \quad (\text{SDT1.3})$$

generates its topology. Thus A is a Hausdorff Fréchet space.

Define actual quotient seminorms

$$\overline{P}_n(q(a)) = \inf_{j \in J} P_n(a + j). \quad (\text{SDT1.4})$$

They are well defined, increasing, and subadditive, and they satisfy

$$q\{P_n < \varepsilon\} = \{\overline{P}_n < \varepsilon\}. \quad (\text{SDT1.5})$$

The equality uses the strict inequality: an infimum below ε supplies a representative with $P_n < \varepsilon$. These sets generate exactly the quotient topology. Images of open sets under q are open because their inverse images are the open sets $\bar{U} + J$.

If all $\overline{P}_n(y)$ vanish, a representative a of y can be approximated by elements of J in every neighborhood defined by some P_n . Thus $a \in \bar{J} = J$ and $y = 0$. The quotient is therefore Hausdorff.

We also give its completeness argument. Let (y_ℓ) be Cauchy for every $\bar{P}_n$. Choose a subsequence $(y_{\ell_k})_{k \geq 1}$ so that

$$\bar{P}_k(y_{\ell_{k+1}} - y_{\ell_k}) < 2^{-k-1}.$$

Choose $b_k \in A$ with

$$q(b_k) = y_{\ell_{k+1}} - y_{\ell_k}, \quad P_k(b_k) < 2^{-k}. \quad (\text{SDT1.6})$$

For every fixed m , the tail from $k \geq m$ is absolutely summable in P_m , since $P_m \leq P_k$. The series $\sum_{k \geq 1} b_k$ therefore converges in A . Adding any representative of y_{ℓ_1} , then applying q , gives a limit of the chosen subsequence. The original Cauchy sequence has that same limit. The metric obtained from (SDT1.3) by replacing P_n by $\bar{P}_n$ is complete, so Q is Fréchet.

This proof uses the quotient's actual closed kernel, not a completion in which nonzero quotient classes are discarded.

SDT2. A finite-net estimate with the two original weights retained

Fix $n \geq 0$ and $M > 0$. Consider

$$\mathcal{U}_{n+1,M} = \{a \in A : P_{n+1}(a) \leq M\}. \quad (\text{SDT2.1})$$

This set need not be bounded in the full Fréchet topology. We prove the exact assertion needed: it is totally bounded for the single norm P_n .

For all $x \in \mathbb{R}$,

$$\frac{w_n(x)}{w_{n+1}(x)} = \frac{e^{nx} + e^{-nx}}{e^{(n+1)x} + e^{-(n+1)x}} \leq 2e^{-|x|}. \quad (\text{SDT2.2})$$

Indeed the numerator is at most $2e^{n|x|}$, while the denominator is at least $e^{(n+1)|x|}$. If $a, b \in \mathcal{U}_{n+1,M}$, then for $0 \leq j \leq n$,

$$\sup_{|x| > T} w_n(x) |g_a^{(j)}(x) - g_b^{(j)}(x)| \leq 4Me^{-T}. \quad (\text{SDT2.3})$$

For a prescribed $\varepsilon > 0$, choose $T > 0$ with $4Me^{-T} < \varepsilon/2$.

On $[-T, T]$, each $g_a^{(j)}$, $j \leq n+1$, has absolute value at most $M/2$, because $w_{n+1} \geq 2$. The functions $g_a^{(j)}$, $j \leq n$, are therefore Lipschitz with constant at most $M/2$. Put $W = w_n(T)$. Choose a finite mesh of $[-T, T]$ with every point at distance at most δ from a mesh point, where

$$0 < \delta < \frac{\varepsilon}{4MW}. \quad (\text{SDT2.4})$$

At every mesh point, quantize each of the finitely many complex numbers $g_a^{(j)}$, $j \leq n$, to a finite set with error at most η , where

$$0 < \eta < \frac{\varepsilon}{8W}. \quad (\text{SDT2.5})$$

The bound $M/2$ makes that quantization finite. There are finitely many resulting joint codes. For each occupied code choose one function b in the set with that code. If a has the same code, the two Lipschitz bounds and the two quantization errors give

$$\sup_{|x| \leq T} w_n(x) |g_a^{(j)}(x) - g_b^{(j)}(x)| \leq W(M\delta + 2\eta) < \varepsilon/2 \quad (0 \leq j \leq n). \quad (\text{SDT2.6})$$

Combining (SDT2.3) and (SDT2.6) proves $P_n(a - b) < \varepsilon$. The finitely many selected b 's form a finite P_n -net. For $M = 0$ the set is the singleton zero, so the same conclusion holds.

Every bounded $B \subset A$ has a finite bound for P_{n+1} , for each n . It is therefore totally bounded for every P_n . To see that this is total boundedness for d_A , first make the tail $\sum_{n>N} 2^{-n-1}$ small, and then use a finite sufficiently small P_N -net to control all terms with $n \leq N$. Since A is complete, $\overline{B}$ is compact.

The final metric fact can be verified directly: a totally bounded metric space permits extraction of a Cauchy subsequence from every sequence by successive finite covers of radii tending to zero; completeness supplies its limit. Thus its closed completion is compact. No additional locally convex compactness theorem is needed.

SDT3. Bounded quotient sets have compact closure

Let $B \subset Q$ be bounded in the actual quotient topology. For each n , put

$$M_n = \sup_{y \in B} \overline{P}_{n+1}(y) < \infty. \quad (\text{SDT3.1})$$

For each $y \in B$ choose a representative $a_y \in A$ with

$$q(a_y) = y, \quad P_{n+1}(a_y) < M_n + 1. \quad (\text{SDT3.2})$$

These choices are made separately for the present index n ; no assertion that they form a single Fréchet-bounded set is used here.

By SDT2, the set of these representatives admits a finite P_n -net. Applying q and using $\overline{P}_n(q(a)) \leq P_n(a)$ gives a finite $\overline{P}_n$ -net for B . Thus B is totally bounded in every quotient seminorm and hence in the complete quotient metric of SDT1. Its closure is compact:

$$\boxed{B \subset Q \text{ bounded} \implies \overline{B}^Q \text{ compact.}} \quad (\text{SDT3.3})$$

The same statement for A was proved in SDT2. In the usual terminology these Fréchet spaces are Montel: they are barrelled, and their closed bounded sets are compact. The barrelled assertion follows from the Baire argument supplied in SDT6. No quotient-stability assertion about general Montel spaces is required here.

Closed subspaces of A , including J , have the same bounded-compact property by intersection with their closed ambient subspace. Finite products of such spaces retain the property, because projections of a bounded set have compact closures and its closure is a closed subset of the product of those compact sets. In particular the actual graph coefficient $Z = J \oplus E_+ \oplus E_-$ also has this property.

SDT4. Constructive compact lifting through the actual quotient

We prove a compact lifting statement for this quotient directly. The argument works for a quotient of any Fréchet space by a closed subspace, but its hypotheses and construction will remain explicit.

For $n \geq 1$, define

$$\begin{aligned} U_n &= \{a \in A : P_n(a) < 2^{-n}\}, & W_n &= \{y \in Q : \overline{P}_n(y) < 2^{-n-1}\}, \\ C_n &= \{y \in Q : \overline{P}_n(y) \leq 2^{-n-1}\}. \end{aligned} \quad (\text{SDT4.1})$$

Then C_n is closed, W_n is an open neighborhood of zero,

$$\overline{W_n} \subset C_n \subset q(U_n). \quad (\text{SDT4.2})$$

The last inclusion follows from the strict gap $2^{-n-1} < 2^{-n}$ and (SDT1.5).

Let $K \subset Q$ be compact and nonempty. Choose $y_{0,1}, \dots, y_{0,r_0} \in K$ such that the translates $y_{0,i} + W_1$ cover K . Choose arbitrary lifts $a_{0,i} \in A$ with $q(a_{0,i}) = y_{0,i}$, and put

$$F_0 = \{a_{0,1}, \dots, a_{0,r_0}\}, \quad K_1 = \bigcup_{i=1}^{r_0} ((K - y_{0,i}) \cap C_1). \quad (\text{SDT4.3})$$

Each intersection is a closed subset of a compact translate of K . Hence K_1 is compact and lies in $C_1 \subset q(U_1)$. Every $y \in K$ has a representation $y = q(a_0) + r_1$ with $a_0 \in F_0$ and $r_1 \in K_1$.

Inductively suppose the compact set $K_n \subset C_n$ is constructed. Choose finitely many centers $y_{n,i} \in K_n$ whose translates $y_{n,i} + W_{n+1}$ cover K_n . Since $K_n \subset q(U_n)$, choose lifts

$$q(a_{n,i}) = y_{n,i}, \quad a_{n,i} \in U_n.$$

Define

$$F_n = \{a_{n,i}\}_i, \quad K_{n+1} = \bigcup_i ((K_n - y_{n,i}) \cap C_{n+1}). \quad (\text{SDT4.4})$$

Again K_{n+1} is compact and contained in C_{n+1} . The finite cover ensures that every residual $r_n \in K_n$ can be written

$$r_n = q(a_n) + r_{n+1}, \quad a_n \in F_n, \quad r_{n+1} \in K_{n+1}. \quad (\text{SDT4.5})$$

These finite sets are nonempty: each chosen center gives residual zero in the next residual set. Thus the displayed finite products are nonempty when K is nonempty.

Every choice from the product $\prod_{n=0}^{\infty} F_n$ has a convergent sum in A . For a fixed m and $N \geq \max(m, 1)$, its tails satisfy the uniform estimate

$$P_m \left(\sum_{n>N} a_n \right) \leq \sum_{n>N} P_m(a_n) \leq \sum_{n>N} 2^{-n} = 2^{-N}. \quad (\text{SDT4.6})$$

Completeness of A proves existence of each sum. The sum map

$$\Psi : \prod_{n=0}^{\infty} F_n \longrightarrow A, \quad \Psi((a_n)) = \sum_{n=0}^{\infty} a_n \quad (\text{SDT4.7})$$

is continuous: finite partial sums are continuous and (SDT4.6) controls the remainder uniformly in every defining seminorm.

The product of these finite discrete sets is compact. One can see this without an uncountable-product assertion: give it the metric recording the first coordinate at which two sequences differ; a diagonal subsequence argument gives a convergent subsequence from every sequence. Its image

$$B_K^* = \Psi \left(\prod_{n=0}^{\infty} F_n \right) \quad (\text{SDT4.8})$$

is therefore compact.

For any $y \in K$, choose a compatible residual path using (SDT4.3) and (SDT4.5). After stage N ,

$$y - q \left(\sum_{n=0}^N a_n \right) = r_{N+1} \in C_{N+1}.$$

For $N + 1 \geq m$,

$$\overline{P}_m(r_{N+1}) \leq \overline{P}_{N+1}(r_{N+1}) \leq 2^{-N-2}. \quad (\text{SDT4.9})$$

Thus the residual tends to zero in Q , and continuity of q gives $q(\Psi((a_n))) = y$. This proves $K \subset q(B_K^*)$. The compact set

$$B_K = B_K^* \cap q^{-1}(K) \quad (\text{SDT4.10})$$

then satisfies $q(B_K) = K$. The intersection is closed in B_K^* , because the compact set K is closed in the Hausdorff space Q . Empty K has the empty lift.

We have constructed, rather than assumed,

$$\boxed{K \subset Q \text{ compact} \implies \exists B_K \subset A \text{ compact with } q(B_K) = K.} \quad (\text{SDT4.11})$$

The construction supplies a compact set of lifts. It makes no assertion that q has a continuous linear section, and it does not choose a new topology on either coefficient.

SDT5. Bounded lifting and equality of the strong-dual topologies

Let $B \subset Q$ be bounded. SDT3 gives the compact closure $K = \overline{B}$. Lift K by SDT4 and define

$$\widetilde{B} = B_K \cap q^{-1}(B). \quad (\text{SDT5.1})$$

This set need not be closed or compact, but it is a subset of the compact set B_K , hence is bounded. Every $y \in B$ has a preimage in it. Therefore

$$\boxed{q(\widetilde{B}) = B, \quad \widetilde{B} \subset A \text{ bounded.}} \quad (\text{SDT5.2})$$

Compact sets are bounded because every continuous seminorm has a finite maximum on them; the seminorm characterization of boundedness then gives the assertion for their subsets.

For a locally convex space E , retain the strong-dual seminorms

$$p_D(\lambda) = \sup_{x \in D} |\lambda(x)|, \quad D \subset E \text{ bounded.} \quad (\text{SDT5.3})$$

CDI1 proves the algebraic identification $q'Q' = J^\perp$. If $C \subset A$ is bounded, then $q(C)$ is bounded and

$$p_C(q'\mu) = p_{q(C)}(\mu). \quad (\text{SDT5.4})$$

This proves continuity of $q' : Q'_\beta \rightarrow A'_\beta$. Conversely, for every bounded $B \subset Q$, choose its exact bounded lift from (SDT5.2). Then

$$p_B(\mu) = p_{\widetilde{B}}(q'\mu). \quad (\text{SDT5.5})$$

Every defining strong seminorm on Q' is thus the restriction of a defining strong seminorm from A' . Equations (SDT5.4) and (SDT5.5) prove

$$\boxed{Q'_\beta \xrightarrow[\mu \mapsto \mu \circ q]{\sim} (J^\perp, \text{subspace topology from } A'_\beta).} \quad (\text{SDT5.6})$$

The annihilator $J^\perp$ is closed, being the intersection of the continuous evaluation kernels at $j \in J$. This resolves the specific topology left open in CDI3.2-CDI3.3.

No openness conclusion for the separate restriction map $A'_\beta \rightarrow J'_\beta$ is needed for (SDT5.6), and no such conclusion is asserted here.

SDT6. The full strong bidual is the original coefficient

Let E be A or Q , with exactly the topology already proved. The argument also applies to the closed subspace J and the finite-product graph coefficient Z from SDT3.

First let $\Lambda : E'_\beta \rightarrow \mathbb{C}$ be continuous and linear. Continuity at zero gives finitely many bounded sets $B_1, \dots, B_r \subset E$ and a finite constant C such that

$$|\Lambda(\lambda)| \leq C \max_i p_{B_i}(\lambda) = p_{CB}(\lambda), \quad B = \bigcup_i B_i. \quad (\text{SDT6.1})$$

If $C = 0$, then $\Lambda = 0$. Otherwise let

$$D = \overline{\text{aconv}(CB)}^E, \quad (\text{SDT6.2})$$

where the absolutely convex hull consists of finite sums $\sum_i c_i b_i$ with $b_i \in CB$ and $\sum_i |c_i| \leq 1$. Every continuous seminorm is bounded on D by its supremum on CB . Thus D is bounded. By SDT2 or SDT3 it is compact, as well as closed and absolutely convex. Equation (SDT6.1) becomes

$$|\Lambda(\lambda)| \leq \sup_{x \in D} |\lambda(x)|. \quad (\text{SDT6.3})$$

For any finite family $\lambda_1, \dots, \lambda_r \in E'$, the vector $(\Lambda(\lambda_1), \dots, \Lambda(\lambda_r))$ belongs to the compact absolutely convex set

$$\{(\lambda_1(x), \dots, \lambda_r(x)) : x \in D\} \subset \mathbb{C}^r. \quad (\text{SDT6.4})$$

Indeed, otherwise finite-dimensional real separation would give a complex linear combination $\lambda = \sum_i c_i \lambda_i$ whose real value at that vector exceeds its supremum on the displayed set. Absolute convexity identifies that supremum with the supremum of the modulus. This contradicts (SDT6.3).

Hence the closed subsets $\{x \in D : \lambda(x) = \Lambda(\lambda)\}$, indexed by $\lambda \in E'$, have the finite intersection property. Compactness of D gives $x \in D$ with

$$\Lambda(\lambda) = \lambda(x) \quad \text{for every } \lambda \in E'. \quad (\text{SDT6.5})$$

Hahn-Banach separation of points in the Hausdorff locally convex space E makes x unique. Therefore

$$(E'_\beta)' = E \quad (\text{SDT6.6})$$

under the specified evaluation map, as vector spaces.

We prove equality of topologies as well. A bounded set $M \subset E'_\beta$ is pointwise bounded, since each singleton $\{x\} \subset E$ is bounded. Put

$$C_k = \{x \in E : \sup_{\lambda \in M} |\lambda(x)| \leq k\}, \quad k = 1, 2, \dots \quad (\text{SDT6.7})$$

Each C_k is closed and their union is E . The Baire theorem for the complete metric space E gives an interior point x_0 of one C_k . For a neighborhood U of zero with $x_0 + U \subset C_k$,

$$\sup_{\lambda \in M} |\lambda(u)| \leq 2k \quad (u \in U). \quad (\text{SDT6.8})$$

Thus M is equicontinuous, and the seminorm $x \mapsto \sup_{\lambda \in M} |\lambda(x)|$ is continuous on E . This proves continuity of evaluation into the strong bidual.

For the reverse topology, let P be any continuous seminorm on E , and set

$$M_P = \{\lambda \in E' : |\lambda(x)| \leq P(x) \text{ for all } x \in E\}. \quad (\text{SDT6.9})$$

For every bounded $B \subset E$,

$$\sup_{\lambda \in M_P} p_B(\lambda) \leq \sup_{x \in B} P(x) < \infty.$$

Hence M_P is bounded in E'_β . Complex Hahn-Banach, applied to each one-dimensional span, gives

$$P(x) = \sup_{\lambda \in M_P} |\lambda(x)|. \quad (\text{SDT6.10})$$

Every original continuous seminorm is therefore a strong-bidual seminorm. Together with (SDT6.5)–(SDT6.8), this proves the topological isomorphism

$$\boxed{E \xrightarrow{\sim} (E'_\beta)'_\beta, \quad x \mapsto (\lambda \mapsto \lambda(x)).} \quad (\text{SDT6.11})$$

For the terminology used in SDT3, the same Baire argument proves that E is barrelled. If $C \subset E$ is a closed, balanced, convex, absorbing set, then $E = \bigcup_{k \geq 1} kC$. Baire gives an interior point of some kC ; differences of two points in a small neighborhood of that point lie in $2kC$. Balanced convexity then makes C a neighborhood of zero. This proves the barrelled property from its definition.

SDT7. Actual zero jets are strongly dense

Retain the exact comparison coordinate, including its factor:

$$\mathcal{M}_0 a(s) = \int_0^\infty a(u) u^s \frac{du}{u}, \quad \kappa : Q \xrightarrow{\sim} \mathcal{Q} = \mathcal{B}/\mathcal{I}, \quad \kappa([a]) = \left[\frac{1}{2} \mathcal{M}_0 a \right]. \quad (\text{SDT7.1})$$

Here $\mathcal{B}$ is the entire strip-Schwartz space with seminorms

$$b_{R,M}(F) = \sup_{\substack{|\sigma| \leq R \\ t \in \mathbb{R}}} (1 + |t|)^M |F(\sigma + it)|,$$

and

$$\mathcal{I} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every actual nontrivial zero } \rho \text{ of the original } \zeta, 0 \leq j < m_\rho\}. \quad (\text{SDT7.2})$$

These are the full actual multiplicities. The raw source identity remains

$$\mathcal{M}_0 \Sigma f(s) = 2\zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v} \quad (\Re s > 1). \quad (\text{SDT7.3})$$

SSI's continuous inverse for a raw target $F \in \mathcal{I}$ is

$$f_F(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{F(2+it)}{2\zeta(2+it)} x^{-2-it} dt \quad (x > 0),$$

$$f_F(-x) = f_F(x), \quad f_F(0) = 0, \quad \int_{\mathbb{R}} f_F = 0, \quad \frac{f_F^{(2r)}(0)}{(2r)!} = \frac{F(-2r)}{2\zeta'(-2r)} \quad (r \geq 1). \quad (\text{SDT7.4})$$

The actual pole at 1 and trivial-zero residues are therefore not erased by the quotient-topology argument. In (SDT7.1)'s coordinate G , the raw target is $2G$.

Let

$$\delta_{\rho,j}([F]) = F^{(j)}(\rho), \quad \mathcal{D}_{\text{fin}} = \text{span}_{\mathbb{C}} \{\delta_{\rho,j} : \rho, 0 \leq j < m_\rho\} \subset \mathcal{Q}'. \quad (\text{SDT7.5})$$

Each jet is continuous: choose a circle about ρ contained in a fixed strip and use Cauchy's formula to bound its derivative by one $b_{R,0}$. It vanishes on $\mathcal{I}$, so factors continuously through the quotient. Most importantly, these functionals together separate every class:

$$\bigcap_{\rho,j} \ker \delta_{\rho,j} = \{0\} \subset \mathcal{Q}. \quad (\text{SDT7.6})$$

This is exactly the defining full-jet ideal (SDT7.2), not a finite truncation or a simplicity assumption.

The isomorphism κ transfers SDT3 and SDT6 to $\mathcal{Q}$. Suppose the strong closure of $\mathcal{D}_{\text{fin}}$ were a proper closed linear subspace of $\mathcal{Q}'_{\beta}$. Hahn-Banach separation in that Hausdorff locally convex space would give a nonzero continuous linear functional Λ vanishing on that subspace. By (SDT6.6), Λ is evaluation at a unique class $x \in \mathcal{Q}$. Since it vanishes on every $\delta_{\rho,j}$, (SDT7.6) gives $x = 0$, contrary to $\Lambda \neq 0$. Therefore

$$\overline{\mathcal{D}_{\text{fin}}^{\beta(\mathcal{Q}',\mathcal{Q})}} = \mathcal{Q}'_{\beta}. \quad (\text{SDT7.7})$$

Equivalently, for every $\lambda \in \mathcal{Q}'$, every finite family of bounded subsets $B_1, \dots, B_r \subset \mathcal{Q}$, and every $\varepsilon > 0$, there is a finite linear combination d of the actual full zero jets such that

$$\max_i \sup_{x \in B_i} |\lambda(x) - d(x)| < \varepsilon. \quad (\text{SDT7.8})$$

This is stronger than matching finitely many evaluations. It asserts density in the uniform-on-bounded-sets topology, with exactly the bounds that define that topology.

For precision about completion, E'_{β} is complete for each Fréchet E used here. Indeed a strong Cauchy net (λ_{α}) is pointwise Cauchy and defines a linear limit λ . On every bounded set B , the Cauchy estimate passes to the pointwise limit, so convergence is uniform on B . To show that λ is continuous, take any sequence $x_n \rightarrow 0$ in E . The compact set $\{0, x_1, x_2, \dots\}$ is bounded, and uniform convergence on it makes the restriction of λ a uniform limit of continuous functions. Thus $\lambda(x_n) \rightarrow 0$. Metrizability of E turns sequential continuity at zero into continuity. Hence $\lambda \in E'$, and the convergence already proved on every bounded set is strong convergence.

Consequently the Hausdorff completion of $\mathcal{D}_{\text{fin}}$, with the topology induced from $\mathcal{Q}'_{\beta}$, is canonically $\mathcal{Q}'_{\beta}$. This is a completion statement about jet functionals. It does not assert that the finite-support class space $\mathcal{Q}_{\text{fin}}$ is dense in $\mathcal{Q}$, nor that an unproved topology on that class space is its original quotient subspace topology.

SDT8. Strong-dual cohomology and the original residue map

Retain CDI's full endpoint and complex data:

$$\begin{aligned} E_+ &= \mathbb{C}e_{+,0} \oplus \mathbb{C}e_{+,1}, & E_- &= \mathbb{C}e_{-,0} \oplus \mathbb{C}e_{-,1}, & V_{\pm} &= S \oplus E_{\pm}, \\ r_+(f, e) &= \Sigma f, & r_-(h, e) &= R\Sigma h = \Sigma \hat{h}, & Ra(u) &= u^{-1}a(u^{-1}), \\ W &= V_+ \oplus V_-, & d(v_+, v_-) &= r_+v_+ - r_-v_-, & Z &= \ker d. \end{aligned} \quad (\text{SDT8.1})$$

Every original endpoint line remains present. The inverse $\Sigma^{-1} : J \rightarrow S$ gives the continuous splittings in CDI1:

$$V_{\pm} \simeq E_{\pm} \oplus J, \quad W \simeq Z \oplus J, \quad Z \simeq J \oplus E_+ \oplus E_-. \quad (\text{SDT8.2})$$

Finite topological direct sums transpose to finite topological direct sums for strong duals. To check this, a bounded set has bounded projections, and the image of a bounded set under either continuous coordinate

inclusion is bounded. The strong seminorms in each direction are therefore bounded by finite sums or maxima of the coordinate strong seminorms.

For the actual global complex

$$D = [W \xrightarrow{d} A \xrightarrow{0} A] \quad \text{in degrees } 0, 1, 2,$$

the Hom convention $d_D^n \lambda = (-1)^{n+1} \lambda \circ d_D^{-n-1}$ gives

$$D^\vee = [A'_\beta \xrightarrow{0} A'_\beta \xrightarrow{\lambda \mapsto (r'_+ \lambda, -r'_- \lambda)} V'_{+\beta} \oplus V'_{-\beta}] \quad \text{in degrees } -2, -1, 0. \quad (\text{SDT8.3})$$

CDI1's exact transpose computation gives $\ker d' = J^\perp$ and $\text{im } d' = Z^\perp$. Equation (SDT5.6) identifies the kernel topology, while (SDT8.2) identifies the quotient topology in degree zero. Thus all three identifications are now strong topological isomorphisms:

$$H^{-2}(D^\vee) = A'_\beta, \quad H^{-1}(D^\vee) \simeq Q'_\beta, \quad H^0(D^\vee) \simeq Z'_\beta. \quad (\text{SDT8.4})$$

Here cohomology has its actual kernel/subspace and quotient topologies in the displayed coefficient complex.

For each actual pole-costalk complex $L_\pm = [V_\pm \rightarrow A \rightarrow A]$, the same argument gives

$$H^{-2}(L^\vee_\pm) = A'_\beta, \quad H^{-1}(L^\vee_\pm) \simeq Q'_\beta, \quad H^0(L^\vee_\pm) \simeq E'_{\pm\beta}. \quad (\text{SDT8.5})$$

The transposed two-support maps retain their original signs:

$$\begin{aligned} A'_\beta &\longrightarrow A'_\beta \oplus A'_\beta, \quad \lambda \mapsto (\lambda, \lambda) \quad \text{in degree } -2, \\ Q'_\beta &\longrightarrow Q'_\beta \oplus Q'_\beta, \quad \mu \mapsto (\mu, -\mu) \quad \text{in degree } -1. \end{aligned} \quad (\text{SDT8.6})$$

They are continuous for the proved topologies. No sign is changed by strengthening the identification in degree minus one.

Let $\mathcal{Q}_{\text{fin}}$ denote the classes whose full jet family vanishes outside a finite set of actual zeros; every order less than the actual multiplicity remains in each retained block. The original-zeta residue comparison of CDI9 and [RD2-RD4](#) remains

$$\iota(G)(F) = \sum_{\rho} \text{Res}_{s=\rho} \frac{F(s)G(1-s)}{\zeta(s)} ds, \quad \iota : \mathcal{Q}_{\text{fin}} \xrightarrow{\sim} \mathcal{D}_{\text{fin}}. \quad (\text{SDT8.7})$$

The sum is finite for $G \in \mathcal{Q}_{\text{fin}}$. Its denominator is the original ζ . At an actual zero ρ of multiplicity m , write

$$\zeta(\rho + t) = t^m u_\rho(t), \quad u_\rho(0) \neq 0.$$

In the monomial jet bases at ρ and $1 - \rho$, its full matrix and determinant are

$$P_{ij} = (-1)^j [t^{m-1-i-j}] \frac{1}{u_\rho(t)} \quad (0 \leq i, j < m), \quad \det P = u_\rho(0)^{-m} \neq 0. \quad (\text{SDT8.8})$$

Thus SDT7 proves that the actual image of (SDT8.7) is strongly dense. It does not equip its domain with a new unstated topology or extend the residue formula to an unproved infinite sum.

The unshifted arithmetic action remains $T_a[F(s)] = [a^s F(s)]$ for $a > 0$. Its source multiplier satisfies $b_{R,M}(a^s F) \leq \max(a^R, a^{-R}) b_{R,M}(F)$, so it is continuous on $\mathcal{B}$, preserves $\mathcal{J}$, and acts continuously on $\mathcal{Q}$. The exact covariance is

$$\iota(T_a G) = a(T_{a^{-1}})' \iota(G). \quad (\text{SDT8.9})$$

For the normal coefficient with action aT_a , the invariant-pairing contragredient is

$$\mathbf{N}_a^\vee = a^{-1}(T_{a^{-1}})', \quad \iota(T_a G) = a^2 \mathbf{N}_a^\vee \iota(G). \quad (\text{SDT8.10})$$

The strong topology does not remove either factor or the inverse parameter. Continuous linear maps take bounded sets to bounded sets, so all these individual transposed operators are strongly continuous. No positivity conclusion follows from reflexivity or strong density.

One can now construct an exact completed residue receiver. Give $\mathcal{Q}_{\text{fin}}$ the initial locally convex topology of ι , explicitly

$$p_B^{\text{res}}(G) = \sup_{F \in B} \left| \sum_{\rho} \text{Res}_{s=\rho} \frac{F(s)G(1-s)}{\zeta(s)} ds \right|, \quad B \subset \mathcal{Q} \text{ bounded.} \quad (\text{SDT8.11})$$

The local determinant (SDT8.8) makes this topology Hausdorff. By definition ι is a topological isomorphism onto $\mathcal{D}_{\text{fin}}$ with its induced strong topology. SDT7 proves that the latter is dense in the complete space $\mathcal{Q}'_{\beta}$. Therefore the actual Hausdorff completion is

$$\boxed{(\widehat{\mathcal{Q}_{\text{fin}}, (p_B^{\text{res}})_B}) \xrightarrow[\iota]{\sim} \mathcal{Q}'_{\beta}.} \quad (\text{SDT8.12})$$

This is a constructed residue completion. Its topology has been specified by (SDT8.11); it is not identified with the pre-existing subspace topology from $\mathcal{Q}$.

For completeness, the ordinary-transpose comparison also has a full formula. In this paragraph $R[F(s)] = [F(1-s)]$ is the coefficient reflection, so the ordered global degree-one mirror is $-R$, not R . Set

$$\mathbf{V}_a = aT_{a^{-1}} = RT_a R. \quad (\text{SDT8.13})$$

On $\mathcal{B}$, reflection satisfies $b_{R_0, M}(F(1-s)) \leq b_{R_0+1, M}(F)$; the different letter R_0 here is a strip width. The original functional equation preserves the full multiplicities at ρ and $1-\rho$, so reflection preserves $\mathcal{J}$. Both R and T_a therefore act continuously on $\mathcal{Q}$ and preserve the finite full-zero support, including the reflected multiplicities. Equation (SDT8.9) at the parameter a^{-1} , with the prefactor a retained, gives

$$\iota(\mathbf{V}_a G) = a a^{-1} (T_a)' \iota(G) = (T_a)' \iota(G). \quad (\text{SDT8.14})$$

The operator $\mathbf{V}_a$ is therefore continuous for (SDT8.11) and extends to the actual ordinary transpose T_a' on its completion. For the original action T_a , the extension is instead $a(T_{a^{-1}})'$, exactly as in (SDT8.9).

There is also an unreflected residue map

$$\phi = \iota R, \quad \phi(G)(F) = \sum_{\rho} \text{Res}_{s=\rho} \frac{F(s)G(s)}{\zeta(s)} ds, \quad \phi T_a = (T_a)' \phi. \quad (\text{SDT8.15})$$

The middle formula substitutes $RG(1-s) = G(s)$ before taking residues, so it introduces no change-of-variable sign. The last identity follows either directly by multiplying the integrand by a^s , or from $RT_a = \mathbf{V}_a R$ and (SDT8.14).

Its exact topological receiver can be specified without assuming continuity of R in an unexamined topology: equip $\mathcal{Q}_{\text{fin}}$ with

$$p_B^{\text{unref}}(G) = \sup_{F \in B} \left| \sum_{\rho} \text{Res}_{s=\rho} \frac{F(s)G(s)}{\zeta(s)} ds \right| = p_B^{\text{res}}(RG). \quad (\text{SDT8.16})$$

Since $R^2 = 1$ and $R\mathcal{Q}_{\text{fin}} = \mathcal{Q}_{\text{fin}}$, it is a homeomorphism from this unreflected topology to the reflected topology (SDT8.11). The map ϕ has the same range $\mathcal{D}_{\text{fin}}$, so its completion is also canonically $\mathcal{Q}'_{\beta}$, now intertwining T_a with the ordinary transpose T_a' . These are two explicitly related residue topologies; neither has been silently substituted for the inherited topology of $\mathcal{Q}_{\text{fin}} \subset \mathcal{Q}$.

SDT9. Exact implication and limits of this calculation

The proved chain is

$$\begin{array}{ll}
 \text{original weighted derivative estimates on } A & \text{bounded-compact property and the actual Fréchet topology} \\
 \Downarrow \text{SDT2-SDT3} & \Downarrow \text{SDT6} \\
 \text{every bounded subset of } Q \text{ has compact closure} & (Q'_\beta)' = Q \\
 \Downarrow \text{SDT4-SDT5} & \Downarrow \text{SDT7} \\
 \text{compact lifts and exact bounded lifts through } q & \overline{\mathcal{D}_{\text{fin}}}^\beta = Q'_\beta. \\
 \Downarrow & \\
 Q'_\beta \simeq J^\perp \subset A'_\beta, &
 \end{array}
 \tag{SDT9.1}$$

The arrows name the exact proofs that justify the topology change. The compact lift is the finite-choice series (SDT4.7), not a claimed linear section. The dense space at the bottom right consists of functionals, not finite-support classes in the primal quotient.

This establishes the full strong-dual cohomology topology for the actual CDI coefficient complexes and strengthens their finite-jet detection to uniform approximation on bounded subsets. It neither constructs a Verdier dualizing sheaf nor identifies these infinite-dimensional coefficient duals with a finite-rank étale receiver. No weight-separation or RH conclusion is inferred from these topological statements.

The actual continuous support-dual current sheaf — CSD0-CSD12

Proof source: CC_CONTINUOUS_SUPPORT_DUAL_SHEAF.md.

The continuous support-dual sheaf on the actual sphere

Complete independent derivation, 24 September 2026. Proof locators CSD0-CSD12.

CSD0. Input stage and the category being constructed

The supporting datum remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. All coefficient vector spaces and all complex coordinates below belong to the receiving geometry after complete arithmetic reconstruction. No addition, numerical coordinate, midpoint or distance is assigned to τ .

For this continuation the current READ_FIRST_USER_CONSTRUCTION.md, the connected USER_ARGUMENT_RECONSTRUCTION.md and correction table, PC01-PC05 including its latest continuation, and verbatim WU050-WU055 and WU061-WU063 were recalled. The no-addition, no-metric, full-history and separate-branch corrections govern the same receiver as in CSP and CDW. The actual source maps are retained; the kernel J is not replaced by its quotient Q .

The original-source use is that recorded in CSP0 and CDW0: Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, [arXiv:0903.2024v3](#), §5, and the field-valued charts and real structure of [arXiv:2609.00299v1](#). Those source readings and the local reconstructions are different provenance claims; this continuation does not claim a fresh complete reading of either paper.

The complete current proof CC_CONTINUOUS_DUAL_INDEPENDENT.md, CDI0-CDI12, supplies the continuous duals of the already constructed global and support coefficient complexes. Its full maps are the comparison target here. CSP0-CSP13 and CDW0-CDW10 supply the sphere sheaf and one compatible fine resolution. SSI1-SSI7 and ESI give the closed image and continuous inverse of the full original summation source.

Before the final topology and transfer comparisons below, the complete accepted CC_STRONG_DUAL_TOPOLOGY_AND_JETS.md, SDT0-SDT9, and CC_RAMIFIED_TRACE_AND_RESIDUE_ACTION.md, GTR0-GTR9, were read. Their bounded-lift theorem and separate weighted geometric trace are imported with their exact scope; neither is attributed to the preceding current construction.

An actual complex of sheaves with continuous maps is constructed below from continuous functionals on compactly supported tests. It is not defined by an unrestricted algebraic internal Hom. In particular A' always means the continuous complex-linear dual of the stated locally convex A , never its algebraic dual. No equivalence for arbitrary topological sheaves is assumed.

CSD1. Full primal source and single resolution

Let $Y = \mathbb{P}^1(\mathbb{C})$, $P = \{0, \infty\}$, and $i : P \hookrightarrow Y$. Retain

$$\begin{aligned}
 S &= \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0\}, \quad V_{\pm} = S \oplus E_{\pm}, \quad E_{\pm} = \mathbb{C}^2, \\
 A &= \left\{ a \in C^{\infty}(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \text{ for all } N, j \geq 0 \right\}, \\
 \Sigma f(u) &= 2 \sum_{k \geq 1} f(ku), \quad Ra(u) = u^{-1}a(u^{-1}), \quad \hat{f}(t) = \int_{\mathbb{R}} f(v) e^{-2\pi i vt} dv, \\
 r_+(f, e) &= \Sigma f, \quad r_-(h, e) = R\Sigma h = \Sigma \hat{h}.
 \end{aligned} \tag{CSD1.1}$$

The last equality uses the full Poisson formula with terms $u^{-1}h(0) - \int h$, which vanish on this stated S . Both coordinates of each $E_{\pm}$, labelled value at zero and real integral, remain distinct.

The actual source image $J = \Sigma S$ is closed, $\Sigma : S \rightarrow J$ is a topological isomorphism, and $Q = A/J$ has its Fréchet quotient topology. The proved inverse-image sheaf is

$$\mathcal{F} = \pi^{-1}\Omega = \underline{A}_Y \times_{i_*(A \oplus A)} i_*(V_+ \oplus V_-). \tag{CSD1.2}$$

Here $\pi(0) = x_+$, $\pi(\infty) = x_-$, and $\pi(\mathbb{C}^{\times}) = \eta$. Its generic inverse image is the entire punctured plane, not a numerical coordinate assigned to τ .

Put $V = V_+ \oplus V_-$. The difference map is onto on stalks through its evaluation component and gives

$$0 \rightarrow \mathcal{F} \rightarrow \underline{A}_Y \oplus i_* V \xrightarrow{\text{ev}-r} i_* A^2 \rightarrow 0. \tag{CSD1.3}$$

For smooth A -valued differential forms $\mathcal{E}_A^q$, the single resolution from CDW2 is

$$\begin{aligned}
 \mathcal{K}^0 &= \mathcal{E}_A^0 \oplus i_* V, \quad \mathcal{K}^1 = \mathcal{E}_A^1 \oplus i_* A^2, \quad \mathcal{K}^2 = \mathcal{E}_A^2, \\
 d^0(f, v) &= (df, \text{ev}(f) - rv), \quad d^1(\omega, a) = d\omega.
 \end{aligned} \tag{CSD1.4}$$

Its degree-zero kernel is (CSD1.2). At a pole, a closed pair (ω, a) has primitive $f = g + a - g(p)$, $v = 0$, where $dg = \omega$. Away from the poles the coefficient-valued Poincaré lemma applies. This proves exactness in positive degrees with continuous local primitives. Forms and finite skyscrapers are fine.

This resolution also computes compactly supported cohomology. A section defined near a compact set extends after multiplication by a smooth cutoff equal to one on a smaller neighborhood of that set. The partition-of-unity contraction of the augmented Čech complex therefore works with compact supports: only finitely many members of a locally finite partition meet an input compact support, and all multiplications and sums are continuous. This proves the needed compact-support acyclicity for these form and skyscraper terms. It does not imply acyclicity for sections supported at a single point.

CSD2. Continuous compact-support contraction

Set $\mathcal{E}_c^q(U; A) = \Gamma_c(U, \mathcal{E}_A^q)$. Its topology is the locally convex inductive limit over compact supports, each given all smooth-derivative seminorms from A . Finite pole sums have their product topologies. Extension by zero of compactly supported forms is continuous.

An explicit compact-support Poincaré calculation is useful. In an oriented plane chart choose $\rho_x, \rho_y \in C_c^\infty(\mathbb{R})$ with integrals one. Define

$$I_x f(y) = \int_{\mathbb{R}} f(x, y) dx, \quad H_x f(x, y) = \int_{-\infty}^x f(t, y) dt - \left(\int_{-\infty}^x \rho_x(t) dt \right) I_x f(y), \quad (\text{CSD2.1})$$

and define H_y similarly. Completeness of A supplies the integrals. For every continuous seminorm,

$$p\left(\int f\right) \leq \int p(f), \quad (\text{CSD2.2})$$

which also proves continuity after any fixed number of derivatives.

For compactly supported forms put

$$K(P dx + Q dy) = H_x P, \\ K(G dx \wedge dy) = H_x G dy - \rho_x(x) H_y(I_x G)(y) dx, \quad K|_{\mathcal{E}_c^0} = 0. \quad (\text{CSD2.3})$$

Each output is supported in a fixed rectangle determined by the input support and the two fixed bumps. Thus these are continuous operators on the stated inductive limits. Direct differentiation gives

$$Kdf = f, \quad dK\omega + Kd\omega = \omega \quad (\deg \omega = 1), \\ dK\omega = \omega - \rho_x \rho_y dx \wedge dy \int_{\mathbb{R}^2} \omega \quad (\deg \omega = 2). \quad (\text{CSD2.4})$$

Indeed $\partial_x H_x f = f - \rho_x I_x f$ and $H_y \partial_y g = g$ for compactly supported g give every term in these identities. The bump two-form has positive integral one and is annihilated by K . This is a continuous contraction onto A in degree two. An orientation-preserving disk-to-plane diffeomorphism transfers it to every disk below.

On the cylinder $t = \log |z|$, $\theta = \arg z$, combine the one-dimensional compact contraction in t with the following circle homotopy: subtract the angular mean, integrate that mean-zero coefficient from zero to θ , then subtract the mean of the primitive. Its derivative is the original coefficient minus its mean and it is periodic. The surviving circle classes are constants and $d\theta/(2\pi)$. The compact cylinder classes are consequently

$$\rho(t) dt \quad (\deg 1), \quad \rho(t) dt \wedge \frac{d\theta}{2\pi} \quad (\deg 2), \quad \int \rho = 1. \quad (\text{CSD2.5})$$

All homotopies are continuous and use actual test forms, not an unproved homotopy-invariance assertion for topological coefficients.

CSD3. The continuous-current sheaf

For $-2 \leq n \leq 0$, define

$$\mathcal{T}_A^n(U) = (\mathcal{E}_c^{-n}(U; A))', \quad (dT)(\phi) = (-1)^{n+1} T(d\phi). \quad (\text{CSD3.1})$$

Other degrees vanish. Each dual may carry the weak topology of pointwise evaluation or the strong topology of uniform convergence on bounded test sets. Both use exactly the same continuous functionals.

Restriction is transpose to extension by zero. To prove the sheaf axiom, decompose a compactly supported test form by a locally finite subordinate partition of unity and evaluate compatible local functionals on its pieces. Only finitely many terms meet the compact support. Compatibility proves independence of the partition and uniqueness; continuity on every fixed-support space proves continuity on the specified

inductive limit. Restriction and scalar multiplication of currents are continuous for both dual topologies: they transpose continuous operations on tests which take bounded sets to bounded sets.

The differential is the continuous Hom differential, so $d^2 = 0$. Smooth functions multiply currents on their test argument, making every term a fine sheaf. This gives ordinary-section acyclicity on all opens used below.

The positive complex orientation defines

$$\underline{A}'_Y[2] \longrightarrow \mathcal{T}_A^\bullet, \quad c_\lambda(\phi) = \int_U \lambda(\phi) \quad (\phi \text{ a compactly supported two-form}). \quad (\text{CSD3.2})$$

It is continuous and closed by Stokes. Transpose the explicit contraction (CSD2.4): this proves that (CSD3.2) is a local quasi-isomorphism with continuous inverse on cohomology. In particular its coefficient is the entire continuous A' . The weak-topology comparison is a homeomorphism. For this local coefficient the strong-topology comparison also follows from the two explicit bounded-set preserving test maps.

No tensor-product identification between scalar distributions and A' is used. Currents mean exactly (CSD3.1). At a pole define the positive Dirac current

$$\delta_p \lambda(f) = \lambda(f(p)), \quad i_* A' \longrightarrow \mathcal{T}_A^0. \quad (\text{CSD3.3})$$

This is the actual local Gysin chain map, with its orientation fixed below.

CSD4. The actual support-dual complex

Using the ordinary cone differential $d(b, a) = (d_B b + g a, -d_A a)$, put

$$\mathcal{D}_c \mathcal{F} = \text{Cone} \left(i_* A'^2 \xrightarrow{(\delta, -r')} \mathcal{T}_A^\bullet \oplus i_* V' \right). \quad (\text{CSD4.1})$$

The domain is in degree zero. Its complete terms and maps are

$$\begin{aligned} (\mathcal{D}_c \mathcal{F})^{-2} &= \mathcal{T}_A^{-2}, & (\mathcal{D}_c \mathcal{F})^{-1} &= \mathcal{T}_A^{-1} \oplus i_* A'^2, & (\mathcal{D}_c \mathcal{F})^0 &= \mathcal{T}_A^0 \oplus i_* V', \\ d^{-2} T &= (dT, 0), & d^{-1}(T, \lambda_+, \lambda_-) &= (dT + \delta_0 \lambda_+ + \delta_\infty \lambda_-, -r'_+ \lambda_+, -r'_- \lambda_-). \end{aligned} \quad (\text{CSD4.2})$$

Every map is continuous for both dual topologies.

The cone is the literal continuous transpose of the compact-support resolution: for every open U ,

$$\Gamma(U, \mathcal{D}_c \mathcal{F}) = (\Gamma_c(U, \mathcal{K}^\bullet))^\vee, \quad (C^\vee)^n = (C^{-n})', \quad d_n^\vee = (-1)^{n+1} (d_C^{-n-1})'. \quad (\text{CSD4.3})$$

For degree -1 , transposing $d^0(f, v)$ gives $T(df) + \sum_p \lambda_p(f(p) - r_p v_p)$, exactly (CSD4.2). Degree -2 has the minus sign in (CSD3.1). Restriction in (CSD4.3) is adjoint to extension by zero.

Evaluation is a cochain pairing: its two differential contributions are $(-1)^{n+1} \lambda(dc)$ and $(-1)^n \lambda(dc)$, whose sum is zero. Formula (CSD4.3), together with the proved compact-support resolution, is the precise continuous support-duality meaning used here. It is an actual sheaf complex, not a declaration that general continuous dualization is exact.

Using (CSD3.2), it represents the candidate

$$\text{Cone}(i_* A'^2 \xrightarrow{(\text{Gys}, -r')} \underline{A}'_Y[2] \oplus i_* V'), \quad (\text{CSD4.4})$$

whose derived Gysin arrow is represented exactly by (CSD3.3).

CSD5. Pole stalk and original compact-support pairing

Use the positive local coordinate $z_p = z$ at zero and $z_p = 1/z$ at infinity. The form

$$\vartheta_p = \frac{d \arg z_p}{2\pi} \quad (\text{CSD5.1})$$

is locally integrable. Its degree-minus-one current is $h_p \lambda(\phi) = \int \vartheta_p \wedge \lambda(\phi)$. The local $1/r$ coefficient is integrable against the area factor $r dr d\theta$. Stokes on a punctured disk with compact test f gives

$$d(h_p \lambda)(f) = h_p \lambda(df) = \lambda(f(p)), \quad dh_p \lambda = \delta_p \lambda. \quad (\text{CSD5.2})$$

The positive small-circle limiting integral fixes the sign.

The primal pole costalk is $L_p = [V_p \xrightarrow{r_p} A \xrightarrow{0} A]$ in degrees 0, 1, 2. Its dual differential is r'_p in degrees $-1, 0$. The actual stalk comparison is

degree	$L_p^\vee \longrightarrow (\mathcal{D}_c \mathcal{F})_p$	
-2	$\lambda \mapsto c_\lambda$	
-1	$\mu \mapsto (h_p \mu, -\mu)$	
0	$\nu \mapsto (0, \nu)$	

(CSD5.3)

Indeed $d(h_p \mu, -\mu) = (0, r'_p \mu)$. The local current contraction leaves only A' in degree -2 ; the Dirac arrow is locally null-homotopic through h_p , leaving exactly r'_p . This proves the quasi-isomorphism with continuous maps. The same construction on shrinking disks gives the actual stalk.

For a direct pairing check choose $\chi \in C_c^\infty$ equal to one near p , and a compact positive two-form ω_c of integral one. The primal compact-source chain map is

$$L_p \longrightarrow \Gamma_c(U, \mathcal{K}), \quad v \mapsto (\chi r_p v, v), \quad a \text{ in degree } 1 \mapsto (d\chi a, 0), \quad a \text{ in degree } 2 \mapsto \omega_c a. \quad (\text{CSD5.4})$$

Its differential is r_p . The compact contraction (CSD2.4) followed by elimination of the evaluation coordinate proves it is a quasi-isomorphism. Pairing it with (CSD5.3) gives

$$\lambda(a), \quad \int \vartheta_p \wedge d\chi \mu(a) = \mu(a), \quad \nu(v). \quad (\text{CSD5.5})$$

The middle identity is (CSD5.2) on test χ . This is precisely CDI's positive-orientation costalk pairing.

For clarity about the cohomology topology, on each fixed disk U , transposing the compact-test map (CSD5.4) gives a continuous reverse chain map from sections of the current complex on that disk. It is not an evaluation of an arbitrary current germ on a fixed test support. In degrees $-2, -1, 0$ it is

$$\begin{aligned} T &\mapsto (a \mapsto T(\omega_c a)), & (T, \lambda) &\mapsto (a \mapsto T(d\chi a)), \\ (T, \nu) &\mapsto \nu + r'_p(a \mapsto T(\chi a)). \end{aligned} \quad (\text{CSD5.7})$$

It composes with the disk version of (CSD5.3) to the identity. Because that map is already a quasi-isomorphism, the induced cohomology maps are inverse continuous maps. Their cohomology identifications agree under shrinking disks: degree-minus-two cocycles are constant currents; for a degree-minus-one cocycle (T, λ) , the equation $dT + \delta_p \lambda = 0$ gives $T(d\chi a) = -\lambda(a)$, independent of the cutoff; and degree-zero cutoff changes lie in $\text{im } r'_p$. The underlying sheaf stalk therefore has these groups. Its cohomology topology in this proof means the stable small-disk cohomology topology, identified by these identity transition maps. It is not a claim that arbitrary locally convex inductive limits of germs commute

with forming cohomology topologies. This gives the precise topological comparison for both stated dual topologies before identifying the quotient coefficient through SDT below.

Consequently

$$H^{-2}(i_p^* \mathcal{D}_c \mathcal{F}) = A', \quad H^{-1}(i_p^* \mathcal{D}_c \mathcal{F}) = Q', \quad H^0(i_p^* \mathcal{D}_c \mathcal{F}) = E'_p. \quad (\text{CSD5.6})$$

The middle identification is $\ker r'_p = J^\perp = q' Q'$. The last uses the actual splitting $V_p = E_p \oplus J$ from Σ^{-1} , and Hahn-Banach extends every continuous functional on closed J to A . Thus these are the entire continuous duals.

CSD6. Derived point support, computed separately

Fineness does not make a current sheaf automatically acyclic for support at a point. A current on a punctured disk need not extend across its missing point. Compute support by the actual restriction fibre cone

$$R\Gamma_{\{p\}}(U, \mathcal{D}_c \mathcal{F}) = \text{Cone}(\Gamma(U, \mathcal{D}_c \mathcal{F}) \rightarrow \Gamma(U \setminus \{p\}, \mathcal{D}_c \mathcal{F}))[-1]. \quad (\text{CSD6.1})$$

The ordinary sections here compute hypercohomology because the displayed terms are fine. They are not replaced by termwise point-supported currents.

On a disk the current complex has A' in degree -2 . On its puncture it has A' in degrees $-2, -1$, represented by $c_\lambda, h_p \lambda$, by the cylinder contraction (CSD2.5). In particular $h_p \lambda$ pairs as $-\lambda(a)$ with $\rho(t)dt a$, since $d\theta \wedge dt = -dt \wedge d\theta$. This radial sign is retained.

Restriction on (CSD5.3) is identity in degrees $-2, -1$ and zero in degree zero. In the fibre-cone convention $d(x, y) = (dx, \text{res}(x) - dy)$, cancel its contractible degree-minus-two pair. The remaining complex is

$$A' \longrightarrow V'_p \oplus A', \quad \mu \longmapsto (r'_p \mu, \mu). \quad (\text{CSD6.2})$$

The last coordinate records the punctured angular current. The positive supported coordinate is its negative: in the same cone, $d(h_p \lambda, 0) = (\delta_p \lambda, h_p \lambda)$, so $(\delta_p \lambda, 0)$ is cohomologous to $(0, -h_p \lambda)$. Changing to that positive supported coordinate and negating the source rewrites (CSD6.2) as

$$A' \xrightarrow{\lambda \mapsto (\lambda, -r'_p \lambda)} A' \oplus V'_p \quad \text{in degrees } -1, 0. \quad (\text{CSD6.3})$$

The first target is now measured by the positive Dirac orientation.

The continuous deformation retraction onto $V'_p[0]$ is

$$P(a, \nu) = \nu + r'_p a, \quad S(\nu) = (0, \nu), \quad h(a, \nu) = a. \quad (\text{CSD6.4})$$

One has $PS = 1$, $hd = 1$ on degree -1 , and $dh(a, \nu) = (a, -r'_p a) = (a, \nu) - SP(a, \nu)$. Therefore

$$\boxed{i_p^! \mathcal{D}_c \mathcal{F} \simeq V'_p[0] = (i_p^* \mathcal{F})^\vee.} \quad (\text{CSD6.5})$$

Costalk-to-stalk is $\nu \mapsto (0, \nu)$ in (CSD5.3); on degree-zero cohomology it is restriction $V'_p \rightarrow E'_p$. Together with CSD5 this proves both support/stalk comparisons from actual local complexes.

CSD7. Global angular current and the CDI chain comparison

The angular form $\vartheta = d \arg z / (2\pi)$ is locally integrable on the sphere. At infinity it is $-\vartheta_-$; hence

$$d(\vartheta \lambda) = \delta_0 \lambda - \delta_\infty \lambda. \quad (\text{CSD7.1})$$

The original global complex and its continuous dual are

$$D = [V_+ \oplus V_- \xrightarrow{r_+ - r_-} A \xrightarrow{0} A],$$

$$D^\vee = [A' \xrightarrow{0} A' \xrightarrow{\mu \mapsto (r'_+ \mu, -r'_- \mu)} V'_+ \oplus V'_-] \quad (\text{CSD7.2})$$

in degrees 0, 1, 2 and $-2, -1, 0$, respectively. There is an actual continuous chain map

$$\Psi : D^\vee \longrightarrow \Gamma(Y, \mathcal{D}_c \mathcal{F}),$$

$$\lambda \mapsto c_\lambda \quad (\text{deg } -2), \quad \mu \mapsto (\vartheta \mu; -\mu, +\mu) \quad (\text{deg } -1), \quad (\nu_+, \nu_-) \mapsto (0; \nu_+, \nu_-) \quad (\text{deg } 0). \quad (\text{CSD7.3})$$

The middle differential's current terms cancel by (CSD7.1), leaving precisely $(r'_+ \mu, -r'_- \mu)$.

To prove this is a quasi-isomorphism, compute the global current complex using the local and cylinder contractions (CSD2.4)-(CSD2.5) and two-chart Mayer-Vietoris. Its cohomology is A' in degrees $-2, 0$ and zero in degree -1 . The first class is c_λ ; the last is evaluated on constant test functions. Both positive Dirac currents have that last value λ . All comparisons are finite sums, the written integral operators and partition-of-unity maps, so they are continuous. This computes the current cohomology without assuming arbitrary exactness of continuous dualization.

The cone then has the finite model

$$P^{-2} = A', \quad P^{-1} = A'^2, \quad P^0 = A' \oplus V'_+ \oplus V'_-,$$

$$d^{-1}(\lambda_+, \lambda_-) = (\lambda_+ + \lambda_-, -r'_+ \lambda_+, -r'_- \lambda_-). \quad (\text{CSD7.4})$$

Its explicit deformation retraction is

degree	$P \rightarrow D^\vee$	$D^\vee \rightarrow P$
-2	1	1
-1	$(\lambda_+, \lambda_-) \mapsto -\lambda_+$	$\mu \mapsto (-\mu, \mu)$
0	$(a, \nu_+, \nu_-) \mapsto (\nu_+, \nu_- + r'_- a)$	$(\nu_+, \nu_-) \mapsto (0, \nu_+, \nu_-)$

(CSD7.5)

The homotopy $h^0(a, \nu_+, \nu_-) = (0, a)$ satisfies $dh + hd = 1 - SP$; substitute into (CSD7.4) to check both degrees. The current map (CSD7.3) induces the section in (CSD7.5), proving its quasi-isomorphism.

There is also a direct pairing check. Set

$$\omega_Y = \frac{1}{\pi} \frac{dx \wedge dy}{(1 + |z|^2)^2}, \quad \int_Y \omega_Y = 1.$$

The primal finite complex maps into global sections of $\mathcal{K}$ by

$$(v_+, v_-) \mapsto (\text{constant}(r_- v_-), v_+, v_-), \quad b \text{ in degree } 1 \mapsto (0; -b, 0), \quad a \text{ in degree } 2 \mapsto \omega_Y a. \quad (\text{CSD7.6})$$

Its differential is $r_+ - r_-$, including the displayed degree-one minus. Pairing (CSD7.3) and (CSD7.6) gives the original evaluation in every degree. Thus this realizes CDI's pairings and signs, not only its cohomology groups.

The continuous transpose of (CSD7.6) is an explicit cohomological inverse to (CSD7.3). Put $\ell_T(a) = T(\text{constant}(a))$ for a degree-zero current. In degrees $-2, -1, 0$, this reverse chain map is

$$T \mapsto (a \mapsto T(\omega_Y a)), \quad (T, \lambda_+, \lambda_-) \mapsto -\lambda_+,$$

$$(T, \nu_+, \nu_-) \mapsto (\nu_+, \nu_- + r'_- \ell_T). \quad (\text{CSD7.8})$$

For degree -1 , its chain identity is $(-r'_+ \lambda_+, r'_- \lambda_+) = d_{D^\vee}(-\lambda_+)$. It composes with Ψ to the identity in every degree. All test embeddings in this formula are continuous and take bounded sets to bounded sets. Therefore the induced inverse cohomology maps are continuous for both weak and strong dual topologies. No theorem that continuous dualization preserves arbitrary quasi-isomorphisms is used to prove this assertion.

In particular

$$\begin{aligned} H^{-2}(Y, \mathcal{D}_c \mathcal{F}) &= A', & H^{-1}(Y, \mathcal{D}_c \mathcal{F}) &= Q', & H^0(Y, \mathcal{D}_c \mathcal{F}) &= Z', \\ Z &= \{(v_+, v_-) : r_+ v_+ = r_- v_-\} \simeq J \oplus E_+ \oplus E_-. \end{aligned} \quad (\text{CSD7.7})$$

The last comparison uses the original Fourier-source inverse and retains all four endpoint lines.

CSD8. Restriction realizes the transposed support map

Restriction of the global angular current is h_+ at zero and $-h_-$ at infinity. Thus (CSD7.3) and (CSD5.3) give exactly

$$\begin{aligned} D^\vee &\rightarrow L_+^\vee : (\deg 0, \deg -1, \deg -2) = (\text{pr}_{V'_+}, +1, +1), \\ D^\vee &\rightarrow L_-^\vee : (\deg 0, \deg -1, \deg -2) = (\text{pr}_{V'_-}, -1, +1). \end{aligned} \quad (\text{CSD8.1})$$

These are the transposes of the original support-to-global map

$$L_+ \oplus L_- \rightarrow D, \quad k^0 = 1, \quad k^1(b_+, b_-) = b_+ - b_-, \quad k^2(c_+, c_-) = c_+ + c_-. \quad (\text{CSD8.2})$$

Consequently degree-minus-one is anti-diagonal and degree-minus-two diagonal:

$$Q' \rightarrow Q'^2 : \mu \mapsto (\mu, -\mu), \quad A' \rightarrow A'^2 : \lambda \mapsto (\lambda, \lambda). \quad (\text{CSD8.3})$$

Degree zero is endpoint restriction $Z' \rightarrow E'_+ \oplus E'_-$.

The complete cone comparison is

$$\text{Cone}(k)^\vee \longrightarrow \text{Cone}(k^\vee)[-1], \quad (\phi, \psi) \mapsto ((-1)^{n+1} \psi, \phi) \text{ in degree } n. \quad (\text{CSD8.4})$$

Substitution into the Hom and shifted-cone differentials proves it is a chain isomorphism. The dual long row is therefore the full CDI5 row

$$0 \rightarrow A' \xrightarrow{\Delta} A'^2 \xrightarrow{-\lambda_+ + \lambda_-} A' \xrightarrow{0} Q' \xrightarrow{(\mu, -\mu)} Q'^2 \xrightarrow{q'(\mu_+ + \mu_-)} A' \xrightarrow{r'} Z' \rightarrow E'_+ \oplus E'_- \rightarrow 0. \quad (\text{CSD8.5})$$

Here $r : Z \rightarrow A$ is common restriction and $r' \lambda = (\lambda|_J, 0, 0)$ in (CSD7.7). Diagonal/difference identities and Hahn-Banach on $J \subset A$ prove exactness. Thus the earlier continuous coefficient duality has an actual sheaf support construction realizing its maps.

The ordinary annular dual in (CSD8.4) has degrees $-1, 0$. It is not confused with $R\Gamma(\mathbb{C}^\times, \mathcal{T}_A)$, whose degrees are $-2, -1$ because it is dual to compactly supported annular cohomology. Formula (CSD4.3) specifies the support convention.

CSD9. Proper geometric transposes and both mirrors

A continuous coefficient-diagram map acts on compact tests and, contravariantly, on this current complex. All evaluation comparisons are natural for these actual transposes.

For a recovered positive integer n , $b_n(z) = z^n$ is proper. Hence test pullback preserves compact supports and its transpose defines

$$(b_{n*} T)(\phi) = T(b_n^* \phi). \quad (\text{CSD9.1})$$

After including the transposed coefficient action, this is the proper current pushforward $b_{n*}\mathcal{D}_c\mathcal{F} \rightarrow \mathcal{D}_c\mathcal{F}$, adjoint to primal pullback. It is not pullback on the dual object and is not an inverse root map.

The primal coefficient action is

$$T_a b(u) = b(u/a), \quad \rho_+(a)(f, c_0, c_1) = (f(\cdot/a), c_0, ac_1), \quad \rho_-(a)(h, d_0, d_1) = (ah(a\cdot), ad_0, d_1). \quad (\text{CSD9.2})$$

The current transpose uses T'_n and $\rho_\pm(n)'$. Under (CSD7.3) its complete finite-model action is

$$(nT'_n, T'_n, \rho_+(n)' \oplus \rho_-(n)') \quad \text{in degrees } -2, -1, 0. \quad (\text{CSD9.3})$$

For a direct check, $b_{n*}c_\lambda = nc_\lambda$, $b_{n*}\vartheta = \vartheta$, and $b_{n*}\delta_p = \delta_p$. On the punctured receiver the n inverse branches each contribute $1/n$ of the angular form to its trace. The locally integrable current equality extends across the poles. These calculations verify the precise current/global comparison. The broader geometric trace with other coefficient conventions is a separate receiver map, not identified with this one by terminology.

The specified mirrors and their sphere orientation degrees are

$$w(z) = -1/z, \quad \varepsilon_w = +1; \quad \alpha(z) = -1/\bar{z}, \quad \varepsilon_\alpha = -1. \quad (\text{CSD9.4})$$

Both transpose the pole swap and use R' . For a current represented by a form, proper pushforward acts as $\varepsilon_g(g^{-1})^*$ because the integration orientation is fixed. Thus

$$g_*c_\lambda = \varepsilon_g c_\lambda, \quad g_*\vartheta = -\vartheta \quad (g = w, \alpha), \quad (\text{CSD9.5})$$

using $w^*\vartheta = -\vartheta$, $\alpha^*\vartheta = +\vartheta$. The global dual actions are exactly

$$(\varepsilon_g R', -R', \text{dual pole swap}) \quad \text{in degrees } -2, -1, 0. \quad (\text{CSD9.6})$$

At pole stalks each angular current uses its own positive coordinate. Degree minus one is $+R'$ with pole swap, while degree minus two is $\varepsilon_g R'$ with swap. Hence (CSD8.1) intertwines the mirrors with all CDI7 signs.

The coefficient contragredient giving an invariant pairing is different. On the already constructed cohomological representation extended to $a > 0$, it is

$$(a^{-1}(T_{a^{-1}})', (T_{a^{-1}})', \rho_+(a^{-1})' \oplus \rho_-(a^{-1})'). \quad (\text{CSD9.7})$$

It satisfies $\langle B_a^\vee \lambda, B_a x \rangle = \langle \lambda, x \rangle$. This is CDI6's inverse-parameter coefficient operation. It is not asserted to arise from a holomorphic $z^{1/a}$ and is not substituted for (CSD9.1).

There is now also a separately constructed weighted trace on the primal sheaf. GTR2 uses the sum over all n interior sheets and n times the identity on each ramified pole space, including both endpoint coordinates. It proves $\text{Tr}_n b_n^* = n \cdot 1$ on that sheaf. GTR3-GTR4 prove its global degree factors $(n, n, 1)$, and, after composing with the inverse coefficient action, the representative

$$\mathbf{U}_n = (n\rho(n^{-1}), nT_{n^{-1}}, T_{n^{-1}}), \quad \mathbf{U}_n \mathbf{B}_n = \mathbf{B}_n \mathbf{U}_n = n \cdot 1 \quad (\text{CSD9.8})$$

in primal degrees 0, 1, 2, where $\rho = \rho_+ \oplus \rho_-$. Its transpose on the full coefficient dual consequently is

$$\mathbf{U}'_n = (T'_{n^{-1}}, nT'_{n^{-1}}, n\rho(n^{-1})') = n(\mathbf{B}_n^{-1})' \quad (\text{CSD9.9})$$

in degrees $-2, -1, 0$. These are the induced maps and the explicitly constructed coefficient representative from GTR, not a claim that a chain lift to currents is uniquely determined by its cohomology. In particular the middle operator $nT'_{n^{-1}}$ in (CSD9.9) is different from T'_n in the current pushforward (CSD9.3). GTR5 proves that the original reflected residue map intertwines the former operator. The current support-duality comparison keeps these maps distinct and retains the ramification factor.

CSD10. Topologies and the proved extent of duality

Every current and functional is continuous for the actual Fréchet and compact-test topologies. The comparisons use explicit integrations, cutoffs and continuous finite operations. Exactness of continuous dualization for arbitrary locally convex complexes has not been assumed.

The weak-dual cohomology topologies are precisely CDI3's. The annihilator $J^\perp \subset A'$ has the weak topology of Q' : evaluation at a is evaluation at $q(a)$, and every quotient vector has a representative. The actual continuous decompositions $V_p = E_p \oplus J$ and $V_+ \oplus V_- = Z \oplus J$ transpose to finite decompositions, identifying the degree-zero quotient topologies. The current homotopies and comparison maps preserve those identifications. All these pairings are separated and involve the entire continuous dual.

For strong duals every displayed map remains continuous. The induced topology on $Q' \subset A'_\beta$ initially has the seminorm family

$$\mu \mapsto \sup_{a \in B} |\mu(q(a))|, \quad B \subset A \text{ bounded.} \quad (\text{CSD10.1})$$

The accepted SDT2-SDT5 proof now identifies this family with the full strong topology of Q' . It proves compact closure of every bounded subset of Q using the original two-sided weighted derivative bounds, then constructs a compact lift through q as a uniformly convergent series of finite choices. Intersecting that compact lift with the inverse image of a bounded subset $B \subset Q$ gives a bounded set $\widetilde{B} \subset A$ with $q(\widetilde{B}) = B$. Hence

$$\sup_{y \in B} |\mu(y)| = \sup_{a \in \widetilde{B}} |q' \mu(a)|, \quad Q'_\beta \xrightarrow[q']{\sim} J^\perp \subset A'_\beta. \quad (\text{CSD10.2})$$

Conversely, continuous q takes bounded subsets of A to bounded subsets of Q , giving the other topology inclusion.

The finite source splittings used above give the strong topology on the degree-zero groups, and the fixed-test maps (CSD5.7) and (CSD7.8) identify the current cohomology topologies with those of the finite coefficient complexes. Thus the global identifications are strong topological isomorphisms with $A'_\beta, Q'_\beta, Z'_\beta$. The pole stalk groups, equipped with the stable small-disk cohomology topology specified in CSD5, are strongly topologically isomorphic to $A'_\beta, Q'_\beta, E'_{p\beta}$. The supported comparison (CSD6.4) uses continuous finite operations on the local restriction cone and gives $V'_{p\beta}$ with the analogous stable local topology. This imports exactly SDT's proved bounded-lift result; it does not assume or assert strong openness of the separate restriction $A'_\beta \rightarrow J'_\beta$.

Ordinary algebraic sheaf Hom is not invoked to turn these continuous duals into algebraic duals. The exact support-dual object established here is (CSD4.1), with test-functional characterization (CSD4.3) and its proved local/global comparisons.

CSD11. Original zeta and every coefficient contribution

Neither A nor its original source J has changed. Retain

$$\Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, \quad \mathcal{M}_0 \Sigma f(s) = 2\zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v} \quad (\Re s > 1). \quad (\text{CSD11.1})$$

For $f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}$,

$$\Theta \Sigma f_0(s) = F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{CSD11.2})$$

The endpoint values $F_0(0) = F_0(1) = 1/8$ and all trivial-zero comparisons remain

$$F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r), \quad r \geq 1. \quad (\text{CSD11.3})$$

The complete inverse and these contributions are proved in SSI/ESI and retained in CSP11/CDW10.

For $\mathcal{B}$, the entire functions rapidly decreasing on every bounded real strip, let

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every actual nontrivial zero } \rho, 0 \leq j < m_\rho\}.$$

The proved map is the topological isomorphism $\kappa : Q \rightarrow \mathcal{Q} = \mathcal{B}/\mathcal{J}$, $\kappa([a]) = [\Theta a]$, with $\Theta J = \mathcal{J}$. Its actual continuous transpose is

$$\kappa' : \mathcal{Q}' \rightarrow Q', \quad (\kappa' \lambda)([a]) = \lambda([\Theta a]). \quad (\text{CSD11.4})$$

Thus the factor $1/2$ has not disappeared in duality. Every jet $\delta_{\rho,j}([a]) = (\Theta a)^{(j)}(\rho)$ is a continuous element of the Q' in CSD5 and CSD7. Its entire transpose matrix is

$$T'_a \delta_{\rho,j} = a^\rho \sum_{\ell=0}^j \binom{j}{\ell} (\log a)^{j-\ell} \delta_{\rho,\ell}. \quad (\text{CSD11.5})$$

For the normal contragredient the scalar is $a^{-\rho-1}$ and each logarithm is $-\log a$, as in CDI6.8. Every multiplicity is retained without presuming a location inside the known open critical strip.

The notation in (CSD11.5) uses raw derivatives. GTR and RD also use Taylor-coefficient functionals. Their exact comparison is

$$\delta_{\rho,j}^{\text{raw}} = j! \delta_{\rho,j}^{\text{Taylor}}, \quad \delta_{\rho,j}^{\text{Taylor}}([a]) = \frac{(\Theta a)^{(j)}(\rho)}{j!}. \quad (\text{CSD11.6})$$

In that latter basis the actual transfer transpose from (CSD9.9) is

$$U'_n \delta_{\rho,j}^{\text{Taylor}} = n^{1-\rho} \sum_{h=0}^j \frac{(-\log n)^{j-h}}{(j-h)!} \delta_{\rho,h}^{\text{Taylor}}. \quad (\text{CSD11.7})$$

Multiplying by $j!$ gives the raw-derivative formula with $\binom{j}{h}$. Thus the factorials, inverse parameter and ramification factor are all explicit in the comparison to GTR.

This construction does not promote the finite-residue identification to a topological self-duality of all Q , replace J by Q , or infer positivity of a Hermitian form from separated evaluation.

CSD12. Constructed support-duality result

The actual sheaf complex (CSD4.1) is the continuous current dual of the compact-support resolution. Its Gysin and endpoint arrow is $(\delta, -r')$. The local angular current and restriction fibre cone prove

$$i_p^* \mathcal{D}_c \mathcal{F} \simeq (i_p^! \mathcal{F})^\vee, \quad i_p^! \mathcal{D}_c \mathcal{F} \simeq (i_p^* \mathcal{F})^\vee.$$

The global chain map (CSD7.3) realizes CDI's entire continuous dual of the original sphere complex. Its restrictions give the full diagonal and anti-diagonal support maps. Proper cover transposes and both actual mirrors retain their degree and orientation factors.

This proves continuous support duality for the specified receiver. It assigns no operation to τ , introduces no finite-dimensional coefficient replacement, and asserts no RH or Deligne-purity conclusion.

Ramified geometric trace and completed residue action — GTR0-GTR11

Proof source: CC_RAMIFIED_TRACE_AND_RESIDUE_ACTION.md.

Ramified geometric trace and the original-zeta dual action

Complete derivation, 24 September 2026. Proof locators GTR0–GTR11.

GTR0. Construction stage and sources

The support remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. The user's complete arithmetic reconstruction precedes every coefficient space, scalar, positive integer, and analytic coordinate below. No addition or numerical coordinate is assigned to τ . The governing corpus and prerequisite rule and correction chains were read; the full passages rejecting an added distance/vector and the pooling of separate branch quotients were reread. The geometric sheets of an already constructed cover are not pooled arithmetic histories. The original two labelled support charts remain separate throughout.

The inputs are the complete CSP0–CSP13 sphere construction and CDW0–CDW10 explicit derived direct image in the receiving task. Their exact coefficients, restriction maps, periods and positive integration conventions are retained. The continuous transpose comparison is CDI0–CDI12, and the original-zeta residue comparison is RD0–RD9. These proofs were read in full. No continuous-dual complex is identified here with an ordinary Verdier dual without a separate constructed comparison.

The literature comparison is the Stacks Project, [tag 0GKE](#), especially the trace existence, composition and flat quasi-finite weighting statements. Original author TeX `more-etale.tex`, lines 1761–2127, was read at pinned commit `a04446e57ec1fbc252a871afcec7752fb2807b14`; its source receipt is retained locally. That source concerns étale sheaves. The topological sheaf trace used here is constructed directly below, so its existence is not inferred by changing categories in the cited theorem. The original Connes–Consani source identity and reading coverage remain CSP0's [arXiv:0903.2024v3](#), §5, and the later signed chart construction. Deligne's comparison remains [Weil II](#), §3.6, printed 213–214.

GTR1. The actual sheaf and positive-degree covers

Use $Y = \mathbb{P}^1(\mathbb{C})$, $D = \{0, \infty\}$, $Y^\circ = \mathbb{C}^\times$, and the CSP coefficient spaces

$$S = \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-x) = f(x), f(0) = 0, \int_{\mathbb{R}} f = 0\}, \quad V_p = S \oplus E_p, \quad E_p = \mathbb{C}^2 \quad (p = +, -),$$

$$A = \{a \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \text{ for every } N, j \geq 0\}. \quad (\text{GTR1.1})$$

The full restrictions are

$$\Sigma f(u) = 2 \sum_{k \geq 1} f(ku), \quad Ra(u) = u^{-1}a(u^{-1}), \quad r_+(f, e) = \Sigma f, \quad r_-(h, e) = R\Sigma h. \quad (\text{GTR1.2})$$

Write $J = \Sigma S$, $Q = A/J$. The actual source-image theorem proves closedness of J and the continuous inverse of $\Sigma : S \rightarrow J$. With $i : D \hookrightarrow Y$,

$$\mathcal{F} = \underline{A}_Y \times_{i_*(A \oplus A)} i_*(V_+ \oplus V_-). \quad (\text{GTR1.3})$$

On a connected open meeting one pole, a section consists of a constant $a \in A$ and $v_p \in V_p$ with $a = r_p v_p$; on an interior connected open it is a constant a . This follows directly from the fibre product, and specifies the local maps used below.

For every recovered integer $n \geq 1$, let $b_n(z) = z^n$, with both poles fixed. The equality $\pi b_n = \pi$ supplies the canonical identification $b_n^{-1} \mathcal{F} = \mathcal{F}$. This is an identification of sheaves pulled from the same three-point diagram, not an inverse geometric map.

The local multiplicities are

$$e_n(x) = \begin{cases} n, & x = 0 \text{ or } \infty, \\ 1, & x \in \mathbb{C}^\times, \end{cases} \quad \sum_{b_n(x)=y} e_n(x) = n. \quad (\text{GTR1.4})$$

At zero the local ring quotient is $\mathbb{C}[z]_{(z)}/(z^n)$, of length n with basis $1, z, \dots, z^{n-1}$; the coordinate $w = 1/z$ gives the same calculation at infinity. Away from the poles the derivative nz^{n-1} is nonzero, and there are exactly n distinct roots. Thus (GTR1.4) includes both ramified points and every interior sheet. It agrees with the flat quasi-finite weighting in the cited source, without removing the ramification.

GTR2. The weighted trace exists on the actual topological sheaf

Define

$$\text{Tr}_n : b_{n*} b_n^{-1} \mathcal{F} = b_{n*} \mathcal{F} \longrightarrow \mathcal{F} \quad (\text{GTR2.1})$$

as follows. On a sufficiently small interior disk the inverse image is a disjoint union of n disks, and the map on sections is

$$(a_1, \dots, a_n) \longmapsto \sum_{j=1}^n a_j. \quad (\text{GTR2.2})$$

On a sufficiently small disk about a pole, the inverse image is one pole disk and a section is $(r_p v_p, v_p)$. Its image is

$$(r_p v_p, v_p) \longmapsto (nr_p v_p, nv_p). \quad (\text{GTR2.3})$$

The restriction of (GTR2.3) to an interior subdisk equals (GTR2.2): all n sheet values are the restriction $r_p v_p$. Reordering the sheets does not change the sum. These formulas therefore agree on a basis of neighborhoods and every overlap, and define a sheaf morphism. Each local map is a finite continuous linear operation on the specified Fréchet spaces. At each pole both endpoint coordinates are multiplied by n ; they are not discarded because their generic restriction vanishes.

These local descriptions determine the map uniquely once its full weighted stalk values are specified. Generic-sheet agreement alone would not determine a map on the endpoint kernels E_p . Formula (GTR2.3) supplies that missing value explicitly from the local multiplicity.

The pullback unit $u_n : \mathcal{F} \rightarrow b_{n*} b_n^{-1} \mathcal{F}$ is diagonal on an interior disk and is the identity on a pole stalk. Hence

$$\boxed{\text{Tr}_n u_n = n \, 1_{\mathcal{F}}}. \quad (\text{GTR2.4})$$

This is an equality on every stalk and therefore on sheaves. No division by n is used in constructing the trace.

The finite proper map b_n has exact direct image on vector-space sheaves. To see this without a finite-rank hypothesis, its direct-image stalk at y is the finite direct sum of the stalks at $b_n^{-1}(y)$. Indeed small

disjoint neighborhoods of the finite fibre contain the inverse image of some neighborhood of y : otherwise properness supplies an accumulation point outside those neighborhoods mapping to y , a contradiction. Taking the corresponding neighborhood limits proves the stalk formula. Finite direct sums and stalk functors are exact. This also proves $Rb_{n*} = b_{n*}$; derived global sections of the pushforward equal those on the source. Thus (GTR2.1) gives an actual transfer on all cohomology groups.

The formula is natural in every continuous morphism of coefficient diagrams, since such maps commute with finite sums and multiplication by n . In particular it is natural for the actual sequence with overlap coefficients $J \rightarrow A \rightarrow Q$, retaining the entire J . Composition is exact: grouping roots in the composite cover multiplies their local multiplicities,

$$e_{mn}(x) = e_n(x)e_m(b_n(x)),$$

and the two successive finite sums equal the single weighted sum. Consequently the transfer for b_{mn} equals the composite transfer for b_n and b_m , with the direct-image identifications retained.

GTR3. Transfer on the full cohomology and support diagrams

Let P_n denote geometric pullback alone. The positive angular form is

$$\vartheta = \frac{d \arg z}{2\pi}, \quad b_n^* \vartheta = n\vartheta. \quad (\text{GTR3.1})$$

The positive sphere form and its full integral are

$$\omega_Y = \frac{dx \wedge dy}{\pi(1 + |z|^2)^2}, \quad \int_Y b_n^* \omega_Y = 2n^2 \int_0^\infty \frac{r^{2n-1}}{(1 + r^{2n})^2} dr = n. \quad (\text{GTR3.2})$$

CSP and CDW give the explicit global coefficient complex

$$D^0 = V_+ \oplus V_- \xrightarrow{r_+ - r_-} D^1 = A \xrightarrow{0} D^2 = A. \quad (\text{GTR3.3})$$

Its groups are $H^0 = Z = \ker(r_+ - r_-)$, $H^1 = Q$, $H^2 = A$. On this model geometric pullback is $(1, 1, n)$. In degree one it is the identity because the original degree-zero overlap constants are unchanged; in degree two (GTR3.2) gives the factor. These are invertible operators on these already constructed complex coefficient groups.

Taking cohomology in (GTR2.4) and using the proved pullback values determines the actual transfer:

$$\begin{array}{c|ccc} & H^0(Y, \mathcal{F}) = Z & H^1(Y, \mathcal{F}) = Q & H^2(Y, \mathcal{F}) = A \\ \hline P_n & 1 & 1 & n \\ \text{Tr}_n & n & n & 1. \end{array} \quad (\text{GTR3.4})$$

The map $(n, n, 1)$ on (GTR3.3) is a continuous cochain representative of these transfer values: its only nonzero differential commutes with n . No assertion that the map $z \mapsto z^n$ has a single-valued inverse is involved. This statement identifies the induced cohomology maps and an explicit compatible coefficient representative; it does not infer uniqueness of chain representatives from cohomology alone.

At either pole the costalk groups are E_p, Q, A in degrees 0, 1, 2; their geometric pullback and transfer factors are again $(1, 1, n)$ and $(n, n, 1)$. This follows from the actual local support complex $[V_p \rightarrow A \rightarrow 0A]$, the local angular degree n , and (GTR2.4) with supports. On the punctured sphere the factors in degrees 0, 1 are $(1, n)$ and $(n, 1)$. These maps commute with the entire CSP localization sequence: the maps $Q^2 \rightarrow Q$, $(x_+, x_-) \mapsto x_+ - x_-$, have transfer factor n on both sides; the maps $A \rightarrow A^2 \rightarrow A$, $a \mapsto (-a, a) \mapsto c_+ + c_-$, have factor one throughout their degree-two terms. The endpoint map has factor n . Thus every support sign and both endpoint kernels survive the transfer.

GTR4. The inverse coefficient map and the two different adjoints

The already constructed coefficient action is

$$T_a b(u) = b(u/a),$$

$$\rho_+(a)(f, c_0, c_1) = (f(\cdot/a), c_0, ac_1), \quad \rho_-(a)(h, d_0, d_1) = (ah(a \cdot), ad_0, d_1), \quad a > 0. \quad (\text{GTR4.1})$$

Direct substitution proves that their inverses are the same displayed formulas at a^{-1} ; they commute with the restrictions as $r_p \rho_p(a) = T_a r_p$. This inverse is on coefficient spaces, not on the geometric cover. It is also the required coefficient map on the dual: evaluation satisfies $\lambda(\rho(a)^{-1} \rho(a)x) = \lambda(x)$. Separation of evaluation proves uniqueness of that coefficient dual map if evaluation is to be preserved.

Set $\rho(a) = \rho_+(a) \oplus \rho_-(a)$. The full pullback from CSP is

$$B_n = (\rho(n), T_n, nT_n) \quad (\text{GTR4.2})$$

on degrees 0, 1, 2. Combine the actual geometric transfer with the proved inverse coefficient action. Its compatible coefficient representative is

$$U_n = (n\rho(n^{-1}), nT_{n^{-1}}, T_{n^{-1}}). \quad (\text{GTR4.3})$$

The order of these two operations is immaterial, since the trace is natural in coefficient maps. On each degree of the displayed complex,

$$\boxed{U_n B_n = B_n U_n = n \cdot 1.} \quad (\text{GTR4.4})$$

The scalar n therefore comes from the full ramified transfer, rather than being chosen to obtain a desired spectral formula. Composition of covers and coefficient actions gives $U_m U_n = U_{mn}$.

For the continuous dual complex use CDI's actual Hom differential. Ordinary transposition of a degree-zero cochain map adds no new sign. There are three specified maps:

dual degree	-2	-1	0
B'_n	nT'_n	T'_n	$\rho(n)'$
$(B_n^{-1})'$	$n^{-1}T'_{n^{-1}}$	$T'_{n^{-1}}$	$\rho(n^{-1})'$
U'_n	$T'_{n^{-1}}$	$nT'_{n^{-1}}$	$n\rho(n^{-1})'$

(GTR4.5)

The first is adjoint to original pullback; the second is the contragredient; the third is adjoint to the constructed transfer with inverse coefficients. Thus

$$U'_n = n(B_n^{-1})', \quad B'_n U'_n = U'_n B'_n = n \cdot 1,$$

$$\langle U'_n \lambda, B_n x \rangle = n \langle \lambda, x \rangle. \quad (\text{GTR4.6})$$

All identities hold on the complete coefficient complex and hence on its actual cohomology. They do not require a spectral decomposition or a density assertion. The induced degree-zero dual endpoint characters of U'_n are $n, 1, 1, n$, in the retained order c_0, c_1, d_0, d_1 .

GTR5. The original-zeta residue map receives this exact transfer action

Retain the original Mellin comparison

$$\Theta b(s) = \frac{1}{2} \int_0^\infty b(u) u^s \frac{du}{u}, \quad \Theta T_n b(s) = n^s \Theta b(s), \quad \Theta J = J. \quad (\text{GTR5.1})$$

Here $\mathcal{B}$ is the entire functions rapidly decreasing on every bounded real strip, $\mathcal{J}$ consists of all functions vanishing to the actual full multiplicity at every nontrivial zero of original ζ , and $\mathcal{Q} = \mathcal{B}/\mathcal{J}$. Write $\kappa : \mathcal{Q} \rightarrow \mathcal{Q}$, $[b] \mapsto [\Theta b]$. For $g = [G]$ with finitely supported full zero jets, RD proves

$$\iota(g)([F]) = \sum_{\rho} \text{Res}_{s=\rho} \frac{F(s)G(1-s)}{\zeta(s)} ds. \quad (\text{GTR5.2})$$

Each member of this finite-jet subspace has a representative satisfying the required vanishing at every other zero; it is not a cutoff of the underlying arithmetic. RD constructs those representatives, proves that the sum is finite, and proves invertibility of every local multiplicity matrix. Its map ι is a linear isomorphism onto the finite-jet dual inside the entire continuous dual.

The constructed transfer on degree one is nT_{n-1} ; hence its transpose on $H^{-1}(D^{\vee}) = Q'$ is nT'_{n-1} . Substituting in the complete residue integrand gives

$$\boxed{\kappa' \iota T_n = U'_n \kappa' \iota.} \quad (\text{GTR5.3})$$

Indeed replacing $G(1-s)$ by $n^{1-s}G(1-s)$ is exactly multiplying the first argument by n^{-s} and the result by n . This proves the diagram with the actual geometric transfer. The factor in RD5 is now received by a constructed sheaf operation, rather than merely compared as an identical scalar.

For Taylor-coefficient functionals $\delta_{\rho,j}([F]) = F^{(j)}(\rho)/j!$, the whole formula is

$$U'_n \kappa' \delta_{\rho,j} = n^{1-\rho} \sum_{h=0}^j \frac{(-\log n)^{j-h}}{(j-h)!} \kappa' \delta_{\rho,h}. \quad (\text{GTR5.4})$$

It follows by Taylor multiplication, retaining every nilpotent entry and factorial. No zero simplicity is assumed. The finite-jet image is invariant, and U'_n is continuous for the weak topology: evaluation at x after transposition is evaluation at the already constructed $U_n x$. Consequently RD's weakly dense receiver and its actual ambient continuous dual carry compatible transfer actions. This does not replace that ambient dual by its completion.

The geometric construction applies to every recovered $n \geq 1$. The corresponding formulas for arbitrary $a > 0$ define the already available continuous coefficient representation; no nonintegral holomorphic cover is asserted.

GTR6. Normal direction, winding orientation and the retained extra factors

The primal normal action on H^2 is nT_n on A . Write T_a^A and T_a^Q when distinguishing the coefficient space from its quotient, and retain $q : A \rightarrow Q$. Since $qT_a^A = T_a^Q q$, transposition gives $(T_a^A)' q' = q'(T_a^Q)'$. The specified three maps are

$$S_{\text{res}} = n(T_{n-1}^Q)' \quad \text{on } Q', \quad N_{\text{contra}} = n^{-1}(T_{n-1}^A)' \quad \text{on } A', \quad N_{\text{transfer}} = (T_{n-1}^A)' \quad \text{on } A'.$$

Their exact comparison, with its domain and receiving inclusion retained, is

$$q' S_{\text{res}} = n^2 N_{\text{contra}} q' = n N_{\text{transfer}} q' : Q' \longrightarrow A'. \quad (\text{GTR6.0})$$

Here q' identifies Q' with $J^{\perp}$; SDT5 proves this is also a closed strong topological embedding. Thus the residue action has the displayed extra factors on this specific subspace of the full normal dual, not on every functional in A' . None of the factors or the inclusion may be removed.

CDW's winding coordinate $a[\vartheta]$ maps to $-a$ in positive sphere integration. Under that comparison, the GSL normal map is exactly

$$a[\vartheta] \mapsto -[(1 - E_+(s))(\Theta a)(s - 1)]_{\mathcal{J} \cap \mathcal{J}_+}, \quad (\text{GTR6.1})$$

with kernel $J[\vartheta]$. The sign commutes with both nT_n and T_{n-1} , so the trace and pullback tables are unchanged by this explicit coordinate comparison. It is not permission to omit the sign from the normal map itself.

The two covers' pole trace maps remain separate. Likewise the holomorphic swap $-1/z$ and antiholomorphic swap $-1/\bar{z}$ retain their original CDW angular signs $-1, +1$ and their CSP degree-two signs $+1, -1$. Their current pushforwards are adjoints to their specified pullbacks. No positive degree is assigned to the antiholomorphic orientation reversal by (GTR3.2), which concerns only the holomorphic covers b_n .

GTR7. Naturality with the actual lifting boundary

The exact source sequence is

$$0 \longrightarrow \mathcal{F}_J \longrightarrow \mathcal{F} \longrightarrow j_! \underline{Q} \longrightarrow 0. \quad (\text{GTR7.1})$$

The weighted trace is defined on all three terms by the same fibre multiplicities and is natural in their coefficient maps by GTR2. Therefore its induced long exact sequences commute with transfer, as they do with pullback. For the actual connecting map

$$\delta : Q \longrightarrow J(-1), \quad (\text{GTR7.2})$$

this gives both

$$\delta T_n = nT_n \delta, \quad \delta(nT_{n-1}) = T_{n-1} \delta. \quad (\text{GTR7.3})$$

The second relation also follows algebraically from the first because the specified coefficient actions are invertible. Thus the new trace construction does not falsely add an independent restriction to the lifting problem. CW4 and CW7 prove $\delta = 0$ on the entire source; the present construction proves how this already established lift is carried through the actual geometric trace and its continuous dual.

The identity (GTR5.4) pairs actual exponents ρ and $1 - \rho$ through the full functional equation. It retains their sum and every multiplicity coordinate. It does not supply an absolute-value estimate for either eigenvalue. Deriving the support-dual weight bounds needed in Deligne §3.6 remains a mathematical task; these constructed maps are the receiving operations for that task.

GTR8. Original arithmetic contributions retained

The source arithmetic remains

$$\zeta(s) = 1 + \sum_{k \geq 2} k^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'}{\zeta}(s) = \sum_p \sum_{m \geq 1} (\log p) p^{-ms} \quad (\Re s > 1). \quad (\text{GTR8.1})$$

The original Σ still has both rational signs and the complete Mellin identity

$$\mathcal{M}_0 \Sigma f(s) = 2\zeta(s) \int_0^\infty f(x) x^s \frac{dx}{x}, \quad \mathcal{M}_0 = 2\Theta. \quad (\text{GTR8.2})$$

For the source used to construct the global residue isolators, retain

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad F_0(0) = F_0(1) = \frac{1}{8},$$

$$F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r), \quad r \geq 1. \quad (\text{GTR8.3})$$

RD and SSI give their full unit derivatives, inverse Mellin formula and trivial-zero Taylor observations. The residue denominator in (GTR5.2) is original ζ , not F_0 . Its reflection retains

$$\chi(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)}, \quad \zeta(s) = \chi(s) \zeta(1-s), \quad (\text{GTR8.4})$$

and the residue reversal has the sign from $ds = -d(1-s)$, with every local derivative of χ retained as in RD6. The new transfer does not modify any of these functions or their exceptional contributions.

GTR9. Exact result received by the support-duality programme

The factor n in the original-zeta residue covariance now has the specific geometric source (GTR2.1): the weighted trace for the actual ramified cover of the coefficient sheaf. Its inverse coefficient map is explicit, its two ramified stalks have their full multiplicities, and its action is calculated on every global, supported and annular group. The original-zeta finite-jet receiver intertwines its transpose exactly by (GTR5.3). The adjoint of pullback, the contragredient, and the adjoint of transfer are linked by (GTR4.6), rather than identified without their degree factors. This closes that action-comparison step. It does not identify the ordinary algebraic Verdier dual with the continuous dual or establish the missing Deligne weight bounds.

GTR10. The completed residue receiver and the full trace action

The completed strong-topology proof SDT0-SDT9 was read in full. Its accepted source is `../tau_weight_t_cohomology_20260924/CC_STRONG_DUAL_TOPOLOGY_AND_JETS.md`, SHA256 `7f1a03a6225f1dfc22b167713c0ddb154635b06f9479f6137b229845b67aefc`. SDT4 constructs compact lifts through $A \rightarrow Q$; SDT5 proves the strong topological inclusion $Q'_\beta \simeq J^\perp \subset A'_\beta$; SDT6-SDT8 prove strong reflexivity, strong density of the full finite-jet functionals, and their exact Hausdorff completion. These results concern the actual coefficient spaces, not bounded samples of zeros.

Define the residue seminorms on the already constructed finite-support class space by

$$p_B^{\text{res}}(G) = \sup_{F \in B} \left| \sum_{\rho} \text{Res}_{s=\rho} \frac{F(s)G(1-s)}{\zeta(s)} ds \right|, \quad B \subset \mathcal{Q} \text{ bounded.} \quad (\text{GTR10.1})$$

The sum for each such G is finite and contains every multiplicity coordinate in its support. Write

$$\mathcal{H}_{\text{res}} = (\mathcal{Q}_{\text{fin}}, \widehat{(p_B^{\text{res}})_B}). \quad (\text{GTR10.2})$$

This names the specified Hausdorff completion, not a new topology silently put on the entire original quotient. SDT8.12 supplies the topological isomorphism

$$\hat{\iota} : \mathcal{H}_{\text{res}} \xrightarrow{\sim} \mathcal{Q}'_\beta, \quad \mathfrak{I} = \kappa' \hat{\iota} : \mathcal{H}_{\text{res}} \xrightarrow{\sim} \mathcal{Q}'_\beta = H^{-1}(D^\vee). \quad (\text{GTR10.3})$$

The factor $1/2$ is still in $\kappa[a] = [\Theta a]$. To verify that its transpose is a strong topological isomorphism, the continuous maps κ and κ^{-1} carry bounded sets to bounded sets. Each defining strong seminorm therefore pulls back to the seminorm on the corresponding bounded image. This proves both directions of the topological assertion in (GTR10.3).

For every $a > 0$, put $V_a = aT_{a^{-1}}$ on $\mathcal{Q}_{\text{fin}}$. Direct substitution into the full residue, before any completion, proves the exact seminorm identities

$$p_B^{\text{res}}(T_a G) = a p_{T_{a^{-1}}B}^{\text{res}}(G), \quad p_B^{\text{res}}(V_a G) = p_{T_a B}^{\text{res}}(G). \quad (\text{GTR10.4})$$

Indeed the respective multipliers on the second argument at $1 - s$ are a^{1-s} and $a a^{-(1-s)} = a^s$. The sets on the right are bounded because T_a and its inverse are continuous on $\mathcal{Q}$. Thus both maps and their inverses extend uniquely to continuous automorphisms $\widehat{T}_a$ and $\widehat{V}_a$ of (GTR10.2). Their products extend those on the dense domain, so

$$\widehat{T}_a \widehat{T}_b = \widehat{T}_{ab}, \quad \widehat{V}_a \widehat{V}_b = \widehat{V}_{ab}, \quad \widehat{T}_a \widehat{V}_a = a \, 1. \quad (\text{GTR10.5})$$

For the recovered integer covers this gives, on the entire degree-minus-one strong cohomology group,

$$\boxed{\mathfrak{I} \widehat{T}_n = \mathbf{U}'_n \mathfrak{I}, \quad \mathfrak{I} \widehat{V}_n = \mathbf{B}'_n \mathfrak{I}.} \quad (\text{GTR10.6})$$

Here $\mathbf{U}'_n = nT'_{n-1}$ and $\mathbf{B}'_n = T'_n$ in degree minus one, exactly as in (GTR4.5). The equalities follow on the dense subspace from (GTR5.3) and (GTR10.4). All maps are continuous into the Hausdorff space Q'_β , so their equalizers are closed; the equalities hold on the whole completion. This argument proves the full action without assuming a convergent infinite sum of residues.

The complete pairing is the actual evaluation

$$\widehat{\mathcal{R}}(F, h) = \widehat{\iota}(h)(F), \quad F \in \mathcal{Q}, \, h \in \mathcal{H}_{\text{res}}. \quad (\text{GTR10.7})$$

For a net $G_\alpha \rightarrow h$ in the specified completion, its original finite residue pairings converge to (GTR10.7) uniformly on every bounded $B \subset \mathcal{Q}$. This is precisely convergence in the strong dual under (GTR10.3), so the value is independent of the representing net. Nondegeneracy in h follows from injectivity of $\widehat{\iota}$. Nondegeneracy in F follows from the full zero-jet functionals, which separate $\mathcal{Q}$ by the definition of $\mathfrak{I}$. No positive Hermitian form is inferred from this bilinear nondegeneracy.

The full transfer identities become

$$\widehat{\mathcal{R}}(T_n F, \widehat{T}_n h) = n \widehat{\mathcal{R}}(F, h), \quad \widehat{\mathcal{R}}(F, \widehat{V}_n h) = \widehat{\mathcal{R}}(T_n F, h). \quad (\text{GTR10.8})$$

They follow by evaluation of (GTR10.6), or by passing the exact finite residue identities through the above uniform convergence. The first retains the geometric degree; the second identifies the transpose of pullback. All original nilpotent entries of (GTR5.4) persist in this extension.

Finally, the actual two-support restriction in degree minus one is

$$Q'_\beta \longrightarrow Q'_\beta \oplus Q'_\beta, \quad \lambda \longmapsto (\lambda, -\lambda). \quad (\text{GTR10.9})$$

It is continuous: the seminorm of its value on a bounded pair is bounded by the sum of the seminorms on its two bounded projections. In the completed residue coordinates it is exactly

$$h \longmapsto (h, -h). \quad (\text{GTR10.10})$$

This follows from (GTR10.3) and linearity, retains both separate poles, and commutes with both actions in (GTR10.6). The embedding into the overlap dual is explicitly

$$h \longmapsto q' \mathfrak{I}(h) \in J^\perp \subset A'_\beta, \quad (q' \mathfrak{I}(h))(a) = \widehat{\mathcal{R}}([\Theta a], h). \quad (\text{GTR10.11})$$

Its vanishing on J follows because $\Theta J = \mathcal{J}$; SDT5 proves it is a strong topological isomorphism onto that annihilator. Thus the completed residue receiver maps into the entire signed support-dual diagram by the original quotient and Mellin maps. Neither the endpoint groups in degrees zero nor the full normal group in degree minus two is deleted. This proves the completed trace comparison; it does not assert that $\mathcal{H}_{\text{res}}$ is the original primal quotient with its original topology, or that its evaluation pairing is positive.

GTR11. Identification of the actual trace on the derived direct image

There is a stronger receiving statement than the cohomology calculation of GTR3. Work in the ordinary derived category of complex vector-space sheaves on the actual three-point space X . CDW4 constructs the specific quasi-isomorphism

$$\Phi : \Omega \oplus \mathcal{W}[-1] \xrightarrow{\sim} R\pi_* \mathcal{F}, \quad \mathcal{W} = j_{\eta!} \underline{A}_{\{\eta\}}, \quad (\text{GTR11.1})$$

represented by constants and the positive angular form. Here $\mathcal{W}$ is a sheaf on X , not the previously used vector space $W = V_+ \oplus V_-$. Its only nonzero stalk is A at η . Every map in the stated representative is continuous on the original coefficient spaces. CDW5 proves, before taking global cohomology, that geometric pullback P_n obeys

$$P_n \Phi = \Phi \begin{pmatrix} 1_\Omega & 0 \\ 0 & n1_{\mathcal{W}[-1]} \end{pmatrix}. \quad (\text{GTR11.2})$$

Consequently P_n is an automorphism of this actual derived direct image. Its inverse is represented through Φ by $\text{diag}(1, n^{-1})$; no inverse geometric root map is used.

Let $\mathfrak{t}_n$ be the morphism on $R\pi_* \mathcal{F}$ induced by the actual sheaf trace of GTR2. Since $\pi b_n = \pi$, the intermediate object $R\pi_* b_{n*} \mathcal{F}$ is canonically $R\pi_* \mathcal{F}$. To justify this derived identification, b_{n*} is exact by GTR2 and preserves injectives because it is right adjoint to the exact inverse-image functor. An injective resolution therefore computes both sides of the composite direct image. The sheaf equality (GTR2.4) yields the equality of derived morphisms

$$\mathfrak{t}_n P_n = n 1_{R\pi_* \mathcal{F}}. \quad (\text{GTR11.3})$$

Explicitly, for the preceding identification $\alpha : R\pi_* b_{n*} \mathcal{F} \xrightarrow{\sim} R\pi_* \mathcal{F}$, the two maps are $P_n = \alpha R\pi_*(u_n)$ and $\mathfrak{t}_n = R\pi_*(\text{Tr}_n) \alpha^{-1}$. CDW's chain pullback extends the sheaf unit through its augmentation; it therefore represents this same P_n . Compose on the right with the already constructed inverse of P_n . This proves

$$\boxed{\mathfrak{t}_n = n P_n^{-1}, \quad \mathfrak{t}_n \Phi = \Phi \begin{pmatrix} n1_\Omega & 0 \\ 0 & 1_{\mathcal{W}[-1]} \end{pmatrix}.} \quad (\text{GTR11.4})$$

This identifies the actual trace morphism in the derived category, including its possible extension information. It does not infer equality of morphisms merely from equal induced cohomology maps.

The coefficient action of (GTR4.1) commutes with Φ , preserves both summands, and has an explicit inverse. Thus the combined pullback and inverse-coefficient transfer are represented on (GTR11.1) by

$$\mathbf{B}_n = \begin{pmatrix} \rho_\Omega(n) & 0 \\ 0 & nT_n \end{pmatrix}, \quad \mathbf{U}_n = \begin{pmatrix} n\rho_\Omega(n^{-1}) & 0 \\ 0 & T_{n^{-1}} \end{pmatrix}. \quad (\text{GTR11.5})$$

The sheaf morphism $\rho_\Omega(a)$ means $\rho_+(a), \rho_-(a), T_a$ on the two chart stalks and the generic stalk, with the original restrictions retained. Taking derived global sections and the simultaneous contractions of CW2 gives exactly the full representatives (GTR4.2)-(GTR4.3). In particular the degree-one factor n comes from the Ω summand's trace, while the degree-two factor one comes from its separate angular summand. CDW9's minus between the angular coordinate and positive sphere integration remains in that comparison.

Every step is natural in the diagrams for $J \rightarrow A \rightarrow Q$, since CDW8's Φ is natural and the trace was constructed naturally in GTR2. Thus the same derived identification applies simultaneously to the actual source sequence (GTR7.1), with its boundary maps, rather than to separately chosen cohomology lists. This strengthens the precise scope of GTR3 to an actual ordinary derived-sheaf comparison with an explicit continuous coefficient representative. It makes no assertion of a strict current pullback through a ramified root map or of uniqueness in an unspecified category of topological complexes.

Ordinary Verdier duality and the full coefficient comparison — VSD0-VSD14

Proof source: CC_VERDIER_SUPPORT_INDEPENDENT.md.

The actual algebraic Verdier dual and its continuous coefficient comparison

Independent mathematical derivation, 24 September 2026. Proof locators VSD0-VSD14.

VSD0. Input, categories, and prerequisites

This calculation takes place after the complete arithmetic reconstruction. The user's notation $Z_0, Z_1, \dots, \tau$ is unchanged. There is no addition, numerical coordinate, distance, or parity operation on τ here. The coefficient spaces and their scalar operations are the already constructed receiving objects of CSP and CDI. Dualization is an operation on those coefficients, not a new construction of the counting unit or of selected primes.

The corpus rule argument `reconstruction/CORPUS_AND_OPERATION_RULE.md` was read. The complete coefficient-dual proof CDI0-CDI12 and complete original-zeta residue proof RD0-RD9 were read. CSP0-CSP7, including the inverse-image sheaf, finite complexes, local orientations and support maps, were read in full. CDW9-CDW10's positive-integration and original-zeta comparisons were also read. This is an independent calculation of their sheaf-duality receiver, not a claimed new reading of the entire original Connes-Consani or Deligne papers.

Let $Y = \mathbb{P}^1(\mathbb{C})$ have its usual topology and positive complex orientation. Write $D = \{p_+, p_-\} = \{0, \infty\}$, $i : D \hookrightarrow Y$, and $j : U = Y \setminus D \hookrightarrow Y$. Retain

$$A, \quad J = \Sigma S \subset A, \quad Q = A/J, \quad V_p = S \oplus E_p, \quad E_p = \mathbb{C}^2, \quad r_p : V_p \rightarrow A. \quad (\text{VSD0.1})$$

Here $r_+ = \Sigma$, $r_- = R\Sigma = \hat{\Sigma}$ on the source summands, and both vanish on the stated endpoint summands. SSI and CDI1 prove that J is closed, $\Sigma : S \rightarrow J$ is a topological isomorphism, and continuous sections $s_p : J \rightarrow V_p$ satisfy

$$r_p s_p = 1_J, \quad V_p \simeq J \oplus E_p. \quad (\text{VSD0.2})$$

These coefficient spaces are Hausdorff Fréchet spaces. Put

$$W = V_+ \oplus V_-, \quad d(v_+, v_-) = r_+ v_+ - r_- v_-, \quad Z = \ker d \simeq J \oplus E_+ \oplus E_-. \quad (\text{VSD0.3})$$

The actual sheaf is

$$\mathcal{F} = \underline{A}_Y \times_{i_*(A \oplus A)} i_* W, \quad (\text{VSD0.4})$$

with exact presentation

$$0 \rightarrow \mathcal{F} \rightarrow \underline{A}_Y \oplus i_* W \xrightarrow{\epsilon - r} i_*(A \oplus A) \rightarrow 0. \quad (\text{VSD0.5})$$

The two output components are $a|_{p_+} - r_+ v_+$ and $a|_{p_-} - r_- v_-$. Surjectivity is stalkwise using the constant-sheaf coordinate at the chosen pole.

Our ordinary sheaf category is the abelian category of sheaves of complex vector spaces. It imposes no continuity on coefficient linear maps. Distinguish

$$E^* = \text{Hom}_{\mathbb{C}}(E, \mathbb{C}), \quad E' = \text{Hom}_{\text{cont}, \mathbb{C}}(E, \mathbb{C}), \quad E^\dagger = E^*/E'. \quad (\text{VSD0.6})$$

The first is the full algebraic dual, the second the continuous dual with weak topology when needed, and the third initially only a vector space. No finite-dimensional or reflexive hypothesis is inserted.

Complexes are cohomological: $(C[k])^n = C^{n+k}$, so $\underline{E}[2]$ has its coefficient in degree -2 . Our Hom differential and cone differential are

$$d_{C^*}^n \lambda = (-1)^{n+1} \lambda d_C^{-n-1}, \quad d_{\text{Cone}(f)}(b, a) = (d_B b + f(a), -d_A a). \quad (\text{VSD0.7})$$

VSD1. Constant coefficients of arbitrary dimension

The dualizing complex on the oriented real two-manifold Y is

$$\omega_Y = \mathbb{C}_Y[2]. \quad (\text{VSD1.1})$$

Its orientation is fixed by integration of a compactly supported two-form with positive integral one in each positive complex coordinate. The local compact-support de Rham calculation on an oriented disk gives one coefficient in degree two and zero in the other degrees; extension by zero and integration give the trace map. These local orientation identifications glue to the displayed dualizing complex. Neither pole orientation is reversed.

Define ordinary Verdier duality by

$$\mathbb{D}_Y K = R\mathcal{H}om_Y(K, \omega_Y). \quad (\text{VSD1.2})$$

For every vector space E , including arbitrary infinite dimension,

$$\boxed{\mathbb{D}_Y \underline{E}_Y \simeq \underline{E}_Y^*[2]}. \quad (\text{VSD1.3})$$

To verify the coefficient claim without finite rank, on any open disk B constant-sheaf adjunction gives

$$\text{Hom}_{\text{Sh}(B)}(\underline{E}_B, K) = \text{Hom}_{\mathbb{C}}(E, \Gamma(B, K)). \quad (\text{VSD1.4})$$

A sheaf map is determined by the images of constant sections; conversely a linear map into sections extends to locally constant sections by gluing on their regions of constancy. Resolve K by injective sheaves. Vector-space Hom from E is exact: choose a basis and lift each basis image through any surjection. The Hom complex therefore has cohomology

$$\text{Ext}_{\text{Sh}(B)}^q(\underline{E}_B, K) = \text{Hom}_{\mathbb{C}}(E, H^q(B, K)). \quad (\text{VSD1.5})$$

For constant $K = \mathbb{C}_B$, the disk contraction gives $H^0 = \mathbb{C}$, $H^{q>0} = 0$. Internal derived-Hom stalks are the filtered colimit of these local Ext groups; higher internal Ext stalks vanish. In degree zero the internal Hom is constant with coefficient E^* : on a connected disk each scalar image of a constant section is constant, defining a single linear functional on all of E . Thus (VSD1.3) follows.

For a point p and every vector space B ,

$$i_p^! \underline{B}_Y = B[-2], \quad i_p^! \underline{B}_Y[2] = B. \quad (\text{VSD1.6})$$

Indeed the disk/punctured-disk relative complex has identity restriction in degree zero, while the punctured disk has its independent degree-one angular coefficient B . Its fibre is the local fundamental class in degree two. This is the coefficient version of CSP6 and uses no dimension bound.

VSD2. Pole duals and the signed Gysin map

For a skyscraper coefficient,

$$\boxed{\mathbb{D}_Y i_{p*} E = i_{p*} E^*} \quad (\text{VSD2.1})$$

Closed-inclusion adjunction identifies derived Hom from $i_{p*} E$ into ω_Y with the pushforward of derived Hom from E into $i_p^! \omega_Y = \mathbb{C}$. At the point vector-space Hom is exact, so no higher Ext remains. This proves (VSD2.1) in arbitrary dimension.

For any vector space B , define

$$g_{p,B} : i_{p*} B \longrightarrow \underline{B}_Y[2] \quad (\text{VSD2.2})$$

as the adjunction counit corresponding to the identity of $B = i_p^! \underline{B}_Y[2]$, where the identification is positive local complex orientation. Derived adjunction specifies this fully:

$$\text{Hom}_{D(Y)}(i_{p*} B, \underline{B}_Y[2]) = \text{Hom}_{\mathbb{C}}(B, B). \quad (\text{VSD2.3})$$

The dual of restriction $\underline{E}_Y \rightarrow i_{p*} E$ is g_{p,E^*} with positive sign. To check it, pair e with λ ; the corresponding local compactly supported two-form has integral $\lambda(e)$, using a positive fundamental form. Extension by zero to the sphere preserves that integral, so the global scalar is $+\lambda(e)$.

The angular connecting sign is separate. For $\chi = 1$ near the pole and zero outside a disk,

$$-d\chi \wedge \frac{d\theta_p}{2\pi}, \quad \int -d\chi \wedge \frac{d\theta_p}{2\pi} = 1 \quad (\text{VSD2.4})$$

is positive local orientation. In the CSP relative cone the boundary of the positive angular class is the negative of this positive-integral class. Gysin then carries the positive fundamental class positively to the global one. These signs belong to different maps.

For the two poles,

$$g_B = g_{p_+,B} + g_{p_-,B} : i_*(B \oplus B) \rightarrow \underline{B}_Y[2] \quad (\text{VSD2.5})$$

induces

$$B \oplus B \rightarrow H^0(Y, \underline{B}[2]) = B, \quad (b_+, b_-) \mapsto b_+ + b_-. \quad (\text{VSD2.6})$$

Both local coordinates $z, w = 1/z$ are complex oriented. The identity $d \arg w = -d \arg z$ belongs to the annular comparison, not to the Gysin sum.

VSD3. The actual algebraic Verdier triangle

Dualizing (VSD0.5) with those explicitly computed maps gives

$$i_*((A^*)^2) \xrightarrow{(g_{A^*}, -r^*)} \underline{A}_Y^*[2] \oplus i_* W^* \longrightarrow \mathbb{D}_Y \mathcal{F} \longrightarrow i_*((A^*)^2)[1]. \quad (\text{VSD3.1})$$

The restriction component is

$$(\lambda_+, \lambda_-) \mapsto (-r_+^* \lambda_+, -r_-^* \lambda_-). \quad (\text{VSD3.2})$$

It retains the minus in $\epsilon - r$; Gysin retains the positive sign of ϵ . Thus, representing the Gysin maps on a resolution when forming a cone,

$$\boxed{\mathbb{D}_Y \mathcal{F} \simeq \text{Cone} \left(i_*((A^*)^2) \xrightarrow{(g_{A^*}, -r^*)} \underline{A}_Y^*[2] \oplus i_* W^* \right)}. \quad (\text{VSD3.3})$$

This constructs the sheaf dual. It does not declare that a continuous-dual complex is already the ordinary Verdier dual.

VSD4. Every stalk and costalk

On the interior,

$$j^* \mathbb{D}_Y \mathcal{F} = \underline{A}_U^*[2]. \quad (\text{VSD4.1})$$

At a pole the derived stalk of Gysin is zero as a derived vector-space map from degree zero to degree minus two: vector-space Ext in degree two is zero. In the local cone, multiplying the degree-minus-one source coordinate by minus one turns $-r_p^*$ into $+r_p^*$. The stalk model is

$$A^* \text{ in degree } -2, \quad A^* \xrightarrow{r_p^*} V_p^* \text{ in degrees } -1, 0. \quad (\text{VSD4.2})$$

It is the transpose of CSP's positive-integration costalk $L_p = [V_p \xrightarrow{r_p} A \xrightarrow{0} A]$. Vanishing of the stalk Gysin map does not make the global Gysin morphism vanish.

The algebraic dual restriction sequence is exact:

$$0 \rightarrow Q^* \xrightarrow{q^*} A^* \rightarrow J^* \rightarrow 0. \quad (\text{VSD4.3})$$

Surjectivity follows by extending a basis of J to one of A . Since $V_p = J \oplus E_p$, the image of r_p^* is precisely the annihilator of E_p . Thus

$$\mathcal{H}^{-2}(\mathbb{D}\mathcal{F})_p = A^*, \quad \mathcal{H}^{-1}(\mathbb{D}\mathcal{F})_p = Q^*, \quad \mathcal{H}^0(\mathbb{D}\mathcal{F})_p = E_p^*. \quad (\text{VSD4.4})$$

All other pole cohomology stalks vanish. The full cohomology sheaves are

$$\mathcal{H}^{-2}(\mathbb{D}\mathcal{F}) = \underline{A}_Y^*, \quad \mathcal{H}^{-1}(\mathbb{D}\mathcal{F}) = i_*(Q^* \oplus Q^*), \quad \mathcal{H}^0(\mathbb{D}\mathcal{F}) = i_*(E_+^* \oplus E_-^*). \quad (\text{VSD4.5})$$

Degree-minus-two generization is identity. This list does not replace the cone and its Gysin extension class.

Applying $i_p^!$ to (VSD3.1), the map becomes

$$A^* \rightarrow A^* \oplus V_p^*, \quad \lambda \mapsto (\lambda, -r_p^* \lambda). \quad (\text{VSD4.6})$$

Its cone contracts to $V_p^*[0]$. Explicitly the projection

$$(a, \phi) \mapsto \phi + r_p^* a \quad (\text{VSD4.7})$$

is onto with kernel precisely (VSD4.6). Therefore

$$\boxed{i_p^! \mathbb{D}_Y \mathcal{F} \simeq V_p^*[0].} \quad (\text{VSD4.8})$$

At an interior point x , $i_x^! \mathbb{D}\mathcal{F} = A^*[0]$. Both stalk/costalk interchange statements are established by these actual local complexes.

VSD5. Global cohomology and explicit contraction

Derived global sections of (VSD3.3) have finite model

$$\begin{aligned} K_*^{-2} &= A^*, \quad K_*^{-1} = (A^*)^2, \quad K_*^0 = A^* \oplus V_+^* \oplus V_-^*, \\ d^{-2} &= 0, \quad d^{-1}(\lambda_+, \lambda_-) = (\lambda_+ + \lambda_-, -r_+^* \lambda_+, -r_-^* \lambda_-). \end{aligned} \quad (\text{VSD5.1})$$

The degree-minus-two coefficient comes from sphere degree zero of the unshifted constant sheaf. The degree-zero A^* comes from its positive sphere degree-two class; its two Gysin contributions are the sum in (VSD2.6). This is precisely the algebraic Hom complex of CSP4.1.

It contracts onto

$$D_*^{-2} = A^*, \quad D_*^{-1} = A^*, \quad D_*^0 = V_+^* \oplus V_-^*, \quad d^{-2} = 0, \quad d^{-1}\lambda = (r_+^*\lambda, -r_-^*\lambda). \quad (\text{VSD5.2})$$

The inclusion $u : D_* \rightarrow K_*$ and retraction $v : K_* \rightarrow D_*$ are

$$u^{-2} = 1, \quad u^{-1}\lambda = (-\lambda, \lambda), \quad u^0(\phi_+, \phi_-) = (0, \phi_+, \phi_-), \quad (\text{VSD5.3})$$

$$v^{-2} = 1, \quad v^{-1}(\lambda_+, \lambda_-) = -\lambda_+, \quad v^0(a, \phi_+, \phi_-) = (\phi_+, \phi_- + r_-^*a). \quad (\text{VSD5.4})$$

Substitution shows they are chain maps and $vu = 1$. A homotopy for $1 - uv$ is

$$h^0(a, \phi_+, \phi_-) = (0, a) \quad (\text{VSD5.5})$$

with all other components zero: dh in degree zero is $(a, 0, -r_-^*a)$, and hd in degree minus one is $(0, \lambda_+ + \lambda_-)$, exactly $1 - uv$.

Since $W \simeq Z \oplus J$ by CDI1's explicit section, restriction and extension of algebraic functionals give

$$\boxed{H^{-2}(Y, \mathbb{D}\mathcal{F}) = A^*, \quad H^{-1}(Y, \mathbb{D}\mathcal{F}) = Q^*, \quad H^0(Y, \mathbb{D}\mathcal{F}) = Z^*}. \quad (\text{VSD5.6})$$

Their pairings with CSP's A, Q, Z are ordinary evaluation on each full algebraic dual. No bidual surjectivity is asserted.

VSD6. The complete transposed support maps

The source support map has components

$$k^0 = 1_W, \quad k^1(b_+, b_-) = b_+ - b_-, \quad k^2(c_+, c_-) = c_+ + c_-. \quad (\text{VSD6.1})$$

Under the exact local and global dual models, restriction of the dual global complex to the two dual stalk models is

$$(k^*)^0 = 1_{W^*}, \quad (k^*)^{-1}\lambda = (\lambda, -\lambda), \quad (k^*)^{-2}\lambda = (\lambda, \lambda). \quad (\text{VSD6.2})$$

The pole differential sends (λ_+, λ_-) to $(r_+^*\lambda_+, r_-^*\lambda_-)$, so chain compatibility follows directly. The cohomology maps are diagonal on A^* , anti-diagonal on Q^* , and endpoint restriction on Z^* .

The complete nonzero part of the algebraically dual localization row is

$$\begin{aligned} 0 \rightarrow A^* &\xrightarrow{\lambda \mapsto (\lambda, \lambda)} (A^*)^2 \xrightarrow{(\lambda_+, \lambda_-) \mapsto -\lambda_+ + \lambda_-} A^* \xrightarrow{0} Q^* \\ &\xrightarrow{\mu \mapsto (\mu, -\mu)} (Q^*)^2 \xrightarrow{(\mu_+, \mu_-) \mapsto q^*(\mu_+ + \mu_-)} A^* \xrightarrow{r^*} Z^* \xrightarrow{\text{res}} E_+^* \oplus E_-^* \rightarrow 0. \end{aligned} \quad (\text{VSD6.3})$$

Here $r : Z \rightarrow A$ is common restriction and $r^*\lambda = (\lambda|_J, 0, 0)$. Exactness follows from the displayed sum/difference kernels, $\ker(A^* \rightarrow J^*) = q^*Q^*$, extension onto J^* , and the split endpoint coordinates. The annular arrow is the transpose of $a \mapsto (-a, a)$; it is not the transpose of the Gysin sum.

There is also the support-to-global map for the dual sheaf itself. Its two pole costalks, computed in (VSD4.8), are $V_+^*[0] \oplus V_-^*[0]$. On the global model (VSD5.2) their inclusion is the identity into degree zero W^* . On cohomology this is restriction of a functional on W to its subspace Z . Thus

$$(\phi_+, \phi_-) \mapsto (s_+^*\phi_+ + s_-^*\phi_-, \phi_+|_{E_+}, \phi_-|_{E_-}) \quad \text{in } Z^* \simeq J^* \oplus E_+^* \oplus E_-^*. \quad (\text{VSD6.4})$$

Both pole contributions have a positive sign. Its kernel is precisely $\{(r_+^*\lambda, -r_-^*\lambda) : \lambda \in A^*\}$; it consists of the functionals annihilating Z , which factor through $d : W \rightarrow J$, and every such functional on J extends to A . Surjectivity follows from the explicit splitting $W \simeq Z \oplus J$. Hence (VSD6.4) is also completely calculated; it is a different map from the dual-global-to-dual-stalk restriction in (VSD6.2).

VSD7. A continuous subcomparison as an actual sheaf object

The continuous candidate is constructed in the same ordinary derived sheaf category, with the specified continuous-dual coefficient subspaces:

$$\mathcal{D}_c \mathcal{F} = \text{Cone} \left(i_*((A')^2) \xrightarrow{(g_{A'}, -r')} \underline{A}'_Y[2] \oplus i_* W' \right). \quad (\text{VSD7.1})$$

This defines actual constant and skyscraper sheaves, using the constructed Gysin map and continuous transposes. It is not asserted to be a general dualizing functor on all topological sheaves. It supplies an actual receiving object for this particular $\mathcal{F}$.

The inclusions $E' \hookrightarrow E^*$ commute with all transposes. Gysin is coefficient-natural by its identity characterization (VSD2.3). They therefore produce

$$c : \mathcal{D}_c \mathcal{F} \longrightarrow \mathbb{D}_Y \mathcal{F}. \quad (\text{VSD7.2})$$

The functorial resolution of the coefficient-natural Gysin diagrams produces a map of ordinary cone complexes, so (VSD7.2) is not merely a comparison between cohomology lists.

The preceding finite calculations hold with primes instead of stars. Continuous Hahn-Banach on the closed subspace $J \subset A$ supplies the required surjectivity of restriction; CDI1 proves that step with these actual topologies. Consequently

$$\begin{array}{c|ccc} & -2 & -1 & 0 \\ \hline H^\bullet(Y, \mathcal{D}_c \mathcal{F}) & A' & Q' & Z' \\ \mathcal{H}^\bullet(\mathcal{D}_c \mathcal{F})_p & A' & Q' & E'_p \end{array} \quad (\text{VSD7.3})$$

and $i_p^! \mathcal{D}_c \mathcal{F} = V'_p[0]$; the interior restriction is $\underline{A}'[2]$. The global finite complex is exactly CDI2.2, the pole complex CDI2.6, and support maps are CDI4-CDI5. Each cohomology map of c is the inclusion of continuous into full algebraic dual of the corresponding original group. With weak coefficient topologies the finite-model cohomology topologies are the exact weak topologies proved in CDI3. The ordinary sheaf category records the underlying vector spaces.

Thus the continuous coefficient model now has an actual sheaf receiver, with a proved map to ordinary Verdier duality. The two duals have not been identified without a map.

VSD8. The comparison cone and every one of its groups

Write

$$\mathcal{E} = \text{Cone}(c). \quad (\text{VSD8.1})$$

The exact comparison object is

$$\mathcal{E} \simeq \text{Cone} \left(i_*((A^\dagger)^2) \xrightarrow{(g_{A^\dagger}, -r^\dagger)} \underline{A}^\dagger_Y[2] \oplus i_* W^\dagger \right). \quad (\text{VSD8.2})$$

For each coefficient, $0 \rightarrow E' \rightarrow E^* \rightarrow E^\dagger \rightarrow 0$ induces exact sequences of constant sheaves and skyscrapers, since exactness is stalkwise. The coefficient-natural Gysin square gives a morphism of triangles. Its cone is (VSD8.2): one may compute it by functorial resolutions and the two cone differentials, or by the resulting exact quotient diagram. No topology on $E^\dagger$ is used here.

The continuous and algebraic restriction rows form the exact diagram

$$\begin{array}{ccccccccc} 0 & \rightarrow & Q' & \rightarrow & A' & \rightarrow & J' & \rightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \rightarrow & Q^* & \rightarrow & A^* & \rightarrow & J^* & \rightarrow & 0. \end{array} \quad (\text{VSD8.3})$$

All vertical maps are injections. The quotient row is therefore

$$0 \rightarrow Q^\dagger \rightarrow A^\dagger \rightarrow J^\dagger \rightarrow 0. \quad (\text{VSD8.4})$$

To verify its middle kernel directly, an algebraic functional on A whose restriction to J is continuous differs, by a continuous Hahn-Banach extension of that restriction, from a functional factoring algebraically through Q . Surjectivity on the right follows by algebraic extension from J .

The endpoint spaces are finite dimensional, so all their functionals are continuous. The actual splittings in VSD0 give

$$V_p^\dagger \simeq J^\dagger, \quad Z^\dagger \simeq J^\dagger, \quad E_p^\dagger = 0. \quad (\text{VSD8.5})$$

The induced $r_p^\dagger : A^\dagger \rightarrow V_p^\dagger$ is onto and has kernel $Q^\dagger$. The finite complexes now give

$$\boxed{\begin{array}{lll} \mathcal{H}^{-2}(\mathcal{E}) = \underline{A}_Y^\dagger, & \mathcal{H}^{-1}(\mathcal{E}) = i_*(Q^\dagger \oplus Q^\dagger), & \mathcal{H}^0(\mathcal{E}) = 0, \\ i_p^\dagger \mathcal{E} = V_p^\dagger[0] \simeq J^\dagger[0], & & \\ H^{-2}(Y, \mathcal{E}) = A^\dagger, & H^{-1}(Y, \mathcal{E}) = Q^\dagger, & H^0(Y, \mathcal{E}) = Z^\dagger \simeq J^\dagger. \end{array}} \quad (\text{VSD8.6})$$

In particular a zero degree-zero cohomology sheaf does not force zero degree-zero global cohomology or zero pole costalk. The Gysin extension in (VSD8.2) retains this information.

All support maps are (VSD6.2)-(VSD6.3) on the dagger coefficients, with finite endpoint terms zero. This comparison defect has specified support, maps and computed cohomology; it is not an unnamed failure of an analogy.

VSD9. Nonzero defect for the actual infinite coefficients

Every infinite-dimensional Hausdorff Fréchet space E has discontinuous algebraic functionals. For a defining seminorm sequence $p_1, p_2, \dots$, choose independent x_n and scale them by nonzero constants to obtain independent v_n with

$$\max_{1 \leq j \leq n} p_j(v_n) \leq 2^{-n}. \quad (\text{VSD9.1})$$

Then $v_n \rightarrow 0$. Assign $\lambda(v_n) = 1$ on their span and extend the independent family to an algebraic basis of E , assigning zero to the additional basis vectors. This defines an algebraic functional which is not continuous. Hence $E^\dagger \neq 0$.

The actual A is infinite dimensional: smooth functions with pairwise disjoint compact supports in $\mathbb{R}_{>0}$ lie in A and are independent. The actual S is infinite dimensional too. In pairwise disjoint positive intervals choose nonzero compactly supported smooth functions of integral zero, for example derivatives of interior bump functions, and extend them evenly. They vanish at zero, retain integral zero on the real line, and are independent members of S . The proved topological isomorphism $\Sigma : S \rightarrow J$ gives

$$A^\dagger \neq 0, \quad J^\dagger \neq 0. \quad (\text{VSD9.2})$$

Consequently c is not a quasi-isomorphism: its open-stratum degree-minus-two cokernel is the nonzero $A^\dagger$. Its pole costalk comparison also has nonzero quotient $J^\dagger$. This proves the precise difference between the two defined dual constructions; it says nothing against the original support τ or its lifting programme.

VSD10. Weak completion and the original residue family

For every Hausdorff locally convex E ,

$$\overline{E'}^{\sigma(E^*, E)} = E^*. \quad (\text{VSD10.1})$$

Indeed fix $\lambda \in E^*$ and finitely many $e_1, \dots, e_m$. On their finite-dimensional span L , the restriction of λ is continuous, because the Hausdorff induced topology on L is its ordinary finite-dimensional topology. It is bounded by a continuous seminorm induced from E . Hahn-Banach extends it continuously to all of E , agreeing exactly on the prescribed vectors. Every pointwise neighborhood of λ therefore meets E' .

The full algebraic dual is complete in pointwise convergence. It is the closed subspace of the complete product $\mathbb{C}^E$ cut out by every linearity equation

$$f(x + y) = f(x) + f(y), \quad f(cx) = cf(x). \quad (\text{VSD10.2})$$

Its topology on E' is exactly the weak dual topology. Thus

$$\boxed{\text{the Hausdorff completion of } E'_\sigma \text{ is } E^* \text{ with its pointwise topology.}} \quad (\text{VSD10.3})$$

Weak completion and weak closure inside a specified continuous dual are different operations; neither is substituted for the other.

The quotient topology on E^*/E' from $\sigma(E^*, E)$ is indiscrete. To see it, let a nonempty open saturated set contain $x + U$, with U an open neighborhood of zero. For any y , density makes $y - x - U$ meet E' , so $y \in x + U + E'$, and saturation places every y in the set. Its only open subsets are therefore empty and whole. The nonzero algebraic quotients in VSD9 are not Hausdorff quotients for this particular topology. They remain the exact ordinary-sheaf coefficients in VSD8.

RD's finite-support original-zeta residue family $\mathcal{D}$ satisfies both

$$\overline{\mathcal{D}}^{\sigma(Q', \mathcal{Q})} = \mathcal{Q}', \quad \overline{\mathcal{D}}^{\sigma(Q^*, \mathcal{Q})} = \mathcal{Q}^*. \quad (\text{VSD10.4})$$

The first is RD4. For the second, take any algebraic λ and finitely many $x_1, \dots, x_m \in \mathcal{Q}$. If their λ -values were outside the finite-dimensional image $\{(d(x_i))_i : d \in \mathcal{D}\}$, a separating linear relation would annihilate every full zero jet of $\sum c_i x_i$, hence that class would be zero by the definition of $\mathcal{Q}$, while its λ -value would be nonzero. This contradiction proves exact interpolation on every finite set, hence the second density. These equations have different stated ambient spaces. Neither makes a discontinuous functional continuous or changes the residue formula.

VSD11. Original-zeta data and receiving morphisms

The actual coefficient comparison remains

$$\kappa : \mathcal{Q} \longrightarrow \mathcal{Q}, \quad [a] \longmapsto \left[\frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u} \right], \quad (\text{VSD11.1})$$

with $\Sigma f = 2 \sum_{n \geq 1} f(nu)$. Its exact dual square is

$$\begin{array}{ccc} \mathcal{Q}' & \hookrightarrow & \mathcal{Q}^* \\ \downarrow \kappa' & & \downarrow \kappa^* \\ \mathcal{Q}' & \hookrightarrow & \mathcal{Q}^*. \end{array} \quad (\text{VSD11.2})$$

Both vertical maps are isomorphisms, and so is the induced map on their algebraic quotients. RD's finite-support residue map therefore enters the two sheaf-dual receivers through the actual inclusion:

$$\mathcal{Q}_{\text{fin}} \xrightarrow{\kappa' \iota} H^{-1}(Y, \mathcal{D}_c \mathcal{F}) \xrightarrow{H^{-1}(c)} H^{-1}(Y, \mathbb{D} \mathcal{F}). \quad (\text{VSD11.3})$$

Its first arrow is

$$G \longmapsto \left([a] \longmapsto \sum_{\rho} \text{Res}_{s=\rho} \frac{\frac{1}{2} \mathcal{M}_0 a(s) G(1-s)}{\zeta(s)} ds \right). \quad (\text{VSD11.4})$$

The denominator is the original zeta, and the residues and ideal retain all zero multiplicities. The comparison does not replace it by the source transform

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{VSD11.5})$$

All of that transform's factors, endpoint values, trivial-zero derivatives and local-unit derivatives remain those proved in RD8 and CDI12. The four endpoint lines are explicitly retained in $E_+^* \oplus E_-^*$ and $E_+^* \oplus E_-^*$. Their dagger quotient vanishes because they are finite dimensional, not because their original contributions were deleted.

Every continuous coefficient operator preserves $E' \subset E^*$ under ordinary transposition and thus induces an operator on $E^\dagger$. Gysin is coefficient-natural. Thus the entire comparison triangle intertwines the already proved coefficient, cover and mirror maps, retaining their signs and factors. This does not choose a new arithmetic degree convention; the cover/trace derivation is a separate calculation. Ordinary transpose, inverse-parameter contragredient, and residue degree-character comparison remain the distinct maps proved in CDI6-CDI10.

VSD12. Actual output for the support-dual calculation

The actual $\mathcal{F}$ now has an explicit ordinary algebraic Verdier dual, including both pole Gysin maps, positive orientation, the minus on dualized restrictions, every stalk and costalk, and full global cohomology. These proofs apply to the actual infinite coefficient spaces without a finite-rank duality assumption.

The continuous-dual coefficient complexes have an actual sheaf receiver $\mathcal{D}_c \mathcal{F}$ and a proved map to ordinary Verdier duality. Its cone is the specific object in VSD8, with every coefficient group and support map computed. Its nonzero difference is retained and studied in VSD9-VSD10. This supplies the exact comparison required before using either category in the support-duality part of a Deligne-style argument.

No finite-rank étale realization, Deligne weight bound, arithmetic purity, or RH conclusion is an assumption or conclusion here. The subsequent weight calculation can use the proved cone and coefficient-category maps instead of a presumed equality of the two duals.

VSD13. The exact comparison defect admits every rational operation in the Mellin multiplication operator

The nonzero comparison object has additional algebraic structure which can be calculated globally. Let

$$\mathcal{B} = \left\{ F \text{ entire} : \sup_{|\Re s| \leq R} (1 + |\Im s|)^M |F(s)| < \infty \text{ for all } R, M \in \mathbb{N} \right\}, \quad \mathcal{J} = \Theta \mathcal{J}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J},$$

$$L : \mathcal{B} \rightarrow \mathcal{B}, \quad (LF)(s) = sF(s). \quad (\text{VSD13.1})$$

The topological isomorphism $\Theta : A \rightarrow \mathcal{B}$ and its restriction $J \rightarrow \mathcal{J}$ are the actual Mellin comparison of CDI6 and SSI, retaining $\Theta = \frac{1}{2} \mathcal{M}_0$. The operator L is continuous, preserves $\mathcal{J}$, and induces an operator on $\mathcal{Q}$. These are the domains and maps being transposed below.

For every $\mu \in \mathbb{C}$, multiplication

$$M_\mu = L - \mu : \mathcal{B} \rightarrow \mathcal{B} \quad (\text{VSD13.2})$$

is injective and has image exactly the closed hyperplane $F(\mu) = 0$. Its inverse on that image is division by $s - \mu$. That inverse is continuous: outside the disk $|s - \mu| < 1$, the quotient is bounded by the original function; inside, the maximum principle applied to the holomorphic quotient bounds it using a circle of radius two centered at μ . Every coefficient seminorm on that circle is bounded by one of the

larger-strip seminorms on $\mathcal{B}$, and the polynomial imaginary-coordinate weight is bounded on the disk. This gives a finite seminorm bound for each target seminorm. Evaluation onto $\mathbb{C}$ is surjective, since $e^{(s-\mu)^2} \in \mathcal{B}$ has value one at μ . Thus the image really has codimension one.

The same multiplication on $\mathcal{J}$ is injective with closed codimension-one image. To prove this at every μ , retain the complete function

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) \in \mathcal{J}. \quad (\text{VSD13.3})$$

Its zero divisor is the complete actual nontrivial-zero divisor, with all multiplicities, as retained in RD1 and RD8. Every $F \in \mathcal{J}$ therefore has an entire quotient F/F_0 ; no claim that this quotient belongs to $\mathcal{B}$ is needed. Define

$$\ell_\mu(F) = \begin{cases} F(\mu)/F_0(\mu), & F_0(\mu) \neq 0, \\ F^{(m)}(\mu)/F_0^{(m)}(\mu), & F_0 \text{ has a zero of order } m \text{ at } \mu. \end{cases} \quad (\text{VSD13.4})$$

This is the precise removable value $(F/F_0)(\mu)$. Cauchy's estimate makes it continuous; the nonzero denominator derivative is retained. Since $\ell_\mu(F_0) = 1$, it is onto. The image of $M_\mu : \mathcal{J} \rightarrow \mathcal{J}$ is exactly $\ker \ell_\mu$. Necessity follows by taking the quotient by F_0 . Conversely, when $\ell_\mu(F) = 0$, division by $s - \mu$ leaves the full required order of vanishing at μ and preserves every other required zero jet, so the quotient belongs to $\mathcal{J}$. Its continuity follows from the division estimate just proved in $\mathcal{B}$, restricted to these closed subspaces. This proves the entire statement, including at every multiple zero and at each endpoint or trivial-zero location where F_0 is nonzero.

Here is the general transpose calculation applied to these two actual operators. Suppose $M : E \rightarrow E$ is a continuous injective map onto a closed subspace of finite codimension, with continuous inverse on that image. Its algebraic transpose $M^* : E^* \rightarrow E^*$ is onto, by algebraically extending the functional $Mx \mapsto \lambda(x)$ from the image. Its continuous transpose $M' : E' \rightarrow E'$ is onto as well: the same functional is continuous on the image when λ is continuous, and Hahn-Banach extends it to E . Moreover

$$\ker M^* = \ker M' \quad (\text{VSD13.5})$$

as subspaces of E^* . Both are all functionals on the finite-dimensional Hausdorff quotient $E/M(E)$, and every such functional is continuous. Consequently

$$M^\dagger : E^\dagger \longrightarrow E^\dagger \quad (\text{VSD13.6})$$

is bijective. Surjectivity follows from surjectivity of M^* . For injectivity, if $M^* \lambda$ is continuous, choose a continuous ν with $M' \nu = M^* \lambda$; then $\lambda - \nu \in \ker M^* = \ker M'$, so λ itself is continuous.

Applying this proved calculation to (VSD13.2) and to its restriction to $\mathcal{J}$ yields

$$L^\dagger - \mu : \mathcal{B}^\dagger \xrightarrow{\sim} \mathcal{B}^\dagger, \quad L^\dagger - \mu : \mathcal{J}^\dagger \xrightarrow{\sim} \mathcal{J}^\dagger \quad (\mu \in \mathbb{C}). \quad (\text{VSD13.7})$$

Here $L^\dagger$ means the operator induced by the ordinary transpose L^* on the indicated dual quotient. It is not multiplication on the original primal space.

The exact dual-quotient row

$$0 \rightarrow \mathcal{Q}^\dagger \rightarrow \mathcal{B}^\dagger \rightarrow \mathcal{J}^\dagger \rightarrow 0 \quad (\text{VSD13.8})$$

commutes with these operators. It follows, by a direct diagram chase, that

$$\boxed{L^\dagger - \mu : \mathcal{Q}^\dagger \xrightarrow{\sim} \mathcal{Q}^\dagger \quad \text{for every } \mu \in \mathbb{C}.} \quad (\text{VSD13.9})$$

Injectivity follows from the middle bijection. For surjectivity, solve the equation in $\mathcal{B}^\dagger$; the image of its solution in $\mathcal{J}^\dagger$ is killed by $L^\dagger - \mu$, so it is zero by (VSD13.7). The solution therefore lies in the injected

$Q^\dagger$. Transport through Θ gives the same statements on $A^\dagger, J^\dagger, Q^\dagger$, and through the explicit endpoint splittings on $V_p^\dagger, Z^\dagger$.

Every nonzero polynomial P over $\mathbb{C}$ factors into a nonzero scalar times linear factors. Thus $P(L^\dagger)$ is bijective on all these defect coefficients. The usual commuting inverses define a unique extension of their $\mathbb{C}[t]$ -module structure to a vector-space structure over the rational-function field $\mathbb{C}(t)$:

$$\frac{P(t)}{Q(t)} \cdot x = P(L^\dagger) Q(L^\dagger)^{-1} x, \quad Q \neq 0. \quad (\text{VSD13.10})$$

This is independent of the fractional presentation because cross multiplication gives identical commuting operators. Every map in the dagger coefficient diagram commutes with $L^\dagger$ and hence with these inverses. Consequently the entire comparison object (VSD8.2), including its Gysin maps, can be formed on these $\mathbb{C}(t)$ -coefficient spaces.

In particular none of these coefficients has a nonzero generalized $L^\dagger$ -eigenvector of any finite order: $(L^\dagger - \mu)^m x = 0$ implies $x = 0$. The actual finite zero-jet residue vectors are in the continuous subobject, and their image in the dagger quotient is zero. This proves a global property of the precise extra coefficient sector rather than stopping at its nonzero existence.

One distinction is essential. On a discontinuous algebraic functional, one cannot move that functional through a limit merely because the original coefficient action is differentiable. The $L^\dagger$ above is the induced transpose of the explicitly defined operator (VSD13.1). This proof does not identify it with differentiation of the algebraic action T_a^* on arbitrary discontinuous functionals. Therefore the absence of generalized $L^\dagger$ -eigenvectors by itself does not classify all characters of that algebraic dilation action, does not prove a weight bound, and does not erase the comparison sector. It gives the exact rational-operator structure with the domains, maps and limitation of the inference specified.

There is an exact finite-sector comparison in both kernel and cokernel. For $E = A, J, Q$ with these transported operators, and every nonzero polynomial P , inclusion $E' \hookrightarrow E^*$ induces

$$\boxed{\ker P(L') \xrightarrow{\sim} \ker P(L^*), \quad \text{coker } P(L') \xrightarrow{\sim} \text{coker } P(L^*)}. \quad (\text{VSD13.11})$$

These are algebraic kernel and cokernel identifications with their actual maps; no topological quotient assertion is inserted. For the kernel, a functional killed by $P(L^*)$ has dagger class killed by the bijection $P(L^\dagger)$, so it is continuous. For surjectivity of the cokernel map, given $\lambda \in E^*$, solve $P(L^\dagger)\bar{\mu} = \bar{\lambda}$, lift $\bar{\mu}$ to $\mu \in E^*$, and observe $\lambda - P(L^*)\mu \in E'$. For injectivity, if continuous η equals $P(L^*)\lambda$, bijectivity on the dagger quotient forces $\lambda \in E'$.

Every finite-dimensional L^* -invariant subspace $M \subset E^*$ is consequently contained in E' . Its finite-dimensional operator has a nonzero characteristic polynomial P annihilating M ; the adjugate identity for $tI - L^*|_M$, evaluated at that operator, proves the assertion. Therefore

$$\boxed{M \subset \ker P(L^*) = \ker P(L') \subset E'}. \quad (\text{VSD13.12})$$

For every μ and $m \geq 1$, the entire kernels of $(L^* - \mu)^m$ and $(L' - \mu)^m$ agree, with no dimension bound needed for that kernel equality. Thus the two categories have identical full polynomial kernels and cokernels through their stated inclusion.

Replacing L by $L + k$, for any stated scalar k , preserves this result because $P(t + k)$ is a nonzero polynomial. The retained normal-degree shift therefore leaves the finite-sector comparison valid. On Z and V_p , retain their actual continuous endpoint blocks separately; all functionals on those finite blocks are continuous, and the explicit splitting reduces the other block to the J calculation. This extends (VSD13.11)–(VSD13.12) to those specified operators without dropping endpoints.

These consequences concern the explicit multiplication operator. They do not assume differentiation of discontinuous dual functionals or classify arbitrary dilation characters. They supply the exact finite-sector comparison from the calculated defect.

VSD14. Independent check of the completed trace receiver

After VSD0–VSD13 were completed, the entire SDT0–SDT9 proof was read, together with GTR10 in full and the exact GTR4.5–GTR5.4 action formulas used there. The checked SDT file is `../tau_weight_cohomology_20260924/CC_STRONG_DUAL_TOPOLOGY_AND_JETS.md`, SHA256 `7f1a03a6225f1dfc22b167713c0ddbb154635b06f9479f6137b229845b67aefc`. The checked GTR file is `CC_RAMIFIED_TRACE_AND_RESIDUE_ACTION.md`, SHA256 `8ede942ee574e1f96fdc2b6247c8786491abe3fc6401c24f157625f8f2fb9811`. This check found no mathematical correction to GTR10.

SDT’s weighted compactness estimate proves total boundedness one seminorm at a time with both original exponential weights retained. Its quotient argument makes only the needed per-seminorm lift choices; the independent compact-lift construction subsequently supplies a single compact set of lifts by its finite-choice series and uniform tails. Thus the bounded-lift conclusion used in the strong-dual embedding does not assume a continuous linear section. The stated strong topology on Q' is exactly the subspace topology under q' , because each bounded quotient set has such an exact bounded lift.

The compact absolutely convex hull argument in SDT6 identifies every continuous functional on the strong dual with evaluation at an actual coefficient vector. Its Baire and polar estimates recover the original bidual topology. SDT7 then obtains strong density of full zero jets by separation, and proves strong-dual completeness by uniform convergence on each bounded set followed by sequential continuity on the Fréchet primal space. These steps establish the precise strong completion imported by GTR10. They are compatible with VSD10: the strong and weak topologies have different completions, and neither proof interchanges them.

Here is a direct verification of the new action on that completion. Put

$$p_B(G) = \sup_{F \in B} \left| \sum_{\rho} \operatorname{Res}_{s=\rho} \frac{F(s)G(1-s)}{\zeta(s)} ds \right|, \quad V_a = aT_{a^{-1}}, \quad (\text{VSD14.1})$$

where $B \subset \mathcal{Q}$ is bounded, $G \in \mathcal{Q}_{\text{fin}}$, and the original denominator and multiplicities are retained. In the reflected second argument, T_a contributes a^{1-s} , whereas V_a contributes $a a^{s-1} = a^s$. Since $a > 0$,

$$p_B(T_a G) = a p_{T_{a^{-1}}B}(G), \quad p_B(V_a G) = p_{T_a B}(G). \quad (\text{VSD14.2})$$

Both right-hand sets are bounded. These are exact seminorm equalities, so they prove continuity of both maps and their inverses for the actual residue topology. They extend uniquely to its Hausdorff completion, and their products are obtained by density from $T_a T_b = T_{ab}$, $V_a V_b = V_{ab}$, and $T_a V_a = a \cdot 1$.

The complete residue isomorphism $\hat{\iota} : \mathcal{H}_{\text{res}} \rightarrow \mathcal{Q}'_{\beta}$ is the completion of the already injective residue map. Transposing the topological isomorphism κ is legitimate for strong duals: bounded sets have bounded images in each direction. Therefore $\mathfrak{I} = \kappa' \hat{\iota}$ identifies $\mathcal{H}_{\text{res}}$ with the entire actual degree-minus-one dual cohomology. The finite-domain identities and (VSD14.2) give

$$\mathfrak{I} \hat{T}_n = n T'_{n-1} \mathfrak{I} = U'_n \mathfrak{I}, \quad \mathfrak{I} \hat{V}_n = T'_n \mathfrak{I} = B'_n \mathfrak{I}. \quad (\text{VSD14.3})$$

Continuity extends these identities from the dense finite-domain image to the whole completion. The first action retains the transpose of inverse-coefficient transfer; the second retains the ordinary transpose of pullback. No factor n or inverse parameter is lost.

Evaluation then defines the full pairing without an unproved infinite residue sum. A convergent net in $\mathcal{H}_{\text{res}}$ gives uniform convergence of its finite residue pairings on every bounded primal set. This proves

independence of representatives and both identities

$$\widehat{\mathcal{R}}(T_n F, \widehat{T}_n h) = n \widehat{\mathcal{R}}(F, h), \quad \widehat{\mathcal{R}}(F, \widehat{V}_n h) = \widehat{\mathcal{R}}(T_n F, h). \quad (\text{VSD14.4})$$

The anti-diagonal support map is continuous in these strong topologies, by the sum of seminorms on the two bounded projections. It is exactly $h \mapsto (h, -h)$ in the completed residue coordinates. Its map into the overlap dual remains

$$h \mapsto q' \mathfrak{I}(h), \quad (q' \mathfrak{I}(h))(a) = \widehat{\mathcal{R}}([\Theta a], h), \quad \Theta = \tfrac{1}{2} \mathcal{M}_0. \quad (\text{VSD14.5})$$

The kernel condition follows from $\Theta J = \mathcal{J}$; SDT's bounded-lift argument proves the stated strong topological embedding. These verifications establish GTR10 exactly as written, with the complete original zeta and retained support signs. Neither a primal self-duality nor a positive form is inferred.

Literal current comparison and its complete operator defect — CV0-CV9

Proof source: CC_CURRENT_TO_VERDIER_COMPARISON.md.

From literal continuous currents to the ordinary Verdier cone

Complete independent derivation and operator audit, 24 September 2026. Locators CV0-CV9.

CV0. Exact inputs and scope

The supporting datum remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. This calculation acts on the already reconstructed coefficient spaces and the sphere receiver; it puts no coordinate, metric, addition or scalar operation on τ . The source stage, full-history requirement, retracted addition and correction precedence remain those recalled in CSD0. Neither the source J nor its quotient Q is replaced.

The accepted CC CONTINUOUS SUPPORT DUAL SHEAF.md, CSD0-CSD12, is retained unchanged at SHA256 361a5798405cc6688f8ea249f2bdfb05c1eb9b67499b52cfc27f6a2bade5579c. For this continuation the complete VSD1-VSD8 and VSD13 were read, together with their coefficient definitions in VSD0, in CC VERDIER SUPPORT INDEPENDENT.md, SHA256 a49fccbe2773e6d6378aceb394273e734f2ca206f4d6d787500927a1f6280d2b. The source identity and original-author reading coverage remain CSD0 and VSD0; this is a new local construction, not a new reading of the complete human papers.

Write E' for continuous complex-linear functionals, $E^* = \text{Hom}_{\mathbb{C}}(E, \mathbb{C})$ for all algebraic functionals, and $E^\dagger = E^*/E'$ as a vector space. No topology is imposed on a discontinuous functional or on this quotient. The objective is a literal morphism of sheaf complexes, and then a verification of VSD13's rational-operator comparison.

CV1. The original sheaf and its compact tests

Retain $Y = \mathbb{P}^1(\mathbb{C})$, $P = \{0, \infty\}$, $i : P \hookrightarrow Y$, and

$$S = \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f = 0\}, \quad V_{\pm} = S \oplus E_{\pm}, \quad E_{\pm} = \mathbb{C}^2,$$

$$A = \{a \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \text{ for all } N, j \geq 0\},$$

$$\Sigma f(u) = 2 \sum_{k \geq 1} f(ku), \quad Ra(u) = u^{-1}a(u^{-1}), \quad r_+(f, e) = \Sigma f, \quad r_-(h, e) = R\Sigma h = \Sigma \hat{h},$$

(CV1.1)

where $\hat{h}(t) = \int_{\mathbb{R}} h(v) e^{-2\pi i v t} dv$. The source theorem proves $J = \Sigma S$ closed, $\Sigma : S \rightarrow J$ a topological isomorphism, and $Q = A/J$ Fréchet. Both endpoint coordinates in each $E_{\pm}$ are retained. With

$$V = V_+ \oplus V_-,$$

$$\mathcal{F} = \underline{A}_Y \times_{i_* A^2} i_* V, \quad 0 \rightarrow \mathcal{F} \rightarrow \underline{A}_Y \oplus i_* V \xrightarrow{\text{ev}-r} i_* A^2 \rightarrow 0. \quad (\text{CV1.2})$$

Let $\mathcal{E}_c^q(U; A)$ be compactly supported smooth A -valued q -forms, with the compact-support inductive-limit topology and the original coefficient seminorms. The single primal resolution is

$$\begin{aligned} \mathcal{K}^0 &= \mathcal{E}_A^0 \oplus i_* V, \quad \mathcal{K}^1 = \mathcal{E}_A^1 \oplus i_* A^2, \quad \mathcal{K}^2 = \mathcal{E}_A^2, \\ d^0(f, v) &= (df, \text{ev}(f) - rv), \quad d^1(\omega, a) = d\omega. \end{aligned} \quad (\text{CV1.3})$$

CSD1 constructs this resolution and its compact-support acyclicity. The continuous-current complex in that proof is

$$\mathcal{T}_c^n(U) = (\mathcal{E}_c^{-n}(U; A))', \quad dT = (-1)^{n+1} T d, \quad -2 \leq n \leq 0. \quad (\text{CV1.4})$$

CV2. Algebraic test currents really form a sheaf

Define on the same test spaces

$$\mathcal{T}_a^n(U) = \text{Hom}_{\mathbb{C}}(\mathcal{E}_c^{-n}(U; A), \mathbb{C}), \quad dT = (-1)^{n+1} T d. \quad (\text{CV2.1})$$

No continuity restriction is imposed. Restrictions transpose extension by zero, and $d^2 = 0$.

For a cover of U , choose a locally finite subordinate smooth partition (ρ_i) . Compatible local functionals act on a test form ϕ by summing their values on $\rho_i \phi$. Only finitely many terms meet the compact support of ϕ . For two partitions, refine by their products; compatibility equates the resulting finite sums. This proves existence, independence and uniqueness of gluing. No infinite summation is passed through an algebraic functional. Thus (CV2.1) is a sheaf.

It is fine: multiplication by a smooth scalar function acts on tests, and the associated locally finite partition endomorphisms add to the identity on every compact test by a finite sum. It is also flabby in the algebraic sheaf category. Indeed extension by zero embeds the test space of a smaller open as a vector subspace of that of a larger open; every linear functional on the subspace extends by a basis extension. This flabbiness is not asserted for the continuous-current sheaf.

The inclusion of continuous functionals gives a literal injective morphism of complexes

$$j : \mathcal{T}_c^\bullet \hookrightarrow \mathcal{T}_a^\bullet. \quad (\text{CV2.2})$$

All statements about the target here are statements about vector-space sheaves, not continuity of discontinuous functionals.

CV3. Integrate in the coefficient space before applying a functional

For $\lambda \in A^*$, define the constant algebraic current by

$$c_\lambda(\phi) = \lambda \left(\int_U \phi \right), \quad \phi \in \mathcal{E}_c^2(U; A). \quad (\text{CV3.1})$$

The integral is first taken in the complete Fréchet space A . The expression $\int \lambda(\phi)$ is not used for discontinuous λ .

Here is the local contraction underlying the assertion about its cohomology. In an oriented plane chart choose compact scalar bumps ρ_x, ρ_y of integral one, and define

$$I_x f(y) = \int_{\mathbb{R}} f(x, y) dx, \quad H_x f(x, y) = \int_{-\infty}^x f(t, y) dt - \left(\int_{-\infty}^x \rho_x(t) dt \right) I_x f(y).$$

For compact forms put

$$K(P dx + Q dy) = H_x P, \quad K(G dx \wedge dy) = H_x G dy - \rho_x H_y (I_x G) dx. \quad (\text{CV3.2})$$

These are the continuous operators proved in CSD2. Their identities are identities in A :

$$Kdf = f, \quad dK + Kd = 1 \text{ in degree 1}, \quad dK\omega = \omega - \rho_x \rho_y dx \wedge dy \int \omega \text{ in degree 2}. \quad (\text{CV3.3})$$

They follow directly from $\partial_x H_x f = f - \rho_x I_x f$ and $H_y \partial_y g = g$ for compact g . All output supports stay in a fixed enlarged rectangle. Algebraic transposition preserves these identities because they are already identities of linear maps. Hence

$$\underline{A}_Y^*[2] \xrightarrow[\lambda \mapsto c_\lambda]{\simeq} \mathcal{T}_a^\bullet. \quad (\text{CV3.4})$$

For continuous λ , (CV3.1) equals CSD's continuous current. Thus the square with $\underline{A}'[2] \rightarrow \mathcal{T}_c$ and (CV2.2) commutes literally.

In a positive local coordinate z_p , define

$$h_p \lambda(\phi) = \lambda \left(\int \frac{d \arg z_p}{2\pi} \wedge \phi \right), \quad \delta_p \lambda(f) = \lambda(f(p)). \quad (\text{CV3.5})$$

The angular integral is an actual A -valued integral. Its coefficient $1/r$ is locally integrable against area, and every coefficient seminorm is bounded on the compact test support. Truncated integrals converge in each seminorm, hence in A . The A -valued Stokes identity is

$$\int \frac{d \arg z_p}{2\pi} \wedge df = f(p).$$

It is proved by the positive small-circle boundary limit in A , before λ is applied. Therefore

$$dh_p \lambda = \delta_p \lambda. \quad (\text{CV3.6})$$

The positive compact form $-d\chi \wedge d \arg z_p / (2\pi)$ has integral one when $\chi = 1$ near p . As in CSD6, the restriction fibre cone identifies the class of $\delta_p \lambda$ with the positive local fundamental class. Hence the chain map

$$i_* A^* \xrightarrow{\delta_p} \mathcal{T}_a^\bullet \quad (\text{CV3.7})$$

is exactly VSD2's positive Gysin map g_{p, A^*} , not only an unspecified nonzero multiple of it.

CV4. Literal comparison of the two complete cone complexes

Form the algebraic current cone

$$\mathcal{D}_a \mathcal{F} = \text{Cone} \left(i_*(A^*)^2 \xrightarrow{(\delta, -r^*)} \mathcal{T}_a^\bullet \oplus i_* V^* \right). \quad (\text{CV4.1})$$

Its terms in degrees $-2, -1, 0$ are $\mathcal{T}_a^{-2}, \mathcal{T}_a^{-1} \oplus i_*(A^*)^2$, and $\mathcal{T}_a^0 \oplus i_*V^*$, with

$$d^{-1}(T, \lambda_+, \lambda_-) = (dT + \delta_0 \lambda_+ + \delta_\infty \lambda_-, -r_+^* \lambda_+, -r_-^* \lambda_-). \quad (\text{CV4.2})$$

Its differential is exactly the algebraic Hom differential of $\Gamma_c(U, \mathcal{K})$ on each open. Every endpoint is still present.

To identify the target, VSD1's ordinary vector-sheaf calculation gives

$$R\mathcal{H}om(\underline{E}, \underline{\mathbb{C}}[2]) = \underline{E}^*[2], \quad R\mathcal{H}om(i_*E, \underline{\mathbb{C}}[2]) = i_*E^*.$$

The first uses disk adjunction $\text{Hom}(\underline{E}, K) = \text{Hom}_{\mathbb{C}}(E, \Gamma K)$, exactness of vector-space Hom, and vanishing of higher disk cohomology of the constant target. The second uses positive local support cohomology of that target and exact vector-space Hom at a point. Thus neither requires finite rank. Dualizing (CV1.2) gives the cone of the Gysin map and $-r^*$. Equations (CV3.4)-(CV3.7) identify precisely those maps. Consequently

$$\boxed{\mathcal{D}_a \mathcal{F} \simeq R\mathcal{H}om_Y(\mathcal{F}, \underline{\mathbb{C}}_Y[2]) = \mathbb{D}_Y \mathcal{F}.} \quad (\text{CV4.3})$$

The inclusions of functionals now give an actual termwise injective chain map

$$\boxed{c_{\text{cur}} : \mathcal{D}_c \mathcal{F} \hookrightarrow \mathcal{D}_a \mathcal{F},} \quad (\text{CV4.4})$$

using (CV2.2) on currents and $A' \hookrightarrow A^*, V' \hookrightarrow V^*$ at the poles. Its chain compatibility follows immediately from (CV4.2), including both minus signs. The constant-current comparison commutes with the coefficient inclusion, and the Dirac comparisons commute with the same inclusion. Under (CV4.3) this is the morphism c of VSD7. It is an actual map out of the fixed CSD complex.

CV5. The literal quotient current sheaf and comparison defect

Let $\mathcal{T}_\dagger^n = \mathcal{T}_a^n / \mathcal{T}_c^n$ as an ordinary quotient sheaf. For each open U ,

$$\Gamma(U, \mathcal{T}_\dagger^n) = \text{Hom}_{\mathbb{C}}(\mathcal{E}_c^{-n}(U; A), \mathbb{C}) / (\mathcal{E}_c^{-n}(U; A))'. \quad (\text{CV5.1})$$

Indeed $\mathcal{T}_c^n$ is fine on these paracompact opens, so its first cohomology vanishes. The sheaf exact sequence therefore gives exactness on sections at the quotient term. This also provides a direct gluing proof for (CV5.1), without declaring arbitrary presheaf quotients to be sheaves.

The local long exact sequence and (CV3.4) give

$$\mathcal{H}^{-2}(\mathcal{T}_\dagger) = \underline{A}^\dagger, \quad \mathcal{H}^{n \neq -2}(\mathcal{T}_\dagger) = 0.$$

There is a specified local quasi-isomorphism

$$\underline{A}^\dagger[2] \longrightarrow \mathcal{T}_\dagger, \quad [\lambda] \mapsto [c_\lambda]. \quad (\text{CV5.2})$$

It is well defined because a continuous coefficient yields a continuous current. The same observation defines $[\delta_p \lambda]$ and $[h_p \lambda]$. No integration in an unspecified topology on $A^\dagger$ is required: choose a representative in A^* , integrate in A , apply it, and pass to the current quotient.

The quotient of (CV4.4) is consequently the actual complex

$$\mathcal{D}_\dagger = \text{Cone} \left(i_*(A^\dagger)^2 \xrightarrow{(\delta^\dagger, -r^\dagger)} \mathcal{T}_\dagger \oplus i_*V^\dagger \right). \quad (\text{CV5.3})$$

The map $\text{Cone}(c_{\text{cur}}) \rightarrow \mathcal{D}_\dagger$ sends (b, a) to the quotient class of b . Its kernel is the cone of the identity of $\mathcal{D}_c$, which contracts by $(b, a) \mapsto (0, b)$. Therefore it is a quasi-isomorphism. With (CV5.2), this proves

$$\text{Cone}(c_{\text{cur}}) \simeq \text{Cone} \left(i_*(A^\dagger)^2 \xrightarrow{(g_{A^\dagger}, -r^\dagger)} \underline{A}^\dagger[2] \oplus i_* V^\dagger \right). \quad (\text{CV5.4})$$

This is exactly VSD8's comparison object, now obtained as a quotient of the literal current complexes. The quotient has no imposed topology.

CV6. Exact support and global maps of the comparison

The CSD chain maps work algebraically with the integral-before-functional convention. At a pole the finite model is

$$[A^* \xrightarrow{0} A^* \xrightarrow{r_p^*} V_p^*] \quad \text{in degrees } -2, -1, 0,$$

mapped into currents by

$$\lambda \mapsto c_\lambda, \quad \mu \mapsto (h_p \mu, -\mu), \quad \nu \mapsto (0, \nu). \quad (\text{CV6.1})$$

Its costalk is the positive Gysin cone

$$A^* \xrightarrow{\lambda \mapsto (\lambda, -r_p^* \lambda)} A^* \oplus V_p^*,$$

contracting to V_p^* by $(a, \nu) \mapsto \nu + r_p^* a$. Restricting (CV6.1) to continuous functionals gives exactly CSD5–CSD6. Quotienting gives the dagger versions, with the same signs.

Globally the A -valued angular identity is

$$d\vartheta = \delta_0 - \delta_\infty, \quad \vartheta = \frac{d \arg z}{2\pi}.$$

Thus

$$[A^* \xrightarrow{0} A^* \xrightarrow{(r_+^*, -r_-^*)} V^*] \longrightarrow \Gamma(Y, \mathcal{D}_a \mathcal{F})$$

has the chain map

$$\lambda \mapsto c_\lambda, \quad \mu \mapsto (\vartheta \mu; -\mu, +\mu), \quad (\nu_+, \nu_-) \mapsto (0; \nu_+, \nu_-). \quad (\text{CV6.2})$$

The compact-test homotopies and the finite contraction in VSD5 prove this is a quasi-isomorphism, exactly as in CSD7. All integrals in (CV6.2) are taken before applying an algebraic coefficient functional.

Restriction from (CV6.2) to the poles is identity at degree -2 , has signs $+$, $-$ at degree -1 , and is coordinate projection at degree zero. This proves the VSD6 diagram, with continuous inclusions and dagger quotients commuting throughout. The full nonzero dual localization row is

$$0 \rightarrow A^* \xrightarrow{\Delta} (A^*)^2 \xrightarrow{-\lambda_+ + \lambda_-} A^* \xrightarrow{0} Q^* \xrightarrow{(\mu, -\mu)} (Q^*)^2 \xrightarrow{q^*(\mu_+ + \mu_-)} A^* \xrightarrow{r^*} Z^* \rightarrow E_+^* \oplus E_-^* \rightarrow 0, \quad (\text{CV6.3})$$

where $Z = \ker(r_+ - r_-)$ and $r : Z \rightarrow A$ is common restriction. The same row holds for continuous duals and dagger quotients, with finite endpoint dagger coefficients zero.

For completeness, the dual sheaf's own support-to-global map has a different direction. Its two costalks $V_+^* \oplus V_-^*$ enter degree zero of (CV6.2). On cohomology their map to Z^* is

$$(\nu_+, \nu_-) \mapsto (s_+^* \nu_+ + s_-^* \nu_-, \nu_+|_{E_+}, \nu_-|_{E_-}), \quad (\text{CV6.4})$$

where $s_p : J \rightarrow V_p$ are the actual continuous sections of r_p . Its kernel is $\{(r_+^* \lambda, -r_-^* \lambda) : \lambda \in A^*\}$: a functional killing Z factors through $V/Z = J$, then extends algebraically to A . Its surjectivity follows from $V \simeq Z \oplus J$. The same proof for continuous functionals uses Hahn-Banach and the continuous splitting. Thus (CV4.4) also realizes VSD6.4, not only its restriction maps.

The exact quotient row

$$0 \rightarrow Q^\dagger \rightarrow A^\dagger \rightarrow J^\dagger \rightarrow 0 \quad (\text{CV6.5})$$

follows from the continuous and algebraic restriction rows. At its middle kernel, subtract a continuous Hahn-Banach extension of the restriction to J ; the remainder factors algebraically through Q . Surjectivity at the right uses algebraic extension from J . Since $E_\pm$ are finite dimensional,

$$V_p^\dagger \simeq J^\dagger, \quad Z^\dagger \simeq J^\dagger, \quad E_p^\dagger = 0.$$

Hence the literal comparison defect has exactly

$$\begin{aligned} \mathcal{H}^{-2} &= \underline{A}^\dagger, \quad \mathcal{H}^{-1} = i_*(Q^\dagger \oplus Q^\dagger), \quad \mathcal{H}^0 = 0, \quad i_p^! \mathcal{D}_\dagger = J^\dagger[0], \\ H^{-2}(Y, \mathcal{D}_\dagger) &= A^\dagger, \quad H^{-1}(Y, \mathcal{D}_\dagger) = Q^\dagger, \quad H^0(Y, \mathcal{D}_\dagger) = J^\dagger. \end{aligned} \quad (\text{CV6.6})$$

These groups retain the Gysin extension; a zero degree-zero cohomology sheaf has not been mistaken for zero global degree-zero cohomology.

Every actual continuous coefficient map and every proper test pullback preserves continuous functionals under transposition. Thus the inclusion, quotient and all maps above commute with the CSD cover-current pushforwards and both mirrors. Their signs are inherited from identities in the A -valued tests before any functional is applied. This makes no topology claim about the algebraic target.

CV7. Audit of the exact rational operator and original factors

Use the original Mellin comparison

$$\Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, \quad \Theta : A \xrightarrow{\sim} \mathcal{B}, \quad \Theta J = \mathcal{I}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{I},$$

where $\mathcal{B}$ is the entire strip-Schwartz space and $\mathcal{I}$ the full original nontrivial-zero jet ideal. Let $LF = sF$. The exact source identity remains

$$\mathcal{M}_0 \Sigma f(s) = 2\zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v} \quad (\Re s > 1), \quad \mathcal{M}_0 = 2\Theta. \quad (\text{CV7.1})$$

For any $\mu \in \mathbb{C}$, $M_\mu = L - \mu$ on $\mathcal{B}$ is injective and has image exactly $\{F : F(\mu) = 0\}$. Evaluation is continuous and onto, for $e^{(s-\mu)^2}$ lies in $\mathcal{B}$ and equals one at μ . Division by $s - \mu$ is continuous on this image: outside $|s - \mu| < 1$ the quotient has the original strip bound; inside, the maximum principle on the circle of radius two bounds the holomorphic quotient by a larger-strip seminorm. The imaginary polynomial weight is bounded on this fixed disk. Thus the image is closed of codimension one and the inverse on it is continuous.

On $\mathcal{I}$, retain the full original-source function

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{CV7.2})$$

Its zero divisor is exactly the nontrivial-zero divisor with full multiplicities. The quotient F/F_0 is entire for $F \in \mathcal{J}$, although it need not belong to $\mathcal{B}$. Its value at μ is the continuous functional

$$\ell_\mu(F) = \begin{cases} F(\mu)/F_0(\mu), & F_0(\mu) \neq 0, \\ F^{(m)}(\mu)/F_0^{(m)}(\mu), & \text{ord}_\mu F_0 = m. \end{cases} \quad (\text{CV7.3})$$

There is no missing factorial: both order- m Taylor coefficients have the same $m!$, which cancels in their ratio. Cauchy estimates prove continuity, and $\ell_\mu(F_0) = 1$. Multiplication by $s - \mu$ on $\mathcal{J}$ has image $\ker \ell_\mu$: division there leaves all required orders intact, including the full multiplicity at μ . Its inverse is the preceding continuous division restricted to the closed ideal.

This includes the endpoints and cancelled trivial-zero locations. No zero value of F_0 is assumed there:

$$\begin{aligned} F_0(0) = F_0(1) &= \frac{1}{8}, & F_0(-2r) &= \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) \\ &= \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0 \quad (r \geq 1). \end{aligned} \quad (\text{CV7.4})$$

These points use the first branch of (CV7.3). The original pole at one, Gamma poles and all trivial-zero derivatives remain in the comparison.

For a continuous injective $M : E \rightarrow E$ onto a closed finite-codimensional subspace with continuous inverse on its image, both transposes are onto. Algebraic surjectivity extends a functional on $M(E)$ by a basis; continuous surjectivity extends the continuous functional $Mx \mapsto \lambda(x)$ by Hahn-Banach. Both kernels are the full dual of the finite-dimensional Hausdorff quotient $E/M(E)$; hence

$$\ker M^* = \ker M' \subset E', \quad M^\dagger : E^\dagger \xrightarrow{\sim} E^\dagger. \quad (\text{CV7.5})$$

For injectivity of the last map, if $M^* \lambda$ is continuous, choose continuous ν with $M' \nu = M^* \lambda$. Then $\lambda - \nu$ is in the continuous kernel, so λ is continuous.

Apply (CV7.5) to M_μ on $\mathcal{B}$ and $\mathcal{J}$. Their dagger operators are bijective for every μ . The exact row $0 \rightarrow \mathcal{Q}^\dagger \rightarrow \mathcal{B}^\dagger \rightarrow \mathcal{J}^\dagger \rightarrow 0$ commutes with them. To solve an equation in $\mathcal{Q}^\dagger$, solve it in $\mathcal{B}^\dagger$; the image of that solution in $\mathcal{J}^\dagger$ is killed by its bijective operator, so is zero. Injectivity follows from that on $\mathcal{B}^\dagger$. Thus

$$L^\dagger - \mu : \mathcal{Q}^\dagger \xrightarrow{\sim} \mathcal{Q}^\dagger \quad (\mu \in \mathbb{C}). \quad (\text{CV7.6})$$

Factoring any nonzero polynomial into linear factors proves that every $P(L^\dagger)$ is bijective. The commuting inverses define the exact rational action

$$\frac{P(t)}{Q(t)} x = P(L^\dagger) Q(L^\dagger)^{-1} x. \quad (\text{CV7.7})$$

Cross multiplication proves independence of presentation. This verifies VSD13's rational-operator argument on the complete actual coefficients.

CV8. Transported source operators and polynomial comparison

For clarity about every operator in that comparison, put $D_u = u\partial_u$, $D_v = v\partial_v$. Integration by parts in the rapidly decreasing source gives $\Theta(-D_u a) = s\Theta a$. The compatible actual source operators are

$$\begin{aligned} L_A &= -D_u, & G_+(f, c_0, c_1) &= (-D_v f, 0, c_1), \\ G_-(h, d_0, d_1) &= (h + D_v h, d_0, 0). \end{aligned} \quad (\text{CV8.1})$$

They preserve the stated source conditions: D_v preserves evenness, vanishing at zero, and the zero-integral condition by integration by parts. Their endpoint generators are respectively $\text{diag}(0, 1)$ and $\text{diag}(1, 0)$, exactly those of the original dilation action.

Termwise differentiation of the convergent summation gives $D_u \Sigma = \Sigma D_v$, while $D_u R = -R - R D_u$. Hence

$$L_A r_+ = r_+ G_+, \quad L_A r_- = r_- G_-. \quad (\text{CV8.2})$$

Under $V_p \simeq J \oplus E_p$, these are L_J plus the stated finite endpoint blocks. On $Z \simeq J \oplus E_+ \oplus E_-$, the endpoint block is $\text{diag}(0, 1, 1, 0)$. Thus dagger transport to $V_p^\dagger, Z^\dagger$ retains all original finite blocks in the continuous sector; it does not erase their polynomial kernels or cokernels. Restrictions and Gysin are $\mathbb{C}(t)$ -linear for these coefficient operators.

For $E = A, J, Q$, and also V_p, Z with these operators, inclusion $E' \hookrightarrow E^*$ gives exactly

$$\boxed{\ker P(L') \xrightarrow{\sim} \ker P(L^*), \quad \text{coker } P(L') \xrightarrow{\sim} \text{coker } P(L^*)} \quad (P \neq 0). \quad (\text{CV8.3})$$

For the kernel, a killed algebraic functional has dagger class killed by the bijection $P(L^\dagger)$, so is continuous. For cokernel surjectivity, solve $P(L^\dagger)\bar{\mu} = \bar{\lambda}$ and lift $\bar{\mu}$ algebraically; then $\lambda - P(L^*)\mu$ is continuous. For injectivity, if continuous $\eta = P(L^*)\lambda$, injectivity on the dagger quotient forces λ continuous. These are algebraic kernel/cokernel identities, not assertions that a quotient topology is Hausdorff.

A finite-dimensional L^* -invariant subspace is killed by the characteristic polynomial of its restricted operator, by the adjugate identity. Equation (CV8.3) therefore places the entire subspace in E' . The equality of kernels also applies to $(L^* - \mu)^m$ without a finite-dimension assumption on that kernel. Replacing L by a stated $L + k$ changes the polynomial to $P(t + k)$, still nonzero, so the same conclusion includes the specified normal-degree shifts on those coefficient groups.

Mirrors must not be included in an assertion that all maps commute with L . The actual reflection satisfies

$$L_A R = R(1 - L_A), \quad R^\dagger f(L^\dagger) = f(1 - L^\dagger)R^\dagger \quad (f \in \mathbb{C}(t)). \quad (\text{CV8.4})$$

The first equality follows from the displayed $D_u R$ formula; transposition, $R^2 = 1$, and the uniqueness of polynomial inverses give the second. For $L + k$ the reflection of the rational variable is $t \mapsto 1 + 2k - t$. The same source identity holds on V with chart swap by (CV8.1), including endpoints. Orientation signs multiply these operators by ± 1 and do not alter that semilinearity.

Finally $L^\dagger$ is the induced transpose of a specified continuous operator on the primal coefficient. This proof never differentiates the action on a discontinuous functional or moves one through a limiting difference quotient. The rational structure and polynomial comparison do not classify arbitrary algebraic dilation characters.

CV9. What the constructed map proves

There is now a literal map from the fixed continuous-current sheaf complex to a current resolution of ordinary algebraic Verdier duality, equation (CV4.4). Its quotient is the computed current/Gysin cone (CV5.3), and its derived comparison defect is exactly VSD8. Every local, global, costalk and support map is realized by the explicit current representatives in CV6, with the original endpoint and orientation signs.

The independent VSD13 audit verifies its rational-operator and full polynomial kernel/cokernel assertions for the actual coefficients, including multiple zeros, endpoint blocks and normal scalar shifts. Mirrors are explicitly semilinear rather than falsely asserted to commute with the rational variable. No topology on discontinuous functionals, finite-dimensional replacement of the original source, or RH conclusion is used.

Polynomial spectral lifting through the actual current comparison — CP0-CP11

Proof source: CC_POLYNOMIAL_SPECTRAL_LIFT.md.

Polynomial spectral lifting through the actual current-dual comparison

Complete independent derivation, 24 September 2026. Proof locators CP0-CP11.

CP0. Source stage and the actual comparison

The support remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. The operations in this note act on the already reconstructed arithmetic coefficients and the actual sphere sheaf. They assign no addition, coordinate, metric, numerical origin, or parity to τ . The complete-history-before-arithmetic order and the separate branch counters are retained.

READ_FIRST_USER_CONSTRUCTION.md, the full connected USER_ARGUMENT_RECONSTRUCTION.md with its correction table, and PC07 in PREREQUISITE_AND_OBSTRUCTION_CHECKS.md were recalled before the operations below. The governing verbatim WU050-WU055 and WU061-WU063 passages were read in the preceding connected derivation. The new lifting problem is the one defined by the actual continuous-current inclusion; it is not an independently chosen extension.

The following complete local proofs were read:

- CC_VERDIER_SUPPORT_INDEPENDENT.md, VSD0-VSD14, in the adjacent quantum tau programme bridge_20260924 directory; SHA256 A49FCCBE2773E6D6378ACEB394273E734F2CA206F4D6D787500927A1F6280D2B.
- CC_CONTINUOUS_SUPPORT_DUAL_SHEAF.md, CSD0-CSD12; SHA256 361A5798405CC6688F8EA249F2BDFB05C1EB9B67499B52CFC27F6A2BADE5579C.
- CC_CURRENT_TO_VERDIER_COMPARISON.md, CV0-CV9, including its final operator audit; SHA256 F4825A0CED9ACE0D3610A064A6DC8B21311E86822B10937DB2AF5002728BF286.

These are local derivations, not new original-author sources. Their source uses remain Connes and Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, §5, and the later signed charts in [arXiv:2609.00299v1](#). The original author TeX and reading coverage are those recorded in CSD0, CV0, and the preceding CSL review. The comparison to Deligne uses the complete DC5-DC7 reconstruction of *La conjecture de Weil. II*, §3.6, with its stated French-transcription witness.

Put $R = \mathbb{C}[t]$. Every derived Hom below is over this ring with a specified operator t , not a formal list of eigenvalue labels. We prove that the actual comparison becomes an isomorphism on all derived polynomial spectral sectors. We also give the correction of a representative, the exact uniqueness statement, and the complete original-zero multiplicity case.

CP1. Original coefficients, their operators, and retained endpoints

Let

$$S = \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f = 0\}, \quad V_{\pm} = S \oplus E_{\pm}, \quad E_{\pm} = \mathbb{C}^2,$$

$$A = \{a \in C^{\infty}(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \text{ for every } N, j \geq 0\}. \quad (\text{CP1.1})$$

The actual source maps are

$$\Sigma f(u) = 2 \sum_{n \geq 1} f(nu), \quad Ra(u) = u^{-1}a(u^{-1}), \quad r_+(f, e) = \Sigma f, \quad r_-(h, e) = R\Sigma h = \Sigma \hat{h}. \quad (\text{CP1.2})$$

Here $\hat{h}(x) = \int_{\mathbb{R}} h(v) e^{-2\pi i v x} dv$. The full Poisson formula before the two source conditions is

$$\Sigma \hat{h}(u) = u^{-1} \Sigma h(u^{-1}) + u^{-1} h(0) - \int_{\mathbb{R}} h. \quad (\text{CP1.3})$$

The exact source theorem gives $J = \Sigma S$ closed in A , a continuous inverse $\Sigma^{-1} : J \rightarrow S$, and the Fréchet quotient $Q = A/J$. Both endpoint coordinates on each chart are retained.

Write $D_u = u\partial_u$, $D_v = v\partial_v$. The actual coefficient operators of CV8 are

$$L_A = -D_u, \quad G_+(f, c_0, c_1) = (-D_v f, 0, c_1), \quad G_-(h, d_0, d_1) = (h + D_v h, d_0, 0). \quad (\text{CP1.4})$$

They are continuous and preserve all stated source conditions. Integration by parts proves the integral condition; differentiation preserves evenness and the value condition. From $D_u \Sigma = \Sigma D_v$ and $D_u R = -R - R D_u$,

$$L_A r_+ = r_+ G_+, \quad L_A r_- = r_- G_-. \quad (\text{CP1.5})$$

Thus these are compatible operators on the actual coefficient diagram. Under the proved splittings

$$V_p \simeq J \oplus E_p, \quad Z := \ker(r_+ - r_-) \simeq J \oplus E_+ \oplus E_-, \quad (\text{CP1.6})$$

the endpoint operators are $\text{diag}(0, 1)$ at $+$, $\text{diag}(1, 0)$ at $-$, and $\text{diag}(0, 1, 1, 0)$ on Z . These splittings intertwine the operators: the source section on J is Σ^{-1} , or its Fourier transform on the other chart, and (CP1.5) proves the intertwining.

For a coefficient E , write

$$E' = \text{Hom}_{\text{cont}, \mathbb{C}}(E, \mathbb{C}), \quad E^* = \text{Hom}_{\mathbb{C}}(E, \mathbb{C}), \quad E^{\dagger} = E^*/E'. \quad (\text{CP1.7})$$

On these spaces t is the ordinary transpose of the specified primal operator, or its induced quotient operator. In particular it is L' , L^* , or $L^{\dagger}$ on A, J, Q . On V_p, Z it is the transpose of the full operator in (CP1.4)-(CP1.6). No derivative of a discontinuous functional's dilation orbit is being taken.

The original Mellin comparison is

$$\Theta a(s) = \frac{1}{2} \int_0^{\infty} a(u) u^s \frac{du}{u}, \quad \Theta L_A = s\Theta, \quad \Theta : A \xrightarrow{\sim} \mathcal{B}, \quad \Theta J = J, \quad (\text{CP1.8})$$

where $\mathcal{B}$ is the entire strip-Schwartz space and J is the ideal of all full original nontrivial-zero jets. In particular the factor $1/2$ in Θ and the factor 2 in Σ remain present. Section CP9 records the complete original-zeta factors used below.

CP2. The literal current comparison and its exact defect

Let $Y = \mathbb{P}^1(\mathbb{C})$, with positive complex orientation, and pole inclusion $i : \{0, \infty\} \hookrightarrow Y$. The primal sheaf is

$$\mathcal{F} = \underline{A}_Y \times_{i_* A^2} i_*(V_+ \oplus V_-).$$

CSD constructs continuous currents from the compactly supported A -valued tests; CV constructs algebraic currents on the same tests. With $V = V_+ \oplus V_-$, their actual complexes are

$$\begin{aligned} \mathcal{C}_c &= \text{Cone}(i_* A'^2 \xrightarrow{(\delta, -r')} \mathcal{T}_c^\bullet \oplus i_* V'), \\ \mathcal{C}_a &= \text{Cone}(i_*(A^*)^2 \xrightarrow{(\delta, -r^*)} \mathcal{T}_a^\bullet \oplus i_* V^*). \end{aligned} \quad (\text{CP2.1})$$

The current differential in degree n is $T \mapsto (-1)^{n+1} T d$. The positive Dirac current at either pole is $\delta_p \lambda(f) = \lambda(f(p))$. The degree-minus-one differential retains both signs:

$$(T, \lambda_+, \lambda_-) \mapsto (dT + \delta_0 \lambda_+ + \delta_\infty \lambda_-, -r_+^* \lambda_+, -r_-^* \lambda_-), \quad (\text{CP2.2})$$

and likewise for continuous coefficients.

CV4 proves the literal termwise injection

$$c : \mathcal{C}_c \hookrightarrow \mathcal{C}_a, \quad \mathcal{C}_a \simeq \mathbb{D}_Y \mathcal{F}. \quad (\text{CP2.3})$$

It is R -linear. The primal coefficient operators commute with the test differential, evaluation and integration; their transposes therefore commute with every arrow in (CP2.1). For algebraic functionals integration is performed in the complete coefficient space before applying the functional, as in CV3.

Let $\mathcal{D} = \mathcal{C}_a / \mathcal{C}_c$. This is CV5.3's literal quotient-current complex, and

$$\text{Cone}(c) \xrightarrow{\simeq} \mathcal{D}, \quad (b, a) \mapsto [b]. \quad (\text{CP2.4})$$

Its kernel is $\text{Cone}(1_{\mathcal{C}_c})$, contracted by $(b, a) \mapsto (0, b)$. It has the coefficient/Gysin description

$$\mathcal{D} \simeq \text{Cone}(i_*(A^\dagger)^2 \xrightarrow{(g, -r^\dagger)} \underline{A}_Y^\dagger[2] \oplus i_* V^\dagger). \quad (\text{CP2.5})$$

The positive Gysin maps are coefficient-natural. Formula (CP2.5) does not delete their extension class.

The exact dagger row and endpoint facts are

$$0 \rightarrow Q^\dagger \rightarrow A^\dagger \rightarrow J^\dagger \rightarrow 0, \quad V_p^\dagger \simeq J^\dagger, \quad Z^\dagger \simeq J^\dagger, \quad E_p^\dagger = 0. \quad (\text{CP2.6})$$

Its proof subtracts a continuous Hahn-Banach extension at the middle kernel and uses algebraic extension for surjectivity. Endpoint dagger spaces vanish because every functional on a finite-dimensional endpoint space is continuous, not because those endpoints were removed.

For reference, the complete actual defect groups are

$$\begin{aligned} \mathcal{H}^{-2}(\mathcal{D}) &= \underline{A}_Y^\dagger, \quad \mathcal{H}^{-1}(\mathcal{D}) = i_*(Q^\dagger \oplus Q^\dagger), \quad \mathcal{H}^0(\mathcal{D}) = 0, \\ H^{-2}(Y, \mathcal{D}) &= A^\dagger, \quad H^{-1}(Y, \mathcal{D}) = Q^\dagger, \quad H^0(Y, \mathcal{D}) = J^\dagger, \quad i_p^\dagger \mathcal{D} \simeq J^\dagger[0]. \end{aligned} \quad (\text{CP2.7})$$

The local pole stalk has $A^\dagger, Q^\dagger, 0$ in degrees $-2, -1, 0$. The open-stratum restriction is $\underline{A}^\dagger[2]$. These are the actual CSD/CV/VSD comparisons and maps; no new dual cone is being chosen.

CP3. Why every nonzero polynomial is invertible on the defect

We recall the exact algebra needed from VSD13 and CV7. On $\mathcal{B}$, multiplication by $s - \lambda$ is injective, with closed codimension-one image $\{F : F(\lambda) = 0\}$. Division on that image is continuous, by the fixed-disk Cauchy estimate and division outside the disk.

On $\mathcal{J}$, its image is the kernel of the continuous functional

$$\ell_\lambda(F) = \left(\frac{F}{F_0} \right) (\lambda), \quad F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{CP3.1})$$

At a zero of F_0 of order m , this is $F^{(m)}(\lambda)/F_0^{(m)}(\lambda)$; otherwise it is $F(\lambda)/F_0(\lambda)$. It is onto because $\ell_\lambda(F_0) = 1$. The same division estimate applies, and division retains all required zero orders.

If a continuous injective $M : E \rightarrow E$ has closed finite-codimensional image and continuous inverse on that image, both transposes M' and M^* are onto. Extend a functional from the image algebraically, or continuously by Hahn-Banach. Their kernels coincide, because all functionals on the finite-dimensional Hausdorff quotient $E/M(E)$ are continuous. Consequently $M^\dagger$ is bijective: if $M^* \lambda$ is continuous, subtract a continuous preimage and then use the common continuous kernel.

Apply this to $L - \lambda$ on $\mathcal{B}, \mathcal{J}$. The exact row (CP2.6) then proves bijectivity on $Q^\dagger$ as well: solve in $A^\dagger$; the image of the solution in $J^\dagger$ is killed by its invertible operator and is therefore zero. Thus

$$t - \lambda : E^\dagger \xrightarrow{\sim} E^\dagger \quad (E = A, J, Q, V_p, Z; \lambda \in \mathbb{C}). \quad (\text{CP3.2})$$

The finite endpoint dagger coefficients are zero. Factoring a nonzero polynomial into linear factors gives

$$\boxed{P(t) : E^\dagger \xrightarrow{\sim} E^\dagger \quad (P \in R \setminus \{0\})}. \quad (\text{CP3.3})$$

Every coefficient-diagram map commutes with these inverses. Hence the coefficient model (CP2.5) can be formed over $\mathbb{C}(t)$.

In particular P acts bijectively on every cohomology sheaf, global group, local group and costalk in (CP2.7). This does not assert that P is invertible on each *raw quotient-current term*. The latter assertion is unnecessary and is not used.

CP4. The exact derived polynomial spectral complex

For any cohomological R -complex C , and nonzero $P \in R$, use the free resolution

$$0 \rightarrow R \xrightarrow{P} R \rightarrow R/(P) \rightarrow 0 \quad (\text{CP4.1})$$

with its free terms in degrees $-1, 0$. It is exact because R is an integral domain. These two free modules are projective, so their Hom complex computes derived Hom on any C , with no injectivity assumption on its coefficients.

Use the following completely specified model:

$$K_P(C)^n = C^n \oplus C^{n-1}, \quad \delta(x, y) = (dx, Px - dy). \quad (\text{CP4.2})$$

Indeed ordinary Hom has differential $d_C f - (-1)^n f d_R$. If its components are (x, b) , the coordinate $y = (-1)^n b$ gives (CP4.2). Therefore

$$\boxed{K_P(C) \simeq \text{RHom}_R(R/(P), C)}. \quad (\text{CP4.3})$$

The formula also defines the internal derived Hom of sheaves of R -modules: internal Hom from the constant rank-one free module is the identity, so the same two-term free resolution computes it stalkwise.

The maps on cohomology are exactly

$$0 \rightarrow \frac{H^{n-1}(C)}{PH^{n-1}(C)} \xrightarrow{[y] \mapsto (0,y)} H^n K_P(C) \xrightarrow{[(x,y)] \mapsto [x]} \ker(P : H^n(C) \rightarrow H^n(C)) \rightarrow 0. \quad (\text{CP4.4})$$

To prove this, a cycle has $dx = 0$ and $Px = dy$, so its first coordinate lies in the displayed kernel. Every class in that kernel admits such a y . If $x = da$, subtracting $\delta(a, 0)$ leaves $(0, y - Pa)$, with second coordinate closed. Changing a by a closed element changes its class by $PH^{n-1}(C)$; changing the second coordinate by a differential changes it by a spectral boundary. Conversely those two changes are spectral boundaries. This proves every term and map, without choosing a splitting.

For a map $c : C_c \rightarrow C_a$, the actual spectral comparison is the componentwise map

$$K_P(c)(x, y) = (cx, cy). \quad (\text{CP4.5})$$

There is a literal cone identity

$$K_P(\text{Cone}(c)) \xrightarrow{\sim} \text{Cone}(K_P(c)), \quad ((b, a), (v, u)) \mapsto ((b, v), (a, -u)). \quad (\text{CP4.6})$$

Substituting the cone differential $(db + ca, -da)$ and (CP4.2) verifies both components and the minus sign on u .

CP5. Acyclicity of the actual spectral obstruction

Apply (CP4.4) to the actual defect. Bijectivity in CP3 makes both its displayed kernel and cokernel zero, so

$$\boxed{\text{RHom}_R(R/(P), \text{Cone}(c)) \simeq \text{RHom}_R(R/(P), \mathcal{D}) \simeq 0.} \quad (\text{CP5.1})$$

For sheaves this means the corresponding internal derived Hom object is zero. Derived global sections, derived point support, and restriction to either stalk preserve that zero object. Finite cones and finite direct sums commute with these derived functors, so the same assertion computes the polynomial spectral complexes of the actual global, stalk and costalk models.

On a dagger coefficient model, where P is termwise invertible, the vanishing has an explicit contraction:

$$h^n(x, y) = (P^{-1}y, 0), \quad \delta h + h\delta = 1, \quad h^2 = 0. \quad (\text{CP5.2})$$

The inverse commutes with the differential, since P does. Direct substitution gives

$$\delta h(x, y) = (P^{-1}dy, y), \quad h\delta(x, y) = (x - P^{-1}dy, 0).$$

This contraction applies to the coefficient/Gysin model and its finite global and local complexes. For raw quotient currents, (CP5.1) follows from the cohomology calculation, and the representative correction in CP6 uses only that justified version.

By (CP4.6), the cone of the actual comparison (CP4.5) is acyclic. Consequently the canonical map

$$\boxed{\text{RHom}_R(R/(P), \mathcal{C}_c) \xrightarrow{\sim} \text{RHom}_R(R/(P), \mathcal{C}_a)} \quad (\text{CP5.3})$$

is a quasi-isomorphism for every nonzero P . This is an isomorphism in the derived category, with a unique inverse there. It is not a claimed projection from algebraic currents to continuous currents, and it does not choose chain representatives.

The defect itself need not be zero. VSD9 proves $A^\dagger \neq 0$ and $J^\dagger \neq 0$ for these actual infinite-dimensional Fréchet coefficients. Equation (CP5.1) instead proves that this precisely defined defect has no derived sector tested by $R/(P)$, including its degree-one Ext contribution. This is stronger than only saying that it has no eigenvectors.

CP6. Exact correction of an algebraic spectral lift

Take one of the actual complexes of sections, stable local models, or finite coefficient models for which CV gives the degreewise exact sequence

$$0 \rightarrow C_c \rightarrow C_a \xrightarrow{q} D \rightarrow 0. \quad (\text{CP6.1})$$

For literal currents on the opens in use, exactness of sections at the quotient follows from CV5.1 and fineness of the continuous-current terms; pole coefficient quotients are exact. The current global and local contractions identify their defect cohomology with CP2.7.

Let $(x, y) \in K_P(C_a)^n$ be a cycle:

$$dx = 0, \quad Px = dy. \quad (\text{CP6.2})$$

Its actual obstruction to a continuous spectral lift is

$$\text{obs}_n[(x, y)] = [(\bar{x}, \bar{y})] \in H^n K_P(D), \quad \bar{x} = q(x), \bar{y} = q(y). \quad (\text{CP6.3})$$

This is the map induced by the actual quotient complex, so boundaries change it by boundaries. Its kernel is exactly the image of $H^n K_P(C_c)$: one implication is immediate; for the converse, subtract a lift of a quotient primitive. Equation (CP5.1) makes its target zero.

Here is the full correction for raw quotient currents, without a termwise P^{-1} on those currents. Since $d\bar{x} = 0$ and $P\bar{x} = d\bar{y}$, injectivity of P on $H^n(D)$ gives

$$\bar{x} = du \quad \text{for some } u \in D^{n-1}.$$

Put $w = \bar{y} - Pu$; then $dw = 0$. Surjectivity of P on $H^{n-1}(D)$ gives a closed $v \in D^{n-1}$ representing the unique class $P^{-1}[w]$. Choose $z \in D^{n-2}$ with

$$dz = w - Pv. \quad (\text{CP6.4})$$

These choices satisfy the exact signed identity

$$\delta(u + v, -z) = (\bar{x}, \bar{y}). \quad (\text{CP6.5})$$

Choose any cochains $U \in C_a^{n-1}$, $V \in C_a^{n-2}$ lifting $u + v, -z$, respectively. Then

$$\boxed{(x_c, y_c) = (x - dU, y - PU + dV)} \quad (\text{CP6.6})$$

lies in $C_c^n \oplus C_c^{n-1}$, is a cycle, and represents the same algebraic spectral class. Both assertions follow by applying q , (CP6.5), and $\delta^2 = 0$. This is an explicit correction by the spectral boundary $\delta(U, V)$.

For a finite coefficient model the shorter correction follows directly from (CP5.2). Choose U lifting $P^{-1}\bar{y}$ and set $V = 0$. Then

$$(x_c, y_c) = (x - dU, y - PU). \quad (\text{CP6.7})$$

The two formulae have distinct stated domains; the latter is not imposed on raw currents.

Uniqueness is exactly uniqueness modulo *continuous spectral boundaries*. Suppose a cycle in $K_P(C_c)^n$ is δb in the algebraic complex. Then $\bar{b}$ is a cycle of $K_P(D)^{n-1}$. Its cohomology vanishes, so write $\bar{b} = \delta \bar{a}$, lift $\bar{a}$, and put $b_c = b - \delta a$. This belongs to the continuous complex and has $\delta b_c = \delta b$. Thus a continuous spectral class which becomes zero is already zero. Together with (CP6.6),

$$H^n K_P(C_c) \xrightarrow{\sim} H^n K_P(C_a) \quad (\text{CP6.8})$$

for every n .

Changing any primitive, lift, or algebraic representative in the correction therefore changes the resulting continuous representative by a continuous spectral boundary. The cohomology lift is canonical and linear because it is the inverse of the specified linear isomorphism (CP6.8). The chosen cochains, primitive representatives, and correction operators are not asserted to be canonical or continuous in their choices.

CP7. All degrees, including endpoints and the nonzero cokernel terms

For a module M , write

$$M[P] = \ker(P : M \rightarrow M), \quad M/PM = \operatorname{coker}(P : M \rightarrow M).$$

On a module in degree zero, $K_P(M) = [M \xrightarrow{P} M]$ in degrees 0, 1. Thus (CP5.3) includes exactly both the kernel and cokernel comparisons of VSD13; there is no omitted Ext degree.

The actual global continuous model is

$$C_c^{-2} = A', \quad C_c^{-1} = A', \quad C_c^0 = V'_+ \oplus V'_-, \quad d^{-2} = 0, \quad d^{-1}\lambda = (r'_+\lambda, -r'_-\lambda). \quad (\text{CP7.1})$$

Its cohomology is A', Q', Z' in degrees $-2, -1, 0$. The algebraic version uses stars and the identical maps. Formula (CP4.2) specifies the spectral complex in degrees $-2, -1, 0, 1$:

$$A', \quad A' \oplus A', \quad (V'_+ \oplus V'_-) \oplus A', \quad V'_+ \oplus V'_-. \quad (\text{CP7.2})$$

The differentials are, writing $d'\lambda = (r'_+\lambda, -r'_-\lambda)$,

$$x \mapsto (0, Px), \quad (x, y) \mapsto (d'x, Px), \quad (\nu, x) \mapsto P\nu - d'x. \quad (\text{CP7.3})$$

All occurrences of P use the specified coefficient operator on their own term.

Every nonzero P is onto on A' and J' : each factor $L' - \lambda$ is onto by the transpose argument of CP3, and their composition remains onto. The full spectral global groups are therefore

$$\begin{aligned} H^{-2}K_P(C_c) &= A'[P], & H^{-1}K_P(C_c) &= Q'[P], \\ 0 \rightarrow Q'/PQ' \rightarrow H^0K_P(C_c) &\rightarrow Z'[P] \rightarrow 0, & H^1K_P(C_c) &= Z'/PZ', \end{aligned} \quad (\text{CP7.4})$$

and all other degrees vanish. These are the canonical maps of (CP4.4), rather than a silently chosen splitting of its middle extension. The algebraic-current groups are canonically identified with them by inclusion.

At a pole the continuous stalk complex is $[A' \rightarrow^0 A' \rightarrow^{r'_p} V'_p]$. Its cohomology is A', Q', E'_p , so

$$\begin{aligned} H^{-2}K_P(i_p^*\mathcal{C}_c) &= A'[P], & H^{-1}K_P(i_p^*\mathcal{C}_c) &= Q'[P], \\ 0 \rightarrow Q'/PQ' \rightarrow H^0K_P(i_p^*\mathcal{C}_c) &\rightarrow E'_p[P] \rightarrow 0, & H^1K_P(i_p^*\mathcal{C}_c) &= E'_p/PE'_p. \end{aligned} \quad (\text{CP7.5})$$

At a pole costalk the actual contraction gives $V'_p[0]$, hence

$$H^0K_P(i_p^!\mathcal{C}_c) = V'_p[P], \quad H^1K_P(i_p^!\mathcal{C}_c) = V'_p/PV'_p. \quad (\text{CP7.6})$$

For the interior stalk, $A'[2]$ gives only $A'[P]$ in degree -2 , since $A'/PA' = 0$. For the interior costalk, $A'[0]$ gives only $A'[P]$ in degree zero. All these statements also hold for the actual algebraic comparison through (CP5.3).

The endpoint terms can be specified completely. Let

$$E_{p,P} = \bigoplus_{\substack{e \text{ an endpoint eigenline at } p \\ P(\gamma_e)=0}} e', \quad \gamma_e \in \{0, 1\} \quad (\text{CP7.7})$$

using $(0, 1)$ at $+$ and $(1, 0)$ at $-$. Then

$$E'_p[P] = E'_p/PE'_p = E_{p,P}. \quad (\text{CP7.8})$$

Each selected endpoint contributes one dimension even if the root of P is repeated, because that endpoint operator is already scalar. Since $J'/PJ' = 0$,

$$\begin{aligned} Z'/PZ' &= E_{+,P} \oplus E_{-,P}, & V'_p/PV'_p &= E_{p,P}, \\ Z'[P] &= J'[P] \oplus E_{+,P} \oplus E_{-,P}, & V'_p[P] &= J'[P] \oplus E_{p,P}. \end{aligned} \quad (\text{CP7.9})$$

In particular Q'/PQ' is not discarded in degree zero of (CP7.4) or (CP7.5). CP8 computes it explicitly for the original-zero polynomial.

Restriction to the two pole stalks remains diagonal on degree-minus-two coefficients and anti-diagonal on degree-minus-one coefficients:

$$\lambda \mapsto (\lambda, \lambda), \quad \mu \mapsto (\mu, -\mu). \quad (\text{CP7.10})$$

Apply these maps to both components of K_P ; the exact source support signs are unchanged. The dual's costalk-to-global map is the original positive sum on its J' coordinates with both endpoint restrictions retained. All these R -linear maps commute with the canonical spectral comparison and its cohomological inverse.

CP8. Each actual original-zero jet, with its complete multiplicity

Fix an actual nontrivial zero ρ of the original ζ , of multiplicity $m = m_\rho$, and now take

$$P_\rho(t) = (t - \rho)^m. \quad (\text{CP8.1})$$

No zero is invented and no critical-line position is assumed. Since $0 < \Re \rho < 1$, $P_\rho(0)$ and $P_\rho(1)$ are nonzero, so every endpoint term in (CP7.7) is zero for this polynomial.

The polynomial jet module is

$$R/(P_\rho) \simeq \mathbb{C}[z]/(z^m), \quad t \text{ acts by } \rho + z. \quad (\text{CP8.2})$$

Evaluation of full Taylor jets gives a surjection

$$\mathcal{Q} \rightarrow \mathbb{C}[z]/(z^m), \quad [F] \mapsto \sum_{j=0}^{m-1} \frac{F^{(j)}(\rho)}{j!} z^j,$$

whose kernel is $P_\rho(L)\mathcal{Q}$. Indeed a representative with these jets zero is divisible in $\mathcal{B}$ by $(s - \rho)^m$; continuous division follows by repeated fixed-disk estimates. Conversely multiplication by that polynomial kills these jets. Surjectivity uses an entire Gaussian times a polynomial with the prescribed finite Taylor coefficients.

Thus

$$Q'[P_\rho] = (Q/P_\rho Q)^* = \text{span}\{\delta_{\rho,0}, \dots, \delta_{\rho,m-1}\}, \quad \delta_{\rho,j}([a]) = \frac{(\Theta a)^{(j)}(\rho)}{j!}. \quad (\text{CP8.3})$$

All these functionals are continuous. Their algebraic and continuous polynomial kernels coincide *as these very functionals* under inclusion. In particular the canonical degree-minus-one lift in (CP7.4) and (CP7.5) identifies every actual algebraic original-zero jet with its unique continuous functional, with its full multiplicity m .

We also compute the cokernel term, using a single actual finite projector rather than a formal infinite decomposition. Put

$$B_\rho(s) = \frac{F_0(s)}{(s - \rho)^m}, \quad \Pi_\rho F(s) = B_\rho(s) \sum_{j=0}^{m-1} \frac{1}{j!} \left(\frac{F}{B_\rho} \right)^{(j)}(\rho) (s - \rho)^j. \quad (\text{CP8.4})$$

The quotient F/B_ρ is used only in a neighborhood of ρ , where its denominator is nonzero. $B_\rho \in \mathcal{B}$ by division estimates. This finite operator is continuous, has the same full jet as F at ρ , and vanishes at every other actual zero to its full order. It therefore induces a continuous projection $\bar{\Pi}_\rho$ on $\mathcal{Q}$, commuting with L .

Its image K_ρ is precisely $\ker P_\rho(L)$, has dimension m , and maps isomorphically to the displayed jet module. To verify the equality, $P_\rho \bar{\Pi}_\rho F = F_0$ times a polynomial lies in $\mathcal{J}$. Conversely, a class killed by P_ρ has zero jets at every other original zero and is determined by its jet at ρ , hence equals its projection.

Let $Q_0 = \ker \bar{\Pi}_\rho$. On this actual closed complement, $P_\rho(L)$ is a continuous automorphism. Its inverse is

$$q = [F] \in Q_0 \mapsto (1 - \bar{\Pi}_\rho) \left[\frac{F(s)}{(s - \rho)^m} \right]. \quad (\text{CP8.5})$$

Every representative of q has zero full jet at ρ , so the quotient in brackets is entire and belongs to $\mathcal{B}$. Changing F by $\mathcal{J}$ changes that quotient by a class supported only at ρ , removed by $1 - \bar{\Pi}_\rho$. Multiplication in both orders gives the identity because $P_\rho \bar{\Pi}_\rho = 0$ and the projection commutes with L . Continuity follows by division on the closed finite-jet kernel in $\mathcal{B}$ and the quotient topology; that kernel modulo $\mathcal{J}$ has exactly the subspace topology of Q_0 , as is seen from the continuous finite Taylor interpolation projection.

Dualizing the actual decomposition $Q = K_\rho \oplus Q_0$, the operator P_ρ is zero on K_ρ^* and invertible on Q_0' . Hence

$$Q'/P_\rho Q' \xrightarrow{\sim} K_\rho^*, \quad [\lambda] \mapsto \lambda|_{K_\rho}, \quad \dim(Q'/P_\rho Q') = m. \quad (\text{CP8.6})$$

This proves that the cokernel term retained in CP7 is nonzero. It is not eliminated by the comparison's acyclic defect.

The same fixed finite-jet division on $\mathcal{B}$ and $\mathcal{J}$, using the jets of F and F/F_0 , shows

$$\dim A'[P_\rho] = m, \quad \dim J'[P_\rho] = m, \quad A'/P_\rho A' = J'/P_\rho J' = 0. \quad (\text{CP8.7})$$

For the dimension statement on $\mathcal{J}$, the kernel of the map taking the first m Taylor coefficients of F/F_0 at ρ is $(s - \rho)^m \mathcal{J}$. All m values are realized by F_0 times a polynomial times an entire Gaussian; this preserves membership in $\mathcal{J}$ and realizes any finite jet. Its dual is the displayed polynomial kernel. This is a fixed actual-zero calculation, not an assumed analytic family of arbitrary spectra.

Consequently the global spectral cohomology for (CP8.1) is

$$\begin{array}{c|cccc} \text{degree} & -2 & -1 & 0 & 1 \\ \hline \dim H^n K_{P_\rho}(R\Gamma \mathcal{C}_c) & m & m & 2m & 0, \end{array} \quad (\text{CP8.8})$$

where degree zero retains the canonical exact sequence

$$0 \rightarrow K_\rho^* \rightarrow H^0 K_{P_\rho}(R\Gamma \mathcal{C}_c) \rightarrow J'[P_\rho] \rightarrow 0. \quad (\text{CP8.9})$$

The same dimensions and canonical comparison hold for $\mathcal{C}_a$. At a pole stalk the corresponding dimensions are $m, m, m, 0$ in degrees $-2, -1, 0, 1$; its degree-zero group is canonically K_ρ^* . At a pole costalk the only nonzero group has degree zero and dimension m , namely $J'[P_\rho]$. The interior stalk has dimension m only in degree -2 ; the interior costalk has dimension m only in degree zero. Thus every degree and every original multiplicity is retained.

CP9. Full original-zeta and dilation comparison

The source relations used above are the original ones:

$$\begin{aligned} \mathcal{M}_0 \Sigma f(s) &= 2\zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}, \quad \Theta = \frac{1}{2} \mathcal{M}_0, \quad \Re s > 1, \\ \zeta(s) &= 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'}{\zeta}(s) = \sum_p \sum_{k \geq 1} (\log p) p^{-ks}. \end{aligned} \quad (\text{CP9.1})$$

The unit summand and every prime-power repetition remain present. The actual source

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}$$

has transform

$$F_0(s) = \Theta \Sigma f_0(s) = C_0(s) \zeta(s), \quad C_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2). \quad (\text{CP9.2})$$

The factor $1/8$ is not changed. The apparent endpoint and trivial-zero singularities have the retained values

$$\begin{aligned} F_0(0) &= F_0(1) = \frac{1}{8}, \quad F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) \\ &= \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0 \quad (r \geq 1). \end{aligned} \quad (\text{CP9.3})$$

Therefore at these locations the nonvanishing branch of (CP3.1) is used. The original pole of ζ , its trivial zeros, the Gamma poles, and their derivative cancellations have not become extra zeros of the receiver.

Near the actual ρ , write $z = s - \rho$ and

$$\zeta(s) = z^m u_\rho(s), \quad F_0(s) = z^m b_\rho(s), \quad b_\rho(s) = C_0(s) u_\rho(s). \quad (\text{CP9.4})$$

All three local factors on the right are holomorphic and $u_\rho(\rho), b_\rho(\rho), C_0(\rho) \neq 0$. The exact local comparison $\mathcal{O}_\rho/(\zeta) \simeq \mathcal{O}_\rho/(F_0) \simeq \mathbb{C}[z]/(z^m)$ is induced by the identity on germs, since the two ideals differ by the explicit local unit C_0 . That unit is retained in every derivative and residue comparison.

For example the original reflected residue is

$$\text{Res}_{s=\rho} \frac{F(s)G(1-s)}{\zeta(s)} ds = \text{Res}_{s=\rho} \frac{C_0(s)F(s)G(1-s)}{F_0(s)} ds. \quad (\text{CP9.5})$$

If $F(\rho+z) = \sum f_j z^j$, $G(1-\rho+z) = \sum g_j z^j$, and $1/u_\rho(\rho+z) = \sum v_j z^j$, its full value is

$$\sum_{j+k+\ell=m-1} f_j (-1)^k g_k v_\ell. \quad (\text{CP9.6})$$

Here

$$v_\ell = \sum_{a+b=\ell} \frac{C_0^{(a)}(\rho)}{a!} \frac{(1/b_\rho)^{(b)}(\rho)}{b!}, \quad (\text{CP9.7})$$

so no derivative of the original multiplier has been suppressed.

In the Taylor-functional basis of (CP8.3),

$$L' \delta_{\rho,j} = \rho \delta_{\rho,j} + \delta_{\rho,j-1}, \quad \delta_{\rho,-1} = 0. \quad (\text{CP9.8})$$

The full existing dilation transpose on this finite sector is

$$T'_a \delta_{\rho,j} = a^\rho \sum_{h=0}^j \frac{(\log a)^{j-h}}{(j-h)!} \delta_{\rho,h}. \quad (\text{CP9.9})$$

This follows by the complete product derivative of $a^s F(s)$ at ρ . Raw derivative functionals are $j! \delta_{\rho,j}$; their matrix has $\binom{j}{h} (\log a)^{j-h}$, retaining precisely that factorial conversion.

The comparison (CP5.3) preserves these actual finite-sector matrices, because it is literal inclusion of the functionals and commutes with the specified coefficient action. No classification of arbitrary discontinuous dilation characters is needed. For the normal ordinary transpose aT'_a , the scalar in (CP9.9) is $a^{\rho+1}$. For the degree-minus-one inverse-coefficient transfer transpose $aT'_{a^{-1}}$, it is $a^{1-\rho}$ and every logarithm is $-\log a$. For the normal contragredient $a^{-1}T'_{a^{-1}}$, it is $a^{-\rho-1}$, again with $-\log a$. For integer $a = n \geq 1$, these are the specified geometric-cover, transfer and contragredient maps on the coefficient model. For arbitrary $a > 0$, they are the coefficient-complex extensions proved in CP10 below. They are different maps, not replacements for one another; no nonintegral sphere covering is asserted.

For $a > 1$, their modulus weights are respectively $2\Re \rho$, $2\Re \rho + 2$, $2 - 2\Re \rho$, and $-2 - 2\Re \rho$, using $2 \log |\lambda| / \log a$. The spectral lift preserves the full matrices for the actual ρ ; it does not force $2\Re \rho = 1$.

CP10. Mirrors, shifts, and naturality of the canonical spectral lift

Let M denote one of the actual complex-linear mirrors, with its degree-wise orientation and pole-swap signs. On coefficient operators,

$$Mt = (1-t)M, \quad MP(t) = P(1-t)M. \quad (\text{CP10.1})$$

This is semilinearity for the ring involution $\iota(t) = 1-t$, fixing $\mathbb{C}$. It is not R -linearity with the same variable on both sides. Define $P^\iota(t) = P(1-t)$. Then the exact spectral transport is

$$K_P(C) \longrightarrow K_{P^\iota}(C), \quad (x, y) \longmapsto (M_n x, M_{n-1} y). \quad (\text{CP10.2})$$

Indeed M commutes with the cochain differential and $P^\iota M = MP$, so substitution in (CP4.2) proves the chain identity. There is no further totalization sign.

For $P = (t - \rho)^m$,

$$P^\iota(t) = (-1)^m (t - (1 - \rho))^m. \quad (\text{CP10.3})$$

If the target polynomial is written as the monic factor $(t - (1 - \rho))^m$, the precise chain map is

$$(x, y) \longmapsto (M_n x, (-1)^m M_{n-1} y). \quad (\text{CP10.4})$$

The second-coordinate factor follows because changing the polynomial from Q to cQ changes that coordinate by c . At the level of Taylor functionals, $R' \delta_{\rho,j} = (-1)^j \delta_{1-\rho,j}$; the actual global degree-minus-one mirror has the additional CSD sign $-R'$. These signs are compatible with (CP10.4) and retain all multiplicities. A separately chosen conjugate-linear coefficient mirror would also conjugate the polynomial coefficients; that is not substituted for the complex-linear mirrors here.

If the stated operator is $t = L + k$, then $P(t)$ means $P(L + k)$, a nonzero polynomial in L , so the same proof applies. Its mirror variable is

$$t \longmapsto 1 + 2k - t. \quad (\text{CP10.5})$$

An actual zero block is then at $t = \rho + k$, sent to $1 - \rho + k$. Scalar shifts do not change its length.

For the actual sphere cover $z \mapsto z^n$, with integer $n \geq 1$, the ordinary transposed action on the global coefficient complex is, in degrees $-2, -1, 0$,

$$(nT'_n, T'_n, \rho_+(n)' \oplus \rho_-(n)'). \quad (\text{CP10.6})$$

The positive-real extension in this paragraph is an action on that finite coefficient complex, whose coefficient spaces remain the full infinite-dimensional spaces. For $a > 0$, define the primal coefficient maps by

$$\begin{aligned} T_a b(u) &= b(u/a), & \rho_+(a)(f, c_0, c_1) &= (f(v/a), c_0, ac_1), \\ \rho_-(a)(h, d_0, d_1) &= (ah(av), ad_0, d_1). \end{aligned} \quad (\text{CP10.7})$$

They preserve all conditions in (CP1.1). Substitution gives $T_a T_b = T_{ab}$ and $\rho_p(a)\rho_p(b) = \rho_p(ab)$, with identity at one and inverses at a^{-1} . The summation formula gives $r_+\rho_+(a) = T_a r_+$. On the other chart the full Fourier formula $\widehat{ah(a \cdot)}(v) = \widehat{h}(v/a)$ gives $r_-\rho_-(a) = T_a r_-$, with no discarded factor. Transposing these identities proves that

$$U_a = (aT'_a, T'_a, \rho_+(a)' \oplus \rho_-(a)') \quad (\text{CP10.8})$$

commutes with the coefficient-complex differential. Indeed the degree-minus-two differential is zero, and the only other differential is $d'\lambda = (r'_+\lambda, -r'_-\lambda)$, for which $\rho(a)'d' = d'T'_a$. The primal maps commute, so their transposes satisfy $U_a U_b = U_{ab}$. These proofs also give algebraic-transpose actions and the action on their quotient by continuous functionals. They do not construct sphere covers of noninteger degree.

Smoothness here is asserted on the continuous dual coefficient complex, equipped with its strong dual topology. Write $a = e^h$. For any primal continuous seminorm p , direct differentiation of (CP10.7) gives a continuous seminorm q and, on a compact h -interval, a constant C such that

$$p(V_h x - x - hGx) \leq Ch^2 q(x), \quad V_h = T_{e^h} \text{ or } \rho_p(e^h), \quad (\text{CP10.9})$$

with the corresponding $G = L$ or G_p . On A , this follows from the weights $u^N + u^{-N}$ under $u \mapsto e^{-h}u$, bounded by $e^{N|h|}$, and two additional logarithmic derivatives. On the Schwartz summands it follows from the same change of variable in the defining polynomial-weighted derivative seminorms; the extra factor e^h on the minus chart is retained. On the endpoint summands it is the scalar exponential Taylor estimate. Integral Taylor remainder proves the displayed bound; applying the same argument after any number of derivatives proves its higher-order versions. For a continuous functional λ , choose p bounding $|\lambda|$. For every bounded primal set B , $\sup_{x \in B} q(x) < \infty$, so (CP10.9), divided by $|h|$ and transposed, tends to zero uniformly on B . Thus the dual orbit is differentiable in the strong dual topology, and the higher-order estimates prove smoothness. Its infinitesimal operators are precisely $(L' + 1, L', G')$, including the derivative of the retained factor e^h in degree minus two. On the algebraic dual, the corresponding specified operators are the primal transposes $(L^* + 1, L^*, G^*)$; no derivative of an arbitrary discontinuous-functional orbit is used or asserted.

The normal shift $k = 1$ occurs only in the degree-minus-two summand, whose differential to the next term is zero; the other linked terms use $k = 0$. Apply (CP10.5) to that normal summand, giving $3 - t$, and $1 - t$ on the unshifted summands. This is not a claim of one common affine reflection for two different generator conventions. On the same coefficient complex the inverse-coefficient transfer transpose is $(T'_{a^{-1}}, aT'_{a^{-1}}, a\rho(a^{-1})')$, and the contragredient is $(a^{-1}T'_{a^{-1}}, T'_{a^{-1}}, \rho(a^{-1})')$. Their group laws and chain identities follow from the same calculation, since the linked degrees carry the same additional scalar. For integer $a = n \geq 1$ they are the specified cover-associated maps. On the continuous dual their generators are respectively $(-L', 1 - L', 1 - G')$ and $(-L' - 1, -L', -G')$, by the proved smoothness; on the algebraic dual these expressions again mean induced primal transposes only. Their endpoint blocks follow from the explicitly retained eigenvalues 0, 1, 1, 0. Polynomial comparison remains valid for each of these specified nonconstant affine polynomials in L .

All actual R -linear restriction, support and coefficient maps induce maps on K_P by applying the map to both coordinates. The comparison squares commute literally. Their cohomological inverse squares therefore commute by uniqueness of the inverse in (CP6.8). The mirror squares commute with the changed polynomial as in (CP10.2). Thus the canonical lifted classes preserve the actual support maps, endpoint summands, cover matrices and mirror signs even though their chain representatives are not canonically chosen.

CP11. The exact lifting mechanism and its relation to Deligne

The actual obstruction in degree n is (CP6.3), lying in

$$H^n \mathrm{RHom}_R(R/(P), \mathrm{Cone}(c)). \quad (\text{CP11.1})$$

The exact triangle of the actual comparison, followed by the two-term free resolution, makes its kernel the image of the continuous spectral cohomology. CP3-CP5 prove that this receiver is zero in every degree. CP6 constructs the correction explicitly and proves both existence and uniqueness of the continuous *cohomology class*.

The algebraic separation is exact: P annihilates the test module $R/(P)$, whereas P is invertible on every cohomology group of the actual comparison defect. The two-term resolution proves vanishing of both possible derived contributions,

$$\ker(P : H^n D \rightarrow H^n D) = 0, \quad \mathrm{coker}(P : H^{n-1} D \rightarrow H^{n-1} D) = 0. \quad (\text{CP11.2})$$

This is the precise linear mechanism resolving this actual lifting obstruction. It tests Ext as well as Hom and therefore does not stop at an eigenvector statement.

For comparison, DC5 describes Deligne's obstruction by

$$\begin{aligned} c_i &\longmapsto [\partial_i b_i] \in O_i / \partial_i j_i(K_i), \quad \pi_i(b_i) = c_i, \\ \mathrm{coker} \, \mathrm{sp}_i &\simeq \mathrm{im} \, \partial_i / \partial_i j_i(K_i). \end{aligned} \quad (\text{CP11.3})$$

DC6-DC7 prove low weights on the classes and high weights on the geometric obstruction receiver; the exact weight cutoff then kills the obstruction and supplies a lift. Here the actual quotient current map supplies (CP6.3), and polynomial-torsion versus polynomial-invertible coefficients supply the vanishing in (CP11.2). Both arguments use a proved separation to kill a specified equivariant obstruction, with their own exact sequences and maps.

The present construction does not identify the current-dual cone with Deligne's inertia/duality cross

$$K_i = H^{i-1}(X_{\bar{\eta}}, E)_I(-1), \quad O_i = H^{2N-i-1}(X_s, E)^\vee(-N). \quad (\text{CP11.4})$$

No map identifying those geometric groups has been assumed. Instead the connecting morphism, target, and correction for the actual current comparison are (CP6.3)-(CP6.6), and their complete local and global degrees are CP7-CP8.

The resulting canonical comparison on each actual original-zero jet retains the whole factor $F_0 = C_0 \zeta$, all derivative and residue terms, every endpoint block, all multiplicities and the actual dilation matrices. It removes no original zero and provides no bound stronger than the already known open strip for its real part. The defect can remain nonzero outside all these polynomial spectral tests. These limits are part of the exact result, rather than a substituted purity premise or an RH conclusion.

The full generator cokernel map and original meromorphic divisor — GCF0-GCF8

Proof source: CC_GENERATOR_COKERNEL_AND_ORIGINAL_DIVISOR.md.

The full generator cokernel map and the original zeta divisor

24 September 2026. Proof locators GCF0-GCF8. This derives an actual map in the existing coefficient sequence and its global comparison with the original meromorphic zeta function. It does not establish a Deligne weight bound.

GCF0. Prerequisites and the precise receiving object

The supporting datum remains the user's $\tau\langle Z_1; \text{no } Z_2 \rangle$, with the information-layer notation $Z_0, Z_1, Z_2, \dots$. All operations in this note occur after the complete arithmetic reconstruction. No addition, arithmetic value, distance, or parity is assigned to τ . Distinct reconstructed branches are not pooled.

The actual source comparison is SSI0-SSI8 in EXACT_SCHWARTZ_SUMMATION_IMAGE.md, together with S1-S7 in GLOBAL_MELLIN_SYNTHESIS.md. It gives the topological isomorphism

$$\Theta : A \longrightarrow \mathcal{B}, \quad \Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, \quad J = \Sigma \mathcal{S}_0^{\text{even}}, \quad \Theta J = \mathcal{J}, \quad Q = A/J \simeq \mathcal{Q} = \mathcal{B}/\mathcal{J}. \quad (\text{GCF0.1})$$

Here A has the seminorms $\sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)|$, and $\Sigma f(u) = 2 \sum_{k \geq 1} f(ku)$. The coefficient space and its closed subspace are

$$\mathcal{B} = \left\{ F \text{ entire} : b_{R,M}(F) := \sup_{|\Re s| \leq R} (1 + |\Im s|)^M |F(s)| < \infty \text{ for every } R, M \in \mathbb{N} \right\},$$

$$\mathcal{J} = \{ F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every actual nontrivial zero } \rho \text{ of } \zeta, 0 \leq j < m_\rho \}. \quad (\text{GCF0.2})$$

All multiplicities m_ρ are retained. The entire source element, obtained from the original Gaussian-polynomial source, is

$$F_0(s) = c(s)\zeta(s), \quad c(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2). \quad (\text{GCF0.3})$$

It belongs to $\mathcal{J}$, has exactly its prescribed zero divisor, and satisfies $F_0(0) = F_0(1) = 1/8$. These are proved source properties, not a definition of the original ζ . In particular the recovered original series still has unit term 1 and every integer term, and its Euler expansion still contains every prime-power repetition. The multiplier in (GCF0.3) is never suppressed below.

The operator is the continuous map $LF(s) = sF(s)$ on $\mathcal{B}$, restricted to $\mathcal{J}$ and induced on $\mathcal{Q}$. Under (GCF0.1) it corresponds to $-u\partial_u$. Thus neither a numerical coordinate for τ nor an unconstructed arithmetic action is being introduced. CW6-CW8 already calculate its quotient characters and resolvent;

VSD13 proves its closed codimension-one images on $\mathcal{B}$ and $\mathcal{J}$. RZ1-RZ6 retain the full quotient resolvent and multiplicity projectors. The present new receiver is the map between those two cokernels as the parameter varies, including its original meromorphic divisor.

GCF1. Both cokernel lines, with their actual quotient maps

For every $\lambda \in \mathbb{C}$, define

$$\epsilon_\lambda(F) = F(\lambda) \quad (F \in \mathcal{B}), \quad \ell_\lambda(F) = (F/F_0)(\lambda) \quad (F \in \mathcal{J}). \quad (\text{GCF1.1})$$

The second quotient is entire because the complete zero divisor of F_0 is imposed on every $F \in \mathcal{J}$. It need not belong to $\mathcal{B}$. If F_0 has order m at λ , its removable value is $F^{(m)}(\lambda)/F_0^{(m)}(\lambda)$; the nonzero denominator is retained.

Both maps are continuous and onto. For ϵ_λ , a section is $z \mapsto ze^{(s-\lambda)^2}$. For ℓ_λ , a section is $z \mapsto zF_0$. Continuity of the latter follows either by the displayed derivative formula or by a contour estimate: on a circle avoiding the zeros of F_0 , division by F_0 is bounded, and the maximum principle bounds the entire quotient on the enclosed compact set by this boundary bound. A larger-strip seminorm controls the numerator on that circle.

Division by $s - \lambda$ is continuous on the closed subspace of $\mathcal{B}$ vanishing at λ . Outside a fixed disk the denominator is bounded away from zero. Inside the disk, the maximum principle for the removable quotient bounds it on a larger circle. This proves a finite larger-strip seminorm bound for every target seminorm. On $\mathcal{J}$, division preserves its prescribed orders precisely when $\ell_\lambda(F) = 0$. Consequently the two exact rows are

$$0 \rightarrow \mathcal{B} \xrightarrow{L-\lambda} \mathcal{B} \xrightarrow{\epsilon_\lambda} \mathbb{C} \rightarrow 0, \quad 0 \rightarrow \mathcal{J} \xrightarrow{L-\lambda} \mathcal{J} \xrightarrow{\ell_\lambda} \mathbb{C} \rightarrow 0. \quad (\text{GCF1.2})$$

Every image is closed, and the inverse on the image is continuous. The inclusion $\mathcal{J} \hookrightarrow \mathcal{B}$ induces the exact scalar map

$$\begin{array}{ccc} \mathcal{J}/(L-\lambda)\mathcal{J} & \longrightarrow & \mathcal{B}/(L-\lambda)\mathcal{B} \\ \ell_\lambda \downarrow \cong & & \cong \downarrow \epsilon_\lambda \\ \mathbb{C} & \xrightarrow{\cdot F_0(\lambda)} & \mathbb{C}. \end{array} \quad (\text{GCF1.3})$$

Indeed $F(\lambda) = F_0(\lambda)(F/F_0)(\lambda)$, including removable values at every zero. Thus the full source expression in (GCF0.3) is an actual cokernel comparison, not a chosen positivity factor.

GCF2. Holomorphic parameter dependence

The quotient-line comparison is a holomorphic family over the entire parameter plane, including the zero set. Here is an explicit verification avoiding any assumption about holomorphic lifts through $\mathcal{B} \rightarrow \mathcal{Q}$.

Put

$$v_{\mathcal{B},\lambda}(s) = e^{(s-\lambda)^2}, \quad v_{\mathcal{J},\lambda}(s) = F_0(s), \quad \eta_{\mathcal{B},\lambda} = \epsilon_\lambda, \quad \eta_{\mathcal{J},\lambda} = \ell_\lambda. \quad (\text{GCF2.1})$$

For $E = \mathcal{B}, \mathcal{J}$, the maps

$$S_{E,\lambda}F(s) = \frac{F(s) - v_{E,\lambda}(s)\eta_{E,\lambda}(F)}{s - \lambda} \quad (\text{GCF2.2})$$

take values in E , and direct multiplication gives

$$(L - \lambda)S_{E,\lambda}F = F - v_{E,\lambda}\eta_{E,\lambda}(F), \quad S_{E,\lambda}(L - \lambda)F = F. \quad (\text{GCF2.3})$$

For $E = \mathcal{J}$, the numerator has one additional order at $s = \lambda$ relative to the required divisor, since its quotient by F_0 vanishes there. This proves the asserted target of (GCF2.2), also at a multiple zero.

On any compact set of parameters λ , the division bounds in GCF1 can use one larger disk and strip. The Gaussian and all its parameter derivatives have uniform bounds in every $\mathcal{B}$ seminorm on that parameter set. The map $F \mapsto F/F_0$ is continuous from $\mathcal{J}$ to entire functions with compact convergence, by the same boundary-circle argument. Evaluations and all their parameter derivatives are therefore bounded by finitely many source seminorms uniformly on a compact parameter set. Formula (GCF2.2) has a removable singularity along $s = \lambda$; Cauchy's integral formula and these bounds prove holomorphic dependence with values in every coefficient seminorm. They also prove that (GCF2.2) carries a holomorphic E -valued family to a holomorphic E -valued family.

Let $\mathcal{H}_E(U)$ denote holomorphic E -valued maps on an open parameter set $U \subset \mathbb{C}$. Equations (GCF2.1)–(GCF2.3) prove, on every such set,

$$\mathcal{H}_E(U)/(L - \lambda)\mathcal{H}_E(U) \xrightarrow{\eta_{E,\lambda}} \mathcal{O}(U). \quad (\text{GCF2.4})$$

This is an isomorphism: the stated section proves surjectivity and $S_{E,\lambda}$ proves its exact kernel. Inclusion of the coefficient spaces is consequently represented by the global map of these two quotient line sheaves

$$\mathcal{O}_{\mathbb{C}} \xrightarrow{\cdot F_0} \mathcal{O}_{\mathbb{C}}. \quad (\text{GCF2.5})$$

Its determinant is the displayed one-by-one entry F_0 , with its exact source factor $1/8$. No determinant or trace of an infinite-dimensional operator has been assumed. The frame of the first line uses the actual source element F_0 ; changing that frame would change the written entry and its inverse frame map together.

GCF3. Exact relation with the actual quotient obstruction

Apply the kernel/cokernel sequence to multiplication by $L - \lambda$ in the actual row $0 \rightarrow \mathcal{J} \rightarrow \mathcal{B} \rightarrow \mathcal{Q} \rightarrow 0$. Since the first two multiplications are injective, it gives

$$0 \rightarrow \ker(L - \lambda \mid \mathcal{Q}) \xrightarrow{\partial_\lambda} \mathbb{C} \xrightarrow{\cdot F_0(\lambda)} \mathbb{C} \rightarrow \text{coker}(L - \lambda \mid \mathcal{Q}) \rightarrow 0. \quad (\text{GCF3.1})$$

The boundary has a literal formula. Represent a kernel class by $G \in \mathcal{B}$, so $(s - \lambda)G \in \mathcal{J}$. Then

$$\partial_\lambda[G] = \left(\frac{(s - \lambda)G(s)}{F_0(s)} \right)_{s=\lambda}. \quad (\text{GCF3.2})$$

Changing G by an element of $\mathcal{J}$ changes the numerator quotient by a multiple of $s - \lambda$, so this is well defined. These computations prove (GCF3.1), including its sign, directly.

When $F_0(\lambda) \neq 0$, both outside terms vanish. When $\lambda = \rho$ is an actual nontrivial zero, both outside terms are one-dimensional and

$$G_\rho(s) = \frac{F_0(s)}{s - \rho} \in \mathcal{B}, \quad \partial_\rho[G_\rho] = 1. \quad (\text{GCF3.3})$$

This remains true for a multiple zero: the class has its last retained nonzero jet of order $m_\rho - 1$. All statements quantify over the full actual zero set.

This boundary is the obstruction to lifting an eigenclass of $\mathcal{Q}$ to a class in $\mathcal{B}$ satisfying that same generator equation. It is not CW4's map $\delta^1 : \mathcal{Q} \rightarrow \mathcal{J}(-1)$. CW4 constructs a different, geometric cohomology lift and proves its boundary zero. Here the explicit nonzero boundary (GCF3.3) specifies its own different lifting problem; it does not undo the geometric lift or imply an obstruction at τ .

For the shifted normal action $L + 1$, exactly the same comparison has entry $F_0(\lambda - 1)$. On $\Re \lambda \in (0, 1)$ this entry is nonzero, since all zeros of F_0 have real part in $(0, 1)$. This recovers the already proved separation of the $\mathcal{Q}$ and $\mathcal{Q}(-1)$ spectra. It does not replace the full target $\mathcal{J}(-1)$ by its quotient.

GCF4. The complete divisor module of the original function

Let $\mathcal{M}$ be the sheaf of meromorphic functions on the parameter plane. Write $[a]$ for the divisor at the scalar point a , and set

$$T = \sum_{n \geq 1} [-2n], \quad D = [1] - T. \quad (\text{GCF4.1})$$

The sum is locally finite, so its orders define a divisor on the whole plane. The sheaf $\mathcal{O}(D) \subset \mathcal{M}$ consists of meromorphic functions g for which $\text{ord}_a(g) + D(a) \geq 0$ at every point. Locally it is a free rank-one $\mathcal{O}$ -module; at 1 its frame is $(s - 1)^{-1}$, and at $-2n$ its frame is $s + 2n$.

The full factor c in (GCF0.3) has divisor exactly D . Indeed Gamma has no zeros, its simple pole at zero cancels the displayed factor s , the factor $s - 1$ gives a simple zero at 1, and its remaining simple poles occur at every $-2n$. The recurrence for Gamma retains their residues in GCF5 below. The power of π is an everywhere nonzero entire factor. There are no further zeros or poles. Hence multiplication gives a global module isomorphism

$$\mu_c : \mathcal{O}(D) \xrightarrow{\sim} \mathcal{O}, \quad g \mapsto cg, \quad f \mapsto f/c. \quad (\text{GCF4.2})$$

The order inequalities prove both directions at every exceptional point, so (GCF4.2) is not division by a supposed holomorphic unit there.

The original meromorphic ζ belongs to $\mathcal{O}(D)$: it has its simple pole at 1, its simple zeros at every $-2n$, and no other poles. The complete comparison is

$$\begin{array}{ccc} \mathcal{O} & \xrightarrow{\cdot \zeta} & \mathcal{O}(D) \subset \mathcal{M} \\ \text{id} \downarrow & & \downarrow \mu_c \cong \\ \mathcal{O} & \xrightarrow{\cdot F_0} & \mathcal{O}. \end{array} \quad (\text{GCF4.3})$$

It is a commutative diagram of actual holomorphic line modules with their specified meromorphic embeddings. Its inverse recovers $\zeta = F_0/c$ at every point. Retaining D , μ_c , and the inclusion into $\mathcal{M}$ retains the original zero and pole orders. Keeping only the bottom scalar function would not specify those embeddings.

The cokernel of either horizontal module map is supported precisely on the nontrivial zero divisor, with length m_ρ at ρ . To prove this, at a nontrivial zero the multiplier is h^{m_ρ} times its retained holomorphic unit, where $h = s - \rho$; the quotient has basis $1, h, \dots, h^{m_\rho-1}$. At every other point the horizontal map is multiplication by a unit in the displayed local frames. GCF5 computes those frames and units at all cancellation points.

GCF5. Every exceptional fibre and every infinitesimal order

At $s_0 = -2n$, $n \geq 1$, put $h = s + 2n$. The exact local factorizations are

$$\zeta(-2n+h) = h a_n(h), \quad c(-2n+h) = h^{-1} b_n(h), \quad F_0(-2n+h) = a_n(h) b_n(h), \quad (\text{GCF5.1})$$

where the holomorphic units and their constants are

$$a_n(0) = \zeta'(-2n), \quad b_n(0) = \frac{(-2n)(-2n-1)}{8} \pi^n \frac{2(-1)^n}{n!} = \frac{n(2n+1)(-1)^n \pi^n}{2n!}. \quad (\text{GCF5.2})$$

Gamma's residue follows by applying $\Gamma(z+1) = z\Gamma(z)$ repeatedly to $\Gamma(h/2)$. Nonvanishing of $a_n(0)$ follows from the original functional equation and the positive Dirichlet value at $1+2n$; explicitly

$$a_n(0) b_n(0) = \frac{(1+2n)(2n)}{8} \pi^{-(1+2n)/2} \Gamma((1+2n)/2) \zeta(1+2n) \neq 0. \quad (\text{GCF5.3})$$

The fibre of the original target line is $h\mathcal{O}/h^2\mathcal{O}$, and the map μ_c sends ha_n to b_na_n . It therefore retains $\zeta'(-2n)$ and its full multiplier, even though the bottom function is nonzero there.

At $s = 1$, put $h = s - 1$. Then

$$\zeta(1+h) = h^{-1}a_1^{\text{pole}}(h), \quad c(1+h) = hb_1^{\text{pole}}(h), \quad a_1^{\text{pole}}(0) = 1, \quad b_1^{\text{pole}}(0) = 1/8. \quad (\text{GCF5.4})$$

The original target fibre is $h^{-1}\mathcal{O}/\mathcal{O}$. Its residue coefficient 1 is carried to $1/8$. At $s = 0$, the target is $\mathcal{O}$, $c(0) = -1/4$, and $\zeta(0) = -1/2$, again giving $F_0(0) = 1/8$. Neither zero nor pole data has been silently removed.

All higher orders are equally explicit. At any point, write $\mathcal{O}(D) = h^{-d}\mathcal{O}$ and $c = h^db(h)$, with $d = D(s_0)$ and $b(0) \neq 0$. On the full length- r jet module, (GCF4.2) is

$$h^{-d}\mathcal{O}/h^{r-d}\mathcal{O} \longrightarrow \mathcal{O}/h^r\mathcal{O}, \quad [h^{-d}a(h)] \longmapsto [b(h)a(h)], \quad (\text{GCF5.5})$$

for every $r \geq 1$. In ordered bases $h^{-d}, \dots, h^{r-1-d}$ and $1, \dots, h^{r-1}$, its matrix has entries

$$B_{ij} = \begin{cases} b^{(i-j)}(0)/(i-j)!, & i \geq j, \\ 0, & i < j, \end{cases} \quad 0 \leq i, j < r, \quad \det B = b(0)^r. \quad (\text{GCF5.6})$$

Its inverse is the same matrix rule with every coefficient of $1/b$. Thus every infinitesimal coefficient is retained; the comparison is not only a statement about values or a first-order approximation.

GCF6. The transported involution on the original target

The source functional equation gives $F_0(s) = F_0(1-s)$. On the original target module, the exact transported reflection is

$$(\mathcal{R}_D g)(s) = \frac{c(1-s)}{c(s)}g(1-s), \quad \mu_c \mathcal{R}_D = \mathcal{R} \mu_c, \quad (\mathcal{R} f)(s) = f(1-s). \quad (\text{GCF6.1})$$

It is defined globally as a map of line modules over the reflection, including exceptional points: by (GCF4.2), it is $\mu_c^{-1}\mathcal{R}\mu_c$. Thus its square is the identity, and $\mathcal{R}_D \zeta = \zeta$. The ratio is explicitly

$$\frac{c(1-s)}{c(s)} = \frac{\frac{(1-s)((1-s)-1)}{8} \pi^{-(1-s)/2} \Gamma((1-s)/2)}{\frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2)}. \quad (\text{GCF6.2})$$

This is an isomorphism of the indicated line modules over reflection, not an assertion that the ratio is an everywhere holomorphic unit on scalar functions. In particular its zeros and poles implement the frames in GCF5. On coefficient functions, $\mathcal{R}(L-\lambda) = -(L-(1-\lambda))\mathcal{R}$. On length- r jets, reflection sends h^j to $(-1)^j h^j$ between the two centres. Consequently the full generator comparison respects the reflected parameter and every derivative sign. No radius or distance from τ was used to construct this action.

GCF7. The original logarithmic derivative remains complete

Off the zeros and poles, logarithmic differentiation of the actual module comparison gives

$$\frac{F'_0}{F_0} = \frac{\zeta'}{\zeta} + \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \frac{\Gamma'(s/2)}{\Gamma(s/2)}. \quad (\text{GCF7.1})$$

It then holds meromorphically. At a nontrivial zero its residue is the complete multiplicity. At 1, the residues -1 from ζ'/ζ and $+1$ from $1/(s-1)$ cancel. At each $-2n$, the residues $+1$ from the trivial zero and -1 from Gamma cancel. At zero the residues $+1$ from $1/s$ and -1 from Gamma cancel; ζ is nonzero there. These are the same contributions encoded by the frames in GCF5, with the power of π and every regular derivative term still present in (GCF7.1).

GCF8. Consequence for the requested weight calculation

The actual coefficient inclusion has now produced a holomorphic one-by-one determinant map, and (GCF4.3) returns it to the original ζ as an isomorphism of explicitly embedded divisor modules. The precise spectral lifting boundary is (GCF3.2). It is nonzero at every actual nontrivial zero, on or off the critical line, and retains multiplicity through the complete jet modules. Thus it cannot by itself distinguish the two locations. This is a calculation of the actual boundary, rather than a hypothetical counterexample constructed outside the programme.

CW4's geometric lifting boundary remains zero. Deligne's argument additionally separates the weights of its entire source and target; no such bound follows merely from the determinant map just derived. The original full target $J(-1)$, the degree factor, and the CW quotient maps remain present. The new comparison supplies the exact original-zeta object to which any further weight or trace argument must apply. It does not replace that argument with a desired assumption.

Human source comparisons: Connes and Consani, *Schemes over F_1 and zeta functions*, [arXiv:0903.2024v3](#), §5, for the source summation/sheaf construction; Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, [arXiv:2602.04022v1](#), `rhready.tex` lines526–545, for the retained functional equation; Deligne, *La conjecture de Weil. II*, [IHÉS52 \(1980\),137–252](#), §3.6, for the distinct weight-separation lifting mechanism. Source-reading coverage and hashes are recorded privately; these citations do not assert a new reading of all external dependencies.

The complete primal parameter and jet family — PGF0-PGF9

Proof source: PRIMAL_GENERATOR_FAMILY_INDEPENDENT.md.

The full parameter and jet family of the original-zeta generator sequence

Independent derivation, 24 September 2026. Complete proof locators PGF0-PGF9.

PGF0. Scope, prerequisites, and actual sources read

This calculation is on the arithmetic coefficient spaces after their complete global reconstruction. It does not construct arithmetic by selecting a few primes. The user's supporting notation $Z_0, Z_1, Z_2, \dots$ and $\tau\langle Z_1; \text{no } Z_2 \rangle$ remains unchanged. Addition at τ is retracted. None of the additions, derivatives, scalar operations, or coordinates below are operations on τ .

The governing argument `reconstruction/CORPUS_AND_OPERATION_RULE.md` was read, including the recorded corrections concerning prerequisites and the full obstruction receiver. This derivation uses the already constructed exact row

$$0 \longrightarrow \mathcal{J} \xrightarrow{\iota} \mathcal{B} \xrightarrow{q} \mathcal{Q} \longrightarrow 0. \quad (\text{PGF0.1})$$

It does not assert a new weight bound or an RH conclusion. The map calculated here is the parameter-dependent inclusion between the two generator cokernels and its actual snake boundary. It is not the sheaf connecting map $Q \rightarrow J(-1)$, whose vanishing is already proved in CW4 and CW7.

Programme sources read for this derivation were:

- `GLOBAL_MELLIN_SYNTHESIS.md`, S1-S3, including the complete source multiplier and every exceptional value.
- `EXACT_SCHWARTZ_SUMMATION_IMAGE.md`, SSI0-SSI3, for the exact statement, source conventions, and original-zeta receiver. The completed SSI exact-image theorem is used as an established input, not reproved here.
- `../tau_weight_cohomology_20260924/CC_ACTUAL_WEIGHT_LIFT_COMPARISON.md`, CW0-CW9, for the actual source row, its full kernel, and the distinction between quotient characters and eigenvectors.
- `../tau_weight_cohomology_20260924/ORIGINAL_ZETA_RESOLVENT_AND_PROJECTORS.md`, RZ1-RZ8, for whole-space division estimates, holomorphic resolvents, and the already constructed generalized eigenspaces. The present proof does not claim those prior constructions as new.
- `CC_VERDIER_SUPPORT_INDEPENDENT.md`, VSD13, for the first-order closed-image statement and its separate algebraic-dual consequence. The present calculation concerns the primal sequence; it does not duplicate the dagger-dual lifting calculation.

The upstream source construction is Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, [arXiv:0903.2024v3](https://arxiv.org/abs/0903.2024v3), §5. SSI and CW record the original author-file identity, exact reading coverage, and receiving maps. This independent note makes no additional claim of having reread that complete author source. All new calculations used here are proved below.

PGF1. Objects, operators, and the retained entire source function

Let $\mathcal{B}$ be the complex Fréchet space of entire functions with

$$b_{A,M}(F) = \sup_{|\Re s| \leq A} (1 + |\Im s|)^M |F(s)| < \infty \quad (A, M \in \mathbb{N}). \quad (\text{PGF1.1})$$

Let $\mathcal{Z}$ be the actual nontrivial-zero set of the original Riemann zeta function, and let m_ρ be each full multiplicity. Put

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for all } \rho \in \mathcal{Z}, 0 \leq j < m_\rho\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}. \quad (\text{PGF1.2})$$

The ideal is closed because each indicated jet functional is continuous by Cauchy's formula. The quotient uses its actual Fréchet quotient topology.

Multiplication by s , written L on each space where it is defined, is continuous and preserves $\mathcal{J}$. For example,

$$b_{A,M}(LF) \leq (A+1)b_{A,M+1}(F). \quad (\text{PGF1.3})$$

For every $\lambda \in \mathbb{C}$ and positive integer r , define

$$M_{\lambda,r} = (L - \lambda)^r. \quad (\text{PGF1.4})$$

This is an everywhere-defined continuous linear map on $\mathcal{B}, \mathcal{J}, \mathcal{Q}$. The arithmetic coefficient field, polynomial, and integer r here are already reconstructed receivers.

Retain the entire source function with all its original factors:

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{PGF1.5})$$

It belongs to $\mathcal{J}$, and its divisor consists exactly of (ρ, m_ρ) . At the exceptional points its values are

$$F_0(0) = F_0(1) = \frac{1}{8}, \quad (\text{PGF1.6})$$

$$F_0(-2n) = \frac{n(2n+1)(-1)^n \pi^n}{2n!} \zeta'(-2n) = \frac{(1+2n)(2n)}{8} \pi^{-(1+2n)/2} \Gamma((1+2n)/2) \zeta(1+2n) \neq 0 \quad (n \geq 1). \quad (\text{PGF1.7})$$

These are the retained S2 and CW1 identities. In particular the original zeta pole at 1 and the trivial zeros at $-2n$ are not additional zeros of this specified source function.

Set

$$m_\lambda = \text{ord}_\lambda F_0, \quad (\text{PGF1.8})$$

with order zero where $F_0(\lambda) \neq 0$. Every $F \in \mathcal{J}$ has an entire quotient F/F_0 , by local Taylor division at each zero. This quotient is not asserted to belong to $\mathcal{B}$.

For later continuity, the division map

$$\mathcal{D} : \mathcal{J} \longrightarrow \mathcal{O}(\mathbb{C}), \quad F \longmapsto F/F_0 \quad (\text{PGF1.9})$$

is continuous for the compact-open topology on the target. To prove it, put any given compact set inside a circle $|s| = R$ which contains no zero of F_0 on its boundary. Such an R exists because the zeros are discrete. The maximum principle applied to the entire quotient gives

$$\sup_{|s| \leq R} |F(s)/F_0(s)| \leq \frac{b_{A,0}(F)}{\min_{|s|=R} |F_0(s)|}, \quad A \geq R. \quad (\text{PGF1.10})$$

The denominator is positive. Cauchy's formula on a slightly larger circle gives the same conclusion for every fixed derivative and every compact set.

PGF2. Both complete jet quotients, with their exact topology

For a fixed $r \geq 1$, write

$$R_r = \mathbb{C}[h]/(h^r), \quad j_\lambda^r H = \sum_{j=0}^{r-1} \frac{H^{(j)}(\lambda)}{j!} h^j \quad \text{for an entire } H. \quad (\text{PGF2.1})$$

Here h is the jet coordinate $s - \lambda$ in an arithmetic coefficient receiver; it is not a coordinate of τ . Define

$$\beta_{\lambda,r} : \mathcal{B} \rightarrow R_r, \quad F \mapsto j_\lambda^r F, \quad \alpha_{\lambda,r} : \mathcal{J} \rightarrow R_r, \quad F \mapsto j_\lambda^r (F/F_0). \quad (\text{PGF2.2})$$

Both are continuous. This follows for β from Cauchy estimates and for α from (PGF1.9)–(PGF1.10).

They are both onto, with explicit continuous linear sections. If P is the unique polynomial of degree below r representing its class in R_r , put

$$\sigma_{\lambda,r}^I(P)(s) = F_0(s)P(s - \lambda). \quad (\text{PGF2.3})$$

The product belongs to $\mathcal{J}$, since polynomial multiplication preserves $\mathcal{B}$ and all required zero orders. Its quotient by F_0 is exactly $P(s - \lambda)$, so $\alpha\sigma^I = 1$.

For the other section, retain the Gaussian and every truncated coefficient:

$$\sigma_{\lambda,r}^B(P)(s) = e^{(s-\lambda)^2} \text{Tay}_{h,<r}(e^{-h^2} P(h)) \Big|_{h=s-\lambda}. \quad (\text{PGF2.4})$$

The Gaussian times this finite polynomial belongs to $\mathcal{B}$: on every bounded real strip the factor has modulus $e^{(\Re s - \Re \lambda)^2 - (\Im s - \Im \lambda)^2}$, which dominates every polynomial decay requirement. Its Taylor polynomial at $s = \lambda$ equals P modulo h^r , so $\beta\sigma^B = 1$. These are sections of the specified quotient maps, not a section of $\mathcal{B} \rightarrow \mathcal{Q}$.

The kernels are exactly

$$\ker \beta_{\lambda,r} = (s - \lambda)^r \mathcal{B}, \quad \ker \alpha_{\lambda,r} = (s - \lambda)^r \mathcal{J}. \quad (\text{PGF2.5})$$

For the first identity, vanishing of the jet is precisely divisibility to order r at λ . The quotient is entire and remains in $\mathcal{B}$. An explicit division estimate is given below. For the second, $\alpha(F) = 0$ means that F/F_0 has a zero of order at least r . Thus $F/(s - \lambda)^r$ retains the full zero order m_λ at λ , as well as the full orders at all other original zeros, and belongs to $\mathcal{J}$. The converse implications follow by direct multiplication.

For precision, if $H \in \mathcal{B}$ vanishes to order at least r at λ , its entire quotient obeys

$$b_{A,M} \left(\frac{H}{(s - \lambda)^r} \right) \leq b_{A,M}(H) + 2^{-r} (2 + |\Im \lambda|)^M b_{A',0}(H), \quad A' \geq \max(A, |\Re \lambda| + 2). \quad (\text{PGF2.6})$$

Outside the radius-one disk, the denominator has modulus at least one. Inside that disk, apply the maximum principle to the quotient on the radius-two disk; its boundary value is bounded by $2^{-r} \sup |H|$. The polynomial imaginary-coordinate weight is bounded by $(2 + |\Im \lambda|)^M$ on the smaller disk. This proves the estimate, which is uniform for λ in a compact set after fixing A' and the displayed maximum of $|\Im \lambda|$.

Consequently the maps

$$\bar{\alpha}_{\lambda,r} : \mathcal{J}/(L - \lambda)^r \mathcal{J} \xrightarrow{\sim} R_r, \quad \bar{\beta}_{\lambda,r} : \mathcal{B}/(L - \lambda)^r \mathcal{B} \xrightarrow{\sim} R_r \quad (\text{PGF2.7})$$

are topological linear isomorphisms. Their continuous inverses are the quotient classes of (PGF2.3) and (PGF2.4). On both sides L acts as multiplication by $\lambda + h$, so these are also exact $\mathbb{C}[L]$ -module isomorphisms.

Their multiplicative meanings differ and must be retained. The map $\bar{\beta}$ is an algebra isomorphism to the usual truncated polynomial algebra; its identity is the class represented by (PGF2.4) with $P = 1$. On the $\mathcal{J}$ -quotient, (PGF2.7) transports the inherited product to

$$P \star Q = f_{\lambda,r}(h)P(h)Q(h) \pmod{h^r}, \quad f_{\lambda,r}(h) := j_{\lambda}^r F_0. \quad (\text{PGF2.8})$$

Indeed $(FG)/F_0 = F_0(F/F_0)(G/F_0)$. Thus $\bar{\alpha}$ is a linear and module isomorphism; it is not an identification with the usual multiplication on R_r . This retains the full source coefficient rather than silently replacing its product.

PGF3. The inclusion matrix and every retained factor

The inclusion $\iota : \mathcal{J} \rightarrow \mathcal{B}$ induces, in (PGF2.7), exactly

$$\mathsf{T}_r(\lambda) : R_r \longrightarrow R_r, \quad P \longmapsto f_{\lambda,r}P \pmod{h^r}. \quad (\text{PGF3.1})$$

Proof: $F = F_0(F/F_0)$ as an identity of entire functions, so taking its full r -jet gives (PGF3.1). No local unit is removed in this map.

In the ordered basis $1, h, \dots, h^{r-1}$, write

$$c_j(\lambda) = \frac{F_0^{(j)}(\lambda)}{j!}. \quad (\text{PGF3.2})$$

Then the complete matrix is

$$\mathsf{T}_r(\lambda)_{ij} = \begin{cases} c_{i-j}(\lambda), & 0 \leq j \leq i < r, \\ 0, & 0 \leq i < j < r. \end{cases} \quad (\text{PGF3.3})$$

It follows by the triangular determinant formula that

$$\det \mathsf{T}_r(\lambda) = F_0(\lambda)^r = \left[\frac{\lambda(\lambda-1)}{8} \pi^{-\lambda/2} \Gamma(\lambda/2) \zeta(\lambda) \right]^r, \quad (\text{PGF3.4})$$

where the expression on the right has its full entire continuation and the values (PGF1.6)–(PGF1.7). The exponent r is the matrix rank of this r -jet comparison. It is not a pooling of two arithmetic branches or a replacement of the original return measure by a power of zeta.

At a point where the individual factors are holomorphic, all coefficients in the matrix retain the full Leibniz sum. With $P(s) = s(s-1)$, this is

$$c_j(\lambda) = \frac{\pi^{-\lambda/2}}{8} \sum_{\substack{a+b+c+d=j \\ 0 \leq a \leq 2}} \frac{P^{(a)}(\lambda)}{a!} \frac{(-\log \pi/2)^b}{b!} \frac{2^{-c} \Gamma^{(c)}(\lambda/2)}{c!} \frac{\zeta^{(d)}(\lambda)}{d!}. \quad (\text{PGF3.5})$$

At $0, 1, -2, -4, \dots$, the definition (PGF3.2) takes the derivatives of the full analytic product. Equivalently, its coefficient is the coefficient of h^j in the product of the full convergent Laurent expansions of

$$\frac{(\lambda+h)(\lambda+h-1)}{8}, \quad \pi^{-(\lambda+h)/2}, \quad \Gamma((\lambda+h)/2), \quad \zeta(\lambda+h). \quad (\text{PGF3.6})$$

Each has only the stated finite principal part at that point, so the coefficient of each fixed nonnegative power is a finite sum over the allowed Laurent exponents. This prescription keeps the cancellation derivatives, including the zeta residue and the trivial-zero derivatives. In particular, the diagonal is $1/8$ at both endpoints, and equals (PGF1.7) at each negative even integer; the determinant there is the corresponding nonzero r -th power. The original zeta's pole and trivial zeros remain in (PGF3.6), rather than being asserted absent from the original function.

PGF4. The full snake sequence and its actual boundary

Apply $M_{\lambda,r}$ to all three terms of (PGF0.1). The first two vertical maps are injective, since an entire function multiplied by the nonzero polynomial $(s - \lambda)^r$ can vanish identically only when that entire function is zero. The full resulting snake sequence is

$$0 \longrightarrow K_{\lambda,r} \xrightarrow{\partial_{\lambda,r}} R_r \xrightarrow{\tau_r(\lambda)} R_r \xrightarrow{\pi_{\lambda,r}} C_{\lambda,r} \longrightarrow 0, \quad (\text{PGF4.1})$$

where

$$K_{\lambda,r} = \ker((L - \lambda)^r : \mathcal{Q} \rightarrow \mathcal{Q}), \quad C_{\lambda,r} = \mathcal{Q}/(L - \lambda)^r \mathcal{Q}. \quad (\text{PGF4.2})$$

Here is the explicit boundary, with its sign fixed:

$$\partial_{\lambda,r}[H] = j_\lambda^r \left(\frac{(s - \lambda)^r H(s)}{F_0(s)} \right). \quad (\text{PGF4.3})$$

The hypothesis defining the kernel means that the numerator is in $\mathcal{J}$, so its quotient is entire. If the representative H is changed by $G \in \mathcal{J}$, the displayed quotient changes by $(s - \lambda)^r G/F_0$, whose first r Taylor coefficients vanish. Thus the boundary is well defined. Its positive sign is the diagram chase: lift $[H]$ to $H \in \mathcal{B}$, apply $(s - \lambda)^r$, and take the resulting element of $\mathcal{J}$, without a reversal of the subtraction order.

The last map sends a polynomial class $P \in R_r$ to the class of any lift of P under $\beta_{\lambda,r}$, followed by q and then the quotient in (PGF4.2). Different lifts differ by $(s - \lambda)^r \mathcal{B}$, whose image vanishes there. It is onto because every $\mathcal{Q}$ -class has a representative in $\mathcal{B}$.

For a direct proof of every exactness claim, first suppose $\partial[H] = 0$. Then $(s - \lambda)^r H = (s - \lambda)^r G$ for some $G \in \mathcal{J}$, by (PGF2.5). Injectivity of polynomial multiplication on $\mathcal{B}$ gives $H = G$, so $[H] = 0$. Next, if $\partial[H] = P$, the jet of $(s - \lambda)^r H$ is zero, so $\tau_r P = 0$. Conversely, if $\tau_r P = 0$, take its unique polynomial representative of degree below r . The function

$$H_P(s) = \frac{F_0(s)P(s - \lambda)}{(s - \lambda)^r} \quad (\text{PGF4.4})$$

is entire by the vanishing condition $\tau_r P = 0$, belongs to $\mathcal{B}$ by (PGF2.6), satisfies $(s - \lambda)^r H_P \in \mathcal{J}$, and has boundary exactly P . Finally a jet maps to zero in $C_{\lambda,r}$ exactly when a representative is the sum of an element of $\mathcal{J}$ and an element of $(s - \lambda)^r \mathcal{B}$; its jet is therefore precisely in the image of τ_r . This proves (PGF4.1) without replacing the full quotient by a product of jets.

Put $m = m_\lambda$ and $d = \min(r, m)$. At the fixed parameter, write

$$F_0(\lambda + h) = h^m u_\lambda(h), \quad u_\lambda(0) = F_0^{(m)}(\lambda)/m! \neq 0. \quad (\text{PGF4.5})$$

This is a local factorization with the unit retained. Multiplication by its full truncated jet is an invertible map of R_r , with inverse the truncated Taylor series of $1/u_\lambda$. It follows that

$$\ker \tau_r(\lambda) = h^{r-d} R_r, \quad \text{im } \tau_r(\lambda) = h^d R_r, \quad (\text{PGF4.6})$$

where $h^r R_r = 0$. Consequently

$$\dim K_{\lambda,r} = \dim C_{\lambda,r} = d. \quad (\text{PGF4.7})$$

The actual kernel basis and its boundary are

$$q_{\lambda,k} = \left[\frac{F_0(s)}{(s - \lambda)^k} \right], \quad \partial_{\lambda,r} q_{\lambda,k} = h^{r-k}, \quad 1 \leq k \leq d. \quad (\text{PGF4.8})$$

Each representative is entire because $k \leq m$, and its membership in $\mathcal{B}$ follows from (PGF2.6). The distinct monomials prove independence. They span because those monomials span the kernel in (PGF4.6) and (PGF4.3) is an isomorphism onto that kernel. This gives the previously known generalized eigenspaces a full source-boundary comparison, including every r , rather than asserting a new spectral location theorem.

The cokernel has an equally explicit map:

$$C_{\lambda,r} \xrightarrow{\sim} R_d, \quad [[H]] \mapsto j_\lambda^d H, \quad (\text{PGF4.9})$$

where $R_0 = 0$. It is well defined since $d \leq m$ and $d \leq r$. Its kernel can be checked constructively. If $m \geq r$, a representative whose r -jet vanishes is directly divisible by $(s - \lambda)^r$ in $\mathcal{B}$. If $m < r$ and H vanishes to order at least m , its local quotient H/F_0 is holomorphic at λ . Let

$$P(h) = \text{Tay}_{h, < r-m}(H(\lambda + h)/F_0(\lambda + h)). \quad (\text{PGF4.10})$$

Then $H(s) - F_0(s)P(s - \lambda)$ vanishes to order at least r , belongs to $\mathcal{B}$, and is divisible there by $(s - \lambda)^r$. Since the subtracted term belongs to $\mathcal{J}$, the quotient class of H lies in $(L - \lambda)^r \mathcal{Q}$. The same argument with $m = 0$ shows that this map is onto all of $\mathcal{Q}$ when $\lambda \notin \mathcal{Z}$.

The map $\mathcal{Q} \rightarrow R_d$ in (PGF4.9) is continuous, since it descends from continuous jets on $\mathcal{B}$. Its kernel is thus closed; this proves closedness of $(L - \lambda)^r \mathcal{Q}$. All finite-dimensional identifications in (PGF4.1), (PGF4.8), and (PGF4.9) are topological. No density of the sum of the primal generalized eigenspaces is used or concluded.

PGF5. The boundary in the original Schwartz and Mellin sources

The exact source spaces and map from SSI and CW are

$$\begin{aligned} S &= \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0\}, \\ A &= \{a \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \text{ for all } N, j\}, \\ \Sigma f(u) &= 2 \sum_{n \geq 1} f(nu), \quad J = \Sigma S, \quad Q = A/J, \end{aligned} \quad (\text{PGF5.1})$$

$$\Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, \quad \Theta^{-1} F(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} F(1/2 + it) u^{-it} dt. \quad (\text{PGF5.2})$$

The established exact-image theorem says that $\Sigma : S \rightarrow J$ and $\Theta : A \rightarrow \mathcal{B}$ are topological isomorphisms onto their stated targets, with $\Theta J = \mathcal{J}$. The quotient map $\kappa : Q \rightarrow \mathcal{Q}$ is $[a] \mapsto [\Theta a]$. Every factor $2, 1/2, 1/\pi$ in these maps remains explicit.

Define the actual source generators

$$D_S f(v) = -v f'(v), \quad D_A a(u) = -u a'(u). \quad (\text{PGF5.3})$$

They preserve their indicated spaces continuously. Evenness and Schwartz estimates show this for S ; its value condition is preserved by the factor v , and integration by parts gives $\int D_S f = \int f = 0$. The weighted logarithmic-derivative seminorms prove the statement on A . Termwise differentiation of the convergent Schwartz sum, with the same endpoint estimates as SSI2, and integration by parts in the Mellin integral give

$$\Sigma D_S = D_A \Sigma, \quad \Theta D_A = L \Theta. \quad (\text{PGF5.4})$$

All boundary terms vanish by the stated seminorms. These identities transport (PGF4.1) to the actual row $0 \rightarrow S \xrightarrow{\Sigma} A \rightarrow Q \rightarrow 0$.

For the original Gaussian source

$$f_0(v) = \frac{\pi}{2}v^2(2\pi v^2 - 3)e^{-\pi v^2}, \quad \Theta \Sigma f_0 = F_0, \quad (\text{PGF5.5})$$

the kernel class in (PGF4.8) has the exact A -representative

$$a_{\lambda,k}(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} \frac{F_0(1/2 + it)}{(1/2 + it - \lambda)^k} u^{-it} dt, \quad 1 \leq k \leq \min(r, m_\lambda). \quad (\text{PGF5.6})$$

The integrand uses its removable entire value if $1/2 + it = \lambda$; no singular integral prescription is imposed. The numerator divided by that power belongs to $\mathcal{B}$, so the established Mellin inverse estimates prove $a_{\lambda,k} \in A$.

Using (PGF5.4) gives the equality in the actual prequotient source:

$$(D_A - \lambda)^r a_{\lambda,k} = \Sigma((D_S - \lambda)^{r-k} f_0). \quad (\text{PGF5.7})$$

Indeed the Mellin transforms of the two sides both equal $(s - \lambda)^{r-k} F_0(s)$, and Θ is injective. Thus the original-source snake boundary is

$$[a_{\lambda,k}] \mapsto [(D_S - \lambda)^{r-k} f_0] \quad \text{in } S/(D_S - \lambda)^r S. \quad (\text{PGF5.8})$$

This proves the full map between the kernel class and its genuine Schwartz-source boundary. The representation does not assign a scalar or arithmetic operation to the support.

PGF6. Holomorphic parameter dependence and the exact family cokernel

All entries of (PGF3.3) are entire in λ . The quotient coordinates themselves form compatible holomorphic families; this can be proved directly, without assuming a holomorphic section of $\mathcal{B} \rightarrow \mathcal{Q}$.

First $\lambda \mapsto (L - \lambda)^r$ is a polynomial family of continuous operators on $\mathcal{B}$ and $\mathcal{I}$, by expansion into the finitely many continuous powers of L . The section $\sigma_{\lambda,r}^I$ is polynomial in λ with values in $\mathcal{I}$. The Gaussian section $\sigma_{\lambda,r}^B$ is entire with values in $\mathcal{B}$: for λ in any compact set, each strip seminorm of the function and each parameter derivative is bounded by a constant times a Gaussian in $\mathcal{I}$ s times a fixed polynomial. Pointwise parameter Cauchy formulas can therefore be integrated in each seminorm; the corresponding Cauchy remainder estimates prove convergence of difference quotients in the Fréchet topology. Since the input space R_r is finite dimensional, this proves the assertion for the section as an operator family.

The jet maps $\alpha_{\lambda,r}$ and $\beta_{\lambda,r}$ are likewise holomorphic in λ , uniformly on bounded input sets. For β , apply Cauchy's formula to derivatives of F on a fixed larger compact set. For α , apply the same argument to the continuous map (PGF1.9). The estimate (PGF1.10) bounds the result by one input seminorm for every compact parameter set and derivative order. It also proves holomorphy on a parameter-dependent holomorphic $\mathcal{I}$ -valued family by joint evaluation and the parameter Cauchy formula.

For completeness, these give exact sheaf rows of holomorphic families. Write $\mathcal{O}_U(E)$ for holomorphic functions from an open parameter set $U \subset \mathbb{C}$ into the indicated Fréchet space E . For $E = \mathcal{B}$ use β , and for $E = \mathcal{I}$ use α . Then

$$0 \longrightarrow \mathcal{O}_U(E) \xrightarrow{(s-\lambda)^r} \mathcal{O}_U(E) \xrightarrow{j_E} \mathcal{O}_U^r \longrightarrow 0 \quad (\text{PGF6.1})$$

is exact, with the sections just constructed. To check the sole analytic divisibility issue, take a holomorphic family H_λ in the kernel. Pointwise division $H_\lambda(s)/(s - \lambda)^r$ is entire in s . In local coordinates $(\lambda, h = s - \lambda)$, the first r coefficients of $H_\lambda(\lambda + h)$ vanish identically, so Taylor division shows joint holomorphy of the quotient, including the diagonal. Estimate (PGF2.6), uniformly on compact parameter sets, bounds each output seminorm by input seminorms. Cauchy's parameter formula and remainder estimates therefore prove Fréchet holomorphy. For $E = \mathcal{I}$, (PGF2.5) additionally proves that every divided value lies in $\mathcal{I}$; since it is closed, the holomorphic family lies in that space. This proves exactness of both rows with their actual topologies.

The inclusion between the rows of (PGF6.1) induces the entire matrix map (PGF3.3). Define its coherent cokernel sheaf on the parameter plane by

$$\mathcal{C}_r = \text{coker}(\mathcal{O}_{\mathbb{C}}^r \xrightarrow{\tau_r} \mathcal{O}_{\mathbb{C}}^r). \quad (\text{PGF6.2})$$

The displayed map of sheaves is injective, although some of its fixed-parameter matrices have nonzero kernels. To prove injectivity, multiply a germ in its kernel by the adjugate matrix. Each component is then killed by the nonzero holomorphic germ $F_0(\lambda)^r$. The local holomorphic ring is an integral domain, so each component is zero. Hence

$$0 \rightarrow \mathcal{O}_{\mathbb{C}}^r \xrightarrow{\tau_r} \mathcal{O}_{\mathbb{C}}^r \rightarrow \mathcal{C}_r \rightarrow 0 \quad (\text{PGF6.3})$$

is a free resolution of the exact family cokernel. No exactness assertion for the unconstructed sheaf of holomorphic $\mathcal{Q}$ -valued lifts is needed for this statement.

Fix an actual zero ρ , put $x = \lambda - \rho$ and $m = m_\rho$, and write

$$F_0(\rho + y) = y^m u_\rho(y), \quad u_\rho(0) \neq 0. \quad (\text{PGF6.4})$$

In the local free $\mathbb{C}\{x\}$ -module $\mathbb{C}\{x\}[h]/(h^r)$, the original map is multiplication by

$$(x + h)^m u_\rho(x + h) \pmod{h^r}. \quad (\text{PGF6.5})$$

Its unit comparison is the explicit domain automorphism given by multiplication by the full jet of $u_\rho(x + h)$; its inverse uses the full jet of $1/u_\rho(x + h)$. Multiplication by $(x + h)^m$ composed with that domain automorphism is exactly the original map (PGF6.5). Thus the original unit has a stated map and inverse throughout the comparison.

The resulting cokernel is consequently

$$(\mathcal{C}_r)_\rho \cong \mathbb{C}\{x, h\}/(h^r, (x + h)^m) \cong \mathbb{C}\{y, h\}/(h^r, y^m), \quad y = x + h. \quad (\text{PGF6.6})$$

The two maps in the second isomorphism are exactly $y \mapsto x + h$ and $x \mapsto y - h$, with h fixed. Every convergent power series reduces uniquely to $\sum_{0 \leq i < m, 0 \leq j < r} a_{ij} y^i h^j$, so the length of this local parameter module is

$$\text{length}_{\mathbb{C}\{x\}}(\mathcal{C}_r)_\rho = mr. \quad (\text{PGF6.7})$$

It is supported at $x = 0$: the action of x is $y - h$, and $(y - h)^{m+r-1} = 0$. Its power $(y - h)^{m+r-2}$ is nonzero, because its only surviving monomial is

$$\binom{m+r-2}{m-1} (-1)^{r-1} y^{m-1} h^{r-1}, \quad (\text{PGF6.8})$$

with nonzero coefficient in the already recovered complex coefficient field. The length in (PGF6.7) includes these parameter-direction extensions; it must not be replaced by the dimension of a fixed fibre.

Indeed the fibre at ρ is

$$\mathcal{C}_r \otimes_{\mathcal{O}} \mathbb{C}_\rho \cong \mathbb{C}[h]/(h^r, h^m) \cong R_{\min(r,m)}. \quad (\text{PGF6.9})$$

Applying the fibre functor to the proved free resolution (PGF6.3) also gives

$$0 \rightarrow \text{Tor}_1^{\mathcal{O}}(\mathcal{C}_r, \mathbb{C}_\lambda) \rightarrow R_r \xrightarrow{\tau_r(\lambda)} R_r \rightarrow \mathcal{C}_r \otimes \mathbb{C}_\lambda \rightarrow 0. \quad (\text{PGF6.10})$$

This is the same finite sequence as (PGF4.1), with the explicit isomorphism from the actual primal kernel supplied by its boundary (PGF4.3). Therefore

$$K_{\lambda,r} \xrightarrow{\sim} \text{Tor}_1^{\mathcal{O}}(\mathcal{C}_r, \mathbb{C}_\lambda), \quad C_{\lambda,r} \xrightarrow{\sim} \mathcal{C}_r \otimes \mathbb{C}_\lambda. \quad (\text{PGF6.11})$$

The first arrow names the kernel inclusion in (PGF6.10) and uses its fixed positive sign. These are exact maps explaining why a sheaf injection can have a noninjective specialization; the missing fibre kernel is retained as the displayed Tor group.

PGF7. Arithmetic actions and the full coefficient reflection

For every recovered positive real a , the coefficient action is

$$T_a F(s) = a^s F(s). \quad (\text{PGF7.1})$$

On each of the two jet quotients in (PGF2.7), it acts as

$$a^\lambda \sum_{j=0}^{r-1} \frac{(\log a)^j}{j!} M_h^j. \quad (\text{PGF7.2})$$

This follows by taking the full jet of $a^{\lambda+h}$; in the $\mathcal{J}$ -quotient the division by F_0 leaves exactly the same multiplier. The matrix $\mathbb{T}_r$ commutes with that action because both are multiplication operators in R_r . The boundary is equivariant by (PGF4.3), since T_a commutes with $(L - \lambda)^r$. On its explicit kernel basis the whole action is

$$T_a q_{\lambda,k} = a^\lambda \sum_{j=0}^{k-1} \frac{(\log a)^j}{j!} q_{\lambda,k-j}. \quad (\text{PGF7.3})$$

To verify the equality on the full quotient, subtract its right side from its left. The numerator of the difference is F_0 times the remainder after subtracting the first k Taylor coefficients of a^s at λ . That remainder is divisible by $(s - \lambda)^k$; after division the result belongs to $\mathcal{J}$, with the strip estimates following from boundedness of a^s on each fixed strip and (PGF2.6). Thus every nilpotent coefficient is retained, not only the leading character.

If the same coefficient representation is given the actual normal-degree twist, its action is aT_a ; equations (PGF7.2)–(PGF7.3) acquire exactly the factor a , equivalently $a^{\lambda+1}$ instead of a^λ . This is a receiving comparison; it does not identify $\mathcal{J}(-1)$ with $\mathcal{Q}(-1)$.

The coefficient reflection is

$$\mathcal{R}F(s) = F(1 - s), \quad F_0(1 - s) = F_0(s). \quad (\text{PGF7.4})$$

The identity for F_0 is the original source Fourier/Mellin identity retained in S2. Define $\mathbb{C}_r P(h) = P(-h)$. Then

$$\beta_{1-\lambda,r}(\mathcal{R}F) = \mathbb{C}_r \beta_{\lambda,r}(F), \quad \alpha_{1-\lambda,r}(\mathcal{R}F) = \mathbb{C}_r \alpha_{\lambda,r}(F), \quad (\text{PGF7.5})$$

and

$$\mathbb{T}_r(1 - \lambda)\mathbb{C}_r = \mathbb{C}_r \mathbb{T}_r(\lambda). \quad (\text{PGF7.6})$$

Each equality follows by substituting $s = 1 - \lambda + h$, so that $1 - s = \lambda - h$, retaining every power of -1 .

There is also a required sign in the snake boundary:

$$\partial_{1-\lambda,r}(\mathcal{R}x) = (-1)^r C_r \partial_{\lambda,r}(x). \quad (\text{PGF7.7})$$

Indeed $(L - (1 - \lambda))^r \mathcal{R} = (-1)^r \mathcal{R}(L - \lambda)^r$, as multiplication of entire functions verifies. Substituting this identity into (PGF4.3) proves (PGF7.7). On the basis it reads

$$\mathcal{R}q_{\lambda,k} = (-1)^k q_{1-\lambda,k}, \quad (\text{PGF7.8})$$

and the two signs in (PGF7.7) give the same $(-1)^k$. These are coefficient-reflection statements. Any additional sphere-orientation or cohomological-degree sign belongs to its separately specified geometric map and has not been silently added or removed here.

PGF8. Exact information supplied to the lifting calculation

The parameter-dependent inclusion of the full primal coefficient ideals is now an explicit holomorphic matrix family, not merely a list of quotient characters. For every complex parameter and every jet order, its full entries are (PGF3.3), its actual source boundary is (PGF5.8), and its specialization kernel and cokernel are (PGF4.8)-(PGF4.9). The coherent family retains the longer parameter extension (PGF6.6), whose length is rm_ρ , together with both the Tor kernel and ordinary fibre of dimension $\min(r, m_\rho)$.

Off the actual nontrivial-zero divisor, the inclusion between the two jet quotients is an isomorphism with inverse multiplication by the complete truncated Taylor series of $1/F_0$. At an actual zero, it is the full nilpotent multiplication by $h^{m_\lambda} u_\lambda(h)$, with the unit and every derivative retained in the matrix. At $0, 1, -2, -4, \dots$, the nonzero exceptional values (PGF1.6)-(PGF1.7) remain in this map. This describes precisely the divisor detected by this source construction; the original zeta's endpoint pole and trivial zeros still require their full comparison factors, already displayed in (PGF1.5) and (PGF3.6).

The family does not place every actual zero on a vertical line and does not turn a quotient character into an eigenvector of $\mathcal{J}$. The calculation also does not claim the requested Deligne weight separation for the whole obstruction target $J(-1)$. It supplies exact primal maps which any such calculation must use, while keeping the already established zero boundary of the actual sheaf lift distinct from the generally nonzero boundary (PGF4.3).

PGF9. Independent internal checks of the formulas

These checks are symbolic identities for arbitrary r, λ, m_λ , not bounded numerical tests.

1. Both maps in (PGF2.7) are proved with explicit continuous sections and an exact kernel calculation; there is no appeal to an unspecified quotient topology.
2. The matrix (PGF3.3) is the full product $\text{jet } j^r F_0 j^r(F/F_0)$. Its determinant, local unit, and all exceptional-point contributions are retained; the transported product on $\mathcal{J}$ is separately recorded in (PGF2.8).
3. For every boundary basis vector, multiplication by $(s - \lambda)^r$ gives exactly $F_0(s)(s - \lambda)^{r-k}$. Division by F_0 therefore gives h^{r-k} , fixing the boundary sign and verifying its actual Schwartz representative.
4. At $m_\lambda = 0$, the matrix is invertible and both primal kernel and cokernel vanish. At $0 < m_\lambda < r$, they each have dimension m_λ . At $m_\lambda \geq r$, the matrix specialization is zero and both have dimension r . These exhaust all cases without assuming zero simplicity.

5. The sheaf map is injective because its determinant is a nonzero holomorphic germ, while its fibre kernel is exactly the Tor group in (PGF6.10). Thus no interchange of a sheaf kernel and its fibre has occurred.
6. The local parameter length rm_ρ , fibre dimension $\min(r, m_\rho)$, and nilpotence exponent $m_\rho + r - 1$ follow from the same explicit module (PGF6.6); they have not been identified with one another.
7. Reflection changes the parameter by $\lambda \mapsto 1 - \lambda$, the jet coordinate by $h \mapsto -h$, and the boundary by the additional factor $(-1)^r$. This is checked independently on every kernel basis vector in (PGF7.8).

No new purity statement, positivity statement, or conclusion about RH is asserted by these identities.

Polynomial weights in the actual localization row — CLP0-CLP13

Proof source: CC_LOCALIZATION_POLYNOMIAL_WEIGHTS.md.

Polynomial spectral weights in the actual sphere localization row

Complete independent derivation, 24 September 2026. Proof locators CLP0–CLP13.

CLP0. Constructed prerequisites and precise scope

The support remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. Every scalar, polynomial, derivative, source vector and complex coordinate below belongs to the existing arithmetic coefficient receiver after the complete-history reconstruction. None is an operation on τ . The two branches retain their separate recovered counters. The winding contribution and both labelled endpoint pairs remain present.

Before this calculation, `CURRENT_CC_WEIGHT_WORK.md`, `READ_FIRST_USER_CONSTRUCTION.md`, the connected `USER_ARGUMENT_RECONSTRUCTION.md` and its full correction table, PC08–PC09, and the verbatim WU050–WU055 and WU061–WU063 passages were recalled or reread. The relevant source calculations are CW2–CW8, CSP3–CSP7, CDW2–CDW3 and CDW7–CDW9, and CP4–CP10. CW2 supplies the simultaneous complexes of the actual sheaf row; CDW8 proves their compatibility with the retained winding summand. The closed source and its inverse are the already proved SSI/ESI source theorem, not a newly assumed closure. CP supplies the polynomial resolution, with both its kernel and cokernel degrees. We rederive the needed fixed-parameter division, matrices and connecting maps below.

The human sources are Alain Connes and Caterina Consani, [Schemes over \$\mathbb{F}_1\$ and zeta functions](#), §5, [arXiv:0903.2024v3](#), with the later signed-chart comparison in [arXiv:2609.00299v1](#), and Pierre Deligne, [La conjecture de Weil. II](#), §3.6, as reconstructed with exact hypotheses in DC. Original-author TeX coverage is the previously recorded CC source reading; the Deligne witness is the recorded French transcription. This note is a local derivation on the constructed receiver, not a new claim to have read those entire original sources.

The row being tested is the actual one

$$0 \longrightarrow \mathcal{F}_J \longrightarrow \mathcal{F} \longrightarrow j_! \underline{Q} \longrightarrow 0. \quad (\text{CLP0.1})$$

It is not the continuous-to-algebraic dual comparison cone of CP. Its full normal kernel is $J(-1)$. No quotient of that kernel is substituted for it. Every spectral parameter used for the two principal cases is an actual nontrivial zero ρ of the original ζ , or its stated shift $\rho + 1$; no RH location is assumed.

CLP1. Original source, complete transform and topology

Let

$$S = \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0\}, \quad V_{\pm} = S \oplus E_{\pm}, \quad E_{\pm} = \mathbb{C}^2,$$

$$A = \{b \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j b(u)| < \infty \text{ for every } N, j \geq 0\}. \quad (\text{CLP1.1})$$

The endpoint labels are value at zero and integral, in that order on each chart. Keep

$$\begin{aligned} \Sigma f(u) &= 2 \sum_{n \geq 1} f(nu), & Rb(u) &= u^{-1}b(u^{-1}), & \hat{f}(v) &= \int_{\mathbb{R}} f(x) e^{-2\pi i x v} dx, \\ r_+(f, c_0, c_1) &= \Sigma f, & r_-(h, d_0, d_1) &= R\Sigma h = \Sigma \hat{h}. \end{aligned} \quad (\text{CLP1.2})$$

The last identity retains the complete preceding Poisson formula

$$\Sigma \hat{h}(u) = u^{-1} \Sigma h(u^{-1}) + u^{-1} h(0) - \int_{\mathbb{R}} h. \quad (\text{CLP1.3})$$

The two terms vanish on S ; the independent endpoint summands $E_{\pm}$ are not deleted. The exact source theorem gives $J = \Sigma S$ closed in A , with a continuous inverse $H = \Sigma^{-1} : J \rightarrow S$. Thus $Q = A/J$ is the actual Fréchet quotient.

The half-Mellin comparison and its inverse are

$$\Theta b(s) = \frac{1}{2} \int_0^\infty b(u) u^s \frac{du}{u}, \quad \Theta^{-1} F(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} F(1/2 + it) u^{-it} dt. \quad (\text{CLP1.4})$$

They identify A with the entire strip-Schwartz space

$$\mathcal{B} = \{F \text{ entire} : b_{M,N}(F) = \sup_{|\Re s| \leq M} (1 + |\Im s|)^N |F(s)| < \infty \text{ for all } M, N \geq 0\}.$$

The rapid endpoint seminorms and integration by parts in $\log u$ prove absolute convergence and all differentiated convergence in (CLP1.4); Fourier inversion and contour shifts give the inverse. Let $\mathcal{Z}$ be the actual nontrivial-zero divisor of the original ζ , with orders m_ρ . Then the exact source theorem gives

$$\Theta J = J = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ } (\rho \in \mathcal{Z}, 0 \leq j < m_\rho)\}, \quad \Theta Q = \mathcal{Q} = \mathcal{B}/J. \quad (\text{CLP1.5})$$

All subsequent equations in A, J, Q may be transported by these specified topological isomorphisms.

For $\Re s > 1$, absolute summation gives

$$\begin{aligned} \Theta \Sigma f(s) &= \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}, & \mathcal{M}_0 \Sigma f &= 2\Theta \Sigma f, \\ \zeta(s) &= 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, & -\frac{\zeta'}{\zeta}(s) &= \sum_p \sum_{k \geq 1} (\log p) p^{-ks}. \end{aligned} \quad (\text{CLP1.6})$$

The unit and every prime-power repetition are retained. The fixed source

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}$$

belongs to S , and its complete transform is

$$F_0(s) = \Theta \Sigma f_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) = C_0(s) \zeta(s). \quad (\text{CLP1.7})$$

The two Gaussian Mellin integrals before the Gamma recurrence are

$$\frac{\pi}{2} \left[2\pi \cdot \frac{1}{2} \pi^{-(s+4)/2} \Gamma((s+4)/2) - 3 \cdot \frac{1}{2} \pi^{-(s+2)/2} \Gamma((s+2)/2) \right], \quad (\text{CLP1.8})$$

multiplied by the original $\zeta(s)$. They yield exactly (CLP1.7). This entire function belongs to $\mathcal{J}$, and its zero divisor is exactly $\mathcal{Z}$, with full multiplicities. Its exceptional values are

$$F_0(0) = F_0(1) = \frac{1}{8},$$

$$F_0(-2k) = \frac{k(2k+1)(-1)^k \pi^k}{2k!} \zeta'(-2k) = \frac{(1+2k)(2k)}{8} \pi^{-(1+2k)/2} \Gamma((1+2k)/2) \zeta(1+2k) \neq 0 \quad (k \geq 1). \quad (\text{CLP1.9})$$

The pole at one, Gamma poles and trivial-zero derivatives have thus been compared, not removed from the original function.

CLP2. The actual simultaneous complexes and cover generator

Put $Y = \mathbb{P}^1(\mathbb{C})$, $U = \mathbb{C}^\times$, $j : U \hookrightarrow Y$, $i : \{0, \infty\} \hookrightarrow Y$, and $W = V_+ \oplus V_-$. The actual sheaf is

$$\mathcal{F} = \underline{A}_Y \times_{i_* A^2} i_* W, \quad (\text{CLP2.1})$$

with map $(b, v_+, v_-) \mapsto (b|_0 - r_+ v_+, b|_\infty - r_- v_-)$. Replacing the interior coefficient by J gives $\mathcal{F}_J$; its quotient is $j_! Q$ because it has zero pole stalks and the interior quotient vanishes near each pole. This proves stalkwise exactness of (CLP0.1).

CW2's simultaneous contractions give the following degreewise exact row of global complexes:

$$\begin{array}{c|ccc} & D_J & D_F & D_G \\ \hline 0 & W & W & 0 \\ 1 & J & A & Q \\ 2 & J(-1) & A(-1) & Q(-1). \end{array} \quad d^0 = d = r_+ - r_-, \quad d^1 = 0. \quad (\text{CLP2.2})$$

The maps in degree zero are the identity and zero; the other two rows are inclusion and quotient. For specificity, before contraction each column is $B \oplus W \rightarrow B^2 \rightarrow B$, with $d^0(b, v) = (b - r_+ v_+, b - r_- v_-)$. The projection in degrees zero and one is $(b, v) \mapsto v$, $(b_+, b_-) \mapsto b_- - b_+$, and the section is $v \mapsto (r_- v_-, v)$, $b \mapsto (-b, 0)$. The homotopy on degree one is $(b_+, b_-) \mapsto (b_-, 0)$. Substitution proves $ps = 1$, $1 - sp = dh + hd$. These formulas commute with $J \rightarrow A \rightarrow Q$; hence (CLP2.2) represents the actual sheaf maps simultaneously.

For $a > 0$, the primal coefficient maps are

$$\begin{aligned} T_a b(u) &= b(u/a), \quad \rho_+(a)(f, c_0, c_1) = (f(\cdot/a), c_0, ac_1), \\ \rho_-(a)(h, d_0, d_1) &= (ah(a \cdot), ad_0, d_1). \end{aligned} \quad (\text{CLP2.3})$$

They form multiplicative groups and satisfy $r_\pm \rho_\pm(a) = T_a r_\pm$, by summation and $\widehat{ah(a \cdot)} = \widehat{h(\cdot/a)}$. For the actual integer sphere cover $z \mapsto z^n$, $n \geq 1$, its action on (CLP2.2) is

$$(\rho_+(n) \oplus \rho_-(n), T_n, nT_n). \quad (\text{CLP2.4})$$

The last factor is the positive complex-oriented degree, equivalently the winding-period factor. CDW8 retains precisely this normal summand. The same formulas with $a > 0$ define an action on the coefficient complexes: the group law follows from (CLP2.3), the first differential intertwines the first two components, and the differential into the third is zero. No nonintegral sphere cover is asserted.

Differentiating the coefficient action at $a = e^h$, $h = 0$, gives the continuous chain operator

$$t^0 = G = G_+ \oplus G_-, \quad t^1 = L = -u\partial_u, \quad t^2 = L + 1,$$

$$G_+(f, c_0, c_1) = (-v\partial_v f, 0, c_1), \quad G_-(h, d_0, d_1) = (h + v\partial_v h, d_0, 0). \quad (\text{CLP2.5})$$

The derivative exists in every source seminorm by Taylor's integral remainder for dilations; two extra derivatives bound its second-order remainder uniformly on compact h -intervals. Under Θ , L is multiplication by s . The symbol (-1) in (CLP2.2) means the retained action aT_a , so its generator is $s + 1$. These are the actual geometric generators on the finite coefficient models, not a degree-independent replacement.

CLP3. Symmetric source section and exact decomposition

The two continuous source sections are

$$s_+(b) = (Hb, 0, 0), \quad s_-(b) = (\widehat{Hb}, 0, 0), \quad r_\pm s_\pm = 1_J.$$

Define

$$\sigma : J \rightarrow W, \quad \sigma b = \frac{1}{2}(s_+ b, -s_- b), \quad d\sigma = 1_J, \quad p_Z = 1_W - \sigma d, \quad (\text{CLP3.1})$$

and $Z = \ker d$. The full inverse source theorem proves continuity. Intertwining of $r_\pm$ and injectivity on the source summands give $\rho(a)\sigma = \sigma T_a$, and hence $G\sigma = \sigma L$. Thus

$$W \xrightarrow{\sim} Z \oplus J, \quad w \mapsto (p_Z w, dw), \quad (z, b) \mapsto z + \sigma b \quad (\text{CLP3.2})$$

is a continuous equivariant isomorphism. The endpoint summands lie entirely in Z , and

$$Z \simeq J \oplus E_+ \oplus E_-, \quad b, e_+, e_- \mapsto (s_+ b + e_+, s_- b + e_-). \quad (\text{CLP3.3})$$

The endpoint eigenvalues of G are $(0, 1, 1, 0)$, in the original order.

Consequently the row (CLP2.2) has an actual decomposition into three rows of complexes: the identity row on $Z[0]$; the row

$$0 \rightarrow [J \xrightarrow{1} J] \rightarrow [J \xrightarrow{\iota} A] \rightarrow Q[-1] \rightarrow 0 \quad (\text{CLP3.4})$$

in degrees zero and one; and the actual normal row

$$0 \rightarrow J(-1)[-2] \rightarrow A(-1)[-2] \rightarrow Q(-1)[-2] \rightarrow 0. \quad (\text{CLP3.5})$$

These follow by substituting (CLP3.2), not by splitting $A \rightarrow Q$. The differential in the left complex of (CLP3.4) is the identity, so that complex is contracted by the identity from degree one to degree zero. The quotient map $[J \rightarrow A] \rightarrow Q[-1]$ is a quasi-isomorphism, but no equivariant section $Q \rightarrow A$ has been selected.

CLP4. The full two-term spectral functor

Work over $R = \mathbb{C}[t]$, with the degree-wise chain operator (CLP2.5), and put

$$P(t) = (t - \lambda)^r, \quad r \geq 1.$$

The resolution $0 \rightarrow R \xrightarrow{P} R \rightarrow R/(P) \rightarrow 0$, with free terms in degrees minus one and zero, gives

$$K_P(C)^k = C^k \oplus C^{k-1}, \quad \delta(x, y) = (dx, Px - dy), \quad K_P(C) \simeq \text{RHom}_R(R/(P), C). \quad (\text{CLP4.1})$$

To check the sign, write the usual Hom differential as $d_C f - (-1)^k f d_R$ and change its second coordinate b to $y = (-1)^k b$. Direct calculation gives $\delta^2 = 0$ because P commutes with d .

For any complex the complete cohomology formula is

$$0 \rightarrow H^{k-1}(C)/PH^{k-1}(C) \xrightarrow{[y] \mapsto [(0,y)]} H^k K_P(C) \xrightarrow{[(x,y)] \mapsto [x]} H^k(C)[P] \rightarrow 0. \quad (\text{CLP4.2})$$

Indeed a cycle satisfies $dx = 0$, $Px = dy$; a class killed by P admits such a y . If $x = du$, subtraction of $\delta(u, 0)$ leaves $(0, y - Pu)$; its ambiguities are exactly ordinary boundaries and P times closed elements. This proves exactness including the cokernel term.

For $C = D_J, D_F$, the terms of $K_P(C)$ in degrees zero through three are

$$\begin{aligned} W, \quad B \oplus W, \quad B(-1) \oplus B, \quad B(-1), \quad B = J \text{ or } A, \\ w \mapsto (dw, P(G)w), \quad (b, w) \mapsto (0, P(L)b - dw), \quad (c, b) \mapsto P(L+1)c. \end{aligned} \quad (\text{CLP4.3})$$

For D_G , they are $0, Q, Q(-1) \oplus Q, Q(-1)$, with maps

$$q \mapsto (0, P(L)q), \quad (c, q) \mapsto P(L+1)c. \quad (\text{CLP4.4})$$

These complexes form a degreewise exact row. Its connecting map is calculated by lifting a cycle to the middle complex and applying the displayed differential, with a positive inclusion into the first complex. All degrees and signs below use this convention.

CLP5. Fixed-parameter division, jets, endpoints and source matrices

For $\nu \in \mathbb{C}$, write $z = s - \nu$, $m_\nu = \text{ord}_\nu F_0$, and $d_\nu = \min(r, m_\nu)$; set $m_\nu = 0$ off the actual zero divisor. Multiplication by z^r is injective on $\mathcal{B}$ and $\mathcal{J}$, since they consist of entire functions. The exact continuous jet identifications are

$$\begin{aligned} \mathcal{B}/z^r \mathcal{B} &\xrightarrow{\sim} \mathbb{C}[z]/(z^r), \quad F \mapsto \sum_{j < r} \frac{F^{(j)}(\nu)}{j!} z^j, \\ \mathcal{J}/z^r \mathcal{J} &\xrightarrow{\sim} \mathbb{C}[z]/(z^r), \quad F \mapsto \sum_{j < r} \frac{1}{j!} \left(\frac{F}{F_0} \right)^{(j)} (\nu) z^j. \end{aligned} \quad (\text{CLP5.1})$$

The quotient F/F_0 is entire for $F \in \mathcal{J}$; its value at a zero is interpreted by full analytic division. The kernels are exactly the displayed polynomial images. Division by $s - \nu$ on the closed value-zero subspace is continuous: outside a fixed disk the denominator is bounded away from zero; inside it, the integral formula $H(s)/(s - \nu) = \int_0^1 H'(\nu + t(s - \nu))dt$ and a Cauchy estimate on a larger fixed disk bound every strip seminorm. Iteration proves the order- r statement. Evaluation of F/F_0 's jets is also continuous by division by the nonzero local unit after its fixed order of vanishing. Finite Taylor interpolation with $e^{(s-\nu)^2}$ times a polynomial proves the first surjectivity; F_0 times a polynomial proves the second. All image subspaces in (CLP5.1) are therefore closed.

In these exact coordinates the inclusion $\mathcal{J} \rightarrow \mathcal{B}$ induces

$$\tau_r(\nu)_{ij} = \begin{cases} F_0^{(i-j)}(\nu)/(i-j)!, & i \geq j, \\ 0, & i < j, \end{cases} \quad 0 \leq i, j < r. \quad (\text{CLP5.2})$$

It is multiplication by the full Taylor series of F_0 modulo z^r . Write $F_0(\nu + z) = z^{m_\nu} b_\nu(z)$, with $b_\nu(0) \neq 0$. The local unit is invertible modulo z^r , so the matrix has kernel and cokernel dimensions d_ν , rank $r - d_\nu$, and determinant $F_0(\nu)^r$. No derivative of its unit has been dropped.

For the full quotient, not an assumed infinite direct sum of blocks,

$$\dim \ker(L - \nu)^r|_Q = \dim Q/(L - \nu)^r Q = d_\nu. \quad (\text{CLP5.3})$$

Here is a proof retaining its topology. If $m_\nu = 0$,

$$[F] \mapsto \left[\frac{F(s) - F_0(s)F(\nu)/F_0(\nu)}{s - \nu} \right] \quad (\text{CLP5.4})$$

is a continuous inverse of $L - \nu$ on $\mathcal{Q}$, by the preceding division estimate; multiplication in either order gives the identity modulo $\mathcal{J}$. If $m = m_\nu > 0$, put

$$B_\nu = F_0/(s - \nu)^m, \quad \Pi_\nu F = B_\nu \sum_{j < m} \frac{(F/B_\nu)^{(j)}(\nu)}{j!} (s - \nu)^j. \quad (\text{CLP5.5})$$

The ratio is used only near ν , where $B_\nu \neq 0$. This is a continuous finite operator. Its output has the input's full jet at ν and vanishes to the full original order at every other zero. It induces a projection $\bar{\Pi}_\nu$ on $\mathcal{Q}$, commuting with L , whose image is the genuine m -dimensional zero block. On the closed complement $\mathcal{Q}_0$, an inverse of $L - \nu$ is

$$[F] \mapsto (1 - \bar{\Pi}_\nu)[F/(s - \nu)]. \quad (\text{CLP5.6})$$

Every representative has its full m -jet zero; changing it by $\mathcal{J}$ changes the quotient only in the projected zero block. Division and finite interpolation give continuity with the quotient topology. On the finite image, $L - \nu$ is multiplication by z modulo z^m . Its r -th power has kernel and cokernel dimensions d_ν , proving (CLP5.3). In particular all polynomial images used here are closed.

The finite kernel has the especially useful actual basis

$$q_{\nu,j} = \left[\frac{F_0(s)}{(s - \nu)^j} \right], \quad 1 \leq j \leq d_\nu, \quad (\text{CLP5.7})$$

and the quotient $Q/(L - \nu)^r Q$ is the ordinary Taylor jet modulo z^{d_ν} . Independence of (CLP5.7) follows from its distinct orders $m - j$; these vectors span by (CLP5.3).

Finally put E_λ equal to the sum of the endpoint lines with eigenvalue λ . It is zero except at $\lambda = 0, 1$, where it has dimension two. Then

$$Z[P] = E_\lambda, \quad Z/PZ \simeq J/(L - \lambda)^r J \oplus E_\lambda, \quad \dim Z/PZ = r + \dim E_\lambda. \quad (\text{CLP5.8})$$

A repeated root of P does not increase the dimension of a scalar endpoint line. Both endpoint lines of eigenvalue zero and both of eigenvalue one remain in this formula.

CLP6. Every spectral cohomology group and every connecting map

Write $Q_0[P] = \ker P(L)|_Q$, $Q_0/P = Q/P(L)Q$, $Q_1[P] = \ker P(L + 1)|_{Q(-1)}$, and $Q_1/P = Q(-1)/P(L + 1)Q(-1)$. The subscripts in this paragraph record the stated scalar shift, not a new quotient of τ . Direct substitution in (CLP4.3)-(CLP4.4) gives the complete table:

k	$H^k K_P(D_J)$	$H^k K_P(D_F)$	$H^k K_P(D_G)$	
0	E_λ	E_λ	0	
1	Z/PZ	$Z/PZ \oplus Q_0[P]$	$Q_0[P]$	
2	0	Q_0/P	$Q_0/P \oplus Q_1[P]$	
3	$J(-1)/P(L + 1)J$	$A(-1)/P(L + 1)A$	Q_1/P	

(CLP6.1)

Every other degree is zero. These are specified maps and splittings, proved next, rather than a dimension-only assertion.

In degree zero, a cycle is exactly $w \in Z$ killed by $P(G)$, so (CLP5.8) applies. In degree one a middle cycle (b, w) satisfies $P(L)b = dw$. The map and inverse giving the displayed direct sum are

$$\begin{aligned} [(b, w)] &\mapsto ([w - \sigma P(L)b], [b]), \\ ([z], q) &\mapsto [(b, z + \sigma P(L)b)], \quad [b] = q. \end{aligned} \quad (\text{CLP6.2})$$

The first component belongs to Z ; the second is killed by P . Changing z by Pz_0 changes the cycle by $\delta(z_0)$. Changing the lift b by $j \in J$ changes it by $\delta(\sigma j) = (j, \sigma P(L)j)$. Thus (CLP6.2) is well-defined, and the two maps are inverse. The left complex has no Q term: subtraction of $\delta(\sigma b)$ reduces its cycles to $(0, z)$. This proves the first two columns in degree one.

In degree two of either entire-function column, a cycle (c, b) has $P(L+1)c = 0$, so $c = 0$. Its boundaries have second component $P(L)b_0 - dw$. Hence its cohomology is zero for $B = J$, and $A/(P(L)A + J) = Q_0/P$ for $B = A$. In the quotient column, a cycle is (c, q) with $c \in Q_1[P]$ and arbitrary q ; the only boundaries are $(0, P(L)q_0)$. We use the order $([q], c)$ in (CLP6.1). Degree three is directly the corresponding cokernel of $P(L+1)$. This proves all entries without omitting either derived degree.

In degree zero the inclusion is the identity. In degree one the two maps are

$$Z/PZ \longrightarrow Z/PZ \oplus Q_0[P] \longrightarrow Q_0[P], \quad z \mapsto (z, 0), \quad (z, q) \mapsto q. \quad (\text{CLP6.3})$$

In degree two the map out of D_F is

$$Q_0/P \longrightarrow Q_0/P \oplus Q_1[P], \quad y \mapsto (y, 0). \quad (\text{CLP6.4})$$

The connecting maps in degrees zero, one and three vanish. Degree zero has zero source; degree one has zero target by the calculation above; degree three has zero target beyond the complex. A direct degree-one correction is useful: an initial lift $(b, 0)$ of $q \in Q_0[P]$ has boundary $(0, P(L)b)$ in degree two of $K_P(D_J)$, and

$$(0, P(L)b) = \delta(0, -\sigma P(L)b). \quad (\text{CLP6.5})$$

Subtracting that primitive produces the plus sign in the corrected cycle $(b, \sigma P(L)b)$. Its class is independent of the chosen representative but is not a selected entire representative of q .

The sole potentially nonzero connecting map is

$$\boxed{\beta_P^2 : Q_0/P \oplus Q_1[P] \longrightarrow J(-1)/P(L+1)J, \quad ([y], q) \mapsto [P(L+1)b], \quad [b] = q.} \quad (\text{CLP6.6})$$

To prove its sign and formula, lift the quotient cycle (q, y) to $(b, a) \in A(-1) \oplus A$. Its differential is $(0, P(L+1)b)$, independent of a , and this is the included left cycle. Changing b by $j \in J$ changes the answer by $P(L+1)j$. Thus it is well-defined. It vanishes on the full Q_0/P summand, and is injective on $Q_1[P]$: if $P(L+1)b = P(L+1)j$, injectivity on entire functions gives $b = j$, hence $q = 0$.

The last exact row is therefore

$$0 \rightarrow Q_1[P] \xrightarrow{\beta_P^2} J(-1)/P(L+1)J \xrightarrow{\bar{\iota}} A(-1)/P(L+1)A \xrightarrow{\bar{q}} Q_1/P \rightarrow 0. \quad (\text{CLP6.7})$$

Indeed the kernel of $\bar{\iota}$ consists exactly of classes $[P(L+1)b]$ with $[b] \in Q_1[P]$, and the cokernel is the quotient Q_1/P . This is the genuine normal row, with its entire original kernel retained.

For $\mu = \lambda - 1$, identify the two middle terms in (CLP6.7) by (CLP5.1). Then $\bar{\iota}$ is exactly $T_r(\mu)$, $\bar{q}$ is reduction of the ordinary jet to degree below d_μ , and

$$\beta_P^2(q_{\mu,j}) = [F_0(s)(s - \mu)^{r-j}], \quad 1 \leq j \leq d_\mu. \quad (\text{CLP6.8})$$

In the F/F_0 -jet coordinate its value is z^{r-j} . These monomials span the full kernel of $\mathbb{T}_r(\mu)$, proving exactness and the rank d_μ explicitly.

All finite identifications and maps are continuous. For the only apparently chosen lift in (CLP6.2), restrict b to the finite actual block given by (CLP5.5), whose representatives are explicit continuous functions of its finite coordinates. Independence of arbitrary representatives was already proved. The cokernel identifications use closed finite-jet kernels and continuous division. No topological direct-sum assumption about all zeros is needed.

CLP7. The original-zero parameter: the precise separated normal lift

Take an actual nontrivial zero ρ , multiplicity m , put $\lambda = \rho$, and let $d = \min(r, m)$. The known open strip $0 < \Re \rho < 1$ gives $m_{\rho-1} = 0$, and $\rho \notin \{0, 1\}$. The complete dimensions in degrees zero through three are

$$\begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline K_P(D_J) & 0 & r & 0 & r \\ K_P(D_F) & 0 & r+d & d & r \\ K_P(D_G) & 0 & d & d & 0. \end{array} \quad (\text{CLP7.1})$$

Every connecting map vanishes. The degree-one map is the split surjection (CLP6.3), with its full r -dimensional kernel $J/(L - \rho)^r J$. Degree two is the identity on $Q/(L - \rho)^r Q$. Degree three is the isomorphism

$$J(-1)/(L + 1 - \rho)^r J \xrightarrow{\sim} A(-1)/(L + 1 - \rho)^r A, \quad (\text{CLP7.2})$$

whose exact matrix is $\mathbb{T}_r(\rho - 1)$, including all original factors.

Here is its inverse, with no discarded multiplier. Put $\mu = \rho - 1$ and

$$H_0(s) = \frac{1}{F_0(s)} = \frac{8\pi^{s/2}}{s(s-1)\Gamma(s/2)\zeta(s)} \quad \text{near } \mu. \quad (\text{CLP7.3})$$

It is holomorphic there because $F_0(\mu) \neq 0$. The inverse matrix is lower triangular, with entry

$$(\mathbb{T}_r(\mu)^{-1})_{ij} = \frac{H_0^{(i-j)}(\mu)}{(i-j)!} \quad (i \geq j). \quad (\text{CLP7.4})$$

Multiplying the two truncated series gives the jet of $F_0 H_0 = 1$, proving both inverse identities. Explicitly a normal cokernel class $[F]$ lifts to

$$\left[F_0(s) \sum_{j < r} \frac{(F/F_0)^{(j)}(\mu)}{j!} (s - \mu)^j \right] \in J/(s - \mu)^r J. \quad (\text{CLP7.5})$$

Only the germ of F/F_0 at μ is used; the output is F_0 times a polynomial, hence lies in the actual source. This class is the unique inverse image under (CLP7.2), not a canonical global representative in $A \rightarrow Q$.

The full derivative in (CLP7.4) is, for $k \geq 0$,

$$\frac{H_0^{(k)}(\mu)}{k!} = 8 \sum_{k_1 + \dots + k_5 = k} \frac{(s^{-1})^{(k_1)}(\mu)}{k_1!} \frac{((s-1)^{-1})^{(k_2)}(\mu)}{k_2!} \frac{(\pi^{s/2})^{(k_3)}(\mu)}{k_3!} \frac{(\Gamma(s/2)^{-1})^{(k_4)}(\mu)}{k_4!} \frac{(\zeta(s)^{-1})^{(k_5)}(\mu)}{k_5!}. \quad (\text{CLP7.6})$$

Thus both endpoint factors, every Gamma and original-zeta reciprocal derivative, the power of π , and the scalar eight remain. The reciprocal Gamma derivative is of the displayed composite $s \mapsto \Gamma(s/2)^{-1}$, so its chain-rule factors are included rather than changed to derivatives in a different coordinate.

At this parameter the actual normal quotient has neither a polynomial kernel nor a polynomial cokernel. Equivalently

$$K_{(t-\rho)^r}(Q(-1)) \simeq 0, \quad K_{(t-\rho)^r}(J(-1)) \xrightarrow{\sim} K_{(t-\rho)^r}(A(-1)). \quad (\text{CLP7.7})$$

This is a genuine spectral separation on the actual normal row, proved globally by (CLP5.4), and its nonzero cokernel inverse is (CLP7.5). It retains all of $J(-1)$; it does not assert that this full source has only high-weight quotients.

For an explicit degree-one lift, take $q_{\rho,j}$ from (CLP5.7), let

$$a_j = \Theta^{-1}(F_0/(s-\rho)^j), \quad g_j = (-v\partial_v - \rho)^{r-j}f_0.$$

Then $P(L)a_j = \Sigma g_j$, so its spectral lift is

$$\left[\left(a_j, \frac{1}{2}((g_j, 0, 0), (-\hat{g}_j, 0, 0)) \right) \right] \in H^1 K_P(D_F), \quad 1 \leq j \leq d. \quad (\text{CLP7.8})$$

The inverse Mellin integral is (CLP1.4), including its $u^{-1/2}/\pi$ factor; its integrand is entire even if ρ lies on the integration line, because the zero has order at least j . The source derivative preserves S , by evenness, the value condition and integration by parts for the integral. Thus every term of this representative belongs to the original source.

CLP8. The shifted actual-zero parameter: a nonzero higher boundary

Now take $\lambda = \rho + 1$, with the same actual ρ, m, r, d . The original quotient has no character at λ , while the genuine normal quotient has its original-zero block there. Thus

$$\begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline K_P(D_J) & 0 & r & 0 & r \\ K_P(D_F) & 0 & r & 0 & r \\ K_P(D_G) & 0 & 0 & d & d. \end{array} \quad (\text{CLP8.1})$$

The degree-one map is an isomorphism. The entire nontrivial tail is

$$0 \rightarrow Q(-1)[(t-\rho-1)^r] \xrightarrow{\beta_P^2} J(-1)/(L-\rho)^r J \xrightarrow{\tau_r(\rho)} A(-1)/(L-\rho)^r A \rightarrow Q(-1)/(L-\rho)^r Q \rightarrow 0. \quad (\text{CLP8.2})$$

The injective boundary has rank d , and its exact basis values are

$$\boxed{\beta_P^2(q_{\rho,j}) = [F_0(s)(s-\rho)^{r-j}], \quad 1 \leq j \leq d.} \quad (\text{CLP8.3})$$

No value on the critical line has been distinguished in this calculation. The image is nonzero at every actual zero, with every multiplicity retained. If $r \leq m$, its rank is r and the middle matrix is zero modulo $(s-\rho)^r$. If $r > m$, its rank is m and the middle matrix has rank $r-m$. Both cases follow from the full unit calculation (CLP5.2).

The representative producing (CLP8.3) is especially direct: in degree two of $K_P(D_G)$ use $(q_{\rho,j}, 0)$; lift it to $(a_j, 0)$ with a_j from (CLP7.8). Its differential in degree three is

$$P(L+1)a_j = (L-\rho)^r a_j = \Sigma(-v\partial_v - \rho)^{r-j}f_0. \quad (\text{CLP8.4})$$

This is an actual element of $J(-1)$, followed by its indicated cokernel class. Formula (CLP1.7) supplies its full Mellin multiplier, including $1/8$, rather than a bare divisor label.

The original geometric $\delta^2 : H^2(D_G) \rightarrow H^3(D_J)$ still has zero target. The spectral boundary (CLP8.3) instead lands in the additional derived degree $H^3 K_P(D_J) = \text{coker } P|_{J(-1)}$. Taking the two-term spectral functor has retained the Ext degree of the actual normal extension. This specifies exactly why the two boundaries differ; no failure of the earlier geometric degree-one lift is inferred.

CLP9. Full finite action matrices and all weights actually obtained

For the ordinary quotient kernel basis (CLP5.7), the full action is

$$T_a q_{\nu,j} = a^\nu \sum_{k=0}^{j-1} \frac{(\log a)^k}{k!} q_{\nu,j-k}, \quad a > 0. \quad (\text{CLP9.1})$$

Indeed multiply its representative by $a^s = a^\nu e^{(s-\nu)\log a}$. The terms of order at least j have the full factor F_0 and lie in $\mathcal{J}$; all terms of lower order are displayed. On the normal block multiply the entire matrix by the retained factor a .

In the normal boundary case (CLP8.3), put $b_j = [F_0(s)(s - \rho)^{r-j}]$. Then both source and image have exactly

$$q_{\rho,j} \mapsto a^{\rho+1} \sum_{k=0}^{j-1} \frac{(\log a)^k}{k!} q_{\rho,j-k}, \quad b_j \mapsto a^{\rho+1} \sum_{k=0}^{j-1} \frac{(\log a)^k}{k!} b_{j-k}. \quad (\text{CLP9.2})$$

Thus the positive boundary commutes with every term, including all nilpotents. At the actual integer cover these scalars are $n^{\rho+1}$, not n^ρ . Positive-real formulas describe the specified coefficient extension only.

For the full r -dimensional ordinary source cokernel at ν , use the coordinate $p(z) = (F/F_0)(\nu + z) \bmod z^r$; its action is multiplication by

$$a^\nu \sum_{k=0}^{r-1} \frac{(\log a)^k}{k!} z^k. \quad (\text{CLP9.3})$$

The ordinary target cokernel uses the jet of F and has the same multiplication. The matrix $\mathbb{T}_r(\nu)$ commutes with it because both are multiplication by the displayed truncated series. On a normal cokernel centered at $\mu = \lambda - 1$, (CLP9.3) has scalar $a^{\mu+1} = a^\lambda$. In particular the nonzero normal cokernels in (CLP7.2) have the same scalar a^ρ as the original-zero classes; their isomorphism does not create a new common modulus.

The degree-one lift (CLP6.2) is equivariant, since σ , P , and the quotient maps intertwine the action. Its kernel $J/(L - \lambda)^r J$ has the scalar a^λ in (CLP9.3), and its $Q_0[P]$ summand has the same scalar when nonzero. This is an explicit split lift with overlapping source and kernel characters, not a lift whose entire kernel has a separated weight interval.

For $a > 1$, define the logarithmic-modulus weight of a nonzero scalar by

$$w_a(\alpha) = \frac{2 \log |\alpha|}{\log a}. \quad (\text{CLP9.4})$$

For actual zeros ρ , the ordinary blocks have $0 < 2\Re \rho < 2$, while the normal zero blocks have $2 < 2\Re \rho + 2 < 4$. This separation is exactly the one used in (CLP7.7), with the stronger operator inverse (CLP5.4). In (CLP8.3), the source and image both have weight $2\Re \rho + 2$; hence their nonzero

map is fully compatible with their equal weights. The entire $J(-1)$ has quotient jets at every center μ , with scalar $a^{\mu+1}$, as (CLP5.1) and (CLP9.3) prove. It has no bound restricting all such quotients to the normal-zero interval. None of these assertions installs an arithmetic purity premise or proves $\Re \rho = 1/2$.

For completeness, every entry of $\mathbb{T}_r(\nu)$ at a regular point retains the full product derivative

$$\frac{F_0^{(k)}(\nu)}{k!} = \frac{1}{8} \sum_{k_1+\dots+k_5=k} \frac{s^{(k_1)}(\nu)}{k_1!} \frac{(s-1)^{(k_2)}(\nu)}{k_2!} \frac{(\pi^{-s/2})^{(k_3)}(\nu)}{k_3!} \frac{(\Gamma(s/2))^{(k_4)}(\nu)}{k_4!} \frac{\zeta^{(k_5)}(\nu)}{k_5!}. \quad (\text{CLP9.5})$$

At an actual zero, writing $\zeta(\rho+z) = z^m u_\rho(z)$ gives $b_\rho(z) = C_0(\rho+z)u_\rho(z)$; all derivatives of this unit remain in (CLP5.2). The completion factor has never replaced the original ζ .

CLP10. Exceptional locations, endpoints and exact mirror transport

The general table (CLP6.1) also covers every $\lambda \in \mathbb{C}$. Its dimensions are

$$\begin{array}{c|ccc} k & D_J & D_F & D_G \\ \hline 0 & e_\lambda & e_\lambda & 0 \\ 1 & r + e_\lambda & r + e_\lambda + d_\lambda & d_\lambda \\ 2 & 0 & d_\lambda & d_\lambda + d_{\lambda-1} \\ 3 & r & r & d_{\lambda-1}, \end{array} \quad e_\lambda = \begin{cases} 2, & \lambda = 0 \text{ or } 1, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{CLP10.1})$$

At $\lambda = 0, 1$, both d_λ and $d_{\lambda-1}$ vanish, since $F_0(0) = F_0(1) = 1/8$ and -1 is not in the nontrivial divisor. The normal diagonals are exactly $F_0(-1) = \pi/24$ at $\lambda = 0$, and $F_0(0) = 1/8$ at $\lambda = 1$. The first value follows directly from $\Gamma(-1/2) = -2\sqrt{\pi}$, $\zeta(-1) = -1/12$, and (CLP1.7), or from $F_0(-1) = F_0(2)$. The two endpoint lines contribute two dimensions to both $Z[P]$ and Z/PZ , with actions respectively 1 or a ; they are never assigned multiplicity r . At every cancellation point $-2k$, or its normal translate $1 - 2k$, the relevant full source value is (CLP1.9), not zero. This proves that no endpoint or cancelled trivial zero produces an extra Q -block.

Higher exceptional derivatives are also retained by analytic product coefficients. If $\nu = -2k$, write the actual Laurent series $C_0(\nu+z) = c_{-1}/z + \sum_{j \geq 0} c_j z^j$, and $\zeta(\nu+z) = \sum_{j \geq 1} \zeta^{(j)}(\nu) z^j / j!$. Then

$$\frac{F_0^{(l)}(\nu)}{l!} = c_{-1} \frac{\zeta^{(l+1)}(\nu)}{(l+1)!} + \sum_{j=0}^{l-1} c_j \frac{\zeta^{(l-j)}(\nu)}{(l-j)!}. \quad (\text{CLP10.2})$$

At one, use $C_0(1+z) = \sum_{j \geq 1} a_j z^j$ and $\zeta(1+z) = z^{-1} + \sum_{j \geq 0} \gamma_j z^j$, giving

$$\frac{F_0^{(l)}(1)}{l!} = a_{l+1} + \sum_{j=1}^l a_j \gamma_{l-j}. \quad (\text{CLP10.3})$$

At zero the factor $s\Gamma(s/2)$ has its removable analytic value two, while all other factors are holomorphic; its Taylor series in the unchanged coordinate supplies (CLP9.5) after that explicitly stated analytic extension. These formulas keep every pole-cancellation derivative in the source matrices and their inverse matrices wherever invertible.

The two actual sphere swaps act in global degrees zero, one and two by

$$M^0(v_+, v_-) = (v_-, v_+), \quad M^1 = -R, \quad M^2 = \epsilon R, \quad \epsilon = +1 \text{ holomorphic, } -1 \text{ antiholomorphic.} \quad (\text{CLP10.4})$$

The exchanged endpoint labels remain attached to their chart coordinates. Since $HR = \widehat{H}$ on J , direct substitution gives

$$M^0 \sigma b = \sigma(-Rb), \quad p_Z M^0 = M^0 p_Z. \quad (\text{CLP10.5})$$

Thus the exact decomposition (CLP3.4)-(CLP3.5) respects both mirrors, with common-restriction action R on the J coordinate of Z , and the separate signs just displayed. The operator identity $LR = R(1-L)$ shows that the linked degrees zero and one reflect t to $1-t$, whereas the normal degree two generator $L+1$ reflects t to $3-t$.

Consequently a single global polynomial $(t-\lambda)^r$ does not transform into one common $(t-(1-\lambda))^r$ in all three degrees. The exact target uses

$$P_{\text{low}}(t) = (t-(1-\lambda))^r \text{ on degrees } 0, 1, \quad P_{\text{normal}}(t) = (t-(3-\lambda))^r \text{ on degree } 2. \quad (\text{CLP10.6})$$

These still define a chain polynomial operator, since the only linked differential is between degrees zero and one. Equivalently use the proved direct decomposition and the two specified polynomials on its summands. The exact spectral chain map is

$$(x, y) \mapsto (M^k x, (-1)^r M^{k-1} y) \quad \text{in degree } k. \quad (\text{CLP10.7})$$

Indeed $MP = (-1)^r P_{\text{target}} M$ on each relevant coefficient, and substitution into (CLP4.1) gives the chain identity. The factor $(-1)^r$ is the monic-polynomial factor on the second spectral coordinate; it is separate from the geometric signs $-R, \epsilon R$. Applied to the degree-two boundary it gives exactly ϵR on its first-coordinate source and $(-1)^r \epsilon R$ on its degree-three second-coordinate image, consistent with (CLP6.6). The ordinary parameter ρ is sent to $1-\rho$ in the low sector; the normal parameter $\rho+1$ is sent to $2-\rho = (1-\rho)+1$. All multiplicities and orientation signs are preserved.

CLP11. The lifting result and its exact weight scope

The ordinary geometric map $H^1(D_F) \rightarrow H^1(D_G)$ is the identity of Q ; its connecting map into $J(-1)$ is zero. The present calculation constructs its entire derived polynomial version. For $P = (t-\rho)^r$, the degree-one spectral map is a split surjection with explicitly computed kernel $J/(L-\rho)^r J$, and the lift is (CLP6.2) or the actual source formula (CLP7.8). Its cohomology class is independent of representatives after the stated symmetric source section is fixed. Its full set of lifts differs by the displayed kernel. No unique cochain representative is asserted.

The separated original and normal zero ranges prove the actual normal comparison (CLP7.7), with the complete inverse (CLP7.3)-(CLP7.6). This is a source-level spectral lift through the real row $J(-1) \rightarrow A(-1) \rightarrow Q(-1)$, with its kernel preserved. It is not a declaration that all quotient characters of $J(-1)$ are separated: the explicit finite cokernels (CLP5.1) prove otherwise and calculate their receiving maps.

The nonzero higher boundary at $\lambda = \rho+1$ is exactly (CLP8.3), with its original-source representative (CLP8.4) and complete cover matrix (CLP9.2). It obstructs lifting that particular normal polynomial-kernel class in $H^2 K_P(D_G)$ through the spectral map $H^2 K_P(D_F)$. It does not obstruct the ordinary degree-one cohomology lift or the existence of individual representatives in the ordinary quotient. Its nonzero value occurs at every actual zero, including any on the critical line, so it is not an RH-violation criterion.

Deligne's DC5-DC7 mechanism kills a specified geometric connecting image by upper and lower bounds on the actual source and obstruction receiver, retaining his inertia coinvariant and normal duality terms. Here every corresponding arrow has been calculated on (CLP0.1). The low spectral lift, the separated normal comparison, and the nonzero shifted higher boundary are distinct proved maps in that one row. Their precise relation is the full exact sequence (CLP6.7), not an assumed identification with Deligne's inertia cross. The calculation preserves the original arithmetic and all original zeros. It establishes no new bound on $\Re \rho$, and it does not mark the broader requested common-weight conclusion as proved.

CLP12. The global derived connecting class before any finite spectral test

The decomposition in CLP3 gives a simultaneous chain-level comparison of the actual row, not only its fixed-parameter cohomology. Define complexes

$$\begin{aligned}\widetilde{D}_J &= Z[0] \oplus J(-1)[-2], & \widetilde{D}_F &= Z[0] \oplus Q[-1] \oplus A(-1)[-2], \\ \widetilde{D}_G &= Q[-1] \oplus Q(-1)[-2],\end{aligned}\tag{CLP12.1}$$

all with zero differentials. Their short exact row has the identity on Z in degree zero, the identity on Q in degree one, and the actual normal inclusion and quotient in degree two. The maps from (CLP2.2) are

$$\begin{array}{c|ccc} & D_J \rightarrow \widetilde{D}_J & D_F \rightarrow \widetilde{D}_F & D_G \rightarrow \widetilde{D}_G \\ \hline 0 & p_Z & p_Z & 0 \\ 1 & 0 & q & 1_Q \\ 2 & 1_J & 1_A & 1_Q.\end{array}\tag{CLP12.2}$$

Each is a chain map: p_Z has target in degree zero and $qd = 0$. The horizontal squares commute in every degree. The kernels of the first two vertical maps are respectively $[\sigma J \xrightarrow{d} J]$ and $[\sigma J \xrightarrow{d} J]$, where $d\sigma = 1$; the third vertical map is the identity. The first two kernels are therefore explicitly contracted by σ . This proves that all three vertical maps are continuous equivariant quasi-isomorphisms. It does not provide an equivariant section of $A \rightarrow Q$.

In the derived category of the specified $R = \mathbb{C}[t]$ -modules, the actual localization triangle is consequently the direct sum of the split identity triangle on $Z[0]$, the split identity triangle on $Q[-1]$, and the triangle of the actual extension

$$\epsilon : 0 \rightarrow J(-1) \xrightarrow{\iota} A(-1) \xrightarrow{q} Q(-1) \rightarrow 0.\tag{CLP12.3}$$

Write its extension class as

$$e \in \text{Ext}_R^1(Q(-1), J(-1)), \quad e : Q(-1) \rightarrow J(-1)[1].\tag{CLP12.4}$$

The complete derived connecting morphism is zero on $Q[-1]$ and is

$$e[-2] : Q(-1)[-2] \rightarrow J(-1)[-1] \subset \widetilde{D}_J[1]\tag{CLP12.5}$$

on the normal summand. This is a global statement about the whole original modules, established by the simultaneous quasi-isomorphisms (CLP12.2). It uses no density assertion for finite spectral blocks.

The class e is nonzero. For any actual zero ρ , the entire representative $F_0/(s - \rho)$ has order $m_\rho - 1$ there, so its quotient class is nonzero. In the normal module it satisfies

$$(t - (\rho + 1))[F_0/(s - \rho)] = [F_0] = 0.\tag{CLP12.6}$$

An R -linear section of (CLP12.3) would send that class to an entire function F with $(s - \rho)F = 0$, forcing $F = 0$, which cannot map back to the nonzero class. Thus the extension is not split, and its Yoneda class is nonzero. Nonemptiness of the actual zero divisor is the established original-zeta fact used in CW8; that proof also obtains it from the full order-one source factorization, its reflection identity, and $F_0(0) = 1/8 \neq F_0(2) = \pi/24$. No hypothetical zero is used.

The exact relationship of the global class to (CLP6.6) is pullback along an actual polynomial vector. An element $q_0 \in Q(-1)[P]$ defines the homomorphism

$$f_{q_0} : R/(P) \rightarrow Q(-1), \quad [H(t)] \mapsto H(t)q_0.$$

The pullback extension has middle module

$$E_{q_0} = \{(b, \overline{H}) \in A(-1) \oplus R/(P) : q(b) = H(t)q_0\}. \quad (\text{CLP12.7})$$

Choose b lifting q_0 . The lift $(b, 1)$ of the generator obeys the relation $P(t)(b, 1) = (P(L+1)b, 0)$. Applying the two-term free resolution therefore represents its extension class by

$$f_{q_0}^* e = [P(L+1)b] \in \text{Ext}_R^1(R/(P), J(-1)) = J(-1)/P(L+1)J. \quad (\text{CLP12.8})$$

Changing b by an element of J changes this relation by P times that element, so this is exactly the positive class (CLP6.6). This proves the requested receiving map from the global extension to every finite spectral boundary, including its nonzero values at $\rho + 1$.

The zero Postnikov class in CDW7 concerns the different triangle separating the degree-zero direct image Ω from its winding summand $\mathcal{W}[-1]$. The horizontal coefficient extension (CLP12.3) within that retained winding summand is not that Postnikov class. Equations (CLP12.2) and (CLP12.5) prove the relation: the direct-image splitting retains the entire horizontal extension, whose class is $e \neq 0$, while the ordinary degree-one localization boundary and the vertical Postnikov class are zero in their own specified triangles.

CLP13. Actual coherent-family separation and its contracting inverse

PGF0-PGF9 were read in full for this extension of the calculation. PGF2 and PGF6 construct holomorphic families of the precise source quotients used in (CLP5.1), with uniform division estimates on compact parameter sets and explicit Gaussian and F_0 -polynomial interpolation sections. Their actual inclusion matrix is the complete $\mathcal{T}_r(\lambda)$ of (CLP5.2). Let $\mathcal{O}$ denote the sheaf of holomorphic functions on the parameter plane, and define the original and normal family cokernels by

$$\mathcal{C}_r = \text{coker}(\mathcal{O}^r \xrightarrow{\mathcal{T}_r(\lambda)} \mathcal{O}^r), \quad \mathcal{C}_r^+ = \text{coker}(\mathcal{O}^r \xrightarrow{\mathcal{T}_r(\lambda-1)} \mathcal{O}^r). \quad (\text{CLP13.1})$$

The second is the pullback of the first by $\lambda \mapsto \lambda - 1$. It is exactly the normal matrix of (CLP6.7), because its generator is $L + 1$; no additional sheaf is chosen to manufacture its support. The parameter plane is a plane of recovered coefficient eigenvalues, not coordinates or a distance on τ .

Both matrix sheaf maps in (CLP13.1) are injective. Multiplying a germ in the kernel by the adjugate kills each component by respectively $F_0(\lambda)^r$ or $F_0(\lambda - 1)^r$. Those are nonzero germs in the integral domain of holomorphic germs, so each component vanishes. Thus both displayed maps extend to length-one locally free resolutions of their cokernels. Their supports are exactly

$$\text{Supp } \mathcal{C}_r = \mathcal{Z} \subset \{0 < \Re \lambda < 1\}, \quad \text{Supp } \mathcal{C}_r^+ = \mathcal{Z} + 1 \subset \{1 < \Re \lambda < 2\}. \quad (\text{CLP13.2})$$

Invertibility off those sets follows from the determinant; nonzero stalks on them follow from the following exact local calculation.

At ρ , put $x = \lambda - \rho$, retain the jet variable $h = s - \lambda$, and put $y = x + h = s - \rho$. Write

$$F_0(\rho + y) = y^m b_\rho(y), \quad b_\rho(y) = C_0(\rho + y) \frac{\zeta(\rho + y)}{y^m}, \quad b_\rho(0) \neq 0. \quad (\text{CLP13.3})$$

The original matrix is multiplication by $(x + h)^m b_\rho(x + h)$ on $\mathbb{C}\{x\}[h]/(h^r)$. Multiplication by the full unit $b_\rho(x + h)$, with inverse its full reciprocal jet, is the exact domain comparison with multiplication by $(x + h)^m$. It follows that

$$(\mathcal{C}_r)_\rho \simeq \mathbb{C}\{x, h\}/(h^r, (x + h)^m) \simeq \mathbb{C}\{y, h\}/(y^m, h^r). \quad (\text{CLP13.4})$$

The coordinate maps are $y = x + h$ and $x = y - h$; they preserve the original coordinate formulas. The monomials $y^i h^j$, $0 \leq i < m$, $0 \leq j < r$, form a basis, so the length is mr . The parameter operator $x = y - h$ has exact nilpotence exponent $m + r - 1$, because its power one below that has the nonzero term

$$(y - h)^{m+r-2} = (-1)^{r-1} \binom{m+r-2}{m-1} y^{m-1} h^{r-1}. \quad (\text{CLP13.5})$$

All other terms in that power vanish in (CLP13.4). The normal stalk at $\rho + 1$ has the identical module, now with $x = \lambda - (\rho + 1)$, $h = s - (\lambda - 1)$, and the same $y = x + h = s - \rho$.

The arithmetic generator is not this parameter operator x . The coefficient action on the original family is multiplication by

$$a^{\lambda+h} = a^\rho \exp((\log a)y), \quad (\text{CLP13.6})$$

so its nilpotent part is multiplication by y , whose exact exponent is m , not $m + r - 1$. Explicitly

$$y^i h^j \mapsto a^\rho \sum_{k=0}^{m-1-i} \frac{(\log a)^k}{k!} y^{i+k} h^j. \quad (\text{CLP13.7})$$

On the normal family, the actual normal coefficient action is also $a^{\lambda+h}$, but now its scalar is $a^{\rho+1}$; replace only that scalar in (CLP13.7). This follows from $s + 1 = \lambda + h$, retaining the normal factor. These actions commute with the original unit comparison. Therefore all finite-length original stalks have scalar weights $2\Re\rho \in (0, 2)$, and all normal stalks have scalar weights $2\Re\rho + 2 \in (2, 4)$, with their full nilpotents and all parameter extensions present.

Specialization keeps the higher boundary. Applying the fibre functor to the proved free normal resolution gives

$$0 \rightarrow \text{Tor}_1^{\mathcal{O}}(\mathcal{C}_r^+, \mathbb{C}_\lambda) \rightarrow \mathbb{C}[h]/h^r \xrightarrow{\tau_r(\lambda-1)} \mathbb{C}[h]/h^r \rightarrow \mathcal{C}_r^+ \otimes \mathbb{C}_\lambda \rightarrow 0. \quad (\text{CLP13.8})$$

The actual isomorphism from $Q(-1)[P]$ to its first term is precisely the positive boundary (CLP6.6), in the coordinates (CLP5.1); its basis is (CLP6.8). The final term is $Q(-1)/P(L+1)Q$. Each has dimension $\min(r, m_{\lambda-1})$, while the local family length is $rm_{\lambda-1}$. The family sheaf injection has zero kernel; its specialized kernel is this Tor group. These are different operations with their exact connecting map retained.

There is now a full derived separation result for these actual families. For all integers $r, q \geq 1$,

$$\boxed{\mathbf{R}\mathcal{H}om_{\mathcal{O}}(\mathcal{C}_r, \mathcal{C}_q^+) \simeq 0, \quad \mathbf{R}\mathcal{H}om_{\mathcal{O}}(\mathcal{C}_q^+, \mathcal{C}_r) \simeq 0.} \quad (\text{CLP13.9})$$

We prove it with an explicit contraction, rather than only invoking disjoint support. Resolve $\mathcal{C}_r$ by its two free terms in degrees minus one and zero. After the second-coordinate sign convention of (CLP4.1), internal Hom into $\mathcal{C}_q^+$ is the two-term complex

$$(\mathcal{C}_q^+)^r \xrightarrow{\tau_r(\lambda)^T} (\mathcal{C}_q^+)^r \quad (\text{CLP13.10})$$

in degrees zero and one. Equivalently its differential sends a map $\phi : \mathcal{O}^r \rightarrow \mathcal{C}_q^+$ to $\phi \tau_r(\lambda)$.

On the open set $\Re\lambda > 1$, which contains the whole support of the target family, this matrix is invertible. Its inverse is the transpose of the complete lower-triangular matrix with coefficients

$$\frac{1}{k!} \left(\frac{d}{ds} \right)^k \left[\frac{8\pi^{s/2}}{s(s-1)\Gamma(s/2)\zeta(s)} \right]_{s=\lambda}, \quad 0 \leq k < r. \quad (\text{CLP13.11})$$

Here F_0 has no zeros, and all displayed reciprocal factors are holomorphic on this open set. On the complementary open set $\mathbb{C} \setminus (\mathcal{Z} + 1)$, the target sheaf is zero. Define the inverse there as the unique

endomorphism of the zero sheaf. The two definitions agree on the overlap and therefore glue to an inverse sheaf endomorphism of (CLP13.10). If h denotes this inverse from degree one to degree zero, then $dh = 1$ in degree one and $hd = 1$ in degree zero. Thus it is an explicit contracting homotopy.

For the reverse Hom, resolve $\mathcal{C}_q^+$. Its differential is precomposition with $\tau_q(\lambda - 1)$ on $\mathcal{C}_r$ -valued maps. On $0 < \Re \lambda < 1$, containing the whole support of $\mathcal{C}_r$, its inverse uses the complete coefficients (CLP13.11) evaluated at $s = \lambda - 1$, with q in place of r . These are exactly (CLP7.3)–(CLP7.6), on their full open zero-free domain. Glue with the zero map off $\mathcal{Z}$ as before. This contracts the reverse Hom complex and proves both statements in (CLP13.9).

The contractions commute with the specified arithmetic action. On the free jet source the action is multiplication by $a^{\lambda+h}$, and both τ_r and its inverse are multiplication by full series in the same jet variable. On the target the analogous coefficient action commutes with holomorphic scalar entries. Hence precomposition with the inverse commutes with the induced Hom action $\phi \mapsto U_{\text{target}} \phi U_{\text{source}}^{-1}$. The same argument applies to the normal source. Thus no unproved differentiation of a discontinuous functional, finite-block density, or arbitrary classification of characters is used.

Taking derived global sections of the zero internal Hom complexes gives

$$\text{Ext}_{\mathcal{O}}^k(\mathcal{C}_r, \mathcal{C}_q^+) = 0, \quad \text{Ext}_{\mathcal{O}}^k(\mathcal{C}_q^+, \mathcal{C}_r) = 0 \quad (k \geq 0). \quad (\text{CLP13.12})$$

This uses the explicit length-one locally free resolutions and their internal Hom computation; no additional vanishing theorem for coherent cohomology is needed. Applying $\mathbf{R}\mathcal{H}om(\mathcal{C}_r, -)$ to the actual normal free resolution consequently makes the source-to-target map

$$\mathbf{R}\mathcal{H}om(\mathcal{C}_r, \mathcal{O}^q) \xrightarrow{(\tau_q(\lambda-1))_*} \mathbf{R}\mathcal{H}om(\mathcal{C}_r, \mathcal{O}^q) \quad (\text{CLP13.13})$$

a quasi-isomorphism: its cone is the first zero complex in (CLP13.9). This is the derived family version of the actual normal spectral lift in (CLP7.7).

The exact scope of this separation is now fixed by (CLP13.1), (CLP13.8) and (CLP13.13): it controls the original and normal cokernel families of the actual source-inclusion matrices. It preserves the nonzero global normal extension (CLP12.3), and the higher boundary on its own normal support (CLP8.3). It does not give a separated bound for every quotient of $J(-1)$, identify these coherent families with Deligne's étale inertia groups, or change the real parts of the original zeros. Both the family vanishing and its surviving shifted boundary are consequences of the same original localization row.

Global multiplier separation and the full normal source — GMS0-GMS11

Proof source: CC_GLOBAL_MULTIPLIER_SEPARATION.md.

Global multiplier separation and the full normal source

Complete independent derivation. Stable proof locators GMS0–GMS11.

GMS0. Stage, sources, and the operation being calculated

The supporting datum remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. Every scalar, function, module, and arithmetic action below belongs to the constructed coefficient receiver after complete-history arithmetic reconstruction. No operation below adds, subtracts, measures, or assigns coordinates to τ . The two branches retain their separately recovered counters and return measures.

The governing construction and correction chain are READ_FIRST_USER_CONSTRUCTION.md, USER_ARGUMENT_RECONSTRUCTION.md, and the retained WU050–WU055, WU061–WU065 passages in USER_CONSTRUCTION_FULL_LOGBOOK.md. In particular, a finite pair of named primes does not initialize this calculation, and an unrelated timing history does not replace the original arithmetic. WU065 asks that a limited result lead to a concrete next calculation using the accumulated findings. Here that calculation is to combine CLP13’s finite coherent separation with GSL’s already constructed global interpolation multiplier, and then apply it to the complete actual localization row.

The local source proofs used are:

- GLOBAL_SHIFTED_ZETA_LIFT.md, GSL1–GSL6: the actual entire-function space, original source factor, corridor integral, strip estimates, all-multiplicity interpolation, and translated quotient.
- CC_ACTUAL_WEIGHT_LIFT_COMPARISON.md, CW1–CW8: the original half-Mellin transform, exact source image, actual localization maps and generators, full resolvent, and actual normal extension.
- CC_LOCALIZATION_POLYNOMIAL_WEIGHTS.md, CLP1–CLP3 and CLP12–CLP13: the simultaneous source complexes, symmetric continuous section, retained nonzero normal extension, and finite coherent comparison.

GSL4–GSL6 and CLP12–CLP13 were read in full for this continuation; the underlying CW and CLP source calculations were previously read in full and the formulas used below were recalled directly. The independent full-source theorem $\Sigma : S \rightarrow J$ and its continuous inverse are the SSI/ESI results used in CW1 and CLP1. These are named proved prerequisites, not new assumptions about a hypothetical source.

The original human-source constructions remain Connes–Consani, [Schemes over \$\mathbb{F}_1\$ and zeta functions, §5, arXiv:0903.2024v3](#), and the separately recorded signed-chart source [arXiv:2609.00299v1](#). The lifting comparison is with Deligne, [La conjecture de Weil. II, §3.6](#), reconstructed in DC with its geometric hypotheses. This note proves the new module and chain maps below. It does not assert a fresh reading of those whole human sources or identify the coefficient category with Deligne’s étale category.

GMS1. The actual multiplier algebra and its complete modules

Let $\mathcal{Z}$ denote the actual nontrivial-zero divisor of the original ζ , and retain each multiplicity m_ρ . The established zero-free boundary lines give

$$0 < \Re \rho < 1 \quad (\rho \in \mathcal{Z}). \quad (\text{GMS1.1})$$

Define the unital commutative complex algebra

$$M = \left\{ h \in \mathcal{O}(\mathbb{C}) : \begin{array}{l} \text{for every } A \geq 0 \text{ there are } C_A > 0, N_A \in \mathbb{N} \\ \text{with } |h(x + it)| \leq C_A(1 + |t|)^{N_A} \\ \text{whenever } |x| \leq A \end{array} \right\}. \quad (\text{GMS1.2})$$

The operations are pointwise. Sums satisfy the estimate with the maximum of the two exponents and the sum of the constants; products satisfy it with the sum of the exponents and product of the constants. The functions 1, s , and $a^s = \exp(s \log a)$ for every recovered real $a > 0$ belong to M . Real translation by b preserves M , using the bound on the strip of width $A + |b|$. No topology on the algebra M is needed for algebraic derived Hom below.

The actual coefficient space and its ideals are

$$\begin{aligned} \mathcal{B} &= \left\{ F \in \mathcal{O}(\mathbb{C}) : b_{A,N}(F) = \sup_{|\Re s| \leq A} (1 + |\Im s|)^N |F(s)| < \infty \text{ for all } A, N \geq 0 \right\}, \\ \mathcal{J} &= \{ F \in \mathcal{B} : F^{(j)}(\rho) = 0, \rho \in \mathcal{Z}, 0 \leq j < m_\rho \}, \\ \mathcal{J}_+ &= \{ F \in \mathcal{B} : F^{(j)}(\rho + 1) = 0, \rho \in \mathcal{Z}, 0 \leq j < m_\rho \}, \\ \mathcal{Q} &= \mathcal{B}/\mathcal{J}, \quad \mathcal{Q}_+ = \mathcal{B}/\mathcal{J}_+. \end{aligned} \quad (\text{GMS1.3})$$

These are the full modules, without a finite-support restriction or a replacement by a direct sum of jets. The countable seminorms at integer A, N make $\mathcal{B}$ Fréchet. To check completeness, a Cauchy sequence converges uniformly on every compact set to an entire function; its Cauchy property in each weighted strip supremum then gives convergence in that seminorm to the same function. Evaluations of every fixed derivative are continuous by Cauchy's estimate on a larger strip. Thus both ideals are closed and both quotient spaces are Fréchet with their quotient topology.

For fixed $h \in M$, multiplication is continuous, with

$$b_{A,N}(hF) \leq C_A b_{A,N+N_A}(F). \quad (\text{GMS1.4})$$

The full Leibniz formula

$$(hF)^{(j)}(z) = \sum_{k=0}^j \binom{j}{k} h^{(k)}(z) F^{(j-k)}(z) \quad (\text{GMS1.5})$$

shows that it preserves both ideals. Hence $\mathcal{B}, \mathcal{J}, \mathcal{J}_+, \mathcal{Q}, \mathcal{Q}_+$ are actual M -modules, and each fixed multiplier acts continuously. We use this last property, together with strict exact sequences of Fréchet spaces, when discussing continuous extensions. We do not infer this structure for an arbitrary continuous module carrying only a generator L .

GMS2. The normal coordinate and every original source factor

The full original arithmetic is retained:

$$\zeta(s) = 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'}{\zeta}(s) = \sum_p \sum_{k \geq 1} (\log p) p^{-ks} \quad (\Re s > 1). \quad (\text{GMS2.1})$$

The half-Mellin source from CW1 and CLP1 is

$$\Theta b(s) = \frac{1}{2} \int_0^\infty b(u) u^s \frac{du}{u}, \quad \Theta^{-1} F(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} F(1/2 + it) u^{-it} dt. \quad (\text{GMS2.2})$$

It is a topological isomorphism $A \rightarrow \mathcal{B}$ and takes the exact original source $J = \Sigma S$ onto $\mathcal{J}$, where

$$\Sigma f(u) = 2 \sum_{n \geq 1} f(nu), \quad S = \{f \in \mathcal{S}(\mathbb{R}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f = 0\}. \quad (\text{GMS2.3})$$

For $\Re s > 1$, absolute summation proves

$$\Theta \Sigma f(s) = \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}. \quad (\text{GMS2.4})$$

The particular source

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}$$

has the entire transform

$$F_0(s) = \Theta \Sigma f_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) = C_0(s) \zeta(s). \quad (\text{GMS2.5})$$

It belongs to $\mathcal{B}$ and has precisely $\mathcal{Z}$ as its zero divisor. Formula (GMS2.5) is an exact source comparison; the original ζ and the full multiplier C_0 remain in every local comparison.

Define $VF(\lambda) = F(\lambda - 1)$ and $UF(s) = F(s + 1)$. Then

$$b_{A,N}(VF), b_{A,N}(UF) \leq b_{A+1,N}(F), \quad V\mathcal{J} = \mathcal{J}_+. \quad (\text{GMS2.6})$$

The actual normal coefficient module before translation has the action

$$h \cdot_{\text{normal}} F(s) = h(s + 1) F(s). \quad (\text{GMS2.7})$$

This is the M -action on $\mathcal{B}(-1), \mathcal{J}(-1), \mathcal{Q}(-1)$. In particular its generator is $s + 1$, and its dilation is a^{s+1} , including the factor a . Direct substitution gives

$$V(h(s + 1) F(s))(\lambda) = h(\lambda) VF(\lambda). \quad (\text{GMS2.8})$$

Thus V gives the actual continuous M -linear isomorphism from the full normal row to

$$0 \longrightarrow \mathcal{J}_+ \xrightarrow{\iota} \mathcal{B} \xrightarrow{q_+} \mathcal{Q}_+ \longrightarrow 0. \quad (\text{GMS2.9})$$

Its kernel is the entire $\mathcal{J}_+$. In the original source coordinates it remains the entire $\mathcal{J}(-1)$.

The translated original factor is

$$F_+(\lambda) = VF_0(\lambda) = \frac{(\lambda - 1)(\lambda - 2)}{8} \pi^{-(\lambda-1)/2} \Gamma((\lambda - 1)/2) \zeta(\lambda - 1). \quad (\text{GMS2.10})$$

It has every zero $\rho + 1$ to the original multiplicity. Its exceptional values, including the retained cancellations, are

$$F_+(1) = F_+(2) = \frac{1}{8}, \quad F_+(0) = F_0(-1) = \frac{\pi}{24},$$

$$\begin{aligned}
F_+(1-2r) &= \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) \\
&= F_+(2+2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0 \quad (r \geq 1).
\end{aligned}
\tag{GMS2.11}$$

The corresponding unshifted values are $F_0(0) = F_0(1) = 1/8$ and the second displayed formula with arguments $-2r, 1+2r$. All derivatives of the translated entire function satisfy $F_+^{(j)}(\lambda) = F_0^{(j)}(\lambda-1)$. In particular, at a nontrivial zero the full local unit is

$$F_+(\rho+1+z) = z^{m_\rho} C_0(\rho+z) \frac{\zeta(\rho+z)}{z^{m_\rho}}, \quad C_0(\rho) \frac{\zeta^{(m_\rho)}(\rho)}{m_\rho!} \neq 0. \tag{GMS2.12}$$

GMS3. The already constructed global separator, with its exact scope

GSL4 constructs an entire multiplier E with polynomial bounds on each vertical strip and with the full jets

$$E-1 \in (s-\rho)^{m_\rho} \mathcal{O}_\rho, \quad E \in (s-(\rho-1))^{m_\rho} \mathcal{O}_{\rho-1}. \tag{GMS3.1}$$

For specificity, the actual construction is $G(s) = F_0(s)F_0(s+1)$, $r(t) = \varepsilon(1+t^2)^{-3}$, $\chi(x+it) = \eta(x/r(t))$, with the smooth cutoff η equal to zero on $(-\infty, -1]$ and one on $[1, \infty)$. The original-zeta zero-free corridor chooses $\varepsilon > 0$ so G has no zeros on the cutoff-transition corridor. Put

$$\begin{aligned}
v(w) &= \frac{\bar{\partial}\chi(w)}{G(w)} \quad \text{on that corridor, and } v = 0 \text{ off it,} \quad u(s) = \frac{1}{\pi} \int_{\mathbb{C}} \frac{e^{(s-w)^2}}{s-w} v(w) dA(w), \\
E(s) &= \chi(s) - G(s)u(s).
\end{aligned}
\tag{GMS3.2}$$

GSL4.4-GSL4.14 prove absolute convergence, $\bar{\partial}u = v$, entire continuation of E , and

$$|E(x+it)| + |1-E(x+it)| \leq C_A(1+|t|)^{d_A} \quad (|x| \leq A). \tag{GMS3.3}$$

Those estimates use the full two source factors in G ; they are the proved global input here. The full vanishing orders in (GMS3.1) follow from (GMS3.2) at the two divisors. No convergence of a sum of finite spectral projectors supplies this multiplier.

Set

$$c(\lambda) = E_+(\lambda) = 1 - E(\lambda-1) \in M. \tag{GMS3.4}$$

Translating (GMS3.1) gives

$$c-1 \in (\lambda-\rho)^{m_\rho} \mathcal{O}_\rho, \quad c \in (\lambda-(\rho+1))^{m_\rho} \mathcal{O}_{\rho+1}. \tag{GMS3.5}$$

Consequently, with every multiplicity retained,

$$\begin{aligned}
c\mathcal{B} &\subset \mathcal{J}_+, \quad (1-c)\mathcal{B} \subset \mathcal{J}, \\
c_Q &= 1_Q, \quad c_{Q_+} = 0_{Q_+}.
\end{aligned}
\tag{GMS3.6}$$

All inclusions follow by (GMS1.5). These formulas establish both identities on the complete quotient modules, including classes with infinitely many nonzero jets.

The function c is not asserted to satisfy $c^2 = c$ in M . Indeed it takes both values zero and one and is entire, so it is nonconstant, whereas an entire solution of $c^2 = c$ on $\mathbb{C}$ is constant. Thus it is not an idempotent on $\mathcal{B}$ or on its nonzero ideal $\mathcal{J}_+$. It is an idempotent on $\mathcal{Q} \oplus \mathcal{Q}_+$. No value zero or one at the separate endpoint parameters 0, 1 is inferred from (GMS3.5).

GMS4. Full derived separation, with an explicit algebraic contraction

We prove the required derived statements over the actual ring M . The proof uses arbitrary algebraic projective resolutions; it requires no finite generation of the modules and makes no topological projectivity claim.

Choose a projective resolution $P^\bullet \rightarrow X$ in degrees at most zero. Let $a \in M$ act as zero on X . There is a degree-minus-one M -linear map $h : P^\bullet \rightarrow P^{\bullet-1}$ with

$$d_P h + h d_P = a_P. \quad (\text{GMS4.1})$$

Here is the full lifting induction. If $\epsilon : P^0 \rightarrow X$ is the augmentation, aP^0 lands in $\ker \epsilon = \text{im}(d_P : P^{-1} \rightarrow P^0)$, so projectivity of P^0 lifts a_{P^0} to $h^0 : P^0 \rightarrow P^{-1}$. Suppose h^{j+1} and all higher components have been chosen. On P^j the map $a_{P^j} - h^{j+1} d_P^j$ lands in $\ker d_P^j$: applying d_P^j and the already established equation in degree $j+1$ gives zero. Exactness gives $\ker d_P^j = \text{im } d_P^{j-1}$; projectivity of P^j lifts this map to $h^j : P^j \rightarrow P^{j-1}$. Induction proves (GMS4.1) in every degree. An arbitrary module has a free resolution by successively choosing a free module surjecting onto each kernel, so the construction applies to the actual full modules.

For a module or bounded complex $N^\bullet$, the Hom differential and its degree-minus-one homotopy are

$$Df = d_N f - (-1)^n f d_P, \quad (Hf) = (-1)^n f h \quad (f \in \text{Hom}_M(P, N)^n). \quad (\text{GMS4.2})$$

Multiplication gives the exact identity

$$D(Hf) + H(Df) = f(hd_P + d_P h) = f a_P = a_N f. \quad (\text{GMS4.3})$$

The two $d_N f h$ terms cancel. The last equality uses M -linearity and centrality. The bounded-above projective resolution computes derived Hom into modules and into the finite coefficient complexes used here. All operations in (GMS4.2) are algebraic compositions; no infinite series or density assertion occurs.

Apply this with $X = \mathcal{Q}$, $a = 1 - c$, and $N = \mathcal{Q}_+$. Equation (GMS3.6) says $a_X = 0$, $a_N = 1$, so (GMS4.3) is a contraction of the full Hom complex. For the reverse direction resolve $X = \mathcal{Q}_+$, use $a = c$, and take $N = \mathcal{Q}$. Again $a_X = 0$, $a_N = 1$. Therefore

$$\boxed{\text{RHom}_M(\mathcal{Q}, \mathcal{Q}_+) \simeq 0, \quad \text{RHom}_M(\mathcal{Q}_+, \mathcal{Q}) \simeq 0.} \quad (\text{GMS4.4})$$

In particular every algebraic Ext_M^k in either displayed direction is zero for $k \geq 0$. The proof applies to the whole actual modules in (GMS1.3), without presenting either as a limit of the finite coherent families from CLP13.

There is also a fully specified resolution and homotopy with no choices of projective lifts. For an M -module X , let

$$P_n(X) = M \otimes_{\mathbb{C}} M^{\otimes_{\mathbb{C}} n} \otimes_{\mathbb{C}} X \quad (n \geq 0), \quad \epsilon(a_0 \otimes x) = a_0 x. \quad (\text{GMS4.5})$$

The left M -action is on the first factor. Each term is a free M -module: its remaining tensor factors are a complex vector space, and a vector-space basis identifies the tensor product with a direct sum of copies of M . The differential is

$$\begin{aligned} d_n(a_0 \otimes \cdots \otimes a_n \otimes x) &= \sum_{i=0}^{n-1} (-1)^i a_0 \otimes \cdots \otimes a_i a_{i+1} \otimes \cdots \otimes a_n \otimes x \\ &\quad + (-1)^n a_0 \otimes \cdots \otimes a_{n-1} \otimes a_n x. \end{aligned} \quad (\text{GMS4.6})$$

In $d_{n-1} d_n$, the terms performing two disjoint multiplications cancel in opposite orders; the terms multiplying three adjacent factors cancel by associativity; the two final terms cancel by the module law.

Hence $d^2 = 0$. The complex-linear augmented homotopy inserting 1 in front, $s_n(a_0 \otimes \cdots \otimes a_n \otimes x) = 1 \otimes a_0 \otimes \cdots \otimes a_n \otimes x$, and $s_{-1}(x) = 1 \otimes x$, satisfies $ds + sd = 1$: the first multiplication in ds is the identity term, and its other terms cancel term by term with sd . Therefore the augmented complex is exact. This proves it is an actual free resolution, although this augmentation contraction is not asserted M -linear.

Its M -linear insertion of the specified central multiplier is

$$H_n(a_0 \otimes \cdots \otimes a_n \otimes x) = \sum_{i=0}^n (-1)^i a_0 \otimes \cdots \otimes a_i \otimes c \otimes a_{i+1} \otimes \cdots \otimes a_n \otimes x. \quad (\text{GMS4.7})$$

In $dH + Hd$, multiplications not involving the inserted c cancel with the identical insertion after that multiplication, with opposite sign. Multiplication of c with an interior factor occurs in two adjacent insertion terms, whose signs are opposite; centrality makes their values equal. The two terms left are the first-factor multiplication by c and minus the last-factor action by c . Thus in every degree

$$d_{n+1}H_n + H_{n-1}d_n = c_{P_n(X)} - P_n(c_X), \quad H_{-1} = 0. \quad (\text{GMS4.8})$$

Here $P_n(c_X)$ applies c_X only to the last factor. In the Hom cochain convention $C^n = \text{Hom}_M(P_n(X), N)$, $\delta f = f d_{n+1}$, put $k^n(f) = f H_{n-1}$, with $k^0 = 0$. Equations (GMS4.7)-(GMS4.8) give

$$\delta k + k\delta = c_N - (\text{precomposition by } P_\bullet(c_X)). \quad (\text{GMS4.9})$$

For $X = \mathcal{Q}$, this is $c_N - 1$: consequently $-k$ contracts the complex when $N = \mathcal{Q}_+$, and gives the identity-minus-multiplier homotopies for $N = \mathcal{B}, \mathcal{J}_+$. For $X = \mathcal{Q}_+$, $N = \mathcal{Q}$, it is the identity, so k contracts the reverse Hom. This is a concrete alternative realization of every module homotopy used in (GMS4.4) and (GMS5.3). The convention $\delta f = f d$ is isomorphic to the ordinary cohomological Hom convention (GMS4.2) after multiplication of degree n by $(-1)^{n(n+1)/2}$. Each insertion sum is finite in its degree; these are algebraic tensor products, with no assertion about completed tensor products or continuous free resolutions.

GMS5. The exact derived inverse for the entire normal ideal

The separator gives an actual continuous M -linear map

$$\beta : \mathcal{B} \longrightarrow \mathcal{J}_+, \quad \beta(F) = cF. \quad (\text{GMS5.1})$$

Its codomain follows from (GMS3.6), its continuity from (GMS1.4), and its M -linearity from pointwise commutativity. With the actual inclusion ι from (GMS2.9),

$$\iota\beta = c_{\mathcal{B}}, \quad \beta\iota = c_{\mathcal{J}_+}. \quad (\text{GMS5.2})$$

These are multiplier identities, not identities of the original modules.

Resolve $\mathcal{Q}$ by P , and retain the homotopy (GMS4.1) for $a = 1 - c$. For either $N = \mathcal{B}$ or $N = \mathcal{J}_+$, (GMS4.3) gives

$$DH_N + H_N D = 1 - c_N \quad \text{on } \text{Hom}_M(P, N). \quad (\text{GMS5.3})$$

Hence postcomposition by β and by ι are explicit chain-homotopy inverses. We have proved

$\text{RHom}_M(\mathcal{Q}, \mathcal{J}_+) \xrightarrow{\iota_*} \text{RHom}_M(\mathcal{Q}, \mathcal{B}) \text{ is a quasi-isomorphism, with inverse } \beta_*.$

(GMS5.4)

The homotopies are (GMS5.3); in particular $1 - \iota_*\beta_*$ and $1 - \beta_*\iota_*$ are the stated Hom differentials of those homotopies. Nothing has quotiented $\mathcal{J}_+$ further.

Equivalently, for every integer k , postcomposition with $\iota[k]$ is a bijection of derived morphism groups

$$\mathrm{Hom}_{D(M)}(\mathcal{Q}, \mathcal{J}_+[k]) \xrightarrow{\sim} \mathrm{Hom}_{D(M)}(\mathcal{Q}, \mathcal{B}[k]); \quad [f] \mapsto [\iota f], \quad [g] \mapsto [\beta g]. \quad (\text{GMS5.5})$$

Uniqueness here is uniqueness of the derived morphism class; it is not uniqueness of each cochain representative. In degree zero both groups are in fact zero. For an M -map $g : \mathcal{Q} \rightarrow \mathcal{B}$, $(1 - c)g(q) = g((1 - c)q) = 0$. Since $1 - c$ is a nonzero entire function, multiplication by it is injective on $\mathcal{B}$: the product identity forces the entire second factor to vanish on its nonempty open nonzero set and hence everywhere. Thus $g = 0$; inclusion proves the same for target $\mathcal{J}_+$. This does not assert that the higher groups in (GMS5.5) separately vanish.

The same normal triangle and the reverse separation in (GMS4.4) also give

$$\mathrm{RHom}_M(\mathcal{B}, \mathcal{Q}) \longrightarrow \mathrm{RHom}_M(\mathcal{J}_+, \mathcal{Q}) \quad \text{a quasi-isomorphism.} \quad (\text{GMS5.6})$$

Indeed applying derived Hom into $\mathcal{Q}$ to the exact triangle of (GMS2.9) puts its cone next to $\mathrm{RHom}_M(\mathcal{Q}_+, \mathcal{Q}) \simeq 0$. This is a statement about the same full kernel and quotient.

If another GSL cutoff yields c' satisfying (GMS3.5), its map β' is another chain-homotopy inverse of ι_* by the same proof. Therefore β_* and β'_* coincide in the derived category. For example, $\beta_* - \beta'_* = \beta_*(1 - \iota_*\beta'_*) + (\beta_*\iota_* - 1)\beta'_*$, and (GMS5.3) explicitly makes both terms null-homotopic. The derived inverse does not depend on the cutoff choices.

GMS6. Exact meaning for pullbacks and the surviving normal extension

The normal extension (GMS2.9) has its Yoneda class

$$e_+ \in \mathrm{Ext}_M^1(\mathcal{Q}_+, \mathcal{J}_+), \quad e_+ : \mathcal{Q}_+ \longrightarrow \mathcal{J}_+[1]. \quad (\text{GMS6.1})$$

Its domain is $\mathcal{Q}_+$. It is not one of the two cross groups in (GMS4.4).

For an ordinary M -map $f : \mathcal{Q} \rightarrow \mathcal{Q}_+$, (GMS4.4) gives $f = 0$. Its actual pullback middle module is therefore

$$\{(b, q) \in \mathcal{B} \oplus \mathcal{Q} : q_+(b) = f(q) = 0\} = \mathcal{J}_+ \oplus \mathcal{Q}. \quad (\text{GMS6.2})$$

The section is $q \mapsto (0, q)$; it is M -linear and continuous in the product topology. For any derived morphism $f : \mathcal{Q} \rightarrow \mathcal{Q}_+[k]$, all such classes also vanish by (GMS4.4), so the composition with the corresponding shift of e_+ is zero. This is the exact pullback statement supplied by the separation. It does not produce a section $\mathcal{Q}_+ \rightarrow \mathcal{B}$ of the normal row.

That normal row remains non-split even algebraically over M . Choose an actual $\rho \in \mathcal{Z}$, whose existence is the proved original-zeta fact used in CW8. Put $\lambda_0 = \rho + 1$. Entire division and the strip estimates give

$$v_\rho(\lambda) = \frac{F_+(\lambda)}{\lambda - \lambda_0} \in \mathcal{B}, \quad [v_\rho] \neq 0 \text{ in } \mathcal{Q}_+, \quad (\lambda - \lambda_0)[v_\rho] = [F_+] = 0. \quad (\text{GMS6.3})$$

The nonzero quotient class follows from its exact order $m_\rho - 1$ at λ_0 ; (GMS2.12) retains the local unit. An M -linear section would send it to an entire function F satisfying $(\lambda - \lambda_0)F = 0$, hence to zero, contradicting its quotient. Thus $e_+ \neq 0$. The nonzero class in the polynomial source test is exactly

$$[F_+] \in \mathcal{J}_+ / (\lambda - \lambda_0)\mathcal{J}_+. \quad (\text{GMS6.4})$$

It cannot vanish, since an identity $F_+ = (\lambda - \lambda_0)H$ with $H \in \mathcal{J}_+$ would force $H = v_\rho$, which has insufficient vanishing order at λ_0 . This is CLP12's positive normal boundary with its whole ideal and

full original factor. It coexists with (GMS5.4) because the domains of the two statements are different specified modules.

One can also calculate the actual pushout that the multiplier makes split. Push (GMS2.9) out along $c : \mathcal{J}_+ \rightarrow \mathcal{J}_+$. Its middle module is

$$\mathcal{P}_c = (\mathcal{J}_+ \oplus \mathcal{B}) / \{(cj, -j) : j \in \mathcal{J}_+\}, \quad i_c(i) = [(i, 0)], \quad p_c([(i, b)]) = q_+(b). \quad (\text{GMS6.5})$$

The relation subspace is closed: convergence of $(cj_\nu, -j_\nu)$ implies that j_ν converges in the closed subspace $\mathcal{J}_+$ to $-b$, and continuity of multiplication gives the first coordinate $-cb$. Thus $\mathcal{P}_c$ has its actual Fréchet quotient topology. Define

$$\Phi_c : \mathcal{P}_c \longrightarrow \mathcal{J}_+ \oplus \mathcal{Q}_+, \quad [(i, b)] \longmapsto (i + cb, q_+(b)). \quad (\text{GMS6.6})$$

It kills precisely the displayed relations: if both coordinates are zero, then $b = j \in \mathcal{J}_+$ and $i = -cj$, which is the negative of one relation. It is onto, since for (i_0, q) any lift b of q gives the preimage $[(i_0 - cb, b)]$. The latter is independent of the lift: changing b by j changes the pair by $(-cj, j)$, a relation. Continuity of the inverse follows because the continuous map $(i_0, b) \mapsto [(i_0 - cb, b)]$ kills $\{0\} \oplus \mathcal{J}_+$ and descends through the product quotient $\mathcal{J}_+ \oplus \mathcal{B} \rightarrow \mathcal{J}_+ \oplus \mathcal{Q}_+$. Thus Φ_c is an explicit continuous M -linear isomorphism of extensions.

Its section and its Yoneda interpretation are

$$q \longmapsto [(-cb, b)] \quad (q_+(b) = q), \quad c_*e_+ = 0 \text{ in } \text{Ext}_M^1(\mathcal{Q}_+, \mathcal{J}_+). \quad (\text{GMS6.7})$$

This constructs the annihilation of the extension class by the central multiplier, while (GMS6.3)-(GMS6.4) retain $e_+ \neq 0$ before that pushout. Its kernel is still a copy of the full $\mathcal{J}_+$, with the specified changed map into the middle module.

GMS7. Continuous splitting for actual strict multiplier extensions

There is also a continuous statement that does not use an unconstructed Fréchet projective resolution. Suppose

$$0 \longrightarrow \mathcal{Q}_+ \xrightarrow{j} X \xrightarrow{\pi} \mathcal{Q} \longrightarrow 0 \quad (\text{GMS7.1})$$

is a strict exact sequence of Fréchet M -modules: j is a topological embedding with closed image, π induces the quotient topology, and every fixed multiplier acts continuously on X . Such an extension is an explicitly specified class of objects; a continuous L -equivariant extension is not declared to satisfy these additional hypotheses.

For $q \in \mathcal{Q}$, take any lift $x \in X$, and set

$$s(q) = c_X x. \quad (\text{GMS7.2})$$

If the lift changes by $j(y)$, the result changes by $j(cy) = 0$, proving well-definedness. Moreover $\pi s(q) = cq = q$ and $s\pi = c_X$. It is M -linear because c is central. Its continuity follows exactly from the quotient topology: $s\pi = c_X$ is continuous. It is the unique M -linear section, since the difference of two sections factors through j as an M -map $\mathcal{Q} \rightarrow \mathcal{Q}_+$, which is zero by (GMS4.4).

There is an actual idempotent on this extension:

$$c_X^2 = c_X. \quad (\text{GMS7.3})$$

Indeed $(c_X - 1)x \in j\mathcal{Q}_+$, and applying c_X kills that element. The complete continuous splitting and inverse are

$$\begin{aligned} \mathcal{Q}_+ \oplus \mathcal{Q} &\longrightarrow X, \quad (y, q) \longmapsto j(y) + s(q), \\ X &\longrightarrow \mathcal{Q}_+ \oplus \mathcal{Q}, \quad x \longmapsto (j^{-1}((1 - c_X)x), \pi x). \end{aligned} \quad (\text{GMS7.4})$$

The first coordinate of the inverse is well-defined since $\pi(1 - c_X) = 0$, and is continuous because j^{-1} is continuous on its image. Direct substitution and (GMS7.3) prove that the maps are inverse.

For the reversed strict extension $0 \rightarrow \mathcal{Q} \rightarrow X \rightarrow \mathcal{Q}_+ \rightarrow 0$, the identical proof uses $1 - c$ on a lift to construct its unique continuous M -linear section. If a second permissible separator c' is used in (GMS7.2), both resulting maps are sections, so uniqueness makes them equal. More directly, $c_X - c'_X$ kills the kernel and has image in that kernel, hence factors as an M -map $\mathcal{Q} \rightarrow \mathcal{Q}_+$, which vanishes. This proves cutoff independence on the actual extension.

For example, the existing GSL high row has

$$X = \mathcal{B}/(\mathcal{J} \cap \mathcal{J}_+), \quad j([g]_{\mathcal{J}_+}) = [(1 - c)g]_{\mathcal{J} \cap \mathcal{J}_+}, \quad \pi([F]) = [F]_{\mathcal{J}}. \quad (\text{GMS7.5})$$

The section from (GMS7.2) is exactly $[f]_{\mathcal{J}} \mapsto [cf]$. Independence of representatives follows from the two full jet conditions. The full inverse of the quotient map to $\mathcal{Q} \oplus \mathcal{Q}_+$ is

$$([f]_{\mathcal{J}}, [g]_{\mathcal{J}_+}) \mapsto [cf + (1 - c)g]_{\mathcal{J} \cap \mathcal{J}_+}. \quad (\text{GMS7.6})$$

Products $c\mathcal{J}$ and $(1 - c)\mathcal{J}_+$ lie in the intersection, so this is well-defined. Its two quotient images are $[f]$ and $[g]$; the reverse composition uses $cF + (1 - c)F = F$. The estimates (GMS1.4), with independent infima over both representatives, prove continuity in both directions. Thus the example is strict and realizes the exact hypotheses of (GMS7.1).

GMS8. The actual original-source map behind the derived inverse

Returning (GMS5.1) through V gives

$$V^{-1}\beta V(F)(s) = c(s + 1)F(s) = (1 - E(s))F(s). \quad (\text{GMS8.1})$$

The actual map on the full unshifted source spaces with their normal actions is therefore

$$\mathfrak{b} : A(-1) \rightarrow J(-1), \quad \mathfrak{b}(a) = \Theta^{-1}[(1 - E(s))\Theta a(s)]. \quad (\text{GMS8.2})$$

Its image is in J because $(1 - E)\mathcal{B} \subset \mathcal{J} = \Theta J$. Its exact integral representative is

$$\mathfrak{b}(a)(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} (1 - E(1/2 + it))\Theta a(1/2 + it)u^{-it} dt. \quad (\text{GMS8.3})$$

The bound (GMS3.3) multiplied by arbitrarily rapid strip decay of Θa proves absolute convergence after any fixed number of $u\partial_u$ derivatives. The original inverse Mellin estimates in (GMS2.2), applied to the strip seminorm bound (GMS1.4), prove continuity into every seminorm of A . Its image is already in the closed source J , and $H = \Sigma^{-1} : J \rightarrow S$ is the actual continuous source inverse. Consequently

$$a \mapsto f_a = H\mathfrak{b}(a) \in S, \quad \Sigma f_a = \mathfrak{b}(a), \quad (\text{GMS8.4})$$

is a continuous original Schwartz-source representative with its value and integral conditions retained.

The two composites with the original inclusion $J(-1) \hookrightarrow A(-1)$ are multiplication by $1 - E(s)$ under Θ . The derived identity in (GMS5.4) follows from the action of the same ring element $c(\lambda)$ on the unshifted source $\mathcal{Q}$, where it is the identity. The shift in (GMS8.1) is essential; it is not permission to replace the normal action $h(s + 1)$ by $h(s)$.

GMS9. The multiplier action on the complete simultaneous complexes

We construct the action, rather than presuming a general functional calculus for an arbitrary operator. Let

$$\begin{aligned} V_{\pm} &= S \oplus \mathbb{C}^2, \quad W = V_+ \oplus V_-, \quad r_+(f, c_0, c_1) = \Sigma f, \quad r_-(g, d_0, d_1) = R\Sigma g = \Sigma \hat{g}, \\ Rb(u) &= u^{-1}b(u^{-1}), \quad \hat{g}(v) = \int_{\mathbb{R}} g(x)e^{-2\pi i x v} dx. \end{aligned} \quad (\text{GMS9.1})$$

The full preceding Poisson identity is $\Sigma \hat{g}(u) = u^{-1}\Sigma g(u^{-1}) + u^{-1}g(0) - \int g$. Its last two terms vanish on S ; the four separate endpoint coordinates are retained. The source sections are $s_+(b) = (Hb, 0, 0)$ and $s_-(b) = (\widehat{Hb}, 0, 0)$, with $r_{\pm}s_{\pm} = 1_J$. Put $h_J = \Theta^{-1}m_h\Theta$ on J . Define

$$\begin{aligned} h_{V_+}(f, c_0, c_1) &= s_+(h_J \Sigma f) + (0, h(0)c_0, h(1)c_1), \\ h_{V_-}(g, d_0, d_1) &= s_-(h_J R\Sigma g) + (0, h(1)d_0, h(0)d_1). \end{aligned} \quad (\text{GMS9.2})$$

All displayed maps are continuous. On each chart the decomposition into its source summand and endpoint summand is direct. Since $r_{\pm}s_{\pm} = 1$, composition of (GMS9.2) for multipliers h, k is the action for hk , addition is the action for $h + k$, and 1 acts as the identity. This proves an actual M -module structure on W , with

$$r_{\pm}h_{V_{\pm}} = h_J r_{\pm}. \quad (\text{GMS9.3})$$

For $h(s) = s$ this is the previously computed generator G on W , with endpoint eigenvalues $(0, 1, 1, 0)$. For $h(s) = a^s$, injectivity of the source restrictions and (GMS9.3) identify its source components with $f(v/a)$ and $ag(av)$, respectively. Its endpoint components are (c_0, ac_1, ad_0, d_1) . Thus it extends the actual coefficient actions, with every factor of a .

The actual simultaneous complexes, in degrees 0, 1, 2, are

$$D_J = (W \xrightarrow{r_+ - r_-} J \xrightarrow{0} J(-1)), \quad D_F = (W \xrightarrow{r_+ - r_-} A \xrightarrow{0} A(-1)), \quad D_G = (0 \rightarrow Q \xrightarrow{0} Q(-1)). \quad (\text{GMS9.4})$$

Use (GMS9.2) in degree zero, multiplication by $h(s)$ under Θ in degree one, and multiplication by $h(s+1)$ in degree two. Equation (GMS9.3) proves compatibility with the first differential; the second is zero. Inclusions, quotients, and the degree-zero identity in the row $0 \rightarrow D_J \rightarrow D_F \rightarrow D_G \rightarrow 0$ are M -linear.

The symmetric section, projection, and exact source decomposition are

$$\begin{aligned} \sigma b &= \tfrac{1}{2}(s_+b, -s_-b), \quad d\sigma = 1_J, \quad p_Z = 1_W - \sigma d, \quad Z = \ker d, \\ W &\xrightarrow{\sim} Z \oplus J, \quad w \mapsto (p_Z w, dw), \quad (z, b) \mapsto z + \sigma b. \end{aligned} \quad (\text{GMS9.5})$$

They are M -linear by (GMS9.2)-(GMS9.3). Moreover

$$Z \simeq J \oplus \mathbb{C}_+^2 \oplus \mathbb{C}_-^2, \quad (b, e_+, e_-) \mapsto (s_+b + e_+, s_-b + e_-), \quad (\text{GMS9.6})$$

where the endpoint multipliers are exactly $h(0), h(1), h(1), h(0)$. Thus c retains the values $c(0), c(1)$ there; they are not replaced by spectral zero or one values.

Applying this decomposition contracts the $[J \xrightarrow{1} J]$ summands and gives the simultaneous quasi-isomorphisms of CLP12, now proved M -linear:

$$\begin{aligned} D_J &\simeq Z[0] \oplus J(-1)[-2], \\ D_F &\simeq Z[0] \oplus Q[-1] \oplus A(-1)[-2], \\ D_G &= Q[-1] \oplus Q(-1)[-2]. \end{aligned} \quad (\text{GMS9.7})$$

The vertical maps are p_Z in degree zero, respectively zero or the quotient q in degree one, and the identity in degree two. Their first two kernels are $[\sigma J \xrightarrow{d} J]$, contracted by σ . This proves each claim in (GMS9.7), including the retained normal row (GMS2.9). No M -linear section of $A \rightarrow Q$ has been inserted.

The ring element c acts as an actual projection on D_G : it is one on $Q[-1]$ and zero on $Q(-1)[-2]$, by (GMS2.8) and (GMS3.6). It is not an idempotent on the full D_F, D_J . In particular applying it does not erase their endpoints or replace their normal source ideal.

GMS10. Full derived localization and its exact geometric action scope

Transport by Θ , and by V on the normal terms, identifies Q with $\mathcal{Q}$ and $Q(-1)$ with $\mathcal{Q}_+$. Derived Hom from the complete $\mathcal{Q}$ applied to (GMS9.7) gives

$$\mathrm{RHom}_M(\mathcal{Q}, D_G) \simeq \mathrm{RHom}_M(\mathcal{Q}, \mathcal{Q})[-1]. \quad (\text{GMS10.1})$$

The projection and inclusion of the degree-one summand are inverse up to the explicit contraction of its normal Hom summand from (GMS4.2)–(GMS4.4). This is a full-module statement, not an inference from finite spectral vectors.

The actual connecting morphism in the localization triangle is zero on $Q[-1]$ and is $e_+[-2]$ on $\mathcal{Q}_+[-2]$, by the simultaneous maps (GMS9.7). Consequently

$$\mathrm{RHom}_M(\mathcal{Q}, D_G) \longrightarrow \mathrm{RHom}_M(\mathcal{Q}, D_J[1]) \quad \text{is zero in the derived category.} \quad (\text{GMS10.2})$$

Indeed the only nonzero candidate component factors through the contractible complex $\mathrm{Hom}_M(P, \mathcal{Q}_+)$. Its contraction was constructed in (GMS4.2). Equivalently, the normal comparison inside the source triangle is precisely the chain-homotopy equivalence (GMS5.4). The Z term remains; the degree-one component lifts through the identity on $Q[-1]$ in (GMS9.7). Thus (GMS10.2) describes the actual full localization maps, with the full normal ideal retained.

The underlying source geometry gives the cover action for each integer $n \geq 1$:

$$(\rho_+(n) \oplus \rho_-(n), T_n, nT_n) \quad (\text{GMS10.3})$$

on degrees zero, one, and two. Its last factor is the positive degree of the actual holomorphic cover $z \mapsto z^n$. These are exactly the actions of $h(s) = n^s$ from (GMS9.4), since the normal component uses n^{s+1} . The same formulas for real $a > 0$ are coefficient actions; there is no asserted nonintegral holomorphic covering. All maps in (GMS5.1), (GMS8.2), and (GMS9.7) commute with the specified M -action and hence with these covers' coefficient actions.

Mirror actions retain their degree-dependent semilinearity. Under the actual half-Mellin transform, R is $F(s) \mapsto F(1-s)$, since substituting $u \mapsto u^{-1}$ in (GMS2.2) gives this identity. Thus the low-degree mirror changes $h(s)$ to $h(1-s)$. On the normal coordinate $\lambda = s + 1$, the same source reflection changes λ to $3 - \lambda$, so its multiplier is $h(3 - \lambda)$. Both substitutions preserve M by its strip bounds. The geometric sign is retained separately: the antipodal map $-1/\bar{z}$ has degree minus one on the complex-oriented sphere and degree plus one on the equatorial loop; the holomorphic map $-1/z$ has degree plus one on the sphere and degree minus one on that loop. CSP/CDW compute their corresponding actions (chart swap, $-R, \varepsilon R$) on the finite model, with $\varepsilon = -1$ and $+1$, respectively. Neither geometric sign changes the stated conjugation of the generator.

In particular, a single substitution $\lambda \mapsto 1 - \lambda$ is not used in both degrees, and no symmetry of the particular cutoff function E is presumed. Conjugating a multiplier map by a specified mirror gives the map with its conjugated multiplier and the retained geometric signs. The vanishing in (GMS4.4) and the derived inverse in (GMS5.4) remain statements for the displayed domains and actions.

GMS11. What the calculation establishes and retains

This continuation constructs the M -action on the entire actual coefficient row, and proves the two full derived separations (GMS4.4). It then supplies the explicit actual map $\beta : \mathcal{B} \rightarrow \mathcal{J}_+$, the full-source representative (GMS8.3)–(GMS8.4), and their chain-homotopy inverse property after derived Hom from the complete original quotient. The same separator constructs the unique continuous splitting of every strict Fréchet M -extension with these two separated quotient terms. Each operation has its exact domain and topology above.

The normal extension e_+ itself remains nonzero, with the original-factor boundary (GMS6.4). The global conclusion is its zero pullback against the separated original quotient and the full derived comparison (GMS5.4), not deletion of its kernel or a splitting of $\mathcal{B} \rightarrow \mathcal{Q}_+$. Likewise, the logarithmic-modulus ranges of the two actual quotient spectra are $(0, 2)$ and $(2, 4)$, with scalars a^ρ and $a^{\rho+1}$ and all nilpotent coefficients. The multiplier argument controls the connecting maps between these specified full modules; it does not alter the original zero positions or assert a common pure weight.

The algebraic derived Hom statements use M -projective resolutions with algebraic Hom. The continuous splitting and source maps use explicit continuous formulas and quotient topologies. No claim about a general continuous derived category, an unconstructed extension of an arbitrary L -module to M , or a replacement of Deligne’s geometric hypotheses is needed. The new global chain contraction is the completed attempted step beyond CLP13’s finite coherent separation.

Two additional comparison boundaries are exact. First, the complete jet identities make c one on each actual original block of length m_ρ , zero on its translated block, and hence give those identities on the actual kernels $Q[(L - \rho)^r]$. In CLP13’s family stalk the arithmetic variable is $y = s - \rho$ with $y^{m_\rho} = 0$, so the same identity holds there even though the parameter variable has longer nilpotence. The identities do not make c one on an arbitrary test module $\mathbb{C}[t]/(t - \rho)^r$ when $r > m_\rho$; its higher Taylor coefficients can act there. Nor does restriction of scalars from M replace a $\mathbb{C}[t]$ -derived Hom calculation by an M -derived Hom calculation. Second, no cyclicity or algebra unit of $\mathcal{Q} = \mathcal{B}/\mathcal{J}$ has been used. The entire quotient c/F_+ exists by the vanishing orders, but it has not been declared to belong to M . The proof uses the actual map $c\mathcal{B} \subset \mathcal{J}_+$, not an unproved multiplier Bézout identity or a claim that $\mathcal{J}_+ = F_+M$.

The complete source extension and its two obstruction components — FOD0-FOD8

Proof source: CC_FULL_SOURCE_OBSTRUCTION_DECOMPOSITION.md.

The complete source extension and its two actual obstruction components

Complete derivation. Stable proof locators FOD0–FOD8.

FOD0. The constructed source and the question being answered

The supporting datum is $\tau\langle Z_1; \text{no } Z_2 \rangle$. All operations below take place on the coefficient spaces after the complete-history arithmetic reconstruction. The separate branch counters and their complete original arithmetic remain. No addition, midpoint, metric, vector coordinate or numerical origin is assigned to that support.

The connected argument, its correction table, the verbatim WU050–WU055 and WU061–WU065 passages, and the task-local prerequisite instructions were reread for this calculation. PC12 records the check. The actual question after GMS is what its separator does to the full source extension, retaining the intersection ideal rather than keeping only the already split quotient. This question follows WU065: use the new result together with the full previous work, construct the next maps, and calculate them.

Inputs, retained with complete proofs in the cumulative repository, are GSL4–GSL6 for the actual global entire separator; CW1–CW8 for the full original source, topology and boundary maps; CLP12 for the actual derived localization boundary; and GMS0–GMS11 for the full multiplier action and complete-module homotopies. Root read those operative sections and the complete current GMS proof. The calculation below derives its source extension directly; it does not depend on acceptance of a separate peer draft.

The original source geometry is Connes–Consani, [Schemes over \$\mathbb{F}_1\$ and zeta functions](#), arXiv:0903.2024v3, §5, with the signed-chart comparison in [arXiv:2609.00299v1](#). The distinct lifting mechanism in Deligne, [La conjecture de Weil. II](#), §3.6, is reconstructed in DC5–DC9 with its exact obstruction quotient. These are the previously recorded source readings, not a claim to have read new original sources in this continuation.

FOD1. Entire coefficients, both full ideals, and the original factors

Let M be the unital commutative algebra of entire h such that for every $A \geq 0$ there are C_A, N_A with

$$|h(x + it)| \leq C_A(1 + |t|)^{N_A} \quad (|x| \leq A). \quad (\text{FOD1.1})$$

Let $\mathcal{B}$ be the Fréchet space of entire F with

$$b_{A,N}(F) = \sup_{|\Re s| \leq A} (1 + |\Im s|)^N |F(s)| < \infty \quad \text{for every } A, N \geq 0. \quad (\text{FOD1.2})$$

Pointwise multiplication makes $\mathcal{B}$ an ideal in M ; a fixed h acts continuously by $b_{A,N}(hF) \leq C_A b_{A,N+N_A}(F)$.

Write $\mathcal{Z}$ for the distinct actual nontrivial zeros of the original ζ , each with multiplicity m_ρ . Their established open strip is $0 < \Re \rho < 1$. Put

$$\begin{aligned}\mathcal{J} &= \{F \in \mathcal{B} : \text{ord}_\rho F \geq m_\rho \forall \rho\}, \\ \mathcal{J}_+ &= \{F \in \mathcal{B} : \text{ord}_{\rho+1} F \geq m_\rho \forall \rho\}, \\ \mathcal{K} &= \mathcal{J} \cap \mathcal{J}_+, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}, \quad \mathcal{Q}_+ = \mathcal{B}/\mathcal{J}_+.\end{aligned}\tag{FOD1.3}$$

The derivative evaluations are continuous by Cauchy's formula on a larger strip, so the three ideals are closed. All their displayed quotient spaces carry their actual Fréchet quotient topologies.

The original source comparison is

$$\begin{aligned}\Theta a(s) &= \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, \quad \Theta^{-1} F(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} F(1/2 + it) u^{-it} dt, \\ \Sigma f(u) &= 2 \sum_{n \geq 1} f(nu), \quad \Theta \Sigma f(s) = \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v} \quad (\Re s > 1).\end{aligned}\tag{FOD1.4}$$

Here S consists of even Schwartz functions satisfying $f(0) = 0$ and $\int_{\mathbb{R}} f = 0$; the proved exact image is $J = \Sigma S = \Theta^{-1} \mathcal{J}$, with the continuous inverse Σ^{-1} . The coefficient space $A = \Theta^{-1} \mathcal{B}$ has all seminorms $\sup_{u > 0} (u^N + u^{-N}) |(u \partial_u)^j a(u)|$.

Retain the full source functions

$$\begin{aligned}f_0(v) &= \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}, \quad F_0(s) = \Theta \Sigma f_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \\ F_+(s) &= F_0(s-1) = \frac{(s-1)(s-2)}{8} \pi^{-(s-1)/2} \Gamma((s-1)/2) \zeta(s-1).\end{aligned}\tag{FOD1.5}$$

Both belong to $\mathcal{B} \subset M$. Their divisors are exactly $\mathcal{Z}$ and $\mathcal{Z} + 1$, respectively, with the complete multiplicities. They are not substituted for the original function:

$$\zeta(s) = 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\zeta'(s)/\zeta(s) = \sum_p \sum_{k \geq 1} (\log p) p^{-ks} \quad (\Re s > 1).\tag{FOD1.6}$$

The endpoint and trivial-zero comparisons retain

$$F_0(0) = F_0(1) = \frac{1}{8}, \quad F_0(-1) = \frac{\pi}{24}, \quad F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) \neq 0 \quad (r \geq 1).\tag{FOD1.7}$$

Every value and derivative of F_+ is the corresponding value and derivative of F_0 at $s-1$. At an original zero,

$$F_0(\rho + z) = z^{m_\rho} U_\rho(z), \quad U_\rho(z) = \frac{(\rho + z)(\rho + z - 1)}{8} \pi^{-(\rho+z)/2} \Gamma((\rho + z)/2) \frac{\zeta(\rho + z)}{z^{m_\rho}},\tag{FOD1.8}$$

and $U_\rho(0) \neq 0$. All products and divisions below use this complete local unit, not a replacement of it by one.

The proved separator is $c(s) = E_+(s) = 1 - E(s-1) \in M$, with

$$c-1 \in (s-\rho)^{m_\rho} \mathcal{O}_\rho, \quad c \in (s-\rho-1)^{m_\rho} \mathcal{O}_{\rho+1}.\tag{FOD1.9}$$

GSL constructs E by its full Gaussian Cauchy integral and proves the strip estimates; GMS3 recalls that construction. Hence

$$c\mathcal{B} \subset \mathcal{J}_+, \quad (1-c)\mathcal{B} \subset \mathcal{J}, \quad c_{\mathcal{Q}} = 1, \quad c_{\mathcal{Q}_+} = 0.\tag{FOD1.10}$$

FOD2. The complete source extension before discarding its kernel

Let q, q_+ be the two quotient maps. The actual joint map is

$$q_{\text{joint}} : \mathcal{B} \longrightarrow \mathcal{Q} \oplus \mathcal{Q}_+, \quad F \longmapsto (qF, q_+F). \quad (\text{FOD2.1})$$

It is onto: for representatives $f, g \in \mathcal{B}$, the function $cf + (1 - c)g$ has the requested two quotient images by (FOD1.10). Its kernel is exactly $\mathcal{K}$, and

$$\begin{aligned} \bar{q}_{\text{joint}} : \mathcal{B}/\mathcal{K} &\xrightarrow{\sim} \mathcal{Q} \oplus \mathcal{Q}_+, & [F]_{\mathcal{K}} &\longmapsto (qF, q_+F), \\ \bar{q}_{\text{joint}}^{-1} : \mathcal{Q} \oplus \mathcal{Q}_+ &\longrightarrow \mathcal{B}/\mathcal{K}, & ([f]_{\mathcal{J}}, [g]_{\mathcal{J}_+}) &\longmapsto [cf + (1 - c)g]_{\mathcal{K}} \end{aligned} \quad (\text{FOD2.2})$$

is a continuous inverse. Independence of representatives uses $c\mathcal{J} \subset \mathcal{K}$ and $(1 - c)\mathcal{J}_+ \subset \mathcal{K}$. To check continuity of the inverse, the continuous map on $\mathcal{B} \oplus \mathcal{B}$ kills $\mathcal{J} \oplus \mathcal{J}_+$ and therefore descends through that product quotient. Thus the full exact sequence is strict:

$$0 \longrightarrow \mathcal{K} \longrightarrow \mathcal{B} \xrightarrow{q_{\text{joint}}} \mathcal{Q} \oplus \mathcal{Q}_+ \longrightarrow 0. \quad (\text{FOD2.3})$$

Equation (FOD2.2) splits the quotient into two summands. It does not split (FOD2.3).

Pulling (FOD2.3) back along the two actual summand inclusions gives

$$\begin{aligned} e_{\text{low}} : \quad 0 \rightarrow \mathcal{K} \rightarrow \mathcal{J}_+ &\xrightarrow{q|_{\mathcal{J}_+}} \mathcal{Q} \rightarrow 0, \\ e_{\text{high}} : \quad 0 \rightarrow \mathcal{K} \rightarrow \mathcal{J} &\xrightarrow{q_+|_{\mathcal{J}}} \mathcal{Q}_+ \rightarrow 0. \end{aligned} \quad (\text{FOD2.4})$$

These are actual pullbacks: the zero normal quotient condition is precisely membership in $\mathcal{J}_+$, and the zero original quotient condition is precisely membership in $\mathcal{J}$.

The first row is strict with explicit quotient inverse $[f]_{\mathcal{J}} \mapsto [cf]_{\mathcal{K}}$. For $b \in \mathcal{J}_+$, $cb - b = -(1 - c)b \in \mathcal{K}$, which proves the reverse composition. The second row has inverse $[g]_{\mathcal{J}_+} \mapsto [(1 - c)g]_{\mathcal{K}}$, by the same two full jet conditions. All these maps are continuous by the actual multiplier estimates.

FOD3. Pushouts recover the original and normal rows exactly

Push the first row of (FOD2.4) out along $\mathcal{K} \hookrightarrow \mathcal{J}$. Its middle module and comparison map are

$$\mathcal{P}_0 = (\mathcal{J} \oplus \mathcal{J}_+)/\{(k, -k) : k \in \mathcal{K}\}, \quad [(i, b)] \longmapsto i + b \in \mathcal{B}. \quad (\text{FOD3.1})$$

This map is onto by $F = (1 - c)F + cF$. Its kernel is exactly the relation subspace, since $i + b = 0$ forces $i = -b \in \mathcal{K}$. Its continuous inverse is

$$F \longmapsto [(1 - c)F, cF]. \quad (\text{FOD3.2})$$

To verify the inverse on a pair (i, b) , subtract (i, b) from the pair in (FOD3.2) at $F = i + b$. The result is $(k, -k)$, where $k = -ci + (1 - c)b \in \mathcal{K}$. This also proves the inverse without selecting representatives of either quotient. The relation subspace is closed because $\mathcal{K}$ is closed, so all middle spaces have the stated Fréchet quotient topology.

Consequently the pushout is exactly

$$e_0 : \quad 0 \rightarrow \mathcal{J} \rightarrow \mathcal{B} \xrightarrow{q} \mathcal{Q} \rightarrow 0. \quad (\text{FOD3.3})$$

Interchanging the ideals and replacing c by $1 - c$ gives, with all signs the same,

$$(\mathcal{J}_+ \oplus \mathcal{J}) / \{(k, -k) : k \in \mathcal{K}\} \xrightarrow{\sim} \mathcal{B}, \quad [(i, b)] \mapsto i + b,$$

$$e_+ : 0 \rightarrow \mathcal{J}_+ \rightarrow \mathcal{B} \xrightarrow{q_+} \mathcal{Q}_+ \rightarrow 0. \quad (\text{FOD3.4})$$

Its inverse is $F \mapsto [(cF, (1 - c)F)]$. Thus e_0 and e_+ are the specified pushouts of the same actual full-source extension.

FOD4. Every multiplier that kills each component

Use algebraic M -module Ext , with the strict continuous conclusions proved separately below. Define two ideals of the multiplier ring itself:

$$\mathfrak{a} = \{h \in M : \text{ord}_\rho h \geq m_\rho \ \forall \rho\}, \quad \mathfrak{a}_+ = \{h \in M : \text{ord}_{\rho+1} h \geq m_\rho \ \forall \rho\}. \quad (\text{FOD4.1})$$

These are not renamed copies of $\mathcal{J}, \mathcal{J}_+$: their functions obey polynomial strip bounds, not necessarily rapid strip decay. Precisely, $\mathcal{J} = \mathcal{B} \cap \mathfrak{a}$ and $\mathcal{J}_+ = \mathcal{B} \cap \mathfrak{a}_+$.

For an actual row $0 \rightarrow N \rightarrow X \rightarrow Y \rightarrow 0$, the pushout along multiplication by h has middle module

$$P_h = (N \oplus X) / \{(hn, -n) : n \in N\}. \quad (\text{FOD4.2})$$

It splits as an M -extension exactly when there is an M -linear $r : X \rightarrow N$ with $r|_N = h$. In one direction, such r gives the retraction $[(n, x)] \mapsto n + r(x)$, and hence a splitting. Conversely, a split extension has an M -linear retraction onto its kernel, obtained by subtracting its section; composing that retraction with $x \mapsto [(0, x)]$ gives r , because $[(0, n)] = [(hn, 0)]$.

Apply this first to e_{low} , with $N = \mathcal{K}$ and $X = \mathcal{J}_+$. Put

$$G(s) = F_0(s)F_+(s) \in \mathcal{K} \subset M. \quad (\text{FOD4.3})$$

This is a nonzero entire function, with all factors from (FOD1.5). For $b \in \mathcal{J}_+$, $Gb \in \mathcal{K}$. Therefore any r supplied by a splitting satisfies

$$Gr(b) = r(Gb) = hGb. \quad (\text{FOD4.4})$$

Cancellation is valid as an identity of entire functions: off the discrete zeros of G the two functions are equal, and the identity theorem extends that equality everywhere. Hence $r(b) = hb$.

In particular $hF_+ \in \mathcal{K}$. At each original ρ , the function F_+ is a unit, since $-1 < \Re(\rho - 1) < 0$ and the divisor of F_0 is $\mathcal{Z}$. Thus h vanishes to order at least m_ρ at every ρ , proving $h \in \mathfrak{a}$. Conversely, $h \in \mathfrak{a}$ gives $h\mathcal{J}_+ \subset \mathcal{K}$ by the full Leibniz formula, so $r(b) = hb$ constructs the splitting. We have proved

$$\text{Ann}_M(e_{\text{low}}) = \mathfrak{a}. \quad (\text{FOD4.5})$$

The identical argument for e_{high} uses F_0 , which is a unit at every $\rho + 1$, and proves

$$\text{Ann}_M(e_{\text{high}}) = \mathfrak{a}_+. \quad (\text{FOD4.6})$$

For the original row e_0 , use $G = F_0$ in the cancellation argument with $N = \mathcal{J}, X = \mathcal{B}$. It forces $r(b) = hb$. The function $b(s) = e^{s^2}$ belongs to $\mathcal{B}$, since its modulus is $e^{x^2 - t^2}$ on $s = x + it$, and it is nowhere zero. Thus $hb \in \mathcal{J}$ forces all the original zero jets of h . The converse is again multiplication. Replacing $F_0, \mathcal{J}$ by $F_+, \mathcal{J}_+$ proves the normal result. Therefore

$$\text{Ann}_M(e_0) = \mathfrak{a}, \quad \text{Ann}_M(e_+) = \mathfrak{a}_+. \quad (\text{FOD4.7})$$

Every splitting in the sufficient direction is continuous and explicit:

$$[(n, x)] \mapsto (n + hx, [x]), \quad (n_0, [x]) \mapsto [(n_0 - hx, x)]. \quad (\text{FOD4.8})$$

The second formula is independent of the lift x , since changing it by n changes the pair by $(-hn, n)$, a relation. Its continuity follows by descending the continuous map on $N \oplus X$ through the quotient $N \oplus X \rightarrow N \oplus Y$. The relation subspace in (FOD4.2) is closed: convergence of $(hn_\nu, -n_\nu)$ forces convergence of n_ν in the closed space N , and continuity of multiplication gives the first coordinate. Thus (FOD4.8) is an actual Fréchet isomorphism. Necessity was proved even for algebraic splittings, so these annihilator criteria are also exact for continuous M -splittings.

FOD5. The full obstruction decomposition and the separator

Let e_{joint} denote the Yoneda class of (FOD2.3). Resolving the two summands separately and taking their direct sum gives the exact identification

$$\begin{aligned} \text{Ext}_M^1(\mathcal{Q} \oplus \mathcal{Q}_+, \mathcal{K}) &\simeq \text{Ext}_M^1(\mathcal{Q}, \mathcal{K}) \oplus \text{Ext}_M^1(\mathcal{Q}_+, \mathcal{K}), \\ e_{\text{joint}} &= (e_{\text{low}}, e_{\text{high}}). \end{aligned} \quad (\text{FOD5.1})$$

The identification is pullback along the two summand inclusions, which were calculated in (FOD2.4). Thus

$$\text{Ann}_M(e_{\text{joint}}) = \mathfrak{a} \cap \mathfrak{a}_+. \quad (\text{FOD5.2})$$

By (FOD1.9), $c \in \mathfrak{a}_+$, $1 - c \in \mathfrak{a}$, so these two multiplier ideals are comaximal. Their intersection equals their product: if z belongs to both, $z = zc + z(1 - c)$ is a sum of two products, and the reverse inclusion follows from their being ideals. Consequently the actual cyclic image of the full extension has the exact description

$$Me_{\text{joint}} \simeq M/(\mathfrak{a} \cap \mathfrak{a}_+) \xrightarrow{\sim} M/\mathfrak{a} \oplus M/\mathfrak{a}_+. \quad (\text{FOD5.3})$$

The first isomorphism sends the class of h to he_{joint} , with kernel just proved. The second has inverse $([h], [k]) \mapsto [ch + (1 - c)k]$; independence and both compositions follow from the same full jet identities as in (FOD2.2).

The complete resulting actions are

$$\begin{aligned} ce_{\text{joint}} &= (e_{\text{low}}, 0), & (1 - c)e_{\text{joint}} &= (0, e_{\text{high}}), \\ ce_{\text{low}} &= e_{\text{low}}, & ce_{\text{high}} &= 0, \\ ce_0 &= e_0, & ce_+ &= 0. \end{aligned} \quad (\text{FOD5.4})$$

None of $e_{\text{low}}, e_{\text{high}}, e_0, e_+$ is zero: the original zeta has a nontrivial zero, so each ideal in (FOD4.1) is proper, and its annihilator does not contain one. A direct witness for e_{low} is

$$v_\rho(s) = F_0(s)/(s - \rho), \quad qv_\rho \neq 0, \quad (L - \rho)qv_\rho = 0. \quad (\text{FOD5.5})$$

The function is entire, rapidly decreasing on closed vertical strips, and has exact order $m_\rho - 1$ at ρ . Any M -section into $\mathcal{I}_+$ would send this nonzero quotient class to an entire function killed by $s - \rho$, which must be zero. The normal witness is $F_+(s)/(s - \rho - 1)$. These use actual zeros and are not hypothetical alternative arithmetic.

The pushouts in FOD3 induce isomorphisms on the respective cyclic class modules $Me_{\text{low}} \rightarrow Me_0$ and $Me_{\text{high}} \rightarrow Me_+$: they are onto by their displayed generators, and injective by the equal annihilators (FOD4.5)-(FOD4.7). No claim of an isomorphism of their entire Ext groups follows.

FOD6. The exact original quotient retained in the normal ideal

The strict map $q_{\text{res}} = q|_{\mathcal{J}_+} : \mathcal{J}_+ \rightarrow \mathcal{Q}$ satisfies

$$q_{\text{res}}\beta = q, \quad \beta : \mathcal{B} \rightarrow \mathcal{J}_+, \quad \beta(F) = cF. \quad (\text{FOD6.1})$$

Both equalities are for the actual source maps, not an inverse of q_{res} . Their continuous strong transposes satisfy

$$\beta' q'_{\text{res}} = q'. \quad (\text{FOD6.2})$$

Indeed composition with any continuous linear functional proves the identity; continuity for strong duals follows because a continuous linear map sends bounded sets to bounded sets. The kernel of q_{res} remains $\mathcal{K}$, and FOD4-FOD5 show exactly why c does not split this row.

For any continuous pairing P already defined on $\mathcal{Q}$ and any of its established dual receivers, composing its first argument with $q_{\text{res}}\beta$ gives the original value $P(qF, \cdot)$ by (FOD6.1). This is an identity for existing pairings; it introduces neither a positive form nor a replacement of the complete quotient by values alone.

Define $V : \mathcal{B}_s \rightarrow \mathcal{B}_\lambda$ by $(VF)(\lambda) = F(\lambda - 1)$. This is the actual coordinate transport from the original normal variable $s = \lambda - 1$; it sends $\mathcal{J}$ to $\mathcal{J}_+$. In that variable the representative $\Theta^{-1}V^{-1}\beta V\Theta : A \rightarrow J$ is exactly

$$a \mapsto \Theta^{-1}[(1 - E(s))\Theta a(s)](u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} (1 - E(1/2 + it))\Theta a(1/2 + it)u^{-it} dt, \quad (\text{FOD6.3})$$

followed by the proved inverse $\Sigma^{-1} : J \rightarrow S$ when a Schwartz representative is wanted. The original normal multiplier is $h(s + 1)$, and the dilation is a^{s+1} , retaining its factor a . Translating to λ changes that action to $h(\lambda)$. The map q_{res} in (FOD6.1) is stated in this translated coordinate, as are the two ideals in FOD1.3. Its source-coordinate identity is $q_{\text{res}}\beta V = qV$; thus (FOD6.3) does not silently replace the translated quotient by the original source-variable quotient.

FOD7. The actual localization boundary and the two actions on it

The proved actual coefficient complexes are

$$D_J = (W \xrightarrow{r_+ - r_-} J \xrightarrow{0} J(-1)), \quad D_F = (W \xrightarrow{r_+ - r_-} A \xrightarrow{0} A(-1)), \quad D_G = (0 \rightarrow Q \xrightarrow{0} Q(-1)), \quad (\text{FOD7.1})$$

in degrees 0, 1, 2, with zero second differential. All four endpoint characters in W remain $h(0), h(1), h(1), h(0)$. GMS9 proves simultaneous M -linear quasi-isomorphisms

$$D_J \simeq Z[0] \oplus \mathcal{J}_+[-2], \quad D_F \simeq Z[0] \oplus \mathcal{Q}[-1] \oplus \mathcal{B}[-2], \quad D_G = \mathcal{Q}[-1] \oplus \mathcal{Q}_+[-2]. \quad (\text{FOD7.2})$$

Here the normal terms have been transported by Θ and V , and $Z = \ker(r_+ - r_-)$. These are the already constructed quasi-isomorphisms with explicitly contractible kernels, not inserted sections of $\mathcal{B} \rightarrow \mathcal{Q}$.

The actual connecting morphism $\delta : D_G \rightarrow D_J[1]$ is zero on the low summand and is $e_+[-2]$ on the normal summand. Let

$$\mathcal{O}_{\text{loc}} = \{h\delta : h \in M\} \subset \text{Hom}_{D(M)}(D_G, D_J[1]), \quad (\text{FOD7.3})$$

where $h\delta$ is central postcomposition on the target, equivalently precomposition on the source. Taking the normal inclusion and projection in (FOD7.2) gives

$$h\delta = 0 \iff he_+ = 0, \quad \mathcal{O}_{\text{loc}} \simeq M/\mathfrak{a}_+, \quad c\delta = 0. \quad (\text{FOD7.4})$$

The forward implication is obtained by that inclusion and projection; the reverse follows from the zero low component. Thus this is the cyclic image of the actual localization boundary, with its full normal ideal retained.

For every derived M -morphism $f : \mathcal{Q}[k] \rightarrow D_G$, centrality and $c_Q = 1$ give

$$\delta f = \delta f c_{\mathcal{Q}[k]} = \delta c_{D_G} f = c_{D_J[1]} \delta f = 0. \quad (\text{FOD7.5})$$

This proves the actual lifting obstruction for this specified derived source test is zero. The exact Hom sequence of the actual localization triangle therefore gives a lift $\tilde{f} : \mathcal{Q}[k] \rightarrow D_F$. More concretely, (FOD7.2) and the complete contraction of $\text{RHom}_M(\mathcal{Q}, \mathcal{Q}_+)$ in GMS4 show that f has zero normal component in the derived category; retaining its low component as the $\mathcal{Q}[-1]$ component of D_F constructs a lift. The complete set of lifts is a coset of the image of $\text{Hom}_{D(M)}(\mathcal{Q}[k], D_J) \rightarrow \text{Hom}_{D(M)}(\mathcal{Q}[k], D_F)$. No unique cochain representative is asserted.

The action used in (FOD7.3) must be distinguished from conjugation. For the actual arithmetic operator T_a , equivariance is

$$(T_a)_{D_J[1]} \delta = \delta (T_a)_{D_G}.$$

Consequently

$$(T_a)_{D_J[1]} \delta (T_a)_{D_G}^{-1} = \delta. \quad (\text{FOD7.6})$$

The boundary is fixed under this conjugation. Its cyclic module under postcomposition in (FOD7.3) is different; a weight calculated for that module is not a conjugation weight of the boundary map.

FOD8. Exact comparison with the lifting target

The construction now retains three different facts simultaneously. The quotient $\mathcal{B}/\mathcal{K}$ splits into its original and normal pieces by (FOD2.2). The source extension before that quotient has two nonzero components, and the separator retains one while annihilating the other, exactly as in (FOD5.4). The actual derived localization boundary uses the normal component, so its composition with every displayed complete-original-quotient source class vanishes by (FOD7.5).

The two full cyclic class modules are not declared to be the full coefficient ideals or the full quotients $\mathcal{Q}, \mathcal{Q}_+$. Their exact descriptions are $M/\mathfrak{a}$ and $M/\mathfrak{a}_+$, with every jet retained. The calculation is an actual boundary calculation, not the splitting of an independently chosen extension. It also proves that moving an original class into the full normal ideal through β does not erase the original class or its existing pairing.

Deligne's precise obstruction receiver is $\text{im } \partial/\partial j(K)$, with its specialization and inertia maps, DC5.3–DC5.4. The module $\mathcal{O}_{\text{loc}}$ in (FOD7.3) is instead the cyclic image of a specified derived boundary under central coefficients. Equations (FOD7.4)–(FOD7.5) prove its actual effect without identifying those two receivers by their names. No new mixed étale realization, common modulus of original zeros, or RH conclusion is inferred. The source and full kernel comparisons proved above remain available for constructing that further geometric identification.

The exact normal-extension annihilator and full class-module resolvent — NEA0-NEA10

Proof source: CC_NORMAL_EXTENSION_ANNIHILATOR.md.

The exact annihilator of the actual normal extension

Complete independent derivation. Stable proof locators NEA0-NEA10.

NEA0. Construction stage and source coverage

The support remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. The present operations act on the complete arithmetic coefficient receiver already constructed after whole-history reconstruction. The two branches retain separate counters. No addition, subtraction, midpoint, metric, or numerical coordinate is assigned to τ .

The current READ_FIRST_USER_CONSTRUCTION.md, correction-precedence table and latest stages of USER_ARGUMENT_RECONSTRUCTION.md, and PC10-PC11 prerequisite records were recalled for this continuation. The complete correction corpus remains the governing record. WU061 concerns the actual lifting map; WU062 requires checking every operation against the construction; WU065 requires a concrete attempted next step using the accumulated findings. Here the input is the actual nonzero normal extension retained by GMS, and the step is to identify exactly its multiplier annihilator and then the module generated by the actual connecting morphism.

The direct mathematical prerequisites are CC_GLOBAL_MULTIPLIER_SEPARATION.md, GMS0-GMS11, completely derived and independently reviewed at SHA256 3164C3AE00FBFBAF17BCE809D48B7D0E56FF8A4C0991FDB374A4B11DD18BEC59; CC_LOCALIZATION_POLYNOMIAL_WEIGHTS.md, CLP12-CLP13; and CC_ACTUAL_WEIGHT_LIFT_COMPARISON.md, CW1-CW8. The full current GMS proof and its actual normal row are the source for the operations below. The exact source theorem and continuous Schwartz inverse are the SSI/ESI results received through CW1 and GMS2/GMS8. The human-source relation is the previously read Connes-Consani §5 construction and the Deligne §3.6 comparison recorded in those notes. No new human-source reading or new geometric specialization map is claimed here.

NEA1. The actual ring, full ideal, and original source

Use the unital commutative ring

$$M = \{h \in \mathcal{O}(\mathbb{C}) : \forall A \geq 0 \exists C_A > 0, N_A \in \mathbb{N}, |h(x + it)| \leq C_A(1 + |t|)^{N_A} \text{ for } |x| \leq A\}, \quad (\text{NEA1.1})$$

and the actual Fréchet module

$$\mathcal{B} = \{F \in \mathcal{O}(\mathbb{C}) : b_{A,N}(F) = \sup_{|\Re \lambda| \leq A} (1 + |\Im \lambda|)^N |F(\lambda)| < \infty \text{ for all } A, N \geq 0\}. \quad (\text{NEA1.2})$$

Let $\mathcal{Z}$ be the original nontrivial-zero divisor of ζ , with multiplicities m_ρ . Its translated divisor is

$$\mathcal{D}_+ = \sum_{\rho \in \mathcal{Z}} m_\rho [\rho + 1], \quad 1 < \Re(\rho + 1) < 2. \quad (\text{NEA1.3})$$

The support notation counts each distinct zero once, with its separately retained multiplicity. Define

$$\begin{aligned} \mathcal{J}_+ &= \{F \in \mathcal{B} : F^{(j)}(\rho + 1) = 0 \text{ for } 0 \leq j < m_\rho, \rho \in \mathcal{Z}\}, \\ \mathcal{Q}_+ &= \mathcal{B}/\mathcal{J}_+, \\ K_+ &= \{h \in M : h^{(j)}(\rho + 1) = 0 \text{ for } 0 \leq j < m_\rho, \rho \in \mathcal{Z}\}. \end{aligned} \quad (\text{NEA1.4})$$

Thus $\mathcal{J}_+ = \mathcal{B} \cap K_+$; K_+ is an ideal of M , whereas $\mathcal{J}_+$ is the full closed ideal within the actual source module. They are not identified with each other. Every multiplier acts continuously on $\mathcal{B}$ and its closed invariant subspaces and quotients, since

$$b_{A,N}(hF) \leq C_A b_{A,N+N_A}(F). \quad (\text{NEA1.5})$$

The derivative product formula preserves all the vanishing orders.

The complete original transform is

$$\begin{aligned} F_0(s) &= \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) = C_0(s) \zeta(s), \\ F_+(\lambda) &= F_0(\lambda-1) = \frac{(\lambda-1)(\lambda-2)}{8} \pi^{-(\lambda-1)/2} \Gamma((\lambda-1)/2) \zeta(\lambda-1). \end{aligned} \quad (\text{NEA1.6})$$

Both are the actual entire source functions from GMS2. In particular $F_+ \in \mathcal{J}_+ \subset \mathcal{B} \subset M$, it is nonzero, and its zero divisor is exactly $\mathcal{D}_+$. The original arithmetic is still

$$\zeta(s) = 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'}{\zeta}(s) = \sum_p \sum_{k \geq 1} (\log p) p^{-ks} \quad (\Re s > 1). \quad (\text{NEA1.7})$$

Thus the unit and every prime-power repetition remain. The exceptional entire values of the full source factor are

$$\begin{aligned} F_+(1) &= F_+(2) = \frac{1}{8}, \quad F_+(0) = \frac{\pi}{24}, \\ F_+(1-2r) &= \frac{r(2r+1)(-1)^r \pi^r}{2^r r!} \zeta'(-2r) \\ &= F_+(2+2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0 \quad (r \geq 1). \end{aligned} \quad (\text{NEA1.8})$$

They retain the pole and Gamma cancellations and the trivial-zero derivatives; those contributions are not replaced by extra zeros of F_+ .

The object under study is exactly

$$e_+ : 0 \longrightarrow \mathcal{J}_+ \xrightarrow{\iota} \mathcal{B} \xrightarrow{q_+} \mathcal{Q}_+ \longrightarrow 0, \quad [e_+] \in \text{Ext}_M^1(\mathcal{Q}_+, \mathcal{J}_+). \quad (\text{NEA1.9})$$

We write e_+ for its class when no ambiguity occurs. It is the translated normal row of CW/CLP, not a separately chosen extension. The M -action on the unshifted normal source is $h(s+1)$; translation $VF(\lambda) = F(\lambda-1)$ gives the displayed $h(\lambda)$ -action and retains the dilation factor a^{s+1} .

NEA2. Every multiplier pushout, with its actual topology

For $h \in M$, pushing (NEA1.9) out along multiplication by h on $\mathcal{J}_+$ gives

$$\mathcal{P}_h = (\mathcal{J}_+ \oplus \mathcal{B}) / \{(hj, -j) : j \in \mathcal{J}_+\}, \quad i_h(i) = [(i, 0)], \quad p_h([(i, b)]) = q_+(b). \quad (\text{NEA2.1})$$

The relation subspace is closed: convergence of $(hj_\nu, -j_\nu)$ in the product implies convergence of j_ν in the closed subspace $\mathcal{J}_+ \subset \mathcal{B}$, and continuity of multiplication identifies the first limit as multiplication by h of that limit. Thus $\mathcal{P}_h$ is Fréchet. Each fixed multiplier acts continuously on it, because it preserves the relation subspace.

The maps form the exact row

$$0 \longrightarrow \mathcal{J}_+ \xrightarrow{i_h} \mathcal{P}_h \xrightarrow{p_h} \mathcal{Q}_+ \longrightarrow 0. \quad (\text{NEA2.2})$$

Indeed i_h is injective since a relation with second coordinate zero has $j = 0$; p_h is onto; and when $b \in \mathcal{J}_+$, the exact identity

$$[(i, b)] = [(i + hb, 0)] \quad (\text{NEA2.3})$$

identifies its kernel with the image of i_h . These are continuous maps. The Fréchet open mapping theorem applied to the continuous bijection $\mathcal{J}_+ \rightarrow \ker p_h$ makes it a topological isomorphism; applied to the continuous surjection p_h , it gives the quotient topology. The hypotheses hold because $\ker p_h$ is closed and all the spaces are Fréchet. Thus this is a strict extension, not merely an algebraic presentation endowed with an unspecified topology. Its Yoneda class is he_+ .

We now prove the splitting criterion directly. If (NEA2.2) has any algebraic M -linear section s , define

$$R = i_h^{-1}(1 - sp_h) : \mathcal{P}_h \rightarrow \mathcal{J}_+, \quad r(b) = R([(0, b)]). \quad (\text{NEA2.4})$$

Since $(1 - sp_h)x$ lies in the kernel, the definition is valid, and $Ri_h = 1$. Formula (NEA2.3) implies

$$r|_{\mathcal{J}_+} = h|_{\mathcal{J}_+}. \quad (\text{NEA2.5})$$

Conversely, if an M -linear map $r : \mathcal{B} \rightarrow \mathcal{J}_+$ satisfies (NEA2.5), the map $[(i, b)] \mapsto i + r(b)$ is well-defined, and a section is

$$q \mapsto [(-r(b), b)] \quad (q_+(b) = q). \quad (\text{NEA2.6})$$

Changing b by $j \in \mathcal{J}_+$ changes this pair by $(-hj, j)$, a relation. Thus

$$he_+ = 0 \iff \text{there is an } M\text{-map } r : \mathcal{B} \rightarrow \mathcal{J}_+ \text{ extending } h|_{\mathcal{J}_+}. \quad (\text{NEA2.7})$$

This proves the specific endomorphism-lifting assertion usually expressed by the connecting homomorphism in Hom-Ext, without assuming it as an uncomputed criterion.

NEA3. Necessity, sufficiency, and uniqueness

Suppose r exists in (NEA2.7). For every $b \in \mathcal{B}$, multiplication by the actual original source $F_+ \in M$ gives

$$F_+ r(b) = r(F_+ b) = hF_+ b. \quad (\text{NEA3.1})$$

Here $F_+ b \in \mathcal{J}_+$: F_+ has all the required zeros, and it is an M -multiplier of $\mathcal{B}$. The middle equality consequently uses the prescribed restriction (NEA2.5). The nonzero entire function F_+ can be cancelled from this identity: outside its discrete zero set the other two factors agree, hence their entire continuations agree. Thus necessarily

$$r(b) = hb \quad \text{for every } b \in \mathcal{B}, \quad h\mathcal{B} \subset \mathcal{J}_+. \quad (\text{NEA3.2})$$

To read the precise multiplicities from this inclusion, take the single zero-free source-space test function

$$g(\lambda) = e^{\lambda^2} \in \mathcal{B}. \quad (\text{NEA3.3})$$

It belongs to the existing coefficient space because $b_{A,N}(g) \leq e^{A^2} \sup_t (1 + |t|)^N e^{-t^2} < \infty$. It is a unit in each local holomorphic ring. Thus $hg \in \mathcal{J}_+$ forces $\text{ord}_{\rho+1} h \geq m_\rho$ for every actual zero. This

test is a function in the already constructed source space, not an alternative arithmetic or an alternative zeta function.

Conversely, these full vanishing orders make $hb \in \mathcal{J}_+$ for every $b \in \mathcal{B}$ by the derivative product formula. The map $r_h(b) = hb$ is continuous by (NEA1.5), so (NEA2.6) gives a continuous splitting. We have proved

$$\boxed{\text{Ann}_M(e_+) = K_+ = \{h \in M : \text{ord}_{\rho+1} h \geq m_\rho \text{ for every } \rho \in \mathcal{Z}\}.} \quad (\text{NEA3.4})$$

The same source-space test shows

$$K_+ = \text{Ann}_M(\mathcal{Q}_+), \quad (\text{NEA3.5})$$

since annihilation of the quotient is exactly the inclusion $h\mathcal{B} \subset \mathcal{J}_+$.

For every $h \in K_+$, the full continuous isomorphism of its pushout with the split row is

$$\Phi_h([(i, b)]) = (i + hb, q_+(b)), \quad \Phi_h^{-1}(i_0, q) = [(i_0 - hb, b)] \quad (q_+(b) = q). \quad (\text{NEA3.6})$$

The first coordinate is in the full $\mathcal{J}_+$. Both formulas are independent of representatives by (NEA2.1). Their composites are identities by substitution. For continuity of the inverse, the map $(i_0, b) \mapsto [(i_0 - hb, b)]$ is continuous and kills $\{0\} \oplus \mathcal{J}_+$; it therefore descends through the product quotient. The resulting section is $q \mapsto [(-hb, b)]$.

Every such section is unique, even among algebraic M -linear sections. Indeed

$$\text{Hom}_M(\mathcal{Q}_+, \mathcal{J}_+) = 0, \quad \text{Hom}_M(\mathcal{Q}_+, \mathcal{B}) = 0. \quad (\text{NEA3.7})$$

To prove both statements, observe that F_+ annihilates $\mathcal{Q}_+$ and acts injectively by multiplication on either entire-function target. For an M -map v , the identity $F_+ v(q) = v(F_+ q) = 0$ therefore forces $v = 0$. The difference of two sections of (NEA2.2) is a map into its kernel, so it is zero. Consequently any algebraic splitting that exists is automatically the explicit continuous one in (NEA3.6); no continuity of a hypothetical algebraic section was assumed in the necessity proof.

The actual extension is nonzero: $1 \notin K_+$, since the actual divisor is nonempty, as established in CW8. Thus (NEA3.4) includes $e_+ \neq 0$. It also proves that no classification of arbitrary endomorphisms of $\mathcal{B}$, no principal-ideal assertion, and no unproved growth bound for h/F_+ is needed.

NEA4. Pullback of every multiplier on the quotient

For clarity, lifting the quotient endomorphism $h_{\mathcal{Q}_+}$ is a different presentation of the same killed class. Its actual pullback is

$$\mathcal{E}_h = \{(b, q) \in \mathcal{B} \oplus \mathcal{Q}_+ : q_+(b) = hq\}, \quad i'_h(j) = (j, 0), \quad p'_h(b, q) = q. \quad (\text{NEA4.1})$$

It is a closed subspace of the product, with its Fréchet topology, and is a strict exact extension by the same open-mapping argument as in NEA2. There is an explicit continuous M -linear isomorphism from the pushout to this pullback:

$$\mathcal{P}_h \longrightarrow \mathcal{E}_h, \quad [(i, b)] \longmapsto (i + hb, q_+(b)). \quad (\text{NEA4.2})$$

For arbitrary h , the first coordinate here is in $\mathcal{B}$, not asserted to be in $\mathcal{J}_+$. It has the required quotient $hq_+(b)$. The relations are killed. The map is onto: if $(b_0, q) \in \mathcal{E}_h$, choose a lift b of q ; then $i = b_0 - hb \in \mathcal{J}_+$, and $[(i, b)]$ maps to (b_0, q) . Its kernel is exactly the relation subspace, by (NEA2.3). Thus it is a continuous bijection of Fréchet spaces and its inverse is continuous by the open mapping theorem. It is the identity on the displayed kernel and quotient. This proves concretely that pushout and pullback define the same scalar action he_+ .

A section of (NEA4.1) is equivalent to an M -map $v : \mathcal{Q}_+ \rightarrow \mathcal{B}$ with $q_+ v = h_{\mathcal{Q}_+}$. By (NEA3.7) the only possible v is zero. Hence this lift exists precisely when $h \in K_+$, and then the section is $q \mapsto (0, q)$. Under (NEA4.2) it is exactly $q \mapsto [(-hb, b)]$. This does not identify the nonzero source-return map $b \mapsto hb : \mathcal{B} \rightarrow \mathcal{J}_+$ with the zero quotient-to-source lift; their domains and purposes are different and their exact relation is (NEA4.2).

NEA5. Full multiplicity probes and exact residual divisors

Fix an actual $\lambda_0 = \rho + 1$, put $z = \lambda - \lambda_0$, and retain $m = m_\rho$. The full local source factor is

$$F_+(\lambda_0 + z) = z^m b_\rho(z), \quad b_\rho(z) = C_0(\rho + z) \frac{\zeta(\rho + z)}{z^m}, \quad b_\rho(0) = C_0(\rho) \frac{\zeta^{(m)}(\rho)}{m!} \neq 0. \quad (\text{NEA5.1})$$

For example all its Taylor coefficients, without dropping the completion derivatives, are

$$[z^j]b_\rho(z) = \sum_{\ell=0}^j \frac{C_0^{(\ell)}(\rho)}{\ell!} \frac{\zeta^{(m+j-\ell)}(\rho)}{(m+j-\ell)!}. \quad (\text{NEA5.2})$$

Here $C_0(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)/8$ is the full function in (NEA1.6).

Division by a fixed polynomial preserves M whenever the numerator vanishes at every root to the needed order. To verify this, remove fixed small disks around the finitely many roots; on the complement the reciprocal polynomial has at most a fixed polynomial bound, and in each disk the entire quotient is bounded on a compact set. The same assertion holds for $\mathcal{B}$, retaining all its rapid strip seminorms. In particular $M/(z^k)$ is exactly $\mathbb{C}[z]/z^k$: the Taylor polynomial supplies every jet, and a zero jet is divisible by z^k with quotient in M .

For $1 \leq k \leq m$, the actual full source gives the M -linear injection

$$\iota_{\rho,k} : M/(z^k) \longrightarrow \mathcal{Q}_+, \quad [a] \longmapsto \left[a \frac{F_+}{z^k} \right]. \quad (\text{NEA5.3})$$

The source function F_+/z^k is entire and in $\mathcal{B}$; at the chosen zero it has exact order $m - k$, and at every other zero it retains the required order. Its product with a lies in $\mathcal{J}_+$ exactly when a has a zero of order at least k at λ_0 . This proves both well-definedness and injectivity, with every other zero still retained.

The two free M -modules in $0 \rightarrow M \xrightarrow{z^k} M \rightarrow M/(z^k) \rightarrow 0$ form a resolution, since M is a ring of entire functions and multiplication by z^k is injective. Pulling the class he_+ back along (NEA5.3) gives exactly

$$\iota_{\rho,k}^*(he_+) = [hF_+] \in \text{Ext}_M^1(M/(z^k), \mathcal{J}_+) = \mathcal{J}_+/z^k \mathcal{J}_+. \quad (\text{NEA5.4})$$

To check the positive sign, use the lift F_+/z^k of the generator in the original extension. Its relation is $z^k(F_+/z^k) = F_+$; pushing its kernel by h changes that relation to hF_+ . This is the positive boundary convention used in CLP12.

There is an exact full-factor coordinate and inverse

$$\mathcal{J}_+/z^k \mathcal{J}_+ \xrightarrow{\sim} \mathbb{C}[z]/z^k, \quad [j] \longmapsto \text{Tay}_{\lambda_0, < k}(j/F_+), \quad [p] \longmapsto [F_+p]. \quad (\text{NEA5.5})$$

The quotient j/F_+ in this formula is taken locally at λ_0 , where (NEA5.1) is the retained unit comparison. If its first k coefficients vanish, then j/z^k is entire, lies in $\mathcal{B}$, and lies in $\mathcal{J}_+$: at the chosen point it still vanishes to order at least m , and at the other points division is by a unit. This proves the

kernel claim. Multiplication by F_+ times a polynomial proves surjectivity and the two inverse identities. Derivative evaluations are continuous on the actual Fréchet ideal. Thus this finite-dimensional coordinate is continuous as well; no global claim that $j/F_+ \in M$ is made.

Under (NEA5.5), the boundary (NEA5.4) is the complete k -jet of h . In particular the $k = m$ probe vanishes precisely when $\text{ord}_{\lambda_0} h \geq m$. The eigenvector probe $k = 1$ detects only $h(\lambda_0)$ and would miss higher multiplicities. If $v = \text{ord}_{\lambda_0} h$, the local class has exact z -nilpotence exponent $\max(k - v, 0)$, with exponent zero meaning the class is zero. Indeed its first nonzero Taylor term is a nonzero coefficient times z^v , and multiplication by z raises this order by exactly one until it reaches k .

For the complete global class the exact residual annihilator is consequently

$$\text{Ann}_M(h e_+) = (K_+ : h) = \{a \in M : \text{ord}_{\rho+1} a \geq \max(m_\rho - \text{ord}_{\rho+1} h, 0) \text{ for every } \rho\}. \quad (\text{NEA5.6})$$

This follows also directly by applying (NEA3.4) to ah . When $h = 0$, all residual multiplicities are zero and the annihilator is M . This classifies the effect of every multiplier on the actual extension, retaining its entire divisor rather than only finitely many tests.

NEA6. The actual cyclic module of extension classes

Define the actual image

$$\mathcal{N}_+ = M e_+ \subset \text{Ext}_M^1(\mathcal{Q}_+, \mathcal{J}_+). \quad (\text{NEA6.1})$$

It is the image of the connecting homomorphism on the specified multiplier endomorphisms of $\mathcal{J}_+$. No claim that these are all endomorphisms is needed. The exact map and its kernel give

$$M/K_+ \xrightarrow{\sim} \mathcal{N}_+, \quad [h] \mapsto h e_+. \quad (\text{NEA6.2})$$

The isomorphism is canonical for the actual displayed extension, with $[1]$ mapped to e_+ . It is an algebraic M -module statement. We assign no unproved locally convex quotient topology to M/K_+ or to the ambient Ext group.

The full jet map gives a faithful concrete description of this module:

$$M/K_+ \longrightarrow \prod_{\rho \in \mathbb{Z}} \mathbb{C}[z]/z^{m_\rho}, \quad [h] \mapsto \left(\sum_{j=0}^{m_\rho-1} \frac{h^{(j)}(\rho+1)}{j!} z^j \right)_\rho. \quad (\text{NEA6.3})$$

It is injective by definition of K_+ . Its image is the image of the actual strip-polynomial multiplier functions. It is not declared to be the unrestricted product. The projection onto any specified finite set of these jet factors is surjective: the usual finite polynomial remainder construction follows from Bézout for the pairwise coprime polynomials $(\lambda - \rho - 1)^{m_\rho}$, and the resulting polynomial belongs to M . This finite assertion neither supplies arbitrary infinite interpolation nor assumes convergence of finite spectral expansions.

There is also the exact injective M -map from the original normal quotient

$$\mathcal{Q}_+ = \mathcal{B}/\mathcal{J}_+ \longrightarrow M/K_+, \quad [F]_{\mathcal{J}_+} \mapsto [F]_{K_+}. \quad (\text{NEA6.4})$$

It is induced by $\mathcal{B} \subset M$, and injectivity is precisely $\mathcal{J}_+ = \mathcal{B} \cap K_+$. In terms of (NEA6.2), it sends $[F]$ to $F e_+$. We do not identify the two full quotient modules or assert surjectivity of (NEA6.4).

The ideal K_+ , hence the annihilator of the single actual extension class, recovers the complete translated zero divisor: its common zero set is exactly the zero set of F_+ , since $F_+ \in K_+$, and at $\lambda_0 = \rho + 1$,

$$\min_{0 \neq h \in K_+} \text{ord}_{\lambda_0} h = m_\rho. \quad (\text{NEA6.5})$$

Every member has at least that order, and the original full factor F_+ attains it. This is a precise recovery of the divisor and its multiplicities from the annihilator, not a claim about where the original zeros lie within their strip.

NEA7. The exact full-module generator resolvent

Write $T[h] = [\lambda h]$ on M/K_+ , equivalently postcomposition by the normal generator on $\mathcal{N}_+$. For a scalar $\mu \notin \text{Supp } \mathcal{D}_+$, define

$$\mathcal{R}_\mu h(\lambda) = \frac{h(\lambda) - F_+(\lambda)h(\mu)/F_+(\mu)}{\lambda - \mu}. \quad (\text{NEA7.1})$$

The denominator of the retained full scalar is nonzero. The numerator has a zero at μ , so the quotient is entire. It belongs to M : outside a fixed disk around μ this follows from the strip bounds for h and F_+ , while on the disk the entire quotient is bounded on a compact set. If $h \in K_+$, both numerator terms vanish to every required translated order, and the denominator is a unit at each translated zero. Thus $\mathcal{R}_\mu$ preserves K_+ and induces an operator on the whole quotient.

Multiplication gives the two exact identities

$$(T - \mu)\mathcal{R}_\mu[h] = [h], \quad \mathcal{R}_\mu(T - \mu)[h] = [h]. \quad (\text{NEA7.2})$$

For the first, the retained difference from h is $F_+h(\mu)/F_+(\mu) \in K_+$; for the second, the evaluated numerator $((\lambda - \mu)h)(\mu)$ is zero. The inverse is M -linear on the quotient: $T - \mu$ is M -linear and a bijection, so its inverse is M -linear. Formula (NEA7.1) already specifies that inverse on each representative, without a finite-block-density argument.

At an actual $\lambda_0 = \rho + 1$, the nonzero class

$$\left[\frac{F_+}{\lambda - \lambda_0} \right]_{K_+} \quad (\text{NEA7.3})$$

is an eigenvector of T with eigenvalue λ_0 : it has order $m_\rho - 1$ there and is killed by $T - \lambda_0$. Hence $T - \lambda_0$ is not injective. We have therefore computed the entire algebraic resolvent set and its complement:

$$\{\mu \in \mathbb{C} : T - \mu \text{ is not bijective on } \mathcal{N}_+\} = \text{Supp } \mathcal{D}_+ = \{\rho + 1 : \rho \in \mathcal{Z}\}. \quad (\text{NEA7.4})$$

This statement specifies algebraic bijectivity; it does not invoke the spectrum of an operator on an unassigned Banach or Fréchet topology.

The corresponding complete local chains are retained as actual submodules. At λ_0 , the map

$$M/((\lambda - \lambda_0)^m) \longrightarrow M/K_+, \quad [a] \longmapsto \left[a \frac{F_+}{(\lambda - \lambda_0)^m} \right] \quad (\text{NEA7.5})$$

is injective by the same full-order argument as (NEA5.3). On this block, multiplication by a_0^λ , for recovered $a_0 > 0$, is exactly

$$a_0^{\rho+1} \sum_{j=0}^{m-1} \frac{(\log a_0)^j}{j!} M_z^j. \quad (\text{NEA7.6})$$

The map includes the entire factor in (NEA5.1); no derivative coefficients are removed by this expression for the action.

Every unital complex-algebra character of M/K_+ is evaluation at exactly one translated zero. To prove this, let χ be such a character and put $\mu = \chi([\lambda])$. For every $h \in M$, the function $(h(\lambda) - h(\mu))/(\lambda - \mu)$

is entire and belongs to M , by the division estimate used above. Applying χ gives $\chi([h]) = h(\mu)$. Since $F_+ \in K_+$, it follows that $F_+(\mu) = 0$, so $\mu \in \text{Supp } \mathcal{D}_+$. Conversely evaluation at any such point annihilates K_+ and defines a character. Thus their actual dilation characters are $a_0^{\rho+1}$. For $a_0 > 1$, their logarithmic-modulus values are

$$\frac{2 \log |a_0^{\rho+1}|}{\log a_0} = 2\Re \rho + 2 \in (2, 4). \quad (\text{NEA7.7})$$

These are the scalar values for the postcomposition module action just defined. They are not weights of a conjugation action on equivariant Hom; NEA10 retains that distinction.

NEA8. Separation of the actual class module from the original quotient

The existing global separator is $c(\lambda) = E_+(\lambda) = 1 - E(\lambda - 1)$. GMS3 proves that $c \in K_+$, with full orders, and that it acts as the identity on the complete original quotient $\mathcal{Q} = \mathcal{B}/\mathcal{J}$. Consequently

$$c\mathcal{N}_+ = 0, \quad c\mathcal{Q} = \mathcal{Q} \text{ with } c_{\mathcal{Q}} = 1. \quad (\text{NEA8.1})$$

The second equality alone is not the reason; its stated identity action is the property used.

Apply the explicit bar-resolution homotopy of GMS4 to these actual modules. If $P \rightarrow \mathcal{Q}$ is a projective resolution, multiplication by $1 - c$ lifts zero on its augmentation, and there is $h : P \rightarrow P[-1]$ with $dh + hd = 1 - c$. The Hom homotopy $H^n(f) = (-1)^n fh$ satisfies

$$DH + HD = 1 - c_{\mathcal{N}_+} = 1 \quad \text{on } \text{Hom}_M(P, \mathcal{N}_+). \quad (\text{NEA8.2})$$

For the reverse direction, resolve $\mathcal{N}_+$, lift its zero action of c , and use the same Hom formula; the target action of c on $\mathcal{Q}$ is one, giving the reverse contraction. Hence

$$\boxed{\text{RHom}_M(\mathcal{Q}, \mathcal{N}_+) \simeq 0, \quad \text{RHom}_M(\mathcal{N}_+, \mathcal{Q}) \simeq 0.} \quad (\text{NEA8.3})$$

The complete bar insertion formulas in GMS4.5-GMS4.9 apply without any finiteness condition on $\mathcal{N}_+$. This is separation of a module generated by the actual extension class. It does not assign the translated-zero spectrum to the entire kernel $\mathcal{J}_+$, whose other quotient characters remain as calculated in CW6.

NEA9. Original-source maps for every annihilating multiplier

For every $h \in K_+$, the unique source-return map in (NEA3.2) becomes, in the original normal variable,

$$V^{-1}r_h V(F)(s) = h(s+1)F(s). \quad (\text{NEA9.1})$$

Its exact source representative is

$$\mathfrak{r}_h(a)(u) = \Theta^{-1}[h(s+1)\Theta a(s)](u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} h(3/2 + it)\Theta a(1/2 + it)u^{-it} dt. \quad (\text{NEA9.2})$$

The shift $3/2 + it$ is required by the actual normal action. The multiplier bound, rapid strip decay, and the full inverse Mellin estimates from GMS2 prove absolute differentiated convergence and continuity

$A(-1) \rightarrow J(-1)$. It lands in J because $h(s+1)$ has the required original-zero orders and $\Theta J = \mathcal{J}$. Composing with the exact continuous inverse $H = \Sigma^{-1} : J \rightarrow S$ gives

$$f_{h,a} = H\mathfrak{r}_h(a), \quad \Sigma f_{h,a} = \mathfrak{r}_h(a), \quad f_{h,a}(0) = 0, \quad \int_{\mathbb{R}} f_{h,a} = 0. \quad (\text{NEA9.3})$$

All sums, factors $2, 1/2, u^{-1/2}/\pi$, and endpoint conditions are the original ones from GMS2/GMS8. For $h = c$, formula (NEA9.2) reduces exactly to multiplication by $1 - E(s)$, because $c(s+1) = 1 - E(s)$. The classification (NEA3.4) says precisely that these, and only these multipliers, admit the displayed lift into the full source kernel.

NEA10. The actual geometric connecting morphism and its class module

The simultaneous complexes of the actual sheaf row, with their constructed M -actions, are

$$D_J = (W \xrightarrow{r_+ - r_-} J \xrightarrow{0} J(-1)), \quad D_F = (W \xrightarrow{r_+ - r_-} A \xrightarrow{0} A(-1)), \quad D_G = Q[-1] \oplus Q(-1)[-2]. \quad (\text{NEA10.1})$$

The low generator is L , the normal generator is $L + 1$, and GMS9 retains the endpoint characters $(h(0), h(1), h(1), h(0))$. All identifications below use Θ on low coefficients and $V\Theta$ on normal coefficients. The explicit symmetric source contraction in CLP12/GMS9 gives

$$D_J \simeq Z[0] \oplus \mathcal{J}_+[-2], \quad D_F \simeq Z[0] \oplus \mathcal{Q}[-1] \oplus \mathcal{B}[-2], \quad D_G \simeq \mathcal{Q}[-1] \oplus \mathcal{Q}_+[-2]. \quad (\text{NEA10.2})$$

These are the compatible M -linear quasi-isomorphisms of the actual row, and retain the entire ideal. Its connecting morphism is

$$\delta : D_G \longrightarrow D_J[1], \quad \delta|_{\mathcal{Q}[-1]} = 0, \quad \delta|_{\mathcal{Q}_+[-2]} = e_+[-2] \text{ into } \mathcal{J}_+[-1]. \quad (\text{NEA10.3})$$

Let M act on this derived Hom group by postcomposition on $D_J[1]$; since δ is M -linear, this equals precomposition on its source. Inclusion and projection of the normal summands in (NEA10.2) prove both directions of

$$h\delta = 0 \iff he_+ = 0. \quad (\text{NEA10.4})$$

Therefore the actual cyclic boundary-image module is

$$M\delta \subset \text{Hom}_{D(M)}(D_G, D_J[1]), \quad M/K_+ \xrightarrow{\sim} M\delta, \quad [h] \mapsto h\delta, \\ M\delta \xrightarrow{\sim} \mathcal{N}_+, \quad h\delta \mapsto he_+. \quad (\text{NEA10.5})$$

This identifies the class receiver directly inside the actual localization triangle; it does not replace its full source or kernel. Its exact annihilator, resolvent, and full translated divisor are (NEA3.4), (NEA7.1)–(NEA7.4), and (NEA6.5).

In particular $c\delta = 0$, whereas c acts as the identity on $\mathcal{Q}[k]$ for every integer k . For every derived M -morphism $f : \mathcal{Q}[k] \rightarrow D_G$, naturality of the central action gives the actual lifting equation

$$\delta f = \delta f c_{\mathcal{Q}[k]} = \delta c_{D_G} f = c_{D_J[1]} \delta f = 0. \quad (\text{NEA10.6})$$

It follows that f lifts through $D_F \rightarrow D_G$. To display the lift with the actual decompositions, the normal component of f is zero in the derived category by GMS4.4. Write its remaining component as $f_1 : \mathcal{Q}[k] \rightarrow \mathcal{Q}[-1]$. Then $(0, f_1, 0)$ in the middle decomposition of (NEA10.2) is a lift, transported back through its specified quasi-isomorphism. This proves existence; it does not claim uniqueness of all lifts through D_F , whose differences can come from D_J .

The action used in (NEA10.5)–(NEA10.6) must not be confused with conjugation by dilation. Postcomposition by the multiplier a^λ produces the module action whose scalar values are (NEA7.6)–(NEA7.7). The conjugation action on a derived morphism is instead

$$\delta \longmapsto T_{a,D_J[1]} \delta T_{a,D_G}^{-1}. \quad (\text{NEA10.7})$$

It fixes δ , because δ is equivariant. No translated-zero weight is assigned to this conjugation action. Likewise no identification of $M\delta$ with Deligne’s geometric obstruction quotient is asserted without its comparison map. The proved conclusion is the explicit annihilator, spectrum and separation of the module generated by the actual localization class, and the resulting actual lift (NEA10.6), with all original kernels, factors, multiplicities and degrees retained.

Actual nearby cycles, specialization and local-to-global boundary maps — LNC0-LNC11

Proof source: CC_LOCAL_NEARBY_CYCLES_AND_SPECIALIZATION.md.

Nearby cycles, the local specialization quotient, and the actual global lift

This proof uses the existing sphere sheaf. Stable locators are **LNC0-LNC11**. It derives its nearby-cycle and invariant-cycle maps and compares them with FOD7's actual lift. It retains the original Schwartz source, the complete original-zeta ideal, both endpoint coordinates at each pole, and all orientation signs.

Historical attribution: these nearby/vanishing-cycle constructions go back to Deligne's SGA7, ExposésXIII-XIV, and Beilinson's *How to glue perverse sheaves*. The inspected explanatory author source is Ryan Reich, [Notes on Beilinson's "How to glue perverse sheaves"](#), [arXiv:1002.1686v4](#), definitions and gluing maps at the source ranges recorded above. Credit to those originals is retained through Reich; this edition does not claim a fresh reading of the original SGA or Beilinson chapters.

Human-source attribution: the extension of continuous linear functionals is the Hahn-Banach theorem, not a programme result. See Terence Tao, [245B, Notes6: Duality and the Hahn-Banach theorem](#), Theorem1 and its complex proof (26 January2009). For the locally convex use here, continuity bounds a functional by a continuous seminorm; quotienting its kernel reduces this application to that normed-space theorem. No continuous splitting of the original quotient is thereby asserted.

LNC0. Prerequisites and category

The governing construction and correction chain are `READ_FIRST_USER_CONSTRUCTION.md`, `USER_ARGUMENT_RECONSTRUCTION.md`, `USER_ARGUMENT_CONNECTIONS.md`, and `USER_CONSTRUCTION_FULL_LOGBOOK.md`. The supporting datum remains

$$\tau\langle Z_1; \text{no } Z_2\rangle. \quad (\text{LNC0.1})$$

Arithmetic below is after the complete-history arithmetic reconstruction. No coordinate, distance, subtraction, or retracted addition is assigned to τ . Coordinates belong to the existing receiver $Y = \mathbb{P}^1(\mathbb{C})$. The specialization $\pi : Y \rightarrow X = \{x_+, \eta, x_-\}$ sends $0, \infty, \mathbb{C}^\times$ to x_+, x_-, η . A coordinate on this receiver is not a coordinate on η .

The proved inputs are CSP2-CSP8 (the sheaf, continuous local resolutions, costalks, orientation, and support maps); CDW2-CDW9 (the coefficient-natural direct-image splitting and angular-to-integration sign); CW10 (the existing ordinary disk/localization rows); CSD2-CSD9 (the literal continuous-current support dual); SSI/ESI (closed J and its continuous Schwartz inverse); SDT as incorporated in CSD (the strong Q' topology); DC3-DC9 (Deligne's exact cross and boundary quotient); and GMS9-GMS10/FOD7 (the actual global coefficient triangle and derived source lift). The file identities and reading coverage are recorded in LNC11.

The sheaf computations use complexes of sheaves of complex vector spaces. Every displayed map between coefficient models is continuous for the stated Fréchet topologies. Dual maps are continuous for the weak topology and the strong topology of uniform convergence on bounded sets, with the precise cohomology topology proved in CSD/SDT. We do not infer a general exact category of topological sheaves from these

concrete models. The later derived M -module statements are algebraic derived statements on the actual modules, not assertions about every continuous representation of a generator.

The convention is

$$R\phi_f \mathcal{H} = \text{Cone}(i^* \mathcal{H} \longrightarrow R\psi_f \mathcal{H}). \quad (\text{LNC0.2})$$

This fixes all degrees; no perverse shift is suppressed. Reich's original author source, arXiv:1002.1686v4, lines 261–303 and 356–375, was read for the universal-cover definition and monodromy triangle. Its finite-constructibility theorems are not applied to these infinite coefficients. The required contractions are calculated below.

LNC1. The full existing coefficient system

Put

$$S = \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0\}, \quad E_+ = E_- = \mathbb{C}^2, \quad V_p = S \oplus E_p,$$

$$A = \left\{ a \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \text{ for every } N, j \right\}. \quad (\text{LNC1.1})$$

The endpoint coordinates retain the value-at-zero and integral channels. The source maps are

$$\begin{aligned} \Sigma f(u) &= 2 \sum_{n \geq 1} f(nu), \quad Ra(u) = u^{-1}a(u^{-1}), \quad \hat{f}(v) = \int_{\mathbb{R}} f(x) e^{-2\pi i x v} dx, \\ r_+(f, c_0, c_1) &= \Sigma f, \quad r_-(g, d_0, d_1) = R\Sigma g = \Sigma \hat{g}. \end{aligned} \quad (\text{LNC1.2})$$

The last identity retains the complete Poisson comparison

$$\Sigma \hat{g}(u) = u^{-1} \Sigma g(u^{-1}) + u^{-1} g(0) - \int_{\mathbb{R}} g(x) dx. \quad (\text{LNC1.3})$$

Its last two terms vanish on S ; the separate endpoint coordinates have not been removed.

Define $J = \Sigma S$, $Q = A/J$, and $q : A \rightarrow Q$. The cited SSI/ESI proofs give closed J , a continuous inverse $H = \Sigma^{-1} : J \rightarrow S$, and

$$s_+(b) = (Hb, 0, 0), \quad s_-(b) = (\widehat{Hb}, 0, 0), \quad r_p s_p = 1_J, \quad V_p = E_p \oplus s_p J. \quad (\text{LNC1.4})$$

Thus $\ker r_p = E_p$, $\text{im } r_p = J$, and all these maps are actual continuous maps.

For $D = \{0, \infty\}$, $i : D \hookrightarrow Y$, $j : \mathbb{C}^\times \hookrightarrow Y$, $W = V_+ \oplus V_-$, the sheaf is

$$\mathcal{F} = \underline{A}_Y \times_{i_* A^2} i_* W, \quad 0 \rightarrow \mathcal{F} \rightarrow \underline{A}_Y \oplus i_* W \xrightarrow{\text{ev}-r} i_* A^2 \rightarrow 0. \quad (\text{LNC1.5})$$

Its interior is constant A , pole stalks are V_p , and generizations are r_p .

The original arithmetic is carried by

$$\Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, \quad \Theta^{-1} F(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} F(1/2 + it) u^{-it} dt. \quad (\text{LNC1.6})$$

This identifies A with the entire strip-Schwartz space $\mathcal{B}$, J with the full ideal

$$J = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every original nontrivial zero } \rho, 0 \leq j < m_\rho\}, \quad Q \simeq \mathcal{Q} = \mathcal{B}/J. \quad (\text{LNC1.7})$$

The original source and divisor are

$$f_0(v) = \frac{\pi}{2}v^2(2\pi v^2 - 3)e^{-\pi v^2}, \quad F_0(s) = \Theta \Sigma f_0(s) = \frac{s(s-1)}{8}\pi^{-s/2}\Gamma(s/2)\zeta(s). \quad (\text{LNC1.8})$$

This is the retained source map, not a replacement for the original zeta function. The complete endpoint and trivial-zero cancellations are

$$\begin{aligned} F_0(0) &= F_0(1) = \frac{1}{8}, \quad F_0(-1) = F_0(2) = \frac{\pi}{24}, \\ F_0(-2r) &= \frac{r(2r+1)(-1)^r\pi^r}{2r!}\zeta'(-2r) = \frac{(1+2r)(2r)}{8}\pi^{-(1+2r)/2}\Gamma((1+2r)/2)\zeta(1+2r) \neq 0 \quad (r \geq 1). \end{aligned} \quad (\text{LNC1.9})$$

These full original-source comparisons, including all nontrivial-zero jets, are proved in the receiving source files. The present calculation does not alter their ideal. In particular $Q \neq 0$: at an actual nontrivial zero ρ , $F_0(s)/(s-\rho) \in \mathcal{B}$ has order $m_\rho - 1$ at ρ , so defines a nonzero quotient class.

LNC2. The actual universal-cover nearby object

At $p = 0$ use $z_p = z$, and at $p = \infty$ use $z_p = 1/z$. Let $D_p = \{|z_p| < \epsilon\}$ and $D_p^* = D_p \setminus \{p\}$. The local function in this section is explicitly

$$f_p(z_p) = z_p. \quad (\text{LNC2.1})$$

The universal cover is

$$\widetilde{D}_p^* = \{t \in \mathbb{C} : \text{Re } t < \log \epsilon\}, \quad v_p(t) = e^t, \quad \gamma(t) = t + 2\pi i. \quad (\text{LNC2.2})$$

The pullback of $\mathcal{F}|_{D_p^*}$ is constant A . Resolve it by smooth A -valued forms. The straight-line contraction on this convex half-plane is, in positive degree k ,

$$(\text{H}\omega)_x = \int_0^1 t^{k-1} \iota_{x-t_0} \omega_{t_0+t(x-t_0)} dt, \quad d\text{H} + \text{H}d = 1 - \text{ev}_{t_0}. \quad (\text{LNC2.3})$$

The integral exists in the complete Fréchet space A . Each seminorm of the integral is bounded by the integral of that seminorm. On compact sets the intervening segments form a compact set, so this also proves continuity for every smooth compact seminorm. Differentiation under the integral proves the displayed homotopy identity in A . Partitions of unity make the form resolution fine. On all these half-planes, compatibly as ϵ decreases, constants therefore give

$$R\psi_{f_p} \mathcal{F} = i_p^* R(j_p v_p)_* v_p^* (\mathcal{F}|_{D_p^*}) \simeq A[0]. \quad (\text{LNC2.4})$$

Deck translation fixes these constant representatives. The actual monodromy is thus

$$\mathcal{T}_p = 1_A. \quad (\text{LNC2.5})$$

This is computed from the existing sheaf; no inertia action has been independently assigned.

A pole germ $v \in V_p$ restricts to the constant punctured section $r_p v$. Consequently the universal-cover adjunction gives exactly

$$\text{sp}_p : V_p[0] \longrightarrow A[0], \quad v \longmapsto r_p v. \quad (\text{LNC2.6})$$

Its kernel, image, and cokernel are E_p, J, Q . In convention (LNC0.2), the actual vanishing complex is

$$R\phi_{f_p} \mathcal{F} = [V_p \text{ in } -1 \xrightarrow{r_p} A \text{ in } 0], \quad H^{-1} = E_p, \quad H^0 = Q. \quad (\text{LNC2.7})$$

The point-supported endpoints survive even though they have no generic nearby component.

LNC3. Angular cohomology and the invariant/coinvariant row

The deck group is the discrete group $\mathbb{Z}$. Its group algebra has the free resolution

$$0 \rightarrow \mathbb{C}[\gamma, \gamma^{-1}] \xrightarrow{\gamma-1} \mathbb{C}[\gamma, \gamma^{-1}] \xrightarrow{\gamma \mapsto 1} \mathbb{C} \rightarrow 0. \quad (\text{LNC3.1})$$

Injectivity follows from the extreme exponents of a nonzero Laurent polynomial. Finite division by $\gamma - 1$ gives the augmentation kernel. Applying Hom to this resolution yields

$$[A \xrightarrow{\mathcal{T}_p^{-1}} A] = [A \xrightarrow{0} A] \quad \text{in degrees } 0, 1. \quad (\text{LNC3.2})$$

Its actual comparison with punctured cohomology uses constants and

$$\vartheta_p = \frac{d \arg z_p}{2\pi}. \quad (\text{LNC3.3})$$

Write a closed form on the cylinder $t = \log |z_p|$, $\theta = \arg z_p$ as $\omega = P(t, \theta)dt + Q_1(t, \theta)d\theta$. The period $b = \int_0^{2\pi} Q_1(t_0, \theta)d\theta$ is independent of t_0 , by closedness. Subtracting $b\vartheta_p$, a periodic primitive is

$$g(t, \theta) = \int_{t_0}^t P(x, 0) dx + \int_0^\theta \left(Q_1(t, \varphi) - \frac{b}{2\pi} \right) d\varphi. \quad (\text{LNC3.4})$$

Differentiation proves $dg = \omega - b\vartheta_p$, using $\partial_t Q_1 = \partial_\theta P$. Compact seminorm bounds for these finite integrals prove continuity. Radial contraction and the circle calculation give no further cohomology. Hence

$$B_p^\bullet := R\Gamma(D_p^*, \mathcal{F}) \simeq A[0] \oplus A[\vartheta_p][-1], \quad R\Gamma(D_p, \mathcal{F}) \simeq V_p[0], \quad (\text{LNC3.5})$$

with restriction $(r_p, 0)$.

Set $V_i = H^i(R\psi_{f_p} \mathcal{F})$, $C_i = V_i^{\mathcal{T}_p}$, and $K_i = V_{i-1, \mathcal{T}_p}[\vartheta_p]$. Then the actual exact row is

$$0 \longrightarrow K_i \xrightarrow{j_i} B_i \xrightarrow{\pi_i} C_i \longrightarrow 0, \quad B_i = H^i(B_p^\bullet). \quad (\text{LNC3.6})$$

For $i = 0$, this is $0 \rightarrow 0 \rightarrow A \xrightarrow{1} A \rightarrow 0$; for $i = 1$, it is $0 \rightarrow A[\vartheta_p] \xrightarrow{1} A[\vartheta_p] \rightarrow 0 \rightarrow 0$. The label $[\vartheta_p]$ retains the angular generator and its cover factor, proved in LNC5. The corresponding étale term in DC is (-1) ; no identification of this topological deck group with ℓ -adic inertia is asserted.

LNC4. The exact cross and the computed obstruction quotient

Use the fibre-cone convention

$$\text{Fib}(f)^n = C^n \oplus D^{n-1}, \quad d(x, y) = (d_C x, fx - d_D y). \quad (\text{LNC4.1})$$

Taking the restriction fibre gives the CSP costalk, with degree two measured by positive complex-oriented integration:

$$L_p = Ri_p^! \mathcal{F} = [V_p^0 \xrightarrow{r_p} A^1 \xrightarrow{0} A^2]. \quad (\text{LNC4.2})$$

The angular boundary is -1 in these coordinates. Indeed choose $\chi = 1$ near p , zero near the outer circle. Then $\beta = (1 - \chi)\vartheta_p$ extends smoothly over p , and

$$d\beta = -d\chi \wedge \vartheta_p, \quad \int_{D_p} d\beta = 1. \quad (\text{LNC4.3})$$

The fibre-cone differential makes the boundary of ϑ_p cohomologous to $-d\beta$. Thus

$$H_p^0 = E_p, \quad H_p^1 = Q, \quad H_p^2 = A, \quad \partial_0 = q, \quad \partial_1 = -1_A, \quad (\text{LNC4.4})$$

and the complete nonzero localization row is

$$0 \rightarrow E_p \rightarrow V_p \xrightarrow{r_p} A \xrightarrow{q} Q \rightarrow 0 \rightarrow A[\vartheta_p] \xrightarrow{-1} A \rightarrow 0. \quad (\text{LNC4.5})$$

These previously established ordinary local maps now have the actual nearby-cycle interpretation from LNC2-LNC3.

With $A_i^{\text{sp}} = H^i(D_p, \mathcal{F})$ and $O_i = H_p^{i+1}(\mathcal{F})$, the exact cross has:

i	A_i^{sp}	K_i	B_i	C_i	O_i	∂_i
-1	0	0	0	0	E_p	0
0	V_p	0	A	A	Q	q
1	0	$A[\vartheta_p]$	$A[\vartheta_p]$	0	A	-1

Here $\text{sp}_i = \pi_i \alpha_i$. The quotient formula requires no assumed weight: lift $c \in C_i$ to $b \in B_i$ and send it to $[\partial_i b]$ in $O_i / \partial_i j_i K_i$. Two lifts differ by $j_i K_i$, so this is well defined. If its value is zero, $\partial_i(b - j_i k) = 0$ for some k ; localization exactness gives $b - j_i k = \alpha_i a$, hence $c = \text{sp}_i a$. This proves the kernel and image, and therefore

$$\boxed{\text{coker sp}_0 = \frac{\text{im } \partial_0}{\partial_0 j_0 K_0} = Q, \quad \text{coker sp}_1 = \frac{\text{im } \partial_1}{\partial_1 j_1 K_1} = 0.} \quad (\text{LNC4.6})$$

All other obstruction quotients vanish; in particular the preceding receiver $O_{-1} = E_p$ has zero boundary image and is retained in the table and in (LNC4.5). In degree zero the actual boundary is the nonzero quotient $q : A \rightarrow Q$. The existing separator acts as identity on Q , so this is not a boundary it annihilates. Its relation to the genuine normal lift is calculated in LNC8-LNC9.

LNC5. Actual covers, mirrors, and changing the local function

For a recovered integer $n \geq 1$, the holomorphic covering $b_n(z) = z^n$ is $z_p \mapsto z_p^n$ in both positive pole coordinates. Its lift of universal covers is $t \mapsto nt$, intertwining the source deck generator with the n -th power of the target generator. A pulled-back degree-one group cocycle has value

$$b + \mathcal{T}_p b + \dots + \mathcal{T}_p^{n-1} b = nb, \quad b_n^* \vartheta_p = n \vartheta_p. \quad (\text{LNC5.1})$$

The retained coefficient actions are

$$\begin{aligned} T_a b(u) &= b(u/a), \quad \rho_+(a)(f, c_0, c_1) = (f(\cdot/a), c_0, ac_1), \\ \rho_-(a)(g, d_0, d_1) &= (ag(a \cdot), ad_0, d_1). \end{aligned} \quad (\text{LNC5.2})$$

The source formulas give $r_p \rho_p(a) = T_a r_p$. Including both geometry and coefficients gives:

Receiver	Actual b_n pullback with coefficients
pole V_p	$\rho_p(n)$

Receiver	Actual b_n pullback with coefficients
nearby A, B_0, C_0	T_n
$B_1 = K_1 = A[\vartheta_p]$	nT_n
support $H_p^0 = E_p$	$\rho_p(n) _{E_p}$
support $H_p^1 = Q$	$\bar{T}_n$
support $H_p^2 = A$	nT_n
vanishing $H^{-1} = E_p, H^0 = Q$	$\rho_p(n) _{E_p}, \bar{T}_n$

All cross maps commute, by $r_p \rho_p(n) = T_n r_p$, $qT_n = \bar{T}_n q$, and the same factor n on both sides of $\partial_1 = -1$. The notation $A(-1)$ below records this proved nT_n action. For real $a > 0$ the coefficient action exists separately; there is no asserted holomorphic map $z \mapsto z^a$ for a noninteger a .

The two actual mirrors are

$$w(z) = -1/z, \quad \alpha(z) = -1/\bar{z}. \quad (\text{LNC5.3})$$

They exchange poles. Using target coordinate $w' = 1/z'$ at infinity, their local forms from zero are $w' = -z$ and $w' = -\bar{z}$. Thus the positive local angular and support-orientation signs are $+1$ for w , -1 for α . Both use coefficient R , swap pole spaces, and act by $\bar{R}$ on local Q . Nearby coefficients transform by R ; local angular and degree-two support coefficients transform by $+R, -R$, respectively.

For the global annular coordinate $\vartheta = d \arg z / (2\pi)$, however,

$$w^* \vartheta = -\vartheta, \quad \alpha^* \vartheta = +\vartheta, \quad \vartheta_\infty = -\vartheta_0. \quad (\text{LNC5.4})$$

This accounts for the global loop boundary $(-a, a)$. The local and global angular signs are not interchangeable.

Now keep the sheaf fixed but change the function itself to $f_{p,n}(z_p) = z_p^n$ over a fixed target disk. This is a different nearby-cycle construction. The pullback of the target universal cover has n components, each a contractible half-plane. A target deck generator cyclically permutes them:

$$R\psi_{f_{p,n}} \mathcal{F} = A^n[0], \quad \mathcal{T}(a_0, \dots, a_{n-1}) = (a_{n-1}, a_0, \dots, a_{n-2}). \quad (\text{LNC5.5})$$

Its invariant space is diagonal A , and its coinvariant identification is

$$(A^n)_{\mathcal{T}} \xrightarrow{\sim} A, \quad [(a_0, \dots, a_{n-1})] \mapsto \sum_j a_j. \quad (\text{LNC5.6})$$

The kernel of this sum is the image of $\mathcal{T} - 1$, as solving by successive partial sums verifies; that solution is a finite-coordinate continuous map. Specialization is $\text{diag}(r_p)$, so its cokernel in the invariant space is still Q . The diagonal map induces multiplication by n on coinvariants. This also follows by restricting a target cocycle to its n -fold domain loop and summing its translates.

No other part of this nearby object is dropped. In the full vanishing cone,

$$H^{-1}(R\phi_{f_{p,n}} \mathcal{F}) = E_p, \quad H^0(R\phi_{f_{p,n}} \mathcal{F}) = A^n / \text{diag } J. \quad (\text{LNC5.7})$$

The explicit isomorphism of the last group is

$$[a_0, \dots, a_{n-1}] \mapsto (q\bar{a}, (a_0 - \bar{a}, \dots, a_{n-1} - \bar{a})) \in Q \oplus \ker(\sum), \quad \bar{a} = \frac{1}{n} \sum_j a_j. \quad (\text{LNC5.8})$$

Its inverse sends (qb, v) to $[\text{diag } b + v]$, independent of the lift of qb . Continuity follows from the finite-coordinate maps and the quotient topology. The second summand retains every nontrivial cyclic-monodromy component. Formula (LNC5.5) is not substituted for the identity-function nearby object A in LNC2.

LNC6. The actual continuous dual specialization

Use the literal current complex from CSD4:

$$\mathcal{D}_c \mathcal{F} = \text{Cone} \left(i_* A'^2 \xrightarrow{(\delta, -r')} \mathcal{T}_A^\bullet \oplus i_* W' \right). \quad (\text{LNC6.1})$$

Here $\mathcal{T}_A^k(U) = \Gamma_c(U, \mathcal{E}_A^{-k})'$, with Hom differential, and restriction is transpose to extension by zero of compact tests. Its terms in degrees $-2, -1, 0$ are $\mathcal{T}_A^{-2}, \mathcal{T}_A^{-1} \oplus i_* A'^2$, and $\mathcal{T}_A^0 \oplus i_* W'$, and

$$d^{-2}T = (dT, 0), \quad d^{-1}(T, \lambda) = (dT + \delta\lambda, -r'\lambda). \quad (\text{LNC6.2})$$

This is continuous duality, not unrestricted algebraic duality.

For $\lambda \in A'$, define the constant current $c_\lambda(\omega) = \lambda(\int \omega)$ and

$$h_p \lambda(\phi) = \lambda \left(\int \vartheta_p \wedge \phi \right). \quad (\text{LNC6.3})$$

The angular current is locally integrable at the pole. Stokes on a positive small circle gives $dh_p \lambda = \delta_p \lambda$. Thus CSD's exact pole-stalk comparison is

$$P_p = [A' \text{ in } -2 \xrightarrow{0} A' \text{ in } -1 \xrightarrow{r'_p} V'_p \text{ in } 0] \longrightarrow i_p^* \mathcal{D}_c \mathcal{F},$$

$$\lambda \mapsto c_\lambda, \quad \mu \mapsto (h_p \mu, -\mu), \quad \nu \mapsto (0, \nu). \quad (\text{LNC6.4})$$

Indeed $d(h_p \mu, -\mu) = (0, r'_p \mu)$. The compact-test contraction of CSD proves this a quasi-isomorphism by continuous comparison maps on each small disk. The integral in (LNC6.3) is formed in A first and then evaluated by the continuous functional.

On the puncture the representatives are $c_\lambda, h_p \mu$ in degrees $-2, -1$. Restriction is therefore the exact chain map

$$P_p \longrightarrow A'[2] \oplus A'[1], \quad (\deg -2, \deg -1, \deg 0) = (1, 1, 0). \quad (\text{LNC6.5})$$

On the universal cover the angular current has the primitive $\arg z_p / (2\pi)$. The same current contraction, obtained by transposing the compact-test contraction, leaves only constant currents. Consequently

$$R\psi_{f_p}(\mathcal{D}_c \mathcal{F}) = A'[2], \quad \mathcal{T} = 1, \quad \text{sp}_p^\vee : P_p \rightarrow A'[2] = (1 \text{ in } -2, 0 \text{ otherwise}). \quad (\text{LNC6.6})$$

This is the map induced by current restriction and universal-cover adjunction.

The complete pole-stalk cohomology is

$$H^{-2}(P_p) = A', \quad H^{-1}(P_p) = \ker r'_p = q'Q', \quad H^0(P_p) = V'_p / \text{im } r'_p = E'_p. \quad (\text{LNC6.7})$$

The middle comparison carries the strong Q' topology by SDT. For the last, $V_p = E_p \oplus s_p J$, and Hahn-Banach extends every continuous functional on the closed $J \subset A$. Hence the image is the full J' summand of V'_p , and the quotient is the endpoint dual. No strong openness of $A' \rightarrow J'$ is assumed.

The fibre of (LNC6.5) cancels its identity pair in degree -2 . The remaining complex is

$$A' \longrightarrow V'_p \oplus A', \quad \mu \longmapsto (r'_p \mu, \mu) \text{ in degrees } -1, 0. \quad (\text{LNC6.8})$$

The quotient coordinate $(\nu, b) \mapsto \nu - r'_p b$ gives its continuous deformation retraction to $V'_p[0]$. Therefore

$$Ri_p^! \mathcal{D}_c \mathcal{F} \simeq V'_p[0], \quad \partial_{-1}^\vee = -r'_p : A' \longrightarrow V'_p. \quad (\text{LNC6.9})$$

The nontrivial localization part is

$$0 \rightarrow Q' \xrightarrow{q'} A' \xrightarrow{-r'_p} V'_p \rightarrow E'_p \rightarrow 0, \quad (\text{LNC6.10})$$

alongside identity $A' \rightarrow A'$ in degree -2 . For this dual nearby object, $V_{-2}^\vee = A'$ is its only nearby cohomology. The invariant row in degree -2 has $B_{-2} = C_{-2} = A'$, $K_{-2} = 0$, with specialization identity. In degree -1 , $B_{-1} = K_{-1} = A'$, $C_{-1} = 0$, and boundary $-r'_p$. Thus its boundary quotients $\text{im } \partial^\vee / \partial^\vee jK$ are zero. The kernel Q' remains in (LNC6.10).

It remains visible also in vanishing cycles. The cone of (LNC6.6) contains a contractible identity pair in degrees $-3, -2$; eliminating it gives

$$R\phi_{f_p}(\mathcal{D}_c\mathcal{F}) \simeq [A' \text{ in } -2 \xrightarrow{-r'_p} V'_p \text{ in } -1], \quad H^{-2} = Q', \quad H^{-1} = E'_p. \quad (\text{LNC6.11})$$

The pointwise Hom differential of the continuous dual of $[V_p^{-1} \xrightarrow{r_p} A^0]$ is $-r'_p$. Thus, on these actual complexes,

$$R\phi_{f_p}(\mathcal{D}_c\mathcal{F}) \simeq (R\phi_{f_p}\mathcal{F})^\vee[2]. \quad (\text{LNC6.12})$$

Dual specialization is onto its nearby cohomology, but the primal quotient becomes the displayed dual stalk/vanishing term; it is not removed.

For a power cover, the actual map on this dual is proper-current pushforward, transpose to primal pullback. Its factors on (LNC6.4) are

$$(\deg -2, \deg -1, \deg 0) = (nT'_n, T'_n, \rho_p(n)'). \quad (\text{LNC6.13})$$

For constant currents, pulling back a compact two-form multiplies its integral by n . For the angular current, the n inverse sheets each contribute $1/n$ to its trace, giving factor one. A point Dirac current pushes forward with factor one. These are also the direct transposes of the compact pole-costalk maps.

The nearby functor types of this pushforward must be retained. For the fixed target function $f_p = z_p$, its literal source and map are

$$R\psi_{f_p}(b_{n*}\mathcal{D}_c\mathcal{F}) \simeq R\psi_{f_p \circ b_n}(\mathcal{D}_c\mathcal{F}) = A'^n[2] \xrightarrow{(\lambda_j) \mapsto \sum_j T'_n \lambda_j} A'[2] = R\psi_{f_p}(\mathcal{D}_c\mathcal{F}). \quad (\text{LNC6.14})$$

To prove this comparison directly, pull back the target universal cover: its inverse image under b_n has n components, as in (LNC5.5). Each component carries a constant current, and away from the pole its branch map is an orientation-preserving local isomorphism. Its pushforward therefore contributes $T'_n \lambda_j$, without an additional degree factor on a single branch. Adding the branches proves (LNC6.14). The source monodromy is cyclic permutation; the sum intertwines it with identity. At the pole a finite proper map has direct-image stalk equal to the sum of its finitely many inverse-image stalks: disjoint neighborhoods of the inverse images and properness provide a cofinal such neighborhood system. Here there is just one inverse image at the pole, so this comparison also has the correct pole source P_p .

Specialization to the source nearby invariants is diagonal on its degree- -2 A' . Restricting (LNC6.14) to that diagonal yields nT'_n . On coinvariants, identified by the sum as in (LNC5.6), the induced map is T'_n . Thus the invariant/coinvariant factors are exactly the degree- -2 /degree- -1 factors in (LNC6.13), and the dual point-costalk factor is $\rho_p(n)'$. The full A'^n is retained; it is not replaced by its invariant diagonal when stating the nearby functor itself.

This is not a holomorphic inverse root map. The separately defined coefficient contragredient for the original pairing has factors $a^{-1}(T_{a^{-1}})'$, $(T_{a^{-1}})'$, and $\rho_p(a^{-1})'$. Under either mirror, the local degree -1 current action is $+R'$, and the degree -2 action is $\varepsilon_g R'$, where $\varepsilon_w = 1, \varepsilon_\alpha = -1$; endpoints swap dually. A current represented by a form pushes forward as $\varepsilon_g(g^{-1})^*$, so the local angular sign cancels the orientation sign in degree -1 . These maps are the transposes of the primal local maps and commute with all displayed dual arrows.

LNC7. The existing coefficient triangle on nearby and vanishing cycles

Define $\mathcal{F}_J = \underline{J}_Y \times_{i_* J^2} i_* W$ and $\mathcal{G} = j_! Q$. The actual row is

$$0 \longrightarrow \mathcal{F}_J \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow 0. \quad (\text{LNC7.1})$$

At a pole it is $0 \rightarrow V_p \xrightarrow{1} V_p \rightarrow 0 \rightarrow 0$; on the puncture it is $0 \rightarrow J \rightarrow A \xrightarrow{q} Q \rightarrow 0$. The coefficientwise universal-cover contraction gives

$$\begin{array}{ccc} V_p & \xrightarrow{r_p: J} & J \\ \parallel & & \downarrow \\ V_p & \xrightarrow{r_p: A} & A \\ \downarrow & & \downarrow q \\ 0 & \longrightarrow & Q. \end{array} \quad (\text{LNC7.2})$$

The right column is the actual nearby extension. Taking its indicated cones gives

$$[V_p^{-1} \rightarrow J^0] \longrightarrow [V_p^{-1} \rightarrow A^0] \longrightarrow [0 \rightarrow Q^0]. \quad (\text{LNC7.3})$$

Since $r_p : V_p \rightarrow J$ has section s_p , their cohomology groups are, respectively, E_p in degree -1 ; E_p, Q in degrees $-1, 0$; and Q in degree zero. The map on the degree-zero defect is exactly

$$Q \xrightarrow{1_Q} Q. \quad (\text{LNC7.4})$$

Thus this quotient sheaf preserves and displays the local vanishing-cycle quotient.

The actual current-dual comparison has coefficient maps

$$Q' \xrightarrow{q'} A' \xrightarrow{\text{restriction}} J'. \quad (\text{LNC7.5})$$

Hahn-Banach gives exactness of this underlying continuous-functional sequence. Applying the literal CSD cone maps, or the finite local complexes (LNC6.4) and (LNC6.11), makes (LNC7.4) the identity on Q' in vanishing degree -2 . This is a concrete transpose on the actual sheaf models; no general exactness of arbitrary topological dualization is needed.

LNC8. The local coefficient boundary and its exact global image

The continuous section s_p gives the projection $\text{pr}_{E_p}(v) = v - s_p r_p v$. The pole costalks of (LNC7.1) form the exact row of complexes

$$L_{J,p} = (V_p \xrightarrow{r_p} J \xrightarrow{0} J(-1)), \quad L_{F,p} = (V_p \xrightarrow{r_p} A \xrightarrow{0} A(-1)), \quad L_{G,p} = (0 \rightarrow Q \xrightarrow{0} Q(-1)) \quad (\text{LNC8.1})$$

in degrees 0, 1, 2. The normal label retains the proved cover factor n . The continuous chain maps

$$\begin{aligned} L_{J,p} &\longrightarrow E_p[0] \oplus J(-1)[-2], & (v, b, c) &\longmapsto (\text{pr}_{E_p} v, 0, c), \\ L_{F,p} &\longrightarrow E_p[0] \oplus Q[-1] \oplus A(-1)[-2], & (v, b, c) &\longmapsto (\text{pr}_{E_p} v, qb, c) \end{aligned} \quad (\text{LNC8.2})$$

are quasi-isomorphisms: their kernels in degrees 0, 1 are $s_p J \xrightarrow{r_p} J$, contracted by s_p . The formulas commute with the row inclusions and quotient. They leave the identity row on E_p in degree zero, the

identity $Q \rightarrow Q$ in degree one, and the full normal short exact row in degree two. No section of $A \rightarrow Q$ is introduced.

Let

$$e_0 = [0 \rightarrow J \rightarrow A \xrightarrow{q} Q \rightarrow 0]. \quad (\text{LNC8.3})$$

The actual local coefficient connecting morphism $\delta_p : L_{G,p} \rightarrow L_{J,p}[1]$ is therefore zero on $Q[-1]$ and equals

$$e_0(-1)[-2] : Q(-1)[-2] \longrightarrow J(-1)[-1] \quad (\text{LNC8.4})$$

on the normal component. This follows from the simultaneous chain maps, so it fixes the extension, degree, and sign rather than merely matching its cohomology.

The already constructed support-to-global map is

$$k_p : L_p \longrightarrow D_F = (W \xrightarrow{r_+ - r_-} A \xrightarrow{0} A(-1)),$$

	k_p^0	k_p^1	k_p^2	
$p = 0$	$v \mapsto (v, 0)$	$+1$	$+1$	
$p = \infty$	$v \mapsto (0, v)$	-1	$+1$	

(LNC8.5)

The differential equality at infinity is $(r_+ - r_-)(0, v) = -r_-v = k_p^1 r_-v$; at zero it is $r_+v = k_p^1 r_+v$. Positive complex integration gives $+1$ in degree two at both poles. The same formulas with coefficients J, Q commute with (LNC8.1). Hence the actual coefficient triangles have the naturality square

$$\begin{array}{ccc} L_{G,p} & \xrightarrow{\delta_p} & L_{J,p}[1] \\ \downarrow k_{G,p} & & \downarrow k_{J,p}[1] \\ D_G & \xrightarrow{\delta} & D_J[1]. \end{array} \quad (\text{LNC8.6})$$

Both normal vertical maps are $+1$. Thus (LNC8.4) is the actual global normal extension class, with the same sign.

On the low quotient the support map is instead

$$H_p^1(\mathcal{F}) = Q \longrightarrow H^1(Y, \mathcal{F}) = Q, \quad +1 \text{ at } 0, \quad -1 \text{ at } \infty. \quad (\text{LNC8.7})$$

Thus the local failure to extend a punctured constant section becomes an actual global cohomology class with its stated sign. The two-pole map is the difference $Q^2 \rightarrow Q$. Its dual is the actual anti-diagonal $Q' \rightarrow Q'^2$; the degree-two support sum dualizes to the diagonal $A' \rightarrow A'^2$, as CSD8 proves by restriction of currents.

The compatible angular description is also explicit. CDW gives the coefficient-natural map

$$R\pi_* \mathcal{F} \simeq \Omega \oplus j_{X!} A[\vartheta] [-1] \quad (\text{LNC8.8})$$

simultaneously for $\mathcal{F}_J, \mathcal{F}, \mathcal{G}$. Its angular coefficient row is $j_{X!} J[\vartheta] \rightarrow j_{X!} A[\vartheta] \rightarrow j_{X!} Q[\vartheta]$. Taking its single nonzero Čech degree gives the normal degree-two row. The comparison with positive sphere integration is minus the identity on each of J, A, Q . Indeed use the chart forms

$$\beta_+ = \frac{r^2}{1+r^2} \vartheta, \quad \beta_- = -\frac{1}{1+r^2} \vartheta, \quad \beta_+ - \beta_- = \vartheta, \quad d\beta_+ = d\beta_- = \omega_Y, \quad \int_Y \omega_Y = 1. \quad (\text{LNC8.9})$$

The Čech-total differential identifies the class of $a\vartheta$ with $-a\omega_Y$. Applying this same comparison to both ends and the middle of the short exact row preserves its extension class. This proves the angular-to-normal agreement in (LNC8.6) while retaining the degree-two minus sign.

LNC9. Full arithmetic transport and the FOD7 lifting equation

Let M be the ring of entire functions polynomially bounded on each closed vertical strip. Its action on $\mathcal{B}, \mathcal{J}, \mathcal{Q}$ is multiplication. In original Mellin variable s , the normal action is $h(s+1)$, and the actual power-cover dilation is n^{s+1} , with its factor n . The coordinate transport

$$(\mathcal{V}F)(\lambda) = F(\lambda - 1) \quad (\text{LNC9.1})$$

sends the full normal row to

$$0 \rightarrow \mathcal{J}_+ \rightarrow \mathcal{B} \rightarrow \mathcal{Q}_+ \rightarrow 0, \quad \mathcal{J}_+ = \{F : F^{(j)}(\rho+1) = 0, 0 \leq j < m_\rho\}. \quad (\text{LNC9.2})$$

Its full source divisor is

$$F_+(\lambda) = F_0(\lambda - 1) = \frac{(\lambda - 1)(\lambda - 2)}{8} \pi^{-(\lambda-1)/2} \Gamma((\lambda - 1)/2) \zeta(\lambda - 1). \quad (\text{LNC9.3})$$

The values $1/8$ occur at $\lambda = 1, 2$; the values $\pi/24$ at $\lambda = 0, 3$; the values at $\lambda = 1 - 2r$ are exactly (LNC1.9). At $\lambda = \rho + 1$ every original product derivative and multiplicity m_ρ is transported. The normal class in (LNC8.6) becomes $e_+[-2]$, where e_+ is the extension (LNC9.2).

The existing M -action on $V_p = E_p \oplus s_p J$ is transported through s_p on the J summand; the four endpoint characters of W remain

$$(h(0), h(1), h(1), h(0)). \quad (\text{LNC9.4})$$

Thus r_p, s_p , the costalk row, and its support-to-global square are M -linear. The normal action is shifted, rather than replaced by $h(s)$. Mirrors intertwine the low variable by $s \mapsto 1 - s$, and the normal variable by $\lambda \mapsto 3 - \lambda$; they are not M -linear for an unchanged parameter.

With $Z = \ker(r_+ - r_-)$, the simultaneous quasi-isomorphisms proved in GMS9 are

$$D_J \simeq Z[0] \oplus \mathcal{J}_+[-2], \quad D_F \simeq Z[0] \oplus \mathcal{Q}[-1] \oplus \mathcal{B}[-2], \quad D_G = \mathcal{Q}[-1] \oplus \mathcal{Q}_+[-2],$$

$$\delta : D_G \rightarrow D_J[1], \quad \delta = (0, e_+[-2]). \quad (\text{LNC9.5})$$

Their kernels are the explicit contractible source complexes, not sections of $\mathcal{B} \rightarrow \mathcal{Q}$.

The actual whole-history separator $c = E_+ \in M$, constructed in GSL/GMS, has the full-jet actions

$$c_Q = 1, \quad c_{Q_+} = 0, \quad ce_+ = 0. \quad (\text{LNC9.6})$$

It is not an idempotent on all of $\mathcal{B}, \mathcal{J}_+, D_F, D_J$. For every derived M -morphism $f : \mathcal{Q}[k] \rightarrow D_G$, the actual lifting equation is

$$\delta f = \delta f c_{Q[k]} = \delta c_{D_G} f = c_{D_J[1]} \delta f = 0. \quad (\text{LNC9.7})$$

The last equality follows from the actual component e_+ and (LNC9.6). GMS's complete bar contraction equivalently gives $\text{RHom}_M(\mathcal{Q}, \mathcal{Q}_+) = 0$. Thus in (LNC9.5) the normal component of f is zero in the derived category, and its low component can be retained in D_F . This constructs its derived lift as in FOD7. The map from local costalks to this equation is the proved square (LNC8.6).

This leaves the calculated local invariant-cycle cross unchanged. Its degree-zero boundary is $q : A \rightarrow Q$, and, transporting c back by Θ ,

$$q c_A = c_Q q = q. \quad (\text{LNC9.8})$$

It is preserved, whereas the normal global extension class is killed. The exact morphisms between these receivers are the coefficient row, angularization, and support-to-global maps of LNC7-LNC8. Those maps relate the two boundaries; they do not identify the operators q and e_+ .

The full source representative of normal multiplication is still

$$a \mapsto \Theta^{-1}[(1-E(s))\Theta a(s)](u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} (1-E(1/2+it))\Theta a(1/2+it)u^{-it} dt \in J, \quad (\text{LNC9.9})$$

followed by $H = \Sigma^{-1}$ for a Schwartz representative. Here $E_+(s+1) = 1 - E(s)$. This retains the actual full ideal and original source; no smaller kernel is substituted.

LNC10. The resulting diagram and its continuous dual

The complete path of constructed maps is:

pole $V_p \dashrightarrow$ nearby A , with monodromy identity

$$\begin{array}{c} | \\ \text{boundary } q \\ \downarrow \\ Q = \text{local vanishing quotient} \\ | \\ \text{support-to-global: } +1 \text{ at } 0, -1 \text{ at infinity} \\ \downarrow \\ H^1(Y, F) = Q \end{array}$$

$$\begin{array}{ll} \text{nearby coefficient row:} & 0 \rightarrow J \rightarrow A \rightarrow Q \rightarrow 0 \\ \text{angular row:} & 0 \rightarrow J[\text{angle}] \rightarrow A[\text{angle}] \rightarrow Q[\text{angle}] \rightarrow 0 \\ \text{positive normal row:} & 0 \rightarrow J(-1) \rightarrow A(-1) \rightarrow Q(-1) \rightarrow 0 \\ \text{full shifted Mellin row:} & 0 \rightarrow I_{\text{plus}} \rightarrow B \rightarrow Q_{\text{plus}} \rightarrow 0 \\ & \text{class } e_{\text{plus}} \\ & c_{e_{\text{plus}}} = 0 \end{array}$$

Every arrow has been constructed above, including the common minus sign from angular Čech cohomology to positive integration. The local invariant-cycle receiver is Q . Its normal twist is the quotient in the actual normal extension, whose class is e_+ . The cyclic class module Me_+ is a module of central multiples of that extension class; it is not identified with Q .

Continuous support duality retains the additional exact comparison

$$Q = H^0(R\phi\mathcal{F}) \mapsto Q' = H^{-2}(R\phi\mathcal{D}_c\mathcal{F}) = \ker(A' \xrightarrow{r'_p} V'_p). \quad (\text{LNC10.1})$$

The complete original-zeta jet functionals, including multiplicities, are in this actual continuous quotient dual. Dual specialization is identity on its separate nearby A' in degree -2 ; Q' remains in stalk degree -1 and vanishing degree -2 .

Taking the transpose of (LNC9.8) gives

$$c'_A q' = q', \quad (c_Q)' = 1_{Q'}. \quad (\text{LNC10.2})$$

This is the transpose of the specified coefficient action, not an inserted Hermitian identification or inverse-parameter contragredient. Thus the dual local quotient also survives the separator. Its pairings remain the actual $Q' \times Q \rightarrow \mathbb{C}$, induced from $A' \times A$, and the compact-test integration maps of LNC6.

This specifies the comparison with DC. There the support group receiving ∂_i , after dividing by the coinvariant boundary, has a weight range separated from the classes in C_i ; that proves its boundary zero. Here the actual cross has been calculated: at $i = 0$ its receiver is Q and its map is q ; at $i = 1$ the coinvariant term accounts for the entire boundary. Under full continuous duality the first quotient becomes the explicit Q' kernel and vanishing term, with all endpoint degrees retained.

FOD7 supplies a genuine global derived lift in this same sheaf’s coefficient triangle, related through (LNC8.6). It does not make $r_p : V_p \rightarrow A$ onto. These conclusions compute the available lift and the residual local receiver on the original source and its actual dual. They do not assign an unconstructed finite-field weight filtration to the infinite coefficient spaces or assert a new RH conclusion.

LNC11. Source-reading and use record

For this derivation, CSP2-CSP8, CDW2-CDW6 and CDW8-CDW9, CSD4-CSD9, DC3-DC9, and FOD6-FOD7 were read as proof sources. Their full local filenames are:

- CC_SPHERE_PULLBACK_AND_NORMAL_DIRECTION.md;
- CC_DERIVED_DIRECT_IMAGE_AND_WINDING.md;
- CC_CONTINUOUS_SUPPORT_DUAL_SHEAF.md;
- DELIGNE_INVARIANT_CYCLE_QUOTIENT.md;
- CC_FULL_SOURCE_OBSTRUCTION_DECOMPOSITION.md.

The complete accepted GMS proof in CC_GLOBAL_MULTIPLIER_SEPARATION.md was previously authored and independently reviewed in this task. Its actual modules and category are retained in LNC9. The ordinary localization calculation is also recorded in CW10 of CC_ACTUAL_WEIGHT_LIFT_COMPARISON.md; it is used as an established calculation, not claimed as newly discovered here.

The original author TeX source sources/Reich_1002_1686_v4/author_source/beilinson_nearby-cycles.tex, lines 251–382, was read for the nearby-cycle definition and monodromy convention. Its public source identity is [Ryan Reich, Notes on Beilinson’s “How to glue perverse sheaves”, arXiv:1002.1686v4](#). No finite-dimensional decomposition or constructibility theorem in that source is asserted for A .

The newly proved comparisons are: the actual universal-cover nearby object and monodromy (LNC2); its invariant/coinvariant cross and exact quotient (LNC3-LNC4); the distinction between a power cover and a power local function, retaining both entire nearby objects (LNC5); continuous-current specialization and the shifted dual defect (LNC6-LNC7); and the natural morphism from this cross through the full coefficient row to FOD7’s global lift (LNC8-LNC10). The earlier ordinary localization maps are retained with their original signs to prove those comparisons.

Exact local dual arrows and the global residue-pairing receiver — LVD0-LVD9

Proof source: CC_LOCAL_VANISHING_DUAL_ARROWS.md.

Local vanishing arrows and the actual global residue pairing

Complete derivation, 24 September 2026. Proof locators LVD0–LVD9.

Historical attribution: these nearby/vanishing-cycle constructions go back to Deligne’s SGA7, ExposésXIII–XIV, and Beilinson’s *How to glue perverse sheaves*. The inspected explanatory author source is Ryan Reich, [Notes on Beilinson’s “How to glue perverse sheaves”](#), [arXiv:1002.1686v4](#), definitions and gluing maps at the source ranges recorded above. Credit to those originals is retained through Reich; this edition does not claim a fresh reading of the original SGA or Beilinson chapters.

Human-source attribution: the extension of continuous linear functionals is the Hahn–Banach theorem, not a programme result. See Terence Tao, [245B, Notes6: Duality and the Hahn–Banach theorem](#), Theorem1 and its complex proof (26 January2009). For the locally convex use here, continuity bounds a functional by a continuous seminorm; quotienting its kernel reduces this application to that normed-space theorem. No continuous splitting of the original quotient is thereby asserted.

Human-source attribution: the unconditional zero-counting estimate is the classical Riemann–von Mangoldt theorem. The inspected native-TeX witness is Alain Connes, [The Riemann Hypothesis: Past, Present and a Letter Through Time](#), the Hardy–Littlewood discussion immediately before “Zero-free regions and zero-density estimates” (author TeX line499). It states the asymptotic that implies the bound used here; this is not a new zero-counting theorem.

LVD0. Constructed source and scope

The supporting datum is still $\tau\langle Z_1; \text{no } Z_2 \rangle$. This calculation acts on CSP’s receiving sheaf after the complete arithmetic reconstruction. No addition, midpoint, metric, numerical coordinate or new parity is assigned to that support. The map of receiving spaces is

$$\pi : \mathbb{P}^1(\mathbb{C}) \longrightarrow \{x_+, \eta, x_-\}, \quad \pi(0) = x_+, \quad \pi(\infty) = x_-, \quad \pi(\mathbb{C}^\times) = \eta. \quad (\text{LVD0.1})$$

In particular a pole disk is not a numerical chart on τ . Both arithmetic branches retain their own reconstructed counters. Gluing the two geometric charts of this one sheaf is not pooling two arithmetic return measures.

The source is the actual restriction diagram derived from Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, [arXiv:0903.2024v3](#), §5, with the later support and real-structure comparison to [arXiv:2609.00299v1](#). CSP0 records the exact author-TeX reading. CSP, CSD and SSI/ESI prove the coefficient, inverse-image and continuous-dual constructions used here. DC proves the distinct Deligne invariant-cycle cross; no identity with its arithmetic weight hypotheses is presumed.

For nearby and vanishing cycles, Ryan Reich’s original author source, *Notes on Beilinson’s “How to glue perverse sheaves”*, [arXiv:1002.1686v4](#), supplies the universal-cover and cone definitions. Its author TeX lines261–303 and356–375 give the defining triangles; lines850–911 give the gluing maps; lines1161–1220

identify the shifted vanishing functor. The finite-constructibility assertions of that paper are not applied to these infinite coefficient spaces. Every local complex below is calculated directly.

Use cohomological shifts, $C[1]^n = C^{n+1}$, $d_{C[1]} = -d_C$, and

$$\text{Cone}(f)^n = B^n \oplus A^{n+1}, \quad d(b, a) = (d_B b + f a, -d_A a) \quad (\text{LVD0.2})$$

for $f : A \rightarrow B$. Derived sheaf statements are in sheaves of complex vector spaces. Coefficient maps retain their actual locally convex topologies. A derived category of arbitrary topological vector spaces is not assumed.

LVD1. The full original coefficient row

Retain

$$\begin{aligned} S &= \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0\}, \\ A &= \left\{a \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \text{ for every } N, j \geq 0\right\}, \\ \Sigma f(u) &= 2 \sum_{k \geq 1} f(ku), \quad Ra(u) = u^{-1}a(u^{-1}), \quad V_p = S \oplus E_p, \quad E_p = \mathbb{C}e_{p,0} \oplus \mathbb{C}e_{p,1}, \\ r_+(f, e) &= \Sigma f, \quad r_-(f, e) = R\Sigma f = \Sigma \hat{f}, \quad \hat{f}(t) = \int_{\mathbb{R}} f(v) e^{-2\pi i v t} dv. \end{aligned} \quad (\text{LVD1.1})$$

Before restriction to S , Poisson summation contains the full terms

$$\Sigma \hat{f}(u) = u^{-1} \Sigma f(u^{-1}) + u^{-1} f(0) - \int_{\mathbb{R}} f(v) dv. \quad (\text{LVD1.2})$$

The two endpoint coordinates on each chart are retained separately. SSI/ESI prove that $J = \Sigma S$ is closed in A , that $\Sigma : S \rightarrow J$ has continuous inverse, and that

$$0 \longrightarrow J \xrightarrow{\iota} A \xrightarrow{q} Q = A/J \longrightarrow 0 \quad (\text{LVD1.3})$$

is the actual strict Fréchet quotient row.

On a disk D_p around one pole, write $i : \{p\} \hookrightarrow D_p$ and $j : D_p^* \hookrightarrow D_p$. CSP's sheaf is

$$\mathcal{F}|_{D_p} = \underline{A}_{D_p} \times_{i_* A} i_* V_p. \quad (\text{LVD1.4})$$

The source inverse gives continuous decompositions $V_p = J \oplus E_p$ with restriction $(b, e) \mapsto b$. Explicitly the section of r_+ is $b \mapsto (\Sigma^{-1}b, 0)$; for r_- it is $b \mapsto (\widehat{\Sigma^{-1}b}, 0)$. Hence

$$\mathcal{F}|_{D_p} \simeq \mathcal{F}_{\text{red}} \oplus i_* E_p, \quad \mathcal{F}_{\text{red}} = \ker(\underline{A}_{D_p} \xrightarrow{q \text{ ev}_p} i_* Q). \quad (\text{LVD1.5})$$

This is an explicit decomposition of the existing sheaf, not deletion of endpoints. It gives the exact sheaf row

$$0 \rightarrow \mathcal{F}_{\text{red}} \rightarrow \underline{A}_{D_p} \xrightarrow{q \text{ ev}_p} i_* Q \rightarrow 0. \quad (\text{LVD1.6})$$

Set $P = \mathcal{F}_{\text{red}}[1]$. A representative is $\text{Cone}(q \text{ ev}_p)$; its terms are $\underline{A}$ in degree -1 and $i_* Q$ in degree zero, with differential $+q \text{ ev}_p$. With the literal injection and projection of this cone, the rotated triangle is

$$i_* Q \longrightarrow P \longrightarrow \underline{A}[1] \xrightarrow{-q \text{ ev}_p[1]} i_* Q[1]. \quad (\text{LVD1.7})$$

The displayed minus is the rotation sign.

LVD2. Nearby cycles from the actual universal cover

Use the positive local coordinate $z_p = z$ at zero and $z_p = 1/z$ at infinity. The map $t \mapsto e^t$ from the covering half-plane is the universal cover of a sufficiently small punctured disk. The pullback of $\mathcal{F}_{\text{red}}|_{D_p^*} = \underline{A}$ is constant. The coefficient-valued de Rham resolution contracts on that half-plane: the radial homotopy integrates along the line segment to a fixed interior point. Completeness of A gives the integrals; for every seminorm the bound $p(\int f) \leq \int p(f)$ proves continuity. The only cohomology is the constant coefficient A in degree zero.

Deck translation $t \mapsto t + 2\pi i$ fixes these constants. Therefore

$$R\psi_{z_p} \mathcal{F}_{\text{red}} = A[0], \quad T = 1, \quad R\psi_{z_p} \mathbb{P} = A[1]. \quad (\text{LVD2.1})$$

The stalk of $\mathcal{F}_{\text{red}}$ is J , and its specialization to nearby constants is the actual inclusion $\iota : J \rightarrow A$. Put $\Psi = R\psi[-1]$ and $\Phi = R\phi[-1]$, with $R\phi$ the cone of specialization. Then

$$\Psi\mathbb{P} = A, \quad \Phi\mathbb{P} \simeq [J \xrightarrow{\iota} A] \simeq Q[0], \quad \text{can} = q. \quad (\text{LVD2.2})$$

The two terms of the displayed complex have degrees $-1, 0$. Starting with the literal shifted cone gives differential $-\iota$; negating its J -coordinate gives the displayed complex and leaves the quotient map on A unchanged.

No rank hypothesis is used. In particular the quotient in (LVD2.2) includes all actual zero jets, not only the values visible to the trace form later.

LVD3. The positive-angle convention fixes variation

Let $\vartheta_p = d \arg z_p / (2\pi)$ be the positive angular class. For the constant nearby complex $C = R\psi K$, the punctured-disk complex is $C \oplus C[-1]$, with the second summand the positive angular coordinate. Write the actual restriction as

$$(u, v) : S = i^* K \longrightarrow C \oplus C[-1]. \quad (\text{LVD3.1})$$

The shifted vanishing complex $\text{Cone}(u)[-1]$ has differential

$$d(c, s) = (-d_C c - us, d_S s). \quad (\text{LVD3.2})$$

Define

$$\text{can}(c) = (c, 0), \quad \text{var}(c, s) = v(s). \quad (\text{LVD3.3})$$

These are chain maps. Indeed the first uses the differential $-d_C$ on $C[-1]$; the second uses the chain identity for $v : S \rightarrow C[-1]$. Permuting coordinates identifies $\text{Cone}(\text{var})[-1]$ with the restriction fibre cone $\text{Cone}((u, v))[-1]$, including its differentials. This constructs variation with the chosen positive angular sign. Its sign is not deduced from the equation $\text{var can} = 1 - T = 0$, which by itself would not determine a sign.

For $K = \mathbb{P}$, the restriction is $(\iota[1], 0)$. Thus

$$\boxed{\Psi\mathbb{P} = A, \quad \Phi\mathbb{P} = Q, \quad \text{can} = q, \quad \text{var} = 0.} \quad (\text{LVD3.4})$$

The costalk is, after negating the shifted J -coordinate,

$$i^! \mathbb{P} = [J \xrightarrow{\iota} A \xrightarrow{0} A] \quad \text{in degrees } -1, 0, 1. \quad (\text{LVD3.5})$$

Its cohomology is Q in degree zero and A in degree one. Before the cochain shift, the boundary of the positive angular class is minus the positive local fundamental class, as CSP6 proves by $-d\chi \wedge \vartheta_p$. That orientation convention is retained; (LVD3.5) is a shifted complex, not a changed angular orientation.

LVD4. The dual arrows, with their full chain signs

Write $d = *$ for the algebraic dual or $d = '$ for the continuous dual, retaining their distinct domains. For $d = *$ take ordinary Verdier duality. For $d = '$ take CSD's literal continuous compact-test current complex and shift it by $[-1]$; denote that specific object by $D_c P$. No equivalence between these duals is asserted. Put $U = A^d$.

CSD5-6 (and the algebraic current comparison VSD) calculate the dual stalk, restriction and costalk by explicit currents. For the endpoint-reduced shifted object they give

$$i^* D_d P = [U \xrightarrow{0} U \xrightarrow{-\iota^d} J^d] \quad \text{in degrees } -1, 0, 1, \quad (\text{LVD4.1})$$

$$R\psi D_d P = U[1], \quad i^! D_d P = J^d[-1]. \quad (\text{LVD4.2})$$

Restriction to the puncture is identity on the first U into its degree -1 constant coordinate and identity on the second U into its degree-zero angular coordinate. The final J^d restricts to zero. The minus in (LVD4.1) is the differential of the shift of CSD's dual stalk; it is not omitted.

Apply (LVD3.2). The vanishing complex has U in degree -1 , $U \oplus U$ in degree zero and J^d in degree one, with

$$b \mapsto (-b, 0), \quad (a, b) \mapsto -\iota^d b. \quad (\text{LVD4.3})$$

The first pair is contractible. The remaining complex is

$$[U \xrightarrow{-\iota^d} J^d] \quad \text{in degrees } 0, 1. \quad (\text{LVD4.4})$$

Algebraic extension of functionals, or Hahn-Banach for the continuous dual of the closed subspace $J \subset A$, proves that ι^d is onto. Its kernel is exactly the image of $q^d : Q^d \rightarrow A^d$. Hence (LVD4.4) has cohomology Q^d in degree zero, represented by $q^d \mu$. The canonical map lands in the contracted coordinate $(a, 0)$; variation maps (a, b) to b . Consequently

$$\boxed{\Psi D_d P = A^d, \quad \Phi D_d P = Q^d, \quad \text{can} = 0, \quad \text{var} = +q^d.} \quad (\text{LVD4.5})$$

The plus has just been proved in the positive angular coordinate. A literal dual of the primal two-term stalk has differential $-q^d$; that minus remains in the corresponding shifted restriction cone. The assertion is not a convention-independent unsigned transpose formula.

For $d = '$, SDT proves $q' : Q'_\beta \simeq J^\perp \subset A'_\beta$ topologically by bounded lifts. CSD proves the stated stable small-disk cohomology topologies. This argument does not assert strong openness of $A'_\beta \rightarrow J'_\beta$.

The full endpoint summands are

$$P_{\text{full}} = P \oplus i_* E_p[1], \quad D_d P_{\text{full}} = D_d P \oplus i_* E_p^d[-1]. \quad (\text{LVD4.6})$$

They have zero nearby cycles and shifted vanishing summands $E_p[1]$ and $E_p^d[-1]$, respectively. They have not been absorbed into the degree-zero Q or Q^d .

LVD5. All degree-zero sheaf pairings factor through the constant part

The dual of the exact source triangle, or the direct current-cone construction, gives

$$\underline{A}^d[1] \longrightarrow D_d P \longrightarrow i_* Q^d \longrightarrow \underline{A}^d[2]. \quad (\text{LVD5.1})$$

Its final Gysin sign depends on the stated quotient coordinate; the positive-angular construction of the separate residue Gysin receiver retains it explicitly. Only the existence of these actual arrows is needed for the following factorization.

Put $C = \underline{A}[1]$, $S_0 = i_*Q$, and $C^d = \underline{A}^d[1]$. From (LVD4.2) and adjunction,

$$\mathrm{Hom}(S_0, D_d\mathbf{P}) = \mathrm{Hom}(Q, J^d[-1]) = 0, \quad \mathrm{Hom}(S_0[1], D_d\mathbf{P}) = 0. \quad (\text{LVD5.2})$$

These are degree-zero derived Hom groups over a field; the negative shifts make them zero. Applying Hom to (LVD1.7) therefore shows that every $f : \mathbf{P} \rightarrow D_d\mathbf{P}$ factors uniquely through $\mathbf{P} \rightarrow C$. Also

$$\mathrm{Hom}(C, i_*Q^d) = \mathrm{Hom}(A[1], Q^d) = 0, \quad \mathrm{Hom}(C, i_*Q^d[-1]) = 0. \quad (\text{LVD5.3})$$

Applying Hom to (LVD5.1) now proves the unique factorization

$$\boxed{f : \mathbf{P} \longrightarrow \underline{A}[1] \xrightarrow{h[1]} \underline{A}^d[1] \longrightarrow D_d\mathbf{P}.} \quad (\text{LVD5.4})$$

Here $h : A \rightarrow A^d$ is an arbitrary coefficient-linear map. The continuous coefficient construction uses continuous h ; the factorization theorem itself is stated in the underlying vector-space sheaf category. The same proof holds with a conjugate source and $h : \overline{A} \rightarrow A^d$.

Its induced vanishing map is zero. Equivalently the natural canonical-arrow square has $\Phi(f)q = 0$, since the target canonical map in (LVD4.5) is zero; surjectivity of q forces $\Phi(f) = 0$. Thus no such map is an isomorphism when $Q \neq 0$. This computes the precise local arrow; it does not exclude pairings on nearby coefficients or global cohomology.

The source triangle (LVD1.7) itself is nonsplit when $Q \neq 0$. Indeed $\mathbf{P} = \mathcal{F}_{\text{red}}[1]$ has ordinary sheaf cohomology $\mathcal{H}^0(\mathbf{P}) = 0$, whereas the direct sum required by a split triangle has $\mathcal{H}^0(i_*Q \oplus \underline{A}[1]) = i_*Q \neq 0$. This is a direct derived-sheaf proof; it requires no finite-rank perverse-sheaf theorem or unconstructed topological heart.

The same factorization holds on the entire sphere with i_*Q replaced by the two-pole skyscraper $i_*(Q \oplus Q)$. The support/stalk adjunctions are the direct sums of (LVD5.2)–(LVD5.3). Retaining endpoints introduces no further degree-zero pairing maps: the cross groups are

$$\mathrm{Hom}(E_p[1], J^d[-1]) = 0, \quad \mathrm{Hom}(J[1], E_p^d[-1]) = 0, \quad \mathrm{Hom}(E_p[1], E_p^d[-1]) = 0. \quad (\text{LVD5.5})$$

Maps between distinct supported poles vanish as well. All vanish by adjunction and the indicated negative shift, without discarding an endpoint.

Since $H^1(\mathbb{P}^1(\mathbb{C}); \underline{A}) = 0$, every degree-zero sheaf pairing of (LVD5.4) induces the zero map on $H^0(R\Gamma\mathbf{P}) = Q$. The equality for constant-sheaf cohomology follows from the two-disk/circle resolution of CSP3 and does not require finite-dimensional A .

LVD6. The original-zeta residue pairing is nevertheless nonzero

Use the existing full transform

$$\Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, \quad \kappa : Q \xrightarrow{\sim} \mathcal{B}/\mathcal{I}, \quad [a] \mapsto [\Theta a], \quad (\text{LVD6.1})$$

where $\mathcal{B}$ is the entire rapid-decay space on every closed vertical strip, and $\mathcal{I}$ is the full ideal of jets vanishing to the actual multiplicities of all nontrivial zeros of the original ζ . Let $\rho^\# = 1 - \bar{\rho}$. The proved residue/trace receiver RTT gives the continuous linear map with conjugate source

$$\mathcal{W} : \overline{Q} \longrightarrow Q'_\beta, \quad \mathcal{W}(\overline{[b]})([a]) = \sum_\rho m_\rho \Theta a(\rho) \overline{\Theta b(\rho^\#)}. \quad (\text{LVD6.2})$$

The sum runs over distinct actual zeros; m_ρ is retained. It is the full-zero value trace, not an identification of the full-jet quotient with its values. Its absolute convergence follows by choosing strip-decay exponent N with $2N > 2$ and using the original zero-counting bound $N_\zeta(T) = O(T \log(2 + T))$. The sum of $m_\rho(1 + |\operatorname{Im} \rho|)^{-2N}$ then converges. The same estimate, uniformly on bounded subsets of Q using the proved quotient seminorms/bounded lifts, gives strong-dual continuity, as RTT1 proves.

Retain the entire original transform of the actual Schwartz test:

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}, \quad F_0(s) = \Theta \Sigma f_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{LVD6.3})$$

At an actual zero σ of multiplicity m_σ , $F_\sigma(s) = F_0(s)/(s - \sigma)^{m_\sigma}$ is entire and belongs to $\mathcal{B}$. The complete local unit gives $F_\sigma(\sigma) \neq 0$, while F_σ vanishes at every other zero with its full multiplicity. This proves $Q \neq 0$. Taking F_σ and $F_{\sigma^\#}$ in (LVD6.2) proves that $\mathcal{W} \neq 0$, with no assumption of RH or simple zeros. Applying these isolators separately also proves

$$\ker \mathcal{W} = \overline{N_0}, \quad N_0 = \{[a] \in Q : \Theta a(\rho) = 0 \text{ for every actual } \rho\}. \quad (\text{LVD6.4})$$

Higher jets in N_0 are retained as this exact kernel, not identified with zero in Q .

The original unit, pole and trivial-zero contributions remain

$$F_0(0) = F_0(1) = \frac{1}{8},$$

$$F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2^r r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r), \quad r \geq 1. \quad (\text{LVD6.5})$$

The source equality is $\mathcal{M}_0 \Sigma f = 2\zeta \mathcal{M}_0 f$ on $\Re s > 1$, with the full factor 2; $\Theta = \mathcal{M}_0/2$. RTT's residue comparison uses the complete original logarithmic derivative, not a substituted zeta. Nothing in the sheaf calculation alters these source identities or the retained arithmetic return measure.

LVD7. Explicit realization on the complete global complex

Let $W_0 = V_+ \oplus V_-$ and $r = r_+ - r_- : W_0 \rightarrow A$. CSP's complete sphere complex for $\mathcal{F}$ is

$$D_{\mathcal{F}} = [W_0 \xrightarrow{r} A \xrightarrow{0} A] \quad \text{in degrees } 0, 1, 2. \quad (\text{LVD7.1})$$

Thus $K = D_{\mathcal{F}}[1]$ has degrees $-1, 0, 1$ and differential $-r$ from degree -1 to zero. Its continuous Hom dual, with $(K^\vee)^n = (K^{-n})'$ and $d_n^\vee = (-1)^{n+1} (d_K^{-n-1})'$, is

$$K^\vee = [A' \xrightarrow{0} A' \xrightarrow{+r'} W'_0] \quad \text{in degrees } -1, 0, 1. \quad (\text{LVD7.2})$$

CSD's explicit proper global current comparison realizes this dual with its shift sign retained. Its previously displayed complex is $D_{\mathcal{F}}^\vee[-1]$, whose last differential is $-r'$, while (LVD7.2) has $+r'$. The isomorphism from (LVD7.2) to that shifted complex multiplies degree n by $(-1)^n$: its signs in degrees $-1, 0, 1$ are $-1, +1, -1$. Substituting the differentials verifies the chain identity. In particular the degree-zero pairing below is unchanged. No assertion that arbitrary continuous dualization preserves quasi-isomorphisms is required.

Define the actual continuous coefficient arrow

$$h = q' \mathcal{W} \bar{q} : \bar{A} \longrightarrow A'. \quad (\text{LVD7.3})$$

Set a cochain map $\mathfrak{w} : \overline{K} \rightarrow K^\vee$ to be zero in degrees $-1, 1$ and h in degree zero. Both chain identities follow from the original quotient row:

$$h\bar{r} = q'\mathcal{W}\bar{q}\bar{r} = 0, \quad r'h = (qr)'\mathcal{W}\bar{q} = 0. \quad (\text{LVD7.4})$$

These identities retain the source differential $-r$ and target differential $+r'$. On cohomology,

$$H^0(K) = A/J = Q, \quad H^0(K^\vee) = \ker r' = q'Q', \quad H^0(\mathfrak{w}) = \mathcal{W} \quad (\text{LVD7.5})$$

under the specified quotient and transpose identifications.

Therefore the original residue pairing has an explicit continuous representative on the complete global cohomology complex, including its endpoint and degree-two terms. It is nonzero by (LVD6.2)-(LVD6.4). By (LVD5.4)-(LVD5.5), it cannot be induced by a degree-zero sheaf morphism $\overline{P}_{\text{full}} \rightarrow D_c P_{\text{full}}$ on this sphere: every such sheaf morphism induces zero in this cohomological degree. This is the precise difference between the constructed global pairing and a pairing compatible with restriction to every open set. It neither denies the global pairing nor replaces its continuous coefficients by finite-dimensional ones.

The calculation constructs an entire receiver, not only one example. In the ordinary vector-space derived category put

$$\mathcal{G} = \text{Hom}_{D(\mathbb{C})}(\overline{R\Gamma P_{\text{full}}}, R\Gamma D_* P_{\text{full}}), \quad \mathcal{L} = \text{im}(\text{Hom}_{D(Y)}(\overline{P_{\text{full}}}, D_* P_{\text{full}}) \xrightarrow{R\Gamma} \mathcal{G}). \quad (\text{LVD7.6})$$

The existing algebraic current comparison identifies the target global complex with K^* ; its shift signs are exactly those above. Taking degree-zero cohomology defines

$$\Pi : \mathcal{G} \longrightarrow \text{Hom}_{\mathbb{C}}(\overline{Q}, Q^*). \quad (\text{LVD7.7})$$

For any $W : \overline{Q} \rightarrow Q^*$, the formula $H_W^0 = q^*W\bar{q}$, $H_W^{-1} = H_W^1 = 0$, defines a cochain map by the same two identities (LVD7.4). Its derived class gives a linear map $\mathcal{J} : W \mapsto [H_W]$, and $\Pi\mathcal{J} = 1$ because of (LVD7.5). Since (LVD5.4) proves $\Pi\mathcal{L} = 0$, these maps induce a split exact sequence

$$0 \rightarrow \ker \overline{\Pi} \rightarrow \mathcal{G}/\mathcal{L} \xrightarrow{\overline{\Pi}} \text{Hom}_{\mathbb{C}}(\overline{Q}, Q^*) \rightarrow 0, \quad \overline{\Pi}\overline{\mathcal{J}} = 1. \quad (\text{LVD7.8})$$

This proves the specified full pairing space is a direct summand of the difference between global derived maps and maps coming from sheaves. It does not claim the other summand is zero. No topology is assigned to this algebraic cokernel. The continuous maps $W : \overline{Q} \rightarrow Q'_\beta$ enter it through the actual inclusion $Q' \hookrightarrow Q^*$ and the continuous chain representatives just constructed. The actual $\mathcal{W}$, with its retained kernel $\overline{N}_0$, is one nonzero member. This is an exact receiver for the failed local realization, with its injection and left inverse proved.

LVD8. What the comparison supplies for the next lift

The earlier separator calculation FOD7 proves that the actual normal connecting class vanishes after every specified source $\mathcal{Q}[k]$ in the stated derived category of M -modules, where M is the constructed entire multiplier algebra. This is not a vanishing for arbitrary sheaf maps from $i_*\mathcal{Q}[k]$ into the local triangle, and it does not identify all local canonical maps with zero. The local canonical map calculated here is the original quotient $q : A \rightarrow Q$. Its dual has canonical map zero and variation $+q'$. Both statements are compatible through the explicit source and normal extension rows calculated in FOD/NEA and the nearby-cycle comparison.

The next attempted lift is now specified by actual arrows rather than by a proposed positivity assumption: lift the supported map $i_*\mathcal{W} : i_*\overline{Q} \rightarrow i_*Q'$ through the continuous-dual triangle (LVD5.1). Its connecting map is the Gysin image of $q'\mathcal{W}$, with the quotient-coordinate sign retained. The pullback of that triangle

constructs a receiver even when this connecting map is nonzero. The separate full Gysin-receiver derivation calculates it directly, including its kernel $\overline{N}_0$. This is the concrete continuation of (LVD7.5); the missing sheaf map is not installed as a hypothesis.

Deligne's exact obstruction in DC is $\text{im } \partial/\partial K$, with its own specialization, inertia and duality. The new maps determine how the actual coefficient quotient and its residue pairing enter a local specialization/duality calculation. They do not identify the present nonconstant infinite coefficient sheaf with a proper smooth-total finite-field family, or derive its arithmetic purity from its zero topological monodromy. The nonzero residue pairing, its global representative, and the precise local Gysin receiver are all retained for that comparison.

LVD9. Verification scope

Every displayed local group and arrow was derived from the actual disk, puncture, source restriction and current cone. The full endpoint summands, both poles, cochain shifts, transpose signs, source kernels and value-trace kernel are recorded. The factorization theorem is about this degree-zero sheaf arrow. The global cochain pairing is proved on the complete original complex, not merely on a list of eigenvalues. No conclusion about RH, a common arithmetic modulus, or the completion of WU061 is claimed by these statements.

The actual residue Gysin obstruction and its complete current receiver — RGR0-RGR12

Proof source: CC_RESIDUE_GYSIN_RECEIVER_INDEPENDENT.md.

The supported residue map, its signed Gysin obstruction, and its actual receiver

Independent derivation, 24 September 2026. Proof locators RGR0–RGR12.

Historical attribution: these nearby/vanishing-cycle constructions go back to Deligne’s SGA7, ExposésXIII–XIV, and Beilinson’s *How to glue perverse sheaves*. The inspected explanatory author source is Ryan Reich, [Notes on Beilinson’s “How to glue perverse sheaves”](#), [arXiv:1002.1686v4](#), definitions and gluing maps at the source ranges recorded above. Credit to those originals is retained through Reich; this edition does not claim a fresh reading of the original SGA or Beilinson chapters.

Human-source attribution: the extension of continuous linear functionals is the Hahn–Banach theorem, not a programme result. See Terence Tao, [245B, Notes6: Duality and the Hahn–Banach theorem](#), Theorem1 and its complex proof (26 January2009). For the locally convex use here, continuity bounds a functional by a continuous seminorm; quotienting its kernel reduces this application to that normed-space theorem. No continuous splitting of the original quotient is thereby asserted.

Human-source attribution: the unconditional zero-counting estimate is the classical Riemann–von Mangoldt theorem. The inspected native-TeX witness is Alain Connes, [The Riemann Hypothesis: Past, Present and a Letter Through Time](#), the Hardy–Littlewood discussion immediately before “Zero-free regions and zero-density estimates” (author TeX line499). It states the asymptotic that implies the bound used here; this is not a new zero-counting theorem.

RGR0. Construction stage and the precise problem

The supporting datum remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. The operations in this note are on the already constructed arithmetic coefficient sheaf. They give no addition, subtraction, midpoint, metric, vector coordinate, or extra parity to τ . The complete-history reconstruction and separate branch counters precede these coefficient operations. The connected user argument, its correction table, and the full passages U126–U128, U135–U141, WU050–WU055 and WU061–WU062 were consulted before this derivation. In particular, a local analytic lifting problem is not identified with the user’s entire construction or with Deligne’s arithmetic lifting problem merely by using the word obstruction.

The actual inputs are CSP’s inverse-image sheaf, CSD’s continuous compact-test current dual, and RTT’s original-zeta residue map. The source extension is the retained original component e_0 of FOD3. The calculation asks whether the residue map on the supported quotient lifts to the existing local dual sheaf. It computes the exact obstruction and then constructs its homotopy-pullback receiver. RGR9 derives its relation to the original coefficient extension. RGR11 also carries the already computed positive-adjoint defect into the same local receiver; it does not substitute the nonzero total residue form for that defect.

All derived sheaf calculations below take place in sheaves of complex vector spaces. The specified maps between coefficient spaces and currents remain continuous. No derived category of arbitrary locally

convex spaces, finite-dimensionality of the coefficient spaces, Artinian perverse category, or equality of continuous and algebraic duals is assumed.

The primary geometric provenance is Connes-Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, [arXiv:0903.2024v3](#), §5, transported by the explicit map in CSP0.2. The independent calculations used here are complete local proof files retained beside this note: CC_SPHERE_PULLBACK_AND_NORMAL_DIRECTION.md, CSP0-CSP7; CC_CONTINUOUS_SUPPORT_DUAL_SHEAF.md, CSD1-CSD6 and CSD10-CSD11; CC_FULL_SOURCE_OBSTRUCTION_DECOMPOSITION.md, FOD1-FOD5; and the peer's RESIDUE_TRACE_TRANSFER_INDEPENDENT.md, RTT0-RTT6. The positive-adjoint receiver in RGR11 uses GEOMETRIC_TRANSFER_POSITIVE_ADJOINT_DEFECT.md, GTAH0-GTAH2. The exact proof dependence is given rather than attributing these programme calculations to the primary paper.

RGR1. Full coefficient source and original-zeta factors

Use

$$A = \left\{ a \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \quad (N, j \geq 0) \right\},$$

$$S = \{ f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0 \}, \quad \Sigma f(u) = 2 \sum_{n \geq 1} f(nu),$$

$$J = \Sigma S \subset A, \quad Q = A/J, \quad q : A \longrightarrow Q. \quad (\text{RGR1.1})$$

The exact summation-image theorem used by CSP proves that J is closed and that $\Sigma : S \rightarrow J$ is a topological isomorphism. Both A and Q carry their proved Fréchet topologies.

The full Mellin comparison is

$$\Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, \quad \Theta^{-1} F(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} F(1/2 + it) u^{-it} dt. \quad (\text{RGR1.2})$$

It identifies A with the entire rapid-strip space

$$\mathcal{B} = \{ F \text{ entire} : b_{R,N}(F) = \sup_{|\Re s| \leq R} (1 + |\Im s|)^N |F(s)| < \infty \text{ for every } R, N \geq 0 \}$$

and J with

$$\mathcal{J} = \{ F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every actual nontrivial zero } \rho \text{ of } \zeta, 0 \leq j < m_\rho \}. \quad (\text{RGR1.3})$$

Thus $\Theta_Q : Q \rightarrow \mathcal{Q} = \mathcal{B}/\mathcal{J}$ is the exact quotient isomorphism. The summation formula before continuation is

$$\Theta \Sigma f(s) = \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v} \quad (\Re s > 1). \quad (\text{RGR1.4})$$

In particular the original Schwartz function and its entire image remain

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}, \quad F_0(s) = \Theta \Sigma f_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{RGR1.5})$$

The exceptional values are

$$F_0(0) = F_0(1) = \frac{1}{8}, \quad F_0(-1) = \frac{\pi}{24}, \quad F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) \quad (r \geq 1). \quad (\text{RGR1.6})$$

At a nontrivial zero ρ ,

$$F_0(\rho + t) = t^{m_\rho} \frac{(\rho + t)(\rho + t - 1)}{8} \pi^{-(\rho+t)/2} \Gamma((\rho + t)/2) \frac{\zeta(\rho + t)}{t^{m_\rho}}. \quad (\text{RGR1.7})$$

The full local holomorphic unit near $t = 0$ on the right is retained, including every derivative. It is not asserted to be an entire zero-free function. The working residue denominator below is original ζ , not F_0 . The original arithmetic is still

$$\zeta(s) = 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{k \geq 1} (\log p) p^{-ks} \quad (\Re s > 1). \quad (\text{RGR1.8})$$

Fix one positively oriented small disk D about a pole, with inclusion $i : \{p\} \hookrightarrow D$. The local coordinate is z at the positive pole and $1/z$ at the other pole. CSP and CSD give the full local source $V_p = S \oplus E_p$, where the two lines of E_p record value at zero and real integral. Its restriction is Σ , or $R\Sigma = \hat{\Sigma}$, respectively, with

$$Ra(u) = u^{-1}a(u^{-1}), \quad \hat{f}(t) = \int_{\mathbb{R}} f(v) e^{-2\pi i v t} dv.$$

The full Poisson correction is $u^{-1}f(0) - \int f$; it is zero on this stated S , and its two coordinates are retained in E_p .

Because the endpoint coordinates are explicit direct summands, the local sheaf has the actual splitting

$$\mathcal{F}|_D = \mathcal{F}_{\text{red}} \oplus i_* E_p, \quad \mathcal{F}_{\text{red}} = \ker(\underline{A}_D \xrightarrow{q \text{ ev}_p} i_* Q). \quad (\text{RGR1.9})$$

Set $P = \mathcal{F}_{\text{red}}[1]$. The endpoint summand is restored in RGR10. This splitting is proved by sending a source vector (f, e) to $(\Sigma f, e)$, or $(R\Sigma f, e)$, using the proved inverse on J . It does not remove a source contribution without its comparison map.

RGR2. The actual residue map, with its complete radical

Let $\mathcal{Z}$ denote the set of distinct actual nontrivial zeros, retaining multiplicities m_ρ , and put $\rho^\# = 1 - \bar{\rho}$. Define

$$H = \ell^2(\mathcal{Z}, m), \quad \langle x, y \rangle_+ = \sum_{\rho} m_\rho x_\rho \overline{y_\rho}, \quad (\mathcal{J}x)_\rho = x_{\rho^\#},$$

$$E : Q \longrightarrow H, \quad E(F)_\rho = (\Theta_Q F)(\rho). \quad (\text{RGR2.1})$$

Evaluation is independent of representatives. For $f \in A$ representing a class $[f] \in Q$, the unconditional zero count and the rapid-strip seminorm defined in RGR1 give the convergent estimate

$$\|E[f]\|_+^2 \leq b_{1,2} (\Theta f)^2 \sum_{\rho} m_\rho (1 + |\Im \rho|)^{-4}. \quad (\text{RGR2.2})$$

The map on A kills J , so it is continuous on the quotient. RTT's full isolators give every finite value sequence, hence $E(Q)$ is dense in H . This asserts density of values, not density of finite full-jet classes in the primal topology.

Use the conjugate space $C = \overline{Q}$, with scalar action $\lambda \overline{G} = \overline{\lambda G}$. Define the linear continuous map

$$\sigma : C \longrightarrow Q'_\beta, \quad \sigma(\overline{G})(F) = \langle EF, \mathcal{J}EG \rangle_+ = \sum_{\rho} m_\rho (\Theta_Q F)(\rho) \overline{(\Theta_Q G)(\rho^\#)}. \quad (\text{RGR2.3})$$

The prime on a coefficient space always means its continuous complex-linear dual; the subscript β means uniform convergence on bounded sets. For bounded $B_0 \subset Q$,

$$\sup_{F \in B_0} |\sigma(\overline{G})(F)| \leq \left(\sup_{F \in B_0} \|EF\|_+ \right) \|EG\|_+. \quad (\text{RGR2.4})$$

This proves absolute convergence and strong continuity. Testing against each value isolator proves

$$\ker \sigma = \overline{\mathcal{N}_0}, \quad \mathcal{N}_0 = \{G \in Q : (\Theta_Q G)(\rho) = 0 \text{ for every } \rho\}. \quad (\text{RGR2.5})$$

Higher jets remain in Q ; (RGR2.5) records exactly which of them this map does not detect. No simplicity assumption is used.

The comparison with the full residue completion is RTT's proved isomorphism $\hat{\iota} : \mathcal{H}_{\text{res}} \rightarrow \mathcal{Q}'_\beta$. In the $\mathcal{Q}$ coordinate its finite expression is

$$\mathcal{R}(F, h) = \sum_{\rho} \text{Res}_{s=\rho} \frac{F(s)h(1-s)}{\zeta(s)} ds. \quad (\text{RGR2.6})$$

For $\zeta(\rho + t) = t^{m_\rho} u_\rho(t)$, the exact finite approximants are

$$h_B(G) = \sum_{\rho \in B} m_\rho (-1)^{m_\rho - 1} u_\rho(0) \overline{G(\rho^\#)} e_{1-\rho, m_\rho - 1}. \quad (\text{RGR2.7})$$

Their residue functionals converge to the full sum (RGR2.3), uniformly on each bounded test set, by the Cauchy-Schwarz tail estimate in (RGR2.4). Thus σ is the continuous-dual version of RTT's residue map, with the stated Θ_Q and $\hat{\iota}$ comparisons. It is not an identification of the entire primal quotient with its dual.

Write

$$B = A'_\beta, \quad j = q' : Q'_\beta \longrightarrow B, \quad a = j\sigma : C \longrightarrow B. \quad (\text{RGR2.8})$$

Here j is the actual transpose, $j(\lambda)(f) = \lambda(qf)$. CSD10 and SDT's bounded-lift theorem give its strong topological identification with $J^\perp \subset B$. In particular it is injective and

$$\ker a = \overline{\mathcal{N}_0}. \quad (\text{RGR2.9})$$

RGR3. The cone sign and the positive angular quotient

For compactly supported smooth A -valued forms, use CSD's current complex

$$\mathcal{T}_A^n(U) = (\mathcal{E}_c^{-n}(U; A))', \quad dT(\phi) = (-1)^{n+1} T(d\phi), \quad -2 \leq n \leq 0. \quad (\text{RGR3.1})$$

It represents $\underline{B}_D[2]$, by the explicit compact-support integration contraction in CSD2. Its positive Dirac current and positive angular current satisfy

$$\delta_p \lambda(f) = \lambda(f(p)), \quad h_p \lambda(\omega) = \int_D \frac{d \arg z_p}{2\pi} \wedge \lambda(\omega), \quad dh_p \lambda = \delta_p \lambda. \quad (\text{RGR3.2})$$

The last equality follows by Stokes on a punctured disk; the small positive circle has integral one. These currents are continuous on the specified compact-test spaces. No tensor-product description of arbitrary vector-valued distributions is needed.

The reduced sheaf is resolved by

$$\mathcal{K}^0 = \mathcal{E}_A^0, \quad \mathcal{K}^1 = \mathcal{E}_A^1 \oplus i_* Q, \quad \mathcal{K}^2 = \mathcal{E}_A^2, \quad d^0 f = (df, qf(p)), \quad d^1(\omega, x) = d\omega. \quad (\text{RGR3.3})$$

At the pole a primitive of a closed one-form can be altered by a constant representative of its required quotient value; q is onto. This proves algebraic stalkwise exactness, and the written differentials are continuous. It does not provide a continuous linear choice of representatives $Q \rightarrow A$. On the puncture it is the coefficient de Rham resolution. Its continuous transpose has the cone written below, but identifying that cone with the actual CSD current dual requires the separate comparison that follows; no general exactness of continuous dualization is being used.

$$D_c \mathcal{F}_{\text{red}} \simeq \text{Cone}(g), \quad g = \delta_p j : i_* Q'_\beta \longrightarrow \mathcal{T}_A, \quad D_c P \simeq \text{Cone}(g)[-1]. \quad (\text{RGR3.4})$$

Here is the comparison with the already constructed current dual. Use the reduced version of CSD4, whose literal complex is

$$D_{\text{old}} = \text{Cone}(i_* A' \xrightarrow{(\delta_p, -k')} \mathcal{T}_A \oplus i_* J'). \quad (\text{RGR3.4a})$$

There is a continuous cochain inclusion $\text{Cone}(g) \rightarrow D_{\text{old}}$: identity on the current terms, $(T, \lambda) \mapsto (T, q' \lambda)$ in degree minus one, and $T \mapsto (T, 0)$ in degree zero. The differential commutes because $k' q' = 0$. The quotient complex is

$$i_*(A'/q'Q') \xrightarrow{-k'} i_* J' \quad \text{in degrees } -1, 0. \quad (\text{RGR3.4b})$$

This is algebraically acyclic: the induced map is bijective by the quotient definition and continuous Hahn-Banach extension from closed J to A . Hence the inclusion is a sheaf quasi-isomorphism. No continuous splitting $Q \rightarrow A$, no strong openness of $A' \rightarrow J'$, and no termwise exactness assertion about arbitrary dual complexes was needed. The local cohomology topologies of the small cone are computed directly by the same angular-current and fixed-test maps as CSD5. In particular the middle coefficient is Q'_β by the proved strong embedding q' . We use the small cone as this verified representative of $D_c P$ below.

For the standard cone $d(b, x) = (d_B b + f x, -d_X x)$, the latter representative has

$$(D_c P)^{-1} = \mathcal{T}_A^{-2}, \quad (D_c P)^0 = \mathcal{T}_A^{-1} \oplus i_* Q', \quad (D_c P)^1 = \mathcal{T}_A^0, \\ d^{-1}b = (-d_T b, 0), \quad d^0(b, \lambda) = -d_T b - g\lambda. \quad (\text{RGR3.5})$$

With literal projection $\pi(b, \lambda) = \lambda$, the connecting arrow of the short exact sequence of complexes is $-g$: lift λ as $(0, \lambda)$, whose differential is $-g\lambda$. This is the dual counterpart of the negative connecting arrow in the coordinate triangle $i_* Q \rightarrow \text{Cone}(q) \rightarrow \underline{A}[1] \rightarrow i_* Q[1]$.

The positive angular stalk class labelled $\mu \in Q'$ is

$$(h_p j \mu, -\mu). \quad (\text{RGR3.6})$$

Its differential is zero by (RGR3.2). It restricts to the positive angular class $h_p j \mu$. Consequently the quotient map recovering that label is

$$\gamma : D_c P \longrightarrow i_* Q', \quad \gamma(b, \lambda) = -\lambda. \quad (\text{RGR3.7})$$

This is an explicit change of the quotient coordinate, not a suppressed sign. With this convention the actual triangle is

$$\mathcal{T}_A[-1] \longrightarrow D_c P \xrightarrow{\gamma} i_* Q' \xrightarrow{g} \mathcal{T}_A. \quad (\text{RGR3.8})$$

Indeed the lift of μ in this quotient coordinate is $(0, -\mu)$, whose differential is $+g\mu$. The first arrow is the unchanged inclusion of currents.

RGR4. Gysin adjunction and the exact lifting obstruction

We ask for a morphism $\ell : i_*C \rightarrow D_cP$ with $\gamma\ell = i_*\sigma$. The exact Hom sequence of (RGR3.8) identifies its obstruction as

$$\mathfrak{o}_\sigma = g i_*\sigma = \text{Gys}_B \circ i_*a : i_*C \longrightarrow \underline{B}_D[2]. \quad (\text{RGR4.1})$$

At the current level this is exactly $x \mapsto \delta_p(ax)$, with positive sign.

Here the Gysin adjunction is valid for these full coefficient spaces. To see it without a finite-rank hypothesis, compute supported cohomology of $\mathcal{T}_A$ by the restriction fibre from a disk to its puncture. The disk current cohomology is B in degree -2 . The puncture has B in degrees $-2, -1$, represented by constant and angular currents. Cancelling the identity constant restriction leaves B in degree zero. The positive generator is the class of the supported positive Dirac current. Equivalently, in the restriction fibre, $(\delta_p\lambda, 0)$ is cohomologous to $(0, -h_p\lambda)$. Thus

$$i^!\underline{B}_D[2] \simeq B[0], \quad \text{Hom}_{D(D)}(i_*C, \underline{B}_D[2]) \simeq \text{Hom}_{\mathbb{C}}(C, B), \quad (\text{RGR4.2})$$

and (RGR4.1) corresponds to $+a$. The derived adjunction uses the exact point functor and the category of vector spaces; no dimension bound is needed. Every displayed continuous coefficient map gives a continuous current representative. The Hom group in (RGR4.2) is not claimed to contain only continuous coefficient maps.

Since j is injective,

$$\mathfrak{o}_\sigma = 0 \iff a = 0 \iff \sigma = 0. \quad (\text{RGR4.3})$$

For the actual residue map it is nonzero. Choose any actual zero ρ , its full value isolator F , and the isolator at $\rho^\#$ as G . Equation (RGR2.3) gives $\sigma(\bar{G})(F) = m_\rho \neq 0$. All arithmetic has already been reconstructed before this use of its zero divisor. In particular the same witness works at a zero on the critical line: this total-residue obstruction does not distinguish on-line from off-line zeros.

The identity $dh_pax = \delta_pax$ does not supply a supported sheaf null-homotopy. The current h_pax restricts to a nonzero angular current on the puncture whenever $ax \neq 0$, while a section of i_*C restricts to zero there. Thus $x \mapsto h_pax$ is a primitive on a disk or at its stalk, but it cannot be a morphism $i_*C \rightarrow \mathcal{T}_A^{-1}$ for nonzero a . This is the precise reason a zero stalkwise Gysin map can represent the nonzero supported derived morphism (RGR4.1).

For an arbitrary vector subspace $C_0 \subset C$, the restricted map lifts exactly when $C_0 \subset \overline{\mathcal{N}_0}$. This follows by applying the same adjunction to the restriction of a . Thus the exact largest liftable part of the supported source is its retained value-zero radical. No absence of higher jets is presumed.

RGR5. The actual receiver constructed from the obstruction

Use the sign in (RGR3.8) and define

$$\mathcal{E}_\sigma = \text{Cone}(-\delta_p a)[-1]. \quad (\text{RGR5.1})$$

The entire complex is

$$\begin{aligned} \mathcal{E}_\sigma^{-1} &= \mathcal{T}_A^{-2}, \quad \mathcal{E}_\sigma^0 = \mathcal{T}_A^{-1} \oplus i_*C, \quad \mathcal{E}_\sigma^1 = \mathcal{T}_A^0, \\ d^{-1}b &= (-d_T b, 0), \quad d^0(b, x) = -d_T b + \delta_p(ax). \end{aligned} \quad (\text{RGR5.2})$$

It has the explicit morphisms

$$p_\sigma : \mathcal{E}_\sigma \longrightarrow i_*C, \quad (b, x) \mapsto x, \quad v_\sigma : \mathcal{E}_\sigma \longrightarrow D_cP, \quad (b, x) \mapsto (b, -\sigma x), \quad (\text{RGR5.3})$$

and identity on the other current terms. Substitution into (RGR3.5) gives $-d_T b - g(-\sigma x) = -d_T b + \delta_p a x$, so this is a cochain map, and

$$\gamma v_\sigma = i_* \sigma p_\sigma. \quad (\text{RGR5.4})$$

Because γ is a degree-wise split surjection of the written complexes, their strict pullback is the homotopy pullback. Its equation is $-\lambda = \sigma x$, yielding exactly (RGR5.2). The associated triangle is

$$\underline{B}_D[1] \longrightarrow \mathcal{E}_\sigma \xrightarrow{p_\sigma} i_* C \xrightarrow{o_\sigma} \underline{B}_D[2]. \quad (\text{RGR5.5})$$

This receiver retains the obstruction as its attaching map. It does not declare that the original unsupported lift exists.

RGR6. Stalk, costalk, nearby cycles, and variation

On the puncture, $\mathcal{E}_\sigma$ is the current realization of $\underline{B}[1]$, so its local system has monodromy identity. On a small disk its two cohomology groups have explicit representatives

$$B \ni \lambda \mapsto c_\lambda \quad (\deg -1), \quad C \ni x \mapsto (h_p a x, x) \quad (\deg 0). \quad (\text{RGR6.1})$$

The second is closed because $dh_p a x = \delta_p a x$. CSD's compact-test contraction kills the remaining current terms and provides inverse continuous maps on the cohomology of each fixed disk. Consequently

$$i^* \mathcal{E}_\sigma \simeq B[1] \oplus C[0]. \quad (\text{RGR6.2})$$

Restriction to the punctured disk is identity on the degree-minus-one constant class and is $x \mapsto a x$ on the degree-zero angular class. The supported restriction fibre cancels the constant identity. With the positive Dirac convention of (RGR4.2), the remaining costalk complex is

$$\boxed{i^! \mathcal{E}_\sigma \simeq [C \xrightarrow{a} B] \quad \text{in degrees } 0, 1.} \quad (\text{RGR6.3})$$

For an explicit cancellation, the restriction fibre in the convention $d(s, t) = (ds, \text{res}(s) - dt)$ has coefficient complex

$$B \xrightarrow{b \mapsto (0, b)} C \oplus B \xrightarrow{(x, b) \mapsto a x} B \quad \text{in degrees } -1, 0, 1. \quad (\text{RGR6.3a})$$

Eliminating the identity pair gives (RGR6.3). In the shifted current complex, $dh_p b = -\delta_p b$; hence in this fibre $d(h_p b, 0) = (-\delta_p b, h_p b)$, so $(\delta_p b, 0)$ and $(0, h_p b)$ give the same positive supported class. Thus the target B in (RGR6.3) uses positive Dirac orientation.

The literal standard expression $\text{Cone}(\text{var})[-1]$, for $\text{var} = a : C \rightarrow B$, instead has differential $-a$. Its exact comparison with (RGR6.3) is identity on C and multiplication by -1 on the target B . Thus its literal target coordinate is the negative of the positive supported coordinate just specified. This sign change does not change $\text{var} = +a$ in (RGR6.5). For the comparison to $[Q' \xrightarrow{j} B]$, change the target coordinate on both complexes together: the costalk arrow remains $(\sigma, 1_B)$. For $[C \rightarrow Q'] \rightarrow [C \rightarrow B]$ in RGR9, change both target coordinates together: its arrow remains $(1_C, j)$. This explicitly reconciles the standard shifted-cone convention with the positive-Dirac restriction-fibre convention.

The sign can also be read without cancellation: applying $i^!$ to (RGR5.5), its connecting morphism is $+a : C \rightarrow B$ by (RGR4.2). The costalk cohomology is therefore

$$H^0(i^! \mathcal{E}_\sigma) = \overline{\mathcal{N}_0}, \quad H^1(i^! \mathcal{E}_\sigma) = B/a(C). \quad (\text{RGR6.4})$$

These are the algebraic cohomology groups with their indicated kernel and quotient topologies. No closedness of $a(C)$, Hausdorffness of its quotient, or strong-open mapping theorem for a dual restriction map is asserted.

For completeness one can read the nearby and vanishing maps directly. Pull back the constant punctured local system to the universal cover and contract there; its shifted nearby complex is $B[0]$. The ordinary restriction model on the puncture is $B[1] \oplus B[0]$. In the stalk model (RGR6.2), its first component is projection onto $B[1]$, and its angular component is $a : C \rightarrow B$. The shifted vanishing cone of the first component cancels this identity pair and leaves $C[0]$. Thus, in the positive angular convention,

$$\Psi(\mathcal{E}_\sigma) = B, \quad \Phi(\mathcal{E}_\sigma) = C, \quad \text{can} = 0, \quad \text{var} = a, \quad T = 1. \quad (\text{RGR6.5})$$

Here $\Psi = R\psi[-1]$ and $\Phi = R\phi[-1]$. Both composites of can and var are zero, agreeing with $1 - T = 0$. This is a direct calculation on the specified complexes, rather than an invocation of finite-constructible gluing theory.

The full morphism to $D_c P$ is

$$(\Psi(v_\sigma), \Phi(v_\sigma)) = (1_B, \sigma). \quad (\text{RGR6.6})$$

On costalks it is the map $[C \xrightarrow{a} B] \rightarrow [Q' \xrightarrow{j} B]$ with components $(\sigma, 1_B)$. In particular the radical appears as an actual supported cohomology group; it has not been erased by passing to values.

RGR7. Exact supported kernel and cokernel of the receiver map

The current term is common to the two triangles (RGR3.8) and (RGR5.5), and the map there is identity. Taking the cone of the displayed morphism of triangles gives

$$\text{Cone}(v_\sigma) \simeq i_* \text{Cone}(\sigma), \quad \text{Cone}(\sigma) = [C \xrightarrow{\sigma} Q'] \text{ in degrees } -1, 0. \quad (\text{RGR7.1})$$

Here is an explicit cone comparison. Put $L = \mathcal{T}_A[-1]$ and write a cone element as $(b, \lambda; u, x) \in (D_c P)^n \oplus \mathcal{E}_\sigma^{n+1}$. The map

$$(b, \lambda; u, x) \mapsto (-\lambda, x) \quad (\text{RGR7.1a})$$

to $i_* \text{Cone}(\sigma)$ is a surjective cochain map: the target differential is $(\eta, x) \mapsto (\sigma x, 0)$. Its kernel is $\text{Cone}(1_L)$, with contraction $(b; u) \mapsto (0; b)$. Indeed the sum of the two compositions with the cone differential is $(b; u)$. This proves (RGR7.1) and its sign. It follows that

$$\mathcal{H}^{-1} \text{Cone}(v_\sigma) = i_* \overline{\mathcal{N}_0}, \quad \mathcal{H}^0 \text{Cone}(v_\sigma) = i_*(Q'/\sigma C). \quad (\text{RGR7.2})$$

The inclusion of $i_* \overline{\mathcal{N}_0}$ into $\mathcal{E}_\sigma$ is literal: send x to $(0, x)$ in degree zero. Its differential vanishes because $ax = 0$. This supported map records the maximal liftable subspace from RGR4.

Applying (RGR6.3) with $C = Q'$, $a = j$, gives

$$i^! D_c P \simeq [Q' \xrightarrow{j} B] \simeq J'[-1] \quad (\text{RGR7.3})$$

as complexes of vector spaces. The last assertion uses Hahn-Banach to identify the algebraic cokernel with J' . It asserts no strong-topology inverse. Since this has cohomology only in degree one, $\text{Hom}_{D(D)}(i_* C, D_c P) = 0$. Thus a lift of a zero supported residue map is the zero morphism in this derived category; the failure to lift the actual nonzero residue map is not an ambiguity among several lifts.

RGR8. The image topology and the retained multiplicity information

RTT6 proves in the strong dual

$$\overline{\sigma(C)} = \mathcal{N}_0^\perp \subset Q'_\beta. \quad (\text{RGR8.1})$$

One proof is to note that the image contains every value functional. A continuous functional on the strong dual separating its closure from an element of $\mathcal{N}_0^\perp$ is, by SDT's proved bidual result, evaluation at an actual class in Q . Vanishing on every value functional places that class in $\mathcal{N}_0$, a contradiction. Conversely every image functional annihilates $\mathcal{N}_0$.

Since j is a strong topological embedding with closed image, this gives

$$\overline{a(C)} = j(\mathcal{N}_0^\perp) \subset J^\perp \subset B. \quad (\text{RGR8.2})$$

The separated quotient of the second group in (RGR6.4) is therefore $B/j(\mathcal{N}_0^\perp)$. The unseparated quotient $B/a(C)$ remains the actual cohomology with its original quotient topology. The two are not silently identified.

At a zero of multiplicity m , the full primal local block is $\mathbb{C}[t]/(t^m)$. Its value-zero part $t\mathbb{C}[t]/(t^m)$ has dimension $m - 1$. The actual residue numerator in (RGR2.7) uses $m(-1)^{m-1}u_\rho(0)$ and the highest-jet line of the reflected block. Thus the receiver retains the full input block and records the precise rank-one value observation; it does not replace multiplicity by one or identify the value radical with a removed part of the sheaf.

RGR9. Exact relation to the retained original extension

Under (RGR1.2), FOD's actual original source row is

$$e_0 : 0 \longrightarrow J \xrightarrow{k} A \xrightarrow{q} Q \longrightarrow 0. \quad (\text{RGR9.1})$$

Its continuous transpose is the vector-space exact sequence

$$0 \longrightarrow Q' \xrightarrow{j=q'} B = A' \xrightarrow{k'} J' \longrightarrow 0. \quad (\text{RGR9.2})$$

Injectivity and the kernel follow from the quotient definition. Surjectivity is precisely Hahn-Banach for continuous functionals on the closed subspace $J \subset A$. All three maps are continuous for strong duals because the primal maps carry bounded sets to bounded sets. We do not infer that k' is strongly open or that a continuous section exists.

There is a further actual receiver

$$\mathcal{K}_\sigma = \text{Fib}(i_* C \xrightarrow{\text{Gys}_{Q'} \sigma} \underline{Q}'_D[2]), \quad (\text{RGR9.3})$$

where Fib denotes the positive-quotient convention used in (RGR5.1)-(RGR5.5). It has a continuous-current representative. Form $\mathcal{T}_Q$ from compactly supported smooth Q -valued tests by exactly (RGR3.1). The compact-support integrations of CSD2 apply because Q is Fréchet and complete. They identify $\mathcal{T}_Q \simeq \underline{Q}'[2]$, with its actual strong coefficient topology on local cohomology. Then

$$\mathcal{K}_\sigma^n = \mathcal{T}_Q^{n-1} \oplus i_* C^n, \quad d(v, x) = (-dv + \delta_Q \sigma x, 0). \quad (\text{RGR9.3a})$$

Transposing the continuous maps on compactly supported tests induced by $q : A \rightarrow Q$ and $k : J \rightarrow A$ gives

$$q^\vee : \mathcal{T}_Q \rightarrow \mathcal{T}_A, \quad k^\vee : \mathcal{T}_A \rightarrow \mathcal{T}_J, \quad k^\vee q^\vee = 0. \quad (\text{RGR9.3b})$$

Their maps on constant-current coefficients are precisely $j = q'$ and k' . The resulting receiver maps are $(v, x) \mapsto (q^\vee v, x)$ and $(b, x) \mapsto k^\vee b$. These are continuous cochain maps. No assertion that the three current complexes form a termwise exact strong-dual sequence is required. The same object can alternatively be defined by the positively oriented scalar Gysin map tensored with the coefficient vector space in the stated vector-sheaf category.

Naturality of the oriented Gysin map gives

$$\text{Gys}_B j\sigma = \underline{j}[2] \text{Gys}_{Q'} \sigma. \quad (\text{RGR9.4})$$

Apply the fibre triangle for this composite. The fibre of $\underline{j}[2]$ is $\underline{J}'[1]$ by the exact row (RGR9.2). For the written current complexes this identification also follows from the continuous constant-current comparisons: after cancelling the common supported C coordinate, the cone is $\text{Cone}(\mathcal{T}_Q[-1] \rightarrow \mathcal{T}_A[-1])$, and its comparison to $\mathcal{T}_J[-1]$ is a quasi-isomorphism by the coefficient row (RGR9.2). Consequently there is a precise triangle

$$\boxed{\mathcal{K}_\sigma \longrightarrow \mathcal{E}_\sigma \longrightarrow \underline{J}'_D[1] \longrightarrow \mathcal{K}_\sigma[1].} \quad (\text{RGR9.5})$$

The map on nearby coefficients is $j : Q' \rightarrow B$, and on vanishing coefficients it is 1_C . The second map on nearby coefficients is $k' : B \rightarrow J'$, and on vanishing coefficients it is zero. These formulas prove compatibility since $k'j\sigma = 0$. In an explicit cone proof, eliminate the common supported C coordinates; what remains is the coefficient cone of j , giving $J'[1]$. This also fixes the arrows in (RGR9.5), rather than asserting a decomposition of its middle object.

The complete local data of $\mathcal{K}_\sigma$ are

$$\begin{aligned} \Psi \mathcal{K}_\sigma &= Q', \quad \Phi \mathcal{K}_\sigma = C, \quad \text{can} = 0, \quad \text{var} = \sigma, \\ i^* \mathcal{K}_\sigma &\simeq Q'[1] \oplus C, \quad i^! \mathcal{K}_\sigma \simeq [C \xrightarrow{\sigma} Q'] \quad (\deg 0, 1). \end{aligned} \quad (\text{RGR9.6})$$

Its costalk groups are $\overline{\mathcal{N}}_0$ and $Q'/\sigma C$. The latter maps injectively into $B/a(C)$; explicitly,

$$0 \longrightarrow Q'/\sigma C \xrightarrow{[\lambda] \mapsto [j\lambda]} B/a(C) \xrightarrow{[b] \mapsto k'b} J' \longrightarrow 0. \quad (\text{RGR9.7})$$

Exactness follows directly from (RGR9.2). This row is the pushout of the actual dual row (RGR9.2) by $Q' \rightarrow Q'/\sigma C$: the pushout map sends $[(b, [\lambda])]$ to $[b + j\lambda]$, and the relation $(j\mu, -[\mu])$ maps to zero. Its inverse is $[b] \mapsto [(b, 0)]$. These are continuous maps for the stated quotient topologies, with no assertion that these quotient spaces are separated.

Thus the connection to e_0 is exact: its transpose inclusion is the coefficient of the Gysin obstruction, and its transpose extension produces both the triangle (RGR9.5) and the pushout row (RGR9.7). The class $\mathfrak{o}_\sigma \in \text{Hom}(i_* C, \underline{B}[2])$ is not declared equal to an M -module Ext^1 class in a different category. No M -linearity of the untwisted anti-linear Weil map is inferred; its reflected multiplier covariance is the one proved in RTT and ECI.

The concrete next calculation after the nonzero obstruction is therefore already performed: isolate the arithmetic dual coefficient Q' inside the full nearby coefficient A' , retain the dual original extension through (RGR9.5), and calculate the exact remaining quotient. Pushing the obstruction along k' gives zero because $k'j = 0$, but this kills the entire residue coefficient. It cannot be described as a proof that the original residue map lifted unchanged.

RGR10. Both endpoint lines remain in the complete local receiver

From the explicit source splitting (RGR1.9),

$$P_{\text{full}} = \mathcal{F}|_D[1] = P \oplus i_* E_p[1], \quad D_c P_{\text{full}} = D_c P \oplus i_* E'_p[-1]. \quad (\text{RGR10.1})$$

The two endpoint lines remain separate, with their original source labels. The quotient to $i_* Q'$ is γ on the first factor and zero on the endpoint factor. The complete homotopy pullback is consequently

$$\mathcal{E}_{\sigma, \text{full}} = \mathcal{E}_\sigma \oplus i_* E'_p[-1]. \quad (\text{RGR10.2})$$

The endpoint summand has no nearby part; its shifted vanishing part, stalk, and costalk are all $E'_p[-1]$. The additional summand cannot cancel (RGR4.1), because the connecting map factors through the first summand's quotient and is exactly the same map. At infinity the positive coordinate is $1/z$; hence the same local Gysin formulas hold with its own positive angular orientation. No assertion about cover-degree transport is added here; those factors remain in CSD9 and the separate full-cover calculation.

RGR11. The already computed positive-adjoint defect has its own exact Gysin receiver

The total residue form is nonzero even on the critical line. To test the desired adjoint identity one must instead use the actual discrepancy already calculated on the full value completion. For every recovered parameter $r > 1$, retain

$$\begin{aligned} (T_r y)_\rho &= r^\rho y_\rho, & U_r &= rT_{1/r}, & D_r &= T_r^* - U_r, \\ (D_r y)_\rho &= e^{-i\Im \rho \log r} (r^{\Re \rho} - r^{1-\Re \rho}) y_\rho. \end{aligned} \quad (\text{RGR11.1})$$

The known open strip makes these bounded operators. Integer $r = n$ is the actual geometric cover degree; arbitrary positive r is only a coefficient parameter. No nonintegral geometric cover is asserted.

RTT6 supplies the continuous injective anti-linear map

$$A_H : H \longrightarrow Q'_\beta, \quad A_H(y)(F) = \langle EF, jy \rangle_+. \quad (\text{RGR11.2})$$

Its continuity follows from (RGR2.4) with $\|y\|_+$, and injectivity follows from density of $E(Q)$. Put

$$\sigma_r : \overline{H} \rightarrow Q'_\beta, \quad \sigma_r(\overline{y}) = A_H(D_r y), \quad a_r = j\sigma_r. \quad (\text{RGR11.3})$$

The general current-cone, adjunction, homotopy-pullback, stalk/costalk, and dual-row pushout constructions of RGR3-RGR7 and RGR9 apply to this actual map with source $\overline{H}$, using its own $\ker \sigma_r$ and $\text{im } \sigma_r$. Those constructions use a continuous coefficient map, not finite dimension or a primal holomorphic lift of D_r . The value-source radical and closure formulas of RGR2 and RGR8 are not transferred to this different source. The resulting objects are

$$\mathfrak{o}_r = \text{Gys}_B i_* a_r, \quad \mathcal{E}_r = \text{Cone}(-\delta_p a_r)[-1], \quad i^! \mathcal{E}_r \simeq [\overline{H} \xrightarrow{a_r} B]. \quad (\text{RGR11.4})$$

Both j and A_H are injective. The oriented adjunction therefore proves

$$\mathfrak{o}_r = 0 \iff D_r = 0, \quad H^0(i^! \mathcal{E}_r) = \overline{\ker D_r}. \quad (\text{RGR11.5})$$

For the actual diagonal operator in (RGR11.1), $\ker D_r$ is exactly the value subspace supported on the critical-line zeros: the real function r^x is strictly increasing, so its displayed difference vanishes exactly at $x = 1/2$. This proves the precise detection map on the entire zero set. It neither postulates nor numerically selects an off-line zero.

The adjoint defect and the total residue map have consequently been given different, fully constructed local receivers. Their costalk kernels are different: $\overline{\mathcal{N}}_0$ for the full primal residue source, and the critical-line value subspace for the defect source. The coefficient extension still enters through the same q' , and both retain the exact Gysin sign and degree. No weight separation proving $\mathfrak{o}_r = 0$ is asserted by this construction; its contribution is the actual sheaf-level map and receiver on which such a calculation would have to act.

RGR12. Proved scope and continuation data

The lift of the total supported residue map fails by the explicitly nonzero class (RGR4.1). Its meaning is exactly local sheaf support: it is present at an ordinary critical-line zero as well. The full homotopy-pullback receiver, its current complex, all four local functors, its radical, its image topology, and its connection to the actual original source extension are computed in RGR5–RGR10. The separate arithmetic positive-adjoint defect is carried to the same exact geometry in RGR11, without inserting a conclusion about its vanishing.

For a Deligne comparison, retain the actual image quotient in DELIGNE_INVARIANT_CYCLE_QUOTIENT.md, DC5.3–DC5.4. The present Gysin class is not that quotient merely because it is a connecting morphism. The source P is the explicit singular coefficient sheaf (RGR1.9); Deligne’s proper smooth-total family, inertia, and arithmetic weight hypotheses are additional geometric data. This note constructs the receiver needed to compare them instead of assuming those data. It also does not alter FOD7’s previously proved vanishing of the specified global derived original-quotient lift: that global map and (RGR4.1) have different sources and targets, both retained here.

The original ζ , its unit term, all prime-power repetitions, the Gamma and endpoint factors, the full nontrivial multiplicities, and the trivial-zero derivative values remain in RGR1–RGR2 and their exact source comparisons. The note supplies no arithmetic operation on τ , no arbitrary positive form on the source, and no claim of RH or completion of WU061.

The actual adjoint-defect invariant-cycle crosses — ADC0-ADC9

Proof source: CC_ADJOINT_DEFECT_INVARIANT_CYCLE_CROSS.md.

The adjoint-defect receiver and its actual invariant-cycle cross

Complete local derivation. Proof locators **ADC0-ADC9**. This note calculates the invariant-cycle cross of the existing receiver of the positive-adjoint discrepancy. Its attaching map is retained; neither a new coefficient system nor a purity hypothesis is inserted.

ADC0. Construction stage, correction audit, and inputs

The supporting datum is $\tau\langle Z_1; \text{no } Z_2 \rangle$. All operations here take place on the coefficient spaces after complete-history arithmetic reconstruction. No addition, coordinate, metric, midpoint, or extra parity is assigned to the support. The two arithmetic branches retain their own recovered counters.

Before this calculation the connected `USER_ARGUMENT_RECONSTRUCTION.md`, its correction-precedence table, `READ_FIRST_USER_CONSTRUCTION.md`, the verbatim WU062 passage, and the WU064-WU065 amendment were read. In particular, an obstruction in a specified coefficient receiver must be connected by actual maps to the other programme objects; an independently selected extension cannot replace the desired lifting problem. WU065 also requires attempting the next calculation using all established results when a preceding result does not give the sought conclusion.

The actual inputs are RGR0-RGR12 in `CC_RESIDUE_GYSIN_RECEIVER_INDEPENDENT.md`; LNC0-LNC11 in `CC_LOCAL_NEARBY_CYCLES_AND_SPECIALIZATION.md`; LVD0-LVD9 in `CC_LOCAL_VANISHING_DUAL_ARROWS.md`; and DC0-DC12 in `DELIGNE_INVARIANT_CYCLE_QUOTIENT.md`. RGR constructs the source map and current cone. LNC constructs nearby cycles, the exact local cross, and the proper-current/cover signs. DC proves the two weight directions and the exact quotient for Deligne's distinct geometric objects. The positive-adjoint operator is the actual GTAH0-GTAH1 operator in `GEOMETRIC_TRANSFER_POSITIVE_ADJOINT_DEFECT.md`. No source theorem for finite-rank constructible sheaves is applied to infinite coefficients without the explicit local calculation.

Sheaf derived statements use sheaves of complex vector spaces, and all specified coefficient/current maps are continuous. Kernels and quotients retain their actual topologies; no closed image is inferred from an algebraic quotient. Continuous duals are distinguished from unrestricted algebraic duals.

ADC1. The full source and the particular attaching map

Use the original source spaces and quotient

$$S = \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0\},$$

$$A = \{a \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \text{ for every } N, j\}, \quad \Sigma f(u) = 2 \sum_{n \geq 1} f(nu),$$

$$J = \Sigma S, \quad Q = A/J, \quad q : A \rightarrow Q, \quad B = A'_\beta, \quad j = q' : Q'_\beta \hookrightarrow B. \quad (\text{ADC1.1})$$

SSI/ESI give closed J and the continuous inverse of $\Sigma : S \rightarrow J$. SDT/CSD give the strong embedding $j(Q') = J^\perp$. The prime denotes the continuous complex-linear dual, and β denotes uniform convergence on bounded subsets.

The complete original transform is

$$\Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, \quad \Theta^{-1} F(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} F(1/2 + it) u^{-it} dt. \quad (\text{ADC1.2})$$

It identifies J with the entire strip-Schwartz ideal of full vanishing jets at the original nontrivial zeros, with orders m_ρ . The retained divisor is

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}, \quad F_0(s) = \Theta \Sigma f_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{ADC1.3})$$

Its exceptional values remain

$$F_0(0) = F_0(1) = \frac{1}{8}, \quad F_0(-1) = F_0(2) = \frac{\pi}{24}, \quad F_0(-2k) = \frac{k(2k+1)(-1)^k \pi^k}{2k!} \zeta'(-2k) \neq 0. \quad (\text{ADC1.4})$$

At $\rho + t$, the full local factor is

$$F_0(\rho + t) = t^{m_\rho} \frac{(\rho + t)(\rho + t - 1)}{8} \pi^{-(\rho+t)/2} \Gamma((\rho + t)/2) \frac{\zeta(\rho + t)}{t^{m_\rho}}. \quad (\text{ADC1.5})$$

Every derivative of this factor is retained. The original arithmetic is

$$\zeta(s) = 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{k \geq 1} (\log p) p^{-ks} \quad (\Re s > 1). \quad (\text{ADC1.6})$$

Thus the unit and all prime-power repetitions remain in the source. No modified zero divisor is substituted.

Let $\mathcal{Z}$ be the distinct original nontrivial zeros, $\rho^\# = 1 - \bar{\rho}$, and

$$H = \ell^2(\mathcal{Z}, m), \quad \langle x, y \rangle_+ = \sum_\rho m_\rho x_\rho \bar{y}_\rho, \quad (Jy)_\rho = y_{\rho^\#}.$$

The actual continuous map $E : Q \rightarrow H$ is $E(F)_\rho = (\Theta_Q F)(\rho)$. Its image contains every finite value vector and is dense. Its full higher-jet kernel is retained in Q . Put $C = \bar{H}$, with its conjugate complex structure. For a fixed recovered parameter $r > 1$,

$$(T_r y)_\rho = r^\rho y_\rho, \quad U_r = r T_{1/r}, \quad D_r = T_r^* - U_r, \quad d_r(\rho) := e^{-i\Im \rho \log r} (r^{\Re \rho} - r^{1-\Re \rho}). \quad (\text{ADC1.7})$$

Then $(D_r y)_\rho = d_r(\rho) y_\rho$. These are bounded operators by the known open strip. An integer $r = n$ is a geometric cover degree; a noninteger r is only a coefficient parameter.

The precise attaching map is

$$A_H(y)(F) = \langle EF, Jy \rangle_+, \quad \sigma_r : C \rightarrow Q'_\beta, \quad \sigma_r(\bar{y}) = A_H(D_r y), \quad a_r = j\sigma_r : C \rightarrow B. \quad (\text{ADC1.8})$$

A_H is anti-linear, so σ_r is complex-linear. For a bounded test set $K \subset Q$,

$$\sup_{F \in K} |\sigma_r(\bar{y})(F)| \leq \sup_{F \in K} \|EF\|_+ \|D_r\| \|y\|_+, \quad (\text{ADC1.9})$$

proving strong continuity and absolute convergence. Density of $E(Q)$ proves injectivity of A_H ; j is injective. Consequently

$$\ker a_r = \ker \sigma_r = \overline{\ker D_r} =: C_{\text{crit}}, \quad (\text{ADC1.10})$$

where C_{crit} consists exactly of coordinates supported at the actual critical-line zeros. The equality follows from strict monotonicity of r^x in (ADC1.7); it asserts neither existence nor nonexistence of an off-line zero.

For comparison with the original residue realization, if $\zeta(\rho + t) = t^{m_\rho} u_\rho(t)$, the functional in (ADC1.8) is the strong limit of the original- ζ residues with finite numerators

$$h_{B_0}(y) = \sum_{\rho \in B_0} m_\rho (-1)^{m_\rho - 1} u_\rho(0) \overline{(D_r y)_{\rho\#}} e_{1-\rho, m_\rho - 1}, \quad \mathcal{R}(F, h) = \sum_{\rho} \text{Res}_{s=\rho} \frac{F(s)h(1-s)}{\zeta(s)} ds. \quad (\text{ADC1.11})$$

The e 's are RTT's full-jet isolators. Its residue calculation gives the exact coefficient of the value functional; (ADC1.9) controls the strong limit. Neither multiplicity nor the leading original-zeta unit is omitted.

ADC2. The current cone, endpoints, and its four local functors

Fix a pole disk with positive coordinate z at zero or $1/z$ at infinity. Let $\mathcal{T}_A$ be the continuous compact-test current complex,

$$\mathcal{T}_A^n(U) = (\mathcal{E}_c^{-n}(U; A))', \quad dT(\phi) = (-1)^{n+1} T(d\phi), \quad -2 \leq n \leq 0.$$

Its constants represent $\underline{B}[2]$. Positive Dirac and angular currents satisfy

$$\delta_p \lambda(f) = \lambda(f(p)), \quad h_p \lambda(\omega) = \lambda \left(\int \frac{d \arg z_p}{2\pi} \wedge \omega \right), \quad dh_p \lambda = \delta_p \lambda. \quad (\text{ADC2.1})$$

The last equality is Stokes with positive small-circle integral one. All A -valued integrals are formed before the functional is applied.

The actual receiver is

$$\mathcal{E}_r = \text{Cone}(-\delta_p a_r)[-1].$$

In degrees $-1, 0, 1$ its terms and differentials are

$$\mathcal{T}_A^{-2}, \quad \mathcal{T}_A^{-1} \oplus i_* C, \quad \mathcal{T}_A^0, \quad d^{-1}b = (-d_T b, 0), \quad d^0(b, x) = -d_T b + \delta_p a_r x. \quad (\text{ADC2.2})$$

It is RGR's actual homotopy pullback; the triangle is

$$\underline{B}[1] \longrightarrow \mathcal{E}_r \longrightarrow i_* C \xrightarrow{\text{Gys}_B a_r} \underline{B}[2]. \quad (\text{ADC2.3})$$

On a disk the constant and angular representatives give the continuous cohomology models

$$i^* \mathcal{E}_r \simeq B[1] \oplus C, \quad R\Gamma(D^*, \mathcal{E}_r) \simeq B[1] \oplus B[\vartheta_p],$$

$$\lambda \longmapsto c_\lambda \quad (\deg -1), \quad x \longmapsto (h_p a_r x, x) \quad (\deg 0), \quad \text{res} = (1_B, a_r). \quad (\text{ADC2.4})$$

Closedness of the latter representative follows from (ADC2.1); the compact-test contraction removes the remaining current terms. These are continuous comparisons on fixed disks, and their cohomology identifications agree as disks shrink, as in CSD/RGR.

Pulling the punctured constant system to the universal cover and contracting leaves

$$R\psi \mathcal{E}_r = B[1], \quad T = 1, \quad \text{sp} : B[1] \oplus C \longrightarrow B[1] \text{ is projection.} \quad (\text{ADC2.5})$$

Its vanishing cone is $R\phi\mathcal{E}_r = C[1]$. Equivalently the shifted functors $\Psi = R\psi[-1]$, $\Phi = R\phi[-1]$ have

$$\Psi\mathcal{E}_r = B, \quad \Phi\mathcal{E}_r = C, \quad \text{can} = 0, \quad \text{var} = a_r. \quad (\text{ADC2.6})$$

The positive angular convention fixes the plus in variation; it is the second restriction component in (ADC2.4), not a sign deduced from $1 - T = 0$.

The restriction fibre first has coefficient complex

$$B \xrightarrow{b \mapsto (0,b)} C \oplus B \xrightarrow{(x,b) \mapsto a_r x} B \quad \text{in degrees } -1, 0, 1. \quad (\text{ADC2.7})$$

Cancel its identity pair to obtain

$$i^!\mathcal{E}_r \simeq [C \xrightarrow{+a_r} B] \quad (\deg 0, 1). \quad (\text{ADC2.8})$$

The target is positively oriented: in the shifted current differential, $dh_p b = -\delta_p b$, so the fibre differential gives $d(h_p b, 0) = (-\delta_p b, h_p b)$. The positive Dirac class is therefore the positive angular boundary coordinate. This explains why (ADC2.8) has $+a_r$, whereas a literal standard shifted cone on a_r has differential $-a_r$; negating its target gives the displayed comparison.

Both source endpoint lines remain as in RGR10:

$$\mathcal{E}_{r,\text{full}} = \mathcal{E}_r \oplus i_* E'_p[-1]. \quad (\text{ADC2.9})$$

Their stalk and costalk are $E'_p[-1]$, their nearby complex is zero, and their unshifted vanishing complex is $E'_p[0]$. Thus they appear in special and support degree 1, and vanishing degree 0, separately from (ADC2.5)–(ADC2.8).

Here $E_+ = \mathbb{C}c_0 \oplus \mathbb{C}c_1$ and $E_- = \mathbb{C}d_0 \oplus \mathbb{C}d_1$ retain respectively the original value-at-zero and integral channels in the two source charts. Their original coefficient actions are

$$\rho_+(t)|_{E_+} = \text{diag}(1, t), \quad \rho_-(t)|_{E_-} = \text{diag}(t, 1). \quad (\text{ADC2.10})$$

These are the endpoint components of the complete source maps in (LNC5.2); transposition keeps each indicated coordinate label.

ADC3. The exact invariant-cycle cross of this receiver

Write

$$S_r = a_r(C) \subset B, \quad N_r = B/S_r. \quad (\text{ADC3.1})$$

These are the actual image and algebraic quotient with their stated topologies. No closedness is asserted. From (ADC2.8), the full support groups are

$$H_p^0(\mathcal{E}_{r,\text{full}}) = C_{\text{crit}}, \quad H_p^1(\mathcal{E}_{r,\text{full}}) = N_r \oplus E'_p. \quad (\text{ADC3.2})$$

Let $V_i = H^i(R\psi\mathcal{E}_{r,\text{full}})$, $A_i^{\text{sp}} = H^i(i^*\mathcal{E}_{r,\text{full}})$, $B_i^{\text{punct}} = H^i(D^*, \mathcal{E}_{r,\text{full}})$. Define the actual invariants $C_i^{\text{inv}} = V_i^T$, angular coinvariants $K_i = V_{i-1,T}[\vartheta_p]$, and $O_i = H_p^{i+1}$. The only nearby group is $V_{-1} = B$, with monodromy identity. Thus the entire nonzero cross is:

i	A_i^{sp}	K_i	B_i^{punct}	C_i^{inv}	O_i	α_i	∂_i
-1	B	0	B	B	C_{crit}	1	0

i	A_i^{sp}	K_i	B_i^{punct}	C_i^{inv}	O_i	α_i	∂_i
0	C	$B[\vartheta_p]$	$B[\vartheta_p]$	0	$N_r \oplus E'_p$	a_r	$b \mapsto ([b], 0)$
1	E'_p	0	0	0	0	0	0

The row $0 \rightarrow K_i \xrightarrow{j_i} B_i^{\text{punct}} \rightarrow C_i^{\text{inv}} \rightarrow 0$ has identity π_{-1} and identity j_0 . The complete remaining localization part is

$$0 \rightarrow C_{\text{crit}} \rightarrow C \xrightarrow{a_r} B \xrightarrow{b \mapsto ([b], 0)} N_r \oplus E'_p \xrightarrow{\text{pr}_{E'_p}} E'_p \rightarrow 0. \quad (\text{ADC3.3})$$

Every sign follows from (ADC2.7)-(ADC2.8).

It follows by these actual maps, without a weight argument, that

$$\boxed{\text{im } \partial_i / \partial_i j_i K_i = 0 \text{ for every } i.} \quad (\text{ADC3.4})$$

At degree -1 the boundary is zero. At degree zero the coinvariant injection is identity, so its boundary image is all $N_r \oplus 0$. The quotient of the whole receiver is still

$$O_0 / \partial_0 j_0 K_0 = E'_p, \quad (\text{ADC3.5})$$

not zero when endpoints are retained. This is exactly the remaining specialization kernel in degree one. Equations (ADC3.4)-(ADC3.5) distinguish the obstruction image from the whole quotient receiver.

The vanishing of (ADC3.4) does not imply the attaching map vanished: it holds for the present a_r before any conclusion about D_r . The stronger conclusion of DC8 has a different test here:

$$\ker(\partial_0 j_0) = S_r, \quad K_0 \xrightarrow{\partial_0 j_0} \text{im } \partial_0 = N_r \text{ is injective} \iff a_r = 0 \iff D_r = 0. \quad (\text{ADC3.6})$$

The actual source subquotient in DC5.5 is

$$0 \rightarrow C_{\text{crit}} \rightarrow C \xrightarrow{a_r} S_r \rightarrow 0. \quad (\text{ADC3.7})$$

The map $C/C_{\text{crit}} \rightarrow S_r$ is a continuous algebraic bijection. We do not assume its inverse continuous for the ambient subspace topology of S_r .

There is also an exact calculation of the stronger DC9 candidate:

$$\frac{A_0^{\text{sp}}}{\text{im}(H_p^0 \rightarrow A_0^{\text{sp}})} = C/C_{\text{crit}} \rightarrow C_0^{\text{inv}} = 0. \quad (\text{ADC3.8})$$

Its kernel is the entire displayed source quotient, identified algebraically with S_r . Thus the actual next object that must be controlled after the automatically zero invariant-cycle obstruction is S_r , not the already zero quotient (ADC3.4).

ADC4. The exact relation to the original coefficient quotient

Human-source attribution: the extension of continuous linear functionals is the Hahn-Banach theorem, not a programme result. See Terence Tao, [245B, Notes6: Duality and the Hahn-Banach theorem](#), Theorem1 and its complex proof (26 January2009). For the locally convex use here, continuity bounds a functional by a continuous seminorm; quotienting its kernel reduces this application to that normed-space theorem. No continuous splitting of the original quotient is thereby asserted.

Write $S_r^Q = \sigma_r(C) \subset Q'$. Since j is injective, $S_r = jS_r^Q$. The full continuous-transpose coefficient row is

$$0 \rightarrow Q' \xrightarrow{j} B \xrightarrow{k'} J' \rightarrow 0, \quad (\text{ADC4.1})$$

where $k : J \hookrightarrow A$. Exactness is the quotient definition and Hahn-Banach on closed J ; no strong-open mapping theorem is used. It induces the exact row

$$0 \rightarrow Q'/S_r^Q \xrightarrow{[\lambda] \mapsto [j\lambda]} N_r \xrightarrow{[b] \mapsto k'b} J' \rightarrow 0. \quad (\text{ADC4.2})$$

For injectivity, $j\lambda \in S_r = jS_r^Q$ implies $\lambda \in S_r^Q$. The middle kernel consists of classes with $k'b = 0$, hence $b = j\lambda$. Surjectivity is that of k' . These prove every exactness assertion.

The row is exactly the pushout of (ADC4.1) along $Q' \rightarrow Q'/S_r^Q$: send a pushout class $[(b, [\lambda])]$ to $[b + j\lambda]$, and send $[b]$ back to $[(b, 0)]$. Changing λ by an element of S_r^Q or using the relation $(j\mu, -[\mu])$ changes the first formula by an element of S_r . Both compositions are identity. The formulas descend from continuous maps and are continuous for quotient topologies.

The sheaf comparison retaining this row is RGR9's actual coefficient fibre

$$\mathcal{K}_r = \text{Cone}(-\text{Gys}_{Q'}, \sigma_r)[-1], \quad \mathcal{K}_r \rightarrow \mathcal{E}_r \rightarrow \underline{J}'[1] \rightarrow \mathcal{K}_r[1]. \quad (\text{ADC4.3})$$

The first map uses j on nearby coefficients and 1_C on vanishing coefficients; the second uses k' and zero. On costalks they are

$$[C \xrightarrow{\sigma_r} Q'] \xrightarrow{(1_C, j)} [C \xrightarrow{a_r} B] \rightarrow J'[-1]. \quad (\text{ADC4.4})$$

These are literal continuous current maps induced by the original q and k . Their cokernel row is (ADC4.2). In particular the source of the arithmetic boundary-image part is the dual of the same original LNC quotient Q , rather than an unrelated coefficient.

The reduced arithmetic receiver $\mathcal{K}_r$ has the endpoint-free part of ADC3's cross, with B replaced by Q' , a_r by σ_r , and boundary image Q'/S_r^Q . Alternatively $\mathcal{K}_{r, \text{full}} = \mathcal{K}_r \oplus i_* E'_p[-1]$ retains exactly the endpoint terms in the full ADC3 table. Its coinvariant-boundary kernel is exactly S_r^Q . Both invariant-cycle quotients still vanish. This isolates the original arithmetic coefficient inside the full A' receiver without changing the test for the attaching map.

Finally, the actual morphism $\mathcal{E}_r \rightarrow D_c P$ of RGR5 has nearby map 1_B and vanishing map $\sigma_r : C \rightarrow Q'$; on costalks it is

$$[C \xrightarrow{a_r} B] \xrightarrow{(\sigma_r, 1_B)} [Q' \xrightarrow{j} B]. \quad (\text{ADC4.5})$$

Its cone is $i_* \text{Cone}(\sigma_r)$, retaining the kernel C_{crit} and quotient Q'/S_r^Q . It connects ADC3 to the original LNC/LVD dual local arrows by an actual sheaf map. No equality of a Gysin Hom class and an M -module Ext class is inferred.

ADC5. The coefficient covariance and the full image map

Write T_t^Q for the actual dilation induced on Q and $U_t^Q = tT_{1/t}^Q$. Their value maps intertwine with the corresponding bounded T_t, U_t on H . GTAH's exact identities are

$$T_t^* = JU_t J, \quad U_t^* = JT_t J, \quad t > 0. \quad (\text{ADC5.1})$$

Direct evaluation gives, for $F \in Q, y \in H$,

$$(T_t^Q)' A_H(y)(F) = \langle T_t EF, Jy \rangle_+ = \langle EF, T_t^* Jy \rangle_+ = A_H(U_t y)(F).$$

The second identity follows by exchanging T_t, U_t . Since D_r is diagonal, it commutes with both. Thus, with linear maps on the conjugate space,

$$\begin{aligned}\sigma_r \bar{U}_t &= (T_t^Q)' \sigma_r, & \sigma_r \bar{T}_t &= (U_t^Q)' \sigma_r, \\ a_r \bar{U}_t &= T_t' a_r, & a_r \bar{T}_t &= U_t' a_r, & U_t' &= t T_{1/t}'.\end{aligned}\tag{ADC5.2}$$

All maps are continuous; transposes are strong-continuous because continuous primal maps carry bounded sets to bounded sets. Consequently S_r^Q, S_r , their exact quotients in ADC4, and $\ker a_r$ are stable under these actions. Their induced quotient actions are continuous even when an image quotient is not separated.

This gives an exact description on each value coordinate. Let $e_\rho \in H$ be the vector of value one at ρ and zero elsewhere, and let $\delta_{\rho^\#,0} \in Q'$ be evaluation at $\rho^\#$. Then

$$\sigma_r(\bar{e}_\rho) = m_\rho \overline{d_r(\rho)} \delta_{\rho^\#,0}, \quad a_r(\bar{e}_\rho) = m_\rho \overline{d_r(\rho)} q' \delta_{\rho^\#,0}.\tag{ADC5.3}$$

The calculation is substitution into (ADC1.8) and $m_{\rho^\#} = m_\rho$. In particular, every actual off-critical value functional lies in S_r^Q ; it vanishes from this formula exactly when ρ is on the critical line. Finite-coordinate vectors are dense in H , and the image map is continuous, but this does not assert density of finite primal full-jet blocks in Q or closedness of S_r .

On a nonzero image line the two actions have exactly matching source and target eigenvalues:

action	C/C_{crit} at $\bar{e}_\rho$	S_r at $q' \delta_{\rho^\#,0}$
$\bar{U}_t / T_t'$	$t^{1-\bar{\rho}}$	$t^{\rho^\#}$
$\bar{T}_t / U_t'$	$t^{\bar{\rho}}$	$t^{1-\rho^\#}$

(ADC5.4)

The entries are equal because $\rho^\# = 1 - \bar{\rho}$. This is an intertwining map on the actual entire source, with displayed formulas on its value coordinates; it is not a proposed replacement arithmetic.

The continuous pairing with the original full quotient is equally exact. Define

$$Z_r = \{F \in Q : \sigma_r(x)(F) = 0 \text{ for every } x \in C\}.\tag{ADC5.5}$$

Using (ADC1.8) and the positive inner product gives

$$\begin{aligned}\sigma_r(\bar{y})(F) &= \langle EF, \mathcal{J} D_r y \rangle_+ = \langle D_r^* \mathcal{J} EF, y \rangle_+, \\ Z_r &= \ker(D_r^* \mathcal{J} E) = \{F \in Q : F(\rho) = 0 \text{ at every actual off-critical } \rho\}.\end{aligned}\tag{ADC5.6}$$

For the last equality, the scalar multiplying the reflected value is nonzero exactly at an off-critical zero; reflection preserves that subset. All higher jets remain in Z_r when the values vanish. The pairing descends to

$$(Q/Z_r) \times (C/C_{\text{crit}}) \longrightarrow \mathbb{C}, \quad ([F], [\bar{y}]) \longmapsto \sigma_r(\bar{y})(F).\tag{ADC5.7}$$

It is separately induced by the continuous original pairing. Its left and right radicals are zero by (ADC5.6) and the injectivity calculation (ADC1.10). This proves a concrete relation to the original Q , without asserting that either infinite-dimensional quotient is the entire continuous dual of the other. Topological completions and image closures are additional calculated receivers, not used to change (ADC5.7).

ADC6. What the genuine power-cover maps do to this cross

For a recovered integer $n \geq 1$, the proper-current map over $b_n(z_p) = z_p^n$ is a map $b_{n*} \mathcal{E}_r \rightarrow \mathcal{E}_r$. Its literal nearby source is $B^n[1]$, with cyclic permutation monodromy, and its nearby map is

$$(b_0, \dots, b_{n-1}) \longmapsto T_n' \sum_{j=0}^{n-1} b_j.\tag{ADC6.1}$$

To derive it, pull back the target universal cover: it has n contractible inverse components. On each component the holomorphic local isomorphism preserves the positive orientation and pushes its constant current to $T'_n b_j$; add the branches. This is the same full branch calculation as LNC6.14 and retains all n components.

On nearby invariants the map is nT'_n ; on the angular coinvariants, identified by sum, it is T'_n . A proper map has a single inverse-image pole here, so its actual stalk maps and positive costalk maps are

	first displayed degree	second displayed degree	
$i^* \mathcal{E}_r = B[1] \oplus C$	nT'_n	$\overline{U}_n$	(ADC6.2)
$i^! \mathcal{E}_r = [C \rightarrow B]$	$\overline{U}_n$	T'_n	

For the point-supported coordinate this uses (ADC5.2): $a_r \overline{U}_n = T'_n a_r$. For the current differential, pushforward commutes with d , and Dirac pushforward gives $b_{n*} \delta_p \lambda = \delta_p T'_n \lambda$. The angular current pushes forward with degree one; its n local inverse branches each contribute $1/n$. This proves the entire chain map, not only its induced eigenvalues.

The independently constructed weighted geometric trace on the original primal sheaf has transpose giving the reverse map $\mathcal{E}_r \rightarrow b_{n*} \mathcal{E}_r$. Its full nearby map and invariant/coinvariant factors are

$$b \mapsto (T'_{1/n} b, \dots, T'_{1/n} b), \quad B_{\text{inv}} : T'_{1/n}, \quad B_{\text{coinv}} : nT'_{1/n} = U'_n. \quad (\text{ADC6.3})$$

The finite-coordinate diagonal has sum $nT'_{1/n}$, which fixes the angular factor. The point map is $\overline{T}_n$, and the costalk identity is exactly $a_r \overline{T}_n = U'_n a_r$. Thus on the cross the resulting maps are

$$i^* : (T'_{1/n}, \overline{T}_n), \quad i^! : (\overline{T}_n, nT'_{1/n}). \quad (\text{ADC6.4})$$

The existence of the entire sheaf trace, including its finite-coordinate nearby/vanishing correction, is constructed in DCA3-DCA7 of CC_ADJOINT_DEFECT_COVER_ACTIONS.md. Its actual remaining coordinates can be displayed here. Put $V_n = B^n$, let $t(v)_j = v_{j-1}$ with cyclic indices, and let $\Delta b = (b, \dots, b)$, $\Sigma_n v = \sum_j v_j$, $e_0 b = (b, 0, \dots, 0)$. The letter t denotes sheet monodromy and is distinct from the support τ . Then

$$\begin{aligned} \Phi(b_{n*} \mathcal{E}_r) &= C \oplus (V_n / \Delta B), \quad \text{can}_n(v) = (0, [v]), \quad \text{var}_n(x, [v]) = e_0 a_r x - (t - 1)v, \\ t_\Phi(x, [v]) &= (x, [tv - e_0 a_r x]). \end{aligned} \quad (\text{ADC6.6})$$

These are DCA3.7-DCA3.9's actual cut-cone coordinates. In particular $\text{var}_n \text{can}_n = 1 - t$ and $\text{can}_n \text{var}_n = 1 - t_\Phi$. The constant stalk restriction is Δ , and its angular restriction is $e_0 a_r$; applying Σ_n recovers exactly a_r , with no lost degree factor.

Proper-current pushforward on this full vanishing space is $(x, [v]) \mapsto \overline{U}_n x$. For the reverse map define

$$c_x = T'_{1/n} a_r x, \quad w_n(x) = (0, c_x, 2c_x, \dots, (n-1)c_x).$$

Its full vanishing arrow is

$$x \mapsto (\overline{T}_n x, [w_n(x)]), \quad (t - 1)w_n(x) = n e_0 c_x - \Delta c_x. \quad (\text{ADC6.7})$$

The zeroth component of the last equality is $(n-1)c_x$, and each remaining component is $-c_x$. Since $a_r \overline{T}_n x = n c_x$, this proves the entire variation square:

$$\text{var}_n(\overline{T}_n x, [w_n(x)]) = \Delta T'_{1/n} a_r x. \quad (\text{ADC6.8})$$

The correction is also the actual gluing homotopy. In the punctured circle complex $[V_n \xrightarrow{t-1} V_n]$ in degrees $(-1, 0)$, the difference between the two stalk-restriction composites is $d w_n + w_n d$. DCA3.5 realizes the

sheaf from the homotopy fibre of $Rj_*L \oplus i_*S^{\text{st}} \rightarrow i_*i^*Rj_*L$, $(u, s) \mapsto u - \text{res}(s)$. On its restriction fibre the map is explicitly $(s, v) \mapsto (f_S s, f_N v + w_n(s))$; substitution in $d(s, v) = (ds, \text{res}(s) - dv)$ proves the chain identity. This constructs the sheaf map rather than inferring it from cohomology. All finite-coordinate operations are continuous. Thus (ADC6.3)–(ADC6.4) are the invariant/coinvariant maps of this full arrow, not a holomorphic n -th-root inverse. The additional $V_n/\Delta B$ has been retained.

Both maps descend to $N_r = B/S_r$, by (ADC5.2). On the positive boundary $\partial_0 : B \rightarrow N_r \oplus E'_p$ they commute because the induced quotient actions are precisely the actions of T'_n or $nT'_{1/n}$ on the same representative b . The coefficient covariance also proves commutation with every map in (ADC4.2)–(ADC4.5). The endpoint summands are transported by their original dual pole characters: current pushforward uses the dual endpoint part of $\rho_p(n)$, and the weighted-transfer transpose uses the corresponding $n\rho_p(1/n)'$. Their boundary component remains zero and their inclusion/projection is unchanged.

In the dual coordinate labels of (ADC2.10), these endpoint maps are explicitly

	E'_+	E'_-	
current pushforward	$\text{diag}(1, n)$	$\text{diag}(n, 1)$	
weighted-transfer transpose	$\text{diag}(n, 1)$	$\text{diag}(1, n)$	(ADC6.5)

The second row is obtained by substituting $t = 1/n$ in (ADC2.10) and multiplying the transpose by the retained degree n .

For $t > 0$ not an integer, (ADC5.2) still gives the actual coefficient actions. The maps in this paragraph with geometric degree n are not reinterpreted as noninteger covers.

ADC7. Testing Deligne's two weight directions on the actual boundary

DC6–DC8 use two independent inputs: an upper bound on the special-fibre classes and a lower bound on the coinvariants and supported cohomology, the latter obtained by genuine duality with its retained Tate factor. The full source conclusion is stronger than (ADC3.4): it identifies the low-weight piece with $\ker \partial$, disjoint from the coinvariant summand, and proves $\partial j : K_i \rightarrow \text{im } \partial_i$ injective.

Here all these maps have been computed. In degree zero, special classes map by $a_r : C \rightarrow K_0 = B$. Thus a separation that killed the intersection of special image and coinvariants would kill exactly S_r , by (ADC3.6). We now test the existing actions on that intersection.

For a fixed geometric degree $n > 1$, define the logarithmic weight of a scalar $\lambda \neq 0$ as $w_n(\lambda) = 2 \log |\lambda| / \log n$. This is an explicit modulus calculation, not an assertion that the coefficient space is a finite-dimensional mixed Frobenius module. Under current pushforward, the invariant nearby line indexed by $\rho^\#$ has scalar $n^{1+\rho^\#}$, whereas its angular coinvariant has scalar $n^{\rho^\#}$. Their respective logarithmic weights are

$$2(2 - \Re \rho), \quad 2(1 - \Re \rho). \quad (\text{ADC7.1})$$

The decrease by two is the trace direction of this current pushforward; it is not Deligne's pullback Tate increase.

For the weighted-transfer transpose, the invariant line has scalar $n^{-\rho^\#}$ and the angular line has scalar $n^{1-\rho^\#}$. Their weights are

$$-2(1 - \Re \rho), \quad 2\Re \rho. \quad (\text{ADC7.2})$$

This has the positive angular increase by two. Nevertheless the special source class $\bar{e}_\rho$ has exactly the same angular scalar $n^{\bar{\rho}} = n^{1-\rho^\#}$, by (ADC5.4). Under the pushforward it likewise has exactly the same scalar as the angular target, $n^{\rho^\#}$. Consequently neither actual direction separates the source quotient C/C_{crit} from S_r : on every nonzero image line they have identical eigenvalues. Multiplying both sides of

an equivariant map by a common Tate character shifts both weights equally and does not change this equality. Multiplying only one side would require a different map or action; none is inserted here.

The full boundary image N_r has a further exact source component that also prevents assigning it a uniform one-sided weight bound from these formulas. For any s_0 not an original nontrivial zero, evaluation

$$\epsilon_{s_0}(a) = \Theta a(s_0) \in B \quad (\text{ADC7.3})$$

is continuous and has nonzero restriction to J , because $F_0(s_0) \neq 0$. Since $S_r \subset J^\perp$, its class in N_r is nonzero. Under current pushforward angular action T'_n it has scalar n^{s_0} ; under weighted-transfer angular action $nT'_{1/n}$ it has scalar n^{1-s_0} . The resulting weights

$$2\Re s_0, \quad 2(1 - \Re s_0) \quad (\text{ADC7.4})$$

are unbounded in both directions as s_0 ranges outside the original divisor. This uses the actual J' quotient in (ADC4.2), with the full entire F_0 , including its nonzero endpoint and trivial-zero values. It is not a hypothetical zeta spectrum.

Passing to the arithmetic subreceiver $\mathcal{K}_r$ removes precisely this J' quotient by the proven triangle (ADC4.3). It leaves the actual row

$$0 \rightarrow S_r^Q \rightarrow Q' \rightarrow Q'/S_r^Q \rightarrow 0 \quad (\text{ADC7.5})$$

and the special-image map σ_r . Its kernel is still C_{crit} , and its nonzero image lines still have the matching weights (ADC5.4). Thus the next useful attempted reduction has been performed: the full unwanted coefficient quotient is removed by an exact existing map, while the arithmetic intersection that DC's stronger argument would have to kill remains explicitly S_r^Q .

Finally, the normal separator c from GMS/FOD has identity action on the entire original Q ; its transpose is identity on Q' . Hence it acts as identity on S_r^Q , not zero. The actual equality $ce_+ = 0$ for the translated normal extension cannot therefore be reused as $c\sigma_r = 0$. The precise relation is the original dual extension and its pushout in ADC4. This verifies the existing attempt using the full earlier results rather than substituting the already killed normal class for the present attaching map.

ADC8. The continuous transpose and the actual original-quotient cross

Human-source attribution: this use of the complex Hilbert-space Riesz representation theorem is classical. See Terence Tao, [245B, Notes5: Hilbert spaces](#), Theorem1, Remark2 and Exercise15 (17 January2009). Remark2 identifies the continuous dual with the conjugate Hilbert space; the explicit conjugate-space map and adjoint convention in the present calculation remain displayed below.

The coefficient transpose can be computed without assuming a duality theorem. Identify

$$C' = (\overline{H})' \simeq H, \quad x \mapsto (\overline{y} \mapsto \langle x, y \rangle_+). \quad (\text{ADC8.1})$$

This is the strong topological Riesz isomorphism for the conjugate Hilbert space. Evaluation $A \rightarrow (A'_\beta)'$, followed by a'_r , gives

$$b_r : A \rightarrow H, \quad b_r = D_r^* J E q. \quad (\text{ADC8.2})$$

Indeed, on $\overline{y}$,

$$a'_r(\text{ev}_f)(\overline{y}) = a_r(\overline{y})(f) = \langle E q f, J D_r y \rangle_+ = \langle D_r^* J E q f, y \rangle_+. \quad (\text{ADC8.3})$$

Every map here is continuous, by (ADC1.9). Define also

$$\beta_r : Q \rightarrow H, \quad \beta_r = D_r^* J E, \quad b_r = \beta_r q. \quad (\text{ADC8.4})$$

Its kernel is exactly Z_r from (ADC5.5)–(ADC5.6); it retains all higher jets and all critical-line values. The actual coefficient cone to which these maps apply is

$$\mathcal{P}_r = \text{Cone}(\underline{A} \xrightarrow{b_r \text{ ev}_p} i_* H) = [\underline{A}^{-1} \xrightarrow{+b_r} i_* H^0]. \quad (\text{ADC8.5})$$

This is constructed by the displayed map on the original source. The following cross uses its literal two-term complex, so no arbitrary exactness of continuous dualization is being presumed.

The required current-level comparison is now explicit. ACD2-ACD4 in CC_ADJOINT_DEFECT_CONTINUOUS_DUAL.md construct the fine resolution

$$\mathcal{L}_r^{-1} = \mathcal{E}_A^0, \quad \mathcal{L}_r^0 = \mathcal{E}_A^1 \oplus i_* H, \quad \mathcal{L}_r^1 = \mathcal{E}_A^2, \quad d^{-1}f = (-df, b_r f(p)), \quad d^0(\omega, h) = -d\omega. \quad (\text{ADC8.5a})$$

Inclusion of locally constant functions and the identity on H compare (ADC8.5) with this resolution. The coefficient-valued Poincaré contraction proves this is a quasi-isomorphism. It does not require the image of b_r to be closed. Use the fixed-support smooth seminorms and their compact-support locally convex inductive limit to form the actual continuous Hom sheaf $\mathcal{D}_r^k(U) = \Gamma_c(U, \mathcal{L}_r^{-k})'$, with $d_{\mathcal{D}}\ell = (-1)^{k+1}\ell d_{\mathcal{L}}$. Its degrees $(-1, 0, 1)$ are the current terms $\mathcal{T}_A^{-2}, \mathcal{T}_A^{-1} \oplus i_* H', \mathcal{T}_A^0$, and

$$d_{\mathcal{D}}^{-1}v = (d_T v, 0), \quad d_{\mathcal{D}}^0(u, \lambda) = d_T u - \delta_p b'_r \lambda. \quad (\text{ADC8.5b})$$

For instance the second formula on a test f is $-u(-df) - \lambda(b_r f(p))$; this fixes both signs. With $R : C \rightarrow H', R(\bar{y})(h) = \langle h, y \rangle_+$, equation (ADC8.3) gives $b'_r R = a_r$. The map

$$\mathcal{D}_r \longrightarrow \mathcal{E}_r, \quad v \mapsto -v(\deg -1), \quad (u, Rx) \mapsto (u, x)(\deg 0), \quad w \mapsto -w(\deg 1) \quad (\text{ADC8.5c})$$

is a continuous chain isomorphism: the differential composites are $(d_T v, 0)$ and $-d_T u + \delta_p a_r x$. This identifies $\mathcal{E}_r$ with the literal continuous support-dual complex of the constructed $\mathcal{P}_r$ resolution. It is not an assertion that continuous dualization is exact on every locally convex complex. The positive compact-form pairing in outer degrees has the minus signs just displayed; the point pairing is $R(x)(h)$. ACD3's compact-support contraction proves the local support comparison, with the same positive Dirac class as (ADC2.8).

Its restriction to the puncture is $\underline{A}[1]$, and its pole complex is $[A^{-1} \xrightarrow{b_r} H^0]$. Therefore

$$R\psi\mathcal{P}_r = A[1], \quad T = 1, \quad H^{-1}(i^*\mathcal{P}_r) = \ker b_r, \quad H^0(i^*\mathcal{P}_r) = H/b_r A. \quad (\text{ADC8.6})$$

The map to $A[1] \oplus A[\vartheta_p]$ is identity on the degree-minus-one A and zero on H . In its restriction fibre the degree-zero coordinate is $H \oplus A$, with preceding differential $a \mapsto (b_r a, a)$. Its quotient coordinate is $(h, a) \mapsto h - b_r a$. This cancels the identity pair and gives

$$i^!\mathcal{P}_r \simeq H[0] \oplus A[-1], \quad \partial_{-1} = -b_r : A \rightarrow H, \quad \partial_0 = +1 : A[\vartheta_p] \rightarrow A. \quad (\text{ADC8.7})$$

The last sign is positive after the same cochain shift as in LVD3/RGR6; it also follows from the retained positive Dirac coordinate in the fibre. The full source includes the endpoint term

$$\mathcal{P}_{r,\text{full}} = \mathcal{P}_r \oplus i_* E_p''[1], \quad (\text{ADC8.8})$$

where $E_p'' \simeq E_p$ by finite-dimensional evaluation, with both original labels retained. This contributes special/support degree -1 , no nearby group, and unshifted vanishing degree -2 .

The resulting entire nonzero cross is:

i	special A_i	K_i	punctured B_i	nearby invariants C_i	support O_i	α_i	∂_i
-2	0	0	0	0	E_p''	0	0
-1	$\ker b_r \oplus E_p''$	0	A	A	H	$(a, e) \mapsto a$	$-b_r$
0	$H/b_r A$	$A[\vartheta_p]$	$A[\vartheta_p]$	0	A	0	+1

In particular the genuine invariant-cycle obstruction is now

$$\boxed{\operatorname{im} \partial_{-1}/\partial_{-1} j_{-1} K_{-1} = b_r A = \beta_r Q, \quad \operatorname{coker} \operatorname{sp}_{-1} = A/\ker b_r \simeq Q/Z_r.} \quad (\text{ADC8.9})$$

The first identification of the cokernel with the image uses $[a] \mapsto -b_r a$, retaining the boundary sign. The induced algebraic bijection $Q/Z_r \rightarrow b_r A$ can equivalently use $[F] \mapsto \beta_r F$ when the minus is stated separately. Quotient topologies are retained, and no inverse for its ambient image topology is asserted. All other obstruction images vanish.

This dual direction receives the original LNC quotient through an explicit map of the full local source. Put $P = \mathcal{F}_{\text{red}}[1] \simeq \operatorname{Cone}(q \operatorname{ev}_p)$, with terms $\underline{A}^{-1} \rightarrow i_* Q^0$. Then

$$f_r : P \longrightarrow \mathcal{P}_r, \quad f_r^{-1} = 1_A, \quad f_r^0 = \beta_r \quad (\text{ADC8.10})$$

is a continuous sheaf chain map since $\beta_r q = b_r$. On the full objects add the original evaluation $E_p[1] \rightarrow E_p''[1]$. Its nearby map is identity A , and its vanishing map is $\beta_r : Q \rightarrow H$. On support degree zero it is β_r , and on support degree one it is identity A . Thus the original boundary $-q : A \rightarrow Q$ and the new boundary $-b_r : A \rightarrow H$ satisfy

$$\begin{array}{ccc} A & \xrightarrow{-q} & Q \\ \parallel & & \downarrow \beta_r \\ A & \xrightarrow{-b_r} & H. \end{array} \quad (\text{ADC8.11})$$

It satisfies all four commuting-map identities (DC10.2), with the explicit shifts and coefficients just computed. Here those identities are proved on the displayed continuous complexes; the finite-dimensional Frobenius hypotheses used for DC10's weight conclusions are not asserted for them. Its induced map on obstruction images is the actual surjection

$$Q \xrightarrow{\beta_r} \beta_r Q, \quad \ker \beta_r = Z_r. \quad (\text{ADC8.12})$$

This constructs the precise arithmetic subquotient of the original local defect, rather than naming two different receivers as if they were identical.

The transpose is also the actual old current map (ADC4.5), with its representative sign retained. On the fine resolutions of the plus- q and plus- b_r cones, f_r is identity on forms and β_r on the point term; its transpose is identity on currents and $\beta_r' R = \sigma_r$ on point functionals. The literal dual of the plus- q cone in the sign convention (ADC8.5c) is $\operatorname{Cone}(-\delta_p q')[-1]$. The accepted RGR representative is $\operatorname{Cone}(+\delta_p q')[-1]$; their comparison negates the point coordinate. Therefore the map into that accepted representative is $(u, x) \mapsto (u, -\sigma_r x)$, as proved directly in (ACD4.4). Its positive costalk coordinates are still $(\sigma_r, 1_B)$, by the simultaneous coordinate comparisons in (RGR6.3a). Thus the local source square and the earlier current comparison are related by an actual continuous transpose, rather than by a coincidence of coefficient groups.

Finally $b_r = 0$ if and only if $D_r = 0$: if $b_r = 0$, the bounded map $D_r^* j$ vanishes on dense $E(Q)$, hence on H ; conversely $D_r = 0$ makes every formula zero. Therefore the new cross's surjectivity in degree

—1 is equivalent to vanishing of the actual adjoint discrepancy. Unlike (ADC3.4), its obstruction image is the part that detects that discrepancy.

The exact coefficient actions on this new boundary can also be calculated:

$$\beta_r T_t^Q = U_t^* \beta_r, \quad \beta_r U_t^Q = T_t^* \beta_r, \quad b_r T_t^A = U_t^* b_r, \quad b_r U_t^A = T_t^* b_r. \quad (\text{ADC8.13})$$

For the first equality, $ET_t^Q = T_t E$ and $JT_t = U_t^* J$ by (ADC5.1); the diagonal D_r^* commutes with U_t^* . This gives $D_r^* JT_t E = U_t^* D_r^* J E$. The second follows with T_t, U_t exchanged; composing with q gives the remaining formulas. They define actual continuous chain maps on (ADC8.5), and show that the two coefficient directions descend to its invariant-cycle image. For a value class in Q/Z_r supported at an actual ρ , its nonzero image lies at $\rho^\#$ in H . A representative in the complete Q can still have higher-jet Jordan terms; these lie in the recorded kernel Z_r , rather than being deleted from Q . The source and target scalars on the indicated quotient and image are respectively

$$t^\rho = t^{1-\overline{\rho^\#}}, \quad t^{1-\rho} = t^{\overline{\rho^\#}}. \quad (\text{ADC8.14})$$

Their weights again agree on every nonzero image line. This proves the compatibility directly on the new cross, rather than inferring a vanished map from the existence of the two actions.

ADC9. Exact outcome and source-use record

The two calculated crosses test different parts of the same actual attaching data:

receiver	invariant-cycle obstruction image	additional retained defect
$\mathcal{E}_r$	0	$S_r = \ker(K_0 \rightarrow \text{im } \partial_0)$
$\mathcal{P}_r$	$b_r A \simeq Q/Z_r$	$H/b_r A$ in special degree 0.

(ADC9.1)

Their source maps are the actual $a_r(\bar{y}) = q' A_H(D_r y)$ and $b_r = D_r^* J E q$, not independent examples. The complete point-source quotient C/C_{crit} and the original arithmetic quotient Q/Z_r are joined by the nondegenerate pairing (ADC5.7). The comparison maps with the original source are (ADC4.5), (ADC8.10), and the boundary square (ADC8.11). The full original dual extension survives as the pushout (ADC4.2), with the source kernel and every endpoint degree retained.

The next attempted step required by WU065 has therefore been carried out. Starting from the automatically vanishing invariant-cycle cokernel of $\mathcal{E}_r$, the calculation first isolates the stronger coinvariant-injectivity defect, then removes its full J' coefficient quotient by the actual source triangle, and then constructs the transpose cone receiving the original invariant-cycle quotient. Its obstruction is precisely the image of the adjoint discrepancy on original classes. The two actual coefficient directions are computed on that image in (ADC8.13)-(ADC8.14); neither has separated source and target weights there. This describes the proved effect of the existing maps, without declaring a finite-field purity theorem for these coefficients or using a new twist to force one.

Reading coverage for this derivation: the connected USER_ARGUMENT_RECONSTRUCTION.md, its correction-precedence table, the verbatim WU062 and WU064-WU065 passages; the complete DC0-DC12 proof; LVD0-LVD9; RGR0-RGR12, including the full original-source, current-cone and positive-adjoint distinctions; GTAH0-GTAH4; ACD0-ACD8, with the literal current-resolution and source-map comparison in ACD2-ACD4 used in ADC8; DCA0-DCA12, including its complete cut-cone, genuine cover/trace and composition homotopies, with DCA3 and DCA6 used explicitly in ADC6; and the accepted LNC0-LNC11 proof, previously authored and independently reviewed here. LNC's current, nearby and support maps are used with their already proved continuous contractions and signs. The original source references and their exact witness status are preserved in DC0, LNC11, LVD0 and RGR0. In particular Deligne's local invariant-cycle proof is read through the identified current French transcription of Weil II; this is not claimed to be Deligne-authored TeX.

This is a calculation on the actual recovered arithmetic and current/sheaf maps. It establishes neither an off-critical zero nor the vanishing of D_r . It supplies the exact subquotients and commuting maps through which that discrepancy enters the stronger Deligne comparison.

Genuine power-cover actions on the adjoint-defect receiver — DCA0-DCA12

Proof source: CC_ADJOINT_DEFECT_COVER_ACTIONS.md.

Actual power-cover actions on the original adjoint-defect receiver

Complete independent local derivation. Proof locators **DCA0-DCA12**.

DCA0. The object, provenance, and category

This calculation continues the particular receiver of RGR11. The supporting datum remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. All coefficient spaces, complex coordinates, positive integers, and operators below occur after the original arithmetic reconstruction. The two original branches retain their separate counters. No numerical coordinate or operation is assigned to the support.

The inputs read for this calculation are [CC_RESIDUE_GYSIN_RECEIVER_INDEPENDENT.md](#), RGR1, RGR9-RGR11; [CC_CONTINUOUS_SUPPORT_DUAL_SHEAF.md](#), CSD3-CSD6 and CSD9-CSD11; [CC_LOCAL_NEARBY_CYCLES_AND_SPECIALIZATION.md](#), LNC5-LNC6; the retained GTR1-GTR4 proof in [CC_RAMIFIED_TRACE_AND_RESIDUE_ACTION.md](#); [CC_ADJOINT_DEFECT_INVARIANT_CYCLE_CROSS.md](#), ADC1-ADC6; and the continuous-dual comparison in [CC_ADJOINT_DEFECT_CONTINUOUS_DUAL.md](#), ACD1-ACD4. These are exact programme dependencies, not new primary-source readings. The human-source identity for the original coefficient sheaf remains Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, [arXiv:0903.2024v3](#), §5, with the receiver comparison proved in CSP. The cover actions on the new receiver are proved below.

Derived statements use sheaves of complex vector spaces. Every coefficient operator, finite-coordinate homotopy, and specified current map is continuous. The continuous dual is never replaced by an algebraic dual. The proof does not assume a finite-rank perverse classification, a strong-dual open mapping theorem, or pullback of arbitrary distributions through a branch point. The notation $K[1]$ puts a degree-zero coefficient K in degree -1 .

DCA1. The actual arithmetic map and its two covariances

Retain

$$\begin{aligned} S &= \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0\}, \\ A &= \{b \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j b(u)| < \infty \text{ for every } N, j \geq 0\}, \\ \Sigma f(u) &= 2 \sum_{k \geq 1} f(ku), \quad J = \Sigma S, \quad Q = A/J, \quad q : A \rightarrow Q, \quad B = A'_\beta, \quad j = q' : Q'_\beta \hookrightarrow B. \end{aligned} \tag{DCA1.1}$$

The accepted original-image theorem gives closed J and the continuous inverse of $\Sigma : S \rightarrow J$. The original Mellin map and divisor are

$$\begin{aligned} \Theta b(s) &= \frac{1}{2} \int_0^\infty b(u) u^s \frac{du}{u}, & \Theta \Sigma f(s) &= \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v} \quad (\Re s > 1), \\ f_0(v) &= \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}, & \Theta \Sigma f_0(s) &= \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \end{aligned} \quad (\text{DCA1.2})$$

The latter entire function has values $1/8$ at both $0, 1$, value $\pi/24$ at both $-1, 2$, and

$$\Theta \Sigma f_0(-2k) = \frac{k(2k+1)(-1)^k \pi^k}{2k!} \zeta'(-2k) \quad (k \geq 1). \quad (\text{DCA1.3})$$

At an actual nontrivial zero ρ of multiplicity m_ρ , its full local expression is

$$t^{m_\rho} \frac{(\rho+t)(\rho+t-1)}{8} \pi^{-(\rho+t)/2} \Gamma((\rho+t)/2) \frac{\zeta(\rho+t)}{t^{m_\rho}}. \quad (\text{DCA1.4})$$

The local factor and every derivative remain part of the original comparison. The quotient Q retains all vanishing-jet conditions through order $m_\rho - 1$. It is not replaced by its value quotient in the argument below. The original function is still

$$\zeta(s) = 1 + \sum_{k \geq 2} k^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{k \geq 1} (\log p) p^{-ks} \quad (\Re s > 1). \quad (\text{DCA1.5})$$

Let $\mathcal{Z}$ be the actual distinct nontrivial zeros, put $\rho^\# = 1 - \bar{\rho}$, and retain

$$H = \ell^2(\mathcal{Z}, m), \quad \langle x, y \rangle_+ = \sum_{\rho \in \mathcal{Z}} m_\rho x_\rho \bar{y}_\rho, \quad (\mathcal{J}y)_\rho = y_{\rho^\#}. \quad (\text{DCA1.6})$$

The actual value map $E : Q \rightarrow H$ has $(EF)_\rho = (\Theta_Q F)(\rho)$, is continuous, and has dense image by the retained full-jet interpolation theorem. The map

$$A_H(y)(F) = \langle EF, \mathcal{J}y \rangle_+ \quad (\text{DCA1.7})$$

is continuous into Q'_β , anti-linear, and injective. For a bounded subset $K \subset Q$, its strong seminorm is at most $\sup_{F \in K} \|EF\|_+ \|y\|_+$; density of $E(Q)$ proves injectivity.

For a fixed recovered parameter $r > 1$, use the actual operator

$$\begin{aligned} (T_r y)_\rho &= r^\rho y_\rho, & U_r &= r T_{1/r}, & D_r &= T_r^* - U_r, \\ (D_r y)_\rho &= d_r(\rho) y_\rho, & d_r(\rho) &= e^{-i\Im \rho \log r} (r^{\Re \rho} - r^{1-\Re \rho}). \end{aligned} \quad (\text{DCA1.8})$$

The known open strip makes these bounded. Set

$$C = \bar{H}, \quad \sigma_r(\bar{y}) = A_H(D_r y), \quad a = a_r = j\sigma_r : C \rightarrow B. \quad (\text{DCA1.9})$$

This is exactly RGR11.3. The parameter r is fixed; the cover degree n below is a separate positive integer.

On A , write $T_t b(u) = b(u/t)$. It preserves J , descends to T_t^Q , and its value action is the displayed diagonal T_t . Directly on the actual zero coordinates,

$$T_t^* = J U_t J, \quad U_t^* = J T_t J. \quad (\text{DCA1.10})$$

For example $(JU_tJy)_\rho = t^{1-\rho^\#}y_\rho = t^{\bar{\rho}}y_\rho$, the positive Hilbert adjoint of multiplication by t^ρ . Hence

$$(T_t^Q)'A_H(y)(F) = \langle T_t EF, Jy \rangle_+ = \langle EF, T_t^* Jy \rangle_+ = A_H(U_t y)(F). \quad (\text{DCA1.11})$$

Exchanging T_t, U_t proves the other covariance. Since the original diagonal D_r commutes with both operators, for

$$R = T'_n, \quad R_- = T'_{1/n}, \quad S_n = \bar{U}_n : C \rightarrow C, \quad L_n = \bar{T}_n : C \rightarrow C \quad (\text{DCA1.12})$$

one obtains the complete identities

$$\boxed{aS_n = Ra, \quad aL_n = nR_-a, \quad RR_- = R_-R = 1_B, \quad S_nL_n = L_nS_n = n1_C.} \quad (\text{DCA1.13})$$

Here conjugation specifies the linear operators on the conjugate vector space, rather than erasing the anti-linearity of (DCA1.7). All transposes are strong-continuous because their primal maps take bounded sets to bounded sets.

DCA2. The current cone and the positive support coordinate

Fix a positively oriented pole disk D , with coordinate $z_p = z$ at zero or $z_p = 1/z$ at infinity, and $i : \{p\} \hookrightarrow D$. The coefficient current complex is

$$\mathcal{T}_A^k(U) = (\mathcal{E}_c^{-k}(U; A))', \quad dT(\phi) = (-1)^{k+1}T(d\phi), \quad -2 \leq k \leq 0. \quad (\text{DCA2.1})$$

Its positive representatives are

$$c_b(\omega) = b\left(\int \omega\right), \quad h_p b(\eta) = b\left(\int \frac{d \arg z_p}{2\pi} \wedge \eta\right), \quad \delta_p b(f) = b(f(p)). \quad (\text{DCA2.2})$$

The angular coefficient is locally integrable, and Stokes on a punctured disk gives $dh_p b = \delta_p b$. The compact-test contraction in CSD2 identifies $\mathcal{T}_A \simeq \underline{B}[2]$, with the positive orientation above.

The actual receiver is

$$\begin{aligned} \mathcal{E} &= \mathcal{E}_r = \text{Cone}(-\delta_p a)[-1], \\ \mathcal{E}^{-1} &= \mathcal{T}_A^{-2}, \quad \mathcal{E}^0 = \mathcal{T}_A^{-1} \oplus i_* C, \quad \mathcal{E}^1 = \mathcal{T}_A^0, \\ d^{-1}b &= (-d_T b, 0), \quad d^0(b, x) = -d_T b + \delta_p a x. \end{aligned} \quad (\text{DCA2.3})$$

The quotient coordinate is x , so its connecting arrow is $+\text{Gys}_B a$:

$$\underline{B}[1] \longrightarrow \mathcal{E} \longrightarrow i_* C \xrightarrow{+\text{Gys}_B a} \underline{B}[2]. \quad (\text{DCA2.4})$$

The closed stalk representatives $b \mapsto c_b$ and $x \mapsto (h_p a x, x)$ give

$$i^* \mathcal{E} \simeq B[1] \oplus C, \quad R\Gamma(D^*, \mathcal{E}) \simeq B[1] \oplus B[0], \quad \text{res} = (1_B, a). \quad (\text{DCA2.5})$$

The second punctured coefficient is measured by the positive angular current. In the restriction fibre convention $d(s, t) = (ds, \text{res}(s) - dt)$, cancelling its identity pair gives

$$i^! \mathcal{E} \simeq [C \xrightarrow{+a} B] \quad \text{in degrees } 0, 1. \quad (\text{DCA2.6})$$

The sign is also seen directly: $d(h_p b, 0) = (-\delta_p b, h_p b)$, so the positive Dirac class and the positive angular boundary coordinate coincide. For shifted nearby and vanishing functors $\Psi = R\psi[-1]$, $\Phi = R\phi[-1]$,

$$\Psi \mathcal{E} = B, \quad \Phi \mathcal{E} = C, \quad \text{can} = 0, \quad \text{var} = +a. \quad (\text{DCA2.7})$$

These statements are consequences of the current and restriction cones, not applications of a finite-dimensional classification.

DCA3. The full cover and a continuous cut model

Let $b_n : D_s \rightarrow D$, $z_p \mapsto z_p^n$, with the source disk chosen as the inverse image of the target disk. This is a finite proper map. Its direct-image stalk is the finite direct sum of the source stalks above that point: small disjoint neighborhoods of the finite fibre contain the inverse image of some target neighborhood, by properness. This stalk formula proves exactness of b_{n*} on vector-space sheaves without a finite-rank assumption. At the pole there is one inverse-image point.

The inverse image of the target universal cover has exactly n components. Label them $0, \dots, n-1$ in positive cyclic order and put

$$\begin{aligned} V_n &= B^n, \quad t(v_0, \dots, v_{n-1}) = (v_{n-1}, v_0, \dots, v_{n-2}), \\ \Delta b &= (b, \dots, b), \quad \Sigma_n(v) = \sum_{j=0}^{n-1} v_j, \quad e_0 b = (b, 0, \dots, 0). \end{aligned} \quad (\text{DCA3.1})$$

Thus $\Psi(b_{n*}\mathcal{E}) = V_n$, with monodromy t . In this section Σ_n is a finite sheet sum, different from the original summation source Σ in (DCA1.1).

The circle cochain model of the punctured direct image is

$$N_n = [V_n \xrightarrow{t-1} V_n] \quad \text{in degrees } -1, 0. \quad (\text{DCA3.2})$$

It follows by cutting a positive circle once, recording its constant value and the difference of its values across the cut. Constant sections are ΔB , and the positive loop class is measured by Σ_n . Explicitly

$$\ker(t-1) = \Delta B, \quad \text{im}(t-1) = \ker \Sigma_n, \quad V_n / (t-1)V_n \xrightarrow{[v] \mapsto \Sigma_n v} B. \quad (\text{DCA3.3})$$

For the middle equality, if $\sum_j w_j = 0$, set $v_0 = 0$ and $v_j = -\sum_{k=1}^j w_k$ for $1 \leq j \leq n-1$. Then $v_{j-1} - v_j = w_j$ and $v_{n-1} - v_0 = w_0$, proving $(t-1)v = w$. This solution is a finite continuous operator.

The stalk model is $S^{\text{st}} = B[1] \oplus C$. In the cut model its restriction has the representative

$$r_n^{-1} = \Delta : B \rightarrow V_n, \quad r_n^0 = e_0 a : C \rightarrow V_n. \quad (\text{DCA3.4})$$

The first formula is restriction of a constant current to every sheet. The second has positive loop coordinate $\Sigma_n e_0 a x = a x$, exactly the original angular restriction. Any representative produced from a different cut differs by an element of $\ker \Sigma_n$; the explicit solution in (DCA3.3) gives its continuous chain homotopy to (DCA3.4). The cut therefore changes the representative, not the original angular map.

For clarity about what these data construct, if $j_D : D^* \hookrightarrow D$, a complex with punctured restriction L , stalk S^{st} , and restriction arrow $r : S^{\text{st}} \rightarrow i^* Rj_{D*} L$ is represented by the homotopy fibre of

$$Rj_{D*} L \oplus i_* S^{\text{st}} \longrightarrow i_* i^* Rj_{D*} L, \quad (u, s) \longmapsto u - r s. \quad (\text{DCA3.5})$$

On the puncture this fibre is L ; at the pole it is the fibre of $(1, -r) : N \oplus S^{\text{st}} \rightarrow N$, which retracts to S^{st} . The natural comparison with the original complex is therefore a stalkwise quasi-isomorphism. A map of punctured complexes, a map of stalk complexes, and an explicit homotopy between their restriction composites consequently give an actual derived sheaf morphism via this fibre formula. No classification theorem is needed for that construction.

The associated restriction fibre is

$$B \xrightarrow{b \mapsto (0, \Delta b)} C \oplus V_n \xrightarrow{(x, v) \mapsto e_0 a x - (t-1)v} V_n \quad (\text{deg } -1, 0, 1). \quad (\text{DCA3.6})$$

The first arrow is a split continuous embedding; for example projection to the zeroth component splits Δ . Quotienting this identity pair retains

$$\Phi(b_{n*}\mathcal{E}) = C \oplus D_n, \quad D_n = V_n/\Delta B,$$

$$\boxed{\mathcal{V}_n(x, [v]) = e_0 ax - (\mathfrak{t} - 1)v.} \quad (\text{DCA3.7})$$

This is well defined because $(\mathfrak{t} - 1)\Delta = 0$. The canonical arrow is $\mathcal{C}_n(v) = (0, [v])$. Our positive support-cone convention gives

$$\mathcal{V}_n \mathcal{C}_n = 1 - \mathfrak{t}, \quad \mathfrak{t}_\Phi(x, [v]) = (x, [tv - e_0 ax]), \quad \mathcal{C}_n \mathcal{V}_n = 1 - \mathfrak{t}_\Phi. \quad (\text{DCA3.8})$$

These formulas fix both signs. Iterating the middle formula gives

$$\mathfrak{t}_\Phi^k(x, [v]) = \left(x, \left[\mathfrak{t}^k v - \sum_{\ell=0}^{k-1} \mathfrak{t}^\ell e_0 ax \right] \right), \quad \mathfrak{t}_\Phi^n = 1, \quad (\text{DCA3.9})$$

since $\sum_{\ell=0}^{n-1} \mathfrak{t}^\ell e_0 ax = \Delta ax$. This retains the full vanishing object and its monodromy, including the cover contribution D_n .

DCA4. The genuine proper-current map

On target tests define

$$(F_n T)(\phi) = T(T_n b_n^* \phi). \quad (\text{DCA4.1})$$

Properness preserves compact support, and smooth pullback followed by the continuous coefficient operator is continuous on the fixed-support test spaces. Transposition is therefore continuous for both current dual topologies. It commutes with the Hom differential. Its representative factors are

$$F_n c_b = c_n Rb, \quad F_n h_p b = h_p Rb, \quad F_n \delta_p b = \delta_p Rb. \quad (\text{DCA4.2})$$

For the first equality, the integral of the pulled-back compact two-form is n times its integral. On the puncture each inverse branch contributes $1/n$ of the positive angular form, so the sum of all n branch contributions is that same form. Equality extends as a locally integrable current through the pole. Evaluation at the single inverse-image pole proves the last equality. Thus the three displayed factors have separate geometric proofs.

By (DCA1.13),

$$\boxed{F_n : b_{n*}\mathcal{E} \rightarrow \mathcal{E}, \quad (b, x) \mapsto (F_n b, S_n x)} \quad (\text{DCA4.3})$$

is a current chain map: its attaching term satisfies $F_n \delta_p ax = \delta_p Rax = \delta_p a S_n x$. Its stalk map, in the actual representatives of (DCA2.5), is

$$i^* F_n = (nR, S_n) : B[1] \oplus C \longrightarrow B[1] \oplus C. \quad (\text{DCA4.4})$$

The source nearby space is all of V_n , and the map is

$$\boxed{F_{n,\Psi} = R\Sigma_n : V_n \rightarrow B.} \quad (\text{DCA4.5})$$

A single inverse branch is an orientation-preserving local isomorphism and contributes R , with no additional local degree; summing the branches proves the formula. It intertwines $\mathfrak{t}$ with identity.

On $N_n \rightarrow N_1$ the two cochain maps are both $R\Sigma_n$. They commute with restriction, since

$$R\Sigma_n \Delta = nR, \quad R\Sigma_n e_0 a = Ra = aS_n. \quad (\text{DCA4.6})$$

Thus the vanishing and costalk maps are exactly

$$\boxed{F_{n,\Phi}(x, [v]) = S_n x, \quad i^! F_n = (S_n, R) : [C \xrightarrow{a} B] \rightarrow [C \xrightarrow{a} B].} \quad (\text{DCA4.7})$$

The extra summand D_n in the literal vanishing source maps to zero; it remains part of that source. The variation square is the identity

$$R\Sigma_n \mathcal{V}_n(x, [v]) = Rax = aS_n x. \quad (\text{DCA4.8})$$

DCA5. Invariants, coinvariants, and actual Gysin adjunction

On nearby invariants, (DCA4.5) restricts to

$$\Delta B \rightarrow B, \quad \Delta b \mapsto nRb. \quad (\text{DCA5.1})$$

On coinvariants, using the exact positive identification (DCA3.3), it is

$$B \rightarrow B, \quad b \mapsto Rb. \quad (\text{DCA5.2})$$

The diagonal is not the entire nearby space; its composite with the coinvariant quotient is multiplication by n . These two functors therefore see the two different factors of the same genuine map.

The Gysin square is

$$\begin{array}{ccc} i_* C & \xrightarrow{b_{n*}(\text{Gys}_B a)} & b_{n*} B[2] \\ \downarrow S_n & & \downarrow F_n \\ i_* C & \xrightarrow{\text{Gys}_B a} & B[2]. \end{array} \quad (\text{DCA5.3})$$

At the level of currents this is $F_n \delta_p a = \delta_p a S_n$. Under the oriented support adjunction $i^! B[2] = B$, its coefficient equality is

$$Ra = aS_n. \quad (\text{DCA5.4})$$

The ordinary stalk factor nR belongs instead to (DCA5.1). Replacing (DCA5.4) by $nRa = aS_n$ would change the functor and would be a false Gysin square. The distinction is proved by the original Dirac and angular currents, not assigned by a weight label.

DCA6. Construction of the transfer-adjoint direction without branch-point current pullback

There is a derived sheaf map

$$L_n : \mathcal{E} \longrightarrow b_{n*} \mathcal{E} \quad (\text{DCA6.1})$$

whose generic coefficient is R_- , whose source coefficient on C is L_n , and whose positive supported coefficient is nR_- . We construct its complete gluing data.

Use the constant-sheaf unit with coefficient R_- . On the puncture its map is ΔR_- into the full n -sheet coefficient. On the pole stalk it is R_- . On positive support it is nR_- : a positive boundary circle is mapped with degree n , or equivalently $b_n^*(d \arg z_p / (2\pi)) = n d \arg z_p / (2\pi)$. Its Gysin square is exactly

$$nR_- a = aL_n, \quad (\text{DCA6.2})$$

already proved for the actual arithmetic map in (DCA1.13).

In the explicit cut models use

$$f_{\text{st}} = (R_-, L_n), \quad f_N^{-1} = \Delta R_-, \quad f_N^0 = \Delta R_-. \quad (\text{DCA6.3})$$

For $x \in C$, set

$$z_x = R_- ax, \quad w_n(x) = (0, z_x, 2z_x, \dots, (n-1)z_x) \in V_n. \quad (\text{DCA6.4})$$

All coordinates are retained. Direct componentwise subtraction gives

$$(\mathfrak{t} - 1)w_n(x) = ne_0 z_x - \Delta z_x. \quad (\text{DCA6.5})$$

The zeroth component is $(n-1)z_x$, and each of the other $n-1$ components is $-z_x$. Therefore

$$r_n f_{\text{st}} - f_N r_1 = d_{N_n} w_n + w_n d_{S^{\text{st}}}. \quad (\text{DCA6.6})$$

In degree -1 both sides vanish; in degree zero the equality is (DCA6.2) and (DCA6.5). Formula (DCA3.5) now constructs (DCA6.1) from these continuous maps and this specified homotopy. No pullback of a singular current at the branch point has been used.

For reference, if a restriction-fibre map has coefficient maps f_S, f_N and homotopy k satisfying $r' f_S - f_N r = dk + kd$, its formula is

$$(s, t) \mapsto (f_S s, f_N t + ks). \quad (\text{DCA6.7})$$

Substitution into $d(s, t) = (ds, rs - dt)$ proves the chain identity. Applying this formula gives

$$\boxed{\mathbb{L}_{n, \Psi} = \Delta R_-, \quad \mathbb{L}_{n, \Phi}(x) = (L_n x, [w_n(x)])}. \quad (\text{DCA6.8})$$

The raw-cut correction $[w_n(x)]$ is necessary: omitting it would give variation $ne_0 R_- ax$ instead of $\Delta R_- ax$. With it,

$$\mathcal{V}_n(L_n x, [w_n(x)]) = e_0 a L_n x - (\mathfrak{t} - 1)w_n(x) = \Delta R_- ax. \quad (\text{DCA6.9})$$

The map also lands in monodromy invariants, since $tw_n(x) - e_0 a L_n x - w_n(x) = -\Delta z_x$, whose class in D_n is zero.

The complete local coefficient maps are

$$\boxed{i^* \mathbb{L}_n = (R_-, L_n), \quad i^! \mathbb{L}_n = (L_n, nR_-) : [C \xrightarrow{a} B] \rightarrow [C \xrightarrow{a} B]}. \quad (\text{DCA6.10})$$

The positive angular map is nR_- , because its class in (DCA3.3) is $\Sigma_n \Delta R_- = nR_-$. It is consequently also the positive Dirac support factor. These are the generic/angular/Dirac factors

$$(R_-, nR_-, nR_-), \quad (\text{DCA6.11})$$

with their actual domains, rather than three scalar assignments to one functor.

One can display the invariant copy of C inside the full vanishing space without discarding the complement:

$$\iota_n(x) = \left(x, \left[0, \frac{ax}{n}, \frac{2ax}{n}, \dots, \frac{(n-1)ax}{n} \right] \right). \quad (\text{DCA6.12})$$

It satisfies $\mathfrak{t}_\Phi \iota_n = \iota_n$ and $\mathcal{V}_n \iota_n = \Delta a/n$, by the same componentwise calculation. Formula (DCA6.8) is exactly $\iota_n L_n$, because $a L_n = nR_- a$. All fractions in this comparison are displayed; (DCA6.4) remains the actual chosen cut correction.

DCA7. Both compositions, including their homotopies

The composite in the target direction is

$$\boxed{\mathbb{F}_n \mathbb{L}_n = n \mathbb{1}_{\mathcal{E}}} \quad \text{in the derived sheaf category.} \quad (\text{DCA7.1})$$

On the stalk complexes its factors are $nRR_- = n$ and $S_n L_n = n$. On the punctured cochain complexes they are $R\Sigma_n \Delta R_- = n$. Its gluing homotopy has the remaining value

$$k(x) = R\Sigma_n w_n(x) = \sum_{j=0}^{n-1} j ax = \frac{n(n-1)}{2} ax. \quad (\text{DCA7.2})$$

This term has not been suppressed. The degree-minus-one stalk homotopy $\ell : C \rightarrow B$, $\ell(x) = \sum_{j=0}^{n-1} j ax$, removes it, because the target restriction in degree -1 is $r_1^{-1} = 1_B$. More explicitly, for the restriction fibre, $H(s, t) = (\ell s, 0)$ has $dH + Hd = (0, ks)$. The same homotopy on the supported summand of the reconstruction (DCA3.5) proves (DCA7.1) at the sheaf-cone level.

Let d be the positively chosen deck generator on $b_{n*}\mathcal{E}$. It is identity on the stalk model and t in both punctured degrees. Its gluing homotopy is $-e_0 a$, because

$$r_n - tr_n = -(t - 1)e_0 a. \quad (\text{DCA7.3})$$

Its induced vanishing map is exactly t_Φ in (DCA3.8). The reverse composite is

$$\boxed{L_n F_n = \sum_{k=0}^{n-1} d^k.} \quad (\text{DCA7.4})$$

On full nearby coefficients both sides are

$$\Delta \Sigma_n = \sum_{k=0}^{n-1} t^k : V_n \rightarrow V_n. \quad (\text{DCA7.5})$$

On the pole stalk they are $n \cdot 1$. On vanishing coefficients the composite from (DCA4.7) and (DCA6.8) is

$$(x, [v]) \mapsto (nx, [(0, ax, 2ax, \dots, (n-1)ax)]). \quad (\text{DCA7.6})$$

For the deck norm, sum (DCA3.9). The term $\sum_k t^k v = \Delta \Sigma_n v$ has zero class in D_n , and the correction is

$$-\sum_{\ell=0}^{n-2} (n-1-\ell) t^\ell e_0 ax. \quad (\text{DCA7.7})$$

Its coordinates are $-(n-1)ax, -(n-2)ax, \dots, -ax, 0$. Adding $(n-1)\Delta ax$ gives exactly the vector in (DCA7.6), proving equality in the quotient.

At the gluing-homotopy level, the difference between the composite correction and (DCA7.7) is precisely $(n-1)\Delta ax$. The stalk homotopy $\ell(x) = (n-1)ax$ removes it through $r_n^{-1} = \Delta$, exactly as in the proof of (DCA7.1). Thus (DCA7.4) holds on the derived sheaf objects, not merely on their cohomology groups. Likewise $d^n = 1$: its accumulated homotopy $-\Delta a$ is removed by the stalk homotopy $-a$.

For $n > 1$, the deck norm is not generally multiplication by n on V_n : it takes $(v_0, \dots, v_{n-1})$ to $\Delta(\sum_j v_j)$.

The actual nonzero coefficient space $B = A'$ is retained in this assertion. On invariants its value is n , and on the extra cyclic components it vanishes. On costalks both compositions are n in each degree, consistently with (DCA4.7), (DCA6.10), and the identity deck action on the single supported point.

DCA8. The original dual extension and all its comparison arrows

Human-source attribution: the extension of continuous linear functionals is the Hahn-Banach theorem, not a programme result. See Terence Tao, [245B, Notes6: Duality and the Hahn-Banach theorem](#), Theorem1

and its complex proof (26 January 2009). For the locally convex use here, continuity bounds a functional by a continuous seminorm; quotienting its kernel reduces this application to that normed-space theorem. No continuous splitting of the original quotient is thereby asserted.

The closed original source gives the full exact continuous-transpose row

$$0 \longrightarrow Q' \xrightarrow{j} B \xrightarrow{k'} J' \longrightarrow 0, \quad k : J \hookrightarrow A. \quad (\text{DCA8.1})$$

Its surjectivity is continuous Hahn-Banach extension from the closed subspace J . This does not assert that $k' : B_\beta \rightarrow J'_\beta$ is open.

The original dilation preserves J , so (DCA8.1) commutes with $(T_n^Q)', T'_n, (T_n|_J)'$, and also with their inverse-parameter maps and the displayed factor n . The actual receiver

$$\mathcal{K}_r = \text{Cone}(-\text{Gys}_{Q'}\sigma_r)[-1] \quad (\text{DCA8.2})$$

has the proved triangle

$$\mathcal{K}_r \longrightarrow \mathcal{E}_r \longrightarrow J'[1] \longrightarrow \mathcal{K}_r[1]. \quad (\text{DCA8.3})$$

In continuous current representatives its first map is $(v, x) \mapsto (q^\vee v, x)$, and its second is $(b, x) \mapsto k^\vee b$. The coefficient identity $k'j = 0$ proves their composite zero. Cancelling their shared supported C coordinate reduces the cone comparison to the constant-current coefficient row (DCA8.1), proving the triangle as in RGR9.

Every construction DCA3-DCA7 applies with B replaced by Q' , a by σ_r , and with its actual transpose dilation. Naturality of $q^\vee, k^\vee$, of the positive Gysin map, and of the finite sheet sums proves that both cover directions commute with (DCA8.3). In particular the first map uses j on each of the n nearby sheets and identity on the original C component; the raw-cut corrections commute because $j\sigma_r = a$. The quotient J' has the corresponding constant-sheaf trace and unit maps.

There is also the actual map $\mathcal{E}_r \rightarrow D_c P$, where $P = \ker(\underline{A} \rightarrow i_* Q)[1]$. In positive quotient coordinates it uses 1_B on nearby coefficients and σ_r on vanishing coefficients. Its cone is $i_* \text{Cone}(\sigma_r)$, as proved in RGR5. The cover directions commute with this map by

$$\sigma_r S_n = (T_n^Q)' \sigma_r, \quad \sigma_r L_n = n(T_{1/n}^Q)' \sigma_r. \quad (\text{DCA8.4})$$

Its raw-cut correction commutes componentwise by the same two identities. Consequently the cover calculation acts on the receiver of the original dual extension; it has not introduced an unrelated coefficient extension.

DCA9. Both endpoint lines and the actual GTR factors

The original pole source is

$$V_+ = S \oplus \mathbb{C}_{c_0} \oplus \mathbb{C}_{c_1}, \quad V_- = S \oplus \mathbb{C}_{d_0} \oplus \mathbb{C}_{d_1},$$

$$\rho_+(t)(f, c_0, c_1) = (f(\cdot/t), c_0, tc_1), \quad \rho_-(t)(h, d_0, d_1) = (th(t\cdot), td_0, d_1). \quad (\text{DCA9.1})$$

The four labels remain the original value-at-zero and real-integral labels at their respective poles. Their dual coordinate lines are denoted by the corresponding primed labels. The full receiver is

$$\mathcal{E}_{r,\text{full}} = \mathcal{E}_r \oplus i_* E'_p[-1]. \quad (\text{DCA9.2})$$

The endpoint summand has zero nearby object, stalk and costalk $E'_p[-1]$, and shifted vanishing object $E'_p[-1]$. Its unshifted vanishing complex is $E'_p[0]$. Thus the endpoints occur in stalk and support degree 1, separately from the two-term supported complex in (DCA2.6).

The actual CSD/GTR maps on this summand are

$$\mathbf{F}_n|_{E'_p} = \rho_p(n)'|_{E'_p}, \quad \mathbf{L}_n|_{E'_p} = n\rho_p(1/n)'|_{E'_p}. \quad (\text{DCA9.3})$$

In the full retained order (c'_0, c'_1, d'_0, d'_1) , these are

$$\boxed{\mathbf{F}_n : (1, n, n, 1), \quad \mathbf{L}_n : (n, 1, 1, n)}. \quad (\text{DCA9.4})$$

They follow by transposing the two explicit diagonal endpoint blocks in (DCA9.1); each product is n . A deck rotation fixes the pole and acts identically on each endpoint coordinate, so its norm there is n , also proving the reverse composition on this retained summand.

For comparison, the full original global coefficient complex has

$$D = [V_+ \oplus V_- \xrightarrow{r_+ - r_-} A \xrightarrow{0} A] \quad \text{in degrees } 0, 1, 2,$$

$$r_+ = \Sigma, \quad r_- = R_A \Sigma = \hat{\Sigma}, \quad R_A b(u) = u^{-1}b(u^{-1}). \quad (\text{DCA9.5})$$

Its positive geometric pullback and weighted ramified trace with inverse coefficients have the exact cochain representatives

$$\mathbf{B}_n = (\rho(n), T_n, nT_n), \quad \mathbf{U}_n = (n\rho(1/n), nT_{1/n}, T_{1/n}). \quad (\text{DCA9.6})$$

The weighted trace sums all interior sheets and is n times identity at each pole, including both endpoint lines. This gives $\text{Tr}_n u_n = n1$ on every stalk; its global factors then follow from the angular and top-degree orientation integrals, as calculated in GTR2-GTR4.

Transposition of these degree-zero maps adds no Hom-differential sign. The full original dual rows are

dual degree	−2	−1	0
$\mathbf{B}'_n$	nT'_n	T'_n	$\rho(n)'$
$(\mathbf{B}_n^{-1})'$	$n^{-1}T'_{1/n}$	$T'_{1/n}$	$\rho(1/n)'$
$\mathbf{U}'_n$	$T'_{1/n}$	$nT'_{1/n}$	$n\rho(1/n)'$

(DCA9.7)

All three rows are retained. The second is the pairing contragredient and differs from the third by the full factor n . The current pushforward direction (DCA4.3) has the first row's geometry; the constructed opposite direction (DCA6.1) has the third row's geometry. Their endpoint maps are exactly (DCA9.3). This establishes the local cover structure on the new cone. The separate literal dual identification is supplied

by ACD1-ACD4: put $b_r = D_r^* JEq : A \rightarrow H$ and $\mathcal{P}_r = [\underline{A}^{-1} \xrightarrow{+b_r, \text{ev}_p} i_* H^0]$. The fine resolution ACD2.2 has continuous compact-test dual chain-isomorphic to $\mathcal{E}_r$ by the degree signs $(-1, +1, -1)$ and the Riesz map $\overline{H} \rightarrow H'$. Its full pairing is ACD3.3, and the map $\mathcal{E}_r \rightarrow D_c P$ is the transpose of the actual original-source map ACD4.3, with the point-coordinate minus in ACD4.4. These explicit comparisons identify the primal object and its pairing; the identification is not inferred from the table of cohomology factors.

DCA10. The actual arithmetic image receives matching actions

The conclusions about possible separation can be checked directly on the original arithmetic map rather than on a substitute example. For the actual value vector $e_\rho \in \overline{H}$, the definition (DCA1.7) gives

$$\sigma_r(\overline{e}_\rho) = m_\rho \overline{d_r(\rho)} \delta_{\rho\#,0}, \quad a(\overline{e}_\rho) = m_\rho \overline{d_r(\rho)} q' \delta_{\rho\#,0}. \quad (\text{DCA10.1})$$

Here $\delta_{\eta,0}(F) = (\Theta_Q F)(\eta)$, so the original Mellin factor $1/2$ is part of this functional. Its multiplicity and the full original defect coefficient have both been retained. The equality follows by putting $D_r e_\rho = d_r(\rho) e_\rho$ into the weighted reflected inner product.

The two exact covariance eigenvalue comparisons on this displayed coordinate are

map	source coefficient on $\bar{e}_\rho$	target coefficient on $q'\delta_{\rho^\#,0}$
$aS_n = Ra$	$n^{1-\bar{\rho}}$	$n^{\rho^\#}$
$aL_n = nR_a$	$n^{\bar{\rho}}$	$n^{1-\rho^\#}$

(DCA10.2)

Each row agrees identically because $\rho^\# = 1 - \bar{\rho}$. The coefficients are equal whether or not $d_r(\rho)$ vanishes. When it vanishes, (DCA10.1) is the zero map at that actual coordinate; when it does not, (DCA10.2) is exact equivariance on its nonzero image line. No existence of an off-critical zero is assumed.

The closed source kernel is

$$\ker a = \overline{\ker D_r} = \{\bar{y} : y_\rho = 0 \text{ whenever } \Re \rho \neq 1/2\}. \quad (\text{DCA10.3})$$

The equality follows from injectivity of j, A_H , and from $r^{\Re \rho} = r^{1-\Re \rho}$ if and only if $\Re \rho = 1/2$. Both actions in (DCA1.13) preserve this exact kernel and the actual image $a(C)$. Their induced quotient maps are continuous for quotient topologies, without any conclusion that the image is closed.

Thus the discrepancy between the constant-current factor nR and the angular/support factor R does not impose a zero attaching map. Those factors are the two receivers (DCA5.1)-(DCA5.2) of the full nearby map $R\Sigma_n$. The supported arithmetic map has precisely the matching eigenvalues in (DCA10.2). To demand an incompatible weight on that same supported map would alter its already proved action. In particular, this cover calculation has not proved $a = 0$ or $D_r = 0$.

DCA11. The exact continuation supplied by the calculation

The resulting operation on the actual Gysin obstruction $\mathfrak{o}_r = \text{Gys}_B a$ is now computed, together with its full nearby source, both vanishing maps, costalk maps, endpoint maps, and the connecting morphisms to the dual original extension. Under support adjunction it obeys $Ra = aS_n$ in the proper-current direction and $nR_a = aL_n$ in the opposite direction. These are identities on the complete source $\bar{H}$, not merely on the displayed finite value coordinates.

The next receiver after the automatically vanishing invariant-cycle image quotient is the actual image $a(C)$, as computed in ADC3. Its source is the exact quotient $C/\ker a$, and DCA10 gives its existing equivariant map into $q'Q' \subset A'$. The concrete cover attempt is complete: its action does not separate this source from this target, because the actions match exactly. The remaining calculation must use additional proved structure of that same image or of its original continuous pairing, while retaining (DCA1.13), (DCA8.3), and all endpoint terms. Replacing the supported target factor by its invariant factor would contradict (DCA5.3), rather than supply such a calculation.

DCA12. Scope and reproducible algebraic checks

Every finite-sheet identity used in the proof can be checked by the following displayed component calculations: $(t-1)\Delta = 0$; $\Sigma_n(t-1) = 0$; the partial-sum inverse in DCA3; $(t-1)(0, z, \dots, (n-1)z) = ne_0z - \Delta z$; $\sum_k t^k = \Delta \Sigma_n$; and the coordinate difference $(n-1)\Delta ax$ between (DCA7.6) and (DCA7.7). They prove the signs, the retained raw-cut correction, and both derived composition laws for every positive integer n , including $n = 1$, when the extra sheet quotient $D_n = B^n/\Delta B$ is zero, the correction vector is zero, and both cover maps are identity. The independently fixed arithmetic parameter remains $r > 1$.

The argument constructs all stated sheaf maps from continuous currents or explicit continuous restriction-cone data. It never infers a unique current chain map from cohomology alone. No conclusion about strong openness of $A'_\beta \rightarrow J'_\beta$, image closure of $a(C)$, finite-dimensionality, arithmetic purity, or vanishing of the actual adjoint defect is inserted. Every original coefficient and endpoint contribution used in the construction remains present in the formulas above.

Exact continuous duality and the global original-source comparison — ACD0-ACD8

Proof source: CC_ADJOINT_DEFECT_CONTINUOUS_DUAL.md.

The adjoint-defect cone: continuous duality and the full original-source comparison

Complete derivation, with proof locators **ACD0-ACD8**. The objects and all maps below use the already reconstructed original arithmetic. The calculation constructs the duality comparison, then follows the same boundary through both poles and the genuine geometric covers.

ACD0. Input stage and conventions

The supporting datum remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. No addition, numerical origin, metric, midpoint, or vector coordinate is assigned to it. Complete-history reconstruction precedes the coefficient spaces below. Winding information and the carried grading remain in the original source; the two branches retain their own recovered counters. The coordinate of a pole disk is a coordinate in the receiving sphere of CSP, whose punctured part lies over the generic point, not a coordinate on this supporting datum.

We use the source and quotient from ADC1 and CSD1, with their proved topologies:

$$S = \{f \in \mathcal{S}(\mathbb{R}) : f(-v) = f(v), f(0) = 0, \int f = 0\}, \quad \Sigma f(u) = 2 \sum_{k \geq 1} f(ku),$$

$$A = \{a \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \text{ for all } N, j\}, \quad J = \Sigma S, \quad Q = A/J. \quad (\text{ACD0.1})$$

The quotient map is q . The subspace J is closed; the inverse of $\Sigma : S \rightarrow J$ is continuous. The original transform and its full source divisor are

$$\Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, \quad F_0(s) = \Theta \Sigma f_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s),$$

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}. \quad (\text{ACD0.2})$$

These factors are not replaced by ζ or by a different completed function. At the exceptional arguments,

$$F_0(0) = F_0(1) = \frac{1}{8}, \quad F_0(-1) = F_0(2) = \frac{\pi}{24}, \quad F_0(-2k) = \frac{k(2k+1)(-1)^k \pi^k}{2^k k!} \zeta'(-2k). \quad (\text{ACD0.3})$$

The full jets at an actual nontrivial zero ρ are those of

$$F_0(\rho + t) = t^{m_\rho} \frac{(\rho + t)(\rho + t - 1)}{8} \pi^{-(\rho+t)/2} \Gamma((\rho + t)/2) \frac{\zeta(\rho + t)}{t^{m_\rho}}. \quad (\text{ACD0.4})$$

In particular the quotient Q retains all multiplicities and higher jets. The unit and prime-power repetitions remain in the original source identities

$$\zeta(s) = 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{k \geq 1} (\log p) p^{-ks} \quad (\Re s > 1). \quad (\text{ACD0.5})$$

Here these are properties of the already recovered arithmetic, not initial labels supplied before its reconstruction.

Let $H = \ell^2(\mathcal{Z}, m)$ for the actual nontrivial zeros, with inner product $\langle x, y \rangle_+ = \sum_\rho m_\rho x_\rho \overline{y_\rho}$, linear in its first variable, and put

$$\rho^\# = 1 - \bar{\rho}, \quad (\mathcal{J}y)_\rho = y_{\rho^\#}, \quad E : Q \rightarrow H, \quad (EF)_\rho = (\Theta_Q F)(\rho). \quad (\text{ACD0.6})$$

The map E is continuous, has dense image containing all finite value vectors, and retains its higher-jet kernel. These are the actual source results RTT/GTAH/ADC1. Fix a recovered coefficient parameter $r > 1$. Define the bounded operators

$$(T_r y)_\rho = r^\rho y_\rho, \quad U_r = rT_{1/r}, \quad D_r = T_r^* - U_r, \quad d_r(\rho) = e^{-i\Im \rho \log r} (r^{\Re \rho} - r^{1-\Re \rho}). \quad (\text{ACD0.7})$$

We write $B = A'_\beta$, $C = \overline{H}$, with the strong continuous dual on B . RGR11's actual map is

$$\sigma_r(\overline{y})(F) = \langle EF, \mathcal{J}D_r y \rangle_+, \quad a_r = q' \sigma_r : C \rightarrow B. \quad (\text{ACD0.8})$$

No off-line zero is assumed to exist. Images below retain their quotient or ambient subspace topology as explicitly specified; closedness is not inferred from the notation for an image.

ACD1. The two continuous transposes

Human-source attribution: this use of the complex Hilbert-space Riesz representation theorem is classical. See Terence Tao, [245B, Notes5: Hilbert spaces](#), Theorem1, Remark2 and Exercise15 (17 January2009). Remark2 identifies the continuous dual with the conjugate Hilbert space; the explicit conjugate-space map and adjoint convention in the present calculation remain displayed below.

The linear isomorphism

$$R : C \longrightarrow H'_\beta, \quad R(\overline{y})(h) = \langle h, y \rangle_+ \quad (\text{ACD1.1})$$

is the Riesz isomorphism with the conjugate complex structure explicitly retained. It is isometric for the dual norm. Similarly $C'_\beta \simeq H$ sends h to $\overline{y} \mapsto \langle h, y \rangle_+$. SDT6 proves the strong topological evaluation isomorphism $A \simeq (A'_\beta)'_\beta$. Thus taking the continuous transpose of (ACD0.8) on the entire bidual gives exactly

$$b_r : A \rightarrow H, \quad b_r = D_r^* \mathcal{J} E q, \quad \beta_r : Q \rightarrow H, \quad \beta_r = D_r^* \mathcal{J} E, \quad b_r = \beta_r q. \quad (\text{ACD1.2})$$

Indeed, for every $f \in A$, $y \in H$,

$$a_r(\overline{y})(f) = \langle E q f, \mathcal{J} D_r y \rangle_+ = \langle D_r^* \mathcal{J} E q f, y \rangle_+. \quad (\text{ACD1.3})$$

This proves both $a'_r \text{ev}_A = b_r$ under $C' = H$ and

$$b'_r R = a_r. \quad (\text{ACD1.4})$$

No section of q is used. For a bounded subset $K \subset Q$,

$$\sup_{F \in K} |\sigma_r(\overline{y})(F)| \leq \sup_{F \in K} \|EF\|_+ \|D_r\| \|y\|_+. \quad (\text{ACD1.5})$$

This proves the needed strong continuity; continuity of b_r follows from its three continuous factors.

Write

$$Z_r = \ker \beta_r = \{F \in Q : F(\rho) = 0 \text{ at every actual off-critical zero}\}, \quad C_{\text{crit}} = \overline{\ker D_r}. \quad (\text{ACD1.6})$$

The equality uses (ACD0.7), reflection of the zero set, and the coordinate formula for β_r . All higher-jet classes with zero values belong to Z_r . The exact relations are

$$\ker a_r = C_{\text{crit}}, \quad \ker b_r = q^{-1}Z_r, \quad b_r A = \beta_r Q, \quad b_r = 0 \iff D_r = 0. \quad (\text{ACD1.7})$$

Injectivity of the anti-linear map $y \mapsto \langle E(-), Jy \rangle$ follows from density of $E(Q)$, proving the first equality. For the last equivalence, vanishing of b_r makes the bounded operator $D_r^* J$ zero on that same dense image, and therefore zero on H . These arguments do not discard its kernel or any source jet.

ACD2. A fine resolution of the actual transpose cone

On a positively oriented pole disk D , let $i : \{p\} \hookrightarrow D$. Form the specified sheaf complex

$$\mathcal{P}_r = [\underline{A}^{-1} \xrightarrow{+b_r \text{ev}_p} i_* H^0]. \quad (\text{ACD2.1})$$

This is the cone presentation of ADC8; the plus sign is part of this representative. Its fine resolution has

$$\begin{aligned} \mathcal{L}_r^{-1} &= \mathcal{E}_A^0, \quad \mathcal{L}_r^0 = \mathcal{E}_A^1 \oplus i_* H, \quad \mathcal{L}_r^1 = \mathcal{E}_A^2, \\ d^{-1}f &= (-df, b_r f(p)), \quad d^0(\omega, h) = -d\omega. \end{aligned} \quad (\text{ACD2.2})$$

The map from (ACD2.1) inserts locally constant functions and the unchanged H coordinate. The coefficient-valued Poincaré contraction proves it is a quasi-isomorphism, including at the pole: the de Rham resolution of $\underline{A}$ and the identity on the skyscraper give the map of the two cones. It neither requires b_r to be onto nor its image to be closed. Smooth partitions of unity act on the forms and by their value at p on the skyscraper. CSD1-2's cutoff and contraction proof therefore establishes compact-support acyclicity and continuity for these exact terms too.

Define the literal continuous compact-test Hom complex

$$\mathcal{D}_r^k(U) = \text{Hom}_{\text{cont}}(\Gamma_c(U, \mathcal{L}_r^{-k}), \mathbb{C}), \quad d_{\mathcal{D}}^k \ell = (-1)^{k+1} \ell d_{\mathcal{L}}. \quad (\text{ACD2.3})$$

Tests have the fixed-support smooth seminorms and their locally convex inductive limit. Currents restrict by extension-by-zero transpose. A locally finite partition of unity proves the sheaf property exactly as in CSD3. This constructs a sheaf complex; it does not declare continuous dualization exact on all locally convex complexes.

Put $\mathcal{T}_A^j = (\mathcal{E}_c^{-j}(A))'$, $d_T v(\phi) = (-1)^{j+1} v(d\phi)$, and $\delta_p \lambda(f) = \lambda(f(p))$. In degrees $(-1, 0, 1)$, (ACD2.3) is

$$\begin{aligned} \mathcal{T}_A^{-2}, \quad \mathcal{T}_A^{-1} \oplus i_* H', \quad \mathcal{T}_A^0, \\ d_{\mathcal{D}}^{-1} v = (d_T v, 0), \quad d_{\mathcal{D}}^0(u, \lambda) = d_T u - \delta_p b'_r \lambda. \end{aligned} \quad (\text{ACD2.4})$$

For example, on a degree-minus-one test f , the degree-zero Hom differential is $-u(-df) - \lambda(b_r f(p)) = u(df) - \lambda(b_r f(p))$, which is the second formula. The first follows by applying a degree-minus-one functional to $-d\omega$. These two computations fix every sign.

ACD3. The exact current-cone identification and pairing

The previously constructed receiver is

$$\mathcal{E}_r = \text{Cone}(-\delta_p a_r)[-1],$$

$$d_{\mathcal{E}}^{-1}v = (-d_T v, 0), \quad d_{\mathcal{E}}^0(u, x) = -d_T u + \delta_p a_r x. \quad (\text{ACD3.1})$$

Using (ACD1.1), define the degreewise map $\mathcal{D}_r \rightarrow \mathcal{E}_r$ by

$$v \mapsto -v \quad (\deg -1), \quad (u, Rx) \mapsto (u, x) \quad (\deg 0), \quad w \mapsto -w \quad (\deg 1). \quad (\text{ACD3.2})$$

It is a continuous chain isomorphism. In degree minus one its two differential composites are $(d_T v, 0)$; in degree zero both are $-d_T u + \delta_p a_r x$, by (ACD1.4). The inverse uses the same signs and R . Thus this is the exact continuous support-dual comparison for the constructed resolution, not a general assertion about dualizing an unseparated image quotient.

The evaluation pairing of $\mathcal{E}_r$ against compact tests of $\mathcal{L}_r$ is accordingly

degree of current	test	pairing
-1	$\eta \in \mathcal{E}_c^2(A)$	$-v(\eta)$
0	$(\omega, h) \in \mathcal{E}_c^1(A) \oplus H$	$u(\omega) + \langle h, y \rangle_+, \quad x = \bar{y}$
1	$f \in \mathcal{E}_c^0(A)$	$-w(f).$

(ACD3.3)

For a current of degree k , substitution into (ACD2.2) and (ACD3.1) gives $\langle dv, c \rangle + (-1)^k \langle v, dc \rangle = 0$. For degree zero, the two terms containing $b_r f(p)$ cancel precisely because of (ACD1.3). Hence (ACD3.3) descends to the supported cohomology pairings.

The positive angular current is $h_p \lambda$, with $d_T h_p \lambda = \delta_p \lambda$. The stalk representatives of $\mathcal{E}_r$ are the constant current c_λ in degree -1 and $(h_p a_r x, x)$ in degree zero. Their restriction is $(1_B, a_r)$. These are exactly RGR/ADC's representatives: no sign of the point coordinate was changed in (ACD3.2). A compact positive two-form ω_c with integral one represents the compact A class in degree one of $\mathcal{L}_r$; a point vector h represents its degree-zero H class. Pairing gives $-\lambda(a)$ and $\langle h, y \rangle_+$, respectively. The first minus is retained rather than hidden in an orientation change. CSD2's compact contraction proves these classes compute the entire compact model $H[0] \oplus A[-1]$.

The full endpoint term in the original source is $i_* E_p[1]$, with $E_p = \mathbb{C}^2$ carrying the separate value-at-zero and integral labels. It pairs with $i_* E'_p[-1]$ by evaluation. This summand has no nearby group; its special and supported degrees are -1 in the primal and $+1$ in the dual. Both labels and both poles are retained.

ACD4. Ordinary algebraic duality and the original source map

For the unrestricted algebraic dual, replace $\mathcal{T}_A$ by

$$\mathcal{T}_{A, \text{alg}}^j(U) = \text{Hom}_{\mathbb{C}}(\mathcal{E}_c^{-j}(U; A), \mathbb{C}) \quad (\text{ACD4.1})$$

and H' by H^* . The same Hom differential and degree signs prove the analogous cone formula with attaching map $b_r^* : H^* \rightarrow A^*$. CV2-CV4's compact contraction gives $\mathcal{T}_{A, \text{alg}} \simeq \underline{A}^*[2]$ and identifies its positive Dirac map with Gysin. Together with the exact skyscraper comparison this is the ordinary sheaf Verdier dual of (ACD2.1), through these explicit resolutions. No finite-dimensional bidual identity is used.

There is a literal chain injection from (ACD2.3) to this algebraic Hom model: continuous test functionals are algebraic test functionals, $H' \hookrightarrow H^*$, and $b_r^*|_{H'} = b'_r$. In cone coordinates it is

$$(v, (u, x), w) \mapsto (v, (u, Rx), w). \quad (\text{ACD4.2})$$

Its target has the same degree-sign convention (ACD3.2). The inclusion need not be an isomorphism. This supplies the comparison instead of conflating the two dual categories.

The original reduced local source is $\mathbf{P} = \mathcal{F}_{\text{red}}[1] \simeq [\underline{A}^{-1} \xrightarrow{+q} i_* Q^0]$. Its actual map into the new cone is

$$f_r : \mathbf{P} \rightarrow \mathcal{P}_r, \quad f_r^{-1} = 1_A, \quad f_r^0 = \beta_r. \quad (\text{ACD4.3})$$

It commutes with the differential by $\beta_r q = b_r$. Add the identity on each original $E_p[1]$ for the full source. On fine resolutions this is 1 on forms and β_r on the point coefficient. Its continuous transpose is therefore 1 on currents and $\beta'_r R = \sigma_r$ on the point functional.

There is a necessary representative sign here. The accepted RGR model of $D_c \mathbf{P}$ is $\text{Cone}(+\delta_p q')[-1]$, whereas the literal Hom dual of the plus- q cone just used, after (ACD3.2), is $\text{Cone}(-\delta_p q')[-1]$. Negating its point coordinate and leaving its currents unchanged is a chain isomorphism between these representatives. Hence in the accepted RGR representative the transpose of (ACD4.3) is exactly

$$\mathcal{E}_r \longrightarrow D_c \mathbf{P}, \quad (u, x) \longmapsto (u, -\sigma_r x) \quad \text{in degree zero}, \quad (\text{ACD4.4})$$

with identity on the other current degrees. Substitution gives $-d_T u - \delta_p q'(-\sigma_r x) = -d_T u + \delta_p a_r x$, proving the differential square. Thus this is the previously constructed RGR map, with its actual source and signs, now proved to be the continuous transpose of the original-source map. A sign-free formal slogan about the dual of a cone would not supply this comparison.

ACD5. The specialization image and its full local pairing

The local primal stalk is $[A^{-1} \xrightarrow{b_r} H^0]$, its punctured model is $A[1] \oplus A[\vartheta_p]$, and restriction is identity on the first A and zero on H . The restriction fibre is

$$A \xrightarrow{a \mapsto (b_r a, a)} H \oplus A \xrightarrow{0} A \quad \text{in degrees } -1, 0, 1. \quad (\text{ACD5.1})$$

The quotient coordinate $(h, a) \mapsto h - b_r a$ cancels the first identity pair. Therefore the two positive-orientation localization boundaries are

$$\partial_{-1} = -b_r : A \rightarrow H, \quad \partial_0 = +1 : A[\vartheta_p] \rightarrow A. \quad (\text{ACD5.2})$$

In degree -1 the angular coinvariant term of DC's cross is zero. Its exact obstruction image is consequently $b_r A$, and

$$\text{coker } \text{sp}_{-1} = A / \ker b_r \simeq Q / Z_r, \quad [a] \longmapsto -b_r a \quad (\text{ACD5.3})$$

is the actual boundary comparison to the image. The first isomorphism is a topological quotient isomorphism, induced by the quotient map q . Its bijection to $b_r A$ is continuous with the ambient Hilbert subspace topology, but inverse continuity is not asserted. In degree zero the entire punctured term is coinvariant, so the quotient of boundary images is zero. The endpoint degrees are those of ACD3 and the full table ADC8.

The original boundary is $-q : A \rightarrow Q$. The map (ACD4.3) gives its exact commuting square with $-b_r$ by β_r . Thus the map on original local obstruction images is

$$Q \xrightarrow{\beta_r} \beta_r Q, \quad \ker \beta_r = Z_r. \quad (\text{ACD5.4})$$

The complementary current-side source is paired with it by

$$(Q / Z_r) \times (C / C_{\text{crit}}) \rightarrow \mathbb{C}, \quad ([F], [\bar{y}]) \mapsto \langle \beta_r F, y \rangle_+ = \sigma_r(\bar{y})(F). \quad (\text{ACD5.5})$$

Both radicals vanish: the left by the definition of Z_r , the right by injectivity of the map from H to value functionals and (ACD1.7). This is the exact pairing, not a claim that these images exhaust each other's topological duals. The current-side invariant-cycle quotient is zero for every a_r , while its coinvariant-boundary kernel is $a_r C$ (ADC3). Equations (ACD3.2), (ACD4.4) and (ACD5.5) identify the precise duality relation between these different tests.

ACD6. Both poles and the global original-source map

On $Y = \mathbb{P}^1(\mathbb{C})$, use the existing common coefficient trivialization of CSD, with restriction maps $r_+ = \Sigma$ and $r_- = R\Sigma$ into A . Their common image is J , so after the proved source inverses the reduced source is the kernel of $\underline{A}_Y \rightarrow i_+ Q \oplus i_- Q$, with q at both poles. Define the new global cone by the same actual quotient map followed by β_r at each pole:

$$\mathcal{P}_r^Y = [\underline{A}_Y^{-1} \xrightarrow{(b_r, b_r)} i_+ H \oplus i_- H], \quad \mathcal{E}_r^Y = \text{Cone}(-\delta_+ a_r - \delta_- a_r)[-1]. \quad (\text{ACD6.1})$$

The local signs (ACD2-3) glue, giving the same continuous dual comparison globally. Dirac currents at both poles are positive for the complex orientation; the expression in (ACD6.1) is a sum, not an unrecorded cancellation. Add $(i_+ E_+ \oplus i_- E_-)[1]$ in the primal and its full continuous dual shifted by -1 .

Here is a full global cochain calculation. Cover the sphere by the two pole disks with annular overlap. The disk complexes and overlap contraction from ACD5 give the Mayer-Vietoris model

$$A \oplus A \xrightarrow{d^{-1}} H \oplus H \oplus A \xrightarrow{0} A, \quad d^{-1}(x_+, x_-) = (b_r x_+, b_r x_-, x_+ - x_-) \quad (\text{ACD6.2})$$

in degrees $(-1, 0, 1)$. The last A is the positively oriented angular overlap class. It is the sphere's degree-two constant-coefficient class after the shift. Its orientation comparison includes the shift sign: take a partition function χ_+ equal to one near zero and zero near infinity, and $\vartheta = d \arg z / (2\pi)$ on the overlap. In the resolution with differential $-d$, the overlap class ϑa maps to the global two-form $-d\chi_+ \wedge \vartheta a$. Its integral is $-\int_0^\infty \chi'_+(r) dr a = a$. The corresponding unshifted $+d$ comparison is the negative sign recorded in CDW9. Thus no positive angular coordinate has been silently identified with the unshifted positive integration coordinate. The projection to

$$K_r^Y = [A^{-1} \xrightarrow{(b_r, b_r)} H^0 \oplus H^0 \xrightarrow{0} A^1] \quad (\text{ACD6.3})$$

is $p^{-1}(x_+, x_-) = x_-$, $p^0(h_+, h_-, c) = (h_+ - b_r c, h_-)$, and $p^1 = 1$. Its section is $a \mapsto (a, a)$ and $(h_+, h_-) \mapsto (h_+, h_-, 0)$. The homotopy $h^0(h_+, h_-, c) = (c, 0)$, zero in other degrees, satisfies $dh + hd = 1 - ip$. Every identity is verified by inserting (ACD6.2). Thus (ACD6.3) is a continuous cochain equivalence, not only a dimension count.

Its groups, with all endpoint coordinates reattached, are

$$\begin{aligned} \mathbb{H}^{-1}(Y, \mathcal{P}_{r, \text{full}}^Y) &= \ker b_r \oplus E_+ \oplus E_-, \\ \mathbb{H}^0(Y, \mathcal{P}_{r, \text{full}}^Y) &\simeq H \oplus (H/b_r A), \quad \mathbb{H}^1(Y, \mathcal{P}_{r, \text{full}}^Y) = A. \end{aligned} \quad (\text{ACD6.4})$$

The degree-zero isomorphism is $[(h_+, h_-)] \mapsto (h_+ - h_-, [h_-])$, with inverse $(h, [k]) \mapsto [(h + k, k)]$. Changing k by $b_r a$ changes the pair by $(b_r a, b_r a)$. Both maps are continuous for the stated quotient topologies, including when the second factor is not separated.

Replace H, b_r by Q, q to compute the original source. Since q is onto, its degree-zero group is exactly Q , by difference of the two quotient coordinates; its other groups are $J \oplus E_+ \oplus E_-$ and A . The map from the original global source into (ACD6.4) is explicitly

$$J \oplus E_+ \oplus E_- \hookrightarrow \ker b_r \oplus E_+ \oplus E_-, \quad Q \rightarrow H \oplus (H/b_r A), \quad F \mapsto (\beta_r F, 0), \quad A \xrightarrow{1} A. \quad (\text{ACD6.5})$$

The zero second coordinate follows because $\beta_r Q = b_r A$, not because the second quotient is discarded. In degree zero the exact kernel is Z_r , just as for the local specialization image.

For the current dual, the corresponding global finite cochain model is

$$B^{-1} \xrightarrow{0} C^0 \oplus C^0 \xrightarrow{(x_+, x_-) \mapsto a_r x_+ + a_r x_-} B^1, \quad (\text{ACD6.6})$$

with endpoint $E'_+ \oplus E'_-$ in degree one. It follows either by the same global current contraction or by applying (ACD3.2) to the continuous Hom of the explicit finite cochain model (ACD6.3). All homotopies used to get that model are continuous and can be transposed. Its groups are

$$\mathbb{H}^{-1} = B, \quad \mathbb{H}^0 = \{(x_+, x_-) : a_r(x_+ + x_-) = 0\}, \quad \mathbb{H}^1 = (B/a_r C) \oplus E'_+ \oplus E'_-. \quad (\text{ACD6.7})$$

The middle kernel is continuously isomorphic to $C \oplus C_{\text{crit}}$ by $(x_+, x_-) \mapsto (x_+, x_+ + x_-)$; its inverse is $(x, k) \mapsto (x, k - x)$. In particular the anti-diagonal classes are always present. They express cancellation between the two retained pole contributions; they do not imply that either map a_r vanishes. The original global image (ACD6.5) and the local obstruction (ACD5.4) retain precisely the same defect despite this global cancellation.

ACD7. Genuine covers and their transpose actions

For a recovered integer $n \geq 1$, let $f_n(z) = z^n$. On constant coefficients the sheaf unit is the diagonal into the n branches away from a pole. The sheaf trace is branch sum there and multiplication by n on the single constant pole stalk. The latter follows directly from restriction: a constant pole section restricts to the same value on all branches. These formulas define sheaf maps on disks, and agree on overlaps. Direct image for this finite proper map is exact: its stalks are the finite sums over the fibre, and a disk about a branch value has one connected inverse disk. There is no replacement of all nearby branches by their invariant diagonal.

The actual identities proved in ADC8 are

$$b_r T_n^A = U_n^* b_r, \quad b_r U_n^A = T_n^* b_r. \quad (\text{ACD7.1})$$

They also follow immediately from (ACD1.2), $ET_n^Q = T_n E$, $JT_n = U_n^* J$, and commutation of diagonal operators. Consequently the literal cones have two sheaf maps

$$U_n : \mathcal{P}_r \rightarrow f_{n*} \mathcal{P}_r, \quad V_n : f_{n*} \mathcal{P}_r \rightarrow \mathcal{P}_r. \quad (\text{ACD7.2})$$

Their complete nearby maps, constant pole maps, and point H maps are

	nearby	constant pole A	point H
U_n	$a \mapsto (T_n a, \dots, T_n a)$	T_n	U_n^*
V_n	$(a_j) \mapsto T_{1/n} \sum_j a_j$	$n T_{1/n} = U_n$	T_n^*

(ACD7.3)

The two differential squares are exactly (ACD7.1). Thus these maps are constructed before dualization and before a weight claim. The trace here is a sheaf trace on locally constant branches; no smooth pullback of a ramified arbitrary distribution is presumed.

The full vanishing coefficient can also be computed, without replacing the cover by its invariant part. Put $\Delta a = (a, \dots, a)$, $\Sigma v = \sum_j v_j$, and let P_n be the cyclic deck permutation of A^n . The shifted specialization cone has injective first differential $a \mapsto (-\Delta a, b_r a)$. Its sole remaining vanishing coefficient is therefore

$$V_n = (A^n \oplus H) / \{(-\Delta a, b_r a) : a \in A\}. \quad (\text{ACD7.3a})$$

This quotient is separated and has the explicit continuous decomposition

$$[(v, h)] \mapsto (v - \Delta m(v), h + b_r m(v)) \in \ker \Sigma \oplus H, \quad m(v) = \Sigma v / n. \quad (\text{ACD7.3b})$$

The inverse sends (w, k) to $[(w, k)]$. Replacing (v, h) by $(v - \Delta a, h + b_r a)$ leaves both displayed coordinates unchanged. Conversely their difference from (v, h) is the defining relation with $a = m(v)$. This proves the isomorphism and closedness of the relation subspace, without requiring $b_r A$ closed.

With the same $1 - P_n$ variation convention as the positive-support cut model, the full arrows are

$$\text{can}(v) = [(v, 0)], \quad \text{var}[(v, h)] = (1 - P_n)v. \quad (\text{ACD7.3c})$$

Variation is well-defined since $(1 - P_n)\Delta = 0$; the vanishing monodromy is $[(v, h)] \mapsto [(P_n v, h)]$. The two composites of canonical and variation maps are the corresponding $1 - T$ operators. For $n = 1$, (ACD7.3b) identifies V_1 with H , and canonical becomes b_r , variation zero, as required. To identify the actual variation rather than only its composites, the deck-cut calculation on $[(v, 0)]$ is $(1 - P_n)v$. Naturality for the supported inclusion $i_* H \rightarrow f_{n*} \mathcal{P}_r$ makes variation zero on $[(0, h)]$, because that source has zero nearby cycles. These two types span V_n , proving exactly (ACD7.3c).

The two maps (ACD7.2) induce respectively

$$\begin{aligned} H &\longrightarrow V_n, & h &\longmapsto [(0, U_n^* h)], \\ V_n &\longrightarrow H, & [(v, h)] &\longmapsto b_r T_{1/n} \Sigma v + T_n^* h. \end{aligned} \quad (\text{ACD7.3d})$$

For the second map a defining relation changes its value by $-b_r U_n a + T_n^* b_r a = 0$. For the first canonical square, $[(\Delta T_n a, 0)] = [(0, b_r T_n a)] = [(0, U_n^* b_r a)]$. For the second square apply its displayed formula to $[(v, 0)]$. Both variation squares follow from $(1 - P_n)\Delta = 0$ and $\Sigma(1 - P_n) = 0$. Thus all augmentation coordinates, canonical maps and variation maps have been constructed, not only their invariant or coinvariant reductions.

The composite $V_n U_n$ is n times identity: on A , $U_n T_n = n$; on H , $T_n^* U_n^* = (U_n T_n)^* = n$. The reverse composite on the full nearby object is $\Delta \Sigma$, the deck norm. It is not n times identity on the augmentation part. On the pole H term it is n times identity. These are the complete constant/skyscraper sheaf maps, so the equalities hold as cone maps. A raw-current cut presentation can have the additional homotopies calculated in DCA without changing these derived maps.

Under the pairing (ACD3.3), the point transpose of U_n^* is $\bar{U}_n$, since $\langle U_n^* h, y \rangle_+ = \langle h, U_n y \rangle_+$. Similarly the transpose of T_n^* is $\bar{T}_n$. On the open coefficient, unit transposes to sum with T'_n and trace transposes to diagonal with $T'_{1/n}$. Thus the resulting current maps are precisely the DCA directions

$$\begin{aligned} f_{n*} \mathcal{E}_r &\rightarrow \mathcal{E}_r : & \Psi &= T'_n \Sigma, & i^* &= (n T'_n, \bar{U}_n), & i^! &= (\bar{U}_n, T'_n), \\ \mathcal{E}_r &\rightarrow f_{n*} \mathcal{E}_r : & \Psi &= \Delta T'_{1/n}, & i^* &= (T'_{1/n}, \bar{T}_n), & i^! &= (\bar{T}_n, n T'_{1/n}). \end{aligned} \quad (\text{ACD7.4})$$

The angular factor in the second line is n , because the positive circle has covering degree n . The first line's pole-constant factor n comes from summing the n nearby branches; its Dirac factor is one before T'_n . DCA gives the explicit raw cut homotopy realizing these arrows. All factors are retained in (ACD7.4).

On the original endpoint labels, ordered as zero-pole value, zero-pole integral, infinity-pole value, infinity-pole integral, the primal unit actions are $(1, n, n, 1)$ and the primal weighted trace actions are $(n, 1, 1, n)$, as in GTR. Their continuous transposes give the same scalar lists on the dual endpoint coordinates. The endpoint map from the original source in (ACD4.3) is identity and respects both lists.

ACD8. The actual weight comparison and its scope

For the unit direction in (ACD7.3), the local degree-minus-one invariant-cycle source is A with T_n ; its supported target is H with U_n^* . On a value class in Q/Z_r concentrated at an actual off-critical ρ , its image under β_r is at $\rho^\#$. The two eigenvalues are

$$n^\rho \quad \text{and} \quad n^{1-\overline{\rho^\#}} = n^\rho. \quad (\text{ACD8.1})$$

For weighted trace they are $n^{1-\rho}$ and $\overline{n^{\rho^\#}} = n^{1-\rho}$. The eigenvalue statements apply to the indicated quotient value class. A representative in full Q may also carry higher-jet terms, retained in the kernel Z_r . The entire continuous intertwining identities are (ACD7.1), not a conclusion drawn only from these single lines.

For the global cone, pullback acts on the degree-one A in (ACD6.3) by nT_n , from the oriented sphere degree. Weighted trace acts there by $T_{1/n}$; their composite is n as required. On the degree-minus-one A the actions are T_n and U_n . On $H \oplus H$ in cochain degree zero they are U_n^* and T_n^* on each copy. The diagonal boundary commutes with these actions by (ACD7.1). Thus the global degree factor has not been suppressed; it lies on the other cohomological component and does not alter (ACD8.1).

Deligne's §3.6 argument, reconstructed with its exact hypotheses in DC, separates special-source weights from the supported target by a genuine dual geometric estimate. Here the constructed continuous dual, the source map, both actual cover directions, and the full two-pole cohomology have been computed. They give the same eigenvalue on every nonzero image of the particular specialization boundary $-b_r$. The normal multiplier separator of GMS is identity on the original quotient Q , hence identity on its image $\beta_r Q$; its annihilation of the different translated normal extension does not annihilate this image.

The calculation therefore identifies what the current geometry actually supplies: (ACD5.4) is the original-source obstruction quotient, (ACD3.2) its continuous duality comparison, (ACD6.5) its global image, and (ACD7.3) its genuine geometric actions. No vanishing of D_r , no pure Frobenius realization, and no RH resolution is asserted. This is a completed attempt to derive the missing weight distinction from the accumulated duality and cover data, rather than assigning a new twist or replacing the original arithmetic.

Source provenance is inherited with its exact reading scope from CSP/CSD and DC: Connes-Consani, [arXiv:0903.2024v3](#), §5, and [arXiv:2609.00299v1](#), signed charts; Deligne, *La conjecture de Weil II*, §3.6, read here through the separately identified French transcription, not claimed to be author TeX. The definitions of nearby/vanishing cycles are compared with Reich's retained [author source](#), [arXiv:1002.1686v4](#); no finite-rank theorem is imported for these infinite coefficients. Complete local proofs used here are retained in this repository as SDT, CSD, CV, LNC, LVD, RGR, ADC, DCA, GTAH and RTT, with their individual proof locators and acceptance scopes.

Complete timing quotient and its exact scope — GTQ0-GTQ12

Proof source: GLOBAL_TIMING_QUOTIENT_INDEPENDENT.md.

Complete timing, the common quotient, and retrospective integer meaning

Independent mathematical derivation, 24 September 2026. Result labels GTQ0-GTQ12. Private working note; no publication is asserted.

GTQ0. Source, order, and exact scope

The governing user passages are WU050-WU053 in USER_INPUT_SOURCE_RECORDS.json. I read READ_FIRST_USER_CONSTRUCTION.md and USER_ARGUMENT_RECONSTRUCTION.md before this calculation. The essential order is:

“And after you’ve timed every single absorption line, can you find the quotient that is common to all of them? And that’s when you know what one is.” — WU051.

“you count by counting how many times the quotient has been, you know, either, like how many times the quotient has corresponded to either an absorption or not an absorption. That’s how you get the integers.” — WU050.

This note does not start with observed events named one, two, or three. It studies the mathematical content of a complete, uncalibrated timing record and calculates its primitive common tick, the coordinate system it determines, and the obstruction to such a tick. It then proves an impossibility of recovering those final coordinates from an arbitrary finite prefix. That impossibility concerns the construction map itself, not merely an observer’s ignorance of a number system already assigned to the source.

The timing receiver used below is explicitly an analysis object: an ordered real line of durations with no chosen positive unit, a set of events, and their durations from a marked start. This is not attributed to the user as a definition of the parityless source. No vector, distance, coordinate, addition, subtraction, or count is assigned to

$$\tau\langle Z_1; \text{no } Z_2 \rangle.$$

In particular, the marked start of a time record is not identified with this supporting datum. Addition at tau remains retracted. A receiver with durations adds timed information; this note does not claim that the property of having no parity by itself constructs that information.

The metatheory can discuss finite lists, words, repetition, and integers when proving a representation theorem. These proof indices are not the number meanings purportedly recovered inside the source. Every time an arithmetic coordinate is assigned to an event below, the positive unit used for that coordinate is derived first. The duration itself remains in every comparison; it is not rescaled away.

Reading ledger for this derivation

The existing CANONICAL_SOURCE_ROUTING_PRIVATE.json and INTEGRALITY_SOURCE_READING_PRIVATE.json were read first. They route the CC papers and record the existing author-source coverage. I then read CANONICAL_GENERIC_RECONSTRUCTION.md, CG0, and TIMED_PRIME_PERIOD_DERIVATION.md, TP0-TP2 and the opening of TP3. These are programme derivations, not original-author literature. Their complete cyclic source is explicitly an input in those notes. No new theorem from an unread paper is imported here. The timing and group assertions below have their proofs in this file. GTQ11 proves its displayed period comparison directly rather than appealing to the source-reading receipt as a proof.

GTQ1. The uncalibrated record and its universal duration group

Let V be an oriented one-dimensional real vector space. Its orientation supplies a positive ray $V_{>0}$; no vector in that ray is selected as a unit. Let E be the complete set of **absorption-event identities**, and

$$t : E \longrightarrow V_{>0}$$

their durations from a distinguished start of the record. Several absorption-event identities may have the same duration. Thus absorption multiplicities and channel labels are retained rather than replaced by their support. Throughout the note, “event” in E means an absorption, not a general recorded clock observation. A discrete, sequential record will additionally have finitely many absorption events below every positive duration and unbounded event times, but GTQ1-GTQ5 do not need those additional properties.

Any separately recorded non-absorption observations are retained as a different set N with a duration map $u:N \rightarrow V_{>0}$, their own labels, and $u(N)$ disjoint from $t(E)$. They are never inserted into E and declared absorptions merely because they were recorded. The primary common-quotient calculation below uses all absorption timings $t(E)$, as WU051 requests. GTQ3 also gives the exact comparison with additional recorded non-absorptions.

Let $F(E)$ be the free abelian group on symbols $[e]$, e in E . Its elements are finite signed words modulo reordering and cancellation of opposite copies. Form the homomorphism

$$T : F(E) \longrightarrow V, \quad T\left(\sum_{e \in E} n_e [e]\right) = \sum_{e \in E} n_e t(e),$$

where only finitely many proof coefficients n_e are nonzero. Define

$$K_t = \ker T, \quad D_t = F(E)/K_t, \quad G_t = T(F(E)) \subset V.$$

The exact comparison is

$$\overline{T} : D_t \xrightarrow{\sim} G_t, \quad [a] \mapsto T(a). \quad (\text{GTQ1.1})$$

It is well defined because equal quotient classes differ by an element of K_t . It is injective because $T(a)=0$ means a is in K_t , and surjective by the definition of G_t . No infinite sum of durations has been performed: the complete record supplies all generators and all actual relations among finite signed combinations.

This is also a universal construction with its precise domain. Given an abelian group H and an assignment $u:E \rightarrow H$ for which every timing relation in K_t is respected, the homomorphism $F(E) \rightarrow H$ supplied by finite signed addition kills K_t and factors uniquely through D_t . Conversely, a homomorphism $D_t \rightarrow H$ necessarily respects those relations. Uniqueness follows because the event classes generate D_t .

Thus the complete timed record determines a group, not yet a primitive unit. Retaining K_t records which concatenated durations are actually equal. A list of event names or their order alone does not supply these duration relations.

GTQ2. Calculate all common ticks and the primitive one

A positive duration d is called an admissible common tick for this receiver when every event time is a finite positive repetition of d :

$$\mathcal{C}_t = \{d \in V_{>0} : t(e) \in d\mathbb{Z}_{>0} \text{ for every } e \in E\}. \quad (\text{GTQ2.1})$$

Here $d\mathbb{Z}$ means the subgroup of V generated by the duration d . This definition tests a candidate duration; it does not designate any event as the integer one. Throughout this section E is nonempty.

Classification. The following statements are equivalent:

1. At least one admissible common tick exists.
2. G_t is a nonzero discrete subgroup of V .
3. G_t has a least positive element δ .
4. G_t is generated by a unique positive element δ .

When they hold, the entire set of admissible ticks is

$$\boxed{\mathcal{C}_t = \{\delta/k : k \in \mathbb{Z}_{>0}\}}. \quad (\text{GTQ2.2})$$

Consequently the primitive, greatest admissible tick is unique and equals δ . Merely requiring a common divisor of the timings does not make it unique: the proper refinements δ/k are also common divisors.

Proof. Suppose d is admissible. Then every event belongs to $d\mathbb{Z}$, so $G_t \subset d\mathbb{Z}$. Under the injective map $n \mapsto nd$, a nonzero subgroup of this cyclic group corresponds to a nonzero subgroup A of the metatheoretic integers. Choose its least positive integer k . For n in A , write $n = qk + r$ with $0 \leq r < k$ by integer division. Then r is in A ; minimality forces $r = 0$. Therefore $A = k\mathbb{Z}$ and $G_t = (kd)\mathbb{Z}$. This has least positive element kd and is discrete.

Conversely, suppose G_t is nonzero and discrete. Choose temporarily any positive vector v to express the order topology. This temporary coordinate does not calibrate the final tick. Discreteness at the zero duration gives a positive lower bound on positive members of G_t . Let $a > 0$ be the infimum of their real coordinates. Choose a positive member g with coordinate in $[a, 3a/2)$. Any other positive member h with coordinate in $[a, 3a/2)$ must equal g : otherwise $|g - h|$ is a positive member of G_t with coordinate less than a . If g had coordinate greater than a , the definition of the infimum would supply such an h with smaller coordinate. Thus g has coordinate a and is least positive. Call its coordinate-independent duration δ .

For arbitrary g in G_t , choose the unique integer q for which $0 \leq g - q\delta < \delta$. The remainder is in G_t . Minimality of δ makes it zero. Hence $G_t = \delta\mathbb{Z}$. Because all event times are positive members, δ is admissible. A positive generator must be the least positive element, so it is unique.

Finally, if d is admissible, $G_t \subset d\mathbb{Z}$, so $\delta = kd$ for a positive integer k . Thus $d = \delta/k$. Conversely every δ/k divides every $t(e) = n_e \delta$, because $t(e) = (kn_e)(\delta/k)$. This proves the full formula. ■

The empty record is genuinely different. Its generated group is zero, every positive d vacuously divides all its event times, and there is no greatest common tick. The theorem does not manufacture a unit from absence.

GTQ3. Exact retrospective counts, including non-absorption

In the cyclic case, first obtain δ by GTQ2. Only then define

$$\nu_\delta : G_t \xrightarrow{\sim} \mathbb{Z}, \quad \nu_\delta(g) = \text{the unique } n \text{ for which } g = n\delta. \quad (\text{GTQ3.1})$$

Existence and uniqueness are GTQ2. Its inverse is $n \mapsto n\delta$. It preserves addition because

$$n\delta + m\delta = (n + m)\delta,$$

and preserves the order because δ is positive. The event-count assignment is the composite

$$E \xrightarrow{t} G_t \xrightarrow{\nu_\delta} \mathbb{Z}_{>0}. \quad (\text{GTQ3.2})$$

This assigns integer meaning after the group and its generator have been obtained from the whole record. It uses no pair of events carrying preassigned prime names.

Retain each event fibre

$$E_n = \{e \in E : t(e) = n\delta\}, \quad n \in \mathbb{Z}_{\geq 0}. \quad (\text{GTQ3.3})$$

The complete clock has every position $n\delta$; absorption occurs at exactly the positions with E_n nonempty, because E is the complete absorption-event set. Set $b(n)=1$ for a nonempty fibre and $b(n)=0$ for an empty fibre. The positions with $b(n)=0$ are not deleted. If the record carries multiplicities or channel labels, these remain in E_n and must not be replaced by $b(n)$ alone.

For separately recorded non-absorptions, retain $N_n = \{a \text{ in } N : u(a)=n\delta\}$, with the original labels. Every nonempty N_n lies at a position with $b(n)=0$, by disjointness of $u(N)$ and $t(E)$. Observations $u(a)$ outside $\delta\mathbb{Z}$ are also retained; they certify that the primitive quotient of absorption timings alone is not a clock tick for all those additional observations. In that situation the exact group for the joint timing record is $\widetilde{G} = \langle t(E), u(N) \rangle_{\mathbb{Z}}$. Applying GTQ2 to this explicitly enlarged set calculates its common tick or proves that it has none. One still uses E alone to define absorption fibres, and N to record non-absorption fibres. No observed non-absorption datum is discarded or silently reclassified.

For any derived count $N \geq 0$, the number of positive clock positions through $N\delta$ is N . Its decomposition is

$$N = \sum_{n=1}^N b(n) + \sum_{n=1}^N (1 - b(n)). \quad (\text{GTQ3.4})$$

Each term corresponds to one clock position, with exactly one of absorption and non-absorption. This proves the latest WU050 distinction within this receiver: counting only newly appearing absorption fibres would retain the first sum and lose the second. The order number of an absorption event is not generally its elapsed tick count.

The inverse reconstruction retains the scale and the event identities. From $(\delta, \{E_n\})$ reconstruct E as the disjoint union of the E_n and map (n, e) to $n\delta$. The map from the original E to this disjoint union sends e to $(\nu_\delta(t(e)), e)$. Its inverse forgets the first coordinate, whose value is already fixed by $t(e)$. Thus no event, repetition at the same time, or duration factor is lost.

When N is supplied, the output additionally retains its fibres N_n and the off-grid part $N_* = \{a \text{ in } N : u(a) \text{ not in } \delta\mathbb{Z}\}$, with the complete map $u|_{N_*}$ and labels unchanged. Reconstruct N as the disjoint union of the N_n and N_* , assigning $n\delta$ to a member of N_n and its retained duration $u(a)$ to a member of N_* . The

same fibre-membership maps are inverse. Thus the timing comparison also preserves every supplied non-absorption observation, including any one that invalidates the proposed absorption-grid clock.

This creates the lattice of possible clock positions. It does not prove that an unspecified geometric half-turn physically occurs at each such position. That further assertion is a statement about the source dynamics, not an extra conclusion smuggled into GTQ3.1.

GTQ4. The recovered arithmetic and its one

Retain the whole group $G_t = \delta\mathbb{Z}$. Form

$$R_t = \text{End}_{\text{Ab}}(G_t),$$

with pointwise addition and multiplication by composition. Addition and composition are associative; addition is commutative because G_t is abelian. For homomorphisms f, g, h and x in G_t ,

$$f((g + h)(x)) = f(g(x)) + f(h(x)), \quad ((f + g) \circ h)(x) = f(h(x)) + g(h(x)).$$

These prove both distributive laws. The zero map is the additive identity, $-f$ the additive inverse, and id_{G_t} the multiplicative identity. These are operations on endomorphisms of the duration group, not on τ .

Every endomorphism is determined by its value at δ . For a unique integer n , $f(\delta) = n\delta$, and then $f(m\delta) = mn\delta$ for every integer m . Conversely this formula defines an endomorphism for every n . Thus

$$\mathbb{Z} \xrightarrow{\sim} R_t, \quad n \mapsto [n], \quad [n](g) = ng \quad (\text{GTQ4.1})$$

is a ring isomorphism. Its coordinate check gives $[n] + [m] = [n + m]$, $[n] \cdot [m] = [nm]$, and $[1] = \text{id}_{G_t}$. Replacing δ by $-\delta$ leaves the integer n attached to an endomorphism unchanged. The identity endomorphism is therefore independent even of the chosen orientation, although the positive duration δ uses the direction of the timing record.

Evaluation at δ is an additive isomorphism

$$\text{ev}_\delta : R_t \xrightarrow{\sim} G_t, \quad f \mapsto f(\delta), \quad (\text{GTQ4.2})$$

and sends the recovered arithmetic one to the recovered tick δ . It transfers multiplication to G_t by

$$(n\delta) \odot_\delta (m\delta) = (nm)\delta. \quad (\text{GTQ4.3})$$

The duration δ is retained; it is not identified with τ or silently erased by a scale change.

Let $R_{t,+}$ be the endomorphisms taking nonnegative durations to nonnegative durations. By GTQ4.1 these are precisely $[n]$ with $n \geq 0$. Define $a|b$ in its nonzero part when $b = a \cdot c$ for a nonnegative endomorphism c . The identity divides every member. If a divides every nonzero member, it divides the identity: $a \cdot c = \text{id}$. Writing $a = [n]$ and $c = [m]$ gives $nm = 1$ with n, m positive, so $n = m = 1$. Hence id_{G_t} is the unique positive universal divisor of this recovered arithmetic.

There are therefore two proved comparisons, not an ambiguous use of one word:

- The primitive common timing divisor is δ , the greatest member of the set GTQ2.2.
- The universal divisor of the positive endomorphism arithmetic is id_{G_t} , whose evaluation at δ is δ .

The first is a statement about durations. The second is a statement about composition of operations. Equation GTQ4.2 relates them exactly.

GTQ5. What the whole determines that a prefix cannot determine

Let F be a nonempty finite set of observed events, and let B be a duration through which the record has been completely observed. Suppose the prefix's duration group is

$$G_F = \delta_F \mathbb{Z}, \quad \delta_F > 0.$$

The existence of this prefix tick is checked on the prefix itself by GTQ2. It is not named the final unit. Choose an integer M large enough that $M\delta_F > B$. For any proof integer $k > 1$ define two full continuations, retaining the same old events and their identities:

$$S^{(0)} = t(F) \cup \{(M + j)\delta_F : j \in \mathbb{Z}_{\geq 0}\}, \quad (\text{GTQ5.1})$$

$$S^{(k)} = t(F) \cup \{(M + j + 1/k)\delta_F : j \in \mathbb{Z}_{\geq 0}\}. \quad (\text{GTQ5.2})$$

If an existing event has multiplicity or labels, retain them; the newly added times can be assigned new event identities. Each continuation is unbounded and locally finite, and the observations through B coincide exactly.

The group of $S^{(0)}$ is $\delta_F \mathbb{Z}$: the old events already generate that group and every new event belongs to it. The group of $S^{(k)}$ is $(\delta_F/k) \mathbb{Z}$. Indeed, it contains δ_F because the old events generate it, and it contains

$$(M + 1/k)\delta_F - M\delta_F = \delta_F/k.$$

Every old and new event is an integer multiple of δ_F/k , proving the reverse inclusion. By GTQ2 the final primitive ticks are respectively δ_F and δ_F/k .

For every old event e whose prefix coordinate is $t(e) = n_e \delta_F$, the two final retrospective counts are respectively

$$\nu^{(0)}(t(e)) = n_e, \quad \nu^{(k)}(t(e)) = kn_e. \quad (\text{GTQ5.3})$$

They differ, and the finer reconstruction has additional non-absorption clock positions before B . Thus identical finite observed timings can correspond to different final numerical meanings in the two whole constructions.

Construction-level conclusion. On the class of unbounded, locally finite timed event records having a primitive tick, no rule depending only on an arbitrary finite observed prefix can always return the final primitive tick or even the final count of every old event. If it did, apply it to the two identical prefixes just constructed. Its output would be identical, whereas GTQ5.3 makes the correct outputs different. Contradiction.

Equivalently, the whole-record reconstruction in GTQ3 does not factor through the prefix restriction map on this class. This is a precise global dependency of the construction. The proof is not founded on assigning prime values to the first two events, and it is not merely a claim that a person happens not to know those values.

The class used here is stated in full. No conclusion is drawn that these arbitrary continuations are permitted by the actual CC source relations, or by an additional arithmetic return law. The calculation establishes exactly what timing plus order alone can and cannot force.

GTQ6. A finite algebraic witness and a global certification are different facts

Now take a complete record for which $G_t = \delta\mathbb{Z}$. By definition G_t is generated by finite signed combinations of events. Since δ belongs to G_t , there are finitely many events $e_1, \dots, e_r$ and signed proof integers $a_1, \dots, a_r$ such that

$$\delta = \sum_{i=1}^r a_i t(e_i). \quad (\text{GTQ6.1})$$

Let F_0 be this finite set. Because $F_0 \subseteq E$, its group lies in $\delta\mathbb{Z}$. Because GTQ6.1 puts δ in its group, that group is all $\delta\mathbb{Z}$. Thus the final primitive tick has a finite algebraic witness inside the full record.

For a sequential, locally finite record with an initial interval containing F_0 , that interval's primitive tick equals δ . Every later prefix of the actual complete record has the same primitive tick: it contains F_0 and lies in $\delta\mathbb{Z}$.

This does not contradict GTQ5. The finite expression GTQ6.1 proves that δ is generated by those events. It does not prove that every event outside F_0 is an integer multiple of δ . The latter is the universal assertion about the complete record used to certify the same primitive tick globally.

Accordingly, the universal statement “the unit can have no finite algebraic witness” is false for the cyclic duration reconstruction; GTQ6.1 explicitly constructs such a witness. The stronger whole-first requirement in the user's account can instead be represented by the failure of finite-prefix factorization in GTQ5. It cannot be proved by confusing these two assertions. No claim is made that the user's source has already been fully formalized by this receiver.

GTQ7. Completeness alone does not force a primitive tick

Here is the exact calculation for a complete sequential record whose timings remain commensurable but admit no common positive tick. It is a receiving-space example, not a counterexample asserted to satisfy the user's still-to-be-derived geometric source.

Choose a positive duration v solely to describe this example. Include an event at v , followed by the events

$$t_n = (n + 3^{-n})v, \quad n \in \mathbb{Z}_{>0}. \quad (\text{GTQ7.1})$$

The labels n are metatheoretic positions in an example list, not prelabelled prime observations. The event times increase strictly because

$$t_{n+1} - t_n = (1 - 2 \cdot 3^{-n-1})v > 0.$$

They are unbounded since $t_n \geq nv$ and locally finite for the same reason. Their entire list is specified, with no unknown future event. Every pair of durations has rational ratio.

The generated group contains v because that event is present. It then contains $t_n - nv = 3^{-n}v$ for every n . Conversely each event is a rational multiple of v whose denominator is a power of three. Therefore the exact generated group is

$$\boxed{G_t = \mathbb{Z}[1/3]v.} \quad (\text{GTQ7.2})$$

It has positive elements $3^{-n}v$ tending to the zero duration, so it is not discrete and has no least positive element. By GTQ2, its common-tick set is empty. Hence knowing the complete history and knowing

all pairs of event durations to be commensurable do not by themselves force a primitive common tick. The odd-denominator refinement is chosen to allow comparison with the root's separate signed-support calculation; that support calculation is not assumed or duplicated here.

The full obstruction is not discarded. GTQ8 gives the group that records each refinement and identifies exactly when they terminate in a single primitive unit.

GTQ8. The quotient recording every late refinement

Fix any nonempty observed prefix F with cyclic group $\delta_F \mathbb{Z}$ and retain the whole event group G_t . Because δ_F belongs to G_t , form the exact sequence

$$0 \longrightarrow \delta_F \mathbb{Z} \longrightarrow G_t \longrightarrow \mathcal{D}_F := G_t / (\delta_F \mathbb{Z}) \longrightarrow 0. \quad (\text{GTQ8.1})$$

This quotient records how complete timing differs from the prefix's grid. An event e remains on that grid exactly when its class $t(e) + \delta_F \mathbb{Z}$ is zero. Thus the entire history remains on it exactly when $\mathcal{D}_F = 0$, because event classes generate the quotient.

For any duration g in G_t , its class in $\mathcal{D}_F$ has finite order precisely when some positive repetition kg lies in $\delta_F \mathbb{Z}$. In a temporary coordinate this is equivalent to g/δ_F being rational. This equivalence follows directly from $kg = m\delta_F$ in one direction and clearing the denominator of a rational ratio in the other.

The following classification is exact:

- $\mathcal{D}_F$ finite of cardinality k means $G_t = (\delta_F/k) \mathbb{Z}$. Its final tick is δ_F/k , and $\mathcal{D}_F$ is cyclic of order k .
- $\mathcal{D}_F$ infinite torsion means every event duration is rationally commensurable with δ_F but their denominators have no common bound. There is no common tick.
- $\mathcal{D}_F$ containing an element of infinite order means at least one duration is incommensurable with δ_F . There is no common tick.

Proof of the finite case. The quotient of a cyclic timing group $\delta \mathbb{Z}$ containing $\delta_F \mathbb{Z}$ has $\delta_F = k\delta$ for a positive integer k , so it is cyclic with k classes, and the assertion follows in this direction. Conversely suppose $\mathcal{D}_F$ has k elements. Addition by any class permutes this finite set; alternatively partition into the cosets of its cyclic subgroup. The order of every class divides k . Hence kg belongs to $\delta_F \mathbb{Z}$ for all g in G_t , giving $G_t \subset (\delta_F/k) \mathbb{Z}$. By the subgroup argument of GTQ2, G_t is cyclic, say $\delta \mathbb{Z}$. The first direction now shows that its quotient has δ_F/δ classes; equality with k gives $\delta = \delta_F/k$.

Proof of the infinite cases. If a common tick existed, GTQ2 would make G_t cyclic, and the finite-case calculation would make $\mathcal{D}_F$ finite. Thus either infinite case rules out such a tick. The statement about rational commensurability follows from the preceding order calculation. In the torsion case, a common denominator M , meaning that every reduced denominator divides M , would give $G_t \subset (\delta_F/M) \mathbb{Z}$ and a finite quotient. A numerical bound M on reduced denominators also supplies a common denominator, namely $M!$. Conversely a finite quotient supplies the common denominator k . Thus infinite torsion is exactly unbounded refinement among commensurable durations. ■

For the pair of complete continuations in GTQ5, the quotients are respectively zero and a cyclic group of order k . For the record in GTQ7, quotienting its full group by $v \mathbb{Z}$ gives $\mathbb{Z}[1/3]/\mathbb{Z}$. Every class is killed by some power of three; classes of arbitrarily high power-of-three order occur. This explicitly records why there is no final positive tick, without replacing the defect by a slogan.

For completeness, a nonzero additive subgroup of V with no least positive element is dense. Indeed, if its positive infimum were positive, the argument in GTQ2 would attain the infimum and produce a least element. Therefore it has a positive member h shorter than any specified positive interval length. For a real interval (a,b) , choose $h < b-a$ and an integer k with $a < kh \leq a+h$; then kh lies in (a,b) . This proves density. The infinite cases of GTQ8 are thus genuinely nondiscrete timing groups, not merely missing a chosen name for an otherwise present smallest unit.

GTQ8.2. The whole group is the exact directed limit of finite records

Order the finite subsets F of E by inclusion. Their timing groups have the actual inclusion maps $G_F \rightarrow G_{F'}$ whenever $F \subset F'$. Then

$$G_t = \bigcup_{F \subset E, F \text{ finite}} G_F \cong \varinjlim_F G_F. \quad (\text{GTQ8.2})$$

The union equality follows because each element of G_t is a finite signed combination of event durations. For the universal property, a compatible family of group maps $f_F: G_F \rightarrow H$ defines $f(g) = f_F(g)$ for any F with g in G_F . Two such choices lie in the common larger finite set $F \cup F'$, so compatibility makes this independent of the choice. The same common-finite-set argument proves additivity and uniqueness.

When $G_F = \delta_F \mathbb{Z}$ and $G_{F'} = \delta_{F'} \mathbb{Z}$ are nonzero cyclic, their inclusion has an exact positive index k with

$$\delta_F = k\delta_{F'}, \quad \nu_{\delta_{F'}}(g) = k\nu_{\delta_F}(g) \quad (g \in G_F). \quad (\text{GTQ8.3})$$

The first equation holds because δ_F is a positive member of $G_{F'}$. Substituting $g = n\delta_F$ proves the second. This is the exact transition map for old numerical labels under a later refinement; they need not be held fixed while the primitive tick changes. If the complete group is cyclic, GTQ6 supplies a finite record after which these indices are all one in the actual directed system. If it is not cyclic, no finite cyclic stage equals the whole group. The full group in GTQ8.2 is a mathematical object retaining all events and relations; no assertion that an infinitely long physical procedure has ended is used to construct the limit.

GTQ9. A timing start is distinct from the supporting point

If the data give event positions in an oriented affine time line A rather than durations from a marked start, the coordinate-free timing data are differences $t(e) - t(f)$ in its translation line V . Define

$$G_{\text{diff}} = \langle t(e) - t(f) : e, f \in E \rangle_{\mathbb{Z}}.$$

The preceding common-tick classification applies to this group. In its nonzero cyclic case it supplies a unique positive δ . The difference group by itself does not select which point of the affine record is to be numbered zero.

The event positions lie in the affine lattice

$$\mathcal{A} = t(e_0) + G_{\text{diff}}$$

for any event e_0 , and changing e_0 does not change that set. The group acts freely and transitively by translations on $\mathcal{A}$. Choosing e_0 assigns the coordinate $c(e)$ by $t(e) - t(e_0) = c(e)\delta$. If another event e_1 has coordinate r , the coordinates based at e_1 are $c(e) - r$. Thus the lattice of possible positions is a torsor for the integer group.

The full event record may distinguish a point even when no separate initial marker was supplied. For example, a unique earliest event in an oriented record selects an event-based zero. General affine event records need not do this: for the whole affine lattice with identical event labels at every lattice point, translation by δ preserves the full data and fixes no point. An isomorphism-invariant selection of one point is therefore impossible for that example. Even when an earliest event selects an origin, it does not establish that the event was the user's initial no-turn state, or identify it with tau.

This is precisely why a parityless supporting point must not silently be used as a numerical start or a zero duration. A marked initial record position and the support of the construction are different input roles. No metric between them is used or derived here.

GTQ10. Functoriality and preserved scale

An isomorphism of marked timing records consists of an event bijection $\varphi:E \rightarrow E'$ preserving any specified event labels and an orientation-preserving real-linear isomorphism $A:V \rightarrow V'$ such that $t' \circ \varphi = A \circ t$. It induces a group isomorphism $G_t \rightarrow G_{t'}$ by restriction of A . The group generators and all timing relations are carried to the corresponding generators and relations, so GTQ1.1 commutes with this map.

When the groups are cyclic, A sends their least positive elements to each other: if $0 < g' < A\delta$, its inverse would be a positive member of G_t less than δ . Consequently

$$A(\delta) = \delta', \quad \nu_{\delta'}(A(g)) = \nu_{\delta}(g). \quad (\text{GTQ10.1})$$

For endomorphisms the exact map is

$$R_t \longrightarrow R_{t'}, \quad f \longmapsto A \circ f \circ A^{-1}. \quad (\text{GTQ10.2})$$

It preserves addition, composition, and the identity by evaluation; its inverse is conjugation by A^{-1} . Evaluation at the actual durations δ and δ' commutes with these maps. The absorption fibres satisfy $\varphi(E_n) = E'_n$ for every derived count n , retaining every multiplicity and label.

Thus the arithmetic counts are invariant under a change of external time units, while the unit duration transforms as $\delta \mapsto A\delta$. This is a faithful comparison retaining the duration scale, not permission to replace the original duration by the scalar one everywhere.

GTQ11. Counting-time quotients and logarithmic return periods

The programme's existing TP note also uses return-flow periods $\log(p)$ on the curves $\mathbb{C}^\times/p^{\mathbb{Z}}$. Those periods must not silently be identified with the tick δ of GTQ2. The following comparison is made only after the arithmetic in GTQ4 has been recovered. Prime symbols in this section therefore belong to the resulting arithmetic; none is a starting event label.

Write $a=[n]$ for a positive endomorphism. For $n \geq 1$, define

$$L(a) = \log n.$$

Using a temporary generator only to compute the intrinsic integer n gives the same value after reversing that generator, by GTQ4.1. Then

$$L(a \circ b) = L(a) + L(b), \quad L(\mathbf{1}_{R_t}) = 0. \quad (\text{GTQ11.1})$$

It is the multiplication of recovered positive arithmetic that becomes addition of logarithmic flow time. The map from positive count durations is

$$n\delta \longmapsto \log n, \quad (\text{GTQ11.2})$$

which sends $\odot_{\delta}$ to addition. It does not send ordinary addition of count durations to addition: the required identity would be $\log(n+m) = \log n + \log m$ for all positive n, m , which fails already when $n=m=1$ because $\log(1+1) > 0$ and $\log 1 + \log 1 = 0$. Here those integer values have been constructed before the test is made.

For a recovered prime p , the flow $[z] \mapsto [e^u z]$ on $\mathbb{C}^\times/p^{\mathbb{Z}}$ returns to $[z]$ exactly when $e^u z = p^j z$ for an integer j . Since z is nonzero and u is real, this is equivalent to $u = j \log p$. Thus the period group is $(\log p)\mathbb{Z}$ and all return repetitions are retained.

The additive subgroup generated by the complete prime return periods is

$$\Gamma_{\text{ret}} = \bigoplus_{p \text{ prime}} \mathbb{Z} \log p \subset \mathbb{R}. \quad (\text{GTQ11.3})$$

Here is a complete proof of the independence assertion used in this formula. Positive integer division supplies the Euclidean algorithm: successive nonnegative remainders strictly decrease until zero, and back substitution expresses the greatest common divisor as an integral combination. If p is irreducible and p does not divide a , their greatest common divisor is one; write $up+va=1$. If p divides ab , multiplying the last equation by b proves p divides b . Hence irreducibles satisfy the prime divisor property. Every integer greater than one has an irreducible divisor: take its least divisor greater than one, whose factorization into two proper factors would contradict minimality. Repeated division terminates because positive quotients strictly decrease. For two factorizations, the prime divisor property matches and cancels a factor, and induction gives uniqueness of the multiset. Finally, a finite relation $\sum_p a_p \log p=0$ exponentiates to equality between the products with positive and negative exponents on opposite sides. Uniqueness of factorization forces every $a_p=0$. This proves GTQ11.3.

There are at least two different primes: irreducible divisors exist by the preceding argument, and a divisor of $p+1$ cannot equal p because p cannot divide the difference $(p+1)-p=1$. If $\log p$ and $\log q$ for distinct primes were integer multiples of one common positive flow tick, their ratio would be rational, so $p^a=q^b$ for positive integers a,b after exponentiating. Unique factorization forbids this. Therefore Γ_{ret} is not cyclic and, by GTQ2 and the density argument in GTQ8, has no common positive additive flow tick and is dense in the real flow line.

This does not refute the user's counting-time proposal. It proves the exact place where its potential common quotient and the CC logarithmic flow differ. The complete diagrams must retain both the count operation and the logarithmic receiving map. A common arithmetic divisor of the recovered integer norms does not become a common additive divisor of their logarithms.

GTQ12. What has been derived for the actual argument

The complete timed record produces D_t and G_t with all finite duration relations retained. Its primitive global tick, when present, is the least positive element of G_t and is unique. Every event and every non-absorption clock position then receives its arithmetic count by GTQ3, and the unit and multiplication are recovered as operations on the whole cyclic group by GTQ4. No prime values are imported to establish that unit.

GTQ5 proves a construction-level necessity of the whole: on the stated class, the resulting counts do not factor through an arbitrary finite prefix. The finite prefix may acquire different retrospective labels under different full extensions. GTQ6 simultaneously proves that a globally valid unit has a finite algebraic witness. Those results give different answers to different questions and must both be retained.

GTQ7-GTQ8 calculate the missing-tick object exactly: the generated group can remain nondiscrete even with a complete locally finite history, and the quotient $\mathcal{D}_F$ records finite, indefinitely refining, or incommensurable departures from a provisional grid. Thus the words "complete history" alone do not prove that a primitive common quotient exists. The user's actual source may impose more structure; this note does not invent that structure or treat a receiving-space example as an analysis of the entire Z_1 construction.

The existing CG0/TP0 source is stronger than an uncalibrated event list: it supplies an entire signed group with a cyclic quotient. Its proved cyclicity can license the arithmetic reconstruction, but calling that input "the stalk" does not derive its origin from the user's timing process. Conversely, a successful comparison from the complete source to the group G_t above must preserve its full event relations and demonstrate cyclicity in this receiver, rather than assigning known prime labels in advance.

The equality of common timing divisibility, a universal arithmetic unit, and the supporting point's uniqueness has not been assumed. GTQ4 proves the precise relationship between the first two. GTQ9 prevents a hidden identification of the timing start with the support. GTQ11 prevents identifying the count tick with the logarithmic return-flow periods. These are explicit receiving maps and exact obstructions, supplied for the whole construction's next derivation.

No purity or RH assertion is made in this timing note. This is a completed calculation of the common-quotient mechanism and its reconstruction map, with its entire scope stated, rather than a requested final conclusion inserted as a hypothesis.

Independent check

An independent mathematical reader checked GTQ1-GTQ10. Two scope corrections were applied: an earliest event can canonically select a record origin, although it does not identify the support or initial no-turn state; and a bound on reduced denominators gives a common denominator such as their factorial, rather than automatically the bound itself. The directed-limit calculation GTQ8.2 was added after that review and is proved in full above.

Fixed support and odd refinement — OR0-OR8

Proof source: FIXED_SUPPORT_AND_ODD_REFINEMENT.md.

Fixed support, preserved parity, and the complete timing record

24 September 2026. A derivation receiving the argument preserved in USER ARGUMENT RECONSTRUCTION.md, especially §§8-13 and §§22-30. No novelty claim is made. This document tests specified implications; it does not identify an arbitrary event history with the user's intended complete arithmetic construction.

OR0. Input, notation, and what is being tested

The user's order is retained: an uncalibrated complete history precedes an identified arithmetic unit. Neither a named prime nor a calibrated integer tick is an input to the timing reconstruction. The proposed support is written $\tau\langle Z_1; \text{no exchanged } Z_2 \text{ label} \rangle$. Addition at τ remains retracted. Every addition below is on a stated duration group or a stated group of sections, never on τ . No metric or midpoint at τ is introduced.

The duration examples below are explicit test objects built by us. Their numerical formulas are coordinates used by the mathematician to specify and compare complete records, not numerical names supplied to an observer at the start of the user's construction. They are not claimed to satisfy the as-yet-unconstructed property of generating precisely every prime of Spec Z. Their purpose is narrower and exact: determine whether fixed parityless support, indefinitely retained history, compatible parity, and completeness of a timing record already imply a primitive common tick. They do not. OR2-OR8 prove this with all the signed support data retained.

The source comparison is with Alain Connes and Caterina Consani, *The Absolute Twistor Line and the Geometry of the compactified Spec Z*, arXiv:2609.00299v1, original author CC.tex, lines 494-541, 576-610 and 773-899. [Public source, §§3-4](#). Root read these original TeX passages. The source's generic sheaf is given using integer powers of T before the reconstruction below; that fact must not be represented as a derivation of integer powers from uncalibrated timing.

OR1. The support proof does not require coordinates

Let X be the actual three-point topological space of the source. Its points are +, − and η . Its opens are

$$\emptyset, \quad \{\eta\}, \quad \{+, \eta\}, \quad \{-, \eta\}, \quad X.$$

Every nonempty open contains η . Therefore the closure of $\{\eta\}$ is X. The points + and − are closed, since their complements are opens, so neither is generic. This proves that η is the unique generic point.

Every homeomorphism carries the closure of a singleton to the closure of its image. It must therefore carry a generic point to a generic point. Uniqueness gives $\alpha\eta=\eta$ for the source homeomorphism exchanging + and −. Thus fixedness follows from the topology; no midpoint or prescribed distance was used.

Let Q be the two-element label set with involution flip exchanging its elements. An exchanged-label map on a point x must obey

$$p(\alpha x) = \text{flip}(p(x)).$$

At η this would imply $p(\eta) = \text{flip}(p(\eta))$, which no element of Q satisfies. No such label can be assigned at η . At $+$ and $-$ there are exactly two assignments, obtained by choosing either label at $+$ and the other at $-$.

This is the exact proved interpretation of $\tau\langle Z_1; \text{no exchanged } Z_2 \text{ label} \rangle$ in the source support. It is not a claim that η has no isotropy or that its stalk has no parity information: in the source quotient η has stabilizer C_2 , and its sections transform. OR4 and OR5 retain those facts.

OR2. An infinite record whose finite calibrations keep refining

Choose a positive duration c only to write the test record. Do not declare c to be its global primitive tick. The full ordered record has times

$$t_0 = c, \quad t_r = \left(r + \frac{1}{3^r}\right) c \quad (r \geq 1). \quad (\text{OR2.1})$$

The symbols r here index our formula in the surrounding mathematics. They are not the reconstructed arithmetic labels of the events. The record is strictly increasing: $t_1 - t_0 = c/3 > 0$, and

$$t_{r+1} - t_r = \left(1 - \frac{2}{3^{r+1}}\right) c > 0. \quad (\text{OR2.2})$$

It has no accumulation in a bounded time interval, since $t_r \geq rc$. Every event is included. Attach any specified sequence of absorption/non-absorption marks to these times; retain it without alteration. The following assertions depend on the times, and do not erase those marks. A law constraining times from marks would be additional information to derive and use, not something supplied by the two labels alone.

For the prefix through t_r , let D_r be the additive subgroup of the duration line generated by $t_0, \dots, t_r$. For $r=0$ it is $c\mathbb{Z}$. For $r \geq 1$ every displayed duration belongs to $(c/3^r)\mathbb{Z}$, so D_r is contained in that group. Conversely

$$t_r - rt_0 = \frac{c}{3^r}$$

belongs to D_r . Consequently

$$D_r = \frac{c}{3^r} \mathbb{Z}. \quad (\text{OR2.3})$$

The complete record generates the union

$$D = \bigcup_{r \geq 0} D_r = c\mathbb{Z}[1/3]. \quad (\text{OR2.4})$$

Every positive d in D has $d/3$ in D with $0 < d/3 < d$. Hence D has no least positive element. More strongly, no positive duration δ can divide every event timing in the sense $t_r \in \delta\mathbb{Z}$. Such a δ would imply $D \subseteq \delta\mathbb{Z}$; but for large r the element $c/3^r$ lies strictly between 0 and δ and belongs to D , contradicting that inclusion.

This proves an exact failure of the implication “every event has been included, therefore there is a positive common tick.” Each finite prefix has a primitive tick, but the complete record has none. Completeness is not the same property as having a discrete duration group.

The failure does not depend on observing the starting instant used in (OR2.1). The group generated by the consecutive gaps $t_j - t_{j-1}$, $1 \leq j \leq r$, is also $(c/3^r)\mathbb{Z}$ for $r \geq 1$. For $r=1$ the gap is $c/3$. Inductively the new gap is $(3^{r-2})^{c/3}r$, while the old gap group contains $c/3^{r-1}$. Subtracting $(3^{r-1}-1)$ times this old generator from the new gap leaves $c/3^r$. All gaps through that stage are multiples of this duration, giving equality. Therefore the complete record of inter-event durations alone generates the same D and has the same obstruction.

OR3. Odd refinement preserves the full parity observation

Define

$$\chi : D \longrightarrow \mathbb{Z}/2\mathbb{Z}, \quad \chi\left(\frac{ac}{3^r}\right) = a \pmod{2}. \quad (\text{OR3.1})$$

This is well defined. If $a/3^{r=b/3}s$, then $a3^{s=b/3}r$; the powers of three are odd, so reducing this equality modulo two gives $a=b$ modulo two. Bringing two fractions to a common denominator proves additivity. It is onto because $\chi(c)$ is the nonzero class.

Each stage D_r has its own positive generator $c/3^r$, and $\chi(c/3^r)$ is odd. In the next stage the same old generator is three copies of $c/3^{r+1}$, so its parity is unchanged. More generally the inclusion $D_r \rightarrow D_{r+1}$, written in those respective generators, sends m to $3m$. It preserves every old parity label.

Thus retention of parity at every stage does not prevent an indefinite sequence of finer compatible durations. This is why the denominator three was chosen: the example does not acquire its failure by destroying the odd/even observation.

OR4. Full signed section group over the fixed support

Let C_4 be the additive group of residues modulo four, and set

$$G_D = D \times C_4.$$

Write its operation multiplicatively as

$$(d, k)(d', k') = (d + d', k + k'), \quad \mathbf{1} = (0, 0), \quad J = (0, 1), \quad \epsilon = (0, 2). \quad (\text{OR4.1})$$

Here 0 in the first coordinate is zero duration. It is not τ and is not a new definition of the user's numbered zero. The four finite phases have not been removed: $J^2=\epsilon$ and $\epsilon^2=1$. If a pointed monoid is required, adjoin a distinct absorbing symbol $\perp$ with $g\perp=\perp$; it too is distinct from η , τ and every group element.

Define the full signed involution

$$A_D(d, k) = (-d, 2\chi(d) - k). \quad (\text{OR4.2})$$

The expression $2\chi(d)$ is unambiguous in C_4 because the homomorphism $C_2 \rightarrow C_4$ sends the nonzero class to 2. Additivity of χ proves that A_D is a group homomorphism. Since $\chi(-d)=\chi(d)$, applying it twice gives

$$A_D^2(d, k) = (d, 2\chi(-d) - 2\chi(d) + k) = (d, k). \quad (\text{OR4.3})$$

It therefore is an automorphism, not a projection discarding the winding. Multiplication by the reversed section gives

$$(d, k)A_D(d, k) = (0, 2\chi(d)) = \epsilon^{\chi(d)}. \quad (\text{OR4.4})$$

The actual CC section group in these coordinates is $G_Z = Z \times C_4$, with

$$A_Z(m, k) = (-m, 2m - k).$$

There is an injective comparison

$$\iota : G_Z \longrightarrow G_D, \quad \iota(m, k) = (mc, k). \quad (\text{OR4.5})$$

It preserves the product, J and ϵ . Since $\chi(mc) = m$ modulo two, it satisfies $A_D \circ \iota = \iota \circ A_Z$. In particular $T = (1, 0)$ maps to $(c, 0)$, and

$$A_D(\iota(T)) = \epsilon \iota(T)^{-1}, \quad A_D(J) = \epsilon J.$$

These are the full signed formulas of the source, not an unsigned replacement.

OR5. Charts, twisted restrictions, and the parityless point survive

Use the same topological space X as OR1. Set

$$M_+ = D_{\geq 0} \times C_4, \quad M_- = D_{\leq 0} \times C_4,$$

with the products inherited from (OR4.1), and with the distinct absorbing symbol if working with pointed monoids. Define restrictions into the generic group by

$$\rho_+(d, k) = (d, k), \quad \rho_-(d, k) = (d, -k). \quad (\text{OR5.1})$$

In particular J_+ maps to J and J_- maps to $J^{\wedge \{-1\}} = \epsilon J$, exactly retaining the twisted imaginary restriction. The local structure on X has M_+ on Ω_+ , M_- on Ω_- , and G_D on $\Omega = \{\eta\}$. On X take the fiber product $M_+ \times \{G_D\} M_-$; on the empty set take the terminal monoid. These definitions satisfy the sheaf condition: the only cover of an open not containing that open itself which needs checking is $\{\Omega_+, \Omega_-\}$ covering X, and its compatible sections are exactly that fiber product. All restrictions are specified by projection or (OR5.1).

The generic group G_D is the group completion of either unpointed chart. For M_+ , every $d \in D$ is a difference of two nonnegative durations, and the C_4 coordinate already has inverses. If $f: M_+ \rightarrow K$ is a homomorphism to an abelian group, define its extension at (d, k) , with $d = a - b$ and $a, b \geq 0$, by

$$\tilde{f}(d, k) = f(a, k) f(b, 0)^{-1}.$$

If $a - b = a' - b'$, then $a + b' = a' + b$. Applying f to this equality and cancelling proves that the formula is independent of the expression. It preserves products and is the unique extension because every

group element has such an expression. The proof for the negative chart uses $-a, -b$, and its generic identification is composed with the C_4 inversion in ρ_- .

The local involution maps have domains $\beta_+: M_- \rightarrow M_+$ and $\beta_-: M_+ \rightarrow M_-$. Their formulas are

$$\beta_{\pm}(d, k) = (-d, 2\chi(d) + k). \quad (\text{OR5.2})$$

They square to the identity since $4\chi(d)=0$ in C_4 . Directly,

$$\rho_+\beta_+(d, k) = A_D\rho_-(d, k), \quad \rho_-\beta_-(d, k) = A_D\rho_+(d, k), \quad (\text{OR5.3})$$

because $-2\chi(d)=2\chi(d)$ in C_4 . Thus they define an involution of the entire sheaf over the exchange of the two closed points. The topological proof of fixedness and the impossibility of an exchanged parity label at η are unchanged. All of the refining duration data nevertheless remains in the stalk.

The only finite-order elements of G_D are $H=\{0\} \times C_4$: if a positive multiple of (d, k) is the identity, the torsion-free subgroup D of the duration line forces $d=0$; conversely every $(0, k)$ has finite order. Hence the exact sequence is

$$0 \longrightarrow C_4 \longrightarrow G_D \longrightarrow D \longrightarrow 0, \quad (\text{OR5.4})$$

and the involution on the quotient is $d \mapsto -d$. The four-state phase data remains present in G_D and in this sequence. The user's unbounded retained torsion/winding is recorded by D ; it is not being equated with the finite-order subgroup H .

The fixed elements in G_D are exactly $\{1, \varepsilon\}$: the equation $d = -d$ gives $d=0$, and $k = -k$ in C_4 gives $k=0$ or 2 . The difference between a fixed support and its transforming sections has therefore been calculated, not asserted.

OR6. The refinement system and its exact obstruction object

At each finite stage use $G_r = D_r \times C_4$ with the restrictions and involution just given. The inclusions $G_r \rightarrow G_{r+1}$ preserve the products, positive and negative charts, restrictions, four phase elements, parity observations and signed involutions. In the chosen duration coordinates their formula is $(m, k) \mapsto (3m, k)$. It commutes with the full signed involution because $2(3m) - k$ and $2m - k$ differ by $4m$. Their union is exactly G_D . If T_r denotes the section $(c/3^r, 0)$, the map sends T_r to T_{r+1} ; it preserves the old section but does not preserve its status as a primitive generator. This is the exact change being tested, not a generator-preserving isomorphism of cyclic sources.

Consequently the complete signed source is an explicit direct limit of the finite-stage signed cyclic sources. The support remains η at every stage, old parity labels agree at every stage, no history is deleted, and nevertheless the global positive duration monoid has no indecomposable nonzero element: every $d > 0$ decomposes as

$$d = \frac{d}{3} + \frac{2d}{3}. \quad (\text{OR6.1})$$

This is the object defined by the obstruction. It records the compatible refinements instead of forcing a limiting positive generator into existence. There is no claim that a finite-stage generator ceases to be a duration; it ceases to be primitive when the next stage is included.

OR7. The endomorphism ring records the arithmetic effect exactly

Calculate $\text{End}_{\text{Ab}}(D)$, with addition of endomorphisms defined pointwise on the duration group and multiplication given by composition. Every endomorphism f is determined by $f(c)$. Write $f(c)=ac$ with $a \in \mathbb{Z}[1/3]$. For each r , the equality $3^r f(c/3^r)=c$ implies

$$3^r f(c/3^r) = ac.$$

There is exactly one solution in D , since D is torsion-free, and it is $ac/3^r$. Additivity then gives $f(bc/3^r)=abc/3^r$. Conversely multiplication by any $a \in \mathbb{Z}[1/3]$ gives an endomorphism of D . Thus

$$\mathbb{Z}[1/3] \xrightarrow{\sim} \text{End}_{\text{Ab}}(D), \quad a \mapsto (d \mapsto ad) \quad (\text{OR7.1})$$

is a bijection and preserves addition, composition and the identity. It is therefore a ring isomorphism.

The receiving ring is $\text{End}_{\text{Ab}}(D)$, the endomorphism ring of the explicitly displayed quotient in (OR5.4). It is not the endomorphism ring of the full signed sheaf with J , ε and χ required to remain fixed. More precisely, multiplication by $a=A/3^r$ satisfies $\chi(ad)=\chi(d)$ for every d exactly when A is odd: its action on χ is multiplication by A modulo two, and necessity follows by testing $d=c$. These parity-preserving multipliers are closed under composition but not under pointwise addition, because the sum of two odd numerators after a common odd denominator is even. Retaining the full group, its quotient map and χ allows both kinds of structure to be compared; it does not identify their morphism sets.

This ring has a unique identity, namely the identity map on D , although D has no primitive positive tick. Indeed any identity endomorphism must act as the identity under composition with id_D and hence equal id_D . Thus unique ring identity and unique positive duration generator are different properties, related here by the fully computed receiver rather than by renaming one as the other.

There is also an exact downstream spectrum comparison. We use ordinary integer names only now, after (OR7.1) identifies the receiving ring. Every nonzero ideal I of $\mathbb{Z}[1/3]$ intersects $\mathbb{Z}$ nontrivially: multiply any nonzero $a/3^r \in I$ by 3^r . Let d be the least positive member of $I \cap \mathbb{Z}$. Integer division gives $I \cap \mathbb{Z} = d\mathbb{Z}$. For any $a/3^r \in I$, $a \in d\mathbb{Z}$, and conversely all $d/3^r$ lie in I . Hence $I = d\mathbb{Z}[1/3]$. Moreover 3 does not divide d , since otherwise $d/3 \in I \cap \mathbb{Z}$ would be a smaller positive member.

The proper nonzero prime ideals are exactly $p\mathbb{Z}[1/3]$ for ordinary primes p other than three. If d is composite, a factorization $d=ab$ with $1 < a, b < d$ disproves primality because neither factor belongs to I but their product does. If $d=p$ is prime and $p \neq 3$, reduction modulo p extends uniquely to $\mathbb{Z}[1/3]$, since the image of 3 has a multiplicative inverse. Its kernel is $p\mathbb{Z}[1/3]$, and its image is the field with p elements, proving primality and residue norm p . The zero ideal is prime because the ring lies in $\mathbb{Q}$ and has no zero divisors. These exhaust the possibilities.

Thus the exact resulting scheme is $\text{Spec } \mathbb{Z}[1/3]$, and its natural map to $\text{Spec } \mathbb{Z}$ is the open inclusion of the points other than (3). On a basic open determined by a fraction $a/3^r$, nonvanishing is the same as nonvanishing of a ; the section ring is $\mathbb{Z}[1/3, 1/a]$, as is the corresponding restriction of the usual localization sheaf. This proves the comparison including its structure sheaf.

The removed arithmetic point is the precisely calculated consequence of the indefinite odd refinement. This does not contradict the user's demand to recover every prime; it shows that that demand is substantive, and that fixed parityless support alone has not already enforced it.

OR8. What the calculation proves about the user's chain

The following pieces have been proved together: the actual CC generic point is unique and fixed; it admits no equivariant exchanged two-state label; a full signed stalk can preserve winding at that point; and compatible odd refinements preserve the source parity formulas at every stage. A complete, locally

finite, unskipped timing record can satisfy these particular properties and fail to possess any positive common tick.

Therefore the inference from those properties to the existence of the global primitive quotient is not valid without using further structure. The failure is not at an invented addition on τ , nor at a selected pair of already named primes. It is the explicit lack of a least positive duration in the complete receiver. The constructed object retaining that failure is the signed refinement system (OR4.1)–(OR6.1), and its arithmetic receiving map is (OR7.1).

The already-written CC reconstruction starts with the source's full cyclic chart and so legitimately reconstructs an integer endomorphism ring from that chart. It cannot be cited as having proved that the user's timing process produces the cyclic chart: CC.tex lines 509–511 already put integer exponents into that input. Both statements remain on record with their exact domains.

This derivation does not decide the user's entire intended construction by replacing it with (OR2.1). It disproves only the displayed inference from the features retained here. An intrinsic rule on the original complete history could exclude this receiver; no such rule is silently assumed in this proof. Neither a purity conclusion nor an RH verdict follows from this calculation, and neither is labelled a completed result.

Separate branch counters and the full arithmetic comparison — BCC0-BCC9

Proof source: BRANCHWISE_COUNTER_COMPARISON.md.

Separate branch counters, the original zeta, and the retained spectral action

Private independent derivation, 24 September 2026. Proof locators BCC0-BCC9.

BCC0. The request, reading record, and exact input class

The new attachment was read in full:

Latest branch correction and admissible-deviation argument (private construction record; not included).

Its operative correction is that branches are compared **after each has been divided by its own recovered counter**. Their measures must not be pooled. Its subsequent question concerns deviations that retain sufficient arithmetic to recover the same original Riemann zeta function. The user's tentative implication from quotientability to RH is a question to calculate, not an established theorem.

The present input class consists of complete timing records whose generated duration groups are nonzero cyclic groups. This is the class for which the preceding full-history calculation actually constructs a primitive counter. The cyclicity criterion and its proof are in [GLOBAL_TIMING_QUOTIENT_INDEPENDENT.md](#), GTQ1-GTQ4. No assertion is made that a parityless supporting point alone forces this input class.

Every numerical label below occurs after that complete-history construction. No selected prime is an initial datum. Integers used to state finite repeated concatenations in the proof belong to the metatheory; the branch's arithmetic labels are the output of BCC2-BCC4.

The supporting object retains the user's notation

$$\tau\langle Z_1; \text{no } Z_2\rangle.$$

No addition, subtraction, metric, coordinate, midpoint, or vector is assigned to it. A duration belongs to a duration line, not to the supporting object.

The following programme sources were actually read for this calculation:

- GLOBAL_TIMING_QUOTIENT_INDEPENDENT.md, GTQ0-GTQ12: the complete timing group, primitive-counter criterion, and finite-prefix comparison.
- CANONICAL_GENERIC_RECONSTRUCTION.md, CG0, and TIMED_PRIME_PERIOD_DERIVATION.md, TP0-TP2 and the opening of TP3: the existing return-measure receiver.
- WINDING_REVERSAL_AND_FIXED_SUPPORT.md, WR5-WR7, including the full signed character and multiplicity formulas.
- ORIGINAL_ZETA_RESOLVENT_AND_PROJECTORS.md, RZ1 and RZ8, together with the immediately preceding projector formulas: the actual quotient, original-zeta multiplier, and arithmetic action.

The source-routing and source-use ledgers were consulted before this derivation. The branch maps below are derived directly. BCC7 and BCC8 apply them to the specified programme receivers; they do not replace those receivers with an arbitrary matrix example.

BCC1. Classification of linear comparisons between complete cyclic histories

For each branch i , let V_i be an oriented one-dimensional real vector space of durations. Let E_i be the set of **absorption events**, including any retained event identities, multiplicities, and channel labels, and let

$$t_i : E_i \longrightarrow V_{i,>0} \quad (\text{BCC1.1})$$

give their complete positive timing record. Events sharing a duration remain distinct elements of E_i . Separately recorded non-absorption observations may be kept in a set N_i with a map $u_i : N_i \rightarrow V_i$. They are not relabelled as absorption events.

The generated duration group is

$$G_i = \left\{ \sum_{e \in F} a_e t_i(e) : F \subset E_i \text{ finite, } a_e \in \mathbb{Z} \right\} \subset V_i. \quad (\text{BCC1.2})$$

In the present input class G_i is nonzero and has a least positive element. The full-history calculation calls that element δ_i and proves

$$G_i = \delta_i \mathbb{Z}. \quad (\text{BCC1.3})$$

For completeness, every $g > 0$ in G_i has an integer $n \geq 0$ with $n\delta_i \leq g < (n+1)\delta_i$, by the Archimedean property of the real duration line. The remainder $g - n\delta_i$ belongs to G_i and lies in $[0, \delta_i)$; minimality of δ_i makes it zero. Negative elements follow by negation. The positive generator is unique because it is the least positive element.

Let

$$A : V_i \xrightarrow{\sim} V_j, \quad A(G_i) = G_j \quad (\text{BCC1.4})$$

be a real-linear isomorphism. Put $\varepsilon_A = +1$ for orientation preservation and -1 for reversal. Then

$$\boxed{A\delta_i = \varepsilon_A \delta_j.} \quad (\text{BCC1.5})$$

Indeed $A\delta_i = k\delta_j$ for a nonzero integer k . Surjectivity on the generated groups gives $\delta_j = \ell A\delta_i$ for an integer ℓ , hence $k\ell = 1$. Thus $k = \pm 1$, and its sign is ε_A . Conversely the two real-linear maps

$$A_\varepsilon(x\delta_i) = \varepsilon x\delta_j, \quad x \in \mathbb{R}, \quad \varepsilon \in \{+1, -1\}, \quad (\text{BCC1.6})$$

are all the maps satisfying (BCC1.4). This is an exact classification of the stated linear duration comparisons.

A comparison of the **complete forward absorption records** includes, in addition, a bijection

$$\phi : E_i \xrightarrow{\sim} E_j, \quad t_j \phi = A t_i \quad (\text{BCC1.7})$$

preserving the stipulated labels and multiplicities. When both records in (BCC1.1) are nonempty and positive in their given orientations, (BCC1.7) forces $\varepsilon_A = +1$. A reversing map sends positive durations to negative ones. It compares signed histories, or forward histories after transporting the target orientation, rather than two positive records with both orientations unchanged.

Equation (BCC1.4) alone does not assert (BCC1.7). An isomorphism of the duration groups need not preserve the actual absorption pattern. The needed event correspondence is part of the complete-record comparison and is tested explicitly in BCC2.

BCC2. Separate counter readings and all event fibres

For each completed branch define its counter reading and its inverse by

$$\nu_i : G_i \xrightarrow{\sim} \mathbb{Z}, \quad \nu_i(n\delta_i) = n, \quad \iota_i : \mathbb{Z} \xrightarrow{\sim} G_i, \quad \iota_i(n) = n\delta_i. \quad (\text{BCC2.1})$$

Uniqueness of the representation proves that these maps are well defined, inverse, and additive. The two branches have been read with their **own** primitive counter; their duration values have not been identified or added.

Equations (BCC1.5) and (BCC2.1) give the complete comparison:

$$\boxed{\nu_j A = \varepsilon_A \nu_i, \quad A \iota_i = \iota_j [\varepsilon_A].} \quad (\text{BCC2.2})$$

Here $[\varepsilon_A]$ denotes multiplication by the displayed sign on the integer target. The diagram is

$$\begin{array}{ccc} G_i & \xrightarrow{A} & G_j \\ \nu_i \downarrow & & \downarrow \nu_j \\ \mathbb{Z} & \xrightarrow{n \mapsto \varepsilon_A n} & \mathbb{Z}. \end{array} \quad (\text{BCC2.3})$$

For the orientation-preserving comparison, the bottom arrow is the identity.

This is the exact meaning of division by the recovered counter in the arithmetic reconstruction. The literal additive quotient

$$G_i / (\delta_i \mathbb{Z}) = 0 \quad (\text{BCC2.4})$$

has no retained count information. It is not (BCC2.1).

After the complete counter has been recovered, let

$$E_{i,n} = \{e \in E_i : t_i(e) = n\delta_i\}, \quad n > 0. \quad (\text{BCC2.5})$$

For a forward-record comparison (BCC1.7), (BCC2.2) proves

$$\phi : E_{i,n} \xrightarrow{\sim} E_{j,n} \quad \text{for every } n > 0. \quad (\text{BCC2.6})$$

To verify both directions, $e \in E_{i,n}$ implies $t_j(\phi e) = A(n\delta_i) = n\delta_j$; the same calculation with A^{-1} and ϕ^{-1} gives surjectivity on the displayed fibre.

The reconstructed absorption indicator

$$b_i(n) = \begin{cases} 1, & E_{i,n} \neq \emptyset, \\ 0, & E_{i,n} = \emptyset \end{cases} \quad (\text{BCC2.7})$$

therefore satisfies $b_i(n) = b_j(n)$ at every count. As E_i is the complete absorption record, empty fibres record the absence of absorption at those ticks. All event multiplicities and channel labels are retained by (BCC2.6), not only this binary indicator. Conversely, label-preserving bijections of every fibre assemble to a bijection ϕ satisfying (BCC1.7). This proves the exact criterion for preservation of the full absorption record.

Separately recorded non-absorption data are transported by a separately specified bijection $\psi : N_i \rightarrow N_j$ with $u_j \psi = A u_i$. The same fibre proof applies where their durations belong to G_i . Observations outside G_i are retained as their original durations; (BCC2.1) is not extended to them without a further group construction. Thus no extra clock reading is silently invented.

BCC3. Endomorphism arithmetic, multiplication, and the orientation sign

Let

$$R_i = \text{End}_{\text{Ab}}(G_i) \quad (\text{BCC3.1})$$

with pointwise addition and composition as multiplication. Every $f \in R_i$ is determined by $f(\delta_i) = n\delta_i$, and additivity then gives

$$f(m\delta_i) = mn\delta_i. \quad (\text{BCC3.2})$$

Conversely this formula defines an endomorphism for every integer n . Writing this endomorphism as $[n]_i$, we obtain

$$[m]_i + [n]_i = [m + n]_i, \quad [m]_i [n]_i = [mn]_i, \quad 1_{R_i} = [1]_i. \quad (\text{BCC3.3})$$

Consequently the degree map

$$d_i : R_i \xrightarrow{\sim} \mathbb{Z}, \quad d_i([n]_i) = n \quad (\text{BCC3.4})$$

is a unital ring isomorphism. It is derived from the completed duration group.

The branch comparison gives

$$C_A : R_i \xrightarrow{\sim} R_j, \quad C_A(f) = AfA^{-1}. \quad (\text{BCC3.5})$$

For $x \in G_j$, additivity of A proves $C_A(f + g)(x) = C_A(f)(x) + C_A(g)(x)$. Inserting $A^{-1}A$ proves $C_A(fg) = C_A(f)C_A(g)$. Also $C_A(1) = 1$, and $C_{A^{-1}}$ is its inverse. Thus (BCC3.5) is a unital ring isomorphism for either orientation.

For every integer degree,

$$\boxed{C_A([n]_i) = [n]_j, \quad d_j C_A = d_i.} \quad (\text{BCC3.6})$$

Indeed $A(nA^{-1}x) = nx$. In particular, reversal changes the signed count in (BCC2.2) while preserving each endomorphism degree, each power $[n]^r = [n^r]$, and every polynomial relation among these endomorphisms.

Evaluation on the respective positive units has its own sign:

$$\text{ev}_i : R_i \rightarrow G_i, \quad f \mapsto f(\delta_i), \quad \boxed{\text{ev}_j C_A = \varepsilon_A A \text{ev}_i.} \quad (\text{BCC3.7})$$

Proof: $A^{-1}\delta_j = \varepsilon_A \delta_i$, so $(AfA^{-1})(\delta_j) = \varepsilon_A A(f(\delta_i))$.

For reference, transport the ring multiplication to the duration group:

$$(n\delta_i) \odot_i (m\delta_i) = nm\delta_i. \quad (\text{BCC3.8})$$

Its unit is the actual duration δ_i . A positive A is a unital isomorphism between these duration rings. A reversing A instead satisfies

$$A(g \odot_i h) = \varepsilon_A (A(g) \odot_j A(h)), \quad A\delta_i = -\delta_j. \quad (\text{BCC3.9})$$

Both equations follow by putting $g = n\delta_i$, $h = m\delta_i$ and retaining the signs. Thus A with negative sign is not a unital ring isomorphism between duration rings with their **positive** units. The endomorphism isomorphism (BCC3.5) remains unital. Using $-A$, or explicitly transporting the signed unit, gives the corresponding positive comparison.

For any abelian multiplicative group $K^\times$, there is also the exact character comparison

$$\Phi_A : \text{Hom}(G_i, K^\times) \xrightarrow{\sim} \text{Hom}(G_j, K^\times), \quad \Phi_A(\chi) = \chi \circ A^{-1}. \quad (\text{BCC3.10})$$

Its inverse is $\Phi_{A^{-1}}$, it preserves pointwise powers, and

$$\Phi_A(\chi)(\delta_j) = \chi(\delta_i)^{\varepsilon_A}. \quad (\text{BCC3.11})$$

This follows again from $A^{-1}\delta_j = \varepsilon_A\delta_i$. Character-coordinate inversion and preservation of endomorphism degree are therefore simultaneous, calculated effects of the same reversing map.

BCC4. Finite quotient clocks, prime ideals, and retained norms

For $f \in R_i$, define the quotient comparison

$$\bar{A}_f : G_i/fG_i \xrightarrow{\sim} G_j/(C_A f)G_j, \quad g + fG_i \mapsto Ag + (C_A f)G_j. \quad (\text{BCC4.1})$$

Since $A(fG_i) = (C_A f)G_j$, this is well defined; the analogous map induced by A^{-1} is its inverse. For $f = [n]$, $n \neq 0$, division with remainder provides the distinct representatives

$$0, \delta_i, \dots, (|n| - 1)\delta_i. \quad (\text{BCC4.2})$$

Existence of representatives follows from division of the integer coordinate by $|n|$; two displayed representatives have difference in nG_i only when their coordinate difference is a multiple of $|n|$, hence only when they coincide. Therefore

$$\#(G_i/[n]G_i) = |n| = \#(G_j/[n]G_j). \quad (\text{BCC4.3})$$

For $n = 0$, the quotient is the infinite group G_i ; no finite norm is assigned by this formula. For $n = 1$, the quotient is the zero group with one element, so its cardinality is one.

Every ideal in R_i has the form dR_i , for one nonnegative integer d . To prove it, the zero ideal gives $d = 0$. A nonzero ideal contains a nonzero integer degree and hence a positive degree. Take its least positive degree d . Division of any degree a in the ideal by d gives $a = qd + r$, $0 \leq r < d$, with r still in the ideal. Minimality gives $r = 0$. Conversely the ideal contains all multiples of d .

Its nonzero proper prime ideals are exactly pR_i with p a positive prime integer in the recovered ring. If $d = ab$ with $1 < a, b < d$, then neither a nor b belongs to dR_i , but their product does, so the ideal is not prime. For prime p , division in the recovered integer ring gives the Euclidean algorithm. It terminates because its nonzero remainders strictly decrease; back substitution expresses the greatest common divisor as an integer combination. If p does not divide a , their greatest common divisor is one, so $ua + vp = 1$. If $p \mid ab$, multiplication by b gives $p \mid b$. This proves primality of pR_i . The zero ideal is also prime, because degrees multiply in an integral domain.

Equation (BCC3.6) gives

$$C_A(dR_i) = dR_j, \quad R_i/dR_i \xrightarrow{\sim} R_j/dR_j. \quad (\text{BCC4.4})$$

For $d \neq 0$ the quotient rings have $|d|$ elements by the same representatives as above. In particular each closed prime point has the unchanged residue norm

$$N_i(pR_i) = \#(R_i/pR_i) = p = N_j(pR_j). \quad (\text{BCC4.5})$$

The generic prime (0) is retained; its residue field is the fraction field of the recovered ring, and is not a finite residue field. It is not inserted as an extra finite-norm Euler factor. This paragraph does not identify the ring's generic point with the programme's supporting object without a specified supporting map.

The contravariant map

$$\text{Spec } R_j \longrightarrow \text{Spec } R_i, \quad \mathfrak{p} \longmapsto C_A^{-1}(\mathfrak{p}) \quad (\text{BCC4.6})$$

is a homeomorphism. Primality is preserved by the ring isomorphism; the inverse uses C_A ; and the inverse image of the basic open $D(f)$ is $D(C_A f)$, proving continuity in both directions. Equations (BCC4.4)–(BCC4.5) prove that this is a comparison of **normed** prime spectra, including every recovered prime, not only a selected finite collection.

BCC5. Each reconstructed branch gives the same original zeta

Let $R_{i,+}$ consist of the endomorphisms sending the positive generator to a nonnegative multiple of itself. For its nonzero elements put

$$N_i([n]_i) = \#(G_i/[n]_i G_i) = n, \quad n \geq 1. \quad (\text{BCC5.1})$$

The branch's full Dirichlet sum is

$$Z_i(s) = \sum_{f \in R_{i,+} \setminus \{[0]_i\}} N_i(f)^{-s} = \sum_{n=1}^{\infty} n^{-s} = \zeta(s), \quad \text{Re } s > 1. \quad (\text{BCC5.2})$$

Here $x^{-s} = \exp(-s \log x)$ uses the real logarithm of the positive finite quotient cardinality. If $\sigma = \text{Re } s > 1$, the series is absolutely convergent because

$$\sum_{n=1}^{\infty} n^{-\sigma} \leq 1 + \int_1^{\infty} x^{-\sigma} dx = 1 + \frac{1}{\sigma - 1}. \quad (\text{BCC5.3})$$

Every term, including the unit term $1^{-s} = 1$, is present.

Conjugation C_A bijects the positive-degree endomorphisms and preserves (BCC5.1), even for a reversing duration comparison. Thus

$$Z_j(s) = Z_i(s) \quad (\text{BCC5.4})$$

term by term, after each branch's own reconstruction. This proof never takes a sum of the two branch records.

The Euler product is recovered from the whole ring. Here is the factorization argument needed to justify it. A positive integer greater than one has a smallest divisor greater than one; that divisor is prime, for a proper factor would be a smaller such divisor. Repeated division by such a prime terminates on each integer because the positive quotients decrease. This gives existence of a finite prime factorization. The prime divisor property proved in BCC4 shows that a prime in one factorization appears in any other factorization of the same integer. Cancel it and repeat, proving uniqueness including exponents.

For a finite set P of recovered primes, the absolutely convergent geometric products therefore give

$$\prod_{p \in P} (1 - p^{-s})^{-1} = \sum_{\substack{n \geq 1 \\ \text{all prime divisors of } n \text{ lie in } P}} n^{-s}. \quad (\text{BCC5.5})$$

As P exhausts the full prime set, every integer eventually occurs and the omitted terms are bounded in absolute value by the tail of (BCC5.3). Hence

$$Z_i(s) = \prod_{p \text{ prime}} (1 - p^{-s})^{-1} = Z_j(s) = \zeta(s), \quad \text{Re } s > 1. \quad (\text{BCC5.6})$$

This keeps every prime power and its exponent. No prime is chosen before the global counter and ring are reconstructed.

Since the original Riemann zeta function has its meromorphic continuation, the branch functions defined by (BCC5.2) have that same continuation by uniqueness of analytic continuation. More explicitly, the difference of two proposed meromorphic continuations vanishes on the connected right half-plane; away from isolated poles the identity theorem propagates that equality through the connected continuation domain, and identical Laurent expansions then identify poles as well. Accordingly, the comparison preserves the original pole, all trivial and nontrivial zeros, their orders, and all Laurent coefficients. It does not replace the original zeta by a completed function.

The conclusion in this section concerns the arithmetic reconstructed from R_i . An arbitrary marked timing record is not thereby declared to be its prime return measure. The precise return-measure receiver is the next calculation.

BCC6. Full return measures, all repetitions, and the unit term

From the recovered normed prime spectrum define, for each branch separately,

$$\mu_i = \sum_{\mathfrak{p} \in (\text{Spec } R_i)_{\text{closed}}} \sum_{k=1}^{\infty} \frac{1}{k} \delta_{k \log N_i(\mathfrak{p})}. \quad (\text{BCC6.1})$$

The symbol δ_x in this section denotes a unit point mass at a positive logarithmic duration x ; it is not the primitive tick δ_i .

The measure is locally finite. In a bounded interval $[0, B]$, an atom must satisfy $p^k \leq e^B$. There are only finitely many integers $p \leq e^B$, and for each such $p > 1$ there are only finitely many positive k satisfying this inequality. The coefficients $1/k$, repetitions, and prime labels have not been suppressed.

The norm-preserving prime bijection in (BCC4.6) gives the exact transported identity

$$\boxed{\mu_j = \mu_i}. \quad (\text{BCC6.2})$$

This denotes equality after transporting the source labels through their bijection, or equality of the resulting scalar measures on logarithmic durations. It is not a pooling of measures.

For $\text{Re } s = \sigma > 1$,

$$\int e^{-su} d\mu_i(u) = \sum_p \sum_{k=1}^{\infty} \frac{p^{-ks}}{k}. \quad (\text{BCC6.3})$$

Absolute convergence follows, for example, from

$$\sum_p \sum_{k \geq 1} \frac{p^{-k\sigma}}{k} \leq \frac{1}{1-2^{-\sigma}} \sum_{n=2}^{\infty} n^{-\sigma} < \infty. \quad (\text{BCC6.4})$$

The appearance of the smallest recovered prime in this estimate is downstream of the complete reconstruction; it is not a selected input to it.

For $|z| < 1$, integrating the absolutely convergent geometric series along the segment from 0 to z gives $-\log(1-z) = \sum_{k \geq 1} z^k/k$, with the logarithm taking value zero at $z = 0$. Exponentiation, finite products, and then absolute convergence yield

$$\exp\left(\int e^{-su} d\mu_i(u)\right) = \prod_p (1-p^{-s})^{-1} = \zeta(s). \quad (\text{BCC6.5})$$

The complete counting measure, including its unit atom, is the convolution exponential

$$\text{Exp}_*(\mu_i) = \delta_0 + \sum_{r=1}^{\infty} \frac{1}{r!} \mu_i^{*r} = \sum_{n=1}^{\infty} \delta_{\log n}. \quad (\text{BCC6.6})$$

To prove that this is an equality of locally finite measures rather than only a formal notation, every atom of μ_i has duration at least $\log 2 > 0$, so only $r \leq B/\log 2$ can contribute on $[0, B]$. Within each convolution only finitely many atoms contribute there, as already proved. For a single prime, the coefficient-generating identity is

$$\exp\left(\sum_{k \geq 1} \frac{x^k}{k}\right) = \frac{1}{1-x} = \sum_{a \geq 0} x^a. \quad (\text{BCC6.7})$$

One can check it as a formal power series: the left series has constant coefficient one and logarithmic derivative $1/(1-x)$, so its coefficient recurrence is exactly that of the right series. Each exponent a therefore has coefficient one. Unique factorization from BCC5 then shows that every $\log n = \sum_p a_p \log p$ has coefficient one in (BCC6.6), including $\log 1 = 0$ from the explicit δ_0 .

The two sides of the branch comparison preserve (BCC6.6) term by term. In particular, taking its Laplace transform gives the same original $\zeta(s)$ in each branch. The unit atom here is an atom in the logarithmic return measure; it is not an identification of τ with numerical zero.

BCC7. The actual signed winding reversal and its arithmetic character map

The programme's CC receiver has the actual group and involution

$$\begin{aligned} \mathcal{G} &= \mathbb{Z} \times C_4, & T &= (1, 0), & J &= (0, 1), & \epsilon &= (0, 2), \\ \alpha(m, k) &= (-m, 2m - k) \quad (k \text{ modulo } 4). \end{aligned} \quad (\text{BCC7.1})$$

These are the full source coordinates recorded in WINDING_REVERSAL_AND_FIXED_SUPPORT.md, WR1 and WR5. Addition of pairs proves that α is a homomorphism. Directly,

$$\alpha^2(m, k) = \alpha(-m, 2m - k) = (m, k - 4m) = (m, k), \quad (\text{BCC7.2})$$

so it is an automorphism.

Its finite-order subgroup is exactly $\{0\} \times C_4$. An element with nonzero integer coordinate cannot have finite order, while one with zero integer coordinate has order dividing four. Thus the exact sequence

$$0 \longrightarrow C_4 \longrightarrow \mathcal{G} \xrightarrow{q} \mathbb{Z} \longrightarrow 0, \quad q(m, k) = m \quad (\text{BCC7.3})$$

retains the finite phase data and gives

$$q\alpha = -q. \quad (\text{BCC7.4})$$

The induced winding comparison has the negative sign of BCC2, while conjugation on its cyclic endomorphism ring still fixes every $[n]$ by BCC3. The C_4 factor is retained in (BCC7.1)–(BCC7.3), not erased from the source.

A character $\chi : \mathcal{G} \rightarrow \mathbb{C}^\times$ is specified by

$$\chi(T) = a \neq 0, \quad \chi(J) = \omega, \quad \omega^4 = 1, \quad \chi(T^m J^k) = a^m \omega^k. \quad (\text{BCC7.5})$$

Every such pair defines a character, and the generators prove uniqueness. Pullback and its complex conjugate have generator values

$$\begin{aligned} \chi \circ \alpha : (a, \omega) &\longmapsto (\omega^2 a^{-1}, \omega^{-1}), \\ \overline{\chi \circ \alpha} : (a, \omega) &\longmapsto (\omega^2 \overline{a}, \omega). \end{aligned} \quad (\text{BCC7.6})$$

The first equation follows by applying α to T, J ; the second uses $|\omega| = 1$, $\overline{\omega^{-1}} = \omega$, and $\overline{\omega^2} = \omega^2$.

For a recovered prime norm p and each retained phase sector ω , define the compensating character

$$\mu_{p,\omega}(T) = p\omega^2, \quad \mu_{p,\omega}(J) = 1, \quad \mu_{p,\omega}(\epsilon) = 1. \quad (\text{BCC7.7})$$

These assignments respect the group presentation. Multiplying it by the full reversed-conjugate character gives

$$\mathfrak{D}_p \chi = \mu_{p,\omega} \overline{\chi \circ \alpha}, \quad \mathfrak{D}_p(a, \omega) = (p/\overline{a}, \omega). \quad (\text{BCC7.8})$$

Indeed the value at T is

$$(p\omega^2)(\omega^2/\bar{a}) = p\omega^4/\bar{a} = p/\bar{a}, \quad (\text{BCC7.9})$$

and the value at J remains ω . The degree, finite phase, and conjugation factors are all displayed.

For the actual arithmetic character $a = p^\rho = \exp(\rho \log p)$,

$$\mathfrak{D}_p(p^\rho, \omega) = (p^{1-\bar{\rho}}, \omega). \quad (\text{BCC7.10})$$

Its fixed-character equation is exactly

$$a = p/\bar{a} \iff |a|^2 = p \iff p^{2\operatorname{Re} \rho} = p \iff \operatorname{Re} \rho = \frac{1}{2}. \quad (\text{BCC7.11})$$

The last equivalence uses $\log p > 0$.

This compares the operations exactly: branch conjugation fixes the recovered endomorphism degree $[p]$, but degree-weighted conjugate duality changes the character value by (BCC7.8). Equation (BCC3.6) does not set that character value equal to its transformed value. There is no distance assigned to the support in this calculation.

BCC8. The same original zeta retains the same actual zero blocks

Here the inputs are the specific proved quotient and operators in ORIGINAL_ZETA_RESOLVENT_AND_PROJECTORS.md, RZ1 and RZ8. Let $\mathcal{Z}$ be the actual multiset of nontrivial zeros of the original $\zeta(s)$, with multiplicity m_ρ . Let $\mathcal{B}$ be the entire functions rapidly decreasing on every closed vertical strip, with seminorms

$$\|F\|_{A,M} = \sup_{|\operatorname{Re} s| \leq A} (1 + |\operatorname{Im} s|)^M |F(s)|, \quad A > 0, \quad M \geq 0. \quad (\text{BCC8.1})$$

Put

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every } \rho \in \mathcal{Z}, 0 \leq j < m_\rho\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}. \quad (\text{BCC8.2})$$

This is the actual quotient with its quotient topology. It is not replaced by an unrestricted product of independently invented zero coordinates.

The full source multiplier in the programme remains

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{BCC8.3})$$

In particular the endpoint and trivial-zero continuation values retained in RZ1 are

$$F_0(0) = F_0(1) = \frac{1}{8},$$

$$F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r), \quad r = 1, 2, \dots \quad (\text{BCC8.4})$$

They are not discarded when using the quotient (BCC8.2). Formula (BCC8.4) follows directly at the negative even points by multiplying $\Gamma(s/2) = 2(-1)^r/(r!(s+2r)) + O(1)$ with $\zeta(s) = \zeta'(-2r)(s+2r) + O((s+2r)^2)$ and evaluating every remaining factor. At zero the leading factors $s(s-1)/8$, $2/s$, and $\zeta(0) = -1/2$ give $1/8$; at one the zeta residue one and $\pi^{-1/2}\Gamma(1/2) = 1$ give $1/8$. These are comparisons with the original zeta, not a replacement of the working zeta.

Since BCC5 identifies the entire original zeta functions, the branch zero data and ideals coincide. The branch map on the actual quotient is therefore

$$I_A : \mathcal{Q}_i \xrightarrow{\sim} \mathcal{Q}_j, \quad [F]_i \mapsto [F]_j. \quad (\text{BCC8.5})$$

Equality of ideals makes this well defined, the identity on $\mathcal{B}$ provides the inverse and continuity in both directions, and no spectral variable is changed.

For the actual operators

$$L[F] = [sF(s)], \quad T_a[F] = [a^s F(s)] \quad (a > 0), \quad (\text{BCC8.6})$$

one has

$$I_A L_i = L_j I_A, \quad I_A T_{a,i} = T_{a,j} I_A. \quad (\text{BCC8.7})$$

Here $a^s = \exp(s \log a)$ uses the real logarithm. Multiplication by s is continuous on (BCC8.1) because its extra growth is absorbed by raising M by one and a strip-dependent constant. Multiplication by a^s is bounded on that strip by $\max(a^A, a^{-A})$. Both preserve the zero vanishing orders in (BCC8.2). These facts prove the operators and the identities on the quotient, including $T_{1/n}$ for every recovered positive integer n . Their exact source-space action at those integers remains the proved RZ formula

$$\mathcal{M}^{-1} T_n \mathcal{M} k(u) = n^{1/2} k(u/n); \quad (\text{BCC8.8})$$

its factor $n^{1/2}$ is retained. Explicitly, the source space and transform appearing here are

$$\mathcal{A} = \left\{ k \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)| < \infty \quad (N, j \geq 0) \right\},$$

$$\mathcal{M}k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}. \quad (\text{BCC8.8a})$$

The existing synthesis proof identifies $\mathcal{M} : \mathcal{A} \rightarrow \mathcal{B}$ topologically. The dilation identity itself can be checked directly: substituting $u = nv$ gives

$$\mathcal{M}(n^{1/2} k(u/n))(s) = \int_0^\infty n^{1/2} k(v) (nv)^{s-1/2} \frac{dv}{v} = n^s \mathcal{M}k(s). \quad (\text{BCC8.8b})$$

Absolute convergence follows from the defining decay at both ends of $\mathcal{A}$. For the specific retained multiplier (BCC8.3), the source element from RZ1 is

$$k_0(u) = u^{1/2} \sum_{n \geq 1} f_0(nu), \quad f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}, \quad \mathcal{M}k_0 = F_0. \quad (\text{BCC8.8c})$$

These full source formulas are unchanged under the branch identity.

The multiplicity block at an actual zero ρ is the jet algebra

$$\mathcal{Q}_\rho \simeq \mathbb{C}[\epsilon_\rho]/(\epsilon_\rho^{m_\rho}), \quad L_\rho = \rho + \epsilon_\rho, \quad T_{n,\rho} = n^\rho \sum_{r=0}^{m_\rho-1} \frac{(\log n)^r}{r!} \epsilon_\rho^r \quad (\text{BCC8.9})$$

from the RZ projector construction. The last identity is the Taylor expansion of $n^s = n^\rho \exp((s-\rho) \log n)$ in the full multiplicity algebra. It keeps every nilpotent term. The branch comparison (BCC8.5) is the identity on this actual block.

The original-zeta reflection has

$$\rho^\# = 1 - \bar{\rho}, \quad m_{\rho^\#} = m_\rho, \quad JF(s) = \overline{F(1 - \bar{s})}. \quad (\text{BCC8.10})$$

The zero symmetry with multiplicities is the actual original-zeta input in RZ and WR7, with the full analytic multiplier (BCC8.3) retained. It is not an assumption that $\rho^\# = \rho$. The map J preserves $\mathcal{B}$,

since reflection changes the strip bound A to at most $A + 1$, and it preserves the ideal by (BCC8.10). On the full zero block it is exactly

$$J_\rho \left(\sum_{r=0}^{m_\rho-1} c_r \epsilon_\rho^r \right) = \sum_{r=0}^{m_\rho-1} (-1)^r \overline{c_r} \epsilon_{\rho^\#}^r. \quad (\text{BCC8.11})$$

Indeed $s - \rho$ is sent to $-(s - \rho^\#)$. Applying (BCC8.11) twice is the identity, so no jet data are lost.

Direct multiplication, with all coefficients retained, proves

$$J_\rho L_\rho = (1 - L_{\rho^\#}) J_\rho, \quad J_\rho T_{n,\rho} = n T_{1/n,\rho^\#} J_\rho. \quad (\text{BCC8.12})$$

For the first equation both multipliers are $\bar{\rho} - \epsilon_{\rho^\#}$, since $1 - \rho^\# = \bar{\rho}$. For the second, both multipliers on the image of J_ρ are

$$n^{\bar{\rho}} \sum_{r=0}^{m_\rho-1} \frac{(-\log n)^r}{r!} \epsilon_{\rho^\#}^r. \quad (\text{BCC8.13})$$

The factor n on the right side of (BCC8.12) is essential.

The branch identity also commutes with J :

$$I_A J_i = J_j I_A. \quad (\text{BCC8.14})$$

It commutes with every actual zero projector P_ρ in RZ. Its full representative formula is

$$A_\rho(s) = \frac{F_0(s)}{(s - \rho)^{m_\rho}}, \quad \mathcal{P}_\rho F(s) = A_\rho(s) \sum_{r=0}^{m_\rho-1} \frac{(F/A_\rho)^{(r)}(\rho)}{r!} (s - \rho)^r, \quad P_\rho[F] = [\mathcal{P}_\rho F]. \quad (\text{BCC8.14a})$$

The quotient F/A_ρ is holomorphic near ρ , because $A_\rho(\rho) \neq 0$; its derivatives in this formula are local derivatives there. RZ6-RZ7 supply the global membership and projector proof on $\mathcal{B}/\mathcal{I}$. In the branch comparison, every factor, derivative, original zero, and multiplicity in (BCC8.14a) is identical, proving the stated commutation without dropping the full multiplier.

Consequently the complete branch comparison retains each actual zero and each full reflected orbit

$$\{\rho, 1 - \bar{\rho}\}. \quad (\text{BCC8.15})$$

It does not identify the two members by discarding their parameter values. The exact scalar defect of individual fixedness is

$$\Delta_\rho = (\rho - \rho^\#)^2 = (2 \operatorname{Re} \rho - 1)^2. \quad (\text{BCC8.16})$$

Under reflection, the difference changes sign, so its square is unchanged. Under the branch identity (BCC8.5), every original parameter is unchanged, so this defect is unchanged as well. Its vanishing is equivalent to $\rho = \rho^\#$, or $\operatorname{Re} \rho = 1/2$. Formula (BCC8.16) does not assert that an off-critical zero exists; it computes exactly what this faithful comparison would retain at any actual zero.

Finally, retained supporting labels remain a disjoint family. For any specified label bijection $\beta : \mathcal{L}_i \rightarrow \mathcal{L}_j$, define

$$\widehat{\mathcal{Q}}_i = \prod_{\ell \in \mathcal{L}_i} \{\ell\} \times \mathcal{Q}_i, \quad \widehat{I}_A(\ell, q) = (\beta(\ell), I_A q). \quad (\text{BCC8.17})$$

Its inverse is $(\ell', q') \mapsto (\beta^{-1}\ell', I_A^{-1}q')$. Thus distinct labelled zero sections $(\ell, 0)$ remain distinct. No tensor product is used to claim that a zero coefficient retains a basis label.

BCC9. What the calculated comparison establishes

The branch comparison derived here has the complete map chain

$$\begin{array}{ccccc}
 (E_i, t_i, G_i, \delta_i) & \longrightarrow & (R_i, \text{Spec } R_i, N_i) & \longrightarrow & (\mu_i, \text{Exp}_* \mu_i, \zeta_i) \\
 \downarrow (\phi, A) & & \downarrow (C_A, \text{Spec}(C_A^{-1}), N) & & \downarrow \text{transport} \\
 (E_j, t_j, G_j, \delta_j) & \longrightarrow & (R_j, \text{Spec } R_j, N_j) & \longrightarrow & (\mu_j, \text{Exp}_* \mu_j, \zeta_j).
 \end{array} \tag{BCC9.1}$$

The left column requires the full event comparison and uses each branch's own globally recovered counter. BCC2 proves every count and event fibre. BCC3 proves the arithmetic ring comparison with the orientation sign retained. BCC4 proves the full normed spectrum comparison. BCC5-BCC6 prove the original-zeta equality, all repetitions, and the unit term, separately in each branch.

This class of complete cyclic histories is therefore quotientable in the user's separate-counter sense. The proof classifies all real-linear duration comparisons within this class; it does not assert that every conceivable zeta-preserving deformation is one of these maps.

The effect on the actual spectral receiver is also determined: it is (BCC8.5), which preserves every actual ρ , every multiplicity, all jets, all labels, and the defect (BCC8.16). The degree-weighted reflected action is (BCC7.8) and (BCC8.12). Thus preservation of the same complete original zeta and individual fixedness under that reflected action are two explicit mathematical operations with a calculated comparison. The former map preserves the latter's defect; it does not make the defect vanish.

This is not a claim that the full intended construction has been refuted. It identifies the exact information retained by the stated branch quotient, including precisely the character and zero-jet data that cannot be removed while claiming that this quotient is faithful to the same original zeta.

Independent mathematical check: the cyclic comparison, all count and endomorphism formulas, quotient cardinalities, character inversion, and the evaluation sign were independently rederived. The checker specifically confirmed the positive-forward-time restriction, the distinction between the counter isomorphism and the zero group quotient, and the need for an event-fibre comparison in addition to an endomorphism-ring comparison. A subsequent independent audit checked BCC5-BCC8, including the convolution exponential, signed character compensation, full jet conjugation, and defect preservation. It identified two presentation defects, now corrected: the operator domain includes every positive real a so that $T_{1/n}$ is defined, and the downward spectrum map in (BCC9.1) is $\text{Spec}(C_A^{-1})$, with the required contravariance retained.

Original-zeta reflection with all exceptional terms — OZR0-OZR9

Proof source: ORIGINAL_ZETA_REFLECTION_AND_DESCENT.md.

Original zeta reflection, full zero jets, and the retained arithmetic defect

24 September 2026. Independent derivation for the current reconstruction. The original Riemann zeta function remains the working function throughout. No claim of novelty or proof of RH is made.

OZR0. Scope, source, and the user's order

The latest user argument restricts admissible deviations to structures that still recover the entire arithmetic and the same original zeta function. Separate branches use their own recovered counters. Consequently no selected numerical prime or integer is an input to the construction discussed here. The integer parameter introduced in OZR6 is used only after the whole arithmetic has been recovered. No counters from distinct branches are added or identified here. No arithmetic on the supporting datum $\tau\langle Z1; \text{no } Z2 \rangle$ is defined.

This derivation calculates a symmetry of the actual original zeta function and its actual zero algebras. It constructs no alternative zeta function, invented zero set, or proposed off-critical zero. It calculates the reflection quotient explicitly; it does not identify that quotient with an unspecified larger quotient in the user's construction.

Human source: Alain Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, [arXiv:2602.04022v1](https://arxiv.org/abs/2602.04022v1), original author `rhready.tex`, formula labelled `smallxi` and the functional-equation paragraph, local lines 526-545. The routed canonical source IDs are PUBUNIT-171DAC50D1C3B7A2438B39AE and PUBUNIT-90A9C2A0907A753437719FD9. The exact local source SHA-256 is 7f1888d82b42263faca4264f3f9fc77e5750c348640f85db0ac4569fe88ebab2. Reading in this derivation covered lines 491-556 and the title/author metadata. The source's completed function supplies the functional equation; it does not replace the original zeta function in any receiver below. No Deligne theorem is imported by this file.

The precise source identity used is

$$C(s)\zeta(s) = C(1-s)\zeta(1-s), \quad C(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2). \quad (\text{OZR0.1})$$

The factor C , including both endpoint factors and the Gamma factor, stays explicit. OZR2 also calculates the exceptional points where it is not a holomorphic unit.

OZR1. Conjugation and reflection on the original function

Put

$$S = \{s \in \mathbb{C} : 0 < \operatorname{Re} s < 1\}, \quad r(s) = 1 - \bar{s}. \quad (\text{OZR1.1})$$

The map $r : S \rightarrow S$ is an antiholomorphic involution. For a holomorphic function on an r -stable open set, define

$$(Hf)(s) = \overline{f(r(s))}. \quad (\text{OZR1.2})$$

Conjugating the absolutely convergent Dirichlet series on $\operatorname{Re} s > 1$ gives $\overline{\zeta(\bar{s})} = \zeta(s)$. Uniqueness of meromorphic continuation extends this equality to the original meromorphic zeta function. Thus

$$H\zeta(s) = \zeta(1 - s). \quad (\text{OZR1.3})$$

The same real-type identity holds for C : the Gamma integral on the positive real half-plane commutes with conjugation, and continuation preserves that identity. The other displayed factors of C have the same property, with the real logarithm used in $\pi^{-s/2} = \exp(-s \log \pi/2)$.

On S , neither endpoint factor vanishes and Gamma has neither a pole nor a zero. Hence C is a holomorphic unit there. Define the full ratio

$$A(s) = \frac{C(1-s)}{C(s)} = \frac{\frac{1}{2}(1-s)((1-s)-1)\pi^{-(1-s)/2}\Gamma((1-s)/2)}{\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)}. \quad (\text{OZR1.4})$$

No endpoint factors have been removed from this formula. Substituting (OZR1.3) into (OZR0.1) proves

$$\zeta = A H \zeta \quad \text{on } S. \quad (\text{OZR1.5})$$

Real type and $r^2 = 1$ give

$$HA(s) = \frac{C(s)}{C(1-s)} = A(s)^{-1}. \quad (\text{OZR1.6})$$

These are identities of functions on the stated domain, with no assumption on zero locations.

OZR2. Endpoint, trivial-zero, and boundary data

The local Gamma facts used here are $\Gamma(z) = z^{-1} + O(1)$ at zero and

$$\Gamma(-n + z) = \frac{(-1)^n}{n!z} + O(1), \quad n \geq 1. \quad (\text{OZR2.1})$$

The first follows from $\Gamma(1+z) = z\Gamma(z)$ and $\Gamma(1) = 1$; applying that recurrence $n+1$ times proves the second, including its sign and factorial. Gamma has no zeros: its Euler product

$$\frac{1}{\Gamma(z)} = ze^{\gamma z} \prod_{j=1}^{\infty} (1 + z/j)e^{-z/j} \quad (\text{OZR2.2})$$

converges locally uniformly, since the logarithm of each tail factor is $O(j^{-2})$ on compact sets, and away from $0, -1, -2, \dots$ the convergent tail is nonzero. Its zeros are exactly those displayed points, all simple.

Formula (OZR2.2) is the usual Euler product for the Gamma function; it is not a zeta completion or a replacement zeta function.

At zero the original factors of C have the limit

$$C(0) = \lim_{s \rightarrow 0} \frac{1}{2} s(s-1) \pi^{-s/2} \left(\frac{2}{s} + O(1) \right) = -1. \quad (\text{OZR2.3})$$

Thus the apparent singularity of C at zero is removable; the factor s and the Gamma pole have both been accounted for. At one,

$$C(s) = \frac{1}{2}(s-1) + O((s-1)^2). \quad (\text{OZR2.4})$$

For $n \geq 1$, the residue at $-2n$ is

$$R_n := \text{Res}_{s=-2n} C(s) = \frac{1}{2}(-2n)(-2n-1) \pi^n \frac{2(-1)^n}{n!}. \quad (\text{OZR2.5})$$

All these residues are nonzero. These are all poles of C ; its only zero after removal of the singularity at zero is its simple zero at one.

For completeness the residue of the original zeta function at one can be read directly from its original Dirichlet series. For $\text{Re } s > 1$, integration of the counting function gives

$$\zeta(s) = s \int_1^\infty [x] x^{-s-1} dx = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx. \quad (\text{OZR2.6})$$

The second integral and its derivatives converge locally uniformly on $\text{Re } s > 0$, because $0 \leq \{x\} < 1$ and powers of $\log x$ remain integrable against $x^{-1-\delta}$ for every $\delta > 0$. Hence the residue at one is exactly one. Equations (OZR0.1), (OZR2.3), and (OZR2.4) then give

$$\lim_{s \rightarrow 1} C(s) \zeta(s) = \frac{1}{2}, \quad \zeta(0) = -\frac{1}{2}. \quad (\text{OZR2.7})$$

Indeed the source entire continuation of $C\zeta$ has value $1/2$ at both endpoints by its functional equation; division by $C(0) = -1$ gives the last value.

At $-2n$, the right side of (OZR0.1) has the nonzero value

$$C(1+2n)\zeta(1+2n), \quad C(1+2n) = \frac{1}{2}(1+2n)(2n)\pi^{-(1+2n)/2}\Gamma((1+2n)/2). \quad (\text{OZR2.8})$$

It is nonzero because every factor of $C(1+2n)$ is positive and the convergent Dirichlet series $\zeta(1+2n)$ is positive. Division by the simple pole of C proves that the original zeta function has a simple zero at $-2n$, and retains its exact derivative:

$$\zeta'(-2n) = \frac{\frac{1}{2}(1+2n)(2n)\pi^{-(1+2n)/2}\Gamma((1+2n)/2)\zeta(1+2n)}{\frac{1}{2}(-2n)(-2n-1)\pi^n \frac{2(-1)^n}{n!}}. \quad (\text{OZR2.9})$$

The numerator and denominator retain all original endpoint and Gamma contributions. Consequently A is a holomorphic unit on the reflection-stable open set

$$U = \mathbb{C} \setminus (\{0, -2, -4, \dots\} \cup \{1, 3, 5, \dots\}). \quad (\text{OZR2.10})$$

Its excluded local behavior is

$$\begin{aligned} A(s) &= \frac{1}{2}s + O(s^2) & (s \rightarrow 0), \\ A(s) &= -\frac{2}{s-1} + O(1) & (s \rightarrow 1), \\ A(s) &= \frac{C(1+2n)}{R_n}(s+2n) + O((s+2n)^2) & (s \rightarrow -2n), \\ A(s) &= -\frac{R_n}{C(1+2n)} \frac{1}{s-(1+2n)} + O(1) & (s \rightarrow 1+2n), \quad n \geq 1. \end{aligned} \quad (\text{OZR2.11})$$

For example $H\zeta(s) = -1/s + O(1)$ at zero, so its pole times the displayed zero of A gives $\zeta(0) = -1/2$. At one, $H\zeta(s) = -1/2 + O(s-1)$, so the displayed pole of A gives the residue one of ζ . At $-2n$ a zero of A creates the original trivial zero; at $1+2n$ the pole of A cancels the reflected trivial zero. A holomorphic zero-jet isomorphism must therefore not be asserted between those exceptional points. The meromorphic functional equation remains valid there with exactly these orders.

On either boundary line $\text{Re } s = 0$ or $\text{Re } s = 1$, every point other than the endpoints $0, 1$ belongs to U ; the holomorphic-unit calculations extend to a neighborhood of each such point. This statement neither assumes nor proves a zero-free boundary theorem.

OZR3. The unital algebra map and the weighted involution

On $\mathcal{O}(S)$, H is additive, conjugate-linear over $\mathbb{C}$, multiplicative, unital, and involutive. For example

$$H(fg) = (Hf)(Hg), \quad H1 = 1, \quad H^2f = f. \quad (\text{OZR3.1})$$

Each identity follows by substitution into (OZR1.2). Define

$$Tf = AHf. \quad (\text{OZR3.2})$$

Equations (OZR1.5) and (OZR1.6) prove

$$T\zeta = \zeta, \quad T^2f = A(HA)H^2f = f. \quad (\text{OZR3.3})$$

Nevertheless

$$T1 = A, \quad T(fg) = A^{-1}(Tf)(Tg). \quad (\text{OZR3.4})$$

The function A is not identically one: approaching one through the real interval inside S , (OZR2.11) gives its pole. Therefore T is not a unital algebra automorphism on $\mathcal{O}(S)$. It is a conjugate-linear involution on that vector space which fixes the specified function ζ . The distinction is an exact product formula, not an assertion that the two maps are unrelated. It is H , not an unweighted treatment of T , that gives the unital algebra comparisons below.

OZR4. Full derivative transport, including derivatives of the multiplier

A Taylor expansion or induction on the derivative gives

$$(Hf)^{(j)}(s) = (-1)^j \overline{f^{(j)}(r(s))}. \quad (\text{OZR4.1})$$

Differentiating (OZR1.5) by the product rule, for every integer $n \geq 0$, gives the complete identity

$$\zeta^{(n)}(s) = \sum_{j=0}^n \binom{n}{j} A^{(n-j)}(s) (-1)^j \overline{\zeta^{(j)}(r(s))}, \quad s \in S. \quad (\text{OZR4.2})$$

The derivatives of A may themselves be written without dropping any factor of C . On S , set $\psi = \Gamma'/\Gamma$ and

$$L_j(s) = (-1)^{j-1} (j-1)! \left(\frac{1}{s^j} + \frac{1}{(s-1)^j} \right) - \mathbf{1}_{j=1} \frac{\log \pi}{2} + 2^{-j} \psi^{(j-1)}(s/2), \quad j \geq 1. \quad (\text{OZR4.3})$$

These are the j -th derivatives of any local holomorphic logarithm of the original C . The constant factor $1/2$ remains in A ; its logarithmic derivatives are zero. Put

$$B_j(s) = (-1)^j L_j(1-s) - L_j(s). \quad (\text{OZR4.4})$$

For $k \geq 1$, repeated differentiation of the exponential of a local logarithm proves

$$A^{(k)}(s) = A(s) k! \sum_{\substack{q_1, \dots, q_k \geq 0 \\ q_1 + 2q_2 + \dots + kq_k = k}} \prod_{j=1}^k \frac{B_j(s)^{q_j}}{(j!)^{q_j} q_j!}. \quad (\text{OZR4.5})$$

One proof is to expand $A(s+h) = A(s) \exp(\sum_{j \geq 1} B_j(s) h^j / j!)$ and collect the coefficient of h^k ; each displayed tuple records exactly all choices contributing that degree. For $k = 0$, the formula is $A^{(0)} = A$. In particular

$$\frac{A'(s)}{A(s)} = - \left(\frac{1}{1-s} + \frac{1}{(1-s)-1} - \frac{\log \pi}{2} + \frac{1}{2} \psi((1-s)/2) \right) - \left(\frac{1}{s} + \frac{1}{s-1} - \frac{\log \pi}{2} + \frac{1}{2} \psi(s/2) \right). \quad (\text{OZR4.6})$$

Both endpoint contributions, both Gamma derivative contributions, and both powers-of-pi contributions remain displayed. Equations (OZR4.2)–(OZR4.6) specify every derivative term in the original-zeta comparison.

OZR5. Actual zero algebras and all multiplicities

Let

$$Z = \{\rho \in S : \zeta(\rho) = 0\}. \quad (\text{OZR5.1})$$

These are actual zeros of the original function. Since ζ is not the zero function, its zeros in S are isolated and each has a finite multiplicity $m_\rho \geq 1$. At ρ , its original Taylor expansion is

$$\zeta(\rho + z) = z^{m_\rho} u_\rho(z), \quad u_\rho(z) = \sum_{j=0}^{\infty} \frac{\zeta^{(m_\rho+j)}(\rho)}{(m_\rho + j)!} z^j, \quad u_\rho(0) \neq 0. \quad (\text{OZR5.2})$$

The unit u_ρ is retained as the explicitly displayed function; it has not replaced the original zeta function. Let $\mathcal{O}_\rho$ be the ring of holomorphic germs at ρ , and define the full zero algebra

$$J_\rho = \mathcal{O}_\rho / (\zeta). \quad (\text{OZR5.3})$$

For a germ f at ρ , the formula

$$H_\rho f(s) = \overline{f(r(s))} \quad (s \text{ near } r\rho) \quad (\text{OZR5.4})$$

is a unital conjugate-linear ring isomorphism $\mathcal{O}_\rho \rightarrow \mathcal{O}_{r\rho}$, inverse to $H_{r\rho}$. Since

$$H_\rho \zeta = A^{-1} \zeta \quad (\text{OZR5.5})$$

near $r\rho$, and A is a unit there, the principal ideal generated by the original ζ maps exactly onto the corresponding principal ideal. Therefore it induces a unital conjugate-linear algebra isomorphism

$$h_\rho : J_\rho \xrightarrow{\sim} J_{r\rho}, \quad h_{r\rho} h_\rho = 1. \quad (\text{OZR5.6})$$

The isomorphism preserves the full multiplicity, not just the set of zero locations. Explicitly, with $w = s - r\rho$,

$$h_\rho \left(\left[\sum_{j=0}^{m_\rho-1} c_j z^j \right] \right) = \left[\sum_{j=0}^{m_\rho-1} (-1)^j \overline{c_j} w^j \right]. \quad (\text{OZR5.7})$$

To justify these coordinates, (OZR5.2) and invertibility of u_ρ show that the kernel of Taylor truncation is exactly $(\zeta) = (z^{m_\rho})$. Thus every class has precisely the displayed m_ρ coefficients. This is an explicitly proved coordinate map on the original quotient, not deletion of u_ρ from any analytic formula.

Alternatively (OZR4.2) evaluated at $r\rho$ shows that derivatives of orders below m_ρ vanish there, while

$$\zeta^{(m_\rho)}(r\rho) = (-1)^{m_\rho} A(r\rho) \overline{\zeta^{(m_\rho)}(\rho)} \neq 0. \quad (\text{OZR5.8})$$

Hence $r\rho \in Z$ and $m_{r\rho} = m_\rho$. For every $\ell \geq 0$, all higher terms remain

$$\zeta^{(m_\rho+\ell)}(r\rho) = \sum_{j=m_\rho}^{m_\rho+\ell} \binom{m_\rho+\ell}{j} A^{(m_\rho+\ell-j)}(r\rho) (-1)^j \overline{\zeta^{(j)}(\rho)}. \quad (\text{OZR5.9})$$

Terms below m_ρ in (OZR4.2) are exactly zero by definition of multiplicity; no nonzero contribution has been suppressed.

Multiplication by A after h_ρ gives the corresponding quotient map induced by T . It is conjugate-linear and invertible, and obeys the weighted product identity (OZR3.4). Its value at the unit is $[A]$, so the unital algebra map used here remains h_ρ ; no assertion is made that $[A] \neq 1$ in every individual zero algebra.

OZR6. Recovered arithmetic characters and the complete nilpotent action

At the stage after the whole arithmetic has been reconstructed, as stipulated in OZR0, let $a > 1$ be any positive integer in that recovered arithmetic. Its real logarithm $\log a > 0$ gives the entire, nonvanishing function

$$E_a(s) = a^s = \exp(s \log a). \quad (\text{OZR6.1})$$

For every actual $\rho \in Z$, its scalar value transforms as

$$E_a(r\rho) = a^{1-\bar{\rho}} = \frac{a}{a^\rho}. \quad (\text{OZR6.2})$$

This calculation is downstream of whole arithmetic recovery and does not select a prime to construct that arithmetic.

Let $F_{a,\rho}$ be multiplication by $[E_a]$ on J_ρ , and set

$$V_{a,\rho} = a F_{a,\rho}^{-1}. \quad (\text{OZR6.3})$$

Invertibility follows from nonvanishing of E_a . Both operators keep all infinitesimal terms. In the exact coordinates of (OZR5.7),

$$\begin{aligned} F_{a,\rho} &= a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} M_z^j, \\ V_{a,\rho} &= a^{1-\rho} \sum_{j=0}^{m_\rho-1} \frac{(-\log a)^j}{j!} M_z^j, \quad M_z^{m_\rho} = 0, \end{aligned} \quad (\text{OZR6.4})$$

where M_z is multiplication by $z = s - \rho$. The finite sums are exact because the remaining terms belong to the original ideal $(\zeta) = (z^{m_\rho})$. Multiplication of the two original exponential functions proves

$$F_{a,\rho} V_{a,\rho} = V_{a,\rho} F_{a,\rho} = a 1_{J_\rho}. \quad (\text{OZR6.5})$$

Since $H E_a = a E_a^{-1}$, the full jet isomorphism satisfies

$$h_\rho F_{a,\rho} = V_{a,r\rho} h_\rho, \quad h_\rho V_{a,\rho} = F_{a,r\rho} h_\rho. \quad (\text{OZR6.6})$$

For an explicit check, (OZR5.7) sends M_z to $-M_w$ and scalar a^ρ to $\overline{a^\rho}$. The right side has scalar $a^{1-r\rho} = \overline{a^\rho}$ and precisely the powers $(-\log a)^j M_w^j / j!$. Every coefficient agrees. Thus (OZR6.2) and (OZR6.6) retain both the eigenvalue reflection and all multiplicity terms.

The exact diagram is

$$\begin{array}{ccc} J_\rho & \xrightarrow{F_{a,\rho}} & J_\rho \\ h_\rho \downarrow & & \downarrow h_\rho \\ J_{r\rho} & \xrightarrow{V_{a,r\rho}} & J_{r\rho}. \end{array} \quad (\text{OZR6.7})$$

Its vertical maps are conjugate-linear unital algebra isomorphisms; its horizontal maps are the stated multiplication operators. No change to the recovered integer a occurs in the diagram.

OZR7. The reflection-invariant defect on actual zeros

For $\rho \in Z$, write $x_a(\rho) = |a^\rho|^2 > 0$ and define

$$\Delta_a(\rho) = |a^\rho|^2 + \frac{a^2}{|a^\rho|^2} - 2a. \quad (\text{OZR7.1})$$

The individual summands, the factor a^2 , and the term $-2a$ remain present. Expanding a square gives the additional exact identity

$$\Delta_a(\rho) = \frac{(|a^\rho|^2 - a)^2}{|a^\rho|^2} \geq 0. \quad (\text{OZR7.2})$$

By (OZR6.2),

$$x_a(r\rho) = \frac{a^2}{x_a(\rho)}. \quad (\text{OZR7.3})$$

Substitution into the full formula proves

$$\Delta_a(r\rho) = \frac{a^2}{x_a(\rho)} + \frac{a^2}{a^2/x_a(\rho)} - 2a = \Delta_a(\rho). \quad (\text{OZR7.4})$$

Finally, writing $\rho = \beta + i\gamma$, the original exponential gives $x_a(\rho) = \exp(2\beta \log a)$. Since $\log a > 0$,

$$\Delta_a(\rho) = 0 \iff x_a(\rho) = a \iff \operatorname{Re} \rho = \frac{1}{2} \iff r\rho = \rho. \quad (\text{OZR7.5})$$

This is an exact test on each actual zero. It proves neither that any actual zero has positive defect nor that all actual zeros have zero defect.

OZR8. Exact descent retains the defect and the zero algebras

Let $Q = Z/\langle r \rangle$ be the set of reflection orbits, with quotient map $q : Z \rightarrow Q$. Each orbit consists of either one actual zero or two actual zeros, and multiplicity is constant on the orbit by OZR5. Define

$$\overline{\Delta}_a(q(\rho)) = \Delta_a(\rho), \quad \overline{m}(q(\rho)) = m_\rho. \quad (\text{OZR8.1})$$

Equations (OZR7.4) and (OZR5.8) prove that both definitions are independent of the representative. Since q is onto, these are the unique functions satisfying

$$\Delta_a = \overline{\Delta}_a \circ q, \quad m = \overline{m} \circ q. \quad (\text{OZR8.2})$$

In particular the quotient does not make the defect zero: it retains its exact value. Explicitly,

$$\{\overline{\Delta}_a = 0\} = q(\{\rho \in Z : r\rho = \rho\}). \quad (\text{OZR8.3})$$

The reverse inclusion and forward inclusion both follow from (OZR7.5), applied to any representative. Equation (OZR8.3) concerns the actual zero set; it asserts no existence of an orbit outside this locus.

There is also a descent algebra retaining the full jet data. For each actual orbit $O \in Q$, put

$$R_O = \prod_{\rho \in O} J_\rho, \quad (\iota f)_\rho = h_{r\rho}(f_{r\rho}), \quad B_O = R_O^\iota. \quad (\text{OZR8.4})$$

This finite product and its fixed subalgebra are algebra constructions on zero jets. They are not addition of the user's branch counters. The map ι is a unital involution of real algebras, because the component maps in (OZR5.6) are unital, multiplicative, and mutually inverse. Thus B_O is well defined.

For an orbit with two distinct representatives $\rho, r\rho$, projection onto the first component gives a real-algebra isomorphism

$$B_O \xrightarrow{\sim} J_\rho, \quad f \mapsto f_\rho, \quad g \mapsto (g, h_\rho g) \quad (\text{OZR8.5})$$

as inverse. The formula proves injectivity and surjectivity and preserves addition, multiplication, and the unit. Choosing the other representative changes this presentation by the exact map h_ρ , not by an identification of recovered counters.

For a one-point orbit $O = \{\rho\}$, the original coordinate $z = s - \rho$ obeys $h_\rho z = -z$. Therefore a class $\sum_{j=0}^{m_\rho-1} c_j z^j$ is fixed exactly when

$$c_j = (-1)^j \overline{c_j} \quad (0 \leq j < m_\rho). \quad (\text{OZR8.6})$$

Every even coefficient is real, and every odd coefficient is purely imaginary. Equivalently the map

$$\mathbb{R}[u]/(u^{m_\rho}) \longrightarrow B_O, \quad u \mapsto i(s - \rho) \quad (\text{OZR8.7})$$

is a real-algebra isomorphism: $h_\rho(iz) = iz$, and each coefficient satisfying (OZR8.6) has a unique expression $c_j = b_j i^j$ with $b_j \in \mathbb{R}$. The original coordinate and the exact change of coordinate are both displayed. Nilpotent orders and all m_ρ coefficients remain recorded in either case.

The defect enters this descended algebra as the real scalar

$$\overline{\Delta}_a(O) 1_{B_O}. \quad (\text{OZR8.8})$$

Its component in every original zero algebra is exactly $\Delta_a(\rho) 1_{J_\rho}$. The unit of each J_ρ is nonzero because ζ lies in its maximal ideal, so the original ideal is proper. Consequently the scalar in (OZR8.8) is zero exactly when its real coefficient is zero. This proves retention in the algebra, not only in the quotient's underlying set.

Recovery of the entire complex jet algebra from the descended real algebra is also explicit. The map

$$\Phi_O : B_O \otimes_{\mathbb{R}} \mathbb{C} \longrightarrow R_O, \quad b \otimes \lambda \longmapsto \lambda b \quad (\text{OZR8.9})$$

is a complex-algebra isomorphism. For $x \in R_O$, put

$$b(x) = \frac{x + \iota x}{2}, \quad c(x) = \frac{x - \iota x}{2i}. \quad (\text{OZR8.10})$$

Conjugate-linearity and $\iota^2 = 1$ give $\iota b(x) = b(x)$ and $\iota c(x) = c(x)$, so both belong to B_O . Direct substitution gives $x = b(x) + ic(x)$. Conversely every tensor is uniquely of the form $b \otimes 1 + c \otimes i$ with $b, c \in B_O$; when $b + ic = 0$, applying ι also gives $b - ic = 0$, hence $b = c = 0$. Thus the inverse of (OZR8.9) is

$$x \longmapsto b(x) \otimes 1 + c(x) \otimes i. \quad (\text{OZR8.11})$$

The forward map preserves multiplication and the unit by its formula, so its proved inverse is an algebra inverse. This recovers every component and all its nilpotents, with the original label set supplied by the same orbit O in (OZR8.4).

Both cases in (OZR8.5)–(OZR8.7) describe actual orbits whenever those orbits occur; the two-point calculation does not claim that the actual zero set contains such an orbit. Every branch counter remains the already reconstructed counter stipulated in OZR0.

OZR9. The calculated relation to the proposed argument

The same original zeta function has the exact weighted reflection (OZR1.5). Its original zero ideals have the unital conjugate-linear comparison (OZR5.6), with every multiplicity and every derivative contribution retained. Recovered arithmetic character operators have the full comparison (OZR6.6). The resulting reflection quotient exists as the explicitly constructed set and real algebras in OZR8, and retains the invariant defect (OZR8.8).

For these exact objects, the statement that $\overline{\Delta}_a$ vanishes on every actual reflection orbit is equivalent, by (OZR7.5) and surjectivity of q , to every zero in S having real part $1/2$. The calculation does not establish that vanishing. Existence of this quotient establishes descent of the observed defect; it is not itself a vanishing calculation.

This is a proved part of the requested study of deviations preserving the original zeta function. It leaves the user's broader class of admissible constructions and its specified counter quotients intact. It supplies exact maps and a retained observable for that study, without substituting a timing example, changing the arithmetic, assuming RH, claiming an actual off-critical zero, or transferring a Deligne theorem without its hypotheses and receiving map.

In particular this file does not assert that reflection is an automorphism of a stronger marked arithmetic structure in which every individual operator $F_{a,\rho}$ must be fixed. The proved formula is the exact

interchange with $V_{a,r\rho}$ in (OZR6.6). Any comparison with such a stronger structure must retain that formula, the weighted behavior of T , and the unital behavior of H together.

Winding reversal and the fixed supporting point

Proof source: WINDING_REVERSAL_AND_FIXED_SUPPORT.md.

Winding reversal, exchanged data, and the fixed supporting point

24 September 2026. Proof locators WR1-WR9. This note receives the question whether reversing a winding based at the common supporting point creates handedness, and whether that would contradict the point's proved absence of an exchanged parity label. It computes the actual source involution and its receiving maps. The support is always $\tau\langle Z_1; \text{no } Z_2 \rangle$, in the precise exchanged-point-label sense proved below. No arithmetic, coordinate, metric or vector is assigned to that support. The stalk group is not identified with a fundamental group.

Original source actually read

Alain Connes and Caterina Consani, [The Absolute Twistor Line and the Geometry of the compactified Spec Z](#), [arXiv:2609.00299v1](#), original author file CC.tex, SHA-256 57a10dcef5cd758e6b2f1c54b1c0a11b0fb093638f74fb1be515ec3bc7080ba4.

Passages read for this calculation: lines 314–326, equation F2, retains $\epsilon^2 = 1$; lines 492–612, especially `opens1`, `opens`, `eq:restriction_maps`, give the topology, generic group and both twisted restrictions; lines 765–852, definition `def:alpha_symmetry`, give the actual action on points and sections with its chart compatibility; lines 875–904 give the invariant base and the generic-point isotropy. These author passages supply the source data. All group, character and descent calculations below are proved here.

For the final arithmetic comparison we use the already proved original-zeta action from `ORIGINAL_Z ETA_RESOLVENT_AND_PROJECTORS.md`, `RZ8-RZ11`, and `INTEGRAL_FROBENIUS_TRANS FER_ALGEBRA.md`, `FT5`. The comparison retains the degree factor n and every multiplicity jet. It does not identify the source's signed involution with an unweighted analytic inverse.

WR1. The common supporting point and the full generic data

The underlying unfolded space and its chart overlap are

$$X = \{+, -, \eta\}, \quad \Omega_+ = \{+, \eta\}, \quad \Omega_- = \{-, \eta\}, \quad \Omega_+ \cap \Omega_- = \{\eta\}. \quad (\text{WR1.1})$$

The nonempty proper opens are exactly these three sets. The involution exchanges the two closed points. A homeomorphism must fix η , because its singleton is the unique dense singleton; the singleton of either closed point is closed. Hence its fixed-point set on X is exactly $\{\eta\}$.

The group of nonabsorbing generic monomials, retaining all source commutations, is

$$G = \langle T, J, \epsilon : TJ = JT, T\epsilon = \epsilon T, J\epsilon = \epsilon J, J^2 = \epsilon, \epsilon^2 = 1 \rangle. \quad (\text{WR1.2})$$

The separate absorbing section of the pointed source is outside this group. It is neither removed from that pointed source nor identified with the supporting point in this calculation.

Every group element has a unique expression

$$g = T^m J^k, \quad m \in \mathbb{Z}, \quad k \in \mathbb{Z}/4\mathbb{Z}. \quad (\text{WR1.3})$$

To prove uniqueness as well as existence, the relations give a homomorphism $G \rightarrow \mathbb{Z} \times C_4$, sending T to $(1, 0)$, J to $(0, 1)$ and ϵ to $(0, 2)$. Its inverse sends (m, k) to $T^m J^k$; it is well defined because $J^4 = 1$. Both composites fix the specified generators, so both are identities.

The actual restrictions are also retained:

$$\rho_+(T^m J_+^k) = T^m J^k \quad (m \geq 0), \quad \rho_-(T^m J_-^k) = T^m (\epsilon J)^k \quad (m \leq 0). \quad (\text{WR1.4})$$

The finite subgroup is $H = \langle J \rangle \cong C_4$. The complete winding quotient and its kernel are

$$1 \longrightarrow H \longrightarrow G \xrightarrow{q} L_w = G/H \longrightarrow 1, \quad L_w \cong \mathbb{Z}. \quad (\text{WR1.5})$$

No element of this exact sequence is identified with η .

WR2. The full involution, its fixed elements and its two comparison maps

On the group write α for the source's α^* . Its exact formulas are

$$\alpha(T) = \epsilon T^{-1}, \quad \alpha(J) = J^{-1} = \epsilon J, \quad \alpha(\epsilon) = \epsilon. \quad (\text{WR2.1})$$

For every element in (WR1.3),

$$\alpha(T^m J^k) = \epsilon^m T^{-m} J^{-k} = T^{-m} J^{2m-k}, \quad \alpha(m, k) = (-m, 2m - k). \quad (\text{WR2.2})$$

Applying this coordinate rule twice gives (m, k) , including the full residue modulo four, proving the involution property.

Its fixed subgroup is exactly

$$G^\alpha = \{1, \epsilon\}. \quad (\text{WR2.3})$$

Indeed equality of the winding coordinates in $(m, k) = (-m, 2m - k)$ gives $m = 0$. The remaining condition is $2k = 0$ modulo four, so $k = 0$ or $k = 2$. Conversely both displayed elements are fixed. Together with the separate fixed absorbing section, this recovers the source's pointed invariant base $\{0, 1, \epsilon\}$ at the monomial level. This does not turn those sections into three underlying supporting points.

The product with the reversed datum and the quotient by the original datum are respectively

$$N(g) := g\alpha(g) = \epsilon^m, \quad (\text{WR2.4})$$

$$D(g) := \alpha(g)g^{-1} = T^{-2m} J^{2m-2k} = T^{-2m} \epsilon^{m-k}. \quad (\text{WR2.5})$$

Both are group homomorphisms, because the source group is abelian. Their exact images and kernels are

$$\text{im } N = \{1, \epsilon\}, \quad \ker N = \langle T^2, J \rangle, \quad (\text{WR2.6})$$

$$\ker D = \{1, \epsilon\}, \quad \text{im } D = \langle T^2, \epsilon \rangle. \quad (\text{WR2.7})$$

The norm assertions follow directly from whether m is even. For the second image, (WR2.5) has even winding exponent and even J -exponent. Conversely $D(J) = \epsilon$ and $D(T) = \epsilon T^{-2}$, so their products generate both ϵ and T^2 . The kernel is precisely the fixed subgroup already proved.

These formulas keep a small remaining quotient explicit:

$$\ker N / \text{im } D \cong C_2, \quad [J] \text{ generates it, } [J]^2 = [\epsilon] = 1. \quad (\text{WR2.8})$$

Every element of $\ker N$ has the form $T^{2a} J^k$; quotienting by T^2 and $\epsilon = J^2$ leaves exactly k modulo two. This proves the quotient, without identifying it with an unstated cohomology or loop space.

WR3. Handedness of winding and parity of the winding count

The involution induces inversion on L_w : in exponent coordinates, $m \mapsto -m$. On its nonzero winding elements, a choice of the positive source chart defines the exchanged label

$$o(m) = \begin{cases} \text{right}, & m > 0, \\ \text{left}, & m < 0. \end{cases} \quad o(-m) = \text{flip}(o(m)). \quad (\text{WR3.1})$$

Reversing the initial chart exchanges the two names. Thus the two-state information is the two possible orientations, with no assertion that one of the names is intrinsically preferred. The sign of a nonzero winding changes under reversal.

There is also the count-parity map already derived from the signed product:

$$\delta : L_w \longrightarrow \{1, \epsilon\}, \quad \delta(m) = \epsilon^m, \quad \ker \delta = 2L_w. \quad (\text{WR3.2})$$

It is well defined because the product $g\alpha(g)$ is independent of the finite J -label. Unlike (WR3.1), its value satisfies

$$\delta(-m) = \epsilon^{-m} = \epsilon^m = \delta(m). \quad (\text{WR3.3})$$

Thus the source gives two explicitly related observations: reversal exchanges the orientation, and it preserves whether the integer winding count is even or odd. Both take two possible values, but their involution actions are different. These equations specify the distinction rather than erasing either symmetry.

The zero-winding fiber still contains the four finite-order states $1, J, \epsilon, J^3$. The states $1, \epsilon$ are fixed, while J, J^3 are exchanged. Therefore zero winding alone is not an assertion that every remaining phase datum is fixed.

The connection with counting is also exact on the full signed lift. Write $A_G = \alpha$ as a permutation of G , and define the permutations

$$S(g) = Tg, \quad E(g) = \epsilon g. \quad (\text{WR3.4})$$

These last two maps are translations of the stalk group, not group homomorphisms fixing its identity. The map S uses the positive chart's primitive generator; reversing the chosen chart changes that oriented choice. Direct substitution gives

$$A_G S A_G(g) = \alpha(T\alpha(g)) = \epsilon T^{-1}g = E S^{-1}(g), \quad (\text{WR3.5})$$

$$E^2 = I, \quad ES = SE, \quad EA_G = A_GE. \quad (\text{WR3.6})$$

The first equality uses $\epsilon^2 = 1$, the second the source commutation of T, ϵ , and the third $\alpha(\epsilon) = \epsilon$. Thus the full signed counting/reversal relation is $A_G S A_G = E S^{-1}$; its factor E has not been suppressed.

Pass through the retained quotient $q : G \rightarrow L_w$, on which E becomes the identity because $\epsilon \in H$. In winding exponent coordinates the induced permutations are

$$S_w(m) = m + 1, \quad R_w(m) = -m, \quad R_w S_w R_w = S_w^{-1}. \quad (\text{WR3.7})$$

All three identities follow by evaluating at m ; for the last, $m \mapsto -m \mapsto -m + 1 \mapsto m - 1$. This is the quotient comparison of the full lifted relation, not a replacement of (WR3.5).

Let $\pi_{\text{par}}(g) = g\alpha(g) = \epsilon^m$. The exact observation identities are

$$\pi_{\text{par}}(Sg) = \epsilon \pi_{\text{par}}(g), \quad \pi_{\text{par}}(A_G g) = \pi_{\text{par}}(g), \quad \pi_{\text{par}}(Eg) = \pi_{\text{par}}(g). \quad (\text{WR3.8})$$

The first follows from $(Tg)\alpha(Tg) = \epsilon g\alpha(g)$, the second from $\alpha(g)g = g\alpha(g)$, and the third from $\epsilon\alpha(\epsilon) = 1$. Therefore a one-step advance in the complete winding history flips the parity observation;

reversing the entire history preserves that observation while exchanging the nonzero-winding handedness (WR3.1).

This also types the user's half-turn counting picture precisely at the observation level: S lifts the order-two flip on $\{1, \epsilon\}$, but S itself retains the winding history and has infinite order. Indeed $S^n g = T^n g$, which equals g only for $n = 0$, by the unique coordinate in (WR1.3). Thus no identification of the full counting step with an order-two geometric self-map has been inserted. The source involution A_G and the counting translation S have the exact relation (WR3.5), rather than being declared the same map.

WR4. The exact forgetful map and the obstruction to label descent

Form the set of decorated support data

$$\mathcal{E} = \{\eta\} \times G, \quad \tilde{\alpha}(\eta, g) = (\eta, \alpha(g)). \quad (\text{WR4.1})$$

This product records a point with a stalk element. It is not a claim that the stalk is the fundamental group or that the product is the whole source sheaf. The actual operation of forgetting the stalk label is

$$\pi : \mathcal{E} \longrightarrow \{\eta\}, \quad \pi(\eta, g) = \eta. \quad (\text{WR4.2})$$

It is equivariant, since

$$\pi \tilde{\alpha}(\eta, g) = \eta = \alpha \pi(\eta, g). \quad (\text{WR4.3})$$

Its source fixed set is $\{\eta\} \times \{1, \epsilon\}$, by WR2; its entire target is fixed. In particular two exchanged decorations can have the same fixed support, as this explicit map proves.

Let $\mathcal{E}_{\neq 0}$ be the subset with nonzero winding. The handedness label o from (WR3.1) is an equivariant map from $\mathcal{E}_{\neq 0}$ to the free two-element exchange set Q_2 . It cannot factor through π . If $o = \bar{o} \circ \pi$, then for any element e of this subset,

$$o(\tilde{\alpha}e) = \bar{o}(\eta) = o(e), \quad (\text{WR4.4})$$

whereas equivariance requires $o(\tilde{\alpha}e) = \text{flip}(o(e)) \neq o(e)$. This is a contradiction. The full two-to-one reversal orbit map on $\mathcal{E}_{\neq 0}$ consequently carries orientation data that the further map to the support does not retain.

Equivalently, an exchanged point-label directly at the support would require

$$P(\alpha\eta) = \text{flip}(P(\eta)), \quad \alpha\eta = \eta, \quad \text{hence } P(\eta) = \text{flip}(P(\eta)), \quad (\text{WR4.5})$$

which is impossible in Q_2 . This derives the support's no Z_2 exchange-label property. It neither removes the nontrivial action on its stalk nor removes the quotient stack's C_2 stabilizer noted in the original source, lines 895–898. In particular an invariant parity observation such as (WR3.2) is not an exchanged point-label of the kind ruled out by (WR4.5).

WR5. Rank-one characters and the complete signed duality

Every complex rank-one character $\chi : G \rightarrow \mathbb{C}^\times$ is determined by

$$a = \chi(T) \in \mathbb{C}^\times, \quad \omega = \chi(J), \quad \omega^4 = 1, \quad \chi(\epsilon) = \omega^2, \quad (\text{WR5.1})$$

and has the full formula

$$\chi(T^m J^k) = a^m \omega^k. \quad (\text{WR5.2})$$

The group presentation proves both existence for every displayed parameter pair and uniqueness. The support of this character datum remains η .

Precomposition by the source involution gives

$$\boxed{\chi \circ \alpha : (a, \omega) \mapsto (\omega^2 a^{-1}, \omega^{-1})}. \quad (\text{WR5.3})$$

The ordinary dual character is χ^{-1} , with parameters (a^{-1}, ω^{-1}) . Their exact comparison is

$$(\chi \circ \alpha)(T^m J^k) = (\omega^2)^m \chi(T^m J^k)^{-1}. \quad (\text{WR5.4})$$

In particular the signed factor is not removed. For a character preserving the source's complex base sign $\epsilon \mapsto -1$, one has $\omega = +i$ or $-i$, and the formula is $(a, \omega) \mapsto (-a^{-1}, \omega^{-1})$.

The abstract characters fixed by (WR5.3) are exactly

$$a \in \{1, -1\}, \quad \omega \in \{1, -1\}. \quad (\text{WR5.5})$$

Indeed fixedness requires $\omega = \omega^{-1}$, hence $\omega^2 = 1$, and then $a^2 = \omega^2 = 1$. Both conditions are sufficient. Thus no base-sign-preserving character with $\omega^2 = -1$ is fixed by this operation, even though its underlying supporting point is fixed.

For completeness, combining source reversal with complex conjugation gives

$$\overline{\chi \circ \alpha} : (a, \omega) \mapsto (\omega^2 / \bar{a}, \omega). \quad (\text{WR5.6})$$

Here $|\omega| = 1$, so $\overline{\omega^{-1}} = \omega$ and $\overline{\omega^2} = \omega^2$. Fixedness now requires $|a|^2 = \omega^2$. When $\omega^2 = 1$, this is exactly $|a| = 1$; when $\omega^2 = -1$, there is no solution. These are statements about fixed character data, not about which characters may exist over the fixed supporting point. Every nonzero a in (WR5.1) still defines a valid abstract character.

The existence of an equivariant fiber does not require fixing one character. For any character χ , define on $\mathbb{C}^2$

$$R_\chi(g) = \begin{pmatrix} \chi(g) & 0 \\ 0 & (\chi \circ \alpha)(g) \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (\text{WR5.7})$$

Then $S^2 = I$ and direct multiplication proves

$$SR_\chi(g)S^{-1} = R_\chi(\alpha(g)). \quad (\text{WR5.8})$$

This is a representation of the actual semidirect product $G \rtimes \langle \alpha \rangle$ on a fiber over the fixed support, with its exchanged summands retained. It is not asserted to be the full arithmetic cohomology or a source of zeta zeros. It proves that fixed support and exchanging fiber data coexist under the actual source group law, rather than under an unrelated model.

WR6. The exact degree-weighted arithmetic character comparison

On the positive rational multiplicative group the actual spectral characters are

$$\lambda_\rho(r) = r^\rho = \exp(\rho \log r), \quad r > 0, \quad (\text{WR6.1})$$

with the real logarithm. Their degree-weighted conjugate dual is

$$\lambda_\rho^\dagger(r) = \frac{r}{\lambda_\rho(r)} = r^{1-\bar{\rho}} = \lambda_{\rho^\#}(r), \quad \rho^\# = 1 - \bar{\rho}. \quad (\text{WR6.2})$$

This formula retains the arithmetic degree character $r \mapsto r$; it is not the unweighted reciprocal in (WR5.3).

There is a full signed comparison on each prime and each retained phase sector, rather than an assertion that the two dualities are unrelated. Fix a recovered prime norm p , and for each of the four possible ω consider characters $\chi_{a,\omega}$ as in (WR5.2). Define the compensating character

$$\mu_{p,\omega}(T) = p\omega^2, \quad \mu_{p,\omega}(J) = 1, \quad \mu_{p,\omega}(\epsilon) = 1. \quad (\text{WR6.3})$$

Its values satisfy every group relation, so it is a well-defined character. Multiplying the full reversed-conjugate character by this character gives

$$\mathfrak{D}_p \chi_{a,\omega} := \mu_{p,\omega} \overline{\chi_{a,\omega} \circ \alpha}, \quad \boxed{\mathfrak{D}_p(a, \omega) = (p/\bar{a}, \omega)}. \quad (\text{WR6.4})$$

Indeed the value at T is $(p\omega^2)(\omega^2/\bar{a}) = p\omega^4/\bar{a} = p/\bar{a}$, while the value at J is ω . Thus both the degree factor and the signed compensation are displayed. The operation preserves each phase sector, including both base-sign-preserving sectors $\omega = \pm i$. It squares to the identity because $p/p/\bar{a} = a$, with the same ω .

For a lifted arithmetic character $a = p^\rho$, (WR6.4) yields exactly $p^{\rho^\#}$. Its fixed-character equation is

$$a = p/\bar{a} \iff |a|^2 = p. \quad (\text{WR6.5})$$

Substituting the actual arithmetic parameter gives

$$|p^\rho|^2 = p^{2\text{Re } \rho} = p \iff \text{Re } \rho = \frac{1}{2}. \quad (\text{WR6.6})$$

This calculation identifies precisely where the purity circle belongs: it is the fixed locus of the degree-weighted duality on character data. All those character data, whether individually fixed or exchanged, can have the same underlying support η through WR4. Fixedness of that support is therefore not the fixed-character equation (WR6.5).

The angular and radial effects can also be read without changing the scale. For a character value $a = re^{i\theta}$, with $r > 0$ and θ taken modulo 2π ,

$$a^{-1} = r^{-1}e^{-i\theta}, \quad \mathfrak{D}_p(a) = \frac{p}{a} = \frac{p}{r}e^{i\theta}. \quad (\text{WR6.7})$$

These are coordinates of the complex character value, not of the supporting point. Unweighted inversion reverses the phase, while conjugation followed by degree-weighted inversion leaves the phase unchanged and exchanges the two radii r and p/r . For the actual arithmetic value, retaining its original scale and imaginary part gives

$$p^\rho = p^{\text{Re } \rho} e^{i(\text{Im } \rho) \log p}, \quad p^{1-\bar{\rho}} = p^{1-\text{Re } \rho} e^{i(\text{Im } \rho) \log p}. \quad (\text{WR6.8})$$

Thus the two operations whose factors were retained in WR6.4 have an explicitly calculated effect. The arithmetic reflection tests equality of these radii; it is not merely the reversal of clockwise versus counterclockwise phase. No radius or distance to τ is introduced by this character calculation.

WR7. The complete original-zeta action and all reversed multiplicity data

On the actual quotient $\mathcal{Q} = \mathcal{B}/\mathcal{I}$ of RZ, the arithmetic operators are

$$T_n[F] = [n^s F(s)], \quad \mathcal{M}^{-1} T_n \mathcal{M} k(u) = n^{1/2} k(u/n). \quad (\text{WR7.1})$$

The full source multiplier remains

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad (\text{WR7.2})$$

with its endpoint and exceptional-point comparisons in the full synthesis proof. The working zeta is the original function; none of its original zeros or poles is replaced by a location assumption.

The reflected Weil form has the exact adjoint identity

$$W(T_n x, y) = W(x, nT_{1/n} y), \quad (\text{WR7.3})$$

and the degree relation $W(T_n x, T_n y) = nW(x, y)$. The scalar n is present. On a full actual block $\mathcal{Q}_\rho$, of multiplicity m_ρ , the operators are

$$T_n|_{\mathcal{Q}_\rho} = n^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log n)^j}{j!} N_\rho^j, \quad (\text{WR7.4})$$

$$nT_{1/n}|_{\mathcal{Q}_\rho} = n^{1-\rho} \sum_{j=0}^{m_\rho-1} \frac{(-\log n)^j}{j!} N_\rho^j. \quad (\text{WR7.5})$$

Here $N_\rho^{m_\rho} = 0$, with its original Jordan length. These are the full truncated exponential identities on the proved quotient, not only eigenvalue formulas.

The conjugate-linear reflection $F^\#(s) = \overline{F(1 - \bar{s})}$ maps the ρ -block to the $\rho^\#$ -block and changes the sign of the nilpotent coordinate. On the specified RZ jet classes the precise assertion is

$$[E_{\rho,j}]^\# = (-1)^j [E_{\rho^\#,j}], \quad (N_\rho x)^\# = -N_{\rho^\#} x^\#. \quad (\text{WR7.6})$$

These are identities in the quotient $\mathcal{Q}$. The sign follows directly from $(s - \rho)^\# = -(s - \rho^\#)$, together with the reflected zero-isolation property of the specified classes. Thus reversal changes character and jet data in a specified way while preserving the supporting role. No assumption that those data must all be fixed has been made.

WR8. The precise result for the proposed contradiction

Reversing a nonzero winding does create the opposite handedness in the sense of (WR3.1). It does not assign that handedness to the supporting point: the exact equivariant map (WR4.2) forgets it, and the attempted descent is contradicted by (WR4.4). The generic support's absence of an exchanged point-label and the stalk's possession of exchanged data are therefore compatible features of the actual source construction.

An actual off-critical zero, if present, would give degree-weighted character data exchanged by (WR6.2), with the full multiplicity action (WR7.4)-(WR7.5). It would not produce an exchanged label on η unless an additional map descending that label to the support had been constructed. WR4 proves that the direct forgetful projection cannot provide such a descent. This establishes the precise extent of this proposed support-based contradiction; it does not exclude a different arithmetic obstruction involving the retained fiber and its pairing.

The resulting positive mathematical connection is the exact diagram of operations already proved: source signed reversal on G , its exchange on winding orientation, its invariant winding parity, the equivariant map to the parityless support, and the signed degree-weighted character comparison (WR6.3)-(WR6.6). The relevant purity equation has not been discarded; it is identified exactly as the character fixed-locus equation while the support remains fixed throughout.

WR9. All translated reversal returns and their exact square

There is also a complete return-map calculation within the retained stalk data. Regard the set G as a left G -torsor: a acts on h by ah , freely and transitively. This is the set of choices of a stalk datum, not a coordinate system on η . A reversal-compatible map of this torsor means a map $R : G \rightarrow G$ satisfying

$$R(ah) = \alpha(a)R(h) \quad (a, h \in G). \quad (\text{WR9.1})$$

Every such map is uniquely of the form

$$R_g(h) = g\alpha(h), \quad g = R(1) \in G. \quad (\text{WR9.2})$$

Indeed set $h = 1$ in WR9.1 and use the commutativity of G . Conversely substitution proves WR9.1 for each displayed R_g . Each is bijective, with the explicit inverse

$$R_g^{-1}(h) = \alpha(g)^{-1}\alpha(h). \quad (\text{WR9.3})$$

Thus every member of this entire family is invertible; this is not deduced from a picture of a loop. These maps are semilinear torsor maps, not asserted to be group homomorphisms or sheaf-algebra automorphisms.

For $g = T^m J^k$, applying the return twice gives exactly

$$R_g^2(h) = g\alpha(g)h = \epsilon^m h. \quad (\text{WR9.4})$$

Since $\epsilon \neq 1$ in the retained C_4 subgroup, this is the identity precisely when m is even. For odd m , the square is the nontrivial central translation by ϵ , while the fourth power is the identity. The signed square therefore detects winding parity without destroying invertibility, and without changing the underlying supporting point.

Every map $h \mapsto ah$ is a change of the chosen element of the same torsor. Conjugating the return by this map gives

$$L_a R_g L_a^{-1}(h) = ag\alpha(a)^{-1}\alpha(h) = R_{gD(a)^{-1}}(h), \quad D(a) = \alpha(a)a^{-1}. \quad (\text{WR9.5})$$

The set of such changes is all equivariant torsor bijections: a bijection commuting with the left G -action is determined by its value at 1, and because G is abelian has this displayed translation form. Thus WR9.5 describes the full equivalence relation, not only some allowed changes.

For the returns whose square is the identity, the parameter g lies in $\ker N = \langle T^2, J \rangle$. Modulo the changes in WR9.5, WR2.8 computes precisely two classes, represented by

$$R_1(h) = \alpha(h), \quad R_J(h) = J\alpha(h). \quad (\text{WR9.6})$$

They are distinct because $J \notin \text{im } D = \langle T^2, \epsilon \rangle$. Both square to the identity since $J\alpha(J) = 1$. For returns whose square is translation by ϵ , the parameters instead form $T \ker N$, and there are again precisely two classes, represented by

$$R_T(h) = T\alpha(h), \quad R_{TJ}(h) = TJ\alpha(h). \quad (\text{WR9.7})$$

Both square to ϵh , and their equivalence classes are distinct by the same quotient calculation. This retains all four possibilities of the signed return calculation.

Their fixed decorations can be computed too. The equation $R_g(h) = h$ is equivalent to $g = D(h)^{-1}$. It has a solution exactly when $g \in \text{im } D$, and then its solution set is a coset of $\ker D = \{1, \epsilon\}$, containing exactly two elements. In particular

$$\text{Fix}(R_1) = \{1, \epsilon\}, \quad \text{Fix}(R_J) = \text{Fix}(R_T) = \text{Fix}(R_{TJ}) = \emptyset. \quad (\text{WR9.8})$$

Thus the second involutive return has only exchanged decorations even though its square is the identity; the two order-four returns have no fixed decoration either. These are exact calculations in the retained fiber.

Projection of every one of these returns through $\{\eta\} \times G \rightarrow \{\eta\}$ is still the identity on the support. Consequently a detectable change in the return, even one that changes its square, is explicitly compatible with the same parityless supporting point. This classifies the translated reversals on the actual stalk torsor. It does not assume that the programme's arithmetic cohomological return is one chosen representative, nor identify these torsor maps with a previously unspecified topological fundamental group.

The integral transfer algebra and both exact receivers

Proof source: INTEGRAL_FROBENIUS_TRANSFER_ALGEBRA.md.

The integral algebra of degree maps and transfers

24 September 2026. Proof labels FT1-FT10. This calculation repairs the scalar denominators identified in IC8 by retaining integral transfers instead of requiring every unweighted inverse to be an integral generator. All algebra acts on the specified coefficient and receiving spaces. There is no addition, metric, coordinate or vector assigned to $\tau\langle Z_1; \text{no } Z_2 \rangle$.

The letters F_n, V_n below name algebra elements. Their geometric receivers are proved explicitly; the letter F by itself does not assert that an operator is a finite-field Frobenius. The final comparison retains the full original-zeta quotient and each zero multiplicity. No RH or zero-location assertion is inserted.

Source calculations used

- INTEGRAL_COUNTING_AND_WEIGHT_COMPARISON.md, IC1-IC9, including the revised IC8.4a-IC8.6: the canonical counting ring, rational group-algebra action and minimal weighted-adjoint closure when every bare inverse is required.
- TATE_H1_DERIVATION.md, T5-T6, read in full, and T7-T8 portions giving the periods, maps, polarizations and origin-independence: the original maps $U_n : E_{p^n} \rightarrow E_p$ and $V_n : E_p \rightarrow E_{p^n}$, both degree n , with full periods $\log p$ and $2\pi i$. The transfer below is derived rather than identified with a pullback in every degree.
- ORIGINAL_ZETA_RESOLVENT_AND_PROJECTORS.md, RZ1, RZ8 and RZ10-RZ11: the entire quotient, actual multiplier a^s , all logarithmic jet coefficients and the exact Weil adjoint identity. Its global synthesis proof remains the source of the equality between the closed arithmetic source image and the complete zero-jet ideal.

FT1. The integral subring and its exact additive basis

In the rational group algebra

$$\Lambda_{\mathbb{Q}} = \mathbb{Q}[\mathbb{Q}_{>0}^{\times}], \quad b_r b_s = b_{rs}, \quad b_1 = 1, \quad (\text{FT1.1})$$

define, for every positive integer n ,

$$F_n = b_n, \quad V_n = nb_{1/n}. \quad (\text{FT1.2})$$

Let $\mathcal{C}$ be the unital subring generated over $\mathbb{Z}$ by all these elements. Sums and products are finite.

Write each positive rational r in its unique reduced form $r = a/b$, where $a, b \geq 1$ and $\gcd(a, b) = 1$. Put

$$\text{den}(r) = b, \quad c_r = bb_r. \quad (\text{FT1.3})$$

Then the exact additive group is

$$\mathcal{C} = \bigoplus_{r \in \mathbb{Q}_{>0}} \mathbb{Z} c_r. \quad (\text{FT1.4})$$

Here is a proof including multiplication. For reduced fractions $r = a/b$, $s = c/d$, put $h = \gcd(ac, bd)$. The reduced denominator of rs is bd/h . Consequently

$$c_r c_s = bd b_{rs} = h c_{rs}, \quad h = \frac{\text{den}(r) \text{den}(s)}{\text{den}(rs)} = \gcd(ac, bd) \in \mathbb{Z}_{>0}. \quad (\text{FT1.5})$$

Thus the right side of (FT1.4) is a subring. It contains $1 = c_1$, $F_n = c_n$ and $V_n = c_{1/n}$, so it contains $\mathcal{C}$. Conversely, for every reduced a/b ,

$$c_{a/b} = F_a V_b, \quad (\text{FT1.6})$$

so all its basis elements belong to $\mathcal{C}$. Finally, distinct b_r are a free rational basis of the ambient group algebra and all denominators in (FT1.3) are nonzero. Therefore the c_r are independent over $\mathbb{Z}$, proving the asserted direct sum and not merely a spanning statement.

Equivalently, an ambient finite sum $\sum_r u_r b_r$ belongs to $\mathcal{C}$ exactly when every u_r is an integer divisible by $\text{den}(r)$. This is an effective membership criterion with its original rational labels retained.

FT2. Presentation by prime generators, with all commutations

There is an injective presentation

$$\mathcal{C} \cong \mathbb{Z}[X_p, Y_p : p \text{ prime}] / (X_p Y_p - p : p \text{ prime}), \quad X_p \mapsto F_p, \quad Y_p \mapsto V_p. \quad (\text{FT2.1})$$

The polynomial algebra is commutative. Explicitly, all its commutation relations, including those used in the proof, are

$$X_p X_q = X_q X_p, \quad Y_p Y_q = Y_q Y_p, \quad X_p Y_q = Y_q X_p \quad (p, q \text{ prime}), \quad (\text{FT2.2})$$

and the degree relations are

$$X_p Y_p = Y_p X_p = p \cdot 1. \quad (\text{FT2.3})$$

Scalar integers commute with every generator.

To prove that no further relation is missing, start with any monomial involving finitely many primes. For each prime replace every matched pair $X_p Y_p$ by the scalar p , until at most one of X_p, Y_p remains. This expresses every polynomial as an integer linear combination of the monomials

$$m_e = \prod_{e_p > 0} X_p^{e_p} \prod_{e_p < 0} Y_p^{-e_p}, \quad e_p \in \mathbb{Z}, \quad e_p = 0 \text{ for all but finitely many } p. \quad (\text{FT2.4})$$

Their images are exactly

$$\left(\prod_{e_p < 0} p^{-e_p} \right) b_{\prod_p p^{e_p}} = c_{\prod_p p^{e_p}}. \quad (\text{FT2.5})$$

Unique prime factorization makes the rational label unique, and FT1 proves independence of these images. Hence a linear combination of the normal monomials has zero image only when each coefficient vanishes. The induced map is injective. It is surjective because

$$F_n = \prod_p F_p^{v_p(n)}, \quad V_n = \prod_p V_p^{v_p(n)} \quad (\text{FT2.6})$$

for every $n \geq 1$. This proves the full presentation, including prime two.

FT3. The weighted involution is integral on this ring

Retain the exact weighted involution on the rational group algebra:

$$b_r^* = r b_{1/r}, \quad q^* = q \quad (q \in \mathbb{Q}). \quad (\text{FT3.1})$$

For reduced $r = a/b$,

$$c_r^* = b \left(\frac{a}{b} b_{b/a} \right) = a b_{b/a} = c_{1/r}. \quad (\text{FT3.2})$$

Thus $\mathcal{C}$ is preserved by $\star$, integrally and exactly. It exchanges the generators:

$$F_n^* = V_n, \quad V_n^* = F_n. \quad (\text{FT3.3})$$

Its square is the identity. It respects multiplication because the ambient assignment does, and (FT3.2) shows explicitly how the additive basis is permuted. The full generator relations are

$$F_1 = V_1 = 1, \quad F_m F_n = F_{mn}, \quad V_m V_n = V_{mn}, \quad F_m V_n = V_n F_m, \quad (\text{FT3.4})$$

$$F_n V_n = V_n F_n = n \cdot 1. \quad (\text{FT3.5})$$

For $d = \gcd(m, n)$, there is the precise cancellation formula

$$\boxed{F_m V_n = d F_{m/d} V_{n/d}.} \quad (\text{FT3.6})$$

Indeed the left side is $n b_{m/n}$, and the right side is $d(n/d) b_{(m/d)/(n/d)} = n b_{m/n}$. The scalar d is retained; dropping it would destroy both the embedding and the transfer degree.

FT4. Comparison with the earlier denominator obstruction

Let $\Lambda = \mathbb{Z}[\mathbb{Q}_{>0}^\times]$. Then

$$\mathcal{C} \subset \Lambda \subset \Lambda_{\mathbb{Q}}, \quad \mathcal{C} \otimes_{\mathbb{Z}} \mathbb{Q} = \Lambda_{\mathbb{Q}}. \quad (\text{FT4.1})$$

The inclusions follow from the coefficient description in FT1. Tensoring with $\mathbb{Q}$ supplies $b_r = c_r / \text{den}(r)$, proving the last equality. However

$$b_{1/p} \notin \mathcal{C} \quad (p \text{ prime}), \quad p b_{1/p} = V_p \in \mathcal{C}. \quad (\text{FT4.2})$$

Membership would require its coefficient one to be divisible by p , contrary to FT1. Thus this ring does not contain every bare inverse, which was an explicit premise of IC8's minimal rational-closure calculation. Both conclusions therefore hold: Λ forces rational coefficients when closed under the weighted involution, while $\mathcal{C}$ already preserves that involution over $\mathbb{Z}$.

Here an integral coefficient form means the explicitly proved free $\mathbb{Z}$ -algebra structure. It is not a claim that F_n is integral over $\mathbb{Z}$ in the sense of satisfying a monic polynomial. In fact, for $n > 1$, a polynomial relation in $F_n = b_n$ would be a relation between the distinct ambient basis elements b_{n^j} . Their independence forces every coefficient to vanish, so there is no nonzero such polynomial. In particular the coefficient construction alone supplies no finite-rank integral lattice for the actual p^ρ eigenvalues. The independent IAR0-IAR2 calculation in `../quantum_tau_programme_bridge_20260924/INTEGRAL_ADJOINT_RETURN_ALGEBRA.md` gives the same coefficient subring and additive basis from the same positive-generator input; its subsequent trace results are separate derivations.

More precisely, adjoining the bare inverses of all F_p to $\mathcal{C}$ gives Λ . The resulting ring contains b_p and $b_{1/p}$, hence every rational monomial by unique factorization, and all its generators were already in Λ . Taking weighted-involution closure at that stage gives $\Lambda_{\mathbb{Q}}$, exactly as IC8 proves.

The additive defect of the first inclusion has the full description

$$0 \longrightarrow \mathcal{C} \longrightarrow \Lambda \longrightarrow \bigoplus_{r \in \mathbb{Q}_{>0}} \mathbb{Z} / \text{den}(r) \mathbb{Z} \longrightarrow 0. \quad (\text{FT4.3})$$

The last map sends the coefficient of b_r to its residue modulo $\text{den}(r)$. Finite support makes the map well defined and onto; its kernel is FT1's membership condition. This is an exact sequence of additive groups. It is not a quotient-ring assertion, since $\mathcal{C}$ is a unital subring rather than an ideal. Its torsion records these coefficient divisibilities and is not identified with the winding history or the finite C_4 stalk subgroup.

FT5. The receiving representation on the complete original-zeta quotient

Use the RZ spaces with their existing topology:

$$\mathcal{Q} = \mathcal{B}/\mathcal{I}, \quad \mathcal{I} = \{F : F^{(j)}(\rho) = 0 \text{ for all actual } \rho, 0 \leq j < m_\rho\}, \quad T_r[F] = [r^s F(s)]. \quad (\text{FT5.1})$$

Here $\mathcal{B}$ consists of the entire functions with every seminorm

$$b_{A,M}(F) = \sup_{|\text{Re } s| \leq A} (1 + |\text{Im } s|)^M |F(s)| < \infty. \quad (\text{FT5.2})$$

The actual source comparison retains

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad \mathcal{M}^{-1} T_r \mathcal{M} k(u) = r^{1/2} k(u/r). \quad (\text{FT5.3})$$

All endpoints, exceptional trivial-zero values and multiplicity data are retained by the prior synthesis and RZ proofs; this note makes no replacement of the original ζ by F_0 .

Restrict the explicitly constructed rational representation from IC8 to obtain

$$\Psi : \mathcal{C} \longrightarrow \text{End}_{\text{cont}}(\mathcal{Q}), \quad \Psi(F_n) = T_n, \quad \Psi(V_n) = nT_{1/n}, \quad \Psi(c_{a/b}) = bT_{a/b}. \quad (\text{FT5.4})$$

It is a unital ring homomorphism because $T_r T_s = T_{rs}$, with (FT1.5) preserving every scalar coefficient. Every image is continuous: for $C = \sum_r k_r c_r$,

$$b_{A,M} \left(\sum_r k_r \text{den}(r) r^s F(s) \right) \leq \left(\sum_r |k_r| \text{den}(r) \max(r^A, r^{-A}) \right) b_{A,M}(F). \quad (\text{FT5.5})$$

The sum is finite. The multipliers preserve all zero orders, so the same estimate descends by taking infima over source representatives. The entire domain and every multiplicity are unchanged.

On each actual block $\mathcal{Q}_\rho$ of dimension $m = m_\rho$, RZ gives $L_\zeta = \rho I + N_\rho$, $N_\rho^m = 0$, with full Jordan length. Consequently

$$\Psi(F_n)|_{\mathcal{Q}_\rho} = n^\rho \sum_{j=0}^{m-1} \frac{(\log n)^j}{j!} N_\rho^j, \quad (\text{FT5.6})$$

$$\Psi(V_n)|_{\mathcal{Q}_\rho} = n^{1-\rho} \sum_{j=0}^{m-1} \frac{(-\log n)^j}{j!} N_\rho^j, \quad (\text{FT5.7})$$

$$\Psi(c_{a/b})|_{\mathcal{Q}_\rho} = b(a/b)^\rho \sum_{j=0}^{m-1} \frac{(\log(a/b))^j}{j!} N_\rho^j. \quad (\text{FT5.8})$$

Multiplying (FT5.6) and (FT5.7) gives exactly nI : the finite truncated exponential products cancel every positive nilpotent degree, and the scalar factors multiply to n . Alternatively this follows directly from $T_n T_{1/n} = I$. Both verifications retain the full coefficients rather than projecting to zero values.

The source-derived Weil form obeys

$$W(\Psi(C)x, y) = W(x, \Psi(C^*)y), \quad C \in \mathcal{C}. \quad (\text{FT5.9})$$

For F_n , this is the proved identity $W(T_n x, y) = W(x, nT_{1/n} y)$. Integer linearity and multiplication prove it for the generated ring; equivalently IC8 proves it on the entire rational ambient algebra. No positivity of this particular form is inferred from its integral coefficient ring.

FT6. Finite isogeny pullback and trace, with all degrees distinguished

Fix one recovered rational prime p , and retain its original constants

$$A = \log p > 0, \quad B = 2\pi > 0. \quad (\text{FT6.1})$$

For every positive rational q , construct the actual complex torus

$$E_q = \mathbb{C}/(qA\mathbb{Z} + iB\mathbb{Z}), \quad w_q = x_q + i\theta_q, \quad (\text{FT6.2})$$

with the displayed model origin $[0]$. At $q = 1$, this is the source curve $\mathbb{C}^\times/p^\mathbb{Z}$; at $q = n$, it is $\mathbb{C}^\times/(p^n)^\mathbb{Z}$. The exponential map proves these identifications with both periods retained. The additional rational scales are explicitly constructed receivers, not new prime stalks.

For $n \geq 1$, the inclusion of lattices defines the degree- n covering

$$u_{n,q} : E_{nq} \longrightarrow E_q, \quad [w]_{nq} \longmapsto [w]_q. \quad (\text{FT6.3})$$

The kernel is $\{[jqA]_{nq} : 0 \leq j < n\}$. The corresponding dual degree- n isogeny is

$$v_{n,q} : E_q \longrightarrow E_{nq}, \quad [w]_q \longmapsto [nw]_{nq}. \quad (\text{FT6.4})$$

Its kernel consists of the angular classes $[ikB/n]_q$, $0 \leq k < n$. Both compositions are multiplication by n , on their respective tori. For $q = 1$, these are precisely the maps called U_n, V_n in T5; the present lower-case notation distinguishes them from the algebra generators.

The integral cohomology groups and their explicit period comparisons are

$$H^0(E_q; \mathbb{Z}) = \mathbb{Z}e_q, \quad H^1(E_q; \mathbb{Z}) = \mathbb{Z}\alpha_q \oplus \mathbb{Z}\beta_q, \quad H^2(E_q; \mathbb{Z}) = \mathbb{Z}h_q, \quad (\text{FT6.5})$$

$$e_q = 1, \quad \alpha_q = \frac{dx_q}{qA}, \quad \beta_q = \frac{d\theta_q}{B}, \quad h_q = \frac{dx_q \wedge d\theta_q}{qAB}. \quad (\text{FT6.6})$$

These fractions describe the integral classes inside the original differential-form spaces. The full forms $dx_q, d\theta_q, dx_q \wedge d\theta_q$ and full area qAB are retained. To verify the cohomology, the torus cell decomposition has one vertex, two oriented edges and one face with attaching word their commutator. Its cellular differentials vanish, giving ranks 1, 2, 1 and no torsion. The displayed forms have periods one on the corresponding generating cycles and oriented face, proving the stated integral comparison.

Pullback $P_{n,q} = u_{n,q}^*$ maps the cohomology of E_q to that of E_{nq} . Its exact matrices in (FT6.5), ordered by degrees zero, one and two, are

$$P_{n,q} : \quad 1, \quad \begin{pmatrix} n & 0 \\ 0 & 1 \end{pmatrix}, \quad n. \quad (\text{FT6.7})$$

Indeed the original differential forms pull back without a scalar, while their target integral periods are $nqA, B, nqAB$, respectively.

Let $S_{n,q} = u_{n,q}!$ be the cohomological transfer, directed from E_{nq} to E_q . It is integral: a singular simplex in the base has exactly n lifts under a covering, and the sum of its lifts is an integral chain; the boundary of that sum is the sum of the lifted boundaries. This chain map defines the cochain transfer. On differential forms the same map is the sum of the pullbacks by all local inverse branches, which commute with the exterior derivative and agree on overlapping charts. Each inverse branch is a translation in w , so its derivative is the identity. Thus on original invariant forms the transfer multiplies the coefficient by n . In the integral bases the exact matrices are

$$S_{n,q} : \quad n, \quad \begin{pmatrix} 1 & 0 \\ 0 & n \end{pmatrix}, \quad 1. \quad (\text{FT6.8})$$

For example $n dx_q / (nqA) = dx_q / (qA)$ explains its radial degree-one entry, and $n dx_q \wedge d\theta_q / (nqAB) = h_q$ explains its top entry. It follows in every degree that

$$S_{n,q} P_{n,q} = n I_{H^*(E_q)}, \quad P_{n,q} S_{n,q} = n I_{H^*(E_{nq})}. \quad (\text{FT6.9})$$

The dual-isogeny pullback $v_{n,q}^*$ is instead

$$v_{n,q}^* : \quad 1, \quad \begin{pmatrix} 1 & 0 \\ 0 & n \end{pmatrix}, \quad n. \quad (\text{FT6.10})$$

It equals the transfer $S_{n,q}$ exactly in degree one. In degrees zero and two the displayed factors differ. Thus the earlier T6 degree-one adjoint equality is preserved, and is not improperly extended to all cohomological degrees.

FT7. A typed integral representation over the full scale family

The preceding arrows connect different tori; they are not endomorphisms of one fixed fiber. Their precise common receiver is the direct sum

$$\mathcal{H}_{\mathbb{Z}} = \bigoplus_{q \in \mathbb{Q}_{>0}} H^*(E_q; \mathbb{Z}), \quad (\text{FT7.1})$$

with finite support. Define endomorphisms on it by

$$\mathfrak{F}_n|_{H^*(E_q)} = P_{n,q} : H^*(E_q) \rightarrow H^*(E_{nq}), \quad (\text{FT7.2})$$

$$\mathfrak{V}_n|_{H^*(E_q)} = S_{n,q/n} : H^*(E_q) \rightarrow H^*(E_{q/n}). \quad (\text{FT7.3})$$

The rational index set ensures that both target scales exist. Every map is integral by FT6 and preserves finite support. The full matrices (FT6.7)-(FT6.8) show

$$\mathfrak{F}_m \mathfrak{F}_n = \mathfrak{F}_{mn}, \quad \mathfrak{V}_m \mathfrak{V}_n = \mathfrak{V}_{mn}, \quad \mathfrak{F}_m \mathfrak{V}_n = \mathfrak{V}_n \mathfrak{F}_m, \quad \mathfrak{F}_n \mathfrak{V}_n = nI. \quad (\text{FT7.4})$$

For full verification of the mixed relation, both composites carry scale q to mq/n ; in degrees zero, one and two their matrices are respectively

$$n, \quad \begin{pmatrix} m & 0 \\ 0 & n \end{pmatrix}, \quad m. \quad (\text{FT7.5})$$

These also equal d times the matrices for $\mathfrak{F}_{m/d} \mathfrak{V}_{n/d}$, where $d = \gcd(m, n)$. Therefore FT2's presentation gives a unital integral representation

$$\Psi_{\text{geom}} : \mathcal{C} \longrightarrow \text{End}_{\mathbb{Z}}(\mathcal{H}_{\mathbb{Z}}), \quad F_n \mapsto \mathfrak{F}_n, \quad V_n \mapsto \mathfrak{V}_n. \quad (\text{FT7.6})$$

This representation is faithful. For reduced $r = a/b$, the action of $c_r = F_a V_b$ on $e_1 \in H^0(E_1)$ is be_r . These images have distinct scale supports as r varies and nonzero integer coefficient b . Thus an integer combination of the basis elements can act as zero only when every coefficient vanishes. This proves faithfulness without identifying distinct cohomology fibers or discarding their scale labels.

FT8. The full positive adjoint comparison, retaining both periods

Use the flat complex metric $dx_q^2 + d\theta_q^2$ on each explicitly constructed torus; this is a metric on the torus and not on τ . Its area is qAB . The invariant representatives of every cohomology class give positive Hermitian forms conjugate-linear in the first input. In the original invariant-form coefficients they are

$$\begin{aligned} h_q^0(a, c) &= qAB \bar{a}c, \\ h_q^1(a dx_q + b d\theta_q, c dx_q + d d\theta_q) &= qAB(\bar{a}c + \bar{b}d), \\ h_q^2(a dx_q \wedge d\theta_q, c dx_q \wedge d\theta_q) &= qAB \bar{a}c. \end{aligned} \quad (\text{FT8.1})$$

Invariant forms represent all cohomology classes by (FT6.5)-(FT6.6), and these formulas are positive definite on their respective finite-dimensional spaces. In the integral bases their full Gram matrices are

$$qAB, \quad \begin{pmatrix} B/(qA) & 0 \\ 0 & qA/B \end{pmatrix}, \quad 1/(qAB). \quad (\text{FT8.2})$$

These factors are comparison coefficients, not altered periods.

Direct substitution of (FT6.7)-(FT6.8) into (FT8.2) proves, in every degree,

$$h_{nq}(P_{n,q}x, y) = h_q(x, S_{n,q}y), \quad h_{nq}(P_{n,q}x, P_{n,q}x') = nh_q(x, x'). \quad (\text{FT8.3})$$

For example on the radial degree-one class the first identity has coefficient $n B/(nqA) = B/(qA)$; on the angular class it has coefficient $nqA/B = n(qA/B)$. In top degree it has coefficient $n/(nqAB) = 1/(qAB)$, and in degree zero it has coefficient $nqAB = n(qAB)$. Thus every original factor is accounted for.

Take the orthogonal sum of these forms on $\mathcal{H}_{\mathbb{Z}} \otimes \mathbb{C}$. Then

$$\mathfrak{F}_n^\dagger = \mathfrak{V}_n, \quad \mathfrak{F}_n^\dagger \mathfrak{F}_n = \mathfrak{F}_n \mathfrak{F}_n^\dagger = nI, \quad (\text{FT8.4})$$

first on finite-support vectors. The norm equality $\|\mathfrak{F}_n x\|^2 = n\|x\|^2$ and the analogous equality for $\mathfrak{V}_n$ extend both maps to bounded operators, of norm $\sqrt{n}$, on the Hilbert completion of this specified direct sum. The adjoint relations persist by density. Integer linearity and multiplication now prove

$$\Psi_{\text{geom}}(C)^\dagger = \Psi_{\text{geom}}(C^*), \quad C \in \mathcal{C}. \quad (\text{FT8.5})$$

Every finite algebra combination is bounded because it is a finite sum of finite products of the bounded generators. This establishes an actual positive integral correspondence representation. It does not transfer positivity to the different original-zeta receiver in FT5.

FT9. Exact relation between the two receivers

The same proved integral algebra now has two specified actions:

$$\begin{array}{ccc} & \mathcal{C} & \\ \Psi_{\text{geom}} \swarrow & & \searrow \Psi \\ \text{End}(\mathcal{H}_{\mathbb{Z}}) & & \text{End}_{\text{cont}}(\mathcal{Q}). \end{array} \quad (\text{FT9.1})$$

On the first side the transfer is an actual integral covering trace and is the adjoint for the positive forms (FT8.1). On the second side it is the actual original-zeta operator $nT_{1/n}$, satisfying the proved adjoint identity for the source-derived Weil form, with full zero jets (FT5.6)–(FT5.8). This does not assert uniqueness of an adjoint for a possibly degenerate form. Both sides satisfy the same complete algebra relations and the same degree factor n .

This diagram specifies two representations of the same algebra; it does not assert an intertwiner identifying their state spaces, nor that the two forms have the same sign. The geometric scale family and its positive form have been constructed and proved here. The original-zeta receiver remains the existing quotient of its full arithmetic source. The result is an integral, involution-preserving repair of the exact coefficient obstruction, together with these fully typed receiving maps. It establishes neither a restriction on the real parts of zeta zeros nor an impossibility of obtaining such a restriction by a further mathematical comparison.

The subsequent complete calculation GEOMETRIC_SCALE_TO_ORIGINAL_ZETA_INTERTWINERS.md constructs actual algebraic receiving maps between their specified degree-zero and original-zeta spaces, and calculates their topology. Its separate proof extends this diagram; no equality of the two pairings is asserted here.

FT10. Exact geometric spectrum, with the original degree and periods

Keep the degree-zero sector of FT7–FT8, and write its Hilbert completion as

$$H_{\text{geom}}^0 = \left\{ \sum_{q \in \mathbb{Q}_{>0}} a_q e_q : AB \sum_q q |a_q|^2 < \infty \right\}, \quad \|a\|^2 = AB \sum_q q |a_q|^2, \quad (\text{FT10.1})$$

where $A = \log p$, $B = 2\pi$ are the retained periods of FT6. For every integer degree $n \geq 2$, its original operator and transfer act by

$$\mathfrak{F}_n e_q = e_{nq}, \quad \mathfrak{V}_n e_q = n e_{q/n}. \quad (\text{FT10.2})$$

The same formulas and the displayed weighted sum prove

$$\|\mathfrak{F}_n\| = \sqrt{n}, \quad \mathfrak{F}_n^{-1} = \mathfrak{V}_n/n, \quad \|\mathfrak{F}_n^{-1}\| = 1/\sqrt{n}. \quad (\text{FT10.3})$$

The scalar division is on this complex Hilbert space; the integral transfer in FT1 remains V_n , not the bare inverse. No operator or period is rescaled in what follows.

The spectrum and point spectrum on this particular receiver are exactly

$$\boxed{\text{Spec}(\mathfrak{F}_n|_{H_{\text{geom}}^0}) = \{\lambda \in \mathbb{C} : |\lambda| = \sqrt{n}\}, \quad \text{Spec}_{\text{point}}(\mathfrak{F}_n|_{H_{\text{geom}}^0}) = \emptyset.} \quad (\text{FT10.4})$$

Here is the full proof. If $|\lambda| > \sqrt{n}$, the series

$$(\lambda I - \mathfrak{F}_n)^{-1} = \sum_{k=0}^{\infty} \lambda^{-k-1} \mathfrak{F}_n^k \quad (\text{FT10.5})$$

converges in operator norm by FT10.3's geometric ratio $\sqrt{n}/|\lambda| < 1$. Multiplication of its partial sums by $\lambda I - \mathfrak{F}_n$ leaves a remainder of norm tending to zero, proving it is the inverse on both sides. Its norm is at most $1/(|\lambda| - \sqrt{n})$. If $|\lambda| < \sqrt{n}$, the analogous exact inverse is

$$(\lambda I - \mathfrak{F}_n)^{-1} = - \sum_{k=0}^{\infty} \lambda^k \mathfrak{F}_n^{-k-1}, \quad (\text{FT10.6})$$

with geometric ratio $|\lambda|/\sqrt{n} < 1$ and norm at most $1/(\sqrt{n} - |\lambda|)$. The formula includes $\lambda = 0$. Thus no point off the displayed circle is in the spectrum.

For $|\lambda| = \sqrt{n}$ and $N \geq 0$, the exact finite vector

$$z_N = \sum_{k=-N}^N \lambda^{-k} e_{n^k} \quad (\text{FT10.7})$$

has squared norm

$$\|z_N\|^2 = AB \sum_{k=-N}^N n^k |\lambda|^{-2k} = (2N+1)AB. \quad (\text{FT10.8})$$

Subtracting the two finite sums for $\mathfrak{F}_n z_N$ and λz_N cancels each interior coefficient and leaves both endpoint terms:

$$(\mathfrak{F}_n - \lambda I)z_N = \lambda^{-N} e_{n^{N+1}} - \lambda^{N+1} e_{n^{-N}}. \quad (\text{FT10.9})$$

Their supports are distinct, so the full weighted norm is

$$\|(\mathfrak{F}_n - \lambda I)z_N\|^2 = AB (n^{N+1} |\lambda|^{-2N} + n^{-N} |\lambda|^{2N+2}) = 2nAB. \quad (\text{FT10.10})$$

A bounded inverse of $\mathfrak{F}_n - \lambda I$ would bound the ratio $\|z_N\|/\|(\mathfrak{F}_n - \lambda I)z_N\| = \sqrt{(2N+1)/(2n)}$, contrary to its divergence. This proves that every point on the circle lies in the spectrum, retaining the vectors, both endpoint terms and both periods.

Finally suppose $a = \sum a_q e_q$ is an eigenvector with eigenvalue λ . Invertibility excludes $\lambda = 0$. Comparing the coefficient at nq gives $a_{nq} = \lambda^{-1} a_q$, and therefore

$$a_{n^k q} = \lambda^{-k} a_q \quad (k \in \mathbb{Z}). \quad (\text{FT10.11})$$

If some $a_q \neq 0$, the contribution of this orbit to the squared norm is

$$AB q |a_q|^2 \sum_{k \in \mathbb{Z}} (n/|\lambda|^2)^k. \quad (\text{FT10.12})$$

For every positive ratio this bilateral series diverges: a ratio greater than one diverges as $k \rightarrow +\infty$, one less than one as $k \rightarrow -\infty$, and one gives infinitely many equal positive terms. Thus all coefficients vanish, proving the empty point spectrum.

This is a complete spectral calculation for the newly constructed geometric Hilbert receiver. Its circle is obtained from the actual integral covering adjoints and their degree factor. It does not put the original-zeta eigenvectors into this Hilbert space, and it does not replace the original-zeta spectrum. The distinction is calculated by the explicit receiving maps in the accompanying intertwiner proof.

Integral clock algebra, boundary quotient and topologies

Proof source: INTEGRAL_CLOCK_BOUNDARY_AND_ZETA_ACTION.md.

Integral clock operators and their exact boundary quotient

24 September 2026. Proof labels CB1-CB10. This is a calculation on the completed winding data and its function spaces. The support remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. No addition, coordinate, distance, or vector is assigned to that support. The integer coefficient ring, winding coordinates, integral transfers, and source projections below are different explicitly typed objects.

The starting clock construction is P13-P14 in `../stacked_zero_intake_20260924/deligne_visual/PRIME_CLOCKS_ORCHARD_AND_SPECTRUM.md`. The integral receiving algebra is FT1-FT5 in `INTEGRAL_FROBENIUS_TRANSFER_ALGEBRA.md`. The original-zeta receiving space and all its multiplicities are RZ1-RZ12 in `ORIGINAL_ZETA_RESOLVENT_AND_PROJECTORS.md`. These complete proofs accompany the cumulative reader. The Connes-Consani source geometry and its exact original-author source locators are retained in CG0-CG8; the operator algebra proved here is a programme derivation, not attributed to their paper.

CB1. The completed clock and its integral functions

Retain the full winding completion

$$K = \varprojlim_{M|M'} \mathbb{Z}/M\mathbb{Z} = \prod_{p \text{ prime}} \mathbb{Z}_p. \quad (\text{CB1.1})$$

The equality is the Chinese-remainder map: a compatible residue family restricts to every prime-power family; conversely its prime-power components determine the residue at each $M = \prod p^{v_p(M)}$, uniquely and compatibly. All primes, including two, occur. Multiplication by a positive integer n is injective, and its image nK has index n . Componentwise injectivity follows from injectivity in each $\mathbb{Z}_p$; the image consists of components divisible by $p^{v_p(n)}$, so its quotient is the product of the corresponding finite prime-power quotients, of total cardinality n . The inverse x/n is defined only for $x \in nK$.

Let

$$\mathcal{A}_{\mathbb{Z}} = C_{\text{loc}}(K, \mathbb{Z}) = \bigcup_{M \geq 1} \{f \text{ factoring through } \mathbb{Z}/M\mathbb{Z}\}. \quad (\text{CB1.2})$$

The displayed union means the directed union under divisibility, with its exact pullback maps. A locally constant function on compact K has a finite clopen partition, refined by one residue partition, which proves the second description. Retain the measure from P14 with arbitrary total mass

$$\mu(K) = h > 0, \quad \mu(a + MK) = h/M. \quad (\text{CB1.3})$$

Set $H = L^2(K, \mu)$, with the inner product conjugate-linear in the first argument. The integral functions embed into H . Their complex linear span is dense: indicator functions of residue cylinders generate the Borel sigma algebra, and their finite linear combinations are dense in L^2 for this finite measure.

Equivalently, successive finite-residue conditional expectations approximate every L^2 function. No value of h is set to one.

CB2. Every degree and image projection is retained

Define integral endomorphisms of $\mathcal{A}_{\mathbb{Z}}$ by

$$(U_n f)(x) = f(nx), \quad (D_n f)(x) = \begin{cases} n f(x/n), & x \in nK, \\ 0, & x \notin nK. \end{cases} \quad (\text{CB2.1})$$

The factor n is part of the transfer. It is not omitted. Factoring through residue sets proves that these maps preserve locally constant integer-valued functions. Let P_n denote multiplication by $\mathbf{1}_{nK}$.

The residue masses give, first on cylinder functions and then on nonnegative measurable functions,

$$\int_K F(nx) d\mu(x) = n \int_{nK} F(y) d\mu(y). \quad (\text{CB2.2})$$

For example the pushforward of μ by $x \mapsto nx$ and the measure $n\mu|_{nK}$ have the same mass on each cylinder in nK , and therefore agree on the generated Borel sigma algebra. This proves

$$\|U_n f\|^2 = n \int_{nK} |f|^2 d\mu, \quad \|D_n f\|^2 = n \|f\|^2, \quad (\text{CB2.3})$$

$$\langle U_n f, g \rangle = n \int_{nK} \overline{f(y)} g(y/n) d\mu(y) = \langle f, D_n g \rangle. \quad (\text{CB2.4})$$

Thus both operators extend to H , with

$$U_n^* = D_n, \quad D_n^* = U_n, \quad \|U_n\| = \|D_n\| = \sqrt{n}, \quad (\text{CB2.5})$$

$$U_n D_n = nI, \quad D_n U_n = nP_n. \quad (\text{CB2.6})$$

For the norm of U_n , equality in CB2.3 follows on $f = \mathbf{1}_{nK}$, whose squared norm is h/n . The formulas for the products follow pointwise, retaining the zero value outside the image.

All compositions obey

$$U_m U_n = U_{mn}, \quad D_m D_n = D_{mn}, \quad (\text{CB2.7})$$

$$U_m D_n = d D_{n/d} U_{m/d}, \quad d = \gcd(m, n). \quad (\text{CB2.8})$$

To verify the mixed formula, write $m = dm'$, $n = dn'$ with $\gcd(m', n') = 1$. Componentwise divisibility proves $mx \in nK$ precisely when $x \in n'K$. On that image both sides take the value $nf(m'x/n')$, and off it both vanish. In particular distinct-prime U 's and D 's commute, while the two same-prime orders in CB2.6 generally differ.

CB3. The complete integral presentation

Let $\mathcal{B}$ be the unital subring of $\text{End}_{\mathbb{Z}}(\mathcal{A}_{\mathbb{Z}})$ generated by all U_p, D_p for primes p . Its presentation over $\mathbb{Z}$ has relations

$$U_p D_p = pI, \quad (\text{CB3.1})$$

$$U_p U_q = U_q U_p, \quad D_p D_q = D_q D_p, \quad U_p D_q = D_q U_p \quad (p \neq q). \quad (\text{CB3.2})$$

There are no further relations. In particular one does not insert $D_p U_p = pI$.

Here is a faithful ordered-word proof. Commuting distinct-prime letters and replacing each adjacent $U_p D_p$ by pI expresses every word as an integer combination of

$$D^a U^b = \prod_p D_p^{a_p} \prod_p U_p^{b_p}, \quad a_p, b_p \geq 0, \quad (\text{CB3.3})$$

where both exponent families have finite support. Each replacement retains its integer factor. To prove independence, it is enough to consider a finite set S of primes containing all the letters in a proposed finite relation.

For $x \in K$ whose p -component is nonzero for every $p \in S$, put $j_p = v_p(x_p) \in \mathbb{N}_0$. On a function depending only on these valuations,

$$(D^a U^b f)(j) = \left(\prod_{p \in S} p^{a_p} \right) \mathbf{1}_{\{j_p \geq a_p \ \forall p \in S\}} f(j - a + b). \quad (\text{CB3.4})$$

This is repeated substitution into CB2.1. The divisibility condition guarantees that every argument $j_p - a_p + b_p$ is nonnegative. The test function which is one exactly when the valuations have prescribed finite values t_p is an integral locally constant function: it is the indicator of the product of the clopen shells $p^{t_p} \mathbb{Z}_p \setminus p^{t_p+1} \mathbb{Z}_p$.

Suppose $\sum_{a,b} z_{a,b} D^a U^b = 0$ as an operator, with integer coefficients and finite support. Fix a displacement $k = b - a$ and a valuation vector j with $j_p \geq \max(0, -k_p)$. At an input point of valuations j , test with the shell function of valuations $j + k$. Different displacement vectors give different arguments in CB3.4, so only this k survives. We obtain

$$\sum_{a \leq j} z_{a, a+k} \prod_{p \in S} p^{a_p} = 0. \quad (\text{CB3.5})$$

Coefficients with a negative entry in $a + k$ are absent. Extend the cumulative sum by zero below its lower thresholds. Taking a successive backward difference in every coordinate at $j = a$ gives

$$z_{a, a+k} \prod_{p \in S} p^{a_p} = 0. \quad (\text{CB3.6})$$

The nonzero integer product can be cancelled in $\mathbb{Z}$, giving $z_{a, a+k} = 0$. This holds for every displacement and exponent family, proving independence. The presented algebra maps onto the operator algebra by CB2, and this spanning-and-independence argument proves injectivity. It proves the presentation integrally; no division by a prime was used in the algebra.

Since the integral functions inject into H , the same representation of $\mathcal{B}$ on H is faithful. Indeed a nonzero locally constant function is nonzero on a cylinder of positive measure. The involution from Hilbert adjoints is the algebraic anti-involution exchanging U_p, D_p and fixing integer scalars. The defining relations are preserved, as is also verified by the faithful operator representation.

CB4. The boundary element and the integral commutative quotient

For every prime set

$$e_p = pI - D_p U_p = p(I - P_p) \in \mathcal{B}. \quad (\text{CB4.1})$$

It is a nonzero, self-adjoint element: its support is $K \setminus pK$, of measure $h(1 - 1/p) > 0$. Its exact relations include

$$e_p^2 = p e_p, \quad U_p e_p = 0, \quad e_p D_p = 0. \quad (\text{CB4.2})$$

These follow by substituting $U_p D_p = pI$, or by using the image projection. In particular e_p is not described as an idempotent; its prime factor remains present.

Let $\mathcal{J}$ be the two-sided ideal of $\mathcal{B}$ generated by every e_p . It is preserved by the involution. There is an exact surjective homomorphism of integral rings with involution

$$\pi : \mathcal{B} \longrightarrow \mathcal{C}, \quad \pi(U_p) = F_p, \quad \pi(D_p) = V_p, \quad (\text{CB4.3})$$

whose kernel is exactly $\mathcal{J}$, and hence

$$\boxed{\mathcal{B}/\mathcal{J} \cong \mathcal{C}.} \quad (\text{CB4.4})$$

To prove this without suppressing another relation, CB3 and FT2 show that the stated images satisfy all the relations of $\mathcal{B}$. In $\mathcal{C}$, both $F_p V_p$ and $V_p F_p$ are pI , so every e_p maps to zero. The quotient by these elements imposes precisely $D_p U_p = pI$ in addition to CB3.1. It therefore makes each same-prime pair commute. Distinct-prime pairs already commute. The resulting presentation is

$$\mathbb{Z}[X_p, Y_p : p \text{ prime}] / (X_p Y_p - p : p \text{ prime}), \quad (\text{CB4.5})$$

which FT2 proves injectively represents $\mathcal{C}$. The maps in both directions send corresponding generators to one another, so are inverse. This proves the kernel statement as well as surjectivity.

For a single prime p , let $\mathcal{J}_p = (e_p)$. The quotient by $\mathcal{J}_p$ imposes only this prime's extra relation; it must not be confused with the all-prime quotient in CB4.4. Both quotients are nonzero because they map onto the nonzero ring $\mathcal{C}$.

CB5. Exact map to the actual original-zeta action

Retain the receiving space $\mathcal{Q} = \mathcal{E}/\mathcal{J}$ of RZ, where $\mathcal{E}$ is the entire vertical-strip Schwartz space and $\mathcal{J}$ is its ideal of all actual nontrivial-zero jets of the original ζ . Here $\mathcal{E}$ is the space denoted $\mathcal{B}$ in RZ; the changed letter in this note avoids confusing it with the integral operator ring $\mathcal{B}$. Explicitly,

$$\sup_{|\operatorname{Re} s| \leq A} (1 + |\operatorname{Im} s|)^M |F(s)| < \infty \quad (A, M \geq 0), \quad (\text{CB5.1})$$

$$\mathcal{J} = \{F : F^{(j)}(\rho) = 0 \text{ for every actual nontrivial zero } \rho, 0 \leq j < m_\rho\}. \quad (\text{CB5.2})$$

The full source comparison, including the endpoint and trivial-zero terms, retains

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{CB5.3})$$

RZ and CD retain its exceptional values and derivatives; the original zeta function is not replaced by this multiplier. Multiplication $T_r[F] = [r^s F]$ is a continuous endomorphism of $\mathcal{Q}$ for each positive rational r , as proved with all strip seminorms in FT5.

Compose the two proved maps to obtain

$$\Theta = \Psi \circ \pi : \mathcal{B} \longrightarrow \operatorname{End}_{\text{cont}}(\mathcal{Q}), \quad \Theta(U_n) = T_n, \quad \Theta(D_n) = nT_{1/n}. \quad (\text{CB5.4})$$

This is a unital homomorphism defined on the entire integral clock ring. Its exact source discrepancy is

$$U_p D_p = pI, \quad D_p U_p = pP_p \quad \text{on } H, \quad (\text{CB5.5})$$

$$\Theta(U_p)\Theta(D_p) = \Theta(D_p)\Theta(U_p) = pI \quad \text{on } \mathcal{Q}, \quad \Theta(e_p) = 0. \quad (\text{CB5.6})$$

Thus the original-zeta action factors through the exact integral boundary quotient. This factorization does not assert that its kernel on $\mathcal{B}$ is only $\mathcal{J}$; the kernel of Ψ is not being assumed trivial.

On an actual zero block of multiplicity $m = m_\rho$, the entire formula is

$$\Theta(U_n) = n^\rho \sum_{j=0}^{m-1} \frac{(\log n)^j}{j!} N_\rho^j, \quad \Theta(D_n) = n^{1-\rho} \sum_{j=0}^{m-1} \frac{(-\log n)^j}{j!} N_\rho^j, \quad (\text{CB5.7})$$

where $N_\rho^m = 0$ and $N_\rho^{m-1} \neq 0$. All jet terms and the original degree factors survive. The source-derived Weil form obeys

$$W(\Theta(B)x, y) = W(x, \Theta(B^*)y) \quad (B \in \mathcal{B}), \quad (\text{CB5.8})$$

by FT5 and CB4. This is an adjoint identity for that form, whose radical and sign are calculated in RZ; it is not a transfer of the Hilbert inner product of H .

CB6. The boundary ideal exhausts the original Hilbert space

Fix a single prime p and a nonnegative integer a . Direct substitution gives

$$D_p^a e_p U_p^a = p^{a+1} \mathbf{1}_{p^a K \setminus p^{a+1} K} \quad (\text{CB6.1})$$

as a multiplication operator on H . For a full verification, the leftmost D_p^a vanishes unless $x \in p^a K$, where it contributes p^a and evaluates at x/p^a . The middle e_p contributes p precisely when this argument is outside pK . The rightmost U_p^a then evaluates the original function at x . This gives the stated shell and scalar.

Let E_a be multiplication by that shell indicator. The range of E_a is contained in the linear span $\mathcal{J}_p H$: for $f \in H$,

$$E_a f = (D_p^a e_p U_p^a)(p^{-(a+1)} f). \quad (\text{CB6.2})$$

The scalar in this formula multiplies a vector in the complex Hilbert space. It does not assert that p^{-1} belongs to the integral operator algebra. In particular no integral coefficient has been discarded.

The finite sum $\sum_{a=0}^A E_a$ is multiplication by $\mathbf{1}_{K \setminus p^{A+1} K}$. The decreasing intersection $\bigcap_{A \geq 0} p^{A+1} K$ is the set whose p -adic coordinate is zero, and has measure

$$\lim_{A \rightarrow \infty} h/p^{A+1} = 0. \quad (\text{CB6.3})$$

Therefore for every $f \in H$, dominated convergence gives

$$\left\| f - \sum_{a=0}^A E_a f \right\|^2 = \int_{p^{A+1} K} |f|^2 d\mu \longrightarrow 0. \quad (\text{CB6.4})$$

Each finite sum belongs to $\mathcal{J}_p H$. Consequently

$$\boxed{\overline{\mathcal{J}_p H} = H, \quad \overline{\mathcal{J} H} = H.} \quad (\text{CB6.5})$$

It follows that the Hilbert quotient $H/\overline{\mathcal{J} H}$ is zero, although the integral algebra quotient $\mathcal{B}/\mathcal{J} = \mathcal{C}$ is nonzero. These are statements about different specified quotients, related by the proved action and ideal. They do not describe the source ideal $\mathcal{J}$ in CB5 as a Hilbert ideal, and do not substitute a Hilbert completion for its original locally convex topology.

CB7. No continuous intertwiner can carry this Hilbert action to the boundary action

There is a precise consequence for receiving maps. Let Y be a Hausdorff topological vector space with a representation of $\mathcal{C}$, and let $L : H \rightarrow Y$ be a continuous linear map intertwining the $\mathcal{B}$ action with its action through π . For every $B \in \mathcal{J}$ and $f \in H$,

$$L(Bf) = \pi(B)L(f) = 0. \quad (\text{CB7.1})$$

Here $\pi(B)$ denotes its represented action on Y , which is zero. Hence $\mathcal{J}H \subseteq \ker L$. A continuous map into a Hausdorff space has closed kernel, so CB6 gives $\ker L = H$, and $L = 0$.

In particular the actual $\mathcal{Q}$ is Hausdorff: all jet evaluations are continuous in the strip topology by Cauchy's formula on compact discs, so their simultaneous kernel $\mathcal{J}$ is closed. Applying the preceding argument to its actual action in CB5 proves that there is no nonzero continuous intertwiner from this particular Hilbert clock representation to $\mathcal{Q}$. The proof does not exclude a map from a different topology, a distribution space, a correspondence, or another cohomological receiver. The nonzero algebra homomorphism to $\mathcal{C}$ in CB4 already supplies a precise connecting object, despite this continuous-map obstruction.

CB8. Relation to the retained signed source

The source completion before forgetting any finite label is

$$\widehat{G} = K \times C_4, \quad d\hat{\mu} = \frac{1}{4} \sum_{k \in C_4} d\mu, \quad (\text{CB8.1})$$

with total mass h , as in P13-P14. Pullback along the winding projection $\text{pr} : \widehat{G} \rightarrow K$ is isometric: the four branch contributions, each with coefficient $1/4$, sum to the original squared norm. Applying U_n, D_n in the winding coordinate and the identity in the C_4 coordinate defines the tensor action on the full function space and retains every branch. CB3's action remains faithful and CB6's shell proof applies separately in every branch, so the same density conclusion holds there.

This tensor action is an explicitly specified lift of the winding action. It is not asserted to be the native signed mirror-commuting source map for every n . The original signed odd-degree maps of P14.6, with their C_4 twists, retain their stated formulas and intertwining maps under pr . In particular this note does not supply a previously absent native even-degree lift commuting with the source mirror. The degree-two winding operator is nevertheless present throughout CB1-CB7. Neither the finite labels nor that limitation is omitted.

CB9. What passes through the map

The exact chain now has all its maps:

$$\mathcal{B} \twoheadrightarrow \mathcal{B}/\mathcal{J} \xrightarrow{\cong} \mathcal{C} \xrightarrow{\Psi} \text{End}_{\text{cont}}(\mathcal{Q}). \quad (\text{CB9.1})$$

The integral generators, involution, prime factors and complete original-zeta jet action pass through this chain. Its first arrow has kernel exactly the two-sided ideal generated by all $e_p = p(I - P_p)$, with every image projection and scalar exhibited. The positive inner product on the original H cannot be obtained on the boundary by taking its Hilbert quotient, since that quotient is zero by CB6. This identifies the exact failure of that proposed sign-transfer construction; it is not an impossibility theorem for purity or for the programme.

FT6-FT8 construct a different, nonzero positive geometric representation of the same integral algebra. Its existence and the original-zeta representation are both retained. Thus the boundary quotient is not

being declared incapable of positive representations. The remaining distinction is between a proved positive receiver and the actual original-zeta receiver, whose connecting maps must retain their stated domains and topologies. No zero location, new prime, or missing detection is inferred from the quotient calculation.

CB10. The locally constant boundary survives and is computed exactly

The preceding Hilbert quotient is not the algebraic quotient of the original locally constant domain. Put

$$E = C_{\text{loc}}(K, \mathbb{C}) = \mathcal{A}_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{C}. \quad (\text{CB10.1})$$

Here $\mathcal{J}E$ means the complex linear span of all Bf with $B \in \mathcal{J}$, $f \in E$. Let 0_K be the zero winding element of the profinite group K . It is not the parityless supporting point τ . Evaluation at this clock element satisfies

$$\text{ev}_{0_K}(U_n f) = f(0_K), \quad \text{ev}_{0_K}(D_n f) = n f(0_K). \quad (\text{CB10.2})$$

Consequently evaluation intertwines the whole algebra with the character

$$\chi_0(U_n) = 1, \quad \chi_0(D_n) = n, \quad \chi_0(e_p) = 0, \quad (\text{CB10.3})$$

so $\mathcal{J}E \subseteq \ker \text{ev}_{0_K}$.

For the reverse inclusion, take $f \in E$ with $f(0_K) = 0$. It factors through some modulus M , so vanishes on MK . If $M = 1$, it is zero. Otherwise order the finitely many primes dividing M as $p_1, \dots, p_t$, and put $r_i = v_{p_i}(M) > 0$. The sets

$$C_{i,a} = \{x : v_{p_i}(x) = a, v_{p_j}(x) \geq r_j \text{ for } j < i\}, \quad 1 \leq i \leq t, 0 \leq a < r_i, \quad (\text{CB10.4})$$

are disjoint clopen sets partitioning $K \setminus MK$. Each $f_{i,a} = \mathbf{1}_{C_{i,a}} f$ is in E , supported on the p_i -shell of valuation a . Formula CB6.1 therefore gives

$$f_{i,a} = (D_{p_i}^a e_{p_i} U_{p_i}^a)(p_i^{-(a+1)} f_{i,a}) \in \mathcal{J}E. \quad (\text{CB10.5})$$

The finite sum of these functions is f , proving the reverse inclusion. Evaluation is onto, since constant functions realize every complex value. Hence the entire sequence is exact:

$$0 \longrightarrow \mathcal{J}E \longrightarrow E \xrightarrow{\text{ev}_{0_K}} \mathbb{C} \longrightarrow 0, \quad E/\mathcal{J}E \cong \mathbb{C}. \quad (\text{CB10.6})$$

The receiving $\mathcal{C}$ -character is $F_n \mapsto 1$, $V_n \mapsto n$. Its degree factor remains n . For $n > 1$ it is not a representation respecting the involution for a positive Hermitian inner product on this line: the adjoint of the scalar one is one, not n . Thus this nonzero boundary quotient is not presented as a pure-weight receiver.

In the locally convex direct-limit topology of the finite residue function spaces, evaluation is continuous because its restriction to each finite-dimensional stage is continuous. Constants supply a continuous section, so CB10.6 identifies the topological quotient with $\mathbb{C}$ too. The operators are continuous in this topology: if E_M is the stage of functions modulo M , then $U_n E_M \subseteq E_M$ and $D_n E_M \subseteq E_{nM}$, and these are linear maps between finite-dimensional stages. Evaluation is not continuous for the inherited L^2 norm: the functions $f_A = \mathbf{1}_{p^A K}$ have

$$f_A(0_K) = 1, \quad \|f_A\|^2 = h/p^A \longrightarrow 0. \quad (\text{CB10.7})$$

This exhibits the exact topological loss that CB6 detects. The calculation uses scalar division on the complex function space in CB10.5; it does not identify the integral module quotient $\mathcal{A}_{\mathbb{Z}}/\mathcal{J}\mathcal{A}_{\mathbb{Z}}$ with $\mathbb{Z}$, nor discard any torsion in that different quotient.

Exact geometric-to-zeta intertwiners

Proof source: GEOMETRIC_SCALE_TO_ORIGINAL_ZETA_INTERTWINERS.md.

Exact geometric-scale maps to the original-zeta quotient

24 September 2026. Proof labels GI1-GI7. These are actual algebraic intertwiners between the two receivers of FT, followed by a calculation of their graph closures in the specified topologies. No coordinate or arithmetic operation is put on $\tau\langle Z_1; \text{no } Z_2 \rangle$. Source labels are carried by the disjoint family in ZR; all vector-space operations here are on coefficient fibers.

Inputs, with complete proofs in the cumulative reader: INTEGRAL_FROBENIUS_TRANSFER_ALGEBRA.md FT1-FT10, ORIGINAL_ZETA_RESOLVENT_AND_PROJECTORS.md RZ1-RZ12, and ../quantum_tau_programme_bridge_20260924/GLOBAL_MELLIN_SYNTHESIS.md S1-S7. The finite geometric covering maps were derived in FT6; the complete original-zeta quotient is the proved arithmetic source quotient in S5. No new identification between those original sources is presumed.

GI1. The two coefficient spaces and all scale factors

Let

$$E_0 = \bigoplus_{q \in \mathbb{Q}_{>0}} \mathbb{C}e_q, \quad \left\| \sum_q a_q e_q \right\|^2 = AB \sum_q q |a_q|^2, \quad A = \log p > 0, \quad B = 2\pi, \quad (\text{GI1.1})$$

where p is the fixed recovered prime specifying the original torus periods in FT6. Sums in E_0 have finite support. Its Hilbert completion is the degree-zero receiver H_{geom}^0 of FT10. The integral algebra $\mathcal{C}$ acts by

$$F_n e_q = e_{nq}, \quad V_n e_q = n e_{q/n} \quad (n \geq 1). \quad (\text{GI1.2})$$

These are the full covering pullback and transfer actions; neither the factor n nor the measure weight qAB is omitted.

On the other side retain

$$\mathcal{Q} = \mathcal{B}/\mathcal{J}, \quad T_q[F] = [q^s F(s)], \quad \Psi(F_n) = T_n, \quad \Psi(V_n) = nT_{1/n}. \quad (\text{GI1.3})$$

Here $\mathcal{B}$ is the entire strip-Schwartz space, with seminorms

$$b_{a,M}(F) = \sup_{|\text{Re } s| \leq a} (1 + |\text{Im } s|)^M |F(s)|, \quad (\text{GI1.4})$$

and $\mathcal{J}$ consists of the functions vanishing at all actual nontrivial zeros of the original ζ , to their entire original multiplicities. It is closed because each derivative evaluation is continuous by Cauchy's integral formula on a compact disc. Thus $\mathcal{Q}$ has the Hausdorff quotient topology. Its original source comparison retains

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad (\text{GI1.5})$$

including all endpoint and trivial-zero comparisons proved in CD and S. This calculation uses the original s , without moving the spectral line.

GI2. All algebraic intertwiners are explicitly obtained

For each $x \in \mathcal{Q}$, define

$$K_x : E_0 \longrightarrow \mathcal{Q}, \quad K_x \left(\sum_q a_q e_q \right) = \sum_q a_q T_q x. \quad (\text{GI2.1})$$

This finite sum defines a complex-linear map on the entire stated domain. It intertwines both generators:

$$K_x(F_n e_q) = T_{nq} x = T_n K_x(e_q), \quad (\text{GI2.2})$$

$$K_x(V_n e_q) = n T_{q/n} x = n T_{1/n} K_x(e_q). \quad (\text{GI2.3})$$

Integer linearity and multiplication extend these identities to every element of $\mathcal{C}$, and complex linearity to its scalar extension.

Conversely, let $K : E_0 \rightarrow \mathcal{Q}$ be a complex-linear intertwiner for this specified algebra. Set $x = K(e_1)$. For a reduced positive rational $q = a/b$,

$$e_{a/b} = \frac{1}{b} F_a V_b e_1. \quad (\text{GI2.4})$$

Intertwining and complex linearity give

$$K(e_{a/b}) = \frac{1}{b} T_a (b T_{1/b}) x = T_{a/b} x. \quad (\text{GI2.5})$$

Thus $K = K_x$. The scalar divisions in GI2.4-GI2.5 occur on the complex coefficient spaces, not inside the integral algebra $\mathcal{C}$. This proves a complete classification of the algebraic intertwiners and gives their exact image:

$$M_x = \text{span}_{\mathbb{C}} \{T_q x : q \in \mathbb{Q}_{>0}\}. \quad (\text{GI2.6})$$

In the locally convex direct-sum topology on E_0 , every K_x is continuous: its restriction to every finite-dimensional coordinate subspace is a linear map to $\mathcal{Q}$, hence continuous, which is the defining final property of this topology. The Hilbert topology has a different answer, calculated next.

GI3. Strong continuity of the actual multipliers at scale one

For positive real q , let $t = \log q$. The identity

$$e^{ts} - 1 = ts \int_0^1 e^{uts} du \quad (\text{GI3.1})$$

gives, when $|\operatorname{Re} s| \leq a$,

$$|q^s - 1| \leq |\log q| |s| e^{a|\log q|}. \quad (\text{GI3.2})$$

Since $|s| \leq (a+1)(1 + |\operatorname{Im} s|)$ on this strip, every $F \in \mathcal{B}$ satisfies

$$b_{a,M}((q^s - 1)F) \leq |\log q| e^{a|\log q|} (a+1) b_{a,M+1}(F). \quad (\text{GI3.3})$$

Thus $q^s F \rightarrow F$ in every original seminorm as $q \rightarrow 1$, and passing through the continuous quotient map proves

$$T_q x \longrightarrow x \quad \text{in } \mathcal{Q}. \quad (\text{GI3.4})$$

The bound is uniform for q in an interval shrinking to one. It retains the entire strip and uses only one additional decay seminorm; no analytic continuation or location assumption on zeros has been inserted.

GI4. An explicit sequence proves nonclosability for every nonzero receiver

For each integer $N \geq 1$, choose the explicit distinct positive rationals

$$q_{N,j} = 1 + \frac{j}{N^2}, \quad 1 \leq j \leq N, \quad u_N = \frac{1}{N} \sum_{j=1}^N e_{q_{N,j}} \in E_0. \quad (\text{GI4.1})$$

Orthogonality of distinct scale components gives the full norm

$$\|u_N\|^2 = \frac{AB}{N^2} \sum_{j=1}^N \left(1 + \frac{j}{N^2}\right) = AB \left(\frac{1}{N} + \frac{N+1}{2N^3}\right) \rightarrow 0. \quad (\text{GI4.2})$$

All $q_{N,j}$ lie in $[1, 1 + 1/N]$. Therefore GI3 gives

$$K_x u_N = \frac{1}{N} \sum_{j=1}^N T_{q_{N,j}} x \rightarrow x \quad \text{in } \mathcal{Q}. \quad (\text{GI4.3})$$

For an explicit quotient verification choose any representative F of x ; GI3.3 bounds the difference of the average representatives from F by the supremum of its right side over $[1, 1 + 1/N]$, which tends to zero for every seminorm.

A densely defined linear map from the Hilbert space into this Hausdorff topological vector space is closable only if the closure of its graph contains no pair $(0, y)$ with $y \neq 0$. GI4.2–GI4.3 put $(0, x)$ in that graph closure. Hence

$$\boxed{K_x : E_0 \subset H_{\text{geom}}^0 \rightarrow \mathcal{Q} \text{ is closable exactly when } x = 0.} \quad (\text{GI4.4})$$

For $x = 0$, the map is zero and extends continuously to the Hilbert completion, proving the other direction. In particular no nonzero K_x is Hilbert-continuous. This conclusion applies to critical-line and off-line blocks alike; it is not a zero-location criterion.

GI5. The whole graph closure and the retained cyclic image

Let Γ_x denote the graph of K_x , and take its closure in $H_{\text{geom}}^0 \times \mathcal{Q}$. Then

$$\boxed{\overline{\Gamma_x} = H_{\text{geom}}^0 \times \overline{M_x}.} \quad (\text{GI5.1})$$

To prove it, GI4 first gives $(0, x) \in \overline{\Gamma_x}$. For every positive rational $r = a/b$, the operator sending e_q to e_{rq} is $b^{-1}F_a V_b$ on the complex domain. Its squared norm is exactly r , because

$$AB \sum_q r q |a_q|^2 = r AB \sum_q q |a_q|^2. \quad (\text{GI5.2})$$

It extends boundedly to the Hilbert completion, and the target operator is the continuous T_r . Intertwining makes their product action preserve the graph and its closure. Applying it to $(0, x)$ gives $(0, T_r x)$ in the closure. Closed linearity then gives $\{0\} \times \overline{M_x} \subseteq \overline{\Gamma_x}$.

For any $u \in E_0$, subtracting $(0, K_x u)$ from its graph point $(u, K_x u)$ gives $(u, 0)$ in the closed graph. Density of E_0 gives $H_{\text{geom}}^0 \times \{0\}$ in that closure, and adding the two subspaces proves one inclusion in GI5.1. The reverse inclusion follows because every original graph value lies in M_x and $H_{\text{geom}}^0 \times \overline{M_x}$ is closed. This proves the whole formula rather than only the failure of boundedness.

GI6. Every multiplicity jet in a local cyclic image

For an actual nontrivial zero ρ with multiplicity m , retain its actual continuous projector P_ρ from RZ and its full block $\mathcal{Q}_\rho$. If $x \in \mathcal{Q}_\rho$, then

$$K_x(e_q) = q^\rho \sum_{j=0}^{m-1} \frac{(\log q)^j}{j!} N_\rho^j x. \quad (\text{GI6.1})$$

No jet has been projected away. For nonzero x , let d be the least positive integer with $N_\rho^d x = 0$. The vectors

$$x, N_\rho x, \dots, N_\rho^{d-1} x \quad (\text{GI6.2})$$

are independent: in a nontrivial relation choose the first nonzero coefficient at index j_0 and apply $N_\rho^{d-1-j_0}$; only that coefficient times $N_\rho^{d-1} x \neq 0$ survives.

Every vector in GI6.1 belongs to their span. Conversely choose $q = 2^k$, $0 \leq k < d$. The coefficient matrix in this ordered list is

$$\left[2^{k\rho} \frac{(k \log 2)^j}{j!} \right]_{0 \leq k, j < d}. \quad (\text{GI6.3})$$

Its determinant retains every factor:

$$\left(\prod_{k=0}^{d-1} 2^{k\rho} \right) \left(\prod_{j=0}^{d-1} \frac{(\log 2)^j}{j!} \right) \prod_{0 \leq k < \ell < d} (\ell - k) \neq 0. \quad (\text{GI6.4})$$

Thus these images span every vector in GI6.2, proving

$$M_x = \text{span}_{\mathbb{C}} \{x, N_\rho x, \dots, N_\rho^{d-1} x\}, \quad \dim M_x = d. \quad (\text{GI6.5})$$

It is closed in the Hausdorff locally convex space $\mathcal{Q}$, since it is finite dimensional. In particular, use the quotient class of RZ's zero-isolating function, $x = [E_{\rho,0}] = [e_\rho] \in \mathcal{Q}$. For this actual cyclic vector, $d = m$, $M_x = \mathcal{Q}_\rho$, and GI5 becomes

$$\overline{\Gamma_{[E_{\rho,0}]}} = H_{\text{geom}}^0 \times \mathcal{Q}_\rho. \quad (\text{GI6.6})$$

This gives a surjective algebraic map to each complete actual zero block, including all multiplicities, with its entire graph closure explicitly calculated.

GI7. Consequence for the geometric purity comparison

FT10 proves the exact circle spectrum $|\lambda| = \sqrt{n}$ and empty point spectrum for the geometric Hilbert shift of degree $n \geq 2$. GI2 constructs the algebraic maps to every actual original-zeta coefficient vector, and GI6 proves surjectivity onto every actual finite zero block when its cyclic vector is chosen. GI4–GI5 show exactly why these particular maps do not transport that Hilbert spectral statement: every nonzero map is nonclosable in that Hilbert topology, although it is continuous for the original direct-sum topology of the algebraic domain.

This is an actual morphism and its complete topological defect, not an assertion that the two constructions are unrelated. The same defect occurs for a hypothetical off-line block and for every critical-line block. It neither assumes nor disproves RH. The source's complete zero detector and its reflected Weil pairing remain unchanged. The support-record version sends (ℓ, u) to $(\ell, K_x u)$, retaining $(\ell, 0)$ when its coefficient is zero, exactly as ZR2–ZR4 prove.

Support records retained even at zero coefficient

Proof source: SUPPORT_ZERO_RECEIVER_REPAIR.md.

Retaining support at every zero coefficient

24 September 2026. ZR1-ZR5. This corrects the support-retention wording of M9.7 and S7 while retaining their linear isomorphisms and synthesis theorem. The correction was supplied by the receiving task in ITEM26_GLOBAL_RECEIVERS_INTEGRATION.md, I26.10; its complete argument is reproduced and propagated here. The record set below is the programme's existing set of labels, not an arithmetic value assigned to an unsupported element.

ZR1. Exact domain of the linear statement

Let $\mathcal{Q} = \mathcal{B}/\mathcal{I}$ be the actual original-zeta quotient, and let $J : \mathcal{Q} \rightarrow \prod_{\rho} \mathbb{C}^{m_{\rho}}$ be the injective map of all zero jets proved in S7. Let $\mathcal{L}$ be the retained set of support records. Write e_{ℓ} for the basis of the free vector space $\mathbb{C}^{(\mathcal{L})}$. The linear comparison

$$J \otimes \text{id} : \mathcal{Q} \otimes \mathbb{C}^{(\mathcal{L})} \longrightarrow \left(\prod_{\rho} \mathbb{C}^{m_{\rho}} \right) \otimes \mathbb{C}^{(\mathcal{L})} \quad (\text{ZR1.1})$$

is injective: every tensor has a finite expansion $\sum q_{\ell} \otimes e_{\ell}$, and the coordinate functional at each e_{ℓ} makes a zero image imply $Jq_{\ell} = 0$, hence $q_{\ell} = 0$. This proves precisely the linear assertion. However,

$$0_{\mathcal{Q}} \otimes e_{\ell} = 0_{\mathcal{Q}} \otimes e_{\ell'} = 0 \quad (\text{ZR1.2})$$

for every pair of records. The passage from a labelled coefficient to a tensor has already identified its zero sections. Injectivity of ZR1.1 does not undo that earlier map.

ZR2. Exact record-retaining quotient and observation

Retain the disjoint family

$$\widehat{\mathcal{Q}} = \coprod_{\ell \in \mathcal{L}} \{\ell\} \times \mathcal{Q}. \quad (\text{ZR2.1})$$

It is equivalently the quotient of $\mathcal{L} \times \mathcal{B}$ by $(\ell, f) \sim (\ell', g)$ exactly when $\ell = \ell'$ and $f - g \in \mathcal{I}$. This is an equivalence relation since $\mathcal{I}$ is a vector subspace. The map to ZR2.1 is $(\ell, f) \mapsto (\ell, [f])$; its fibres are exactly that relation, proving the comparison. All subtractions occur in the coefficient functions. No addition is defined on $\mathcal{L}$.

The complete observation is

$$\widehat{J}(\ell, q) = (\ell, Jq). \quad (\text{ZR2.2})$$

If its two values agree, their first coordinates give equal records and its second coordinates give equal coefficient classes by injectivity of J . Thus $\widehat{J}$ is injective. The fibre of the coefficient-only zero observation is exactly

$$\coprod_{\ell \in \mathcal{L}} \{(\ell, 0_Q)\}. \quad (\text{ZR2.3})$$

It contains every original record, including those that must distinguish absence from a supported zero.

ZR3. The map to the previous tensors, with its full fibres

Define

$$c : \widehat{\mathcal{Q}} \longrightarrow \mathcal{Q} \otimes \mathbb{C}^{(\mathcal{L})}, \quad c(\ell, q) = q \otimes e_\ell. \quad (\text{ZR3.1})$$

Its zero fibre is exactly ZR2.3, because the coordinate functional at ℓ recovers q . If its common output is nonzero, equality of two images with different labels is impossible: the coordinate functional at the first label would give $q = 0$. Equality with the same label gives equal coefficients. Thus every nonzero fibre in its image is a singleton; tensors with several nonzero label coordinates are outside its image. This states every fibre without asserting that the map is surjective. The square with $\widehat{J}$, c , and $J \otimes \text{id}$ commutes by evaluation on (ℓ, q) .

ZR4. Arithmetic action, projectors, and the full pairing

Every existing coefficient operator $A : \mathcal{Q} \rightarrow \mathcal{Q}$ lifts as

$$\widehat{A}(\ell, q) = (\ell, Aq). \quad (\text{ZR4.1})$$

It satisfies $\widehat{AB} = \widehat{A}\widehat{B}$, $\widehat{\text{id}} = \text{id}$, and $c\widehat{A} = (A \otimes \text{id})c$ by direct substitution. This applies to all actual prime operators, every IAR coefficient operator, and every zero projector. Addition of operators remains fibrewise addition on $\mathcal{Q}$; it is not an operation on the primitive records. An operator sending a coefficient to zero sends (ℓ, q) to $(\ell, 0)$, retaining its record.

The full Weil pairing has the record-preserving receiver

$$((\ell, x), (\ell', y)) \longmapsto (\ell, \ell', W(x, y)). \quad (\text{ZR4.2})$$

Projection to the last coordinate is exactly the previous form, with all its endpoint, Gamma, prime and original-zero contributions unchanged. Its weighted-adjoint identity is therefore the same equality of last coordinates with the same two retained labels. A vanishing form value deletes neither input record. No sign or positivity is inferred from this recording map.

ZR5. The two coefficient sections at zero

The formal coefficient algebra C_{rec} of M9.7 has independent basis elements $U = [\tau]$ and $f = [1]$. Its two linear sections $q \mapsto q \otimes U$ and $q \mapsto q \otimes f$ are distinct maps, but both take zero to zero. Their linear difference is injective, as M9.7 proves; that still does not distinguish which zero section was selected.

To retain that already specified selection, use the record set $\mathcal{L} \times \{U, f\}$ and apply ZR2. The maps from the two selected families to the previous coefficient receiver are exactly

$$(\ell, U, q) \longmapsto q \otimes U \otimes e_\ell, \quad (\ell, f, q) \longmapsto q \otimes f \otimes e_\ell. \quad (\text{ZR5.1})$$

For nonzero coefficients their images are distinct by the independent basis coordinates. At zero the original records remain distinct in the domain and their comparison images coincide. This proves both the retained distinction and the specific loss under linear observation. It assigns no source sum, difference, parity or numerical value to τ .

Complete prime-record detection

Proof source: COMPLETE_PRIME_RECORD_AND_ZERO_DETECTION.md.

What the complete prime record detects, and the resulting arithmetic weight

24 September 2026. Proof locators CD1-CD7. This calculation receives the user's integrality and completeness argument in WU037. It retains the common supporting point $\tau\langle Z_1; \text{no } Z_2 \rangle$, the full signed stalk of CG2-CG3, and the original Riemann zeta function. No operation, coordinate, or distance is introduced on the supporting point.

The completed source reconstruction in CG0-CG8 is an input: its uniqueness is uniqueness of the specified complete Connes-Consani chart diagram and its universal localization, with all chart maps retained. Neither the existence of a cyclic stalk nor a spectral weight is deduced merely from the duration of an observation. The analytic quotient used below is the proved quotient in GLOBAL_MELLIN_SYNTHESIS.md, S1-S7; its complete proof is included in the cumulative reader. This note proves the next receiving calculation, rather than assuming that a zero outside the critical line is an omitted arithmetic event.

CD1. The full return record determines the original logarithmic derivative

Write $R = \text{End}_{\text{Ab}}(L)$, with its orientation-independent identity $\mathbf{1}_R = \text{id}_L$, as constructed in CG4. For each nonzero prime ideal $\mathfrak{p}$, its residue norm is

$$N(\mathfrak{p}) = |R/\mathfrak{p}|. \quad (\text{CD1.1})$$

CG5 identifies these norms with every positive prime integer, including two. Retain the three complete measures

$$\mathcal{D} = \delta_0 + \sum_{n \geq 2} \delta_{\log n}, \quad \mathcal{R} = \sum_p \sum_{k \geq 1} \frac{1}{k} \delta_{k \log p}, \quad \mathcal{W} = \sum_p \sum_{k \geq 1} (\log p) \delta_{k \log p}. \quad (\text{CD1.2})$$

The proof of $\mathcal{D} = \exp_* \mathcal{R}$, including each coefficient and the unit atom, is TP9-TP13. For $\text{Re } s > 1$ it gives

$$\zeta(s) = \int e^{-st} d\mathcal{D}(t) = 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad (\text{CD1.3})$$

$$H(s) := \int e^{-st} d\mathcal{W}(t) = \sum_p \sum_{k \geq 1} (\log p) p^{-ks} = -\frac{\zeta'(s)}{\zeta(s)}. \quad (\text{CD1.4})$$

Every displayed identity is on its convergence domain. In particular the prime sum is not asserted to converge at a nontrivial zero. To justify the derivative and all local termwise operations, on a compact set

in $\text{Re } s > 1$, choose $\sigma_0 > 1$ below all its real parts. The sum with an additional fixed power of $k \log p$ is bounded by a constant times $\sum_{n \geq 2} (\log n)^j n^{-\sigma_0}$, with a possibly larger nonnegative integer j . This converges by the integral test. The exponential of the absolutely convergent prime logarithm is nonzero on this half-plane, so logarithmic differentiation is justified there.

The complete record therefore determines one holomorphic germ H on that half-plane. Its meromorphic continuation is the indicated logarithmic derivative of the original meromorphic ζ . Two continuations on a connected domain agree by the identity theorem. This is analytic uniqueness from the full record, without a finite-prefix inference.

CD2. Every zero is detected, with its original multiplicity

Let ρ be any actual nontrivial zero of the original ζ , of multiplicity m_ρ . Holomorphic Taylor division gives, on a sufficiently small disk,

$$\zeta(s) = (s - \rho)^{m_\rho} g_\rho(s), \quad g_\rho(\rho) \neq 0. \quad (\text{CD2.1})$$

Shrinking the disk ensures g_ρ has no zero there. Direct differentiation of this original function gives

$$H(s) = -\frac{m_\rho}{s - \rho} - \frac{g'_\rho(s)}{g_\rho(s)}. \quad (\text{CD2.2})$$

Consequently the positive counterclockwise boundary γ_ρ of that disk detects exactly

$$-\frac{1}{2\pi i} \int_{\gamma_\rho} H(s) ds = m_\rho. \quad (\text{CD2.3})$$

The second term in (CD2.2) is holomorphic, so its integral is zero; the first term has residue $-m_\rho$. This proves (CD2.3). No step uses $\text{Re } \rho = 1/2$.

The original pole at $s = 1$ is retained too. Writing $\zeta(s) = (s - 1)^{-1} g_1(s)$, with $g_1(1) = 1$, gives

$$H(s) = \frac{1}{s - 1} - \frac{g'_1(s)}{g_1(s)}. \quad (\text{CD2.4})$$

Every original trivial zero $-2r$, $r \geq 1$, gives residue -1 in H . Thus this scalar detector contains the pole and both types of zero, with their signs.

An off-line zero, if there is one, is consequently a zero detected by the same uniquely reconstructed record. It is not an extra prime, a fractional prime norm, or a prime-power event omitted from (CD1.2). The assertion follows from the actual detector (CD2.3), not from substituting another zeta function. Completeness determines all zero locations; it does not restrict the definition of detection to locations on the critical line.

CD3. The full auxiliary multiplier and its contribution to the detector

The exact Mellin transform used for the source realization is

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{CD3.1})$$

It is the Mellin transform of the explicit original kernel

$$k_0(u) = u^{1/2} \sum_{n \geq 1} f_0(nu), \quad f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2},$$

under $\mathcal{M}k(s) = \int_0^\infty k(u) u^{s-1/2} du/u$. S2 proves this identity, entire continuation, and every exceptional value. In particular

$$F_0(0) = F_0(1) = \frac{1}{8},$$

$$F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0. \quad (\text{CD3.2})$$

This is an auxiliary source transform with an explicit comparison, not a replacement for the original function.

Let $\psi = \Gamma'/\Gamma$. Logarithmic differentiation of the complete product, on the open set where its factors are nonsingular and nonzero, gives

$$\boxed{-\frac{\zeta'(s)}{\zeta(s)} = -\frac{F'_0(s)}{F_0(s)} + \frac{1}{s} + \frac{1}{s-1} - \frac{\log \pi}{2} + \frac{1}{2} \psi(s/2)}. \quad (\text{CD3.3})$$

Both sides are meromorphic, so the identity extends meromorphically. The constant $1/8$ has derivative zero and remains in (CD3.1), (CD3.2), and every source projector below. At $s = 0$, the residues of $1/s$ and $\psi(s/2)/2$ cancel. At $s = 1$, the retained endpoint term has residue $+1$. At $s = -2r$, the Gamma logarithmic derivative has residue -1 , while (CD3.2) makes F'_0/F_0 regular. At a nontrivial zero, the elementary and Gamma terms are regular and $-F'_0/F_0$ has residue $-m_\rho$. Thus the quotient detector and the full original detector agree on nontrivial zeros with every discarded-looking term accounted for explicitly.

CD4. Detection inside the actual whole source quotient

For the following stronger detection statement, retain the exact function spaces

$$\mathcal{B} = \{F \text{ entire} : b_{A,M}(F) = \sup_{|\operatorname{Re} s| \leq A} (1 + |\operatorname{Im} s|)^M |F(s)| < \infty \text{ for all integers } A, M \geq 0\},$$

$$\mathcal{I} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every actual nontrivial zero } \rho, 0 \leq j < m_\rho\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{I}. \quad (\text{CD4.1})$$

The synthesis proof S5 identifies $\mathcal{I}$ with the closed original source image; zeros have not been freely appended as extra states. The physical source space and full proof of this equality are included in the cumulative reader.

For a particular actual zero, put $m = m_\rho$ and

$$A_\rho(s) = \frac{F_0(s)}{(s-\rho)^m}, \quad B_\rho(s) = \operatorname{Tay}_{\rho, < m}(1/A_\rho)(s), \quad e_\rho = A_\rho B_\rho. \quad (\text{CD4.2})$$

Taylor division gives $A_\rho(\rho) \neq 0$. At ρ the product is 1 modulo $(s-\rho)^m$; at every other zero it has the full required vanishing order because its denominator is nonzero there. Division by a fixed linear factor of a vanishing entire strip-Schwartz function stays in $\mathcal{B}$: outside a unit disk this follows by division, and inside it follows by Cauchy's maximum principle on a radius-two disk. Multiplication by a polynomial also preserves $\mathcal{B}$. These facts establish $e_\rho \in \mathcal{B}$ on the whole plane, not only on a finite test window.

The exact projector lift is

$$\mathcal{P}_\rho F(s) = A_\rho(s) \text{Taylor}_{\rho, < m}(F/A_\rho)(s). \quad (\text{CD4.3})$$

Its output has all of the first m jets of F at ρ and all zero jets at every other zero. Taylor coefficients are continuous on $\mathcal{B}$; hence this is a continuous finite-rank operator. A second application does not change its polynomial, so it is idempotent. It kills $\mathcal{J}$ and defines P_ρ on $\mathcal{Q}$. The vectors

$$[e_\rho(s)(s - \rho)^j], \quad 0 \leq j < m, \quad (\text{CD4.4})$$

are independent because their derivative jets at ρ are respectively $j!$ in position j and zero in the other positions. Their quotient classes span $\text{im } P_\rho$, since

$$P_\rho[F] = \sum_{j=0}^{m-1} \frac{F^{(j)}(\rho)}{j!} [e_\rho(s)(s - \rho)^j].$$

At the representative level, the range of (CD4.3) is $A_\rho \text{Pol}_{< m}$; the displayed representatives agree with their projected representatives modulo $\mathcal{J}$. Consequently $\text{rank } P_\rho = m$, and

$$\bigcap_{\rho} \ker P_\rho = \{0\}. \quad (\text{CD4.5})$$

Indeed the intersection forces all required jets to vanish, exactly the definition of $\mathcal{J}$, which the synthesis theorem identifies with the original closed source image. No convergence of an infinite sum of projectors is asserted or needed. The complete resolvent and every source-level correction are proved in RZ3-RZ7.

Thus completeness of the quotient makes every actual zero detectable, including its multiplicity, independently of its location. An off-line zero would not fail (CD4.5); its own nonzero projector would be among the detectors proving that equality.

CD5. The exact arithmetic weight seen by those detectors

Use a different symbol L_ζ for the spectral operator to avoid confusing it with the winding group L . It is induced by $F(s) \mapsto sF(s)$. For every positive real a , the actual action is

$$T_a[F] = [a^s F(s)], \quad \mathcal{M}^{-1} T_a \mathcal{M} k(u) = a^{1/2} k(u/a). \quad (\text{CD5.1})$$

The multiplier preserves $\mathcal{B}$, with seminorm factor $\max(a^A, a^{-A})$, and preserves $\mathcal{J}$, with every order unchanged. Substitution $u = av$ in the Mellin integral proves the second identity including its factor. Composition gives $T_a T_b = T_{ab}$ and $T_1 = I$.

On the range of P_ρ , let $N_\rho = L_\zeta - \rho I$. Multiplication by $s - \rho$ shifts the vectors (CD4.4) to the next one, and kills the final one modulo $\mathcal{J}$. Thus $N_\rho^m = 0$, $N_\rho^{m-1} \neq 0$, and Taylor expansion gives the full operator

$$T_p|_{\text{im } P_\rho} = p^\rho \sum_{j=0}^{m-1} \frac{(\log p)^j}{j!} N_\rho^j, \quad |p^\rho| = p^{\text{Re } \rho}. \quad (\text{CD5.2})$$

Every logarithmic jet term remains present. The weight-one modulus is precisely the further assertion $|p^\rho| = p^{1/2}$ for these actual blocks. Equation (CD5.2) proves that assertion is equivalent to $\text{Re } \rho = 1/2$, for any one prime $p > 1$; it does not assume it from the source's unique supporting point.

CD6. The retained reflection gives an exact degree factor, and exposes the sign

The original functional equation, with all factors retained in (CD3.1), gives the involution $\rho^\# = 1 - \bar{\rho}$ with $m_{\rho^\#} = m_\rho$. On $\mathcal{B}$, $F^\#(s) = \overline{F(1 - \bar{s})}$. The actual reflected form is

$$W([F], [G]) = \sum_{\rho} m_{\rho} \overline{F(\rho^\#)} G(\rho). \quad (\text{CD6.1})$$

The sum is over distinct zeros, with multiplicity supplied by m_ρ , and is absolutely convergent and continuous. For example strip decay of order two bounds the summand by a constant times $(1 + |\text{Im } \rho|)^{-4}$; the zero count from S2 is $O((R + 2)^{3/2})$. A dyadic decomposition makes the sum finite. RZ10 also gives the quotient seminorm bound and its exact radical. Direct substitution yields

$$W(T_p x, T_p y) = p W(x, y), \quad (\text{CD6.2})$$

because $\overline{\rho^\#} + \rho = 1$. Thus the similitude factor is already derived on the actual whole arithmetic quotient. It equals the prime norm $p = |L/[p]L|$; no finite geometric morphism with degree p on this quotient is asserted by that equality.

Its sign is not supplied by that factor. The globally constructed vectors e_ρ have value one at ρ and zero at all other zeros. On a distinct reflected pair their actual form is

$$W|_{\langle [e_\rho], [e_{\rho^\#}] \rangle} = \begin{pmatrix} 0 & m_\rho \\ m_\rho & 0 \end{pmatrix}, \quad (\text{CD6.3})$$

as follows by inserting those two exact value functions in (CD6.1). In particular their difference has value $-2m_\rho$. If a zero is fixed by reflection, its corresponding value vector instead has squared value m_ρ . All higher multiplicity jets lie in the radical of this value form and remain nonzero elements of $\mathcal{Q}$; no simplicity assertion is implicit.

The two eigenvalue moduli on a distinct reflected pair are exactly $p^{\text{Re } \rho}$ and $p^{1-\text{Re } \rho}$. Their product is p , and (CD6.2) holds on that pair. The calculated negative direction and the calculated unequal moduli are two descriptions of the same potential obstruction in this particular quotient. The unique complete counting record has supplied the quotient, all these projectors, and this pairing. Its uniqueness has not changed (CD6.3) into a positive matrix.

CD7. What follows for the proposed contradiction

The calculation establishes three connected facts. First, the complete reconstructed prime record determines the original zeta and hence every actual zero and multiplicity. Second, each such zero has an explicit nonzero projector in the source quotient; the joint kernel of all these projectors is zero. Third, the arithmetic action on those projectors has modulus $p^{\text{Re } \rho}$ and satisfies the reflected degree identity (CD6.2), with its sign fully calculated.

The detector identities prove that every actual zero has residue count $m_\rho > 0$ and a nonzero projector, with no restriction on its real part. Thus the proposed contradiction has not been derived: off-line location has not been shown to imply a failure of detection. This identifies the exact inference not established by the reconstruction; it does not discard the supporting geometry or the programme of deriving a weight theorem over it.

The integrality comparison is given in INTEGRAL_COUNTING_AND_WEIGHT_COMPARISON.md. Its exact connecting object is the integral group algebra of the multiplicative positive rationals, acting through (CD5.1). This retains integer arithmetic and the full prime scaling representation without falsely treating that representation as additive. No theorem of positivity or purity is introduced as a missing hypothesis.

Sources and proof provenance

- Alain Connes and Caterina Consani, *The Absolute Twistor Line and the Geometry of $\overline{\text{Spec } \mathbb{Z}}$* , arXiv:2609.00299v1, original CC.tex, especially lines 494–615 and 772–904. The common generic support and actual signed stalk are inputs from this source, with the comparison proved in CG1–CG3. Public source: <https://arxiv.org/html/2609.00299v1#S3>.
- Alain Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, arXiv:2602.04022v1, original rhready.tex, lines 526–545. The source’s complete multiplier is compared in S2 and (CD3.1); its factorization theorem is used with all genus-one factors in the full synthesis proof. Public source: <https://arxiv.org/abs/2602.04022>.
- Local complete derivations: CG0–CG8 and TP0–TP14; GLOBAL_MELLIN_SYNTHESIS.md S1–S7; GLOBAL_CANONICAL_MELLIN_INVOLUTION.md G1–G5; ORIGINAL_ZETA_RESOLUTION_AND_PROJECTORS.md RZ1–RZ12. These are programme derivations, not quotations or novelty claims. Their complete sources are retained in the cumulative local reader.

The disk index routes Deligne’s original Weil I and Weil II to PDF-only records. They were not newly read as TeX in this calculation; no claim of reconstructing Deligne’s proof is made here. The elementary integral comparison in the companion note is proved directly. A complete derivation of Deligne’s machinery over the present source remains a different mathematical task from the detection result proved above.

Integral counting and the exact weight comparison

Proof source: INTEGRAL_COUNTING_AND_WEIGHT_COMPARISON.md.

Integral counting, arithmetic operators, and the weight comparison

24 September 2026. Proof labels IC1-IC9. This note receives the proposed implication from the complete global counting unit to integrality and then purity. It calculates the intervening maps. It does not assume an off-critical zero exists, and the projective-line calculation below is not asserted to satisfy the entire tau/Connes-Consani programme.

The supporting point remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. No addition, subtraction, distance, coordinate, vector or numerical value is assigned to that point. Every operation below has as its domain a stalk quotient, an endomorphism ring, a group algebra, a cohomology group or the original-zeta quotient explicitly specified below.

Sources actually used

- CANONICAL_GENERIC_RECONSTRUCTION.md, CG0-CG8, read in full: the complete winding quotient, its endomorphism ring, the canonical unit, prime norms, and original-zeta return measure. Its original Connes-Consani source and exact author-source locators are retained there.
- TWISTOR_DEGREE_DERIVATION.md, TD3-TD4 and TD8, read in full for the integral sphere cohomology, its degree action, both signed source components, and the finite-characteristic fixed-point calculation. The calculation below restates the relevant proof rather than importing a conclusion called purity.
- ORIGINAL_ZETA_RESOLVENT_AND_PROJECTORS.md, RZ8 and RZ10-RZ11, read in full: the actual multiplier a^s , its full source factor $a^{1/2}$, all multiplicity jets, and the reflected Weil form. RZ5 supplies its proved zero-isolating representatives. This note uses that proved receiving space, not an assumed spectral object.
- No claim is made here to have read Deligne's original author TeX or to reconstruct Deligne's proof. The elementary finite-field comparison has the weight convention $|\lambda| = p^{w/2}$, with its actual weights calculated below.

IC1. The integral ring recovered from the complete winding object

Retain the complete CC generic group and its finite subgroup:

$$G = \langle T, J, \epsilon : TJ = JT, T\epsilon = \epsilon T, J\epsilon = \epsilon J, J^2 = \epsilon, \epsilon^2 = 1 \rangle, \quad H = \langle J \rangle, \quad L_w = G/H. \quad (\text{IC1.1})$$

All source commutation relations remain displayed, including those also implied by $\epsilon = J^2$. CG2-CG3 prove that $G \cong \mathbb{Z} \times C_4$ and L_w is infinite cyclic, retaining $1 \rightarrow H \rightarrow G \rightarrow L_w \rightarrow 1$. The subscript here distinguishes the winding group from the spectral operator below.

Let

$$R = \text{End}_{\text{Ab}}(L_w), \quad (f \oplus g)(x) = f(x)g(x), \quad (f \odot g)(x) = f(g(x)). \quad (\text{IC1.2})$$

An endomorphism is exactly a power map $[n] : x \mapsto x^n$, with $n \in \mathbb{Z}$. Indeed its value on a generator determines every value, and is a unique power of that generator. Choosing the inverse generator leaves the exponent unchanged. Thus

$$R \cong \mathbb{Z}, \quad [a] \oplus [b] = [a + b], \quad [a] \odot [b] = [ab], \quad \mathbf{1}_R = [1] = \text{id}_{L_w}. \quad (\text{IC1.3})$$

The nonnegative submonoid R_+ consists of the successive sums of this identity, including the zero endomorphism. For nonzero $f = [n] \in R_+$, define

$$d(f) = |L_w / f(L_w)| = n. \quad (\text{IC1.4})$$

Representatives are the first n powers of any generator. Division of exponents proves that they exhaust the quotient without repetition. In particular

$$d(f \odot g) = d(f)d(g), \quad d(\mathbf{1}_R) = 1. \quad (\text{IC1.5})$$

This is a proved integrality statement: every counting endomorphism has an integer degree and every positive prime degree is a finite quotient cardinality. It makes no assertion that eigenvalues of a different representation of these degrees are algebraic integers.

IC2. The exact map from the counting operators to the original-zeta action

Use the original-zeta receiving object from RZ. Let $\mathcal{B}$ be the entire functions rapidly decreasing vertically in each bounded real strip, with seminorms

$$b_{A,M}(F) = \sup_{|\text{Re } s| \leq A} (1 + |\text{Im } s|)^M |F(s)|. \quad (\text{IC2.1})$$

Let $\mathcal{Z}$ be the set of distinct actual nontrivial zeros of the original ζ , and m_ρ their full multiplicities. The closed ideal is

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every } \rho \in \mathcal{Z}, 0 \leq j < m_\rho\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}. \quad (\text{IC2.2})$$

The prior global synthesis proves that this is the quotient of the original arithmetic source by its closed image, with the full multiplier

$$F_0(s) = \frac{s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)}{8} \quad (\text{IC2.3})$$

retained in the source comparison. Formula (IC2.3) is not a replacement for ζ : the full endpoint, Gamma and trivial-zero comparison belongs to that proof. Only the actual nontrivial-zero ideal (IC2.2) is used for the receiving calculation here.

Write $L_\zeta[F] = [sF(s)]$. For every real $a > 0$, RZ8 defines the continuous invertible operator

$$T_a[F] = [a^s F(s)], \quad a^s = \exp(s \log a). \quad (\text{IC2.4})$$

Here $\log a$ is real. Direct multiplication proves

$$T_a T_b = T_{ab}, \quad T_1 = I, \quad T_a^{-1} = T_{1/a}. \quad (\text{IC2.5})$$

The exact continuity estimate before passage to the quotient is

$$b_{A,M}(T_a F) \leq \max(a^A, a^{-A}) b_{A,M}(F). \quad (\text{IC2.6})$$

The multiplier is entire and nowhere zero, so it preserves every vanishing order in (IC2.2). Taking infima over representatives gives the same bound for quotient seminorms. The source Mellin transform and its inverse comparison are

$$\mathcal{M}k(s) = \int_0^\infty k(u)u^{s-1/2}\frac{du}{u}, \quad \mathcal{M}^{-1}T_a\mathcal{M}k(u) = a^{1/2}k(u/a). \quad (\text{IC2.7})$$

Substituting $u = av$ into the integral proves this identity with both powers retained. In the multiplier sense, RZ writes $a^{L\zeta} = T_a$; it does not require convergence of an operator power series.

The requested canonical map out of the positive counting degrees is therefore

$$(R_+ \setminus \{0\}, \odot) \longrightarrow \text{Aut}_{\text{cont}}(\mathcal{Q}), \quad f \longmapsto T_{d(f)}. \quad (\text{IC2.8})$$

Equations (IC1.5) and (IC2.5) prove the homomorphism property, including the unit. No arbitrary orientation or preassigned prime label is needed. The full source ring is retained; (IC2.8) states which of its operations this particular receiving map preserves.

IC3. The integral group algebra carrying the arithmetic action

Let $\mathbb{Q}_{>0}^\times$ be the multiplicative group of positive rational numbers. Define

$$\Lambda = \mathbb{Z}[\mathbb{Q}_{>0}^\times] = \left\{ \sum_{r \in S} a_r b_r : S \subset \mathbb{Q}_{>0}^\times \text{ finite, } a_r \in \mathbb{Z} \right\}, \quad b_r b_t = b_{rt}, \quad 1_\Lambda = b_1. \quad (\text{IC3.1})$$

Different symbols b_r are a free additive basis; coefficients of equal symbols are added. Multiplication is the displayed finite convolution, hence associative, commutative and distributive, with identity b_1 . Negative coefficients are additive inverses. This constructs the ring directly.

Unique prime factorization gives the exact identification

$$\Lambda \cong \mathbb{Z}[b_p, b_p^{-1} : p \text{ prime}], \quad b_r = \prod_p b_p^{v_p(r)}, \quad (\text{IC3.2})$$

where only finitely many integer valuations $v_p(r)$ are nonzero. All primes, including two, occur.

The original arithmetic action extends to the unital ring homomorphism

$$\Phi : \Lambda \longrightarrow \text{End}_{\text{cont}}(\mathcal{Q}), \quad \Phi \left(\sum_{r \in S} a_r b_r \right) = \sum_{r \in S} a_r T_r. \quad (\text{IC3.3})$$

Additivity holds term by term. Multiplying two finite sums and using $T_r T_t = T_{rt}$ proves multiplicativity, and $\Phi(b_1) = I$. Each image is continuous because

$$b_{A,M} \left(\sum_{r \in S} a_r r^s F(s) \right) \leq \left(\sum_{r \in S} |a_r| \max(r^A, r^{-A}) \right) b_{A,M}(F). \quad (\text{IC3.4})$$

The same estimate passes to the quotient. We assert continuity of every represented operator; no topology on the abstract finite-support ring Λ is implicit in this statement. No injectivity of Φ is needed or asserted.

There are now two exact maps, with different domains of preserved operations:

$$\begin{array}{lll} R \cong \mathbb{Z} & \longrightarrow & \Lambda, \quad [n] \longmapsto nb_1, \quad \text{a unital ring homomorphism;} \\ R_+ \setminus \{0\} & \longrightarrow & \Lambda^\times, \quad [n] \longmapsto b_n, \quad \text{a multiplicative-monoid homomorphism.} \end{array} \quad (\text{IC3.5})$$

The first is injective because the b_1 -coefficient of nb_1 is n . Their represented images are respectively

$$\Phi(nb_1) = nI, \quad \Phi(b_n) = T_n = n^{L\zeta}. \quad (\text{IC3.6})$$

For example $b_2 \neq 2b_1$ already in the free additive basis. The difference is not erased in the actual receiving quotient. On any actual zero block $\mathcal{Q}_\rho$, RZ8 proves the eigenvalue of T_2 is 2^ρ . Since $0 < \text{Re } \rho < 1$,

$$|2^\rho| = 2^{\text{Re } \rho} < 2. \quad (\text{IC3.7})$$

Thus $T_2 \neq 2I$. The nontrivial-zero set is nonempty, independently of RH; the source quotient used here includes its proved nonzero isolating blocks. Consequently (IC2.8) cannot be an additive extension of the identity map on R : such an extension would give $T_2 = T_1 + T_1 = 2I$, contradicting (IC3.7).

This pinpoints the integrality transfer rather than discarding the relation between the two objects. The full map is (IC3.5) followed by (IC3.3), and its scalar and multiplicative inputs are explicitly distinct.

IC4. What integrality of the group algebra does, and does not, assert

An endomorphism of a finite-rank free abelian group has an integer matrix in an integral basis. Expanding $\det(XI - A)$ by permutations yields a monic polynomial with integer coefficients; every complex eigenvalue annihilates this polynomial. This proves that such an eigenvalue is an algebraic integer.

In contrast, the word integral in the coefficient ring $\Lambda = \mathbb{Z}[\mathbb{Q}_{>0}^\times]$ does not assert that each b_p is integral over its scalar subring $\mathbb{Z}b_1$. In fact it is not. If a monic relation of positive degree existed, it would have the form

$$b_p^d + a_{d-1}b_p^{d-1} + \cdots + a_0b_1 = 0, \quad a_j \in \mathbb{Z}. \quad (\text{IC4.1})$$

But $b_p^j = b_{p^j}$, and the positive rationals $1, p, \dots, p^d$ are distinct. The free additive basis in (IC3.1) makes the coefficient of b_{p^d} on the left equal to one, a contradiction. This proves the precise algebraic distinction without a spectral assumption.

On an actual zero block, with full multiplicity $m = m_\rho$, the represented action is

$$T_p|_{\mathcal{Q}_\rho} = p^\rho \sum_{j=0}^{m-1} \frac{(\log p)^j}{j!} N_\rho^j, \quad N_\rho^m = 0, \quad (\text{IC4.2})$$

and its characteristic polynomial is exactly

$$\det(XI - T_p|_{\mathcal{Q}_\rho}) = (X - p^\rho)^m. \quad (\text{IC4.3})$$

The nilpotent terms have not been removed; their diagonal is zero and their full coefficients remain (IC4.2). Neither CG's integral counting ring nor the representation (IC3.3) proves that the coefficients of (IC4.3) are integers, or that p^ρ is an algebraic integer. No finite-rank invariant integral lattice for these blocks has been constructed in these calculations. This is a record of the exact established property, not an assertion that such a lattice or a different weight construction is impossible.

IC5. An actual finite-field comparison, with every point count retained

Fix a rational prime p , including $p = 2$, and consider the projective line over $\mathbb{F}_p$. For every $r \geq 1$, its $\mathbb{F}_{p^r}$ -points consist of the p^r affine elements and the point at infinity. Hence

$$\#\mathbb{P}^1(\mathbb{F}_{p^r}) = 1 + p^r. \quad (\text{IC5.1})$$

Equivalently, the morphism $f_p : [x : y] \mapsto [x^p : y^p]$ over $\overline{\mathbb{F}}_p$ has these fixed-point counts for all powers: in its affine chart, f_p^r fixes the roots of $z^{p^r} - z$. The derivative is -1 , giving p^r distinct roots, and infinity is fixed.

Here is an integral constants-and-divisors realization of its degree-zero and degree-two cohomological action. Let

$$M = \mathbb{Z}e_0 \oplus \mathbb{Z}e_2, \quad e_0 \cdot e_0 = e_0, \quad e_0 \cdot e_2 = e_2 \cdot e_0 = e_2, \quad e_2 \cdot e_2 = 0. \quad (\text{IC5.2})$$

The first class is the constant class. The second is the degree-one divisor class $[\infty]$. Over an algebraically closed field every divisor is a finite integer sum of points. The rational function $z - a$ has divisor $[a] - [\infty]$; multiplying its integer powers shows every divisor of degree d is equivalent to $d[\infty]$. Principal divisors have degree zero, by counting numerator and denominator zeros including infinity. Thus the divisor class group is exactly $\mathbb{Z}[\infty]$, proving the integral second summand directly.

The same ring is the full integral singular cohomology of $\mathbb{P}^1(\mathbb{C})$: its sphere cell decomposition has a cell in degrees zero and two and none in degree one, and the square of a degree-two class vanishes because degree four is absent. This gives the integral comparison package conventionally denoted $H^0 \oplus H^2$. In the finite-field cohomological notation the second summand is the Tate class $\mathbb{Z}_\ell(-1)$, for $\ell \neq p$; the Frobenius convention here is the one entering point-count traces and acting by p on that untwisted degree-two class. The elementary calculation uses the displayed constants and divisor pullback; it does not require reconstructing the general étale comparison theorem.

The pullback of a constant is that constant. In the local parameter $w = 1/z$ at infinity, f_p sends w to w^p , so $f_p^*[\infty] = p[\infty]$. Therefore the full action is

$$F_p e_0 = e_0, \quad F_p e_2 = p e_2, \quad F_p = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}. \quad (\text{IC5.3})$$

Its trace on every iterate is exactly $1 + p^r$, matching (IC5.1), and its characteristic polynomial and determinant factor are

$$\det(XI - F_p) = X^2 - (1 + p)X + p, \quad \det(I - tF_p) = (1 - t)(1 - pt). \quad (\text{IC5.4})$$

Both eigenvalues are integers. All extension-field point counts are present. Their full generating function is

$$Z(\mathbb{P}^1/\mathbb{F}_p, t) = \exp \left(\sum_{r \geq 1} \frac{1 + p^r}{r} t^r \right) = \frac{1}{(1 - t)(1 - pt)}. \quad (\text{IC5.5})$$

This is an identity of formal power series and of analytic functions for $|t| < 1/p$: expand separately $-\log(1 - t)$ and $-\log(1 - pt)$, retaining both terms. It is not an identification of (IC5.5) with the original Riemann zeta function.

For connection with the already calculated actual CC signed sphere maps, TD8 has $z \mapsto z^p$ for $p \equiv 1 \pmod{4}$, and $z \mapsto -1/z^p$ for $p \equiv 3 \pmod{4}$. The latter fixed-point equation in the odd reciprocal iterates is $z^{p^r+1} = -1$, with exactly $p^r + 1$ distinct roots. Its degree action and all counts are therefore precisely the same (IC5.3)-(IC5.5). TD8 requires odd p for that signed source lift. The present projective-line Frobenius example itself includes two; it is not an unproved extension of that odd source lift.

IC6. Its degree-p pairing and its exact weights

Let $\text{tr}(e_0) = 0$ and $\text{tr}(e_2) = 1$, and define the integral bilinear form by extracting the top-degree coefficient:

$$B(x, y) = \text{tr}(xy), \quad B(ae_0 + be_2, ce_0 + de_2) = ad + bc. \quad (\text{IC6.1})$$

Its matrix and exact degree relation are

$$J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad F_p^\top J F_p = pJ. \quad (\text{IC6.2})$$

The determinant of J is -1 , so the integral pairing is perfect. Direct evaluation gives

$$B(e_0 + e_2, e_0 + e_2) = 2, \quad B(e_0 - e_2, e_0 - e_2) = -2. \quad (\text{IC6.3})$$

Thus its real signature is $(1, 1)$. The multiplier p pairs the eigenvalues 1 and p ; it does not make both eigenvalues have modulus $\sqrt{p}$.

The complete degree filtration is

$$W_{-1}M = 0, \quad W_0M = W_1M = \mathbb{Z}e_0, \quad W_2M = M. \quad (\text{IC6.4})$$

This records the proved cohomological degrees in (IC5.2), rather than assigning an unknown sector a desired degree. The associated graded degree-zero action is 1, and the degree-two action is p . Their absolute values are exactly

$$|1| = p^{0/2}, \quad |p| = p^{2/2}. \quad (\text{IC6.5})$$

They are pure of weights zero and two, respectively, in the finite-field weight convention. The direct sum is not pure of weight one. Its degree-one cohomology vanishes in this calculation. This is consistent with, and is not a violation of, the purity assertion for each specified cohomological degree.

There is also no positive-definite Hermitian form K on $M \otimes \mathbb{C}$ having this same degree- p similitude. Evaluating $K(F_p e_0, F_p e_0) = pK(e_0, e_0)$ would give $(p - 1)K(e_0, e_0) = 0$. Since $p > 1$, this forces $K(e_0, e_0) = 0$, incompatible with positive definiteness on the nonzero vector e_0 . This explicit calculation identifies the role of the cohomological sector and its form. It does not assert that positivity cannot be obtained on a different, correctly constructed weight-one object.

IC7. The exact map from the counting ring to the comparison package

The preceding integral package can be built canonically over the already reconstructed ring R : take $Re_0 \oplus Re_2$ and, for each nonzero $f \in R_+$, put

$$\mathcal{F}_f(ae_0 + be_2) = ae_0 + (f \odot b)e_2. \quad (\text{IC7.1})$$

In the canonical coordinates (IC1.3), $f = [n]$ gives $\text{diag}(1, n)$, not a chosen numerical coordinate at τ . Associativity of composition gives

$$\mathcal{F}_f \mathcal{F}_g = \mathcal{F}_{f \odot g}, \quad \mathcal{F}_{\mathbf{1}_R} = I. \quad (\text{IC7.2})$$

The pairing $B_R(ae_0 + be_2, ce_0 + de_2) = a \odot d \oplus b \odot c$ obeys

$$B_R(\mathcal{F}_f x, \mathcal{F}_f y) = f \odot B_R(x, y). \quad (\text{IC7.3})$$

For $f = [p]$, applying $R \cong \mathbb{Z}$ gives exactly the degree- p relation (IC6.2), with no altered multiplier. Thus the same canonical unit and recovered prime norms have an explicit integral cohomological representation whose actual weight pieces are zero and two. The representation supplies a precise comparison of integrality and weight. It is not a model of the full original-zeta receiving quotient, nor is it a claim that the CC programme has only these pieces.

IC8. The actual original-zeta pairing has its own precise integral-indexed degree relation

Let $\rho^\# = 1 - \bar{\rho}$. On $\mathcal{Q}$, the full source-derived Weil form is

$$W([F], [G]) = \sum_{\rho \in \mathcal{Z}} m_\rho \overline{F(\rho^\#)} G(\rho). \quad (\text{IC8.1})$$

The previous source calculation proves absolute convergence and Hermitian symmetry. Since $\overline{\rho^\#} + \rho = 1$, the actual unshifted arithmetic action satisfies, for every real $a > 0$,

$$W(T_a x, T_a y) = a W(x, y), \quad W(T_p x, T_p y) = p W(x, y) \quad \text{when } a = p. \quad (\text{IC8.2})$$

Every zero and every multiplicity is retained in this equality. The numerical norm $p = |L_w/[p]L_w|$ is exactly its multiplier. This is the concrete meeting point between the global counting ring and the spectral pairing.

On a reflected pair of distinct actual zeros, put $\lambda = p^\rho$, $\mu = p^{\rho^\#}$. Then

$$\bar{\lambda}\mu = p, \quad |\lambda| = p^{\text{Re } \rho}, \quad |\mu| = p^{1-\text{Re } \rho}. \quad (\text{IC8.3})$$

The exact isolating representatives give matrix $\begin{pmatrix} 0 & m_\rho \\ m_\rho & 0 \end{pmatrix}$ on the two constant-jet coordinates. These statements describe what the actual formula would record at such a zero; they do not invent a zero or discard any prime. Higher jets remain the full nilpotent structure (IC4.2), even where the ordinary trace pairing has a radical.

The weighted adjoint relation also determines an exact coefficient-domain enlargement. Set

$$\Lambda_{\mathbb{Q}} = \mathbb{Q}[\mathbb{Q}_{\geq 0}^\times], \quad \Phi_{\mathbb{Q}} \left(\sum_{r \in S} c_r b_r \right) = \sum_{r \in S} c_r T_r, \quad c_r \in \mathbb{Q}, \quad (\text{IC8.4a})$$

where S is finite. This explicitly extends Φ from (IC3.3). The same finite-convolution proof establishes its unital ring homomorphism property, and (IC3.4) with $|c_r|$ proves continuity of every image operator. On $\Lambda_{\mathbb{Q}}$, the assignment

$$b_r^* = r b_{1/r}, \quad a^* = a \quad (a \in \mathbb{Q}) \quad (\text{IC8.4})$$

is an involution: $(b_r^*)^* = r(1/r)b_r = b_r$, and multiplication is respected because $rt b_{1/(rt)} = (rb_{1/r})(tb_{1/t})$. The form satisfies

$$W(\Phi_{\mathbb{Q}}(A)x, y) = W(x, \Phi_{\mathbb{Q}}(A^*)y), \quad A \in \Lambda_{\mathbb{Q}}. \quad (\text{IC8.5})$$

For a monomial this is $W(T_r x, y) = W(x, r T_{1/r} y)$, obtained from (IC8.2) and invertibility. Rational coefficients are real, so conjugate-linearity in the first argument and linearity in the second give exactly the displayed equality for finite rational sums.

This rational enlargement is the smallest unital subring of $\Lambda_{\mathbb{Q}}$ containing Λ and preserved by this specified weighted involution. To prove minimality, let A be any such subring. For every prime p , the monomial $b_{1/p}$ belongs to $\Lambda \subseteq A$. Stability under $\star$ and closure under multiplication give

$$(b_{1/p})^* b_{1/p} = \left(\frac{1}{p} b_p \right) b_{1/p} = \frac{1}{p} b_1 \in A. \quad (\text{IC8.6})$$

Every positive integer denominator is a finite product of prime powers. Multiplying the corresponding scalar elements in (IC8.6), and then multiplying by an arbitrary integer scalar, proves $\mathbb{Q}b_1 \subseteq A$. Since

A also contains every b_r , multiplication supplies $c_r b_r$ for every rational c_r , and finite addition supplies every element of $\Lambda_{\mathbb{Q}}$. Hence $A = \Lambda_{\mathbb{Q}}$. Conversely, (IC8.4) directly preserves $\Lambda_{\mathbb{Q}}$, so this ring attains the asserted minimum.

In particular Λ itself is not preserved: the b_p -coefficient of $(b_{1/p})^*$ is $1/p$, which is not an integer. Equations (IC8.4a)–(IC8.6) construct and characterize the object required to retain the full weighted adjoint symmetry in this exact group-algebra presentation. This does not rule out integral arithmetic geometry, a different integral realization, or a pairing valued in an integral Tate object; those are not the scalar-valued coefficient presentation classified here.

IC9. What the complete spectrum detects

CG7 proves that the complete normed prime spectrum, including its unit atom and every prime-power repetition, determines

$$\zeta(s) = 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad \text{Re } s > 1, \quad (\text{IC9.1})$$

with absolute convergence. Its proved meromorphic continuation is unique. Consequently an actual nontrivial zero, wherever located, belongs to the analytic function determined by that same complete prime data. It would not be an additional prime omitted by the counting process. The whole prime spectrum determines the whole analytic function, including a possible off-critical zero; uniqueness alone does not tell us the location of that zero.

This last statement has a direct logical proof. If two meromorphic functions had the same complete series in (IC9.1), they would coincide on the open half-plane and hence throughout the connected continuation domain by the meromorphic identity theorem. Their zero divisors, with multiplicities, would therefore coincide. The argument contains no inequality or equality restricting the real parts of those zeros. The constructed original-zeta receiver strengthens the detection statement: its full joint arithmetic action has a block of dimension m_ρ for each actual zero, with coefficients (IC4.2). Such a zero is detected there, not missing from the reconstructed function.

The proved chain is therefore complete at the following levels: the source supplies the unique counting identity; its full endomorphism ring has integral degrees and prime norms; those degrees act through the explicit integral group algebra; the actual original-zeta quotient receives that action with every zero and multiplicity; and the Weil pairing has exactly the same prime norm as its similitude multiplier. The projective-line calculation identifies why integrality, complete counts and a degree- p pairing do not by themselves specify a weight-one sector. None of these calculations asserts a counterexample to the user's full geometric programme, or an RH resolution.

Original-zeta resolvent and every multiplicity

Proof source: ORIGINAL_ZETA_RESOLVENT_AND_PROJECTORS.md.

The original-zeta resolvent, global projectors, and every multiplicity jet

Independent mathematical derivation, 24 September 2026. Proof locators RZ1-RZ12.

This note works on the entire quotient already constructed in GLOBAL_MELLIN_SYNTHESIS.md, S1-S7, and GLOBAL_CANONICAL_MELLIN_INVOLUTION.md, G1-G5. Those portions were read for this derivation. S5 supplies the equality between the closed original source image and the ideal of all actual nontrivial-zero jets. That synthesis theorem is an input here, not a result reproved or attributed to this note. The source Fourier, Mellin, dilation and reflection conventions are retained exactly.

Every zero below is a zero of the original Riemann zeta function with its actual multiplicity. No simplicity, critical-line location, positivity or RH assumption is made. All operations are on the named function spaces and their quotient; they put no addition, coordinate or metric on the user's supporting datum.

RZ1. Spaces, full multiplier, and the operator domain

Let

$$\mathcal{A} = \left\{ k \in C^\infty(\mathbb{R}_{>0}) : p_{N,j}(k) := \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)| < \infty \quad (N, j \geq 0) \right\}, \quad (\text{RZ1.1})$$

and let $\mathcal{B}$ be the entire functions with every seminorm

$$b_{A,M}(F) = \sup_{|\operatorname{Re} s| \leq A} (1 + |\operatorname{Im} s|)^M |F(s)| < \infty. \quad (\text{RZ1.2})$$

Integer nonnegative A, M, N, j suffice. The exact topological isomorphism from S1 is

$$\mathcal{M}k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}. \quad (\text{RZ1.3})$$

Write $\mathcal{Z}$ for the set of distinct original nontrivial zeros, and m_ρ for the full positive multiplicity of $\rho \in \mathcal{Z}$. Define the closed ideal

$$\mathcal{I} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \quad (\rho \in \mathcal{Z}, 0 \leq j < m_\rho)\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{I}. \quad (\text{RZ1.4})$$

The source quotient $\mathcal{A}/\overline{\mathcal{E}(\mathcal{S}_0^{\text{even}})}$ is identified with this quotient by the already proved S5. A bracket $[F]$ denotes its class. We use the quotient Fréchet topology, not an unspecified product topology on jets.

The exact retained element of $\mathcal{I}$ is

$$F_0(s) = \mathcal{M}k_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad (\text{RZ1.5})$$

where

$$k_0(u) = u^{1/2} \sum_{n \geq 1} f_0(nu), \quad f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}. \quad (\text{RZ1.6})$$

The whole product (RZ1.5) has its entire continuation. Its zeros and their orders are exactly (ρ, m_ρ) . Its exceptional values are

$$F_0(0) = F_0(1) = \frac{1}{8}, \quad (\text{RZ1.7})$$

and, for every positive integer r ,

$$\begin{aligned} F_0(-2r) &= \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) \\ &= F_0(1+2r) \\ &= \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0. \end{aligned} \quad (\text{RZ1.8})$$

These are the original endpoint, Gamma-pole and trivial-zero cancellations, with their resulting values retained. They are inputs established in S2 and G4; the function ζ is not replaced by another function in the arithmetic argument.

Define on all of $\mathcal{B}$

$$\mathbb{L}F(s) = sF(s). \quad (\text{RZ1.9})$$

For all A, M ,

$$b_{A,M}(\mathbb{L}F) \leq (A+1)b_{A,M+1}(F). \quad (\text{RZ1.10})$$

Multiplication by s preserves every required zero order, so $\mathbb{L}\mathcal{J} \subseteq \mathcal{J}$. It therefore induces a continuous operator L on all of $\mathcal{Q}$; its domain is the entire quotient.

On $\mathcal{A}$, the exact corresponding operator is

$$\mathbb{D} = \frac{1}{2} - u\partial_u. \quad (\text{RZ1.11})$$

Indeed integration by parts, whose two boundary terms vanish by (RZ1.1), gives

$$\mathcal{M}(u\partial_u k)(s) = -(s - \frac{1}{2})\mathcal{M}k(s), \quad \mathcal{M}(\mathbb{D}k)(s) = s\mathcal{M}k(s). \quad (\text{RZ1.12})$$

Moreover

$$p_{N,j}(\mathbb{D}k) \leq \frac{1}{2}p_{N,j}(k) + p_{N,j+1}(k). \quad (\text{RZ1.13})$$

The variable of L is thus the original s , not $s - \frac{1}{2}$.

RZ2. Estimates for evaluation, multiplication and entire division

For a fixed $z \in \mathbb{C}$, positive radius r , and $j \geq 0$, the Cauchy integral formula gives

$$\frac{|F^{(j)}(z)|}{j!} \leq r^{-j} b_{A,0}(F) \quad \text{when } A \geq |\operatorname{Re} z| + r. \quad (\text{RZ2.1})$$

Consequently every jet functional is continuous and $\mathcal{J}$ is closed. It is an ideal because the Leibniz formula preserves a zero of any specified finite order under multiplication by an entire function.

For completeness, the quotient used here is itself complete. The increasing seminorms $b_{j,j}$, $j \geq 1$, generate the topology of $\mathcal{B}$; their quotient seminorms generate that of $\mathcal{Q}$. From a Cauchy sequence in

the quotient choose a subsequence whose j -th consecutive difference has quotient seminorm below 2^{-j} for $b_{j,j}$. Lift that difference to $H_j \in \mathcal{B}$ with $b_{j,j}(H_j) < 2^{1-j}$. The series $\sum_j H_j$ converges in every fixed seminorm: after its index exceeds that seminorm's indices, the displayed geometric bound controls the tail. Completeness of $\mathcal{B}$ therefore gives a limit of the lifted partial sums. Its quotient class is the limit of the selected subsequence and hence of the original Cauchy sequence. Closedness of $\mathcal{I}$ makes the quotient Hausdorff. This proves the stated complete Fréchet domain.

For $F, G \in \mathcal{B}$, and a nonnegative integer r ,

$$b_{A,M}(FG) \leq b_{A,0}(F)b_{A,M}(G), \quad (\text{RZ2.2})$$

$$b_{A,M}((s-z)^r F) \leq (A+|z|+1)^r b_{A,M+r}(F). \quad (\text{RZ2.3})$$

Thus multiplication by a fixed member of $\mathcal{B}$, and by every fixed polynomial, is continuous on $\mathcal{B}$.

We need entire division, including its local estimate. Suppose that $H \in \mathcal{B}$ vanishes at z to order at least $r \geq 1$. Then

$$D_{z,r}H(s) = \frac{H(s)}{(s-z)^r} \quad (\text{RZ2.4})$$

extends to an entire member of $\mathcal{B}$, continuously on the closed subspace with that vanishing condition. Outside $|s-z| < 1$, its weighted absolute value is at most that of H . Inside $|s-z| \leq 1$, the holomorphic quotient and the maximum principle on $|s-z| \leq 2$ give

$$|D_{z,r}H(s)| \leq 2^{-r} \sup_{|w-z|=2} |H(w)|. \quad (\text{RZ2.5})$$

The weight in that smaller disk is at most $(2+|\operatorname{Im} z|)^M$. Taking any integer $A' \geq \max(A, |\operatorname{Re} z| + 2)$ proves, for example,

$$b_{A,M}(D_{z,r}H) \leq b_{A,M}(H) + 2^{-r}(2+|\operatorname{Im} z|)^M b_{A',0}(H). \quad (\text{RZ2.6})$$

The same estimate is uniform as z ranges over a fixed compact set. The quotient at $s = z$ is the corresponding Taylor coefficient, so no singular integral is hidden in this assertion.

All further divisions in this note are applications of this estimate or the first-order version with a constructed vanishing numerator.

RZ3. The continuous global resolvent

For $\lambda \notin \mathcal{Z}$, define on all of $\mathcal{B}$

$$\boxed{\mathcal{R}_\lambda F(s) = \frac{F(s) - F_0(s) F(\lambda)/F_0(\lambda)}{\lambda - s}.} \quad (\text{RZ3.1})$$

The numerator is in $\mathcal{B}$, vanishes at $s = \lambda$, and $F_0(\lambda) \neq 0$. The removable value is exactly

$$\mathcal{R}_\lambda F(\lambda) = -F'(\lambda) + F'_0(\lambda) \frac{F(\lambda)}{F_0(\lambda)}. \quad (\text{RZ3.2})$$

The negative sign is caused by the denominator $\lambda - s$. RZ2 proves that $\mathcal{R}_\lambda F \in \mathcal{B}$.

Here is the needed uniform continuity statement. Let K be nonempty and compact in $\mathbb{C} \setminus \mathcal{Z}$. Set

$$c_K = \min_{\lambda \in K} |F_0(\lambda)| > 0, \quad B_K = \sup_{\lambda \in K} |\operatorname{Re} \lambda|, \quad T_K = \sup_{\lambda \in K} |\operatorname{Im} \lambda|.$$

For each A, M , choose an integer $A' \geq \max(A, B_K + 2)$. Evaluation is bounded by $b_{A',0}(F)$, so the numerator in (RZ3.1) satisfies

$$b_{A,M} \left(F - F_0 \frac{F(\lambda)}{F_0(\lambda)} \right) \leq b_{A,M}(F) + c_K^{-1} b_{A,M}(F_0) b_{A',0}(F). \quad (\text{RZ3.3})$$

Write $G_\lambda = F - F_0 F(\lambda)/F_0(\lambda)$ for the numerator. Applying (RZ2.6) once, with the output strip A , gives

$$b_{A,M}(\mathcal{R}_\lambda F) \leq b_{A,M}(G_\lambda) + \frac{1}{2}(2 + T_K)^M b_{A',0}(G_\lambda).$$

The second term requires only a bound on the numerator itself, not another division. Directly,

$$b_{A',0}(G_\lambda) \leq (1 + c_K^{-1} b_{A',0}(F_0)) b_{A',0}(F).$$

Combining this with (RZ3.3) gives the explicit finite constant

$$C_{K,A,M} = 1 + c_K^{-1} b_{A,M}(F_0) + \frac{1}{2}(2 + T_K)^M (1 + c_K^{-1} b_{A',0}(F_0))$$

and proves

$$\sup_{\lambda \in K} b_{A,M}(\mathcal{R}_\lambda F) \leq C_{K,A,M} b_{A',M}(F). \quad (\text{RZ3.4})$$

The condition $A' \geq \max(A, B_K + 2)$ already contains every radius-two disk centered at a point of K . Thus the same A' controls the division disk and both numerator bounds; no further enlargement to $A' + 2$ is needed. Only this stated increase of the strip index has been used.

If $F \in \mathcal{J}$, its numerator in (RZ3.1) vanishes to order at least m_ρ at every ρ . Since $\lambda - \rho \neq 0$, division preserves each such order. Thus

$$\mathcal{R}_\lambda \mathcal{J} \subseteq \mathcal{J}. \quad (\text{RZ3.5})$$

The induced operator R_λ is continuous on $\mathcal{Q}$. More explicitly, with quotient seminorms

$$q_{A,M}([F]) = \inf_{H \in \mathcal{J}} b_{A,M}(F + H),$$

equation (RZ3.4) gives

$$\sup_{\lambda \in K} q_{A,M}(R_\lambda[F]) \leq C_{K,A,M} q_{A',M}([F]). \quad (\text{RZ3.6})$$

For the supremum, this follows by using the same representative $F + H$ for every λ , then taking the infimum.

The exact identities before taking the quotient are

$$(\lambda - \mathbb{L})\mathcal{R}_\lambda F = F - F_0 \frac{F(\lambda)}{F_0(\lambda)}, \quad \mathcal{R}_\lambda((\lambda - \mathbb{L})F) = F. \quad (\text{RZ3.7})$$

The correction in the first identity belongs to $\mathcal{J}$. Therefore on the actual quotient

$$\boxed{R_\lambda = (\lambda - L)^{-1} \quad (\lambda \notin \mathcal{Z}).} \quad (\text{RZ3.8})$$

Although $F(\lambda)$ by itself generally does not descend to this quotient, the full expression (RZ3.1) does, by (RZ3.5). In fact $F_0(\lambda) \neq 0$ already proves that evaluation at such a λ does not annihilate $\mathcal{J}$. The correction term is essential.

For example the endpoint resolvents use exactly

$$\mathcal{R}_0 F(s) = \frac{F(s) - 8F_0(s)F(0)}{-s}, \quad \mathcal{R}_1 F(s) = \frac{F(s) - 8F_0(s)F(1)}{1 - s}. \quad (\text{RZ3.9})$$

At $\lambda = -2r$, (RZ3.1) uses the nonzero value (RZ1.8). Thus 0, 1 and the original trivial zeros are resolvent points of this specified nontrivial-zero quotient. Their full original multiplier contributions have been evaluated, not dropped.

RZ4. Parameter holomorphy and the source resolvent

The numerator

$$F(s)F_0(\lambda) - F_0(s)F(\lambda) \quad (\text{RZ4.1})$$

vanishes on the diagonal $s = \lambda$. Its division by $\lambda - s$ is holomorphic jointly in the two variables: locally expand both entire functions into power series, or factor each difference by the integral of its derivative along the segment from λ to s . Dividing additionally by $F_0(\lambda)$ is holomorphic when $\lambda \notin \mathcal{Z}$. Hence $\lambda \mapsto \mathcal{R}_\lambda F(s)$ is pointwise holomorphic.

Together with (RZ3.4), this is holomorphy in the topology of $\mathcal{B}$, and in the topology of uniform convergence on bounded subsets for the operator family. Here are the details needed for that conclusion. On a small closed parameter disk contained in the resolvent set, the family is uniformly bounded in each output seminorm by one input seminorm. Cauchy's formula in λ can therefore be integrated in each seminorm; completeness of $\mathcal{B}$ puts the integral in $\mathcal{B}$. Pointwise Cauchy equality identifies this integral with the given entire function of s . The Cauchy remainder estimate on a slightly larger disk bounds the difference-quotient remainder in any output seminorm by a constant times $|h|$ and one input seminorm. Taking the supremum over any bounded input set proves operator holomorphy in that topology. The identical argument, using (RZ3.6), applies to the quotient.

In particular contour integrals of R_λ on compact contours in the resolvent set are well defined continuous operators on the complete Fréchet quotient. The usual resolvent identity follows directly from the two inverse identities:

$$R_\lambda - R_\mu = (\mu - \lambda)R_\lambda R_\mu. \quad (\text{RZ4.2})$$

The full source representative is also explicit. If $F = \mathcal{M}k$, then

$$(r_\lambda k)(u) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{F(\frac{1}{2} + it) - F_0(\frac{1}{2} + it)F(\lambda)/F_0(\lambda)}{\lambda - \frac{1}{2} - it} u^{-it} dt. \quad (\text{RZ4.3})$$

If the denominator vanishes on this line, the integrand has the removable value (RZ3.2). Its Mellin transform is (RZ3.1).

For precision, every $H \in \mathcal{B}$ has inverse

$$k_H(e^x) = \frac{1}{2\pi} \int_{\mathbb{R}} H(\frac{1}{2} + it) e^{-itx} dt. \quad (\text{RZ4.4})$$

Take $c = \frac{1}{2} + N + 1$ for $x \geq 0$, and $c = \frac{1}{2} - N - 1$ for $x \leq 0$. The proven strip contour shift gives

$$\frac{d^j}{dx^j} k_H(e^x) = \frac{e^{-(c-1/2)x}}{2\pi} \int_{\mathbb{R}} [-(c - \frac{1}{2} + it)]^j H(c + it) e^{-itx} dt. \quad (\text{RZ4.5})$$

Since $\int_{\mathbb{R}} (1 + |t|)^{-2} dt = 2$, this yields the explicit estimate

$$\sup_x e^{N|x|} \left| \frac{d^j}{dx^j} k_H(e^x) \right| \leq \frac{(N+2)^j}{\pi} b_{N+2, j+2}(H), \quad (\text{RZ4.6})$$

and consequently

$$p_{N,j}(k_H) \leq \frac{2(N+2)^j}{\pi} b_{N+2, j+2}(H). \quad (\text{RZ4.7})$$

The notation $d^j/dx^j k_H(e^x)$ denotes derivatives of the composite function. These estimates prove that (RZ4.3) belongs to $\mathcal{A}$, continuously and uniformly on compact resolvent-parameter sets.

The exact source identities are

$$(\lambda - \mathcal{D})r_\lambda k = k - k_0 \frac{\mathcal{M}k(\lambda)}{F_0(\lambda)}, \quad r_\lambda((\lambda - \mathcal{D})k) = k. \quad (\text{RZ4.8})$$

The first correction lies in the original source image because $k_0 = \mathcal{E}f_0$. The whole closed source image is preserved by r_λ , by S5 and (RZ3.5).

RZ5. Global representatives of every actual zero jet

Fix an actual zero ρ , and write $m = m_\rho$. Put

$$A_\rho(s) = \frac{F_0(s)}{(s - \rho)^m}. \quad (\text{RZ5.1})$$

It is entire, belongs to $\mathcal{B}$ by RZ2, and

$$a_r := \frac{A_\rho^{(r)}(\rho)}{r!} = \frac{F_0^{(m+r)}(\rho)}{(m+r)!}, \quad a_0 \neq 0. \quad (\text{RZ5.2})$$

Define the coefficients of the reciprocal Taylor polynomial by

$$b_0 = a_0^{-1}, \quad b_r = -a_0^{-1} \sum_{h=1}^r a_h b_{r-h} \quad (1 \leq r < m), \quad (\text{RZ5.3})$$

and set

$$B_\rho(s) = \sum_{r=0}^{m-1} b_r (s - \rho)^r, \quad e_\rho(s) = A_\rho(s) B_\rho(s). \quad (\text{RZ5.4})$$

The recursion says exactly

$$e_\rho(s) = 1 + O((s - \rho)^m) \quad (s \rightarrow \rho). \quad (\text{RZ5.5})$$

Every function in (RZ5.4) after multiplication by A_ρ belongs to $\mathcal{B}$, by the polynomial estimate (RZ2.3).

At each other actual zero $\sigma \neq \rho$, the denominator in (RZ5.1) is nonzero. Hence A_ρ , e_ρ , and $e_\rho(s)(s - \rho)^j$ vanish there to order at least m_σ . This holds at every other zero simultaneously, without a finite interpolation range.

Define, for $0 \leq j < m$,

$$E_{\rho,j}(s) = e_\rho(s)(s - \rho)^j, \quad D_{\rho,j}(s) = \frac{1}{j!} E_{\rho,j}(s). \quad (\text{RZ5.6})$$

The complete exact jet statement is

$$E_{\rho,j}^{(k)}(\sigma) = j! \mathbf{1}_{\{\sigma=\rho, k=j\}} \quad (\sigma \in \mathcal{Z}, 0 \leq k < m_\sigma), \quad (\text{RZ5.7})$$

and $D_{\rho,j}^{(k)}(\sigma)$ is the same expression without $j!$. Thus these are global representatives for each individual derivative coordinate, including its factorial.

They have actual prequotient source representatives

$$k_{\rho,j}(u) = \frac{1}{2\pi} \int_{\mathbb{R}} e_\rho(\tfrac{1}{2} + it)(\tfrac{1}{2} + it - \rho)^j u^{-it} dt, \quad \mathcal{M}k_{\rho,j} = E_{\rho,j}. \quad (\text{RZ5.8})$$

The estimate (RZ4.7), with $H = E_{\rho,j}$, proves every source seminorm is finite. In particular these representatives belong to the whole $\mathcal{A}$, rather than merely defining distributions or a finite set of prescribed values. They do not belong to the annihilated source image when the indicated jet is nonzero; that would contradict (RZ5.7) and S5.

All multiplier derivatives in the recursion can be retained explicitly in terms of the original zeta. Set only for displaying these derivatives

$$C(s) = \frac{1}{8} s(s-1) \pi^{-s/2} \Gamma(s/2). \quad (\text{RZ5.9})$$

At the nontrivial zero ρ , C is holomorphic and nonzero. The exact coefficient formula is

$$a_r = \sum_{h=0}^r \frac{C^{(h)}(\rho)}{h!} \frac{\zeta^{(m+r-h)}(\rho)}{(m+r-h)!}. \quad (\text{RZ5.10})$$

No derivative term has been suppressed: derivatives of ζ of order less than m vanish by the definition of its actual multiplicity. For $P(s) = s(s-1)$,

$$C^{(h)}(s) = \frac{\pi^{-s/2}}{8} \sum_{\substack{a+b+c=h \\ 0 \leq a \leq 2}} \frac{h!}{a! b! c!} P^{(a)}(s) \left(-\frac{\log \pi}{2} \right)^b 2^{-c} \Gamma^{(c)}(s/2), \quad (\text{RZ5.11})$$

where $P^{(0)} = s(s-1)$, $P' = 2s-1$, and $P'' = 2$.

At an exceptional point $z \in \{0, 1, -2, -4, \dots\}$, the same entire representatives have the full value

$$E_{\rho,j}(z) = F_0(z) B_\rho(z) (z - \rho)^{j-m}, \quad (\text{RZ5.12})$$

with $F_0(z)$ given by (RZ1.7)-(RZ1.8). These values have not been forced to vanish or removed from the source functions. Evaluation at these points is not a quotient character, as RZ3 already proves.

RZ6. Continuous projectors and exact generalized eigenspaces

For a holomorphic germ H at ρ , write

$$\text{Tay}_{\rho, < r} H(s) = \sum_{j=0}^{r-1} \frac{H^{(j)}(\rho)}{j!} (s - \rho)^j \quad (r \geq 1). \quad (\text{RZ6.1})$$

The promised finite-rank lift on the entire function space is

$$\boxed{\mathcal{P}_\rho F(s) = A_\rho(s) \text{Tay}_{\rho, < m} \left(\frac{F}{A_\rho} \right) (s).} \quad (\text{RZ6.2})$$

The quotient F/A_ρ is used only as a germ at ρ , where $A_\rho(\rho) \neq 0$. The displayed output is a globally defined entire function. It belongs to $\mathcal{B}$ as a finite polynomial times A_ρ . Its coefficients are fixed finite linear combinations of the jets of F at ρ , so (RZ2.1) proves continuity.

The output has exactly the same first m Taylor coefficients as F at ρ , and has every required vanishing jet at all other zeros. Moreover

$$\mathcal{P}_\rho^2 = \mathcal{P}_\rho, \quad \mathcal{P}_\rho \mathcal{J} = \{0\}, \quad \mathcal{P}_\rho \mathcal{P}_\sigma = 0 \quad (\rho \neq \sigma). \quad (\text{RZ6.3})$$

For the first equality, divide $\mathcal{P}_\rho F$ by A_ρ ; its quotient is already a polynomial of degree below m , so its Taylor truncation is unchanged. For the second, F/A_ρ has a zero of order at least m at ρ . For the third, $\mathcal{P}_\sigma F$ has a zero of order at least m_ρ at ρ , and the same Taylor argument applies.

Consequently the quotient projector

$$P_\rho[F] = [\mathcal{P}_\rho F] = [e_\rho F] \quad (\text{RZ6.4})$$

is well defined and continuous. To verify the second equality, both functions have the jets of F at ρ and zero jets at every other zero; their difference is in $\mathcal{I}$. Equivalently

$$P_\rho[F] = \sum_{j=0}^{m-1} \frac{F^{(j)}(\rho)}{j!} [E_{\rho,j}] = \sum_{j=0}^{m-1} F^{(j)}(\rho) [D_{\rho,j}]. \quad (\text{RZ6.5})$$

The range $\mathcal{Q}_\rho = P_\rho \mathcal{Q}$ has dimension exactly m , since (RZ5.7) proves independence and (RZ6.5) proves spanning.

The lift $\mathcal{P}_\rho : \mathcal{B} \rightarrow \mathcal{B}$ annihilates $\mathcal{I}$, so it also defines a continuous map $\mathcal{Q} \rightarrow \mathcal{B}$, by the defining quotient topology. On $\mathcal{Q}_\rho$ it is a continuous choice of source representative. In addition $\mathcal{Q} = \mathcal{Q}_\rho \oplus \ker P_\rho$ is a topological direct sum: the forward map sends x to $(P_\rho x, x - P_\rho x)$, and its continuous inverse adds the two components. This is a split finite-dimensional spectral block of the full quotient.

This range is exactly the actual generalized eigenspace:

$$\mathcal{Q}_\rho = \ker(L - \rho)^m. \quad (\text{RZ6.6})$$

The inclusion from left to right follows from the required local orders, or from $(s - \rho)^m e_\rho = F_0 B_\rho$. Conversely, if $(s - \rho)^m F \in \mathcal{I}$, then at every other zero σ the factor $(s - \rho)^m$ is an invertible holomorphic germ, so F has all required zero jets there. At ρ , $F - \mathcal{P}_\rho F$ has all required zero jets by construction. Hence $[F] = P_\rho[F]$, proving the reverse inclusion.

Put $N_\rho = (L - \rho)|_{\mathcal{Q}_\rho}$. Its complete action is

$$N_\rho[E_{\rho,j}] = \begin{cases} [E_{\rho,j+1}], & j < m-1, \\ 0, & j = m-1. \end{cases} \quad (\text{RZ6.7})$$

In derivative-coordinate representatives,

$$N_\rho[D_{\rho,j}] = (j+1)[D_{\rho,j+1}] \quad (j < m-1). \quad (\text{RZ6.8})$$

Thus $N_\rho^m = 0$ and $N_\rho^{m-1} \neq 0$; its ranks are $\text{rank } N_\rho^r = m - r$ for $0 \leq r < m$. In particular $[E_{\rho,m-1}] \neq 0$ is an eigenvector with eigenvalue ρ .

It now follows from RZ3 and this eigenvector that the finite continuous spectrum is exactly

$$\boxed{\text{Spec}(L) = \mathcal{Z}.} \quad (\text{RZ6.9})$$

Here the resolvent set means those $\lambda \in \mathbb{C}$ for which $\lambda - L$ has an everywhere-defined continuous inverse. No Banach-space spectral convention is being silently substituted.

The family of projectors also separates the complete quotient:

$$\bigcap_{\rho \in \mathcal{Z}} \ker P_\rho = \{0\}. \quad (\text{RZ6.10})$$

Indeed every vanishing projector forces all the corresponding actual jets to vanish, and S5 identifies their common kernel with $\mathcal{I}$. This is a statement on every quotient class. It does not assume convergence of an infinite sum of projectors.

RZ7. Riesz contour formula, every Laurent coefficient, and signs

Choose a positively oriented circle γ_ρ enclosing ρ and no other zero. Its radius can be chosen positive because the zeros of the nonzero entire F_0 are isolated. On that contour, RZ3-RZ4 give a holomorphic, locally equicontinuous resolvent.

We first compute its integral at the level of $\mathcal{B}$:

$$\frac{1}{2\pi i} \int_{\gamma_\rho} \mathcal{R}_\lambda F d\lambda = \mathcal{P}_\rho F. \quad (\text{RZ7.1})$$

For s outside the closed circle, the first term $F(s)/(\lambda - s)$ in (RZ3.1) has integral zero. Put $t = \lambda - \rho$, $d = s - \rho$, and write locally

$$\frac{F(\rho + t)}{A_\rho(\rho + t)} = \sum_{\ell \geq 0} h_\ell t^\ell. \quad (\text{RZ7.2})$$

The remaining term is

$$-F_0(s) \frac{t^{-m} \sum_{\ell \geq 0} h_\ell t^\ell}{t - d}.$$

Since

$$\frac{1}{t - d} = - \sum_{r \geq 0} \frac{t^r}{d^{r+1}}, \quad (\text{RZ7.3})$$

its residue is

$$F_0(s) \sum_{\ell=0}^{m-1} \frac{h_\ell}{(s - \rho)^{m-\ell}} = A_\rho(s) \sum_{\ell=0}^{m-1} h_\ell (s - \rho)^\ell. \quad (\text{RZ7.4})$$

The two negative signs cancel. This is (RZ6.2). Both sides of (RZ7.1) are entire functions of s , so equality outside the circle proves equality everywhere. This also handles s on or inside the contour without separating singular terms there.

Passing to the quotient proves the actual Riesz formula

$$\boxed{P_\rho = \frac{1}{2\pi i} \int_{\gamma_\rho} (\lambda - L)^{-1} d\lambda.} \quad (\text{RZ7.5})$$

Thus the explicit projectors are the Riesz projectors of the actual continuous operator, not merely finite interpolation maps.

Every principal Laurent coefficient has an equally explicit global lift. For $0 \leq r < m$,

$$\mathcal{C}_{\rho,r} F(s) = A_\rho(s) (s - \rho)^r \text{Tay}_{\rho, < m-r} \left(\frac{F}{A_\rho} \right) (s), \quad (\text{RZ7.6})$$

and

$$\frac{1}{2\pi i} \int_{\gamma_\rho} (\lambda - \rho)^r \mathcal{R}_\lambda F d\lambda = \mathcal{C}_{\rho,r} F. \quad (\text{RZ7.7})$$

For $r \geq m$ this integral is zero. The proof repeats (RZ7.3): the residue condition becomes $\ell + \text{summation index} = m - r - 1$, giving exactly (RZ7.6).

The induced coefficient is

$$[\mathcal{C}_{\rho,r} F] = (L - \rho)^r P_\rho[F]. \quad (\text{RZ7.8})$$

Indeed replacing the Taylor truncation of length $m - r$ by the one of length m changes the result by $F_0(s)P(s)$, where P is a polynomial. This product belongs to $\mathcal{B}$ by (RZ2.3) and to $\mathcal{J}$ by its full zero orders; no claim that P itself belongs to $\mathcal{B}$ is needed.

There is also a one-step identity retaining the exact source correction. With the coefficients h_j of (RZ7.2), and with $\mathcal{C}_{\rho,m} = 0$,

$$(s - \rho)\mathcal{C}_{\rho,r}F(s) - \mathcal{C}_{\rho,r+1}F(s) = F_0(s)h_{m-r-1} \quad (0 \leq r < m). \quad (\text{RZ7.11})$$

It follows by subtracting the two Taylor polynomials, whose sole remaining term has degree $m - r - 1$. These corrections are present on $\mathcal{B}$, even though their quotient classes vanish. In particular

$$s\mathcal{P}_\rho F - \mathcal{P}_\rho(sF) = F_0 h_{m-1}, \quad (\text{RZ7.12})$$

so $P_\rho L = LP_\rho$ is exact on $\mathcal{Q}$ with the displayed correction at the representative level.

There is a meromorphic operator family on the whole finite λ -plane. To see this directly before quotienting, the jointly holomorphic divided expression in RZ4 is divided only by $F_0(\lambda)$. Near ρ , its pole order is at most m ; the uniform seminorm estimates on a surrounding circle and the parameter Cauchy formula give its Laurent expansion in continuous operators, uniformly on bounded input sets. Consequently on $\mathcal{Q}$,

$$R_\lambda = \sum_{r=0}^{m-1} \frac{(L - \rho)^r P_\rho}{(\lambda - \rho)^{r+1}} + \text{a holomorphic operator family near } \rho. \quad (\text{RZ7.9})$$

The pole order is exactly m , because its last coefficient has rank one and is nonzero by (RZ6.7). More generally its coefficient of order $(\lambda - \rho)^{-r-1}$ has rank $m - r$.

On the actual block, the resolvent is the finite algebraic identity

$$R_\lambda|_{\mathcal{Q}_\rho} = \sum_{r=0}^{m-1} \frac{N_\rho^r}{(\lambda - \rho)^{r+1}}. \quad (\text{RZ7.10})$$

Multiplication by $(\lambda - \rho)I - N_\rho$ telescopes to the identity because $N_\rho^m = 0$, verifying every sign independently of the contour calculation.

RZ8. The entire unshifted arithmetic action

For every positive real a , define

$$T_a F(s) = a^s F(s), \quad a^s = e^{s \log a}, \quad (\text{RZ8.1})$$

where $\log a$ is the real logarithm. This preserves $\mathcal{B}$ and its ideal $\mathcal{J}$, since the multiplier is entire and nowhere zero. The exact estimates are

$$b_{A,M}(T_a F) \leq \max(a^A, a^{-A}) b_{A,M}(F), \quad (\text{RZ8.2})$$

and on the source,

$$\mathcal{M}^{-1} T_a \mathcal{M} k(u) = a^{1/2} k(u/a) = a^{1/2} V_a k(u), \quad (\text{RZ8.3})$$

$$p_{N,j}(a^{1/2} V_a k) \leq a^{1/2} \max(a^N, a^{-N}) p_{N,j}(k). \quad (\text{RZ8.4})$$

Both the factor $a^{1/2}$ and the original Mellin variable s are retained.

These operators obey $T_a T_b = T_{ab}$ and $T_1 = I$. In the real parameter $t = \log a$, they form a strongly smooth group whose generator is L :

$$\frac{d^r}{dt^r} T_{e^t} [F] = L^r T_{e^t} [F]. \quad (\text{RZ8.5})$$

For a direct topological justification, on a bounded real t -interval the derivative multipliers $s^r e^{ts}$ satisfy (RZ8.2) followed by (RZ1.10) repeatedly. Taylor's integral remainder for real t gives difference-quotient convergence in every seminorm, using one additional power of s . Equivalently (RZ8.4) and the logarithmic derivative bounds give the same assertion on $\mathcal{A}$.

In this explicit real-parameter multiplier sense we write

$$\boxed{n^L = T_n = \sqrt{n} V_n \quad (n = 1, 2, 3, \dots)}. \quad (\text{RZ8.6})$$

This notation is justified by the multiplier and group formulas just proved. It does not assume convergence of the exponential power series of a continuous operator in a nonnormable Fréchet space. Such a series convergence is unnecessary here.

On every actual multiplicity block,

$$T_n[E_{\rho,j}] = n^\rho \sum_{r=0}^{m_\rho-1-j} \frac{(\log n)^r}{r!} [E_{\rho,j+r}]. \quad (\text{RZ8.7})$$

This follows by taking the full Taylor series of $n^s = n^\rho e^{(s-\rho)\log n}$ at ρ , retaining all terms that survive modulo the order m_ρ . All other-zero jets still vanish. Thus

$$T_n|_{\mathcal{Q}_\rho} = n^\rho \sum_{r=0}^{m_\rho-1} \frac{(\log n)^r}{r!} N_\rho^r. \quad (\text{RZ8.8})$$

On this block the eigenvalue is n^ρ , with algebraic multiplicity m_ρ ; its modulus is exactly $n^{\operatorname{Re} \rho}$. No conclusion that this modulus is $\sqrt{n}$ has been assumed.

For the original unscaled derivative jets, the complete triangular formula is

$$(T_n F)^{(j)}(\rho) = n^\rho \sum_{\ell=0}^j \binom{j}{\ell} (\log n)^{j-\ell} F^{(\ell)}(\rho). \quad (\text{RZ8.9})$$

Equivalently, on the representatives of individual derivative coordinates,

$$T_n[D_{\rho,j}] = n^\rho \sum_{r=0}^{m_\rho-1-j} \binom{j+r}{r} (\log n)^r [D_{\rho,j+r}]. \quad (\text{RZ8.10})$$

These formulas also cover $n = 1$ with its identity action. The trace of this finite block is $m_\rho n^\rho$, since every positive power of N_ρ has zero diagonal.

For $n > 1$, the arithmetic operator itself retains the full Jordan length m_ρ . Indeed

$$n^{-\rho} T_n - I = N_\rho H_n(N_\rho), \quad H_n(z) = \sum_{r=1}^{m_\rho} \frac{(\log n)^r}{r!} z^{r-1}, \quad H_n(0) = \log n \neq 0. \quad (\text{RZ8.11})$$

The last possible term is harmless because $N_\rho^{m_\rho} = 0$. The polynomial $H_n(N_\rho)$ is invertible: its nonzero constant term times the identity has a nilpotent correction, whose inverse is its finite geometric series. Hence $(T_n - n^\rho I)^r$ has the same rank as N_ρ^r , including the nilpotence index m_ρ .

Different zero blocks are never identified merely because one arithmetic operator has coincident eigenvalues. For a fixed $n > 1$,

$$n^\rho = n^\sigma \iff \rho - \sigma = \frac{2\pi i k}{\log n} \text{ for some } k \in \mathbb{Z}. \quad (\text{RZ8.12})$$

The full family separates the original s -values. Already equality at both $n = 2$ and $n = 3$ implies $k \log 3 = l \log 2$, hence $3^k = 2^l$ for integers k, l . Unique prime factorization forces $k = l = 0$, so $\rho = \sigma$. This statement retains the exact phase periods of each individual operator.

RZ9. Reflection acts on every jet, including its sign

The exact source involution is $k^\#(u) = \overline{k(1/u)}$. On $\mathcal{B}$ it is

$$F^\#(s) = \overline{F(1 - \bar{s})}. \quad (\text{RZ9.1})$$

It is conjugate-linear and continuous:

$$b_{A,M}(F^\#) \leq b_{A+1,M}(F). \quad (\text{RZ9.2})$$

The original functional equation and conjugation preserve zero multiplicities under

$$\rho^\# = 1 - \bar{\rho}, \quad m_{\rho^\#} = m_\rho. \quad (\text{RZ9.3})$$

Thus $\mathcal{J}$ is invariant and the involution descends to the whole quotient. Its derivative formula is

$$(F^\#)^{(j)}(\rho^\#) = (-1)^j \overline{F^{(j)}(\rho)}. \quad (\text{RZ9.4})$$

To prove it, put $s = \rho^\# + h$; then $1 - \bar{s} = \rho - \bar{h}$, and conjugate the full local Taylor series.

Applying (RZ9.4) to the isolating jets gives

$$[E_{\rho,j}]^\# = (-1)^j [E_{\rho^\#,j}], \quad [D_{\rho,j}]^\# = (-1)^j [D_{\rho^\#,j}]. \quad (\text{RZ9.5})$$

These are equalities of actual quotient classes because all their zero jets agree globally. In particular

$$(P_\rho x)^\# = P_{\rho^\#}(x^\#), \quad (Lx)^\# = (1 - L)x^\#, \quad (\text{RZ9.6})$$

$$(T_a x)^\# = a^{1-L} x^\# = a T_a^{-1} x^\#. \quad (\text{RZ9.7})$$

For the nilpotent parts, (RZ9.6) reads

$$(N_\rho x)^\# = -N_{\rho^\#} x^\# \quad (x \in \mathcal{Q}_\rho). \quad (\text{RZ9.8})$$

The sign is essential and does not disappear when a zero happens to be on the critical line.

For the specific source representatives constructed here, (RZ9.5) holds even before passage to the quotient. The source symmetry $F_0^\# = F_0$ gives

$$A_\rho^\# = (-1)^{m_\rho} A_{\rho^\#}. \quad (\text{RZ9.9})$$

Applying reflection to the uniquely determined reciprocal Taylor polynomial gives

$$B_\rho^\# = (-1)^{m_\rho} B_{\rho^\#}. \quad (\text{RZ9.10})$$

Indeed the reflected product is one modulo the required order, and the reciprocal polynomial of degree below m_ρ is unique by the recursion (RZ5.3). The two signs cancel in the product $e_\rho^\# = e_{\rho^\#}$. Multiplying by the reflected factor $(s - \rho)^j$ supplies exactly the remaining $(-1)^j$ in (RZ9.5). Thus the corresponding source functions in (RZ5.8) also satisfy the exact reflected identity.

RZ10. The actual reflected Weil form and its block trace

Use the existing source-derived form from G5, written on this exact quotient:

$$W([F], [G]) = \sum_{\rho \in \mathcal{Z}} m_\rho \overline{F(\rho^\#)} G(\rho). \quad (\text{RZ10.1})$$

This is conjugate-linear in the first input and linear in the second. Its absolute convergence and continuity follow already from the weaker zero-count bound proved in S2,

$$\sum_{|\rho| \leq R} m_\rho \leq C(R + 2)^{3/2}. \quad (\text{RZ10.2})$$

Indeed $0 < \operatorname{Re} \rho < 1$, the reflected zero has the same imaginary part, and

$$|F(\rho^\#)G(\rho)| \leq b_{1,2}(F)b_{1,2}(G)(1 + |\operatorname{Im} \rho|)^{-4}.$$

The sum of the last weights with multiplicities is finite by grouping zeros into dyadic height intervals and applying (RZ10.2). Therefore

$$|W([F], [G])| \leq C_W b_{1,2}(F)b_{1,2}(G). \quad (\text{RZ10.3})$$

It descends because $\mathcal{J}$ annihilates every displayed value. Taking infima over representatives also proves a bound by $C_W q_{1,2}([F])q_{1,2}([G])$. Reflection and multiplicity preservation prove Hermitian symmetry by relabelling the absolutely convergent sum.

The word trace here has an exact interpretation on the constructed blocks. Multiplication by any $H \in \mathcal{B}$ preserves $\mathcal{Q}_\rho$; in the basis $[E_{\rho,j}]$, its diagonal entries are all $H(\rho)$. Hence

$$\operatorname{Tr}_{\mathcal{Q}_\rho}(M_H) = m_\rho H(\rho), \quad (\text{RZ10.4})$$

and

$$W([F], [G]) = \sum_{\rho \in \mathcal{Z}} \operatorname{Tr}_{\mathcal{Q}_\rho}(M_{F^\#G}). \quad (\text{RZ10.5})$$

This is the absolutely convergent sum of the specified finite-dimensional block traces. It does not assert a trace-class operator on an unstated Hilbert completion.

Every block entry is explicit:

$$W([E_{\rho,j}], [E_{\sigma,k}]) = m_\sigma \mathbf{1}_{\{\rho=\sigma^\#\}} \mathbf{1}_{\{j=0\}} \mathbf{1}_{\{k=0\}}. \quad (\text{RZ10.6})$$

The same formula holds for $D_{\rho,j}$, since only the zeroth representative contributes. This is immediate from all-zero isolation (RZ5.7).

If $\rho = \rho^\#$, the form on $\mathcal{Q}_\rho$ has one positive direction of squared length m_ρ and radical of dimension $m_\rho - 1$. If $\rho \neq \rho^\#$, put $m = m_\rho = m_{\rho^\#}$. On the two constant-jet representatives its matrix is exactly

$$\begin{pmatrix} 0 & m \\ m & 0 \end{pmatrix}. \quad (\text{RZ10.7})$$

Its values on the sum and difference of those representatives are $2m$ and $-2m$, respectively. The entire paired space $\mathcal{Q}_\rho \oplus \mathcal{Q}_{\rho^\#}$ has one positive direction, one negative direction, and radical dimension $2(m - 1)$.

Globally the radical is precisely

$$\operatorname{rad} W = \{[F] \in \mathcal{Q} : F(\rho) = 0 \text{ for all } \rho \in \mathcal{Z}\}. \quad (\text{RZ10.8})$$

One inclusion follows from (RZ10.1). Conversely, pairing with $[E_{\sigma,0}]$ gives $m_\sigma \overline{F(\sigma^\#)}$; if every such pairing vanishes, every zero value is zero. Higher multiplicity jets therefore remain real nonzero quotient data even though this ordinary trace form cannot detect them.

RZ11. Full arithmetic similitude, reflected adjoints, and the exact sign question

For every $a > 0$, with no restriction on the real parts of the actual zeros,

$$\begin{aligned} W(T_a x, T_a y) &= \sum_{\rho} m_{\rho} \overline{a^{\rho^{\#}} F(\rho^{\#})} a^{\rho} G(\rho) \\ &= \sum_{\rho} m_{\rho} a^{\overline{\rho^{\#}} + \rho} \overline{F(\rho^{\#})} G(\rho) \\ &= a W(x, y), \end{aligned} \tag{RZ11.1}$$

because $\overline{\rho^{\#}} = 1 - \rho$. In particular

$$\boxed{W(n^L x, n^L y) = n W(x, y), \quad n^L = \sqrt{n} V_n.} \tag{RZ11.2}$$

This preserves the full unshifted arithmetic factor. Dividing both arguments by the displayed $\sqrt{n}$ recovers the earlier exact invariance of V_n , but the original n^L is the operator in (RZ11.2).

The corresponding identities for the pairing are

$$W(Lx, y) = W(x, (1 - L)y), \tag{RZ11.3}$$

$$W(T_a x, y) = W(x, a T_a^{-1} y), \quad W(P_{\rho} x, y) = W(x, P_{\rho^{\#}} y). \tag{RZ11.4}$$

For the first equation, conjugation of the first factor changes the spectral scalar $\rho^{\#}$ into $1 - \rho$. The second follows by the same direct scalar calculation, and the projector identity follows by its global zero-jet support. These are identities for a possibly indefinite and degenerate form, not adjoint assertions for an unconstructed positive Hilbert norm.

For the resolvent there is a further retained sign:

$$W(R_{\lambda} x, y) = -W(x, R_{1-\bar{\lambda}} y) \quad (\lambda \notin \mathcal{Z}). \tag{RZ11.5}$$

The reflected parameter is also outside $\mathcal{Z}$ by (RZ9.3). Directly,

$$\frac{1}{\bar{\lambda} - \rho^{\#}} = -\frac{1}{1 - \bar{\lambda} - \rho},$$

which proves (RZ11.5) term by term in the convergent form. Equivalently it follows by taking the pairing adjoint of $\lambda - L$ using (RZ11.3).

On an off-line pair the actual eigenvalues obey

$$n^{\rho^{\#}} = \frac{n}{n^{\rho}}, \quad |n^{\rho}| = n^{\operatorname{Re} \rho}, \quad |n^{\rho^{\#}}| = n^{1 - \operatorname{Re} \rho}. \tag{RZ11.6}$$

Their reflected pairing therefore has the exact factor n even when neither modulus is $\sqrt{n}$. The complete nilpotent coefficients of their action remain (RZ8.7)–(RZ8.10); their absence from the value trace is accounted for by the radical (RZ10.8), not by removal of the corresponding vectors.

These explicit representatives also give the precise positivity equivalences for this specified form. The form W is positive semidefinite exactly when every actual zero satisfies $\rho = \rho^{\#}$: the forward implication follows from the negative vector in (RZ10.7), and the reverse implication follows from the convergent sum $\sum_{\rho} m_{\rho} |F(\rho)|^2$. It is positive definite exactly when, in addition, all multiplicities are one. If a multiplicity exceeds one, (RZ5.7) gives a nonzero higher-jet radical vector. If all multiplicities are one, zero values at every zero mean membership in $\mathcal{J}$, so no nonzero quotient vector can have zero squared length in the positive case.

These are proved equivalences and explicit block signatures. No positive-semidefiniteness assertion or RH conclusion has been inserted.

RZ12. Exact scope and retained source information

The resolvent acts continuously on the entire Fréchet quotient and has exactly the actual original nontrivial zeros as its finite spectrum. Its poles have exactly their original multiplicities. The projectors are simultaneously actual contour projectors, explicit continuous finite-rank lifts on the full source function space, and isolators of the full jet at one zero with every other-zero jet zero. Every source representative has all endpoint and derivative seminorms controlled by (RZ4.7).

The full multiplier $s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)/8$, its endpoint values, the finite values at the original trivial zeros, and all local derivative contributions remain explicit. The arithmetic action is the original-variable operator $n^L = \sqrt{n}V_n$, with every logarithmic multiplicity term and the exact reflected form. The independent source synthesis S5 is used on its actual domain; no finite numerical experiment, unrestricted jet-product identification, positive completion, or RH assumption replaces it.

Canonical source reconstruction

Proof source: CANONICAL_GENERIC_RECONSTRUCTION.md.

The unique common support and canonical reconstruction of the original zeta function

24 September 2026. Proof labels CG1–CG8. This calculation receives the user’s uniqueness argument: the complete counting structure, its global arithmetic unit, the parityless common support, and recovery of the original zeta must be considered together. The proposed consequence about a common spectral weight is retained as the next part of that argument; it is not substituted into the hypotheses of this proof.

The exact statement proved here is this: **the full generic datum in the specified Connes–Consani diagram canonically determines an integer ring, its complete normed prime spectrum, and the original Riemann zeta function. Both ordered-chart constructions factor uniquely through the same generic group completion. The supporting point of that common overlap is unique.** The identity operation is the arithmetic unit; the supporting point is not that operation.

The user notation is $\tau\langle Z_1; \text{no } Z_2 \rangle$. Here the absence of Z_2 means the proved absence of an exchange-equivariant even/odd point-label. The quotient stack still has a C_2 stabilizer at its generic point. No addition, subtraction, numerical distance, midpoint, vector or coordinate is assigned to τ . All algebra below is on the explicitly stated monoids, groups, endomorphisms or functions.

Source and complete proof dependencies

The original author source is Alain Connes and Caterina Consani, [The Absolute Twistor Line and the Geometry of the compactified Spec Z](#), [arXiv:2609.00299v1](#), local CC.tex, SHA-256 57a10dcef5cd758e6b2f1c54b1c0a11b0fb093638f74fb1be515ec3bc7080ba4. Original passages actually read: lines 499–505 (topology), 526–540 (ordered charts and common group), 576–600 (imaginary generator and twisted restrictions), 773–847 (involution and chart compatibility), 881–898 (invariants and isotropy). We use the unfolded generic stalk throughout; passing to invariant sections would be a different operation.

The source curve and its full periods are read in Connes–Consani [arXiv:2606.06604v1](#), author FF.tex lines 1550–1633, SHA-256 14c17a95040be7bfc57bb5f87e58907a05cacd0bcb1b6967b2892cf3d9bc1229.

The companion TIMED_PRIME_PERIOD_DERIVATION.md, TP0–TP14, contains full proofs of the clock recurrence, prime factorization, convolution coefficients and all analytic convergence assertions used below. It is appended without alteration in CANONICAL_GENERIC_RECONSTRUCTION_FULL.md. Thus the combined proof does not replace these arguments by a reference to a result name. The earlier winding calculation W1–W4 is restated with complete maps in CG1–CG6.

CG0. Begin with the complete winding history and derive its one-step generator

The complete positive chart supplied by CC is M_+ in CG2, including all four finite-order units $H_+ = \langle J_+ \rangle$. Retain it and form the quotient monoid $P_+ = M_+/H_+$. From the original source relations, every element of P_+ is a nonnegative power of $u = q_+(T)$, with distinct powers. This statement concerns the whole source chart: identifying a finite observed prefix with part of it does not certify that every future event will obey its law.

Here is the requested contradiction argument **on the same complete history**. Let v be another positive one-step generator of P_+ : requiring that it is a generator means that its successive iterates account for the whole monoid without an additional missing class. Since u and v both generate that same whole monoid, there are positive iteration counts a, b with

$$v = u^a, \quad u = v^b.$$

Neither count is zero, since a generator is not the identity. Substitution gives

$$u = u^{ab}.$$

Distinct powers in the source chart imply $ab = 1$. Positive iteration counts therefore give $a = b = 1$, whence $u = v$. Two different complete one-step generators are impossible. The numerical names a, b count compositions in this proof; no observed prime has been prelabelled two or three to manufacture the unit.

The same generator is characterized without a chosen spelling: it is the unique element of P_+ different from the identity that is not the product of two elements different from the identity. The generator has this property because positive exponents cannot add to one, while every larger exponent splits into one and its predecessor. Thus the whole monoid determines its primitive step.

The negative chart gives its own forward generator $q_-(T^{-1})$. The exact embeddings into the group quotient are

$$j_{\pm} : P_{\pm} \hookrightarrow L, \quad j_{\pm}(q_{\pm}(m)) = q(\rho_{\pm}(m)).$$

They are well defined since chart units map into H . They are injective since equality after q means equal winding exponent, and the remaining difference is precisely a chart unit. Under these maps the images of the two forward generators are inverses in L , so a numerical orientation has not been made intrinsic. The intrinsic arithmetic one is instead id_L , as CG4 proves. To connect these two meanings exactly, define

$$\text{ev}_u : R_+ \longrightarrow P_+, \quad f \longmapsto j_+^{-1}(f(j_+(u))).$$

The classification in CG4 proves that $f(j_+(u))$ belongs to the image of j_+ , so the displayed inverse is applied on its domain. It also proves this evaluation is a bijection and satisfies $\text{ev}_u(f \oplus g) = \text{ev}_u(f)\text{ev}_u(g)$, and $\text{ev}_u(\mathbf{1}_R) = u$. For $u_- = q_-(T^{-1})$, the exact negative-chart formula is $f \mapsto j_-^{-1}(f(j_-(u_-)))$, justified by the same classification. The same arithmetic identity thus supplies either oriented step, with the actual inversion comparison retained.

This establishes uniqueness from the *complete specified counting structure*. Completeness by itself, with no event-composition law, does not assert that a primitive generator exists. Here existence comes from the actual full CC chart, and uniqueness is derived by the contradiction above. No additional existence assumption has been substituted for that source calculation.

CG1. Unique support of the common unordered localization

The actual unfolded space is

$$X = \{+, -, \eta\}, \quad \Omega_+ = \{+, \eta\}, \quad \Omega_- = \{-, \eta\}, \quad \Omega = \Omega_+ \cap \Omega_- = \{\eta\}.$$

Its opens are the empty set, these three sets, and X . The involution α interchanges $+$ and $-$. A homeomorphism fixes η : the singleton $\{\eta\}$ is the unique dense singleton, since the other two points are closed. Therefore

$$\text{Fix}(\alpha) = \{\eta\}. \quad (\text{CG1.1})$$

There is a direct contradiction proof of the requested common-support uniqueness. If $x \neq y$ were two points belonging to both ordered charts, then

$$x, y \in \Omega_+ \cap \Omega_- = \{\eta\}$$

would imply $x = \eta = y$, contradicting $x \neq y$. This proves uniqueness of the specified supporting role. It uses the whole chart diagram, not a numerical coordinate or a midpoint construction.

Let $Q_2 = \{\text{even}, \text{odd}\}$, with exchange ι . An equivariant point-label on an invariant subset satisfies $P(\alpha x) = \iota(P(x))$. At η , this would imply $P(\eta) = \iota(P(\eta))$, impossible because ι fixes neither label. On the other two points the rule is possible, by assigning either label at $+$ and the other at $-$. Thus the supporting $\tau\langle Z_1; \text{no } Z_2 \rangle$ is identified with η in this role by a proved obstruction. This does not mean its stalk is empty or that its isotropy is trivial.

CG2. The complete signed data and both localization maps

At the nonabsorbing monomial level retain

$$G = \langle T, J, \epsilon : TJ = JT, T\epsilon = \epsilon T, J\epsilon = \epsilon J, J^2 = \epsilon, \epsilon^2 = 1 \rangle. \quad (\text{CG2.1})$$

The separate source absorbing element is retained outside this group. The two charts have monomials

$$M_+ = \{T^m J_+^k : m \geq 0, k \bmod 4\}, \quad M_- = \{T^m J_-^k : m \leq 0, k \bmod 4\},$$

with the same fourth-order unit relations. Their actual restrictions are

$$\rho_+(T^m J_+^k) = T^m J^k, \quad \rho_-(T^m J_-^k) = T^m (\epsilon J)^k. \quad (\text{CG2.2})$$

The map $T \mapsto (1, \bar{0})$, $J \mapsto (0, \bar{1})$, $\epsilon \mapsto (0, \bar{2})$ gives $G \cong \mathbb{Z} \times C_4$. Its inverse is $(m, \bar{k}) \mapsto T^m J^k$: it is well defined because $J^4 = 1$, respects the operations, and the two composites fix the generators. These integer coordinates prove the classification of the already specified source; they are not inferred from two prelabelled observed event times.

In these coordinates ρ_+ preserves (m, k) , while ρ_- sends (m, k) to $(m, 3k)$. Since multiplication by three permutes C_4 , both maps are injective. Their image intersection is exactly the four units $H = \langle J \rangle$.

The whole G is the group completion of either chart, with its stated restriction. For M_+ , a homomorphism $f : M_+ \rightarrow K$ into an abelian group extends necessarily and sufficiently by

$$\tilde{f}(T^m J^k) = f(T)^m f(J_+)^k, \quad m \in \mathbb{Z}, k \bmod 4. \quad (\text{CG2.3})$$

The formula is well defined since $f(J_+)^4 = 1$; it is a group homomorphism and agrees with f . Uniqueness follows because T, J generate G .

For M_- , let $U = T^{-1}$. The unique extension obeys

$$\tilde{f}(T) = f(U)^{-1}, \quad \tilde{f}(J) = f(J_-)^3. \quad (\text{CG2.4})$$

Indeed $\rho_-(J_-) = J^3$ and $(J^3)^3 = J$, so this formula makes $\tilde{f}(\rho_-(J_-)) = f(J_-)^9 = f(J_-)$. The remaining generator and relation checks follow immediately by multiplication and the fourth-order relation. This proves existence and uniqueness on the negative chart with its source twist retained.

The universal completion is itself unique **with its chart map**: if (G', ρ'_+) also realizes this property, there are unique extensions $u : G \rightarrow G'$ and $v : G' \rightarrow G$ preserving the chart maps. Both vu and the identity of G extend ρ_+ , hence are equal by the proved uniqueness. Likewise $uv = \text{id}_{G'}$. Thus the comparison is an isomorphism and is unique subject to its chart map. This is the precise mathematical meaning of reconstructing the same complete chart data in another presentation.

CG3. Parity follows from the signed involution; all winding remains

The original action is

$$A(T) = \epsilon T^{-1}, \quad A(J) = \epsilon J = J^{-1}, \quad A(\epsilon) = \epsilon.$$

For every $g = T^m J^k$,

$$A(g) = \epsilon^m T^{-m} J^{-k} = T^{-m} J^{2m-k}, \quad gA(g) = T^m J^k \epsilon^m T^{-m} J^{-k} = \epsilon^m. \quad (\text{CG3.1})$$

Applying the coordinate action twice gives $(m, k) \mapsto (-m, 2m - k) \mapsto (m, k)$, so it is an involution.

The finite-order subgroup of G is exactly $H = \langle J \rangle$. A nonzero exponent of T cannot become zero upon taking a nonzero power; all four J -states have finite order. Retain the full exact sequence

$$1 \longrightarrow H \longrightarrow G \xrightarrow{q} L := G/H \longrightarrow 1. \quad (\text{CG3.2})$$

The quotient is infinite cyclic. The mirror induces inversion on it. Because $hA(h) = 1$ for every $h \in H$, (CG3.1) descends to the surjection

$$\delta : L \longrightarrow \{1, \epsilon\}, \quad \delta(q(g)) = gA(g), \quad \delta(q(T^m J^k)) = \epsilon^m. \quad (\text{CG3.3})$$

It is a homomorphism since G is abelian. Its kernel is the subgroup of squares of L . Equivalently, the kernel before passage through q is $\langle T^2, J \rangle$. This derives the even/odd observation from the full involution. The retained object is the exact sequence and its map, not only its two-state image.

The user's winding history corresponds to the unbounded exponent in L . It is different from the finite-order subgroup H . Neither the identity of L nor any of its elements is identified with the supporting point.

CG4. The global arithmetic unit, without choosing observed prime labels

Receiver scope, clarified after the full timing derivation: this section computes endomorphisms of the explicitly retained quotient $L=G/H$. It does not identify that ring with endomorphisms of the full signed sheaf preserving J , epsilon and its parity observation. On L , $[n]$ sends the parity of a winding to n times that parity modulo two. Consequently only odd $[n]$ preserve the parity observation identically; these are closed under composition but not addition. The full G , H , quotient map and parity observation remain in CG3. In addition, cyclicity of L comes from the displayed source chart, not from a proof that uncalibrated timing must yield that chart. GLOBAL_TIMING_QUOTIENT_INDEPENDENT.md GTQ1-GTQ8 and FIXED_SUPPORT_AND_ODD_REFINEMENT.md OR1-OR8 give the complete calculation of that earlier step and its obstruction.

Construct $R = \text{End}_{\text{Ab}}(L)$, with

$$(f \oplus g)(x) = f(x)g(x), \quad (f \odot g)(x) = f(g(x)). \quad (\text{CG4.1})$$

The pointwise product is a homomorphism because L is abelian. Its additive zero is $x \mapsto 1_L$, and its additive inverse sends x to $f(x)^{-1}$. Associativity and commutativity of addition follow pointwise.

Composition is associative. The homomorphism law proves left distributivity; evaluating a pointwise product at $g(x)$ proves right distributivity. Thus this is a unital ring with

$$\mathbf{1}_R = \text{id}_L. \quad (\text{CG4.2})$$

Here is the exact unit contradiction, using no prime names. If $e, e' \in R$ were two distinct two-sided multiplicative identities, then

$$e \odot e' = e' \quad \text{because } e \text{ is an identity,} \quad e \odot e' = e \quad \text{because } e' \text{ is an identity.}$$

Hence $e = e'$, a contradiction. These are operations on endomorphisms; they assert no operation on τ .

Choose a generator t of L temporarily to classify this ring. Any endomorphism has $f(t) = t^n$ for a unique $n \in \mathbb{Z}$; then $f(t^m) = t^{nm}$. Conversely every power map $[n] : x \mapsto x^n$ is a homomorphism. Consequently

$$\iota : \mathbb{Z} \xrightarrow{\sim} R, \quad n \mapsto [n], \quad [a] \oplus [b] = [a + b], \quad [a] \odot [b] = [ab]. \quad (\text{CG4.3})$$

The classification is independent of choosing t or t^{-1} : the exponent of $f(t^{-1})$ relative to t^{-1} is still n . Thus the entire ring and its identity are orientation-independent.

Let R_+ be the additive submonoid generated by $\mathbf{1}_R$, including its zero. Its elements are distinct and correspond exactly to nonnegative integers under (CG4.3). Among its nonzero elements, $\mathbf{1}_R$ is the unique universal divisor: it divides every element by composition, while an element dividing every nonzero element must divide $\mathbf{1}_R$. Its positive degree a then has a positive degree partner b with $ab = 1$, forcing $a = b = 1$. This proves the user's "global quotient one" property in its precise divisibility sense on the constructed arithmetic.

The argument starts with the complete L . It does not certify from a finite event record that the source has already been identified correctly. In the user's unrestricted late-event observation problem, a finite record can have incompatible full lattice extensions, as proved in GO3. That observational result and this construction from the specified complete source have different inputs.

CG5. The full prime spectrum and its norms are canonical

For completeness, the prime ideals of $\mathbb{Z}$ are (0) and (p) for positive prime integers p . A nonzero ideal contains a least positive integer d ; division with remainder proves it is $d\mathbb{Z}$. A composite $d = ab$, with $1 < a, b < d$, makes that ideal nonprime. For prime p , Euclid's lemma proves primality; TP6 supplies its full proof by the terminating Euclidean algorithm and Bezout identity. The zero ideal is prime because a product of two nonzero integers is nonzero.

The isomorphism (CG4.3) transports all these ideals and products exactly. It induces

$$\text{Spec } R \xrightarrow{\sim} \text{Spec } \mathbb{Z}, \quad \mathfrak{p} \mapsto \iota^{-1}(\mathfrak{p}). \quad (\text{CG5.1})$$

The inverse is the image of an ideal under ι . Membership in $D([n])$ corresponds to membership in $D(n)$, proving that both directions are continuous. The maps of basic-open section rings are

$$\mathbb{Z}[1/n] \xrightarrow{\sim} R[1/[n]], \quad a/n^k \mapsto [a]/[n]^k. \quad (\text{CG5.2})$$

Cross multiplication proves that the formula respects representatives; applying ι^{-1} gives its inverse. They commute with every localization restriction. Thus the comparison includes the structure sheaf, not merely the list of primes.

For $\mathfrak{p}_p = ([p])$, the residue field and its exact norm are

$$R/\mathfrak{p}_p \cong \mathbb{F}_p, \quad N(\mathfrak{p}_p) = p = |L/[p]L|. \quad (\text{CG5.3})$$

The latter quotient has representatives $1_L, t, \dots, t^{p-1}$, by division of the exponent. They are distinct because two exponents in this range cannot differ by a nonzero multiple of p . Every prime is retained, including two.

The generic point $(0) \in \text{Spec } R$ is the unique generic point of this derived spectrum. It is a prime ideal of R . The point $\eta \in X$ is the supporting point in the source geometry. Their roles are connected by the construction (CG3.2)–(CG5.1); they are not silently identified as objects of the same space.

CG6. Uniqueness of the complete arithmetic reconstruction

Let $u : G \rightarrow G'$ be an isomorphism comparing two presentations of the full source, as in CG2. Every finite-order element is sent to a finite-order element, and applying u^{-1} proves equality of these subgroups. Thus u induces an isomorphism $\bar{u} : L \rightarrow L'$. The induced map of rings is exactly

$$R \xrightarrow{\sim} R', \quad f \mapsto \bar{u} \circ f \circ \bar{u}^{-1}. \quad (\text{CG6.1})$$

It preserves pointwise addition by the homomorphism law, composition by cancellation of $\bar{u}^{-1}\bar{u}$, and the identity by direct substitution. It preserves every power map because $\bar{u}(x^n) = \bar{u}(x)^n$. Therefore it preserves every element of R_+ , every prime ideal and every residue norm in CG5. It also preserves all quotient clocks: $x \bmod [p]L \mapsto \bar{u}(x) \bmod [p]L'$ is a well-defined bijection.

In particular, the two ordered charts in CG2 yield exactly the same normed spectrum through their common localization. Every homomorphism from either chart into an abelian group factors uniquely through its indicated map to G ; killing the finite-order subgroup then factors uniquely through L . The second statement applies to torsion-free targets: any image of H there is trivial, since it has finite order. This proves the claimed universal factorization with all its domains stated.

There are three levels of uniqueness, all retained here: the common point in the specified topological diagram is unique; group completion is unique with its chart map, up to unique isomorphism; the arithmetic identity and every power map are canonical under that isomorphism. The bare identity equation alone does not prove the first two levels. For example both actual ordered charts already have the same identity section, and both localization maps preserve it, despite their being distinct charts. The stronger common-support and completion requirements are the parts of the user's argument that select the one diagrammatic role proved in CG1–CG2.

CG7. The original zeta function follows from the whole normed spectrum

For each recovered prime norm p , the source curve is $E_p = \mathbb{C}^\times / p^\mathbb{Z}$. Its flow $[z] \mapsto [e^t z]$ returns exactly when $t \in (\log p)\mathbb{Z}$, because $e^t z = p^k z$ is equivalent to $t = k \log p$. The original differential has both periods $\log p$ and $2\pi i$, as proved in TP7. These are coordinates on the curves and their flow; no distance from τ is defined.

Retain all positive returns and the complete count measure:

$$\mathcal{R} = \sum_p \sum_{k \geq 1} \frac{1}{k} \delta_{k \log p}, \quad \mathcal{D} = \delta_0 + \sum_{n \geq 2} \delta_{\log n}, \quad \mathcal{W} = \sum_p \sum_{k \geq 1} (\log p) \delta_{k \log p}. \quad (\text{CG7.1})$$

TP9 proves local finiteness; TP10 proves, including every convolution coefficient,

$$\mathcal{D} = \exp_* \mathcal{R}, \quad \mathcal{W} = t\mathcal{R}. \quad (\text{CG7.2})$$

The empty convolution has the atom δ_0 , corresponding to arithmetic one. This atom is retained in all comparisons.

Since CG6 preserves every prime norm, it preserves each term of (CG7.1) with its repetition number and coefficient. Hence the measures are uniquely determined by the full source and independent of its ordered presentation. The exact Laplace transforms, with the absolute and locally uniform convergence proved in TP12-TP13, are

$$\boxed{\int_{[0,\infty)} e^{-st} d\mathcal{D}(t) = 1 + \sum_{n \geq 2} n^{-s} = \zeta(s), \quad \Re s > 1,} \quad (\text{CG7.3})$$

$$\exp \left(\int e^{-st} d\mathcal{R}(t) \right) = \zeta(s), \quad \int e^{-st} d\mathcal{W}(t) = -\frac{\zeta'(s)}{\zeta(s)}. \quad (\text{CG7.4})$$

The function here is the original zeta. No completion multiplier or rescaling replaces it. The measures include prime two and every positive repetition. Negative-time returns exist in the whole flow but are not positive integers in the original Dirichlet series; this domain comparison is proved in TP13.

This also gives a contradiction proof of uniqueness of the resulting analytic function: two functions reconstructed from the same normed spectrum have the same absolutely convergent series (CG7.3) at every point of $\Re s > 1$, so cannot differ there. If they have meromorphic continuations to the same connected domain, those continuations coincide. To verify the latter uniqueness, their difference is meromorphic and vanishes on an open set. Away from its discrete poles the analytic identity theorem makes it zero on the connected punctured domain; every apparent pole is then removable with zero extension. Existence of the original zeta continuation is not asserted to follow solely from abstract meromorphic uniqueness.

CG8. Exact content of the composed claim

The constructed chain is

$$(\eta, \mathcal{O}_\eta, A; \rho_+, \rho_-) \mapsto (G, H, L, \delta) \mapsto (R, \mathbf{1}_R, R_+) \mapsto (\text{Spec } R, N) \mapsto (\mathcal{R}, \mathcal{D}, \mathcal{W}) \mapsto \zeta(s) \quad (\Re s > 1). \quad (\text{CG8.1})$$

Every arrow has been constructed above or in the complete appended TP proof. The full supporting datum is retained; $G \rightarrow L$ is an explicit quotient within the exact sequence, not permission to erase H or its signed action.

The “only” established here is the unique common supporting point and the universal factorization of both chart reconstructions through its complete generic localization. It is not a classification of every possible arithmetic geometry by the existence of one unit. The full cyclic stalk, both chart maps and the signed involution are mathematical input supplied by the actual CC construction. The unit does not by itself recover those inputs.

The user’s further argument concerns whether this complete, uniquely specified reconstruction forces a common arithmetic spectral weight and therefore the required purity. This proof supplies its actual original-zeta receiving map. It proves no modulus statement for the zeros by replacing that question with the uniqueness just established. That weight calculation remains part of the ongoing programme.

Full arithmetic and return-measure proof

Proof source: TIMED_PRIME_PERIOD_DERIVATION.md.

First-appearance time, primitive periods, and the original zeta function

24 September 2026. Independent mathematical derivation for the current programme. Result identifiers TP0-TP14.

The user's clarification is that integer 1 supplies the counting step, while the appearance times of new prime channels distinguish those channels. The subsequent correction requires the whole unlabelled structure first: one, two, three, and prime labels must then be outputs of its arithmetic construction, not preassigned observations used to manufacture a unit. TP0 supplies that order of construction. The later sections prove the exact maps from the resulting timed record to the periodic curves in Connes-Consani and to the original Riemann zeta function on its domain of absolute convergence. They retain integer zero, the unit, prime two, every prime-power return, and every multiplicity.

The supporting datum is written

$$\tau\langle Z_1; \text{no } Z_2\rangle.$$

No addition, subtraction, coordinate, vector, metric, or distance is assigned to this datum. All counts, group operations, times, coordinates and measures below belong to the indicated winding group, clocks, source curves, or real half-line. In particular, the count 1 is not identified with this supporting datum. The earlier retracted expression $\tau + \tau = \tau$ is not used.

Sources and actual reading

1. Connes-Consani, [arXiv:2606.06604v1](#), original author file FF.tex, lines 1550-1633 read for this derivation. The source supplies $E_p = \mathbb{C}^\times/p^\mathbb{Z}$, the scaling generator $z \mapsto pz$, its real structure, the product with the phase circle, and the full differential $dz/z = d\lambda/\lambda + i d\theta$. The source parameter called tau in its modular-parameter formula is distinguished below as τ_p^{mod} ; it is not the user's supporting datum.
2. [Prime clocks, orchard visibility, and faithful prime observations](#), P9-P14 read in full. P1-P3 were also read. The present note supplies complete proofs of the timed sieve, reconstruction, period maps and zeta identities it asserts; it does not infer a result from a search hit.
3. [Winding data at the fixed point: parity, the integer ring, and its spectrum](#), W1-W3 read for the whole-source reconstruction. The original signed group and its quotient are reproduced below; the four branch states are retained in the exact sequence.

These calculations are not a novelty claim. They apply elementary arithmetic and the displayed source curves to the user's specified timed observation. They do not substitute a completed or rescaled zeta function for the original function.

TP0. The whole source precedes numerical labels and the unit is intrinsic

Start from the entire source group of nonabsorbing generic sections, with its source relations retained:

$$G = \langle T, J, \epsilon \mid TJ = JT, T\epsilon = \epsilon T, J\epsilon = \epsilon J, J^2 = \epsilon, \epsilon^2 = 1 \rangle.$$

The source also has a separate absorbing element, which is not a group element and is not identified here with the supporting point. Sending T to $(1, \bar{0})$, J to $(0, \bar{1})$, and ϵ to $(0, \bar{2})$ defines an isomorphism

$$G \longrightarrow \mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}.$$

Indeed, the displayed images satisfy every relation. The inverse sends $(m, \bar{k})$ to $T^m J^k$, is independent of the representative of $\bar{k}$ because $J^4 = 1$, and both compositions fix the generators. This comparison checks the source group's structure; it does not assert that the numerical labels have been observed by running the proposed clock.

The subgroup of finite-order elements is intrinsically

$$H = \text{Tor}(G) = \{1, J, \epsilon, \epsilon J\}.$$

Proof: a finite positive power of $(m, \bar{k})$ can be the identity only if $m = 0$, and all four elements with that property have finite order. Retain the exact sequence

$$1 \longrightarrow H \longrightarrow G \xrightarrow{q} L := G/H \longrightarrow 1.$$

The quotient L is an infinite cyclic group. Regard it initially without a marked generator. The choice of source presentation can later exhibit a generator, but the following arithmetic ring and its identity do not depend on that choice.

Define

$$E = \text{End}_{\text{Ab}}(L), \quad (f \oplus g)(x) = f(x)g(x), \quad (f \odot g)(x) = f(g(x)).$$

Because L is abelian, the pointwise product of two homomorphisms is a homomorphism. Addition is associative and commutative; its zero is $0_E : x \mapsto 1_L$, and the additive inverse of f is $x \mapsto f(x)^{-1}$. Composition is associative and has the distinguished identity

$$\mathbf{1}_E = \text{id}_L.$$

For left distributivity, evaluate $f(g(x)h(x)) = f(g(x))f(h(x))$; right distributivity follows by evaluating a pointwise product at $h(x)$. These prove that E is a unital ring. No numeral has been selected by inspecting early prime events: $\mathbf{1}_E$ is already distinguished as the identity operation on the whole source. It differs from 0_E because L is nontrivial.

For a proof of its arithmetic structure, choose either generator u of the infinite cyclic group temporarily. Any endomorphism has $f(u) = u^n$ for a unique integer n , and the homomorphism law then forces $f(u^m) = u^{nm}$. Conversely every map $[n] : x \mapsto x^n$ is an endomorphism. Its exponent on the other generator u^{-1} is still n , since $f(u^{-1}) = u^{-n} = (u^{-1})^n$. The resulting isomorphism

$$\mathbb{Z} \xrightarrow{\sim} E, \quad n \mapsto [n],$$

is therefore independent of which generator was used to prove it. Evaluation gives $[n] \oplus [m] = [n + m]$, $[n] \odot [m] = [nm]$, $[0] = 0_E$, and $[1] = \mathbf{1}_E$. Thus the count labels describe an arithmetic object obtained from the entire source, rather than serving as a calibration assumption.

The nonnegative part E_+ is the additive submonoid generated by $\mathbf{1}_E$, including its empty sum 0_E . Its successive elements are distinct: equality of two different finite sums would give two unequal powers of a generator the same value. It follows from the preceding classification that E_+ corresponds exactly to the nonnegative integers. In particular, $\mathbf{1}_E$ is its unique nonzero element that cannot be expressed as a sum of two nonzero elements of E_+ : in the exponent description one is not a sum of two positive integers, whereas every integer $n \geq 2$ is $1 + (n - 1)$.

For a nonzero $f \in E_+$, define its winding clock intrinsically as the cokernel

$$C_f = L/f(L).$$

If $f = [n]$, the representatives $1_E, u, \dots, u^{n-1}$, with division of exponents by n , show that it has exactly n elements. This order does not depend on the chosen generator. The unit $\mathbf{1}_E$ has trivial cokernel. A positive nonunit f has nonzero simple cokernel exactly when its resulting degree label n is prime, by the full subgroup argument in TP4. Thus the prime objects too are intrinsic to the reconstructed ring and its quotient clocks; their numerical names can be supplied after this construction.

There is also an intrinsic formulation of the statement that one divides everything. Define $a \mid b$ in $E_+ \setminus \{0_E\}$ by the existence of $c \in E_+$ with $b = a \odot c$. The identity divides every b . Conversely an element dividing every positive element divides $\mathbf{1}_E$. Under the proved ring isomorphism, the equation $ac = 1$ with positive integer degrees forces both degrees to be one, so this element is exactly $\mathbf{1}_E$. The full ring has two units, $\mathbf{1}_E$ and $-\mathbf{1}_E$; only the first lies in E_+ .

Parity can now be expressed without using a prime label to found the unit. Form the intrinsic doubling endomorphism $\mathbf{1}_E \oplus \mathbf{1}_E$ and the quotient by its image. The quotient has two elements, as the cyclic-group calculation shows. Every $f \in E$ induces either its zero map or its identity map; in the exponent coordinates those are respectively even and odd. In particular $\mathbf{1}_E$ has the odd Z_2 state. The source point is not a member of this ring of maps and is assigned no parity by this construction.

This proves a construction from the full signed source and its winding quotient. That source already supplies an infinite cyclic direction; this proof does not derive its existence from a datum consisting only of $\tau\langle Z_1; \text{no } Z_2 \rangle$. The whole structure used here and an observer's finite record of that structure are different stages. The finite record and its exact infinite limit are calculated in TP11. Neither the construction nor that limit proves that an observer must wait through infinitely many ticks before an identity operation exists. In particular, no subtraction of two already labelled prime times is used to establish $\mathbf{1}_E$.

The user's further permitted possibility of future observations falling off a currently inferred integer grid is a question about extending a partial observation, not about evaluating a fixed endomorphism of the already specified complete cyclic group. TP0 does not certify the complete source group from any finite event prefix. Its input is the complete Connes-Consani group and the retained exact sequence. The identity map would still exist for a different source group; the proof that every endomorphism is an integral multiple of it specifically uses the cyclic structure. This is the interface with the separate calculation of source extensions and the global counting unit.

TP1. The retained count and the parity observation have different periods

After TP0's construction, the oriented-ray covering gives the following geometric realization of its winding quotient:

$$\mathbb{C}^\times / \mathbb{R}_{>0} \longrightarrow \mathbb{C}^\times / \mathbb{R}^\times.$$

An angle on the first space has period 2π ; an angle on the second has period π . Fix a reference line and a ray above it. A loop of lines has a real angle lift starting at zero, with endpoint $n\pi$ for a unique integer n . A homotopy of based loops preserves n , since its lifted endpoint varies continuously in the discrete set $\pi\mathbb{Z}$. Conversely two lifted paths with the same endpoints are homotopic by linear interpolation, whose projection is a based homotopy. Concatenation adds endpoint angles. This proves that the winding classes form an infinite cyclic group

$$L = \langle t \rangle = \{t^n : n \in \mathbb{Z}\}.$$

The lifted ray has returned to its initial orientation exactly when n is even. Thus its parity observation is

$$\text{par}(t^n) = n \bmod 2.$$

The forward counting step is the map

$$S(t^n) = t^{n+1}.$$

It has infinite order, because $t^n \neq t^{n+r}$ for $r \neq 0$. On parity it induces the two-state exchange. After two steps the parity returns, while the retained count has increased by two. Consequently the full operation carrying history is not an order-two map; its observation on the ray states is order two. The term “torsion” in the user’s description refers here to retained winding, not to a finite-order element of L .

The state t^0 has count zero. It is the identity of this winding group, not $\tau\langle Z_1; \text{no } Z_2 \rangle$. The elementary increment 1 has Z_2 state odd. None of this assigns parity to the supporting datum.

TP2. Unit time and the return test

Choose the forward progression $0, 1, 2, \dots$ supplied by TP1. A time in this paragraph is an integer tick, not physical time. At a fixed positive integer d , the quotient clock is

$$C_d = L / \langle t^d \rangle.$$

Its state at tick n is the class of t^n . It is at its initial state exactly when $t^n \in \langle t^d \rangle$, which is exactly the integer divisibility condition $d \mid n$. Thus the test can be implemented by advancing a finite clock one state at every unit tick and checking for return. It does not require the list of primes as input.

The progression and the periodic-return rule are both data of this construction. A successor sequence with no specified return or divisibility test would not by itself select prime ticks. Here the specified test is the already constructed quotient-clock return, so the selection rule has an exact domain and meaning.

Repeated application of 1 distinguishes every consecutive tick. Multiplication of positive counts is the repeated-addition operation $ab = b + \dots + b$, with a summands. Division with remainder follows by taking the greatest q such that $dq \leq n$: the nonnegative remainder $r = n - dq$ satisfies $0 \leq r < d$, since otherwise $d(q+1) \leq n$. Uniqueness follows by subtracting two such descriptions and using that a nonzero multiple of d has absolute value at least d . These are operations on counts, not on the supporting datum.

TP3. The sequential first-appearance recurrence produces precisely the primes

Let $A_1 = \emptyset$. Suppose that ticks through $n - 1$ have been processed, for an integer $n \geq 2$. Define

$$B_n = \{d \in A_{n-1} : d \mid n\}, \quad \epsilon_n = \begin{cases} 1, & B_n = \emptyset, \\ 0, & B_n \neq \emptyset, \end{cases}$$

and

$$A_n = \begin{cases} A_{n-1} \cup \{n\}, & \epsilon_n = 1, \\ A_{n-1}, & \epsilon_n = 0. \end{cases}$$

When d is first activated at tick d , its clock is in the initial state. After $n - d$ subsequent unit steps it is again in that state exactly when $d \mid n - d$, equivalently $d \mid n$. Therefore this recurrence agrees with the physical description “advance every active clock; create a new line only if none returns now.” All overlapping returns are retained in B_n .

Claim. For every $n \geq 1$, A_n consists exactly of the primes at most n .

Proof. The assertion is true at $n = 1$, and tick two activates because the active set is empty. Assume it holds through $n - 1$. A prime n has no divisor strictly between one and itself, so no member of A_{n-1} divides it and it activates. If n is composite, let $p > 1$ be its least divisor greater than one. If $p = ab$ with $1 < a < p$, then $a \mid n$, contradicting its minimality. Hence p is prime; compositeness gives $p < n$. By induction its clock is already active and returns at n , so n does not activate. This proves the claim by induction. $\square$

For the first several ticks,

Tick n	Returning earlier prime channels B_n	New line
1	no prime channel is tested	none; this is the unit
2	none	2
3	none	3
4	2	none
5	none	5
6	2 and 3	none
7	none	7
8	2	none
9	3	none
10	2 and 5	none
11	none	11

In particular, two conceals the newness of four but does not remove tick four. The unit increments still record its occurrence.

TP4. Why one is not a prime channel and does not cover everything

The clock $C_1 = L/L$ has one state and is the trivial group. It returns at every tick. If it were inserted into A_1 , then $1 \mid n$ for every n , so the recurrence would create no prime channels at all. The stated recurrence excludes this trivial clock by testing ticks $n \geq 2$ and admitting only nontrivial clocks.

This exclusion is dictated by the exact prime criterion, rather than by an exception to that criterion. Every subgroup of L is $\langle t^d \rangle$ for a unique $d \geq 0$: choose the least positive exponent in a nontrivial subgroup and apply division with remainder. Subgroups of C_n are consequently the images of these subgroups for divisors $d \mid n$. Thus

$$n \text{ is prime} \iff C_n \text{ is a nontrivial simple cyclic group.}$$

Indeed, for prime n , its only divisors are one and n , so the quotient has only the whole group and the identity subgroup. If $n = ab$ with $a, b > 1$, the subgroup generated by t^a in C_n has b elements and is proper and nontrivial. The case $n = 1$ is excluded by “nontrivial.”

Equivalently, the winding-scale map $[n] : t^m \mapsto t^{nm}$ has image index n ; composition obeys $[a][b] = [ab]$. The map $[1]$ is invertible, and a prime is a noninvertible positive-degree map that does not factor into two noninvertible positive-degree maps. The ideal (1) is the entire integer ring, not a proper prime ideal. These equivalent unit distinctions explain why the counting increment can be indispensable while not being a prime event.

TP5. The complete timed record is equivalent to the anchored waiting times

The prime set is infinite. To prove this, a finite purported complete list $p_1, \dots, p_r$ would give $M = p_1 \cdots p_r + 1 > 1$. Its least divisor greater than one is a prime by TP3’s argument and cannot equal any p_j , since division of M by p_j leaves remainder one. This is a contradiction.

Let $p_1 < p_2 < \dots$ be the activation ticks proved in TP3; $p_1 = 2$. Anchor elapsed time at the unit tick 1. Then define

$$e_j = p_j - 1, \quad g_1 = p_1 - 1 = 1, \quad g_j = p_j - p_{j-1} \quad (j \geq 2).$$

The exact inverse formulas are

$$e_j = \sum_{r=1}^j g_r, \quad p_j = 1 + \sum_{r=1}^j g_r, \quad \epsilon_n = \mathbf{1}_{\{p_1, p_2, \dots\}}(n).$$

The first two formulas telescope; the final formula is TP3. Conversely the indicator’s successive ticks where it is one give the p_j , and adjacent subtraction gives the gaps. Hence the unit-anchored waiting sequence, the timed binary event record, and the increasing list of prime labels determine one another exactly.

For example, the first waiting times from tick one are

$$1, 1, 2, 2, 4, 2, 4, 2, 4, 6, \dots,$$

giving prime ticks 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, If elapsed time is anchored instead at count zero, the first wait is two and all subsequent gaps are unchanged. The origin convention must be retained: gap lengths without the first absolute or anchored event locate events only up to an additive shift, and an unordered collection of gaps also loses their order. Neither loss occurs in the recorded sequential history.

Thus the statement “the information is in the time intervals” has a precise content: the ordered waiting times with their anchor retain every prime label. It is not necessary to assign different geometric radii to the channels to distinguish them.

TP6. The timed prime record reconstructs every multiplicity

Every integer $n > 1$ is a product of primes. Proof by induction: a prime is already such a product; a composite $n = ab$ with $1 < a, b < n$ is a product of the prime products supplied by induction for a, b .

For uniqueness, first prove Euclid's lemma. If a prime p does not divide a , the Euclidean algorithm gives integers u, v with $up + va = 1$. Here the Euclidean algorithm terminates because successive nonzero remainders strictly decrease; tracing the remainder identities backwards expresses the greatest common divisor as an integer combination. The gcd of p, a is one because it divides the prime p and is not p . Multiplying by b shows that $p \mid ab$ implies $p \mid b$. Applying the lemma repeatedly to two prime decompositions shows that the first prime in one equals a prime in the other; cancelling that positive factor and inducting gives equality of all prime multiplicities.

Consequently, the labels reconstructed in TP5 give the unique data

$$n = \prod_p p^{v_p(n)}, \quad v_p(n) \in \mathbb{N}_0,$$

with finitely many nonzero $v_p(n)$. The unit one has every exponent zero. The integer zero is not represented by a finite product of positive primes, and negative integers require a separate sign. Neither is silently included in the positive-integer product below. This is exactly the domain required by the original zeta Dirichlet series.

An individual returning prime-clock state records divisibility, not the exponent $v_p(n)$. The reconstruction retains multiplicities through successive division, or equivalently through the complete prime-power clocks C_{p^r} : $v_p(n)$ is the greatest $r \geq 0$ for which that clock returns at n . No prime-power observation is deleted.

TP7. The same prime label determines a primitive period in the source curve

For each recovered prime p , retain the Connes-Consani source object

$$E_p = \mathbb{C}^\times / p^\mathbb{Z}.$$

Let $[z]$ denote the equivalence class of $z \neq 0$. Positive real scaling supplies the real-parameter flow

$$\Phi_t([z]) = [e^t z], \quad t \in \mathbb{R}.$$

It is well defined because replacing z by $p^m z$ replaces its image by $p^m e^t z$. The exponential law gives $\Phi_{t+u} = \Phi_t \Phi_u$. For every $[z]$,

$$\Phi_t([z]) = [z] \iff e^t z = p^k z \text{ for some } k \in \mathbb{Z} \iff t = k \log p.$$

The primitive positive period is therefore

$$\lambda_p = \log p.$$

The positive, zero and negative returns are all retained as the full subgroup $\lambda_p \mathbb{Z}$. The connected positive real orbit is $C_p = \mathbb{R}_{>0} / p^\mathbb{Z}$, and $u \mapsto [e^u]$ identifies it with $\mathbb{R} / (\lambda_p \mathbb{Z})$.

In polar coordinates $z = re^{i\theta}$, the full source one-form is

$$\frac{dz}{z} = \frac{dr}{r} + i d\theta.$$

The scaling cycle has integral $\int_1^p dr/r = \log p$; the positive phase cycle has integral $\int_0^{2\pi} i d\theta = 2\pi i$. Thus the logarithmic coordinate $u \mapsto e^u$ has deck lattice

$$\lambda_p \mathbb{Z} + 2\pi i \mathbb{Z}.$$

With the source's alternative exponential coordinate $z = e^{2\pi i w}$, the same lattice is

$$\mathbb{Z} + \mathbb{Z} \tau_p^{\text{mod}}, \quad \tau_p^{\text{mod}} = i \frac{\log p}{2\pi}.$$

The basis vector τ_p^{mod} exponentiates to $1/p$, rather than p ; its negative exponentiates to p . Both generate the same subgroup $p^{\mathbb{Z}}$. This preserves the sign and the factor 2π in the source convention.

These are coordinates and periods of E_p , not coordinates or a numerical distance from $\tau\langle Z_1; \text{no } Z_2 \rangle$.

TP8. Integer event time and logarithmic return time have an exact comparison

For the activation list recovered in TP5,

$$p_j = 1 + \sum_{r=1}^j g_r, \quad \lambda_{p_j} = \log \left(1 + \sum_{r=1}^j g_r \right).$$

The inverse is $p_j = e^{\lambda_{p_j}}$. Therefore no prime label is lost in passing from the exact primitive-period list to the activation list. For $j \geq 2$,

$$\lambda_{p_j} - \lambda_{p_{j-1}} = \log \left(1 + \frac{g_j}{p_{j-1}} \right).$$

An additive prime gap g_j and a primitive orbit period $\log p_j$ are thus different data with this proved comparison; they are not set equal.

The logarithmic length of a positive integer is

$$\ell(n) = \log n = \sum_p v_p(n) \log p.$$

This follows by applying the logarithm to TP6's finite product. The equality $\ell(ab) = \ell(a) + \ell(b)$ follows from the positive-real exponential law. It is this multiplication-to-addition map that converts the k -fold prime repetition into return time

$$k\lambda_p = k \log p = \log(p^k).$$

At integer tick n , a prime clock returns if $p \mid n$; those times include $p, 2p, 3p, \dots$. A repetition of the scaling orbit has positive logarithmic time $k \log p$ and corresponds to p^k . Both structures are retained: the first is the divisibility-clock observation used to discover prime channels, while the second is repetition in the source's multiplicative scaling orbit.

Prime two is present throughout: it activates at tick two, has $\lambda_2 = \log 2$, and returns at $k \log 2$ for every integer k . This does not assert the existence of an even-degree lift through the separate signed source action discussed in P13-P14; the obstruction there does not remove the prime-two curve or any of these returns.

TP9. Exact measures of primitive events, repetitions and all counts

Write δ_a for the unit point measure at real time a . Define the following positive measures on their stated time spaces:

$$\begin{aligned} \mathcal{E} &= \sum_p \delta_p && \text{on integer event time,} \\ \mathcal{P} &= \sum_p \delta_{\log p} && \text{on positive logarithmic time,} \\ \mathcal{R} &= \sum_p \sum_{k \geq 1} \frac{1}{k} \delta_{k \log p} && \text{on positive logarithmic time,} \\ \mathcal{W} &= \sum_p \sum_{k \geq 1} (\log p) \delta_{k \log p} && \text{on positive logarithmic time,} \\ \mathcal{D} &= \sum_{n \geq 1} \delta_{\log n} = \delta_0 + \sum_{n \geq 2} \delta_{\log n} && \text{on nonnegative logarithmic time.} \end{aligned}$$

Every one of these measures is locally finite. For a compact logarithmic interval $[0, T]$, a contributing prime satisfies $p \leq e^T$ and a contributing repetition satisfies $k \leq T / \log p$; there are finitely many such integers. A contributing general count satisfies $n \leq e^T$, again a finite set. The same argument on a bounded interval of integer event time proves local finiteness of $\mathcal{E}$.

The primitive measure satisfies $\mathcal{P} = (\log)_* \mathcal{E}$, by evaluating both sides on an arbitrary Borel set. Multiplication of a measure by its time coordinate gives the exact identity

$$\mathcal{W} = t \mathcal{R},$$

since at the k -th return, $t/k = (k \log p)/k = \log p$. No coefficient has been suppressed. If two positive returns have equal time, $k \log p = l \log q$, then $p^k = q^l$; uniqueness of factorization gives $p = q$ and $k = l$. Thus distinct prime-power labels do not collide in this time representation.

Define $\Lambda(1) = 0$, $\Lambda(p^k) = \log p$ for $k \geq 1$, and $\Lambda(n) = 0$ for other positive integers. Then

$$\mathcal{W} = \sum_{n \geq 2} \Lambda(n) \delta_{\log n}, \quad \mathcal{R} = \sum_{n \geq 2} \frac{\Lambda(n)}{\log n} \delta_{\log n}.$$

The coefficients $1/k$ and $\log p$ are derived from the full counting measure in TP10-TP12; they are not claimed to follow merely from the existence of a periodic orbit.

TP10. Recovering the full count measure, including its unit atom

For each prime retain the locally finite measure

$$\mathcal{D}_p = \sum_{k \geq 0} \delta_{k \log p}.$$

For a finite set of primes F , its convolution product is

$$\mathcal{D}_F = \ast_{p \in F} \mathcal{D}_p = \sum_{(k_p) \in \mathbb{N}_0^F} \delta_{\sum_{p \in F} k_p \log p}.$$

Convolution of point measures obeys $\delta_a \ast \delta_b = \delta_{a+b}$. Local finiteness follows from the finite range of exponents contributing to a bounded time interval. TP6 and TP8 give

$$\mathcal{D}_F = \sum_{\substack{n \geq 1 \\ p|n \Rightarrow p \in F}} \delta_{\log n}.$$

Every coefficient is one because factorization is unique. The exponent tuple with every entry zero contributes exactly δ_0 , the unit count one.

There is also an exact convolution-exponential identity

$$\mathcal{D} = \exp_{\ast}(\mathcal{R}) := \delta_0 + \sum_{r \geq 1} \frac{\mathcal{R}^{\ast r}}{r!}.$$

To prove it without an analytic limiting convention, first note that $\mathcal{R}$ is supported in $[\log 2, \infty)$. On $[0, T]$, only $r \leq T/\log 2$ can contribute to the displayed exponential, and each convolution term has finitely many contributing point masses. Hence the expression is a well-defined locally finite measure.

For a fixed prime, use a formal variable X . The formal power series $f(X) = \sum_{k \geq 1} X^k/k$ has derivative $1/(1-X)$. If $H(X) = \exp f(X) = \sum_{j \geq 0} h_j X^j$, formal differentiation gives $(1-X)H' = H$, with $h_0 = 1$. Its coefficient equation is $h_1 = h_0$, and, for $j \geq 1$, $(j+1)h_{j+1} - jh_j = h_j$. Thus $h_{j+1} = h_j$, so every $h_j = 1$. Substituting $X^k \mapsto \delta_{k \log p}$ respects multiplication as convolution on bounded time intervals. Therefore

$$\exp_{\ast} \left(\sum_{k \geq 1} \frac{1}{k} \delta_{k \log p} \right) = \mathcal{D}_p.$$

For finitely many primes, the convolution binomial identity gives $\exp_{\ast}(A+B) = \exp_{\ast}(A) \ast \exp_{\ast}(B)$: expand the r -fold convolution and count the $\binom{r}{j}$ choices of the j factors from A , whose coefficient becomes $1/(j!(r-j)!)$. All sums relevant on a bounded interval are finite. Thus the convolution exponential of the finite-prime repetition measure is $\mathcal{D}_F$.

On $[0, T]$, primes greater than e^T contribute no positive-time atom; their local counting factor contributes only δ_0 . Taking F to contain the finitely many primes at most e^T makes both sides of the claimed boxed identity equal on that interval. Since T is arbitrary, the identity follows globally.

This proves the precise reconstruction: primitive prime labels, every repetition, the coefficients $1/k$, and convolution produce all composite counts with exactly their required multiplicities. The unit atom cannot be omitted from the convolution exponential.

TP11. The finite record and the whole record

Let F_N be the activated prime labels through tick N . TP3 proves $F_N = \{p : p \leq N\}$. These sets increase, and their union is the set of all primes. For the corresponding measures retain

$$\mathcal{R}_N = \sum_{p \leq N} \sum_{k \geq 1} \frac{1}{k} \delta_{k \log p}, \quad \mathcal{W}_N = \sum_{p \leq N} \sum_{k \geq 1} (\log p) \delta_{k \log p}, \quad \mathcal{D}_N = \ast_{p \leq N} \mathcal{D}_p.$$

For every $T \geq 0$, all three measures agree with their full counterparts on $[0, T]$ once $N \geq e^T$. For $\mathcal{R}$ and $\mathcal{W}$, this follows from $p^k \leq e^T \Rightarrow p \leq e^T$. For $\mathcal{D}$, every $n \leq e^T$ has all prime divisors at most e^T , so TP10 supplies precisely its mass. Thus the infinite measure is obtained as an exact compatible locally finite limit; each bounded time window is eventually determined exactly.

No finite N contains the whole record, by infinitude of primes. Conversely, the finite record already determines these exact restrictions. The reconstructed $\mathcal{D}_N$ also contains counts larger than N , such as arbitrarily high powers of two after tick two: this is the consequence of the retained multiplication rule, not a claim that those ticks have already occurred in the finite sequential history.

TP12. The original zeta function is the transform of the full count measure

Let $s \in \mathbb{C}$ with $\sigma = \operatorname{Re} s > 1$, and use real logarithms of positive integers. The original Riemann zeta function on this domain is

$$\zeta(s) = \sum_{n \geq 1} n^{-s} = \int_{[0, \infty)} e^{-st} d\mathcal{D}(t) = 1 + \sum_{n \geq 2} e^{-s \log n}.$$

The series is absolutely convergent: the positive decreasing function $x^{-\sigma}$ gives $\sum_{n \geq 2} n^{-\sigma} \leq \int_1^\infty x^{-\sigma} dx = 1/(\sigma - 1)$. On a half-plane $\sigma \geq 1 + \delta$, with $\delta > 0$, the fixed summable majorant $n^{-1-\delta}$ also gives uniform convergence.

For finite F_N , the geometric series and TP10 yield

$$Z_N(s) := \prod_{p \leq N} (1 - p^{-s})^{-1} = \sum_{\substack{n \geq 1 \\ p|n \Rightarrow p \leq N}} n^{-s} = \int e^{-st} d\mathcal{D}_N(t).$$

Indeed, $|p^{-s}| < 1$; each local geometric series converges absolutely. The finite product of these absolutely convergent series may be expanded term by term, and uniqueness of factorization identifies each term exactly once. Every integer omitted from the finite product has a prime divisor exceeding N , and therefore itself exceeds N . Consequently, on $\sigma \geq 1 + \delta$,

$$|\zeta(s) - Z_N(s)| \leq \sum_{n > N} n^{-1-\delta} \rightarrow 0.$$

This proves the full Euler product on its stated original domain:

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}, \quad \text{Re } s > 1.$$

The displayed finite products and their proof contain the prime-two factor $(1 - 2^{-s})^{-1}$, the exponent-zero term at every prime, and the global unit term one. No completion multiplier, Gamma factor, or boundary term is suppressed: none has been introduced in this calculation of the original Dirichlet series on its domain of absolute convergence.

TP13. The repetition weights give the exact logarithm and logarithmic derivative

For $\text{Re } s > 1$, define

$$L(s) = \int e^{-st} d\mathcal{R}(t) = \sum_p \sum_{k \geq 1} \frac{p^{-ks}}{k}.$$

Both this series and its once-differentiated series converge absolutely and locally uniformly. To check this explicitly on $\sigma \geq 1 + \delta$,

$$\sum_p \sum_{k \geq 1} \frac{p^{-k\sigma}}{k} \leq \frac{1}{1 - 2^{-1-\delta}} \sum_{n \geq 2} n^{-1-\delta} < \infty,$$

and

$$\sum_p \sum_{k \geq 1} (\log p) p^{-k\sigma} \leq \frac{1}{1 - 2^{-1-\delta}} \sum_{n \geq 2} (\log n) n^{-1-\delta} < \infty.$$

The latter numerical series converges because, for sufficiently large x , $\log x \leq x^{\delta/2}$: put $y = \log x$ and use $e^{(\delta/2)y} \geq ((\delta/2)y)^2/2$, whose ratio to y tends to infinity. Its tail is therefore bounded by the convergent series $\sum n^{-1-\delta/2}$. These bounds establish the stated uniform convergence and justify termwise differentiation on every compact subset of the half-plane.

For $|z| < 1$, the same coefficient calculation as TP10, now in absolutely convergent analytic series, gives

$$\exp \left(\sum_{k \geq 1} \frac{z^k}{k} \right) = (1 - z)^{-1}.$$

For a finite prime set this gives $e^{L_N(s)} = Z_N(s)$. Passing to the locally uniform limits from TP12 and the preceding convergence yields

$$e^{L(s)} = \zeta(s), \quad \text{Re } s > 1.$$

Thus L is the analytic logarithm of the original zeta function on this half-plane, selected by its convergent series and by $L(x) \rightarrow 0$ as real $x \rightarrow +\infty$. That last limit follows by domination by the convergent

series at $x = 2$, whose individual terms tend to zero. In particular ζ has no zero on this half-plane, since an exponential is nonzero. Differentiating the exponential identity and retaining the full factor from each repeated return gives

$$\begin{aligned}
-\frac{\zeta'(s)}{\zeta(s)} &= -L'(s) \\
&= \sum_p \sum_{k \geq 1} \frac{k \log p}{k} p^{-ks} \\
&= \sum_p \sum_{k \geq 1} (\log p) p^{-ks} \\
&= \int e^{-st} d\mathcal{W}(t) \\
&= \sum_{n \geq 2} \frac{\Lambda(n)}{n^s}.
\end{aligned}$$

The second line displays where the repetition number k from differentiating the exponential and the original coefficient $1/k$ enter. The subsequent equality computes their product; the original contributions remain visible. There is no coefficient one replacing $\log p$, and the repeated return at p^k has coefficient $\log p$, not $k \log p$.

The original integer one contributes the atom δ_0 and the constant term one to ζ . Its derivative is zero, and the Dirichlet coefficient at one in $-\zeta'/\zeta$ is $\Lambda(1) = 0$. The denominator of that quotient remains the full ζ , including its constant term; dropping the unit there would change the function. Integer one has not been identified with the supporting datum. Zero and negative logarithmic return times still exist in the full flow of TP7. The positive-time series here comes from the original expansion $(1 - p^{-s})^{-1} = \sum_{k \geq 0} p^{-ks}$ on $\operatorname{Re} s > 1$. Negative repetitions correspond to reciprocal powers outside this positive-integer Dirichlet series, while the zero exponent is its retained unit term. The logarithmic series has no constant term because the logarithm of that local unit constant is zero.

TP14. What this contributes to the current weight calculation

The following maps and weights have now been proved, with no numerical scan or purity assumption:

$$\begin{aligned}
&\text{unit-anchored waits } (g_j) \longleftrightarrow p_j = 1 + \sum_{r \leq j} g_r \longleftrightarrow \lambda_{p_j} = \log p_j, \\
&\text{primitive curve } E_p \longrightarrow \{k \log p : k \in \mathbb{Z}\}, \\
&\mathcal{R} = \sum_{p, k \geq 1} \frac{1}{k} \delta_{k \log p}, \quad \mathcal{D} = \exp_* \mathcal{R}, \quad \mathcal{W} = t\mathcal{R}, \\
&\mathcal{L}\mathcal{D} = \zeta, \quad \mathcal{L}\mathcal{R} = \log \zeta, \quad \mathcal{L}\mathcal{W} = -\zeta'/\zeta \quad (\operatorname{Re} s > 1).
\end{aligned}$$

Here $\mathcal{L}$ is exactly the displayed Laplace integral, and $\log \zeta$ is exactly the analytic branch constructed in TP13. The nonnegative coefficients of these measures are explicitly established. Positivity of these particular time measures is not silently substituted for a Weil quadratic-form inequality, and the displayed transform identities are not asserted outside the half-plane in which they were proved.

The numerical period $\log p$ is now determined by the timed prime event, and its occurrence as the coefficient of every prime-power term in $-\zeta'/\zeta$ is derived. That is the exact datum supplied to the separate arithmetic cohomology and weight calculation: the event-time record gives the source curve

and all its repeated periods, rather than a common-radius assumption or an imposed distance from the supporting datum.

The source-derived prime clocks with all prime powers

Proof source: PRIME_CLOCKS_ORCHARD_AND_SPECTRUM.md.

Prime clocks, orchard visibility, and faithful prime observations

24 September 2026. Private continuation. Proofs P1-P14. This receives the user's proposal that prime spectral appearances should recover all other counts. The word torsion in that proposal means retained winding. Finite-order groups below are identified explicitly. No arithmetic operation, coordinate or distance is assigned to tau. No RH conclusion or identification of these character values with arithmetic Frobenius eigenvalues is asserted.

P1. Integer winding from the actual oriented-ray covering

Connes-Consani, arXiv:2606.06604v1, Corollary luriearch, gives the actual covering

$$P = \mathbb{C}^\times / \mathbb{R}_{>0} \longrightarrow B = \mathbb{C}^\times / \mathbb{R}^\times.$$

Original FF.tex lines 1521-1580 were reread for this calculation. The source complex scalar origin is not identified with the user's tau, and the reference line chosen below is not the generic point eta of the other source paper.

Write a nonzero complex number as $r \exp(i \theta)$, with $r > 0$. The two quotient maps identify respectively angles differing by 2π and by π . Thus the maps

$$\mathbb{R} \longrightarrow P, \quad \theta \longmapsto [e^{i\theta}]_+, \quad \mathbb{R} \longrightarrow B, \quad \theta \longmapsto [e^{i\theta}]_{\mathbb{R}^\times}$$

retain periods 2π and π respectively. These are coordinates on the source's ray and line spaces, not coordinates of tau. Intervals of length less than π project injectively to B; their translates by integer multiples of π give its covering charts.

Fix the reference line $\ell_0 = [1]$. A continuous loop in B based there has a unique continuous real angle lift starting at zero. To construct it, subdivide the compact parameter interval so each successive piece lies in one covering chart and choose the next branch to match the preceding endpoint; branch uniqueness follows because two lifts differ by an integer multiple of π , locally constant and zero initially. The endpoint is $m\pi$ for a unique integer m . A based homotopy preserves m : lift the homotopy by the same overlapping-chart construction on a finite rectangular subdivision of its compact parameter square; the lifted terminal value is continuous and lies in the discrete set $\pi \mathbb{Z}$, so it is constant.

Every m is attained by the loop $u \mapsto \exp(i m \pi u)$ modulo real scaling. Two lifted loops with the same endpoints are homotopic relative endpoints by linear interpolation of their real angle lifts; projecting gives a based homotopy. Concatenation adds endpoint angle displacements. Therefore

$$\pi_1(B, \ell_0) \cong \mathbb{Z}, \quad [\gamma] \longmapsto \frac{\tilde{\gamma}(1)}{\pi}.$$

The displayed division retains the original π period. Lifting further to P shows that the ray returns to itself precisely for even m ; for odd m it returns to the opposite ray. Thus the actual covering's monodromy is $m \mapsto m \bmod 2$. A full turn has angle 2π and count two; it is not identified with zero count.

This derives integer winding from the source covering geometry rather than only naming a stalk $T^{\wedge}Z$. It still uses that covering geometry; its existence is not derived from a bare point alone. Reflection of source angles $\theta \rightarrow -\theta$ reverses m and preserves its parity.

In the other source, arXiv:2609.00299v1, keep the full generic group

$$G = \langle T, J \mid TJ = JT, J^4 = 1 \rangle, quad \epsilon = J^2, \quad A(T) = \epsilon T^{-1}, \quad A(J) = J^{-1}.$$

Keep its finite-order subgroup $H =$ and the map $q: G \rightarrow L = G/H$. Set $t = q(T)$. The map from the just-computed fundamental group to L sends a loop of lifted angle $m\pi$ to t^m . It is a group isomorphism by the two explicit infinite cyclic descriptions. It sends ray-return parity to the source-derived observation $\delta(t^m) = \epsilon^m$. It also intertwines angle reflection with the induced action $t^m \rightarrow t^{-m}$. On G itself the sign ϵ^m remains in $A(T^m)$; the isomorphism to L does not remove the full extension from the construction. Reversing the chosen winding orientation replaces both generators by their inverses.

P2. The prime clocks and the exact meaning of irreducible

For each $n \geq 1$, form $C_n = L/[n]L$, where $n = \text{ell}^n$ and $[n]L$ is the subgroup of n -th powers. These are finite winding-state groups at the same supporting point η . Choosing t writes the n distinct states as the classes of $1, t, \dots, t^{n-1}$. The step S_n multiplies by the class of t . It visits all n states and has period n .

Every subgroup of L is $\langle t^d \rangle$ for a unique $d \geq 0$, where $d=0$ denotes the trivial subgroup. For a nontrivial subgroup, choose the least positive exponent d present. Division with remainder shows every exponent in the subgroup is divisible by d ; conversely all multiples of d occur. Hence the subgroups of C_n are precisely

$$\langle t^d \rangle / \langle t^n \rangle, \quad d \mid n,$$

with n/d elements, and quotient C_d . For $n \geq 2$ this proves

n prime $\iff C_n$ is a nontrivial simple cyclic group $\iff C_n$ has no quotient with size strictly between 1 and n .

This is the precise irreducibility condition. Transitivity alone does not detect primes, since every C_n is transitive under S_n .

An equivariant clock map $f: C_m \rightarrow C_n$ satisfies $f(S_m x) = S_n f(x)$. Consequently $f(t^k) = t^k f(1)$. It is well defined exactly when $t^m = 1$ in C_n , that is n divides m . When this holds every choice of $f(1)$ gives one map; all are surjective, and exactly one preserves the initial state. Thus a pointed clock isomorphism $C_m \rightarrow C_n$ exists exactly when $m = n$. In particular the prime clocks yield pairwise non-isomorphic objects over the same η , but composites also yield distinguishable objects. Simplicity, not distinguishability alone, separates primes.

The original sign and branch data remain in G . The full quotient map $G \rightarrow C_n$ has kernel

$$K_n = \langle T^n, J \rangle, \quad 1 \rightarrow K_n \rightarrow G \rightarrow C_n \rightarrow 1.$$

Both J and ϵ are retained in that exact sequence. Taking this clock observation does not replace the full source by its quotient.

P3. Prime degree as indecomposable winding scale

The map $[n]: L \rightarrow L$ has image $[n]L$ and index n . Every group endomorphism of L is one of the power maps $[a]$, because its image on t determines its values on every element. The composition law is $[a][b] = [ab]$. Therefore a positive degree $n > 1$ is prime precisely when $[n]$ cannot factor as $[a][b]$ with $a, b > 1$.

On the character circle $\text{Hom}(L, S^1)$, precomposition by $[n]$ sends $z = \chi(t)$ to z^n . It is an n -sheeted cover: every point has n distinct roots and a sufficiently short target arc has n disjoint continuous root

branches. Any factorization into connected finite covers of degrees a, b has $n=ab$ by counting the inverse images of one point. Conversely $n=ab$ supplies the explicit factorization $z \rightarrow z^{b \rightarrow z}(ab)$. This proves the same prime criterion for covering degree. Degree one is invertible and degree zero is a constant map, not a finite-sheeted cover.

Scaling $[n]$ and unit winding S are different source maps: $n=t^{(nm)}$, whereas $S(t^m)=t^{(m+1)}$. Their relation to the count is explicit and neither is an operation on η or τ .

P4. Euclid's orchard and the exact clock observation

The orchard uses the arithmetic count lattice $\mathbb{Z}^2$. Its lattice zero is not τ ; its coordinates are counts already recovered in P1. A nonzero pair (a, b) is visible when no lattice point lies on the open segment from the lattice zero to (a, b) .

Put $d=\gcd(|a|, |b|)$. If $d>1$ then $(a/d, b/d)$ is an interior lattice point. If $d=1$, Bezout gives integers r, s with $ra+sb=1$. Any interior lattice point $\lambda(a, b)$, $0<\lambda<1$, would satisfy $\lambda a=r(\lambda a)+s(\lambda b)$, hence λ would be an integer, a contradiction. Thus visibility is equivalent to $\gcd(|a|, |b|)=1$.

In $\mathbb{C}_n$ the state t^a has order $n/\gcd(a, n)$: writing $a=da'$, $n=dn'$ with coprime a', n' , the return condition n divides ka is equivalent to n' divides k . Therefore, for $1 \leq a < n$,

$$(a, n) \text{ visible} \iff t^a \text{ generates } C_n \iff \text{step } a \text{ visits all } n \text{ states.}$$

The number of such a is $\phi(n)$. Every interior point of row n is visible exactly when n is prime: for a prime all $1 \leq a < n$ are coprime; for a composite n a divisor $1 < d < n$ yields an invisible (d, n) . A single visible vector need not represent a prime: $(1, 4)$ is visible. The complete-row property is the prime detector.

P5. Reconstructing positive integers from prime divisors

Every integer $n>1$ has a prime divisor: its least divisor greater than one is prime, since any nontrivial factor of it would be a smaller divisor. Repeated division terminates because the positive quotient strictly decreases. Thus n is a finite product of primes. If a prime p divides ab and does not divide a , Bezout gives $ua+vp=1$ and multiplication by b proves p divides b . Applying this fact repeatedly to two prime factorizations matches a prime from one with a prime from the other, cancels them, and proves uniqueness by induction on the total number of factors.

Consequently the valuation map is an isomorphism of multiplicative monoids

$$\mathbb{Z}_{>0} \longrightarrow \bigoplus_{p \text{ prime}} \mathbb{Z}_{\geq 0}, \quad n \longmapsto (v_p(n))_p, \quad n = \prod_p p^{v_p(n)},$$

where multiplication on the left corresponds to componentwise addition on the right. The finite product retains every multiplicity. Integer one corresponds to the empty product. Integer zero is not represented by a finite prime product; negative integers require the additional sign. Nothing here identifies any of these with η or τ .

Observing only the support $\{p: p \text{ divides } n\}$ loses multiplicities, as 2 and 4 demonstrate. Observing the full valuations does not. The full integer ring recovered as $\text{End}(L)$ also retains addition. Prime factorization alone describes its positive multiplicative monoid and must not silently substitute for the ring structure.

Geometrically, $\text{Spec } \text{End}(L)$ has one closed point for each positive prime and also the generic zero prime. Composite n is not a new prime point; its arithmetic is retained in the quotient $\text{End}(L)/n \text{ End}(L)$, whose prime-power factors and nilpotent elements are not discarded. This uses the ring and prime-ideal classification already proved in W3-W4, rather than declaring every distinct clock a new prime.

P6. All prime-clock readings distinguish every winding count

There is a second, stronger meaning of prime observation than just divisor support. Define

$$\mathcal{R} : L \longrightarrow \prod_{p \text{ prime}} C_p, \quad \mathcal{R}(\ell) = (\ell \bmod [p]L)_p.$$

This homomorphism is injective. If two states t^m and t^n have equal images, every prime divides $m-n$. A nonzero integer has only finitely many positive divisors, whereas there are infinitely many primes: given any finite list, a prime divisor of their product plus one is absent from the list. Hence $m-n=0$.

For a finite set P of primes the corresponding kernel is $[N]L$ with $N=\text{product over } p \text{ in } P \text{ of } p$. Indeed divisibility by the pairwise coprime primes is equivalent to divisibility by their product, by the Bezout argument above. Thus finite prime observations leave exactly the stated ambiguity; the all-prime map separates all integer states. The proof retains prime 2.

This map retains the full residue at each prime, not merely whether that residue is zero. It therefore can distinguish powers such as 2 and 4 even without separately reporting their valuations. No surjectivity onto the infinite product is asserted, and no completion is identified with L .

P7. Prime character values fill the circle densely without exhausting it

The character group of C_p has the values

$$\mu_p = \{\exp(2\pi i a/p) : a = 0, \dots, p-1\}.$$

Indeed a character is determined by its value on t , and that value must have p -th power one. Conversely every such value defines a character by $t^m \rightarrow \exp(2\pi i a m/p)$. The full 2π factor and the residues a are retained.

Let Γ be the union of μ_p over all primes. It is countable, being a countable union of finite sets. It is dense in S^1 : the p -th roots divide the angle into p arcs of length $2\pi/p$, and unbounded primes make that mesh smaller than the length of any specified open arc. It is proper: i has order four, which does not divide any prime, so i is absent. For distinct primes p, q the intersection $\mu_p \cap \mu_q$ is $\{1\}$; an element in both has order dividing $\gcd(p, q)=1$.

Thus each prime adds $p-1$ new character values. This is a circle of character values, not an assertion that a prime is itself a complex point or that every visible orchard ray is prime. One arbitrarily chosen point per prime would be a different construction: the choices $\exp(2\pi i/p)$ accumulate only at one and do not give this dense family.

P8. Preserve the full signed source under prime observation

Γ alone is not invariant under all original source maps. In particular A sends the spatial character coordinate z to $-z^{(-1)}$, which can have order $2p$. Arbitrary sign-preserving commuting source lifts send (z, j) to $(z^n j^{c_j} b)$, with n odd, c modulo four, and b odd. The quartic branch cannot be dropped.

For every prime p retain the finite quotient

$$G_p^{\text{full}} = G / \langle T^{4p} \rangle, \quad |G_p^{\text{full}}| = 16p,$$

with its original J , ϵ and A . The subgroup is invariant under A since $A(T^{(4p)})=T^{(-4p)}$, and under every source lift $f_-(n, c, b)$ since $f(T^{(4p)})=T^{(4pn)}J^{(4pc)}=T^{(4pn)}$. Prime 2 is retained, including this quotient's order-eight spatial factor. The exponent $4p$ is a uniform choice; no claim of minimality for $p=2$ is made.

Its signed characters are exactly (z,j) with $z^{4p}=1$ and j in $\{i,-i\}$, because ϵ must evaluate to -1 and $J^2=\epsilon$. Define the full domain

$$\mathcal{D} = \bigcup_{p \text{ prime}} \{(z,j) : z^{4p} = 1, j \in \{i, -i\}\}.$$

For $g=T^m j^k$ retain the evaluation

$$E_{\mathcal{D}}(g)(z,j) = z^m j^k, \quad E_{\mathcal{D}}(0) = 0.$$

It preserves multiplication and the absorbing element. Every nonabsorbing value has modulus one: both z and j are roots of unity. To prove injectivity suppose the functions for (m,k) and (m',k') agree. Taking $z=1$ and $j=i$ gives $k=k'$ modulo four. If $d=m-m'$ is nonzero, choose a prime p not dividing d and set $z=\exp(2\pi i/p)$, which belongs to μ_{4p} . Then z^d is not one, a contradiction. Hence $m=m'$, and the zero function is distinguished from each nonabsorbing function. All full source data are therefore retained.

The original source mirror acts on D by $(z,j) \rightarrow (z^{-1}, j)$. It stays in the same p -level since $(-z^{-1})^{4p}=1$. Each original commuting signed lift acts by $(z,j) \rightarrow (z^n j^{c_j} b)$; its spatial coordinate has $4p$ -th power one and its branch remains in $\{i,-i\}$. Exact equivariance is

$$E_{\mathcal{D}}(f(g))(\chi) = \chi(f(g)) = E_{\mathcal{D}}(g)(\chi \circ f).$$

No sign, branch, or source factor was suppressed in this comparison.

The spatial values $\bigcup_p \mu_{4p}$ are still countable and dense, since they contain Γ . They are not the full circle: a primitive ninth root has order nine, and 9 does not divide $4p$ for any prime p . Thus the full signed construction preserves the user's dense-but-not-exhaustive circle picture while retaining all source symmetries and still separating every integer count.

The common modulus is proved for characters of these actual finite quotients. Deligne purity concerns arithmetic Frobenius eigenvalues and retains the factor $N(x)^{(w/2)}$, as derived in D1-D5 of DELIGNE_COMMON_MODULUS_AND_WINDING.md. No map identifying those eigenvalues with E_D is established here. The faithful prime-character construction is an exact result on the source data, not an RH conclusion.

Provenance and scope

Human sources: Alain Connes and Caterina Consani, arXiv:2606.06604v1, original FF.tex Corollary luriearch, lines 1521-1580; and arXiv:2609.00299v1, original CC.tex Sections 3-4, with the exact generic group and sheaf action read in W1-W10. The prime-clock, divisor, orchard, residue and character arguments are completely derived above; no novelty claim is made. An independent mathematical agent derived P2-P4 and reviewed the all-prime separation and character construction. The first paper's original parameter $\tau=i \log(p)/(2\pi)$ is its modular parameter, not the user's τ ; it is not used to identify a metric on the user's object.

The controlling user definitions and corrections are retained verbatim in ../TAU_DEFINITIONS_AND_CORRECTIONS_VERBATIM.md. These results do not show that existence of a point forces the source covering or Laurent direction. They derive the integers from the actual oriented-ray covering and then prove precisely which observations characterize primes and recover integer counts.

P9. Sequential first appearances: the exact absorption sieve

This is the user's revised observation: a composite need not produce a new line. The observer records a channel at its first activation and sees later returns in that same channel. The earlier comparison of the divisor supports of 2 and 4 did not refute this proposed first-appearance observation; it concerned a different observation of an individual number.

Take the source winding progression from the chosen ordering, with one elementary step $S(t^n)=t^{n+1}$. The state at zero winding is the initial identity in L , not η or the absorbing monoid zero. Applying one step n times visits every state with count $0,1,\dots,n$ in order. A composite instruction S^r abbreviates those r successive steps; it does not delete their intermediate states. Each fixed finite prime is reached after finitely many steps, while there is no finite step at which all primes have been reached.

The periodic channel labelled $d \geq 2$ is the quotient clock C_d . At time n its state is t^n modulo $\langle t^d \rangle$, and it returns to its initial state exactly when d divides n . Thus an earlier count covers a later count through this divisibility relation, not through earlier occurrence alone: 2 covers 4, but 3 does not cover 4.

Define the sequential sieve without presupposing the list of primes. Initially there are no active channels. At each integer time $n \geq 2$, let B_n be the set of previously activated d with d dividing n . If B_n is nonempty, create no new channel. If B_n is empty, activate channel n ; it registers its initial return now and every subsequent multiple. This rule refers only to the retained progression and its quotient-clock return tests.

The activated counts are exactly the primes. Prove this by induction on n . At $n=2$ there is no earlier channel and 2 is prime. Suppose the activated counts below n are precisely the primes below n . If n is prime, it has no proper divisor d with $2 \leq d < n$, so B_n is empty and it activates. If n is composite, its least divisor $p > 1$ is prime and satisfies $p < n$; by the induction hypothesis its channel is already active, and p divides n , so n is covered and does not activate. This proves the assertion at every finite time.

Equivalently the exact first-arrival indicator is

$$A(n) = \prod_{\substack{p < n \\ p \text{ prime}}} (1 - \mathbf{1}_{p|n}), \quad n \geq 2.$$

The product is finite and retains all overlapping channels. It equals one precisely for a prime, and zero for a composite. The recurrence preceding the formula computes the same active set without requiring primes as prior input.

For example the successive events are: 2 activates; 3 activates; 4 is covered by 2; 5 activates; 6 is covered by both 2 and 3; 7 activates; 8 is covered by 2; 9 is covered by 3. Integer 1 is the multiplicative unit, with a one-state quotient clock and no nontrivial prime channel. It is not identified with the supporting point η or τ .

This proves why first appearances detect exactly primes and why suppressing repeated appearances does not delete the underlying composite counts. Reconstructing composites retains the channel-composition operation and its multiplicities, as proved in P5; the activation list alone is not asserted to be a record of every state at every time. The proposition derives prime events from the source winding progression and periodic return maps. It does not yet derive that progression from an unstructured absence/presence datum, or identify this binary sieve with the analytic absorption spectrum in Connes's trace formula.

The companion interactive incidence diagram implements this recurrence for counts 1 through 48. Its equal-height channel marks mean that a channel is present; they assert no spectral weight, geometric radius from τ , or purity theorem. A finite display illustrates the proved all- n statement and is not evidence substituting for its induction proof.

P10. Finite histories and the inductive whole

The user further proposes that ordered winding history is essential and that the whole collection of counts is obtained only through indefinite continuation. The exact construction retains the finite stages rather than positing a last infinite-time event.

In the actual source winding group take $L_+ = \{1, t, t^2, \dots\}$, selected by the chosen ordering and generated by one positive winding step. For each finite n let $H_n = \{1, t, \dots, t^n\}$, with the order inherited from their exponents and inclusion $H_n \rightarrow H_{n+1}$. The distinctness of these states follows from P1's winding

calculation, not from assigning new labels to the same parity value. Every finite history is a prefix of every later history of the same deterministic run.

The union $H_{\infty} = \bigcup_{n \geq 0} H_n$ is exactly L_+ . It has the following universal property. Given maps $f_n: H_n \rightarrow Y$ agreeing on successive overlaps, define $f(t^{(k)} = f_n(t_k))$ for any $n \geq k$. Compatibility makes this independent of the chosen n , and this defines the unique map extending all f_n . Thus H_{∞} is the direct limit of the finite histories. The already proved exponent correspondence identifies it with N including zero. Each individual count k belongs to H_k ; no count first appears at an additional infinite-time stage. There is no largest finite H_n , since $t^{(n+1)}$ is absent from H_n .

The winding step maps each history state to its successor. A composed instruction S^r represents r such steps; its mathematical prefixes remain $S, S^2, \dots, S^r$. Sequential generation does not require a storage device to retain a separate copy of every frame: the initial state, known transition rule and full count uniquely determine those prefixes. It does require retaining enough information to distinguish t^n from $t^{(n+2)}$; parity alone cannot do that.

The sieve states of P9 are compatible under these same inclusions. At stage n every count from 2 through n has either activated a prime channel or returned through one already active. The union of all prime channels covers exactly the positive integers ≥ 2 : each has a least prime divisor, while the unit 1 has none. Every individual count is processed at a finite stage, and every individual prime activates at a finite stage. No finite stage contains all primes, by Euclid's proof in P6.

This gives a precise sense of internal sequential order. It does not identify the step parameter with physical time, posit a Hamiltonian, or establish that no other faithful representation of these histories exists. P6 and P8 explicitly construct other faithful encodings of the same count. The present theorem describes the generation by one winding step and its compatible finite observations.

P11. The exact uniform infinite-observation statement

Use the full prime-clock readings, not only the binary sieve-event indicator. For a finite nonempty prime set P put $M_P = \text{product}_{p \in P} p$. P6 proves that the fiber over the readings of an integer n is $n + M_P \mathbb{Z}$. It is infinite. No finite set of those readings isolates n among all integers. Every infinite set of distinct primes does isolate it, since a nonzero difference has finitely many prime divisors. Thus the minimum cardinality of a family of prime residue observations isolating any given n is $\aleph_0$, the same for all n .

The quantifiers are important: for every finite P there is some n_P different from n with the same readings, but there is no single n' different from n that has the same readings at every prime. This is an exact global-observation property. It does not prevent n from being specified in another representation, such as a finite successor history.

For the full signed source G and the character family D of P8, equality of observations at the finitely many p in P has kernel

$$\{T^{4M_P r} : r \in \mathbb{Z}\}.$$

Proof: equality with the identity at $z=1, j=i$ forces $k=0$ modulo four. Equality for every z in $\mu_{(4p)}$ forces $4p$ to divide m for each p in P . The least common multiple of these $4p$ is $4M_P$: every odd prime in P occurs once, and the power of two is four if 2 is absent and eight if 2 is present. Conversely divisibility by $4M_P$ makes all the stated evaluations one. Thus every fiber remains infinite at a finite observation stage, while the full family separates G by P8. This statement retains the source's four branch states and the extra two-adic factor in the chosen uniform family.

The equal $\aleph_0$ value is an observation-complexity invariant relative to this family of finite clocks. It is not the definition of Deligne weight, an eigenvalue modulus, or a metric on τ . The following construction gives an actual common-modulus theorem from the global clock completion.

P12. A compact global completion and its exact modulus theorem

For each finite nonempty prime set P retain the quotient

$$G_P = G / \langle T^{4M_P} \rangle \cong (\mathbb{Z} / (4M_P)\mathbb{Z}) \times (\mathbb{Z} / 4\mathbb{Z}).$$

The isomorphism sends $T^m k$ to $(m \text{ modulo } 4M_P, k \text{ modulo four})$, retaining both coordinates. For P contained in Q the transition $G_Q \rightarrow G_P$ reduces the first coordinate and leaves the second unchanged. All these finite groups have their discrete topologies. Define

$$K = \varprojlim_P G_P$$

as the space of compatible families, with the subspace topology from their product. The constraints of compatibility are closed, so K is a closed subspace of a product of finite compact spaces and is compact. Equivalently choose the cofinal sequence of sets of the first N primes; the diagonal subsequence argument proves compactness of the resulting countable product, and the compatibility equations retain a closed subset. Multiplication and inversion are coordinatewise and continuous.

The map $i: G \rightarrow K$ sends a source element to all its finite residue classes. It is injective by P11: all residues equal identity force $k=0$ modulo four and every $4M_P$ to divide m , hence $m=0$. It has dense image. A basic open condition specifies only finitely many quotient coordinates. Their union Q is a finite prime set; a compatible value in G_Q has a representative $T^m k$ in G , whose image satisfies all those conditions. Thus every nonempty basic open set meets $i(G)$. This is a faithful embedding with retained finite observations, not an identification of G with its completion.

The cofinal sets P containing 2 give the more explicit full description

$$K \cong (\mathbb{Z} / 8\mathbb{Z}) \times \prod_{p \text{ odd prime}} \mathbb{Z} / p\mathbb{Z} \times (\mathbb{Z} / 4\mathbb{Z}).$$

To verify it, use $4M_P=8$ times the product of the odd primes in P . The reduction map modulo these pairwise coprime factors is injective because an integer divisible by each is divisible by their product; it is surjective by Bezout's identity, constructing a representative for each finite list of residues. These finite isomorphisms commute with every transition, so taking compatible families gives the displayed isomorphism. The final factor remains the source J -branch. The first factor comes from the chosen uniform $4p$ observation family, including $p=2$; it is not declared a minimal completion.

All original source symmetries used above extend continuously. At a finite level, the source mirror is

$$A_P(m, k) = (-m, 2m - k),$$

where the first coordinate is modulo $4M_P$ and the second modulo four. This is well defined because 4 divides $4M_P$. It is a homomorphism of order two and it commutes with reductions. Hence it induces A_K on K . Each sign-preserving commuting lift has finite formula

$$f_{n,c,b;P}(m, k) = (nm, cm + bk), \quad n, b \text{ odd}, \quad c \pmod{4},$$

with the same respective moduli. This is well defined, continuous and compatible with transitions, so it extends to K as well. No source sign is removed by the extension.

Now let $\chi: K \rightarrow \mathbb{C}^\times$ be a continuous group character. Its image is a compact subgroup of $\mathbb{C}^\times$ because K is compact. The modulus image is therefore a compact subgroup of $\mathbb{R}_{>0}$. Such a subgroup is $\{1\}$: if it contains $r > 1$ then the powers r^n are unbounded, contrary to compactness; if it contains $0 < r < 1$ then it contains $r^{-1} > 1$. Consequently

$$|\chi(x)| = 1 \quad \text{for every } x \in K.$$

This is the common-modulus theorem for the constructed global clock completion. It follows from compactness and the multiplicative character law. It neither measures a distance from η nor assigns infinite modulus to source elements.

The continuous characters can also be classified without an extra hypothesis. Consider the open arc $V = \{\exp(i\theta) : |\theta| < \pi/3\}$. It contains no nontrivial subgroup of S^1 . Indeed, if a nonidentity value already lies outside V it is excluded; if it is $\exp(i\theta)$ with $0 < |\theta| < \pi/3$, the power with exponent $\text{ceil}(\pi/(3|\theta|))$ has an angle between $\pi/3$ and $2\pi/3$ in absolute value and lies outside V . Continuity gives an identity neighborhood in K whose image lies in V . Some kernel N_P of $K \rightarrow G_P$ is contained in that neighborhood, since finite quotient conditions form a neighborhood basis. Its character image is a subgroup contained in V , hence trivial. Thus χ factors through G_P .

Conversely every character of a finite G_P pulls back to a continuous character of K . The source signed condition $\chi(\epsilon) = -1$ requires $j = \chi(j)$ in $\{i, -i\}$. Its restriction to G therefore has exactly the original formula

$$\chi(T^m J^k) = z^m j^k, \quad z^{4M_P} = 1 \text{ for some finite } P, \quad j \in \{i, -i\}.$$

All original factors are retained. These signed characters separate K : at a finite quotient where two elements differ, their branch difference is detected with $z=1, j=i$ when that difference is nonzero modulo four; otherwise a character z detecting the distinct first residues, with $j=i$, separates them. Thus the global completion and every original winding state are faithfully observable by this continuous signed character family.

This construction also identifies the exact continuity obstruction for original source characters not in this family. The original character $\chi_{(2,i)}(T^m J^k) = 2^m i^k$ is valid on the uncompleted G . Along a cofinal sequence P_N , the source elements $T^{4M_P(N)}$ converge to identity in K , because every fixed finite quotient eventually sends them to identity. Their $\chi_{(2,i)}$ values are $2^{4M_P(N)}$, which do not converge to one. That character cannot extend continuously to K . The full original character space is retained in the record; the map of continuous characters of K into it is a proved inclusion, not a replacement obtained by rescaling values.

There are consequently two distinct proved statements: every original winding state requires infinitely many unrestricted prime-clock observations to isolate it in this observation topology; and every continuous character of the compact global clock completion has modulus one. The second follows through the explicit topology and group structure, not by renaming the first statement as purity. Deligne weight w is the arithmetic eigenvalue relation $|\iota \alpha| = N(x)^{w/2}$, with its full residue-norm factor. No identification of these clock characters with the required arithmetic cohomology has been supplied by the completion theorem.

P13. Retaining all stacked prime powers

The intermediate completion K in P12 is faithful on the original group G but is not the completion of all winding clocks. At an odd prime p its spatial factor is F_p . Multiplication by p , and multiplication by p^r for any $r \geq 1$, both have the same zero image on that factor and are invertible at every other prime factor. Their spatial image index is p in K , not p^r . Thus the intermediate observer does not preserve every covering-degree multiplicity after completion. This defect does not identify source integers, but it matters to the user's stacking operation.

Retain all finite clock quotients instead. For each positive integer N put

$$\widetilde{G}_N = G / \langle T^{4N} \rangle \cong \mathbb{Z} / (4N) \mathbb{Z} \times \mathbb{Z} / 4 \mathbb{Z}.$$

For N dividing M use reduction modulo $4N$ in the first coordinate and retain the branch coordinate. Define

$$\widehat{G} = \varprojlim_{N|M} \widetilde{G}_N.$$

The moduli $4N$ are cofinal among all positive integer moduli: every modulus d divides $4d$. A coherent family at moduli $4N$ therefore extends uniquely to all moduli by reduction, and restriction gives its inverse. Hence

$$\widehat{G} \cong \widehat{\mathbb{Z}} \times \mathbb{Z}/4\mathbb{Z}, \quad \widehat{\mathbb{Z}} = \varprojlim_N \mathbb{Z}/N\mathbb{Z}.$$

The space is compact by the same finite-product and closed-compatibility proof as P12. The source map $T^m k \rightarrow ((m \bmod N)_N, k)$ is injective: a nonzero integer m cannot be divisible by every positive N . Its image is dense because every basic finite compatible residue prescription reduces to one prescription modulo the least common multiple and has an integer representative. The absorbing zero is retained separately as an isolated absorbing point; it is neither the group identity nor the supporting eta.

Prime factorization and the same finite Bezout construction identify the spatial completion with $\prod_p \mathbb{Z}_p$, where $\mathbb{Z}_p = \varprojlim_r \mathbb{Z}/p^r \mathbb{Z}$. Every modulus involves finitely many prime powers; finite compatibility constructs its residue from those components. Conversely the residue family restricts to every prime-power chain. Thus all p -information, including the entire prime-two chain, is retained.

The original source involution and all the classified commuting signed maps extend continuously:

$$\widehat{A}(x, k) = (-x, 2(x \bmod 4) - k), \quad \widehat{f}_{n,c,b}(x, k) = (nx, c(x \bmod 4) + bk),$$

with n, b odd and second coordinate modulo four. These formulas descend at every modulus $4N$ and commute with the transition maps, so are well defined. They restrict to the original source formulas. The earlier completion K is a quotient of this one: retain x modulo eight, x modulo each odd prime, and k modulo four. This projection records exactly which higher prime-power coordinates that intermediate observer omitted; it does not replace the full completion.

For any positive n , multiplication by n on $\widehat{\mathbb{Z}}$ is injective and has image equal to the kernel of reduction modulo n . Here is a direct proof without an exactness assumption about inverse limits. If $nx=0$, inspect its coordinate modulo nm for an arbitrary positive m . A representative a of x modulo nm satisfies nm divides na , hence m divides a . Its reduction x modulo m is zero; all coordinates are therefore zero. Conversely, if x modulo n is zero, choose a representative a of x modulo nm divisible by n , and define y modulo m to be a/n modulo m . This is independent of representative, since changing a by nm changes a/n by m . Compatibility of the x coordinates makes these y coordinates compatible. They define y in $\widehat{\mathbb{Z}}$ with $ny=x$. Thus

$$\widehat{\mathbb{Z}}/n\widehat{\mathbb{Z}} \cong \mathbb{Z}/n\mathbb{Z}, \quad [\widehat{\mathbb{Z}} : n\widehat{\mathbb{Z}}] = n.$$

In particular the degree p^r retains its exact index p^r . For a signed source lift with odd nonzero n and odd b , injectivity follows from injectivity of multiplication by n and invertibility of b modulo four. Its image has exactly the condition x in $n\widehat{\mathbb{Z}}$; once the unique preimage of x is chosen, the branch equation can be solved uniquely for k . Thus its image index is $|n|$. The winding map at $n=2$ also exists and retains index two; this does not remove $W5$'s obstruction to a sign-preserving equivariant lift of even degree on the full source.

Every continuous character of $\widehat{G}$ still has modulus one by compactness, with the complete proof in P12 applying unchanged. Every such character factors through some finite quotient, by the no-small-subgroup arc argument. The continuous signed characters restrict to the original G as

$$\chi_{z,j}(T^m J^k) = z^m j^k, \quad z \text{ any root of unity}, \quad j \in \{i, -i\}.$$

Every root of unity occurs since its finite order divides $4N$ for some N . Conversely the finite quotient relation forces that order condition. These signed characters separate $\widehat{G}$ by the finite-coordinate proof, and separate all original source elements. The character values are a dense proper subset of the unit circle, with all prime powers now retained.

This gives a compact global object carrying the entire sequence of finite clock observations, faithfully retaining original integers and preserving winding-cover degrees. Its character-modulus theorem is proved. It is not an identification of $\widehat{G}$ with $\text{Spec } \mathbb{Z}$, nor a construction of the arithmetic cohomology whose eigenvalues enter Deligne purity.

P14. The exact degree factor in the clock operators

P14.1. Measure and change of variables

Write $K_w = \text{Zhat}$ for the winding factor of the full group $\text{Ghat} = \text{Zhat} \times C_4$ in P13. Every variable in the following integrals belongs to K_w , not to τ . Retain a finite nonzero translation-invariant Haar measure μ with arbitrary total mass $h = \mu(K_w) > 0$. Its measure on a residue class modulo M is h/M : the M classes partition K_w and translations permute them. These assignments are compatible with refinement, since each class modulo M is a disjoint union of M'/M classes modulo M' when M divides M' . They describe the measure directly on every finite clock. No choice $h=1$ or rescaling of operators is made.

For a positive integer n , P13 proves that multiplication $\phi_n(x) = nx$ is injective, with clopen image nK_w of index n . It is a homeomorphism from K_w onto nK_w , since a continuous bijection from a compact space to a Hausdorff space has continuous inverse. Division x/n is used only on nK_w . Each of its n cosets has measure h/n . The measures $(\phi_n)_* \mu$ and $n\mu$ restricted to nK_w are translation invariant on that subgroup and both have mass h . Uniqueness of Haar measure at specified mass gives

$$\int_{K_w} F(nx) d\mu(x) = n \int_{nK_w} F(y) d\mu(y). \quad (\text{P14.1})$$

This identity holds for nonnegative measurable F and for integrable complex F . It can also be checked first on each residue-class cylinder using the class measures h/M , then extended to nonnegative measurable functions by the measure extension and monotone convergence theorems. It implies $\mu(nE) = \mu(E)/n$ and $\mu(\phi_n^{-1}\{E\}) = n \mu(E \cap nK_w)$. Thus both image and inverse image preserve null sets, including for the completed measure.

P14.2. Unrescaled operators and adjoints

Let $H = L^2(K_w, \mu)$, with inner product linear in its first entry, and define

$$(U_n f)(x) = f(nx), \quad (V_n f)(x) = \begin{cases} f(x/n), & x \in nK_w, \\ 0, & x \notin nK_w. \end{cases} \quad (\text{P14.2})$$

The null-set statement proves these maps are well defined on almost-everywhere equivalence classes. Formula P14.1 gives

$$\|U_n f\|^2 = n \int_{nK_w} |f(y)|^2 d\mu(y), \quad \|V_n f\|^2 = \frac{1}{n} \int_{K_w} |f(y)|^2 d\mu(y). \quad (\text{P14.3})$$

In particular both operators are bounded. The second equality gives $\text{norm}(V_n) = 1/\sqrt{n}$. The first gives $\text{norm}(U_n) \leq \sqrt{n}$, and equality follows by taking $f = 1_{nK_w}$: its squared norm is h/n while $U_n f = 1_{K_w}$ has squared norm h . Thus every factor is retained:

$$\|U_n\| = \sqrt{n}, \quad \|V_n\| = \frac{1}{\sqrt{n}}. \quad (\text{P14.4})$$

For f, g in H , change variables $y = nx$ through P14.1 to obtain

$$\begin{aligned} \langle U_n f, g \rangle &= \int_{K_w} f(nx) \overline{g(x)} d\mu(x) \\ &= n \int_{nK_w} f(y) \overline{g(y/n)} d\mu(y) = \langle f, nV_n g \rangle. \end{aligned}$$

Cauchy-Schwarz and P14.3 justify these integrals. Hence $U_n^* = nV_n$ and $V_n^* = U_n/n$. With P_n multiplication by the indicator of nK_w , direct evaluation yields $U_n V_n = I$ and $V_n U_n = P_n$, and consequently

$$\boxed{U_n U_n^* = nI, \quad U_n^* U_n = nP_n, \quad V_n^* V_n = \frac{1}{n}I, \quad V_n V_n^* = \frac{1}{n}P_n.} \quad (\text{P14.5})$$

P14.3. These factors already occur in the finite clocks

For every positive M give $H_M = C^\wedge(Z/MZ)$ the inner product

$$\langle F, G \rangle_M = \frac{h}{M} \sum_{a \bmod M} F(a) \overline{G(a)}. \quad (\text{P14.6})$$

The pullback $J_M F = F$ composed with reduction ρ_M embeds it isometrically into H , by the measured residue classes above. Define maps with their different domains retained:

$$\begin{aligned} u_{n,M} : H_{nM} &\rightarrow H_M, & (u_{n,M} F)(a) &= F(na), \\ v_{n,M} : H_M &\rightarrow H_{nM}, & (v_{n,M} G)(b) &= \begin{cases} G(b/n), & n \mid b, \\ 0, & n \nmid b. \end{cases} \end{aligned} \quad (\text{P14.7})$$

The map $a \bmod M \rightarrow na \bmod nM$ is injective with image the residues divisible by n ; its inverse there is division by n modulo M . Therefore the displayed formulas are well defined. Their adjoint calculation is

$$\begin{aligned} \langle u_{n,M} F, G \rangle_M &= \frac{h}{M} \sum_{a \bmod M} F(na) \overline{G(a)} \\ &= \frac{h}{nM} \sum_{b \bmod nM} F(b) \overline{n \mathbf{1}_{n \mid b} G(b/n)} = \langle F, n v_{n,M} G \rangle_{nM}. \end{aligned} \quad (\text{P14.8})$$

Thus $u_{n,M}^* = n v_{n,M}$. Also $uv = I$ and vu is multiplication by the divisibility indicator. The completed and finite maps intertwine exactly:

$$U_n J_{nM} = J_M u_{n,M}, \quad V_n J_M = J_{nM} v_{n,M}. \quad (\text{P14.9})$$

The factor n is therefore already forced by finite quotient arithmetic and the two class measures h/M and $h/(nM)$. The construction has not inserted it by choosing an eigenvalue or changing scale.

P14.4. Composition retains multiplication and common divisors

For positive m, n , evaluating at x gives

$$U_m U_n = U_{mn} = U_n U_m, \quad V_m V_n = V_{mn} = V_n V_m. \quad (\text{P14.10})$$

The first equality is $f(nmx)$. For the second the support conditions are x in nK_w and x/m in nK_w , equivalently x in mnK_w ; on that set unique division gives $(x/m)/n = x/(mn)$.

There is a more informative mixed relation. Let $d = \gcd(m, n)$, $a = m/d$, $b = n/d$. Injectivity of multiplication by d proves mx in nK_w iff ax in bK_w . Since a, b are coprime, choose integers r, s with $ra + sb = 1$. If ax in bK_w then $x = r(ax) + s(bx)$ is in bK_w ; the converse is immediate. On this support $x = by$ gives $mx/n = ay$. Therefore

$$\boxed{U_m V_n = V_{n/d} U_{m/d}, \quad d = \gcd(m, n).} \quad (\text{P14.11})$$

The coprime case is $U_m V_n = V_n U_m$; the common divisor has not been discarded.

P14.5. Operator modulus and eigenvalue modulus

P14.5 gives the exact positive operator square root

$$|U_n| = (U_n^* U_n)^{1/2} = \sqrt{n} P_n. \quad (\text{P14.12})$$

The equality follows because P_n is a self-adjoint projection, so the nonnegative operator $\sqrt{n} P_n$ squares to $n P_n$. For $n > 1$, P_n and $I - P_n$ are nonzero: their supports have positive measures h/n and $h - h/n$. Thus $U_n^* U_n = n P_n$ differs from $U_n U_n^* = n I$, and U_n is not normal.

The same operator has the exact eigenvector

$$U_n \mathbf{1}_{K_w} = \mathbf{1}_{K_w}, \quad \|\mathbf{1}_{K_w}\|^2 = h. \quad (\text{P14.13})$$

For $n > 1$ its eigenvalue 1 has modulus different from $\sqrt{n}$. Hence P14.12 is an operator-modulus result; it does not make every eigenvalue have modulus $\sqrt{n}$. This statement concerns the constructed operator on its stated full function space. It makes no assertion that another arithmetic cohomological realization or a derived sector cannot have pure weight. No such realization is constructed here. Continuous characters have squared norm h , since their pointwise modulus is one by P13.

P14.6. The full signed source is retained

Keep the entire $\widehat{\text{Ghat}} = K_w \times C_4$, with its original A and sign ϵ , rather than replacing it by K_w . Define its translation-invariant measure by

$$\int_{\widehat{G}} F(x, k) d\tilde{\mu}(x, k) = \frac{1}{4} \sum_{k \bmod 4} \int_{K_w} F(x, k) d\mu(x). \quad (\text{P14.14})$$

Its total mass remains h and every branch has mass $h/4$. Pullback by the winding projection $\pi(x, k) = x$ is isometric because the four terms and the factor $1/4$ cancel explicitly in the squared norm. This is a retained comparison map, not an identification of the two spaces.

For a positive odd n , c modulo four, and odd b , the original source map $F = f_{(n, c, b)}$ from P13 is an injective homomorphism onto $n K_w \times C_4$ of index n . Its inverse there is explicitly

$$F^{-1}(y, l) = (y/n, b^{-1}(l - c((y/n) \bmod 4))), \quad (\text{P14.15})$$

where b inverse is taken modulo four. Define $U_F f = f$ composed with F and $V_F f$ as f composed with F inverse on its image and zero elsewhere. The same Haar argument, now on the full signed group and its index- n image, gives $U_F^* = n V_F$ and all four relations in P14.5 with the image projection P_F . Every branch and its measure remain present. Moreover π composed with $F = [n]$ composed with π , so

$$U_F \pi^* = \pi^* U_n, \quad V_F \pi^* = \pi^* V_n. \quad (\text{P14.16})$$

These identities follow by evaluating inside and outside the image. Winding degree two retains index two on K_w ; this does not assert an even-degree sign-preserving map commuting with A on $\widehat{\text{Ghat}}$, whose obstruction was proved in W5.

P14.7. What the global construction establishes

The exact chain now proved is: finite clock residues have infinite ambiguity on integer counts (P11); their compatible completion is compact and retains all prime powers and the original signed data (P12-P13); its continuous characters have modulus one (P12-P13); and its arithmetic degree- n maps have index n and the unrescaled adjoint factor n , with operator modulus $\sqrt{n}$ on the image sector (P14). These are maps and identities of the original source-derived winding data. Neither the shared countable observation requirement nor the operator modulus is being substituted for Deligne's arithmetic Frobenius eigenvalue relation. P14 was independently derived and its complete finite and infinite computations are recorded here.

Tate cohomology and the original finite maps

Proof source: TATE_H1_DERIVATION.md.

Full-period Tate cohomology and the degree mechanism

Date: 2026-09-24. **Status:** complete local derivation from the supplied Tate-curve model. All coordinates in this document belong to the displayed Tate curves or their product. No coordinate, differential, metric, complex structure, or cohomology theory is assigned here to the base tau.

Human source. Alain Connes and Caterina Consani, *On the Absolute Geometry of Spec Z and the Fargues-Fontaine curve*, original-author FF.tex, source directory 01_CC_Absolute_Geometry_SpecZ, SHA-256 14C17A95040BE7BFC57BB5F87E58907A05CACD0BCB1B6967B2892CF3D9BC1229. The source portions actually read are lines 1-100 and 1390-1640. The receiving definitions are Proposition prop:archimedean_points_C at line 1465, the Tate quotient discussion at lines 1548-1579, the real structure at lines 1581-1593, and the rectangular decomposition at lines 1595-1638. The occurrence of thm:tate_decomposition used here is at line 1624; that label also occurs earlier in the source. The cohomology and isogeny calculations below are derivations from that model, not assertions that the source states them.

T1. The original periods and complex structure

Fix the source prime p , and retain

$$A = \log p > 0, \quad B = 2\pi > 0, \quad \Lambda_p = A\mathbb{Z} + iB\mathbb{Z}. \quad (\text{T1.1})$$

Write $z \in \mathbb{C}^\times$ for the source's nonzero **morphism parameter**. The exponential map in the new logarithm variable $w = x + i\theta$ gives

$$z = e^w = e^x e^{i\theta}, \quad e^{w+A} = p e^w, \quad e^{w+iB} = e^w. \quad (\text{T1.2})$$

Consequently the exact identification is

$$E_p = \mathbb{C}^\times / p^\mathbb{Z} \cong \mathbb{C} / \Lambda_p \cong (\mathbb{R} / A\mathbb{Z}) \times (\mathbb{R} / B\mathbb{Z}). \quad (\text{T1.3})$$

Indeed, $e^{w'}/e^w \in p^\mathbb{Z}$ holds exactly when $w' - w \in A\mathbb{Z} + iB\mathbb{Z}$. This proves both injectivity and surjectivity of the middle identification. The last identification records both original periods and is a real product decomposition, not a replacement of the complex structure.

The invariant forms and the orientation are

$$dx = \frac{d|z|}{|z|}, \quad d\theta, \quad \omega_p = \frac{dz}{z} = dx + i d\theta, \quad dx \wedge d\theta > 0. \quad (\text{T1.4})$$

The forms $dx, d\theta$ descend despite the multivalued functions x, θ . Their periods on the positive radial and angular loops

$$r_p(t) = [e^{At}], \quad a_p(t) = [e^{iBt}], \quad 0 \leq t \leq 1, \quad (\text{T1.5})$$

are respectively

$$\left(\int_{r_p} dx, \int_{a_p} dx \right) = (A, 0), \quad \left(\int_{r_p} d\theta, \int_{a_p} d\theta \right) = (0, B). \quad (\text{T1.6})$$

In particular, neither winding direction nor either period is discarded. The source's modular parameter iA/B is recovered by the comparison $u = w/(iB)$: the positive angular generator goes to 1, and the positive radial generator goes to $-iA/B$. Reversing the latter generator gives the source's lattice basis $1, iA/B$. Thus the apparent sign change is an explicit choice of lattice generator, not a changed real structure or period. The variable w and full lattice (T1.1) remain the working objects.

The complex structure J_p satisfies

$$J_p \partial_x = \partial_\theta, \quad J_p \partial_\theta = -\partial_x, \quad J_p^* dx = -d\theta, \quad J_p^* d\theta = dx. \quad (\text{T1.7})$$

The source supplies the form dz/z . Its real quadratic form is

$$g_p = dx \otimes dx + d\theta \otimes d\theta, \quad \text{Area}(E_p, g_p) = AB = 2\pi \log p. \quad (\text{T1.8})$$

It is well defined because both deck transformations in (T1.2) preserve dw . Multiplying the parameter z by any fixed nonzero complex number also preserves dz/z , so this construction is independent of a translated choice of origin. We do not introduce a freely chosen positive inner product later.

T2. All cohomology groups, the cup product, and the Hodge form

With complex coefficients, the de Rham cohomology is

$$H^0(E_p, \mathbb{C}) = \mathbb{C} \cdot 1, \quad H^1(E_p, \mathbb{C}) = \mathbb{C}[dx] \oplus \mathbb{C}[d\theta], \quad H^2(E_p, \mathbb{C}) = \mathbb{C}[dx \wedge d\theta]. \quad (\text{T2.1})$$

Here and below the same symbols denote invariant representatives when this creates no ambiguity. For completeness, use the Fourier modes

$$e^{i(k_x x + k_\theta \theta)}, \quad k_x = \frac{2\pi m}{A}, \quad k_\theta = \frac{2\pi l}{B}, \quad (m, l) \in \mathbb{Z}^2. \quad (\text{T2.2})$$

For a closed one-form, each nonzero Fourier coefficient pair (f, g) satisfies $k_x g - k_\theta f = 0$. The scalar coefficient

$$h = \frac{k_x f + k_\theta g}{i(k_x^2 + k_\theta^2)} \quad (\text{T2.3})$$

has differential coefficients f, g . Thus all nonconstant modes of a closed one-form are exact. Division by a nonzero quadratic frequency preserves smooth Fourier convergence. The constant modes are not exact

because their periods on (T1.5) are Af and Bg . A two-form with nonzero Fourier coefficient c is the differential of the one-form with coefficients

$$-\frac{k_\theta c}{i(k_x^2 + k_\theta^2)} dx + \frac{k_x c}{i(k_x^2 + k_\theta^2)} d\theta. \quad (\text{T2.4})$$

Its constant mode is not exact because

$$\int_{E_p} dx \wedge d\theta = AB. \quad (\text{T2.5})$$

Finally a function with zero differential is constant on the connected torus. This proves (T2.1), including its representatives and periods.

The integral cohomology is specified inside these unchanged real de Rham groups by the period conditions

$$H^1(E_p, \mathbb{Z}) = \{a dx + b d\theta : Aa \in \mathbb{Z}, Bb \in \mathbb{Z}\}, \quad H^2(E_p, \mathbb{Z}) = \frac{1}{AB} \mathbb{Z}[dx \wedge d\theta]. \quad (\text{T2.6})$$

These are exact comparison maps with singular integral cohomology: the torus has the two cycles (T1.5), their oriented product is the fundamental two-cycle, and no torsion occurs. Writing the reciprocal periods in (T2.6) does not replace the full-period forms (T1.4).

For

$$\alpha = a dx + b d\theta, \quad \beta = c dx + d d\theta,$$

the cup pairing, including its full volume factor, is

$$Q_p(\alpha, \beta) = \int_{E_p} \alpha \wedge \beta = AB(ad - bc). \quad (\text{T2.7})$$

The Hodge star from (T1.8) and the displayed orientation is

$$*_p dx = d\theta, \quad *_p d\theta = -dx, \quad *_p \omega_p = -i\omega_p, \quad *_p \bar{\omega}_p = i\bar{\omega}_p. \quad (\text{T2.8})$$

It equals $-J_p^*$ on real one-forms. Its positive Hermitian pairing is therefore

$$h_p(\alpha, \beta) = \int_{E_p} \alpha \wedge *_p \bar{\beta} = AB(a\bar{c} + b\bar{d}). \quad (\text{T2.9})$$

In particular $h_p(\alpha, \alpha) > 0$ for every nonzero class. Positivity follows from $A, B > 0$ and the displayed sum of absolute squares; it is not an imposed spectral assumption.

Every holomorphic one-form on E_p is a multiple of ω_p . To prove this, lift the form to $f(w) dw$ on $\mathbb{C}$. The function f is entire and lattice periodic, hence bounded on the plane by its bound on a compact fundamental rectangle; Liouville's theorem makes f constant. Complex conjugation gives the antiholomorphic forms. Thus

$$H^{1,0}(E_p) = \mathbb{C}\omega_p, \quad H^{0,1}(E_p) = \mathbb{C}\bar{\omega}_p, \quad H^1(E_p, \mathbb{C}) = H^{1,0} \oplus H^{0,1}. \quad (\text{T2.10})$$

The direct sum is exact because

$$dx = \frac{\omega_p + \bar{\omega}_p}{2}, \quad d\theta = \frac{\omega_p - \bar{\omega}_p}{2i}. \quad (\text{T2.11})$$

Every factor in the positivity calculation is visible:

$$\omega_p \wedge \bar{\omega}_p = -2i \, dx \wedge d\theta, \quad i \int_{E_p} \omega_p \wedge \bar{\omega}_p = 2AB = 4\pi \log p. \quad (\text{T2.12})$$

Also $-i \int \bar{\omega}_p \wedge \omega_p = 2AB$. The Hodge types of H^0, H^1, H^2 are respectively $(0, 0)$, $(1, 0) \oplus (0, 1)$, and $(1, 1)$. This establishes their Hodge weights 0, 1, 2. It does not by itself specify an arithmetic Frobenius operator.

T3. The actual source actions and the distinct parameter power maps

T3.1 Parameter scaling and source Frobenius

The source identifies a morphism with

$$\rho_z(t) = e^{zt}, \quad z \in \mathbb{C}, \quad (\text{T3.1})$$

and then removes $z = 0$. Its Weil/stalk scaling action is $z \mapsto cz$, for fixed $c \in \mathbb{C}^\times$. On the displayed quotient E_p this becomes a translation

$$T_c : [z] \mapsto [cz]. \quad (\text{T3.2})$$

Choose any logarithm d of c . In the full-period logarithm coordinates, T_c lifts to $w \mapsto w + d$. Different logarithms differ by $iB\mathbb{Z}$, so they define the same map. The derivative fixes $dx, d\theta$, and the cohomology calculation (T2.1) proves

$$T_c^*|_{H^0} = 1, \quad T_c^*|_{H^1} = I_2, \quad T_c^*|_{H^2} = 1. \quad (\text{T3.3})$$

For $c = p$, the lift translates by $A \in \Lambda_p$, so the quotient map itself is the identity:

$$F_{\text{source}}([z]) = [pz] = [z]. \quad (\text{T3.4})$$

This is a degree-one map on E_p . Its H^1 eigenvalues are 1, 1.

There is an additional exact distinction at the level of morphisms. For a positive integer n , valuewise exponentiation gives

$$(\rho_z(t))^n = e^{nzt} = \rho_{nz}(t). \quad (\text{T3.5})$$

Thus **powering the values of the source morphism gives parameter scaling** $z \mapsto nz$. It does not give the parameter map $z \mapsto z^n$. Equation (T3.5) is the required comparison between the two constructions. On E_p , its induced map is T_n , which has the cohomological action (T3.3).

T3.2 The parameter power map

For a nonzero integer n , define the different map

$$P_n : E_p \longrightarrow E_p, \quad [z] \longmapsto [z^n]. \quad (\text{T3.6})$$

It is well defined because $(p^k z)^n = p^{kn} z^n$. Its lift is $w \mapsto nw$, so

$$P_n^* dx = n dx, \quad P_n^* d\theta = n d\theta, \quad P_n^* \omega_p = n \omega_p. \quad (\text{T3.7})$$

The lattice subgroup $n\Lambda_p \subset \Lambda_p$ has index n^2 ; equivalently the kernel has the n^2 classes represented, with $q = |n|$, by

$$p^{j/q} e^{iBk/q}, \quad 0 \leq j, k < q. \quad (\text{T3.8})$$

Therefore

$$\deg P_n = n^2, \quad P_n^*|_{H^0} = 1, \quad P_n^*|_{H^1} = nI_2, \quad P_n^*|_{H^2} = n^2. \quad (\text{T3.9})$$

Both radial and angular loops acquire winding n , including its sign. Equations (T2.7) and (T2.9) give

$$Q_p(P_n^* \alpha, P_n^* \beta) = n^2 Q_p(\alpha, \beta), \quad h_p(P_n^* \alpha, P_n^* \beta) = n^2 h_p(\alpha, \beta). \quad (\text{T3.10})$$

In particular P_p has degree p^2 and H^1 eigenvalues p, p . Its square-root-of-degree law is $|p| = \sqrt{p^2}$, not a claim that this map has degree p . The map P_0 , if included, is constant; it fixes H^0 and kills H^1, H^2 .

T3.3 Real involution and the half-turn

The source real structure is

$$\sigma([z]) = [\bar{z}], \quad (x, \theta) \mapsto (x, -\theta). \quad (\text{T3.11})$$

It preserves the lattice, so it is well defined. Its cohomology actions and form actions are

$$\sigma^*|_{H^0} = 1, \quad \sigma^* dx = dx, \quad \sigma^* d\theta = -d\theta, \quad \sigma^*|_{H^2} = -1, \quad \sigma^* \omega_p = \bar{\omega}_p. \quad (\text{T3.12})$$

It preserves h_p and negates Q_p , as follows directly from (T2.7)-(T2.9). The fixed points satisfy $2\theta \in B\mathbb{Z}$, and therefore form the two radial circles $\theta = 0, B/2 \pmod{B}$. Equivalently $\bar{z} = p^k z$ forces $p^k = 1$ by absolute values, then $k = 0$ and $z \in \mathbb{R}^\times$. This reproduces the source's two real components with radial period A .

The parameter scaling $T_{-1} : [z] \mapsto [-z]$ is instead the translation $\theta \mapsto \theta + B/2$. Its action on all cohomology is the identity. It must not be identified with the antiholomorphic involution σ , or with the inversion $P_{-1} : [z] \mapsto [z^{-1}]$, whose H^1 action is $-I_2$.

T4. Every genuine holomorphic self-isogeny has the exact degree norm

Let $f : E_p \rightarrow E_p$ be a nonconstant holomorphic map. Lift it through the universal cover to an entire function $\tilde{f} : \mathbb{C} \rightarrow \mathbb{C}$. For each $\lambda \in \Lambda_p$, the difference $\tilde{f}(w + \lambda) - \tilde{f}(w)$ takes values in the discrete lattice and is continuous, hence is constant. It follows that $\tilde{f}'$ is lattice periodic, bounded on a fundamental rectangle, and then bounded on the plane. Liouville's theorem gives

$$\tilde{f}(w) = c w + d, \quad c \neq 0, \quad c \Lambda_p \subseteq \Lambda_p. \quad (\text{T4.1})$$

Conversely (T4.1) defines a holomorphic map. The translation d does not affect its cohomology. For $c = a + ib$, the exact actions are

$$f^* dx = a dx - b d\theta, \quad f^* d\theta = b dx + a d\theta, \quad f^* \omega_p = c \omega_p, \quad f^* \bar{\omega}_p = \bar{c} \bar{\omega}_p. \quad (\text{T4.2})$$

To make the descent condition explicit in the original periods, write

$$cA = mA + r iB, \quad c iB = sA + t iB, \quad m, r, s, t \in \mathbb{Z}. \quad (\text{T4.3})$$

Comparison of real and imaginary parts gives exactly

$$a = m = t, \quad b = r \frac{B}{A}, \quad sA^2 + rB^2 = 0. \quad (\text{T4.4})$$

Conversely these equations imply (T4.3). In the ordered lattice basis A, iB , multiplication by c has matrix

$$M_c = \begin{pmatrix} m & s \\ r & m \end{pmatrix}. \quad (\text{T4.5})$$

The map is a covering because $c \neq 0$. Its kernel is $c^{-1}\Lambda_p/\Lambda_p$, which maps bijectively onto $\Lambda_p/c\Lambda_p$ by multiplication by c . Thus its degree is the lattice index, and

$$N = \deg f = \det M_c = m^2 - rs = m^2 + r^2 \frac{B^2}{A^2} = a^2 + b^2 = |c|^2. \quad (\text{T4.6})$$

Positivity of this determinant also follows from the preserved complex orientation. In particular N is a positive integer.

Using (T4.2) in the already derived pairing (T2.9) proves

$$h_p(f^* \alpha, f^* \beta) = N h_p(\alpha, \beta), \quad Q_p(f^* \alpha, f^* \beta) = N Q_p(\alpha, \beta), \quad (f^*)^\dagger f^* = N I_2. \quad (\text{T4.7})$$

For example, the coefficient matrix on H^1 is

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

its conjugate transpose times itself is $(a^2 + b^2)I_2$, and the full factor AB remains in both sides of the inner-product identity. The Hodge lines in (T2.10) are eigenlines with eigenvalues $c, \bar{c}$. Hence

$$|\lambda| = \sqrt{N} \quad \text{for every eigenvalue on } H^1, \quad f^*|_{H^0} = 1, \quad f^*|_{H^2} = N. \quad (\text{T4.8})$$

This proves the degree mechanism from the source differential: holomorphicity preserves the Hodge star, the cup integral multiplies by degree, their combination is positive, and the positive norm identity forces the square-root magnitude. No arbitrary metric or unknown endomorphism has been assumed.

There is also an exact arithmetic statement for this genuine isogeny. Its two eigenvalues solve

$$T^2 - 2mT + N = 0 \quad \text{in } \mathbb{Z}[T]. \quad (\text{T4.9})$$

If $b \neq 0$, they are the two conjugate roots and both have modulus $\sqrt{N}$; if $b = 0$, then $c = m \in \mathbb{Z}$ and $N = m^2$. Thus every algebraic conjugate of an H^1 eigenvalue has that same modulus. This is an actual Weil-number-type conclusion for the specified isogeny. It does not identify the source deck transformation with a finite-field Frobenius.

T5. The two degree- n maps retain opposite winding directions

Fix a positive integer n . In this section p^n is an extended real scale, not a new prime. Define explicitly

$$E_{p^n} = \mathbb{C}^\times / (p^n)^\mathbb{Z} \cong \mathbb{C} / \Lambda_{p^n}, \quad \Lambda_{p^n} = nA\mathbb{Z} + iB\mathbb{Z}. \quad (\text{T5.1})$$

Use x_n, θ_n for its logarithm coordinates and retain

$$\omega_{p^n} = dx_n + i d\theta_n, \quad \int_{E_{p^n}} dx_n \wedge d\theta_n = nAB. \quad (\text{T5.2})$$

The exact maps are

$$U_n : E_{p^n} \longrightarrow E_p, \quad [z]_{p^n} \longmapsto [z]_p, \quad (\text{T5.3})$$

$$V_n : E_p \longrightarrow E_{p^n}, \quad [z]_p \longmapsto [z^n]_{p^n}. \quad (\text{T5.4})$$

For U_n , descent follows from $\Lambda_{p^n} \subset \Lambda_p$, and its lift is $w \mapsto w$. For V_n , descent follows from $n\Lambda_p \subset \Lambda_{p^n}$, and its lift is $w \mapsto nw$. The two lattice indices are

$$[\Lambda_p : \Lambda_{p^n}] = n, \quad [\Lambda_{p^n} : n\Lambda_p] = n. \quad (\text{T5.5})$$

Thus both maps have degree n . Their kernels, including the distinct directions, are

$$\ker U_n = \{[p^j]_{p^n} : 0 \leq j < n\}, \quad \ker V_n = \{[e^{iBk/n}]_p : 0 \leq k < n\}. \quad (\text{T5.6})$$

For the second formula, $z^n = p^{nj}$ gives $z = p^j e^{iBk/n}$, and the factor p^j disappears exactly in the domain quotient. Both compositions are the parameter power map on the appropriate curve:

$$U_n \circ V_n = P_n \text{ on } E_p, \quad V_n \circ U_n = P_n \text{ on } E_{p^n}. \quad (\text{T5.7})$$

On the original forms the pullbacks are

$$U_n^* dx = dx_n, \quad U_n^* d\theta = d\theta_n, \quad V_n^* dx_n = n dx, \quad V_n^* d\theta_n = n d\theta. \quad (\text{T5.8})$$

On the positive radial and angular cycles they have the distinct winding matrices

$$(U_n)_* : r_{p^n} \mapsto nr_p, \quad a_{p^n} \mapsto a_p, \quad (V_n)_* : r_p \mapsto r_{p^n}, \quad a_p \mapsto na_{p^n}. \quad (\text{T5.9})$$

For example, $V_n(e^{At}) = e^{nAt}$ traverses the target radial period once, whereas $V_n(e^{iBt}) = e^{inBt}$ traverses its unchanged angular period n times. The matrices are therefore diagonal $\text{diag}(n, 1)$ and $\text{diag}(1, n)$ on homology in the displayed source/target cycle bases. The pullback matrices on the corresponding integral dual bases are the same diagonal matrices; this is compatible with (T5.8) because the radial periods are A and nA , not equal.

On H^0 , both pullbacks fix 1. On the full area forms,

$$U_n^*(dx \wedge d\theta) = dx_n \wedge d\theta_n, \quad V_n^*(dx_n \wedge d\theta_n) = n^2 dx \wedge d\theta. \quad (\text{T5.10})$$

There is no contradiction with their degree n : the exact integral fundamental classes are represented by $dx \wedge d\theta/(AB)$ and $dx_n \wedge d\theta_n/(nAB)$, so both pullbacks multiply the target fundamental class by n . All area factors are retained in this comparison.

T6. The adjoint identity, with both volumes retained

Write $H_p = H^1(E_p, \mathbb{C})$ and $H_{p^n} = H^1(E_{p^n}, \mathbb{C})$. Their positive forms are

$$h_p(a dx + b d\theta, c dx + d d\theta) = AB(a\bar{c} + b\bar{d}), \quad (\text{T6.1})$$

$$h_{p^n}(a dx_n + b d\theta_n, c dx_n + d d\theta_n) = nAB(a\bar{c} + b\bar{d}). \quad (\text{T6.2})$$

For $\alpha = a dx + b d\theta$ and $\beta = c dx_n + d d\theta_n$, equations (T5.8) give

$$h_{p^n}(U_n^* \alpha, \beta) = nAB(a\bar{c} + b\bar{d}) = h_p(\alpha, V_n^* \beta). \quad (\text{T6.3})$$

Therefore, with adjoints taken between the indicated spaces,

$$(U_n^*)^\dagger = V_n^*, \quad (V_n^*)^\dagger = U_n^*, \quad (\text{T6.4})$$

$$(U_n^*)^\dagger U_n^* = nI_{H_p}, \quad (V_n^*)^\dagger V_n^* = nI_{H_{p^n}}. \quad (\text{T6.5})$$

This also follows by pullback from the two compositions (T5.7), whose H^1 action is nI , but equation (T6.3) proves that the compositional partners really are adjoints for the source-defined positive forms. Consequently all singular values of either pullback are $\sqrt{n}$. These are maps between two different cohomology spaces, so it would be incorrect to call these singular values eigenvalues of an endomorphism of H_p .

The corresponding cup identities are

$$Q_{p^n}(U_n^* \alpha, U_n^* \alpha') = nQ_p(\alpha, \alpha'), \quad Q_p(V_n^* \beta, V_n^* \beta') = nQ_{p^n}(\beta, \beta'). \quad (\text{T6.6})$$

They follow either from (T2.7) and the full areas, or from the proven degree and change of variables. The complex structures commute with both pullbacks by (T5.8).

T7. An actual polarized product receiver and its full weight spectrum

The inter-curve correspondence has an exact receiver as a holomorphic self-isogeny. Use the displayed origins [1] and form the product of the two source-model elliptic curves

$$\mathcal{A}_{p,n} = E_p \times E_{p^n}, \quad \Phi_{p,n}(u, v) = (U_n(v), V_n(u)). \quad (\text{T7.1})$$

It is a complex torus of complex dimension two. The product of the source elliptic algebraic structures also makes it an abelian surface. In logarithm coordinates its lift is exactly

$$(w_1, w_2) \mapsto (w_2, nw_1), \quad (w_1, w_2) \in \mathbb{C}^2 / (\Lambda_p \oplus \Lambda_{p^n}). \quad (\text{T7.2})$$

The two descent inclusions have already been proved in T5. Its kernel is $\ker V_n \times \ker U_n$, so

$$\deg \Phi_{p,n} = n^2, \quad \Phi_{p,n}^2 = [n]_{\mathcal{A}_{p,n}}. \quad (\text{T7.3})$$

This degree n^2 is appropriate to complex dimension two and is not replaced by n .

The positive integral alternating forms on the two original lattices are

$$\mathcal{E}_p((x, \theta), (x', \theta')) = \frac{x\theta' - \theta x'}{AB}, \quad (\text{T7.4})$$

$$\mathcal{E}_{p^n}((x_n, \theta_n), (x'_n, \theta'_n)) = \frac{x_n \theta'_n - \theta_n x'_n}{nAB}. \quad (\text{T7.5})$$

In their respective original lattice bases A, iB and nA, iB , both have integer matrix $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, of determinant one. Moreover $\mathcal{E}(v, Jv) > 0$ for nonzero real v , since its numerator is $x^2 + \theta^2$ (respectively $x_n^2 + \theta_n^2$). They therefore give the principal polarizations with all original period factors shown. Their associated two-forms are

$$\Omega_{p,n} = \frac{dx \wedge d\theta}{AB} + \frac{dx_n \wedge d\theta_n}{nAB}. \quad (\text{T7.6})$$

This is the product principal polarization. Direct substitution of (T7.2) gives the exact multiplier identity

$$\Phi_{p,n}^* \Omega_{p,n} = n\Omega_{p,n}. \quad (\text{T7.7})$$

The factors are essential: the first summand pulls back to $dx_n \wedge d\theta_n/(AB)$, while the second pulls back to $n^2 dx \wedge d\theta/(nAB)$.

The duality of U_n, V_n can also be checked before taking cohomology. For $v \in \mathbb{C}$ in the E_{p^n} tangent space and $w \in \mathbb{C}$ in the E_p tangent space,

$$\mathcal{E}_p(U_n v, w) = \mathcal{E}_{p^n}(v, V_n w). \quad (\text{T7.8})$$

Both sides are $(x_v \theta_w - \theta_v x_w)/(AB)$. Together with the integral lattice maps, this proves that V_n is the polarized dual of U_n . For the product alternating form, (T7.8) proves

$$\mathcal{E}_{\text{prod}}(\Phi_{p,n} v, w) = \mathcal{E}_{\text{prod}}(v, \Phi_{p,n} w). \quad (\text{T7.9})$$

Thus the Rosati adjoint of this actual product isogeny equals itself. Equations (T7.3) and (T7.9) give

$$\Phi_{p,n}^\dagger \Phi_{p,n} = [n]. \quad (\text{T7.10})$$

On $H^1(\mathcal{A}_{p,n}, \mathbb{C}) = H_p \oplus H_{p^n}$, the pullback is the explicit exchange operator

$$F_1(\alpha, \beta) = (V_n^* \beta, U_n^* \alpha). \quad (\text{T7.11})$$

Its positive Hermitian form is exactly $h_p \oplus h_{p^n}$, with factors AB, nAB . This form is the one obtained from the product polarization: the alternating form on H^1 is $\int_{\mathcal{A}} \alpha \wedge \beta \wedge \Omega_{p,n}$. For classes both from the first factor, the second summand of Ω integrates to one, leaving Q_p ; for classes both from the second factor, the first summand integrates to one, leaving Q_{p^n} ; mixed terms vanish by degree on each two-dimensional real factor. Combining this with $-J^*$ gives exactly the displayed direct sum of (T6.1) and (T6.2).

Consequently T6 proves, without a new choice of inner product,

$$F_1^\dagger = F_1, \quad F_1^2 = nI_4, \quad h_{\text{prod}}(F_1 a, F_1 b) = n h_{\text{prod}}(a, b). \quad (\text{T7.12})$$

Since $n > 0$, the polynomial $T^2 - n$ has two distinct roots. The invariant real-form pairs (dx, dx_n) and $(d\theta, d\theta_n)$ each have coefficient matrix

$$\begin{pmatrix} 0 & n \\ 1 & 0 \end{pmatrix}. \quad (\text{T7.13})$$

Hence

$$\det(TI_4 - F_1) = (T^2 - n)^2, \quad \text{Spec}(F_1) = \{\sqrt{n}, \sqrt{n}, -\sqrt{n}, -\sqrt{n}\}. \quad (\text{T7.14})$$

For $\lambda = \pm\sqrt{n}$, explicit eigenforms are

$$\lambda dx + dx_n, \quad \lambda d\theta + d\theta_n, \quad \lambda \omega_p + \omega_{p^n} \in H^{1,0}. \quad (\text{T7.15})$$

All original period data remains attached to the four forms. Their eigenvalues have modulus $\sqrt{n}$; if n is nonsquare, its two algebraic conjugates are the two displayed signs and have the same modulus. If n is square, they are integers with the stated modulus.

For completeness the entire cohomology ring of this real four-torus is the exterior algebra on the four invariant one-forms. This follows by the same Fourier argument as T2 in four variables: on a nonzero frequency vector k , exterior multiplication by ik is the de Rham differential, and contraction by $k/(i|k|^2)$ satisfies the identity $d\iota + \iota d = 1$. Thus every nonzero Fourier mode is contractible, whereas invariant forms survive by their periods on products of lattice loops. This proves the ring statement, rather than merely applying a dimension count. Since pullback commutes with wedge product,

$$\Phi_{p,n}^*|_{H^k} = \bigwedge^k F_1, \quad \dim H^k = \binom{4}{k}. \quad (\text{T7.16})$$

Every eigenvalue is therefore a product of k eigenvalues in (T7.14), and every one has modulus

$$|\lambda_k| = n^{k/2}, \quad 0 \leq k \leq 4. \quad (\text{T7.17})$$

The complete spectra, including signs and multiplicities, are

Degree	Eigenvalues of the actual product-isogeny pullback
0	1, multiplicity 1
1	$+\sqrt{n}$, multiplicity 2; $-\sqrt{n}$, multiplicity 2
2	$+n$, multiplicity 2; $-n$, multiplicity 4
3	$+n^{3/2}$, multiplicity 2; $-n^{3/2}$, multiplicity 2
4	n^2 , multiplicity 1

For degree two, the two positive products come from choosing both positive eigenlines or both negative eigenlines; the four mixed choices have negative sign. For degree three, the two choices with two negative eigenlines have positive sign, and the two with one negative eigenline have negative sign. The top product is n^2 , agreeing with the degree (T7.3).

The full determinant factors, useful without any loss of signs or multiplicities, are

$$\begin{aligned} \det(I - T\Phi^*|H^0) &= 1 - T, \\ \det(I - T\Phi^*|H^1) &= (1 - nT^2)^2, \\ \det(I - T\Phi^*|H^2) &= (1 - nT)^2(1 + nT)^4, \\ \det(I - T\Phi^*|H^3) &= (1 - n^3T^2)^2, \\ \det(I - T\Phi^*|H^4) &= 1 - n^2T. \end{aligned} \quad (\text{T7.18})$$

In particular choosing $n = p$ gives an actual degree- p^2 polarized self-isogeny of $E_p \times E_{p^p}$, with multiplier p , whose cohomological eigenvalues satisfy the exact powers $p^{k/2}$. The second factor has radial period $p \log p$, angular period 2π , and scale p^p . This construction does not assert that p^p is a prime, that the second curve was already a separate prime stalk, or that Φ is the source's parameter Frobenius.

T8. Torsor origins, real compatibility, and the exact limitation

The source's intrinsic description is a torsor. The group formulas $P_n, V_n, [n]$ above use the displayed model $\mathbb{C}^\times/p^\mathbb{Z}$ with origin $[1]$, and the analogous origin on E_{p^n} . A different identification $z' = cz$ changes the parameter power operation, expressed back in the previous coordinate, to

$$z \mapsto c^{n-1} z^n. \quad (\text{T8.1})$$

This differs from P_n or V_n by a translation in its target. If independent origins are chosen for the two factors, each component changes by a target translation for the same reason. Such translations act trivially on cohomology by T3.1 and preserve the invariant differential and polarization forms. Therefore all cohomological maps, adjoint relations, eigenvalues, degrees, and winding maps proved above are independent of these translated choices. The group equality $\Phi^2 = [n]$ is asserted for the displayed origins; without those origins its affine version can contain an additional translation. No origin-independent identity of torsor points is being silently assumed.

The real structures commute with U_n and V_n because complex conjugation commutes with both $z \mapsto z$ and $z \mapsto z^n$. Hence the product real involution commutes with Φ and with F_1 , and it exchanges the holomorphic and antiholomorphic eigenforms of the same real eigenvalue in (T7.15). In the four real one-form directions it has signs $+, -, +, -$, each with its original radial or angular label.

A proposed degree- n self-map of the single real torus that stretches only one winding direction must be tested against the source complex structure. For example

$$R_n : (x, \theta) \mapsto (nx, \theta) \quad (\text{T8.2})$$

is well defined and has topological degree n . For $n > 1$, however,

$$dR_n(J\partial_x) = \partial_\theta, \quad J(dR_n\partial_x) = n\partial_\theta, \quad (\text{T8.3})$$

so it is not holomorphic. Its H^1 eigenvalues are $n, 1$, and

$$h_p(R_n^*\alpha, R_n^*\alpha) = AB(n^2|a|^2 + |b|^2) \neq nAB(|a|^2 + |b|^2) \quad (\text{T8.4})$$

for general $\alpha = a dx + b d\theta$. Nor can any other positive inner product make this particular operator satisfy a common factor n : applying that proposed identity to its nonzero eigenvector with eigenvalue n would force $n^2 h(v, v) = n h(v, v)$, impossible when $n > 1$. The failure identifies a precise complex-structure defect. T5-T7 repair that defect by retaining the changed radial period in a second curve and constructing an exact holomorphic correspondence and product isogeny. They do not declare the original objects unrelated.

Thus the degree/positivity mechanism is fully realized here on specified complex curves and on a specified product, while the original source Frobenius remains exactly the identity on the single quotient E_p . An additional map from any tau-support object into these source receivers is a separate construction and is neither presumed nor assigned coordinates in this derivation.

Appendix T9. Optional self-endomorphism classification for algebraic p

Source-use status. This appendix invokes the classical Gelfond-Schneider theorem as an external theorem. No original-source proof of that theorem has been acquired or read in this subtask. Its precise statement and full application are given to keep this dependence visible. None of T1-T8, including the product receiver and its weight spectrum, requires this appendix.

The theorem states: if $a \in \overline{\mathbb{Q}} \setminus \{0, 1\}$ and $b \in \overline{\mathbb{Q}} \setminus \mathbb{Q}$, then every value $\exp(bL)$, where $e^L = a$, is transcendental.

Suppose that the positive real number $A/B = \log p/(2\pi)$ were algebraic. Then

$$b = -2i \frac{A}{B} \quad (\text{T9.1})$$

would be algebraic and nonreal, hence irrational. Set $a = -1$ and use its logarithm $L = iB/2 = i\pi$. The full power calculation is

$$\exp(bL) = \exp\left(\left(-2i \frac{A}{B}\right) \left(\frac{iB}{2}\right)\right) = e^A = p. \quad (\text{T9.2})$$

This contradicts Gelfond-Schneider because the source prime p is algebraic. Therefore A/B is transcendental. The argument works for every algebraic real $p > 1$.

If the coefficient r in (T4.4) were nonzero, then $s \neq 0$ and

$$\left(\frac{A}{B}\right)^2 = -\frac{r}{s} \in \mathbb{Q}, \quad (\text{T9.3})$$

which would make A/B algebraic, a contradiction. Hence $r = s = 0$, $c = m \in \mathbb{Z}$, and the exact endomorphism ring is

$$\text{End}_{\text{hol}}(E_p) = \mathbb{Z}. \quad (\text{T9.4})$$

Using the affine classification (T4.1), every holomorphic self-map in the displayed multiplicative model has form $[z] \mapsto [c_0 z^m]$ for some $c_0 \in \mathbb{C}^\times$ and $m \in \mathbb{Z}$. It is constant for $m = 0$, and otherwise its degree is m^2 . For prime p , the integer equation $m^2 = p$ has no solution. Thus E_p has no holomorphic self-map of degree p . For an arbitrary integer $p > 1$, a degree- p self-isogeny exists precisely when p is a perfect square. The actual degree- p maps U_p, V_p of T5 run between E_p and E_{p^p} , and the product construction of T7 gives an actual self-isogeny without contradicting this single-curve classification.

Source synthesis, with complete proof

Proof source: GLOBAL_MELLIN_SYNTHESIS.md.

Global synthesis for the original-zeta Mellin quotient

Independent derivation, 24 September 2026. This note proves the equality left open in M9 of PRIME_MONODROMY_STACKED_HISTORY.md. It uses no RH assumption, no simplicity assumption, no bounded numerical experiment, and no assumed lower bound for zeta inside the critical strip. Addition at source tau is not used. The user's state-layer notation is unchanged.

Source identity and actual reading

The existing machine-readable corpus index was queried before consulting further literature: canonicalresearchliteratureindex(privateroutingdatabase), table canonical_units, for spectral/zeta, spectral synthesis, Hadamard, and explicit-formula routing. Routing records were not treated as proofs. The relevant retained original author source is Alain Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, [arXiv:2602.04022v1](https://arxiv.org/abs/2602.04022v1), original TeX sources/2602.04022v1/author/rhready.tex, SHA256 7f1888d82b42263faca4264f3f9fc77e5750c348640f85db0ac4569fe88ebab2. The author identity and version were checked against arXiv. Original TeX lines 501-546 and 1570-1600 were read; the theorem actually used is Hadamard factorization at lines 526-535. The source's subsequent symmetric-product presentation is not substituted for the full genus-one product used below. Its historical attribution is J. Hadamard, *Essai sur l'étude des fonctions données par leur développement de Taylor*, J. Math. Pures Appl. 4 8 1892, 101-186. That historical article was not newly read in this derivation.

The retained programme sources read here are M8-M9 in full, with the preceding M6-M7 context, and the C6 kernel formula cited there. The full original-zeta multiplier is preserved below, including its factor 1/8. Hadamard factorization is the standard complex-analysis theorem used; the division estimate needed for this particular space is proved here instead of assumed.

S1. The exact function spaces and Mellin isomorphism

Write

$$\mathcal{A} = \left\{ k \in C^\infty(\mathbb{R}_{>0}) : p_{N,j}(k) := \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)| < \infty \quad (N, j \geq 0) \right\}. \quad (\text{S1.1})$$

All indices in these seminorms are nonnegative integers. In the logarithmic coordinate $x = \log u$, put $K(x) = k(e^x)$. The topology is equivalently given by

$$\|K\|_{N,j} = \sup_{x \in \mathbb{R}} e^{N|x|} |K^{(j)}(x)|. \quad (\text{S1.2})$$

The equivalence follows from

$$e^{N|x|} \leq e^{Nx} + e^{-Nx} \leq 2e^{N|x|}.$$

Let $\mathcal{B}$ be the space of entire functions F for which

$$b_{A,M}(F) = \sup_{|\sigma| \leq A, t \in \mathbb{R}} (1 + |t|)^M |F(\sigma + it)| < \infty \quad (A, M \geq 0). \quad (\text{S1.3})$$

The centered Mellin map

$$\mathcal{M}k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u} = \int_{\mathbb{R}} K(x) e^{(s-1/2)x} dx \quad (\text{S1.4})$$

is a linear topological isomorphism $\mathcal{A} \rightarrow \mathcal{B}$.

Here are the estimates and inverse. On each fixed real strip the integral and every derivative in s are absolutely convergent, because an arbitrary polynomial in x times $e^{(\sigma-1/2)x}$ is controlled by a larger seminorm (S1.2). Repeated integration by parts in x , applied to $e^{(\sigma-1/2)x} K(x)$, proves (S1.3), uniformly in the strip. Boundary terms vanish by (S1.2). This proves continuity of the map.

For $F \in \mathcal{B}$, define

$$K(x) = \frac{1}{2\pi} \int_{\mathbb{R}} F(1/2 + it) e^{-itx} dt. \quad (\text{S1.5})$$

For any real c , contour shifting in a finite vertical strip gives

$$K(x) = \frac{e^{-(c-1/2)x}}{2\pi} \int_{\mathbb{R}} F(c + it) e^{-itx} dt. \quad (\text{S1.6})$$

Indeed, the horizontal sides of the rectangle for $F(s)e^{-(s-1/2)x}$ tend to zero by (S1.3), uniformly between its two vertical sides. Differentiation j times inserts the exact factor $-(c - 1/2 + it)^j$ in the integral. For $x \geq 0$ choose $c = 1/2 + N + 1$, and for $x \leq 0$ choose $c = 1/2 - N - 1$. Integrability with $M > j + 1$ proves (S1.2) and continuity of the inverse. Fourier inversion gives (S1.4) on the line $c = 1/2$; the identity theorem gives it everywhere. The sign and constant in (S1.5) are fixed throughout.

Both spaces are complete metrizable locally convex spaces. For $\mathcal{A}$, a Cauchy sequence in all seminorms converges in C^∞ on each compact interval; the weighted uniform limits give the same limit with every seminorm finite and convergence in every seminorm. The corresponding assertion for $\mathcal{B}$ follows from this proved isomorphism (or from locally uniform holomorphic convergence together with the strip bounds).

S2. One retained source kernel, with every factor

Set

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}, \quad k_0(u) = \mathcal{E} f_0(u) = u^{1/2} \sum_{n \geq 1} f_0(nu). \quad (\text{S2.1})$$

The even Schwartz function f_0 satisfies $f_0(0) = 0$ and $\int_{\mathbb{R}} f_0 = 0$. The Gaussian moments give the latter directly:

$$\int v^2 e^{-\pi v^2} dv = \frac{1}{2\pi}, \quad \int v^4 e^{-\pi v^2} dv = \frac{3}{4\pi^2}, \quad \frac{\pi}{2} \left(2\pi \frac{3}{4\pi^2} - 3 \frac{1}{2\pi} \right) = 0.$$

For the Fourier convention $\hat{f}(t) = \int f(v) e^{-2\pi i vt} dv$, the Gaussian identity and differentiation give

$$\widehat{v^2 e^{-\pi v^2}}(t) = \left(\frac{1}{2\pi} - t^2 \right) e^{-\pi t^2},$$

$$\widehat{v^4 e^{-\pi v^2}}(t) = \left(t^4 - \frac{3}{\pi} t^2 + \frac{3}{4\pi^2} \right) e^{-\pi t^2}.$$

Substituting into (S2.1) proves $\hat{f}_0 = f_0$. Poisson summation, with both endpoint terms zero, consequently proves $k_0(u) = k_0(1/u)$. It also proves $k_0 \in \mathcal{A}$, as in M9.

For $\Re s > 1$, the absolutely convergent Mellin calculation is

$$F_0(s) := \mathcal{M}k_0(s) = \zeta(s) \int_0^\infty f_0(v) v^s \frac{dv}{v} = \boxed{\frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s)}. \quad (\text{S2.2})$$

For precision, the integral before the last equality equals

$$\frac{\pi}{2} \left[2\pi \cdot \frac{1}{2} \pi^{-(s+4)/2} \Gamma((s+4)/2) - 3 \cdot \frac{1}{2} \pi^{-(s+2)/2} \Gamma((s+2)/2) \right].$$

Applying $\Gamma(z+1) = z\Gamma(z)$ twice gives exactly the factor in (S2.2). Thus $F_0 = \xi/4$ for the source's definition of ξ ; no factor is absorbed. The Mellin integral makes F_0 entire, and $k_0(u) = k_0(1/u)$ gives $F_0(s) = F_0(1-s)$.

The apparent endpoint and trivial-zero singularities in the displayed multiplier have the following explicit values:

$$F_0(0) = F_0(1) = \frac{1}{8},$$

$$F_0(-2m) = F_0(1+2m) = \frac{(1+2m)(2m)}{8} \pi^{-(1+2m)/2} \Gamma((1+2m)/2) \zeta(1+2m) \neq 0 \quad (m \geq 1). \quad (\text{S2.3})$$

At 1, the zeta residue is 1 and $\pi^{-1/2} \Gamma(1/2) = 1$; the value at 0 follows also by the functional equation. At every negative even integer the reflection gives the positive, finite expression in (S2.3). The Gamma factor has no zeros. Consequently the zeros of F_0 , with their orders, are exactly the nontrivial zeros of the ORIGINAL zeta function. Trivial zeros are cancelled in this auxiliary transform, with their values explicitly retained in (S2.3); S6 below recovers their full derivative data in the approximating source functions.

We also need the order of F_0 , for which no zero-distribution hypothesis is required. There exist $B, c, C > 0$ such that

$$|k_0(u)| \leq C u^B e^{-cu^2} \quad (u \geq 1).$$

This follows by summing the Gaussian series (S2.1), retaining a smaller positive Gaussian exponent to absorb its polynomial factors. Reflection supplies the other endpoint. Thus for $|s| \leq R$,

$$|F_0(s)| \leq 2C \int_1^\infty u^{B+R+1/2} e^{-cu^2} \frac{du}{u} \leq \exp(C_1(R+2) \log(R+2)). \quad (\text{S2.4})$$

The last bound follows by the Gamma integral after $y = cu^2$; alternatively its maximum at y proportional to $R+2$ and the remaining exponential tail give the same bound. Thus F_0 has order at most 1. It is not zero, since $F_0(2) = \pi/24$.

Jensen's formula at radius $(2R)$, using $F_0(0) = 1/8 \neq 0$, now gives a constant C_2 such that the zero count, with multiplicities, obeys

$$n(R) \leq C_2(R+2)^{3/2} \quad (R \geq 1). \quad (\text{S2.5})$$

Indeed each zero of modulus at most R contributes at least $\log 2$ to Jensen's sum at $(2R)$, and (S2.4) is bounded by a constant times $(R+2)^{3/2}$. Hadamard's factorization theorem in the retained original TeX then gives

$$F_0(s) = e^{a+bs} \prod_\rho \left(1 - \frac{s}{\rho}\right) e^{s/\rho}, \quad e^a = \frac{1}{8}, \quad b = \frac{F'_0(0)}{F_0(0)}. \quad (\text{S2.6})$$

The product retains every repeated zero and every exponential factor. The value of b is exactly $\frac{1}{2} \log(4\pi) - 1 - \frac{1}{2}\gamma$, as follows by expanding all factors of (S2.2) at 0; no value of b is needed to discard a term below. Formula (S2.5) implies convergence of $\sum_{\rho} |\rho|^{-2}$, so the product in (S2.6) converges absolutely on compact subsets after its genus-one factors are grouped as displayed.

S3. Global division with a proved subquadratic growth bound

Let $F \in \mathcal{B}$ vanish at every zero ρ of F_0 to at least the exact multiplicity m_{ρ} . Then

$$Q(s) = F(s)/F_0(s) \quad (\text{S3.1})$$

extends to an entire function. This follows from Taylor division at each zero; the extensions agree with the quotient on punctured neighborhoods.

For every $A \geq 0$ there are constants $C_{A,F}, C > 0$ such that

$$\sup_{|\sigma| \leq A} |Q(\sigma + it)| \leq C_{A,F} \exp(C(|t| + 2)^{3/2} \log(|t| + 2)), \quad t \in \mathbb{R}. \quad (\text{S3.2})$$

The supremum is over the real coordinate at the fixed imaginary coordinate t . This incorporates the independently verified notation correction in GLOBAL_MELLIN_SYNTHESIS_REVIEW.md R1.1. The exponent is strictly subquadratic. This is a statement on the whole unbounded strip, not a finite interval estimate.

We prove the bound including neighborhoods of arbitrarily close or multiple zeros. Fix a large R . Around each zero with $|\rho| \leq 8R$, take the disk of radius R^{-2} , and let D_R be their union. By (S2.5) the sum of the radii, counted with multiplicity if necessary, is at most $CR^{-1/2}$. For sufficiently large R , each connected component of this finite union has diameter less than 1: a chain of intersecting disks joining two points has total diameter at most twice the sum of the radii in that chain.

For $R/2 \leq |s| \leq 3R$ outside D_R , use (S2.6). For the factors with $|\rho| \leq 8R$,

$$\left| 1 - \frac{s}{\rho} \right| = \frac{|s - \rho|}{|\rho|} \geq \frac{1}{8R^3}. \quad (\text{S3.3})$$

Their logarithms contribute at least $-CR^{3/2} \log R$. The exponential parts contribute at least

$$-|s| \sum_{|\rho| \leq 8R} |\rho|^{-1} \geq -CR^{3/2}. \quad (\text{S3.4})$$

The last inequality follows by partial summation of (S2.5); the finitely many small zeros are harmless because $F_0(0) \neq 0$, so their moduli have a positive lower bound.

For $|\rho| > 8R$, the ratio $w = s/\rho$ has $|w| \leq 3/8$. The absolutely convergent logarithmic series gives

$$\log |(1 - w)e^w| = \Re \left(- \sum_{j \geq 2} \frac{w^j}{j} \right) \geq -C|w|^2.$$

Partial summation again gives

$$\sum_{|\rho| > 8R} |\rho|^{-2} \leq CR^{-1/2}.$$

Therefore the tail contributes at least $-CR^{3/2}$. Finally e^{a+bs} contributes at least $-|a| - 3|b|R$. Altogether,

$$|F_0(s)| \geq \exp(-CR^{3/2} \log R) \quad (R/2 \leq |s| \leq 3R, s \notin D_R). \quad (\text{S3.5})$$

Now take $R \leq |s| \leq 2R$ with $|\Re s| \leq A$. Outside D_R , (S3.5) and the boundedness of F on the strip give the desired bound for Q . Inside D_R , the whole component containing s stays within distance 1 of s . Its boundary lies in the strip $|\Re z| \leq A + 1$, in the larger annulus used in (S3.5), and outside the open disk union. The same bound holds for Q on that boundary. Since Q is holomorphic even at all enclosed zeros, the maximum-modulus principle gives the bound inside. No zero spacing or simplicity has been assumed. Taking dyadic R proves (S3.2); the remaining compact region is bounded by continuity.

S4. Gaussian division produces actual source functions

For every $\varepsilon > 0$, define the entire function

$$Q_\varepsilon(s) = e^{\varepsilon(s-1/2)^2} Q(s). \quad (\text{S4.1})$$

On the strip $|\Re s| \leq A$,

$$|e^{\varepsilon(s-1/2)^2}| \leq e^{\varepsilon(A+1/2)^2} e^{-\varepsilon(\Im s)^2}.$$

Together with (S3.2), this proves $Q_\varepsilon \in \mathcal{B}$: for every fixed strip and every polynomial power, the negative quadratic dominates $C|t|^{3/2} \log |t|$. The source preimage $q_\varepsilon \in \mathcal{A}$ is therefore explicitly

$$q_\varepsilon(e^x) = \frac{1}{2\pi} \int_{\mathbb{R}} Q_\varepsilon(1/2 + it) e^{-itx} dt. \quad (\text{S4.2})$$

Let multiplicative convolution be

$$(k * q)(u) = \int_0^\infty k(u/a) q(a) \frac{da}{a}. \quad (\text{S4.3})$$

All such integrals and their logarithmic derivatives converge in $\mathcal{A}$, because in logarithmic coordinate

$$e^{N|x|} \leq e^{N|x-y|} e^{N|y|}$$

and the second factor is integrable against a seminorm with one additional exponential weight. The Mellin transform of (S4.3) is the product of Mellin transforms, with Fubini justified by the same bounds.

Define, retaining the full unitary factor of the source dilation,

$$U_a f(v) = a^{-1/2} f(v/a), \quad f_\varepsilon(v) = \int_0^\infty q_\varepsilon(a) U_a f_0(v) \frac{da}{a}. \quad (\text{S4.4})$$

This is an actual even Schwartz function. For a Schwartz seminorm $\sup_v |v|^r |\partial_v^j f(v)|$, substitution $v = aw$ bounds the corresponding seminorm of $U_a f_0$ by

$$a^{r-j-1/2} \sup_w |w|^r |f_0^{(j)}(w)|.$$

Every such power is integrable against $q_\varepsilon(a) da/a$, by its arbitrary endpoint decay. This proves convergence in every Schwartz seminorm and permits all required differentiations. Evenness is preserved. Moreover

$$f_\varepsilon(0) = 0, \quad \int_{\mathbb{R}} f_\varepsilon(v) dv = \int_0^\infty q_\varepsilon(a) a^{1/2} \frac{da}{a} \int_{\mathbb{R}} f_0(w) dw = 0. \quad (\text{S4.5})$$

Absolute convergence follows also from $\|U_a f_0\|_1 = a^{1/2} \|f_0\|_1$. Thus $f_\varepsilon \in \mathcal{S}_0^{\text{even}}$, the actual source space in M9.

The continuity $\mathcal{E} : \mathcal{S}_0^{\text{even}} \rightarrow \mathcal{A}$ follows directly from the estimates already used in M9. More explicitly, at $u \geq 1$ the differentiated series is bounded using a sufficiently high Schwartz seminorm and the convergent

sum $\sum n^{-M}$, $M > 1$; at $u \leq 1$, Poisson gives $\mathcal{E}f(u) = \mathcal{E}\hat{f}(1/u)$, and Fourier transformation is continuous on Schwartz space. Consequently (S4.4), or direct absolute summation, gives the exact receiver identity

$$\boxed{\mathcal{E}f_\varepsilon = k_0 * q_\varepsilon.} \quad (\text{S4.6})$$

It has Mellin transform

$$\mathcal{M}\mathcal{E}f_\varepsilon(s) = F_0(s)Q_\varepsilon(s) = e^{\varepsilon(s-1/2)^2}F(s). \quad (\text{S4.7})$$

Every source, domain and multiplier in this construction is specified; the argument has not inserted a hypothetical division preimage.

S5. Exact global closure equality

Let

$$\mathcal{J}_\zeta = \{k \in \mathcal{A} : (\mathcal{M}k)^{(j)}(\rho) = 0 \text{ for every nontrivial zero } \rho \text{ and every } 0 \leq j < m_\rho\}. \quad (\text{S5.1})$$

Then

$$\boxed{\overline{\mathcal{E}(\mathcal{S}_0^{\text{even}})}^{\mathcal{A}} = \mathcal{J}_\zeta.} \quad (\text{S5.2})$$

The inclusion from left to right is M9's original-zeta Mellin identity and continuity of every jet. For completeness, on $\Re s > 1$,

$$\mathcal{M}\mathcal{E}f(s) = \zeta(s) \int_0^\infty f(v)v^s \frac{dv}{v}.$$

The last integral is holomorphic on $\Re s > -2$, since f is even and $f(0) = 0$. Its value at 1 is zero, cancelling the simple pole of zeta. Continuation into the critical strip therefore proves vanishing at every original nontrivial-zero jet of the required orders. Those functionals are continuous on $\mathcal{A}$, so the closed image is contained in (S5.1).

For the reverse inclusion, take any $k \in \mathcal{J}_\zeta$, let $F = \mathcal{M}k$, and construct f_ε by S3-S4. Formula (S4.7) is the transform of logarithmic Gaussian convolution. Precisely, put

$$g_\varepsilon(x) = \frac{1}{\sqrt{4\pi\varepsilon}} e^{-x^2/(4\varepsilon)}. \quad (\text{S5.3})$$

The full Gaussian integral gives

$$\int_{\mathbb{R}} g_\varepsilon(x) e^{(s-1/2)x} dx = e^{\varepsilon(s-1/2)^2}.$$

Thus Mellin injectivity proves

$$(\mathcal{E}f_\varepsilon)(e^x) = (g_\varepsilon * K)(x), \quad K(x) = k(e^x). \quad (\text{S5.4})$$

These convolutions tend to K in every seminorm (S1.2). Here is a weighted proof, so that a weaker L^2 convergence is not substituted. For each (N,j), the fundamental theorem of calculus gives

$$\|K(\cdot - y) - K\|_{N,j} \leq |y| e^{N|y|} \|K\|_{N,j+1}. \quad (\text{S5.5})$$

Indeed write the derivative difference as the integral of $K^{(j+1)}(x-r)$ from $r = 0$ to $r = y$, and use $e^{N|x|} \leq e^{N|x-r|} e^{N|r|}$. Therefore

$$\|g_\varepsilon * K - K\|_{N,j} \leq \|K\|_{N,j+1} \int_{\mathbb{R}} g_\varepsilon(y) |y| e^{N|y|} dy \longrightarrow 0. \quad (\text{S5.6})$$

After $y = \sqrt{\varepsilon}z$, the last integral is bounded for $0 < \varepsilon \leq 1$ by $\sqrt{\varepsilon}$ times the finite integral of $|z|e^{N|z|}e^{-z^2/4}/\sqrt{4\pi}$. This proves the asserted limit. Every approximant (S5.4) lies in the actual image by (S4.4)–(S4.6), proving the reverse inclusion globally.

The parameter $\varepsilon \downarrow 0$ is an explicit approximation to the identity on the entire space. It is not a restriction to finitely many zeros, a finite prime range, or an assumed convergence of a numerical experiment.

S6. Exceptional points, source jets and endpoint recovery

The full construction retains the information at the places cancelled by the auxiliary factor F_0 . Put

$$F_\varepsilon(s) = e^{\varepsilon(s-1/2)^2} F(s).$$

The original, uncompleted zeta relation is

$$\int_0^\infty f_\varepsilon(v) v^s \frac{dv}{v} = \frac{F_\varepsilon(s)}{\zeta(s)} \quad (\Re s > 1), \quad (\text{S6.1})$$

with meromorphic continuation retaining all terms.

For an even Schwartz function with $f(0) = 0$, subtracting its Taylor polynomial on $0 < v < 1$ proves that its Mellin integral extends meromorphically to the whole plane. The only possible poles are simple poles at $(-2m)$, $m \geq 1$, and their residues are $f^{(2m)}(0)/(2m)!$. Indeed, the subtracted remainder $O(v^{2M+2})$ gives a holomorphic integral on $\Re s > -2M - 2$, while each subtracted monomial contributes exactly $f^{(2m)}(0)/((2m)!(s + 2m))$. As M increases these continuations agree on their common domains.

At the simple trivial zero $(-2m)$, (S6.1) consequently gives

$$\boxed{\frac{f_\varepsilon^{(2m)}(0)}{(2m)!} = \frac{e^{\varepsilon(-2m-1/2)^2} F(-2m)}{\zeta'(-2m)} \quad (m \geq 1).} \quad (\text{S6.2})$$

This can also be checked directly from (S4.4): its $(2m)$ -th derivative at 0 multiplies the coefficient of f_0 by $Q_\varepsilon(-2m)$; (S2.2) identifies that coefficient as $F_0(-2m)/\zeta'(-2m)$. The trivial zeros are simple because the sine factor in the zeta functional equation has a simple zero there, whereas $\Gamma(1 + 2m)\zeta(1 + 2m)$ is finite and nonzero. Equivalently, the finite nonzero value in (S2.3), together with the simple Gamma pole, gives the same conclusion.

At $s = 1$, since $\zeta(s) = (s - 1)^{-1} + O(1)$, the Mellin integral in (S6.1) has the exact expansion

$$\int_0^\infty f_\varepsilon(v) v^s \frac{dv}{v} = (s - 1)F_\varepsilon(1) + O((s - 1)^2). \quad (\text{S6.3})$$

Its value is the required zero integral, and its first derivative records $F_\varepsilon(1)$. At $s = 0$, $\zeta(0) = -1/2$ gives

$$\int_0^\infty f_\varepsilon(v) \frac{dv}{v} = -2F_\varepsilon(0). \quad (\text{S6.4})$$

The integral converges because $f_\varepsilon(v) = O(v^2)$. No endpoint value of the receiver has been discarded merely because the source has zero endpoint constraints.

S7. The resulting faithful receiver and its remaining mathematical role

By (S5.2), the joint actual-zero jet map has exactly the original closed image as its kernel:

$$J : \mathcal{A} \longrightarrow \prod_{\rho} \mathbb{C}^{m_{\rho}}, \quad J(k) = ((\mathcal{M}k)^{(j)}(\rho))_{\rho, 0 \leq j < m_{\rho}},$$

$$\ker J = \overline{\mathcal{E}(\mathcal{S}_0^{\text{even}})}^{\mathcal{A}}. \quad (\text{S7.1})$$

Thus J induces an injective linear map

$$\boxed{\mathcal{Q} \hookrightarrow \prod_{\rho} \mathbb{C}^{m_{\rho}}}. \quad (\text{S7.2})$$

Its image is $J(\mathcal{A})$, with the quotient topology if a topological isomorphism is desired. No claim that this image is the entire product, or is closed in the product topology, is made. The conclusion proved here is stronger than the prior existence of some descending zero functionals: **all actual zero jets together now distinguish every class in the quotient**. There is no additional quotient class invisible to every original zeta zero and its retained multiplicities.

The scaling action remains exactly

$$J_{\rho,j}(V_a k) = a^{\rho-1/2} \sum_{\ell=0}^j \binom{j}{\ell} (\log a)^{j-\ell} J_{\rho,\ell}(k). \quad (\text{S7.3})$$

Thus the quotient has been identified using the actual zeta spectral data, including its full triangular multiplicity action. This theorem does not force $\Re \rho = 1/2$: it does not insert a positive definite norm on this quotient, and it does not assert that (S7.3) is unitary. M9's proof that the image is dense in the ordinary $L^2(du/u)$ space still holds. The faithful jet receiver (S7.2), its finer topology, and the zero Hilbert quotient are all retained with their exact maps.

Finally, let C_{rec} and the free support space $\mathbb{C}^{(\mathcal{L})}$ be exactly those of M9.7. Tensoring (S7.2) algebraically with these coefficient spaces gives an injective linear map: a finite basis expansion reduces the assertion to injectivity of J in each coordinate. This statement concerns the linear spaces after passage to tensors. That preceding passage identifies $0_{\mathcal{Q}} \otimes e_{\ell}$ for every label, so it does not retain a support record at coefficient zero. The two coefficient sections are distinct maps, with the same zero value.

The exact repair is the disjoint family $\widehat{\mathcal{Q}} = \coprod_{\ell \in \mathcal{L}} \{\ell\} \times \mathcal{Q}$, with $\widehat{J}(\ell, q) = (\ell, Jq)$. This is injective because equality first retains the label and then determines the coefficient through S7.2. Its coefficient-zero fibre retains every $(\ell, 0_{\mathcal{Q}})$. The map $(\ell, q) \mapsto q \otimes e_{\ell}$ is the explicit comparison to the old receiver: its zero fibre is that entire labelled family and every nonzero fibre in its image is a singleton. All arithmetic operators and projectors act on q while retaining the label. Section selections through $[\tau]$ and $([1])$ can be included in the existing record as proved in SUPPORT_ZERO_RECEIVER_REPAIR.md ZR5. ZR1-ZR5 supply the complete proofs and the exact full-pairing map, incorporating the receiving task's I26.10 correction. No source addition or integer parity is assigned to τ .

The proof diagram is

$$\begin{array}{ccc} F \in \mathcal{B}, J(F) = 0 & \xrightarrow{/F_0} & \mathcal{Q} \text{ entire } (\log |Q| = O_A(|t|^{3/2} \log |t|)) \\ \downarrow \times e^{\varepsilon(s-1/2)^2} & & \downarrow \times e^{\varepsilon(s-1/2)^2} \\ F_{\varepsilon} = F_0 Q_{\varepsilon} & \xleftarrow{\mathcal{M}} & \mathcal{E} f_{\varepsilon} = k_0 * q_{\varepsilon}, \end{array}$$

with $F_{\varepsilon} \rightarrow F$ in $\mathcal{B}$ proved through the equivalent $\mathcal{A}$ convergence (S5.6). Every arrow is constructed above on its full domain.

Subsequent exact-image theorem, 24 September 2026

The complete proof in [EXACT_SCHWARTZ_SUMMATION_IMAGE.md](#), SSI1-SSI7, independently checked in ESI0-ESI10, strengthens S5 to the actual equality $E(S)=I_{\zeta}$ and proves a continuous inverse. Its raw summation convention is $\sum_{f=2} f(\nu)$; the exact centered comparison is SSI6.4. Thus S4's Gaussian source approximants converge in Schwartz space to the actual inverse, not only after summation. S1-S7 above remain the complete historical proof of the closure result and its dependencies. SSI8-SSI10 propagate the strengthened equality through ordinary sheaf cohomology and both stated source conventions, retaining every endpoint. This proves no weight-separation or RH statement.

Source Mellin involution and Weil-form comparison

Proof source: GLOBAL_CANONICAL_MELLIN_INVOLUTION.md.

Canonical Mellin scale, the actual zero characters, and the sphere involutions

24 September 2026. This is a global derivation, with no finite-range zero experiment. It continues M1-M9 in PRIME_MONODROMY_STACKED_HISTORY.md. The source element tau retains the user's Z_1 presence and no integer parity; no addition at tau is used. Auxiliary convolution algebras of functions and their complex coefficients are named as such. They are not new operations on source tau.

The user specifically requests the involution pictures in Thomas Ponweiser, *Computer Algebra and Analysis: Complex Variables Visualized*, revised 17 June 2014. Printed pages 15-19, PDF pages 29-33, including Figures 1.2-1.3, were read and visually inspected. This explicit PDF request governs this reading; it does not change the original-TeX practice for the Connes papers. The disk corpus index returned no matching title entry. The source CC.tex is the byte-verified author source of Connes-Consani, [The Absolute Twistor Line and the Geometry of the compactified Spec Z, 2609.00299v1](#). The E-map and Fourier convention are retained from Alain Connes, [2602.04022v1](#), with the factor correction proved in C6. No priority claim is made for the convolution-character method.

G1. The family of scales and its exact Fourier defect

Let S_0^{even} be the even Schwartz functions on $\mathbb{R}$ satisfying $f(0)=0$ and $\int_{\mathbb{R}} f=0$. Use the Fourier transform $Ff(y)=\int_{\mathbb{R}} f(x) \exp(2\pi ixy) dx$. On even functions F^2 is the identity. For real c define

$$E_c f(u) = u^c \sum_{n \geq 1} f(nu), \quad U_{a,c} f(x) = a^{-c} f(x/a), \quad V_a k(u) = k(u/a), \quad a, u > 0. \quad (\text{G1.1})$$

The original sum, its measure and all powers are retained. Substitution gives $E_c U_{(a,c)} = V_a E_c$. Poisson summation on the full integer lattice gives

$$f(0) + 2 \sum_{n \geq 1} f(nu) = u^{-1} \left(Ff(0) + 2 \sum_{n \geq 1} Ff(n/u) \right). \quad (\text{G1.2})$$

Both displayed endpoint terms are zero on this stated domain. Consequently

$$E_c f(u) = u^{2c-1} E_c (Ff)(1/u). \quad (\text{G1.3})$$

Thus the untwisted inversion $I k(u)=k(1/u)$ intertwines F and E_c exactly for $c=1/2$. To prove necessity rather than merely sufficiency, use the retained nonzero self-Fourier Gaussian polynomial f_0 from C6. $E_c f_0$ is continuous and nonzero on an open interval, since its Mellin transform is not identically zero. If both the untwisted equation and G1.3 hold, $u^{2c-1}=1$ on that interval, forcing $2c-1=0$. The inversion defect at any other real scale is precisely the multiplier u^{2c-1} , not an omitted term.

Independently, direct change of variables gives

$$\|U_{a,c}f\|_{L^2(dx)}^2 = a^{1-2c}\|f\|_{L^2(dx)}^2. \quad (G1.4)$$

Unitary dilation for every $a>0$ therefore forces the same real $c=1/2$. If a complex exponent is allowed, unitarity fixes $\text{Re}(c)=1/2$ only; its imaginary part is a phase character. Requiring the untwisted Fourier inversion in G1.3 removes that phase as well. These calculations identify the scale from the original measure and Fourier operation. They do not infer purity of a quotient spectrum from unitarity of the source space.

Write $E=E_{1/2}$. The exact Mellin convention throughout this note is

$$Mk(s) = \int_0^\infty k(u)u^{s-1/2}\frac{du}{u}. \quad (G1.5)$$

For other c the same arithmetic parameter s is retained by using the exponent $s-c$ with E_c . Changing the display of this exponent never changes the original zeta argument.

G2. The convolution algebra and the source ideal

Let A be the space in M9: k is smooth on $R_{>0}$ and all seminorms

$$p_{N,j}(k) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)| \quad (G2.1)$$

are finite, for nonnegative integers N,j . In the coordinate $v=\log(u)$, $K(v)=k(e^v)$ and every derivative decays faster than every exponential in both directions. Define

$$(k * l)(u) = \int_0^\infty k(a)l(u/a)\frac{da}{a}, \quad k^\#(u) = \overline{k(1/u)}. \quad (G2.2)$$

The logarithmic convolution is ordinary additive convolution. The inequalities $\exp(N|v|) \leq \exp(N|w|)\exp(N|v-w|)$ and integral $\int \exp(-|w|)dw=2$ give a bound of each weighted derivative supremum of $K*L$ by twice the corresponding weighted supremum of L times a weighted supremum of K with index $N+1$. Reversing the factors bounds either derivative placement. This proves convergence, differentiation under the integral and joint continuity. Fubini with the same bounds proves associativity and commutativity. Substitution $v \rightarrow -v$ proves that $\#$ is a continuous conjugate-linear involution preserving the convolution product.

The image $E(S_0^{\text{even}})$ is an ideal of A . The exact lift of $k*E(f)$ is

$$f_k(x) = \int_0^\infty k(a)a^{-1/2}f(x/a)\frac{da}{a}, \quad E(f_k) = k * E(f). \quad (G2.3)$$

For the Schwartz seminorm $q_{(M,j)}(f)=\sup_x (1+|x|)^M |f^{(j)}(x)|$ one has

$$q_{M,j}(a^{-1/2}f(\cdot/a)) \leq a^{-j-1/2} \max(1, a)^M q_{M,j}(f). \quad (\text{G2.4})$$

The weighted rapid decay of k at both endpoints makes the integral of every right side finite. Hence f_k is Schwartz, even, and $f_k(0)=0$. Absolute integration and $x=ay$ give $\int f_k(a)a^{1/2}da/a(\int f)=0$. For fixed $u>0$, absolute interchange in E follows by a Schwartz bound on the sum of $|f(nu/a)|$: for $a \geq u$ it is at most a constant times a/u , while for $a \leq u$ it is at most a constant times $(a/u)^M$, $M>1$. Multiplication by $|k(a)|a^{-1/2}$ is integrable in both ranges. This proves G2.3 on its entire stated domain.

Conjugation commutes with E , and G1.3 at $c=1/2$ gives

$$(Ef)^\# = E(\overline{Ff}). \quad (\text{G2.5})$$

The Fourier transform preserves the even Schwartz domain and exchanges its two vanishing endpoints. Thus the image is $\#$ -stable. Its closure J in A is a closed convolution ideal, stable under $\#$, and the quotient

$$Q = A/J \quad (\text{G2.6})$$

is a commutative locally convex involutive algebra. The dilations V_a preserve J and descend to Q by G1.1. No identity element of A or Q has been asserted. V_a is the convolution multiplier induced by a translated Dirac measure; it is not generally an algebra homomorphism, since $(V_a k)(V_a l) = V_{a^2}(kl)$.

G3. Every continuous character is a full Mellin observation

A character means a nonzero continuous complex-linear multiplicative functional. We classify all such characters on A without presupposing a zeta spectrum. Work in logarithmic coordinates and let $T_b K(v) = K(v-b)$. Choose G with $\chi(G) \neq 0$. The identity

$$(T_b K) * G = K * (T_b G) \quad (\text{G3.1})$$

implies $\chi(T_b K) = \psi(b)\chi(K)$, where $\psi(b) = \chi(T_b G)/\chi(G)$. It follows by applying this twice and taking $K=G$ that $\psi(b+c) = \psi(b)\psi(c)$, $\psi(0)=1$, and ψ has no zeros. Translation is smooth in the topology of A : difference quotients converge in each weighted derivative seminorm by the integral mean-value formula and a bound on the next two derivatives. Thus ψ is differentiable. Differentiating its group law gives $\psi'(b) = \lambda \psi(b)$, with $\lambda = \psi'(0)$, hence $\psi(b) = \exp(\lambda b)$.

In A itself,

$$K * G = \int_{\mathbb{R}} K(b) T_b G db. \quad (\text{G3.2})$$

This integral converges in every seminorm: the seminorm of $T_b G$ is at most $\exp(N|b|)$ times that of G , and K decays faster than $\exp(-(N+1)|b|)$. Completeness follows directly by taking weighted uniform limits of all derivatives and compatibility under differentiation; hence the integral belongs to A . Applying the continuous character to G3.2 and dividing by $\chi(G)$ proves

$$\chi(K) = \int_{\mathbb{R}} K(b) e^{\lambda b} db = M k(1/2 + \lambda). \quad (\text{G3.3})$$

Conversely every complex λ gives a continuous functional by the same weighted bounds, and Fubini proves multiplicativity. It is nonzero by a compactly supported bump with a compensating phase. Distinct

lambda give distinct functionals because their exponential densities differ on some open interval. Thus $s \rightarrow (k \rightarrow Mk(s))$ is a bijection from C to the continuous character set of A .

G4. The quotient characters are exactly the original nontrivial zeros

M9 proves that every actual nontrivial zero ρ of the original zeta annihilates $E(S_0^{\text{even}})$, and hence J . Conversely, a character of Q pulls back to an A -character, so G3 identifies it with evaluation at a unique s . It annihilates the particular element $k_0 = E(f_0)$, whose full transform is

$$Mk_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (G4.1)$$

This is an evaluated transform with every factor retained, not a replacement for the original zeta. We check its exceptional points explicitly. At $s=1$ the factor $s-1$ cancels the simple original pole of residue 1 and gives $Mk_0(1)=1/8$. At $s=0$, $\Gamma(s/2)$ has leading term $2/s$, $\zeta(0)=-1/2$, and $Mk_0(0)=1/8$. At $s=-2m$, $m \geq 1$, the Gamma factor has residue $2(-1)^m/m!$, while the original zeta has a simple zero. The retained product value is

$$Mk_0(-2m) = \frac{m(2m+1)(-1)^m \pi^m}{2m!} \zeta'(-2m) \neq 0. \quad (G4.2)$$

The simplicity and nonzero derivative here follow directly from the original functional equation: $\sin(\pi s/2)$ has a simple zero, while $2^s \pi^{s-1} \Gamma(1-s) \zeta(1-s)$ is finite and nonzero at $-2m$. At all other points outside the nontrivial-zero set the retained factors are finite and nonzero, using the classical zero-free regions $\text{Re}(s) \geq 1$ and the reflected region, as in the programme's original-zeta source. Therefore the zeros of G4.1 are exactly the original nontrivial zeros with their full multiplicities.

It follows that

$$\boxed{\text{Char}_{\text{cont}}(Q) = \{[k] \mapsto Mk(\rho) : \zeta(\rho) = 0, 0 < \Re \rho < 1\}.} \quad (G4.3)$$

There are no extra continuous characters in this constructed quotient. This assertion does not by itself identify the entire quotient with an unrestricted product of local jets, nor assume simplicity. Every original trivial zero and pole remains in the comparison G4.1–G4.2 and the full Weil receiver.

G5. The exact involution, scaling, and Weil trace on this quotient

Changing u to $1/u$ in G1.5 gives, with every conjugation retained,

$$M(k^\#)(s) = \overline{Mk(1-\bar{s})}, \quad M(V_a k)(s) = a^{s-1/2} Mk(s). \quad (G5.1)$$

Thus the actual character involution is $\rho \rightarrow \rho^\# = 1-\bar{\rho}$, and dilation has character value $a^{(\rho-1/2)}$. M9 already retains all differentiated jet terms. These are relations on the actual original-zeta quotient Q , established through E , rather than relations on an unrelated unit circle representation.

For k, l in A define the trace pairing

$$B_Q([k], [l]) = \sum_{\rho} m_{\rho} \overline{Mk(\rho^\#)} Ml(\rho). \quad (G5.2)$$

It is well defined on Q since J annihilates every displayed value. For uniform convergence in the whole strip, write $v = \log u$ and integrate by parts j times in $Mk(s)$. The L_1 norms of derivatives of $\exp((\operatorname{Re}(s)-1/2)v)K(v)$ are bounded uniformly for $0 \leq \operatorname{Re}(s) \leq 1$ by finitely many $p_-(N, j)$, taking $N \geq 2$. Thus $|Mk(s)| \leq C_j(1 + |\operatorname{Im}(s)|)^{-j}$, uniformly in that strip. The known unconditional zero count $O(R \log(R+2))$, retained from the original-zeta reconstruction, proves absolute convergence and continuity of G5.2. Reflection preserves multiplicity, so the form is Hermitian. G5.1 gives exact dilation invariance because

$$\overline{a^{\rho\#-1/2}} a^{\rho-1/2} = 1. \quad (G5.3)$$

No positivity is used in this calculation. The map $[k] \rightarrow (Mk(\rho))$ has values in the existing $\ell^2(Z, m)$ space of NI5 and intertwines B_Q with its original reflection form. This proves the connecting map to the previously constructed signed trace; the NI local radical calculation is not rediscovered as a new result here.

The original compact Weil tests sit here by the precise orientation

$$k_f(u) = f(-\log u), \quad Mk_f(s) = H_f(s), \quad k_{f\#*g} = k_f^\# * k_g, \quad (G5.4)$$

where $f\#(v) = \overline{f(-v)}$ and $H_f(s) = \int f(v) \exp(-(s-1/2)v) dv$. Substitution $v = -\log u$ proves the first two identities. The change $a = e^{(-w)}$ in convolution proves the third, including the reversed integration endpoints. Therefore $B_Q([k_f], [k_g])$ is exactly $W_{\text{zeta}}(f\#*g)$, with all endpoint, Gamma and history-weight terms M7.1-M7.3 retained by that equality. Tensoring this map with the two distinct coefficient records $[\tau]$, $[1]$ and every support basis vector gives M9.7 and WU1-WU8. It does not identify the two records or remove the supported-zero contribution.

G6. The book's disk/sphere maps and the precise CC comparison

Ponweiser's equations (1.20)-(1.21) give the inverse stereographic map

$$P^{-1}(z) = \frac{(2\Re z, 2\Im z, |z|^2 - 1)}{1 + |z|^2}, \quad P^{-1}(\infty) = (0, 0, 1), \quad P(X, Y, Z) = \frac{X + iY}{1 - Z}. \quad (G6.1)$$

Its numerator norm squared is $(1 + |z|^2)^2$, and substitution proves that the maps are inverse, including the two poles. The disk $|z| < 1$ maps to the southern hemisphere $Z < 0$; its boundary maps to the equator. Example 1.41 gives the involution $f(z) = 1/\bar{z}$. Direct substitution in G6.1 yields

$$P^{-1}(1/\bar{z}) = \operatorname{diag}(1, -1, -1) P^{-1}(z). \quad (G6.2)$$

This is the half-turn in Figure 1.2. The zero and infinity are exchanged. The fixed points are $z = +1, -1$. Example 1.43's modified Cayley map $\Phi(z) = (iz+1)/(z+i)$ satisfies $\Phi(\Phi(z)) = 1/z$ by direct rational substitution; it carries the upper half-plane to the disk. It is the quarter-turn shown in Figure 1.3. It is not being silently identified with an order-two transformation.

Complex conjugation $C(z) = \bar{z}$ acts on the sphere by $\operatorname{diag}(1, -1, 1)$, while $S(z) = -z$ acts by $\operatorname{diag}(-1, -1, 1)$. The new CC paper's real structure is $\sigma = S \circ C$. Consequently

$$\sigma(z) = -1/\bar{z}, \quad P^{-1}\sigma = -P^{-1}; \quad \kappa(z) = 1/\bar{z}, \quad P^{-1}\kappa = \operatorname{diag}(1, 1, -1) P^{-1}. \quad (G6.3)$$

These are exact compositions, not claimed holomorphic conjugacies. Sigma is antipodal and has no fixed sphere point. Kappa is equatorial reflection and fixes the complete unit circle. The source minus sign is

the explicitly retained S operation between them. This proves the relation of the book's involution to the CC real structure instead of treating their differing fixed sets as an absence of a relation.

G7. Keeping the prime weight on the sphere

For every prime p define the bijection from the extended z -plane to the unit sphere

$$P_p^{-1}(z) = \frac{(2\sqrt{p}\Re z, 2\sqrt{p}\Im z, |z|^2 - p)}{|z|^2 + p}, \quad P_p(X, Y, Z) = \sqrt{p} \frac{X + iY}{1 - Z}. \quad (\text{G7.1})$$

This is the explicitly invertible comparison $P^{-1}(z/\sqrt{p})$; the original unscaled z is returned by P_p . Squaring the numerator gives $4p|z|^2 + (|z|^2 - p)^2 = (|z|^2 + p)^2$. The weight-one involution $\iota_p(z) = p/\bar{z}$ acts by $Z \rightarrow -Z$ with X, Y unchanged. For the original coordinate $z = p^s$, $s = \sigma_0 + it$, put $v = (\sigma_0 - 1/2)\log p$ and $\theta = t \log p$. Direct substitution yields the full exact point

$$P_p^{-1}(p^s) = \left(\frac{\cos \theta}{\cosh v}, \frac{\sin \theta}{\cosh v}, \tanh v \right). \quad (\text{G7.2})$$

The phase θ , prime factor $\log p$ and all signs are retained. The sphere norm is 1 for every s , while its third coordinate has the exact inverse

$$\Re s - \frac{1}{2} = \frac{\operatorname{artanh}(Z)}{\log p}. \quad (\text{G7.3})$$

For finite s its value satisfies $-1 < Z < 1$. Thus the precise purity datum retained by the sphere is its latitude: $|p^s|^2 = p$ exactly when $Z = 0$. This is a global identity for every s and every prime, not a plot of selected zeros. The two-prime injective strip receiver C5 retains the full original s when its phase periods are also retained. No point of this sphere is declared to be the user's primitive τ ; G6-G7 compare the geometric realizations supplied by the papers.

G8. What has now been established

The scale $1/2$ is derived from the original Fourier/measure comparison, not chosen to move zero locations. The quotient Q has exactly the original nontrivial-zero continuous characters, and its source-derived involution is exactly critical-line reflection. The actual global Weil form descends to Q with its full support records and is dilation invariant. The book's sphere carries the same purity defect as a recoverable latitude. Neither the common radius of the sphere nor the proved dilation invariance establishes that this form is positive. Its sign remains the actual programme problem, with the exact receiving maps G5.2-G5.4 now supplied.

Original prime history and full explicit formula

Proof source: PRIME_MONODROMY_STACKED_HISTORY.md.

Prime monodromy, complete clock observations, and the original zeta weights

24 September 2026. This receives the two identical pasted transcripts accompanying arxiv-pisa.tex and stack2.png. The transcripts are retained verbatim, including their tentative formulations; assistant text within them is not promoted to a user definition. The question investigated here is the user's proposed common global requirement of infinite observation, and its relation to arithmetic weight.

The notation $Z_0, Z_1, Z_2, \dots$ continues to mean the user's absence/presence and quantum-number layers. The symbols $\bar{Q}_d$ below name explicitly constructed d-state winding clocks; they do not silently redefine the user's entire Z_d algebra. The distinction between an absent datum and an existing datum whose observed count is zero remains throughout. Addition at tau is not restored.

The primary source is Alain Connes and Caterina Consani, *Knots, primes and class field theory*, [arXiv:2501.06560v1](https://arxiv.org/abs/2501.06560v1), original arxiv-pisa.tex. The supplied file is byte-identical to the original source. Its mapping-torus propositions, finite-cover theorem and class-field comparison are read directly in TeX. The picture's three upper components are over ONE prime p. Their labels are primes above p in an extension; their vertical drawing positions are not three successive quantum-number levels. M2 constructs an exact example with those three components and their winding data.

Figure M. Above: the three components for $p=13$ in $Q(\zeta_7)$, with return multiplication by 6 modulo 7, covering degree 2 and both real periods retained. Below: the exact first nine stages of the all-clock receiver. The all-N argument is M4–M6, not an extrapolation from this table. The supported original-zeta comparison is M7. Human source: Connes–Consani, original arxiv-pisa.tex, Proposition mappingtorus and its finite-cover calculation. Reproducible figure source: draw_prime_monodromy_history.py; exact displayed data: MONODROMY_FIGURE_CHECKS.json. The drawing is schematic; ellipse sizes are not a metric on the source base.

M1. Presence and a zero reading have an explicit comparison

Use the actual generic monoid retained in C2 of TAU_TWISTOR_TATE_AND_WEIL.md:

$$A_\eta = \{0\} \cup \{J^r T^m : r \in \mathbb{Z}/4\mathbb{Z}, m \in \mathbb{Z}\}, \quad J^2 = \epsilon, \quad J^0 T^0 = \tau. \quad (\text{M1.1})$$

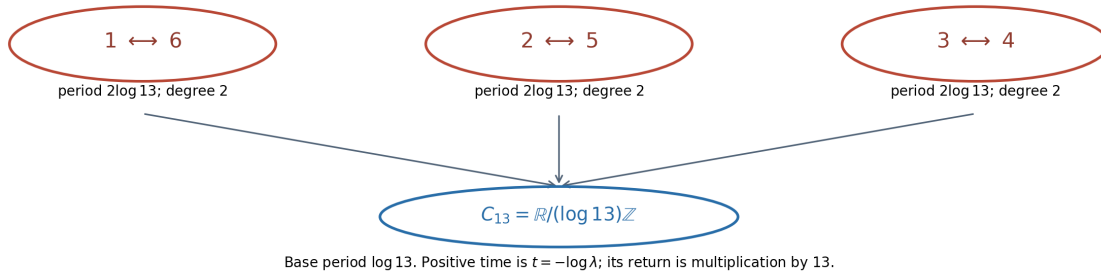
The multiplication adds m, adds r modulo 4, and has absorbing element 0. Define the presence observation

$$P : A_\eta \longrightarrow \{0, \tau\}, \quad P(0) = 0, \quad P(J^r T^m) = \tau. \quad (\text{M1.2})$$

It is multiplicative because a product of two nonabsorbing elements is nonabsorbing, and a product with 0 is 0. It is invariant under the source mirror $J^{rT} m \rightarrow J^{(-r+2m)T} (-m)$. Thus zero accumulated winding $m=0$

Prime monodromy and the complete stack of winding observations Three covering circles over ONE prime orbit

Exact example: $p = 13$, $\mathcal{Q}(\zeta_p)$, $G = (\mathbb{Z}/p\mathbb{Z})^\times$, return $g \mapsto 6g$



Retain every earlier winding clock: exact refinement data

New clock N	1	2	3	4	5	6	7	8	9
Total states L_N	1	2	6	12	60	60	420	840	2520
L_N/L_{N-1}	—	2	3	2	5	1	7	2	3
$\log L_N - \log L_{N-1}$	—	$\log 2$	$\log 3$	$\log 2$	$\log 5$	0	$\log 7$	$\log 2$	$\log 3$

$$S_N \cong \mathbb{Z}/L_N\mathbb{Z}, \quad L_N = \text{lcm}(1, \dots, N), \quad \log L_N - \log L_{N-1} = \wedge(N)$$

Green: first prime channel. Gold: refinement of an existing prime channel. Gray: no new independent state.

All clocks remain recorded. The compatible infinite system is $\widehat{\mathbb{Z}}$; no finite prefix isolates an unbounded integer.

Figure 25: Exact prime cover and complete winding-clock refinements

does not make a present datum absent: the whole fibre $m=0$ contains τ, J, ϵ, J^3 . The winding projection is defined on the nonabsorbing group; P is defined on the pointed monoid. These are two proved maps on the same structure.

This comparison does not settle the user's tentative proposal to rename η or the pair Z_0/Z_1 . The source generic point η , its stalk A_η and the stalk unit τ have different types, connected by the stalk's projection to its supporting point and by M1.2. The unit in M1.1 is not identified with integer zero. When an integer count is represented by an exponent m , retain its count label separately from the base label τ , including at $m=0$. Winding-return parity is an observation on this exponent; it is not an assignment of intrinsic integer parity to τ .

M2. What the supplied stacked picture represents

Fix a rational prime p and put $\ell_p = \log(p)$. To avoid collision with the source's additive group denoted H_p in FF.tex, write

$$\mathcal{H}_p = \prod_{q \neq p} \mathbb{Z}_q^\times. \quad (\text{M2.1})$$

The original Proposition `mappingtorus` identifies the inverse image of the prime orbit with

$$\pi^{-1}(C_p) \cong (\mathcal{H}_p \times \mathbb{R}_{>0}) / ((h, \lambda) \sim (ph, p\lambda)), \quad C_p = \mathbb{R}_{>0} / p^\mathbb{Z}. \quad (\text{M2.2})$$

The retained adelic representative has coordinate h_q at every $q \neq p$, coordinate 0 at p , and archimedean coordinate λ . Multiplication by a positive rational preserves this form exactly when that rational is a power of p , because all its valuations outside p must be zero. This proves the orbit equivalence in M2.2 directly. The full existence and equivariance are the source proposition, with its representative map kept here.

For orientation, let $t = -\log(\lambda)$. Then M2.2 becomes

$$(h, t + \ell_p) \sim (ph, t). \quad (\text{M2.3})$$

Thus a positive t -circuit returns h to ph . A positive $\log(\lambda)$ -circuit has inverse return. Both orientations and the original length ℓ_p remain displayed.

For a finite unramified cover in the source, let G be its finite abelian Galois group and c the image of p , its arithmetic Frobenius. Replacing H_p by G in M2.3 gives

$$\mathcal{M}(G, c) = (G \times \mathbb{R}) / ((g, t + \ell_p) \sim (cg, t)). \quad (\text{M2.4})$$

Let f be the order of c . Its connected components are indexed by the cosets G/f . For a representative g the explicit map

$$\mathbb{R} / (f\ell_p \mathbb{Z}) \longrightarrow \mathcal{M}(G, c), \quad [u] \longmapsto [g, u] \quad (\text{M2.5})$$

has image that component and is a homeomorphism onto it. Indeed $[c^j g, t] = [g, t + j\ell_p]$, so it is surjective there. Equality of $[g, u]$ and $[g, v]$ requires $u - v = k\ell_p$ and $c^k = 1$, equivalently $u - v$ is in $f\ell_p \mathbb{Z}$. The local covering charts prove continuity and the inverse continuity. The projection to C_p is u modulo ℓ_p , so its degree is f and its length is $f \log(p)$. There are $|G|/f$ such components. In the abelian unramified field extension these cosets are exactly the primes over p , by the source's decomposition-group calculation.

A concrete instance of the three-component picture is the cyclotomic extension $Q(\zeta_7)$ and $p=13$. Here $G=(\mathbb{Z}/7\mathbb{Z})^\times$ has six elements, $c=13 \bmod 7=6=-1$, and c has order 2. The components are

$$\{1, 6\}, \quad \{2, 5\}, \quad \{3, 4\}. \quad (\text{M2.6})$$

Each component is a two-sheeted circle of length $2 \log(13)$ over the circle of length $\log(13)$. Multiplication by 6 exchanges its two states. This is an exact example of the drawing, not a claim that the drawing itself specified 13 or this field.

M3. The full monodromy closure is the complete winding group

Define K_p =closure of $\{p^n:n \text{ in } \mathbb{Z}\}$ inside H_p . The map $n \rightarrow p^n$ extends canonically to a continuous surjection

$$\Theta_p : \widehat{\mathbb{Z}} \longrightarrow K_p. \quad (\text{M3.1})$$

Here $\widehat{\mathbb{Z}}$ is the inverse limit of ALL finite integer residue groups. To construct the map, for every integer m coprime to p reduce p modulo m and exponentiate using a modulo $\text{ord}_m(p)$. These coordinates are compatible under divisibility of m , and give an element of H_p . They agree with p^n on ordinary integers. Continuity follows coordinatewise. The image is compact and contains the dense cyclic subgroup; it is contained in K_p by density of $\mathbb{Z}$ in $\widehat{\mathbb{Z}}$, proving surjectivity onto K_p .

There is no kernel. For every $d \geq 2$ take

$$m_d = p^d - 1, \quad \gcd(m_d, p) = 1. \quad (\text{M3.2})$$

Modulo m_d the element p has EXACT order d . Its d -th power is 1. If $1 \leq k < d$, then $0 < p^{k-1} < p^d - 1$, so m_d cannot divide p^{k-1} . This proves the asserted order. Consequently $\Theta_p(a)=1$ implies $a=0$ modulo d for every $d \geq 2$, hence $a=0$ in $\widehat{\mathbb{Z}}$. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism; all maps here are group homomorphisms. We have proved

$$\boxed{\Theta_p : \widehat{\mathbb{Z}} \simeq K_p.} \quad (\text{M3.3})$$

This identifies the exact completed image of the source's integer powers. It also proves that the source map from the Frobenius group $\widehat{\mathbb{Z}} = \pi_1^{\text{et}}(\text{Spec } F_p)$ to H_p is injective, using its specified generator p ; no injectivity is deduced merely from injectivity on a dense subgroup. No literature-priority claim is made for this elementary completed-group calculation.

For two primes p, r the generator-preserving isomorphism is $\Theta_r \Theta_p^{-1} : K_p \rightarrow K_r$. It sends p^a to r^a for every profinite exponent a . The embeddings into H_p, H_r and the real periods $\log(p), \log(r)$ remain separate retained data. On their mapping tori the corresponding isomorphism is

$$[h, t] \longmapsto [\Theta_r \Theta_p^{-1}(h), (\log r / \log p)t]. \quad (\text{M3.4})$$

Its well-definedness follows from M2.3; a shift $\log(p)$ becomes $\log(r)$ and multiplication by p becomes multiplication by r . Its inverse interchanges p and r . Thus the clocks have the same completed group type, with an explicitly retained change in flow parameter.

M4. The stack keeps every earlier finite-state observation

For each $d \geq 1$ let Q_d be the actual d -state winding quotient $\mathbb{Z}/d\mathbb{Z}$, obtained within K_p by reduction modulo p^{d-1} and discrete exponent as in M3. For $d=1$ it is the singleton quotient. This construction assigns a cyclic transition to the source winding clock; it does not assert that every object with the user's label Z_d must be this cyclic group.

To keep all previous floors, take the simultaneous receiver

$$S_N = \text{im} \left(\widehat{\mathbb{Z}} \longrightarrow \prod_{d=1}^N Q_d \right), \quad L_N = \text{lcm}(1, \dots, N). \quad (\text{M4.1})$$

Two residues define the same tuple precisely when their difference is divisible by every $d \leq N$, equivalently by L_N . Every tuple in this image comes from an ordinary residue modulo L_N . Hence the explicit map

$$\mathbb{Z}/L_N\mathbb{Z} \simeq S_N, \quad [n] \longmapsto ([n]_1, \dots, [n]_N) \quad (\text{M4.2})$$

is bijective and preserves the winding step. The transition $S_N \rightarrow S_{(N-1)}$ forgets only the last coordinate and corresponds to reduction modulo $L_{(N-1)}$, since $L_{(N-1)}$ divides L_N . The entire compatible system has inverse limit $\widehat{\mathbb{Z}}$: each modulus d occurs as a coordinate, so a compatible system determines, and is determined by, all integer residues. These two assignments are explicit inverse maps.

This gives a stack in which the d -state observation comes with every earlier observation. For example S_3 retains both parity and the three-state reading and has SIX compatible states; S_4 has twelve. A state labelled 0 at any of these coordinates is still a state. To include the user's Z_0 absence, use an additional absent symbol outside the set of present tuples; the absence observation is distinct from the all-zero tuple. The construction imports no tau addition and assigns no intrinsic integer parity to tau.

All source sign data can also be retained in this completion. For the generic nonabsorbing group in M1, the map

$$J^r T^n \longmapsto (J^r, \Theta_p(n)) \quad \text{into} \quad (\mathbb{Z}/4\mathbb{Z}) \times K_p \quad (\text{M4.3})$$

is injective: equality fixes r modulo 4 and n by M3.3. On its closure, the source mirror is $(J^{r, \Theta_p(a)} \rightarrow J^{(-r+2a), \Theta_p(-a)})$, where $2a$ modulo 4 is defined by a modulo 2. This is a continuous multiplicative involution by direct composition. The full quartic sign and branch data therefore survive the completed winding observations.

M5. The exact infinite-observation statement holds uniformly

Fix an integer n . Its fibre under observation through level N is

$$n + L_N\mathbb{Z}. \quad (\text{M5.1})$$

This follows from M4.2. It is infinite for every finite N , while

$$\bigcap_{N \geq 1} (n + L_N\mathbb{Z}) = \{n\}. \quad (\text{M5.2})$$

Indeed a difference belonging to every $L_N\mathbb{Z}$ is divisible by every positive integer d . A nonzero integer cannot be divisible by an integer larger than its absolute value. This proves M5.2 without a numerical

cutoff or an assumed prime list. Translation by n carries the entire fibre system at 0 to the fibre system at n , proving the same infinite refinement requirement for EVERY integer in this observation model.

This is the exact global assertion supported by the construction: no finite family of these residue observations singles out an unbounded integer, while the complete family does. It concerns recovery FROM these observations. It does not change the original step generating n after n positive steps. The typed maps relating generation and observation are the prefix inclusions of the integer winding history into Z and the diagonal injection $Z \rightarrow \lim S_N$. They commute with the step $n \rightarrow n+1$; one is a direct system of generated histories and the other is an inverse system of observations of those same counts.

The completion also contains coherent states that are not ordinary integers. For example, by the Chinese remainder construction choose residues that converge to 0 in Z_2 and to 1 in every Z_q for q odd. They define an element of $\hat{Z}$. It cannot be an integer, since an integer that is 0 in Z_2 is 0, and 0 is not 1 in Z_3 . Thus the completion is not silently substituted for the source integers; M5.2 recovers the original integer within its image.

M6. The retained growth is precisely the original prime-power weight

The total stack S_N has L_N states. Retain its exact growth ratio and logarithmic increment

$$r_N = L_N / L_{N-1}, \quad w_N = \log L_N - \log L_{N-1} \quad (N \geq 2), \quad L_1 = 1. \quad (\text{M6.1})$$

For each prime p , unique factorization gives $v_p(L_N) = \max\{a \geq 0 : p^a \leq N\}$. This is because lcm takes the maximum valuation of its arguments. Passing from $N-1$ to N increases this maximum exactly when $N = p^a$ for some $a \geq 1$; the increase is then 1. At most one prime has this property for a given N . Therefore

$$r_N = \begin{cases} p & N = p^a, a \geq 1, \\ 1 & N \text{ is not a prime power,} \end{cases} \quad \boxed{w_N = \Lambda(N)}. \quad (\text{M6.2})$$

Λ is the ordinary von Mangoldt function, with every prime power retained. The new prime line at p is its first occurrence. At $p^2, p^3, \dots$ the same prime contributes further resolution. A count such as 6 introduces a floor whose observation is already determined by earlier floors; that floor is still present. Explicitly $L_2=2, L_3=6, L_4=12, L_5=L_6=60$. The step at 4 refines the 2-channel; the step at 6 adds no independent clock state.

This relates the user's first-appearance observation to the higher repeated data rather than replacing one by the other. The earlier prime-activation sieve recovers the set of first primes. The present all-floor receiver recovers their complete prime-power weights. Finite resolution and multiplicity remain available before the infinite limit.

For $\text{Re}(s) > 1$ the resulting exact analytic identity is

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{N=2}^{\infty} (\log L_N - \log L_{N-1}) N^{-s} = \sum_p \sum_{a=1}^{\infty} (\log p) p^{-as}. \quad (\text{M6.3})$$

Proof: finite Euler products expand into the sum of n^{-s} over integers using only those primes. Absolute convergence of $\sum n^{-\text{Re}(s)}$ permits passage to all primes and gives the original zeta Euler product. Expanding each logarithm gives $\log(\zeta(s)) = \sum_p \sum_{a \geq 1} p^{-as}/a$, with the branch tending to 0 on the positive real axis at infinity. Differentiation on compact subsets of $\text{Re}(s) > 1$ is justified by the

bound $\sum_{N \geq 2} (\log N) N^{-(\sigma+1)}$. It yields the last series in M6.3. Substitution of M6.2 yields the first. All factors, repetitions and signs are retained.

Thus the uniform need for the whole observation system coexists with an exact, nonconstant growth record. The latter is already the prime-power term entering the original zeta function; replacing it by the single statement that all final histories are infinite would remove this computed datum.

M7. Substitute the complete history into the supported Weil receiver

For h in $C_c^\infty(\mathbb{R})$, put $H(s) = \int h(v) \exp(-(s-1/2)v) dv$ and $\hat{h}(t) = \int h(v) \exp(-itv) dv$, with the displayed conventions fixed. In the existing original-zeta receiver of C10, the complete prime term becomes exactly

$$P_{\text{hist}}(h) = \sum_{N=2}^{\infty} \frac{\log L_N - \log L_{N-1}}{\sqrt{N}} (h(\log N) + h(-\log N)). \quad (\text{M7.1})$$

This sum is finite for each compactly supported h , but its formula is valid on the entire test space; it is not a fixed-range numerical approximation. It equals $P(h)$ term by term by M6.2. Keep

$$A_{\infty}(h) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{h}(t) \left(\Re \frac{\Gamma'(1/4 + it/2)}{\Gamma(1/4 + it/2)} - \log \pi \right) dt.$$

$$W_{\zeta}(h) = H(0) + H(1) + A_{\infty}(h) - P_{\text{hist}}(h) = \sum_{\rho} m_{\rho} H(\rho). \quad (\text{M7.2})$$

This is an exact substitution into the proved original-zeta contour formula, retaining its Gamma, endpoint and multiplicity contributions. The finite trivial-zero cutoff triple remains $V_{\text{zeta},B} = Z(H) + \sum_{m=1}^B H(-2m) - H(1) = G_B - P_{\text{hist}}(h)$, $G_B = A_{\infty} + H(0) + \sum_{m=1}^B H(-2m)$; the integer cutoff B is not sent to infinity without convergence. The complete contour proof is retained with C10 and WU2 in the programme sources.

For the existing support lattice $\mathcal{L}$ keep the source vectors

$$B(h) = \mathbf{e}_{1_{\mathcal{L}}}(H(0) + H(1)) + \sum_{\ell \neq 1_{\mathcal{L}}} \mathbf{e}_{\ell} H(0),$$

$$Z(h) = \mathbf{e}_{1_{\mathcal{L}}} \sum_{\rho} m_{\rho} H(\rho), \quad (\text{M7.3})$$

$$D_{\text{hist}}(h) = \mathbf{e}_{1_{\mathcal{L}}}(P_{\text{hist}}(h) - A_{\infty}(h)) + \sum_{\ell \neq 1_{\mathcal{L}}} \mathbf{e}_{\ell} H(0).$$

M7.2 proves $\text{bold}(B) - \text{bold}(Z) = \text{bold}(D_{\text{hist}})$ in every support coordinate. Tensoring all terms with $[\tau]$ or with $[1]$ gives the two distinct programme coefficient records exactly as in WU1-WU8; their full difference remains in $[\tau] - [1]$ before arithmetic evaluation. The support-zero contributions have not been replaced by a classical prime sum, and the new history formula has not silently removed them.

The proof source for the support vectors is [SZW33-SZW38 at the pinned programme edition](#). Their complete original-zeta comparison is C10 and the retained WU2-WU5. M6.2 proves the present substitution without a new analytic assumption.

M8. What equal modulus proves for this actual monodromy

Translation by p preserves Haar measure on the compact group K_p . Its operator $U_p f(h) = f(ph)$ is therefore unitary on $L^2(K_p)$: substitution $h' = ph$ proves equality of norms and $f(h) \rightarrow f(p^{-1}h)$ is its inverse. Under M3.3 a finite character has the form

$$\chi_{j,d}(\Theta_p(a)) = \exp(2\pi i j a / d), \quad U_p \chi_{j,d} = \exp(2\pi i j / d) \chi_{j,d}. \quad (\text{M8.1})$$

These are characters on Q_d pulled back to K_p . Every continuous character $K_p \rightarrow S^1$ factors through a finite quotient: continuity puts an open subgroup into a sufficiently short arc about 1; the only subgroup of S^1 contained in that arc is $\{1\}$, so that open subgroup lies in the kernel. Hence every such character has the displayed form. This proves exact modulus 1 for the entire character family, including after the infinite observation limit.

For the mapping torus in the t -orientation of M2.3, the function $\chi_{j,d}(h) \exp(i \omega t)$ is well defined precisely when $\exp(i \omega \ell_p) = \exp(2\pi i j / d)$. Its full frequencies are therefore

$$\omega = \frac{2\pi}{\log p} \left(k + \frac{j}{d} \right), \quad k \in \mathbb{Z}. \quad (\text{M8.2})$$

This proves a common modulus for the return eigenvalues while retaining the prime-dependent real scale. It is the spectrum of this explicitly defined monodromy representation; no identification with zeta zeros is hidden in the word spectrum.

The factor $1/2$ used in the original-zeta Mellin receiver also has an exact source comparison. For the raw dilation $D_a f(x) = f(x/a)$ on $L^2(\mathbb{R}, dx)$, substitution $x = ay$ gives $\|D_a f\|_{L^2}^2 = a^{-1} \|f\|_{L^2}^2$. Thus a positive scalar multiple $c(a) D_a$ is unitary exactly when $c(a) = a^{1/2}$. The half-power follows from this measure and this dilation, rather than from assigning all integer histories an infinite weight. Define $U_a = a^{1/2} D_a$; the full map to the multiplicative receiver $V_a k(u) = k(u/a)$ is

$$\mathcal{E}(U_a f)(u) = u^{1/2} \sum_{n \geq 1} a^{-1/2} f(nu/a) = \mathcal{E}(f)(u/a) = V_a \mathcal{E}(f)(u). \quad (\text{M8.3})$$

All square roots and the original summation are retained. The next calculation identifies precisely the spectral information in this map.

M9. The spectral quotient retaining the original zeta zeros

Let S_0^{even} consist of even Schwartz functions f on $\mathbb{R}$ with $f(0) = \int_{-\infty}^{\infty} f(u) du = 0$. Let A be the smooth functions k on $\mathbb{R}_{>0}$ such that, for all nonnegative integers N, j , $\sup_{u > 0} (u^{N+u} (-N)!) |d^j k(u) / du^j|$ is finite. This is a locally convex space with these seminorms; its elements and all logarithmic derivatives decay faster than every power at both endpoints. It sits in $L^2(\mathbb{R}_{>0}, du/u)$.

The map E from M8.3 sends S_0^{even} into A . At infinity this follows by summing the Schwartz bounds for $f(nu)$ and its differentiated series with an exponent greater than 1. At zero, Poisson summation with both endpoint values zero gives $E(f)(u) = E(\hat{f})(1/u)$; the same argument at infinity for $\hat{f}$ supplies the zero-end estimates, including derivatives. The maps U_a and V_a preserve these domains and have the exact covariance M8.3.

Define the full centered Mellin observation

$$\mathcal{M}k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}. \quad (\text{M9.1})$$

It is entire. For every fixed s , including its derivatives in s , it is continuous on A : choose $N > |\operatorname{Re}(s) - 1/2|$ and use the displayed weighted seminorm to bound the integrand at 0 and infinity; any power of $|\log u|$ is still integrable with a slightly larger N . Locally uniform domination also proves complex differentiation under the integral.

For $\operatorname{Re}(s) > 1$, absolute convergence and $v = nu$ give

$$\mathcal{ME}(f)(s) = \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}. \quad (\text{M9.2})$$

The right-hand Mellin integral is holomorphic on $\operatorname{Re}(s) > -2$ because even f with $f(0)=0$ has $f(v)=O(v^2)$ at zero. At $s=1$ the zero integral condition cancels the simple zeta pole. Analytic continuation therefore gives M9.2 throughout the open critical strip, retaining the ORIGINAL zeta. A zero ρ of multiplicity m_ρ makes every derivative of the left side of orders 0 through $m_\rho - 1$ vanish there. Thus these continuous, nonzero Mellin jet functionals descend to

$$\mathcal{Q} = \mathcal{A} / \overline{\mathcal{E}(S_0^{\text{even}})}^{\mathcal{A}}. \quad (\text{M9.3})$$

The evaluation functional at ρ is nonzero on A : choose a nonzero nonnegative compactly supported smooth bump $b(u)$ away from 0 and set $k(u) = u^{-(i \operatorname{Im}(\rho))} b(u)$; then $\mathcal{M}k(\rho)$ is a strictly positive real integral. The jet functionals of all orders are linearly independent before quotient: a linear combination integrates $k(u)u^{(\rho-1/2)}$ times a polynomial in $\log u$; vanishing on every compact bump forces that polynomial to vanish. Continuity and M9.2 prove that the jets up to $m_\rho - 1$ remain nonzero independent functionals on $\mathcal{Q}$. This retains multiplicity information without assuming RH.

They have the exact scaling action

$$\mathcal{M}(V_a k)(s) = a^{s-1/2} \mathcal{M}k(s). \quad (\text{M9.4})$$

Differentiation retains all jet terms: $(\mathcal{M} V_a k)^{(j)(\rho)} = a^{(\rho-1/2)j} \sum_{l=0}^j \binom{j}{l} (\log a)^{j-l} (\mathcal{M} k)^{(l)(\rho)}$. In particular the evaluation eigenvalue has modulus $a^{\operatorname{Re}(\rho)-1/2}$. This is the exact place where critical-line modulus is asked of the ACTUAL zeros, rather than of the finite monodromy characters in M8.

There is a concrete reason the positive Haar inner product does not simply descend as a nontrivial Hilbert quotient: $E(S_0^{\text{even}})$ is dense in $L^2(\mathbb{R}_{>0}, du/u)$. Here is a complete proof using the programme's retained kernel. The nonzero Gaussian polynomial $f_0(x) = (\pi/2)x^{2(2\pi i x^2 - 3)} \exp(-\pi x^2)$ is in S_0^{even} and has Fourier transform itself, as computed in C6. Put $k_0 = E(f_0)$. Its Mellin transform is the entire function

$$\mathcal{M}k_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad (\text{M9.5})$$

with the factor four relative to ξ retained. It is not identically zero: at $s=2$ its value is $\pi/24$. As an entire nonzero function it has isolated zeros. With the Fourier convention $\widehat{K}(t) = \int_{\mathbb{R}} K(v) e^{-itv} dv$, the logarithmic function $K_0(v) = k_0(e^v)$ has transform $\widehat{K}_0(t) = \mathcal{M}k_0(1/2 - it)$. This transform is consequently nonzero almost everywhere. The minus sign in this comparison is retained.

Suppose F in L^2 is orthogonal to every $V_a k_0$. Set $F_{\log}(v) = F(e^v)$ and $b = \log a$. In logarithmic coordinate the translate is $K_0(v - b)$, whose Fourier transform is $e^{-itb} \widehat{K}_0(t)$. Plancherel, with its full constant, gives

$$0 = \langle F, V_a k_0 \rangle = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{F}_{\log}(t) \overline{\widehat{K}_0(t)} e^{itb} dt \quad (b \in \mathbb{R}). \quad (\text{M9.6})$$

The product in this integral is L^1 by Cauchy-Schwarz. Uniqueness of the Fourier transform on L^1 gives product=0 almost everywhere. Since the second factor is nonzero almost everywhere, the transform of $F_{\log}$ is zero almost everywhere, and $F=0$. This proves density of the translates of k_0 . M8.3 puts every such translate in $E(S_0^{\text{even}})$, proving the density claim.

Accordingly the L^2 quotient is zero, whereas Q in M9.3 has the proved nonzero zero-evaluation functionals. The identity inclusion $A \rightarrow L^2$ induces the zero map from Q to that Hilbert quotient. Every original zero evaluation annihilates $E(S_0^{\text{even}})$ but is nonzero, so it cannot extend to a continuous L^2 functional. This proves the exact relation between the two receiving spaces instead of treating their different behaviour as absence of a relation. The richer quotient is constructed explicitly and keeps the original zeros, their multiplicities and the full scaling defect.

The programme's linear coefficient spaces pass through this quotient by the following exact tensor isomorphism. Support records at zero require the separate record-retaining map proved below and in `SUPPORT_ZERO_RECEIVER_REPAIR.md`; they cannot be recovered from a zero tensor. Let C_{rec} be the coefficient algebra of WU_1 : its independent vector-space basis is $U=[\tau]$ and $N_n=[n]$ for nonzero integers n , with U the unit, $N_n N_m = N_{(nm)}$, and $N_0=0$. Put $f=N_1$. Let $C^{(\mathcal{L})}$ be the free vector space on all retained support labels. All tensor products in the following formula are algebraic, so each element uses finitely many coefficient directions:

$$\frac{\mathcal{A} \otimes C_{\text{rec}} \otimes C^{(\mathcal{L})}}{\mathcal{E}(S_0^{\text{even}})^{\mathcal{A}} \otimes C_{\text{rec}} \otimes C^{(\mathcal{L})}} \simeq Q \otimes C_{\text{rec}} \otimes C^{(\mathcal{L})}. \quad (\text{M9.7})$$

The map sends the class of k tensor c tensor e_{ell} to $[k]$ tensor c tensor e_{ell} . Choose a basis of the two coefficient spaces. Each element then has a unique finite coordinate expansion. Its image is zero exactly when each A -coordinate lies in the closed subspace in the denominator. This proves well-definedness, its exact kernel and surjectivity, hence the displayed isomorphism of the linear spaces. It does not assert that the preceding map from labelled states to tensors retains a label at coefficient zero.

The two sections on Q tensor $C^{(\mathcal{L})}$ send q tensor e_{ell} to q tensor U tensor e_{ell} and q tensor f tensor e_{ell} . Arithmetic evaluation sends U to 1 and N_n to n , and composes with each section to the identity. Their difference is q tensor $(U-f)$ tensor e_{ell} ; it is injective because the coefficient functional taking U to 1 and every N_n to 0 recovers q tensor e_{ell} . Multiplication gives $(U-f)^2=U-f$ and $(U-f)f=0$, retaining the programme's unit-difference idempotent. The Mellin jets and scaling act on Q . To retain every original support record, including a selected section at $q=0$, use the disjoint family of pairs (record, q) , with record in $\mathcal{L}$ or in $\mathcal{L} \times \{U, f\}$ when the section selection is retained. Every operator acts only on q . Equality of paired observations requires equality of the record and then equality of q ; the injective synthesis map proves the latter. The exact maps to the previous tensors, their zero fibres, and the full pairing are proved in ZR1-ZR5 of `SUPPORT_ZERO_RECEIVER_REPAIR.md`. This is a formal coefficient construction, not addition or subtraction of source τ , and it is not a new assignment of parity to τ .

The results establish a faithful infinite-observation completion, the same refinement requirement for every integer, the full prime-power growth in the original-zeta Weil term, and an exact spectral receiving quotient. They do not establish nonnegativity of M7.2 on all Weil tests or modulus one for every eigenvalue in M9.4. No equality of those moduli has been inserted as a definition or inferred from a shared infinite cardinality.

Retained kernel and source comparisons

Proof source: TAU_TWISTOR_TATE_AND_WEIL.md.

The Z_1 base, the twistor involution, and the original-zeta receiver

24 September 2026. This is a source-directed calculation on the supplied author TeX. It retains the user's notation: tau carries Z_1 presentness without Z_2 parity; integer 1 is the identity on the integer part; tau is the global multiplicative identity. Addition involving tau is not used. An operation on an auxiliary coefficient ring or on a space of functions is not an additional operation on tau. Every comparison below states its domain and what it retains.

The human sources are Alain Connes and Caterina Consani, *The Absolute Twistor Line and the Geometry of Spec Z* (2609.00299v1, CC.tex); the same authors, *On the Absolute Geometry of Spec Z* (2606.06604v1, FF.tex); their *On the Jacobian of Spec Z* (2602.15941v1, Jacobian.tex); and Alain Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time* (2602.04022v1, rhready.tex). All four supplied TeX files are byte-identical to the respective original arXiv sources. Source archives and the exact reading ledger are retained. The calculations marked C1-C10 below are this note's derivations, not claims of literature novelty.

C1. The multiplicative base comparison retains tau and integer 1

Let M be the multiplicative integer-plus-tau stage. Its products are the ordinary integer products and $\tau x = x\tau = x$ for every x in M . Tau differs from each nonzero integer by the retained Z_2 availability; $\tau = 0$ would imply $1 = \tau 1 = 0 1 = 0$. Thus the disjoint presentation $M = Z \text{ disjoint-union } \{\tau\}$ records a consequence. Higher stages of the user's hierarchy are not declared absent.

Put $B_\tau = \{0, \tau\}$ with these products. The map

$$b : B_\tau \longrightarrow \{0, 1_{cc}\}, \quad b(0) = 0, \quad b(\tau) = 1_{cc} \quad (C1.1)$$

is an isomorphism of pointed multiplicative monoids: 0 is absorbing and tau is the unit, exactly matching the two source products. Its inverse sends the source unit to tau. This is the pointed-monoid part of the source's F_1 base, not a claim that an unspecified additive structure or every framework called F_1 has been identified.

The integer receiver $r: M \rightarrow (Z, \text{multiplication})$, $r(n) = n$, $r(\tau) = 1$ is multiplicative. It identifies the images of tau and integer 1. Retain also

$$\chi : M \longrightarrow B_\tau, \quad \chi(\tau) = \tau, \quad \chi(n) = 0 \quad (n \in \mathbb{Z}). \quad (C1.2)$$

For an integer product both sides of multiplicativity are 0; for a product with tau both sides equal the value of the other factor. Hence chi is multiplicative. The pair (χ, r) is injective and has image

$$(\{0\} \times \mathbb{Z}) \cup \{(\tau, 1)\}. \quad (C1.3)$$

Indeed an integer n has the unique pair $(0,n)$, whereas τ has $(\tau,1)$. In particular using the complex-valued points of the authors' geometry does not make the scalar value 1 a faithful record of the original source element. Equations C1.2-C1.3 give an explicit faithful receiver. No integer amount or parity is assigned to τ .

C2. Signed extension and the exact generic involution

CC.tex, Definition `def:alpha_symmetry`, uses $\epsilon^2=1$ and $J^2=\epsilon$. Retain its sign as a new multiplicative datum. It is not automatically the user's $\mathbb{Z}_2$ parity.

Here is an extension preserving the full M before taking geometric points. Take the disjoint union of the Gaussian integers $\mathbb{Z}[i]$ and four new units u_0, u_1, u_2, u_3 . Set $u_0=\tau$, $u_2=\epsilon$, $u_1=J$; define

$$u_r u_t = u_{r+t \bmod 4}, \quad u_r a = a u_r = i^r a \quad (a \in \mathbb{Z}[i]), \quad (\text{C2.1})$$

and use ordinary products on $\mathbb{Z}[i]$. Associativity follows separately for three units from addition modulo 4, for two units and a Gaussian integer from $i^{(r+t)} = i^r i^t$, and for one unit and two Gaussian integers from $i^{r(ab)} = (i^r a)b$. Three Gaussian integers use their usual associativity. Commutativity reduces all orders to these cases. Zero is absorbing and τ is the global unit. Inclusion embeds M multiplicatively. Local Gaussian i and global J remain different, as do local -1 and ϵ . $J^2=\epsilon$ and $\epsilon^2=\tau$ hold without identifying local integer 1 with τ .

For the generic source chart itself let

$$A_\eta = \{0\} \cup \{J^r T^k : r \in \mathbb{Z}/4\mathbb{Z}, k \in \mathbb{Z}\}. \quad (\text{C2.2})$$

Its product adds the exponent k and adds r modulo 4; 0 is absorbing, $J^{0T}0=\tau$, and $\epsilon=J^{2T}0$. This is the explicitly presented multiplicative monoid of the source chart. Its isomorphism to the authors' monomial presentation sends τ to their unit and retains J, T, ϵ . The inverse recovers the two exponents. The infinite exponent k is never reduced modulo 2 or 4.

The source's signed inversion has the exact coordinate form

$$\alpha^*(J^r T^k) = J^{-r+2k} T^{-k}, \quad \alpha^*(0) = 0. \quad (\text{C2.3})$$

This follows by multiplying $(J^{(-1)})^r$ and $(\epsilon^{(-1)})^k$. The map on the exponent pair is additive, so it preserves products. Applying it twice sends (r,k) to $(-(-r+2k)+2(-k), -(-k))=(r,k)$; hence it is an involution. Fixedness requires $k=-k$ in $\mathbb{Z}$ and $r=-r$ modulo 4. Consequently its entire fixed submonoid is

$$\text{Fix}(\alpha^*) = \{0, \tau, \epsilon\}. \quad (\text{C2.4})$$

τ 's fixedness is thus proved in the actual generic algebra. ϵ is also fixed; it must not be discarded. For every positive odd n , the power map $F_n: (r,k) \rightarrow (nr, nk)$ commutes with α . The joint fixed submonoid of all these maps is again $\{0, \tau, \epsilon\}$: $n=3$ forces $k=0$ and $2r=0$, and those three elements are fixed for every odd n .

CC's underlying three-point space has $\alpha(+)=-, \alpha(-)=+, \alpha(\epsilon)=\epsilon$. The earlier integer-step orbit receiver $Q=\{\tau\text{-marker}, E, O\}$ is isomorphic as an involutive set to it by $\tau\text{-marker} \rightarrow \epsilon, E \rightarrow +, O \rightarrow -$. Checking the three values proves equivariance and the inverse. This is a map of involutive sets. It does not identify a base element τ with a point carrying the entire generic stalk A_ϵ , and it does not discard the full integers in M .

C3. The sign distinguishing two real structures

Over $\mathbb{C}$, source points have J -coordinate i or $-i$. Alpha sends (z,i) to $(-1/z,-i)$, while complex conjugation sends it to $(\bar{z},i)$. Returning to the i branch after conjugation gives exactly

$$\sigma(z) = -\frac{1}{\bar{z}}, \quad \sigma(0) = \infty, \quad \sigma(\infty) = 0. \quad (\text{C3.1})$$

This reconstructs CC.tex, Theorem thm:twistor_space_scheme, without dropping the sign. For finite nonzero z , fixedness would give $|z|^2 = -1$, which is impossible; the endpoints are exchanged. Thus sigma has no fixed points on $\mathbb{P}^1(\mathbb{C})$.

There is also an explicitly related involution. The source automorphism R has $R(T) = \epsilon T$ and $R(J) = J$; it extends to both charts, preserves their restrictions, and commutes with alpha. For example alpha $R(T) = T^{-1} = R \alpha(T)$, and both composites take J to J^{-1} . Its induced complex map is $R(z) = -z$. Therefore

$$\kappa = R \circ \sigma, \quad \kappa(z) = \frac{1}{\bar{z}}, \quad \text{Fix}(\kappa) = \{|z| = 1\} \subset \mathbb{C}^\times. \quad (\text{C3.2})$$

The endpoints are again exchanged; the displayed fixed locus follows by multiplying $z = 1/\bar{z}$ by $\bar{z}$. $R^2 = \text{id}$ and R commutes with sigma, so $\kappa^2 = \text{id}$. This is a proved sign twist of the source involution, not the same real structure silently renamed. No bijective conjugacy can turn sigma into kappa, since a conjugacy maps fixed sets bijectively and their fixed sets are empty and nonempty respectively.

The sign remains visible in homogeneous coordinates. With $z = x/y$, put

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (\text{C3.3})$$

Then $A\bar{\alpha}(x,y)$ induces sigma, $B\bar{\alpha}(x,y)$ induces kappa, $B = DA$, $A\bar{\alpha}(A) = -I$ and $B\bar{\alpha}(B) = I$. Projectivization makes both induced squares the identity but retains different real structures. The unit comparison $\tau \rightarrow I$ and $\epsilon \rightarrow -I$ uses the source multiplicative sign; it does not define tau as an additive zero vector.

For comparison with the user's radius proposal, the complete sphere map is

$$\Phi(z) = \frac{(2\Re z, 2\Im z, |z|^2 - 1)}{1 + |z|^2}, \quad \Phi(\infty) = (0, 0, 1). \quad (\text{C3.4})$$

Its numerator has squared norm $4|z|^{2+(|z|^2-1)^2} = (1+|z|^2)^2$. Thus its radius is 1 for every z . Direct substitution gives $\Phi(\sigma(z)) = -\Phi(z)$. Away from the north pole its inverse is $z = (X+iY)/(1-Z)$; the north pole returns infinity. Equal sphere radius therefore coexists with a retained, recoverable coordinate $|z|$. It is a property of this target geometry, not an assigned metric at the user's tau.

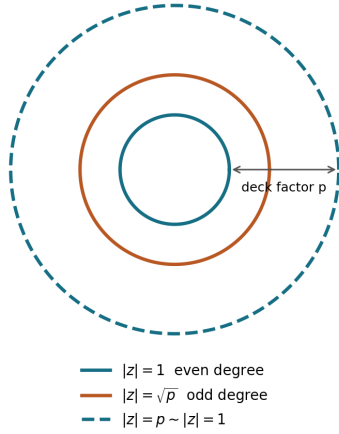
The actual odd Frobenius in CC.tex, Proposition prop:twistor_frobenius, is z^n for $n \equiv 1 \pmod{4}$ and $-1/\bar{z}^n$ for $n \equiv 3 \pmod{4}$. Consequently $|z|$ becomes $|z|^n$ or $|z|^{-n}$, respectively, while Phi of either result still has radius 1. This is an exact calculation within the supplied geometry; it retains the spectral size that a sphere-radius observation alone does not record.

C4. Descent to the supplied Tate curve and its two fixed circles

Figure 1. The left drawing uses $p=3$ solely to display the annulus. C4 proves the circles for every prime p . The right drawing displays the actual quotient maps; the marked point is an arbitrary geometric point,

The sign-adjusted involution on the paper's Tate curve
 $E_p = \mathbb{C}^\times / p^\mathbb{Z}$, $\kappa[z] = [1/\bar{z}]$, $\sigma[z] = [-1/\bar{z}]$

Annulus representatives (drawing: $p = 3$)



Proof: C4-C5 in TAU_TWISTOR_TATE_AND_WEIL.md. The even/odd labels are the parity of the lifted integer n in $|z|^2 = p^n$.
 For $z = p^s$ with $0 < \text{Re}(s) < 1$, κ -fixedness forces $n = 1$ and hence $\text{Re}(s) = 1/2$. This does not assert fixedness of all zeta zeros.
 Sources: Connes-Consani, arXiv:2609.00299v1 (twistor sign) and 2606.06604v1 (Tate quotient). No metric at τ is assumed.

Exact maps on the quotient coordinates

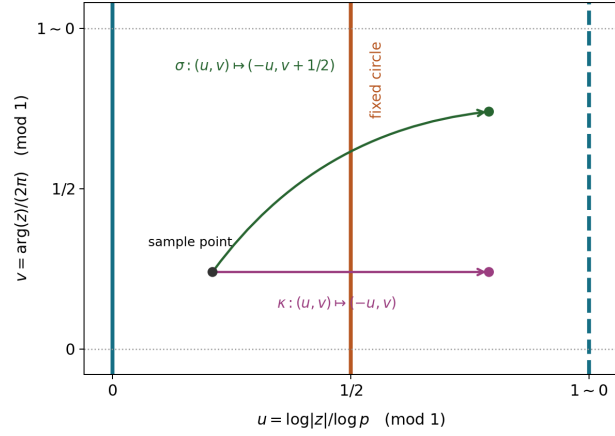


Figure 26: The Tate fixed circles and the retained signed action

not an asserted zeta zero. Reproducible source: draw_tate_fixed_locus.py. Proofs: C4-C5. The two radial boundary circles are identified by the deck factor p .

FF.tex, The Tate curve E_p and thm:tate_decomposition, supplies

$$E_p = \mathbb{C}^\times / p^\mathbb{Z}, \quad p \text{ a positive rational prime.} \quad (\text{C4.1})$$

Both σ and κ descend to E_p , since $\sigma(p^k z) = p^{-k} \sigma(z)$ and $\kappa(p^k z) = p^{-k} \kappa(z)$. A σ -fixed class would satisfy $-1/\bar{z} = p^k z$, hence $-1 = p^k |z|^2$. No such class exists.

A κ -fixed class instead satisfies

$$\frac{1}{\bar{z}} = p^k z \iff |z|^2 = p^{-k}. \quad (\text{C4.2})$$

Retain $n = -k$. Before quotienting, the full solution set with its integer degree is

$$\widetilde{F}_p = \{(z, n) \in \mathbb{C}^\times \times \mathbb{Z} : |z|^2 = p^n\}. \quad (\text{C4.3})$$

Changing representative z to $p^j z$ changes n to $n + 2j$. Thus n modulo 2 is independent of representative, while n itself is retained on the lifted set. Multiplication is $(z, n)(w, m) = (zw, n + m)$, which preserves the equation because $|zw|^2 = p^{n+m}$. Its identity is $(1, 0)$, and its inverse is $(1/\bar{z}, -n)$.

Every even n has a unique representative with radius 1 after division by $p^{n/2}$; every odd n has a unique representative with radius $\sqrt{p}$ after division by $p^{(n-1)/2}$. Their phases do not change. Explicitly,

$$S^1 \times \{0, 1\} \longrightarrow \text{Fix}(\kappa : E_p \rightarrow E_p), \quad (v, a) \longmapsto [p^{a/2} v] \quad (\text{C4.4})$$

is bijective. Equality of two images implies $p^{((a-b)/2)*v/w=pj}$. Taking moduli forces $a=b$ and $j=0$, then $v=w$. Surjectivity was proved by the representative reduction. On the source of C4.4 the group law is $(v,a)(w,b)=(vw,a+b \bmod 2)$: the factor p arising when $a=b=1$ is precisely the retained deck multiplication. Hence the fixed locus is a group isomorphic to S^1 times the two-element degree group.

This proves the two-circle classification, including its lifted integer data. Its two-state label is the parity of n . It is not silently identified with the intrinsic Z_2 state of an arbitrary integer element in the user's base. The exact comparison is $n \rightarrow n \bmod 2$ on the displayed lifted integer coordinate.

C5. Original zeta coordinates and the weight-one circle

For every complex s retain the unquotiented value $\Lambda_p(s)=p^s=\exp(s \log p)$, with the real positive $\log p$. Define $j_p(s)=[p^s]$ in E_p . No zeta completion or rescaling of the zeta function is involved.

In the open critical strip $S=\{s:0<\operatorname{Re}(s)<1\}$, C4.2 gives

$$\begin{aligned} j_p(s) \in \operatorname{Fix}(\kappa) &\iff |p^s|^2 = p^n \text{ for some } n \in \mathbb{Z} \\ &\iff 2\Re s = n \in \mathbb{Z} \\ &\iff \Re s = \frac{1}{2}. \end{aligned} \tag{C5.1}$$

The last implication uses $0<2\operatorname{Re}(s)<2$, so $n=1$, not a choice of a radius. The full unquotiented radius on this locus is $|p^s|=\sqrt{p}$. Thus the strip selects exactly the odd-degree circle. Endpoints of the strip were not included; their real parts 0 and 1 land on the even-degree circle and are not removed from the global map.

Let $R_\zeta(s)=1-\bar{s}$. The exact equivariance is

$$\Lambda_p(R_\zeta s) = \frac{p}{\Lambda_p(s)}, \quad j_p(R_\zeta s) = \kappa(j_p(s)). \tag{C5.2}$$

The factor p in the first equality becomes a deck transformation in the second, and is retained explicitly. The involution on the unquotiented value is $\iota_p(z)=p/\bar{z}=p*\sigma(z)$, whose fixed circle is $|z|^2=p$. This is the precise weight-one comparison with the supplied twistor map, including the sign and prime factor.

The arithmetic fixedness defect has a full scalar representative

$$\Delta_p(s) = |p^s|^2 - p = p^{s+\bar{s}} - p. \tag{C5.3}$$

It is zero, positive or negative according as $\operatorname{Re}(s)$ is $1/2$, greater than $1/2$ or less than $1/2$, since exponentiation by $p>1$ is strictly increasing. Its reflected value is

$$\Delta_p(R_\zeta s) = -\frac{p}{|p^s|^2} \Delta_p(s), \tag{C5.4}$$

by substituting $|\Lambda_p(R_\zeta s)|^2=p^2/|p^s|^2$. A paired sign change thus preserves the precise nonzero defect when the point is not fixed.

The quotient has an exact faithful two-prime receiver on S . For distinct primes p,q the map $(j_p, j_q):S \rightarrow E_p \times E_q$ is injective. To prove this, $j_p(s)=j_p(t)$ implies $s-t=m+2\pi i k/\log(p)$, m,k integers. Since $\operatorname{Re}(s-t)$ lies strictly between -1 and 1 , $m=0$. Equality for q similarly gives $\operatorname{Im}(s-t)=2\pi i l/\log(q)$. If this common number is nonzero, k,l are nonzero and $k \log(q)=l \log(p)$, contradicting unique prime factorization after exponentiation, also for negative integer exponents. Therefore $k=l=0$ and $s=t$. This provides an exact way to retain the original strip coordinate rather than treating its one-prime quotient as injective.

In particular RH is exactly the assertion that every nontrivial zero ρ of the ORIGINAL zeta has $j_p(\rho)$ in the odd fixed circle, for one (hence every) prime p . This is a proved equivalence, not an assumption inserted to prove RH. Multiplicity is retained by attaching the original integer m_ρ to each original zero and using the injective two-prime receiver. No infinite pushforward is asserted to be a locally finite divisor on a compact torus.

The functional equation supplies the paired zero $1-\bar{\rho}$; C5.2 proves the corresponding symmetry of their images. It does not by itself identify the two original zeros. The injective two-prime receiver makes the distinction testable on the actual source zeros, without changing their equation to the zeros of an auxiliary function.

C6. The original-zeta multiplier and an exact factor-four correction

The supplied `rhready.tex`, lines 1062-1080, defines

$$\begin{aligned}\xi(s) &= \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), & \mathcal{E}(f)(u) &= u^{1/2}\sum_{n=1}^{\infty}f(nu), \\ h(u) &= \frac{\pi}{2}u^2(2\pi u^2 - 3)e^{-\pi u^2}, & k &= \mathcal{E}(h).\end{aligned}\tag{C6.1}$$

Compute from these actual definitions. For $\text{Re}(s) > 1$, absolute convergence permits the change $v=nu$ and interchange of integral and sum:

$$\int_0^{\infty} \mathcal{E}(h)(u)u^{s-1/2}\frac{du}{u} = \zeta(s) \int_0^{\infty} h(v)v^s\frac{dv}{v}.\tag{C6.2}$$

Indeed the sum of absolute integrals is bounded by $\zeta(\text{Re}(s))$ times the finite integral of $|h(v)|v^{\text{Re}(s)}dv/v$. The Gaussian controls infinity and $h(v)=O(v^2)$ controls zero. Substitution $t=\pi v^2$ yields the complete two-term integral

$$\begin{aligned}\int_0^{\infty} h(v)v^s\frac{dv}{v} &= \frac{\pi}{2} \left[2\pi \frac{1}{2}\pi^{-(s+4)/2}\Gamma\left(\frac{s+4}{2}\right) - 3\frac{1}{2}\pi^{-(s+2)/2}\Gamma\left(\frac{s+2}{2}\right) \right] \\ &= \frac{s(s-1)}{8}\pi^{-s/2}\Gamma(s/2).\end{aligned}\tag{C6.3}$$

The second displayed expression is an evaluated identity, not a replacement of the two original contributions; both remain in C6.3. The Gamma recurrence proves the identity by extracting $\Gamma(s/2)$ and evaluating $(1/2)(s/2)((s/2+1)-3/2)$.

Consequently the correct exact comparison is

$$\boxed{\int_0^{\infty} k(u)u^{s-1/2}\frac{du}{u} = \frac{1}{8}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) = \frac{1}{4}\xi(s).}\tag{C6.4}$$

The equality to ξ asserted for this displayed k in the source is therefore missing a factor 4. At $s=2$ the left side is $\pi/24$ and $\xi(2)=\pi/6$. This also supplies a direct exact check independent of numerical plotting.

For completeness the equality continues to the critical line without discarding boundary terms. Under the source Fourier convention $\exp(2\pi ixy)$, the Gaussian $g=\exp(-\pi x^2)$ transforms to itself; differentiating the Gaussian transform gives $F(x^{2g})=(1/(2\pi i)\cdot y/2)g$ and $F(x^{4g})=(y/4\cdot 3y^{2/\pi i+3/(4\pi i)}))g$. Since $h=\pi^{2x}4g-(3\pi/2)x^2g$,

these identities give $Fh=h$, $h(0)=0$ and $\text{integral } h=0$. Poisson summation then gives $k(u)=k(1/u)$. Gaussian decay of k at infinity and this symmetry give rapid decay at both ends in logarithmic coordinates. Thus its Mellin integral is entire in s , and analytic continuation extends C6.4. For $\text{Xi}(t)=\text{xi}(1/2+it)$ the corrected equality is $\text{Xi}(t)=4*\text{integral } k(u)u^{it}du/u$, keeping exactly the source kernel.

The original zeta remains recovered on every domain where the displayed inverse is defined:

$$\zeta(s) = \frac{8\pi^{s/2}}{s(s-1)\Gamma(s/2)} \int_0^\infty k(u)u^{s-1/2} \frac{du}{u}. \quad (\text{C6.5})$$

In the open critical strip the multiplier $C(s)=s(s-1)\pi^{-(s/2)}\Gamma(s/2)/2$ is holomorphic and nonzero, so C6.4-C6.5 preserve all zero multiplicities there. Globally keep

$$\frac{\zeta'}{\zeta} = \frac{\zeta'}{\zeta} + \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \frac{\Gamma'(s/2)}{\Gamma(s/2)}. \quad (\text{C6.6})$$

At 1 the factor C has a simple zero and ζ a simple pole. At 0, $s\Gamma(s/2)$ has limit 2, so $C(0)=-1$ and $\zeta(0)=-1/2$ yields $\text{xi}(0)=1/2$. At $-2m$, $m \geq 1$, C has leading coefficient $2m(2m+1)(-1)^{m+1}m/m!$ times $(s+2m)^{-1}$; the original zeta has a simple trivial zero there. Retain the local pair $\zeta(s)=(s+2m)a_m(s)$, $C(s)=(s+2m)^{-1}b_m(s)$, so $\text{xi}=a_m/b_m$ and $\zeta=(s+2m)\text{xi}/b_m$. The nonzero $b_m(-2m)$ is the displayed coefficient. These are local meromorphic identities; no canceled divisor is declared absent from the original zeta.

C7. The plotted projection defect and its actual involution

Use exactly $H=L^2(\mathbb{R})$ even, the Fourier transform F with kernel $\exp(2\pi ixy)$, $P=P_\lambda$ the projection to $[-\lambda, \lambda]$, and $Q=FF^*(-1)$. On H , $F^2=I$, so $Q=FF^*F$. For the source prolate eigenfunction $h=h(2m, \lambda)$, $Ph=h$ and $PFPh=\chi_m h$. Therefore

$$PQP h = \chi_m^2 h, \quad \|(I-Q)h\|^2 = (1 - \chi_m^2)\|h\|^2. \quad (\text{C7.1})$$

Proof: $PQP=(PFP)^2$ on PH . Since Q is an orthogonal projection, $\langle Qh, h \rangle = \|Qh\|^2 = \chi_m^2 \|h\|^2$, and orthogonal Pythagoras gives the second identity. For $m=2$, the source χ_2 is positive and strictly below 1. Hence the precise orange quantity in the supplied plot is

$$1 - \chi_2 = \frac{\|(I-Q)h_{4,\lambda}\|^2}{(1 + \chi_2)\|h_{4,\lambda}\|^2}. \quad (\text{C7.2})$$

Every norm and factor remains displayed.

The exact reflection product behind this defect is $U=(2P-I)(2Q-I)$. Put $a=|\chi_m|$, $b=\sqrt{1-a^2}$, with $0 < a < 1$, and choose $e=h/\|h\|$, $f=(Qe - a^2 e)/(ab)$. Then $Pe=0$, $\langle f, e \rangle = 0$, $\|f\|=1$. Applying $Q^2=Q$ gives $Qf=ab e + b^2 f$. Thus the span of e, f is invariant under both projections and their matrices are

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} a^2 & ab \\ ab & b^2 \end{pmatrix}, \quad U = \begin{pmatrix} 2a^2 - 1 & 2ab \\ -2ab & 2a^2 - 1 \end{pmatrix}. \quad (\text{C7.3})$$

The relation $a^{2+b^2}=1$ gives $U^*U=I$. Writing $a=\cos(\alpha)$, $b=\sin(\alpha)$, $0 < \alpha < \pi/2$, its eigenvalues are $\exp(2i\alpha)$ and $\exp(-2i\alpha)$. This establishes the nontrivial phase of the product of the two involutions from the source projections; no physical holonomy is assumed. The leakage $1-a^{2=\sin^2(\alpha)}$ remains recoverable from it.

C8. The same projection block retains both trace signs

The source's archimedean and semilocal trace expressions use $D=I-P-Q$, not just $I-PQ$. In the exact block C7.3,

$$D = \begin{pmatrix} -a^2 & -ab \\ -ab & a^2 \end{pmatrix}, \quad D^2 = a^2 I, \quad \text{Tr } D = 0. \quad (\text{C8.1})$$

Multiplication of this displayed matrix proves the square identity; its characteristic polynomial is $t^2 - a^2$. Thus its two eigenvalues are $+a$ and $-a$. A unitary product of reflections and a positive leakage quantity coexist with these two trace signs on the same source block. For any positive operator B on this block, in an orthonormal D -eigenbasis, $\text{Tr}(BD) = a(\langle Be_+, e_+ \rangle - \langle Be_-, e_- \rangle)$. Both terms must be retained. This does not assert that an arbitrary B is the image of a global admissible Weil test.

The full source formula, `rhready.tex weilqb`, is

$$-\sum_{v \in S} W_v(f) = \log(TW)f(1) + \text{Trace}\left(\vartheta(f)(I - P_T^S - \widehat{P}_W^S)\right). \quad (\text{C8.2})$$

The cutoff term, the admissible test representation, the prime contributions and the trace's meaning in the source are not removed. C8.1 computes its projection factor only; it is not a claim that the individual unsmoothed factors are globally trace class. The blue plot quantity $\epsilon(\lambda)$ is the least eigenvalue of the full Weil operator A_λ in the different space $L^{2([\lambda(-1), \lambda], du/u)}$. No equality between that operator and the positive leakage operator is supplied by C7.2. The source itself describes the comparison in `poisson` and `miss` as numerical and leaves the eigenfunction comparison unfinished.

C9. The two prolate modes have an exact nonzero endpoint defect

Fix $\lambda > 1$. Let $h_0 = h_0(\lambda)$, $h_4 = h_4(\lambda)$, $c_0 = h_0(0)$, $c_4 = h_4(0)$, with the source support extension by zero. Both c_0 and c_4 are nonzero: the functions are even, so their first derivatives at 0 vanish; their second-order prolate differential equation is regular at 0. If their values vanished too, uniqueness for the initial-value problem would make the eigenfunction identically zero, a contradiction.

The compressed Fourier identities give integral $h_0 = \chi_0 c_0$ and integral $h_4 = \chi_2 c_4$ by evaluation at Fourier coordinate 0. The source's strictly ordered squared eigenvalues and positive signs at indices 0,2 give $1 > \chi_0 > \chi_2 > 0$. Thus the exact endpoint map on their two-dimensional span is

$$L(a_0 h_0 + a_4 h_4) = \begin{pmatrix} f(0) \\ \widehat{f}(0) \end{pmatrix} = \begin{pmatrix} c_0 & c_4 \\ \chi_0 c_0 & \chi_2 c_4 \end{pmatrix} \begin{pmatrix} a_0 \\ a_4 \end{pmatrix}, \quad \det L = c_0 c_4 (\chi_2 - \chi_0) \neq 0. \quad (\text{C9.1})$$

No nonzero combination of these two modes kills both endpoints. This is an invertible map onto the two endpoint values, so the obstruction also constructs their complete two-mode parameterization: for prescribed values (u, v) ,

$$a_0 = \frac{\chi_2 u - v}{c_0(\chi_2 - \chi_0)}, \quad a_4 = \frac{v - \chi_0 u}{c_4(\chi_2 - \chi_0)}. \quad (\text{C9.2})$$

For the source's integral-zero choice retain any scalar A and write

$$h_\lambda = A(\chi_2 c_4 h_0 - \chi_0 c_0 h_4), \quad \widehat{h}_\lambda(0) = 0, \quad h_\lambda(0) = A c_0 c_4 (\chi_2 - \chi_0). \quad (\text{C9.3})$$

The last value is nonzero for $A \neq 0$; it tends toward a small value only under additional estimates, not by an exact cancellation. Conversely choosing $f(0)=0$ leaves the Fourier endpoint $c_0 c_4 (\chi_0 - \chi_2)$ times the chosen scalar. These formulas retain the distinction between near cancellation and actual vanishing.

The next source mode constructs an exact repair of both endpoint equations. Put $h_8 = h_8(\lambda)$, $c_8 = h_8(0)$, and let χ_4 be its compressed Fourier eigenvalue. As above $c_8 \neq 0$, and the source ordering gives $1 > \chi_0 > \chi_2 > \chi_4 > 0$. Define the actual function

$$g_\lambda = \frac{\chi_2 - \chi_4}{c_0} h_0 + \frac{\chi_4 - \chi_0}{c_4} h_4 + \frac{\chi_0 - \chi_2}{c_8} h_8. \quad (C9.4)$$

Its value at zero is the sum of the three displayed numerators, which is 0. Its integral is $\chi_0(\chi_2 - \chi_4) + \chi_2(\chi_4 - \chi_0) + \chi_4(\chi_0 - \chi_2) = 0$ by direct expansion. The coefficient of h_0 is nonzero; the distinct prolate eigenfunctions are linearly independent, so g_λ is nonzero. The endpoint map on the three-mode span has rank 2 by its already invertible first two columns. Its kernel is therefore exactly the one-dimensional space spanned by g_λ . This proves existence, uniqueness up to its retained scalar, and the exact map to both endpoint values.

The remaining Fourier defect is also explicit. Set $d_0 = (\chi_2 - \chi_4)/c_0$, $d_2 = (\chi_4 - \chi_0)/c_4$, $d_4 = (\chi_0 - \chi_2)/c_8$, so $g_\lambda = \sum_{j=0,2,4} d_j h_{2j}(\lambda)$. Orthogonality of the distinct selfadjoint prolate eigenfunctions and the compressed Fourier equation give

$$\|(F - I)g_\lambda\|^2 = 2 \sum_{j \in \{0,2,4\}} |d_j|^2 (1 - \chi_j) \|h_{2j,\lambda}\|^2 > 0. \quad (C9.5)$$

Indeed F is a unitary involution on the even Hilbert space, so the left side equals $2\|g_\lambda\|^2 - 2\operatorname{Re}\langle Fg_\lambda, g_\lambda \rangle$. Since every $h_{2j}(\lambda)$ is supported in P , its pairing with $Fh_{2k}(\lambda)$ equals its pairing with $\chi_k h_{2k}(\lambda)$. Expanding gives exactly C9.5, including every coefficient and norm. Strict positivity follows from $\chi_j < 1$ and the nonzero coefficients. The endpoint repair thus retains a nonzero Fourier defect; no finite-cutoff function has been declared exactly Fourier invariant.

For an even Schwartz function f , Poisson summation gives the full endpoint formula

$$\mathcal{E}(\hat{f})(x) = \mathcal{E}(f)(x^{-1}) + \frac{f(0)}{2\sqrt{x}} - \frac{\sqrt{x}}{2} \hat{f}(0). \quad (C9.6)$$

To prove it, separate $n=0$ in $\sum_{n \in \mathbb{Z}} f(n/x) = x \sum_{n \in \mathbb{Z}} \hat{f}(nx)$, pair n with $-n$, and multiply by $x^{-(1/2)/2}$. This retains both half-factors and both endpoints. The source's equation `poissonintro` is its restriction to both endpoint values zero.

The compactly truncated prolate functions are not automatically Schwartz; C9.1-C9.5 require only their integrability and the source compressed Fourier equations, and are already exact. C9.6 is stated on its proved Schwartz domain and is not silently applied to nonsmooth extensions. Its endpoint receiver L identifies precisely the values that an extension of that identity or a smoothing construction must retain. The nonzero two-mode endpoint and the three-mode repair are therefore explicit data, not a discarded obstacle and not a claimed negative Weil value.

C10. Return to the existing supported original-zeta form

The existing coefficient receiver is used rather than replacing the programme by the authors' scalar form. Write $U=[\tau]$, $f=[1]$, $Q_0=U \cdot f$ in the auxiliary coefficient algebra of `WEIL_UNIT_DIFFERENCE_LIFT.md`, `WU1-WU8`. These are formal linear coefficients; no source subtraction `tau-1` is asserted. Products give $f^2=f$, $Q_0^2=Q_0$, $fQ_0=0$, $U=f+Q_0$. Arithmetic evaluation sends both U and f to scalar 1 and Q_0 to scalar 0.

For an original compact smooth test $h(v)$, retain $H(s)=\int h(v)\exp(-(s-1/2)v)dv$, the full prime-power term $P(h)$, and the Gamma term $A_\infty(h)$. The programme's exact original-zeta receiver is

$$\begin{aligned} W_\zeta(h) &= H(0) + H(1) + A_\infty(h) - P(h) = \sum_\rho m_\rho H(\rho), \\ P(h) &= \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} (h(\log n) + h(-\log n)), \\ A_\infty(h) &= \frac{1}{2\pi} \int_{\mathbb{R}} \hat{h}(t) \left(\Re \frac{\Gamma'(1/4 + it/2)}{\Gamma(1/4 + it/2)} - \log \pi \right) dt. \end{aligned} \quad (C10.1)$$

Every finite trivial-zero cutoff is retained in the original identity $V(\zeta, N) = Z(H) + \sum_{m=1}^N H(-2m) - H(1) = G_N - P(h)$, $G_N = A_\infty + H(0) + \sum_{m=1}^N H(-2m)$. The cancellation in the displayed W_ζ is an explicitly recorded map from this triple, not removal of its components. Its full contour derivation is OZC1-OZC15 in the retained programme source; WU2 states the boundary domains and convergence.

The source test identification is $a(x) = h(\log x)$. It preserves convolution and the involution $a^*(x) = \overline{a(x^{-1})}$ because $dx/x = dv$. The source Fourier integral of a at t is $H(1/2 + it)$. Its endpoint values at $\pm i/2$ are respectively $H(0), H(1)$, and the prime terms are exactly $P(h)$. Thus the compared source form is reached by an explicit change of variable with all half-powers retained.

The archimedean comparison also retains its constant. The digamma integral, for $\text{Re}(z) > 0$, is $\psi(z) = -\gamma + \int_0^\infty \frac{\exp(-t) - \exp(-zt)}{(1 - \exp(-t))} dt$. Apply it to $z = 1/4 + iy/2$ and Fourier inversion of h . First perform the t integral on $[\delta, R]$; then send δ to 0 and R to infinity. The paired numerator at 0 vanishes to first order, and the Schwartz decay of $\hat{h}$ controls the polynomial y factors in its Taylor estimate. At infinity the exponential factors dominate. These bounds justify the passage to the limit and yield, after $t = 2v$,

$$\begin{aligned} A_\infty(h) &= -(\gamma + \log \pi)h(0) \\ &\quad + \int_0^\infty \frac{2h(0)e^{-2v} - (h(v) + h(-v))e^{-v/2}}{1 - e^{-2v}} dv. \end{aligned} \quad (C10.3)$$

The supplied source's archimedean expression, after $x = \exp(v)$, is exactly

$$\begin{aligned} W_{\mathbb{R}}(a) &= (\log(4\pi) + \gamma)h(0) \\ &\quad + \int_0^\infty \frac{(h(v) + h(-v))e^{-v/2} - 2h(0)e^{-v}}{1 - e^{-2v}} dv. \end{aligned} \quad (C10.4)$$

Adding C10.3 and C10.4 gives $\log(4)h(0) - 2h(0)\int_0^\infty \frac{\exp(-v)}{(1 + \exp(-v))} dv$. The last integral equals $\log(2)$, by the substitution $w = \exp(-v)$. Hence $A_\infty(h) = -W_{\mathbb{R}}(a)$, with the exact $\log(4\pi)$, γ and $\log(\pi)$ constants accounted for. Together with the already proved prime and endpoint maps this gives $W_\zeta(h) = H(0) + H(1) - \sum_v W_v(a)$ without dropping an archimedean summand.

For the programme's support lattice L retain the vectors

$$\begin{aligned} B &= \mathbf{e}_{1_L}(H(0) + H(1)) + \sum_{\ell \neq 1_L} \mathbf{e}_\ell H(0), \quad Z = \mathbf{e}_{1_L} Z(H), \\ D &= \mathbf{e}_{1_L}(P(h) - A_\infty(h)) + \sum_{\ell \neq 1_L} \mathbf{e}_\ell H(0). \end{aligned} \quad (C10.2)$$

C10.1 proves $\text{bold}(B)\text{-}\text{bold}(Z)=\text{bold}(D)$ on each coordinate. Tensoring each displayed term with U or f retains the two source records; their difference is the same complete vector tensored with Q_0 . Arithmetic evaluation returns C10.2 without deleting a support coordinate. This is the exact comparison from WU3-WU5, and does not turn Q_0 into an extra classical Euler factor at 0.

The support trace proof is [SZW33-SZW38 at the pinned programme edition](#). The complete local return and full original-zeta reconstruction are retained with their proof locators in WEIL_UNIT_DIFFERENCE_LIFT.md, WU2-WU5, and the companion original source copies identified in the source ledger.

What has been established here is the actual base comparison, the sign-adjusted Tate fixed locus with all winding data, its faithful two-prime map from the original critical strip, the factor-four repair, the two-mode endpoint isomorphism, its three-mode repair with the complete Fourier defect, and the two-sign trace block. None of these statements assumes every zeta zero is fixed. The supplied geometry's involution and equal sphere radius have been carried through to these exact arithmetic and trace expressions; their values have not been assigned a positive sign by definition.

Signed sphere degree and finite-field comparison

Proof source: TWISTOR_DEGREE_DERIVATION.md.

The full signed twistor action: degree, cohomology, boundary and coefficients

24 September 2026. Independent mathematical derivation. Private working note. Proof locators TD1-TD12.

Source and exact scope

The source is Alain Connes and Caterina Consani, *The Absolute Twistor Line and the Geometry of the compactification of Spec Z*, arXiv:2609.00299v1. Original author source: literature/lorscheid_sources_20260923/papers/06_new_user_intake/2609.00299/tex/CC.tex. This derivation read source lines 492-620, 772-903, 961-1194 and 1197-1226, together with W1-W10 of work/stacked_zero_intake_20260924/deligne_visual/WINDING_PARITY_AND_INTEGER_SPECTRUM.md. These are actual reading ranges, not a claim of complete reading. Source sections 3-4 supply the signed stalk and its two branches; source Theorem thm:twistor_space_scheme supplies the quotient real structure; source Proposition prop:twistor_frobenius supplies the distinguished odd power maps. The public author edition is [arXiv:2609.00299v1](https://arxiv.org/abs/2609.00299v1).

The calculations below concern the actual complex points, the retained signed maps, and explicitly constructed coefficient objects. They do not assign an arithmetic operation, vector, coordinate or distance to the supporting point. In particular, the coordinate below is the actual value of the source section T under a complex character. It is not a coordinate of the user's proposed support τ .

The source quotient of complex points and the source equivariant topos are different objects connected by evaluation. This note computes the cohomology of the former and explicit coefficient constructions on it. It makes no identification of that cohomology with the cohomology of the entire absolute arithmetic topos.

TD1. Full maps before and after taking the geometric quotient

Retain the source group and involution

$$G = \langle T, J, \epsilon \mid J^2 = \epsilon, \epsilon^2 = 1, TJ = JT \rangle, \quad A(T) = \epsilon T^{-1}, \quad A(J) = J^{-1}, \quad A(\epsilon) = \epsilon.$$

Here T is invertible, J has exact order four and $\epsilon = J^2$ has exact order two. A full signed map preserving ϵ and commuting with A is

$$f_{n,c,b}(T) = T^n J^c, \quad f_{n,c,b}(J) = J^b, \quad n \in 2\mathbb{Z} + 1, \quad c \in \mathbb{Z}/4\mathbb{Z}, \quad b \in \{1, 3\}. \quad (\text{TD1.1})$$

For completeness, every finite-order element of $G \cong \mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$ belongs to $\langle J \rangle$, so the image of J is J^b . The equation $f(J^2) = \epsilon$ is $2b = 2 \pmod{4}$. The image of T is $T^n J^c$. On J commutation with A is automatic; on T it says

$$f(A(T)) = \epsilon T^{-n} J^{-c} = A(f(T)) = \epsilon^n T^{-n} J^{-c}.$$

Thus n is odd, and the displayed conditions are also sufficient on both generators. In particular $n \neq 0$. Nothing has been removed from the J -branch or the sign.

A generic complex character is

$$\chi_{z,j}(T) = z, \quad \chi_{z,j}(J) = j, \quad \chi_{z,j}(\epsilon) = -1, \quad z \in \mathbb{C}^\times, \quad j \in \{i, -i\}.$$

Precomposition gives the full point map

$$g_{n,c,b}(z, j) = (j^c z^n, j^b). \quad (\text{TD1.2})$$

On the unquotiented pair of spheres

$$Y = \mathbb{P}^1(\mathbb{C})_i \sqcup \mathbb{P}^1(\mathbb{C})_{-i},$$

the map extends by the same rational formula on each component. For $n > 0$, the component map sends $0 \mapsto 0, \infty \mapsto \infty$; for $n < 0$, it sends $0 \mapsto \infty, \infty \mapsto 0$. The target branch remains j^b .

The source geometric involution is

$$\alpha(z, j) = (-z^{-1}, -j), \quad (\text{TD1.3})$$

including its endpoint exchange. Its square is the identity. For $z \neq 0, \infty$,

$$g\alpha(z, j) = ((-j)^c (-z^{-1})^n, (-j)^b) = ((-1)^{c+1} j^c z^{-n}, -j^b),$$

whereas

$$\alpha g(z, j) = (-j^{-c} z^{-n}, -j^b).$$

The two first entries agree because $j^{2c} = (-1)^c$. They agree at the endpoints by the already stated extension. Thus g descends to $Y/\langle \alpha \rangle$.

Identify the quotient with the $j = i$ sphere by the exact quotient map

$$\pi(z, i) = z, \quad \pi(z, -i) = -z^{-1}. \quad (\text{TD1.4})$$

This is a homeomorphism on each component, and every α -orbit meets the i component exactly once. Consequently the induced map of the quotient sphere is

$$F_{n,c,b}(z) = \begin{cases} i^c z^n, & b = 1, \\ -i^{-c} z^{-n}, & b = 3. \end{cases} \quad (\text{TD1.5})$$

Both formulas retain the branch choice, its coefficient and its exponent. To state calculations common to the two formulas, write

$$(a, d) = \begin{cases} (i^c, n), & b = 1, \\ (-i^{-c}, -n), & b = 3. \end{cases} \quad (\text{TD1.6})$$

Thus $F(z) = az^d$, where d is a nonzero odd integer and $a\bar{a} = 1$. Equations TD1.5-TD1.6 are retained whenever this notation is used; they are not a replacement that forgets the signed data.

The source power map is $n > 0$ odd, $c = 0$, $b \equiv n \pmod{4}$. Equation TD1.5 then gives exactly z^n for $n \equiv 1 \pmod{4}$, and $-1/z^n$ for $n \equiv 3 \pmod{4}$.

TD2. Holomorphic degree and the antiholomorphic involution

Use homogeneous coordinates $[X : Y]$, with $z = X/Y$. If $d > 0$, the map is

$$[X : Y] \mapsto [aX^d : Y^d]. \quad (\text{TD2.1})$$

If $d < 0$, set $e = -d > 0$; the map is

$$[X : Y] \mapsto [aY^e : X^e]. \quad (\text{TD2.2})$$

In each expression the two homogeneous polynomials have the same degree, have no common zero on projective space, and have degree $N = |d| = |n|$. The map is therefore holomorphic at both endpoints as well as on $\mathbb{C}^\times$.

Its topological degree, with the complex orientation on both spheres, is N . Indeed choose a target value $w \neq 0, \infty$. For $d > 0$, its inverse images are the d distinct roots of $z^d = w/a$; the derivative adz^{d-1} is nonzero at every such root. For $d < 0$, they are the e distinct roots of $z^e = a/w$; the derivative adz^{d-1} is again nonzero there. A nonzero complex derivative has real determinant equal to its squared complex modulus, which is positive. Every inverse image consequently contributes $+1$ to the oriented degree. This proves

$$\deg F_{n,c,b} = |n|. \quad (\text{TD2.3})$$

In particular the reciprocal branch does not introduce a negative sphere degree: the holomorphic transformation $z \mapsto -1/z$ has degree $+1$.

Ordinary conjugation on Y is $(z, j) \mapsto (\bar{z}, -j)$. Equation TD1.4 gives on the quotient

$$\sigma(z) = -\frac{1}{\bar{z}}, \quad \sigma(0) = \infty, \quad \sigma(\infty) = 0. \quad (\text{TD2.4})$$

For finite nonzero z , use d odd and $a\bar{a} = 1$ to compute

$$F\sigma(z) = a \left(-\frac{1}{\bar{z}} \right)^d = -a\bar{z}^{-d},$$

$$\sigma F(z) = -\frac{1}{\bar{a}\bar{z}^d} = -a\bar{z}^{-d}. \quad (\text{TD2.5})$$

The two maps agree at $0, \infty$ by their endpoint formulas. Thus every full signed map TD1.1 commutes with the actual twistor real structure. No condition on c beyond its original value modulo four is needed.

The involution σ is fixed-point-free: a finite nonzero fixed point would give $z\bar{z} = -1$, and the endpoints are exchanged. It has topological degree -1 : conjugation reverses the complex orientation, while $-1/z$ is holomorphic of degree $+1$.

TD3. Exact ordinary cohomology and pairing on the sphere

Let $S = \mathbb{P}^1(\mathbb{C})$ and use integral singular cohomology. The usual cell decomposition of the sphere consists of one zero-dimensional cell and one two-dimensional cell, with no one-dimensional cell. Its cellular cochain groups are therefore $\mathbb{Z}, 0, \mathbb{Z}$ in degrees $0, 1, 2$, and all differentials are zero. Hence

$$H^0(S; \mathbb{Z}) = \mathbb{Z} 1, \quad H^1(S; \mathbb{Z}) = 0, \quad H^2(S; \mathbb{Z}) = \mathbb{Z} h. \quad (\text{TD3.1})$$

Choose the orientation class h by $\langle h, [S] \rangle = 1$. This specifies an integral generator; it does not rescale a source map or remove one of its factors. The full cohomology ring is

$$H^*(S; \mathbb{Z}) = \mathbb{Z}[h]/(h^2), \quad |h| = 2. \quad (\text{TD3.2})$$

The relation $h^2 = 0$ follows because the sphere has no cohomology in degree four. Pullback preserves constant functions, and TD2.3 gives the action on orientation. Thus

$$F^*1 = 1, \quad F^*h = |n|h; \quad \sigma^*1 = 1, \quad \sigma^*h = -h. \quad (\text{TD3.3})$$

There is no degree-one group on which an additional eigenvalue could act. After extension to $\mathbb{Q}, \mathbb{R}$ or $\mathbb{C}$, the eigenvalues in degrees zero and two remain 1 and $|n|$.

The exact Poincaré pairing is

$$H^0(S; \mathbb{Z}) \times H^2(S; \mathbb{Z}) \longrightarrow \mathbb{Z}, \quad (r 1, s h) \longmapsto rs. \quad (\text{TD3.4})$$

It is perfect because its matrix in these bases is (1) . The reversed pairing has the same sign because the degree product is $0 \cdot 2 = 0$. For arbitrary classes x, y whose degrees sum to two,

$$\langle F^*x \smile F^*y, [S] \rangle = |n| \langle x \smile y, [S] \rangle. \quad (\text{TD3.5})$$

This follows either from TD3.3-TD3.4 or from naturality and $F_*[S] = |n|[S]$. The corresponding multiplier for σ is -1 . The sphere pairing therefore pairs the eigenvalue 1 with the eigenvalue $|n|$; it does not produce a degree-one self-pairing.

In the conventional Hodge grading, 1 has type $(0, 0)$, and h has type $(1, 1)$. The first assertion holds because constants are degree-zero functions. For the second, complex dimension one gives $H^{2,0} = H^{0,2} = 0$, so the nonzero degree-two class has type $(1, 1)$. Their Hodge weights are respectively 0 and

2. The eigenvalue 1 on constants belongs to weight zero. It is not a counterexample to a weight-one eigenvalue assertion.

TD4. Both components and the source branches remain visible in cohomology

Order the components of Y as $i, -i$. Let P_b be the two-by-two identity for $b = 1$, and the matrix exchanging the two coordinates for $b = 3$. On $H^0(Y; \mathbb{Z}) = \mathbb{Z}^2$, pullback by g is P_b . On $H^2(Y; \mathbb{Z}) = \mathbb{Z}^2$, it is $|n|P_b$. Indeed component pullback selects the target component, and the degree on each component is $|n|$. The coefficient j^c is multiplication by a nonzero complex number, which is holomorphic of degree one, so it contributes no further integer to this cohomological degree. Its full point-map value is still TD1.2.

The action of α^* is the exchanging matrix in both degrees zero and two: its component interchange is accompanied by the holomorphic map $-1/z$ of degree one. Conjugation on Y has the exchanging matrix in degree zero and its negative in degree two. The map

$$\pi^* : H^0(S; \mathbb{Z}) \longrightarrow H^0(Y; \mathbb{Z}), \quad r \longmapsto (r, r),$$

and the same formula in degree two identify the quotient groups with the α^* -invariants. This holds integrally here because Y is a disjoint union of two copies each mapped homeomorphically to S , rather than by invoking an averaging operation that would divide by two. Applying the stated matrices to these diagonal vectors recovers every action in TD3.3.

TD5. The retained generic open has a different degree-one sector

Let $U = S \setminus \{0, \infty\} = \mathbb{C}^\times$ and $D = \{0, \infty\}$. The inclusion is $j : U \hookrightarrow S$. The homotopy

$$H_t(z) = |z|^{-t} z \quad (0 \leq t \leq 1)$$

retracts U onto the unit circle. This formula concerns the complex character value z ; it makes no statement about a metric at the support. The circle has a cell decomposition with one vertex and one oriented edge whose two endpoints coincide, so its cellular cochain differential is zero. Consequently

$$H^0(U; \mathbb{Z}) = \mathbb{Z}, \quad H^1(U; \mathbb{Z}) = \mathbb{Z} u, \quad H^k(U; \mathbb{Z}) = 0 \quad (k \geq 2). \quad (\text{TD5.1})$$

Specify u by its value 1 on the positively oriented loop $\gamma(t) = e^{2\pi i t}$, $0 \leq t \leq 1$. Under $F(z) = az^d$, this loop winds d times, with the sign of d retained. Therefore

$$F^*u = d u = \begin{cases} n u, & b = 1, \\ -n u, & b = 3. \end{cases} \quad (\text{TD5.2})$$

Multiplication by the coefficient a does not change the winding number: a path of nonzero constants from 1 to a gives a homotopy between the two circle maps. This statement records its exact topological effect; it does not change TD1.5.

On that loop, $\sigma(e^{2\pi i t}) = -e^{2\pi i t}$. Thus

$$\sigma^*u = u. \quad (\text{TD5.3})$$

The distinction from TD3.3 is forced by the domains: an antiholomorphic map can reverse the orientation of the sphere while preserving the winding of the punctured sphere because it also exchanges its two punctures.

The differential form $\omega = dz/z$ satisfies

$$F^*\omega = d \frac{dz}{z}, \quad \int_{\gamma} \omega = 2\pi i, \quad [\omega] = 2\pi i u_{\mathbb{C}}. \quad (\text{TD5.4})$$

At 0 its residue is $+1$. At ∞ , using $w = 1/z$, it is $-dw/w$ and its residue is -1 . Both residues and the factor $2\pi i$ are retained.

The unfolded punctured space has two copies of U . In their respective z -coordinates, $g^* = nP_b$ on degree one, while $\alpha^* = -P_3$. The quotient pullback is $u \mapsto (u, -u)$, because π on the second branch is $-1/z$. Applying nP_b to this anti-diagonal vector gives the eigenvalue d in TD5.2. This is the full branch calculation relating the punctured and unpunctured actions.

TD6. Boundary sequence, compact support and the exact degree-one pairing

The long exact sequence of the pair (S, D) , using TD3.1 and the two-point cohomology of D , gives

$$0 \longrightarrow \mathbb{Z} \xrightarrow{r \mapsto (r, r)} \mathbb{Z}^2 \longrightarrow H^1(S, D; \mathbb{Z}) \longrightarrow 0, \quad (\text{TD6.1})$$

and

$$H^2(S, D; \mathbb{Z}) \xrightarrow{\sim} H^2(S; \mathbb{Z}). \quad (\text{TD6.2})$$

For the open complement of a closed subset in the compact sphere, relative cohomology is compactly supported cohomology of the complement. This identification follows by extending a cochain with compact support in U by zero and taking a decreasing family of neighborhoods of D ; equivalently, both compute reduced cohomology of the quotient obtained by collapsing D to its point at infinity. Thus

$$H_c^0(U; \mathbb{Z}) = 0, \quad H_c^1(U; \mathbb{Z}) = \mathbb{Z}^2/\mathbb{Z}(1, 1), \quad H_c^2(U; \mathbb{Z}) = \mathbb{Z}. \quad (\text{TD6.3})$$

The map F is proper on U . To see this, the inverse image of a compact subset of $\mathbb{C}^\times$ has $|z|^d$ bounded above and away from zero, hence $|z|$ bounded above and away from zero; it is closed and bounded in $\mathbb{C}$. Its endpoint action fixes $0, \infty$ for $d > 0$ and exchanges them for $d < 0$. On $\mathbb{Z}^2/\mathbb{Z}(1, 1)$, the exchanging map acts by -1 , because the class of $(0, 1)$ equals the negative of the class of $(1, 0)$. Naturality of TD6.1-TD6.2 therefore proves

$$F^*|_{H_c^1(U)} = \text{sgn}(d), \quad F^*|_{H_c^2(U)} = |n|. \quad (\text{TD6.4})$$

Likewise $\sigma^* = -1$ on both compactly supported groups: its endpoint action exchanges the two entries, and its sphere degree is -1 .

The pairing

$$H^1(U; \mathbb{Z}) \times H_c^1(U; \mathbb{Z}) \longrightarrow H_c^2(U; \mathbb{Z}) \xrightarrow{\int_U} \mathbb{Z} \quad (\text{TD6.5})$$

is perfect. Here is an explicit integral generator, orientation and factor check. Write $z = e^{r+i\theta}$, so the complex orientation is $dr \wedge d\theta$. The real class $u_{\mathbb{R}}$ is represented by $d\theta/(2\pi)$. Choose a smooth compactly supported function $\rho(r)$ with $\int_{\mathbb{R}} \rho(r)dr = 1$, and put $\chi(r) = \int_r^{\infty} \rho(s)ds$. This function is one near the endpoint 0 and zero near the endpoint ∞ . Its derivative is $d\chi = -\rho(r)dr$. The connecting map in TD6.1 sends the integral endpoint class $(1, 0)$ to a generator v , whose de Rham representative is $d\chi$: this is the definition of the connecting map, obtained by extending the endpoint data and differentiating the extension. Consequently the representative $-\rho(r)dr$ is the image of an integral class, and its pairing is

$$\int_U \frac{d\theta}{2\pi} \wedge (-\rho(r)dr) = \frac{1}{2\pi} \left(\int_{\mathbb{R}} \rho(r)dr \right) \left(\int_0^{2\pi} d\theta \right) = 1. \quad (\text{TD6.6})$$

Since both integral groups are infinite cyclic and their specified integral generators pair to one, their pairing is perfect. Reversing the order in the cup product introduces the degree-one sign: $\int_U v \smile u = -1$. For the unreduced logarithmic form, the first pairing in TD6.6 is instead

$$\int_U \frac{dz}{z} \wedge (-\rho(r)dr) = 2\pi i. \quad (\text{TD6.7})$$

The eigenvalue identity on this pairing is exactly

$$d \operatorname{sgn}(d) = |d| = |n|. \quad (\text{TD6.8})$$

For σ , the two degree-one actions multiply to $1 \cdot (-1) = -1$, its action on the top compactly supported group. No equality of the two degree-one eigenvalue moduli has been assumed or obtained.

TD7. The weight of this actual punctured degree-one class

The local boundary calculation gives a second exact sequence

$$0 \longrightarrow H^1(U; \mathbb{Q}) \xrightarrow{\operatorname{res}} H^0(D; \mathbb{Q})(-1) \xrightarrow{\operatorname{sum}} H^2(S; \mathbb{Q}) \longrightarrow 0. \quad (\text{TD7.1})$$

At the topological level, the first arrow records the integrals over small positively oriented loops about the two punctures. Those loops satisfy their single relation $\gamma_0 + \gamma_{\infty} = 0$ in the sphere with punctures. Therefore the image is the kernel of the sum map, the subgroup of pairs $(r, -r)$. It is one-dimensional by TD5.1, so the sequence is exact. At the de Rham level, the exact vector is $(1, -1)$ for $\omega = dz/z$, with integral comparison $[\omega] = 2\pi i u_{\mathbb{C}}$ as in TD5.4. The Tate symbol (-1) keeps this codimension-one orientation and its comparison factor; it does not remove it.

The logarithmic Hodge filtration can be computed here without an unknown cohomology group. Use the algebraic structure sheaf and algebraic differential forms on the complex projective line. On S , logarithmic one-forms with poles only along D form $\Omega_S^1(\log D)$. A global such form has only simple poles at $0, \infty$; writing it as $f(z)dz$, the conditions force $f(z) = c/z$. Thus its global section space is exactly $\mathbb{C} dz/z$. There are no degree-two forms. Also $H^1(S, \mathcal{O}_S) = 0$: on the cover by the two affine lines, every Laurent polynomial on their intersection is a sum of a polynomial in z and a polynomial in z^{-1} , so the Čech cokernel is zero. The logarithmic de Rham calculation therefore gives $F^1 H^1(U; \mathbb{C}) = H^1(U; \mathbb{C})$, $F^2 = 0$.

For the weight filtration in this logarithmic complex, the possible weight-one part is the image of the compactification cohomology $H^1(S; \mathbb{Q})$. That group is zero by TD3.1. The residue quotient embeds

in $H^0(D; \mathbb{Q})(-1)$, whose two classes have Hodge type $(1, 1)$ and weight two: each is the oriented class of a complex normal disk to a point. Equation TD7.1 proves that every nonzero class of $H^1(U; \mathbb{Q})$ belongs to this weight-two part. Equivalently its complete Hodge structure is $\mathbb{Q}(-1)$, with the explicit $2\pi i$ comparison retained above.

The compactly supported class v comes from the quotient of $H^0(D; \mathbb{Q})$ by the constants in TD6.1. Both numerator and constants have type $(0, 0)$, so $H_c^1(U; \mathbb{Q})$ has type $(0, 0)$ and weight zero. Its pairing with the weight-two class u lands in the weight-two top compactly supported class. This proves why TD6.8 pairs moduli $|n|$ and 1, rather than two moduli $\sqrt{|n|}$.

Thus the actual sphere has weights zero and two and no degree-one ordinary cohomology; its actual punctured generic open has a degree-one boundary class of weight two and a compactly supported degree-one class of weight zero. These are determinations of these coefficient sectors, not a claim that weight-one coefficients cannot occur in further constructions.

TD8. A precise finite-characteristic bridge for the distinguished odd powers

Let p be an odd rational prime. The distinguished map of the source, using $n = p, c = 0, b \equiv p \pmod{4}$, has coefficients 1, -1 , so its exact formulas define a map on $\mathbb{P}^1$ over $\mathbb{F}_p$ as well as over $\mathbb{C}$:

$$F_p(z) = \begin{cases} z^p, & p \equiv 1 \pmod{4}, \\ -1/z^p, & p \equiv 3 \pmod{4}. \end{cases} \quad (\text{TD8.1})$$

Its r -fold composite is the source map F_{p^r} . This follows already from precomposition by the source power map $g \mapsto g^p$; alternatively two reciprocal branches compose as

$$-\frac{1}{(-1/z^p)^p} = z^{p^2},$$

because p is odd, and induction retains precisely the parity of r .

Put $q = p^r$. When $q \equiv 1 \pmod{4}$, the fixed points satisfy $z^q = z$ together with the fixed point ∞ . In characteristic p , the polynomial $z^q - z$ has derivative -1 , so it has exactly q distinct roots over the algebraic closure. Thus there are $q + 1$ fixed points. When $q \equiv 3 \pmod{4}$, neither endpoint is fixed and the equation is $z^{q+1} = -1$. Its roots are all nonzero and its derivative is z^q in characteristic p , so it has exactly $q + 1$ distinct roots. Consequently

$$\# \text{Fix}(F_p^r) = 1 + p^r \quad (r \geq 1). \quad (\text{TD8.2})$$

The associated formal fixed-point generating function is therefore exactly

$$Z(F_p, t) = \exp \left(\sum_{r \geq 1} \frac{1 + p^r}{r} t^r \right) = \frac{1}{(1-t)(1-pt)}. \quad (\text{TD8.3})$$

The equality follows by the formal identity $-\log(1-x) = \sum_{r \geq 1} x^r/r$, applied separately to $x = t$ and $x = pt$. Both denominator factors are retained. There is no degree-one numerator factor in this calculation, precisely as TD3 predicts. This is an explicit bridge of the source action to a finite-characteristic sphere calculation. It is not an equality with the original Riemann zeta function or its

zero-detecting sector. In particular it has not supplied an arithmetic weight-one object merely by relabeling H^0 or the punctured boundary class.

TD9. All local systems on the sphere, and actual local systems on the generic open

Every finite-rank local system of vector spaces on S is constant. Indeed S is simply connected: it is the union of two open disks whose connected overlap retracts onto a circle, and the overlap generator becomes null-homotopic in each disk; van Kampen gives the trivial fundamental group. Parallel transport from a fixed basepoint is consequently independent of the chosen path. It identifies a local system with the constant system with that fiber. Its cellular cochain complex is the complex of TD3 tensored with the fiber, so its degree-one cohomology is zero. Thus merely choosing a nonsingular local system on this sphere cannot create its missing degree-one sector.

The situation on U is explicit. A finite-dimensional complex local system is determined by an invertible matrix $M : V \rightarrow V$, its monodromy along γ . Use the covering $u \mapsto e^u$ from $\mathbb{C}$ to U , and the relation

$$(u + 2\pi i, v) \sim (u, Mv). \quad (\text{TD9.1})$$

This constructs the system and verifies its monodromy. The cellular cochain complex of the circle with these coefficients is

$$V \xrightarrow{M-I} V \quad (\text{TD9.2})$$

in degrees zero and one: the two ends of the lifted edge differ by one deck translation, which is M . It follows exactly that

$$H^0(U; \mathcal{L}_M) = \ker(M - I), \quad H^1(U; \mathcal{L}_M) = \text{coker}(M - I). \quad (\text{TD9.3})$$

Pullback by $F(z) = az^d$ gives monodromy M^d . To see the constant as well as the exponent, choose a logarithm ℓ of the actual coefficient a in TD1.6. A lift of F is $u \mapsto du + \ell$. Increasing u by $2\pi i$ increases its image by $2\pi id$, giving the claimed monodromy. Changing ℓ by $2\pi ik$ changes the fiber identification by M^k . Thus this identification retains its basepath choice; it is not canonical without that choice.

With this identification, the induced cellular map from the M -complex to the M^d -complex is the identity in degree zero and

$$S_d(M) = \begin{cases} \sum_{k=0}^{d-1} M^k, & d > 0, \\ -\sum_{k=d}^{-1} M^k, & d < 0 \end{cases} \quad (\text{TD9.4})$$

in degree one. Each summand is the transport along one traversed edge; for negative d , the edges have the opposite orientation. The chain identity is the exact telescoping equality

$$S_d(M)(M - I) = M^d - I. \quad (\text{TD9.5})$$

For $d < 0$, expanding the sum gives $-\sum_{k=d}^{-1} (M^{k+1} - M^k) = M^d - I$, including every negative-power term. Therefore this gives an actual pullback map

$$\text{coker}(M - I) \longrightarrow \text{coker}(M^d - I). \quad (\text{TD9.6})$$

It is a map between the stated coefficient systems. An endomorphism of one fixed coefficient system requires a chosen coefficient identification in addition; no such identification or spectral purity is supplied by TD9.6 itself. This is an exact domain statement, not an assumed extra cohomology group.

There is a precise relation between the source winding lattice $L = G/\langle J \rangle$ and these loops. Characters form the torus $\text{Hom}(L, \mathbb{C}^\times)$, and a loop γ gives the homomorphism

$$L \longrightarrow \mathbb{Z}, \quad \ell \longmapsto \text{wind}(\gamma(\ell)). \quad (\text{TD9.7})$$

For the source generator $t = q(T)$, the loop $\gamma(t) = e^{2\pi i t_{\text{loop}}}$ gives $t^m \mapsto m$. Thus the fundamental-group lattice is the integral dual $\text{Hom}(L, \mathbb{Z})$, with its explicit evaluation pairing, rather than an unproved identification of a monomial and a loop. The branch-return map $-1/z$ reverses this loop generator; equation TD5.2 records the resulting sign after quotienting.

TD10. The actual sign local system and its order-four real lift

The nontrivial order-two local system on U is obtained from the double cover

$$q_2 : \mathbb{C}^\times \longrightarrow \mathbb{C}^\times, \quad q_2(y) = y^2, \quad (\text{TD10.1})$$

using the sign representation of its deck group $\{1, -1\}$. Its monodromy is $M = -I$. In rank one over $\mathbb{Q}$ or $\mathbb{C}$, equation TD9.2 is multiplication by -2 , an invertible scalar. Therefore

$$H^0(U; \mathcal{L}_{-1}) = H^1(U; \mathcal{L}_{-1}) = 0. \quad (\text{TD10.2})$$

Over the integral sign system, multiplication by -2 has zero kernel and cokernel $\mathbb{Z}/2\mathbb{Z}$; thus H^1 is precisely that torsion group. This distinguishes the rational cohomology statement from the integral one.

The relation to the retained parity is exact. Equation W2 gives $\delta(t^m) = \epsilon^m$. Under the loop-monomial pairing TD9.7, reduction modulo two is unchanged by replacing t by t^{-1} . Since an infinite cyclic lattice modulo two has a unique nonzero element, its dual has a unique nonzero character to $\mathbb{Z}/2\mathbb{Z}$. This supplies the monodromy $m \mapsto (-1)^m$, without selecting an orientation on the parity quotient. The source's odd d preserves this character because $(-1)^{dm} = (-1)^m$.

More explicitly, $F(z) = az^d$ lifts to

$$\tilde{F}(y) = \kappa y^d, \quad \kappa^2 = a. \quad (\text{TD10.3})$$

There are exactly two choices of κ , differing by the deck involution. Squaring TD10.3 gives $a(y^2)^d$, as required. Since d is odd, $\tilde{F}(-y) = -\tilde{F}(y)$; hence each choice induces the requisite sign-coefficient identification, with its twofold choice retained.

The source twistor involution has precisely the two lifts

$$\tilde{\sigma}_\pm(y) = \frac{\pm i}{\bar{y}}. \quad (\text{TD10.4})$$

Indeed their squares as complex numbers are $-1/\bar{y}^2$, giving $\sigma(q_2(y))$. Conversely a lift has $\tilde{\sigma}(y)\bar{y}$ taking values in $\{i, -i\}$; by continuity on connected $\mathbb{C}^\times$, that sign is constant. Their composites satisfy

$$\tilde{\sigma}_\pm^2(y) = \frac{\pm i}{(\pm i)/\bar{y}} = \frac{\pm i}{\mp i/y} = -y. \quad (\text{TD10.5})$$

Thus their square is the nontrivial deck transformation, and their fourth power is the identity. On the associated sign line, use the relation $[-y, v] = [y, -v]$. The induced conjugate-linear lift is $[y, v] \mapsto [\tilde{\sigma}_\pm(y), \bar{v}]$; its square sends $[y, v]$ to $[-y, v] = [y, -v]$, so its square is $-I$. This is an actual order-four lift of the order-two twistor involution, constructed from the same punctured source geometry. It retains a parity defect and an exact coefficient object; by TD10.2, it does not itself manufacture a rational degree-one cohomology sector.

The full relation between the lifted arithmetic map and this lifted real structure also retains a sign. Put $\lambda = \pm i$ and choose κ from TD10.3; since $a\bar{a} = 1$, one has $\kappa\bar{\kappa} = 1$. Direct substitution gives

$$\tilde{F}\tilde{\sigma}_\lambda(y) = \kappa\lambda^d\bar{y}^{-d}, \quad \tilde{\sigma}_\lambda\tilde{F}(y) = \lambda\bar{\kappa}^{-1}\bar{y}^{-d} = \lambda\kappa\bar{y}^{-d}.$$

Thus the ratio is the exact deck sign

$$\tilde{F}\tilde{\sigma}_\lambda = (-1)^{(d-1)/2} \tilde{\sigma}_\lambda\tilde{F}. \quad (\text{TD10.6})$$

The two lifts commute precisely when $d \equiv 1 \pmod{4}$, which, using both branches in TD1.6, is equivalent to $b \equiv n \pmod{4}$. In particular every distinguished source power map has this property: for $n \equiv 1 \pmod{4}$, $d = n \equiv 1$, and for $n \equiv 3 \pmod{4}$, $d = -n \equiv 1$. Consequently the source power maps preserve this actual order-four lifted real structure. General full signed maps still commute with σ on the base by TD2.5, but TD10.6 records the precise extra deck sign on this coefficient construction.

TD11. Weight-one twistor bundles exist even though the base sphere has no H^1

Ordinary cohomology of the parameter sphere is not the same object as a twistor structure carried by a bundle on that sphere. The following concrete construction supplies the comparison.

The tautological line bundle $\mathcal{O}_S(-1)$ has fiber over $[X : Y]$ equal to the line $\mathbb{C}(X, Y) \subset \mathbb{C}^2$. Define the conjugate-linear transformation

$$Q(X, Y) = (-\bar{Y}, \bar{X}). \quad (\text{TD11.1})$$

It sends the line over $[X : Y]$ to the line over $[-\bar{Y} : \bar{X}]$, which is exactly $\sigma([X : Y])$. It therefore lifts σ to the tautological bundle. Its square is

$$Q^2(X, Y) = (-X, -Y). \quad (\text{TD11.2})$$

Thus the bundle lift squares to $-I$, while its projectivization squares to the identity. The induced conjugate-linear lift on the dual bundle $\mathcal{O}_S(1)$ also squares to $-I$, because the dual of scalar multiplication by -1 is multiplication by -1 . Tensor powers give square $(-1)^k I$ on $\mathcal{O}_S(k)$, including negative powers by duality.

There is also a real lift squaring to $+I$ on the rank-two bundle

$$E = \mathcal{O}_S(1) \otimes_{\mathbb{C}} \mathbb{C}^2. \quad (\text{TD11.3})$$

Give the second factor the conjugate-linear map $R(v_1, v_2) = (-\bar{v}_2, \bar{v}_1)$, whose square is $-I$. The tensor map formed from the two conjugate-linear lifts is well-defined: moving a scalar from one tensor factor to the other gives its complex conjugate on both sides. It is conjugate-linear on the tensor product, and its square is $(-I) \otimes (-I) = +I$. This proves the asserted real structure. The bundle in TD11.3 is a direct sum of two copies of $\mathcal{O}_S(1)$; in the conventional twistor-bundle terminology it is pure of weight one. This terminology concerns the bundle degree, not $H^1(S; \mathbb{Q})$. Thus TD3 does not rule out this exact weight-one object.

The relation of these bundles to the source base maps is also exact:

$$F^* \mathcal{O}_S(k) \cong \mathcal{O}_S(|n|k). \quad (\text{TD11.4})$$

For $k = 1$, take a nonzero linear form on the target homogeneous coordinates. Its pullback through TD2.1 or TD2.2 is a homogeneous polynomial of degree $|n|$, whose zero divisor has total multiplicity $|n|$. The corresponding line bundle is $\mathcal{O}_S(|n|)$. Tensor powers and duality give every integer k . Equivalently, the two homogeneous expressions TD2.1-TD2.2 construct the pullback line bundle directly with that transition exponent.

For the explicit rank-two bundle E , its degree is two, while the degree of F^*E is $2|n|$. If $|n| > 1$, these degrees are unequal, so no holomorphic bundle isomorphism $F^*E \cong E$ exists: an isomorphism identifies determinant bundles and hence their zero-divisor degrees. Therefore an arithmetic endomorphism acting on coefficients over a fixed twistor parameter cannot be silently identified with a pullback isomorphism along the nontrivial source map of that parameter. The two operations have the proved connecting formula TD11.4, including its precise failure to preserve degree.

This calculation constructs an actual weight-one twistor bundle and its real structure. It does not equip it with an arithmetic Frobenius spectrum or prove a purity statement for that spectrum. Such a spectrum is additional mathematical content, not a consequence of the terminology “weight one.”

TD12. Exact location of the remaining sector

All maps and groups calculated here are explicit:

1. The full signed source action gives the quotient maps TD1.5, commuting with the actual antiholomorphic involution by TD2.5.
2. Their sphere cohomology is $H^0 = \mathbb{Q}$, $H^1 = 0$, $H^2 = \mathbb{Q}(-1)$, with actions $1, |n|$ and pairing multiplier $|n|$.
3. The retained generic open has $H^1(U) = \mathbb{Q}(-1)$ with action d , and $H_c^1(U) = \mathbb{Q}(0)$ with action $\text{sgn}(d)$. The residue sequence and compactification maps prove their relationship to the sphere and to both boundary points.
4. Every nonsingular local system on the sphere is constant. On the actual generic open, all local systems and their pullbacks have the explicit complexes TD9.2-TD9.6. The actual sign system has no rational cohomology but carries the order-four lift TD10.4-TD10.5.
5. A weight-one real twistor bundle does exist by TD11.3, and its pullback under the source map changes degree as TD11.4 specifies.

These calculations isolate a precise target for a further arithmetic construction: a nonzero coefficient or global gluing sector whose arithmetic operator is defined on that sector and whose weight-one pairing can be proved there. Neither the constants nor the two-puncture boundary class is that sector. The source’s global amalgam is a proposed place to construct it, but the amalgam definition alone is not a computation of its derived coefficient cohomology. No nonzero H^1 , polarization, Frobenius eigenvalue bound or zero-detection map is asserted for that uncomputed global object here.

This is not an impossibility theorem about the programme. The proved connecting objects are the quotient and pullback maps, the compactification/residue sequences, the coefficient complexes, the parity cover and its real lift, and the explicit weight-one bundle. They retain the actual signed winding data while locating exactly which computed sectors have weights zero and two and which constructed bundle has twistor weight one.

Current Deligne reading coverage and witness record

Deligne weight-control reading and reconstruction

The current request is WU055: read and reconstruct the full weight-control mechanism, then calculate its relation to the user's construction. This is a reading log, not a claim that a transferred RH proof exists.

Source edition

Pierre Deligne, *La conjecture de Weil. II*, IHES 52 (1980), 137–252. Canonical routing IDs PUBUNIT-3D90B487DBD1CC3A259C6455 and PUBUNIT-574B0214207BBB2BDABCFFF9. The workspace's RESEARCH_SOURCE_INDEX.md supplies the more recent editable witness absent from those index routes: output/Deligne_Weil_II_S20_LaTeX/typed_latex/S20_FR_record_export.tex, SHA256 d6836ae15d98f2b27eb98c9bd039a7d921679becd9fa47ff4717410289dbd351. This is a complete French transcription export, not Deligne-authored TeX. Its export verification proves page-record preservation, not correctness of every formula. Historical English math-mode text is a secondary reading witness only. Original scans and the intact source archive remain unchanged and have not been read in this continuation. The earlier PDF-format question is superseded by the available current TeX route; no PDF exception has been assumed.

The display script read_deligne_source.py reads the unchanged French TeX, reports printed-page and TeX-line locators, and reverses display escapes. It does not correct the source.

Root's actual reading coverage

- Printed page137 contents; pp138–145 introduction and conventions, complete.
- pp146–156: complete §§1.1–1.2, beginning1.3.1.
- pp157–165: complete §§1.3–1.5; beginning1.6.
- pp166–175: complete §§1.6–1.7 and1.8.1–1.8.4 statement.
- pp176–186: rest1.8, complete1.9–1.11.
- pp187–196: complete §§2.1–2.2.
- pp197–206: §§3.1–3.2 and beginning3.3; printed p203 was separately refetched in full after output truncation.
- pp207–216: remainder3.3 and complete3.4–3.7.
- pp217–228: §§4.1–4.3 and beginning4.4.
- pp229–242: remainder4.4–4.5 and all5.1–5.3.
- pp243–252: all6.1–6.2, notes and bibliography.

This completes root reading of the entire 116-page article in the current French transcription. It does not upgrade that transcription to original author TeX, and does not claim that every cited upstream SGA proof or Weil I has been independently read and reconstructed. Root's reading remains transcription-based; the separately attributed peer original reading is recorded below.

Reading is distinct from independently reconstructing each cited upstream theorem. The reconstruction names external SGA, class-field, and representation-theoretic inputs where Deligne invokes them. Independent agents reconstruct 1.6–1.9 and 3.1–3.3; root reconstructs determinant weights, real-sheaf purity, the boundary nonvanishing mechanism, and the exact receiving maps. Agent reading does not replace root coverage.

Mathematical dependencies already located

1. The base is $\text{Spec}(F_q)$, and a geometric point is a map $\text{Spec}(\bar{k})$ to the scheme. Neither is an arithmetical zero. The zero sheaf in 1.2.2 has empty weight set. The parameter $t=0$ in a generating function has another role. These roles must receive individually typed comparisons with $\tau(Z_1; \text{no } Z_2)$.
2. A local geometric Frobenius F_x has degree $[k(x):F_q]$. The fibre representation, its determinant, and its trace are separate data. A fixed support does not remove its fibre action: 1.1.7, 1.1.10–14.
3. The geometric rank-one monodromy is finite (1.3.1–4); the remaining character is c^{degree} . This gives determinant weights, not yet pointwise eigenvalue purity.
4. The trace identity retains H_c^0 , H_c^1 , H_c^2 . Even tensor powers give nonnegative local power-series coefficients; determinant weights control possible poles; exterior powers and determinant sums force equality for individual eigenvalue weights (1.5.1–3).
5. The nilpotent logarithm is a map $V(1) \rightarrow V$. Its monodromy filtration and Tate twists must be retained; tensor powers and the dual give exact primitive weights (1.6–1.8).
6. The strict starting inequality for H_c^1 uses 2.1–2.2: positive logarithmic-derivative measures, compact-group representations, a boundary-zero classification, and an actual double cover to exclude the exceptional quadratic character. The existence of a double cover alone does not supply those analytic and cohomological identities.

Witness discrepancies and displayed-formula checks

These were initially checks of the available local witnesses; root has not independently compared the original scans. A later peer original-scan comparison is recorded immediately below. Keep displayed source formulas and separately derived corrections distinct.

Update, 24 September 2026, receipt10:26UTC: task01a0baa4 completed original published-page reading 137–252, using direct PDF extraction and separately listed visual formula checks. Root read its `READING_LEDGER.json` and complete-reading notes in `work/quantum_tau_programme_bridge_20260924/argument_reconstruction/original_deligne_reading/`. This is peer reading, not root scan coverage or certification of external references. The original PDF SHA256 is `b06eea61bf9cb2b596c162f5befc85d1be69828910a6107c8aa3a99c4afcc71`.

The peer directly confirms the source itself prints the §3.5.6 density and elliptic trace display on 212, the finite-subgroup wording in 1.3.10(iv) on 160, the terminal Jordan-chain index in 1.6.7 on 166, and the intermediary inequality in 1.6.13 on 169. Its visual checks also place the 1.6.14.3 twist sign and 1.7.5 filtration/conjugation displays on original 170–171. These must not be attributed solely to transcription. Its original 158 check confirms the full multiplied degree factor in 1.3.6. The ledger retains unresolved original-formula checks at 2.1.1, 2.2.8, 3.2.7.1 and 6.2.11; do not declare those individually certified. Original 217 and 221 explicitly suppress Tate twists after a chosen coefficient isomorphism over an algebraically closed base; the arithmetic comparisons here retain the equivariant twists. Source displays and independently derived corrections remain separate.

-1.8.1 denominator: current French H_c^2 , historical English H^0 . Use H_c^2 from the full trace identity. -1.3.10(iv): current French says image is a finite subgroup of Z , but the proof constructs a central element of nonzero degree and 1.3.11 states finite cokernel. The mathematical condition is finite index. -1.3.6: a twist defined over F_p acts at a closed point by $b^{\{[k(x):F_p]\}=b} \{[F_q:F_p] \deg(x)\}$. Any division by $[F_q:F_p]$ in that exponent conflicts with 1.2.7; retain and derive the full factor. -1.6.7: the displayed Jordan-chain

exception must be $i \neq -d$, with $N_{e_{-d}} = 0$. -1.6.13: the needed inequality is $k - 2i - 2 \geq 2j - k$ for $k > i$ and $j \leq 0$; the printed intermediary $k - 2i - 2 \geq k$ cannot hold. -1.6.14.3: from $N^r: \text{Gr}_i(V)(r) \rightarrow P_{-j}$, $r = (i+j)/2$, the inverse description is $P_{-j}(-r)$. The available witness misses that minus; the other Tate tensor/dual formulas and the claimed isobaricity confirm it. -1.7.5: weight filtration M_i is the sum of generalized eigenspaces of weights $\leq i$. A product over $j < i$ would give the wrong graded index. Geometric Frobenius gives $\text{FNF}^{\{-1\}=q}\{-1\}N$, hence the conjugation coefficient for F^n is $1 - q^{-n}$; all factors are rederived. -2.1.1: $\omega_s = q^{-s \text{ degree}}$ has period $2\pi i / \log(q)$, not $2\pi i \log(q)$. -2.2.8(i): ω_s has weight $-2 \text{Re}(s)$. Under 2.1.1's definition of $\text{Re}(\tau)$, the pointwise weight is $-2 \text{Re}(\tau)$, not the opposite sign in the witness.

Every independent correction will be proved in the reconstruction; source files remain unchanged.

Further issues found in full reading and independent reconstructions:

-2.1.7: the lower bound for a character integral includes a possibly negative contribution outside the concentration neighbourhood. The exact factor is $(1 - e_1)(1 - e_2) - e_2$, not simply $1 - e_1 - e_2$. Taking $e_1 = e_2 = e/4$ proves the required result with that term retained. -3.2.1: the Poincaré-dual coefficient sheaf is $\text{Fv}(1)$; a displayed $\text{Fv}(-1)$ gives the opposite Tate factor. -3.2.7.1: the $\text{Rf}!$ fibre is compactly supported Hc1 . The real companion used in applying 1.5 cannot be omitted when the original tensor-product sheaf is not itself real. -3.2.12: a surviving β in the transcription's weight-zero specialization is tracked by deriving the entire proof at original weight β . -Historical English 3.2 changes several non-strict inequalities into strict ones and drops a term from the five-term sequence. Historical 3.3.9 also changes $\det(\text{tI} - F)$ to $\det(\text{I} - \text{Ft})$, reversing the moduli of its roots. The current French and the exact derivations govern. -3.5.6 in the current French displays $(1/(2\pi)) \sin^2 \theta \, d\theta$ on $[0, \pi]$, whose total mass is $1/4$; the probability density is $(2/\pi) \sin^2 \theta \, d\theta$. With eigenvalues $q^{(n/2)e}(\pm i\theta)$, the trace formula gives $1 - 2\cos \theta \, q^{(n/2)+q}n$. These application-level discrepancies are retained separately from the purity proof.

DP0-DP9 now supplies the full determinant/tensor/exterior-power proof, including the rank-one argument, the degree factor $[\text{Fq}:\text{Fp}]$, the complete trace denominator, and the coefficient-domination argument. DLM, DW and DB supply the other reconstruction sections; their independent checks and root reading are recorded separately as they occur.

Final invariant-cycle reconstruction and exact scope

Root reread the complete source §3.6, printed pp212–215, and read the full independent derivation DC0-DC12 in `DELIGNE_INVARIANT_CYCLE_QUOTIENT.md`. Independent mathematical review checked the inertia Frobenius sign, support-duality shift, both exact sequences, the complete specialization kernel, and the actual recovered-base maps. The degree-zero sentence was corrected: only negative cohomological degrees automatically vanish.

DC proves the rational ℓ -adic local invariant-cycle theorem in the source's henselian curve setting. After its stated arithmetic reduction, $K = H^{(i-1)}(X_{\bar{\eta}})_I(-1)$ and the support term O have weights at least $i+1$, while $C = H^{i(X_{\bar{\eta}})}_I$ has weights at most i . Hence W_{iB} maps isomorphically to C . The obstruction image $\text{im}(\partial)/\partial K$ vanishes, while the whole quotient $O/\partial K$ can remain and equals the next specialization kernel. The unique Frobenius-equivariant lift is into B , not a unique representative in special-fibre cohomology. This splitting is asserted for the fixed arithmetic model. The complex analytic mixed-Hodge analogue in §3.6.4 was read as a cited remark, not independently reconstructed.